Probability From the Ground Up

Chance, Intuition, and the Mathematics of Uncertainty

From Coin Flips to Measure Theory

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Preface

There is a particular moment, familiar to anyone who has really learned a piece of mathematics, when a formula stops being a string of symbols and becomes a *picture*. Before that moment you can plug numbers in and get the right answer; after it, you can close your eyes and *see* why the answer had to come out that way. This book is written entirely in service of that second kind of knowing.

Probability is unusually rich in such moments, and unusually generous about handing them to you, because almost every result in the subject is secretly a statement about something you can imagine doing: flipping a coin, drawing a card, waiting for a bus, letting a rumor spread through a town. The mathematics is a careful bookkeeping of *what tends to happen when chance is in charge*. Our whole approach is to keep that concrete activity in view at all times – to let you watch the dice before we ever write down a distribution.

What this book is for

Most probability courses, and many fine textbooks, are organized around results to be proved and formulas to be applied. You meet the binomial coefficient, then the binomial distribution, then its mean and variance, and you are asked to trust that this is all going somewhere. Often it is – but the trust is doing a lot of the work, and trust is exactly what evaporates under exam pressure or three years later when you actually need the idea.

We have written this book on a different bet: that if you genuinely *understand* where a formula comes from, you will rarely need to memorize it, and you will be able to rebuild it from scratch when you have forgotten the details (you will forget the details; everyone does). Understanding is more robust than memory. It also transfers. The person who truly sees why the law of large numbers works is the same person who, years later, instantly smells something wrong in a viral statistic, a clinical-trial summary, or a backtested trading strategy.

Focus

The single organizing principle of this book: **intuition first, then history or mechanism, then a real proof.** We never lead with the theorem. We lead with the situation that makes the theorem feel inevitable, and only then do we make it precise. If at any point a derivation feels like it appeared out of nowhere, that is a bug in our writing, not a gap in your ability.

This is not a "probability without tears" book that hides the mathematics. The proofs are here, and they are real. But a proof can be rigorous without being dry, and it can be complete without being a wall of symbols. Where a standard text writes "it follows that," we try to write the sentence that tells you *why* it follows – the one line a good teacher says out loud at the board but that somehow

never makes it into print.

How to read this book

Read it with a pencil, and read it slowly. Probability punishes skimming more than almost any other branch of mathematics, because so many of its errors are *plausible*. A wrong answer in calculus usually looks wrong; a wrong answer in probability often looks completely reasonable until you simulate it and watch it fail. (This is why the subject is stuffed with paradoxes, and why we devote so much space to them.)

A few concrete suggestions:

- **When you meet a new claim, guess first.** Before reading our explanation of, say, the birthday problem, write down your gut estimate. Being wrong on purpose, and then seeing exactly where your intuition misfired, is the fastest way to upgrade that intuition permanently.
- **Treat every box as a different "voice" in a conversation.** The colored boxes (described below) are not decoration. Each one answers a specific question you might be asking at that moment.
- **Do not skip the figures.** They are the argument, not an illustration of it. We have drawn each one to make a single idea visible.
- **Simulate.** More on this shortly, but: the surest cure for a shaky probabilistic intuition is twenty lines of code and ten thousand trials.
- **Do the exercises.** Every chapter ends with a set spanning computation, concept, and at least one proof. The conceptual ones are where the real learning hides.

Reading note

You do not have to read linearly, but the early chapters genuinely build on one another. Chapters 1–9 are the load-bearing foundation (sample spaces, conditioning, random variables, expectation, variance). After that the book branches: continuous distributions and limit theorems (Chapters 10–22), processes that unfold in time (Chapters 23–29), inference and modern applications (Chapters 30–35), and a rigorous measure-theoretic capstone (Chapters 36–38). Two appendices collect the calculus, sums, and linear algebra you may want to refresh.

The visual language: a guide to the boxes

Throughout the book, certain kinds of remarks are pulled out into colored boxes. The point is that *different questions deserve different answers*, and once you learn the color code you can read at two speeds – following the main thread, and dipping into exactly the kind of aside you happen to want. Here is the whole vocabulary, with a tiny example of each.

Intuition

Intuition (green). The mental picture before the mathematics. Example: "Variance measures how far a random quantity typically lands from its average – it is the spread of the cloud, not its center."

Why it works

Why it works (blue). The engine room of a proof or formula – the load- bearing reason, stated as plainly as we can manage. Example: "Expectation is linear because summing and averaging commute; no independence is needed, which is why $\mathbb{E}[X+Y] = \mathbb{E}\,X + \mathbb{E}\,Y$ always holds."

What is really happening

What is really happening (deep green). A step back to name the phenomenon underneath the symbols. Example: "The central limit theorem is not about coins or dice in particular; it is about *averaging washing out detail*, so that almost any source of small independent wiggles produces the same bell-shaped fog."

History

History (yellow). Where an idea came from, and the human problem that forced it into existence. Example: probability was born not in a seminar but at a gambling table, when the Chevalier de Méré asked Pascal why a betting rule that had always worked for him suddenly did not.

Paradox

Paradox (red). A result that contradicts the answer almost everyone gives – stated sharply, then resolved. Example: switching doors on a game show doubles your chance of winning, even though, after a door is opened, it *feels* like a coin flip. We will see exactly why the feeling is wrong.

Application

Application (magenta). The idea at work in the world – machine learning, A/B testing, finance, genetics, signal processing. Example: the same Bayes' rule that updates a medical diagnosis is the spam filter quietly sorting your inbox.

Connection

Connection (teal). A bridge to the rest of mathematics, often pointing to the companion *Calculus From the Ground Up* and *Linear Algebra From the Ground Up*. Example: a probability density is just an integrand, and expectation is an integral against it – so everything you know about integration is suddenly about averages.

Common trap

Common trap (red). A specific, tempting mistake – placed exactly where you are most likely to make it. Example: confusing $\mathbb{P}(A \mid B)$ with $\mathbb{P}(B \mid A)$, the error at the heart of countless courtroom and medical blunders.

Walking question

Walking question (orange). A question to chew on while you walk away from the page; we usually answer it shortly after. Example: "If you keep flipping a fair coin, must the number of heads and tails eventually equal out?" (The answer is more surprising than it sounds.)

Work it yourself

Work it yourself (purple). A short, hands-on task – often a simulation – where you produce the insight rather than read it.

Why this matters

Why this matters (cyan). The longer-range payoff, so that no result feels like an isolated trophy.

There are a few more – Focus, Formal, Reading note, and Pause – whose names say what they do. You have already met Focus and Reading note above. Do not worry about memorizing the palette; after a chapter or two the colors will mean something to you without effort, the same way road signs do.

The arc: from a coin flip to measure theory

Seeing the whole journey before taking the first step is worth the detour, because the destination explains the route.

We begin (Chapter 1) with the most elementary object imaginable: a single fair coin. From it we extract the entire conceptual skeleton of the subject – outcomes, events, the idea that probability is a kind of *mass* spread over possibilities. Counting (Chapter 2) lets us handle situations with many equally likely outcomes, and counting alone takes us surprisingly far. Then comes the real engine of probabilistic reasoning: *conditioning* (Chapters 4–5). Learning to update a probability in the light of new information is, more than any formula, the skill that separates fluent probabilists from the rest.

With those foundations we introduce *random variables* (Chapter 6) – numbers attached to chance outcomes – and the two summaries that compress a whole distribution into a few telling numbers: *expectation* (Chapter 8) and *variance* (Chapter 9). From there probability goes continuous (Chapters 10–12), where it becomes area under a curve and the bell curve makes its entrance.

The middle of the book is about *how randomness adds up*. When you combine many random pieces – summing them, averaging them, letting time run – striking regularities emerge from the chaos. This is the territory of the law of large numbers and the central limit theorem (Chapters 19–22), arguably the two most important facts in all of applied mathematics. The figure below is a small preview: as

we average more and more coin flips, the running average is pulled, as if by gravity, toward the true probability 1/2.

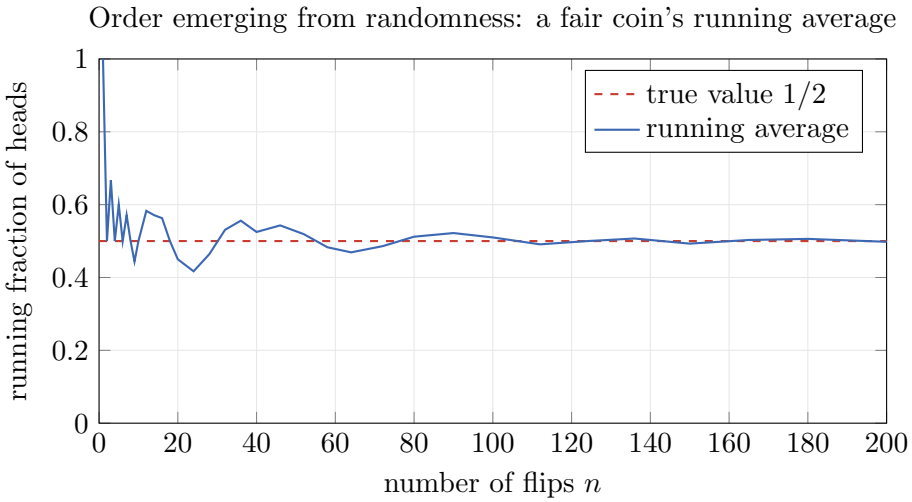


Figure 1: A single simulated run of 200 fair coin flips. Early on, the running fraction of heads swings wildly; as n grows it is reeled toward 1/2. The law of large numbers (Chapter 20) explains the pull, and the central limit theorem (Chapter 21) explains exactly how large the remaining wiggles tend to be.

After the limit theorems we let time enter the picture and study *processes*: Markov chains, the Poisson process, random walks, martingales, and Brownian motion (Chapters 23–29). Here probability stops being a snapshot and becomes a movie. Then we turn the whole apparatus around and use it for *inference* (Chapters 30–35): given data, what can we legitimately conclude about the world that produced it? This is where probability meets statistics, machine learning, and data science.

Finally, in the capstone chapters (36–38), we return to the very beginning – "what is a probability, really?" – and answer it properly, with measure theory. This is the rigorous foundation that makes everything earlier unshakable. We have deliberately placed it last. You will appreciate why Kolmogorov's axioms are shaped the way they are only *after* you have felt, across thirty-five chapters, the places where naive intuition needs a guardrail.

Why this matters

Notice the shape of the journey: it is a spiral, not a line. We begin with the coin flip, travel out through the entire applied subject, and return to the coin flip to ask what it really was all along. Each pass over a familiar idea is deeper than the last. If a concept feels only half-understood the first time you meet it, that is by design – you will meet it again, equipped to see more.

History

That spiral mirrors the history of the field. Probability spent its first two centuries (Cardano, Pascal, Fermat, the Bernoullis, Laplace) as a brilliant but informal art of calculation, applied to games, annuities, and errors of astronomical measurement. Only in 1933 did Andrey Kol-

mogorov give it the measure-theoretic foundation that placed it among the rigorous branches of mathematics. We follow the same path the subject did: intuition and calculation first, foundations once we know what they must support. There is no shame in this order; it is how humanity itself learned probability.

Simulate, then trust, then verify

If we could press one habit on you, it would be this: **when in doubt, simulate.** A computer that flips ten million coins in a second is a remarkably reliable probability tutor. It has no intuition to fool and no formula to misremember; it just runs the experiment and reports what happened.

Work it yourself

Here, in a few lines of pseudocode, is the entire Figure 1 experiment: flip a fair coin and report the running fraction of heads after each flip. We will write code in this spirit throughout the book.

```
import random
heads = 0
for n in range(1, 201):
    heads += (random.random() < 0.5)   # one fair flip
    print(n, heads / n)                # running fraction of heads
```

Run it. Watch the printed fraction swing wildly at first, then creep toward 0.5 and refuse to leave its neighborhood. (Push the loop to two hundred thousand flips and it will hug 0.5 even more tightly.) You have just witnessed the law of large numbers with your own eyes, before we have proved a single thing about it.

Simulation plays three distinct roles in this book, and keeping them separate in your mind pays off.

1. **Discovery.** When you do not yet have a guess, simulate to form one. The numbers will suggest a pattern; your job is then to explain it.

2. **Verification.** When you have derived a formula, simulate to check it. If the theory and the simulation disagree, one of them is wrong, and finding out which is enormously instructive.

3. **Computation.** For problems too hard to solve exactly – and modern applications are full of them – simulation *is* the answer. Chapter 35 (Monte Carlo methods) rests entirely on this idea, and it is one of the quiet workhorses of modern science.

But notice the order in our section title: simulate, *then trust*, then verify. There is a temptation, once you have a fast simulator, to stop thinking and just measure everything. Resist it. A simulation can tell you *that* something is true to high confidence; only a proof tells you *why*, and only the why generalizes. A measured probability of 0.4999 across ten million flips is persuasive, but it is the *argument* that the true value is exactly $1/2$ – by symmetry, with no appeal to any experiment at all – that you can carry into every other symmetric situation you ever meet.

What is really happening

Simulation and proof are not rivals; they are two instruments measuring the same object. The simulation is your laboratory: it shows you what nature does. The proof is your theory: it explains why nature had no choice. A mature probabilist moves fluidly between them – guessing with the laboratory, securing with the theory, and trusting neither blindly. Cultivating that two-handed fluency is, more than any list of distributions, what this book is trying to teach.

Paradox

A warning about "trust," delivered up front so you remember it. Your raw probabilistic intuition is, frankly, unreliable – not because you are bad at it, but because the human brain was simply not built for it. Consider: in a room of just 23 people, the chance that two share a birthday is already better than even. Almost everyone guesses the number must be in the hundreds. The intuition that misleads you here is the same one used by every confident person who has ever been badly wrong about risk. Our remedy is not "trust your gut" – it is "train your gut, then verify it." By the end of this book your intuition will be a far better instrument, precisely because you will have spent so much time catching it in the act of being wrong.

Because that birthday claim is the single best advertisement for the whole enterprise, let us actually do the arithmetic now – not because you need to follow every step on a first reading, but so you can see the shape of the kind of reasoning this book will teach you to perform without flinching.

Example 0.1 (The birthday problem, worked once, slowly)**.** With n people in a room and 365 equally likely birthdays, it is far easier to compute the chance that *nobody* shares a birthday and subtract from one. Line the people up. The first person's birthday is free. The second must dodge one taken day, so it is "safe" with probability 364/365. The third must dodge two, giving 363/365, and so on. Multiplying these step-by-step chances together (each conditioned on the people before it landing on distinct days),

$$\mathbb{P}(\text{all distinct}) = \frac{365}{365} \cdot \frac{364}{365} \cdots \frac{365-n+1}{365} = \prod_{k=0}^{n-1} \Big(1 - \frac{k}{365}\Big).$$

The probability of *at least one match* is then 1 minus this product. Plugging in $n = 23$ gives 0.5073, just over a coin flip. The reason it feels impossible is that we instinctively count the 23 people, when what matters is the number of *pairs* of people, $\binom{23}{2} = 253$ – and 253 chances to collide is a great many. Figure 2 shows the whole curve.

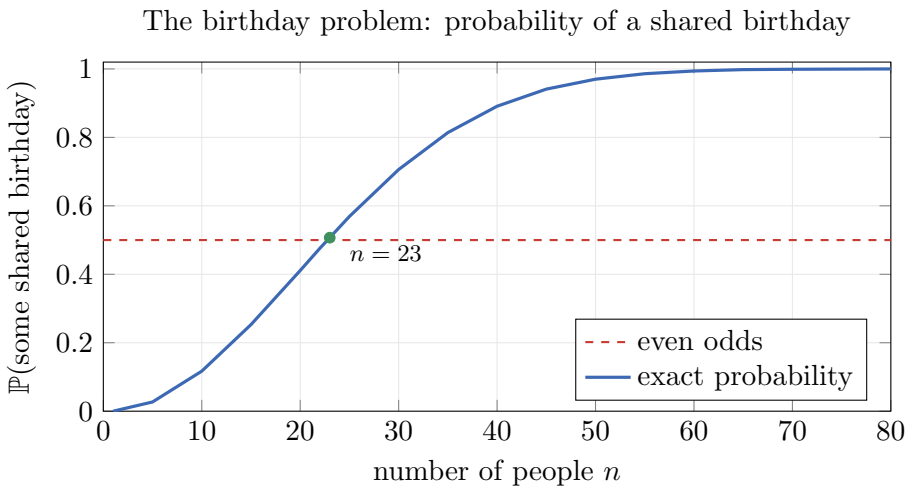


Figure 2: The probability that at least two of n people share a birthday, computed exactly from the formula in the worked example. The curve crosses the even-odds line (dashed) at the famous $n = 23$, marked in green, and is essentially certain by $n = 60$. The steepness early on is the surprise: the count of *pairs* grows like n^2, so the matches pile up far faster than the linear headcount suggests.

Why it works

Notice what just happened, because it is the rhythm of the entire book. We met a claim that offends intuition; we did not argue about it, we *computed* it; and the computation then handed back a *new and better* intuition ("count the pairs, not the people"). Intuition, calculation, and a sharper intuition – that loop, run thousands of times, is what turns a beginner into someone fluent.

A note on rigor

We promise two things about the mathematics. We will not lie to you for simplicity's sake: when a result holds only under conditions, we say so, and when a "proof" is really a plausibility argument, we flag it clearly and point to where the genuine proof lives (often in Chapters 36–38). Nor will we hide behind rigor. A definition that is technically precise but motivationally empty has failed at its real job, which is to capture an idea you already half-see.

Connection

This book leans on calculus (for densities, expectations as integrals, and generating functions) and on linear algebra (for covariance matrices, Markov chains, and Gaussian vectors). We have written it to be self-contained – the appendices gather what you need – but where a deeper treatment helps, we point to the companion *Calculus From the Ground Up* and *Linear Algebra From the Ground Up*, which share this book's philosophy and its box-colored visual language. The three are meant to be read as a family.

Acknowledgments and an invitation

Every probability book stands on the shoulders of a few great ones, and the debts of this one are easy to name. From Blitzstein and Hwang we learned that a rigorous probability text can also be genuinely warm and story-driven. From Ross we inherited the breadth of worked examples and applications. From Feller, the conviction that the deep results deserve deep treatment and that the subject's beauty is part of its content. From Wasserman, the insistence that all of this must ultimately help us reason from data. Where we have succeeded, much credit is theirs; where we have stumbled, the fault is ours alone.

Most of all, this book is an invitation. Probability is the mathematics of the uncertain world – which is to say, of the only world there is. Learning it well does not make the future predictable, but it does something better: it makes the future *intelligible*. You will come to see chance not as the absence of order but as a different, subtler kind of order, with its own laws as firm as gravity. Few intellectual pleasures compare to the moment when a tangle of uncertainty suddenly resolves into a clean, inevitable answer.

One more preview, because it is the result the whole middle of the book builds toward. We claimed that averaging "washes out detail." Here is what that means made visible. Take the humble outcome of summing a handful of dice. A single die is flat – every face equally likely. Sum two and the distribution becomes a tent, peaked at 7. Sum five and it is already a recognizable bell. Nothing about the die changed; the *shape* emerged purely from adding independent copies together. That convergence, dramatized in Figure 3, is the central limit theorem in miniature.

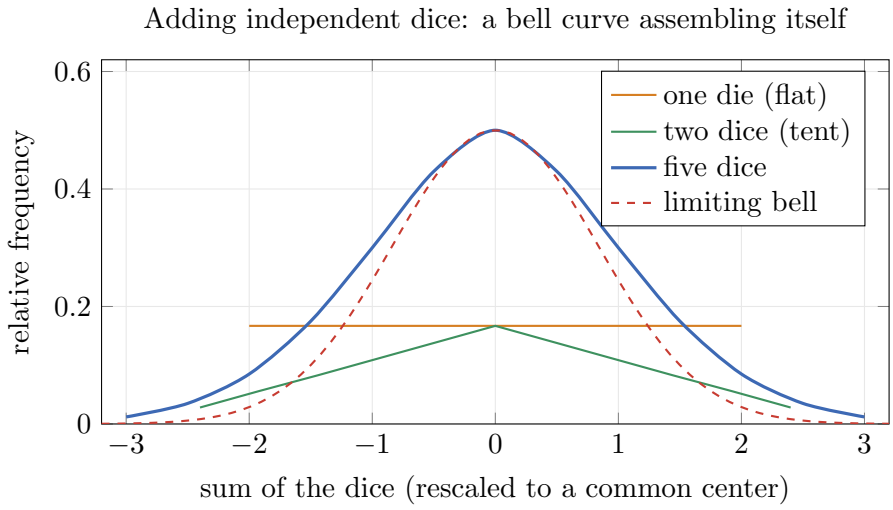


Figure 3: The distribution of a sum of fair dice, each curve rescaled to share a center and width. One die is flat; two dice form a triangular tent peaked in the middle; five dice already hug the smooth limiting bell (dashed). The astonishing fact, proved in Chapter 21, is that the same bell appears no matter what we are averaging – dice, heights, measurement errors, monthly returns – as long as the pieces are independent and none dominates.

Why this matters

If a single picture could justify learning probability, it might be this one. The bell curve is not a fact about dice; it is a fact about *addition under uncertainty*. The reason it governs measurement error in physics, sampling error in polling, and diffusion in finance alike is that all three are sums of many small independent pushes. Learn why the bell appears once, and you have understood it everywhere at once.

A few warm-up questions

You cannot do the real exercises yet – we have not taught you anything. But the spirit of this book is that you should always be *guessing*, so here are five questions to carry into Chapter 1. Do not look anything up. Write a number or a sentence for each, seal it in an envelope (metaphorically), and check it against your answer when the relevant chapter arrives. The gap between your guess now and the truth later is precisely the intuition this book exists to install.

Exercise 0.1 (Guess before you learn)**.** For each, commit to an answer *before* reading on in the book.

1. A fair coin is flipped 10 times. Which exact sequence is more likely: `HHHHHHHHHH` or `HTHHTHTTHH`? (Careful.)
2. In the same 10 flips, are you more likely to get exactly 5 heads, or some number of heads that is *not* 5?
3. A test for a rare disease is "99% accurate." You test positive. Off the top of your head, what is the chance you actually have the disease: closer to 99%, or closer to 10%? (This one ruins careers.)
4. You wait at a bus stop where buses arrive "on average every 10 minutes," at random times. You show up at a random moment. Do you expect to wait about 5 minutes, or longer?
5. Roll a fair die repeatedly until you see a 6. On average, how many rolls will that take?

Walking question

Take just question 1 on your next walk. Most people feel the second sequence is "more likely" because it looks random. But we asked for one *specific* sequence, and every specific sequence of 10 flips has the identical probability $(1/2)^{10}$. The illusion comes from quietly comparing "ten heads" against "the huge *family* of jumbled-looking sequences." Holding that distinction straight – a single outcome versus a class of outcomes – is half of all of probability. Question 2 is its mirror image, and your answers should disagree.

Turn the page. The first thing we are going to do is flip a coin – and we are going to take it absolutely seriously.

Focus

The compact version of everything above. Build intuition first. Distrust your untrained gut, especially when it feels most certain. Simulate to discover and to check, but prove to

understand and to generalize. Read the boxes, draw the pictures, do the exercises. And remember that the whole enormous edifice – from the central limit theorem to Brownian motion to Bayesian machine learning – is built, brick by careful brick, on top of a single fair coin flip.

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Part I

Foundations of Chance

Chapter 1

What Probability Really Is

There is a coin in your hand. You flip it. While it spins in the air, glinting, something strange is true: nobody on Earth knows which way it will land. Not you, not a physicist with a supercomputer, not a prophet. And yet you would happily bet, at even odds, that it comes up heads – and you would be a fool to bet at ten-to-one against. Somehow, in the middle of total ignorance about *this* flip, you have a confident, quantitative opinion. You can put a *number* on your uncertainty.

That number is a probability. This whole book is about where such numbers come from, what they mean, why they obey the rules they do, and how astonishingly far they reach – from card tables in Renaissance Italy to the spam filter in your inbox, from insurance premiums to quantum mechanics to the confidence interval in a clinical trial. But before we compute a single thing, we have to answer a deceptively hard question: what *is* a probability, really?

This chapter is the foundation, so we will go slowly and take the idea apart to see what makes it tick.

1.1 Two Good Answers to "What Does 1/2 Mean?"

Ask ten people what "the probability of heads is 1/2" means and you will get, mostly, two genuinely different answers. Both are correct. They are correct about *different things*, and confusing them is the source of an enormous amount of muddle. Both deserve a hearing.

1.1.1 The frequentist answer: probability is long-run frequency

The first answer points outward, at the world. "Probability 1/2" means: if you flipped this coin a great many times, about half of the flips would be heads. The probability is the *limiting relative frequency* of the outcome in a long run of repetitions.

This is the interpretation most people absorb in school, and it has a bracing, no-nonsense appeal. It says probability is not in your head; it is a measurable property of a repeatable experiment, like the boiling point of water. You do not have to argue about it – you go flip the coin ten thousand times and count.

Intuition

Imagine a long ledger. Each row is one flip: H or T. Walk down the ledger and, at every row, write down the fraction of H's *so far*. Early on that fraction lurches around – one flip in, it is either 0 or 1; ten flips in it might be 0.3 or 0.7. But keep walking. The fraction settles. It homes in on a number and stops wandering far from it. The frequentist says: *that number* is the probability. It is the address the running fraction is heading toward.

We can watch this happen. Figure 1.1 shows the running fraction of heads for two simulated coins, one fair ($p = 0.5$) and one biased ($p = 0.7$), as the number of flips grows. The early chaos is real; so is the eventual calm.

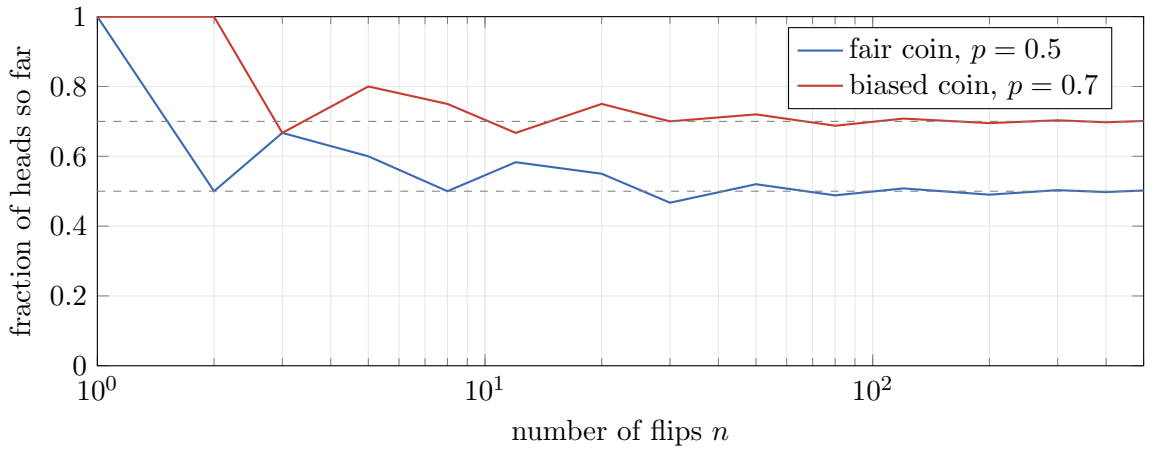


Figure 1.1: Running fraction of heads versus number of flips (horizontal axis on a log scale, so the early flips are not crushed against the left edge). Both curves are violent at first and then settle toward their true probabilities (dashed lines). The frequentist reads probability off the destination, not the journey.

Common trap

The frequentist definition has a famous soft spot, and you should see it now rather than be ambushed by it later. The phrase "in the long run" hides an infinity. You can never actually flip a coin infinitely many times, so the limiting frequency is never literally observed – it is an idealization. Worse, it is circular if you are not careful: to say the fraction "converges" is itself a statement that only holds *with probability one*, so we have used the word probability to define probability. We will untangle this in the next chapter, when we lay down the axioms, and again when we meet the Law of Large Numbers later in the book. For now, hold the frequentist picture as an *interpretation*, a way of attaching meaning to the math – not as the math itself.

The frequentist view is at its most natural when the experiment really is repeatable under stable conditions: dice, cards, manufactured parts coming off a line, radioactive atoms, the click of a Geiger counter. In all these cases "do it again" means something concrete.

1.1.2 The Bayesian answer: probability is degree of belief

Now consider a different kind of question. *What is the probability that it rained in Rome on the day Julius Caesar was assassinated?* Or: *what is the probability that this particular patient has the disease, given this particular test result?* Or: *what is the probability that the missing airliner is in this search box?*

There is no "long run" here. Caesar was assassinated exactly once. This patient is one person. We cannot re-run history under stable conditions and count. The frequentist definition, taken literally, simply has nothing to say – the event either happened or it did not; the frequency is 0 or 1 and we just do not know which.

And yet these are exactly the situations where we most desperately need probability. So there is a second answer, older in spirit and in some ways deeper. *Probability is a measure of how strongly a rational agent believes something, given the information they have.* It lives in the mind, not in the coin. "Probability 0.7" means: I am willing to bet on this at odds of 7-to-3; I would be indifferent between winning a prize if the event happens and winning that same prize if I draw a green ball from an urn that is 70% green.

Intuition

A degree of belief is not a feeling, not a vague hunch. It is pinned down by *action*. Suppose I offer you a ticket that pays \$1 if it rained in Rome on the Ides of March, 44 BC, and \$0 otherwise. How much would you pay for that ticket? If the most you would pay is 30 cents, then – on pain of being a sure loser to a clever bookmaker – your degree of belief in "it rained" is 0.3. Probability, on this view, is the price of a bet, made coherent.

The Bayesian (named for Thomas Bayes, an 18th-century minister whose posthumous essay we will study in depth) takes degree of belief as primary and then insists on *coherence*: your various beliefs must fit together without contradiction. The astonishing fact – which we will prove later in the book via so-called "Dutch book" arguments – is that the only way to keep your betting prices from being exploitable is to make them obey the exact same rules of probability that the frequentist derives from counting. Two completely different starting points, ignorance-priced-as-bets and frequency-in-the-limit, are forced to the same algebra – a coincidence too strong to be a coincidence, and we will not take it for granted.

History

The word "probable" long predates the mathematics. In medieval and early modern usage, an opinion was *probabilis* if it was approvable, worthy of belief – backed by authority or argument. (Echoes survive in "probity" and "approve.") The shift from probability-as-credibility to probability-as-number was gradual, and the two senses, belief and frequency, were tangled together for centuries before anyone realized they could come apart. When you feel the pull of both interpretations at once, you are feeling a genuine three-hundred-year-old ambiguity, not a personal confusion.

1.1.3 When each is natural – and why the fight is overblown

Generations of statisticians waged something close to a holy war over which interpretation is "correct." You can largely sidestep it. Here is the practical truth.

- Use the **frequentist** picture when the experiment is genuinely repeatable and you care about long-run behavior: quality control, casino odds, estimating an error rate, the performance of a procedure over many future cases.
- Use the **Bayesian** picture when you face a *one-off* uncertainty and want to combine evidence with prior knowledge: medical diagnosis for this patient, the location of one lost ship, whether one hypothesis is true given one dataset.

Why this matters

Here is the reconciling secret, and it is worth tattooing on the inside of your eyelids: *the mathematics is the same either way.* Both camps agree that probabilities are numbers between 0 and 1, that the probability of "A or B" for incompatible events is the sum, that conditional probability multiplies along a tree. The interpretations differ only in what those numbers *mean* and hence in what questions they let you ask. So we can build the entire machine – the axioms, the random variables, the distributions, the limit theorems – on neutral ground, and then plug in whichever interpretation fits the problem in front of you. That is exactly what we will do. The philosophy is real and we will honor it, but it never blocks the calculation.

1.2 Why Uncertainty Needs Mathematics

You might reasonably ask: if probability is about not knowing, why do we need *mathematics*? Mathematics is the science of certainty – of proof, of $2 + 2 = 4$ forever. What business does it have with the unknown?

The answer is that uncertainty, left to intuition alone, makes us stupid in predictable, systematic ways. Our gut is a magnificent instrument for many things and a disaster for chance. We see faces in clouds and patterns in lottery numbers. We think a basketball player who hit five shots is "hot" when the streak is well within randomness. We are terrified of shark attacks and blase about car rides. We will spend a whole chapter on these cognitive traps, but here is the headline: *quantitative uncertainty does not yield to common sense; it yields to calculation.*

Intuition

Consider a quick gut-check. A test for a rare disease is "99% accurate." You test positive. What is the probability you have the disease? Most people, including many doctors, blurt out "about 99%." The real answer, when the disease is rare, can be under 10%. Your intuition is not slightly off here; it is off by a factor of ten. Only arithmetic – which we will do carefully when we reach Bayes' theorem – can rescue you. This single example has changed real medical practice.

Mathematics does three things for uncertainty that intuition cannot.

First, it forces consistency. If you believe A has probability 0.6, then you are *committed* to "not

A" having probability 0.4 – you cannot choose otherwise without contradicting yourself. The rules act like the rules of accounting: they do not tell you the numbers, but once you set some, they pin down others, and they catch you when your story does not add up.

Second, it lets us combine uncertainties. Real problems chain together – the probability of a system failing depends on its parts, the probability of a diagnosis depends on several tests. Intuition cannot reliably multiply and add a dozen partial beliefs; a calculus can.

Third, it reveals stable laws hiding inside the chaos. This is the most beautiful part, and the deepest reason the subject exists at all. Individual random events are unpredictable, but *aggregates of them are exquisitely predictable.* You cannot say whether the next customer buys, but a store can forecast next month's sales to within a percent. You cannot predict one molecule, but you can write the gas laws. Out of a swarm of unknowables, sharp regularities crystallize. Probability is the mathematics of that crystallization.

What is really happening

What is really happening when "randomness" produces stable averages? Nothing mystical. Each flip contributes a tiny, bounded nudge to the running fraction, and the nudges are not coordinated – they do not conspire to all push the same way. With many independent nudges, the ups and downs largely cancel, and what is left is the underlying tendency, exposed. The cancellation is the whole story. Later we will make it precise as the Law of Large Numbers (the average converges) and the Central Limit Theorem (the leftover wiggle has a universal bell shape). Keep this image: *randomness in the small, law in the large, with cancellation as the bridge between them.*

1.3 Randomness vs. Determinism

Here is a question that troubles careful people, and it should: is the coin *really* random, or does it just look that way?

Press on the physics. A coin is a rigid body obeying Newton's laws. If you knew its exact initial position, spin, velocity, the air currents, the elasticity of the table – in principle you could compute the outcome exactly. There is no genuine chance in there at all. The same is true of dice, of shuffled cards, of the roulette wheel. These are *deterministic* systems. So in what sense are they random?

Focus

The resolution is that probability is, very often, a confession of *ignorance*, not a claim about the universe's deepest nature. The coin's outcome is determined; it is determined by quantities we do not and practically cannot measure. The system is so sensitive to its starting conditions that a difference too small to see – a hair's width in launch angle – flips the result. This is the hallmark of *chaos*: deterministic dynamics that amplify tiny unknowns into total unpredictability. Probability steps in precisely where determinism becomes useless to us. The coin is not random; *we* are ignorant, in a sharp and quantifiable way, and probability is the mathematics of that ignorance.

This is liberating, because it means we do not have to settle the metaphysics of free will and quantum mechanics before we can do statistics. Whether or not the universe is "really" random at bottom

(quantum mechanics suggests some of it genuinely is), the *practical* situation – a system whose relevant details we cannot pin down – is identical, and the same mathematics applies.

Connection

There is a deep link to dynamical systems and chaos theory here, and through it to calculus and differential equations. A chaotic map like the "doubling map" $x \mapsto 2x \bmod 1$ takes a perfectly deterministic rule and, by stretching and folding the interval, produces sequences that pass every statistical test for randomness. Sensitive dependence on initial conditions – the "butterfly effect" – is exactly the engine that converts unmeasurable precision into effective chance. If you want to see how a single deterministic equation can manufacture apparent randomness, that is the doorway, and it connects this book to the calculus companion's treatment of dynamical systems.

Paradox

Is a fixed-but-unknown digit "random"? Consider the trillionth digit of π. It is a specific, fixed number – there is nothing chancy about it; π does not change. Yet if you have not looked it up, you would happily assign each digit 0 through 9 a probability of $1/10$, and you would be *right* to, in the only sense that matters for betting. This is the cleanest possible case where frequentist language fails (the digit is not the outcome of any repeatable experiment) and Bayesian language thrives (it is a perfect picture of degree of belief under ignorance). The "paradox" – assigning probabilities to a thing with no randomness in it – dissolves the moment you accept that probability can measure *your information*, not just the world's caprice. Many real problems, from cryptography to forecasting, are exactly of this type.

1.4 A First Pass at the Rules

Enough philosophy; let us start computing. We will keep it light and concrete here, and build the formal scaffolding (sample spaces, axioms, the algebra of events) properly in the next chapter. The goal now is to get the *feel* of how probabilities behave, using examples a child could check.

1.4.1 Outcomes, events, and the basic vocabulary

Roll a single fair die. The possible **outcomes** are $1, 2, 3, 4, 5, 6$. The complete list of outcomes is the **sample space**, written Ω (capital omega). An **event** is any collection of outcomes – a question you can ask that the experiment answers yes or no. "The roll is even" is the event $\{2, 4, 6\}$. "The roll is at least 5" is $\{5, 6\}$.

Definition 1.1 (Equally likely outcomes)**.** If a sample space Ω has finitely many outcomes and there is no reason to favor any one over another – by symmetry, as with a fair die or a well-shuffled deck – then we assign each outcome the same probability, and for any event A,

$$\mathbb{P}(A) \;=\; \frac{\text{number of outcomes in } A}{\text{number of outcomes in } \Omega} \;=\; \frac{|A|}{|\Omega|}.$$

Probability becomes counting: favorable cases over total cases.

This is the oldest definition in the book – the "classical" definition of Laplace – and it carries us a long way. The probability the die shows an even number is $|\{2,4,6\}|/|\{1,\dots,6\}| = 3/6 = 1/2$. The probability it is at least 5 is $2/6 = 1/3$.

Common trap

The phrase "no reason to favor any one" is doing heavy lifting, and it can betray you. It is tempting to say "there are two outcomes, rain or no rain, so each has probability 1/2." That is nonsense – the two are not symmetric. Equal likelihood must be *justified*, by a physical symmetry or by genuine ignorance that is itself symmetric, not assumed for free just because you can list two cases. We will meet a sharp version of this trap, Bertrand's paradox, later on.

1.4.2 Three rules you already believe

Even without axioms, you already accept three things about probabilities. Let us name them, because everything else is built from them.

Rule 1 (Bounds). A probability is a number between 0 and 1. An impossible event has probability 0; a certain event has probability 1. You cannot be "140% sure" and you cannot be "−20% sure." In symbols, $0 \le \mathbb{P}(A) \le 1$, with $\mathbb{P}(\Omega) = 1$.

Rule 2 (Addition for incompatible events). If two events cannot both happen – they are *mutually exclusive*, or *disjoint* – then the probability that one or the other happens is the sum of their probabilities. The die cannot be both a 2 and a 5, so

$$\mathbb{P}(\text{roll is 2 or 5}) = \mathbb{P}(2) + \mathbb{P}(5) = \tfrac{1}{6} + \tfrac{1}{6} = \tfrac{1}{3}.$$

Rule 3 (Complement). The probability that A does *not* happen is 1 minus the probability that it does: $\mathbb{P}(\text{not } A) = 1 - \mathbb{P}(A)$. This follows from Rules 1 and 2, since A and "not A" are incompatible and together cover everything. It is the single most useful trick in elementary probability, because "not" is often far easier to count.

Why it works

Why does the complement rule follow? The events A and "not A" are disjoint (an outcome cannot be both in A and not in A) and exhaustive (every outcome is in one of them). So by Rule 2 their probabilities add, and by Rule 1 that total is $\mathbb{P}(\Omega) = 1$:

$$\mathbb{P}(A) + \mathbb{P}(\text{not } A) = \mathbb{P}(\Omega) = 1 \quad \Longrightarrow \quad \mathbb{P}(\text{not } A) = 1 - \mathbb{P}(A).$$

Notice we proved a rule from simpler rules. That is the texture of the whole subject: a tiny set of obvious-sounding axioms, and a tower of consequences. In this section you saw the words; in the next chapter we make these the official axioms (due to Kolmogorov, 1933) and watch the tower rise.

Example 1.1 (The power of looking at the complement)**.** Roll a fair die four times. What is the probability of getting *at least one* six?

Counting "at least one" directly is a headache: exactly one six, exactly two, exactly three, exactly four – four cases to tally and add. But the complement of "at least one six" is "*no* sixes at all,"

which is one clean case. Each roll avoids six with probability 5/6, and (because the rolls do not influence each other) the chance all four avoid six is $(5/6)^4$. Therefore

$$\mathbb{P}(\text{at least one six}) = 1 - \left(\tfrac{5}{6}\right)^4 = 1 - \tfrac{625}{1296} = \tfrac{671}{1296} \approx 0.518.$$

Just over an even chance. We will not formalize the multiplication-for-independence step until the next chapters; here, savor that flipping to the complement turned four cases into one.

1.4.3 Why these rules and not others?

It would be fair to ask whether we could have written down *different* rules. Why addition for disjoint events? Why a ceiling of 1? The frequentist and Bayesian each answer in their own voice, and – as promised – they land in the same place.

Why it works

The frequentist's reason. Probabilities are limiting frequencies, and frequencies obviously obey these rules at every finite stage, so they must in the limit too. In n flips, the fraction of heads is between 0 and 1 – there is Rule 1. If A and B never co-occur, then the count of "A or B" is exactly the count of A plus the count of B, so the fractions add – there is Rule 2. The fraction of "not heads" is 1 minus the fraction of heads – there is Rule 3. The axioms are nothing more than the bookkeeping of counting, carried to the limit.
The Bayesian's reason. Probabilities are coherent betting prices. If you price bets in a way that violates these rules – say you are willing to pay 0.6 for a bet on A *and* 0.6 for a bet on "not A" – then a sharp operator can sell you both bets, collect 1.20, and pay out exactly 1.00 no matter what happens, pocketing 0.20 from you with certainty. The only prices immune to such a guaranteed loss (a "Dutch book") are exactly those satisfying Rules 1–3. Coherence *forces* the axioms.

That two such different stories – counting in the limit, and refusing to be swindled – arrive at the identical algebra is the first sign that we have found something real, not arbitrary. We are not inventing probability; we are discovering the only consistent way to reason under uncertainty.

1.4.4 A tiny worked tree

One more taste before the formalities. Probabilities chain together along sequences of events, and a *tree diagram* is the natural picture. Figure 1.2 shows two flips of a fair coin.

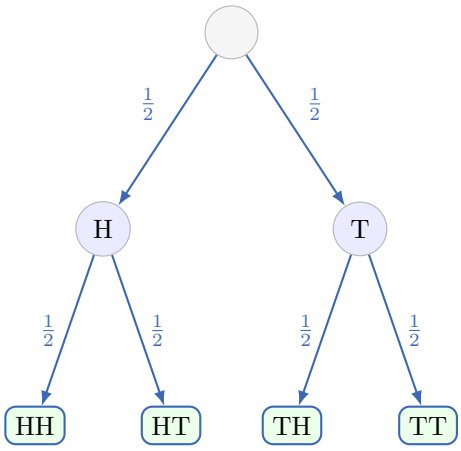


Figure 1.2: Two flips of a fair coin. Each path from root to leaf is one outcome; its probability is the product of the edge probabilities along the way, here $\frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$. The four leaves are equally likely and their probabilities sum to 1, as they must.

Reading the tree: multiply *along* a branch (the two flips happen in sequence), and add *across* branches (different ways to succeed are disjoint). What is the probability of exactly one head? Two leaves qualify, HT and TH, each with probability 1/4, and they are disjoint, so the answer is $1/4 + 1/4 = 1/2$. Multiply along, add across – you will use that rhythm for the rest of your life.

1.5 Where It All Came From: Games of Chance

Mathematics usually grows from one of two roots: a practical problem that will not go away, or a puzzle too pretty to ignore. Probability grew from both at once, and the soil was the gambling table.

History

Gerolamo Cardano (1501–1576): the gambler who counted. Cardano was a physician, astrologer, inveterate gambler, and one of the most colorful figures in the history of mathematics – he solved cubic equations, cast horoscopes (including, notoriously, one for Christ), and by his own account gambled nearly every day for decades. Around 1564 he wrote *Liber de Ludo Aleae* ("The Book on Games of Chance"), the first real attempt to put gambling odds on a mathematical footing. In it he states, more or less, the classical definition: for fair dice the probability of an event is the number of favorable cases over the total. He understood that with two dice there are 36 equally likely outcomes, not 11 (the mistake of counting only the distinguishable sums), and he used this to price bets. The book sat unpublished until 1663, long after his death, so its influence was muted – but Cardano had glimpsed, alone, the central idea of the whole field: *enumerate the equally likely cases and count.*

Cardano's insight about two dice is worth a picture, because it is the single most common counting mistake in the subject. The eleven possible *sums* (2 through 12) are emphatically not equally likely: there is only one way to roll a 2 (namely 1+1) but six ways to roll a 7 (1+6, 2+5, 3+4, 4+3, 5+2, 6+1). The equally likely objects are the 36 *ordered pairs*, and the probability of each sum is just how many of those 36 pairs produce it. Figure 1.3 shows the resulting tent shape.

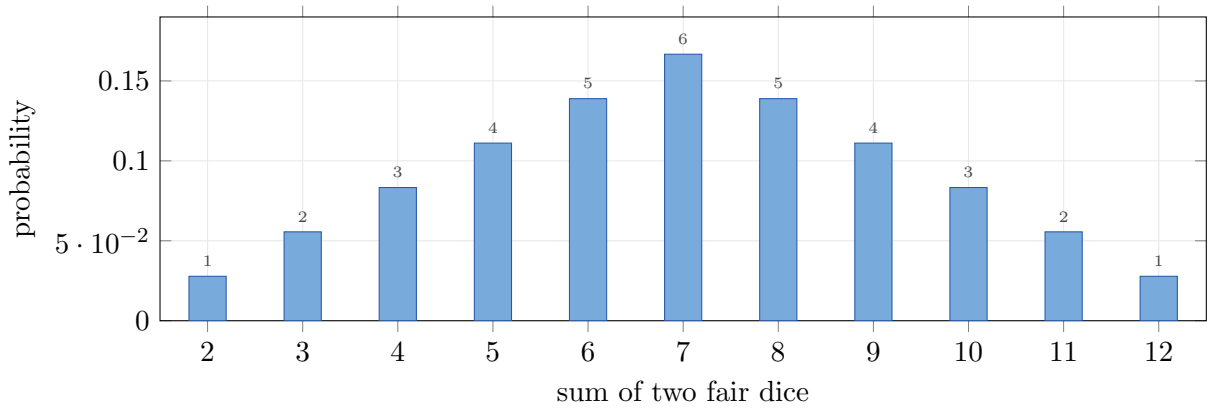


Figure 1.3: The probability of each sum when two fair dice are rolled. The numbers above the bars are the counts of ordered pairs (out of 36) that yield that sum; dividing by 36 gives the probability. The distribution is a symmetric tent peaking at 7 – the most "central" sum, reachable the most ways. Treating the eleven sums as equally likely, the error Cardano warned against, would replace this tent with a flat line.

If Cardano planted a seed in private, the field truly began in public in the summer of 1654, in one of the most consequential exchanges of letters in the history of science.

History

The Pascal–Fermat correspondence (1654). A French nobleman and gambler, the Chevalier de Mere, posed a puzzle to Blaise Pascal. The most famous version is the *problem of points*: two players are partway through a game of chance, each needing some number of rounds to win, when the game is interrupted. How should the stakes be divided *fairly*, given the position? Splitting by who is currently ahead is crude; it ignores how close each player is to victory. The right answer must weigh each player's *chances* of going on to win.
Pascal corresponded with Pierre de Fermat, and over a handful of letters the two hammered out the answer. Their insight – revolutionary at the time – was to divide the stakes in proportion to each player's probability of eventual victory, computed by enumerating the equally likely ways the unfinished game could play out. This is the birth of probability as a mathematical discipline. For the first time, chance was not merely tabulated but *reasoned about*: future possibilities were laid out, weighted, and combined. Within a few years Huygens wrote the first textbook, and within a generation the Bernoullis turned it into a science.

Let us actually solve a clean version of their problem, because it is short, it is beautiful, and it shows the new style of thinking in action.

Example 1.2 (A small problem of points)**.** Alice and Bob play rounds of a fair game; each round, either wins with probability 1/2, independently. The first to win 3 rounds takes the whole pot of \$64. The game is interrupted when Alice has 2 wins and Bob has 1. How should the \$64 be split fairly?

The wrong-but-tempting answer is 2:1, splitting by current score. The right answer asks: if they *had* continued, how likely was each to win? Alice needs just 1 more round; Bob needs 2.

At most 2 more rounds settle it. Enumerate the four equally likely two-round continuations – writing

A for "Alice wins the round" and B for "Bob wins the round":

$$\text{AA}, \quad \text{AB}, \quad \text{BA}, \quad \text{BB}, \qquad \text{each with probability } \tfrac{1}{4}.$$

Alice wins the pot in any continuation containing at least one A among the rounds she needs – that is, in AA, AB, and BA (in each, she reaches 3 wins). Bob wins only in BB, where he takes both remaining rounds to reach 3. So

$$\mathbb{P}(\text{Alice wins}) = \tfrac{3}{4}, \qquad \mathbb{P}(\text{Bob wins}) = \tfrac{1}{4}.$$

Fair division: Alice gets $\frac{3}{4} \cdot \$64 = \$48$, Bob gets $\frac{1}{4} \cdot \$64 = \$16$. Notice the split is 3:1, not the naive 2:1 – being one round from victory is worth far more than the score alone suggests. This is the Pascal–Fermat method entire: enumerate the equally likely futures, weight, and combine.

Remark 1.1. A subtlety worth noticing: we pretended both remaining rounds are played even though the game would actually stop the instant someone reaches 3. This is harmless – playing "phantom" extra rounds does not change who has already won, and it makes every leaf equally likely so we can just count. Pascal and Fermat argued about exactly this point in their letters; Fermat favored the enumeration of all continuations, Pascal a recursive "expected value" bookkeeping. Both are correct and they agree. We will meet both viewpoints again, formalized, when we study expectation.

Application

From the card table to the modern world. It is easy to smile at the idea of probability being born to settle gambling debts, but the through-line to today is direct and unbroken. The "problem of points" is, structurally, the problem an insurance company solves when it prices a policy on an uncertain future, the problem a finance desk solves when it values an option whose payoff depends on where a stock lands, and the problem a clinical trial faces when it decides whether to stop early because one treatment is pulling ahead. Each asks: *given where we are now, weight the possible futures and combine them into a fair present value.* Every time your phone's keyboard predicts the next word, every time a spam filter scores an email, every time a navigation app estimates your arrival time, the machine is enumerating possibilities, weighting them by probability, and combining – the Pascal–Fermat move, executed billions of times a second. The card table was just the first place humans were forced to get it right.

1.6 The Understanding-First Promise

Before the exercises, a word about how to read the rest of this book, since this chapter sets the tone for everything after.

Focus

Probability is unusually punishing to memorize and unusually rewarding to understand. Memorize the formula for conditional probability and you will misapply it within a week; understand the tree picture behind it and you will reconstruct the formula whenever you need it, forever. So throughout this book we will always do the same three things, in this order. *First*, build the idea from a concrete story until it feels obvious. *Second*, make it precise – a real definition, a real derivation, no hand-waving disguised as rigor. *Third*, stress-test it against a paradox, a real application, or a connection to other mathematics, so the idea earns its keep. If at any

point a formula feels like a spell you are reciting, stop – you have skipped step one, and the cure is always to return to a small, concrete example you can hold in your head.

Pause

Where we have been. Probability is a number, between 0 and 1, that measures uncertainty. It has two solid interpretations – long-run frequency (natural for repeatable experiments) and degree of belief (natural for one-off questions) – and, remarkably, both obey the *same* three rules: probabilities lie in $[0, 1]$, disjoint events add, and complements subtract from 1. We need mathematics because intuition fails systematically at chance, because uncertainties must be combined consistently, and because stable laws hide inside aggregate randomness. "Randomness" is often a confession of ignorance about a deterministic but chaotic system. And the whole subject was born at the gambling table, in Cardano's lonely counting and the Pascal–Fermat letters, from a single move: enumerate the equally likely cases, weight them, combine. Next we make Ω, events, and the axioms official.

1.7 Exercises

Exercise 1.1 (Frequency in your hand)**.** Flip a real coin 20 times and record the running fraction of heads after each flip. Plot it (by hand is fine) against the flip number. Does it wander early and settle late, as in Figure 1.1? Now imagine doing this for 2000 flips: describe in one sentence what the plot would look like, and say precisely which interpretation of probability – frequentist or Bayesian – this exercise is illustrating.

Exercise 1.2 (Naming the interpretation)**.** For each statement, say whether it is most naturally frequentist, most naturally Bayesian, or genuinely either, and explain in one sentence why.

1. "The probability this fair die shows a 4 is 1/6."
2. "There is a 30% chance it rains in Lagos tomorrow."
3. "The probability the defendant is guilty, given the evidence, is high."
4. "About 0.7% of manufactured chips from this line are defective."
5. "The probability that the millionth digit of π is a 7 is 1/10."

Exercise 1.3 (Complement practice)**.** Use the complement rule to answer each, and in each case say why "not" is easier than direct counting.

1. Roll a fair die three times. Probability of at least one even number?
2. Flip a fair coin five times. Probability of at least one head?
3. Roll two fair dice. Probability the sum is *not* 7? (There are 36 equally likely outcomes; first count the ways to get 7.)

Exercise 1.4 (A bigger problem of points)**.** Alice and Bob play the fair game of Example 1.2 for a pot of \$128, first to 4 wins. The game is interrupted with Alice on 3 wins and Bob on 1. At most how many more rounds settle it? Enumerate the equally likely continuations, find each player's probability of winning, and give the fair split. (Hint: as in the example, you may play phantom rounds so every leaf is equally likely.)

Exercise 1.5 (Spotting the false symmetry)**.** A friend argues: "A tossed thumbtack lands either point-up or point-down. Two outcomes, so the probability of point-up is 1/2." Explain carefully what is wrong with this reasoning, referencing Definition 1.1. How could you *find* the true probability, and which interpretation of probability does your method rely on?

Exercise 1.6 (The Dutch book, made concrete)**.** A bookmaker offers to sell you, for \$0.60 each, two tickets: ticket A pays \$1 if a certain horse wins, ticket A' pays \$1 if it does not. You buy both.

1. How much do you pay in total, and how much do you collect, no matter what happens?
2. Show that you lose a guaranteed amount, and state it.
3. Which of Rules 1–3 has the bookmaker's pricing violated? What is the largest price he could charge for *each* ticket without exposing himself to a sure loss in the other direction?

Exercise 1.7 (Randomness from determinism)**.** Consider the deterministic "doubling map": start with a number x_0 in $[0, 1)$ and repeatedly apply $x_{n+1} = 2x_n \bmod 1$ (double it, drop the integer part). Write each x_n in binary. Show that applying the map simply *shifts the binary digits left by one*. Use this to explain, in a few sentences, why the sequence of "is $x_n \geq 1/2$?" answers looks like fair coin flips even though the rule is completely deterministic. What role does our ignorance of x_0's exact digits play? Connect this to the discussion in Section 1.3.

Exercise 1.8 (Two dice, the right count)**.** Cardano's contemporaries sometimes treated the sum of two dice as having 11 equally likely values (2 through 12). Explain why this is wrong by listing the 36 equally likely outcomes of an ordered pair of dice. Compute $\mathbb{P}(\text{sum} = 7)$ and $\mathbb{P}(\text{sum} = 2)$, and explain in one sentence the conceptual error in the "11 equally likely sums" view. (This is precisely the kind of mistake Cardano avoided and that the classical definition is designed to prevent.)

Exercise 1.9 (Conceptual: why infinity is uncomfortable)**.** The frequentist defines $\mathbb{P}(\text{heads})$ as the limit of the fraction of heads as the number of flips goes to infinity. Identify two distinct difficulties with taking this as a *definition* (as opposed to an interpretation). One should concern what we can actually observe; the other should concern a hidden circularity in the word "converges." (You are not expected to resolve them here – only to articulate them clearly. The book resolves them when it reaches the Law of Large Numbers.)

Exercise 1.10 (Proof: the complement rule, twice)**.**

1. Using only Rules 1 and 2 from Section 1.4.2, prove that $\mathbb{P}(\text{not } A) = 1 - \mathbb{P}(A)$. State explicitly where you use that A and "not A" are disjoint and where you use that together they are everything.
2. Deduce that the impossible event has probability 0. (Hint: take $A = \Omega$.)
3. Deduce that if A is contained in B – every outcome in A is also in B – then $\mathbb{P}(A) \leq \mathbb{P}(B)$. (Hint: write B as the disjoint union of A and "the part of B outside A," then use Rule 1 on the leftover piece.)

Chapter 2

Counting and Combinatorics

There is a secret hiding in the first probability you ever computed. When you said "a fair coin lands heads with probability 1/2," or "a die shows a four with probability 1/6," you were not really doing probability at all. You were *counting*. You counted the outcomes you cared about (one: heads), you counted all the outcomes there were (two: heads, tails), and you divided. The whole edifice of discrete probability rests on this one humble act.

That is the thesis of this chapter, and I want you to feel its weight before we start. Whenever the outcomes of an experiment are *equally likely* – every card equally probable to be drawn, every seating arrangement equally probable, every poker hand equally probable – the probability of an event is nothing more than

$$\mathbb{P}(\text{event}) \;=\; \frac{\text{number of outcomes in the event}}{\text{total number of outcomes}}.$$

This is sometimes called the *naive definition of probability*, and we will sharpen it in the next chapter. But "naive" is a misleading name. The formula is exactly right whenever the symmetry of the setup justifies it, and an astonishing fraction of real problems – cards, dice, lotteries, random samples, shuffles, hashing, password strength – have exactly that symmetry. The catch is the word *number*. To use the formula you must be able to count, and counting large structured sets is a genuine mathematical art. That art is *combinatorics*, and this chapter builds it up from scratch.

Focus

By the end of this chapter you will be able to count: arrangements (permutations), selections (combinations), things drawn with or without replacement, ordered or unordered. You will know the binomial coefficient inside and out, build Pascal's triangle and read its secrets, resolve the famous birthday paradox, wield inclusion–exclusion to count overlapping things, and use "stars and bars" to count the ways of distributing identical objects. Every technique will be tied back to the counting-equals-probability idea above.

2.1 The Multiplication Principle

Everything in this chapter grows from a single seed, and the seed is so simple it feels like a tautology. Suppose you are getting dressed and you own 3 shirts and 4 pairs of trousers. How many distinct outfits can you assemble? For each of the 3 shirts you may pair any of the 4 trousers, so the outfits

come in 3 groups of 4, giving $3 \times 4 = 12$.

Intuition

The picture to keep in your head is a *tree*. From a single root you draw one branch for each shirt; from the end of each shirt-branch you draw one branch for each pair of trousers. The number of complete root-to-leaf paths is the number of outfits. Because every shirt-branch sprouts the same number of trouser-branches, you do not add, you *multiply*.

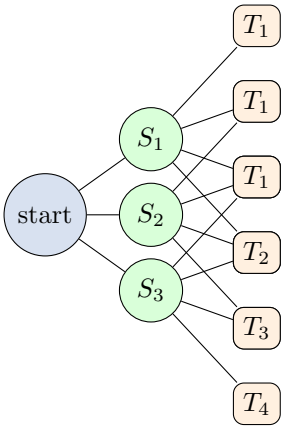


Figure 2.1: The multiplication principle as a tree. Three shirts, each branching into four trousers, give $3 \times 4 = 12$ leaves, one per outfit.

Theorem 2.1 (Multiplication Principle)**.** *Suppose a procedure consists of k successive stages, and suppose that*

- *stage 1 can be carried out in n_1 ways, and*
- *for each outcome of stage 1, stage 2 can be carried out in n_2 ways, and*
- *in general, for each outcome of the first $i-1$ stages, stage i can be carried out in n_i ways.*

Then the whole procedure can be carried out in $n_1 \cdot n_2 \cdots n_k$ ways.

Why it works

Why is the product right, and not, say, the sum? Induct on the number of stages. With one stage there are obviously n_1 ways. Now suppose the claim holds for $k-1$ stages, so the first $k-1$ stages can be completed in $N = n_1 \cdots n_{k-1}$ ways. Fix any one of those N partial completions. The crucial hypothesis is that *no matter which* of the N we fixed, stage k offers exactly n_k continuations. So each of the N partial paths fans out into n_k full paths, and these groups of n_k never overlap (they start from different partial paths). Total full paths: $N \cdot n_k = n_1 \cdots n_k$. The induction closes.

The hypothesis I emphasized in the proof – "n_i ways *regardless of* the earlier choices" – is the only thing that can trip you up. The *number* of options at stage i must be constant; the options themselves may change.

Example 2.1 (License plates)**.** A license plate has the form letter-letter-digit-digit-digit. How many plates are there if repeats are allowed? Five stages: $26 \times 26 \times 10 \times 10 \times 10 = 676{,}000$. Now suppose

no symbol may repeat. The letters: 26 then 25 (any letter but the first). The digits: 10, then 9, then 8. The count is $26 \times 25 \times 10 \times 9 \times 8 = 468{,}000$. Notice the option *count* at the second letter stage is 25 no matter which first letter you chose – so the multiplication principle still applies cleanly even though *which* 25 letters are available changes.

Example 2.2 (Subsets and the power of two)**.** How many subsets does a set of n elements have? Build a subset by walking through the elements one at a time and, for each, deciding *in* or *out*. That is n stages of 2 choices each, so 2^n subsets. This includes the empty set (everyone "out") and the full set (everyone "in"). A set of 10 elements has 1024 subsets; this exponential blow-up is exactly why brute-force search over subsets becomes hopeless fast, and why we will need cleverer counting.

What is really happening

What is really happening is that *independent choices compose by multiplication*. This is the combinatorial shadow of a fact you will meet again and again in probability: when events are independent, their probabilities multiply. The multiplication principle is independence *before* we have even mentioned chance. Keep that resonance in mind; in the next chapter "equally likely outcomes built from independent stages" will turn products of counts into products of probabilities.

2.2 Permutations: Counting Arrangements

An *arrangement* is an ordering. In how many orders can n distinct books be placed on a shelf? Fill the leftmost slot first: n choices. The next slot: $n-1$ books remain. Then $n-2$, and so on down to 1 for the last slot. By the multiplication principle the number of orderings is

$$n \cdot (n-1) \cdot (n-2) \cdots 2 \cdot 1 \;=\; n!,$$

read "n factorial." We define $0! = 1$, the empty arrangement; there is exactly one way to arrange nothing, and as we will see this convention makes every later formula come out right.

Definition 2.1 (Factorial and k-permutations)**.** For an integer $n \geq 0$, $n! = n(n-1)\cdots 1$ with $0! = 1$. The number of ways to choose and arrange k of n distinct objects in order, written $P(n,k)$ or $(n)_k$, is

$$P(n,k) \;=\; n(n-1)\cdots(n-k+1) \;=\; \frac{n!}{(n-k)!}, \qquad 0 \leq k \leq n.$$

The product $n(n-1)\cdots(n-k+1)$ has exactly k factors – one per filled slot – counting down from n. The closed form $n!/(n-k)!$ is the same product with the unused tail $(n-k)(n-k-1)\cdots 1$ canceling top and bottom. When $k = n$ the tail is $0! = 1$ and we recover $P(n,n) = n!$, the full arrangement.

Example 2.3 (Race podium)**.** Ten sprinters race; how many ways can gold, silver, bronze be awarded (no ties)? Order matters – gold then silver is different from silver then gold – so this is $P(10,3) = 10 \cdot 9 \cdot 8 = 720$.

Common trap

The single most common error in counting is failing to ask: **does order matter?** "Arrange three of ten runners on a podium" (order matters: $P(10,3) = 720$) is a different question from "choose three of ten runners for a relay pool" (order does not matter, and the answer will be $720/3! = 120$). Before you compute anything, decide whether two selections that differ only in order should be counted once or twice. Get this wrong and your count is off by a factor of $k!$ – a mistake that scales catastrophically.

History

Factorials and arrangements are ancient. The Indian mathematician and astronomer *Bhaskara II* (12th century), and earlier Jain mathematicians, knew rules for counting permutations and combinations; the medieval Hebrew scholar *Abraham ibn Ezra* computed combinations in an astrological context. The notation "$n!$" is far younger – introduced by Christian Kramp in 1808 precisely because writing out the long product was tedious. The *idea* of counting arrangements long predates probability theory itself; Cardano, Pascal, and Fermat inherited a rich counting tradition and pointed it at games of chance.

2.2.1 Permutations with repeated objects

Suppose some objects are identical. How many distinct "words" can you make by rearranging the letters of `MISSISSIPPI`? There are 11 letters, so if all were distinct there would be 11! arrangements. But the four `S`'s are interchangeable: any of their 4! internal orderings produces the *same* word, so we have overcounted by a factor of 4!. Likewise the four `I`'s overcount by 4! and the two `P`'s by 2!. The correct count is

$$\frac{11!}{4!\,4!\,2!\,1!} = 34650.$$

This is the *multinomial* count, and it embodies a principle worth naming.

Intuition

Overcount, then divide by the symmetry. Few counting strategies are this general: pretend everything is distinguishable and count the easy (larger) number, then divide by the number of ways the indistinguishable items could be permuted among themselves without changing anything. We will use this exact move to derive combinations in a moment.

2.3 Combinations: Counting Selections

Now order stops mattering. A *combination* is an unordered selection – a subset. How many 3-person committees can be formed from 10 people? If order mattered we computed $P(10,3) = 720$ ordered selections. But a committee $\{A, B, C\}$ is the same committee no matter the order we named its members, and there are $3! = 6$ orderings of any given trio. So each committee was counted 6 times among the 720, and the true count is $720/6 = 120$.

Definition 2.2 (Binomial coefficient)**.** The number of k-element subsets of an n-element set is the *binomial coefficient*

$$\binom{n}{k} = \frac{P(n,k)}{k!} = \frac{n!}{k!\,(n-k)!}, \qquad 0 \le k \le n,$$

read "n choose k." By convention $\binom{n}{k} = 0$ when $k < 0$ or $k > n$.

Why it works

The derivation is the overcount-then-divide move in its purest form. To build an ordered selection of k from n you may first *choose* the unordered set of k ($\binom{n}{k}$ ways) and then *order* it ($k!$ ways). By the multiplication principle the ordered count factors as

$$P(n,k) = \binom{n}{k} \cdot k!.$$

Solving for $\binom{n}{k}$ gives $P(n,k)/k! = n!/(k!(n-k)!)$. Every binomial coefficient is therefore a count of ordered things with the order divided back out.

Example 2.4 (Poker hands)**.** A standard deck has 52 cards. A five-card hand is an unordered selection, so the number of distinct hands is

$$\binom{52}{5} = \frac{52!}{5!\,47!} = \frac{52 \cdot 51 \cdot 50 \cdot 49 \cdot 48}{120} = 2{,}598{,}960.$$

Now we can do real probability. How many hands are four-of-a-kind? Choose the rank that appears four times (13 ways), take all four of its cards (one way), then choose the fifth card from the remaining 48 (48 ways): $13 \cdot 48 = 624$ hands. Because all $\binom{52}{5}$ hands are equally likely,

$$\mathbb{P}(\text{four of a kind}) = \frac{624}{2{,}598{,}960} \approx 0.00024.$$

Counting and probability are one motion: count the event, count the whole, divide.

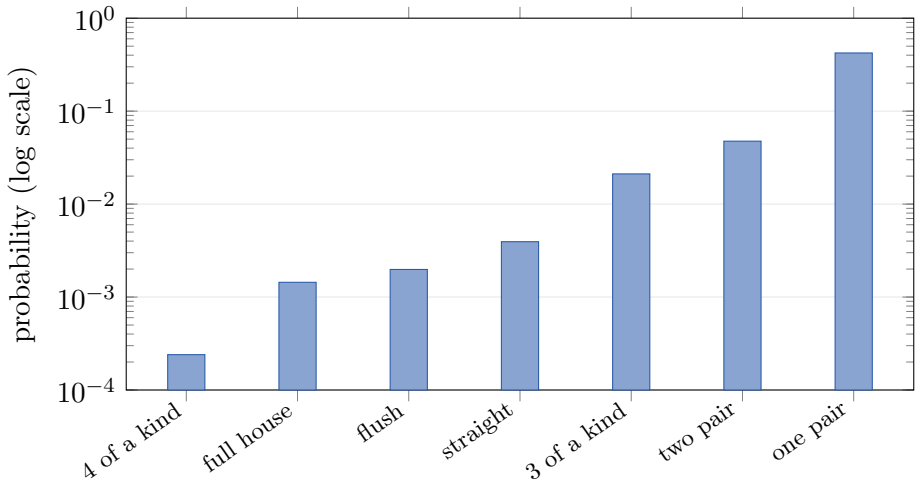


Figure 2.2: Probabilities of seven poker hands, each obtained by the same recipe: count the hands of that type, divide by $\binom{52}{5} = 2{,}598{,}960$. The vertical axis is logarithmic because the rarities span more than three orders of magnitude – four of a kind is roughly 1760 times rarer than one pair. (Here "flush" counts all five-same-suit hands and "straight" all five-in-sequence hands; you will count two of these yourself in the exercises.)

Why this matters

The notation $\binom{n}{k}$ is more telling than "nCk" because it reminds you the answer is a single number attached to a pair (n, k), and because it slots straight into the binomial theorem $(x+y)^n = \sum_{k=0}^{n} \binom{n}{k} x^k y^{n-k}$ that gave it its name. We will return to that theorem; for now notice that the very same numbers count subsets *and* expand powers. When one quantity answers two unrelated-looking questions, mathematics is telling you the questions were secretly the same.

The same divide-by-the-whole recipe ranks every poker hand at once. Figure 2.2 plots the probabilities of seven of them on a logarithmic scale; the steep staircase down to four of a kind is a picture of why rare hands beat common ones.

2.3.1 Symmetry: choosing what to leave out

A small identity carries a large idea:

$$\binom{n}{k} = \binom{n}{n-k}.$$

Algebraically it is immediate from the formula, since swapping k and $n-k$ leaves $k!(n-k)!$ unchanged. But the *reason* is prettier than the algebra. To choose k objects to take is exactly to choose $n-k$ objects to leave behind; each selection of a committee is simultaneously a selection of the non-committee. The two acts are the same act seen from opposite sides, so they must have the same count. This is a *bijective* proof – we exhibited a one-to-one correspondence rather than manipulating symbols – and bijective arguments are the soul of combinatorics. Whenever two counts are equal, ask whether you can *see* why by pairing the things up.

2.4 The Binomial Coefficient and Pascal's Triangle

Arrange the binomial coefficients in a triangle, row n holding $\binom{n}{0}, \binom{n}{1}, \ldots, \binom{n}{n}$:

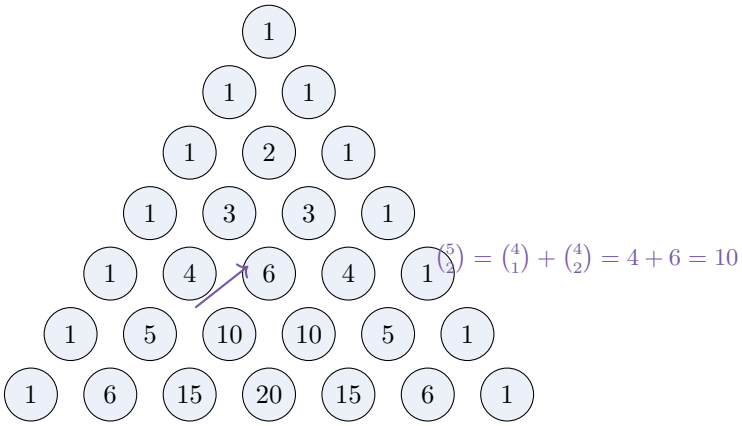


Figure 2.3: Pascal's triangle. Each entry is the sum of the two entries above it: the edges are all 1's ($\binom{n}{0} = \binom{n}{n} = 1$), and each interior number is born from the pair diagonally above. Row n sums to 2^n.

The defining miracle of the triangle is that each number is the sum of the two above it. In symbols this is *Pascal's rule*.

Theorem 2.2 (Pascal's rule)**.** *For* $1 \le k \le n$,

$$\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}.$$

Why it works

You could prove this by grinding the factorials together over a common denominator, and it works. But the *combinatorial* proof tells you why it is true. We are counting k-element subsets of $\{1, 2, \ldots, n\}$, which is $\binom{n}{k}$ by definition. Now sort those subsets by a single question: *does the subset contain the element* n*?*

- If it *does* contain n, the remaining $k-1$ members come from $\{1, \ldots, n-1\}$: there are $\binom{n-1}{k-1}$ such subsets.
- If it *does not* contain n, all k members come from $\{1, \ldots, n-1\}$: there are $\binom{n-1}{k}$ such subsets.

Every k-subset falls into exactly one case (it either has n or it does not), so the two counts add to the total. That is Pascal's rule, and the "two entries above" in the picture are precisely these two cases.

This conditioning move – split a count by whether some particular element is in or out – is a recurring theme, and it foreshadows *conditioning* on an event, the engine of the next chapters. Pascal's rule also gives a fast, multiplication-free way to compute the coefficients: just add your way down the triangle.

History

The triangle is named for Blaise Pascal, who in his 1654 *Traité du triangle arithmétique* organized these numbers and used them to solve the "problem of points" (how to fairly split the stakes of an interrupted game) in his famous correspondence with Pierre de Fermat – correspondence usually marked as the birth of probability theory. But the triangle is much older: it appears in the work of the Persian mathematicians al-Karaji and Omar Khayyam, in China as "Yang Hui's triangle" (13th century, citing earlier sources), and in India centuries before. Pascal's contribution was not discovery but systematic use – turning a numerical curiosity into a tool for reasoning about chance.

2.4.1 Two identities you should be able to see

The row sum. Adding across row n,

$$\sum_{k=0}^{n} \binom{n}{k} = 2^n.$$

Why? The left side counts subsets of an n-set grouped by size; the right side counts all subsets at once (in/out for each element, Example 2.2). Same set, two counts, so they are equal. No algebra needed.

The hockey-stick. Summing down a diagonal,

$$\sum_{i=r}^{n} \binom{i}{r} = \binom{n+1}{r+1}.$$

The picture (a straight diagonal of entries that "bends" at the bottom into the answer) gives the rule its name. We will not belabor the proof, but it follows by repeatedly applying Pascal's rule, and it is a handy closed form for sums that look intimidating.

2.5 The Four Sampling Scenarios

Almost every elementary counting problem is one of four flavors, indexed by two yes/no questions. Are we sampling *with replacement* (can an item be chosen more than once)? Does *order matter*? Here are all four, for choosing k items from n.

	Order matters	**Order does not matter**
Without replacement	$\dfrac{n!}{(n-k)!}$	$\binom{n}{k}$
With replacement	n^k	$\binom{n+k-1}{k}$

Figure 2.4: The four ways to sample k objects from n. The bottom-right cell – ordered *no*, replacement *yes* – is the surprising one and is the subject of the "stars and bars" section below.

Three of the four we have essentially seen. Ordered without replacement is the permutation $P(n,k) = n!/(n-k)!$ (each pick removes an option, and order is recorded). Unordered without replacement is the combination $\binom{n}{k}$. Ordered with replacement is the simplest of all: k independent stages of n choices each, so n^k – this is the license-plate count, the password count, the "rolls of a die" count. The fourth, unordered *with* replacement, is genuinely subtle and we devote a full section to it.

Example 2.5 (Three flavors of "three from a deck")**.** *Deal* 3 *cards, recording order, no card reused:* $52 \cdot 51 \cdot 50 = 132600$. *Deal* 3 *cards as an unordered hand:* $\binom{52}{3} = 22100 = 132600/3!$. *Draw a card, note it, replace, reshuffle, repeat* 3 *times, recording order:* $52^3 = 140608$. The same deck, three different questions, three different counts – and each is the denominator of a different probability calculation.

2.6 The Birthday Problem

Here is a question that has been startling people for a century. In a room of just 23 people, the probability that *some two of them share a birthday* is already better than even. Most people guess you would need around 180 people (half of 365). The gap between intuition and truth is so large that the result is called the *birthday paradox* – though, as with all good paradoxes, nothing is actually contradictory; our intuition is simply miscalibrated.

Paradox

The birthday paradox. Assume 365 equally likely birthdays and ignore leap years and twins. With 23 people in a room, the chance that at least two share a birthday is about 50.7%. With 50 people it is 97%; with 70 people it is 99.9%. How can so few people suffice? The resolution: you are not comparing your birthday to one other person's. You are comparing *every pair* of people, and the number of pairs grows quadratically. With 23 people there are $\binom{23}{2} = 253$ pairs – already comfortably more than half of 365 – and each pair is a fresh chance for a collision.

Now compute it. The trick is to count the *complement*. Asking for "at least one shared birthday" invites messy overlaps (maybe two pairs match, maybe three people all match). The opposite event – "all n birthdays distinct" – is clean: it is just an ordered sampling without replacement. The first person can have any of 365 birthdays, the second any of the remaining 364, and so on, while the total number of birthday assignments (with replacement, since people may repeat) is 365^n. So

$$\mathbb{P}(\text{all distinct}) \;=\; \frac{365 \cdot 364 \cdots (365-n+1)}{365^n} \;=\; \prod_{i=0}^{n-1}\Big(1-\frac{i}{365}\Big),$$

and the probability of at least one match is 1 minus that.

Walking question

Before reading on, predict the shape of the curve. Each factor $(1-i/365)$ is slightly less than 1, and you are multiplying n of them. With many near-1 factors multiplied together, the product drops faster than you would think – a fact captured by $\prod(1-i/365) \approx \exp\big(-\sum i/365\big) = \exp\big(-\frac{n(n-1)}{2\cdot 365}\big)$. The exponent has $n(n-1)/2 = \binom{n}{2}$ in it – the number of pairs! That is the paradox in one line: the collision probability is governed by the *number of pairs*, which is

quadratic in n.

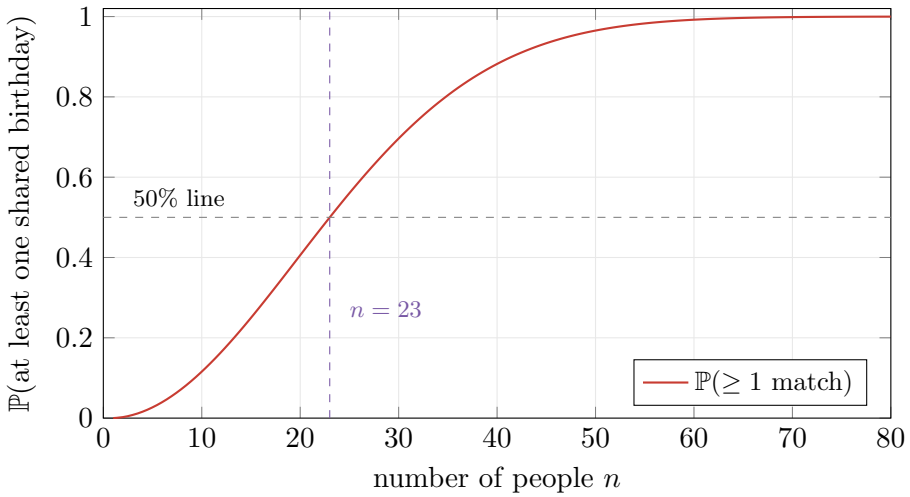


Figure 2.5: The birthday curve (closed-form approximation $1 - e^{-n(n-1)/730}$). It crosses 50% near $n = 23$ and is essentially certain by $n \approx 60$. The steep rise is the signature of the quadratic number of pairs.

Example 2.6 (The numbers)**.** For $n = 23$, the exact product gives $\mathbb{P}(\text{all distinct}) = 0.4927$, so $\mathbb{P}(\text{match}) = 0.5073$ – just over half. For $n = 30$ it is 0.706; for $n = 50$ it is 0.970; for $n = 57$ it has passed 0.99. The crossover at 23 is the headline, but the *speed* of the climb afterwards is just as striking.

Application

The birthday attack in cryptography. This is not a parlor trick. A hash function maps data to one of N possible fingerprints; a "collision" is two different inputs with the same fingerprint, and collisions can break digital signatures. The birthday math says you expect a collision after hashing only about $\sqrt{N}$ random inputs, not N – because, again, it is the number of *pairs* that matters. To make collisions infeasible a hash therefore needs roughly *twice* as many output bits as you might naively demand: a 128-bit security target needs a 256-bit hash. The phrase "birthday bound" is standard in cryptography, and it traces directly to the count $\binom{n}{2}$ we did above. The same $\sqrt{N}$ scaling appears in hash-table load analysis and in distributed systems that assign random IDs.

2.7 Inclusion–Exclusion

The multiplication and combination tools count things that fit cleanly into stages or subsets. But many counts involve *overlap*: "how many integers up to 100 are divisible by 2 or 3?" If you add the multiples of 2 (50 of them) to the multiples of 3 (33), you have double-counted everything divisible by both 2 and 3, i.e. by 6 (16 of them). Subtract those once and the count is $50 + 33 - 16 = 67$.

Intuition

Picture two overlapping circles. To find the area of the union you add the two circle areas and then *subtract the lens* where they overlap, because you counted it twice. With three circles you add the three, subtract the three pairwise overlaps, then add back the triple overlap (which you removed one time too many). The alternating add–subtract–add pattern is inclusion–exclusion.

Theorem 2.3 (Inclusion–Exclusion)**.** *For finite sets $A_1, \dots, A_n$,*

$$\left|A_1 \cup \cdots \cup A_n\right| \;=\; \sum_{i} |A_i| \;-\; \sum_{i<j} |A_i \cap A_j| \;+\; \sum_{i<j<k} |A_i \cap A_j \cap A_k| \;-\; \cdots \;+\; (-1)^{n+1} \left|A_1 \cap \cdots \cap A_n\right|.$$

The same identity holds with $|\cdot|$ replaced by $\mathbb{P}(\cdot)$, giving the inclusion–exclusion formula for the probability of a union.

Why it works

The cleanest proof tracks a single element x and asks: *how many times does x get counted on the right-hand side?* Suppose x lies in exactly $m \geq 1$ of the sets. Then x is counted $\binom{m}{1}$ times in the singles, $\binom{m}{2}$ times in the pairs (once for each pair of the m sets containing it), $\binom{m}{3}$ in the triples, and so on. Its net contribution to the right side is

$$\binom{m}{1} - \binom{m}{2} + \binom{m}{3} - \cdots \;=\; -\sum_{j=1}^{m} (-1)^j \binom{m}{j} \;=\; 1 - \sum_{j=0}^{m} (-1)^j \binom{m}{j} \;=\; 1 - 0 \;=\; 1,$$

where we used the alternating row-sum $\sum_{j=0}^{m} (-1)^j \binom{m}{j} = (1-1)^m = 0$ for $m \geq 1$ (the binomial theorem at $x = -1$, $y = 1$). So every element of the union is counted exactly once on the right, which means the right side equals $|A_1 \cup \cdots \cup A_n|$. Elements in *no* set contribute 0 on both sides. Done.

Example 2.7 (The hat-check problem / derangements)**.** n guests check identical-looking hats; the absent-minded attendant returns them at random. What is the probability that *no one* gets their own hat back? Let A_i be the event that guest i *does* get their own hat. We want $\mathbb{P}(\text{none}) = 1 - \mathbb{P}(A_1 \cup \cdots \cup A_n)$. By symmetry $\mathbb{P}(A_i) = 1/n$ (guest i's hat is equally likely to land anywhere), $\mathbb{P}(A_i \cap A_j) = \frac{(n-2)!}{n!} = \frac{1}{n(n-1)}$, and in general the intersection of any k of them has probability $(n-k)!/n!$. There are $\binom{n}{k}$ such k-fold intersections, and $\binom{n}{k}\frac{(n-k)!}{n!} = \frac{1}{k!}$. Inclusion–exclusion gives

$$\mathbb{P}(A_1 \cup \cdots \cup A_n) = \sum_{k=1}^{n} (-1)^{k+1} \frac{1}{k!}, \qquad \text{so} \qquad \mathbb{P}(\text{no match}) = \sum_{k=0}^{n} \frac{(-1)^k}{k!}.$$

Paradox

The hat-check limit. As $n \to \infty$ the no-match probability is $\sum_{k \geq 0} (-1)^k/k! = e^{-1} \approx 0.3679$. The startling part: the answer barely depends on n. Whether 5 guests or 5 million guests check their hats, the probability that the random return gives *nobody* their own hat hovers near 37%. Adding more guests does not wash out the chance of a total mismatch, because each new guest brings both a new chance to match *and* a new hat to be mismatched, and these

effects almost perfectly cancel. The constant e – the hero of calculus – shows up unbidden in a pure counting problem.

Connection

That appearance of e^{-1} is your first sighting of a bridge we will cross repeatedly: the link between combinatorics and *analysis*. The series $\sum(-1)^k/k!$ is the Taylor expansion of e^x evaluated at $x = -1$ – the very series the companion calculus book derives. Throughout probability, finite counting identities have limits that are calculus objects: factorials smooth into the Gamma function, binomial coefficients smooth into the Beta function, and the binomial distribution we build from $\binom{n}{k}$ will, in the limit, become the bell curve via the Central Limit Theorem. Counting is discrete probability; calculus is its continuous shadow.

2.8 Stars and Bars

We saved the strangest cell of Figure 2.4 for last: counting *unordered selections with replacement*. Equivalently – and this is the framing that unlocks it – how many ways can we distribute k identical objects into n distinct boxes?

Concretely: you buy $k = 7$ identical doughnuts from a shop with $n = 3$ varieties. A purchase is determined entirely by *how many of each variety* you take, say $(3, 2, 2)$ or $(7, 0, 0)$ – order of buying is irrelevant and you may repeat a variety. How many purchases are possible?

Intuition

Represent a purchase as a row of 7 stars (the doughnuts) separated into 3 groups by 2 bars (the dividers between varieties). For instance

$$\star\star\star\,|\,\star\star\,|\,\star\star \qquad \text{means } (3,2,2),$$

$$\star\star\star\star\star\star\star\,||\qquad \text{means } (7,0,0).$$

Every purchase is one arrangement of 7 stars and 2 bars in a row; every such arrangement is one purchase. To count the purchases, just count the arrangements of stars and bars!



Figure 2.6: Stars and bars for the purchase $(3, 2, 2)$: seven stars (doughnuts) and two bars (dividers) in a row of nine positions. Each distinct arrangement of stars and bars is one purchase, so the count is the number of ways to place the 2 bars among 9 slots, $\binom{9}{2}$.

A row has $7 + 2 = 9$ positions, and we choose which 2 of them hold bars (the rest are stars). That is $\binom{9}{2} = 36$ (Figure 2.6). In general:

Theorem 2.4 (Stars and bars)**.** *The number of ways to write a nonnegative integer k as an ordered sum of n nonnegative parts $x_1 + x_2 + \cdots + x_n = k$ – equivalently, the number of ways to place k*

identical objects into n distinct boxes, equivalently the number of unordered samples of size k from n types with replacement – is

$$\binom{n+k-1}{k} = \binom{n+k-1}{n-1}.$$

Why it works

A configuration is a row of k stars and $n-1$ bars: the $n-1$ bars cut the row into n (possibly empty) groups, and the group sizes are exactly the x_i. There are $k+(n-1)$ total symbols, and a configuration is fixed once we decide which $n-1$ positions are bars – that is $\binom{n+k-1}{n-1}$ – or equivalently which k positions are stars, which is $\binom{n+k-1}{k}$ (the two are equal by the choosing-what-to-leave-out symmetry). The map from configurations to solutions $(x_1, \ldots, x_n)$ is a bijection: each star-and-bar row yields exactly one tuple, and each tuple yields exactly one row. Bijection means equal counts.

Common trap

Stars and bars counts *nonnegative* solutions (boxes may be empty). If every box must be nonempty – every variety bought at least once, every $x_i \geq 1$ – first hand one object to each box (n objects spent), then distribute the remaining $k-n$ freely: $\binom{(k-n)+n-1}{n-1} = \binom{k-1}{n-1}$. Forgetting which regime you are in (boxes allowed empty vs. forced nonempty) is a classic slip. Always ask: can a part be zero?

Example 2.8 (Distributing identical gifts)**.** In how many ways can 10 identical candies be split among 4 children (some children may get none)? Stars and bars with $k = 10$, $n = 4$: $\binom{10+4-1}{4-1} = \binom{13}{3} = 286$. If every child must get at least one candy, give one to each first and distribute the remaining 6: $\binom{6+4-1}{4-1} = \binom{9}{3} = 84$.

What is really happening

Why does "unordered with replacement" equal stars and bars? An unordered sample with replacement does not care which doughnut you grabbed first; it cares only about the *multiplicities* – how many of each type. A list of multiplicities $(x_1, \ldots, x_n)$ that sums to k is exactly a stars-and-bars configuration. So the seemingly exotic bottom-right cell of the sampling table is the same object as the down-to-earth "how many of each" question. This is also why this count is $\binom{n+k-1}{k}$ and *not* $\binom{n}{k}$: with replacement the pool effectively grows, and the algebra records that with the $+k-1$ in the top.

2.9 Why Counting Is the Backbone of Discrete Probability

Here, finally, is the payoff promised at the start. Whenever a finite experiment has equally likely outcomes, every probability question becomes two counting questions: count the favorable outcomes, count all outcomes, divide. The techniques of this chapter are therefore not a warm-up *before* probability; for equally-likely models they *are* probability.

Formal

The naive model, stated. Let the sample space Ω be a finite set of outcomes, all assumed equally likely. For any event $A \subseteq \Omega$,

$$\mathbb{P}(A) = \frac{|A|}{|\Omega|}.$$

Every fact you need follows. Nonnegativity: $|A| \geq 0$. Normalization: $\mathbb{P}(\Omega) = |\Omega|/|\Omega| = 1$. Additivity for disjoint events: $|A \cup B| = |A| + |B|$ when $A \cap B = \varnothing$, so the probabilities add. These are exactly the axioms of probability we develop in the next chapter – here they are theorems about counting.

Example 2.9 (Two dice, the long way and the right way)**.** Roll two fair dice. What is $\mathbb{P}(\text{sum} = 7)$? The sample space is the $6 \times 6 = 36$ ordered pairs (multiplication principle), all equally likely. The sum-7 outcomes are $(1,6), (2,5), (3,4), (4,3), (5,2), (6,1)$ – six of them. So $\mathbb{P} = 6/36 = 1/6$. The reason a naive count over *unordered* pairs would mislead you here is that unordered pairs are *not* equally likely (you can roll a $\{3,4\}$ two ways but a $\{3,3\}$ only one way). The art is choosing a sample space whose outcomes truly are equally likely, and ordered pairs are that space. **Choose the equally-likely space first; then count.**

Why this matters

Counting is the backbone for a deeper reason than convenience. The hardest part of applied probability is rarely the arithmetic – it is building an accurate model of the situation: What are the outcomes? Are they equally likely? Does order matter? Can things repeat? Those four questions are the very ones combinatorics forces you to answer precisely. Master counting and you have mastered the discipline of stating a probability problem cleanly, which is most of the battle. Everything later – random variables, expectation, distributions – is built on sample spaces, and sample spaces are counted things.

Work it yourself

Pick a small experiment of your own – drawing 2 cards, flipping 4 coins, rolling 3 dice – and do the following before computing any probability. (1) Write down the sample space explicitly enough to count it. (2) Argue why its outcomes are equally likely (or fix the space until they are). (3) Express your event as a subset and count it with one of this chapter's tools. (4) Only then divide. Doing this slowly a few times builds the reflex that makes hard problems easy.

2.10 Exercises

Exercise 2.1 (Warm-up counts)**.** (a) How many 4-digit PINs are there if digits may repeat? If they may not? (b) How many ways can 7 people sit in a row? Around a round table (rotations counted as the same)? (c) How many distinct rearrangements are there of the letters in `BOOKKEEPER`?

Exercise 2.2 (Order or not)**.** A class has 20 students. (a) In how many ways can a President, Vice-President, and Treasurer be chosen (distinct roles)? (b) In how many ways can a committee of 3 (no roles) be chosen? (c) Explain the exact numerical relationship between your two answers and why it is what it is.

Exercise 2.3 (Poker probabilities)**.** Using the $\binom{52}{5} = 2{,}598{,}960$ hands, count and then find the probability of: (a) a flush (all five cards the same suit, including straight flushes); (b) exactly one pair (one rank twice, the other three cards of three distinct other ranks); (c) compare your one-pair probability to the four-of-a-kind probability from Example 2.4 and comment.

Exercise 2.4 (Birthday variations)**.** (a) Using the approximation $\mathbb{P}(\geq 1 \text{ match}) \approx 1 - e^{-n(n-1)/730}$, estimate the smallest n giving probability at least 0.5, and compare with the exact answer 23. (b) Suppose instead birthdays are spread over a planet with D equally likely "days." Show that the number of people needed for an even-chance collision grows like $\sqrt{D}$, and state the constant in front of $\sqrt{D}$.

Exercise 2.5 (Inclusion–exclusion in the wild)**.** How many integers from 1 to 1000 are divisible by at least one of 2, 3, or 5? Then find how many are divisible by *none* of them. (Hint: count the divisible ones with inclusion–exclusion and subtract from 1000.)

Exercise 2.6 (Stars and bars)**.** (a) How many nonnegative integer solutions does $x_1 + x_2 + x_3 + x_4 = 12$ have? (b) How many with every $x_i \geq 1$? (c) How many with $x_1 \geq 2$ and the rest ≥ 0? (Hint for (c): pre-assign the required minimum, then count freely.)

Exercise 2.7 (A bijective proof)**.** Prove the identity $\sum_{k=0}^{n} \binom{n}{k} = 2^n$ *combinatorially* – that is, by describing a single set that both sides count – rather than by algebra. Then prove $\binom{n}{k} = \binom{n}{n-k}$ the same way, exhibiting an explicit bijection between the two families of subsets.

Exercise 2.8 (Vandermonde's identity)**.** Prove Vandermonde's identity $\binom{m+n}{r} = \sum_{j=0}^{r} \binom{m}{j}\binom{n}{r-j}$ by a counting argument. (Hint: you have m red balls and n blue balls and want to choose r in total; condition on how many of the chosen are red.) Then deduce the special case $\sum_{k=0}^{n} \binom{n}{k}^2 = \binom{2n}{n}$.

Exercise 2.9 (Derangement check)**.** Using Example 2.7, compute the exact probability that no guest gets their own hat for $n = 3, 4, 5$ guests, and verify that the values are already very close to $e^{-1} \approx 0.3679$. How large must n be for the answer to be within 0.001 of e^{-1}?

Exercise 2.10 (Modeling care)**.** A bag has 3 red and 2 blue marbles. You draw 2 without replacement. A student argues: "There are three unordered outcomes – two red, two blue, one of each – so each has probability 1/3." Explain precisely why this reasoning is wrong, identify the correct equally-likely sample space, and compute the right probability of "one of each."

Chapter 3

Sample Spaces, Events, and the Axioms

So far we have spoken of probability the way most people first meet it: a vague feeling that some things are more likely than others, sharpened by counting in the special case where outcomes are "equally likely." That is a fine place to start, but it is not a foundation. If a friend says "the probability is 0.7," what *object* is being assigned the number 0.7? What rules must those numbers obey so that they hang together into a consistent picture instead of contradicting one another? And how do we talk about probability when the outcomes are *not* equally likely, or when there are infinitely many of them, or uncountably many?

This chapter builds the stage on which all the rest of the book performs. The work falls into three parts. We cast every random situation as a *set*: a sample space of possible outcomes, with events as subsets. We write down the three rules — Kolmogorov's axioms — that any consistent assignment of probabilities must satisfy, and we argue carefully *why exactly these three* and not some other list. Then we squeeze those three rules for everything they are worth, deriving the everyday tools (complements, the union bound, monotonicity, inclusion–exclusion) as *theorems* rather than as separate articles of faith. Along the way we meet the gentle idea of an event algebra, which quietly becomes the σ-algebra of the measure-theoretic chapters at the end of the book.

What makes this worth the trouble is a little surprising: from a list of three modest-looking rules, the entire edifice of probability follows by pure logic.

3.1 Outcomes, sample spaces, and events

Intuition

Before any formalism: imagine you are about to do something whose result you cannot predict — flip a coin, roll dice, draw a card, measure tomorrow's rainfall, watch a stock until noon. Probability begins by listing *everything that could happen*. That master list is the sample space. Anything you might later care to bet on — "the coin shows heads," "the dice sum to seven," "it rains more than an inch" — is just a *portion* of that list, a subset. The whole subject is, at bottom, the study of how to put sensible numbers on subsets.

Definition 3.1 (Sample space and outcomes)**.** A **random experiment** is any process whose outcome is uncertain in advance. The **sample space**, written Ω (capital Greek omega), is the set of all possible outcomes of the experiment. A single element $\omega \in \Omega$ is called an **outcome** or a **sample point**.

The only requirement on Ω is that it be *exhaustive* (every possible result is listed) and that its elements be *mutually exclusive* (exactly one ω actually occurs on any given run). These two conditions are the entire content of "list everything that could happen, with no overlaps and no gaps."

Example 3.1 (Three classic sample spaces)**.**

- One coin flip: $\Omega = \{H, T\}$.
- One die roll: $\Omega = \{1, 2, 3, 4, 5, 6\}$.
- Two distinguishable dice: $\Omega = \{(i, j) : i, j \in \{1, \ldots, 6\}\}$, a set of 36 ordered pairs.

A handy way to *generate* a sample space for a multi-stage experiment is to grow it stage by stage as a tree: each level branches over the possibilities at that step, and the leaves are the outcomes. For three coin flips the tree has $2 \times 2 \times 2 = 8$ leaves, which is exactly $|\Omega|$ for $\Omega = \{H, T\}^3$ (Figure 3.1).

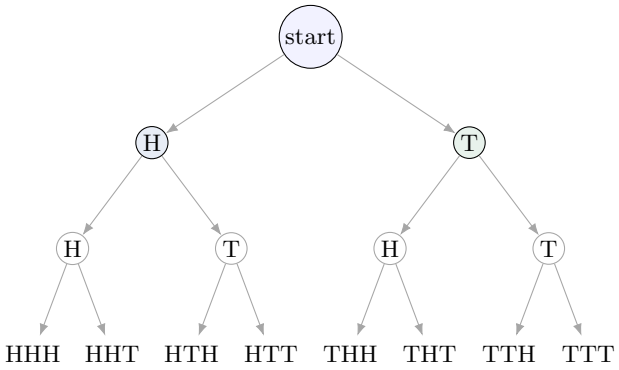


Figure 3.1: A tree for three flips of a coin. Each level branches over the result of one flip; the eight leaves are exactly the outcomes of $\Omega = \{H, T\}^3$. Trees make the multiplication of possibilities visible and are the natural scaffold for staged experiments.

Definition 3.2 (Event)**.** An **event** is a subset $A \subseteq \Omega$. We say the event A **occurs** on a particular run of the experiment if the realized outcome ω lies in A. The whole space Ω is the **sure event** (it always occurs); the empty set $\varnothing$ is the **impossible event** (it never occurs). A one-element event $\{\omega\}$ is sometimes called an **elementary event**.

This identification — *event = subset* — is the quiet hinge on which the whole theory turns, because it lets us translate ordinary spoken language about chance into the precise grammar of set theory. The translation table below repays memorizing.

Focus

The dictionary between English and sets. Let A and B be events.

English	Set notation
"A does not occur"	$A^c = \Omega \setminus A$ (complement)
"A or B occurs" (inclusive)	$A \cup B$ (union)
"both A and B occur"	$A \cap B$ (intersection)
"A but not B"	$A \cap B^c = A \setminus B$
"A implies B"	$A \subseteq B$
"A and B cannot both occur"	$A \cap B = \varnothing$ (disjoint)
"at least one of $A_1, A_2, \ldots$"	$\bigcup_i A_i$
"all of $A_1, A_2, \ldots$"	$\bigcap_i A_i$

Two events with empty intersection, $A \cap B = \varnothing$, are called **disjoint** or **mutually exclusive**: they cannot occur together. A collection of events that are pairwise disjoint and whose union is all of Ω is a **partition** — it carves the sample space into non-overlapping pieces that together cover everything. Partitions will become one of our sharpest tools when we reach the law of total probability.

Example 3.2 (Reading events off two dice)**.** On $\Omega = \{(i, j)\}$ for two dice, consider:

- $A =$ "the sum is 7" $= \{(1,6), (2,5), (3,4), (4,3), (5,2), (6,1)\}$, which has six outcomes.
- $B =$ "the first die is even" $= \{(i, j) : i \in \{2, 4, 6\}\}$, which has 18 outcomes.
- $A \cap B = \{(2,5), (4,3), (6,1)\}$: sum seven *and* first die even.
- $A^c =$ "the sum is not 7," the remaining 30 outcomes.

Notice we have not yet assigned any probabilities. We have only described *which outcomes belong to which events*. Geometry first, numbers later.

History

The word *event* (événement) and the deliberate habit of treating chance problems as questions about *sets of cases* go back to the founding correspondence between Blaise Pascal and Pierre de Fermat in 1654, prompted by gambling questions posed by the Chevalier de Méré. But it was the nineteenth-century rise of set theory under Georg Cantor, and then the twentieth-century work of Émile Borel and Henri Lebesgue on *measure* — on assigning sizes to subsets of the line — that supplied the language in which "event = subset" could be made completely rigorous. Andrey Kolmogorov's 1933 monograph *Grundbegriffe der Wahrscheinlichkeitsrechnung* ("Foundations of the Theory of Probability") brought these threads together and is the direct ancestor of every definition in this chapter.

3.1.1 Sample spaces come in three sizes

The character of a probability problem is set largely by how big Ω is, and there are exactly three regimes worth distinguishing.

- **Finite.** Ω has finitely many outcomes: coins, dice, cards, a poll of 1000 voters. Here we can in principle list every outcome and the only hard part is counting (Chapter on combinatorics).

- **Countably infinite.** Ω is infinite but can be listed $\omega_1, \omega_2, \omega_3, \ldots$, like the natural numbers. Example: flip a coin until the first head and record how many flips it took, giving $\Omega = \{1, 2, 3, \ldots\}$. We can still assign a probability to each outcome individually and add them up — but now the sum is an infinite series, so convergence enters the story.
- **Uncountable.** Ω is a continuum, like an interval of the real line: tomorrow's temperature, the exact time a bus arrives, a dart's landing point. Now the outcomes cannot be listed, and — this is the subtle part — typically *every individual outcome has probability zero*, yet whole intervals have positive probability. Here probability becomes *area under a curve*, and calculus takes over.

Connection

The jump from countable to uncountable is precisely the jump from *sums* to *integrals*. In a discrete problem the probability of an event is a sum of the probabilities of its outcomes, $\mathbb{P}(A) = \sum_{\omega \in A} \mathbb{P}(\{\omega\})$. In a continuous problem it becomes $\mathbb{P}(A) = \int_A f(x)\,\mathrm{d}x$ for a density f. The whole machinery of the calculus — limits, areas, the fundamental theorem — is exactly what we will borrow to handle uncountable sample spaces in the continuous chapters. If the symbol $\int_A f\,\mathrm{d}x$ feels rusty, the companion calculus book is the place to refresh; here we only need to believe that "total area = 1" is the continuous echo of "probabilities sum to one."

Common trap

The sample space is a modeling choice, not a fact of nature. For two dice you could use the 36 ordered pairs, or you could use the 11 possible sums $\{2, 3, \ldots, 12\}$, or the 21 unordered pairs. All three are legitimate sample spaces for "roll two dice"! They differ in what they let you talk about and in whether the outcomes are equally likely. The 36 ordered pairs are equally likely; the 11 sums are emphatically *not* (a seven is six times as likely as a two). Choosing Ω well — usually so that the outcomes are equally likely — is half the art of solving a probability problem.

3.2 The Kolmogorov axioms

We now ask the central question of the chapter: what is a probability, really? We want a function $\mathbb{P}$ that takes an event and returns a number measuring how likely it is. Rather than try to define "likely" from scratch — a swamp of philosophy — we follow Kolmogorov's strategy: write down the handful of properties any reasonable likelihood-measure *must* have, and then declare *anything* satisfying them to be a probability. This is the axiomatic method, the same move Euclid made for geometry.

Intuition

What do we already believe about likelihood, before any theory? Three things, really. **(1)** A likelihood is never negative — it makes no sense for an event to be "minus thirty percent" likely. **(2)** Something is certain to happen, so the whole sample space should carry the maximal likelihood, which we normalize to 1. **(3)** If two events cannot both occur, the chance that *one or the other* happens is just the sum of their chances — likelihoods of separate, non-overlapping possibilities add up. That is the entire intuition. The axioms below are just these

three sentences written carefully.

Definition 3.3 (Probability measure; the Kolmogorov axioms)**.** Let Ω be a sample space and let $\mathcal{F}$ be a collection of events (subsets of Ω) on which we wish to define probabilities. A **probability measure** is a function $\mathbb{P} : \mathcal{F} \to \mathbb{R}$ satisfying:

Axiom 1 (Nonnegativity). $\mathbb{P}(A) \geq 0$ for every event A.

Axiom 2 (Normalization). $\mathbb{P}(\Omega) = 1$.

Axiom 3 (Countable additivity). For any *countable* collection of *pairwise disjoint* events $A_1, A_2, A_3, \ldots$ (so $A_i \cap A_j = \varnothing$ whenever $i \neq j$),

$$\mathbb{P}\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} \mathbb{P}(A_i).$$

The triple $(\Omega, \mathcal{F}, \mathbb{P})$ is called a **probability space**.

That is the whole foundation. Everything else in probability theory — the normal distribution, the central limit theorem, Markov chains, Brownian motion — is a logical consequence of these three lines together with definitions. We assumed remarkably little, and a great deal will follow.

Why it works

Why exactly these three, and why countable additivity in particular?
Axiom 1 is forced: probabilities are sizes, and sizes are not negative.
Axiom 2 is a normalization, a choice of units. We could have used a scale from 0 to 100 (percent) or 0 to 1. Setting $\mathbb{P}(\Omega) = 1$ makes the arithmetic clean: probabilities become fractions of the total certainty, and a probability is literally "the share of the sure thing that this event claims."
Axiom 3 is the real engine, and the choice of *countable* (rather than merely *finite*) additivity is the one genuinely deep decision. Finite additivity — $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B)$ for disjoint A, B — is obvious and would be agreed to by anyone. The strengthening to countably many sets is what lets us take *limits*, and limits are the lifeblood of analysis. Want to compute the probability that a coin eventually shows heads? That is a union of infinitely many disjoint events ("first head on flip 1," "first head on flip 2," ...), and only countable additivity lets us add their probabilities to get the answer. Without it, the continuity properties at the end of this chapter fail, and with them every limit theorem in the book. Kolmogorov chose countable additivity precisely because it is the weakest assumption strong enough to make probability a branch of *measure theory*, where the full power of calculus becomes available.

What is really happening

Probability is normalized area. Picture Ω as a region in the plane with total area 1. An event A is a sub-region, and $\mathbb{P}(A)$ is simply *its area*. Axiom 1 says regions do not have negative area. Axiom 2 says the whole board has area 1. Axiom 3 says that if you chop a region into non-overlapping pieces, the areas of the pieces add up to the area of the whole. Stated this way the axioms are not mysterious at all — they are the elementary school facts

about area, transplanted onto the sample space. Everything we prove below is something you already believe about areas; we are just being careful enough to be sure. This picture is literally correct: a probability measure *is* a measure in the sense of Lebesgue, with the single extra condition that the total measure is 1.

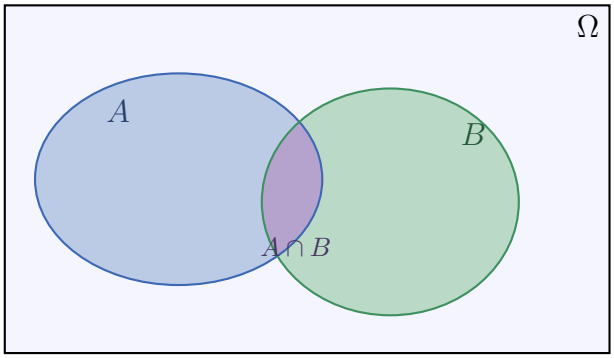


Figure 3.2: The area picture. The sample space Ω is a board of total area 1; events are sub-regions and $\mathbb{P}$ measures their area. When A and B overlap, the overlap $A \cap B$ is counted in both $\mathbb{P}(A)$ and $\mathbb{P}(B)$, which is the seed of inclusion–exclusion.

Formal

A subtle point hides in Definition 3.3: we said $\mathbb{P}$ is defined on a *collection* $\mathcal{F}$ of events, not necessarily on *all* subsets of Ω. For finite or countable Ω we may safely take $\mathcal{F}$ to be the set of all subsets, and you may ignore the distinction. But for uncountable Ω — the real line, say — it turns out to be mathematically *impossible* to assign a sensible probability to every single subset at once. We are forced to restrict $\mathcal{F}$ to a well-behaved family, a *σ-algebra*. We preview this in Section 3.6 and treat it in full in the measure-theory chapter. For now, whenever we write $\mathbb{P}(A)$, trust that A is one of the allowed events.

3.3 Basic consequences of the axioms

Now we harvest. Each result below is a theorem — it *follows* from the three axioms by logic alone. We prove them in full and in an order where each leans on the previous ones. Together they show how much a tiny axiom set yields.

3.3.1 From countable to finite additivity

First a housekeeping step: Axiom 3 is stated for infinitely many sets, but we usually have only two or three. The finite version is a corollary.

Proposition 3.1 (Finite additivity)**.** *If $A_1, \dots, A_n$ are pairwise disjoint events, then $\mathbb{P}(\bigcup_{i=1}^{n} A_i) = \sum_{i=1}^{n} \mathbb{P}(A_i)$. In particular, for disjoint A, B:* $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B)$.

Proof. We first show $\mathbb{P}(\varnothing) = 0$. The empty set is disjoint from itself, and $\varnothing = \varnothing \cup \varnothing \cup \cdots$. Apply

Axiom 3 to the countable disjoint family $A_1 = A_2 = \cdots = \varnothing$: $\mathbb{P}(\varnothing) = \sum_{i=1}^{\infty} \mathbb{P}(\varnothing)$. The only nonnegative number equal to an infinite sum of copies of itself is 0, so $\mathbb{P}(\varnothing) = 0$.

Now take the finite family $A_1, \ldots, A_n$ and pad it with empties: $A_{n+1} = A_{n+2} = \cdots = \varnothing$. This is a countable disjoint family with the same union, so Axiom 3 gives

$$\mathbb{P}\left(\bigcup_{i=1}^{n} A_i\right) = \sum_{i=1}^{\infty} \mathbb{P}(A_i) = \sum_{i=1}^{n} \mathbb{P}(A_i) + \sum_{i>n} \mathbb{P}(\varnothing) = \sum_{i=1}^{n} \mathbb{P}(A_i),$$

since the tail terms are all 0. □

Notice that countable additivity quietly did two jobs: it gave us $\mathbb{P}(\varnothing) = 0$ *and* the finite rule. This is the payoff of choosing the stronger axiom.

3.3.2 The complement rule and the unit interval

Proposition 3.2 (Complement rule)**.** *For any event A, $\mathbb{P}(A^c) = 1 - \mathbb{P}(A)$. Consequently $0 \le \mathbb{P}(A) \le 1$.*

Proof. The events A and A^c are disjoint and their union is Ω. By finite additivity and Axiom 2,

$$1 = \mathbb{P}(\Omega) = \mathbb{P}(A \cup A^c) = \mathbb{P}(A) + \mathbb{P}(A^c),$$

so $\mathbb{P}(A^c) = 1 - \mathbb{P}(A)$. Rearranging and using Axiom 1 ($\mathbb{P}(A^c) \ge 0$) gives $\mathbb{P}(A) = 1 - \mathbb{P}(A^c) \le 1$; combined with $\mathbb{P}(A) \ge 0$ we get $0 \le \mathbb{P}(A) \le 1$. □

So the familiar fact that probabilities live in $[0, 1]$ was *not* an axiom — only $\mathbb{P}(A) \ge 0$ and $\mathbb{P}(\Omega) = 1$ were assumed. The upper bound $\mathbb{P}(A) \le 1$ is a theorem we just earned. This is the kind of economy the axiomatic method buys: assume less, derive more.

Why this matters

The complement rule is the single most useful trick in elementary probability, because "not" is often vastly easier to count than the thing itself. "What is the probability that at least one of these ten components fails?" is a tangle of overlapping cases. "What is the probability that *none* fails?" is a single clean product. Whenever a problem says "at least one," your reflex should be: take the complement. We use exactly this move to crack the birthday problem below.

3.3.3 Monotonicity and the difference rule

Proposition 3.3 (Monotonicity)**.** *If $A \subseteq B$ then $\mathbb{P}(A) \le \mathbb{P}(B)$, and moreover $\mathbb{P}(B \setminus A) = \mathbb{P}(B) - \mathbb{P}(A)$.*

Proof. Split B into the part inside A and the part outside: $B = A \cup (B \setminus A)$, where A and $B \setminus A$ are disjoint (the second is $B \cap A^c$). Finite additivity gives $\mathbb{P}(B) = \mathbb{P}(A) + \mathbb{P}(B \setminus A)$. Since $\mathbb{P}(B \setminus A) \ge 0$ by Axiom 1, we get $\mathbb{P}(B) \ge \mathbb{P}(A)$; and rearranging gives the difference formula. □

Monotonicity is the formal statement of an obvious truth: *a bigger event is at least as likely as a smaller one it contains.* If A implies B (every way A can happen is also a way B happens), then B cannot be less likely than A. Many "paradoxes" in probability are really violations of monotonicity that a careless verbal description sneaks past you — the conjunction fallacy, where people rate "Linda is a feminist bank teller" as more probable than "Linda is a bank teller," is exactly a failure to notice $A \cap B \subseteq B$.

3.3.4 The inclusion–exclusion principle

When two events overlap, we cannot simply add their probabilities — we would count the overlap twice (look again at Figure 3.2). The fix is to subtract the double-counted part.

Theorem 3.1 (Inclusion–exclusion, two events)**.** *For any events A, B,*

$$\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B).$$

Proof. We decompose $A \cup B$ into three disjoint pieces: the part in A only, the part in B only, and the part in both:

$$A \cup B = (A \setminus B) \;\cup\; (B \setminus A) \;\cup\; (A \cap B),$$

a disjoint union. By finite additivity,

$$\mathbb{P}(A \cup B) = \mathbb{P}(A \setminus B) + \mathbb{P}(B \setminus A) + \mathbb{P}(A \cap B).$$

Now use the difference rule (Proposition 3.3) on the pieces of A and of B: writing $A = (A \setminus B) \cup (A \cap B)$ disjointly gives $\mathbb{P}(A \setminus B) = \mathbb{P}(A) - \mathbb{P}(A \cap B)$, and likewise $\mathbb{P}(B \setminus A) = \mathbb{P}(B) - \mathbb{P}(A \cap B)$. Substituting,

$$\mathbb{P}(A \cup B) = \big[\mathbb{P}(A) - \mathbb{P}(A \cap B)\big] + \big[\mathbb{P}(B) - \mathbb{P}(A \cap B)\big] + \mathbb{P}(A \cap B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B). \qquad \square$$

Theorem 3.2 (Inclusion–exclusion, three events)**.** *For any events A, B, C,*

$$\mathbb{P}(A \cup B \cup C) = \mathbb{P}(A) + \mathbb{P}(B) + \mathbb{P}(C) - \mathbb{P}(A \cap B) - \mathbb{P}(A \cap C) - \mathbb{P}(B \cap C) + \mathbb{P}(A \cap B \cap C).$$

The pattern continues for n events: add all single probabilities, subtract all pairwise overlaps, add back all triple overlaps, subtract the quadruples, and so on, alternating signs. The general statement is

$$\mathbb{P}\left(\bigcup_{i=1}^{n} A_i\right) = \sum_{k=1}^{n} (-1)^{k+1} \sum_{1 \le i_1 < \cdots < i_k \le n} \mathbb{P}(A_{i_1} \cap \cdots \cap A_{i_k}).$$

We will derive the general version cleanly in a later chapter using indicator random variables, where it falls out in one line; for now the two- and three-event cases are all we need.

Example 3.3 (Counting with inclusion–exclusion)**.** In a class of 100 students, 40 take French, 30 take German, and 12 take both. A student is chosen uniformly at random. What is the probability the student takes at least one of the two languages?

Let F and G be the events of taking French and German. Then $\mathbb{P}(F) = 0.40$, $\mathbb{P}(G) = 0.30$, and $\mathbb{P}(F \cap G) = 0.12$. By Theorem 3.1,

$$\mathbb{P}(F \cup G) = 0.40 + 0.30 - 0.12 = 0.58.$$

Had we forgotten to subtract the overlap, we would have wrongly reported 0.70, double-counting the 12 students who take both. The probability of taking *neither* language is then $1 - 0.58 = 0.42$ by the complement rule.

3.3.5 The union bound (Boole's inequality)

Inclusion–exclusion is exact but, for many events, a nightmare of alternating terms. Very often we do not need the exact value of $\mathbb{P}(\bigcup A_i)$ — we just need to know it is *small*. For that there is a far cheaper tool.

Theorem 3.3 (Union bound / Boole's inequality)**.** *For any events $A_1, A_2, A_3, \ldots$ (finitely or countably many, disjoint or not),*

$$\mathbb{P}\left(\bigcup_i A_i\right) \leq \sum_i \mathbb{P}(A_i).$$

Proof. The trick is *disjointification*: replace the A_i by disjoint pieces with the same union. Define

$$B_1 = A_1, \qquad B_i = A_i \setminus (A_1 \cup \cdots \cup A_{i-1}) \text{ for } i \geq 2.$$

Each B_i is the "new" part of A_i not already covered by earlier sets. By construction the B_i are pairwise disjoint, $B_i \subseteq A_i$, and $\bigcup_i B_i = \bigcup_i A_i$. Then, using countable additivity on the disjoint B_i and monotonicity ($B_i \subseteq A_i$ so $\mathbb{P}(B_i) \leq \mathbb{P}(A_i)$),

$$\mathbb{P}\left(\bigcup_i A_i\right) = \mathbb{P}\left(\bigcup_i B_i\right) = \sum_i \mathbb{P}(B_i) \leq \sum_i \mathbb{P}(A_i). \qquad \square$$

Intuition

Why is the union bound true "morally"? Because when you add up $\sum_i \mathbb{P}(A_i)$, any outcome lying in several of the A_i gets counted multiple times, whereas in $\mathbb{P}(\bigcup A_i)$ it is counted once. Overcounting can only inflate the sum, never shrink it — so the sum is an *upper bound*. The inequality is tight (an equality) exactly when the events are disjoint, in which case there is no overcounting and the union bound reduces to additivity.

Application

The union bound is the workhorse of computer science and statistics. Suppose you run 20 different hypothesis tests, each with a 5% chance of a false alarm under the null. What is the chance of *at least one* false alarm across all 20? Exact computation requires knowing the dependence structure, but the union bound gives an instant, assumption-free ceiling: $\mathbb{P}(\text{any false alarm}) \leq 20 \times 0.05 = 1.0$ — here useless, but for 5 tests it gives ≤ 0.25, a real guarantee. This is the logic behind the *Bonferroni correction*: to keep the overall error rate below α across m tests, test each at level α/m, since $m \cdot (\alpha/m) = \alpha$. The same one-line bound underlies generalization guarantees in machine learning (bounding the chance that *some* hypothesis in a class looks good by luck) and tail bounds for the maximum of many random quantities. A great deal of theoretical computer science is, at heart, the sustained and clever application of this single inequality.

3.4 Equally likely outcomes

We now connect the abstract axioms back to the concrete counting you already know. In the simplest and oldest model of chance, every outcome is just as likely as every other. The axioms then force a

single rigid conclusion.

Proposition 3.4 (The classical / equally-likely model)**.** *Suppose Ω is finite with $|\Omega| = N$ outcomes, and that all outcomes are equally likely. Then each outcome has probability $1/N$, and for any event A,*

$$\mathbb{P}(A) = \frac{|A|}{|\Omega|} = \frac{\textit{number of outcomes in } A}{\textit{total number of outcomes}}.$$

Proof. Let p be the common probability of each outcome. The N elementary events $\{\omega\}$ are disjoint and their union is Ω, so by finite additivity and normalization, $1 = \mathbb{P}(\Omega) = \sum_\omega \mathbb{P}(\{\omega\}) = Np$. Hence $p = 1/N$. For a general event A with $|A| = k$ outcomes, additivity again gives $\mathbb{P}(A) = \sum_{\omega \in A} \mathbb{P}(\{\omega\}) = k \cdot \frac{1}{N} = |A|/N$. □

This is the formula — "favorable over total" — that probability is often introduced with, and now you see it is not a definition but a *theorem*, valid *only* when outcomes are equally likely. In that regime, computing probabilities reduces entirely to *counting*, which is why combinatorics is the constant companion of elementary probability.

History

This "favorable over total" rule is the original, pre-axiomatic definition of probability, codified by Pierre-Simon Laplace in his *Théorie analytique des probabilités* (1812) as the ratio of favorable to "equally possible" cases. For a century it *was* the definition. Its fatal weakness is the circularity buried in "equally possible" — you need a notion of equal likelihood to define likelihood — and its uselessness when outcomes are not symmetric or when there are infinitely many. Kolmogorov's axioms (1933) dethroned it: now "equally likely" is just one special *assumption* you may or may not make, and the rule of Proposition 3.4 is what that assumption implies. The classical model became a corollary instead of a creed.

Example 3.4 (Two dice, carefully)**.** Roll two fair dice. Use the equally-likely space Ω of 36 ordered pairs. What is $\mathbb{P}(\text{sum} = 7)$? From Example 3.2, the event has six outcomes, so $\mathbb{P} = 6/36 = 1/6$. What about $\mathbb{P}(\text{sum} = 2)$? Only $(1, 1)$ works, so $\mathbb{P} = 1/36$. This is exactly why the alternative "sums" sample space $\{2, \ldots, 12\}$ is *not* equally likely, and why choosing the 36-pair space was the right modeling move (recall the trap box of Section 3.1.1).

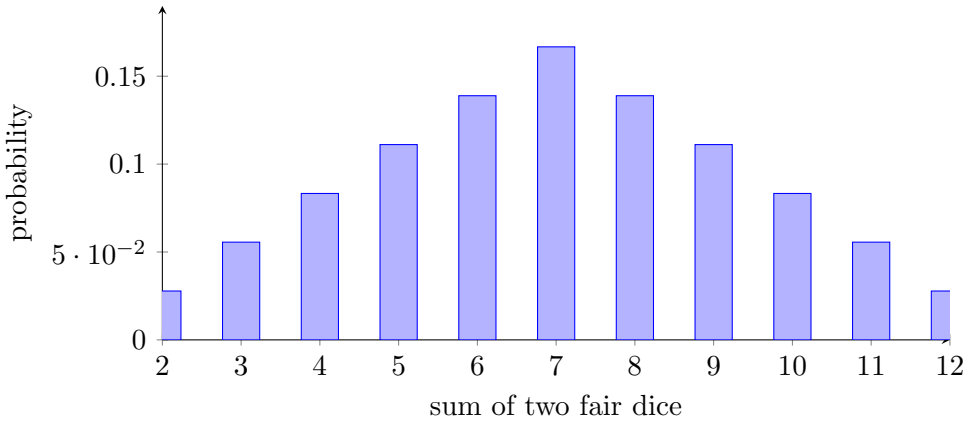


Figure 3.3: The probability mass over the eleven possible sums of two fair dice. Each of the 36 ordered pairs is equally likely, but the *sums* are not: seven (six ways, probability 6/36) is the peak, twos and twelves (one way each) the valleys. The triangular shape is the first hint of how adding independent things builds up toward a bell.

Paradox

The birthday problem. In a room of just 23 people, the probability that two share a birthday is already better than even — about 50.7%. With 50 people it rockets to 97%. People find this shocking because they unconsciously ask the wrong question ("does someone share *my* birthday?") instead of the right one ("does *any* pair share a birthday?"). The number of *pairs* among 23 people is $\binom{23}{2} = 253$, and it is those 253 chances, not 23, that drive the collision.

Here is the clean computation, using the complement trick of Section 3.3.2. Model the n birthdays as a uniform outcome in $\Omega = \{1, \ldots, 365\}^n$, all 365^n sequences equally likely (ignoring leap years and seasonal effects). Let A be "at least one shared birthday." Directly, A is a mess of overlapping coincidences. Its complement A^c = "all n birthdays distinct" is a clean count: the first person is free (365 choices), the second must dodge one date (364), the third must dodge two (363), and so on. By the equally-likely formula,

$$\mathbb{P}(A^c) = \frac{365 \cdot 364 \cdots (365 - n + 1)}{365^n}, \qquad \mathbb{P}(A) = 1 - \mathbb{P}(A^c).$$

For $n = 23$ this gives $\mathbb{P}(A^c) \approx 0.4927$, so $\mathbb{P}(A) \approx 0.5073$. The lesson generalizes far beyond birthdays: coincidences in *any* large set of *pairwise* comparisons — hash collisions in computing, matching DNA profiles, "amazing" shared traits at a party — are far more common than raw intuition suggests, because the number of pairs grows like $n^2/2$.

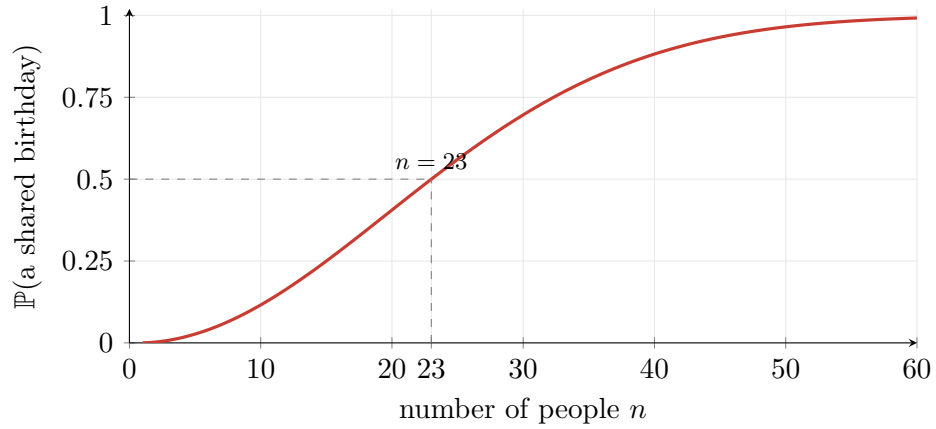


Figure 3.4: The birthday curve. The probability of a shared birthday crosses $1/2$ at just $n = 23$ people (dashed lines) and is near-certain by $n = 50$. The S-shape reflects the quadratic growth in the number of pairs. (Drawn via the accurate approximation $\mathbb{P}(A) \approx 1 - e^{-n(n-1)/730}$.)

3.5 Continuity of probability

We promised that countable additivity buys us the ability to take limits. Here is the cash payment. The idea: if a sequence of events *grows* steadily toward a limiting event, their probabilities should grow toward the limit's probability — and likewise for shrinking sequences. This is so natural you might assume it needs no proof, but in fact it is *equivalent* to countable additivity, and it is the property a merely finitely-additive theory lacks.

Definition 3.4 (Increasing and decreasing sequences of events)**.** A sequence $A_1, A_2, \ldots$ is **increasing** if $A_1 \subseteq A_2 \subseteq A_3 \subseteq \cdots$; its limit is $A = \bigcup_n A_n$. It is **decreasing** if $A_1 \supseteq A_2 \supseteq \cdots$; its limit is $A = \bigcap_n A_n$.

Theorem 3.4 (Continuity of probability)**.** *If $A_n \uparrow A$ (increasing, with union A), then $\mathbb{P}(A_n) \to \mathbb{P}(A)$. If $A_n \downarrow A$ (decreasing, with intersection A), then likewise $\mathbb{P}(A_n) \to \mathbb{P}(A)$.*

Proof. Increasing case. Turn the nested sequence into disjoint shells. Set $B_1 = A_1$ and $B_n = A_n \setminus A_{n-1}$ for $n \geq 2$. The B_n are pairwise disjoint, $\bigcup_{k=1}^{n} B_k = A_n$, and $\bigcup_{k=1}^{\infty} B_k = A$. By countable additivity and then the definition of an infinite sum as the limit of partial sums,

$$\mathbb{P}(A) = \sum_{k=1}^{\infty} \mathbb{P}(B_k) = \lim_{n\to\infty} \sum_{k=1}^{n} \mathbb{P}(B_k) = \lim_{n\to\infty} \mathbb{P}(A_n),$$

where the last equality uses finite additivity, $\sum_{k=1}^{n} \mathbb{P}(B_k) = \mathbb{P}\big(\bigcup_{k=1}^{n} B_k\big) = \mathbb{P}(A_n)$.

Decreasing case. Take complements to reduce to the increasing case. If $A_n \downarrow A$ then $A_n^c \uparrow A^c$, so the increasing case gives $\mathbb{P}(A_n^c) \to \mathbb{P}(A^c)$. By the complement rule $\mathbb{P}(A_n^c) = 1 - \mathbb{P}(A_n)$ and $\mathbb{P}(A^c) = 1 - \mathbb{P}(A)$, hence $1 - \mathbb{P}(A_n) \to 1 - \mathbb{P}(A)$, i.e. $\mathbb{P}(A_n) \to \mathbb{P}(A)$. □

What is really happening

What continuity really lets you do is reason about the infinite future. "The coin eventually shows heads," "the random walk returns to the origin," "the process never escapes this region" — all of these are statements about *limits* of events, not about any single finite stage. Continuity is the bridge that lets us compute the probability of such limiting events as the limit of probabilities we *can* compute at each finite stage. Strip out countable additivity, and that bridge collapses: you could no longer say that the chance of "eventually heads" is the limit of "heads within the first n flips." Every asymptotic theorem in this book — the laws of large numbers, recurrence of random walks, the Borel–Cantelli lemmas — crosses this bridge.

Example 3.5 (Eventually heads)**.** Flip a fair coin repeatedly. Let $A_n =$ "at least one head in the first n flips." These events increase, $A_n \uparrow A$ where $A =$ "a head appears at some point." We can compute each finite term with the complement rule: $\mathbb{P}(A_n) = 1 - \mathbb{P}(\text{all tails in } n) = 1 - (1/2)^n$. By continuity,

$$\mathbb{P}(A) = \lim_{n\to\infty} \mathbb{P}(A_n) = \lim_{n\to\infty} \left(1 - 2^{-n}\right) = 1.$$

A head appears eventually with probability 1 — certain, in the probabilistic sense, even though no individual flip is certain and the all-tails-forever outcome is not logically *impossible*. It is merely *null*: it has probability zero. This distinction between "impossible" and "probability zero" is one of the most important refinements the axioms give us, and it only makes sense because of continuity.

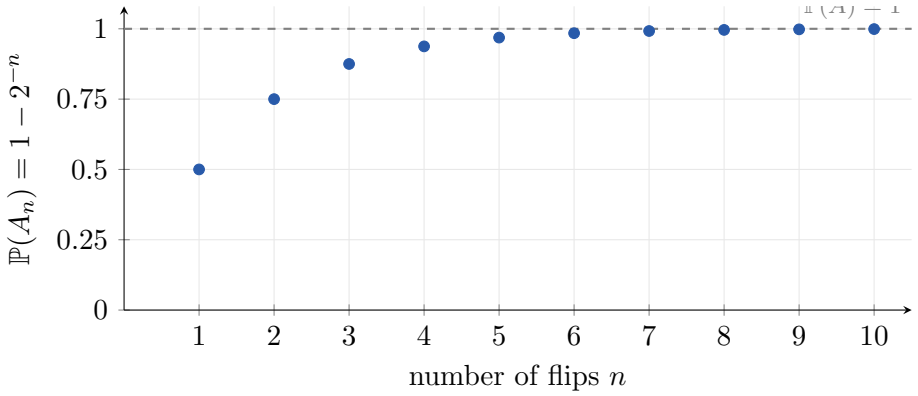


Figure 3.5: Continuity in action for "eventually heads." The finite-stage probabilities $\mathbb{P}(A_n) = 1-2^{-n}$ climb toward the limiting value $\mathbb{P}(A) = 1$ (dashed line). Continuity of probability is precisely the guarantee that the limit of these computable dots equals the probability of the limiting event.

Common trap

Probability zero does not mean impossible (and probability one does not mean certain). In Example 3.5, the outcome "tails forever" is a perfectly well-defined element of the sample space; it just sits inside a set of probability zero. In continuous models this is the norm, not the exception: every *individual* value of a continuous measurement has probability zero, yet one of them does occur. "Probability zero" means *negligible in the limit / of zero area*, which is a strictly weaker notion than "cannot happen." Conflating the two leads to genuine paradoxes; keeping them separate is a mark of real understanding.

3.6 Event algebras: a gentle preview of σ-algebras

We close by returning to the fine print of Definition 3.3: the collection $\mathcal{F}$ of events on which $\mathbb{P}$ is defined. Why not just take *all* subsets of Ω and be done with it?

Intuition

Think about which sets you actually need to be "events." If A is an event, you certainly want "not A" to be an event too — it would be bizarre to be able to bet on A but not on its negation. If A and B are events, you want "A or B" and "A and B" to be events. And from Section 3.5, if $A_1, A_2, \ldots$ are events you want their *countable* union to be an event, so that limiting statements like "eventually heads" are themselves things you can assign a probability to. A collection of sets closed under exactly these operations is what we need — nothing more, nothing less.

Definition 3.5 (σ-algebra / event algebra)**.** A collection $\mathcal{F}$ of subsets of Ω is a **σ-algebra** (or σ-field) if:

1. $\Omega \in \mathcal{F}$;
2. if $A \in \mathcal{F}$ then $A^c \in \mathcal{F}$ (closed under complements);
3. if $A_1, A_2, \cdots \in \mathcal{F}$ then $\bigcup_n A_n \in \mathcal{F}$ (closed under countable unions).

The members of $\mathcal{F}$ are exactly the **events** to which $\mathbb{P}$ assigns probabilities.

These three closure properties are deliberately the set-operation mirror of the three things we want to be able to *say* about events. From them everything else follows: $\varnothing = \Omega^c \in \mathcal{F}$, and countable *intersections* are in $\mathcal{F}$ too, because $\bigcap_n A_n = \left(\bigcup_n A_n^c\right)^c$ by De Morgan's law — complement, union, complement, all of which stay inside $\mathcal{F}$.

Example 3.6 (Two extreme σ-algebras, and a useful one)**.**

- The *trivial* σ-algebra $\{\varnothing, \Omega\}$: you can only ask "did anything happen?" Useless, but legal.
- The *power set* of all subsets: the richest possible, and the right choice whenever Ω is finite or countable.
- For one coin flip with $\Omega = \{H, T\}$, the power set is $\{\varnothing, \{H\}, \{T\}, \{H, T\}\}$ — four events. Check: it contains Ω, it is closed under complement ($\{H\}^c = \{T\}$), and under union. This is a σ-algebra with $2^2 = 4$ members.

Formal

Why can't we always use all subsets? For uncountable Ω, such as $\Omega = [0, 1]$ with "uniformly random point," it is a theorem (relying on the axiom of choice) that there is *no* way to assign a length to every subset of $[0, 1]$ while keeping countable additivity and translation invariance — the so-called non-measurable (Vitali) sets break it. The escape is to define $\mathbb{P}$ only on a σ-algebra rich enough to contain every set we could ever construct in practice — the *Borel* σ-algebra, generated by the intervals — and to simply not ask about the pathological sets. This is not a cop-out; it is the precise boundary between what can and cannot be measured, and mapping that boundary is the achievement of measure theory. We

develop all of this carefully in the measure-theoretic chapters; for now, the takeaway is that $\mathcal{F}$ exists for good reason and that for the finite and countable problems of the next many chapters you may safely take $\mathcal{F}$ to be all subsets and forget the subtlety entirely.

Connection

A σ-algebra is more than a technical fence — it encodes *information*. The events in $\mathcal{F}$ are precisely the questions you are allowed to answer; a smaller σ-algebra is a coarser, more ignorant view of the world. This reading becomes powerful when we study conditional expectation and martingales, where a growing family of σ-algebras (a *filtration*) models information accumulating over time — what you know at each moment as a random process unfolds. The humble closure rules of Definition 3.5 are the seed of that entire theory. The algebraic flavor of "closed under operations" also echoes the subspace and span ideas from the companion linear algebra book: in both cases we ask what a set of objects generates under the allowed moves.

3.7 Chapter summary

We built the formal stage for all of probability:

- A random experiment is modeled by a **sample space** Ω of outcomes; **events** are subsets, and the operations of logic (and, or, not, implies) become the operations of sets (intersection, union, complement, inclusion).
- A **probability measure** $\mathbb{P}$ obeys three Kolmogorov axioms: nonnegativity, normalization ($\mathbb{P}(\Omega) = 1$), and countable additivity over disjoint events. The triple $(\Omega, \mathcal{F}, \mathbb{P})$ is a **probability space**.
- From these three axioms alone we *derived* every elementary rule: $\mathbb{P}(\varnothing) = 0$, the complement rule $\mathbb{P}(A^c) = 1 - \mathbb{P}(A)$ (so $\mathbb{P}(A) \in [0, 1]$), monotonicity, the difference rule, inclusion–exclusion, and the union bound.
- When outcomes are **equally likely**, the axioms force $\mathbb{P}(A) = |A|/|\Omega|$, turning probability into counting — but this is a theorem about a special case, not a definition.
- **Continuity of probability** (equivalent to countable additivity) lets us pass to limits of growing or shrinking events, which is what every asymptotic theorem in the book depends on, and which forces us to distinguish *probability zero* from *impossible*.
- The events form a **σ-algebra** $\mathcal{F}$, closed under complement and countable union — a structure that is invisible for finite problems but indispensable for the continuum, and which secretly encodes *information*.

Everything that follows — conditional probability, random variables, expectation, the great limit theorems — is built on top of this small, sturdy foundation. With the stage set, we can bring on the actors.

3.8 Exercises

Exercise 3.1 (Set translations)**.** Let A, B, C be events. Express each of the following in set notation, and then in words describe its complement.

(a) Exactly one of A, B occurs.

(b) At least two of A, B, C occur.

(c) None of A, B, C occurs.

Exercise 3.2 (Warm-up computations)**.** Suppose $\mathbb{P}(A) = 0.6$, $\mathbb{P}(B) = 0.5$, and $\mathbb{P}(A \cap B) = 0.3$. Compute $\mathbb{P}(A \cup B)$, $\mathbb{P}(A^c)$, $\mathbb{P}(A^c \cap B^c)$, $\mathbb{P}(A \cap B^c)$, and $\mathbb{P}(A^c \cup B^c)$. (Hint: De Morgan turns the last two complements into things you already have.)

Exercise 3.3 (A bound becomes an equality)**.** For events A and B, prove that $\mathbb{P}(A \cap B) \geq \mathbb{P}(A) + \mathbb{P}(B) - 1$. This is *Bonferroni's inequality*; show it follows from $\mathbb{P}(A \cup B) \leq 1$ and inclusion–exclusion. When is it an equality?

Exercise 3.4 (Counting with equally likely outcomes)**.** A fair die is rolled twice. Using the 36-element equally-likely sample space, find the probability that (a) the maximum of the two rolls is 5; (b) the two rolls differ by exactly 2; (c) at least one roll is a 6. For part (c), do it twice — directly and via the complement — and check the answers agree.

Exercise 3.5 (Why not finite additivity alone?)**.** Consider $\Omega = \{1, 2, 3, \ldots\}$ and the geometric assignment $\mathbb{P}(\{k\}) = 2^{-k}$. (a) Verify $\mathbb{P}(\Omega) = 1$ using countable additivity. (b) Compute $\mathbb{P}(\{k : k \text{ even}\})$. (c) Explain in a sentence why finite additivity alone would not even let you conclude $\mathbb{P}(\Omega) = 1$ from the individual outcome probabilities.

Exercise 3.6 (Continuity in action)**.** Let A_n = "the first head in repeated fair-coin flips occurs on or before flip n."

(a) Are the A_n increasing or decreasing? What is their limit event?

(b) Compute $\mathbb{P}(A_n)$ and take the limit using Theorem 3.4.

(c) Now let B_n = "the first n flips are all heads." Are these increasing or decreasing? Find $\bigcap_n B_n$ and its probability, and interpret the result in light of the trap box on probability zero.

Exercise 3.7 (Inclusion–exclusion for three)**.** In a survey, 50% of people own a cat, 40% own a dog, 30% own a fish; 20% own both a cat and a dog, 15% a cat and a fish, 10% a dog and a fish, and 5% own all three. Using Theorem 3.2, find the fraction who own *at least one* of the three pets, and then the fraction who own *none*.

Exercise 3.8 (The union bound is sometimes all you can do)**.** A server runs 100 independent jobs; each fails with probability 0.001. Give an upper bound on the probability that *at least one* job fails using the union bound. Then compute the *exact* probability using independence ($1 - 0.999^{100}$) and compare. By roughly how much does the union bound overshoot, and why is the gap so small here?

Exercise 3.9 (Proving inclusion–exclusion for three events)**.** Prove Theorem 3.2 from the two-event version. (Hint: write $A \cup B \cup C = (A \cup B) \cup C$, apply two-event inclusion–exclusion to the outer union, then distribute $(A \cup B) \cap C = (A \cap C) \cup (B \cap C)$ and apply it again. Keep careful track of the triple-overlap term.)

Exercise 3.10 (A σ-algebra by hand)**.** Let $\Omega = \{1, 2, 3, 4\}$. (a) Write down the smallest σ-algebra containing the single set $\{1\}$. (b) How many members does the power-set σ-algebra of Ω have? (c) Give an example of a collection of subsets of Ω that is *not* a σ-algebra, and state which closure property fails.

Exercise 3.11 (Boole, Bonferroni, and a sandwich)**.** For events $A_1, \dots, A_n$, the union bound gives an upper estimate of $\mathbb{P}(\bigcup_i A_i)$. Prove the matching *lower* estimate $\mathbb{P}(\bigcup_i A_i) \geq \sum_i \mathbb{P}(A_i) - \sum_{i<j} \mathbb{P}(A_i \cap A_j)$. (Hint: this is the first Bonferroni inequality; induct on n using two-event inclusion–exclusion and monotonicity.) Together with Boole's inequality this sandwiches the union probability between two computable bounds.

Exercise 3.12 (Conceptual: choosing the sample space)**.** You flip a fair coin three times. Propose two *different* sample spaces for this experiment: one in which the outcomes are equally likely, and one in which they are not. For each, give the sample space explicitly, state $\mathbb{P}$ of one outcome, and explain which space you would prefer for computing "the probability of getting exactly two heads," and why.

Chapter 4

Conditional Probability and Independence

So far we have learned to assign probabilities to events in a fixed sample space. We chose a model – a list of outcomes and a way of weighting them – and from that model we read off numbers like $\mathbb{P}(A)$. But almost no interesting question about the world is asked in a vacuum. We never really want to know "what is the probability it rains?" We want to know the probability it rains *given* that the sky is grey, *given* that it is November, *given* that the forecast said seventy percent. Probability becomes useful at exactly the moment we learn something and ask how that knowledge should change our beliefs.

This chapter is about that moment. *Conditional probability* is the mathematics of updating, and it may be the most important single idea in the whole subject: Bayes' theorem (next chapter), Markov chains, statistical inference, machine learning – all of them are conditional probability wearing different costumes. We build the idea from a concrete picture, make it precise, and then meet its three companions: the multiplication and chain rules (how to build big probabilities out of small conditional ones), independence (the precise statement that one event tells you *nothing* about another), and the law of total probability (how to reason case by case). We also resolve the classic confusion between "independent" and "disjoint," which trips up nearly everyone the first time.

4.1 Conditioning Is Restricting the Universe

Begin with a picture worth keeping.

Intuition

A probability is a fraction of a universe. When you *condition* on an event B, you throw away the rest of the universe and pretend B *is* the universe. Probabilities of other events get *renormalized* so that, inside the new universe B, the total is again 1. Conditioning is zooming in.

Here is the canonical example. Roll a fair six-sided die. The sample space is $\Omega = \{1, 2, 3, 4, 5, 6\}$, each outcome weight $1/6$. Let $A = \{\text{the roll is a } 2\} = \{2\}$, so $\mathbb{P}(A) = 1/6$. Now a friend who can see the die tells you, truthfully, "the roll is even." Call that event $B = \{2, 4, 6\}$. *Given* this information,

what is the probability the roll is a 2?

Three outcomes remain possible: $2, 4, 6$. They were equally likely before, and learning "even" gives no reason to prefer one even number over another, so they should stay equally likely. Each now carries probability $1/3$. The answer is $1/3$. We write this

$$\mathbb{P}(A \mid B) = \frac{1}{3},$$

read "the probability of A given B." Notice what happened. The single favorable outcome (rolling a 2) used to be one of six; now it is one of three. The information *shrank the universe* from $\{1, \dots, 6\}$ to $\{2, 4, 6\}$, and the probability of A rose accordingly, from $1/6$ to $1/3$.

Where exactly did $1/3$ come from arithmetically? The favorable outcomes are those in A that are still possible, i.e. those in A *and* B: here $A \cap B = \{2\}$, with probability $1/6$. The whole new universe is B, with probability $1/2$. And indeed

$$\frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)} = \frac{1/6}{1/2} = \frac{1}{3}.$$

This little computation is the entire definition, but it pays to see *why* it must take this shape before we write it down formally.

Why it works

Conditioning on B means: among all the probability mass, keep only the part sitting inside B, and rescale it to total 1. The mass of A that survives is the mass in $A \cap B$ – the part of A inside the new universe. Dividing by $\mathbb{P}(B)$ is the rescaling that makes the surviving mass add up to 1 again, because $\mathbb{P}(B \mid B) = \mathbb{P}(B \cap B)/\mathbb{P}(B) = 1$. So the formula is not an arbitrary choice; it is the *only* way to "keep B, renormalize" that respects the original proportions.

Definition 4.1 (Conditional probability)**.** Let A and B be events with $\mathbb{P}(B) > 0$. The *conditional probability of A given B* is

$$\mathbb{P}(A \mid B) \;=\; \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}.$$

If $\mathbb{P}(B) = 0$ the expression is undefined (you cannot condition on something impossible – there is no universe to zoom into).

Here is a subtle but crucial fact. Once you fix B, the map $A \mapsto \mathbb{P}(A \mid B)$ is itself a genuine probability. It is not some lesser, second-class number; it obeys all the same axioms, and we lean on that repeatedly.

Proposition 4.1 (Conditioning yields a probability measure)**.** *Fix B with $\mathbb{P}(B) > 0$. Define $\mathbb{P}_B(A) := \mathbb{P}(A \mid B)$. Then $\mathbb{P}_B$ satisfies the axioms of probability:*

1. *$\mathbb{P}_B(A) \geq 0$ for every event A;*
2. *$\mathbb{P}_B(\Omega) = 1$;*
3. *if $A_1, A_2, \dots$ are disjoint, then $\mathbb{P}_B(\bigcup_i A_i) = \sum_i \mathbb{P}_B(A_i)$.*

Proof. Nonnegativity is immediate: $\mathbb{P}(A \cap B) \geq 0$ and $\mathbb{P}(B) > 0$, so the ratio is ≥ 0. For (2), $\mathbb{P}_B(\Omega) = \mathbb{P}(\Omega \cap B)/\mathbb{P}(B) = \mathbb{P}(B)/\mathbb{P}(B) = 1$. For (3), if the A_i are disjoint then so are the sets

$A_i \cap B$ (subsets of disjoint sets are disjoint), and using countable additivity of the original $\mathbb{P}$,

$$\mathbb{P}_B\Big(\bigcup_i A_i\Big) = \frac{\mathbb{P}\left(\bigcup_i (A_i \cap B)\right)}{\mathbb{P}(B)} = \frac{\sum_i \mathbb{P}(A_i \cap B)}{\mathbb{P}(B)} = \sum_i \frac{\mathbb{P}(A_i \cap B)}{\mathbb{P}(B)} = \sum_i \mathbb{P}_B(A_i). \qquad \square$$

What is really happening

Because $\mathbb{P}_B$ is a bona fide probability measure, *every theorem you ever prove about probabilities automatically holds for conditional probabilities*, as long as you keep the same conditioning event on every term. For instance the complement rule becomes $\mathbb{P}(A^c \mid B) = 1 - \mathbb{P}(A \mid B)$, and inclusion–exclusion becomes $\mathbb{P}(A_1 \cup A_2 \mid B) = \mathbb{P}(A_1 \mid B) + \mathbb{P}(A_2 \mid B) - \mathbb{P}(A_1 \cap A_2 \mid B)$. You get a whole parallel world of identities for free. The one thing you may *not* do is mix conditioning events – $\mathbb{P}(A \mid B)$ and $\mathbb{P}(A \mid C)$ live in different universes and cannot be combined naively.

4.1.1 A picture: areas inside a square

Area makes conditioning concrete. Imagine the unit square as your sample space, total area 1, with two overlapping regions A and B. The probability of an event is its area. Conditioning on B means we crop the picture down to B and ask what fraction of *that* cropped region is also in A.

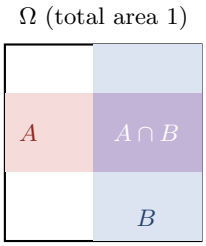


Figure 4.1: Conditioning as cropping. The probability $\mathbb{P}(A \mid B)$ is the area of the purple overlap $A \cap B$ divided by the area of the blue band B: of the part of the world where B happens, what fraction also has A?

If B has area 0.4 and $A \cap B$ has area 0.16, then $\mathbb{P}(A \mid B) = 0.16/0.4 = 0.4$. Note that $\mathbb{P}(A \mid B)$ can be larger or smaller than $\mathbb{P}(A)$ depending on whether A and B "like" each other; conditioning can raise a probability or lower it. That asymmetry of effect is the whole reason information is valuable.

History

The phrase "conditional probability" is modern, but the idea is old. Abraham de Moivre, in *The Doctrine of Chances* (1718), spoke of the probability of an event "upon supposition" that another had happened, and used exactly the multiplication idea to chain dependent draws from urns. Thomas Bayes (whose theorem we meet next chapter) and Pierre-Simon Laplace turned conditioning into a tool for *inference* – reasoning backward from observed effects to unknown causes. It was not until Andrey Kolmogorov's 1933 axiomatization, *Grundbegriffe der Wahrscheinlichkeitsrechnung*, that $\mathbb{P}(A \mid B) = \mathbb{P}(A \cap B)/\mathbb{P}(B)$ was crowned a *definition* rather than a derived fact – and that Kolmogorov showed how to extend it, via conditional

expectation, to the awkward case $\mathbb{P}(B) = 0$ that haunts continuous variables. We will return to that deep version much later in the book.

Walking question

A standard deck has 52 cards, 4 of them aces. You draw one card. Let $A =$ "it is the ace of spades" and $B =$ "it is a spade." Without looking, what is $\mathbb{P}(A \mid B)$? And what is $\mathbb{P}(B \mid A)$? Try both before reading on.
There are 13 spades; exactly one is the ace of spades, so $\mathbb{P}(A \mid B) = 1/13$. Conversely, if you already know it is the ace of spades, then it is certainly a spade, so $\mathbb{P}(B \mid A) = 1$. The two conditional probabilities are wildly different – a first warning that $\mathbb{P}(A \mid B)$ and $\mathbb{P}(B \mid A)$ are not interchangeable. Confusing them is the engine behind a whole family of fallacies, and untangling them is what Bayes' theorem is for.

4.2 The Multiplication Rule and the Chain Rule

The definition of conditional probability has a twin, obtained simply by clearing the denominator. It looks almost too simple to deserve a name, yet it is one of the great workhorses of the subject.

Proposition 4.2 (Multiplication rule)**.** *For events with* $\mathbb{P}(B) > 0$*,*

$$\mathbb{P}(A \cap B) = \mathbb{P}(B)\ \mathbb{P}(A \mid B).$$

By symmetry, if $\mathbb{P}(A) > 0$ *as well, also* $\mathbb{P}(A \cap B) = \mathbb{P}(A)\ \mathbb{P}(B \mid A)$*.*

The reason this is so useful is that real processes happen *in sequence*, and at each step it is usually easy to say what happens next *given* everything so far. The multiplication rule lets us assemble the probability of a whole sequence out of these easy one-step conditional probabilities. Think of drawing cards, drawing balls from an urn, or any multi-stage experiment.

Example 4.1 (Two aces in a row)**.** Draw two cards from a well-shuffled deck without replacement. What is the probability both are aces? Let $A_1 =$ "first card is an ace," $A_2 =$ "second card is an ace." We want $\mathbb{P}(A_1 \cap A_2)$. The first card is an ace with probability $4/52$. *Given* the first was an ace, only 3 aces remain among 51 cards, so $\mathbb{P}(A_2 \mid A_1) = 3/51$. By the multiplication rule,

$$\mathbb{P}(A_1 \cap A_2) = \frac{4}{52} \cdot \frac{3}{51} = \frac{12}{2652} = \frac{1}{221} \approx 0.0045.$$

The key move is that the *second* factor is a conditional probability computed in the smaller, post-draw universe of 51 cards. This is why drawing without replacement is so naturally a conditional-probability problem.

The multiplication rule extends to any number of stages. This generalization is the *chain rule*, and it is the formula that powers everything from sequence models in machine learning to the analysis of card games.

Theorem 4.1 (Chain rule / general multiplication rule)**.** *For events* $A_1, A_2, \ldots, A_n$ *with* $\mathbb{P}(A_1 \cap \cdots \cap A_{n-1}) > 0$*,*

$$\mathbb{P}(A_1 \cap A_2 \cap \cdots \cap A_n) = \mathbb{P}(A_1)\ \mathbb{P}(A_2 \mid A_1)\ \mathbb{P}(A_3 \mid A_1 \cap A_2) \cdots \mathbb{P}(A_n \mid A_1 \cap \cdots \cap A_{n-1}).$$

Proof. This is a telescoping product. Each conditional probability is a ratio:

$$\mathbb{P}(A_1) \cdot \frac{\mathbb{P}(A_1 \cap A_2)}{\mathbb{P}(A_1)} \cdot \frac{\mathbb{P}(A_1 \cap A_2 \cap A_3)}{\mathbb{P}(A_1 \cap A_2)} \cdots \frac{\mathbb{P}(A_1 \cap \cdots \cap A_n)}{\mathbb{P}(A_1 \cap \cdots \cap A_{n-1})}.$$

Every numerator cancels the next denominator, leaving $\mathbb{P}(A_1 \cap \cdots \cap A_n)$. The positivity assumption guarantees no denominator is zero. (Formally one proves it by induction on n, the base case $n = 2$ being the multiplication rule.) □

Intuition

The chain rule says: *to walk a path through a tree of possibilities, multiply the probability of each step given everything that happened before it.* Probability of a whole story = probability of the first chapter × probability of the second chapter given the first × .... This is exactly how you compute the probability of any particular branch in a probability tree, which we draw next.

Example 4.2 (No shared birthday, retold as a chain)**.** Here is the famous birthday problem seen through the chain rule. With k people and 365 equally likely birthdays, what is the probability that all k birthdays are distinct? Line the people up. Person 1 can have any birthday. Person 2 must avoid person 1's day: probability 364/365 *given* person 1's birthday. Person 3 must avoid both: 363/365. In general person j must dodge the $j - 1$ days already used. The chain rule multiplies these:

$$\mathbb{P}(\text{all distinct}) = \frac{365}{365} \cdot \frac{364}{365} \cdot \frac{363}{365} \cdots \frac{365 - k + 1}{365} = \prod_{j=0}^{k-1} \frac{365 - j}{365}.$$

For $k = 23$ this product is about 0.493, so the probability that *some* pair shares a birthday is $1 - 0.493 = 0.507$ – just over half. The famous shock is that 23 feels small. The chain-rule view explains the shock: each new person creates a fresh chance to collide *with every earlier person*, so the number of opportunities for a match grows like $\binom{k}{2}$, not like k.

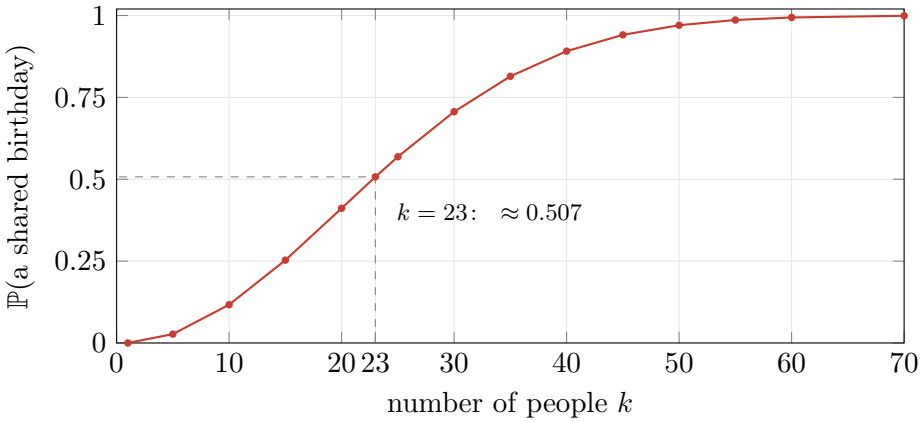


Figure 4.2: The birthday problem. The probability that at least two of k people share a birthday, computed by the chain rule of Example 4.2. It crosses 1/2 already at $k = 23$ and is near-certain by $k = 60$.

Connection

The chain rule is the mathematical backbone of *autoregressive language models*. A sentence $w_1 w_2 \cdots w_n$ is assigned the probability

$$\mathbb{P}(w_1, \ldots, w_n) = \mathbb{P}(w_1)\ \mathbb{P}(w_2 \mid w_1)\ \mathbb{P}(w_3 \mid w_1, w_2) \cdots \mathbb{P}(w_n \mid w_1, \ldots, w_{n-1}),$$

which is exactly Theorem 4.1 with "events" replaced by "the next word is so-and-so." A neural network's only job is to estimate each conditional factor $\mathbb{P}(w_j \mid \text{everything before})$. Generating text means sampling one factor at a time and feeding the result back in. When people say a model "predicts the next token," they are describing one term of a chain-rule product.

4.3 Independence: When Information Tells You Nothing

Conditioning measures how much one event changes the probability of another. Sometimes the answer is: not at all. Learning B leaves the probability of A exactly where it was. That is the precise meaning of *independence*, and it is one of the most important – and most misunderstood – concepts in probability.

Intuition

A and B are *independent* if knowing whether B happened gives you zero information about whether A happened: $\mathbb{P}(A \mid B) = \mathbb{P}(A)$. The two events do not "talk" to each other.

That intuition, $\mathbb{P}(A \mid B) = \mathbb{P}(A)$, is the right mental image, but it has an annoying flaw: it requires $\mathbb{P}(B) > 0$ to even be written down, and it is asymmetric in appearance. Multiplying both sides by $\mathbb{P}(B)$ fixes both problems and gives the clean, symmetric definition we actually use.

Definition 4.2 (Independence of two events)**.** Events A and B are *independent*, written $A \perp\!\!\!\perp B$, if

$$\mathbb{P}(A \cap B) = \mathbb{P}(A)\ \mathbb{P}(B).$$

This version is symmetric (it treats A and B the same) and it makes sense even when $\mathbb{P}(A)$ or $\mathbb{P}(B)$ is zero. When $\mathbb{P}(B) > 0$ it is equivalent to the intuitive form: dividing $\mathbb{P}(A \cap B) = \mathbb{P}(A)\,\mathbb{P}(B)$ by $\mathbb{P}(B)$ gives $\mathbb{P}(A \mid B) = \mathbb{P}(A)$. And equally, dividing by $\mathbb{P}(A)$ gives $\mathbb{P}(B \mid A) = \mathbb{P}(B)$ – so independence really is a two-way street. If B tells you nothing about A, then A tells you nothing about B.

Example 4.3 (Two coin flips)**.** Flip a fair coin twice. Let $A =$ "first flip is heads" and $B =$ "second flip is heads." Each has probability $1/2$. The four outcomes HH, HT, TH, TT are equally likely, so $\mathbb{P}(A \cap B) = \mathbb{P}(HH) = 1/4 = (1/2)(1/2) = \mathbb{P}(A)\,\mathbb{P}(B)$. They are independent, matching physical intuition: a coin has no memory. Now let $C =$ "the two flips agree" $= \{HH, TT\}$. Then $\mathbb{P}(C) = 1/2$, and $\mathbb{P}(A \cap C) = \mathbb{P}(HH) = 1/4 = \mathbb{P}(A)\,\mathbb{P}(C)$, so A and C are *also* independent – even though C is built out of A! Knowing the first flip is heads tells you nothing about whether the flips agree, because the second flip is still a free coin. Independence can be surprising; always check the product, do not trust your gut.

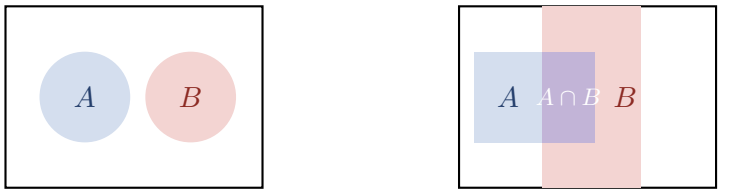
4.3.1 The trap: independent is NOT disjoint

We now confront the single most common error in elementary probability. Students hear "A and B don't affect each other" and reach for the picture of two non-overlapping blobs – disjoint events. This is exactly backward.

Common trap

Disjoint and independent are nearly opposite. Disjoint (mutually exclusive) means $A \cap B = \emptyset$: the two events *cannot both happen*. Independent means $\mathbb{P}(A \cap B) = \mathbb{P}(A)\,\mathbb{P}(B)$: knowing one happened *does not change the odds* of the other. These are not the same idea – and for events of positive probability they actively *conflict*.

Suppose A and B are disjoint with $\mathbb{P}(A) > 0$ and $\mathbb{P}(B) > 0$. Then $\mathbb{P}(A \cap B) = \mathbb{P}(\emptyset) = 0$, but $\mathbb{P}(A)\,\mathbb{P}(B) > 0$. So $\mathbb{P}(A \cap B) \neq \mathbb{P}(A)\,\mathbb{P}(B)$, and they are *not* independent. In fact they are maximally *dependent*: if B happens, A is now *impossible* ($\mathbb{P}(A \mid B) = 0$), which is the largest possible swing from its prior value. Disjoint events scream information at each other; independent events whisper nothing.



Disjoint: $A \cap B = \emptyset$ if B happens, A is impossible – strongly dependent

Independent: they overlap overlap area $= \mathbb{P}(A)\,\mathbb{P}(B)$ – just the right amount

Figure 4.3: Disjoint (left) versus independent (right). Independent events must *overlap*; the overlap has area exactly $\mathbb{P}(A)\,\mathbb{P}(B)$. Disjoint events never overlap, which makes them strongly dependent, not independent.

Why it works

Why is the picture for independence an *overlap of exactly the right size*? Think of B as a vertical band covering a fraction $\mathbb{P}(B)$ of the square (as in Figure 4.1). Independence says A occupies the same fraction $\mathbb{P}(A)$ *inside* the band B as it does in the whole square. Geometrically, A must slice through B in the same proportion as it slices through everything. A horizontal band A does exactly this: it covers fraction $\mathbb{P}(A)$ of every vertical strip, including the strip B. Their overlap is then $\mathbb{P}(A) \cdot \mathbb{P}(B)$ by the area of a rectangle. Disjointness, where A dodges B entirely, is the most violent possible departure from "same proportion inside as outside."

4.3.2 Independence of more than two events

For three or more events, independence is fussier than you might guess. It is not enough that they be independent in pairs; you also need the full product rule to hold for *every* sub-collection.

Definition 4.3 (Mutual independence)**.** Events $A_1, \dots, A_n$ are *(mutually) independent* if for every sub-collection $\{i_1, \dots, i_k\} \subseteq \{1, \dots, n\}$,

$$\mathbb{P}(A_{i_1} \cap \cdots \cap A_{i_k}) = \mathbb{P}(A_{i_1}) \cdots \mathbb{P}(A_{i_k}).$$

If only the two-at-a-time equations hold, the events are merely *pairwise* independent.

Pairwise independence is strictly weaker than mutual independence. The classic example – due to the Russian probabilist Sergei Bernstein – shows three events that are independent in every pair yet badly dependent as a trio.

Example 4.4 (Bernstein's three events: pairwise but not mutually independent)**.** Flip a fair coin twice. Define

$$A = \{\text{first flip is } H\}, \quad B = \{\text{second flip is } H\}, \quad C = \{\text{the flips differ}\}.$$

The equally likely outcomes are HH, HT, TH, TT. Then $\mathbb{P}(A) = \mathbb{P}(B) = \mathbb{P}(C) = 1/2$. Check the pairs:

$$\mathbb{P}(A \cap B) = \mathbb{P}(HH) = \tfrac{1}{4} = \mathbb{P}(A)\,\mathbb{P}(B), \quad \mathbb{P}(A \cap C) = \mathbb{P}(HT) = \tfrac{1}{4} = \mathbb{P}(A)\,\mathbb{P}(C),$$

$$\mathbb{P}(B \cap C) = \mathbb{P}(TH) = \tfrac{1}{4} = \mathbb{P}(B)\,\mathbb{P}(C).$$

All three pairs are independent. But the triple is not: $A \cap B \cap C$ requires the first flip H, the second H, *and* the flips to differ – impossible. So $\mathbb{P}(A \cap B \cap C) = 0$, while $\mathbb{P}(A)\,\mathbb{P}(B)\,\mathbb{P}(C) = 1/8$. The two are unequal, so the three events are *not* mutually independent. Indeed, once you know A and B both happened (both heads), you know with certainty that C did *not*. Any two of them stay quiet; all three together leak everything.

Common trap

Do not report "these events are independent" after checking only the pairs. Mutual independence of n events is $2^n - n - 1$ separate equations, not $\binom{n}{2}$. In practice we usually *build* independence into a model ("assume the flips are independent") rather than verify it; but when you must check, check all the way up.

4.4 Conditional Independence

Independence can appear, vanish, or materialize once you condition on extra information. This relativity is captured by *conditional independence*, and it is the concept on which graphical models, Naive Bayes classifiers, and Markov chains are all built.

Definition 4.4 (Conditional independence)**.** Events A and B are *conditionally independent given* C (with $\mathbb{P}(C) > 0$) if

$$\mathbb{P}(A \cap B \mid C) = \mathbb{P}(A \mid C)\ \mathbb{P}(B \mid C).$$

Equivalently, when the conditional probabilities are defined, $\mathbb{P}(A \mid B \cap C) = \mathbb{P}(A \mid C)$: once you know C, learning B in addition tells you nothing more about A.

The phrase "once you know C" is the heart of it. Conditional independence says C is the *whole story* connecting A and B; with C pinned down, the remaining wobble in A and B is unrelated. Two facts that look correlated in the wild can become independent once you control for a common cause – and, just as strikingly, two genuinely independent facts can become *dependent* once you condition on a common effect.

Example 4.5 (Two coins in a drawer: conditioning creates dependence)**.** A drawer holds two coins. Coin F is fair ($\mathbb{P}(H) = 1/2$); coin D is double-headed ($\mathbb{P}(H) = 1$). You pick a coin at random (each with probability $1/2$) and, without looking at it, flip it twice. Let H_1 and H_2 be the events that the first and second flips land heads.

Given which coin you hold, the flips are independent – that is how physical coins work. Formally, $\mathbb{P}(H_1 \cap H_2 \mid F) = (1/2)(1/2) = 1/4 = \mathbb{P}(H_1 \mid F)\,\mathbb{P}(H_2 \mid F)$, and given D both are certain. So $H_1 \perp\!\!\!\perp H_2$ *conditionally on the coin's identity.*

But *unconditionally* they are dependent. By the law of total probability (next section),

$$\mathbb{P}(H_1) = \tfrac{1}{2}\cdot\tfrac{1}{2} + \tfrac{1}{2}\cdot 1 = \tfrac{3}{4}, \qquad \mathbb{P}(H_1 \cap H_2) = \tfrac{1}{2}\cdot\tfrac{1}{4} + \tfrac{1}{2}\cdot 1 = \tfrac{5}{8}.$$

Now $\mathbb{P}(H_1)\,\mathbb{P}(H_2) = (3/4)^2 = 9/16 = 0.5625$, but $\mathbb{P}(H_1 \cap H_2) = 5/8 = 0.625$. Not equal – the flips are *dependent*. Intuitively: seeing the first flip land heads is *evidence you might be holding the double-headed coin*, which makes the second flip more likely to be heads too. The flips talk to each other through the hidden coin identity. Pin down the coin (condition on F or D) and they fall silent again.

What is really happening

Example 4.5 is the engine of Bayesian learning compressed into two flips. Each observation updates your belief about a hidden quantity (which coin), and that updated belief changes your prediction for the next observation. The flips are independent *given the truth* but dependent *to you*, because you do not know the truth and are learning it from the data. This is precisely why repeated measurements are informative: each one tells you a little about the hidden parameter, which sharpens your forecast of the rest.

Common trap

Conditional independence and ordinary independence are logically separate – neither implies the other.

- Example 4.5: independent given C, but *not* independent unconditionally.
- Reverse case: A and B can be unconditionally independent yet *dependent* given C. Let A, B be two independent fair coin flips and let $C =$ "the flips differ." Marginally $A \perp\!\!\!\perp B$. But given C, knowing $A = H$ forces $B = T$: now they are perfectly dependent. Conditioning on a common *effect* (here, the parity C) can manufacture dependence between independent causes – the phenomenon statisticians call *collider bias* or "explaining away."

Always state *what you are conditioning on*; independence is not a property of two events alone but of two events *relative to a state of knowledge.*

Application

The *Naive Bayes* classifier – still a strong baseline for spam filtering and text categorization – rests entirely on a conditional-independence assumption. To classify an email as spam or not, it assumes that, *given the class* (spam or ham), the presence of each word is independent of the others. That is almost certainly false in reality ("viagra" and "cheap" co-occur), which is why it is called *naive*. Yet conditioning on the class soaks up most of the dependence between words, leaving residual correlations small enough that the classifier works astonishingly well. The assumption turns an intractable joint distribution over thousands of words into a simple product of per-word conditional probabilities – the chain rule with all the cross-terms thrown away.

4.5 The Law of Total Probability

We often cannot compute $\mathbb{P}(A)$ directly, but we *can* compute it on each of several mutually exclusive scenarios. The law of total probability stitches those cases back together: split the world into scenarios, find the probability of A within each, and average, weighting each scenario by how likely it is.

Intuition

To find an overall probability, slice the sample space into cases that cover every possibility and never overlap (a *partition*). The total probability of A is a weighted average of its probability in each case, the weights being the probabilities of the cases. "Reason case by case, then average."

Theorem 4.2 (Law of total probability)**.** *Let $B_1, B_2, \ldots, B_n$ be a* partition *of Ω: they are disjoint ($B_i \cap B_j = \emptyset$ for $i \neq j$) and exhaustive ($\bigcup_i B_i = \Omega$), with $\mathbb{P}(B_i) > 0$ for each i. Then for any event A,*

$$\mathbb{P}(A) = \sum_{i=1}^{n} \mathbb{P}(A \cap B_i) = \sum_{i=1}^{n} \mathbb{P}(A \mid B_i)\ \mathbb{P}(B_i).$$

Proof. Because the B_i partition Ω, the sets $A \cap B_i$ partition A: every outcome of A lands in exactly one B_i. So $A = \bigcup_i (A \cap B_i)$ as a disjoint union, and countable additivity gives $\mathbb{P}(A) = \sum_i \mathbb{P}(A \cap B_i)$. The second equality applies the multiplication rule $\mathbb{P}(A \cap B_i) = \mathbb{P}(A \mid B_i)\,\mathbb{P}(B_i)$ to each term. □

The simplest and most-used version uses a two-case partition, B and its complement B^c:

$$\mathbb{P}(A) = \mathbb{P}(A \mid B)\ \mathbb{P}(B) + \mathbb{P}(A \mid B^c)\ \mathbb{P}(B^c).$$

This one line resolves a huge fraction of word problems. The skill lies in choosing a partition on which the conditional probabilities are easy.

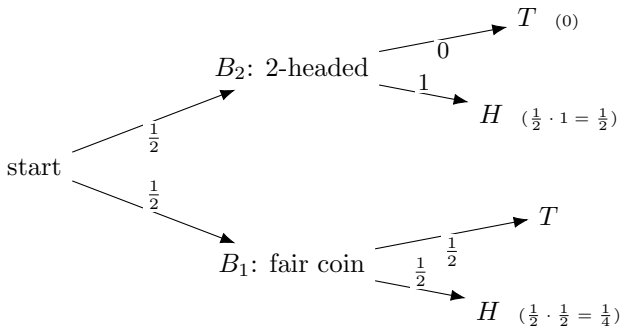


Figure 4.4: A probability tree for the drawer of Example 4.5 (one flip). Multiply along a branch to get the probability of that path (chain rule); add the leaf probabilities that show heads to get $\mathbb{P}(H) = \frac{1}{4} + \frac{1}{2} = \frac{3}{4}$ (law of total probability).

Why this matters

The probability tree in Figure 4.4 is the law of total probability and the chain rule *drawn as a picture*. Going *across* a branch (multiplying edge probabilities) is the chain rule: $\mathbb{P}(\text{path}) = \mathbb{P}(\text{first edge}) \cdot \mathbb{P}(\text{next edge} \mid \text{first}) \cdots$. Going *down* the leaves (summing those whose path ends in your event) is the law of total probability. Two of the most useful theorems in the subject are just "multiply across, add down." Whenever a problem describes a staged experiment, draw the tree first; the algebra writes itself.

4.5.1 Worked examples, gambler style

Conditional reasoning was born at the gambling table, so let us return there. These examples reward the habit of choosing a good partition and then turning a crank.

Example 4.6 (The two-children problem)**.** A family has two children; assume each child is independently a boy or girl with probability 1/2. The four equally likely families are $\{BB, BG, GB, GG\}$ (older child listed first).

1. *"At least one is a boy."* Given this, what is the probability both are boys? The condition rules out GG, leaving $\{BB, BG, GB\}$, each with probability 1/3 in the new universe. Of these, only BB has two boys. So $\mathbb{P}(\text{both boys} \mid \text{at least one boy}) = 1/3$.
2. *"The older child is a boy."* Given this, the universe is $\{BB, BG\}$, and both-boys is BB. So the answer is now 1/2.

The two conditions sound almost identical in English, yet they give 1/3 and 1/2. The reason is that "at least one is a boy" is satisfied by *two* of the mixed families (BG and GB), diluting the both-boys case, whereas "the older is a boy" picks out a single mixed family. The precise wording of the conditioning event is everything – a recurring lesson.

Example 4.7 (Drawing to a hidden color)**.** Urn 1 holds 3 red and 1 white ball; urn 2 holds 1 red and 3 white. You flip a fair coin to choose an urn, then draw one ball. (a) What is the probability the ball is red? (b) If it came out red, what is the probability you had drawn from urn 1?

(a) Partition on the urn. $\mathbb{P}(\text{urn 1}) = \mathbb{P}(\text{urn 2}) = 1/2$, and $\mathbb{P}(\text{red} \mid \text{urn 1}) = 3/4$, $\mathbb{P}(\text{red} \mid \text{urn 2}) = 1/4$.

By the law of total probability,

$$\mathbb{P}(\text{red}) = \tfrac{1}{2}\cdot\tfrac{3}{4} + \tfrac{1}{2}\cdot\tfrac{1}{4} = \tfrac{3}{8} + \tfrac{1}{8} = \tfrac{1}{2}.$$

(b) Now we run the conditioning backward – a foretaste of Bayes' theorem. By the definition of conditional probability,

$$\mathbb{P}(\text{urn } 1 \mid \text{red}) = \frac{\mathbb{P}(\text{urn } 1 \cap \text{red})}{\mathbb{P}(\text{red})} = \frac{\frac{1}{2}\cdot\frac{3}{4}}{\frac{1}{2}} = \frac{3/8}{1/2} = \frac{3}{4}.$$

A red draw shifts your belief from 1/2 (before drawing) to 3/4 in favor of urn 1 – because urn 1 is the redder urn, a red ball is evidence for it. Notice how part (a)'s denominator fed directly into part (b). The forward computation (total probability) and the backward computation (updating) are two readings of the same tree.

Example 4.8 (The gambler's chain: three good cards)**.** From a standard deck, you deal three cards without replacement. What is the probability all three are face cards (J, Q, K – there are 12)? Pure chain rule:

$$\mathbb{P}(\text{all three face}) = \frac{12}{52}\cdot\frac{11}{51}\cdot\frac{10}{50} = \frac{1320}{132600} = \frac{11}{1105} \approx 0.00995.$$

Each factor is conditioned on the cards already gone, shrinking both the count of face cards and the deck. Compare drawing *with* replacement, where each draw is independent and the answer would be $(12/52)^3 \approx 0.01228$. Drawing without replacement makes a second face card slightly less likely (one fewer is left), so the without-replacement probability is the smaller of the two. Dependence, made of nothing but bookkeeping, bends the answer.

Paradox

[The Monty Hall problem] You are on a game show facing three doors. Behind one is a car; behind the other two, goats. You pick a door – say door 1. The host, who knows where the car is, opens a *different* door that he knows hides a goat – say door 3 – and offers you the chance to switch to the remaining door 2. Should you switch?

Intuition shouts "it's 50/50 now, it doesn't matter." Intuition is wrong; you should switch, doubling your winning chance from 1/3 to 2/3. The clean way to see it is the law of total probability conditioned on *where the car actually is*, with the strategy fixed as "always switch." Your initial pick (door 1) hits the car with probability 1/3 and a goat with probability 2/3. *If your initial pick was the car* (prob. 1/3), the host opens one of the two goat doors, the remaining door has a goat, and switching *loses*. *If your initial pick was a goat* (prob. 2/3), the other goat is the only door the host can safely open, so the remaining door is the *car*, and switching *wins*. Therefore

$$\mathbb{P}(\text{win by switching}) = \underbrace{\tfrac{1}{3}\cdot 0}_{\text{picked car}} + \underbrace{\tfrac{2}{3}\cdot 1}_{\text{picked goat}} = \tfrac{2}{3}.$$

What is really happening: the host is not a random door-opener. He *always* reveals a goat, and he uses his knowledge to do so. That action is information, and it flows onto the unopened door. Conditioning on a knowledgeable action is different from conditioning on a random event – exactly the subtlety that makes Monty Hall feel impossible. The 1/3 chance that your first guess was right never changes (nothing the host does can touch the door you already chose); so the other 2/3, formerly spread over two doors, now piles entirely onto the single door the host left closed.

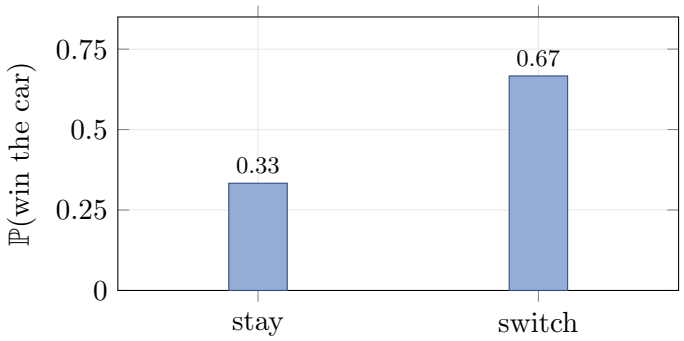


Figure 4.5: Monty Hall, win probability by strategy. Staying wins only when your first guess was the car (1/3); switching wins in the complementary 2/3 of cases. Switching doubles your chance.

History

Monty Hall became a national argument in September 1990, when Marilyn vos Savant gave the switching answer in her *Parade* magazine column. She was deluged with roughly ten thousand letters insisting she was wrong, many from people with PhDs, some from mathematicians; one scolded that "you are the goat." The great Hungarian mathematician Paul Erdos reportedly remained unconvinced until he was shown a computer simulation. The problem is named for the host of the game show *Let's Make a Deal*, and a version had in fact been posed and solved correctly by Steve Selvin in 1975. The episode is a humbling reminder that conditional-probability intuition fails even the well trained – the safe move is always to write down the partition and compute.

Work it yourself

Convince yourself of Monty Hall by simulation in your head or on paper. List the car position $(1, 2, 3$, each $1/3)$, fix your pick as door 1, and for each car position write down which door the host opens and whether switching wins. You will find switching wins in two of the three cases. Then redo it for the variant where the host opens a door *at random* (and the game is voided if he happens to reveal the car). You will find that, in the surviving games, switching is genuinely 50/50. The difference between the two variants is the entire lesson: *whose knowledge generated the data.*

Example 4.9 (Reliability by cases: a noisy channel)**.** A communication channel sends a single bit. The source emits 0 with probability 0.6 and 1 with probability 0.4. The channel is noisy: it flips a transmitted 0 to 1 with probability 0.1, and flips a transmitted 1 to 0 with probability 0.2. You receive a 1. What is the probability of receiving a 1, and what was probably sent given you saw a 1?

First, $\mathbb{P}(\text{receive } 1)$ by total probability, partitioning on what was sent:

$$\mathbb{P}(R{=}1) = \mathbb{P}(R{=}1 \mid S{=}0)\,\mathbb{P}(S{=}0) + \mathbb{P}(R{=}1 \mid S{=}1)\,\mathbb{P}(S{=}1) = (0.1)(0.6) + (0.8)(0.4) = 0.06 + 0.32 = 0.38.$$

(Here $\mathbb{P}(R{=}1 \mid S{=}1) = 1 - 0.2 = 0.8$, the bit surviving uncorrupted.) Then the backward question:

$$\mathbb{P}(S{=}1 \mid R{=}1) = \frac{\mathbb{P}(R{=}1 \mid S{=}1)\,\mathbb{P}(S{=}1)}{\mathbb{P}(R{=}1)} = \frac{0.32}{0.38} \approx 0.842.$$

So a received 1 probably *was* a 1, but only with 84% confidence – the channel's noise plus the source's lean toward 0 leave real doubt. This case-by-case computation is exactly how a digital

receiver decides what was sent; it is conditional probability doing quiet engineering work billions of times a second.

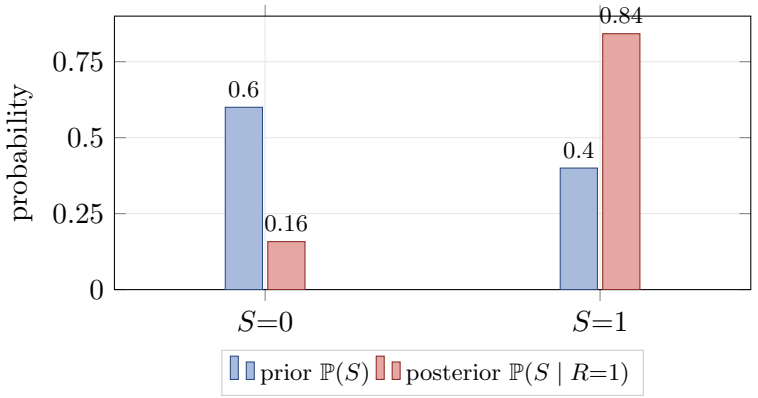


Figure 4.6: The noisy channel of Example 4.9. Seeing $R{=}1$ pulls belief from the prior (which favored $S{=}0$) toward $S{=}1$: the posterior $\mathbb{P}(S{=}1 \mid R{=}1) = 0.32/0.38 \approx 0.842$. Observation reshapes belief, but noise keeps the posterior short of certainty.

4.6 Putting It Together

Every idea in this chapter grew from a single picture: *conditioning is cropping the universe down to what you know, then renormalizing.* From that one move everything followed.

Focus

The five tools of this chapter.

- *Conditional probability:* $\mathbb{P}(A \mid B) = \mathbb{P}(A \cap B)/\mathbb{P}(B)$ – updating a belief on new information; a genuine probability measure in its own right.
- *Multiplication rule:* $\mathbb{P}(A \cap B) = \mathbb{P}(B)\,\mathbb{P}(A \mid B)$ – build joint probabilities from staged conditionals.
- *Chain rule:* $\mathbb{P}(A_1 \cap \cdots \cap A_n) = \prod_j \mathbb{P}(A_j \mid A_1 \cap \cdots \cap A_{j-1})$ – multiply across a path.
- *Independence:* $\mathbb{P}(A \cap B) = \mathbb{P}(A)\,\mathbb{P}(B)$ – B tells you nothing about A; *not* the same as disjoint; for 3+ events check all sub-products. Conditional independence relativizes this to a state of knowledge.
- *Law of total probability:* $\mathbb{P}(A) = \sum_i \mathbb{P}(A \mid B_i)\,\mathbb{P}(B_i)$ over a partition – reason case by case, then average; add down the tree.

The backward direction – inferring the cause from the effect, $\mathbb{P}(B_i \mid A)$ – is so important it gets its own chapter next: Bayes' theorem. Everything you have learned here is the scaffolding it stands on.

4.7 Exercises

Exercise 4.1 (Warm-up conditioning)**.** A fair die is rolled. Let $A = \{\text{roll} \geq 4\}$ and $B = \{\text{roll is even}\}$.

1. Compute $\mathbb{P}(A \mid B)$ and $\mathbb{P}(B \mid A)$. Are they equal? Should you have expected that?
2. Are A and B independent? Justify with the product rule.

Exercise 4.2 (Multiplication rule with an urn)**.** An urn has 5 green and 3 yellow marbles. You draw three without replacement. Find the probability that all three are green using the chain rule, and then find the probability that the first is green and the next two yellow.

Exercise 4.3 (Disjoint vs. independent)**.** Let $\mathbb{P}(A) = 0.3$ and $\mathbb{P}(B) = 0.5$.

1. If A and B are disjoint, find $\mathbb{P}(A \cup B)$ and $\mathbb{P}(A \mid B)$.
2. If instead A and B are independent, find $\mathbb{P}(A \cap B)$ and $\mathbb{P}(A \cup B)$.
3. Explain in one sentence why A and B cannot be both disjoint and independent here.

Exercise 4.4 (Pairwise but not mutual)**.** Reproduce Bernstein's example (Example 4.4) with your own labeling, then construct a *different* triple of events on the two-coin sample space that is pairwise but not mutually independent. Verify all three pairwise products and the failing triple product.

Exercise 4.5 (Law of total probability)**.** A factory runs three machines producing 50%, 30%, and 20% of all widgets, with defect rates 1%, 2%, and 3% respectively. A widget is picked at random.

1. What is the probability it is defective?
2. Given it is defective, what is the probability it came from machine 3? (This is Bayes in disguise – use the definition of conditional probability.)

Exercise 4.6 (The other two-children problem)**.** Reconsider Example 4.6. A family has two children and you learn "at least one is a boy born on a Tuesday." Assuming day-of-week is uniform and independent of sex, compute the probability both children are boys. (You may be surprised; explain why the extra detail moves the answer away from 1/3.)

Exercise 4.7 (Conditional independence, by hand)**.** In the two-coins-in-a-drawer setup (Example 4.5), verify directly that H_1 and H_2 are conditionally independent given the coin's identity, and confirm numerically that they are *not* unconditionally independent by comparing $\mathbb{P}(H_1 \cap H_2)$ with $\mathbb{P}(H_1)\,\mathbb{P}(H_2)$.

Exercise 4.8 (Monty Hall with four doors)**.** Generalize Monty Hall to four doors, one car. You pick a door; the host then opens *one* goat door (from the three you did not pick) and offers a switch to one of the two remaining closed doors. Using the law of total probability conditioned on whether your initial pick was the car, compute the probability of winning by switching to a (randomly chosen) other closed door. Compare to staying.

Exercise 4.9 (A chain-rule proof)**.** Prove the chain rule (Theorem 4.1) carefully by induction on n. State precisely where the positivity hypothesis $\mathbb{P}(A_1 \cap \cdots \cap A_{n-1}) > 0$ is used, and explain why it guarantees that all the earlier conditional probabilities in the product are also well defined.

Exercise 4.10 (Symmetry of independence)**.** Using only Definition 4.2, prove that if $A \perp\!\!\!\perp B$ then $A \perp\!\!\!\perp B^c$, $A^c \perp\!\!\!\perp B$, and $A^c \perp\!\!\!\perp B^c$. (Hint for the first: write $\mathbb{P}(A \cap B^c) = \mathbb{P}(A) - \mathbb{P}(A \cap B)$.) Interpret the result: if B is uninformative about A, so is the news that B did *not* happen.

Chapter 5

Bayes' Theorem and Reasoning Under Uncertainty

There is a single, short equation in this chapter. You can write it on the back of a business card, and it has been written on more than a few. Yet that one line is the engine behind spam filters, medical diagnosis, courtroom reasoning, the search for lost submarines, the calibration of climate models, and most of modern machine learning. It is the rule that tells you, precisely, *how a rational mind should change its beliefs when new evidence arrives.*

That last sentence is the whole chapter in disguise. Probability up to this point has mostly answered the question "how likely is this outcome?" Bayes' theorem answers a different and far more human question: "I observed *something*; what should I now believe about the hidden cause behind it?" A doctor sees a positive test and asks about the unseen disease. A detective sees a fingerprint and asks about the unseen suspect. A phone's autocorrect sees the keystrokes `teh` and asks about the unseen word you meant. In every case we are running a chain of cause and effect *backwards*: from the visible effect to the invisible cause. Bayes' theorem is the mathematics of running that arrow backwards correctly.

The derivation takes one line; the rest of the chapter is about learning to *feel* it. The equation has four parts — prior, likelihood, evidence, posterior — and we will watch what each one does. Along the way we meet the base-rate fallacy, perhaps the most expensive single mistake in all of applied probability, and see exactly how an accurate test can produce mostly-wrong-looking results. We settle the Monty Hall problem with a method that leaves no room for argument, and we learn to update not once but repeatedly, letting evidence accumulate. The chapter ends by building, by hand, the algorithm that for two decades kept the spam out of your inbox.

5.1 Running the Arrow Backwards

Start with a story, because Bayes' theorem is really a story about asymmetry.

Intuition

Imagine two facts about a far-off town. First: *if it is raining, the streets are wet* — this happens almost always, say with probability 0.95. Second: you are told, by a friend who glanced out a window, only that *the streets are wet*. How likely is it that it is raining?
Your instinct might be to reach for that 0.95. But notice the question has flipped. You were given the chance of *wet streets given rain*, and you were asked for the chance of *rain given wet streets*. These are not the same number, and confusing them is the original sin of probabilistic reasoning. Streets can be wet for many reasons — a street cleaner, a burst pipe, melting snow, a parade with water balloons. To answer the question you actually need a third fact you have not been given: *how often does it rain in this town at all?* In a rainforest the answer is "almost surely raining." In a desert, "almost surely not, despite the wet streets." Same evidence, opposite conclusions. The missing ingredient is what we will call the *prior*, and half of all reasoning errors come from forgetting it.

The asymmetry in that story — $\mathbb{P}(\text{wet} \mid \text{rain})$ versus $\mathbb{P}(\text{rain} \mid \text{wet})$ — is exactly the asymmetry Bayes' theorem resolves. Recall from the previous chapter the definition of conditional probability: for events A and B with $\mathbb{P}(B) > 0$,

$$\mathbb{P}(A \mid B) \;=\; \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}.$$

Read it as: "of all the ways B can happen, what fraction also have A?" Now here is the trick. The intersection $A \cap B$ does not care which letter we write first; $\mathbb{P}(A \cap B) = \mathbb{P}(B \cap A)$. So we can expand that same joint probability two different ways using the multiplication rule:

$$\mathbb{P}(A \cap B) \;=\; \mathbb{P}(A \mid B)\,\mathbb{P}(B) \qquad \text{and} \qquad \mathbb{P}(A \cap B) \;=\; \mathbb{P}(B \mid A)\,\mathbb{P}(A).$$

Both equal the same thing, so they equal each other:

$$\mathbb{P}(A \mid B)\,\mathbb{P}(B) \;=\; \mathbb{P}(B \mid A)\,\mathbb{P}(A).$$

Divide both sides by $\mathbb{P}(B)$ and you have arrived.

Formal

Bayes' Theorem (basic form). For events A and B with $\mathbb{P}(B) > 0$,

$$\mathbb{P}(A \mid B) \;=\; \frac{\mathbb{P}(B \mid A)\,\mathbb{P}(A)}{\mathbb{P}(B)}. \tag{5.1}$$

That is the entire derivation. One symmetry of the intersection, one division. It is almost embarrassing how little machinery it takes — and yet the consequences fill this chapter and several careers. The reason it feels deep even though it is shallow to prove is that it tells you something your intuition resists: that to flip a conditional you *must* bring in the unconditional probabilities $\mathbb{P}(A)$ and $\mathbb{P}(B)$, and people chronically leave them out.

Why it works

Why is $\mathbb{P}(A)$ in there at all? Because $\mathbb{P}(B \mid A)$ only tells you how strongly the cause A *predicts* the effect B — it says nothing about how common A is to begin with. A cause can be a perfect

predictor of its effect and still be a terrible explanation for it, simply because the cause is rare. "Being struck by lightning" predicts "hospital visit" very strongly, but almost no hospital visit is from lightning, because lightning strikes are rare. The factor $\mathbb{P}(A)$ is the brake that keeps a rare-but-predictive explanation from running away with your belief. Equation (5.1) is the precise bookkeeping of how predictive power ($\mathbb{P}(B \mid A)$) and prior plausibility ($\mathbb{P}(A)$) trade off.

5.1.1 The four names

Equation (5.1) earns its fame once we attach a story to each piece. Let A be a *hypothesis* H (the hidden cause: it is raining, the patient is sick, the email is spam) and let B be the *evidence* E we actually observe (wet streets, a positive test, the word "Viagra"). Rewriting,

$$\underbrace{\mathbb{P}(H \mid E)}_{\text{posterior}} = \frac{\overbrace{\mathbb{P}(E \mid H)}^{\text{likelihood}}\ \overbrace{\mathbb{P}(H)}^{\text{prior}}}{\underbrace{\mathbb{P}(E)}_{\text{evidence}}}. \tag{5.2}$$

- The **prior** $\mathbb{P}(H)$ is what you believed about the hypothesis *before* seeing this piece of evidence. The rainfall rate of the town. Your starting credence.
- The **likelihood** $\mathbb{P}(E \mid H)$ is how well the hypothesis predicts the evidence: *if* H were true, how probable would E be? Note carefully: the likelihood is a function of H, asking "which hypothesis makes the data look least surprising?" It is *not* a probability of H.
- The **evidence** (or *marginal likelihood*) $\mathbb{P}(E)$ is the total probability of seeing E at all, under every hypothesis weighted by its prior. It is the great normalizer, the thing that makes the posterior add up to one.
- The **posterior** $\mathbb{P}(H \mid E)$ is your updated belief after folding in the evidence. It is the answer. It is what you came for.

What is really happening

Here is the one sentence to remember when the symbols blur:

$$\text{posterior} \ \propto\ \text{likelihood} \times \text{prior}.$$

The denominator $\mathbb{P}(E)$ does not depend on the particular hypothesis H — it is the same number whether you are asking about rain or street cleaners — so it cannot change *which* hypothesis wins. It only rescales all the posteriors so they sum to one. What actually does the work, what actually moves your belief, is the product $\mathbb{P}(E \mid H)\ \mathbb{P}(H)$. New belief is old belief reweighted by how well each hypothesis explained what you just saw. Everything else is normalization. If you internalize only this proportionality, you understand Bayes' theorem.

5.1.2 Computing the evidence: the law of total probability

We will constantly need $\mathbb{P}(E)$, and almost never is it handed to us directly. Instead we compute it by splitting the world into the hypothesis and its complement (or into a full set of competing hypotheses)

and adding up the ways E can occur. If H and its negation H^c together cover all possibilities, then

$$\mathbb{P}(E) \;=\; \mathbb{P}(E \mid H)\,\mathbb{P}(H) \;+\; \mathbb{P}(E \mid H^c)\,\mathbb{P}(H^c). \tag{5.3}$$

This is the *law of total probability*: the evidence can arrive either through the hypothesis being true or through it being false, and we tally both routes. More generally, if $H_1, \dots, H_k$ are mutually exclusive and exhaustive (a *partition* of the sample space), then $\mathbb{P}(E) = \sum_{i=1}^{k} \mathbb{P}(E \mid H_i)\,\mathbb{P}(H_i)$, and Bayes' theorem becomes

$$\mathbb{P}(H_j \mid E) \;=\; \frac{\mathbb{P}(E \mid H_j)\,\mathbb{P}(H_j)}{\sum_{i=1}^{k} \mathbb{P}(E \mid H_i)\,\mathbb{P}(H_i)}. \tag{5.4}$$

The denominator is just the numerator summed over all hypotheses — which is why the posteriors are guaranteed to add to one. Let us put the rain story to rest with real numbers.

Example 5.1 (Wet streets, finally answered)**.** Suppose in our town it rains on a given morning with probability $\mathbb{P}(R) = 0.2$. If it rains, the streets are wet with probability $\mathbb{P}(W \mid R) = 0.95$. If it does not rain, the streets are still wet — cleaners, sprinklers, the occasional spill — with probability $\mathbb{P}(W \mid R^c) = 0.1$. Your friend reports wet streets. What is $\mathbb{P}(R \mid W)$?

First the evidence, by (5.3):

$$\mathbb{P}(W) = (0.95)(0.2) + (0.1)(0.8) = 0.19 + 0.08 = 0.27.$$

Then Bayes:

$$\mathbb{P}(R \mid W) = \frac{(0.95)(0.2)}{0.27} = \frac{0.19}{0.27} \approx 0.70.$$

So wet streets push you from a 20% belief in rain up to a 70% belief — a big update, but nowhere near the 95% that the careless reader grabbed from the likelihood. The other 30% of wet-street mornings are dry-but-wet for mundane reasons. Change the town's rainfall rate to $\mathbb{P}(R) = 0.02$ (a desert) and rerun it: $\mathbb{P}(W) = 0.019 + 0.098 = 0.117$ and $\mathbb{P}(R \mid W) = 0.019/0.117 \approx 0.16$. Same evidence, same likelihoods, but now wet streets leave you *still doubting* the rain. The prior is doing exactly the work the intuition box promised.

History

The theorem is named for the Reverend **Thomas Bayes** (c. 1701–1761), a Presbyterian minister and amateur mathematician in Tunbridge Wells, England. He never published it. The result appeared posthumously in 1763 when his friend Richard Price read Bayes' notes to the Royal Society in "An Essay towards solving a Problem in the Doctrine of Chances." Bayes' own problem was a charming one about a billiard ball rolled on a table: having seen where later balls land relative to the first, what can you infer about the first ball's hidden position? That is inference about a hidden cause from visible effects — exactly our theme.

But the person who truly built the cathedral was **Pierre-Simon Laplace**, who around 1774, apparently unaware of Bayes, stated the rule in full generality and spent decades applying it to astronomy, geodesy, and demography — he used it to estimate, from birth records, that boys are very slightly more likely than girls, and to put error bars on the masses of the planets. For a century the method was simply called "inverse probability." It fell out of fashion in the early twentieth century, attacked by frequentists like R. A. Fisher who distrusted the subjectivity of the prior, and was kept alive in the wilderness by a few — Jeffreys, de Finetti, Savage. It came roaring back at the end of the century when cheap computation finally made

the integrals tractable, and today "Bayesian" is one of the most common adjectives in all of applied science. A two-page note by a country parson became a worldview.

5.2 The Base-Rate Fallacy: When a Good Test Lies

We now meet the single most important application of Bayes' theorem for everyday life, and the place where untrained intuition fails most spectacularly and most expensively. The setting is medical testing, but the lesson reaches into security screening, fraud detection, courtroom DNA evidence, and every alarm system ever built.

Paradox

The disease that the test cannot find. A certain disease afflicts 1 person in 1000. There is a test for it that is, by any normal standard, excellent: if you have the disease, the test is positive 99% of the time (only a 1% false-negative rate); if you do *not* have the disease, the test is negative 95% of the time (a 5% false-positive rate). You take the test. It comes back **positive**. How worried should you be — what is the probability you actually have the disease?

Most people, including a famously large fraction of doctors when surveyed, answer somewhere around 95% or 99%. The test is 95–99% accurate, after all. The true answer is about **2%**. You are almost certainly fine. Read that again, because it is genuinely shocking: a 99%-sensitive, 95%-specific test, returning positive, and you are still 98% likely to be healthy.

How can an excellent test be so misleading? Bayes' theorem will tell us exactly, but first let us see it with our bare hands, by counting people. This "natural frequencies" trick is the single best way to defuse the paradox, and it is worth doing even when you trust the algebra.

Walking question

Imagine 100,000 people, a small city, all getting tested.

Step 1 — the truly sick. The disease rate is 1 in 1000, so about $100{,}000/1000 = 100$ people genuinely have it. Of these 100, the test catches 99%, so 99 of them test positive (and 1 slips through as a false negative).

Step 2 — the healthy many. The other 99,900 people are disease-free. The test wrongly flags 5% of them, so $0.05 \times 99{,}900 \approx 4995$ healthy people also test positive.

Step 3 — who got the bad news? Total positives: $99 + 4995 = 5094$. Of these, only 99 are actually sick. So the probability that a positive result means real disease is

$$\frac{99}{5094} \approx 0.0194 \approx 1.9\%.$$

There is the answer, with no formula in sight. The flood of false positives from the enormous healthy population (nearly 5000 of them) drowns out the trickle of true positives from the tiny sick population (just 99). The test is fine. The problem is that the disease is *rare*, and rarity is the prior.

Now the same computation in the language of Bayes, so you see they agree. Let D be "has disease" and $+$ be "tests positive." We are given the prior $\mathbb{P}(D) = 0.001$, the likelihood (sensitivity) $\mathbb{P}(+ \mid$

$D) = 0.99$, and the false-positive rate $\mathbb{P}(+ \mid D^c) = 0.05$. The evidence:

$$\mathbb{P}(+) = \mathbb{P}(+ \mid D)\,\mathbb{P}(D) + \mathbb{P}(+ \mid D^c)\,\mathbb{P}(D^c) = (0.99)(0.001) + (0.05)(0.999) = 0.00099 + 0.04995 = 0.05094.$$

The posterior:

$$\mathbb{P}(D \mid +) = \frac{\mathbb{P}(+ \mid D)\,\mathbb{P}(D)}{\mathbb{P}(+)} = \frac{0.00099}{0.05094} \approx 0.0194.$$

Identical to the counting, as it must be — the natural-frequencies method *is* Bayes' theorem with the denominators cleared.

Common trap

The fallacy has a precise shape: people silently swap $\mathbb{P}(D \mid +)$ for $\mathbb{P}(+ \mid D)$. They hear "the test is 99% accurate" (a statement about $\mathbb{P}(+ \mid D)$, the likelihood) and report it as the chance of being sick (the posterior $\mathbb{P}(D \mid +)$). These differ by exactly the factor $\mathbb{P}(D)/\,\mathbb{P}(+)$ that Bayes inserts — and when the disease is rare, that factor is tiny. Whenever you catch yourself reasoning "the evidence is very reliable, therefore the conclusion is very likely," stop and ask: *how common was the conclusion to begin with?* A reliable detector of a rare thing is mostly an alarm that cries wolf.

Why this matters

This is not an academic curiosity; it is a matter of public policy and personal anguish. It is the central argument against mass screening of the general population for rare diseases: you generate a torrent of frightened, falsely-positive patients — biopsies, anxiety, follow-up procedures with their own risks — to find a handful of true cases. It is why doctors test *symptomatic* or *high-risk* patients: doing so raises the prior $\mathbb{P}(D)$, which is the only lever that improves the posterior. The same arithmetic governs airport security, mass surveillance for terrorism, and automated fraud flags: when you hunt for needles, almost everything your detector lights up is hay, no matter how good the detector. The remedy is never "a better test" alone — it is a higher prior, achieved by testing a population in which the thing is not so rare.

5.2.1 Watching the prior take over

The whole drama lives in how the posterior depends on the prior, the base rate. Let us hold the test fixed (sensitivity 0.99, specificity 0.95) and sweep the disease prevalence from very rare to fairly common, plotting $\mathbb{P}(D \mid +)$ as we go. The curve in Figure 5.1 says it all.

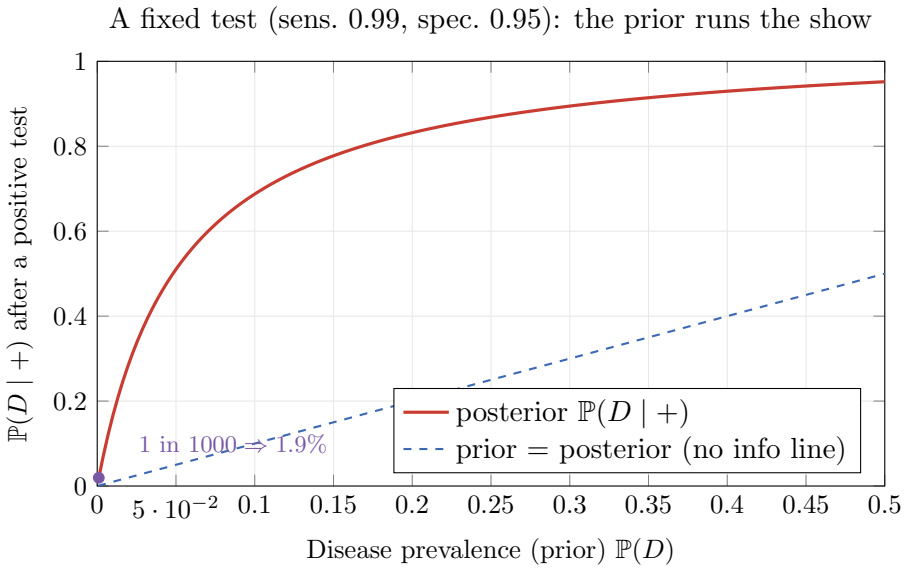


Figure 5.1: For a strong but imperfect test, the post-positive probability of disease climbs steeply with the base rate. At a prevalence of 1 in 1000 (purple dot) a positive result barely moves you off zero; only when the disease is already reasonably common does a positive test become trustworthy. The dashed line marks where evidence would tell you nothing; the gap above it is how much the test *does* help. Same test, wildly different verdicts — the only thing that changed is the prior.

The shape of that curve is the base-rate fallacy made visible. Down in the far left, where real diseases live, the red posterior hugs the floor: a positive test is almost worthless because false positives swamp it. The posterior only becomes large — the test only becomes trustworthy — once the prior is already substantial. The test never changes; the verdict does. This is why the very same positive mammogram means something very different for a 40-year-old with no family history than for a 65-year-old with a suspicious lump: their priors are not the same, so their posteriors cannot be either.

5.3 Sequential Updating: Yesterday's Posterior Is Today's Prior

So far we have updated once. But evidence rarely arrives all at once; it dribbles in. Bayes' theorem handles a stream of evidence with no extra machinery at all. You update, and then update again, treating the belief you just computed as the starting belief for the next round. This is the rhythm of all learning:

$$\text{prior} \xrightarrow{E_1} \text{posterior}_1 = \text{prior}_2 \xrightarrow{E_2} \text{posterior}_2 = \text{prior}_3 \xrightarrow{E_3} \cdots$$

Why it works

Why is it legitimate to feed yesterday's posterior in as today's prior? Because that is literally

what conditioning on more data does. Apply Bayes twice with two pieces of evidence E_1, E_2:

$$\mathbb{P}(H \mid E_1, E_2) = \frac{\mathbb{P}(E_2 \mid H, E_1)\ \mathbb{P}(H \mid E_1)}{\mathbb{P}(E_2 \mid E_1)}.$$

The term $\mathbb{P}(H \mid E_1)$ sitting where the prior usually sits is precisely the posterior from round one. The data simply accumulate inside the conditioning bar. If the pieces of evidence are conditionally independent given H — meaning that once you know the true hypothesis, E_1 tells you nothing further about E_2 — then $\mathbb{P}(E_2 \mid H, E_1) = \mathbb{P}(E_2 \mid H)$, and the whole stream collapses to multiplying likelihoods:

$$\mathbb{P}(H \mid E_1, \ldots, E_n) \ \propto\ \mathbb{P}(H) \prod_{i=1}^{n} \mathbb{P}(E_i \mid H).$$

Each new datum just multiplies in another likelihood factor. Belief compounds like interest.

A spectacular feature of this process is its *commutativity and convergence*: the order in which conditionally-independent evidence arrives does not change the final belief, and as evidence piles up the posterior typically concentrates on the truth almost regardless of where the prior started (as long as the prior didn't dogmatically assign probability zero to the truth — a hypothesis you give zero prior can never be revived, which is a moral as much as a theorem). Let us watch it happen.

Example 5.2 (Two tests are much better than one)**.** Return to the patient from the base-rate paradox, who tested positive and now stands at posterior $\mathbb{P}(D \mid +) \approx 0.0194$. The doctor, sensibly, orders a *second*, independent test of the same quality. It too comes back positive. Now what?

Yesterday's posterior becomes today's prior: $\mathbb{P}(D) = 0.0194$. Run Bayes again with the same likelihoods:

$$\mathbb{P}(+) = (0.99)(0.0194) + (0.05)(0.9806) = 0.01921 + 0.04903 = 0.06824,$$

$$\mathbb{P}(D \mid +, +) = \frac{(0.99)(0.0194)}{0.06824} = \frac{0.01921}{0.06824} \approx 0.281.$$

Two positives lift the patient from 1.9% to about 28%. A *third* positive, updating again with prior 0.281, gives

$$\mathbb{P}(D \mid +, +, +) = \frac{(0.99)(0.281)}{(0.99)(0.281) + (0.05)(0.719)} = \frac{0.2782}{0.2782 + 0.03595} \approx 0.886.$$

Now you should be worried. This is exactly how diagnosis actually works: no single test is conclusive for a rare disease, but independent confirmations compound. Each positive multiplies the odds by the same factor; pile up enough of them and even a tiny prior is overwhelmed.

Figure 5.2 draws that compounding. Each positive test multiplies the *odds* by the constant Bayes factor 19.8, so the posterior probability follows the unmistakable S-shaped march from near-zero toward near-certainty: slow while the rare prior still dominates, explosive once enough evidence has accumulated, then flattening as it saturates against the ceiling of 1.

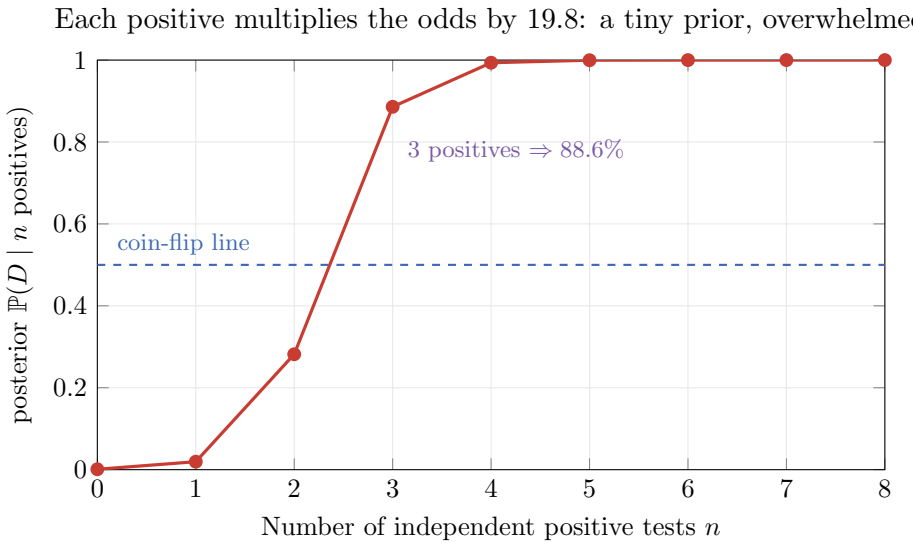


Figure 5.2: Repeated positive results on independent copies of the same imperfect test, starting from a prevalence of 1 in 1000. The first positive barely registers (the base-rate fallacy in action); the second and third do the real work; by the third the patient is past the coin-flip line and climbing toward certainty. Belief compounds like interest, because in log-odds each positive simply *adds* the same fixed weight $\log 19.8$.

5.3.1 The odds form: Bayes' theorem with the friction removed

Example 5.2 hints at something cleaner. Recomputing $\mathbb{P}(E)$ every round is tedious, and it obscures the pattern. There is a reformulation in terms of *odds* that makes sequential updating almost trivial, and that reveals what evidence "weighs." Take Bayes' theorem for H and for H^c and divide one by the other; the shared denominator $\mathbb{P}(E)$ cancels:

$$\underbrace{\frac{\mathbb{P}(H \mid E)}{\mathbb{P}(H^c \mid E)}}_{\text{posterior odds}} = \underbrace{\frac{\mathbb{P}(E \mid H)}{\mathbb{P}(E \mid H^c)}}_{\text{likelihood ratio}} \times \underbrace{\frac{\mathbb{P}(H)}{\mathbb{P}(H^c)}}_{\text{prior odds}}. \tag{5.5}$$

This is the most useful single equation for actually *doing* Bayesian updates by hand. Odds are just probabilities written as a ratio: a probability of 0.2 is odds of 0.2/0.8 = 1:4. The *likelihood ratio* (also called the Bayes factor) is the weight of the evidence — how many times more likely the data are under H than under H^c. And the rule could not be simpler:

$$\text{posterior odds} \;=\; \text{Bayes factor} \times \text{prior odds}.$$

No normalizer to compute. To chain evidence, just keep multiplying by each Bayes factor.

Example 5.3 (The same diagnosis, the easy way)**.** For our test, the Bayes factor of a positive result is

$$\frac{\mathbb{P}(+ \mid D)}{\mathbb{P}(+ \mid D^c)} = \frac{0.99}{0.05} = 19.8.$$

Every positive multiplies the odds of disease by 19.8. The prior odds at prevalence 1 in 1000 are $0.001/0.999 \approx 1/999$. After one positive: $\frac{1}{999} \times 19.8 = \frac{19.8}{999} \approx 0.0198$, which is odds, i.e. probability $0.0198/(1.0198) \approx 0.0194$ — matching before. After three positives: $\frac{1}{999} \times 19.8^3 = \frac{7762}{999} \approx 7.77$, i.e.

probability $7.77/8.77 \approx 0.886$. Three multiplications and we are done, no total-probability sums anywhere. *This* is why practitioners think in odds and log-odds.

Connection

Take the logarithm of the odds form (5.5) and the products become sums:

$$\log \frac{\mathbb{P}(H \mid E_{1:n})}{\mathbb{P}(H^c \mid E_{1:n})} = \log \frac{\mathbb{P}(H)}{\mathbb{P}(H^c)} + \sum_{i=1}^{n} \log \frac{\mathbb{P}(E_i \mid H)}{\mathbb{P}(E_i \mid H^c)}.$$

Each piece of evidence contributes an additive "weight" log(Bayes factor) to a running total of *log-odds*. It is no accident that this looks like a linear model. The logistic-regression classifier of machine learning is exactly this equation: it learns a weight for each feature and *adds them up* to form a log-odds, then squashes back to a probability with the logistic function $\sigma(t) = 1/(1 + e^{-t})$, which is precisely the inverse of the log-odds transform. Information theory will later measure these weights in *bits* and *nats* — the "weight of evidence" is literally a quantity of information. The thread from Reverend Bayes to your linear classifier runs straight through this one logarithm.

The bridge between the additive world of log-odds and the bounded world of probabilities is the logistic function σ, drawn in Figure 5.3. Adding up the weights of evidence is easy and unbounded; the squashing curve then folds that running total back into a probability in $[0, 1]$, gently near the middle and ever more compressed out in the tails — which is exactly why the sequential-test curve of Figure 5.2 flattens as it approaches certainty.

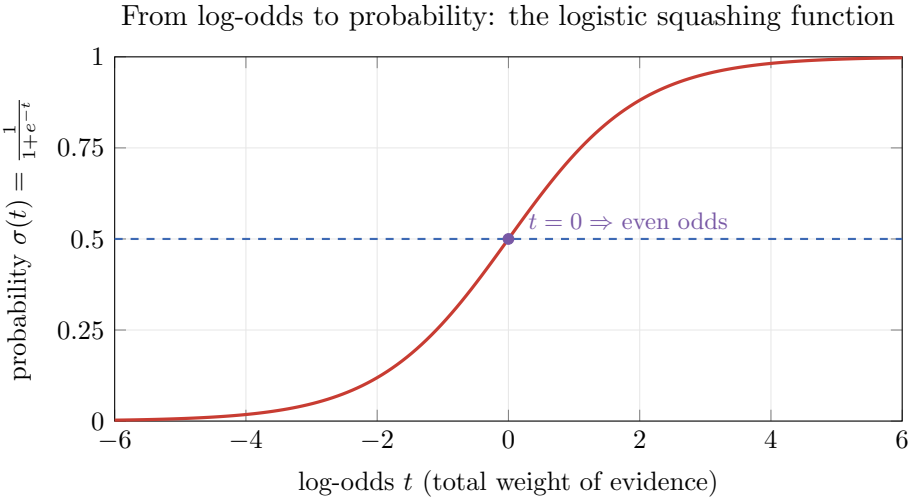


Figure 5.3: The logistic function maps the running total of log-odds back to a probability. Zero log-odds (evidence perfectly balanced) maps to probability $\frac{1}{2}$; each unit of accumulated evidence-weight slides you along the curve, with diminishing effect once you are already near 0 or 1. Sequential Bayesian updating walks rightward along this curve, one Bayes factor at a time.

5.4 The Monty Hall Problem, Finally Resolved

No book on Bayesian reasoning can skip the problem that broke the internet of its day. It is a perfect specimen: a question simple enough for a child, an answer that PhDs got wrong in print, and a resolution that Bayes' theorem delivers without drama.

Paradox

Monty Hall. On a game show there are three doors. Behind one is a car; behind the other two, goats. You pick a door — say Door 1 — but do not open it. The host, Monty, who knows where the car is, opens one of the *other* two doors to reveal a goat — say he opens Door 3. He then offers you a choice: stick with Door 1, or switch to the remaining closed Door 2. *Should you switch?*

The overwhelming intuition is: it doesn't matter. Two doors remain, one has the car, so it's 50/50 — switching is pointless. This intuition is **wrong**. Switching wins the car with probability 2/3; staying wins with only 1/3. You should always switch, and doing so *doubles* your chance of the car. When the columnist Marilyn vos Savant published this answer in 1990, she received thousands of angry letters, many from mathematicians, insisting she was wrong. She was not.

History

The problem is named for **Monty Hall**, host of the American television show *Let's Make a Deal*, though Monty's real game was looser than the puzzle. Its mathematical skeleton dates to Joseph Bertrand's box paradox (1889) and to a 1975 letter by statistician Steve Selvin. It exploded into public consciousness in September 1990 when **Marilyn vos Savant** answered it in her *Parade* magazine column. The ensuing furor — roughly 10,000 letters, a sizable fraction from people with doctorates, nearly all of them wrong and many of them rude — became a landmark in the history of how badly trained intuition handles conditional probability. Even Paul Erdős, one of the great mathematicians of the century, reportedly refused to believe the answer until shown a computer simulation. If it fooled Erdős, you are in good company.

5.4.1 Why the intuition fails, in words

Before any algebra, here is the idea that makes it click for most people. Your original door had a 1/3 chance of hiding the car, and *nothing Monty does afterward can change a probability you already locked in.* So Door 1 stays at 1/3 forever. But the total must be 1, and Door 3 is now revealed to be 0. The entire remaining 2/3 has nowhere to go but onto Door 2. Monty, by deliberately opening a goat door he *chose knowing the layout*, has effectively concentrated the other two doors' combined 2/3 chance onto the single door he left closed.

Intuition

If three doors still feels slippery, scale it up. Imagine 100 **doors**, one car. You pick Door 1 — a 1% shot. Now Monty, who knows where the car is, opens 98 of the other doors, every one a goat, leaving just your Door 1 and one other door, say Door 57, closed. Do you really believe it's now 50/50? Of course not. You almost certainly picked a goat at the start (99% chance),

and Monty has carefully cleared away every other goat, leaving the car sitting behind Door 57 in 99 cases out of 100. Switching wins 99% of the time. The three-door case is the same effect, merely too small to feel: staying wins 1/3, switching wins 2/3. The host's knowledge is the whole trick — he never opens the car, so his choice of door leaks information.

5.4.2 The Bayesian computation, leaving no doubt

Words convince most people; algebra convinces everyone. Let us nail it with Bayes' theorem. Define the hypotheses — the car's true location:

$$C_1 = \{\text{car behind Door 1}\}, \quad C_2 = \{\text{car behind Door 2}\}, \quad C_3 = \{\text{car behind Door 3}\}.$$

By symmetry before anything happens, $\mathbb{P}(C_1) = \mathbb{P}(C_2) = \mathbb{P}(C_3) = 1/3$; these are our priors. You have picked Door 1. The evidence we will condition on is $M_3 = \{\text{Monty opens Door 3 to show a goat}\}$. We need the likelihoods $\mathbb{P}(M_3 \mid C_i)$ — and *here* is where the host's rules, which the careless solver ignores, enter the math.

- $\mathbb{P}(M_3 \mid C_1)$: the car is behind your Door 1, so both other doors hide goats and Monty may open either Door 2 or Door 3. With no reason to prefer one, he opens Door 3 with probability $\frac{1}{2}$. So $\mathbb{P}(M_3 \mid C_1) = \frac{1}{2}$.
- $\mathbb{P}(M_3 \mid C_2)$: the car is behind Door 2. Monty cannot open your Door 1, and he cannot open Door 2 (the car!), so he is *forced* to open Door 3. Thus $\mathbb{P}(M_3 \mid C_2) = 1$.
- $\mathbb{P}(M_3 \mid C_3)$: the car is behind Door 3. Monty will never open the car's door, so he certainly does not open Door 3. Thus $\mathbb{P}(M_3 \mid C_3) = 0$.

Now assemble the evidence by total probability:

$$\mathbb{P}(M_3) = \tfrac{1}{2} \cdot \tfrac{1}{3} + 1 \cdot \tfrac{1}{3} + 0 \cdot \tfrac{1}{3} = \tfrac{1}{6} + \tfrac{1}{3} = \tfrac{1}{2}.$$

(Sensible: by symmetry Monty opens Door 3 half the time.) Then the two posteriors that matter:

$$\mathbb{P}(C_1 \mid M_3) = \frac{\mathbb{P}(M_3 \mid C_1)\,\mathbb{P}(C_1)}{\mathbb{P}(M_3)} = \frac{\frac{1}{2} \cdot \frac{1}{3}}{\frac{1}{2}} = \frac{\frac{1}{6}}{\frac{1}{2}} = \frac{1}{3},$$

$$\mathbb{P}(C_2 \mid M_3) = \frac{\mathbb{P}(M_3 \mid C_2)\,\mathbb{P}(C_2)}{\mathbb{P}(M_3)} = \frac{1 \cdot \frac{1}{3}}{\frac{1}{2}} = \frac{\frac{1}{3}}{\frac{1}{2}} = \frac{2}{3}.$$

Staying with Door 1 wins with probability 1/3; switching to Door 2 wins with probability 2/3. The paradox is dissolved, exactly, with no appeal to intuition.

The whole computation is laid bare by the probability tree in Figure 5.4, which simply enumerates the worlds and weights each by its probability. Conditioning on "Monty opened Door 3" means keeping only the branches consistent with that observation and renormalizing — and when you do, the surviving weight on "car behind Door 2" is twice the weight on "car behind Door 1."

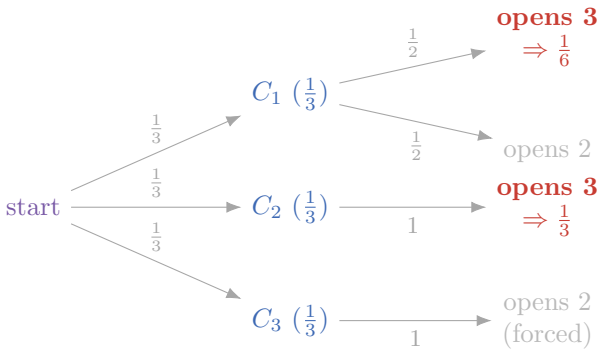


Figure 5.4: The Monty Hall probability tree (you have chosen Door 1). Each path's probability is the product of the edges along it. Only the two red leaves match the evidence "Monty opened Door 3": the C_2 world contributes weight $\frac{1}{3}$, the C_1 world only $\frac{1}{6}$. Renormalizing these two ($\frac{1}{3} : \frac{1}{6} = 2 : 1$) gives posterior $\mathbb{P}(C_2 \mid M_3) = \frac{2}{3}$ for switching against $\mathbb{P}(C_1 \mid M_3) = \frac{1}{3}$ for staying. The asymmetry is born entirely in the forced "1" edge under C_2 versus the free "$\frac{1}{2}$" edge under C_1.

What is really happening

The crux is the asymmetric likelihood: $\mathbb{P}(M_3 \mid C_2) = 1$ but $\mathbb{P}(M_3 \mid C_1) = \frac{1}{2}$. When the car is behind Door 2, Monty is *compelled* to open Door 3; when the car is behind your door, he had a free choice and opened Door 3 only half the time. So the observation "Monty opened Door 3" is *twice as likely* in the world where the car is behind Door 2. That factor of two is the Bayes factor, and it is exactly what shifts your belief from 1/3 to 2/3. The information did not come from *which* door Monty opened so much as from the *constraints he was under* when he chose — constraints created by his knowledge of the car's location.

Common trap

Everything hinges on Monty's protocol, and changing it changes the answer. Suppose Monty does *not* know where the car is and opens one of the other doors *at random*, and it happens to reveal a goat. Now $\mathbb{P}(M_3 \mid C_2) = \frac{1}{2}$ instead of 1 — a clueless Monty could have blundered into the car. Redo the arithmetic and you get $\mathbb{P}(C_1 \mid M_3) = \mathbb{P}(C_2 \mid M_3) = \frac{1}{2}$: switching no longer helps! The "50/50" answer is correct — for a *different game*. The lesson is permanent: *a likelihood is a statement about the data-generating process*, and you cannot write one down until you know the rules by which the evidence was produced. Most people get Monty wrong because they never asked how Monty decides which door to open.

5.5 A First Taste of the Bayesian Worldview

We have been using Bayes' theorem as a calculating device. But it is also the seed of an entire philosophy of inference — a way of doing statistics that differs from the classical, "frequentist" approach you will meet later, and that has its own deep chapter ahead (it returns in force when we study Bayesian inference). Here is the seed.

The Bayesian stance is this: *probability is a measure of belief*, and unknown quantities — not just future coin flips, but fixed-but-unknown facts like a drug's true effectiveness or a physical constant

— can carry probability distributions that describe your uncertainty about them. You start with a prior distribution over the unknown, you observe data through a likelihood, and Bayes' theorem hands you a posterior distribution that is your complete, updated state of knowledge. Inference is not a yes/no verdict; it is a reshaping of a distribution.

Why this matters

What makes this worldview so attractive — and so useful in machine learning — is that it produces *calibrated uncertainty*, not just point answers. A frequentist estimate might say "the click-through rate is 4.1%." A Bayesian posterior says "here is the full curve of how plausible each possible rate is, given what we've seen," and from that single object you can read off a best guess, a credible interval, and the probability of any statement you care about ("what's the chance the new design beats the old by more than a point?"). When data are scarce, the prior gently regularizes your conclusions; when data are abundant, the likelihood overwhelms the prior and Bayesians and frequentists shake hands. The posterior is a complete accounting of what you know and how sure you are — which is exactly what a decision-maker needs.

Application

The naive Bayes spam filter — building one by hand. For two decades the first line of defense in your inbox was a direct application of this chapter. The hypotheses are $S =$ spam and $H =$ ham (legitimate mail); the evidence is the set of words in the email. By the odds form (5.5), and *assuming words are conditionally independent given the class* (this is the "naive" assumption — crude, since "free" and "money" surely co-occur, but astonishingly effective),

$$\underbrace{\frac{\mathbb{P}(S \mid \text{words})}{\mathbb{P}(H \mid \text{words})}}_{\text{posterior odds}} = \underbrace{\frac{\mathbb{P}(S)}{\mathbb{P}(H)}}_{\text{prior odds}} \prod_{\text{word } w} \frac{\mathbb{P}(w \mid S)}{\mathbb{P}(w \mid H)}.$$

Each word contributes a Bayes factor $\mathbb{P}(w \mid S)/\mathbb{P}(w \mid H)$, estimated by counting: out of all words in known spam, what fraction were "w"? Suppose training data give $\mathbb{P}(\text{"free"} \mid S) = 0.05$ and $\mathbb{P}(\text{"free"} \mid H) = 0.001$. Then the word "free" carries a Bayes factor of 50 — seeing it multiplies the spam odds fifty-fold. A word like "meeting" might have a factor of $1/20$, pulling toward ham. The filter simply multiplies all the factors (in practice it *adds the logs*, as in the connection box, to avoid numerical underflow), starts from the base-rate prior odds of spam, and flags the email if the posterior odds clear a threshold.

This is *naive Bayes*, and beyond your inbox it classifies documents by topic, diagnoses faults from sensor readings, and serves as the dependable baseline against which fancier classifiers are measured. Two practical wrinkles complete it: *Laplace smoothing* (add a token count of 1 to every word so a never-before-seen word doesn't multiply your probability to a hard zero — the dogmatic-prior trap again), and working entirely in log-space. With those, a hundred lines of code and the theorem from page one of this chapter gave the world a usable spam filter.

Example 5.4 (A two-word spam score)**.** Take prior odds of spam $\mathbb{P}(S)/\mathbb{P}(H) = 0.4/0.6 = 2/3$ (your inbox is 40% junk). An email contains exactly the salient words "free" (Bayes factor 50) and "meeting" (Bayes factor $1/20 = 0.05$). The posterior odds:

$$\frac{\mathbb{P}(S \mid \text{words})}{\mathbb{P}(H \mid \text{words})} = \frac{2}{3} \times 50 \times 0.05 = \frac{2}{3} \times 2.5 = \frac{5}{3} \approx 1.67.$$

Posterior odds of 1.67:1 means a posterior probability of $1.67/(1+1.67) \approx 0.625$. The "free" shouted spam, the "meeting" whispered ham, and on balance the message lands at about 63% likely spam — borderline, exactly the kind of case a threshold has to adjudicate. Add one more spammy word and it tips over; that is updating, word by word, the whole chapter in miniature.

Work it yourself

Before reading on, try these three to make sure the machinery is in your hands.

1. In Example 5.1, suppose the streets are reported *dry*. Without peeking, is $\mathbb{P}(R \mid \text{dry})$ bigger or smaller than the prior 0.2? Compute it.
2. In the base-rate paradox, what disease prevalence $\mathbb{P}(D)$ would make a single positive test a coin flip, i.e. $\mathbb{P}(D \mid +) = 0.5$? (Use the odds form: you need the prior odds to equal $1/19.8$.)
3. In Monty Hall, redo the posterior if Monty, when he has a free choice (car behind your door), prefers the *lowest-numbered* available door. If he opens Door 3, what is $\mathbb{P}(C_1 \mid M_3)$ now? (This shows how a quirk in Monty's habits leaks extra information.)

5.6 The Shape of What We Learned

Step back and notice how much reach came from how little. A single line of algebra — the symmetry of an intersection, divided through — became a complete theory of how to revise belief in the face of evidence.

- **Bayes' theorem** flips a conditional probability: $\mathbb{P}(H \mid E) = \mathbb{P}(E \mid H)\,\mathbb{P}(H)/\,\mathbb{P}(E)$, turning the forward arrow "cause → effect" into the inferential arrow "effect → likely cause."
- Its four parts are **prior** (belief before), **likelihood** (how the hypothesis predicts the data), **evidence** (the normalizer, computed by the law of total probability), and **posterior** (belief after). The slogan that carries it all: *posterior* $\propto$ *likelihood* $\times$ *prior*.
- The **base-rate fallacy** is the failure to multiply by the prior: a reliable detector of a rare thing fires mostly false alarms. Always ask how common the conclusion was to begin with; count natural frequencies when the algebra feels slippery.
- **Sequential updating** makes yesterday's posterior today's prior; with conditionally independent evidence, beliefs compound by multiplying likelihoods — or, in the **odds form**, posterior odds = Bayes factor $\times$ prior odds, which in log-space is just addition.
- **Monty Hall** is won by writing the likelihoods carefully — which forces you to ask how the evidence was generated. The host's knowledge is the whole secret.
- The **Bayesian worldview** treats probability as belief and returns whole posterior distributions, not just point estimates — the foundation of **naive Bayes** and a thread that runs all the way to modern machine learning.

What you should carry forward is less a formula than a habit of mind: when evidence arrives, do not ask "is this consistent with my hypothesis?" — almost everything is consistent with almost everything. Ask instead, "how much *more* likely is this evidence under my hypothesis than under

its rivals, and how plausible was my hypothesis to begin with?" That question, asked carefully and repeatedly, is the entire art of reasoning under uncertainty. The rest of this book gives it ever sharper tools.

5.7 Exercises

Exercise 5.1 (Warming up the flip)**.** A factory has two machines. Machine A makes 60% of the widgets and 2% of its output is defective; Machine B makes 40% and 5% is defective. A widget is drawn at random and found defective. What is the probability it came from Machine B? (Find $\mathbb{P}(E)$ first, then apply Bayes.)

Exercise 5.2 (The base rate bites)**.** A new rapid test for a virus has sensitivity 0.98 and specificity 0.96. In a town where 0.5% of people are currently infected, a random resident tests positive. (a) Find the posterior probability of infection. (b) The town doctor only tests people with symptoms, among whom the prevalence is 30%. Recompute the posterior for a symptomatic patient who tests positive, and explain in one sentence why the same test is now so much more informative.

Exercise 5.3 (Compounding evidence)**.** Using the odds form, show that for the test of Exercise 5.2(a) (prevalence 0.5%, Bayes factor $0.98/0.04 = 24.5$), it takes how many *independent* positive tests to push the posterior probability of infection above 0.95? (Hint: posterior odds $= (1/199) \times 24.5^{\,n}$; solve for the smallest integer n giving odds ≥ 19.)

Exercise 5.4 (Monty's quirk, in full)**.** Generalize the biased-host variant. When the car is behind your chosen Door 1, Monty opens Door 2 with probability p and Door 3 with probability $1-p$ (when forced, he opens whichever goat door he must). You see him open Door 3. (a) Write the three likelihoods $\mathbb{P}(M_3 \mid C_i)$ in terms of p. (b) Show that $\mathbb{P}(C_2 \mid M_3) = 1/(2-p)$ and $\mathbb{P}(C_1 \mid M_3) = (1-p)/(2-p)$. (c) Verify that for *every* $p \in [0,1]$, switching is at least as good as staying, and find the p that makes staying *least* bad.

Exercise 5.5 (Three boxes, two coins)**.** Box 1 holds two gold coins, Box 2 one gold and one silver, Box 3 two silver. You pick a box at random and draw one coin; it is gold. What is the probability the *other* coin in that box is also gold? (This is Bertrand's box paradox; the naive answer 1/2 is wrong — find the right one and explain the error.)

Exercise 5.6 (Build the filter)**.** You are training a tiny naive Bayes spam filter. Your corpus has 200 spam and 300 ham emails. The word "winner" appears in 40 of the spam and in 3 of the ham. (a) Estimate $\mathbb{P}(\text{"winner"} \mid S)$ and $\mathbb{P}(\text{"winner"} \mid H)$ and the word's Bayes factor. (b) Using the prior odds from the corpus, find the posterior probability of spam for an email whose only informative word is "winner." (c) Explain why Laplace smoothing (adding 1 to each count) would matter for a word that appears in spam but *never* in your ham sample, and recompute the Bayes factor for such a word that appears in 10 spam and 0 ham, with smoothing.

Exercise 5.7 (The prosecutor's fallacy)**.** At a crime scene, DNA is recovered. A suspect's DNA matches, and an expert testifies that "the probability of such a match in an innocent person is 1 in a million." The prosecutor concludes the suspect is guilty with probability 0.999999. Suppose the relevant population of possible perpetrators is 2 million people, exactly one of whom is guilty. (a) Using Bayes' theorem with a uniform prior over the 2 million, find the posterior probability of guilt given the match. (b) Explain in terms of this chapter exactly which probabilities the prosecutor confused.

Exercise 5.8 (Proof: order does not matter)**.** Let H be a hypothesis and let E_1, E_2 be two pieces of evidence that are conditionally independent given H and given H^c. Prove, starting from the definition of conditional probability, that

$$\mathbb{P}(H \mid E_1, E_2) = \mathbb{P}(H \mid E_2, E_1),$$

and more strongly that the posterior depends on (E_1, E_2) only through the product of their individual Bayes factors. (This is the formal statement that conditionally independent evidence can be processed in any order with the same result.)

Exercise 5.9 (Conceptual: zero is forever)**.** Explain, with a short argument from Bayes' theorem, why a hypothesis assigned prior probability exactly 0 has posterior probability 0 no matter what evidence arrives, and why a hypothesis assigned prior 1 stays at 1. What practical advice for assigning priors follows? (This is sometimes called Cromwell's rule.)

Exercise 5.10 (A continuous preview)**.** A coin has unknown bias $\theta \in [0, 1]$ of landing heads. Before flipping, you regard all values of θ as equally likely (a uniform prior). You flip once and get heads. Argue informally — you may use the proportionality slogan — that your posterior for θ should now be proportional to θ (i.e. larger biases become more plausible), and sketch its shape on $[0, 1]$. (You have just done your first Bayesian update on a *continuous* parameter; the full treatment with the Beta distribution awaits a later chapter.)

Part II

Random Variables

Chapter 6

Random Variables: The Idea

So far, the objects we have reasoned about have been *events* – subsets of a sample space, things that either happen or do not. "The coin lands heads." "At least one of the three dice shows a six." "The drawn card is a spade." Events are yes/no questions about the outcome of a random experiment, and probability assigns each of them a number between 0 and 1.

But most of the questions we actually care about are not yes/no. We do not just want to know *whether* a gambler wins; we want to know *how much* they win. We do not just want to know whether a clinical trial shows an effect; we want the *number* of patients who improved. We do not just want to know whether a packet arrives; we want the *delay* in milliseconds. The world speaks to us in *numbers attached to random outcomes*, and the mathematical object that captures "a number attached to a random outcome" is the **random variable**.

This chapter is about an idea rather than a pile of formulas. The idea looks slight, yet once it clicks a great deal of probability organizes itself around it. A random variable is a function. That is the whole of it. Much of the rest of the book amounts to working out what we can do once we accept that sentence.

6.1 From outcomes to numbers

Start with a story you already understand and watch the new idea grow out of it. Roll two fair six-sided dice. The sample space – the set of all possible outcomes – is the 36 ordered pairs

$$\Omega = \{(1,1), (1,2), \ldots, (6,6)\},$$

each equally likely with probability $1/36$. So far this is Chapter-3 material.

Now ask a natural question: *what is the sum of the two dice?* The sum is not an outcome in Ω; the outcomes are pairs. The sum is a *number we compute from* an outcome. If the dice come up $(3,5)$, the sum is 8. If they come up $(6,6)$, the sum is 12. There is a rule – a machine – that eats an outcome and spits out a number.

That machine is a random variable. Call it S. Formally, S is a function from the sample space to the real numbers:

$$S : \Omega \to \mathbb{R}, \qquad S\big((a,b)\big) = a + b.$$

The word "variable" is a little misleading and the word "random" is downright sneaky – there is nothing random about the function S itself. S is a fixed, deterministic rule. The randomness lives entirely in *which outcome* $\omega \in \Omega$ *gets fed in.* Once nature picks ω, the value $S(\omega)$ is completely determined. The randomness comes in the front door (which ω?) and the function carries it straight through to the number $S(\omega)$.

Intuition

Think of a random variable as a *measuring instrument* pointed at a random experiment. The experiment produces an outcome ω – the full, detailed state of the world. The instrument reads off one number from that state. A thermometer in a turbulent room, a scoreboard tracking a game, the meter on a taxi: each is a fixed device, but because the world it measures is uncertain, the reading is uncertain too. The random variable *is* the device, not the reading.

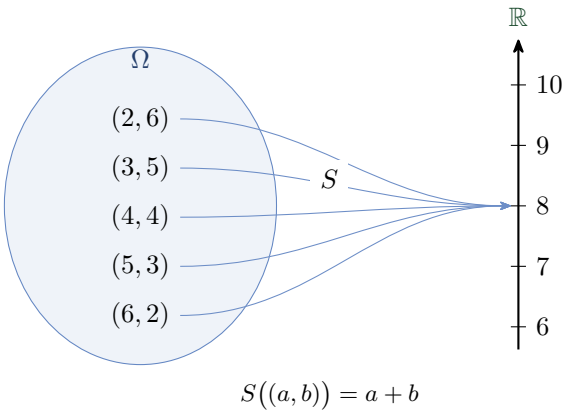


Figure 6.1: A random variable is a function. The instrument S reads each detailed outcome $\omega \in \Omega$ and reports a single number $S(\omega) \in \mathbb{R}$. All five pairs summing to 8 are collapsed onto the value 8; the event $\{S = 8\}$ is exactly their pre-image. The randomness enters through which ω nature selects; the function itself is fixed.

Definition 6.1 (Random variable, discrete first pass)**.** Let Ω be the sample space of a random experiment. A **random variable** is a function $X : \Omega \to \mathbb{R}$ that assigns a real number $X(\omega)$ to each outcome $\omega \in \Omega$. We write X, Y, S, N, ... (capital letters) for the function itself, and lowercase x, y, s, n for particular numerical values it might take.

The capital/lowercase convention is not bureaucratic fussiness; it is the single most important piece of notation in the subject, and getting it into your bones now will save you endless confusion later. X is the random thing – the device. x is a plain number – a possible reading. The expression $\{X = x\}$ then means something precise: it is the *event*

$$\{X = x\} \;=\; \{\omega \in \Omega : X(\omega) = x\},$$

the collection of all outcomes that the device would read as x. For our dice, $\{S = 8\}$ is the set of pairs summing to 8:

$$\{S = 8\} = \{(2, 6), (3, 5), (4, 4), (5, 3), (6, 2)\},$$

five outcomes out of thirty-six. And here is the bridge from the old world to the new: because

$\{S = 8\}$ is just an ordinary event, we can ask for its probability using everything we already know.

$$\mathbb{P}(S = 8) = \frac{|\{S = 8\}|}{|\Omega|} = \frac{5}{36}.$$

What is really happening

A random variable is a *relabeling* of the sample space. The original outcomes might be pairs of dice, sequences of coin flips, points in a square, or whatever messy thing the experiment naturally produces. The random variable collapses all of that detail down onto a number line. Outcomes $(2,6)$, $(3,5)$, $(4,4)$, $(5,3)$, $(6,2)$ are five genuinely different states of the world, but S glues them together: as far as S is concerned they are all just "8," as Figure 6.1 shows. This deliberate forgetting is where the leverage comes from. We throw away the information we do not need and keep the number we do.

6.1.1 Why bother with the function?

You might reasonably object: I have been computing things like "probability the sum is 8" since grade school without ever uttering the word "function." Why introduce the formal machine $S : \Omega \to \mathbb{R}$ at all?

Three reasons, each of which will pay off repeatedly.

First, it lets one experiment feed many questions. From the *same* roll of two dice we can define $S = a + b$ (the sum), $M = \max(a, b)$ (the larger die), $D = |a - b|$ (the absolute difference), and $T = \mathbf{1}\{a = b\}$ (an indicator of doubles). These are four different functions on the *one* sample space Ω. Each is a different instrument reading the same underlying randomness. Keeping the sample space fixed and letting the functions multiply is enormously economical – and, crucially, it lets us ask how these readings relate to each other (Are S and M related? Does knowing D tell us anything about S?), which is the entire content of later chapters on joint distributions and correlation.

Second, it lets us do arithmetic with randomness. If X and Y are random variables on the same Ω, then $X + Y$, XY, $3X - 2$, $\max(X, Y)$, and e^X are *also* random variables – they are just new functions on Ω, built pointwise: $(X + Y)(\omega) = X(\omega) + Y(\omega)$. Treating randomness as functions means we inherit all of algebra for free. "The total winnings over ten games" becomes a sum of ten random variables, and we can manipulate it like any other algebraic expression.

Third, it is the abstraction that lets us forget the sample space entirely. This is the deepest reason. Once we know how a random variable distributes its probability over the number line, we almost never need to look back at Ω again. Whether "the number of heads in 10 flips" came from coins, from a sequence of yes/no survey answers, or from defective items on an assembly line, the *numbers* behave identically. The random variable is the interface; Ω is the hidden implementation. The next section makes this concrete.

Connection

If you have read the companion book on linear algebra, the move here will feel familiar. There, a linear map is the central object and a matrix is merely one representation of it in a chosen basis. Here, the *distribution* (what comes next) is the central object, and the sample space Ω is merely one underlying "basis" that realizes it. Mathematicians habitually push the concrete

scaffolding into the background and reason about the structure it supports. A random variable is the first big instance of that habit in probability.

6.2 The distribution of a random variable

Now comes the conceptual leap that makes random variables indispensable. We have a function $X : \Omega \to \mathbb{R}$. The function induces, on the number line, a *spread of probability*: each possible value x gets the probability of the event that lands on it. This spread is called the **distribution** of X. It answers, in one package, every question of the form "how likely is X to fall here?"

Definition 6.2 (Distribution)**.** The **distribution** of a random variable X is the rule that assigns to each (reasonable) set $B \subseteq \mathbb{R}$ the number

$$\mathbb{P}(X \in B) \;=\; \mathbb{P}\big(\{\omega : X(\omega) \in B\}\big).$$

It tells us how the total probability mass of 1 is distributed along the real line.

For a discrete random variable – one whose possible values form a finite or countable set – the distribution is captured completely by listing how much probability sits on each individual value. That list is the **probability mass function**.

Definition 6.3 (Probability mass function (PMF))**.** Let X be a discrete random variable. Its **probability mass function** is

$$p_X(x) \;=\; \mathbb{P}(X = x) \;=\; \mathbb{P}\big(\{\omega : X(\omega) = x\}\big), \qquad x \in \mathbb{R}.$$

The PMF is positive only on the (countable) set of values X actually takes, called the *support* of X.

A PMF cannot be just any function. To be a legitimate distribution of probability it must obey two rules, and these rules are simply the probability axioms, transported through the function X.

Proposition 6.1 (Valid PMF)**.** *A function p_X is the PMF of some discrete random variable if and only if*

1. $p_X(x) \geq 0$ *for every x (probabilities are never negative), and*
2. $\displaystyle\sum_x p_X(x) = 1$ *(the total probability is one),*

where the sum runs over the support of X.

Why it works

Why must the masses sum to 1? Because the events $\{X = x\}$, as x ranges over the support, are *disjoint* (an outcome cannot make the instrument read both 3 and 5 at once) and *exhaustive* (every outcome reads *some* value). They partition Ω. By countable additivity – the third

probability axiom from Chapter 3 – the probabilities of a partition add up to $\mathbb{P}(\Omega) = 1$:

$$\sum_x p_X(x) = \sum_x \mathbb{P}(X = x) = \mathbb{P}\left(\bigcup_x \{X = x\}\right) = \mathbb{P}(\Omega) = 1.$$

Nonnegativity is just the first axiom passed along. So a PMF is nothing more exotic than the probability measure on Ω, repackaged on the number line. No new axioms are needed; the old ones are wearing new clothes.

6.2.1 The dice sum, fully worked

Let us build the entire distribution of the dice sum S and look at it. We need $\mathbb{P}(S = s)$ for each s from 2 through 12. Each is a count of pairs summing to s, divided by 36.

Example 6.1 (Distribution of the sum of two dice)**.** Counting the pairs (a, b) with $a + b = s$:

s	2	3	4	5	6	7	8	9	10	11	12
$\#\{S = s\}$	1	2	3	4	5	6	5	4	3	2	1

so the PMF is

$$p_S(s) = \frac{\#\{S = s\}}{36}, \qquad s = 2, 3, \ldots, 12.$$

As a sanity check, the counts add to $1 + 2 + \cdots + 6 + \cdots + 2 + 1 = 36$, so $\sum_s p_S(s) = 36/36 = 1$. The distribution is a perfect triangle: it rises to a peak at $s = 7$, the most likely sum, and falls off symmetrically. There is exactly one way to get a 2 (snake eyes) or a 12 (boxcars), but six ways to get a 7, which is why 7 rules at the craps table.

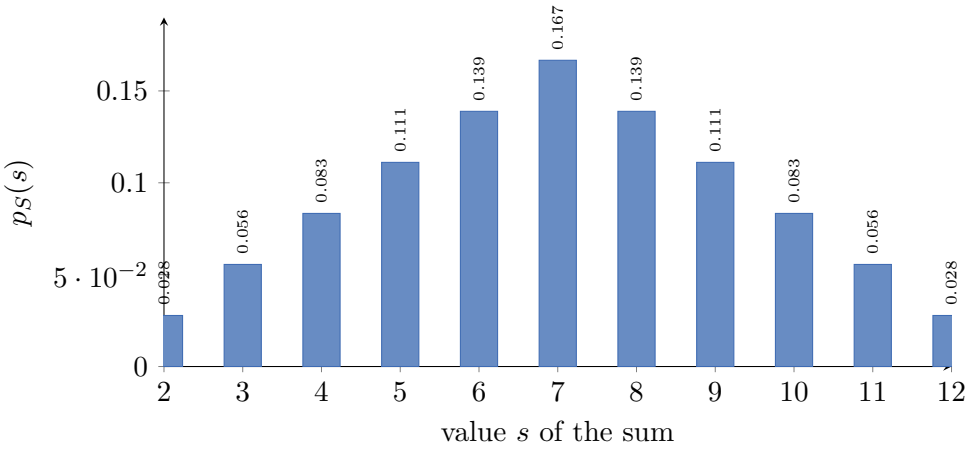


Figure 6.2: The PMF of S, the sum of two fair dice. Each bar's height is the probability mass on that value. The masses are nonnegative and add to 1; the shape is the famous craps triangle, peaking at 7.

Figure 6.2 is your first *distribution plot*, and it deserves a moment. The bar chart *is* the distribution – it shows, at a glance, where the random variable likes to land. Everything we will eventually

compute about S – its average value, its spread, the chance it exceeds 9 – is already encoded in these eleven bars. The 36-element sample space has dropped out of sight entirely; the triangle is all we need.

Walking question

Use Figure 6.2 to read off $\mathbb{P}(S \geq 10)$ without any new computation. (Add the masses on 10, 11, 12.) Then find $\mathbb{P}(S \text{ is even})$. Which is bigger, $\mathbb{P}(S = 7)$ or $\mathbb{P}(S \in \{2, 3, 11, 12\})$? Answers: $\mathbb{P}(S \geq 10) = (3{+}2{+}1)/36 = 1/6$; the even sums $2, 4, 6, 8, 10, 12$ carry $(1{+}3{+}5{+}5{+}3{+}1)/36 = 18/36 = 1/2$; and the four extreme values carry $(1 + 2 + 2 + 1)/36 = 6/36 = 1/6 = \mathbb{P}(S = 7)$ exactly – a tie.

6.2.2 Same distribution, different worlds

Here is the payoff promised earlier. Consider three completely different experiments.

Example 6.2 (Three experiments, one distribution)**. (A)** Flip a fair coin three times; let X be the number of heads. **(B)** A factory line produces three independent items, each defective with probability $1/2$; let Y be the number of defectives. **(C)** Three independent voters each answer "yes" with probability $1/2$; let Z be the number of yeses.

The sample spaces are different in flavor – coins, widgets, ballots – but each X, Y, Z counts "successes" in three independent fair trials. For X (and identically for Y, Z):

$$p_X(0) = \tfrac{1}{8}, \quad p_X(1) = \tfrac{3}{8}, \quad p_X(2) = \tfrac{3}{8}, \quad p_X(3) = \tfrac{1}{8}.$$

(There are $2^3 = 8$ equally likely outcomes; $\binom{3}{k}$ of them have k successes.) All three random variables have *exactly the same PMF*. They live on different sample spaces, but their distributions are identical.

This is why probabilists give distributions their own names. We say X, Y, and Z are each *Binomial* with parameters $n = 3$ and $p = 1/2$, written $X \sim \text{Binomial}(3, 1/2)$. The symbol $\sim$ reads "is distributed as." From this point forward, the moment you hear "count of successes in n independent trials, each succeeding with probability p," you may forget the coins and widgets and ballots entirely and reach for the Binomial(n, p) distribution. The named distribution is a reusable template; the next chapter is devoted to building a small library of such templates.

Why this matters

This is the moment probability becomes a science rather than a bag of puzzles. By focusing on the *distribution* – the spread of probability on the number line – we discover that wildly different situations share the same mathematical heart. The same bell curve governs measurement errors in astronomy, heights in a population, and the noise in a financial return. The same Poisson law counts radioactive decays, typos per page, and customers per minute. Recognizing the distribution beneath the story is much of what a working scientist actually does, and the random-variable abstraction is what makes that recognition possible.

History

The word "random variable" is younger than you might expect. Through the seventeenth and eighteenth centuries – Cardano, Pascal, Fermat, the Bernoullis, de Moivre, Laplace – probabilists computed expectations and chances of events with great virtuosity but without ever naming the function we now call a random variable; they reasoned directly about outcomes and bets. The Russian school, especially Chebyshev and his students Markov and Lyapunov in the late nineteenth century, made "random quantities" (slučajnaya velichina) the central object of study, proving limit theorems about them. The fully modern definition – a measurable function from a probability space to the reals – crystallized in Andrey Kolmogorov's 1933 *Grundbegriffe der Wahrscheinlichkeitsrechnung*, the little book that put the whole subject on rigorous set-theoretic footing. What you are learning in this chapter took nearly three centuries to come into focus.

6.3 A first look at the CDF

The PMF answers "how much mass sits exactly at x?" Often the more useful question is "how much mass sits at or below x?" – the running total. That accumulated probability is the **cumulative distribution function**, or CDF, and it is the one description of a distribution that works for *every* random variable, discrete or continuous, without modification. We will lean on it heavily once continuous variables arrive, so meet it now in the friendly discrete setting.

Definition 6.4 (Cumulative distribution function (CDF))**.** The **CDF** of any random variable X is

$$F_X(x) \;=\; \mathbb{P}(X \le x), \qquad x \in \mathbb{R}.$$

For a discrete X it is a running sum of the PMF:

$$F_X(x) \;=\; \sum_{t \le x} p_X(t).$$

For the dice sum S, the CDF climbs in jumps. Below 2 it is flat at 0; at $s = 2$ it jumps up by $p_S(2) = 1/36$; at $s = 3$ it jumps by $2/36$; and so on, reaching 1 at $s = 12$ and staying there. The *size of each jump is exactly the PMF mass at that point* – so the CDF secretly contains the PMF, and vice versa. They are two views of the same distribution.

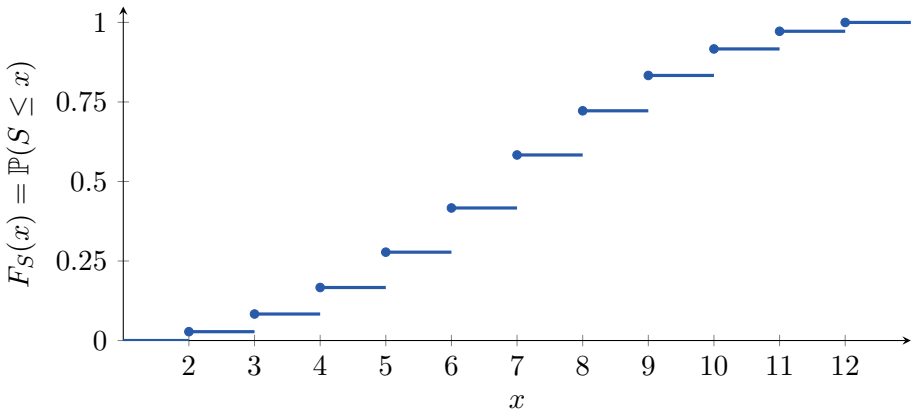


Figure 6.3: The CDF of the dice sum S. It is a right-continuous staircase: flat between the integers, jumping up at each value of the support. The height of the jump at s equals the PMF mass $p_S(s)$. The function starts at 0 on the far left and reaches 1 on the far right.

Three properties leap out of Figure 6.3, and they hold for the CDF of *any* random variable, not just this one. We state them now and prove them carefully; they will be old friends by the end of the book.

Proposition 6.2 (Defining properties of a CDF)**.** *Every CDF F_X satisfies:*

1. ***Monotone:*** *if $a \leq b$ then $F_X(a) \leq F_X(b)$.*
2. ***Limits:*** *$F_X(x) \to 0$ as $x \to -\infty$ and $F_X(x) \to 1$ as $x \to +\infty$.*
3. ***Right-continuous:*** *$F_X(x) = \lim_{t \downarrow x} F_X(t)$ for all x.*

Why it works

Monotonicity is just monotonicity of probability. If $a \leq b$ then the event $\{X \leq a\}$ is a subset of $\{X \leq b\}$ (anything at most a is certainly at most b), and probability respects containment, so $F_X(a) = \mathbb{P}(X \leq a) \leq \mathbb{P}(X \leq b) = F_X(b)$. The staircase can only go up.
The limits say that pushing the threshold to $-\infty$ excludes everything (probability $\to 0$) and pushing it to $+\infty$ includes everything (probability $\to 1$). Formally these follow from continuity of probability along the shrinking events $\{X \leq -n\} \downarrow \varnothing$ and the growing events $\{X \leq n\} \uparrow \Omega$.
Right-continuity is the subtle one, and it is the reason we define the CDF with $\leq$ rather than $<$. Because $\{X \leq x\}$ includes the value x itself, the function attains its new, higher level *exactly at* the jump point – the solid dots in the figure sit on top of each riser. As we approach x from the right, the shrinking events $\{X \leq x + 1/n\}$ decrease down to $\{X \leq x\}$, so by continuity of probability the values come back down to $F_X(x)$. The jump in F_X at a point x is precisely $\mathbb{P}(X = x)$, which recovers the PMF as the size of the jump.

Common trap

A maddeningly common error is sloppiness about $<$ versus $\leq$ at the jump points. For the dice sum, $\mathbb{P}(S \leq 7) = F_S(7) = 21/36$, but $\mathbb{P}(S < 7) = F_S(6) = 15/36$; the difference is exactly $p_S(7) = 6/36$, the mass that sits *at* 7. For a *discrete* variable, where to land matters because individual points carry mass. (For a continuous variable in later chapters, single points

carry zero mass and the distinction evaporates – but never assume that before you know your variable is continuous.) When in doubt, draw the staircase and ask whether the dot at the jump is included.

The CDF also makes interval probabilities a one-line subtraction, which is its great practical convenience. For any $a < b$,

$$\mathbb{P}(a < X \le b) = F_X(b) - F_X(a).$$

This holds because $\{X \le b\}$ splits into the disjoint pieces $\{X \le a\}$ and $\{a < X \le b\}$, and probabilities of disjoint events add. For the dice, $\mathbb{P}(4 < S \le 9) = F_S(9) - F_S(4) = \frac{30}{36} - \frac{6}{36} = \frac{24}{36} = \frac{2}{3}$.

6.4 Indicator random variables: small but mighty

Of all random variables, the simplest is the one that can only be 0 or 1. It attaches the number 1 to outcomes in some event A and the number 0 to everything else. It is the random-variable incarnation of a yes/no question, and despite – in fact because of – its simplicity it does an astonishing amount of work throughout probability.

Definition 6.5 (Indicator random variable)**.** For an event $A \subseteq \Omega$, the **indicator** of A is the random variable

$$\mathbf{1}_A(\omega) = \begin{cases} 1 & \text{if } \omega \in A, \\ 0 & \text{if } \omega \notin A. \end{cases}$$

Its distribution is concentrated on two points: $\mathbb{P}(\mathbf{1}_A = 1) = \mathbb{P}(A)$ and $\mathbb{P}(\mathbf{1}_A = 0) = 1 - \mathbb{P}(A)$. A random variable with this two-point distribution is called *Bernoulli* with parameter $p = \mathbb{P}(A)$, written $\mathbf{1}_A \sim \text{Bernoulli}(\mathbb{P}(A))$.

Why does such a trivial object matter? Because indicators convert *logic* into *arithmetic*. The set operations on events – and, or, not – become ordinary algebra on their indicators, and "counting how many things happen" becomes "adding up indicators." Watch.

Proposition 6.3 (The indicator dictionary)**.** *For events $A, B \subseteq \Omega$, as functions on Ω:*

$$\mathbf{1}_{A^c} = 1 - \mathbf{1}_A, \qquad \mathbf{1}_{A \cap B} = \mathbf{1}_A \, \mathbf{1}_B, \qquad \mathbf{1}_{A \cup B} = \mathbf{1}_A + \mathbf{1}_B - \mathbf{1}_A \mathbf{1}_B.$$

Why it works

Check each pointwise – that is, plug in an arbitrary ω and verify both sides agree. The complement: $\mathbf{1}_{A^c}(\omega) = 1$ exactly when $\omega \notin A$, i.e. when $\mathbf{1}_A(\omega) = 0$, which is exactly when $1 - \mathbf{1}_A(\omega) = 1$. The intersection: $\mathbf{1}_A(\omega)\mathbf{1}_B(\omega)$ is a product of two numbers in $\{0, 1\}$, so it equals 1 if and only if *both* factors are 1, i.e. ω is in both A and B – exactly the membership rule for $A \cap B$. The union: by inclusion–exclusion at the level of single outcomes, ω is in $A \cup B$ iff it is in A *or* B; the formula $\mathbf{1}_A + \mathbf{1}_B - \mathbf{1}_A\mathbf{1}_B$ counts A, adds B, and subtracts the double-counted overlap, landing on 1 precisely when at least one holds. Logic in, algebra out.

The real magic is the **counting trick**: if you want to count how many of several events occur, write the count as a sum of indicators.

Example 6.3 (Counting by indicators)**.** Suppose we run n independent trials and A_i is the event "trial i is a success." Let N be the total number of successes. Then

$$N = \mathbf{1}_{A_1} + \mathbf{1}_{A_2} + \cdots + \mathbf{1}_{A_n} = \sum_{i=1}^{n} \mathbf{1}_{A_i}.$$

At any outcome ω, the right-hand side adds a 1 for each successful trial and a 0 for each failure, totaling precisely the number of successes – which is what N means. So a complicated counting random variable has been disassembled into n tiny, identical, easy-to-understand pieces.

This decomposition looks almost too cheap to be useful, but it is the single most productive idea in elementary probability. The reason – which we will only fully cash in two chapters from now, when expectation arrives – is that the *average* of a sum is always the sum of the averages, even when the pieces are tangled together in complicated ways. Counting problems that are hopeless by brute force become one-liners once you write the count as $\sum \mathbf{1}_{A_i}$ and average term by term. Keep this in your back pocket; we will return to it again and again.

Application

Indicators are the bridge from probability to data. In a spreadsheet of survey responses, a column "owns a car (yes/no)" is literally a column of indicator values. The *fraction* of 1s in that column – the sample mean of the indicator – is your estimate of the underlying probability $\mathbb{P}(\text{owns a car})$. In machine learning, the 0/1 *loss* of a classifier is the indicator of the event "the prediction is wrong," and the model's error rate is the average of that indicator over the test set. One-hot encoding – the standard way to feed a categorical variable ("color $\in$ {red, green, blue}") into a model – replaces the category with a little vector of indicators, one per possible value. Whenever you see a column of 0s and 1s, you are looking at an indicator random variable, and its average is a probability estimate.

6.5 Functions of a random variable

We noted earlier that $X + Y$, X^2, and e^X are again random variables. Now we make that precise and work out how a transformation reshapes a distribution – a skill we will need constantly.

If $X : \Omega \to \mathbb{R}$ is a random variable and $g : \mathbb{R} \to \mathbb{R}$ is an ordinary function, then the composition $Y = g(X)$ is the random variable defined by

$$Y(\omega) = g\big(X(\omega)\big).$$

First the experiment produces ω, then X reads off a number, then g processes that number. It is two machines wired in series, and the result is one new instrument. "Apply g" is just feeding the reading through a second device.

The crucial question is: given the distribution of X, what is the distribution of $Y = g(X)$? For discrete variables there is a clean recipe – *gather the masses*. The value $Y = y$ happens exactly when X lands on some value that g sends to y. So we sum the PMF of X over all such pre-images.

Proposition 6.4 (PMF of a function of a discrete variable)**.** *If X is discrete with PMF p_X and $Y = g(X)$, then*

$$p_Y(y) \;=\; \mathbb{P}(Y = y) \;=\; \sum_{x\,:\,g(x)=y} p_X(x).$$

That is: to find the mass Y puts on y, find every x with $g(x) = y$ and add up the mass X already put on those x.

Why it works

The event $\{Y = y\}$ is, unwound through the definition, $\{\omega : g(X(\omega)) = y\}$, which is the union over all x with $g(x) = y$ of the events $\{X = x\}$. These events are disjoint (different x's), so their probabilities add – giving the sum over the pre-image $g^{-1}(\{y\})$. No mass is created or destroyed; it is merely relabeled by g and then pooled. When g is one-to-one, each y has a single pre-image and the sum has one term: $p_Y(y) = p_X(g^{-1}(y))$. When g collapses several x's onto one y, those masses *merge*.

Example 6.4 (A transformation that merges mass)**.** Let X be uniform on $\{-2, -1, 0, 1, 2\}$, so $p_X(x) = 1/5$ for each. Let $Y = X^2$. The map $g(x) = x^2$ is not one-to-one: it sends both -2 and 2 to 4, and both -1 and 1 to 1. Gathering masses:

$$p_Y(0) = p_X(0) = \tfrac{1}{5}, \qquad p_Y(1) = p_X(-1) + p_X(1) = \tfrac{2}{5}, \qquad p_Y(4) = p_X(-2) + p_X(2) = \tfrac{2}{5}.$$

The support shrank from five values to three, and the masses on the merged values doubled. Total is still $\frac{1}{5} + \frac{2}{5} + \frac{2}{5} = 1$, as it must be. Squaring folded the symmetric distribution in half.

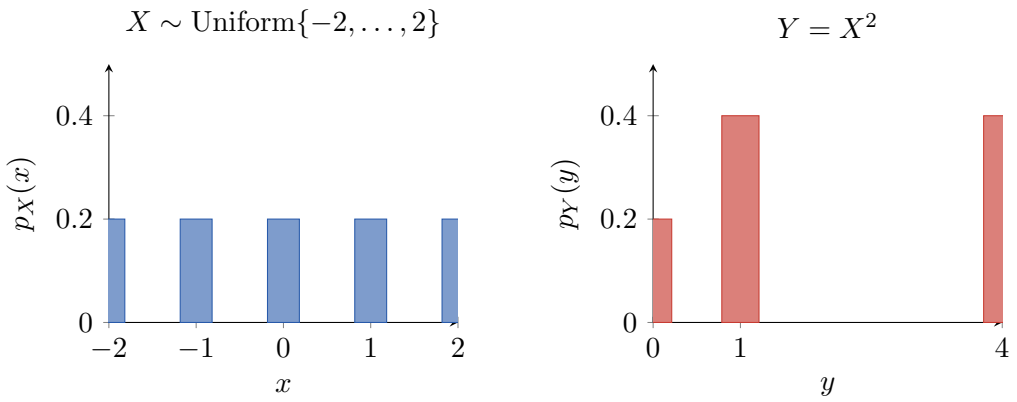


Figure 6.4: A non-invertible transformation pools probability. On the left, X is uniform on five points. Squaring (right) merges the symmetric pairs ± 1 and ± 2, doubling their masses and collapsing the support to $\{0, 1, 4\}$.

Common trap

The number-one beginner mistake with transformations is to assume $p_Y(y) = p_X(g^{-1}(y))$ in general – "just substitute." That is only valid when g is one-to-one on the support. When g merges values (like squaring, or taking absolute value, or rounding), you *must* sum over all pre-images. Forgetting the second term turned $p_Y(4) = 2/5$ into a wrong $1/5$ in the example above, and the masses would no longer sum to 1 – always a red flag that a transformation was done carelessly. The summing-to-one check is your friend.

A special, important case is the **affine** (linear-plus-shift) transformation $Y = aX + b$ with $a \neq 0$. Here $g(x) = ax + b$ is one-to-one, so no merging occurs; the distribution is merely relocated and

rescaled. Each value x moves to $ax + b$, carrying its mass along untouched:

$$p_Y(y) = p_X\left(\frac{y-b}{a}\right).$$

Geometrically, multiplying by a stretches the number line (and flips it if $a < 0$), while adding b slides it. The *shape* of the distribution is preserved; only its location and scale change. This is exactly what happens when we convert Celsius to Fahrenheit, dollars to euros, or – a move we will make constantly – when we *standardize* a variable by subtracting its center and dividing by its spread.

Connection

Affine transformations of random variables are the probabilistic shadow of the affine maps you met in linear algebra: a scaling (the a) composed with a translation (the b). When we reach the normal distribution, the entire family of bell curves will turn out to be the affine images of a single "standard" bell curve $\mathcal{N}(0,1)$ – one shape, infinitely relocated and rescaled. The habit of asking "what does this transformation do to the distribution?" is, in disguise, the habit of asking "how does this map move space around?" For continuous variables the same question requires a derivative correction (the change-of-variables / Jacobian factor), which the later chapter on transformations develops in full. The discrete mass-gathering recipe here is the clean prototype of that idea.

6.6 A subtle point: which functions are allowed?

We have been saying "X is any function $\Omega \to \mathbb{R}$." For finite or countable sample spaces – coins, dice, cards, counts – that is literally true and you may take it at face value; nothing can go wrong. But for the uncountable sample spaces that appear once we model continuous quantities (a uniformly random point in $[0,1]$, say), a fully arbitrary function can be pathological, and we need a mild restriction. This subsection is a signpost, not a detour; you can skim it and return after meeting continuous variables.

The whole point of a random variable was to let us ask "$\mathbb{P}(X \in B)$." For that question to have an answer, the set $\{\omega : X(\omega) \in B\}$ must be an *event* – something our probability measure actually knows how to weigh. A function for which this holds (for every reasonable target set B) is called **measurable**, and the official definition of a random variable includes this word.

Formal

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space, where $\mathcal{F}$ is the collection of events the measure $\mathbb{P}$ is defined on. A function $X : \Omega \to \mathbb{R}$ is a **random variable** if it is *measurable*: for every $x \in \mathbb{R}$,

$$\{\omega : X(\omega) \leq x\} \in \mathcal{F}.$$

Equivalently, $X^{-1}(B) \in \mathcal{F}$ for every Borel set $B \subseteq \mathbb{R}$. The condition guarantees that $\mathbb{P}(X \leq x)$ – and hence $\mathbb{P}(X \in B)$ for all the sets we care about – is well defined, which is precisely what makes the CDF and the distribution exist.

Do not let this worry you. Every function you will ever write down by an explicit formula – sums, products, maxima, polynomials, exponentials, indicators of sensible events, limits of these – is measurable. Measurability is a safety rail that rules out artificially constructed monsters, not a constraint

on real modeling. We mention it only so that the word does not ambush you later, and so you understand *why* the careful definition exists: it is the price of guaranteeing that the question "$\mathbb{P}(X \in B)$?" always has an answer. The full machinery is the subject of the measure-theoretic chapters at the end of the book; until then, "a random variable is a number-valued reading of a random outcome" will serve you perfectly.

Paradox

The two-envelope puzzle, and why "a random variable" is not enough. Here is a puzzle that quietly abuses the random-variable idea. Two envelopes contain money; one holds twice as much as the other, but you do not know which. You pick one and find \$100. Should you switch? A tempting argument: the other envelope holds either \$50 or \$200, each "equally likely," so its expected contents are $\frac{1}{2}(50) + \frac{1}{2}(200) = \$125 > \$100$ – always switch! But by symmetry the same argument applies *before* you open anything, suggesting you should switch endlessly, which is absurd.

The resolution lives exactly in the ideas of this chapter. The flawed step smuggles in a *distribution* for the amount of money that does not actually exist. To say "the other envelope is \$50 or \$200 with probability 1/2 each" you must specify a genuine random variable – a real distribution over how much money was put in the envelopes in the first place. The instant you write down any *valid* PMF for the smaller amount (one whose masses are nonnegative and sum to 1, as Proposition 6.1 demands), the seductive "always \$125" symmetry breaks: for some observed values switching helps and for others it hurts, and they balance. No legitimate distribution makes switching always better. The paradox is not about envelopes; it is about pretending a random variable exists when no valid distribution backs it up. *Always ask whether your masses sum to one.*

6.7 Putting it together

One example can exercise every idea in the chapter at once: a random variable as a function, its PMF, its CDF, an indicator decomposition, and a transformation. Here it is.

Example 6.5 (Coupons in a tiny collection)**.** A cereal company hides one of two equally likely toys, A or B, in each box. You buy three boxes. Model the outcome as a string like ABA; there are $2^3 = 8$ equally likely strings. Let X be the number of *distinct* toys you collect (so $X \in \{1, 2\}$).

As a function. $X(\omega) = 1$ if all three boxes match (AAA or BBB) and $X(\omega) = 2$ otherwise. The event $\{X = 1\}$ contains 2 of the 8 strings.

PMF. $p_X(1) = 2/8 = 1/4$ and $p_X(2) = 6/8 = 3/4$. Masses are nonnegative and sum to 1. (So in fact $X \sim \text{Bernoulli}(3/4) + 1$, a shifted coin – or just a two-point distribution.)

CDF. $F_X(x) = 0$ for $x < 1$; $F_X(x) = 1/4$ for $1 \le x < 2$; and $F_X(x) = 1$ for $x \ge 2$. A two-step staircase, jumping by $p_X(1)$ then by $p_X(2)$.

Indicator view. Let C be the event "you collected both toys." Then $\mathbf{1}_C = \mathbf{1}\{X = 2\}$, a Bernoulli(3/4) variable, and $X = 1 + \mathbf{1}_C$. We have written the count as "one (you always get at least one toy) plus an indicator for having gotten the second." This is the counting trick in miniature.

Transformation. Suppose completing the set (getting both toys) wins a \$10 prize and otherwise

you win nothing: $W = 10\,\mathbf{1}_C$. This is a function of X (indeed $W = g(X)$ with $g(1) = 0, g(2) = 10$). Gathering masses, W has PMF $p_W(0) = 1/4$, $p_W(10) = 3/4$. The whole problem – function, distribution, running total, indicator, payoff – has been narrated in the language of this chapter without a single new formula.

Pause

Take a breath and assemble the picture. A *random variable* is a function from outcomes to numbers – a measuring device. Its *distribution* is how it spreads probability along the number line, summarized for discrete variables by the *PMF* (mass at each point, nonnegative, summing to one) and for any variable by the *CDF* (running total $\mathbb{P}(X \le x)$, a monotone right-continuous climb from 0 to 1, whose jumps recover the PMF). *Indicators* are the 0/1 atoms that turn logic into arithmetic and counting into summing. *Functions* of a random variable are new devices wired in series, whose distribution you find by gathering masses over pre-images. Everything that follows – expectation, variance, the named distributions, the limit theorems – is built on these five ideas. You now own the noun that the rest of the subject is about.

6.8 Exercises

Exercise 6.1 (Reading a PMF)**.** A random variable X has PMF given by $p_X(0) = 0.1$, $p_X(1) = 0.3$, $p_X(2) = c$, $p_X(3) = 0.2$, and $p_X(x) = 0$ otherwise. (a) Find c. (b) Compute $\mathbb{P}(X \ge 2)$. (c) Compute the CDF value $F_X(2)$. (d) Compute $\mathbb{P}(1 \le X < 3)$ and $\mathbb{P}(1 < X \le 3)$, and explain why they differ.

Exercise 6.2 (Building a random variable)**.** Three fair coins are flipped. Let H be the number of heads and let L be the length of the longest run of consecutive heads (so $L = 0$ if no heads, $L = 3$ for HHH). (a) Write out the eight equally likely outcomes and the values of H and L on each. (b) Give the PMF of L. (c) Are H and L the same random variable? Explain what it means for two random variables on the same Ω to be different even when they share a sample space.

Exercise 6.3 (Max of two dice)**.** Roll two fair dice and let $M = \max(a, b)$. (a) Show $\mathbb{P}(M \le m) = (m/6)^2$ for $m = 1, \dots, 6$ by a direct counting argument, and use it to find the CDF. (b) Obtain the PMF as the jumps of the CDF, i.e. $p_M(m) = F_M(m) - F_M(m-1)$, and verify the masses sum to 1. (c) Which is more likely, $M = 6$ or $M \le 3$?

Exercise 6.4 (Indicator algebra)**.** Let A, B, C be events. (a) Using only the indicator dictionary (Proposition 6.3), expand $\mathbf{1}_{A\cup B\cup C}$ in terms of $\mathbf{1}_A, \mathbf{1}_B, \mathbf{1}_C$ and their products. (b) Read off the inclusion–exclusion formula for $\mathbb{P}(A \cup B \cup C)$ by taking the expectation of each side (you may assume, as we will prove in Chapter 8, that the expectation of a sum of indicators is the sum of their probabilities). (c) Explain in one sentence why this derivation is cleaner than a Venn-diagram argument.

Exercise 6.5 (Transforming a distribution)**.** Let X be uniform on $\{-3, -2, -1, 0, 1, 2, 3\}$ (seven values, each mass 1/7). (a) Find the PMF of $Y = |X|$. (b) Find the PMF of $Z = X^2 - 2X$. (Watch for values of x that collide.) (c) Find the PMF of $W = 2X + 1$ and explain why no masses merge here.

Exercise 6.6 (Counting fixed points – a preview)**.** Three people check their coats and the coats are returned in a uniformly random order (one of the $3! = 6$ permutations). Let N be the number of

people who get their own coat back. (a) For $i = 1, 2, 3$, let A_i be the event "person i gets their own coat"; write $N = \mathbf{1}_{A_1} + \mathbf{1}_{A_2} + \mathbf{1}_{A_3}$ and find $\mathbb{P}(A_i)$ for each. (b) By listing all six permutations, find the full PMF of N. (Notice $N = 2$ is impossible – why?) (c) Verify your PMF sums to 1.

Exercise 6.7 (CDF properties, conceptual)**.** Decide whether each function F can be the CDF of some random variable; if not, say which property of Proposition 6.2 fails. (a) $F(x) = 0$ for $x < 0$, $F(x) = x$ for $0 \le x \le 1$, $F(x) = 1$ for $x > 1$. (b) $F(x) = 0$ for $x < 0$, $F(x) = 1 - e^{-x}$ for $x \ge 0$. (c) The function that is 0 for $x < 0$, jumps to 0.6 at $x = 0$, drops to 0.4 at $x = 1$, and is 1 for $x \ge 2$. (d) $F(x) = \frac{1}{2} + \frac{1}{\pi}\arctan(x)$ for all $x \in \mathbb{R}$.

Exercise 6.8 (Same distribution, different functions)**.** On the sample space of a single fair die ($\Omega = \{1, 2, 3, 4, 5, 6\}$), define $X(\omega) = \omega$ and $Y(\omega) = 7 - \omega$. (a) Show X and Y have the *same* distribution. (b) Show that nevertheless X and Y are different functions, e.g. they never agree on any outcome. (c) Compute the PMF of $X + Y$ and explain why it is degenerate (puts all its mass on one value). What does this teach you about the difference between "same distribution" and "same random variable"?

Exercise 6.9 (A proof: distributions determine each other)**.** Let X be a discrete random variable taking values in $\{x_1 < x_2 < x_3 < \cdots\}$. (a) Prove that the PMF determines the CDF via $F_X(x) = \sum_{x_k \le x} p_X(x_k)$. (b) Prove the converse: the CDF determines the PMF via $p_X(x_k) = F_X(x_k) - F_X(x_k^-)$, where $F_X(x_k^-) = \lim_{t \uparrow x_k} F_X(t)$ is the left limit. (c) Conclude that two discrete random variables with the same CDF have the same PMF, and hence the same distribution. Where in your argument did you use right-continuity?

Exercise 6.10 (Modeling open-ended)**.** You roll a fair die repeatedly until the first time a 6 appears; let T be the number of rolls (including the final 6). (a) Argue that the sample space here is naturally *countably infinite*, and describe it. (b) Find $\mathbb{P}(T = k)$ for $k = 1, 2, 3, \ldots$ and verify $\sum_{k=1}^{\infty} \mathbb{P}(T = k) = 1$ using the geometric series. (c) This is your first encounter with a random variable on an infinite sample space. Explain why the "masses sum to one" requirement now genuinely needs the *infinite* sum, and why that is exactly what countable additivity (Chapter 3) was built to handle.

Chapter 7

Discrete Distributions

In the last chapter you met the random variable: a number whose value is decided by chance, with a probability mass function (PMF) that tells you how much weight sits on each possible value. So far each PMF has been a one-off, invented for the problem at hand. But a small handful of PMFs show up again and again, across coin games, genetics, call centers, quality control, web traffic, radioactivity, and machine learning, and they are not arbitrary. Each one is the answer to a *story* – a particular way of generating randomness – and once you know the story, you can recognize the distribution when it appears and write down its mean, its variance, and its shape without re-deriving anything.

We will meet six named families: **Bernoulli**, **Binomial**, **Geometric**, **Negative Binomial**, **Hypergeometric**, and **Poisson**. The trick to remembering them forever is to never memorize the formula first. Memorize the *story* first, and let the formula fall out of the story. By the end you should be able to read a word problem, hear which story it is telling, and name the distribution before you pick up a pen.

Focus

The single most useful skill in this chapter is *model recognition*: given a real situation, deciding *which* of these families fits. The questions to ask are always the same. Is each trial yes/no? Are the trials independent? Is the number of trials fixed in advance, or are we waiting for something? Are we sampling with or without replacement? Are we counting rare events in a window of time or space? Each answer points at a distribution.

7.1 Bernoulli: the atom of randomness

Everything in this chapter is built out of one indivisible piece: a single trial with two outcomes, "success" and "failure." Flip a coin. Did a customer buy? Did the patient recover? Did the bit arrive corrupted? Strip away the story and you have a number that is 1 with probability p and 0 with probability $1 - p$. That is the **Bernoulli** random variable, named for Jacob Bernoulli, and it is the atom from which the binomial, geometric, and negative binomial are all assembled.

Definition 7.1 (Bernoulli distribution)**.** A random variable X has the **Bernoulli** distribution with parameter $p \in [0, 1]$, written $X \sim \text{Bernoulli}(p)$, if it takes the value 1 with probability p and 0 with

probability $1-p$:

$$\mathbb{P}(X=1)=p, \qquad \mathbb{P}(X=0)=1-p=q.$$

We call the outcome $X=1$ a *success* and $X=0$ a *failure*; p is the *success probability* and we often abbreviate $q=1-p$.

The choice to code success as the *number* 1 and failure as 0 is not cosmetic. It is exactly what makes the algebra work, because then X is an *indicator*: $X=\mathbf{1}_A$ where A is the event "success." Its mean and variance are immediate.

Proposition 7.1 (Mean and variance of Bernoulli)**.** *If* $X \sim \text{Bernoulli}(p)$ *then* $\mathbb{E}[X]=p$ *and* $\text{Var}(X)=p(1-p)=pq$.

Proof. By the definition of expectation for a discrete variable,

$$\mathbb{E}[X]=1\cdot p+0\cdot(1-p)=p.$$

Since X only takes the values 0 and 1, we have $X^2=X$, so $\mathbb{E}[X^2]=\mathbb{E}[X]=p$. Therefore

$$\text{Var}(X)=\mathbb{E}[X^2]-(\mathbb{E}[X])^2=p-p^2=p(1-p)=pq. \qquad \square$$

Intuition

The variance $p(1-p)$ is a little parabola, zero at $p=0$ and $p=1$, peaking at $p=\frac{1}{2}$ where it equals $\frac{1}{4}$. This matches intuition perfectly: a coin that always lands heads ($p=1$) or always tails ($p=0$) has *no* randomness, so zero variance. The most unpredictable coin is the fair one, and that is exactly where the variance is largest.

Bernoulli variance: a parabola peaking at the fair coin

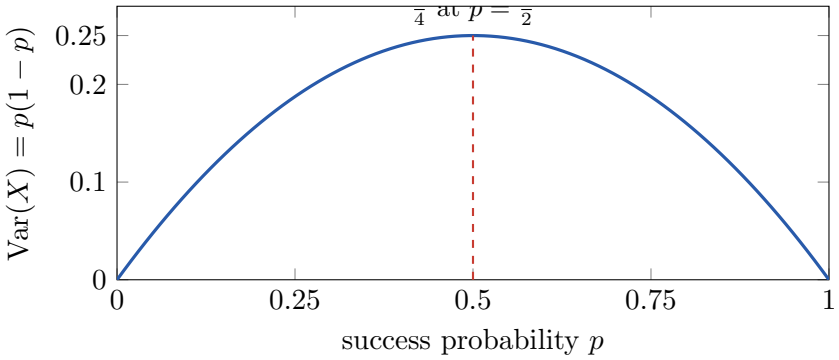


Figure 7.1: The Bernoulli variance $p(1-p)$ as a function of p. It vanishes at the deterministic extremes $p=0$ and $p=1$ and is largest, equal to $\frac{1}{4}$, at the fair coin $p=\frac{1}{2}$.

Why it works

Why does the trick $X^2=X$ work? Because the only values X can take are 0 and 1, and both satisfy $x^2=x$. Squaring a 0/1 variable changes nothing. This is the single most useful identity for indicators, and you will use it constantly: the variance of an indicator of an event

A is $\mathbb{P}(A)\big(1-\mathbb{P}(A)\big)$, full stop.

A Bernoulli on its own is almost too simple to be interesting. What makes it matter is that it is a building block. The next move, which drives the rest of this chapter, is to line up many independent Bernoulli trials and ask different questions about them. Each question gives a different named distribution.

7.2 Binomial: counting successes in a fixed number of trials

Here is the first compound story. Fix the number of trials at n. Run n independent Bernoulli(p) trials – n coin flips, n patients, n shots on goal. Now ask: *how many successes did I get?* That count, anywhere from 0 to n, is a **Binomial** random variable.

Intuition

The keyword is *fixed n, count the successes*. You decided in advance how many trials to run, the trials don't influence each other, each has the same success chance p, and your random quantity is the tally of successes. Whenever a problem matches that template – 20 free throws, 500 users shown an ad, 8 births at a hospital – reach for the binomial.

Definition 7.2 (Binomial distribution)**.** Let X be the number of successes in n independent Bernoulli(p) trials. Then X has the **Binomial** distribution with parameters n and p, written $X \sim \text{Binomial}(n, p)$, with PMF

$$\mathbb{P}(X = k) = \binom{n}{k} p^k (1-p)^{n-k}, \qquad k = 0, 1, \ldots, n.$$

Why it works

Where does the formula come from? Build it from the story. One *specific* sequence of k successes and $n-k$ failures – say SSFSF… – has probability $p^k(1-p)^{n-k}$, because the trials are independent so we multiply, and there are k factors of p and $n-k$ factors of q. But we don't care *which* trials succeeded, only *how many*. The number of distinct sequences with exactly k successes is the number of ways to choose which k of the n slots are successes, namely $\binom{n}{k}$. Each such sequence has the same probability $p^k q^{n-k}$, and they are disjoint, so we add them up: $\binom{n}{k} p^k q^{n-k}$. The two ingredients are: a *counting* factor $\binom{n}{k}$ (how many arrangements) times a *probability* factor $p^k q^{n-k}$ (the chance of any one arrangement).

Check that these probabilities sum to 1; the check reveals where the name "binomial" comes from. Summing over all k,

$$\sum_{k=0}^{n} \binom{n}{k} p^k q^{n-k} = (p+q)^n = 1^n = 1,$$

by the binomial theorem. The PMF is literally the binomial expansion of $(p+q)^n$, term by term. That is the whole reason for the name.

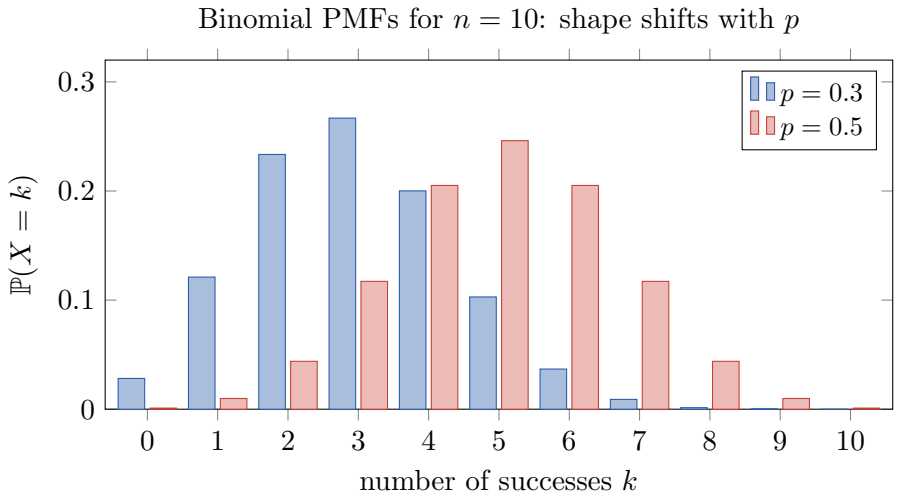


Figure 7.2: Two binomial PMFs with $n = 10$. With $p = 0.5$ the bars are symmetric and centered at $5 = np$. With $p = 0.3$ the whole pile shifts left to center near $3 = np$ and develops a right skew. The expected count np is always where the mass piles up.

7.2.1 Mean and variance the easy way

You could compute $\mathbb{E}[X] = \sum_k k\binom{n}{k}p^k q^{n-k}$ directly, and it can be done, but it is a slog. The elegant route is to remember the story: a binomial is a *sum* of Bernoullis. Write $X = X_1 + X_2 + \cdots + X_n$ where X_i is the indicator that trial i was a success. Each $X_i \sim \text{Bernoulli}(p)$ and they are independent.

Proposition 7.2 (Mean and variance of Binomial)**.** *If* $X \sim \text{Binomial}(n, p)$ *then*

$$\mathbb{E}[X] = np, \qquad \text{Var}(X) = np(1-p) = npq.$$

Proof. Write $X = \sum_{i=1}^{n} X_i$ with $X_i \sim \text{Bernoulli}(p)$ independent. Expectation is linear, *always*, even without independence, so

$$\mathbb{E}[X] = \sum_{i=1}^{n} \mathbb{E}[X_i] = \sum_{i=1}^{n} p = np.$$

For the variance we use that the variance of a sum of *independent* variables is the sum of the variances:

$$\text{Var}(X) = \sum_{i=1}^{n} \text{Var}(X_i) = \sum_{i=1}^{n} p(1-p) = np(1-p). \qquad \square$$

What is really happening

Notice the division of labor in that proof. The mean came from *linearity of expectation*, which never needs independence – it would hold even if the trials were correlated. The variance, on the other hand, leaned on independence: only because the X_i are independent do the variances simply add with no covariance terms. This is a recurring theme. Means are robust and add up no matter what; variances are delicate and add up only when things don't interfere. If your trials were positively correlated, the count would be *more* spread out than npq; if negatively correlated, less. The hypergeometric, coming up, is exactly a case of negatively correlated

trials, and you will see its variance come out *smaller* than the binomial's for that reason.

Example 7.1 (Free throws)**.** A player makes each free throw independently with probability $p = 0.8$ and takes $n = 12$ shots. Let X be the number made. Then $X \sim \text{Binomial}(12, 0.8)$.

- Expected made: $\mathbb{E}[X] = np = 12 \times 0.8 = 9.6$.
- Standard deviation: $\text{SD}(X) = \sqrt{npq} = \sqrt{12 \cdot 0.8 \cdot 0.2} = \sqrt{1.92} \approx 1.39$.
- Probability of making exactly 10: $\binom{12}{10}(0.8)^{10}(0.2)^2 = 66 \cdot 0.10737 \cdot 0.04 \approx 0.2835$.
- Probability of making all 12: $(0.8)^{12} \approx 0.0687$.

So even an 80% shooter sinks all twelve only about 7% of the time – a nice gut-check against the intuition that "a good shooter should hit them all."

Example 7.2 (A/B test conversions)**.** You show a new checkout button to $n = 500$ visitors. Historically the conversion rate is $p = 0.06$. If the button doesn't change behavior, the number of conversions is $X \sim \text{Binomial}(500, 0.06)$, with $\mathbb{E}[X] = 30$ and $\text{SD}(X) = \sqrt{500 \cdot 0.06 \cdot 0.94} \approx 5.31$. If you actually observe 45 conversions, that is $(45 - 30)/5.31 \approx 2.8$ standard deviations above the mean – a result this extreme by chance alone is unusual, which is the seed of a hypothesis test. The binomial is the null-hypothesis model that A/B testing rests on.

History

The Bernoulli trial and the binomial both trace to Jacob Bernoulli's *Ars Conjectandi*, published posthumously in 1713. There Bernoulli proved the first law of large numbers: as you run more and more independent trials, the observed fraction of successes converges to the true p. He saw clearly that the binomial count $\text{Binomial}(n, p)$ concentrates ever more tightly around np as n grows – our variance $np(1-p)$ grows only like n, while the count itself grows like n, so the *proportion* has variance $p(1-p)/n \to 0$. That single observation, that averages stabilize, is the foundation on which all of statistics, polling, and empirical science rests.

Common trap

The classic binomial mistake is forgetting the counting factor $\binom{n}{k}$. "The probability of 10 makes out of 12 is $0.8^{10} \cdot 0.2^{2}$" is wrong; that is the probability of *one particular* sequence (the first ten in, last two missed). There are $\binom{12}{10} = 66$ such sequences, and you want any of them. A second trap: applying the binomial when trials are *not* independent or p *changes* between trials. Sampling people without replacement from a small group, for instance, is hypergeometric, not binomial.

7.3 Geometric: waiting for the first success

The binomial fixed n and counted successes. Now flip the question on its head. Don't fix the number of trials; instead keep going until you get your first success, and ask: *how many trials did that take?* You are waiting for something – the first defective part, the first "yes," the first six on a die. The waiting time (in trials) is a **Geometric** random variable.

Intuition

The keyword is *wait for the first success*. The number of trials is now random, not fixed; it is the thing you are measuring. Where the binomial answers "how many successes in n tries," the geometric answers "how many tries until the first success."

Definition 7.3 (Geometric distribution)**.** Run independent Bernoulli(p) trials with $p > 0$ until the first success. Let X be the number of trials needed, including the successful one. Then X has the **Geometric** distribution with parameter p, written $X \sim \text{Geometric}(p)$, with PMF

$$\mathbb{P}(X = k) = (1-p)^{k-1}p, \qquad k = 1, 2, 3, \ldots$$

The story makes the formula transparent: to have your first success on trial k, you need the first $k-1$ trials to fail (probability q^{k-1}, multiplying because they're independent) and then the k-th to succeed (probability p). Hence $q^{k-1}p$. These sum to 1 as they should, since it is a geometric series:

$$\sum_{k=1}^{\infty} q^{k-1}p = p\sum_{j=0}^{\infty} q^j = p \cdot \frac{1}{1-q} = \frac{p}{p} = 1.$$

That geometric series is, of course, where the distribution gets its name.

Reading note

Beware of two conventions. Some books (and software) define the geometric as the number of *failures before* the first success, $Y = X - 1$, supported on $0, 1, 2, \ldots$ with PMF $q^k p$. Both are called "geometric." We use the *number of trials* version (support starting at 1) throughout this book, so its mean is $1/p$, not q/p. Always check which convention a source uses.

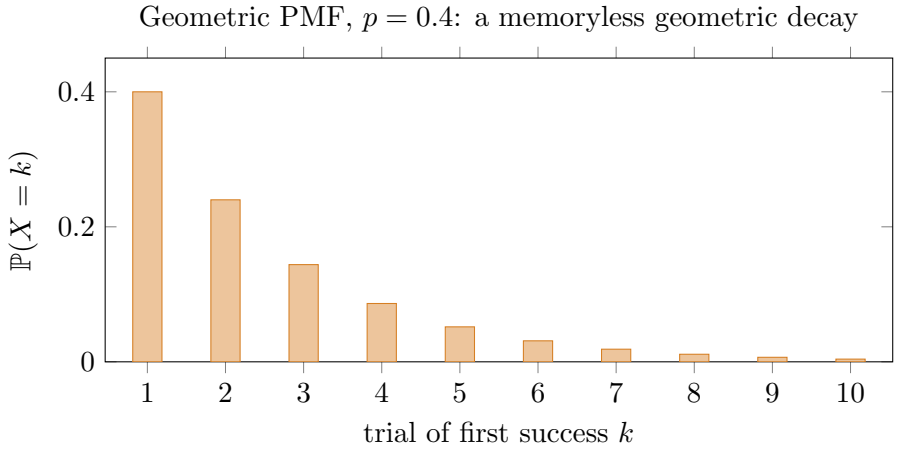


Figure 7.3: Geometric PMF with $p = 0.4$. The bars decay by a constant factor $q = 0.6$ at each step – each bar is 60% of the one before. The mode is always at $k = 1$: the single most likely outcome is success on the very first try, no matter how small p is.

7.3.1 Mean, variance, and the memoryless property

Proposition 7.3 (Mean and variance of Geometric)**.** *If* $X \sim \text{Geometric}(p)$ *then*

$$\mathbb{E}[X] = \frac{1}{p}, \qquad \text{Var}(X) = \frac{1-p}{p^2}.$$

Proof. We give the slick first-step argument for the mean; it is more illuminating than grinding the series. Condition on the first trial. With probability p it succeeds and we are done in one trial. With probability q it fails, and then – here is the key – we are starting over, having already used one trial. The *remaining* number of trials is again Geometric(p) by independence. So

$$\mathbb{E}[X] = p \cdot 1 + q \cdot (1 + \mathbb{E}[X]).$$

Solve: $\mathbb{E}[X] = p + q + q\,\mathbb{E}[X]$, hence $\mathbb{E}[X](1-q) = 1$, so $\mathbb{E}[X] = 1/p$. The variance follows from a similar first-step relation on $\mathbb{E}[X^2]$ (Exercise), giving $\text{Var}(X) = (1-p)/p^2$. □

Intuition

The mean $1/p$ is wonderfully intuitive. If a success happens one time in ten ($p = 0.1$), you expect to wait about ten trials to see one. If $p = 1/6$ (rolling a six), you expect 6 rolls. The rarer the success, the longer the wait, in exact reciprocal proportion. This $\mathbb{E}[X] = 1/p$ is one of the most quoted facts in all of applied probability.

The geometric distribution has a famous and almost magical property that no other discrete waiting-time distribution shares.

Theorem 7.1 (Memorylessness)**.** *If* $X \sim \text{Geometric}(p)$*, then for all integers* $s, t \geq 0$*,*

$$\mathbb{P}(X > s + t \mid X > s) = \mathbb{P}(X > t).$$

Proof. First, $\mathbb{P}(X > m) = q^m$: to exceed m trials means the first m trials all failed, probability q^m. Then

$$\mathbb{P}(X > s+t \mid X > s) = \frac{\mathbb{P}(X > s+t,\ X > s)}{\mathbb{P}(X > s)} = \frac{\mathbb{P}(X > s+t)}{\mathbb{P}(X > s)} = \frac{q^{s+t}}{q^s} = q^t = \mathbb{P}(X > t),$$

using that $\{X > s+t\} \subseteq \{X > s\}$. □ □

Paradox

[The gambler's fallacy, formalized] Memorylessness says: *the dice have no memory.* Suppose you have rolled a die 20 times without a six. How long must you now expect to wait? The seductive wrong answer is "a six is overdue, it should come soon." The correct answer is that the future is statistically identical to a fresh start: you still expect 6 more rolls, exactly as if the 20 misses never happened. The coin, the die, the roulette wheel – none of them are keeping a ledger that they owe you a win. The geometric distribution is the precise mathematical statement that past failures carry zero information about when the next success arrives. The "hot hand" intuition and the "law of averages" intuition are both dismantled by this one equation. (The continuous cousin of this property belongs to the exponential distribution,

which is why exponential and geometric are spiritual siblings – one in continuous time, one in discrete trials.)

Why it works

Why is the geometric memoryless, intuitively? Because the trials are independent and identical. The process has no internal state that records what has happened; each new trial is a clean coin flip with the same p. "How many more trials until success" depends only on the same coin you have been flipping all along, not on how long you have been flipping it. Any waiting-time distribution built from *independent identical* trials *must* be memoryless – and in fact the geometric is the *only* discrete distribution on $\{1, 2, 3, \dots\}$ with this property, which is a clean characterization.

Example 7.3 (Hiring search)**.** Suppose each candidate you interview is a good hire with probability $p = 0.15$, independently. Let X be the number of interviews until your first good hire. Then $X \sim \text{Geometric}(0.15)$.

- Expected interviews: $\mathbb{E}[X] = 1/0.15 \approx 6.67$.
- Probability the first good hire is the third candidate: $\mathbb{P}(X = 3) = (0.85)^2(0.15) \approx 0.1084$.
- Probability you need more than 10 interviews: $\mathbb{P}(X > 10) = (0.85)^{10} \approx 0.1969$ – about a 20% chance of a long search.

7.4 Negative Binomial: waiting for the r-th success

The geometric waits for the *first* success. The natural generalization waits for the r-th. Keep running independent Bernoulli(p) trials until you have accumulated r successes, and ask how many trials that took. That count is a **Negative Binomial** random variable. Setting $r = 1$ recovers the geometric exactly, so the geometric is the special case "wait for one success."

Intuition

Keyword: *wait for the r-th success.* Collecting r defects before shutting down a line; selling r subscriptions; a pitcher recording the r-th strikeout. The number of trials is random, and you stop the instant the r-th success lands.

Definition 7.4 (Negative Binomial distribution)**.** Run independent Bernoulli(p) trials until the r-th success ($r \geq 1$ an integer). Let X be the total number of trials. Then X has the **Negative Binomial** distribution with parameters r and p, with PMF

$$\mathbb{P}(X = k) = \binom{k-1}{r-1} p^r (1-p)^{k-r}, \qquad k = r, r+1, r+2, \dots$$

Why it works

Build the formula from the story, and watch how it differs from the binomial in one crucial way. To have the r-th success land on *exactly* trial k, two things must happen. First, among

the first $k-1$ trials there must be exactly $r-1$ successes (and $k-r$ failures); second, trial k itself must be a success. The last trial is *pinned* – it is forced to be a success, because that's what stops the process. So we only get to arrange the other $r-1$ successes among the first $k-1$ slots, giving $\binom{k-1}{r-1}$ ways, not $\binom{k}{r}$. Each such full sequence has r successes and $k-r$ failures, probability $p^r q^{k-r}$. Multiply: $\binom{k-1}{r-1}p^r q^{k-r}$. The pinned last trial is exactly what distinguishes the negative binomial from a binomial; forgetting it gives the wrong binomial coefficient.

Now the easy-way insight pays off again. A negative binomial is a *sum of r independent geometrics*: wait for the first success (Geometric(p) trials), then from scratch wait for the second (Geometric(p) more, by memorylessness), and so on r times. The total trial count is $X = G_1 + G_2 + \cdots + G_r$ with each $G_i \sim$ Geometric(p) independent. Mean and variance fall out instantly by adding.

Proposition 7.4 (Mean and variance of Negative Binomial)**.** *If X is negative binomial with parameters r and p, then*

$$\mathbb{E}[X] = \frac{r}{p}, \qquad \mathrm{Var}(X) = \frac{r(1-p)}{p^2}.$$

Proof. Write $X = \sum_{i=1}^{r} G_i$ with $G_i \sim$ Geometric(p) independent. Then by linearity $\mathbb{E}[X] = r \cdot (1/p) = r/p$, and by independence $\mathrm{Var}(X) = r \cdot (1-p)/p^2$. □ □

Intuition

If one success takes $1/p$ trials on average, then r successes take r/p – you just stack r independent waits end to end. There is nothing to memorize beyond the geometric; the negative binomial is geometrics in series.

Example 7.4 (Door-to-door sales)**.** Each house you visit results in a sale with probability $p = 0.2$, independently. You want $r = 3$ sales. Let X be the number of houses visited.

- Expected houses: $\mathbb{E}[X] = r/p = 3/0.2 = 15$.
- Probability the third sale comes exactly at the 10th house: $\mathbb{P}(X = 10) = \binom{9}{2}(0.2)^3(0.8)^7 = 36 \cdot 0.008 \cdot 0.2097 \approx 0.0604$.
- Standard deviation: $\mathrm{SD}(X) = \sqrt{3 \cdot 0.8/0.04} = \sqrt{60} \approx 7.75$ houses – a wide spread, so plan for a long day.

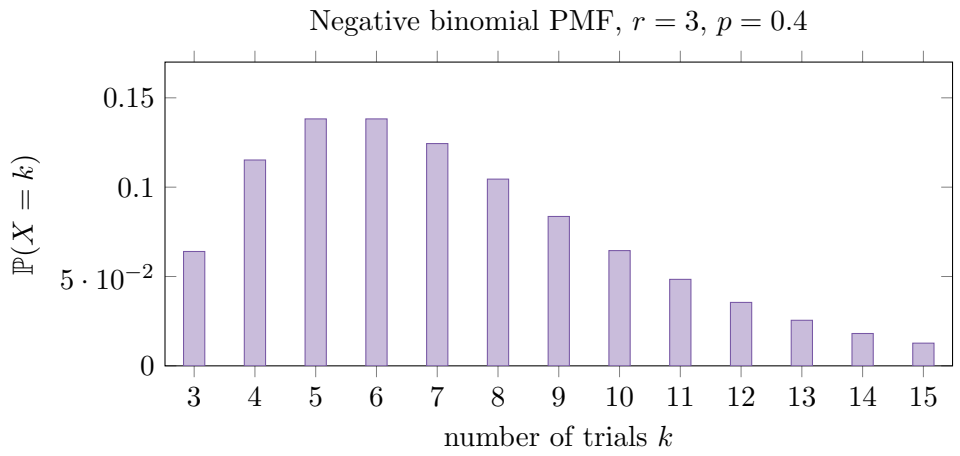


Figure 7.4: Negative binomial PMF for $r = 3$ successes at $p = 0.4$. The support starts at $k = r = 3$ (you cannot finish before the third success), the mass peaks near $\mathbb{E}[X] = r/p = 7.5$, and the long right tail is the signature of a waiting time – unlucky runs can drag on. Compare to the geometric ($r = 1$) of Figure 7.3, which is this picture with its mode pinned to the first trial.

Application

[Overdispersed counts in data science] There is a second, statistical reason the negative binomial matters enormously. When you model count data – numbers of website visits, insurance claims, reads of a gene in RNA-sequencing – the default model is the Poisson (next section), which forces variance = mean. Real count data is almost always *overdispersed*: its variance exceeds its mean. The negative binomial, re-parameterized to allow any mean and a separately tunable variance, is the standard fix. In fact a negative binomial arises exactly when the Poisson rate itself varies randomly (a Gamma-distributed rate), a so-called Gamma–Poisson mixture. This is why "negative binomial regression" is a workhorse in epidemiology, genomics (the tool DESeq2 is built on it), and actuarial science. The waiting-time story and the overdispersed-count story are two faces of the same distribution.

7.5 Hypergeometric: sampling without replacement

Every distribution so far assumed the trials were *independent* with a *fixed* success probability p. That assumption quietly breaks the moment you sample from a finite population *without replacement*. Pull a card from a deck and don't put it back, and the composition of what's left has changed: the probabilities on the next draw depend on what you already drew. The trials are now dependent. The count of successes in such a sample follows the **Hypergeometric** distribution.

Intuition

Keyword: *sampling without replacement from a finite pool.* There is a box with N items, of which K are "special" (defective, red, tagged, aces). You grab a handful of n without replacement. How many specials did you get? That count is hypergeometric. The tell-tale sign is a *finite population* and *no replacement* – if you replaced each item, or if the population were effectively infinite, you'd be back to the binomial.

Definition 7.5 (Hypergeometric distribution)**.** A population of N items contains K successes and $N-K$ failures. Draw a sample of n items uniformly at random without replacement. Let X be the number of successes drawn. Then X has the **Hypergeometric** distribution with parameters N, K, n, and PMF

$$\mathbb{P}(X=k) = \frac{\binom{K}{k}\binom{N-K}{n-k}}{\binom{N}{n}}, \qquad \max(0,\, n-(N-K)) \le k \le \min(n, K).$$

Why it works

This is pure counting – no products of p's at all, because every possible sample of size n is equally likely. The denominator $\binom{N}{n}$ is the total number of ways to choose the sample. The numerator counts the favorable samples: to get exactly k successes, choose which k of the K successes you took ($\binom{K}{k}$ ways) *and* which $n-k$ of the $N-K$ failures you took ($\binom{N-K}{n-k}$ ways), and multiply. Divide favorable by total and you have the probability. The hypergeometric is the binomial's combinatorial twin: the binomial multiplies probabilities of independent trials, the hypergeometric counts equally likely subsets.

Proposition 7.5 (Mean and variance of Hypergeometric)**.** *Let $p = K/N$ be the population success fraction. If X is hypergeometric with parameters N, K, n, then*

$$\mathbb{E}[X] = np = n\frac{K}{N}, \qquad \mathrm{Var}(X) = np(1-p)\cdot\frac{N-n}{N-1}.$$

Proof sketch. Write $X = \sum_{i=1}^{n} X_i$ where X_i indicates that the i-th draw is a success. By symmetry, every draw is equally likely to land on any item, so $\mathbb{P}(X_i = 1) = K/N = p$ for *every* i – even though the draws are dependent. Linearity of expectation (which never cares about dependence!) gives $\mathbb{E}[X] = np$, the same mean as a binomial with the same p. The variance is where dependence bites. The draws are *negatively* correlated: drawing a success leaves fewer successes for later draws. The covariance terms are negative, shrinking the variance by exactly the factor $(N-n)/(N-1)$ relative to the binomial $np(1-p)$. □ □

What is really happening

That extra factor $\dfrac{N-n}{N-1}$ is called the **finite population correction**, and it is the whole personality of the hypergeometric. Stare at it:

- If $n = 1$ (one draw) it equals 1: a single draw is just a Bernoulli, identical to the binomial case. Dependence hasn't had a chance to act.
- If $n = N$ (sample the entire population) it equals 0: the variance collapses to zero, which is exactly right – if you draw *everything*, you get *all* K successes with certainty, no randomness left.
- When N is huge compared to n, the factor is ≈ 1 and the hypergeometric becomes indistinguishable from Binomial$(n, K/N)$. Removing a few fish from a vast lake barely changes the proportions, so with-vs-without replacement stops mattering.

The correction is the precise mathematical statement of when "without replacement" matters and when you can ignore it.

Example 7.5 (Quality control)**.** A shipment has $N = 50$ chips, of which $K = 5$ are defective. An inspector samples $n = 10$ without replacement and rejects the shipment if any are defective. Let X be the number of defectives found, $X \sim \text{Hypergeometric}(50, 5, 10)$.

- Probability of finding zero defectives (and wrongly accepting):

$$\mathbb{P}(X = 0) = \frac{\binom{5}{0}\binom{45}{10}}{\binom{50}{10}} = \frac{\binom{45}{10}}{\binom{50}{10}} \approx 0.3106.$$

- Expected defectives found: $\mathbb{E}[X] = 10 \cdot \frac{5}{50} = 1$.

So there is roughly a 31% chance the sampling plan misses all defectives – a sobering number for anyone designing acceptance sampling.

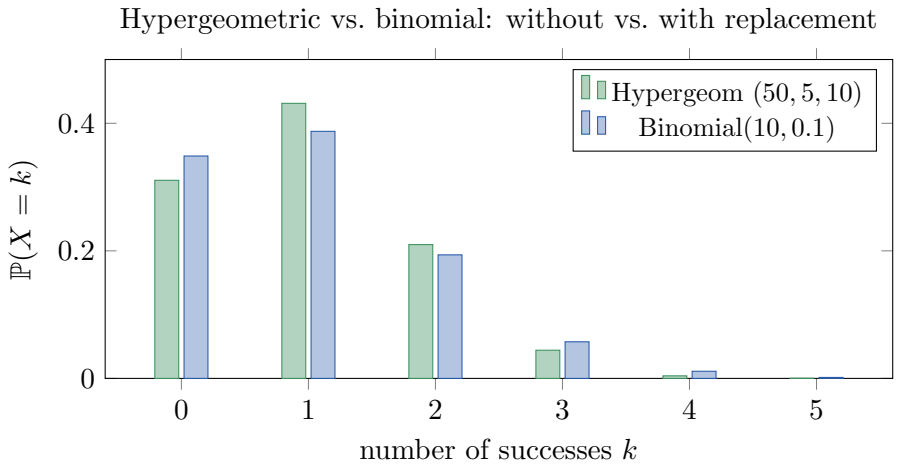


Figure 7.5: Drawing $n = 10$ from a pool of $N = 50$ with $K = 5$ successes. Both distributions share the mean $\mathbb{E}[X] = 1$, but the hypergeometric (without replacement) is more concentrated – its variance carries the finite-population factor $(N - n)/(N - 1) = 40/49 \approx 0.82$, pulling probability toward the center and away from the tails relative to the binomial (with replacement).

Example 7.6 (Lottery)**.** A lottery draws 6 numbers from 1 to 49 without replacement; you hold one ticket with 6 chosen numbers. The number of your numbers that match is hypergeometric with $N = 49$, $K = 6$ (your picks are the "successes"), $n = 6$ (the drawn numbers). The chance of matching all six:

$$\mathbb{P}(X = 6) = \frac{\binom{6}{6}\binom{43}{0}}{\binom{49}{6}} = \frac{1}{13{,}983{,}816} \approx 7.15 \times 10^{-8}.$$

One in fourteen million. The hypergeometric is the exact model of every draw-without-replacement game of chance.

History

The word "hypergeometric" comes not from probability but from the *hypergeometric series* of the eighteenth and nineteenth centuries (Euler, Gauss), a vast generalization of the geometric series whose coefficients are ratios of products much like the ones in this PMF. The distribution inherited the name because its probabilities are terms of such a series. It is a reminder that the named distributions did not arrive labeled and gift-wrapped; their names are historical accidents, dragged in from the analytic machinery mathematicians happened to be using. Don't read deep meaning into the names – read it into the stories.

7.6 Poisson: counting rare events in a continuum

We come to the most surprising and arguably the most useful distribution in the chapter. The Poisson describes the *number of events that occur in a fixed interval* of time, space, or volume, when the events happen at some average rate and don't collide with one another: calls arriving at a switchboard per minute, raindrops hitting a tile per second, mutations along a strand of DNA, typos per page, customers entering a shop per hour, photons striking a detector. It has a single parameter λ, the average number of events expected in the interval.

Definition 7.6 (Poisson distribution)**.** A random variable X has the **Poisson** distribution with rate parameter $\lambda > 0$, written $X \sim \text{Poisson}(\lambda)$, if

$$\mathbb{P}(X = k) = e^{-\lambda}\frac{\lambda^k}{k!}, \qquad k = 0, 1, 2, \ldots$$

These sum to 1 courtesy of the exponential series, which is the source of the $e^{-\lambda}$ normalizer:

$$\sum_{k=0}^{\infty} e^{-\lambda}\frac{\lambda^k}{k!} = e^{-\lambda}\sum_{k=0}^{\infty}\frac{\lambda^k}{k!} = e^{-\lambda}\cdot e^{\lambda} = 1.$$

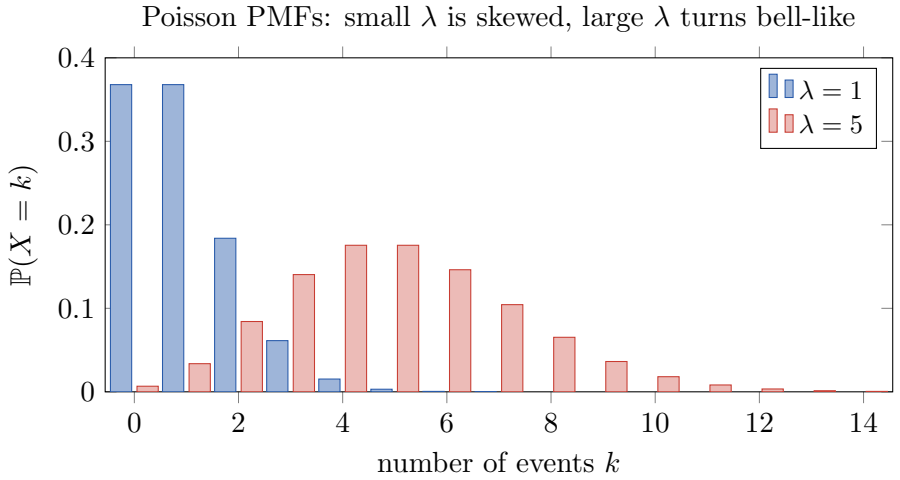


Figure 7.6: Poisson PMFs. For $\lambda = 1$ the mass piles near 0 and is strongly right-skewed (note $k = 0$ and $k = 1$ are equally likely). For $\lambda = 5$ the distribution has shifted right, centered near $\lambda = 5$, and is starting to look like the bell curve – a preview of the normal approximation.

Proposition 7.6 (Mean and variance of Poisson)**.** *If* $X \sim \text{Poisson}(\lambda)$ *then*

$$\mathbb{E}[X] = \lambda, \qquad \text{Var}(X) = \lambda.$$

The mean and the variance are equal.

Proof. For the mean, drop the $k = 0$ term (it contributes nothing) and shift the index:

$$\mathbb{E}[X] = \sum_{k=0}^{\infty} k\, e^{-\lambda} \frac{\lambda^k}{k!} = \sum_{k=1}^{\infty} e^{-\lambda} \frac{\lambda^k}{(k-1)!} = \lambda \sum_{j=0}^{\infty} e^{-\lambda} \frac{\lambda^j}{j!} = \lambda \cdot 1 = \lambda,$$

substituting $j = k - 1$ and recognizing the full PMF summing to 1. For the variance, the cleanest route computes $\mathbb{E}[X(X-1)]$ by the same index-shift trick:

$$\mathbb{E}[X(X-1)] = \sum_{k=2}^{\infty} k(k-1) e^{-\lambda} \frac{\lambda^k}{k!} = \lambda^2 \sum_{j=0}^{\infty} e^{-\lambda} \frac{\lambda^j}{j!} = \lambda^2.$$

Then $\mathbb{E}[X^2] = \mathbb{E}[X(X-1)] + \mathbb{E}[X] = \lambda^2 + \lambda$, so $\text{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2 = \lambda^2 + \lambda - \lambda^2 = \lambda$. □ □

Why this matters

$\mathbb{E}[X] = \text{Var}(X) = \lambda$ is the Poisson's fingerprint. It gives you a free diagnostic: compute the sample mean and sample variance of any count data, and if they are wildly different, the data is *not* Poisson. Variance much bigger than the mean signals overdispersion (reach for the negative binomial); variance smaller signals underdispersion. This one-number summary – a distribution whose spread is locked to its center – is both the Poisson's great convenience and its great limitation.

7.6.1 The law of rare events: Poisson as the limit of the binomial

Where does this distribution *come from*? The Poisson is not handed down from on high; it emerges as a limit of the binomial, and that limit is the deepest idea in the chapter. It explains why so many unrelated phenomena are Poisson.

Intuition

Imagine counting events in an interval by chopping the interval into a huge number n of tiny sub-intervals, each so small that at most one event can occur in it. Each sub-interval is then a Bernoulli trial: an event happens (probability p) or doesn't. The total count is Binomial(n, p). As you slice finer and finer, $n \to \infty$ while the chance of an event in any one tiny slice $p \to 0$, but the *expected total* $np = \lambda$ stays fixed at the average rate. The binomial in this limit *becomes* the Poisson. That is why the Poisson governs "many opportunities, each rarely taken."

Theorem 7.2 (Poisson limit theorem / law of rare events)**.** *Fix* $\lambda > 0$*. Suppose* $X_n \sim \text{Binomial}(n, p_n)$ *with* $np_n \to \lambda$ *as* $n \to \infty$*. Then for each fixed* $k = 0, 1, 2, \ldots,$

$$\mathbb{P}(X_n = k) \longrightarrow e^{-\lambda} \frac{\lambda^k}{k!}.$$

Proof. Set $p_n = \lambda/n$ for clarity (the general case is the same with a vanishing correction). Then

$$\mathbb{P}(X_n = k) = \binom{n}{k}\left(\frac{\lambda}{n}\right)^k\left(1-\frac{\lambda}{n}\right)^{n-k}.$$

Expand the binomial coefficient and regroup the powers of n:

$$\binom{n}{k}\frac{1}{n^k} = \frac{n(n-1)\cdots(n-k+1)}{k!\,n^k} = \frac{1}{k!}\underbrace{\frac{n}{n}\frac{n-1}{n}\cdots\frac{n-k+1}{n}}_{\to 1}.$$

For fixed k, each of the k fractions tends to 1 as $n\to\infty$, so this whole piece tends to $1/k!$. Meanwhile $\left(1-\lambda/n\right)^n \to e^{-\lambda}$ (the defining limit of the exponential), and $\left(1-\lambda/n\right)^{-k}\to 1$. Collecting the surviving pieces,

$$\mathbb{P}(X_n = k) \;\longrightarrow\; \frac{1}{k!}\cdot\lambda^k\cdot e^{-\lambda} = e^{-\lambda}\frac{\lambda^k}{k!}. \quad \square$$

$\square$

History

Siméon Denis Poisson published this distribution in 1837 in a treatise on the probability of judicial verdicts, but it languished as a mathematical curiosity for decades. Its fame came in 1898 when Ladislaus Bortkiewicz studied a famously morbid dataset: the number of Prussian cavalry soldiers killed each year by being kicked by a horse, recorded across 14 corps over 20 years. Deaths-by-horse are the perfect Poisson scenario: enormously many soldiers (n huge), each with a tiny chance of this particular fate (p tiny), giving a modest expected count ($\lambda \approx 0.7$ per corps-year). Bortkiewicz showed the yearly counts matched the Poisson almost exactly, and his little book *The Law of Small Numbers* gave the law of rare events its name. The deep point: *the mechanism, not the subject matter, makes something Poisson.* Horse-kicks, radio decays, and typos obey the same law because they share the same structure.

Example 7.7 (Poisson approximation in action)**.** A factory produces widgets with a 0.1% defect rate. In a batch of $n = 2000$, let X be the number of defective widgets. Exactly, $X \sim \text{Binomial}(2000, 0.001)$, which is painful to compute by hand. But $np = \lambda = 2$, and p is tiny, so $X \approx \text{Poisson}(2)$. The probability of zero defectives:

$$\mathbb{P}(X = 0) \approx e^{-2} \approx 0.1353.$$

The exact binomial value is $(0.999)^{2000} \approx 0.1352$ – agreement to four decimals. The probability of at most one defective: $e^{-2}(1+2) = 3e^{-2} \approx 0.406$. The Poisson turns a brutal binomial into a one-line calculation, and the law of rare events guarantees the answer is excellent whenever n is large and p small.

Example 7.8 (Call center staffing)**.** Calls arrive at a help line at an average rate of $\lambda = 4$ per minute. Model the count in a given minute as Poisson(4).

- Exactly 4 calls: $\mathbb{P}(X = 4) = e^{-4}\dfrac{4^4}{4!} = e^{-4}\dfrac{256}{24} \approx 0.1954$.
- No calls in a minute: $\mathbb{P}(X = 0) = e^{-4} \approx 0.0183$.
- More than 6 calls: $\mathbb{P}(X > 6) = 1 - \sum_{k=0}^{6} e^{-4}\frac{4^k}{k!} \approx 1 - 0.889 = 0.111$. Roughly an 11% chance any given minute is a surge – the kind of number that drives staffing decisions.

Connection

The Poisson does not live alone. It is the count side of the **Poisson process**, the canonical model of random arrivals in continuous time, which we build carefully in a later chapter. In that process, the *number* of arrivals in an interval is Poisson, while the *waiting time between* consecutive arrivals is Exponential – the continuous, memoryless cousin of the geometric. So the discrete waiting-time/count pair (geometric, binomial) has a continuous mirror image (exponential, Poisson), and the bridge between them is exactly the limiting argument of this section. When you later meet the exponential and gamma distributions, remember that they are the continuous-time reincarnations of the geometric and negative binomial you learned here. The whole landscape rhymes.

Application

[Poisson in modern data science] Poisson assumptions are everywhere in applied work. *Poisson regression* models count outcomes (hospital admissions, goals scored, clicks) as a function of predictors, and is the count-data analogue of linear regression. In natural language processing, simple word-count models lean on Poisson assumptions. In genomics, read counts in sequencing are Poisson under idealized sampling (and negative binomial once biological variation is added). In sports analytics, goals in soccer are famously well modeled as Poisson, which lets you compute the probability of any score line from two rate parameters. Whenever you find yourself counting independent rare events in a window, the Poisson is your first model and your null hypothesis.

7.7 Putting it together: which distribution fits?

The whole point of naming these families is to recognize them in the wild. Here is the decision tree, distilled. Run a problem through these questions in order.

Focus

1. Single yes/no trial? ⇒ Bernoulli(p).
2. Fixed number n of independent trials, counting successes? ⇒ Binomial(n, p).
3. Independent trials, waiting for the *first* success (count the trials)? ⇒ Geometric(p).
4. Independent trials, waiting for the r-*th* success? ⇒ Negative Binomial(r, p).
5. Drawing a sample *without replacement* from a finite pool, counting successes? ⇒ Hypergeometric(N, K, n).
6. Counting rare events in an interval of time/space at a fixed average rate? ⇒ Poisson(λ).

The two perennial points of confusion are worth isolating.

Common trap

Binomial vs. Hypergeometric. Both count successes. The difference is *replacement*. With replacement (or an effectively infinite population) the trials are independent and p is constant:

binomial. Without replacement from a finite pool, the trials are dependent and p drifts as you draw: hypergeometric. When the population is much larger than the sample ($N \gg n$), the two coincide, and the binomial is the easy approximation.

Binomial vs. Negative Binomial. Both involve trials and successes. The difference is *what is fixed and what is random*. Binomial: n is fixed, the number of successes is random. Negative binomial: the number of successes r is fixed, and the number of trials is random. "I ran 20 trials, how many wins?" is binomial. "I kept going until 5 wins, how many trials?" is negative binomial.

Walking question

A test has 30 multiple-choice questions, each with 4 options, and a student guesses every one. What is the distribution of the number correct, and what is its expected value? Pause and run it through the tree before reading on.

Answer: fixed $n = 30$ independent trials, each a success (correct guess) with probability $p = 1/4$, counting successes. That is question 2 on the tree: Binomial$(30, 1/4)$, with $\mathbb{E}[X] = 30 \cdot 1/4 = 7.5$. A pure guesser averages 7.5 out of 30.

Distribution	**PMF**	**Mean**	**Variance**
Bernoulli(p)	$p^k(1-p)^{1-k}$	p	$p(1-p)$
Binomial(n,p)	$\binom{n}{k}p^k(1-p)^{n-k}$	np	$np(1-p)$
Geometric(p)	$(1-p)^{k-1}p$	$\frac{1}{p}$	$\frac{1-p}{p^2}$
NegBin(r,p)	$\binom{k-1}{r-1}p^r(1-p)^{k-r}$	$\frac{r}{p}$	$\frac{r(1-p)}{p^2}$
Hypergeom(N,K,n)	$\frac{\binom{K}{k}\binom{N-K}{n-k}}{\binom{N}{n}}$	$n\frac{K}{N}$	$n\frac{K}{N}\frac{N-K}{N}\frac{N-n}{N-1}$
Poisson(λ)	$e^{-\lambda}\frac{\lambda^k}{k!}$	λ	λ

Table 7.1: The six discrete families at a glance. Notice the family resemblances: the binomial mean np and the hypergeometric mean $n(K/N)$ are the same with $p = K/N$, differing only by the finite-population factor in the variance; the negative binomial mean r/p is just r copies of the geometric mean $1/p$.

What is really happening

Step back and admire the architecture. The Bernoulli is the atom. Sum n of them and ask "how many successes" $\to$ binomial. Ask instead "how long until the first success" $\to$ geometric; "until the r-th" $\to$ negative binomial. Break the independence by sampling without replacement $\to$ hypergeometric. Push the binomial to its rare-event limit $\to$ Poisson. Every distribution in this chapter is a different *question* asked of the same underlying coin-flipping process, or a different *limit* of it. You are not memorizing six unrelated formulas; you are reading six chapters of one story. That is the sense in which you can now *see* discrete probability rather than recite it.

7.8 Exercises

Exercise 7.1 (Bernoulli warm-up)**.** A biased coin has $\mathbb{P}(\text{heads}) = 0.7$. Let $X = 1$ if heads. Compute $\mathbb{E}[X]$, $\text{Var}(X)$, and $\text{SD}(X)$. For what value of p is the variance of a Bernoulli maximized, and what is that maximum?

Exercise 7.2 (Binomial computations)**.** A multiple-choice quiz has 8 questions, each with 5 choices; a student guesses randomly. Let X be the number correct. (a) Name the distribution of X with its parameters. (b) Compute $\mathbb{P}(X = 2)$. (c) Compute $\mathbb{E}[X]$ and $\text{Var}(X)$. (d) Compute $\mathbb{P}(X \geq 1)$, the chance of getting at least one right.

Exercise 7.3 (Geometric waiting)**.** You roll a fair die repeatedly until the first six. (a) What is the probability the first six appears on the fourth roll? (b) What is the expected number of rolls? (c) Given that you have rolled five times with no six, what is the expected number of *additional* rolls until the first six? Explain how memorylessness makes this trivial.

Exercise 7.4 (Negative binomial)**.** A basketball player makes each free throw with probability 0.6. She shoots until she has made 4 baskets. Let X be the number of shots taken. (a) Identify the distribution and its parameters. (b) Compute $\mathbb{P}(X = 6)$. (c) Compute $\mathbb{E}[X]$ and $\text{SD}(X)$. (d) Explain, using the "sum of geometrics" picture, why $\mathbb{E}[X] = 4/0.6$.

Exercise 7.5 (Hypergeometric)**.** A standard 52-card deck contains 4 aces. You are dealt a 5-card poker hand. Let X be the number of aces in your hand. (a) Identify the distribution and its parameters. (b) Compute $\mathbb{P}(X = 2)$. (c) Compute $\mathbb{E}[X]$. (d) If instead you drew 5 cards *with* replacement (reshuffling each time), what distribution would X follow, and how would $\mathbb{E}[X]$ compare?

Exercise 7.6 (Poisson modeling)**.** A website receives an average of 3 sign-ups per hour, modeled as Poisson. (a) Probability of exactly 5 sign-ups in a given hour. (b) Probability of no sign-ups in a given hour. (c) Probability of at least 2 sign-ups in a given hour. (d) What is the distribution of sign-ups in a *half*-hour, and its mean? (Hint: the rate scales with the length of the interval.)

Exercise 7.7 (Poisson approximation)**.** Each of 10,000 independent visitors clicks a rare banner ad with probability 0.0003. (a) Give the exact distribution of the number of clicks X. (b) Justify approximating X by a Poisson and give λ. (c) Use the Poisson approximation to compute $\mathbb{P}(X = 0)$, $\mathbb{P}(X = 3)$, and $\mathbb{P}(X \geq 1)$.

Exercise 7.8 (Conceptual: which model?)**.** For each scenario, name the distribution and its parameters, and justify in one sentence by referencing the decision tree. (a) The number of defective bulbs in a random sample of 20 drawn without replacement from a crate of 200 containing 15 defectives. (b) The number of emails you receive between 9 and 10 a.m., averaging 12. (c) The number of coin flips until you see two heads in total. (d) The number of heads in 50 flips of a fair coin. (e) Whether a single randomly chosen voter supports a measure.

Exercise 7.9 (Proof: geometric variance)**.** Let $X \sim \text{Geometric}(p)$. Using a first-step argument analogous to the one for the mean, derive $\mathbb{E}[X^2]$ and hence show $\text{Var}(X) = (1-p)/p^2$. (Hint: condition on the first trial. If it fails, the remaining count X' satisfies $X = 1 + X'$ with $X' \sim \text{Geometric}(p)$ independent, so $\mathbb{E}[X^2] = p \cdot 1 + q\,\mathbb{E}[(1+X')^2]$. Expand and solve, using $\mathbb{E}[X] = 1/p$.)

Exercise 7.10 (Proof: binomial mode)**.** Let $X \sim \text{Binomial}(n, p)$. By examining the ratio $\dfrac{\mathbb{P}(X = k)}{\mathbb{P}(X = k-1)}$, show that this ratio exceeds 1 exactly when $k < (n+1)p$. Conclude that the PMF increases then decreases, and that the most likely value (the mode) is $\lfloor (n+1)p \rfloor$ (with a tie when $(n+1)p$ is an integer). Verify for the free-throw example $\text{Binomial}(12, 0.8)$ that the mode is 10.

Exercise 7.11 (Proof: Poisson sum)**.** Let $X \sim \text{Poisson}(\lambda)$ and $Y \sim \text{Poisson}(\mu)$ be independent. Prove that $X+Y \sim \text{Poisson}(\lambda+\mu)$. (Hint: $\mathbb{P}(X+Y=n) = \sum_{k=0}^{n} \mathbb{P}(X=k)\,\mathbb{P}(Y=n-k)$. Factor out $e^{-(\lambda+\mu)}/n!$ and recognize the binomial theorem $\sum_k \binom{n}{k}\lambda^k\mu^{n-k} = (\lambda+\mu)^n$ inside the sum.) Interpret the result through the law of rare events: merging two independent streams of rare events gives one rare-event stream at the combined rate.

Exercise 7.12 (Connecting the families)**.** (a) Show that the negative binomial with $r = 1$ is exactly the geometric. (b) Explain in words why the hypergeometric variance is always $\leq$ the binomial variance with matching $p = K/N$, and identify the case where they are equal. (c) Two friends each model the number of typos on a page. One uses Binomial(n, p) with n characters and tiny per-character error rate p; the other uses Poisson(λ). Under what relationship between the parameters do they essentially agree, and which theorem of this chapter justifies it?

Chapter 8

Expectation

There is a single number that, more than any other, summarizes a random quantity: its *expectation*. If you remember nothing else from this chapter, remember that expectation is just an average – but an average taken in two different and equally vivid ways. It is the value you would converge to if you repeated an experiment forever and averaged the outcomes. And it is the balance point of a distribution, the place where the picture of the probabilities would sit level on a knife edge. Both pictures are correct, both are useful, and holding them together is most of what it means to "understand" expectation.

After that comes the workhorse of elementary probability: *linearity of expectation*. It is so simple that it looks like it cannot possibly do much work, yet it does an astonishing amount – it solves counting problems that would otherwise require heroic combinatorics, and it keeps working even when the pieces you are adding up are tangled together in complicated, dependent ways. We will end with a famous puzzle, the St. Petersburg paradox, where expectation gives an answer of $+\infty$ and almost nobody would pay even a few dollars to play. Resolving that tension teaches you exactly what expectation does and does not promise.

8.1 Two pictures of the average

8.1.1 The long-run-average picture

Imagine a game. You roll a fair six-sided die, and you win in dollars whatever number comes up. Some rounds you win \$1, some rounds \$6, most rounds something in between. If you play many, many rounds, what is your average winning per round?

Intuitively, each face shows up about one-sixth of the time, so in 6000 rounds you would see roughly 1000 of each face. Your total winnings would be about

$$1000 \cdot 1 + 1000 \cdot 2 + \cdots + 1000 \cdot 6 = 1000 \cdot 21 = 21000,$$

and dividing by the 6000 rounds gives an average of $21000/6000 = 3.5$ dollars per round. Notice what happened: the 1000's cancelled, and we were left with

$$\frac{1+2+3+4+5+6}{6} = \frac{21}{6} = 3.5.$$

The fraction of rounds with each face – the probability – is what survives. That is the long-run-average picture. The expectation is the per-round payoff you approach as the number of rounds grows.

Intuition

Expectation answers the question: *if I could repeat this exactly the same way an enormous number of times and average the results, what number would I land on?* You do not need the experiment to actually be repeatable in real life. The expectation is a property of the distribution, computed as if it were repeatable.

8.1.2 The center-of-mass picture

Here is the second picture, the one that makes expectation feel geometric and inevitable. Draw the probability mass function of your random variable as little weights sitting on a number line: at each possible value x you place a lump of mass equal to $\mathbb{P}(X = x)$. The total mass is 1 (the probabilities sum to one). Now ask a physics question: where is the balance point of this arrangement of weights?

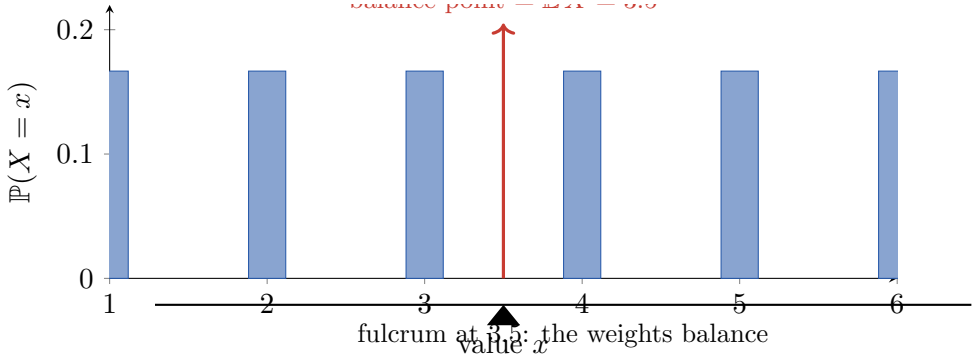


Figure 8.1: The fair die as a center of mass. Six equal weights sit at $1, \ldots, 6$; the arrangement balances exactly at 3.5, which is the expectation. The fulcrum sits under the mean, not under any actual outcome.

For the die, six equal weights at 1 through 6 obviously balance at the middle, 3.5. The expectation *is* the balance point. And notice – crucially – the balance point need not be a value the variable can ever take. You never roll a 3.5. The expected value is not the "most likely" value, and it need not even be a *possible* value. It is the center of mass of the probabilities.

History

The word "expectation" is older than the modern theory. In 1654 the Chevalier de Méré posed gambling puzzles to Blaise Pascal, who corresponded with Pierre de Fermat about how to divide the stakes in an interrupted game – the famous *problem of points*. Their solution was, in effect, "give each player the amount they could *expect* to win." In 1657 Christiaan Huygens turned this into the first published treatise on probability, *De Ratiociniis in Ludo Aleae*, and his very first proposition is a calculation of *expectatio* – the fair price of a gamble. So historically expectation came *first*, before a clean notion of probability was even nailed down. People knew what a fair price was before they knew what a probability was.

8.1.3 The definition

We can now write down the definition that both pictures point to. For a discrete random variable X taking values $x_1, x_2, \dots$ with probabilities $p_i = \mathbb{P}(X = x_i)$, weight each value by its probability and add:

Definition 8.1 (Expectation, discrete case)**.** Let X be a discrete random variable with possible values $x_1, x_2, \dots$. Its *expectation* (or *expected value*, or *mean*) is

$$\mathbb{E}\,X \;=\; \sum_i x_i\, \mathbb{P}(X = x_i),$$

provided the sum converges absolutely, i.e. $\sum_i |x_i|\, \mathbb{P}(X = x_i) < \infty$. We also write μ or μ_X for $\mathbb{E}\,X$.

The absolute-convergence proviso is not pedantry: if the positive and negative parts both blow up, no single number deserves to be called "the average," and rearranging the sum could give you anything. We will meet a clean example (St. Petersburg) where the sum genuinely diverges and the expectation is $+\infty$.

Focus

Both pictures are the *same formula*, read two ways.

- **Long-run average:** $\mathbb{P}(X = x_i)$ is the fraction of trials landing at x_i, and $\sum_i x_i\, \mathbb{P}(X = x_i)$ is the limiting average payoff.
- **Center of mass:** $\mathbb{P}(X = x_i)$ is the mass at position x_i, and $\sum_i x_i\, \mathbb{P}(X = x_i)$ is the formula for the balance point (the first moment).

Example 8.1 (A loaded bet)**.** A carnival game: you pay nothing to play. With probability 0.7 you win \$1, with probability 0.25 you win \$5, and with probability 0.05 you lose \$20. Let X be your net gain. Then

$$\mathbb{E}\,X = (1)(0.7) + (5)(0.25) + (-20)(0.05) = 0.7 + 1.25 - 1.0 = 0.95.$$

On average you make 95 cents per play. Most plays you win a little; the rare \$20 loss is real but does not eat the whole edge. If the carnival ran this game it would slowly go broke – which is exactly why real carnival games have $\mathbb{E}\,X < 0$ for the player.

Example 8.2 (A constant is its own mean)**.** If $X = c$ always (a degenerate "random" variable), then $\mathbb{E}\,X = c \cdot 1 = c$. Obvious, but worth stating: averaging something that never varies gives back the thing. This is the base case that linearity will lean on.

8.2 Expectation of an indicator is a probability

Before linearity, we need one small fact that turns out to be the secret engine behind half the clever arguments in probability. Recall the *indicator* random variable of an event A:

$$\mathbf{1}_A = \begin{cases} 1 & \text{if } A \text{ occurs,} \\ 0 & \text{otherwise.} \end{cases}$$

It is the simplest non-constant random variable there is: it just flags whether A happened. What is its expectation? By the definition, $\mathbf{1}_A$ takes value 1 with probability $\mathbb{P}(A)$ and value 0 with probability $1 - \mathbb{P}(A)$, so

$$\mathbb{E}[\mathbf{1}_A] = 1 \cdot \mathbb{P}(A) + 0 \cdot \big(1 - \mathbb{P}(A)\big) = \mathbb{P}(A).$$

Formal

The bridge. For any event A,

$$\boxed{\mathbb{E}[\mathbf{1}_A] = \mathbb{P}(A).}$$

Probability is a special case of expectation. Every probability you have ever computed is secretly an expected value – the expected value of a yes/no flag.

This little identity is the reason expectation is more fundamental than it first appears. It lets us *translate counting questions into averaging questions*, and averaging questions are exactly where the next tool earns its keep.

What is really happening

Why is this so useful? Because we are about to learn that expectation splits over sums effortlessly, even when the pieces depend on each other. So if you can write the quantity you care about as a *sum of indicators* – "how many of these events happened?" – then its expectation is just the *sum of the probabilities*, $\mathbb{E}\big[\sum_i \mathbf{1}_{A_i}\big] = \sum_i \mathbb{P}(A_i)$. A hard combinatorial count collapses into an easy addition of probabilities. Keep this phrase in mind: *count by summing indicators.*

8.3 Linearity of expectation

Here is the theorem that earns its place as the single most useful fact in this book.

Theorem 8.1 (Linearity of expectation)**.** *Let X and Y be random variables on the same probability space with finite expectations, and let a, b be constants. Then*

$$\mathbb{E}[aX + bY] = a\,\mathbb{E}\,X + b\,\mathbb{E}\,Y.$$

More generally, for any random variables $X_1, \ldots, X_n$ with finite means,

$$\mathbb{E}\left[\sum_{i=1}^{n} X_i\right] = \sum_{i=1}^{n} \mathbb{E}\,X_i.$$

No independence is required. *The X_i may depend on each other in any way whatsoever.*

Read that last sentence twice. It is the part people find hard to believe, and it is the part that gives the theorem its reach. The expectation of a sum equals the sum of expectations *always*, dependence be damned.

Why it works

Why is linearity true, with no independence assumption? Go back to the most basic level: a random variable is a function $X(\omega)$ of the underlying outcome ω, and its expectation is an average over outcomes weighted by their probabilities:

$$\mathbb{E}\,X = \sum_{\omega} X(\omega)\ \mathbb{P}(\omega).$$

Now for the sum $X + Y$, at *each individual outcome* ω we simply add the two numbers: $(X + Y)(\omega) = X(\omega) + Y(\omega)$. There is no interaction – addition of two numbers is done pointwise. So

$$\mathbb{E}[X + Y] = \sum_{\omega} \big(X(\omega) + Y(\omega)\big)\,\mathbb{P}(\omega) = \sum_{\omega} X(\omega)\,\mathbb{P}(\omega) + \sum_{\omega} Y(\omega)\,\mathbb{P}(\omega) = \mathbb{E}\,X + \mathbb{E}\,Y.$$

Independence never entered because we never had to split a probability of a *joint* event. We only rearranged a sum – and sums of numbers can always be regrouped. Dependence is about how X and Y *co-occur*, i.e. about joint probabilities; but linearity never touches joint probabilities. It just adds two real numbers at each outcome and then totals up. *That* is why it does not care about dependence.

Common trap

Linearity is about *sums*, not products. In general

$$\mathbb{E}[XY] \neq (\mathbb{E}\,X)(\mathbb{E}\,Y),$$

and the discrepancy is exactly the covariance (next chapter). Equality of the product holds when $X \perp\!\!\!\perp Y$, but not otherwise. Likewise $\mathbb{E}[g(X)] \neq g(\mathbb{E}\,X)$ for a nonlinear g – that is Jensen's inequality, also coming later. Linearity is a statement about $aX + bY$, and only that.

8.3.1 Why linearity is the single most useful tool

When you want to find $\mathbb{E}\,X$ directly from the definition, you need the full distribution of X – every value and its probability. For complicated X that distribution can be a nightmare. Linearity offers an escape hatch: *break X into a sum of simple pieces*, find the easy expectation of each piece, and add. The pieces are allowed to be hopelessly entangled. You never have to untangle them. You never have to find the distribution of X at all.

The recipe is almost always the same three steps.

Focus

The indicator method (the most useful trick in the book).

1. Write the quantity of interest as a sum of indicators: $X = \sum_i \mathbf{1}_{A_i}$ ("count how many A_i happened").
2. Use $\mathbb{E}[\mathbf{1}_{A_i}] = \mathbb{P}(A_i)$ on each piece.

3. Add up by linearity: $\mathbb{E}\,X = \sum_i \mathbb{P}(A_i)$.

The dependence between the A_i is irrelevant. Just sum the marginal probabilities.

8.3.2 Warm-up: the mean of a binomial without any algebra

How many heads do you expect in n flips of a coin that lands heads with probability p? Let $X \sim \text{Binomial}(n, p)$ be the number of heads. The direct approach is

$$\mathbb{E}\,X = \sum_{k=0}^{n} k \binom{n}{k} p^k (1-p)^{n-k},$$

a sum you *can* grind out with a binomial-identity trick, but it is fiddly. Now do it the easy way. Let $X_i = \mathbf{1}\{\text{flip } i \text{ is heads}\}$. Then

$$X = X_1 + X_2 + \cdots + X_n,$$

and each X_i is an indicator with $\mathbb{E}\,X_i = \mathbb{P}(\text{heads on flip } i) = p$. By linearity,

$$\mathbb{E}\,X = \sum_{i=1}^{n} \mathbb{E}\,X_i = \sum_{i=1}^{n} p = np.$$

Three lines, no binomial coefficients. The flips here happen to be independent, but – and this is the point we will hammer next – we never used that.

8.4 The astonishing power: hat-matching and dependence

Now the showcase – the example that tends to convert skeptics.

Example 8.3 (The hat-check / matching problem)**.** n people check their hats at a restaurant. A careless attendant returns the hats in a completely random order – a uniformly random permutation. On average, how many people get their *own* hat back?

Let X be the number of people who receive their own hat. Finding the distribution of X is genuinely hard: it involves *derangements* (the number of permutations with no fixed point) and an inclusion–exclusion sum. But we do not want the distribution. We want $\mathbb{E}\,X$, and linearity makes it trivial.

Define the indicator $X_i = \mathbf{1}\{\text{person } i \text{ gets their own hat}\}$. Then $X = X_1 + \cdots + X_n$. For each person, the hat they receive is equally likely to be any of the n hats, so

$$\mathbb{E}\,X_i = \mathbb{P}(\text{person } i \text{ gets own hat}) = \frac{1}{n}.$$

By linearity,

$$\mathbb{E}\,X = \sum_{i=1}^{n} \mathbb{E}\,X_i = n \cdot \frac{1}{n} = 1.$$

On average exactly one person gets their own hat back, no matter how large n is. Ten people or ten million, the expected number of matches is 1.

What is really happening

Look at what we sailed past. The indicators X_i are *not* independent: if the first $n-1$ people all got their own hats, then the last person is *forced* to get theirs too (only one hat remains). The events are intricately linked. Yet linearity did not blink, because – as the proof showed – it only ever adds the marginal probabilities $\mathbb{P}(X_i = 1) = 1/n$, and the marginal probability of person i's match does not care about anyone else. The dependence would matter enormously if we wanted the *variance* of X (it involves $\mathbb{E}[X_i X_j]$, a joint quantity) – but it is invisible to the mean. This is the whole lesson of the chapter in one example.

Work it yourself

Before reading on, try this with the same one-line method. You shuffle a standard deck and deal the cards face up one at a time while reciting "ace, two, three, ..., king, ace, two, ..." through the 52 cards in a fixed order. A "match" is when the card you name equals the card you flip. What is the expected number of matches? *Hint:* indicator per position, each matches with probability... what?

Answer: each of the 52 positions matches its named rank with probability $\frac{4}{52} = \frac{1}{13}$ (four cards of the named rank among 52), so the expected number of matches is $52 \cdot \frac{1}{13} = 4$. Dependence between positions is irrelevant.

Example 8.4 (Expected number of records)**.** A sequence of n distinct numbers arrives in random order (all $n!$ orders equally likely). Position i is a *record* if its value exceeds everything before it. How many records do you expect?

Let $R_i = \mathbf{1}\{\text{position } i \text{ is a record}\}$. Position i is a record exactly when the largest of the first i values happens to sit in slot i. By symmetry every one of those i values is equally likely to be the largest, so $\mathbb{P}(R_i = 1) = 1/i$. Hence

$$\mathbb{E}[\text{records}] = \sum_{i=1}^{n} \frac{1}{i} = H_n \approx \ln n + 0.577.$$

The expected number of records grows only *logarithmically*. After a thousand observations you expect about $\ln 1000 + 0.58 \approx 7.5$ record-breaking values. This H_n (the harmonic number) is the same constant that governs the coupon-collector problem; records and coupons are secretly cousins.

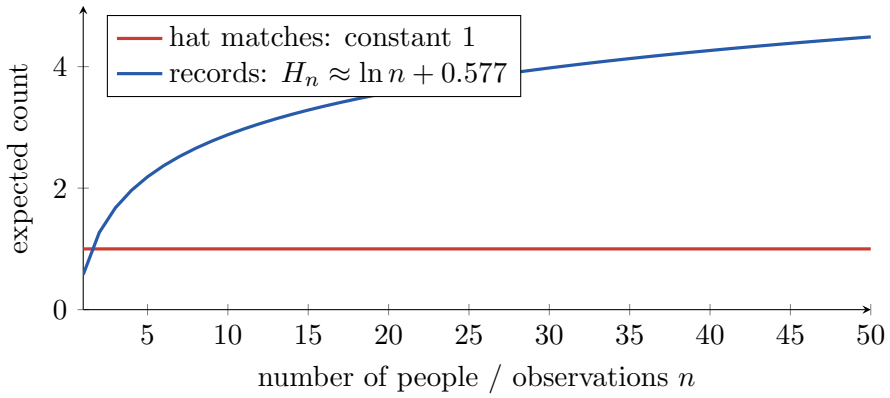


Figure 8.2: Two indicator-method results. The expected number of self-matched hats stays pinned at 1 for every n, while the expected number of records crawls upward like $\ln n$. Both were one-line linearity computations over dependent indicators.

Application

Linearity in computer science and data science. The indicator method is the bread and butter of *randomized algorithm analysis*.

- **Hashing / load balancing.** Throw m items into n hash buckets uniformly. The expected number of items in a given bucket is, by indicators, $m \cdot \frac{1}{n} = m/n$ – and the expected number of *empty* buckets is $n\,(1-\frac{1}{n})^m \approx n\,e^{-m/n}$.
- **Expected comparisons in QuickSort.** Let A_{ij} be the event that the ith and jth smallest elements are ever compared. One shows $\mathbb{P}(A_{ij}) = \frac{2}{j-i+1}$, and summing these (highly dependent!) indicators gives the famous $\mathbb{E}[\#\text{comparisons}] \approx 2n \ln n$.
- **Count-distinct sketches** (HyperLogLog and friends) are built on expectations of indicator-like statistics over hashed streams.

In each case nobody computes a full distribution. They sum marginal probabilities, because expectation lets them.

8.5 LOTUS: the expectation of a function

So far we have averaged X itself. But usually we care about some *function* of X – your payoff is $g(X)$, your loss is $(X-\mu)^2$, your utility is $\sqrt{X}$. How do you compute $\mathbb{E}[g(X)]$?

The tempting wrong move is to first find the distribution of the new variable $Y = g(X)$, then apply the definition to Y. That works but is usually painful: deriving the distribution of $g(X)$ can be hard, and it is wasted effort. There is a shortcut so handy it has a nickname.

Theorem 8.2 (Law of the Unconscious Statistician, discrete case)**.** *Let X be a discrete random variable with pmf p_X, and let g be a function. Then*

$$\mathbb{E}[g(X)] \;=\; \sum_{x} g(x)\;\mathbb{P}(X=x),$$

provided $\sum_x |g(x)|\;\mathbb{P}(X=x) < \infty$.

In words: to average $g(X)$, just apply g to each value of X and weight by the *original* probabilities of X. You never have to find the distribution of $g(X)$ at all. The nickname pokes fun at the statistician who uses this formula "unconsciously," as if it were the definition – but it is a theorem, and it needs a (short) proof.

Why it works

Why is LOTUS true? The real worry is that g might collapse several different x-values onto the same y-value, so the probabilities should pile up. They do, and the bookkeeping works out. Group the x's by the value $y = g(x)$ they produce:

$$\begin{aligned}
\mathbb{E}[g(X)] &= \sum_y y\, \mathbb{P}\big(g(X) = y\big) && \text{(definition, applied to } Y = g(X)\text{)} \\
&= \sum_y y \sum_{x:\, g(x)=y} \mathbb{P}(X = x) && \text{(the event } g(X) = y \text{ is the union of those } x\text{)} \\
&= \sum_y \sum_{x:\, g(x)=y} g(x)\, \mathbb{P}(X = x) && \text{(inside this group } y = g(x)\text{)} \\
&= \sum_x g(x)\, \mathbb{P}(X = x). && \text{(the double sum reassembles all } x\text{)}
\end{aligned}$$

The collapsing is handled automatically: every x contributes $g(x)\,\mathbb{P}(X = x)$ exactly once, whether or not it shares its g-value with other x's.

Connection

LOTUS is the discrete cousin of a change-of-variables you already know from calculus. In the continuous chapters $\mathbb{E}[g(X)] = \int_{-\infty}^{\infty} g(x)\, f_X(x)\, \mathrm{d}x$, integrating against the density instead of summing against the pmf. The structure is identical: *apply g, weight by how much probability sits at each input, accumulate.* Sums become integrals, the pmf becomes the density, but the idea – average over inputs, not outputs – is the same. See the companion calculus reader for why $\int$ is the continuous limit of $\sum$.

Example 8.5 (LOTUS in action)**.** Let X be the result of a fair die, and let your payoff be $g(X) = X^2$ dollars. By LOTUS,

$$\mathbb{E}[X^2] = \sum_{x=1}^{6} x^2 \cdot \frac{1}{6} = \frac{1 + 4 + 9 + 16 + 25 + 36}{6} = \frac{91}{6} \approx 15.17.$$

Compare with $(\mathbb{E}\,X)^2 = 3.5^2 = 12.25$. They differ – and the gap $15.17 - 12.25 = 2.92$ is precisely the variance of the die, which is the subject of the next chapter. The fact that $\mathbb{E}[X^2] > (\mathbb{E}\,X)^2$ is no accident; it is Jensen's inequality for the convex function x^2, and it is why squaring inflates the average.

8.6 A nonnegative-integer shortcut (the tail-sum formula)

Sometimes the indicator trick wears a different costume. If X takes values in $\{0, 1, 2, \dots\}$, there is a beautiful alternative formula for its mean in terms of *tail probabilities*:

$$\mathbb{E}\,X = \sum_{k=1}^{\infty} \mathbb{P}(X \geq k).$$

Why it works

Why? Use the indicator method on the integer line itself. The clean identity for a nonnegative integer X is

$$X = \sum_{k=1}^{\infty} \mathbf{1}\{X \geq k\},$$

which just says "count up to X by ticking off 1 for each level k you have reached." (If $X = 3$, the indicators for $k = 1, 2, 3$ are all 1 and the rest are 0, summing to 3.) Take expectations and swap the (nonnegative) sum with the expectation:

$$\mathbb{E}\,X = \sum_{k=1}^{\infty} \mathbb{E}\big[\mathbf{1}\{X \geq k\}\big] = \sum_{k=1}^{\infty} \mathbb{P}(X \geq k).$$

Example 8.6 (Mean of a geometric variable, painlessly)**.** Let $X \sim \text{Geometric}(p)$ count the number of trials up to and including the first success, so $\mathbb{P}(X \geq k) = (1-p)^{k-1}$ (the first $k-1$ trials all failed). Then

$$\mathbb{E}\,X = \sum_{k=1}^{\infty} (1-p)^{k-1} = \frac{1}{1-(1-p)} = \frac{1}{p},$$

a clean geometric series, no differentiation needed. If a success has probability $p = 1/6$ (rolling a six), you expect 6 rolls to get your first six. Reassuringly intuitive.

8.7 The St. Petersburg paradox

We close with the most famous cautionary tale about expectation – a puzzle that forced mathematicians to think hard about what an "expected value" actually buys you.

Paradox

The game. A fair coin is flipped until it first comes up heads. If the first head appears on flip n, the casino pays you 2^n dollars. So heads on the first flip pays \$2, heads first on the second flip pays \$4, then \$8, \$16, and so on, doubling each time you survive another tail.
The question. What is the fair price to play – how much should you be willing to pay for one round?

Expectation's answer. The first head lands on flip n with probability $(1/2)^n$, and that outcome pays 2^n. So

$$\mathbb{E}[\text{payout}] = \sum_{n=1}^{\infty} \underbrace{\left(\tfrac{1}{2}\right)^n}_{\mathbb{P}(\text{first head on } n)} \cdot \underbrace{2^n}_{\text{payout}} = \sum_{n=1}^{\infty} 1 = 1 + 1 + 1 + \cdots = +\infty.$$

Every term contributes exactly 1 dollar of expected value, and there are infinitely many of them. The expected payout is *infinite*. So by the logic of expected value you should be willing to pay any finite amount – a million dollars, a billion – for a single play. **Yet essentially nobody would pay even $20.** That is the paradox: a mathematically infinite expectation meeting a flatly finite human willingness to pay.

What is going on? Several things at once, and untangling them is genuinely illuminating.

What is really happening

First resolution: the infinity is real but lives in tiny tails. The expectation is infinite only because of vanishingly rare, astronomically large payoffs. To even *collect* $1024 = 2^{10}$ you need the first head to land on flip 10 – nine tails then a head, probability $2^{-10} = 1/1024$. The median payout is just $2. With probability 3/4 you win $4 or less. The mean is dragged to infinity by a sliver of probability promising unimaginable jackpots. Expectation reports the long-run average exactly; it just so happens that the long-run average is dominated by events you will almost never see, and "almost never" here means you would have to play for longer than the age of the universe to make the average behave. The *law of large numbers fails to kick in on any human timescale.*

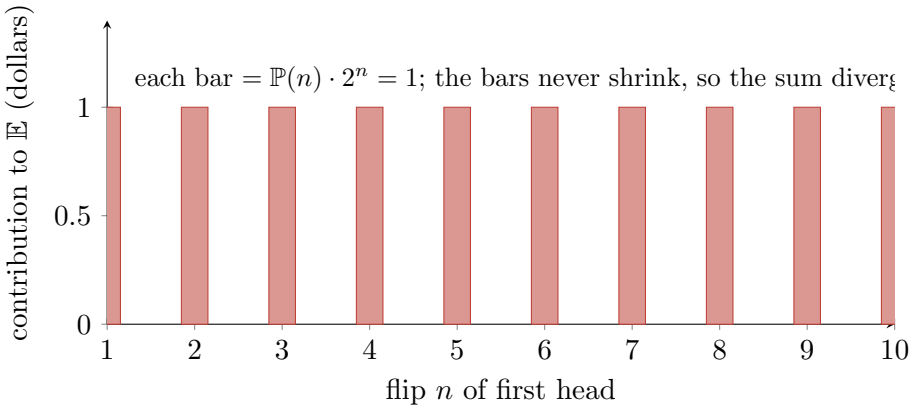


Figure 8.3: St. Petersburg: every outcome contributes the same $1 of expected value. A normal payout distribution has contributions that die off, letting the sum settle; here they refuse to, so $\mathbb{E} = \infty$. The divergence is structural, not a rounding artifact.

What is really happening

Second resolution: finite banks cap the game. An infinite expected value requires the casino to be able to pay unbounded sums. If the bank holds only 2^{40} dollars (about a

trillion) and caps the prize there, the expectation collapses. Each of the first 40 outcomes still contributes \$1, and every later outcome now pays only the capped 2^{40}, contributing $2^{40}\sum_{n>40} 2^{-n} = 2^{40}\cdot 2^{-40} = 1$ in total:

$$\mathbb{E}[\text{capped payout}] = \sum_{n=1}^{40} 1 \;+\; 1 = 41 \text{ dollars.}$$

A bank of size 2^N yields an expected payout of about $N+1$ dollars. Suddenly \$20 is a perfectly sensible price against a realistically finite casino. The infinity was an artifact of pretending an infinitely rich opponent exists.

History

The paradox is named for the city, not a person. Nicolaus Bernoulli posed it in 1713; his cousin **Daniel Bernoulli** published the influential analysis in 1738 in the *Commentaries* of the St. Petersburg Academy – hence the name. Daniel's resolution launched a whole field. He argued that people do not maximize expected *money*; they maximize expected *utility*, and the utility of money has diminishing returns – the joy of your second million is less than the joy of your first. He proposed logarithmic utility, $u(x) = \ln x$. With log utility the expected utility of the game is

$$\sum_{n=1}^{\infty} \left(\tfrac{1}{2}\right)^n \ln(2^n) = \ln 2 \sum_{n=1}^{\infty} \frac{n}{2^n} = 2\ln 2 < \infty,$$

using $\sum_{n\ge 1} n/2^n = 2$. This expected *utility* of $2\ln 2 = \ln 4$ is the utility of a sure \$4, so a log-utility player values the whole game at exactly \$4 – a finite, modest price in place of the infinite expected payout. This 1738 idea – *expected utility* – became the foundation of modern economics, decision theory, and the theory of risk aversion two centuries later.

Why this matters

The St. Petersburg paradox is not a flaw in expectation; it is a lesson about what expectation *means*. Expectation is the right summary when (a) you can repeat the gamble many times, (b) the average actually stabilizes on a relevant timescale, and (c) you care about money linearly. St. Petersburg quietly breaks all three. The takeaway is mature, not cynical: a single number can be exactly correct and still fail to tell you what to do. Expectation is the beginning of decision-making under uncertainty, not the end of it – which is why the next several chapters add *variance* (how spread out), *concentration* (how fast averages stabilize), and the laws of large numbers (when, precisely, the long-run picture is allowed to take over).

8.8 A gallery of worked examples

Example 8.7 (Expected sum of two dice – the slow way and the fast way)**.** Roll two fair dice; let S be their sum. *Slow way:* write out the distribution of S (values 2 through 12 with the familiar triangular probabilities $1/36, 2/36, \ldots, 6/36, \ldots, 1/36$, drawn in Figure 8.4) and compute $\sum_s s\,\mathbb{P}(S=s) = 7$. *Fast way:* $S = D_1 + D_2$, and by linearity $\mathbb{E}\,S = \mathbb{E}\,D_1 + \mathbb{E}\,D_2 = 3.5 + 3.5 = 7$. The fast way needs no

distribution at all, and it generalizes instantly: the expected sum of ten dice is 35.

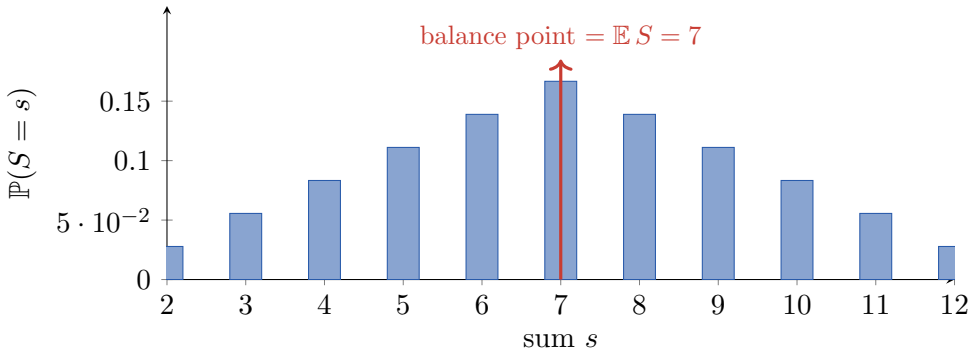


Figure 8.4: The triangular pmf of the sum of two fair dice. The peak (mode) and the balance point (mean) coincide here at 7 only because the picture is symmetric; linearity gives $\mathbb{E}\,S = 7$ with no need to draw the triangle at all.

Example 8.8 (Coupon collector: expected time to complete the set)**.** A cereal company hides one of n equally likely coupons in each box. How many boxes do you expect to buy to collect all n distinct coupons? Split the wait into stages: T_i is the number of boxes bought while you hold exactly $i-1$ distinct coupons and are waiting for a new one. When you already have $i-1$ coupons, each box is new with probability $(n-(i-1))/n$, so T_i is geometric with that success probability and (by Example 8.6)

$$\mathbb{E}\,T_i = \frac{n}{n-(i-1)}.$$

The total time is $T = T_1 + \cdots + T_n$, and by linearity

$$\mathbb{E}\,T = \sum_{i=1}^{n} \frac{n}{n-i+1} = n \sum_{j=1}^{n} \frac{1}{j} = n\,H_n \approx n \ln n.$$

For $n = 50$ coupons, $\mathbb{E}\,T = 50\,H_{50} \approx 225$ boxes. The stages T_i are chained together in time, but linearity sums their means without a second thought.

Example 8.9 (Expected value of a max via the tail-sum formula)**.** Roll a fair die twice and let M be the larger of the two values. Using $\mathbb{E}\,M = \sum_{k=1}^{6} \mathbb{P}(M \geq k)$ and $\mathbb{P}(M \geq k) = 1 - \mathbb{P}(\text{both} < k) = 1 - \left(\frac{k-1}{6}\right)^2$:

$$\mathbb{E}\,M = \sum_{k=1}^{6} \left[1 - \left(\frac{k-1}{6}\right)^2\right] = 6 - \frac{0+1+4+9+16+25}{36} = 6 - \frac{55}{36} = \frac{161}{36} \approx 4.47.$$

The maximum of two dice averages about 4.47, comfortably above the single-die mean of 3.5 – taking the better of two tries pulls you upward, as it should.

Walking question

You are about to flip a fair coin 100 times. A friend bets you that the number of times the pattern `HH` appears (in overlapping windows – positions 1-2, 2-3, . . . , 99-100) will be more than 30. Should you take the bet? Use indicators: there are 99 overlapping windows, and

each window shows HH with probability $\frac{1}{4}$. By linearity the expected count is $99 \cdot \frac{1}{4} = 24.75$, regardless of the heavy overlap between consecutive windows. Since $24.75 < 30$, betting against "more than 30" is the favorable side – and again, the dependence among overlapping windows never entered the mean calculation.

8.9 What to carry forward

Focus

The five things that matter.

1. **Two pictures.** $\mathbb{E}\,X = \sum_x x\ \mathbb{P}(X = x)$ is both a long-run average and a center of mass. It need not be a possible value.
2. **Indicators bridge to probability.** $\mathbb{E}[\mathbf{1}_A] = \mathbb{P}(A)$. Every probability is an expectation in disguise.
3. **Linearity is unconditional.** $\mathbb{E}[\sum_i X_i] = \sum_i \mathbb{E}\,X_i$ with *no* independence assumption – the most useful tool in the book.
4. **The indicator method.** Write a count as $\sum_i \mathbf{1}_{A_i}$, then $\mathbb{E} = \sum_i \mathbb{P}(A_i)$. It cracked hats, records, coupons, QuickSort.
5. **LOTUS.** $\mathbb{E}[g(X)] = \sum_x g(x)\ \mathbb{P}(X = x)$: average over inputs, weighting by the original pmf, never deriving the distribution of $g(X)$.

And one humility: an infinite or huge-tailed expectation (St. Petersburg) can be mathematically right yet practically useless. The mean is a summary, and a summary hides things – which is exactly why the next chapter asks how far from the mean a random variable tends to stray.

Exercises

Exercise 8.1 (Warm-up means)**.** Compute $\mathbb{E}\,X$ directly from the definition for each: (a) X is the number on a fair four-sided die $(1, 2, 3, 4)$; (b) X pays \$10 with probability 0.1 and \$0 otherwise; (c) $X \sim \text{Bernoulli}(p)$.

Exercise 8.2 (Center of mass is not the mode)**.** A random variable takes value 0 with probability 0.9 and value 100 with probability 0.1. Find $\mathbb{E}\,X$. Explain in one sentence why the expectation lies in a region the variable visits only 10% of the time, using the center-of-mass picture.

Exercise 8.3 (Linearity, no independence)**.** Draw 3 cards without replacement from a standard 52-card deck. Let A be the number of aces drawn. Using indicators $A = A_1 + A_2 + A_3$ (one per drawn position), find $\mathbb{E}\,A$. Then explain why the same answer holds even though the three draws are dependent.

Exercise 8.4 (Hat-matching variations)**.** n couples (so $2n$ people) are seated at a round table by a random permutation. (a) What is the expected number of couples seated next to each other? *Hint:*

indicator per person for "my right neighbor is my partner." (b) Why is the dependence between these indicators irrelevant to the mean?

Exercise 8.5 (Fixed points of a random function)**.** A function $f : \{1, \dots, n\} \to \{1, \dots, n\}$ is chosen so that each $f(i)$ is an independent uniform value in $\{1, \dots, n\}$ (note: not necessarily a permutation). Let $X = \#\{i : f(i) = i\}$ be the number of fixed points. Find $\mathbb{E}\, X$. Compare with the permutation (hat) case and comment.

Exercise 8.6 (LOTUS practice)**.** Let X be uniform on $\{1, 2, 3, 4, 5\}$. (a) Compute $\mathbb{E}\left[\frac{1}{X}\right]$ using LOTUS. (b) Compute $\mathbb{E}[(X-3)^2]$. (c) Verify that $\mathbb{E}[(X-3)^2] \neq (\mathbb{E}\, X - 3)^2$ and explain which inequality (Jensen) is at work.

Exercise 8.7 (Tail-sum formula)**.** (a) Use $\mathbb{E}\, X = \sum_{k \geq 1} \mathbb{P}(X \geq k)$ to find the expected number of rolls of a fair die needed to first see a number ≥ 5. (b) Let X be the minimum of three independent fair die rolls. Using $\mathbb{P}(X \geq k) = \left(\frac{7-k}{6}\right)^3$ (all three rolls must be $\geq k$) for $k = 1, \dots, 6$, compute $\mathbb{E}\, X$. *Answer:* $\mathbb{E}\, X = \sum_{k=1}^{6} \left(\frac{7-k}{6}\right)^3 = \frac{6^3+5^3+4^3+3^3+2^3+1^3}{216} = \frac{441}{216} \approx 2.04$.

Exercise 8.8 (St. Petersburg with a cap)**.** Modify the St. Petersburg game so the casino pays $\min(2^n, 2^{20})$ – the prize is capped at about a million dollars. (a) Show the expected payout is finite and find it (approximately). (b) Explain in words how a finite bank turns an infinite expectation into a small one.

Exercise 8.9 (Proof: linearity from first principles)**.** Let X and Y be discrete random variables on a finite sample space Ω with outcome probabilities $\mathbb{P}(\omega)$. Starting from $\mathbb{E}\, Z = \sum_{\omega \in \Omega} Z(\omega)\, \mathbb{P}(\omega)$, prove that $\mathbb{E}[aX + bY] = a\, \mathbb{E}\, X + b\, \mathbb{E}\, Y$ for constants a, b. Identify the single step where you use the fact that we are adding numbers *pointwise at each outcome*, and explain why no assumption about independence is needed.

Exercise 8.10 (Proof: LOTUS)**.** Prove the discrete LOTUS, $\mathbb{E}[g(X)] = \sum_x g(x)\, \mathbb{P}(X = x)$, by grouping the x-values according to the value $y = g(x)$ they map to. Where in the argument does the possibility that g is *not* one-to-one get correctly handled?

Exercise 8.11 (Indicator method in the wild)**.** You throw n balls independently and uniformly into n bins. (a) Find the expected number of empty bins (exactly, then as an approximation for large n). (b) Find the expected number of bins containing exactly one ball. (c) These two indicator families overlap and are dependent. State precisely why that does not affect either expected count.

Exercise 8.12 (Conceptual: when is the mean a bad summary?)**.** Give two qualitatively different real-world situations in which the expectation of a quantity is a misleading thing to report, and say what you would report instead. One of your examples should involve a heavy/long tail (St. Petersburg flavored) and the other should involve a strongly bimodal distribution.

Chapter 9

Variance, Moments, and Spread

In the last chapter you learned to find the *center* of a random variable: its expectation, the long-run average value, the balance point of its distribution. The expectation answers the question "where does this quantity sit, on average?" But two random quantities can share the same center and still behave in wildly different ways. A casino that wins a dollar half the time and loses a dollar the other half is, on average, exactly as profitable as one that wins a million dollars half the time and loses a million the other half. The average is the same. The *experience* is not.

This chapter is about that difference. It is about *spread*: how far a random variable typically wanders from its center. Spread is what separates a calm process from a wild one, a reliable estimate from a noisy guess, a safe investment from a gamble. Once you have a number for the center and a number for the spread, you have captured the two most important facts about almost any random quantity you will ever meet. The number for the spread is called the *variance*, and its square root, the *standard deviation*, is the unit in which uncertainty is measured throughout statistics, science, and machine learning.

We will build variance from the ground up: first the intuition of average squared distance, then the algebra that makes it computable, then the single most useful fact in all of applied probability — that variances of *independent* things add. From there we climb to higher moments (skewness and kurtosis), meet an inequality due to Jensen, and end by seeing why variance turns up almost everywhere in applied work.

9.1 The Problem: Two Distributions, One Center

Let us make the opening story concrete. Consider two games, each costing nothing to play, each with expected payoff zero.

- **Game A (mild).** You win \$1 or lose \$1, each with probability 1/2.
- **Game B (wild).** You win \$100 or lose \$100, each with probability 1/2.

Call the payoffs X and Y. Both have expectation zero:

$$\mathbb{E}[X] = \tfrac{1}{2}(1) + \tfrac{1}{2}(-1) = 0, \qquad \mathbb{E}[Y] = \tfrac{1}{2}(100) + \tfrac{1}{2}(-100) = 0.$$

A summary that only reports the mean would call these games identical. Your stomach knows better. Game B is a hundred times more nerve-wracking, and any accurate description of it must say so. We need a number that grows when the outcomes are flung far from the center and stays small when they huddle close.

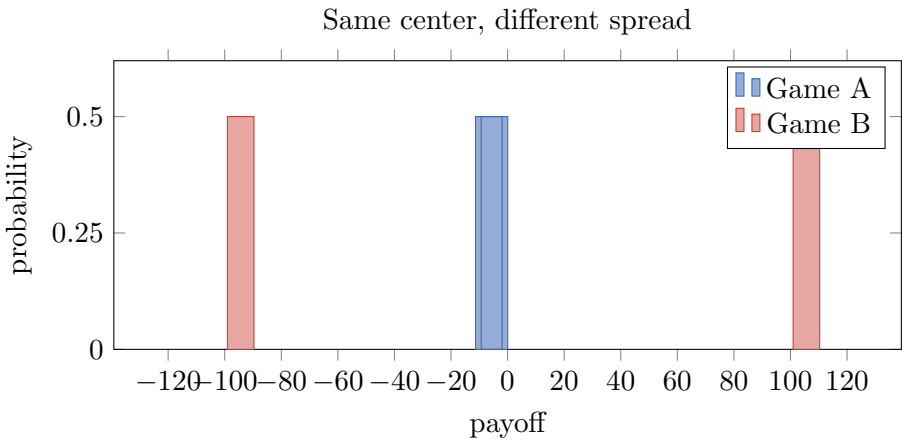


Figure 9.1: Two distributions with identical mean (zero) but very different spread. Game A (blue) places its mass right next to the center; Game B (red) flings the same probability far away. Variance is the number that distinguishes them.

Intuition

Think of a distribution as a pile of sand sitting on a seesaw. The expectation is the point where the seesaw balances. The *spread* is how spread out the sand is around that balance point. You could pile it all in a narrow heap at the center, or smear it thin out to the edges — the balance point can be identical in both cases. We want a single number that measures "how smeared out."

9.1.1 Why not just average the distance?

The natural first idea is: measure each outcome's distance from the mean, and average those distances. If $\mu = \mathbb{E}[X]$, look at $|X - \mu|$ and take its expectation. This quantity, $\mathbb{E}\,|X - \mu|$, is called the *mean absolute deviation*, and it is a perfectly sensible measure of spread. For Game A it equals 1; for Game B it equals 100. So far so good.

But the absolute value has a sharp corner at zero, and corners are the enemy of calculus. The function $|x|$ is not differentiable at the origin, which makes the mean absolute deviation awkward to optimize, awkward to combine across several random variables, and awkward to push through the machinery of sums and limits we will build in later chapters. Mathematicians and statisticians, faced with this corner, made a fateful and enormously productive choice: instead of penalizing distance by $|x - \mu|$, penalize it by $(x - \mu)^2$. Squaring kills the corner — it is smooth everywhere — and, as we will see, it interacts cleanly with sums, with independence, and with the geometry of the normal distribution. The price is that squaring exaggerates large deviations and gives answers in squared units. We pay that price gladly, and then take a square root at the end to get back to the original units.

9.2 Variance and Standard Deviation

Definition 9.1 (Variance)**.** Let X be a random variable with finite mean $\mu = \mathbb{E}[X]$. The *variance* of X is the expected squared deviation from the mean,

$$\mathrm{Var}(X) \;=\; \mathbb{E}\big[(X-\mu)^2\big].$$

Its non-negative square root is the *standard deviation*,

$$\mathrm{SD}(X) \;=\; \sqrt{\mathrm{Var}(X)},$$

often written σ, with σ^2 for the variance.

Three properties follow at once.

First, **variance is never negative.** It is the average of squares, and squares are never negative, so $\mathrm{Var}(X) \geq 0$ always. It equals zero exactly when $(X-\mu)^2 = 0$ with probability one — that is, when X is a constant. A constant has no spread, and variance correctly reports zero. This is your first sanity check on any variance calculation: a negative answer is always an arithmetic mistake.

Second, **variance lives in squared units.** If X is measured in dollars, $\mathrm{Var}(X)$ is in dollars-squared, which is meaningless to a human. The standard deviation $\mathrm{SD}(X)$ undoes the squaring and returns to dollars. That is why, when reporting results to people, we almost always quote the standard deviation, not the variance — "the returns have a standard deviation of \$3" is something you can feel, while "a variance of 9 dollars-squared" is not.

Third, **the standard deviation is a genuine yardstick.** It tells you the typical scale of fluctuation. A standard deviation of \$3 means that draws of X routinely land a few dollars away from the mean; landing \$30 away would be extraordinary. We will sharpen "routinely" and "extraordinary" into a precise statement in Section 9.6 (Chebyshev's inequality) but the slogan to carry is: *the standard deviation is the natural unit of surprise.*

Example 9.1 (The two games, measured)**.** Return to Games A and B from Section 9.1, both with mean $\mu = 0$. For Game A,

$$\mathrm{Var}(X) = \tfrac{1}{2}(1-0)^2 + \tfrac{1}{2}(-1-0)^2 = \tfrac{1}{2} + \tfrac{1}{2} = 1, \qquad \mathrm{SD}(X) = 1.$$

For Game B,

$$\mathrm{Var}(Y) = \tfrac{1}{2}(100)^2 + \tfrac{1}{2}(-100)^2 = 10{,}000, \qquad \mathrm{SD}(Y) = 100.$$

The standard deviations are 1 and 100: Game B is exactly a hundred times wilder, which is precisely what your stomach said. Notice that the *variances* differ by a factor of 10,000, the square of the scale factor. Squaring magnifies; the square root in the standard deviation tames it back to a linear, human scale.

Example 9.2 (A single fair die)**.** Let X be the result of one roll of a fair six-sided die. Its mean is $\mu = (1+2+3+4+5+6)/6 = 3.5$. The variance is the average of the squared deviations:

$$\mathrm{Var}(X) = \frac{1}{6}\sum_{k=1}^{6}(k-3.5)^2 = \frac{1}{6}\big(6.25 + 2.25 + 0.25 + 0.25 + 2.25 + 6.25\big) = \frac{17.5}{6} \approx 2.92.$$

So $\mathrm{SD}(X) = \sqrt{2.9167} \approx 1.71$. A typical roll lands about 1.7 away from 3.5, which matches the feel of the numbers 1 through 6 spread evenly around 3.5.

9.3 The Computational Formula

Computing variance straight from the definition means first finding μ, then forming $(X-\mu)^2$ for every outcome, then averaging. That is two passes over the distribution. There is a slicker route that needs only the first two *moments* — $\mathbb{E}[X]$ and $\mathbb{E}[X^2]$ — and it is the formula you will use ninety percent of the time.

Theorem 9.1 (Computational formula for variance)**.** *For any random variable X with finite second moment,*

$$\operatorname{Var}(X) \;=\; \mathbb{E}[X^2] - \big(\mathbb{E}[X]\big)^2.$$

Proof. Expand the square inside the definition and use linearity of expectation (Chapter "Expectation"). Writing $\mu = \mathbb{E}[X]$, which is a constant,

$$\operatorname{Var}(X) = \mathbb{E}\left[(X-\mu)^2\right] = \mathbb{E}\left[X^2 - 2\mu X + \mu^2\right].$$

Linearity lets us split the expectation over the sum, and pull the constants 2μ and μ^2 outside:

$$= \mathbb{E}[X^2] - 2\mu\,\mathbb{E}[X] + \mu^2 = \mathbb{E}[X^2] - 2\mu\cdot\mu + \mu^2 = \mathbb{E}[X^2] - \mu^2.$$

That is exactly $\mathbb{E}[X^2] - (\mathbb{E}[X])^2$. □

Why it works

Why does "mean of the square minus square of the mean" measure spread? Read it as a competition between two ways of averaging. $\mathbb{E}[X^2]$ is the average of the squares; $(\mathbb{E}[X])^2$ is the square of the average. These two are equal only when X never varies. The moment X spreads out, the big values get *extra* weight under squaring (because squaring is convex — more on this when we meet Jensen's inequality), so $\mathbb{E}[X^2]$ pulls ahead of $(\mathbb{E}[X])^2$. The gap between them is exactly the spread. Variance is the toll that randomness charges for the privilege of squaring.

Common trap

A perennial error is to compute $(\mathbb{E}[X])^2$ and call it $\mathbb{E}[X^2]$. These are almost never equal! For the fair die, $\mathbb{E}[X] = 3.5$ so $(\mathbb{E}[X])^2 = 12.25$, while $\mathbb{E}[X^2] = \frac{1}{6}(1+4+9+16+25+36) = \frac{91}{6} \approx 15.17$. The difference $15.17 - 12.25 = 2.92$ is the variance — matching Example 9.2. If you ever find $\mathbb{E}[X^2] < (\mathbb{E}[X])^2$, you have made an arithmetic slip, because variance cannot be negative.

Example 9.3 (Bernoulli variance)**.** Let $X \sim \text{Bernoulli}(p)$, so $X = 1$ with probability p and $X = 0$ with probability $1-p$. Then $\mathbb{E}[X] = p$. Because X only takes the values 0 and 1, we have $X^2 = X$, so $\mathbb{E}[X^2] = \mathbb{E}[X] = p$ as well. The computational formula gives

$$\operatorname{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2 = p - p^2 = p(1-p).$$

This little formula $p(1-p)$ is one of the most important in all of statistics. Plotted against p it is a downward parabola, zero at $p = 0$ and $p = 1$ (a coin that always lands the same way has no spread) and largest at $p = 1/2$, where uncertainty is maximal.

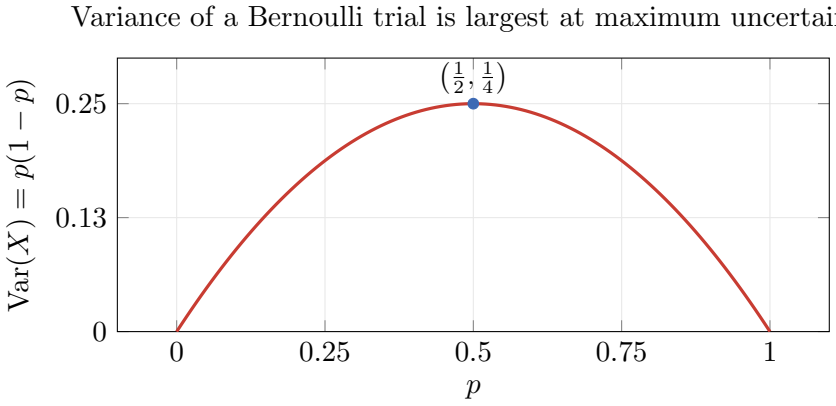


Figure 9.2: The Bernoulli variance $p(1-p)$. A fair coin ($p = 1/2$) is the most unpredictable and carries the most variance; a coin guaranteed to land one way ($p = 0$ or 1) carries none.

9.4 How Variance Behaves Under Shifts and Scalings

To work fluently with variance you need to know what happens when you slide a random variable along the number line or stretch it. These two rules come up constantly and follow directly from the definition.

Proposition 9.1 (Affine transformation)**.** *For any constants a and b,*

$$\mathrm{Var}(aX + b) = a^2 \,\mathrm{Var}(X), \qquad \textit{equivalently} \qquad \mathrm{SD}(aX + b) = |a| \,\mathrm{SD}(X).$$

Proof. Let $\mu = \mathbb{E}[X]$, so $\mathbb{E}[aX + b] = a\mu + b$ by linearity. Then

$$\mathrm{Var}(aX + b) = \mathbb{E}\left[\big((aX + b) - (a\mu + b)\big)^2\right] = \mathbb{E}\left[\big(a(X - \mu)\big)^2\right] = a^2 \,\mathbb{E}\left[(X - \mu)^2\right] = a^2 \,\mathrm{Var}(X). \quad \square$$

Two messages live in this one line.

Shifting does nothing. The constant b vanished. Adding a fixed amount to every outcome slides the whole distribution rigidly along the axis without changing how spread out it is. If everyone in a room gets a $10 raise, the inequality between their salaries is untouched. Variance, being a measure of spread, ignores the location entirely.

Scaling squares. Multiplying by a multiplies the standard deviation by $|a|$ and the variance by a^2. Double every outcome and you double the spread (standard deviation $\times 2$) but quadruple the variance. This is the same factor-of-square we saw between Games A and B: stretching the axis by 100 multiplied the variance by 10,000.

What is really happening

The appearance of a^2 (not a) is the clearest sign that variance is a *quadratic* quantity — it behaves like an area, not a length. Standard deviation is the corresponding length. This is exactly the relationship between a square's area and its side. When you later see variances adding like the squares of legs in the Pythagorean theorem (Section 9.7), remember that variance was a squared quantity from birth.

9.5 Standardization and the z-Score

Proposition 9.1 hands us a useful construction. Given any random variable X with mean μ and standard deviation $\sigma > 0$, we can manufacture a new variable that is centered at zero and has spread exactly one. Subtract the mean (to recenter) and divide by the standard deviation (to rescale):

$$Z \;=\; \frac{X - \mu}{\sigma}.$$

This is the *standardization* of X, and Z is called the *z-score*. Let us verify its two defining properties. By linearity,

$$\mathbb{E}[Z] = \frac{\mathbb{E}[X] - \mu}{\sigma} = \frac{\mu - \mu}{\sigma} = 0,$$

and by the affine rule with $a = 1/\sigma$,

$$\mathrm{Var}(Z) = \frac{1}{\sigma^2}\,\mathrm{Var}(X) = \frac{\sigma^2}{\sigma^2} = 1.$$

So every X, no matter its original units — dollars, kilograms, exam points — can be converted into a dimensionless number Z whose value is "how many standard deviations above or below the mean am I?"

Intuition

A z-score is a universal translator for surprise. A student who scores 650 on one exam and 34 on another cannot be compared until you know the mean and spread of each test. But if both scores correspond to $z = +1.5$, you know instantly that the student did equally well on both relative to peers — one-and-a-half standard deviations above average in each case. The z-score strips away the arbitrary units and leaves pure standing.

Example 9.4 (Two exams)**.** On a midterm with mean 70 and standard deviation 8, you score 82. On a final with mean 65 and standard deviation 20, you score 90. Which is the better performance relative to the class?

$$z_{\text{mid}} = \frac{82 - 70}{8} = 1.5, \qquad z_{\text{fin}} = \frac{90 - 65}{20} = 1.25.$$

The raw final score (90) is higher, but the midterm z-score (1.5) is larger: you stood out more on the midterm, because that test had much less spread, so 12 points above its mean was rarer than 25 points above the final's mean.

Standardization is not a mere convenience; it is the gateway to the normal distribution and the central limit theorem in later chapters. There, the standardized version of a sum of many independent pieces converges to a fixed universal shape — the standard normal — regardless of the messy details of the originals. Z-scores are the coordinates in which that miracle becomes visible.

9.6 Chebyshev's Inequality: Making "Surprise" Precise

We have twice promised to turn the slogan "the standard deviation is the unit of surprise" into a theorem. Here it is. The remarkable thing is that a single inequality, demanding *nothing* about the shape of the distribution — not symmetry, not unimodality, not even a name — already forces most of the probability mass to live within a few standard deviations of the mean.

Theorem 9.2 (Chebyshev's inequality)**.** *Let X have finite mean μ and finite variance σ^2. For any $k > 0$,*

$$\mathbb{P}\left(|X - \mu| \geq k\sigma\right) \;\leq\; \frac{1}{k^2}.$$

In words: the probability of landing k or more standard deviations away from the mean is at most $1/k^2$.

Proof. Let $D = (X-\mu)^2$, a non-negative random variable with $\mathbb{E}[D] = \sigma^2$. The event $\{|X-\mu| \geq k\sigma\}$ is exactly the event $\{D \geq k^2\sigma^2\}$. Now use the elementary bound: whenever $D \geq 0$ and $t > 0$, the indicator $\mathbf{1}\{D \geq t\}$ satisfies $\mathbf{1}\{D \geq t\} \leq D/t$, because if the indicator is 0 the inequality is $0 \leq D/t$ (true), and if it is 1 then $D \geq t$ makes $D/t \geq 1$. Taking expectations of both sides with $t = k^2\sigma^2$,

$$\mathbb{P}\left(D \geq k^2\sigma^2\right) = \mathbb{E}\left[\mathbf{1}\{D \geq k^2\sigma^2\}\right] \;\leq\; \frac{\mathbb{E}[D]}{k^2\sigma^2} = \frac{\sigma^2}{k^2\sigma^2} = \frac{1}{k^2}. \qquad \square$$

Setting $k = 2$ shows at least $1 - 1/4 = 75\%$ of any distribution's mass lies within two standard deviations of the mean; $k = 3$ guarantees at least $1 - 1/9 \approx 89\%$ within three. These bounds are deliberately *loose* — they must hold for every distribution at once, including adversarial ones engineered to push mass outward — so for any particular well-behaved distribution the truth is far tighter.

Example 9.5 (Chebyshev versus the normal rule)**.** For the bell-shaped normal distribution the famous "empirical rule" says about 68% of mass lies within one standard deviation, 95% within two, and 99.7% within three. Compare these to what Chebyshev guarantees for *any* distribution:

k	Chebyshev bound on $\mathbb{P}(\lvert X-\mu\rvert < k\sigma)$	normal (empirical rule)
2	$\geq 75\%$	$\approx 95\%$
3	$\geq 88.9\%$	$\approx 99.7\%$

The normal does far better than Chebyshev demands — that is the reward for having thin, light tails. Chebyshev is the universal floor; the normal rule is what a particularly tame distribution actually achieves.

The normal density with its tails beyond $\pm 2\sigma$ shaded

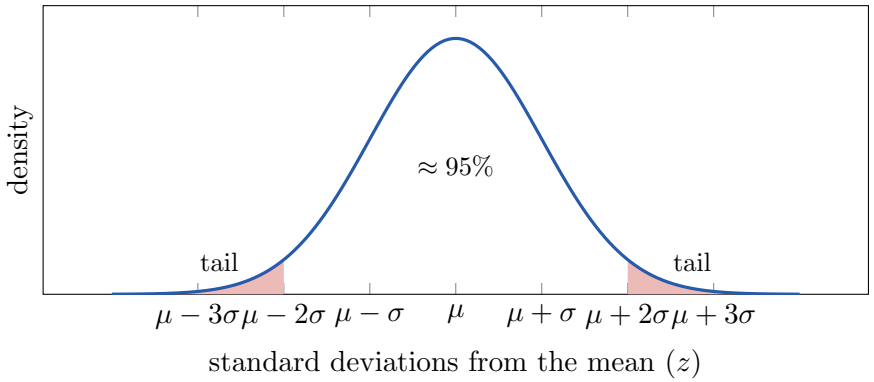


Figure 9.3: The standard normal density. The shaded tails beyond $\pm 2\sigma$ together hold only about 5% of the mass, so roughly 95% sits within two standard deviations. Chebyshev's inequality guarantees *at least* 75% for any distribution whatsoever; the bell curve, with its light tails, comfortably beats that floor. The standard deviation is the ruler along the horizontal axis.

Why this matters

Chebyshev is the reason variance is more than bookkeeping. A small variance does not merely *describe* a tidy distribution — it *forces* one. Once you know σ, you have a guaranteed, distribution-free leash on how far the variable can roam: the probability of straying k standard deviations out can never exceed $1/k^2$, no matter how exotic the underlying randomness. This is what lets you attach reliable error bars to estimates whose exact distribution you do not know, and it is the inequality that powers the first proof of the law of large numbers in a later chapter.

9.7 The Variance of a Sum: Where Independence Earns Its Keep

Expectation is gloriously simple under addition: $\mathbb{E}[X+Y] = \mathbb{E}[X]+\mathbb{E}[Y]$, *always*, with no conditions whatsoever. You might hope variance is just as easy. It is not — and the way it fails is one of the most illuminating facts in probability.

Theorem 9.3 (Variance of a sum of two variables)**.** *For any two random variables X and Y with finite variances,*

$$\operatorname{Var}(X+Y) = \operatorname{Var}(X) + \operatorname{Var}(Y) + 2\,\operatorname{Cov}(X,Y),$$

where the covariance *is* $\operatorname{Cov}(X,Y) = \mathbb{E}\left[(X-\mu_X)(Y-\mu_Y)\right] = \mathbb{E}[XY] - \mathbb{E}[X]\,\mathbb{E}[Y]$.

Proof. Let $\mu_X = \mathbb{E}[X]$, $\mu_Y = \mathbb{E}[Y]$, so $\mathbb{E}[X+Y] = \mu_X + \mu_Y$. Then, abbreviating the centered variables $\tilde{X} = X - \mu_X$ and $\tilde{Y} = Y - \mu_Y$,

$$\operatorname{Var}(X+Y) = \mathbb{E}\left[(\tilde{X}+\tilde{Y})^2\right] = \mathbb{E}\left[\tilde{X}^2 + 2\tilde{X}\tilde{Y} + \tilde{Y}^2\right].$$

By linearity this splits into $\mathbb{E}[\tilde{X}^2] + 2\,\mathbb{E}[\tilde{X}\tilde{Y}] + \mathbb{E}[\tilde{Y}^2] = \operatorname{Var}(X) + 2\operatorname{Cov}(X,Y) + \operatorname{Var}(Y)$. □

The cross term $2\operatorname{Cov}(X,Y)$ is the whole story. Covariance measures how X and Y move *together*: it is positive when they tend to be large at the same time, negative when one being large signals the other being small, and zero when their fluctuations are unrelated in the averaged sense. We devote a full later chapter to covariance and its normalized cousin correlation; here we need only one consequence.

Corollary 9.1 (Variances of independent variables add)**.** *If X and Y are independent, then* $\operatorname{Cov}(X,Y) = 0$*, and therefore*

$$\operatorname{Var}(X+Y) = \operatorname{Var}(X) + \operatorname{Var}(Y).$$

More generally, if $X_1, \dots, X_n$ are pairwise independent (or merely pairwise uncorrelated), then

$$\operatorname{Var}\Big(\sum_{i=1}^{n} X_i\Big) = \sum_{i=1}^{n} \operatorname{Var}(X_i).$$

Proof. If $X \perp\!\!\!\perp Y$ then $\mathbb{E}[XY] = \mathbb{E}[X]\,\mathbb{E}[Y]$ (a fact proved in the chapter on expectation), so $\operatorname{Cov}(X,Y) = \mathbb{E}[XY] - \mathbb{E}[X]\,\mathbb{E}[Y] = 0$. The cross terms in the general expansion $\operatorname{Var}(\sum_i X_i) = \sum_i \operatorname{Var}(X_i) + \sum_{i \neq j} \operatorname{Cov}(X_i, X_j)$ all vanish under pairwise uncorrelatedness, leaving only the diagonal sum. □

Focus

Independence is exactly the condition that lets variances add. The mean adds for free, no strings attached. Variance adds *only when the parts do not conspire.* This single asymmetry between mean and variance is the engine behind an enormous amount of statistics — error bars, the $\sqrt{n}$ law, portfolio diversification, and the central limit theorem all run on it.

Connection

Theorem 9.3 is the Pythagorean theorem in disguise. Picture each centered random variable as a vector, with the "length-squared" of a vector being its variance and the "dot product" of two vectors being their covariance. Independent (uncorrelated) variables are then *perpendicular* vectors, and for perpendicular vectors the squared length of the sum is the sum of squared lengths — exactly $\mathrm{Var}(X+Y) = \mathrm{Var}(X) + \mathrm{Var}(Y)$. The companion linear-algebra reader develops this inner-product picture; here it is enough to know that the geometry is real, and that standard deviations combine like the legs of a right triangle: $\sigma_{X+Y} = \sqrt{\sigma_X^2 + \sigma_Y^2}$.

9.7.1 The $\sqrt{n}$ law and why averaging works

Here is the payoff that justifies all the machinery. Let $X_1, \ldots, X_n$ be independent draws from the same distribution, each with mean μ and variance σ^2 — think of n repetitions of the same noisy measurement. Consider their sum $S_n = \sum_i X_i$ and their average $\bar{X}_n = S_n/n$.

For the **sum**, variances add (Corollary 9.1):

$$\mathrm{Var}(S_n) = n\sigma^2, \qquad \mathrm{SD}(S_n) = \sqrt{n}\,\sigma.$$

For the **average**, apply the scaling rule with $a = 1/n$:

$$\mathrm{Var}(\bar{X}_n) = \frac{1}{n^2}\,\mathrm{Var}(S_n) = \frac{n\sigma^2}{n^2} = \frac{\sigma^2}{n}, \qquad \mathrm{SD}(\bar{X}_n) = \frac{\sigma}{\sqrt{n}}.$$

Why this matters

This last line, $\mathrm{SD}(\bar{X}_n) = \sigma/\sqrt{n}$, is one of the most consequential formulas in all of science. It says that averaging n independent measurements shrinks the spread of your estimate by a factor of $\sqrt{n}$. Average 4 readings and the noise halves; average 100 and it drops by a factor of 10. This is *why repeated measurement works*, why larger samples give tighter conclusions, and why every poll and every clinical trial reports a margin of error that shrinks with the square root of the sample size. The mean of the average stays put at μ (no bias), while its spread melts away as $1/\sqrt{n}$ — so the average homes in on the truth.

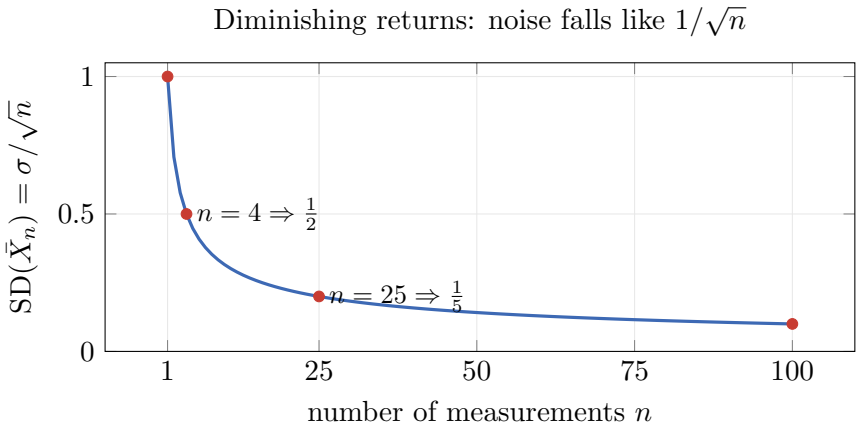


Figure 9.4: The standard deviation of a sample average decays as $\sigma/\sqrt{n}$. The curve is steep at first — the first few extra measurements help a lot — then flattens, so cutting the error in half always costs four times the data. This diminishing return is a permanent feature of estimation.

Common trap

The $\sqrt{n}$ scaling has a sober flip side. To halve your error you must *quadruple* your sample, and to cut it to a tenth you need a hundred times the data. Precision is expensive, and it gets more expensive the more you already have. Beware anyone who promises that "just collecting more data" will cheaply rescue a noisy study; past a point, the $1/\sqrt{n}$ curve in Figure 9.4 is nearly flat.

Example 9.6 (Variance of the binomial)**.** A binomial random variable $X \sim \text{Binomial}(n, p)$ counts the successes in n independent Bernoulli(p) trials, so $X = \sum_{i=1}^{n} B_i$ with each $B_i \sim \text{Bernoulli}(p)$ independent. Each B_i has variance $p(1-p)$ (Example 9.3), and because the trials are independent the variances simply add:

$$\text{Var}(X) = \sum_{i=1}^{n} \text{Var}(B_i) = n\,p(1-p).$$

Deriving this straight from the binomial pmf would mean wrestling with sums of $k^2\binom{n}{k}p^k(1-p)^{n-k}$. Decomposing into independent indicators makes it a one-liner. This "break a complicated variable into a sum of simple independent pieces" move is one of the sharpest tactics in the subject.

Application

[Why diversification reduces risk] A portfolio's value is a weighted sum of its assets' returns, and a portfolio's *risk* is the standard deviation of that sum. If you put all your money in one stock, your risk is that stock's full standard deviation. If instead you split it across many assets whose returns are roughly independent, the cross-covariances stay small and the variances add as in Corollary 9.1 — but you have also *divided* each position, so each scaled variance shrinks. The net effect is the $\sqrt{n}$ law again: spreading a fixed sum across n uncorrelated assets cuts the portfolio's standard deviation toward $\sigma/\sqrt{n}$. This is the mathematical core of Harry Markowitz's portfolio theory, and the reason "do not put all your eggs in one basket" is not folk wisdom but a theorem. The catch sits in the covariance term: in a crash, assets fall *together*, covariances turn sharply positive, and the diversification benefit evaporates exactly

when you need it most.

Paradox

[The independence assumption that wasn't] The variance-adds rule is so convenient that people apply it reflexively — and forget its one hypothesis. Suppose an insurer writes 10,000 flood policies and reasons: each policy has small variance, variances add, so the total variance is 10,000 times small, and the standard deviation of the payout is only $\sqrt{10{,}000} = 100$ times small — a comfortably tiny *relative* risk by the $\sqrt{n}$ argument. Then a single flood soaks every policyholder at once. The losses were never independent; they shared a common cause. With perfect positive correlation the standard deviations *add* instead of the variances, so the spread grows like n, not $\sqrt{n}$, and the insurer is wiped out. The moral: before you let variances add, ask hard whether the parts can fail together. The $2\,\mathrm{Cov}$ term in Theorem 9.3 is not a footnote — it is sometimes the entire risk.

9.8 Moments: A Family of Summaries

The mean and variance are the first two members of an infinite family. The k-th *moment* of X is $\mathbb{E}[X^k]$, and the k-th *central moment* is $\mathbb{E}[(X-\mu)^k]$, the moment taken about the mean.

Definition 9.2 (Moments)**.** For a positive integer k, the *k-th raw moment* of X is $m_k = \mathbb{E}[X^k]$, and the *k-th central moment* is $\mu_k = \mathbb{E}[(X-\mu)^k]$, where $\mu = \mathbb{E}[X]$.

What each moment captures follows a clear pattern:

- **First moment** $\mathbb{E}[X] = \mu$ — the *center*.
- **Second central moment** $\mathbb{E}[(X-\mu)^2] = \mathrm{Var}(X)$ — the *spread*.
- **Third central moment** measures *asymmetry* (skewness): is the distribution lopsided?
- **Fourth central moment** measures *tail weight* (kurtosis): how prone is it to extreme values?

Each higher moment refines the silhouette of the distribution. The first locates it, the second sizes it, the third tilts it, the fourth fattens or thins its tails. In principle, knowing all the moments often pins down the distribution completely — a theme we return to with moment generating functions in a later chapter.

9.8.1 Skewness: the third moment, made scale-free

The raw third central moment $\mathbb{E}[(X-\mu)^3]$ has units of X cubed, which makes it impossible to compare across different variables. We standardize it by dividing by σ^3, producing a pure number.

Definition 9.3 (Skewness)**.** The *skewness* of X is

$$\mathrm{Skew}(X) = \mathbb{E}\left[\left(\frac{X-\mu}{\sigma}\right)^3\right] = \frac{\mathbb{E}[(X-\mu)^3]}{\sigma^3}.$$

The cube preserves sign, and that is the point. When X exceeds its mean the cubed deviation is positive; when it falls short it is negative. Skewness asks which side wins on average. A symmetric distribution contributes equal positive and negative cubes that cancel, giving skewness zero. A distribution with a long *right* tail — a few very large values — has its big positive cubes dominate, giving *positive* skew. A long left tail gives negative skew.

Intuition

The cleanest mental image: where is the long tail pointing? *Right*-skewed means the tail stretches toward large values, dragging the mean to the right of the bulk. Incomes are the classic case — most people earn modestly while a few earn enormously, so the average income sits well above the median. Whenever the mean noticeably exceeds the median, suspect a long right tail and positive skew.

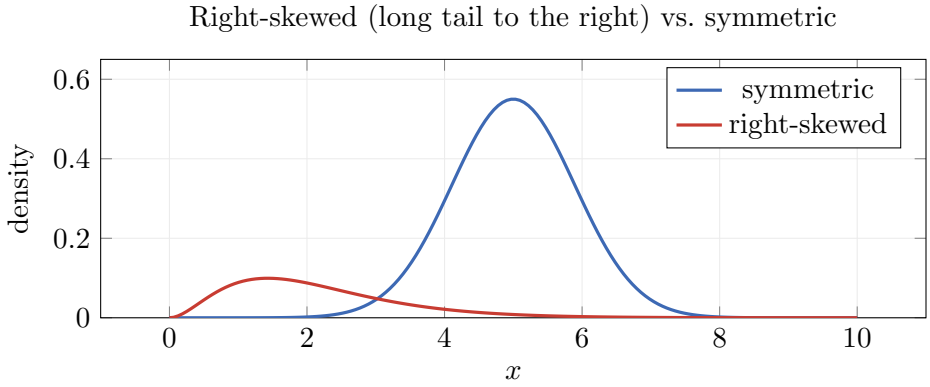


Figure 9.5: A symmetric density (blue) has zero skewness: equal mass on each side of its peak. A right-skewed density (red) rises quickly then trails off in a long right tail, pulling its mean rightward and giving positive skewness.

9.8.2 Kurtosis: the fourth moment and the weight of the tails

The fourth standardized central moment measures how much of a distribution's variance comes from rare, extreme deviations rather than ordinary middle-sized ones.

Definition 9.4 (Kurtosis)**.** The *kurtosis* of X is

$$\operatorname{Kurt}(X) = \mathbb{E}\left[\left(\frac{X-\mu}{\sigma}\right)^4\right] = \frac{\mathbb{E}[(X-\mu)^4]}{\sigma^4}.$$

The *excess kurtosis* is $\operatorname{Kurt}(X) - 3$, defined so that the normal distribution has excess kurtosis zero.

Because the deviations are raised to the fourth power, an outlier four standard deviations out contributes $4^4 = 256$ times as much as a typical one-standard-deviation wobble. Kurtosis is therefore a tail-sensitivity meter. High kurtosis ("heavy tails") means the distribution, while perhaps calm most of the time, produces occasional violent excursions. Low kurtosis means the outcomes stay obediently near a fixed band.

Common trap

Kurtosis is *not* a measure of "peakedness," despite a stubborn old textbook tradition that says so. Two distributions can have nearly identical central peaks and vastly different kurtosis purely because of what happens far out in the tails. The fourth power is overwhelmingly dominated by extreme values, so read kurtosis as a statement about rare large events, not about the shape of the hump in the middle.

Application

[Heavy tails and the limits of the bell curve] The normal distribution has excess kurtosis exactly zero, and for decades financial models assumed asset returns were normal. Real markets disagree loudly: daily stock returns are sharply *leptokurtic* (positive excess kurtosis), meaning crashes and spikes occur far more often than a normal model predicts. A "six-sigma" daily move should, under normality, happen roughly once in several million years; in real markets such moves arrive every few years. Underestimating kurtosis — pretending the tails are thin when they are fat — has bankrupted funds and helped fuel financial crises. The fourth moment is where the bell curve's optimism goes to die, and why modern risk management leans on heavy-tailed models instead.

History

The vocabulary of moments is borrowed wholesale from mechanics. In physics the "first moment" of a mass distribution locates its center of gravity, and the "second moment" — the moment of inertia — measures resistance to rotation, computed as mass times distance *squared*. Probabilists in the nineteenth century, led by figures like Pafnuty Chebyshev and later Karl Pearson, saw that a probability distribution is just a unit mass smeared along the line, and lifted the terminology intact. Variance *is* the moment of inertia of the probability mass about its center of gravity. The word "kurtosis" (Greek for "bulging") and the systematic study of skewness are due to Pearson around 1900, as he tried to classify the zoo of non-normal distributions appearing in biology and economics. When you compute a variance, you are, quite literally, computing how hard the distribution would be to spin.

9.9 A Preview of Jensen's Inequality

We have leaned repeatedly on the fact that squaring is *convex* — that it bends upward. The general principle behind that fact deserves a name, because it explains in one stroke why $\mathbb{E}[X^2] \geq (\mathbb{E}[X])^2$, and far more besides. We will prove it carefully in the chapter on inequalities; here is the idea and the consequence.

A function g is *convex* if its graph lies below every chord and above every tangent line — it curves upward like a bowl, as $g(x) = x^2$ does. Convexity has a probabilistic shadow.

Theorem 9.4 (Jensen's inequality, preview)**.** *If g is a convex function and X a random variable with finite mean, then*

$$g\big(\mathbb{E}[X]\big) \;\leq\; \mathbb{E}\,\big[g(X)\big].$$

If g is concave, the inequality reverses. Equality holds (for strictly convex g) only when X is a constant.

Why it works

Why is the function of the average no larger than the average of the function, for a bowl-shaped g? Picture the points $(X, g(X))$ scattered along the upward-curving graph. Averaging the X-values lands you at $\big(\mathbb{E}[X], g(\mathbb{E}[X])\big)$ — a point *on* the curve. But averaging the *heights* $g(X)$ lands you somewhere among the points, and because the curve bows downward away from any chord, the cloud of points sits *above* the curve between them. So the averaged height $\mathbb{E}[g(X)]$ is pulled up, above the on-curve height $g(\mathbb{E}[X])$. Convexity systematically rewards spreading out: the more X varies, the higher $\mathbb{E}[g(X)]$ climbs above $g(\mathbb{E}[X])$.

Take $g(x) = x^2$, which is convex. Jensen says $(\mathbb{E}[X])^2 \le \mathbb{E}[X^2]$, and the gap between the two sides is precisely $\mathrm{Var}(X) \ge 0$. So the non-negativity of variance is just Jensen's inequality for the squaring function, and variance *measures the size of the Jensen gap*. That is a satisfying unification: the fact that spread is non-negative and the fact that bowls bend upward are the same fact.

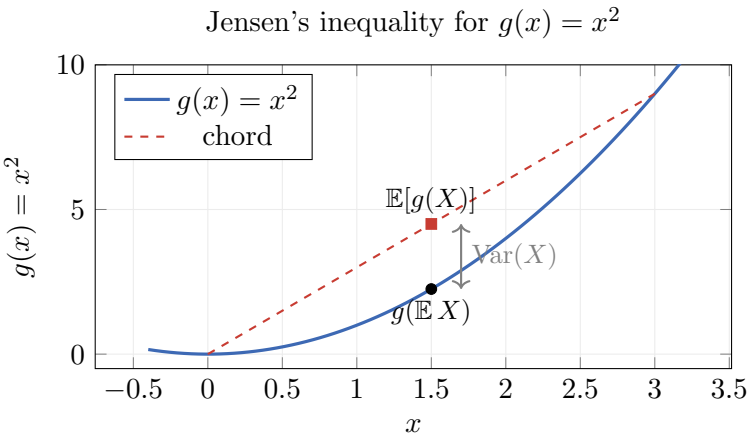


Figure 9.6: For X equal to 0 or 3 with equal probability, $\mathbb{E}[X] = 1.5$. The on-curve height $g(\mathbb{E}[X]) = 2.25$ sits below the chord height $\mathbb{E}[g(X)] = 4.5$. The gap, 2.25, is exactly $\mathrm{Var}(X) = p(1-p)\cdot 3^2$ with $p = 1/2$. Jensen's gap is the variance.

9.10 Why Variance Is Everywhere

A short tour shows how widely variance runs through quantitative work. None of the following needs more than the ideas of this chapter, yet together they explain why "minimize the variance" is practically a mantra across the data sciences.

- **Least squares.** The workhorse of regression fits a line by minimizing the sum of *squared* residuals — that is, by minimizing a variance. The choice of squares (rather than absolute errors) traces directly back to the same smoothness and additivity that made variance the canonical measure of spread.
- **The bias–variance tradeoff.** A model's expected prediction error decomposes into bias squared plus variance plus irreducible noise. Underfitting means high bias; overfitting means high variance. Every learning algorithm is, at heart, negotiating this tradeoff, and the variance term is the one this chapter taught you to read.

- **Error bars and confidence.** Every reported uncertainty — the ± on a measurement, the margin of error in a poll — is a standard deviation (or a multiple of one) in disguise, almost always shrunk by the $\sigma/\sqrt{n}$ law of Section 9.7.
- **Signal processing.** The "power" of a noise source is its variance; the signal-to-noise ratio compares the variance of the signal to the variance of the noise. Filtering is the art of removing variance you do not want.
- **Analysis of variance (ANOVA).** An entire branch of statistics is built on partitioning the total variance of a dataset into components attributable to different causes — a direct application of the variance-of-a-sum decomposition.

Why this matters

If expectation answers "what will happen on average," variance answers "how much should I trust that average." The first number without the second is a forecast with no confidence attached — a point with no error bar, a model that says nothing about its own uncertainty. Mature quantitative reasoning always carries both. That is why, having learned to find centers in the last chapter, you have just doubled your power by learning to measure spread: together they are the irreducible minimum for describing any uncertain quantity.

Summary

- **Variance** $\mathrm{Var}(X) = \mathbb{E}[(X-\mu)^2]$ is the average squared distance from the mean; **standard deviation** $\mathrm{SD}(X) = \sqrt{\mathrm{Var}(X)}$ returns it to the original units and is the natural "unit of surprise."
- **Computational formula:** $\mathrm{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2$. Variance is never negative; it is zero only for a constant.
- **Affine rule:** $\mathrm{Var}(aX+b) = a^2\,\mathrm{Var}(X)$. Shifting does not change spread; scaling by a scales the standard deviation by $|a|$.
- **Standardization:** $Z = (X-\mu)/\sigma$ has mean 0 and variance 1; z-scores compare across scales.
- **Chebyshev:** for any distribution, $\mathbb{P}(|X-\mu| \geq k\sigma) \leq 1/k^2$ — variance alone forces most mass to stay within a few standard deviations of the mean.
- **Sums:** $\mathrm{Var}(X+Y) = \mathrm{Var}(X) + \mathrm{Var}(Y) + 2\,\mathrm{Cov}(X,Y)$. For independent (or uncorrelated) variables the covariance vanishes and variances add, giving the $\sigma/\sqrt{n}$ law for averages.
- **Higher moments:** skewness (third) measures asymmetry, kurtosis (fourth) measures tail weight.
- **Jensen:** $\mathbb{E}[X^2] \geq (\mathbb{E}[X])^2$ is the convexity of squaring, and the gap is exactly the variance.

9.11 Exercises

Exercise 9.1 (Warm-up computations)**.** For each of the following, compute $\mathbb{E}[X]$, $\mathbb{E}[X^2]$, $\mathrm{Var}(X)$, and $\mathrm{SD}(X)$.

1. X takes values $2, 4, 6$ each with probability $1/3$.
2. $X \sim \text{Bernoulli}(0.2)$.
3. X is the number of heads in two independent fair coin flips.

Exercise 9.2 (Shifting and scaling)**.** A temperature C measured in Celsius has mean 20 and standard deviation 5. The Fahrenheit reading is $F = \frac{9}{5}C + 32$. Find $\mathbb{E}[F]$, $\text{Var}(F)$, and $\text{SD}(F)$. Explain in words why the $+32$ did not affect the spread.

Exercise 9.3 (The computational formula in reverse)**.** A random variable has mean $\mathbb{E}[X] = 3$ and variance $\text{Var}(X) = 4$. Find $\mathbb{E}[X^2]$. Then find $\mathbb{E}[(2X - 1)^2]$. (Hint: expand and reuse the first two moments.)

Exercise 9.4 (z-scores in practice)**.** A standardized test has mean 500 and standard deviation 100; a second test has mean 24 and standard deviation 4. Alice scores 640 on the first; Bob scores 30 on the second. Convert both to z-scores and say who performed better relative to peers. By how many points above the mean did each score, in units of their own test's standard deviation?

Exercise 9.5 (Variance of a sum, with and without independence)**.** Let X and Y each have variance 9.

1. If X and Y are independent, find $\text{Var}(X + Y)$ and $\text{SD}(X + Y)$.
2. If instead $\text{Cov}(X, Y) = 6$, find $\text{Var}(X + Y)$.
3. If $Y = X$ (perfect dependence), find $\text{Var}(X + Y)$ directly and confirm it matches the formula with the appropriate covariance.

Exercise 9.6 (The $\sqrt{n}$ law)**.** A single measurement of a length has standard deviation $\sigma = 2$ mm.

1. How many independent measurements must you average so that the standard deviation of the average is at most 0.25 mm?
2. Suppose you have reached an average error of 0.5 mm with some number of measurements. How many *times* more measurements are needed to reach 0.1 mm?

Exercise 9.7 (Chebyshev in action)**.** A machine fills bags with a mean weight of $\mu = 500$ g and a standard deviation of $\sigma = 4$ g; the distribution of weights is unknown.

1. Use Chebyshev's inequality to give a guaranteed lower bound on the probability that a bag weighs between 492 g and 508 g.
2. What is the largest possible probability that a bag deviates from 500 g by 12 g or more?
3. If the weights were known to be normal, roughly what fraction would fall within 492–508 g? Compare to your Chebyshev bound and explain the gap.

Exercise 9.8 (Binomial variance two ways)**.** Let $X \sim \text{Binomial}(n, p)$.

1. Using the decomposition $X = \sum_{i=1}^{n} B_i$ into independent Bernoulli trials, state $\text{Var}(X)$.
2. For $n = 100$, find the value of p that maximizes $\text{Var}(X)$, and give the resulting standard deviation.
3. Interpret: for which p is a binomial outcome hardest to predict?

Exercise 9.9 (Skewness of a simple variable)**.** Let X take the value 0 with probability 0.9 and the value 10 with probability 0.1.

1. Compute $\mu = \mathbb{E}[X]$ and $\sigma = \mathrm{SD}(X)$.
2. Compute the third central moment $\mathbb{E}[(X-\mu)^3]$ and the skewness.
3. Is the distribution skewed left or right? Sketch why this matches the sign you found.

Exercise 9.10 (Jensen and the variance gap)**.** Let X take values 1 and 9 with equal probability.

1. Compute $g(\mathbb{E}[X])$ and $\mathbb{E}[g(X)]$ for $g(x) = x^2$, and verify that their difference equals $\mathrm{Var}(X)$.
2. The square-root function $h(x) = \sqrt{x}$ is concave. Compute $h(\mathbb{E}[X])$ and $\mathbb{E}[h(X)]$ and check that the inequality runs the other way, as Jensen predicts for concave functions.

Exercise 9.11 (Proof: variance is translation invariant and quadratic in scale)**.** Starting from the definition $\mathrm{Var}(X) = \mathbb{E}[(X-\mathbb{E}[X])^2]$ and using only linearity of expectation, prove that $\mathrm{Var}(aX+b) = a^2\,\mathrm{Var}(X)$ for all constants a, b. Then prove the computational formula $\mathrm{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2$ as a special consequence of your expansion.

Exercise 9.12 (Proof: the variance of a sum)**.** Let $X_1, \ldots, X_n$ be random variables with finite variances. Prove that

$$\mathrm{Var}\Big(\sum_{i=1}^{n} X_i\Big) = \sum_{i=1}^{n} \mathrm{Var}(X_i) + \sum_{i \neq j} \mathrm{Cov}(X_i, X_j).$$

Deduce the special case for pairwise uncorrelated variables. Where exactly in your proof did you use (or not use) independence?

Exercise 9.13 (Conceptual: when does averaging *not* help?)**.** The $\sigma/\sqrt{n}$ law assumed the measurements were independent. Describe a realistic scenario — in measurement, polling, or finance — where averaging many observations fails to reduce the error nearly as fast as $1/\sqrt{n}$. Identify which term in the variance-of-a-sum formula is responsible, and state what its sign must be for the failure to occur.

Chapter 10

Continuous Random Variables and Densities

So far our random variables have been *discrete*: they take values in a list you can write down, like $\{0, 1, 2, \ldots\}$ or $\{1, 2, 3, 4, 5, 6\}$, and each value x carries a clump of probability $\mathbb{P}(X = x)$. The whole theory of expectation and variance was built out of sums over those clumps.

But the world is full of quantities that are not like that. How long until a radioactive atom decays? The exact height of a randomly chosen adult? The precise voltage on a noisy wire at the instant you measure it? The fraction of a website's visitors who click a button, in the limit of infinitely many visitors? None of these live on a tidy list. They live on a *continuum* – an interval of real numbers, with no gaps and no smallest step.

This chapter is about how probability copes with the continuum. The headline is strange enough to be worth staring at: for a genuinely continuous quantity, the probability of *any single exact value* is zero. The chance that a dart lands at exactly $x = 0.5000000\ldots$ on a ruler, infinitely many digits all correct, is 0. And yet some regions are clearly more likely than others. The fix is to stop measuring probability at points and start measuring it as *area*, a job done by the *probability density function*. Along the way we will see why $\mathbb{P}(X = x) = 0$ is a feature rather than a contradiction, how densities encode all the probability, how the cumulative distribution function ties the discrete and continuous worlds together, and how expectation and variance become *integrals* – the calculus analogue of the sums you already know.

10.1 From bars to curves: the central picture

Start where your intuition already is. Suppose you measure the waiting time X (in minutes) for a bus that is "equally likely to come any time in the next 10 minutes." You cannot list the outcomes – X could be 3, or 3.5, or 3.14159, or anything in $[0, 10]$. So let us approximate.

First, round X to the nearest whole minute. Now you have a discrete variable on $\{0, 1, \ldots, 10\}$, and you can draw a familiar bar chart of probabilities. Each bar has height roughly $1/10$. Now refine: round to the nearest half-minute. You get twice as many bars, each half as tall, because each old minute splits into two sub-intervals sharing its probability. Refine again to tenths of a minute: ten times as many bars, each a tenth as tall. The bars are getting shorter and shorter – the *height*

$\mathbb{P}(X \in \text{tiny bin})$ shrinks toward zero – but the overall *shape* is settling down to a flat plateau of constant height.

Intuition

The probability in a single bin shrinks as the bins get finer, so heights are the wrong thing to track. The right thing is height *per unit width* – the bar's area divided by its width. That ratio does *not* collapse to zero; it converges to a stable curve. That stable curve is the density. Density is to a continuous variable what the PMF was to a discrete one, but it lives on the vertical axis as *probability-per-length*, not probability.

Figure 10.1 shows the refinement in action for a bell-shaped example, where the limiting curve is more interesting than a flat line.

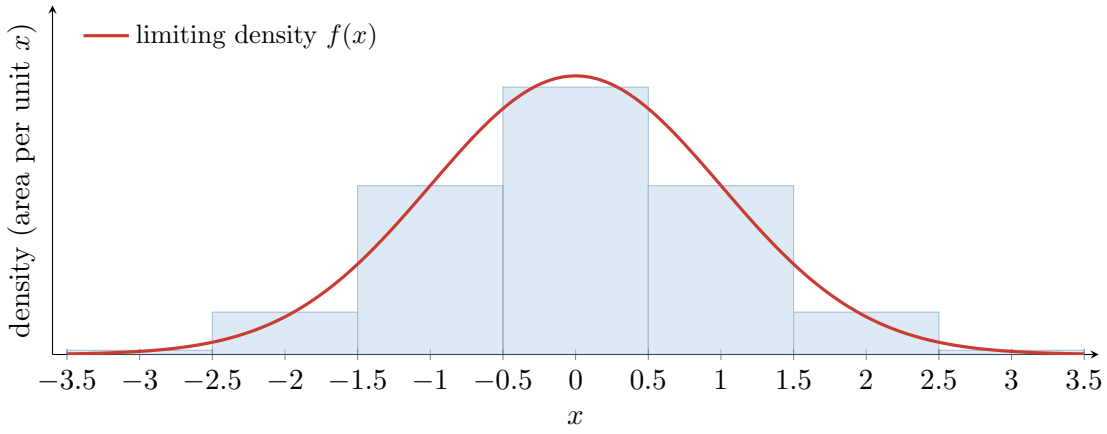


Figure 10.1: Coarse probability bars (width 1) with their heights equal to the probability in each bin. As bins shrink, the bars hug a smooth curve when you plot *probability per unit length*. That curve is the probability density.

The picture suggests the whole plan. A density f is a curve such that the *area* under it over an interval equals the probability of landing in that interval:

$$\mathbb{P}(a \le X \le b) \;=\; \text{area under } f \text{ from } a \text{ to } b \;=\; \int_a^b f(x)\,\mathrm{d}x.$$

Probabilities are areas – that one equation carries most of continuous probability. Figure 10.2 shows it directly: the probability of landing in $[a, b]$ is the shaded region under the curve.

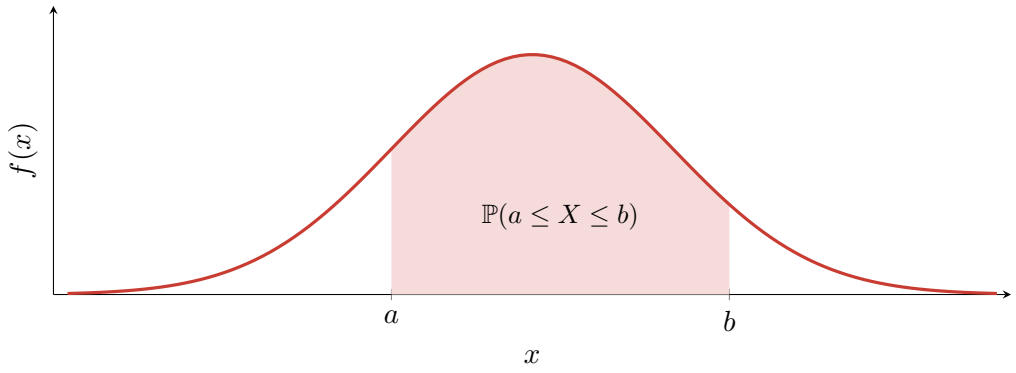


Figure 10.2: Probability as area. For a continuous variable the chance of landing in $[a, b]$ is the shaded area under the density f between a and b, i.e. $\int_a^b f(x)\,\mathrm{d}x$. Total area under the whole curve is 1.

10.2 Why $\mathbb{P}(X = x) = 0$, and why that is fine

Here is the fact that trips up everyone the first time. If X is continuous, then for *every* particular value x,

$$\mathbb{P}(X = x) = 0.$$

Not "very small." Exactly zero. Yet X does take *some* value every time you sample it. How can every outcome have probability zero, while one of them always happens?

What is really happening

The trap is importing intuition from the discrete world, where "probability zero" means "impossible." In the continuum the two notions come apart. Probability is *area*, and a single point is a line segment of width zero, so it bounds zero area no matter how tall the density is there. Think of probability as a *liquid* of total volume 1 poured over the real line. Ask how much liquid sits exactly over the single point $x = 0.5$. None – a point has no width to hold any. But the liquid over the interval $[0.49, 0.51]$ is a real, positive amount. In the continuum, *probability lives in intervals, not in points.* "Probability 0" no longer means "cannot happen"; it means "occupies no length."

We can prove $\mathbb{P}(X = x) = 0$ directly from the area picture, and the proof is illuminating. Pin down a value x and trap it in a shrinking interval. For any $\varepsilon > 0$,

$$\mathbb{P}(X = x) \;\le\; \mathbb{P}\big(x - \tfrac{\varepsilon}{2} \le X \le x + \tfrac{\varepsilon}{2}\big) \;=\; \int_{x-\varepsilon/2}^{x+\varepsilon/2} f(t)\,\mathrm{d}t.$$

If f is bounded near x, say $f \le M$ there, this integral is at most $M\varepsilon$, which we can drive to 0 by shrinking ε. A nonnegative number smaller than every positive number must be 0. So $\mathbb{P}(X = x) = 0$.

Common trap

Because individual points carry no probability, the difference between $<$ and $\le$ *vanishes* for

continuous variables:

$$\mathbb{P}(a \le X \le b) = \mathbb{P}(a < X < b) = \mathbb{P}(a \le X < b) = \mathbb{P}(a < X \le b).$$

Including or excluding an endpoint changes the answer by $\mathbb{P}(X = a)$ or $\mathbb{P}(X = b)$, which are both 0. This is a genuine convenience – but it is *false* for discrete variables, where endpoints can carry real mass. Always know which world you are in.

10.3 The probability density function, precisely

We can now state the definition the picture has been pointing at.

Definition 10.1 (Continuous random variable and density)**.** A random variable X is *continuous* if there is a function $f : \mathbb{R} \to [0, \infty)$, called its *probability density function* (pdf), such that for every interval $[a, b]$,

$$\mathbb{P}(a \le X \le b) = \int_a^b f(x)\,\mathrm{d}x.$$

A pdf must satisfy two requirements, the continuous analogues of the PMF's rules:

1. **Nonnegativity:** $f(x) \ge 0$ for all x (areas of probability cannot be negative);
2. **Normalization:** $\displaystyle\int_{-\infty}^{\infty} f(x)\,\mathrm{d}x = 1$ (all the probability is somewhere).

Common trap

A density is *not* a probability, and it is not capped at 1. The value $f(x)$ has units of probability *per unit* x. It can be huge – a density sharply concentrated on $[0, 0.01]$ might reach a height of 100 – as long as the total area stays 1. Never say "$f(x)$ is the probability that $X = x$." The probability of any exact value is 0. The right reading is local: $f(x)\,\mathrm{d}x$ is approximately the probability that X lands in a tiny window of width $\mathrm{d}x$ around x.

That last sentence is worth promoting to a working tool, because it is how practitioners actually think.

Focus

The infinitesimal reading of a density. For a small width $\mathrm{d}x$,

$$\mathbb{P}(x \le X \le x + \mathrm{d}x) \;\approx\; f(x)\,\mathrm{d}x.$$

Read $f(x)$ as "how densely probability is packed near x." Where f is tall, outcomes cluster; where f is short, they are rare; where $f = 0$, they never occur. Ratios are meaningful even though absolute values are not: if $f(2) = 3f(7)$, then a tiny window near 2 is three times as likely as an equally tiny window near 7.

Example 10.1 (Checking and using a density)**.** Suppose X has the candidate density

$$f(x) = \begin{cases} c\,x^2, & 0 \le x \le 1, \\ 0, & \text{otherwise.} \end{cases}$$

First find c so that f is a legitimate density, then compute $\mathbb{P}(X \le \frac{1}{2})$.

Normalize. We need the total area to be 1:

$$1 = \int_{-\infty}^{\infty} f(x)\,\mathrm{d}x = \int_0^1 c\,x^2\,\mathrm{d}x = c \cdot \frac{x^3}{3}\Big|_0^1 = \frac{c}{3} \quad \Longrightarrow \quad c = 3.$$

So $f(x) = 3x^2$ on $[0, 1]$. It is nonnegative there, so it is a valid density.

Compute the probability. Probability is area:

$$\mathbb{P}\big(X \le \tfrac{1}{2}\big) = \int_0^{1/2} 3x^2\,\mathrm{d}x = x^3\Big|_0^{1/2} = \big(\tfrac{1}{2}\big)^3 = \tfrac{1}{8}.$$

Notice the answer is much less than $1/2$, even though $1/2$ is the midpoint of the range. The density $3x^2$ grows toward $x = 1$, so probability piles up on the right.

10.4 The cumulative distribution function: the bridge

There is one object that works for *every* random variable – discrete, continuous, or a mix – and ties the whole subject together. It is the *cumulative distribution function* (CDF).

Definition 10.2 (CDF)**.** The CDF of any random variable X is

$$F(x) = \mathbb{P}(X \le x), \qquad x \in \mathbb{R}.$$

It accumulates all the probability lying at or below x.

For a continuous X, accumulating probability means accumulating area, so the CDF is the running integral of the density:

$$F(x) = \mathbb{P}(X \le x) = \int_{-\infty}^{x} f(t)\,\mathrm{d}t.$$

Every CDF, regardless of type, obeys three structural rules that come straight from the axioms of probability:

1. F is **nondecreasing** (more territory, more accumulated probability);
2. $\lim_{x \to -\infty} F(x) = 0$ and $\lim_{x \to +\infty} F(x) = 1$ (eventually we have collected none, then all, of the probability);
3. F is **right-continuous**.

The defining contrast between the two worlds is visual. A discrete CDF is a *staircase*: flat stretches punctuated by vertical jumps, and the height of the jump at x is exactly $\mathbb{P}(X = x)$. A continuous CDF has *no jumps* – it is a smooth ramp – precisely because every point mass is 0. Figure 10.3 puts the two side by side.

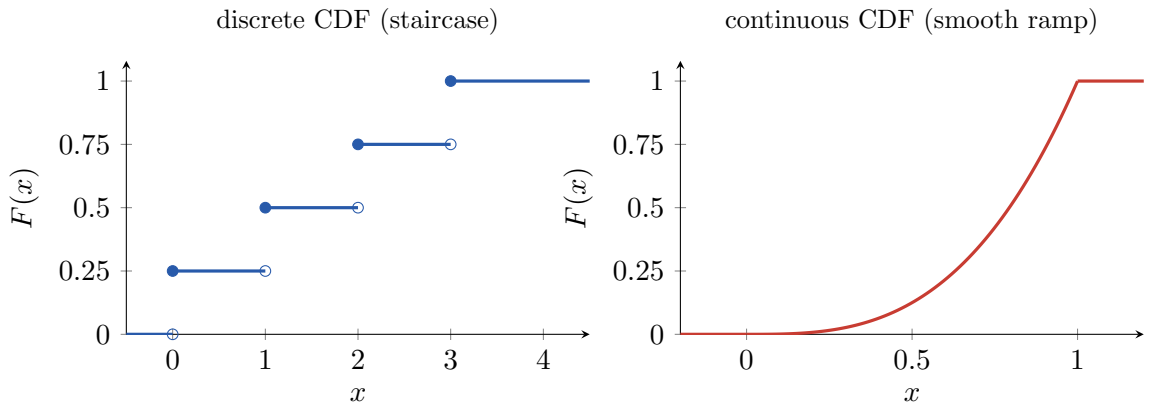


Figure 10.3: Left: a discrete CDF is a staircase, with jump heights equal to point masses. Right: a continuous CDF (here $F(x) = x^3$ for the density $f = 3x^2$) is a jumpless ramp – every point carries probability 0, so there is nothing to jump.

10.4.1 Getting the density back from the CDF

The CDF integrates the density. The Fundamental Theorem of Calculus runs that machine in reverse: *differentiate the CDF to recover the density.*

Theorem 10.1 (Density is the derivative of the CDF)**.** *If X has CDF F that is differentiable at x, then its density is*

$$f(x) = F'(x).$$

Why it works

This is not a coincidence; it is the Fundamental Theorem of Calculus wearing a probabilistic costume. Start from the definition of the derivative of F and read each piece probabilistically:

$$F'(x) = \lim_{h \to 0} \frac{F(x+h) - F(x)}{h} = \lim_{h \to 0} \frac{\mathbb{P}(x < X \le x + h)}{h}$$
$$= \lim_{h \to 0} \frac{\text{probability in a window of width } h}{h}.$$

That final ratio is exactly *probability per unit length* near x – which is the very thing we have been calling density. So $F'(x) = f(x)$ says: the slope of the accumulated-probability curve at x *is* the local density. Steep ramp = probability arriving fast = tall density. Flat ramp = no probability here = density 0. The CDF and pdf are two views of one object, related by the same integrate/differentiate duality you met for area and slope in calculus.

Connection

This is the cleanest place in all of probability to *see* the Fundamental Theorem of Calculus at work. $F = \int f$ and $f = F'$ are an integrate–differentiate pair: the CDF is the antiderivative (the "area-so-far" function) of the density, and the density is the slope of the CDF. Everything you know about antiderivatives – that they are determined up to a constant, that definite integrals are computed by evaluating an antiderivative at the endpoints – transfers directly.

In particular

$$\mathbb{P}(a \le X \le b) = \int_a^b f = F(b) - F(a),$$

which is just "evaluate the antiderivative at the endpoints." See the companion calculus volume for the FTC; here it is the engine that converts between the two faces of a distribution.

Example 10.2 (Round trip: pdf → CDF → pdf)**.** Take again $f(x) = 3x^2$ on $[0, 1]$. Build the CDF by integrating:

$$F(x) = \int_0^x 3t^2 \, \mathrm{d}t = x^3 \quad \text{for } 0 \le x \le 1,$$

with $F(x) = 0$ for $x < 0$ and $F(x) = 1$ for $x > 1$. Check the endpoints: $F(0) = 0$, $F(1) = 1$, good. Now differentiate to return home: for $0 < x < 1$, $F'(x) = 3x^2 = f(x)$. The loop closes. As a bonus, the CDF makes Example 10.1 instant: $\mathbb{P}(X \le \frac{1}{2}) = F(\frac{1}{2}) = (\frac{1}{2})^3 = \frac{1}{8}$, no integral required at the point of use.

Walking question

You are told X has CDF $F(x) = 1 - e^{-2x}$ for $x \ge 0$ (and 0 for $x < 0$). What is the density? What is $\mathbb{P}(X > 1)$?

Density: differentiate. $f(x) = F'(x) = 2e^{-2x}$ for $x \ge 0$. (This is the Exponential(2) density – the exponential distribution of a later chapter.) *Tail probability:* the CDF gives it directly, $\mathbb{P}(X > 1) = 1 - F(1) = 1 - (1 - e^{-2}) = e^{-2} \approx 0.135$. Carrying the CDF around is often faster than re-integrating the density.

10.5 Expectation and variance as integrals

For a discrete variable you computed the mean by summing value × probability: $\mathbb{E}[X] = \sum_x x \, \mathbb{P}(X = x)$. In the continuum the point masses $\mathbb{P}(X = x)$ are all 0, so that sum is hopeless. But recall the infinitesimal reading: the probability in a sliver of width $\mathrm{d}x$ near x is $f(x)\,\mathrm{d}x$. Replace the discrete mass $\mathbb{P}(X = x)$ by the sliver mass $f(x)\,\mathrm{d}x$, and replace the sum by an integral. The recipe "value times probability, then add up" becomes:

Definition 10.3 (Expectation and variance of a continuous variable)**.** For a continuous X with density f,

$$\mathbb{E}[X] = \int_{-\infty}^{\infty} x \, f(x) \, \mathrm{d}x, \qquad \mathbb{E}[g(X)] = \int_{-\infty}^{\infty} g(x) \, f(x) \, \mathrm{d}x,$$

the latter being the continuous Law of the Unconscious Statistician. The variance is the mean squared deviation, exactly as before:

$$\mathrm{Var}(X) = \mathbb{E}\big[(X - \mu)^2\big] = \int_{-\infty}^{\infty} (x - \mu)^2 f(x) \, \mathrm{d}x = \mathbb{E}[X^2] - \big(\mathbb{E}[X]\big)^2,$$

where $\mu = \mathbb{E}[X]$, and $\mathrm{SD}(X) = \sqrt{\mathrm{Var}(X)}$.

Connection

The transition $\sum \rightsquigarrow \int$ is not an analogy – it is the literal definition of the integral. The Riemann integral $\int x f(x)\,\mathrm{d}x$ *is* the limit of sums $\sum_i x_i\, f(x_i)\,\Delta x$ over finer and finer partitions. Each term $x_i \cdot \big(f(x_i)\Delta x\big)$ is "a value times the probability of a bin," precisely the discrete expectation of the binned-up variable. Letting the bins shrink turns the discrete mean into the integral. So continuous expectation is exactly discrete expectation in the limit of vanishing bin width. (The fully general definition, valid even when no density exists, uses the Lebesgue integral against the distribution; that is the subject of the measure-theory chapters.)

Intuition

The mean $\mathbb{E}[X]$ is the *balance point* of the density, the place where a cardboard cutout of the region under f would balance on a knife edge. The variance is its *rotational inertia* (moment of inertia) about that balance point: mass far from center contributes as the square of its distance, so Var measures spread, heavily weighting the tails. These mechanical pictures hold identically in the discrete and continuous cases; only the bookkeeping (sum vs. integral) changes.

Example 10.3 (Mean and variance of $f = 3x^2$)**.** With $f(x) = 3x^2$ on $[0, 1]$:

$$\mathbb{E}[X] = \int_0^1 x \cdot 3x^2\,\mathrm{d}x = \int_0^1 3x^3\,\mathrm{d}x = \frac{3x^4}{4}\Big|_0^1 = \frac{3}{4}.$$

This sits to the right of the midpoint $\frac{1}{2}$, confirming that the rising density tips the balance point toward 1. For the variance, get the second moment first:

$$\mathbb{E}[X^2] = \int_0^1 x^2 \cdot 3x^2\,\mathrm{d}x = \int_0^1 3x^4\,\mathrm{d}x = \frac{3x^5}{5}\Big|_0^1 = \frac{3}{5}.$$

Then

$$\mathrm{Var}(X) = \mathbb{E}[X^2] - \big(\mathbb{E}[X]\big)^2 = \frac{3}{5} - \left(\frac{3}{4}\right)^2 = \frac{3}{5} - \frac{9}{16} = \frac{48-45}{80} = \frac{3}{80} = 0.0375,$$

so $\mathrm{SD}(X) = \sqrt{3/80} \approx 0.194$.

Why this matters

Every theorem you proved about expectation in the discrete chapters – linearity $\mathbb{E}[aX+b] = a\,\mathbb{E}[X]+b$, the variance scaling $\mathrm{Var}(aX+b) = a^2\,\mathrm{Var}(X)$, the identity $\mathrm{Var}(X) = \mathbb{E}[X^2]-\mathbb{E}[X]^2$ – holds *verbatim* for continuous variables. Their proofs used only the linearity of the operation that combines values with weights, and integration is linear exactly as summation is. That is the payoff of a unified theory: you do not relearn expectation, you reuse it, swapping $\sum$ for $\int$.

10.6 The uniform distribution: the base case

Every subject has a simplest nontrivial example worth knowing cold. For continuous probability it is the *uniform* distribution, the formal version of "a number chosen completely at random from an interval, with no part favored."

Definition 10.4 (Uniform distribution)**.** $X \sim \text{Uniform}(a, b)$ has constant density on $[a, b]$:

$$f(x) = \begin{cases} \dfrac{1}{b-a}, & a \le x \le b, \\ 0, & \text{otherwise.} \end{cases}$$

The height $1/(b-a)$ is forced by normalization: a rectangle of width $b-a$ and area 1 must have that height.

Its CDF is a straight ramp – probability accumulates at a constant rate:

$$F(x) = \begin{cases} 0, & x < a, \\ \dfrac{x-a}{b-a}, & a \le x \le b, \\ 1, & x > b. \end{cases}$$

Figure 10.4 shows the flat density and its ramp CDF together.

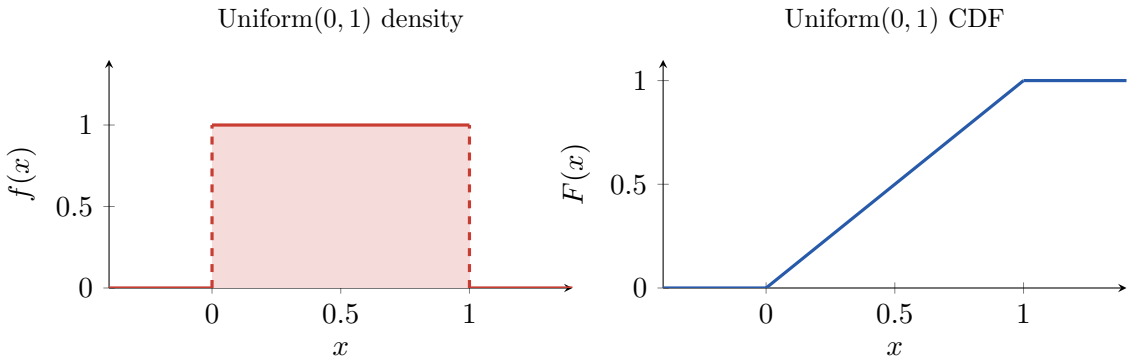


Figure 10.4: The uniform Uniform$(0, 1)$: a flat density (left) of constant height 1, and its CDF (right), a straight ramp from 0 to 1. Constant density means probability accumulates at a constant rate.

Example 10.4 (Mean and variance of Uniform(a, b))**.** By symmetry the balance point is the midpoint, $\mathbb{E}[X] = (a+b)/2$, but let us see the integral confirm it:

$$\mathbb{E}[X] = \int_a^b x \, \frac{1}{b-a} \, \mathrm{d}x = \frac{1}{b-a} \cdot \frac{x^2}{2}\Big|_a^b = \frac{b^2 - a^2}{2(b-a)} = \frac{a+b}{2}.$$

For the variance, first get the second moment,

$$\mathbb{E}[X^2] = \int_a^b x^2 \, \frac{1}{b-a} \, \mathrm{d}x = \frac{1}{b-a} \cdot \frac{x^3}{3}\Big|_a^b = \frac{b^3 - a^3}{3(b-a)} = \frac{a^2 + ab + b^2}{3},$$

using $b^3 - a^3 = (b-a)(a^2 + ab + b^2)$. Then subtract the squared mean:

$$\text{Var}(X) = \mathbb{E}[X^2] - \big(\mathbb{E}[X]\big)^2 = \frac{a^2 + ab + b^2}{3} - \frac{(a+b)^2}{4} = \frac{4(a^2 + ab + b^2) - 3(a+b)^2}{12} = \frac{(b-a)^2}{12}.$$

The factor $1/12$ is worth remembering – it is a recurring constant in noise modeling and in the analysis of rounding error.

Application

Quantization noise and the $1/12$ rule. When an analog signal is rounded to the nearest digital level spaced Δ apart, the rounding error is, to good approximation, Uniform$(-\Delta/2, \Delta/2)$. By Example 10.4 its mean is 0 and its variance is $\Delta^2/12$. That single fact is the foundation of the signal-processing rule that each extra bit of resolution (halving Δ) improves the signal-to-noise ratio by about 6 dB: halving Δ divides the noise variance by 4, i.e. the noise standard deviation by 2, and $20\log_{10} 2 \approx 6$. A toy model of rounding, made precise, drives real engineering decisions about how many bits a converter needs.

History

The idea that probability could be *geometric* – an area or a length rather than a count – predates the formal theory of densities. In 1733 Georges-Louis Leclerc, Comte de Buffon, posed his famous *needle problem*: drop a needle on a floor of parallel planks; what is the chance it crosses a crack? The answer involves π, and the underlying model is a pair of *uniform* continuous variables (the needle's position and angle). Buffon's needle is arguably the first continuous probability problem ever solved, and it already contained the modern picture: probability as area in a continuous sample space. It would take until the twentieth century, and Kolmogorov's measure-theoretic axioms (1933), for "area" to be made fully rigorous as a measure – but the geometric instinct was right from the start.

10.7 Quantiles: inverting the CDF

The CDF answers "given a value x, what fraction of probability lies below it?" Quantiles answer the reverse question, which is often the one you actually want: "given a fraction p, what value has that fraction of probability below it?" The 90th percentile of a test, the median income, the 0.975 critical value behind a confidence interval – all are quantiles.

Definition 10.5 (Quantile function)**.** For $0 < p < 1$, the p-th *quantile* of X is the value x_p with $F(x_p) = p$, i.e.

$$x_p = F^{-1}(p),$$

when F is continuous and strictly increasing so the inverse exists. The 0.5 quantile is the *median*; the quartiles are $p = 0.25, 0.5, 0.75$; percentiles use $p = k/100$.

Geometrically, the median splits the area under the density into two equal halves of 0.5 each. The quartiles cut it into four equal-area quarters. Quantiles are the "area coordinates" of the distribution. Figure 10.5 shows the median as the vertical line that balances area, not the line that balances width.

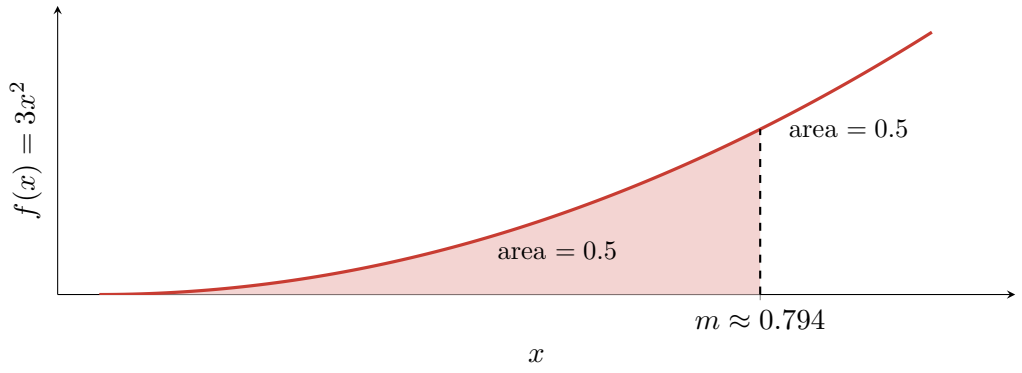


Figure 10.5: The median m of $f(x) = 3x^2$ on $[0, 1]$ splits the area into two equal halves of 0.5. Because the density piles up on the right, the equal-area line sits well to the right of the midpoint 0.5 of the interval.

Example 10.5 (Quantiles of $f = 3x^2$)**.** We found $F(x) = x^3$ on $[0, 1]$. Invert it: $F(x) = p$ means $x^3 = p$, so

$$x_p = p^{1/3}.$$

The median is $x_{0.5} = 0.5^{1/3} \approx 0.794$ – notably larger than the mean $\mathbb{E}[X] = 0.75$ from Example 10.3. *Mean below median* is the signature of a left-skewed shape: here the density rises toward 1 and has a long thin tail toward 0, which drags the mean down more than the median. The lower and upper quartiles are $0.25^{1/3} \approx 0.630$ and $0.75^{1/3} \approx 0.909$.

Common trap

The mean and the median are different summaries and can disagree a lot. The mean is the balance point and is sensitive to far-away tail mass; the median is the equal-area split and shrugs off the tails. For symmetric densities they coincide. For skewed ones – incomes, waiting times, file sizes – they can be far apart, which is exactly why "median household income" is reported instead of the mean: a few billionaires move the mean but barely budge the median.

The quantile function is not just a reporting convenience; it is a machine for *making* random variables, which is how simulators conjure samples from any distribution out of plain uniform randomness.

Theorem 10.2 (Inverse-transform / probability integral transform)**.** *Let F be a continuous, strictly increasing CDF with inverse F^{-1}.*

1. ***(Generation.)*** *If $U \sim \text{Uniform}(0, 1)$, then $X = F^{-1}(U)$ has CDF F.*
2. ***(Flattening.)*** *Conversely, if X has CDF F, then $F(X) \sim \text{Uniform}(0, 1)$.*

Why it works

Both halves are one short computation. For generation, compute the CDF of $X = F^{-1}(U)$ directly, using that F is increasing (so it preserves the inequality) and that U is uniform (so $\mathbb{P}(U \le u) = u$):

$$\mathbb{P}(X \le x) = \mathbb{P}\big(F^{-1}(U) \le x\big) = \mathbb{P}\left(U \le F(x)\right) = F(x).$$

That is exactly the CDF we wanted, so $X \sim F$. For flattening, run it backward:

$$\mathbb{P}\big(F(X) \le p\big) = \mathbb{P}\big(X \le F^{-1}(p)\big) = F\big(F^{-1}(p)\big) = p,$$

which is the CDF of Uniform$(0,1)$. The deep content is that *a CDF maps its own distribution to the flat uniform*: pushing X through its accumulated-probability function evens out all the lumps, because $F(x)$ measures exactly how much probability sits below x. Generation simply reverses that flattening.

Application

Simulating any distribution from coin-flip randomness. A computer's random-number generator hands you Uniform$(0,1)$ samples and nothing else. The inverse-transform theorem turns that single source into samples from *any* distribution whose quantile function you can write down. Want an exponential waiting time with rate λ? Its CDF is $F(x) = 1 - e^{-\lambda x}$, so $F^{-1}(u) = -\frac{1}{\lambda}\ln(1-u)$. Draw $U \sim$ Uniform$(0,1)$ and report $X = -\frac{1}{\lambda}\ln(1-U)$; the result is exactly Exponential(λ). This is the workhorse behind Monte Carlo simulation, and we will lean on it heavily in the chapter on Monte Carlo methods.

10.8 A cautionary paradox

Continuous probability rewards careful thinking and punishes sloppy phrases like "choose a random chord." Here is the classic warning.

Paradox

Bertrand's chord paradox (1889). Draw a circle and inscribe an equilateral triangle. Pick a chord "at random." What is the probability the chord is longer than a side of the triangle? Three perfectly reasonable people give three different answers:

- *Random endpoints.* Fix one endpoint, choose the other uniformly on the circle. The chord beats the side when the second point lands on the far arc: probability $\frac{1}{3}$.
- *Random radius.* Choose a radius, then a point uniformly along it for the chord's midpoint. The chord is long enough when the midpoint is within half a radius of the center: probability $\frac{1}{2}$.
- *Random midpoint.* Choose the chord's midpoint uniformly over the disk's *area*. The chord is long when the midpoint falls inside the inscribed circle of half the radius, whose area is one-quarter of the disk: probability $\frac{1}{4}$.

All three are correct – for the procedure they describe. *Resolution:* "at random" is meaningless on a continuum until you specify *which density* you are sampling from. The three answers correspond to three genuinely different distributions on the space of chords. In the discrete world "equally likely" has one obvious meaning (count and divide); in the continuous world there is no single canonical uniform measure on a curved or higher-dimensional space, and you must name your density before any probability is even defined.

What is really happening

The moral generalizes far beyond circles. A continuous probability question is *not well posed* until the density (equivalently, the CDF) is on the table. "Pick a random point," "a random line," "a random triangle" – each hides a choice of measure. Whenever you meet "at random" in a continuous setting, your first job is to ask: *random according to what density?* Naming f is not a technicality; it is what makes the probability exist at all.

10.9 Mixed and degenerate cases (a brief caveat)

Not every random variable is purely discrete or purely continuous. Consider an insurance payout that is 0 with probability 0.7 (no claim filed) and otherwise a continuous positive amount. Its CDF jumps at 0 (a genuine point mass of 0.7) and then rises smoothly – it is a *mixture*. Such variables have no ordinary density, because no integrable curve can produce a jump.

Reading note

This is exactly why the CDF, not the pdf, is the truly universal object: *every* random variable has a CDF, but only purely continuous ones have a density and only discrete ones have a PMF. The CDF handles staircases, ramps, and any blend of the two in one uniform language. When we build measure-theoretic probability later, the CDF (and the measure it induces) becomes the bedrock, with PMFs and pdfs as two special cases distinguished by whether the CDF is a sum of jumps or a smooth integral. For the rest of this book, "continuous" will mean "has a density" unless we say otherwise.

10.10 Chapter summary

Focus

The continuous toolkit.

- A continuous X has a **density** $f \geq 0$ with $\int_{-\infty}^{\infty} f = 1$, and probability is **area**: $\mathbb{P}(a \leq X \leq b) = \int_a^b f$.
- **Points carry no probability:** $\mathbb{P}(X = x) = 0$, so $<$ and $\leq$ are interchangeable; $f(x)$ is probability *per unit length*, not a probability, and may exceed 1.
- The **CDF** $F(x) = \mathbb{P}(X \leq x) = \int_{-\infty}^{x} f$ is universal; it is a jumpless ramp for continuous X. Recover the density by differentiating: $f = F'$ (the FTC in disguise).
- **Expectation and variance are integrals:** $\mathbb{E}[X] = \int xf$, $\mathbb{E}[g(X)] = \int g\, f$, $\mathrm{Var}(X) = \int (x-\mu)^2 f = \mathbb{E}[X^2] - \mu^2$. All the old identities survive, with $\sum \to \int$.
- The **uniform** Uniform(a, b) is the base case: flat density $1/(b-a)$, mean $(a+b)/2$, variance $(b-a)^2/12$.
- **Quantiles** invert the CDF, $x_p = F^{-1}(p)$; the median is the equal-area split. The inverse-transform $X = F^{-1}(U)$ turns uniform noise into any distribution.

- "At random" in the continuum is **undefined until you name the density** (Bertrand's paradox).

10.11 Exercises

Exercise 10.1 (Normalize and use a density)**.** Let $f(x) = c(1-x^2)$ for $-1 \le x \le 1$ and 0 otherwise. (a) Find c. (b) Compute $\mathbb{P}(X > 1/2)$. (c) By symmetry, what is $\mathbb{E}[X]$? Verify with the integral.

Exercise 10.2 (CDF to pdf and back)**.** A variable has CDF $F(x) = \sqrt{x}$ for $0 \le x \le 1$ (with 0 below, 1 above). (a) Find the density f. (b) Sketch why f blows up near 0 even though it is a valid density. (c) Compute $\mathbb{P}(0.25 \le X \le 0.81)$ two ways: via F, and via $\int f$.

Exercise 10.3 (Exponential basics)**.** Let X have density $f(x) = \lambda e^{-\lambda x}$ for $x \ge 0$. (a) Find the CDF. (b) Show $\mathbb{E}[X] = 1/\lambda$ (integrate by parts). (c) Show $\mathrm{Var}(X) = 1/\lambda^2$. (d) Find the median and explain why it is smaller than the mean.

Exercise 10.4 (Why $\mathbb{P}(X = x) = 0$ in your own words)**.** Using the area picture, explain to a skeptical friend how every outcome of a continuous variable can have probability 0 while one of them certainly occurs. Then give the ε-trapping argument from Section 10.2 in two or three lines.

Exercise 10.5 (Uniform computations)**.** Let $X \sim \mathrm{Uniform}(2, 8)$. (a) Write f and F. (b) Find $\mathbb{P}(3 \le X \le 5)$. (c) Find $\mathbb{E}[X]$ and $\mathrm{Var}(X)$ from the formulas, then re-derive Var by direct integration as a check. (d) Find the 0.9 quantile.

Exercise 10.6 (Mean versus median under skew)**.** For $f(x) = 4x^3$ on $[0, 1]$: (a) find the CDF; (b) find the mean and the median; (c) state which is larger and connect your answer to the direction of the skew, as in Example 10.5.

Exercise 10.7 (Simulating by inverse transform)**.** You can only draw $U \sim \mathrm{Uniform}(0, 1)$. (a) Give an explicit formula $X = g(U)$ that produces $X \sim \mathrm{Uniform}(2, 8)$. (b) Give a formula that produces a variable with CDF $F(x) = x^4$ on $[0, 1]$. (c) Verify (b) by computing the CDF of your $g(U)$ directly, using $\mathbb{P}(U \le u) = u$.

Exercise 10.8 (Density can exceed one)**.** Let $X \sim \mathrm{Uniform}(0, 0.01)$. (a) What is the height of its density? (b) Reconcile a density value of 100 with the rule that probabilities never exceed 1. (c) Compute $\mathbb{P}(X \le 0.001)$ and confirm it is a legitimate probability.

Exercise 10.9 (Proof: variance shortcut)**.** Prove for a continuous X with mean μ that $\int_{-\infty}^{\infty}(x-\mu)^2 f(x)\,\mathrm{d}x = \int_{-\infty}^{\infty} x^2 f(x)\,\mathrm{d}x - \mu^2$. State exactly which two properties of the integral you use, and where the fact $\int f = 1$ enters.

Exercise 10.10 (Conceptual: Bertrand revisited)**.** In your own words, explain why Bertrand's chord paradox is *not* a flaw in probability theory but a flaw in the problem statement. What single piece of information, if supplied, would make the question have exactly one correct answer?

Chapter 11

The Major Continuous Distributions

In the last chapter you learned the grammar of continuous random variables: the density f, the cumulative distribution function F, the rule that probabilities are *areas* under the density, and the slogan that for a continuous variable any single point has probability zero. Grammar without vocabulary says nothing, though, so this chapter supplies the vocabulary. We will meet the handful of continuous distributions that come up again and again – in physics, in queueing, in genetics, in finance, in machine learning – and we will do more than state their formulas. We will tell the *story* each one is built to model, watch the story force the density into a particular shape, and then read the mean and variance off that shape.

There are four families to learn here, and they form a small, tightly knit clan:

- the **Uniform** – "I know it is somewhere in this interval and I have no further opinion";
- the **Exponential** – "how long until the next event?", the continuous cousin of the geometric, and the carrier of the famous *memorylessness* property;
- the **Gamma** – "how long until the k-th event?", the sum of independent exponentials, and the workhorse of waiting-time models;
- the **Beta** – "what is a plausible value for an unknown *probability*?", a distribution that lives on $[0, 1]$ and that will become the secret hero of Bayesian inference.

We deliberately set the *Normal* aside for its own chapter; it deserves one. Everything here is connected: the exponential is a special gamma, the gamma is a sum of exponentials, the beta is built from two gammas, and the uniform is the simplest beta of all. These are not four formulas to memorize but four chapters of one continuous story.

11.1 The Uniform Distribution: Pure Ignorance

Intuition

Suppose I tell you a bus will arrive sometime in the next ten minutes, and that is genuinely *all* I know – no schedule, no tendency to be early or late. What is your belief about the arrival time? You should spread your belief evenly over the interval $[0, 10]$: no minute is more likely than any other. That even spreading, made precise, is the uniform distribution. It is what "I have no reason to prefer any point over any other in this range" looks like when written as a density.

Definition 11.1 (Uniform distribution)**.** A random variable X is *uniform on* $[a, b]$, written $X \sim \text{Uniform}(a, b)$, if it has the constant density

$$f(x) = \begin{cases} \dfrac{1}{b-a}, & a \le x \le b, \\ 0, & \text{otherwise.} \end{cases}$$

Why must the constant be $1/(b-a)$ and not something else? Because a density must integrate to 1. A flat rectangle of height h sitting over an interval of width $b-a$ has area $h(b-a)$, and setting that equal to 1 forces $h = 1/(b-a)$. The density does not care *where* the interval sits, only how *wide* it is: a wider interval means a shorter rectangle, because the same unit of total probability must be smeared over more room.

The CDF is just the running area of that rectangle. For $a \le x \le b$,

$$F(x) = \int_a^x \frac{1}{b-a}\,\mathrm{d}t = \frac{x-a}{b-a},$$

a straight line climbing from 0 at $x = a$ to 1 at $x = b$ (Figure 11.1). The fraction $\frac{x-a}{b-a}$ is exactly "how far along the interval are we?" – which is precisely the chance of landing to the left of x when all points are equally likely.

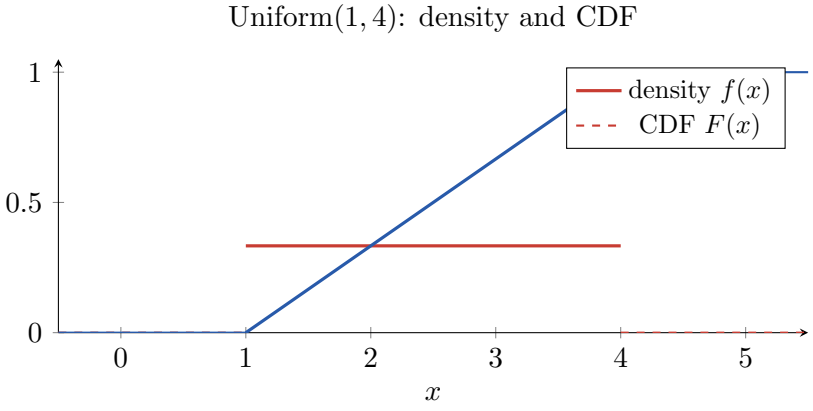


Figure 11.1: The uniform density is a flat rectangle; its CDF is the straight ramp that accumulates that rectangle's area. Height $1/(4-1) = 1/3$ makes the total area equal 1.

11.1.1 Mean and variance: the center and the spread of a rectangle

By symmetry the average value of a uniform on $[a, b]$ must be the midpoint $(a+b)/2$ – the rectangle balances on its center. Let us confirm it and then work out the variance.

$$\mathbb{E}[X] = \int_a^b x \cdot \frac{1}{b-a}\,\mathrm{d}x = \frac{1}{b-a}\cdot\frac{b^2-a^2}{2} = \frac{(b-a)(b+a)}{2(b-a)} = \frac{a+b}{2}.$$

For the variance, the cleanest route is $\mathrm{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}\,X)^2$. First,

$$\mathbb{E}[X^2] = \int_a^b x^2 \cdot \frac{1}{b-a}\,\mathrm{d}x = \frac{1}{b-a}\cdot\frac{b^3-a^3}{3} = \frac{a^2+ab+b^2}{3},$$

using $b^3 - a^3 = (b-a)(a^2+ab+b^2)$. Subtracting the square of the mean,

$$\mathrm{Var}(X) = \frac{a^2+ab+b^2}{3} - \frac{(a+b)^2}{4} = \frac{4(a^2+ab+b^2) - 3(a+b)^2}{12} = \frac{a^2-2ab+b^2}{12} = \frac{(b-a)^2}{12}.$$

Why it works

Why $(b-a)^2/12$ and not $(b-a)^2/4$ or some other fraction? The variance can depend only on the *width* $w = b-a$, never on where the interval sits, since shifting a distribution leaves its spread unchanged. Dimensionally it must be w^2 times a pure number. That pure number is $1/12$. You can feel why it is small and not, say, $1/4$: most of the probability is *not* out at the extreme ends, so the typical squared distance from the center is well under the maximum possible $(w/2)^2 = w^2/4$. The average of those squared distances works out to a third of that maximum, $w^2/12$. The standard deviation is $w/\sqrt{12} \approx 0.289\,w$ – about 29% of the interval width, a number worth remembering.

Example 11.1 (Rounding error)**.** When you round a measurement to the nearest unit, the error – true value minus rounded value – is well modeled as Uniform$(-\frac{1}{2}, \frac{1}{2})$: you have no reason to think the discarded part favors any sub-part of the rounding window. Its mean is 0 (rounding is unbiased on average) and its variance is $\frac{1^2}{12} = \frac{1}{12}$, so its standard deviation is $1/\sqrt{12} \approx 0.289$ units. This single fact – the "*quantization noise has variance* $1/12$" rule – underlies error budgets in digital audio, sensors, and analog-to-digital converters.

Why this matters

The uniform is the distribution that admits the most ignorance: among all distributions supported on $[a, b]$, it is the one with the largest entropy, the one that assumes the least. We will make "entropy" precise in the information-theory chapter, but the intuition is already here. To say $X \sim \mathrm{Uniform}(a, b)$ is to refuse to smuggle in any preference you do not actually have.

Connection

The uniform is the gateway to *simulating* every other distribution. If $U \sim \text{Uniform}(0,1)$ and F is any continuous, strictly increasing CDF, then $X = F^{-1}(U)$ has CDF exactly F. This is the *inverse-transform* or *probability integral transform* method, and it is why a single stream of Uniform$(0,1)$ numbers from your computer can be reshaped into exponentials, normals, anything. We prove it carefully in the chapter on transformations; for now, hold the picture: the flat U is the raw clay every other shape is sculpted from. We will use it in a moment to build an exponential.

11.2 The Exponential Distribution: How Long Until It Happens?

Intuition

A Geiger counter sits next to a lump of uranium. Clicks come at random. You start a stopwatch. *How long until the next click?* You cannot say exactly – it might be an instant, it might be a while – but the longer you wait the more it feels "overdue." Or does it? The exponential distribution is the mathematics of this waiting time, and it hides a shock: a decaying atom does not age. The wait, no matter how long you have already waited, looks brand new. That is memorylessness, and it is the most surprising thing in this chapter.

11.2.1 From the geometric to the exponential

Recall the geometric distribution from the discrete chapter: flip a biased coin with success probability p on each trial; the number of trials until the first success is geometric. The exponential is what the geometric *becomes* when trials happen continuously in time rather than one per tick.

Here is the limiting argument; doing it slowly shows where the formula comes from. Chop time into tiny slices of width Δt. In each slice, suppose an event happens with probability $\lambda\,\Delta t$, where $\lambda > 0$ is a *rate* (events per unit time), and slices are independent. The chance of *surviving* (no event) for a whole stretch of length t, i.e. through $n = t/\Delta t$ slices, is

$$\mathbb{P}(T > t) = (1 - \lambda\,\Delta t)^n = \left(1 - \frac{\lambda t}{n}\right)^n \xrightarrow[n\to\infty]{} e^{-\lambda t},$$

the famous limit $(1-x/n)^n \to e^{-x}$. So in the continuous limit the survival function is $\mathbb{P}(T > t) = e^{-\lambda t}$. Everything else follows. The CDF is $F(t) = 1 - e^{-\lambda t}$, and differentiating gives the density.

Definition 11.2 (Exponential distribution)**.** A random variable $T \geq 0$ is *exponential with rate* $\lambda > 0$, written $T \sim \text{Exponential}(\lambda)$, if it has density

$$f(t) = \lambda e^{-\lambda t}, \qquad t \geq 0,$$

equivalently survival function $\mathbb{P}(T > t) = e^{-\lambda t}$ and CDF $F(t) = 1 - e^{-\lambda t}$ for $t \geq 0$.

The density starts at its highest point λ at $t = 0$ and decays exponentially: short waits are the most likely, and very long waits are rare but never impossible (Figure 11.2). A bigger rate λ means events

come faster, so the density is taller near zero and falls off more steeply – you do not wait long when things happen often.

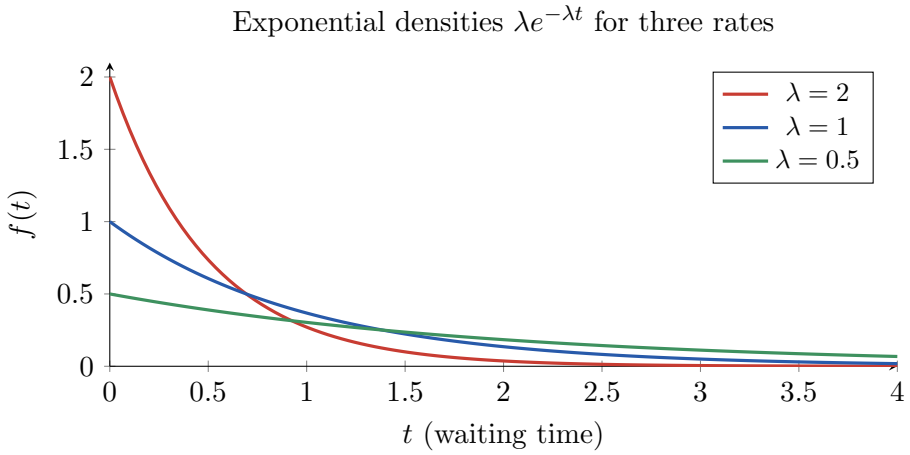


Figure 11.2: Every exponential density is a pure decay starting at height λ. Faster rates (larger λ) crowd the probability toward short waits.

11.2.2 Mean and variance: the rate sets the clock

The mean wait should be the reciprocal of the rate: if events come at 3 per minute, you wait about 1/3 of a minute on average. Here is the derivation. With integration by parts (or by recognizing a gamma integral, which we do below),

$$\mathbb{E}[T] = \int_0^\infty t\,\lambda e^{-\lambda t}\,\mathrm{d}t = \frac{1}{\lambda}.$$

A slicker derivation uses the survival-function formula for nonnegative variables, $\mathbb{E}[T] = \int_0^\infty \mathbb{P}(T > t)\,\mathrm{d}t$:

$$\mathbb{E}[T] = \int_0^\infty e^{-\lambda t}\,\mathrm{d}t = \frac{1}{\lambda}.$$

For the second moment, $\mathbb{E}[T^2] = \int_0^\infty t^2 \lambda e^{-\lambda t}\,\mathrm{d}t = 2/\lambda^2$, so

$$\mathrm{Var}(T) = \frac{2}{\lambda^2} - \frac{1}{\lambda^2} = \frac{1}{\lambda^2}, \qquad \mathrm{SD}(T) = \frac{1}{\lambda}.$$

Notice something elegant: the standard deviation *equals* the mean. The exponential is exactly as spread out as it is large. This is a fingerprint – whenever you see data whose standard deviation roughly equals its mean and which is supported on the positive reals, suspect an exponential.

Walking question

The mean of Exponential(λ) is $1/\lambda$, but the *median* – the point splitting probability in half – is smaller. Solve $1 - e^{-\lambda t} = 1/2$ to get $t = (\ln 2)/\lambda \approx 0.693/\lambda$. So the median is about 69% of the mean. Why is the median below the mean? Because the long right tail drags the average up while leaving the halfway point behind. This right-skew is typical of all positive waiting-time distributions.

11.2.3 Memorylessness: the property that defines the exponential

Common trap

Here is the trap nearly everyone falls into. You have been waiting 10 minutes for the bus, and the wait is exponential. Surely the bus is now "due" – your remaining wait should be shorter than it was at the start? **No.** For an exponential wait, the time you have already spent waiting tells you *nothing* about the time remaining. The distribution of your remaining wait is identical to the distribution of a fresh wait. The clock does not remember.

Let us prove it. The *memoryless property* is the statement

$$\mathbb{P}(T > s + t \mid T > s) = \mathbb{P}(T > t) \qquad \text{for all } s, t \geq 0.$$

In words: given you have already survived past time s, the chance of surviving an *additional* t is the same as the chance a brand-new wait survives t.

Why it works

The proof is one line, and it is the cleanest demonstration of why exponentials are special. By the definition of conditional probability, and because $\{T > s + t\} \subseteq \{T > s\}$,

$$\mathbb{P}(T > s + t \mid T > s) = \frac{\mathbb{P}(T > s + t)}{\mathbb{P}(T > s)} = \frac{e^{-\lambda(s+t)}}{e^{-\lambda s}} = e^{-\lambda t} = \mathbb{P}(T > t).$$

The whole thing turns on the algebraic miracle $e^{-\lambda(s+t)} = e^{-\lambda s}\, e^{-\lambda t}$: the survival function *factors*, so the "already waited" part cancels exactly. Memorylessness is nothing but the multiplicative law of exponents wearing a probabilistic costume.

What is really happening

What is really happening is that the exponential is the *only* continuous distribution with this property, and that is no coincidence – it is forced. If a positive variable satisfies $\mathbb{P}(T > s+t) = \mathbb{P}(T > s)\,\mathbb{P}(T > t)$ for all $s, t \geq 0$, then writing $g(t) = \mathbb{P}(T > t)$ we have $g(s + t) = g(s)g(t)$. The only measurable solutions of this functional equation (Cauchy's exponential equation) that also decrease from 1 are $g(t) = e^{-\lambda t}$. So "the wait does not age" and "the wait is exponential" are two names for the same fact. Memorylessness is not a quirky bonus feature of the exponential; it is its defining DNA.

Paradox

[The waiting-time paradox / the bus that is always late] Memorylessness leads to a genuinely disorienting consequence. Buses arrive according to a process whose gaps are exponential with mean 10 minutes. You show up at the stop at a random moment. How long do you expect to wait for the next bus? Naively, "I arrive in the middle of a gap, so about 5 minutes." But memorylessness says: from the instant you arrive, the wait to the next bus is a *fresh* exponential with mean 10 minutes – you expect to wait 10, not 5! And by the same logic the time *since* the last bus also averages 10. So the gap you landed in averages 20 minutes, twice the average gap of 10.

The resolution is *length-biased sampling*: a randomly chosen *moment* is more likely to fall

inside a long gap than a short one, simply because long gaps cover more of the timeline. You are not sampling gaps uniformly; you are sampling them in proportion to their length (Figure 11.3). The gaps you tend to land in are the big ones, which is exactly why your wait feels worse than the "average gap" suggests. This is not a flaw in the model; it is a real effect, and it is why the line at the bus stop, or the queue you join, so often feels longer than the advertised average.

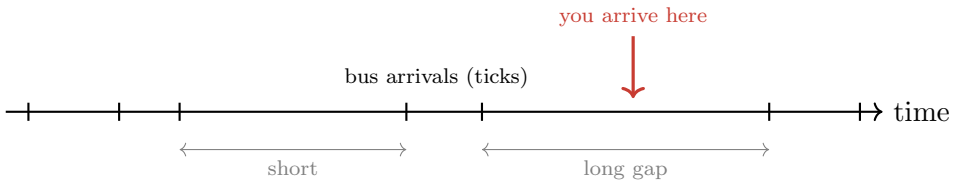


Figure 11.3: Length-biased sampling. Bus arrivals (vertical ticks) carve the timeline into gaps of unequal length. A uniformly random arrival instant is more likely to land inside a wide gap than a narrow one, simply because wide gaps occupy more of the line – so the gap you experience is longer than the average gap, and your expected wait is larger than half the mean gap.

History

The exponential's memorylessness mirrors a physical fact discovered by Ernest Rutherford and Hans Geiger around 1908: radioactive decay has no memory. An atom that has not yet decayed is, statistically, indistinguishable from a freshly minted one – it has no internal "odometer." This is why we describe radioactivity by a *half-life*: the time for half a sample to decay is the median of an exponential, $(\ln 2)/\lambda$, and it is constant no matter how old the sample is. The same constant-hazard idea reappears wherever failure is "random" rather than driven by wear – which is why electronic components in their useful-life phase, before wear-out sets in, are modeled exponentially.

Connection

[The hazard rate viewpoint] Engineers think in terms of the *hazard rate*: given survival to time t, the instantaneous rate of failing right now, $h(t) = f(t)/(1 - F(t))$. For the exponential, $h(t) = \lambda e^{-\lambda t}/e^{-\lambda t} = \lambda$, a *constant*. Constant hazard *is* memorylessness: the object is no more likely to fail in the next instant after a year than on day one. Distributions with *increasing* hazard (the Weibull with shape > 1) model wear-out – things that age. Decreasing hazard models "infant mortality." The exponential sits exactly on the boundary, the unique no-aging case.

Example 11.2 (Exponential by inverse transform)**.** Now we cash in the connection from the uniform section. To simulate $T \sim \text{Exponential}(\lambda)$, set $U \sim \text{Uniform}(0, 1)$ and solve $F(T) = U$, i.e. $1 - e^{-\lambda T} = U$. Then $e^{-\lambda T} = 1 - U$ and

$$T = -\frac{1}{\lambda} \ln(1 - U).$$

Since $1 - U$ is also $\text{Uniform}(0, 1)$, programmers often write $T = -\frac{1}{\lambda} \ln U$. One uniform draw, one logarithm, and you have a perfect exponential waiting time. This little formula runs inside every discrete-event simulation – network packet arrivals, customers at a teller, particles in a detector.

Example 11.3 (Minimum of exponentials: the race)**.** Two independent components have lifetimes $T_1 \sim \text{Exponential}(\lambda_1)$ and $T_2 \sim \text{Exponential}(\lambda_2)$. The system fails when the *first* one does, so its lifetime is $M = \min(T_1, T_2)$. When does M exceed t? Only if *both* survive past t:

$$\mathbb{P}(M > t) = \mathbb{P}(T_1 > t)\ \mathbb{P}(T_2 > t) = e^{-\lambda_1 t} e^{-\lambda_2 t} = e^{-(\lambda_1+\lambda_2)t}.$$

That is the survival function of an exponential with rate $\lambda_1 + \lambda_2$. So the minimum of independent exponentials is again exponential, and *rates add*. Two independent Geiger counters pointed at the same source behave like one counter with the combined click rate. This "competing exponentials" fact is the engine of the Poisson process and of continuous-time Markov chains in later chapters.

11.3 The Gamma Distribution: Waiting for the k-th Event

Intuition

The exponential answers "how long until the *next* click?" Now ask: how long until the *tenth* click? You must wait for click one, then – starting fresh, by memorylessness – for click two, and so on, ten times. The total wait is a *sum* of ten independent exponentials. That sum has its own shape: it can no longer pile up at zero (ten waits can hardly all be tiny), so the density rises from zero, humps in the middle, and trails off to the right. That humped, right-skewed shape is the gamma distribution.

11.3.1 The gamma function, briefly

The gamma distribution borrows its normalizing constant from the *gamma function*, the continuous extension of the factorial:

$$\Gamma(\alpha) = \int_0^\infty x^{\alpha-1} e^{-x}\, \mathrm{d}x, \qquad \alpha > 0.$$

Two facts are all we need. First, integration by parts gives the recursion $\Gamma(\alpha+1) = \alpha\,\Gamma(\alpha)$, the same rule that defines the factorial. Second, $\Gamma(1) = \int_0^\infty e^{-x}\mathrm{d}x = 1$. Together these force $\Gamma(n) = (n-1)!$ for positive integers n. So Γ interpolates the factorial to all positive reals – it is the factorial with the gaps filled in.

Definition 11.3 (Gamma distribution)**.** A random variable $X \geq 0$ has the *gamma distribution* with shape $\alpha > 0$ and rate $\lambda > 0$, written $X \sim \text{Gamma}(\alpha, \lambda)$, if

$$f(x) = \frac{\lambda^\alpha}{\Gamma(\alpha)}\, x^{\alpha-1} e^{-\lambda x}, \qquad x \geq 0.$$

The factor $\lambda^\alpha/\Gamma(\alpha)$ is exactly what makes the area equal 1; you can check it by substituting $u = \lambda x$ in the integral and recognizing the definition of $\Gamma(\alpha)$.

Two pieces multiply to make the shape: the rising power $x^{\alpha-1}$ pulls the density up away from zero, and the decaying $e^{-\lambda x}$ pulls it back down. Their tug-of-war produces a hump. When $\alpha = 1$ the rising factor is $x^0 = 1$ and the gamma collapses to a pure exponential – the exponential is just $\text{Gamma}(1, \lambda)$. As α grows the hump moves right and the shape grows more symmetric (Figure 11.4); we will see in the central-limit chapter that this is the gamma slowly turning into a bell.

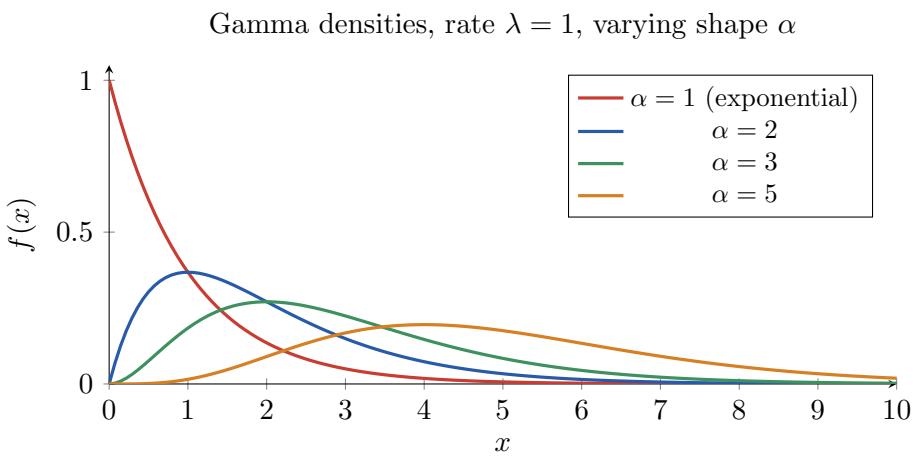


Figure 11.4: Increasing the shape α pushes the hump rightward and makes it more symmetric. At $\alpha = 1$ the gamma is exactly an exponential; for integer $\alpha = k$ it is the wait for the k-th event.

Reading note

Beware: there are *two* common parametrizations of the gamma. We use the *rate* λ (so the exponential rate carries over cleanly). Others use the *scale* $\theta = 1/\lambda$, writing the density with $e^{-x/\theta}$ and dividing by θ^{α}. Software defaults differ – always check whether a library wants rate or scale, or you will be off by powers of λ.

11.3.2 The gamma as a sum of exponentials

The promised story – "wait for the k-th event" – is a theorem.

Theorem 11.1 (Sum of i.i.d. exponentials is gamma)**.** *If* $T_1, \dots, T_k$ *are independent, each* Exponential(λ)*, then* $S_k = T_1 + \cdots + T_k \sim \text{Gamma}(k, \lambda)$.

Why it works

Here is the idea without the full convolution grind (we develop convolutions properly in the sums chapter). The wait S_k for the k-th event in a stream arriving at rate λ exceeds t exactly when *fewer than* k events have arrived by time t. The number of events by time t is Poisson with mean λt (we build this carefully in the Poisson-process chapter). So

$$\mathbb{P}(S_k > t) = \mathbb{P}(\text{at most } k-1 \text{ events by } t) = \sum_{j=0}^{k-1} \frac{(\lambda t)^j}{j!} e^{-\lambda t}.$$

Differentiate the CDF $1 - \mathbb{P}(S_k > t)$ with respect to t: the sum telescopes beautifully (each term's derivative cancels against the next), leaving only

$$f_{S_k}(t) = \frac{\lambda^k}{(k-1)!} t^{k-1} e^{-\lambda t} = \frac{\lambda^k}{\Gamma(k)} t^{k-1} e^{-\lambda t},$$

which is exactly Gamma(k, λ). The shape parameter $\alpha = k$ *is* the number of exponentials you summed. The gamma is the exponential's story told k times over.

This instantly gives the mean and variance for free, by linearity and independence:

$$\mathbb{E}[S_k] = k \cdot \frac{1}{\lambda} = \frac{k}{\lambda}, \qquad \mathrm{Var}(S_k) = k \cdot \frac{1}{\lambda^2} = \frac{k}{\lambda^2}.$$

And the formulas extend, by the same integral computation as before, to *any* real shape α:

$$\boxed{\mathbb{E}[X] = \frac{\alpha}{\lambda}, \qquad \mathrm{Var}(X) = \frac{\alpha}{\lambda^2}} \qquad \text{for } X \sim \mathrm{Gamma}(\alpha, \lambda).$$

Both scale the obvious way: α exponentials stacked give α times the mean and α times the variance, while a faster rate λ shrinks both.

Example 11.4 (How long until the 5th customer?)**.** Customers walk into a shop at rate $\lambda = 2$ per hour, with exponential gaps. How long until the fifth customer, and how variable is that wait? The wait is Gamma$(5, 2)$, so $\mathbb{E} = 5/2 = 2.5$ hours and Var $= 5/4 = 1.25$, giving SD $= \sqrt{1.25} \approx 1.12$ hours. Compare a *single* exponential wait, whose SD (0.5 h) equals its mean (0.5 h): the five-event wait has SD that is *less than half* its mean. Summing independent waits averages out their fluctuations – the relative spread shrinks like $1/\sqrt{k}$. This shrinking is the first whisper of the law of large numbers and the central limit theorem.

Connection

[The chi-squared distribution is a gamma] You will meet the *chi-squared* distribution constantly in statistics – it governs sample variances and goodness-of-fit tests. It is secretly a gamma. A chi-squared with k degrees of freedom is exactly Gamma$(\alpha = k/2, \lambda = 1/2)$. In particular the square of a single standard normal is Gamma$(1/2, 1/2)$, and summing k of them gives Gamma$(k/2, 1/2)$ because shapes add (next paragraph). When the statistics chapters invoke χ^2, remember: it is just a gamma in disguise, and everything you know here applies.

Proposition 11.1 (Gammas with a common rate add)**.** *If $X \sim \mathrm{Gamma}(\alpha, \lambda)$ and $Y \sim \mathrm{Gamma}(\beta, \lambda)$ are independent and share the same rate λ, then $X + Y \sim \mathrm{Gamma}(\alpha + \beta, \lambda)$.*

This is immediate from the sum-of-exponentials picture when α, β are integers (concatenate the two stacks of waits), and holds for all positive shapes by the convolution computation. *Shapes add; the rate stays fixed.* The exponential's "rates add for minima" (Example 11.3) and the gamma's "shapes add for sums" are the two fundamental closure properties of this family.

11.4 The Beta Distribution: A Distribution Over Probabilities

Intuition

Every distribution so far has lived on an unbounded stretch of the line. The beta lives on the unit interval $[0, 1]$ – and that domain is the whole point. What naturally lives in $[0, 1]$? A *probability*, or a *proportion*, or a *fraction*. If you do not know the bias p of a coin, or the click-through rate of an ad, or the fraction of voters favoring a candidate, your uncertainty about that unknown number is itself a distribution on $[0, 1]$. The beta is the natural language for "what do I believe the probability is?" – a probability distribution *over* probabilities.

Definition 11.4 (Beta distribution)**.** A random variable X on $[0,1]$ has the *beta distribution* with parameters $\alpha > 0$ and $\beta > 0$, written $X \sim \text{Beta}(\alpha, \beta)$, if

$$f(x) = \frac{1}{B(\alpha,\beta)}\, x^{\alpha-1}(1-x)^{\beta-1}, \qquad 0 \le x \le 1,$$

where the *beta function* $B(\alpha,\beta) = \int_0^1 x^{\alpha-1}(1-x)^{\beta-1}\mathrm{d}x$ is the constant that makes the area 1.

The shape is a tug-of-war between two factors. The term $x^{\alpha-1}$ pulls mass toward 1 (it grows as $x \to 1$ when $\alpha > 1$), while $(1-x)^{\beta-1}$ pulls mass toward 0. Raising α favors large values; raising β favors small values. When $\alpha = \beta$ the two pulls balance and the density is symmetric about $1/2$. And when $\alpha = \beta = 1$ both exponents vanish, the density is the flat constant 1, and the beta is exactly the uniform on $[0,1]$: the uniform is the simplest beta, the one that says "every probability is equally plausible" (Figure 11.5).

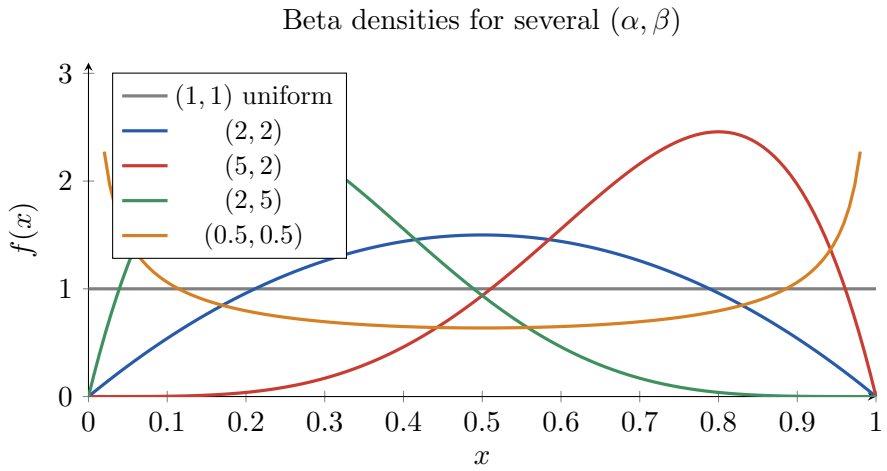


Figure 11.5: The beta family covers an astonishing range of shapes on $[0,1]$: flat $(1,1)$, hump-in-the-middle $(2,2)$, skewed toward 1 $(5,2)$ or toward 0 $(2,5)$, and even U-shaped $(0.5,0.5)$ piling mass at both ends. This flexibility is why it is the go-to model for an unknown proportion.

11.4.1 The beta function and the beta–gamma bridge

The beta function is not a new mystery; it is two gammas in a trench coat:

$$B(\alpha,\beta) = \frac{\Gamma(\alpha)\,\Gamma(\beta)}{\Gamma(\alpha+\beta)}.$$

Why it works

This identity is where the four distributions of this chapter snap together. Take independent $X \sim \text{Gamma}(\alpha, 1)$ and $Y \sim \text{Gamma}(\beta, 1)$. Consider the *fraction* $R = X/(X+Y)$ – the share of the total contributed by X. It is a number in $[0,1]$. A change of variables (done in full in the transformations chapter) shows two clean facts: $R \sim \text{Beta}(\alpha,\beta)$, and R is *independent* of the total $X+Y \sim \text{Gamma}(\alpha+\beta, 1)$. Matching the normalizing constants in that computation is exactly the statement $B(\alpha,\beta) = \Gamma(\alpha)\Gamma(\beta)/\Gamma(\alpha+\beta)$. So the beta *is* the distribution of "what

fraction of two competing gamma quantities came from the first one." If X is the time you wait for α red events and Y for β blue events, the beta tells you what fraction of the total waiting was red.

11.4.2 Mean, variance, and how to read the parameters

With the beta–gamma bridge, the mean is almost guessable: if X contributes a shape-α portion and Y a shape-β portion, the expected fraction from X should be $\alpha/(\alpha+\beta)$. It is:

$$\mathbb{E}[X] = \frac{\alpha}{\alpha+\beta}, \qquad \mathrm{Var}(X) = \frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}.$$

The mean is just the ratio of α to the total $\alpha+\beta$. The variance deserves a closer look: write $s = \alpha+\beta$ for the "total strength." Then $\mathrm{Var}(X) = \mathbb{E}[X](1-\mathbb{E}[X])/(s+1)$. The numerator $\mathbb{E}[X](1-\mathbb{E}[X])$ is the spread you would get from a single coin flip with that mean; the denominator $s+1$ *shrinks* it. Larger $\alpha+\beta$ means a tighter, more confident distribution clustered near its mean. This is the key to reading betas: the *ratio* $\alpha:\beta$ sets *where* the belief sits, and the *sum* $\alpha+\beta$ sets *how concentrated* it is.

Example 11.5 (Reading a beta)**.** Beta$(2,2)$ has mean $2/4 = 1/2$ and variance $\frac{4}{16\cdot 5} = 1/20 = 0.05$ (SD ≈ 0.224): a gentle hump centered at $1/2$, mild confidence. Beta$(20,20)$ has the same mean $1/2$ but variance $\frac{400}{1600\cdot 41} \approx 0.0061$ (SD ≈ 0.078): same center, much sharper – ten times the strength, far more confident. And Beta$(8,2)$ has mean $8/10 = 0.8$: belief concentrated up near 80%. The two knobs do exactly what their names promise.

11.4.3 Conjugacy: a preview of Bayes

Here is the property that makes the beta a superstar, and the reason it earns a preview now even though Bayesian inference gets its own chapter later.

Application

[The beta is the conjugate prior for a probability] You want to learn the unknown success probability p of a coin (or an ad, or a drug). Before seeing data, encode your belief about p as a beta: $p \sim \mathrm{Beta}(\alpha,\beta)$. Now flip the coin n times and observe h heads and $t = n-h$ tails. Bayes' rule (next part of the book) multiplies your prior density by the likelihood of the data, which for h heads and t tails is proportional to $p^h(1-p)^t$. So the updated (posterior) density is proportional to

$$\underbrace{p^{\alpha-1}(1-p)^{\beta-1}}_{\text{prior}} \cdot \underbrace{p^h(1-p)^t}_{\text{likelihood}} = p^{(\alpha+h)-1}(1-p)^{(\beta+t)-1}.$$

But that is just another beta! The posterior is

$$p \mid \text{data} \;\sim\; \mathrm{Beta}(\alpha+h,\ \beta+t).$$

The beta goes in, a beta comes out: this is *conjugacy*. And the update rule could not be more intuitive – **add your heads to α, add your tails to β**. You can even read the prior as "pretend experience": Beta(α,β) behaves like having already seen $\alpha-1$ heads and $\beta-1$ tails

before the real data arrived. Starting from the uniform Beta$(1, 1)$ ("no prior opinion") and seeing 7 heads in 10 flips lands you at Beta$(8, 4)$, with posterior mean $8/12 \approx 0.67$ – pulled a touch toward $1/2$ from the raw 0.7, exactly the gentle regularization a sensible prior should provide. Figure 11.6 shows a broad prior visibly sharpening into a tighter posterior once data arrive.

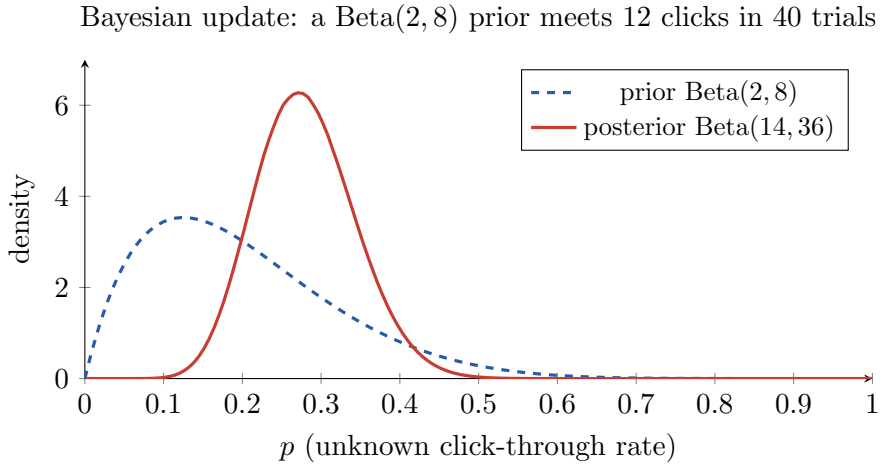


Figure 11.6: Data sharpens belief. The broad prior Beta$(2, 8)$ (mean 0.2) is updated by 12 clicks and 28 non-clicks into the much tighter posterior Beta$(14, 36)$ (mean 0.28). Conjugacy means the update is just "add 12 to α and 28 to β"; the posterior is pulled from the prior toward the data proportion $12/40 = 0.30$.

What is really happening

Why does conjugacy work? Look at the two ingredients. The beta density is $p^{\square}(1-p)^{\square}$ in shape, and the coin-flip likelihood is also $p^{\square}(1-p)^{\square}$ in shape. Multiplying functions of this form just *adds the exponents* – the algebraic family $p^a(1-p)^b$ is closed under multiplication. Conjugacy is not a deep coincidence; it is the statement that the beta was *designed* (or rather, discovered) to be the same shape as the thing it multiplies. Counting heads and tails becomes literal arithmetic on the parameters. This is why, in online systems that must update beliefs millions of times a second – A/B testing platforms, recommendation engines, multi-armed-bandit ad servers – the beta-binomial pair is the default: each new click or non-click is one cheap increment of a counter.

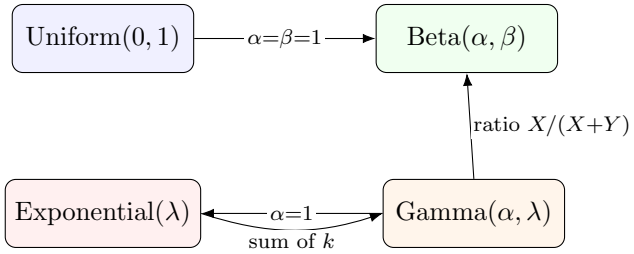
Connection

The same conjugacy idea, transplanted to waiting times, makes the gamma the conjugate prior for an exponential or Poisson *rate* λ: a gamma prior on λ, updated by observed event counts and total time, yields a gamma posterior, where you add the event count to the shape and the elapsed time to the rate. So three of our four families – beta for probabilities, gamma for rates, and (in the next chapter) normal for means – are precisely the conjugate priors that make Bayesian updating into simple bookkeeping. The continuous distributions you are learning here are the alphabet of Bayesian statistics.

11.5 The Family Portrait

Step back and see the clan as a whole. Every relationship in this chapter is a sentence about how one story becomes another.

- **Uniform = Beta$(1,1)$.** Flat ignorance is the simplest beta.
- **Exponential = Gamma$(1,\lambda)$.** One waiting time is a one-step gamma.
- **Sum of k Exponentials = Gamma(k,λ).** Stacking waits adds shapes.
- **Min of Exponentials = Exponential, rates add.** Racing waits combine rates.
- **Gamma+Gamma (same rate) = Gamma, shapes add.**
- **Beta = Gamma/(Gamma + Gamma).** A beta is the fraction of a total contributed by one of two gammas.
- **Chi-squared = Gamma$(k/2, 1/2)$.** Statistics' favorite is a gamma in costume.



Why this matters

Notice what organizes the whole picture: the exponent patterns. The gamma is "power times decaying exponential," $x^{\alpha-1}e^{-\lambda x}$. The beta is "power times power," $x^{\alpha-1}(1-x)^{\beta-1}$. Both are built from the same atom, x^{power}, tamed on the right either by an exponential (unbounded support) or by another power (bounded support). Seen this way the families are not arbitrary; they are the two simplest ways to take a rising power and make it integrable. That is the quiet aesthetic unity behind the zoo.

11.6 A Quick Reference

Distribution	Support	Density	Mean	Variance
Uniform(a,b)	$[a,b]$	$\dfrac{1}{b-a}$	$\dfrac{a+b}{2}$	$\dfrac{(b-a)^2}{12}$
Exponential(λ)	$[0,\infty)$	$\lambda e^{-\lambda x}$	$\dfrac{1}{\lambda}$	$\dfrac{1}{\lambda^2}$
Gamma(α,λ)	$[0,\infty)$	$\dfrac{\lambda^\alpha}{\Gamma(\alpha)}x^{\alpha-1}e^{-\lambda x}$	$\dfrac{\alpha}{\lambda}$	$\dfrac{\alpha}{\lambda^2}$
Beta(α,β)	$[0,1]$	$\dfrac{x^{\alpha-1}(1-x)^{\beta-1}}{B(\alpha,\beta)}$	$\dfrac{\alpha}{\alpha+\beta}$	$\dfrac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}$

Pause

Before the exercises, the one-sentence summary of each story. *Uniform*: equally likely across an interval – pure ignorance, variance $(b-a)^2/12$. *Exponential*: memoryless waiting time, constant hazard, mean = SD = $1/\lambda$. *Gamma*: the wait for the α-th event, a sum of exponentials, mean α/λ. *Beta*: a flexible distribution over a probability on $[0,1]$, the conjugate prior whose update is "add heads to α, tails to β." Four shapes, one connected family.

11.7 Exercises

Exercise 11.1 (Uniform basics)**.** Let $X \sim \text{Uniform}(2,8)$. (a) Write down the density and CDF. (b) Compute $\mathbb{P}(3 \le X \le 5)$. (c) Find $\mathbb{E}[X]$ and $\text{Var}(X)$. (d) Find the value m with $\mathbb{P}(X \le m) = 0.9$.

Exercise 11.2 (The quantization-noise rule)**.** An analog-to-digital converter rounds voltages to the nearest step of size Δ. Model the rounding error as $\text{Uniform}(-\Delta/2, \Delta/2)$. (a) Show the error has mean 0 and variance $\Delta^2/12$. (b) If the signal itself has power (variance) σ^2, the signal-to-noise ratio is $\sigma^2/(\Delta^2/12)$. Explain in words why halving the step size Δ improves the SNR by a factor of 4 (about 6 decibels).

Exercise 11.3 (Exponential computations)**.** Calls arrive at a help desk at rate $\lambda = 4$ per hour with exponential inter-arrival times. (a) What is the probability the next call takes more than 30 minutes to arrive? (b) What is the expected wait, and the median wait? (c) Given that no call has come in the first 20 minutes, what is the expected *additional* wait? Justify your answer with the memoryless property.

Exercise 11.4 (Memorylessness, proved both ways)**.** (a) Using only $\mathbb{P}(T > t) = e^{-\lambda t}$, verify directly that $\mathbb{P}(T > s+t \mid T > s) = \mathbb{P}(T > t)$. (b) Conversely, suppose a continuous positive variable T satisfies $g(s+t) = g(s)g(t)$ for $g(t) = \mathbb{P}(T > t)$, all $s, t \ge 0$, with g decreasing and $g(0) = 1$. By considering $g(t)$ for rational t first, argue that $g(t) = e^{-\lambda t}$ for some $\lambda > 0$, so the exponential is the *only* memoryless continuous distribution.

Exercise 11.5 (Minimum of exponentials and who wins the race)**.** Let $T_1 \sim \text{Exponential}(\lambda_1)$ and $T_2 \sim \text{Exponential}(\lambda_2)$ be independent. (a) Show $\min(T_1, T_2) \sim \text{Exponential}(\lambda_1 + \lambda_2)$. (b) Show that the probability that component 1 fails first is $\mathbb{P}(T_1 < T_2) = \lambda_1/(\lambda_1 + \lambda_2)$. *Hint: condition on T_1 and integrate, or set up the double integral over $\{t_1 < t_2\}$.* (c) Interpret (b): why is the faster component more likely to win?

Exercise 11.6 (Gamma as a sum)**.** Machine parts arrive at rate $\lambda = 3$ per hour, exponential gaps. (a) Name the distribution of the wait until the 4th part. (b) Give its mean, variance, and standard deviation. (c) Compare the ratio SD/mean for this wait against the ratio for a single arrival, and explain why summing waits reduces relative variability.

Exercise 11.7 (Verifying a gamma normalizing constant)**.** Using the substitution $u = \lambda x$ and the definition $\Gamma(\alpha) = \int_0^\infty u^{\alpha-1} e^{-u} \mathrm{d}u$, show that $\int_0^\infty \frac{\lambda^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\lambda x} \mathrm{d}x = 1$. Then use $\Gamma(\alpha+1) = \alpha\Gamma(\alpha)$ to derive $\mathbb{E}[X] = \alpha/\lambda$ directly from the integral $\int_0^\infty x\, f(x)\, \mathrm{d}x$.

Exercise 11.8 (Reading and updating a beta)**.** You are testing a new web button. Your prior on its click-through rate p is $\text{Beta}(2,8)$. (a) What is your prior mean and standard deviation for p? (b) You show the button to 40 visitors; 12 click. Write down the posterior distribution of p. (c) Give the posterior mean. Is it closer to the prior mean or to the raw data proportion 12/40? Explain why.

Exercise 11.9 (The uniform is a beta)**.** (a) Show that Beta$(1, 1)$ has density equal to 1 on $[0, 1]$, i.e. it is Uniform$(0, 1)$. (b) Using the general beta mean and variance formulas, recover the Uniform$(0, 1)$ mean $1/2$ and variance $1/12$. (c) Conceptually: explain why starting Bayesian updating from a Beta$(1, 1)$ prior corresponds to "assuming nothing" about the unknown probability.

Exercise 11.10 (The length-biased waiting paradox)**.** Buses have i.i.d. exponential gaps with mean 10 minutes. (a) Using memorylessness, explain why a passenger arriving at a random time waits, on average, 10 minutes for the next bus – not 5. (b) Argue that the time since the previous bus also averages 10 minutes, so the observed gap averages 20. (c) In one or two sentences, explain the resolution: why does sampling by a random *instant* over-represent long gaps?

Exercise 11.11 (Proof: scaling an exponential)**.** Let $T \sim$ Exponential(λ) and let $c > 0$. Using CDFs, prove that $cT \sim$ Exponential(λ/c). Interpret: rescaling time (e.g. measuring in seconds instead of minutes) just rescales the rate, as it must.

Exercise 11.12 (Inverse-transform simulation, by hand)**.** You have a uniform draw $U = 0.2$. (a) Using $T = -\frac{1}{\lambda}\ln(1 - U)$ with $\lambda = 2$, compute the corresponding exponential sample. (b) Explain why $-\frac{1}{\lambda}\ln(1 - U)$ and $-\frac{1}{\lambda}\ln U$ give samples from the *same* distribution. (c) Why does the inverse-transform method turn a flat uniform into a right-skewed exponential – what is the logarithm doing geometrically to the unit interval?

Chapter 12

The Normal Distribution

There is one curve in all of probability that you will meet more often than any other. It is the smooth, symmetric hump that statisticians draw on napkins, that shows up in physics labs measuring the same length over and over, that describes the spread of human heights, the noise in an electrical signal, the errors of a telescope, the fluctuations of a stock index over a day, and the average of almost anything you can average. It has a name that sounds like a value judgment – the *normal* distribution – and a second name that honors the man who made it famous, the *Gaussian*.

This chapter is a slow, careful walk around that curve. We will not just write down its formula and move on. We will ask: where does that strange-looking $e^{-x^2/2}$ come from? Why is there a $\sqrt{2\pi}$ bolted to the front? What do the two parameters really control? Why does adding normals give you a normal, and scaling a normal give you a normal, when almost no other distribution is so well-behaved? And, most importantly, why is this one curve *everywhere*?

That last question has a deep answer – the Central Limit Theorem – which gets its own chapter later. Here we lay the groundwork: we get to know the normal distribution as an object, intimately, so that when the grand theorem arrives you already understand the shape it is reaching for.

12.1 A Curve Born From Errors

Begin with a story, because the normal distribution was discovered through a very human problem: *nobody can measure anything exactly.*

Imagine an eighteenth-century astronomer trying to pin down the position of a star. He looks through his telescope and writes down an angle. Then he looks again and writes down a slightly different angle. Again – different again. His hand trembles, the air shimmers, his eye tires. Each measurement is the true position plus a little *error*, and the errors are different every time.

What can he say about these errors? A few things feel obvious once you say them out loud:

- Small errors are more common than large errors. A measurement off by a hair happens constantly; a measurement off by a mile almost never.
- Errors are symmetric: overshooting by 0.1 is just as likely as undershooting by 0.1. There is no built-in bias toward "too big" or "too small" if the instrument is unbiased.

- Errors pile up smoothly. There is no magic threshold where suddenly an error becomes impossible; the chance just fades away as you go further out.

Intuition

The astronomer's instinct – "small errors common, big errors rare, symmetric about the truth, fading smoothly" – is not yet a formula. But it is already a *shape*: a single hump centered on the true value, tallest in the middle, falling away on both sides, never quite touching zero. Every detail of the normal density is just this picture made precise.

The genius of Gauss (and before him, in pieces, of de Moivre and Laplace) was to turn that intuition into an exact curve by adding one more natural-sounding assumption – and then asking what function is *forced* on you. We will see the resulting form in Section 12.2. But first, let us simply meet the curve.

History

The curve has a tangled birth. Abraham de Moivre found it around 1733 as an approximation to the binomial distribution – he was computing odds for gamblers and noticed that the staircase of binomial probabilities, when there are many trials, traces out a smooth bell. Pierre-Simon Laplace generalized the idea. But it was Carl Friedrich Gauss who, in 1809, derived the curve from the theory of measurement errors and tied it to the method of least squares, and it is his name that stuck. The word "normal" is even later and more accidental: it was used in the late nineteenth century (by Galton, Peirce, and others) to mean something like "the typical, standard case," not "correct" or "healthy." Many statisticians regret the name, because it makes every other distribution sound abnormal – which is exactly backwards.

12.2 The Standard Normal: The Simplest Bell

The cleanest version of the curve has no free knobs at all. It is centered at zero, has its width fixed at one, and is called the **standard normal**. Its density – the function whose area under pieces gives probability – is

Formal

Standard normal density. A random variable Z is *standard normal*, written $Z \sim \mathcal{N}(0,1)$, if it has density

$$\varphi(z) \;=\; \frac{1}{\sqrt{2\pi}}\, e^{-z^2/2}, \qquad z \in \mathbb{R}.$$

We write φ (lowercase phi) for this density and Φ (uppercase phi) for its cumulative distribution function, $\Phi(z) = \mathbb{P}(Z \le z) = \int_{-\infty}^{z} \varphi(t)\,\mathrm{d}t$.

Let us read this formula slowly, because every symbol is doing a job.

- The $e^{-z^2/2}$ is the *heart*. The z^2 makes it symmetric (since $(-z)^2 = z^2$, the curve looks the same on both sides). The minus sign in the exponent makes it *decrease* as you move away from zero. The squaring makes it decrease *fast* – faster than any straight line, faster even than a simple $e^{-|z|}$ – so the tails are thin.

- The $\frac{1}{2}$ inside the exponent is a choice of scale that, as we will see, makes the variance come out to exactly 1. Without it the curve would still be a bell, just a different width.
- The $\frac{1}{\sqrt{2\pi}}$ out front is a *normalizing constant*. It is there for exactly one reason: to make the total area under the curve equal 1, as every density must. It carries no shape information at all.

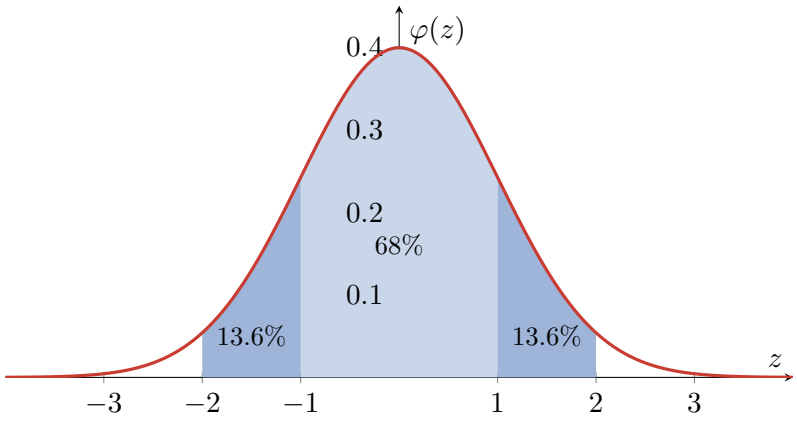


Figure 12.1: The standard normal density $\varphi(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}$. The shaded regions are the chance of landing within one and two standard deviations of the center. The peak height is $1/\sqrt{2\pi} \approx 0.399$.

Notice in Figure 12.1 that the curve is tallest at $z = 0$, where its height is $1/\sqrt{2\pi} \approx 0.399$, and that it has already become visually flat by $z = \pm 3$. By $z = \pm 4$ it is essentially glued to the horizontal axis, though in truth it never reaches it – the curve is positive for *every* real number, no matter how far out. There is, in principle, a nonzero chance of an arbitrarily large error. It is just astronomically small.

Walking question

The peak of φ is about 0.399. A narrow normal, say $\mathcal{N}(0, 0.1^2)$, has a peak height of $1/(0.1\sqrt{2\pi}) \approx 3.99$ – a density value well above 1. But probabilities can never exceed 1. How can a density take a value bigger than any probability, and yet areas under it still come out as proper probabilities? Pause and recall the difference between a density and a probability. (Answer: a density is not a probability; it is probability *per unit length*. You multiply by a width to get probability, and the widths near a tall narrow peak are tiny. The total area, not the peak height, is what must equal 1.)

12.2.1 Why the constant is $1/\sqrt{2\pi}$

Here is the single most famous integral in all of mathematics, and the reason π – a constant about *circles* – shows up in a curve about *errors*:

Theorem 12.1 (The Gaussian integral)**.**

$$\int_{-\infty}^{\infty} e^{-z^2/2}\,\mathrm{d}z \;=\; \sqrt{2\pi}.$$

Why it works

The function $e^{-z^2/2}$ has no elementary antiderivative – you cannot write its integral in terms of polynomials, exponentials, logs, and trig functions. So the usual "find the antiderivative and plug in the limits" is hopeless. The trick, due to Poisson, is to compute the *square* of the integral and pass to two dimensions, where the rotational symmetry of $e^{-(x^2+y^2)/2}$ lets polar coordinates rescue you.
Let $I = \int_{-\infty}^{\infty} e^{-z^2/2}\,\mathrm{d}z$. Then

$$I^2 = \left(\int_{-\infty}^{\infty} e^{-x^2/2}\,\mathrm{d}x\right)\left(\int_{-\infty}^{\infty} e^{-y^2/2}\,\mathrm{d}y\right) = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} e^{-(x^2+y^2)/2}\,\mathrm{d}x\,\mathrm{d}y.$$

Now switch to polar coordinates: $x^2 + y^2 = r^2$, and the area element $\mathrm{d}x\,\mathrm{d}y$ becomes $r\,\mathrm{d}r\,\mathrm{d}\theta$. That extra factor of r is the miracle – it is exactly what makes the integral doable:

$$I^2 = \int_0^{2\pi}\int_0^{\infty} e^{-r^2/2}\,r\,\mathrm{d}r\,\mathrm{d}\theta = \int_0^{2\pi}\Big[-e^{-r^2/2}\Big]_0^{\infty}\mathrm{d}\theta = \int_0^{2\pi} 1\,\mathrm{d}\theta = 2\pi.$$

Therefore $I = \sqrt{2\pi}$, and dividing $e^{-z^2/2}$ by this number makes its total area exactly 1. That is the entire reason for the $\sqrt{2\pi}$.

Connection

The substitution $u = r^2/2$ (so $\mathrm{d}u = r\,\mathrm{d}r$) is where the lonely r from the polar area element gets absorbed. If you have read the companion calculus book, this is the cleanest possible illustration of why the Jacobian of the polar change of variables matters: drop the r and the integral is impossible; keep it and it falls out in one line. The appearance of π is geometry leaking into analysis – it comes from the 2π of a full rotation, the angular sweep of the plane.

12.3 Adding Two Knobs: Mean and Variance

The standard normal is centered at 0 with spread 1. Real data are centered wherever they happen to be centered and spread however they spread. So we give the curve two adjustable knobs: a location μ and a scale σ.

Formal

General normal density. A random variable X is *normal* with mean μ and variance σ^2, written $X \sim \mathcal{N}(\mu, \sigma^2)$, if it has density

$$f(x) \;=\; \frac{1}{\sigma\sqrt{2\pi}}\,\exp\!\left(-\frac{(x-\mu)^2}{2\sigma^2}\right), \qquad x \in \mathbb{R}.$$

Here $\mu \in \mathbb{R}$ is the *mean* and $\sigma > 0$ is the *standard deviation*; σ^2 is the *variance*.

This is the same bell, repositioned and rescaled. Compare it term by term with the standard normal and you can read off what each knob does:

- Replacing z by $(x-\mu)/\sigma$ *slides* the curve so its peak sits at $x=\mu$ instead of 0. The mean μ is the center of symmetry, the peak, the median, and (because the curve is symmetric) the average – all at once.
- Dividing the deviation by σ *stretches* the horizontal axis. Large σ spreads the curve wide and flat; small σ squeezes it tall and narrow.
- The leading $\frac{1}{\sigma\sqrt{2\pi}}$ keeps the total area at 1 even as you stretch: a wider curve must be shorter so the area is conserved. That is why a narrow normal is tall and a wide one is squat.

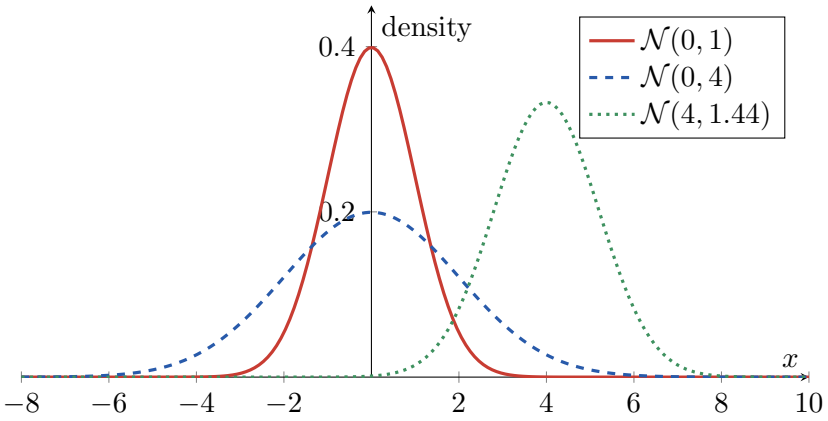


Figure 12.2: Three normals. Changing μ slides the bell left or right; changing σ makes it narrow-and-tall or wide-and-short. The area under each is always exactly 1.

What is really happening

Every normal distribution is the *same* curve in disguise. The two knobs μ and σ do nothing to the *shape* – they only choose where to put the curve and how big to draw it. This is why a single table or a single function Φ is enough to answer any question about any normal: you just translate and rescale. The technical name for this is that all normals form a single *location-scale family*, and the next subsection makes the translation explicit.

12.3.1 Standardizing: the universal translator

The link between the general normal and the standard one is a single, enormously useful operation: subtract the mean, divide by the standard deviation.

Theorem 12.2 (Standardization)**.** *If* $X \sim \mathcal{N}(\mu, \sigma^2)$ *then*

$$Z = \frac{X-\mu}{\sigma} \sim \mathcal{N}(0,1).$$

Conversely, if $Z \sim \mathcal{N}(0,1)$ *then* $X = \mu + \sigma Z \sim \mathcal{N}(\mu, \sigma^2)$.

Why it works

This is a change of variables. Let $X \sim \mathcal{N}(\mu, \sigma^2)$ and set $Z = (X - \mu)/\sigma$, so $X = \mu + \sigma Z$ and $\mathrm{d}x = \sigma\,\mathrm{d}z$. The density of Z is the density of X at the point $x = \mu + \sigma z$, multiplied by the stretch factor $|\mathrm{d}x/\mathrm{d}z| = \sigma$:

$$f_Z(z) = f_X(\mu + \sigma z) \cdot \sigma = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(\mu + \sigma z - \mu)^2}{2\sigma^2}\right) \cdot \sigma = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}.$$

The σ in the denominator cancels the σ from the stretch, the $(\mu + \sigma z - \mu)^2 = \sigma^2 z^2$ collapses, and out drops $\varphi(z)$ exactly. So Z is standard normal.

Standardization is what lets you answer *any* normal question with one table. To find $\mathbb{P}(X \le x)$ for $X \sim \mathcal{N}(\mu, \sigma^2)$, you convert the threshold x into a number of standard deviations away from the mean – a *z-score* – and look that up:

$$\mathbb{P}(X \le x) = \mathbb{P}\left(\frac{X - \mu}{\sigma} \le \frac{x - \mu}{\sigma}\right) = \Phi\left(\frac{x - \mu}{\sigma}\right).$$

Example 12.1 (A test-score calculation)**.** Scores on a standardized exam are modeled as $X \sim \mathcal{N}(500, 100^2)$. What fraction of test-takers score above 650?

The z-score of 650 is

$$z = \frac{650 - 500}{100} = 1.5.$$

So a score of 650 is exactly 1.5 standard deviations above the mean. Then

$$\mathbb{P}(X > 650) = \mathbb{P}(Z > 1.5) = 1 - \Phi(1.5) \approx 1 - 0.9332 = 0.0668.$$

About 6.7% of test-takers score above 650. Notice we never needed a special table for "mean 500, sd 100" – standardizing reduced it to a single lookup of $\Phi(1.5)$.

12.4 Why μ and σ^2 Really Are the Mean and Variance

We have been *calling* μ the mean and σ^2 the variance. We owe you a proof that the names are accurate – that if you compute $\mathbb{E}[X]$ and $\mathrm{Var}(X)$ from the density, you actually get μ and σ^2.

Proposition 12.1. *If* $X \sim \mathcal{N}(\mu, \sigma^2)$ *then* $\mathbb{E}[X] = \mu$ *and* $\mathrm{Var}(X) = \sigma^2$.

Why it works

By standardization it is enough to handle $Z \sim \mathcal{N}(0, 1)$ and then transfer the result, because $X = \mu + \sigma Z$ gives $\mathbb{E}[X] = \mu + \sigma\,\mathbb{E}[Z]$ and $\mathrm{Var}(X) = \sigma^2\,\mathrm{Var}(Z)$ by linearity of expectation and the scaling rule for variance.

The mean of Z is 0. The integrand $z\,\varphi(z)$ is an *odd* function: replacing z by $-z$ flips its sign, because z is odd and $\varphi(z) = \varphi(-z)$ is even. The integral of an odd function over a symmetric interval $(-\infty, \infty)$ is zero (the left half exactly cancels the right half). Hence $\mathbb{E}[Z] = \int z\,\varphi(z)\,\mathrm{d}z = 0$. So $\mathbb{E}[X] = \mu$.

The variance of Z is 1. Since $\mathbb{E}[Z] = 0$, $\mathrm{Var}(Z) = \mathbb{E}[Z^2] = \int_{-\infty}^{\infty} z^2 \varphi(z)\,\mathrm{d}z$. Compute it by integration by parts. Write $z^2 e^{-z^2/2} = z \cdot \left(z e^{-z^2/2}\right)$ and note that $z\,e^{-z^2/2}$ is the derivative of $-e^{-z^2/2}$. With $u = z$ and $\mathrm{d}v = z e^{-z^2/2}\mathrm{d}z$, so $v = -e^{-z^2/2}$:

$$\int_{-\infty}^{\infty} z^2 e^{-z^2/2}\,\mathrm{d}z = \Big[-z\,e^{-z^2/2} \Big]_{-\infty}^{\infty} + \int_{-\infty}^{\infty} e^{-z^2/2}\,\mathrm{d}z.$$

The boundary term vanishes (the exponential crushes the linear z at both ends), and the remaining integral is the Gaussian integral $\sqrt{2\pi}$ from Theorem 12.1. Dividing by the normalizer $\sqrt{2\pi}$,

$$\mathrm{Var}(Z) = \frac{1}{\sqrt{2\pi}} \cdot \sqrt{2\pi} = 1.$$

Therefore $\mathrm{Var}(X) = \sigma^2 \cdot 1 = \sigma^2$, and the parameter names are vindicated.

Common trap

A persistent source of confusion: $\mathcal{N}(\mu, \sigma^2)$ is parameterized by the *variance* σ^2, not the standard deviation σ. So $\mathcal{N}(0, 9)$ has standard deviation 3, not 9. Some books and software (notably some R functions) instead take σ directly. Always check which convention you are in before plugging numbers, or you will be off by a square.

12.5 The 68–95–99.7 Rule

Because every normal is the same curve rescaled, the probability of landing within a given number of standard deviations of the mean is *always the same*, no matter what μ and σ are. Three of these numbers are worth memorizing for life; they are sometimes called the *empirical rule*.

Focus

For *any* $X \sim \mathcal{N}(\mu, \sigma^2)$:

$$\begin{aligned} \mathbb{P}(\mu - 1\sigma \le X \le \mu + 1\sigma) &\approx 0.6827 \quad \text{(about 68\%)}, \\ \mathbb{P}(\mu - 2\sigma \le X \le \mu + 2\sigma) &\approx 0.9545 \quad \text{(about 95\%)}, \\ \mathbb{P}(\mu - 3\sigma \le X \le \mu + 3\sigma) &\approx 0.9973 \quad \text{(about 99.7\%)}. \end{aligned}$$

These come straight from the standard normal CDF: $\mathbb{P}(|Z| \le 1) = \Phi(1) - \Phi(-1) = 2\Phi(1) - 1 \approx 2(0.84134) - 1 \approx 0.6827$, and likewise for 2 and 3.

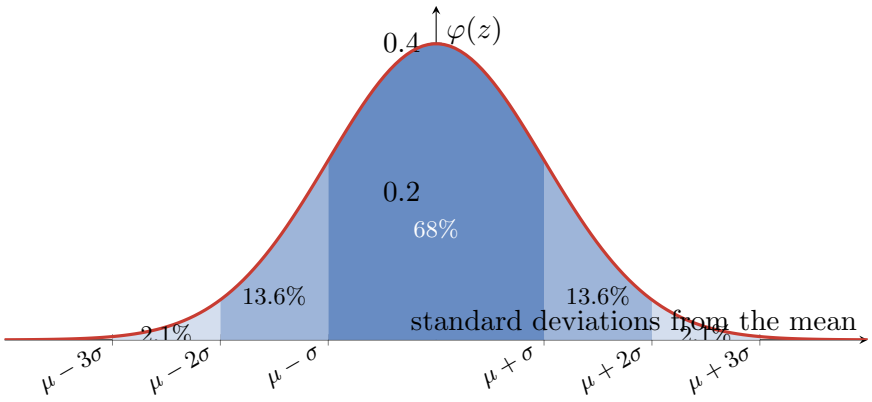


Figure 12.3: The 68–95–99.7 rule. Each shell of width one standard deviation adds less and less probability; almost everything lives within three deviations of the mean.

Intuition

The empirical rule is the normal distribution's "feel." Once it is in your bones, you can sanity-check claims instantly. "IQ is $\mathcal{N}(100, 15^2)$, and someone scored 145" – that is three standard deviations above the mean, so they are out past the 99.85th percentile, rarer than 1 in 700. "A part is supposed to be 50mm with sd 0.5mm, and yours measures 52mm" – four standard deviations out, which for a normal process should essentially never happen, so your machine is probably broken. You did both of those in your head.

Application

Six Sigma. The manufacturing-quality movement named "Six Sigma" takes its name directly from this rule. The goal is to make a process so tightly controlled that the nearest specification limit is six standard deviations away from the mean. For a perfectly centered normal process, being outside $\pm 6\sigma$ has probability about 2×10^{-9} – two defects per billion. (In practice the program allows for a 1.5σ drift of the mean over time, which is why the famous quoted figure is 3.4 defects per million rather than the pure-normal two per billion. The arithmetic of those tail areas is exactly what you are learning to do here.)

12.6 The Tails: Reading the Standard Normal Table

Because Φ has no closed form, for centuries people read its values off a printed table. Understanding the *idea* of the table is worth more than any particular software call, because it forces you to think in terms of z-scores and tail areas.

A standard normal table simply lists $\Phi(z) = \mathbb{P}(Z \le z)$ for a grid of z-values. A small slice:

z	0.00	0.50	1.00	1.50	2.00	2.50
$\Phi(z)$	0.5000	0.6915	0.8413	0.9332	0.9772	0.9938

The table is really a picture of one curve: the cumulative distribution function Φ, which climbs from

0 on the far left to 1 on the far right, passing through $\frac{1}{2}$ at the symmetric center $z = 0$.

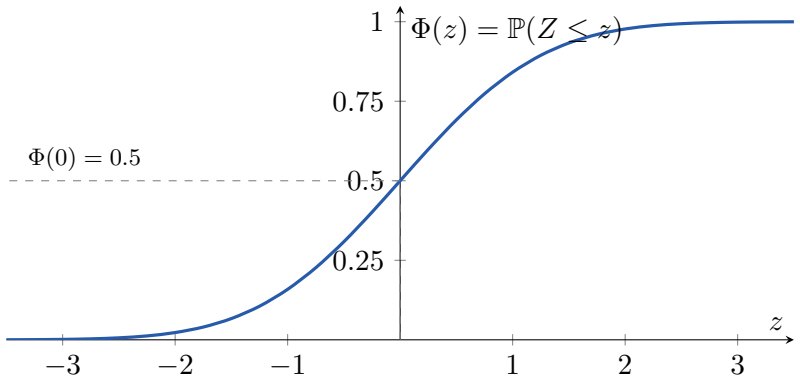


Figure 12.4: The standard normal CDF Φ. It is the running area under the bell of Figure 12.1: an S-shaped (sigmoid) curve rising from 0 to 1, steepest where the density is tallest (at $z = 0$) and flattening into the thin tails. Every table value is one point on this curve.

Everything else you compute from three symmetry-and-complement facts:

Focus

1. **Total area:** $\mathbb{P}(Z \le z) + \mathbb{P}(Z > z) = 1$, so $\mathbb{P}(Z > z) = 1 - \Phi(z)$.
2. **Symmetry:** $\Phi(-z) = 1 - \Phi(z)$, because the curve is a mirror image about 0.
3. **Intervals:** $\mathbb{P}(a \le Z \le b) = \Phi(b) - \Phi(a)$.

Example 12.2 (Working the table both ways)**.** Let $Z \sim \mathcal{N}(0, 1)$.

(a) A right tail. $\mathbb{P}(Z > 1.5) = 1 - \Phi(1.5) = 1 - 0.9332 = 0.0668$.

(b) A left tail by symmetry. $\mathbb{P}(Z \le -1) = \Phi(-1) = 1 - \Phi(1) = 1 - 0.8413 = 0.1587$.

(c) A symmetric middle. $\mathbb{P}(-2 \le Z \le 2) = \Phi(2) - \Phi(-2) = 0.9772 - (1 - 0.9772) = 0.9544$ – the 95% of the empirical rule, recovered from the table.

(d) Inverse lookup. "What cutoff c has 5% of the mass above it?" We need $\Phi(c) = 0.95$, and the table (read backwards) gives $c \approx 1.645$. This 1.645 is the famous one-sided 5% critical value you will meet again in hypothesis testing.

Connection

Modern software replaces the table with two functions: a CDF call (`Phi`, e.g. `scipy.stats.norm.cdf`) and its inverse, the *quantile function* or *probit* (`scipy.stats.norm.ppf`). The inverse lookup in Example 12.2(d) is exactly $\Phi^{-1}(0.95)$. The probit function also reappears as a link function in statistics (probit regression) and as the core of one common method for generating normal random numbers from uniform ones.

Remark 12.1 (The error function). You will sometimes see Φ written through the *error function* $\operatorname{erf}(x) = \frac{2}{\sqrt{\pi}} \int_0^x e^{-t^2}\, dt$. The relationship is $\Phi(z) = \frac{1}{2}\big(1 + \operatorname{erf}(z/\sqrt{2})\big)$. They are the same idea wearing different clothes; the factor $\sqrt{2}$ is just the difference between the e^{-t^2} and $e^{-z^2/2}$ conventions.

12.7 The Magic Property: Normality Survives Linear Operations

Here is what makes the normal distribution not merely common but *convenient*. Two operations that destroy the shape of most distributions leave the normal completely unharmed.

12.7.1 Linear transformations stay normal

We already saw half of this in standardization. The full statement:

Theorem 12.3 (Linear closure)**.** *If $X \sim \mathcal{N}(\mu, \sigma^2)$ and $a \neq 0$, b are constants, then*

$$aX + b \sim \mathcal{N}\big(a\mu + b,\ a^2\sigma^2\big).$$

The mean shifts and scales the way any mean does ($\mathbb{E}[aX + b] = a\mu + b$); the variance scales by a^2 ($\mathrm{Var}(aX + b) = a^2\sigma^2$); and – this is the special part – the *shape stays normal*. Shifting and stretching a bell gives another bell. For almost any other distribution this fails: stretch an exponential and you get another exponential only if you do not shift it; the family is far more fragile. The normal family is closed under *every* non-degenerate linear map.

12.7.2 Sums of independent normals are normal

The deeper and more surprising closure property:

Theorem 12.4 (Additive closure)**.** *If $X \sim \mathcal{N}(\mu_1, \sigma_1^2)$ and $Y \sim \mathcal{N}(\mu_2, \sigma_2^2)$ are* independent*, then*

$$X + Y \sim \mathcal{N}\big(\mu_1 + \mu_2,\ \sigma_1^2 + \sigma_2^2\big).$$

The means add (always true for sums); the variances add (always true for *independent* sums); and again the shape stays normal – which is special. The sum of two independent normals is not some new lumpy thing; it is just a wider bell.

Why it works

Why do the variances add but not the standard deviations? Variance is the natural additive quantity for independent sums:

$$\mathrm{Var}(X + Y) = \mathrm{Var}(X) + \mathrm{Var}(Y) \quad \text{(independence)},$$

because the cross-term $2\,\mathrm{Cov}(X, Y)$ vanishes when X and Y are independent. Standard deviations therefore add "in quadrature": $\sigma_{X+Y} = \sqrt{\sigma_1^2 + \sigma_2^2}$, like the hypotenuse of a right triangle with legs σ_1 and σ_2. Two equal-sized independent errors of size σ do *not* combine into an error of size 2σ; they combine into $\sqrt{2}\,\sigma \approx 1.41\sigma$. Random errors partly cancel.

That the result is again *normal* is the genuinely deep fact. The cleanest proof uses the moment generating function (see the chapter on generating functions): for $X \sim \mathcal{N}(\mu, \sigma^2)$,

$$M_X(t) = \mathbb{E}[e^{tX}] = \exp\Big(\mu t + \tfrac{1}{2}\sigma^2 t^2\Big).$$

For independent X, Y the MGF of the sum is the product of the MGFs:

$$M_{X+Y}(t) = M_X(t)\, M_Y(t) = \exp\Big((\mu_1 + \mu_2)t + \tfrac{1}{2}(\sigma_1^2 + \sigma_2^2)t^2\Big),$$

which is precisely the MGF of a $\mathcal{N}(\mu_1+\mu_2, \sigma_1^2+\sigma_2^2)$. Since the MGF determines the distribution, the sum *is* that normal. Adding the quadratic-in-t exponents is what adds the variances; the form "exp(linear + quadratic)" is closed under multiplication, and that algebraic closure is the secret behind the statistical closure.

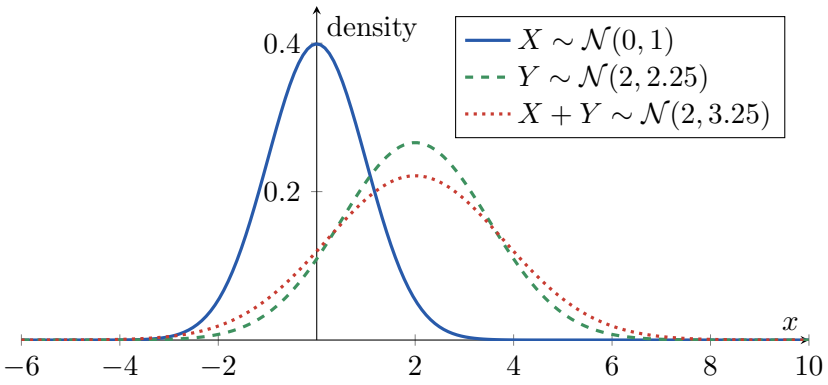


Figure 12.5: Adding two independent normals gives a third normal: means add ($0+2=2$), variances add ($1+2.25=3.25$), and the sum is wider than either piece but still a clean bell.

Example 12.3 (Stacking tolerances)**.** A factory stacks two metal plates. Plate A has thickness $\mathcal{N}(10, 0.2^2)$ mm and plate B has thickness $\mathcal{N}(15, 0.3^2)$ mm, independently. The total thickness is

$$T = A + B \sim \mathcal{N}\big(10+15,\ 0.2^2+0.3^2\big) = \mathcal{N}(25,\ 0.13).$$

So the stack averages 25 mm with standard deviation $\sqrt{0.13} \approx 0.36$ mm. What is the chance the stack exceeds 25.5 mm?

$$\mathbb{P}(T > 25.5) = \mathbb{P}\left(Z > \frac{25.5-25}{0.36}\right) = \mathbb{P}(Z > 1.39) \approx 1 - 0.9177 = 0.0823.$$

About 8%. Note the combined sd 0.36 is less than $0.2+0.3=0.5$: errors partly cancel, just as the in-quadrature rule promised.

Corollary 12.1 (Averages of normals)**.** *If $X_1, \dots, X_n$ are independent $\mathcal{N}(\mu, \sigma^2)$, then their sum is $\mathcal{N}(n\mu, n\sigma^2)$ and their average is*

$$\bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i \sim \mathcal{N}\left(\mu, \frac{\sigma^2}{n}\right).$$

This little corollary is the engine of all of statistics. Averaging n normal measurements gives a normal answer centered at the true μ with the spread shrunk by a factor $\sqrt{n}$. Average four measurements and you halve the error; average a hundred and you cut it tenfold. The astronomer from Section 12.1 was right to take many readings and average them.

12.8 Why Is the Normal Everywhere? A Preview of the CLT

We can now address the mystery in the chapter's introduction. Why does this one curve describe heights, errors, noise, test scores, and the daily wiggles of a stock index – things with no obvious connection to each other?

The answer is that all of those quantities are *sums of many small, independent effects*. A person's height is the cumulative result of thousands of genes and environmental nudges, each contributing a tiny push up or down. A measurement error is the sum of many small disturbances – air, vibration, the observer's nerves. And the punchline, which the next chapters make rigorous, is:

Why this matters

The Central Limit Theorem (preview). When you add up many independent random contributions, no single one of them dominating, the distribution of the sum – after centering and rescaling – looks like a normal distribution, *almost regardless of what the individual contributions look like*. The ingredients can be coin flips, dice, skewed payouts, anything with a finite variance. Their accumulated sum forgets its origins and converges to the bell.

This is the real reason the normal is "normal." It is not that nature has a fondness for $e^{-x^2/2}$. It is that nature is full of *accumulation*, and accumulation of many small independent pieces forces the bell to appear. The normal is the universal shape of *averages and aggregates*.

We can already see the convergence happening with the humblest of building blocks: sums of coin flips. Add up n independent ± 1 steps and the histogram of the result, once n is even modestly large, is visually indistinguishable from a normal curve.

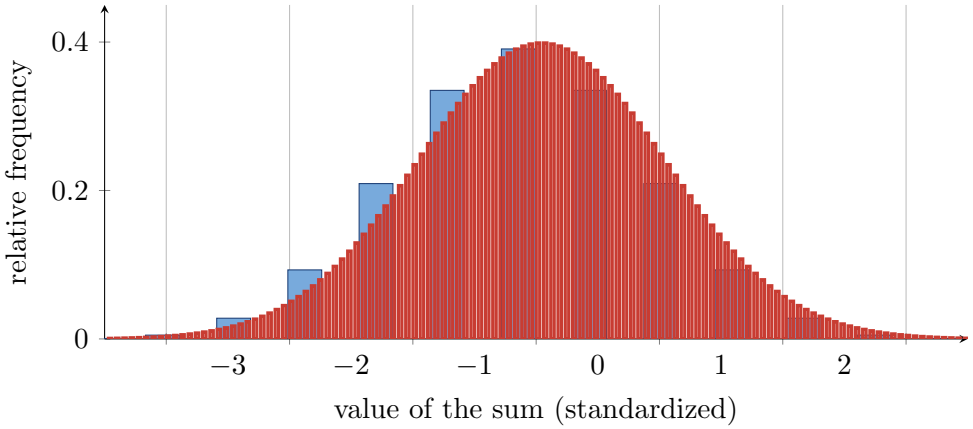


Figure 12.6: The standardized distribution of a sum of 12 fair coin steps (bars, each ± 1) against the standard normal density (curve). With only twelve flips the bell is already an excellent fit – this is de Moivre's original discovery and a glimpse of the Central Limit Theorem.

History

This figure is essentially the picture de Moivre drew around 1733. He was studying the binomial distribution – the number of heads in n tosses – and realized that for large n the discrete bars were hugging a smooth curve he could write down. He had, without quite knowing its future importance, discovered the normal approximation to the binomial, the very first instance of a central limit theorem. It took another century and Gauss and Laplace before the curve became the centerpiece of statistics.

12.9 A Paradox and a Caution

The normal distribution is so useful that it is over-used, and its thin tails are the most common place where the model and reality part ways.

Paradox

The "impossible" market crash. Suppose daily returns of a stock index are modeled as normal, and the historical daily standard deviation is about 1%. On October 19, 1987 ("Black Monday"), the index fell about 20% in a single day – a 20-sigma event.
How rare is a 20-sigma event under a normal model? The tail probability $\mathbb{P}(Z < -20)$ is on the order of 10^{-89}. To put that in perspective: if the market had been open every day since the Big Bang, roughly 5×10^{12} days, the expected number of such days would still be about 10^{-76} – effectively zero, short of one by some seventy orders of magnitude. Under the normal model, Black Monday simply cannot happen.
But it *did* happen – and so did several other "impossible" days. The resolution is not that probability is broken. It is that the *model is wrong*: real financial returns are *not* normal. They have *fat tails* – extreme moves are far more common than a bell predicts. The normal's $e^{-z^2/2}$ decay is breathtakingly fast, and that very feature, which makes it mathematically lovely, makes it a dangerous model for anything where rare catastrophes are the whole point. The lesson: the normal is the right model for *averages of many small independent effects*, and crashes are the opposite – rare, correlated, dominant shocks.

Common trap

"Normal" does not mean "default." Before assuming a quantity is normal, ask the CLT question: *is this an average or sum of many small independent pieces?* Heights, measurement errors, and sample means: yes, plausibly normal. Incomes (a few people earn enormous amounts): no, heavily skewed. Waiting times: no, often exponential. City sizes and word frequencies: no, power-law. Reaching for the bell out of habit is one of the most common modeling mistakes in applied statistics.

Application

Where the normal genuinely rules in machine learning. Even with the caution above, the Gaussian is the workhorse of modern statistics and ML for good reasons. Least-squares regression is exactly maximum-likelihood estimation under the assumption of normal noise – minimizing squared error *is* assuming a bell-shaped error. Gaussian mixture models cluster data as overlapping bells. Gaussian processes put a normal distribution over entire functions. The "reparameterization trick" in variational autoencoders relies on $X = \mu + \sigma Z$ with $Z \sim \mathcal{N}(0, 1)$ – exactly the standardization identity of Theorem 12.2, run backwards, so that gradients can flow through the randomness. In every one of these, the linear-closure and additive-closure properties of this chapter are what make the math tractable.

12.10 The Bivariate Normal, Briefly

Normals generalize to several dimensions, and the two-dimensional case is worth a glimpse because it reveals what the curve looks like when two normal quantities travel together. If X and Y are jointly normal, their density is a bell over the plane, and its level sets – the curves of equal height – are *ellipses*. When X and Y are independent the ellipses are axis-aligned; when they are correlated the ellipses tilt, leaning along the direction of the correlation.

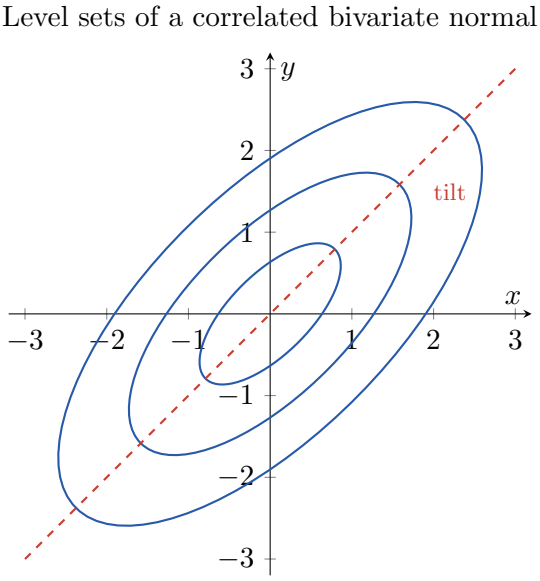


Figure 12.7: Contours of a bivariate normal with positive correlation. The tilted, elongated ellipses show that large x tends to come with large y. Independent normals would give circles or axis-aligned ellipses.

Connection

The full multivariate normal is governed by a mean *vector* $\boldsymbol{\mu}$ and a *covariance matrix* Σ, with density proportional to $\exp\big(-\frac{1}{2}(\mathbf{x}-\boldsymbol{\mu})^\top \Sigma^{-1}(\mathbf{x}-\boldsymbol{\mu})\big)$. The quadratic form in the exponent is the multidimensional cousin of $(x-\mu)^2/\sigma^2$, and the ellipses of Figure 12.7 are its level sets – their shape is dictated by the eigenvectors and eigenvalues of Σ. This is where probability and linear algebra meet most beautifully; the companion linear algebra book's treatment of symmetric matrices and their spectral decomposition is exactly the machinery you need, and we develop it fully in the later chapters on joint distributions and covariance.

12.11 What to Carry Forward

Pause

The essential picture:

- The standard normal density is $\varphi(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}$. The $e^{-z^2/2}$ is the bell; the $\sqrt{2\pi}$ is just the area-normalizer, and it is $\sqrt{2\pi}$ because of the polar-coordinates Gaussian integral.
- The general normal $\mathcal{N}(\mu, \sigma^2)$ is the same curve slid to μ and stretched by σ. Standardizing, $Z = (X - \mu)/\sigma$, turns every normal question into one lookup of Φ.
- 68–95–99.7: the chance of being within one, two, or three standard deviations of the mean, for *any* normal.
- Linear maps of a normal stay normal; sums of independent normals stay normal, with means adding and variances adding. These closure properties are what make the normal so easy to compute with.
- The normal is everywhere because of the Central Limit Theorem: it is the universal shape of sums and averages of many small independent effects. But its thin tails make it a poor model for rare, extreme, correlated shocks.

In the chapters ahead, the normal will be your constant companion. The Central Limit Theorem will explain its ubiquity rigorously; confidence intervals and hypothesis tests will lean on it at every step; and the multivariate version will carry the whole theory of covariance and regression. You now know the curve from the inside.

Exercises

Exercise 12.1 (Reading the curve)**.** Let $Z \sim \mathcal{N}(0, 1)$. Using the table values in Section 12.6 and the symmetry/complement rules, compute: (a) $\mathbb{P}(Z > 2)$; (b) $\mathbb{P}(Z < -0.5)$; (c) $\mathbb{P}(-1 \le Z \le 2)$; (d) $\mathbb{P}(|Z| > 1)$.

Exercise 12.2 (Standardizing)**.** Adult systolic blood pressure is modeled as $X \sim \mathcal{N}(120, 12^2)$ mmHg. (a) What fraction of adults have pressure above 140 (a common "high" cutoff)? (b) Below 100? (c) Between 108 and 132? Relate part (c) to the empirical rule.

Exercise 12.3 (Inverse lookups)**.** For $Z \sim \mathcal{N}(0, 1)$, find the cutoffs (using the table or known critical values): (a) the value c with $\mathbb{P}(Z > c) = 0.025$; (b) the value c with $\mathbb{P}(Z \le c) = 0.90$; (c) the symmetric interval $[-c, c]$ containing exactly 99% of the mass.

Exercise 12.4 (Adding normals)**.** Let $X \sim \mathcal{N}(3, 4)$ and $Y \sim \mathcal{N}(-1, 5)$ be independent. Find the distribution of (a) $2X - 1$; (b) $X + Y$; (c) $X - Y$; (d) $\frac{1}{2}(X + Y)$. In each case give the mean and variance and state why the result is still normal.

Exercise 12.5 (The shrinking error)**.** A single thermometer reading is $\mathcal{N}(\mu, 0.4^2)$ degrees, where μ is the true temperature. (a) How many independent readings must you average so that the standard deviation of the average is below 0.1 degrees? (b) With that many readings, what is the chance the average is within 0.1 of the truth? Use Corollary 12.1.

Exercise 12.6 (Why $\sqrt{2\pi}$, your turn)**.** Show that $\int_{-\infty}^{\infty} e^{-ax^2}\,\mathrm{d}x = \sqrt{\pi/a}$ for $a > 0$, by the polar-coordinates square trick of Theorem 12.1. Then set $a = 1/(2\sigma^2)$ and confirm that $\frac{1}{\sigma\sqrt{2\pi}}\int_{-\infty}^{\infty} e^{-(x-\mu)^2/(2\sigma^2)}\,\mathrm{d}x = 1$, i.e. that the general normal density really integrates to 1.

Exercise 12.7 (A proof: oddness gives the mean)**.** Prove carefully that $\mathbb{E}[Z] = 0$ for $Z \sim \mathcal{N}(0,1)$ by splitting $\int_{-\infty}^{\infty} z\,\varphi(z)\,\mathrm{d}z$ into $\int_{-\infty}^{0}$ and $\int_{0}^{\infty}$ and using the substitution $z \mapsto -z$ on one piece. Where exactly does the symmetry (evenness of φ) get used?

Exercise 12.8 (Variance by parts)**.** Reproduce the integration-by-parts computation of $\mathrm{Var}(Z) = 1$ from Proposition 12.1, filling in the missing step: explain in your own words why the boundary term $\big[-z\,e^{-z^2/2}\big]_{-\infty}^{\infty}$ is zero. (Hint: compare the growth of z to the decay of $e^{-z^2/2}$.)

Exercise 12.9 (The fat-tail check)**.** A risk model assumes daily losses are $\mathcal{N}(0, (1\%)^2)$. (a) Under this model, what is the probability of a daily loss exceeding 5%? (b) Exceeding 10%? (c) Empirically, large markets see 5%+ down days every few years. Roughly how many years between 5% down days does the normal model predict (assume 250 trading days per year)? Comment on what this says about the normal as a model for financial risk, connecting to the paradox box in Section 12.9.

Exercise 12.10 (Conceptual: shape versus parameters)**.** Explain, in a few sentences each and without heavy algebra: (a) why the peak of a $\mathcal{N}(\mu, \sigma^2)$ density gets *lower* as σ grows; (b) why standard deviations of independent normals add "in quadrature" rather than straight; (c) why a single function Φ suffices to answer questions about every normal distribution, regardless of μ and σ.

Exercise 12.11 (The maximum-entropy view, challenge)**.** It is a deep fact that among all distributions on $\mathbb{R}$ with a given mean μ and variance σ^2, the normal is the one with the *largest entropy* – the "most spread out" or "least committal" such distribution. Without proving it, explain in words why this is an appealing justification for using the normal as a default model of a quantity when you know only its mean and variance, and connect it to the error-curve story of Section 12.1. (We return to entropy in the information-theory chapter.)

Part III

Multiple Random Variables

Chapter 13

Joint Distributions

Up to now we have mostly followed one random quantity at a time. We rolled a die and watched the single number that came up; we measured one person's height; we counted the number of heads in a string of coin flips. A random variable, in that picture, is a single needle on a single dial, twitching from trial to trial.

But one dial is rarely what a real problem hands you. The interesting questions almost always involve *several* random quantities at once, and – this is the whole point – the quantities are usually *tangled together*. A person's height and weight are both random, but tall people tend to weigh more. The temperature tomorrow and the temperature the day after are both uncertain, but they move in sympathy. A patient's blood pressure and their risk of stroke are both random, and the first tells you something about the second. To reason about any of this you cannot study the variables one at a time and then staple the answers together. You need the *joint* distribution: a single object that describes how two (or more) random variables behave *together*, including how they lean on one another.

This chapter builds that object from the ground up. We start with the discrete case, where everything is a table of numbers you can literally point at. Then we cross into the continuous case, where the joint distribution becomes a *surface* hovering over the plane and probability becomes *volume* under that surface. Along the way we will extract the pieces we already know how to handle – the *marginal* distribution of each variable alone, and the *conditional* distribution of one given the other – and we will pin down exactly what it means for two random variables to be *independent*. By the end you should be able to look at a contour plot and read off the story of two intertwined quantities.

13.1 Two Dials at Once: The Joint PMF

Let us begin where intuition is safest: with discrete random variables, where a probability distribution is just a list of outcomes and their probabilities.

Intuition

Think of two random variables X and Y as two dials read off the *same* experiment. One spin of the experiment produces one reading on each dial, i.e. one pair (x, y). The joint distribution is nothing more than the answer to: "how often does each *pair* of readings occur?" If you

have ever made a two-way table – rows for one category, columns for another, counts in the cells – you have already met a joint distribution.

Definition 13.1 (Joint PMF)**.** Let X and Y be discrete random variables defined on the same sample space. Their *joint probability mass function* is

$$p_{X,Y}(x, y) \;=\; \mathbb{P}(X = x,\; Y = y),$$

where the comma means "and": the event that $X = x$ *and* $Y = y$ simultaneously. A valid joint PMF satisfies $p_{X,Y}(x, y) \geq 0$ for all (x, y) and

$$\sum_{x} \sum_{y} p_{X,Y}(x, y) \;=\; 1.$$

The single most important mental move in this chapter is to stop thinking of x and y as separate axes you process in turn, and to start thinking of the *pair* (x, y) as the basic atom. The joint PMF lives on the grid of all possible pairs.

Example 13.1 (A weather two-way table)**.** A meteorologist records two things each morning: X, whether it is cloudy ($X = 1$) or clear ($X = 0$); and Y, whether it rains that day ($Y = 1$) or not ($Y = 0$). Over many mornings she estimates the joint PMF in Table 13.1.

	$Y = 0$ (dry)	$Y = 1$ (rain)	row total
$X = 0$ (clear)	0.55	0.05	0.60
$X = 1$ (cloudy)	0.15	0.25	0.40
col. total	0.70	0.30	1.00

Table 13.1: Joint PMF of cloud cover X and rain Y. The four interior cells are $p_{X,Y}(x, y)$; the margins (literally the edges of the table) are the marginal PMFs.

The four interior numbers are the whole joint distribution. They sum to 1. Notice $p_{X,Y}(1, 1) = 0.25$: a cloudy, rainy day happens about a quarter of the time. And $p_{X,Y}(0, 1) = 0.05$: rain out of a clear sky is rare. The table already encodes a relationship – rain clusters with clouds – and our whole job is to learn how to read that relationship off precisely.

What is really happening

A joint PMF is not two distributions glued together; it is *one* distribution over pairs. The reason this matters: knowing how X behaves alone and how Y behaves alone is generally *not enough* to reconstruct the table. In Table 13.1 the row totals (how clouds behave) and column totals (how rain behaves) are fixed, but you could rearrange the four interior cells, keeping those totals, and get a completely different relationship. The interior holds information the margins cannot see. That extra information – the *dependence* – is the entire reason joint distributions deserve their own chapter.

13.1.1 Reading probabilities of events off the joint PMF

Once you have the joint PMF, the probability of *any* event described in terms of X and Y is just a sum over the matching cells. Want $\mathbb{P}(X \leq Y)$? Sum $p_{X,Y}(x, y)$ over all pairs with $x \leq y$. Want

$\mathbb{P}(X+Y=1)$? Sum over the cells on the anti-diagonal. The joint PMF is a complete description: every probabilistic question about the pair reduces to adding up the right cells.

Example 13.2 (Two dice, sum and max)**.** Roll two fair dice. Let X be the value on the first die and Y the value on the second. Because the dice are physically separate and fair, every pair (x, y) with $x, y \in \{1, \dots, 6\}$ is equally likely, so

$$p_{X,Y}(x, y) = \tfrac{1}{36}, \qquad x, y \in \{1, \dots, 6\}.$$

This is the discrete uniform distribution on a 6×6 grid. Now ask: what is $\mathbb{P}(\max(X, Y) \le 4)$? The event $\max(X, Y) \le 4$ means *both* dice are at most 4, i.e. all pairs in the 4×4 lower-left block: 16 cells, each of weight $1/36$, giving $16/36 = 4/9$. And $\mathbb{P}(X + Y = 7)$? The anti-diagonal $(1, 6), (2, 5), \dots, (6, 1)$ has 6 cells, so $6/36 = 1/6$. Every question is a cell count divided by 36.

Figure 13.1 draws this joint PMF as what it literally is: a grid of equal-probability points, with the two events above shaded. Reading a probability is reading off how many dots fall in the shaded set.

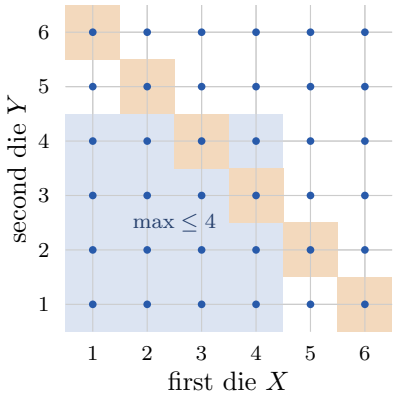


Figure 13.1: The joint PMF of two fair dice: 36 equally likely points, each of weight $1/36$. Blue block: $\max(X, Y) \le 4$ (16 points, probability $4/9$). Orange anti-diagonal: $X + Y = 7$ (6 points, probability $1/6$). For a discrete joint distribution, a probability is just a weighted dot count.

13.2 Marginals: Collapsing the Table

Suppose you have the joint PMF but you only care about X. You want $p_X(x) = \mathbb{P}(X = x)$, ignoring Y entirely. How do you get it?

Intuition

"$X = x$" happens whenever $X = x$ *and* Y takes *some* value – any value at all. So to find how often $X = x$, you sweep across every possible y and add up the probabilities. On the two-way table this is exactly summing along a row (or column). That is why these are called *marginal* distributions: historically you wrote the row and column sums in the *margins* of the table.

Definition 13.2 (Marginal PMFs)**.** From the joint PMF $p_{X,Y}$, the *marginal PMFs* of X and of Y are

$$p_X(x) = \sum_y p_{X,Y}(x, y), \qquad p_Y(y) = \sum_x p_{X,Y}(x, y).$$

Why it works

Why does summing out Y give the right answer for X? Because the events $\{X = x, Y = y\}$, as y ranges over all values, are *disjoint* (you cannot have two different y readings on one trial) and *exhaustive* of the event $\{X = x\}$ (some y always occurs). By the addition rule for disjoint events,

$$\mathbb{P}(X = x) = \mathbb{P}\Big(\bigcup_y \{X = x, Y = y\}\Big) = \sum_y \mathbb{P}(X = x, Y = y).$$

This is the law of total probability applied to the partition of $\{X = x\}$ by the value of Y. Nothing more mysterious is happening than "slice the event into non-overlapping pieces and add the pieces."

In Table 13.1, the marginal of X is the column of row totals, $p_X(0) = 0.60$, $p_X(1) = 0.40$; the marginal of Y is the row of column totals, $p_Y(0) = 0.70$, $p_Y(1) = 0.30$. So it is clear 60% of the time and dry 70% of the time.

Common trap

You can always go from the joint to the marginals (just add along rows/columns). You can *almost never* go the other way. Given only p_X and p_Y, there are usually many joint PMFs consistent with them. Marginals throw away the relationship; collapsing the table is a one-way street. Whenever someone hands you "the distribution of X" and "the distribution of Y" and asks about both together, your first question must be: *what is their joint behavior?*

13.3 Conditionals: Zooming Into One Row

Marginals forget about Y. Conditionals do the opposite: they *fix* a value of one variable and ask how the other is distributed in that slice of the world.

Intuition

Suppose you walk outside and see that it is cloudy ($X = 1$). You have learned something. Among cloudy days, how often does it rain? You are no longer looking at the whole table – only at the cloudy row. But that row's entries (0.15 and 0.25) sum to 0.40, not 1. To turn a row of a joint table into a genuine probability distribution, you must *renormalize*: divide by the row total so the conditional probabilities sum to 1.

Definition 13.3 (Conditional PMF)**.** For any x with $p_X(x) > 0$, the *conditional PMF* of Y given $X = x$ is

$$p_{Y|X}(y \mid x) \;=\; \mathbb{P}(Y = y \mid X = x) \;=\; \frac{p_{X,Y}(x, y)}{p_X(x)}.$$

For each fixed x this is a valid PMF in y: it is nonnegative and, summing over y, the numerator sums to $p_X(x)$, which cancels the denominator to give 1.

This is just the ordinary definition of conditional probability, $\mathbb{P}(A \mid B) = \mathbb{P}(A \cap B)/\mathbb{P}(B)$, with $A = \{Y = y\}$ and $B = \{X = x\}$. The denominator $p_X(x)$ is exactly the marginal we computed by summing the row – which is why marginals and conditionals are two sides of one coin.

Example 13.3 (Rain given clouds)**.** From Table 13.1, given a cloudy day ($X = 1$, row total 0.40):

$$p_{Y|X}(1 \mid 1) = \frac{0.25}{0.40} = 0.625, \qquad p_{Y|X}(0 \mid 1) = \frac{0.15}{0.40} = 0.375.$$

So once you know it is cloudy, the rain probability jumps to 62.5% – far above the unconditional 30%. Given a clear day ($X = 0$, row total 0.60):

$$p_{Y|X}(1 \mid 0) = \frac{0.05}{0.60} \approx 0.083.$$

Clouds raise the rain probability from 8.3% to 62.5%. The variables clearly inform each other; that is dependence, made quantitative.

Figure 13.2 contrasts the marginal of Y with the two conditional distributions. Conditioning rebuilds the bars: a clear sky pushes almost all the mass onto "dry," a cloudy sky shifts the balance toward "rain." If X and Y were independent, all three panels would be identical.

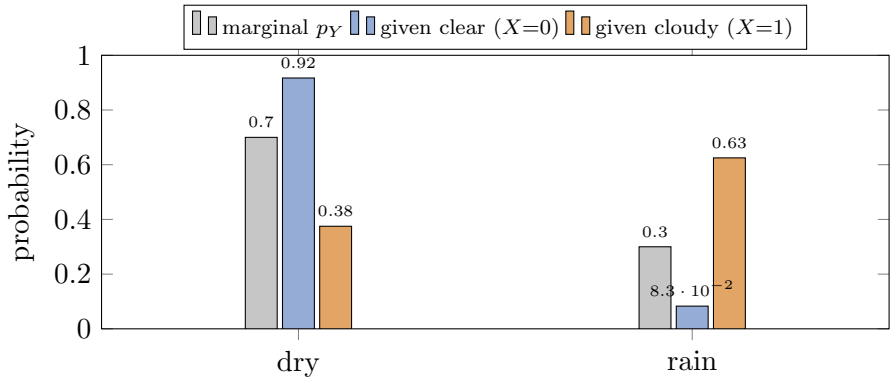


Figure 13.2: The marginal distribution of rain (gray) versus the two conditional distributions. Each set of bars sums to 1: conditioning fixes a row of Table 13.1 and renormalizes it. The bars move because the variables are dependent.

Focus

Three objects, one table:

- **Joint** $p_{X,Y}(x, y)$: a single cell – "both at once."
- **Marginal** $p_X(x)$: a row total – "one variable, forget the other."
- **Conditional** $p_{Y|X}(y \mid x)$: a row *rescaled to sum to one* – "the other variable, given this one is fixed."

And the bridge that ties them is one identity, the *chain rule*:

$$p_{X,Y}(x, y) = p_X(x)\, p_{Y|X}(y \mid x) = p_Y(y)\, p_{X|Y}(x \mid y).$$

Joint = marginal × conditional. Read it left to right and it builds the joint from a margin and a slice; read it right to left and it is the definition of the conditional. You will use this constantly.

13.4 Independence of Random Variables

We can now say precisely what it means for two random variables to have *nothing to do with each other*.

Intuition

X and Y are independent if learning the value of one tells you *nothing* about the distribution of the other. Concretely: the conditional distribution of Y given $X = x$ does not depend on x at all – every row of the table, once renormalized, looks identical. If knowing X never changes your beliefs about Y, the two carry no information about each other.

Definition 13.4 (Independence, discrete case)**.** Discrete random variables X and Y are *independent*, written $X \perp\!\!\!\perp Y$, if their joint PMF factors into the product of the marginals:

$$p_{X,Y}(x, y) = p_X(x)\, p_Y(y) \qquad \text{for every pair } (x, y).$$

Equivalently, $p_{Y|X}(y \mid x) = p_Y(y)$ for all x with $p_X(x) > 0$: the conditional equals the marginal, so conditioning on X changes nothing.

Why it works

Why is the product rule the right definition? Independence of *events* means $\mathbb{P}(A \cap B) = \mathbb{P}(A)\,\mathbb{P}(B)$. Random-variable independence demands this for *every* pair of events of the form $A = \{X = x\}$ and $B = \{Y = y\}$:

$$\mathbb{P}(X = x, Y = y) = \mathbb{P}(X = x)\ \mathbb{P}(Y = y) \quad \text{for all } x, y.$$

That is exactly the factorization $p_{X,Y} = p_X p_Y$. And it is genuinely stronger than independence of just one chosen pair of events – it must hold across the *whole* grid. If even one cell fails to equal the product of its margins, the variables are dependent.

Example 13.4 (Two dice are independent; sum and max are not)**.** For the two fair dice of Example 13.2, $p_{X,Y}(x, y) = 1/36$ while $p_X(x) = p_Y(y) = 1/6$, and indeed $\frac{1}{6} \cdot \frac{1}{6} = \frac{1}{36}$. So $X \perp\!\!\!\perp Y$: the two physical dice know nothing about each other.

But now let $S = X + Y$ and $M = \max(X, Y)$. These are *not* independent. Knowing $S = 2$ forces $M = 1$ (only $(1, 1)$ gives sum 2), whereas knowing $S = 12$ forces $M = 6$. Learning the sum sharply constrains the max, so $S \not\perp\!\!\!\perp M$. The moral: independence is a property of *specific* variables, not of the experiment. Functions built from independent variables are usually *not* independent of each other.

Common trap

Independence requires the factorization at *every* cell. In Table 13.1, $p_X(1)p_Y(1) = 0.40 \times 0.30 = 0.12$, but the actual joint cell is $p_{X,Y}(1, 1) = 0.25 \neq 0.12$. One mismatch is enough: clouds and rain are dependent. You never need to check all cells to *disprove* independence – a single counterexample does it.

History

The word "independence" is older than its modern definition. Early probabilists – Cardano in the 1500s tallying dice, then Pascal and Fermat in their 1654 letters on games of chance – multiplied probabilities together whenever the trials "did not interfere," but they had no formal notion of a random variable, let alone a joint distribution. It was the systematization of the subject in the 19th and early 20th centuries, culminating in Kolmogorov's 1933 axioms, that turned the intuitive "does not interfere" into the crisp algebraic statement $p_{X,Y} = p_X p_Y$. What had been a modeling instinct became a checkable equation.

13.5 Crossing to the Continuum: Joint Densities

Now let X and Y be continuous – say a person's height and weight, each able to take any real value in a range. We can no longer ask $\mathbb{P}(X = x, Y = y)$, because for a continuous variable any single value has probability zero, and so does any single point in the plane. The fix is exactly the one-dimensional fix, lifted to two dimensions: probability is no longer mass sitting on points, but *volume* sitting under a surface.

Intuition

In one dimension, a density $f_X(x)$ is a curve, and $\mathbb{P}(a \le X \le b)$ is the *area* under the curve between a and b. In two dimensions, a joint density $f_{X,Y}(x, y)$ is a *surface* floating above the plane, and the probability that (X, Y) lands in a region R is the *volume* of the solid under the surface and above R. Height of the surface = how densely probability is packed near that point. Tall surface = likely neighborhood; flat ground = unlikely.

Definition 13.5 (Joint PDF)**.** A function $f_{X,Y} : \mathbb{R}^2 \to \mathbb{R}$ is a *joint probability density function* of (X, Y) if $f_{X,Y}(x, y) \ge 0$ everywhere,

$$\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f_{X,Y}(x, y)\,\mathrm{d}x\,\mathrm{d}y = 1,$$

and for every (reasonable) region $R \subseteq \mathbb{R}^2$,

$$\mathbb{P}\big((X, Y) \in R\big) = \iint_R f_{X,Y}(x, y)\,\mathrm{d}x\,\mathrm{d}y.$$

Connection

This is the multivariable calculus you may have met as the double integral, now wearing a probabilistic hat. The double integral was *invented* to compute volumes under surfaces; here that volume *is* a probability. The same toolkit transfers wholesale: to integrate over a non-rectangular region you set up limits carefully (often the inner limits depend on the outer variable); switching the order of integration is Fubini's theorem; and for circular regions you change to polar coordinates, picking up the Jacobian factor r. If those phrases are rusty, the companion calculus book's chapter on double integrals is the exact prerequisite. Probability gives you a vivid reason to care: the answers are chances.

13.5.1 The surface picture

A joint density really is a landscape. Figure 13.3 shows the density of a pair of independent standard normals – the two-dimensional bell, a single smooth hill centered at the origin. The probability of any region is how much "dirt" sits over it.

The 2D bell: density of two independent standard normals

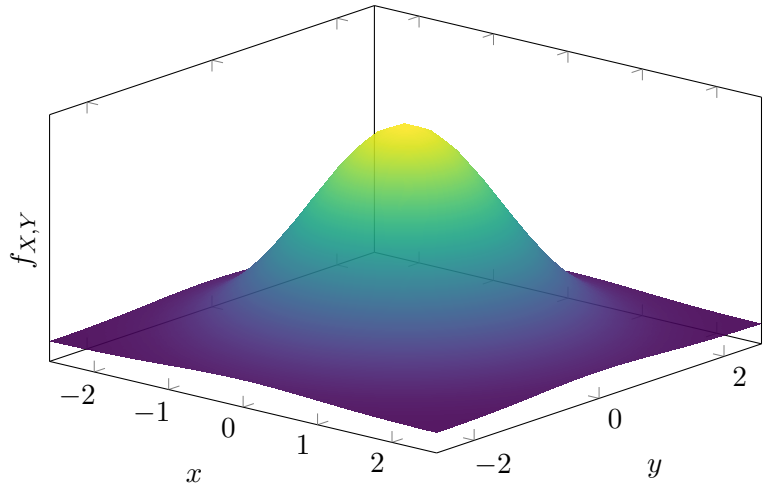


Figure 13.3: The joint density $f_{X,Y}(x,y) = \frac{1}{2\pi}e^{-(x^2+y^2)/2}$ as a surface. Probability is volume under this hill. It is tallest at the origin (the most likely neighborhood) and decays smoothly in every direction.

Surfaces are vivid but hard to read quantitatively. The flat-map version is the *contour plot*: curves of constant density, exactly like elevation lines on a topographic map. Figure 13.4 draws them. Concentric circles mean the density depends only on distance from the center, and equal spacing of the probability mass in every direction – a signature, as we will see, of independence plus rotational symmetry.

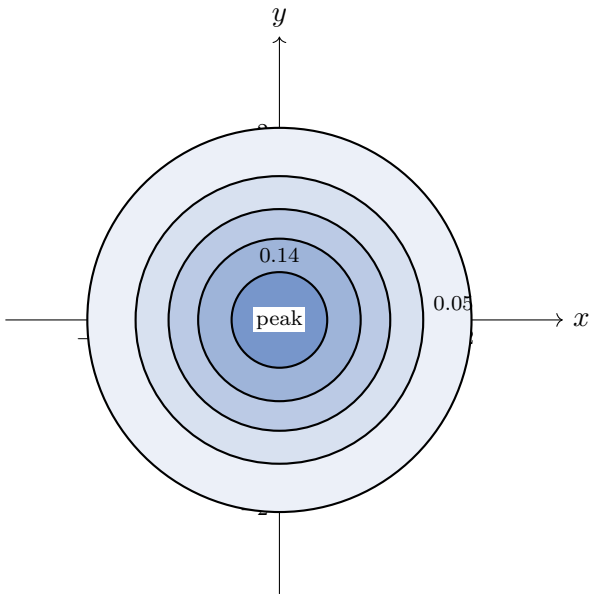


Figure 13.4: Contours (level curves) of the 2D bell, drawn as the exact circles $r = \sqrt{-2\ln(2\pi L)}$ for densities $L = 0.02, 0.05, 0.08, 0.11, 0.14$. Each ring is a set of points with equal density; the rings crowd together where the surface falls off fastest. Reading a joint density as contours is the everyday tool of data scientists looking at scatterplots of two variables.

What is really happening

What is the height $f_{X,Y}(x, y)$, really? It is *not* a probability – it can exceed 1. It is a probability *per unit area*, a density. The precise statement is the small-rectangle approximation:

$$\mathbb{P}\left(x < X \le x + \mathrm{d}x,\ y < Y \le y + \mathrm{d}y\right) \approx f_{X,Y}(x, y)\,\mathrm{d}x\,\mathrm{d}y.$$

Density times area equals (approximate) probability. This is the continuous echo of "$p_{X,Y}(x, y)$ is the probability of that cell": a cell of the grid has become an infinitesimal tile of the plane, and the density is the probability of the tile divided by its area.

13.6 Marginals and Conditionals in the Continuum

Every idea from the discrete table survives the crossing, with sums replaced by integrals. To get the marginal of X, you "sum out" Y – but summing out a continuous variable means integrating it away.

Definition 13.6 (Marginal PDFs)**.**

$$f_X(x) = \int_{-\infty}^{\infty} f_{X,Y}(x, y)\,\mathrm{d}y, \qquad f_Y(y) = \int_{-\infty}^{\infty} f_{X,Y}(x, y)\,\mathrm{d}x.$$

Intuition

Picture the surface again. To find $f_X(x)$ – how dense X is near a particular x, ignoring Y – you stand at that x, slice the hill with a vertical plane parallel to the y-axis, and *add up the entire slice*: the integral $\int f_{X,Y}(x,y)\,dy$ is the area of that cross-section. Tall fat slices mean x is a likely value of X. You are flattening the surface onto the x-axis by letting all the y-probability rain straight down.

Definition 13.7 (Conditional PDF)**.** For any x with $f_X(x) > 0$, the conditional density of Y given $X = x$ is

$$f_{Y|X}(y \mid x) = \frac{f_{X,Y}(x,y)}{f_X(x)}.$$

For fixed x this is a valid density in y: it is the cross-section slice of the surface at x, *rescaled* (divided by its own area $f_X(x)$) so that it integrates to 1.

Notice the perfect parallel. The conditional density is a vertical slice of the joint surface, renormalized – just as the discrete conditional was a row of the table, renormalized. And conditioning on the measure-zero event $\{X = x\}$, which looked illegal (you cannot divide by $\mathbb{P}(X = x) = 0$), becomes legitimate the moment you work with densities instead of probabilities: you divide by the *density* $f_X(x)$, which is generally positive.

Definition 13.8 (Independence, continuous case)**.** X and Y are independent if and only if their joint density factors:

$$f_{X,Y}(x,y) = f_X(x)\,f_Y(y) \quad \text{for all } (x,y),$$

equivalently $f_{Y|X}(y \mid x) = f_Y(y)$. For the 2D bell, $\frac{1}{2\pi}e^{-(x^2+y^2)/2} = \big(\frac{1}{\sqrt{2\pi}}e^{-x^2/2}\big)\big(\frac{1}{\sqrt{2\pi}}e^{-y^2/2}\big)$: it splits into an x-part times a y-part, so the two coordinates are independent standard normals. The exponent splitting as a sum $x^2 + y^2$ is exactly what makes the density a product.

Why this matters

The contour plot carries a clean tell for independence. A joint density factors into $g(x)h(y)$ exactly when the surface is a "product landscape" – the same y-profile at every x, up to vertical scaling. For such a density the contours are symmetric about lines parallel to the axes and never tilt. Dependence shows up as contours that *tilt* or *bend*: an ellipse slanted at 45° says "large x tends to go with large y." You will meet that tilted ellipse again in the next chapter under the name *correlation*. The geometry of the contours *is* the relationship.

13.7 Uniform on a Region

The cleanest continuous joint distribution is the one where the surface is perfectly flat over some region and zero outside it: all of probability is spread evenly, like jam over toast cut to the shape of R.

Definition 13.9 (Uniform on a region)**.** (X,Y) is *uniform on a region* $R \subseteq \mathbb{R}^2$ of finite area $|R|$ if

$$f_{X,Y}(x,y) = \begin{cases} \dfrac{1}{|R|}, & (x,y) \in R, \\ 0, & \text{otherwise.} \end{cases}$$

The constant $1/|R|$ is forced: the volume under a flat sheet of height h over a region of area $|R|$ is $h \cdot |R|$, and that must equal 1.

What is really happening

When the density is constant, *probability becomes geometry.* For any region $A \subseteq R$,

$$\mathbb{P}\big((X,Y) \in A\big) = \iint_A \frac{1}{|R|}\,\mathrm{d}x\,\mathrm{d}y = \frac{|A|}{|R|} = \frac{\text{area of } A}{\text{area of } R}.$$

Computing a probability is computing an area ratio – nothing more. This is the modern, rigorous form of the old "geometric probability" that Buffon used in 1733 when he dropped needles on a floor of parallel lines and turned the chance of a crossing into an integral involving π.

Example 13.5 (Uniform on the unit square is independent; on the triangle is not)**. Square.** Let (X,Y) be uniform on the unit square $R = [0,1] \times [0,1]$, so $f_{X,Y} = 1$ on the square. The marginal of X is

$$f_X(x) = \int_0^1 1\,\mathrm{d}y = 1, \quad 0 \le x \le 1,$$

i.e. $X \sim \text{Uniform}(0,1)$, and by symmetry $Y \sim \text{Uniform}(0,1)$. Since $f_{X,Y}(x,y) = 1 = f_X(x) f_Y(y)$, the coordinates are *independent.*

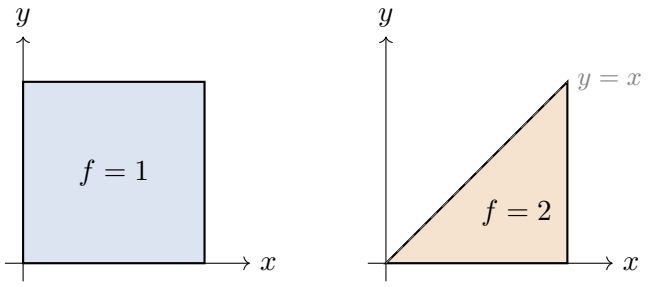
Triangle. Now let (X,Y) be uniform on the triangle $T = \{(x,y) : 0 \le x \le 1,\ 0 \le y \le x\}$ below the diagonal. Its area is $\frac{1}{2}$, so $f_{X,Y} = 2$ on T and 0 elsewhere. The marginal of X depends on where you are, because the height of the triangle grows with x:

$$f_X(x) = \int_0^x 2\,\mathrm{d}y = 2x, \quad 0 \le x \le 1.$$

So X is *not* uniform – larger x values are more likely, because there is more triangle above them. And the conditional of Y given $X = x$ is

$$f_{Y|X}(y \mid x) = \frac{2}{2x} = \frac{1}{x}, \quad 0 \le y \le x,$$

which is $\text{Uniform}(0,x)$: given $X = x$, Y is uniform on the segment up to x. The conditional distribution depends on x, so X and Y are *dependent.* Same flat density, totally different story – because the *shape of the region* couples the variables. The region's geometry creates dependence even though the density is constant.



square: $X \perp\!\!\!\perp Y$ triangle: $X \not\perp\!\!\!\perp Y$

Figure 13.5: Two uniform distributions. Over the square the coordinates are independent; over the triangle the very shape of the support ties them together. Support shape alone can manufacture dependence.

Common trap

A fast independence test for *any* joint density: X and Y can be independent only if the support (the set where $f_{X,Y} > 0$) is a *rectangle* (possibly infinite), aligned with the axes *and* the density factors there. If the region where the density lives is a triangle, a disk, or anything where the allowed range of y *depends on* x, the variables are automatically dependent – because the conditional support changes with x. The triangle above fails this test on sight, before you compute a single integral.

13.8 A Fully Worked Continuous Example

Let us run the entire machine on one nontrivial density, computing the normalizing constant, both marginals, a conditional, and a region probability.

Example 13.6 (A coupled density on the unit square)**.** Let

$$f_{X,Y}(x,y) = c\,(x+y), \qquad 0 \le x \le 1,\ 0 \le y \le 1,$$

and zero elsewhere. This surface tilts upward toward the corner $(1,1)$.

Step 1: find c. The total volume must be 1:

$$\int_0^1\!\!\int_0^1 c(x+y)\,\mathrm{d}x\,\mathrm{d}y = c\int_0^1 \Big[\tfrac{x^2}{2} + xy\Big]_{x=0}^{1} \mathrm{d}y = c\int_0^1 \Big(\tfrac{1}{2} + y\Big)\mathrm{d}y = c\Big(\tfrac{1}{2} + \tfrac{1}{2}\Big) = c.$$

So $c = 1$, and $f_{X,Y}(x,y) = x + y$ on the square.

Step 2: marginal of X. Integrate out y:

$$f_X(x) = \int_0^1 (x+y)\,\mathrm{d}y = \Big[xy + \tfrac{y^2}{2}\Big]_{y=0}^{1} = x + \tfrac{1}{2}, \qquad 0 \le x \le 1.$$

A quick sanity check: $\int_0^1 (x + \frac{1}{2})\,\mathrm{d}x = \frac{1}{2} + \frac{1}{2} = 1$. Good – it is a valid density. By the symmetry of $x + y$ in its two arguments, $f_Y(y) = y + \frac{1}{2}$ as well.

Step 3: are they independent? Test the factorization:

$$f_X(x)f_Y(y) = \big(x+\tfrac{1}{2}\big)\big(y+\tfrac{1}{2}\big) = xy + \tfrac{x}{2} + \tfrac{y}{2} + \tfrac{1}{4},$$

which is *not* equal to $x+y$. So X and Y are *dependent*. (The support is a nice square, so the rectangle test does not rule out independence – but the density still fails to factor, and that settles it.)

Step 4: a conditional. Given $X = x$,

$$f_{Y|X}(y \mid x) = \frac{x+y}{x+\frac{1}{2}}, \qquad 0 \le y \le 1.$$

For $x = 0$ this is $f_{Y|X}(y \mid 0) = \frac{y}{1/2} = 2y$, a density that *rises* with y – given $X = 0$, larger Y is favored. For $x \to \infty$... well, x is capped at 1, but at $x = 1$ we get $\frac{1+y}{3/2} = \frac{2}{3}(1+y)$, which is much flatter. So the shape of Y's distribution genuinely shifts as x changes; that shifting *is* the dependence, seen up close.

Step 5: a region probability. What is $\mathbb{P}(X+Y \le 1)$? Integrate the density over the lower-left triangle $\{x \ge 0,\, y \ge 0,\, x+y \le 1\}$:

$$\mathbb{P}(X+Y \le 1) = \int_0^1 \int_0^{1-x} (x+y)\,\mathrm{d}y\,\mathrm{d}x.$$

The inner integral:

$$\int_0^{1-x} (x+y)\,\mathrm{d}y = \Big[xy + \tfrac{y^2}{2}\Big]_0^{1-x} = x(1-x) + \tfrac{(1-x)^2}{2}.$$

Expand: $x - x^2 + \frac{1}{2}(1 - 2x + x^2) = \frac{1}{2} - \frac{1}{2}x^2$. Then

$$\mathbb{P}(X+Y \le 1) = \int_0^1 \Big(\tfrac{1}{2} - \tfrac{1}{2}x^2\Big)\mathrm{d}x = \tfrac{1}{2} - \tfrac{1}{6} = \tfrac{1}{3}.$$

So the pair falls in the lower triangle a third of the time. (Sensible: the density leans toward $(1,1)$, the *upper* corner, so it should favor the upper triangle $X+Y > 1$, which gets the remaining 2/3.)

Work it yourself

Try the twin computation yourself with $f_{X,Y}(x,y) = c\,x\,y$ on the unit square. (a) Show $c = 4$. (b) Find f_X and f_Y. (c) Show that this one *does* factor, so here $X \perp\!\!\!\perp Y$ – a useful contrast with the $x+y$ example, where the same square support gave dependence. Multiplication factors; addition does not.

13.9 A Discrete Worked Example with Real Dependence

Example 13.7 (Drawing without replacement)**.** A box holds 3 red and 2 blue balls. Draw two balls *without* replacement. Let $X = 1$ if the first ball is red (else 0), and $Y = 1$ if the second is red. Because the first draw changes what is left, X and Y should be dependent – let us build the joint PMF and confirm.

By the chain rule, $p_{X,Y}(x,y) = p_X(x)\,p_{Y|X}(y \mid x)$. The first draw: $p_X(1) = 3/5$, $p_X(0) = 2/5$. After a red first draw, 2 red and 2 blue remain, so $p_{Y|X}(1 \mid 1) = 2/4$. After a blue first draw, 3 red and 1 blue remain, so $p_{Y|X}(1 \mid 0) = 3/4$. Multiplying:

	$Y=0$	$Y=1$	p_X
$X=0$	$\frac{2}{5}\cdot\frac{1}{4} = \frac{1}{10}$	$\frac{2}{5}\cdot\frac{3}{4} = \frac{3}{10}$	$\frac{2}{5}$
$X=1$	$\frac{3}{5}\cdot\frac{2}{4} = \frac{3}{10}$	$\frac{3}{5}\cdot\frac{2}{4} = \frac{3}{10}$	$\frac{3}{5}$
p_Y	$\frac{2}{5}$	$\frac{3}{5}$	1

Table 13.2: Joint PMF for two draws without replacement.

The four cells sum to 1. Now read off the lessons.

The marginals are equal. Summing the columns gives $p_Y(1) = \frac{3}{10} + \frac{3}{10} = \frac{3}{5}$, the same as $p_X(1)$! The second ball is red with probability 3/5 too. This is the (initially surprising) "symmetry of draws": *before you look*, the second draw is just as likely to be red as the first. Marginally, Y behaves exactly like X.

But they are dependent. Check a cell: $p_{X,Y}(1,1) = 3/10$, while $p_X(1)p_Y(1) = \frac{3}{5}\cdot\frac{3}{5} = \frac{9}{25} = 0.36 \neq 0.30$. The factorization fails, so $X \not\perp\!\!\!\perp Y$. Equivalently, $p_{Y|X}(1 \mid 1) = \frac{1}{2}$ but $p_{Y|X}(1 \mid 0) = \frac{3}{4}$: knowing the first ball *does* change the odds on the second. Identical marginals, genuine dependence – a clean reminder that marginals do not determine the joint.

Paradox

Two coins, the "other child," and why the joint is everything. A family has two children; you learn "at least one is a boy." What is the chance both are boys? Let X, Y be the sexes of the (ordered) children, each independent and equally likely $\{B, G\}$, so the four pairs BB, BG, GB, GG each have probability 1/4. "At least one boy" removes only GG, leaving $\{BB, BG, GB\}$, of which one is BB:

$$\mathbb{P}(\text{both boys} \mid \text{at least one boy}) = \frac{1/4}{3/4} = \frac{1}{3},$$

not 1/2. The trap is to reason about one child at a time; the resolution is to reason on the *joint* sample space of pairs and condition there. Now change the information to "the *older* child is a boy" – this fixes $X = B$ and leaves $\{BB, BG\}$, giving 1/2. Same family, different conditioning event, different answer. The paradox dissolves the instant you insist on conditioning inside the joint distribution rather than juggling marginals. The joint distribution is not a luxury; it is the only place these questions are well-posed.

13.10 Beyond Two: Joint Distributions of Many Variables

Everything generalizes to n random variables. The joint PMF or PDF lives on n-tuples, and marginals come from summing or integrating out all the variables you do not want. The chain rule telescopes:

$$p(x_1, \ldots, x_n) = p(x_1)\,p(x_2 \mid x_1)\,p(x_3 \mid x_1, x_2) \cdots p(x_n \mid x_1, \ldots, x_{n-1}).$$

And *mutual independence* means the joint is the product of all the marginals:

$$f(x_1, \ldots, x_n) = f_1(x_1)\, f_2(x_2) \cdots f_n(x_n).$$

The most important special case in all of statistics is a sequence of variables that are *independent and identically distributed* (i.i.d.): they share one marginal f and their joint is simply $\prod_i f(x_i)$. That single product is the foundation of the laws of large numbers, the central limit theorem, and the likelihood function of nearly every statistical model.

Application

The likelihood function is a joint density read backwards. In machine learning and statistics you observe data $x_1, \ldots, x_n$ and posit a model with an unknown parameter θ. If the data are i.i.d. from a density $f(x; \theta)$, their *joint* density is

$$L(\theta) = \prod_{i=1}^{n} f(x_i; \theta).$$

Read as a function of the *data* with θ fixed, this is an ordinary joint density. Read as a function of θ with the data fixed, it is the *likelihood*, and the value of θ that maximizes it is the maximum likelihood estimate – the workhorse of fitting models, from logistic regression to neural networks (where the log-likelihood becomes the cross-entropy loss). The product structure – which is to say, the assumption of independence – is what turns a hairy joint object into a sum of log-terms you can differentiate. Whole fields rest on the factorization $f = \prod f_i$.

Connection

When all the variables are jointly normal, the joint density is governed by a *covariance matrix* Σ, and the contours of the density are ellipses whose axes are the eigenvectors of Σ. Independence corresponds to a diagonal Σ (axis-aligned ellipses, like our 2D bell); dependence tilts the ellipses. This is where probability shakes hands with linear algebra: the geometry of a Gaussian cloud is the spectral theory of its covariance matrix. We develop covariance in the next chapter, and the multivariate normal in full later; the companion linear algebra book's chapter on symmetric matrices and the spectral theorem is the natural partner.

13.11 Summary

Pause

The whole chapter in one breath:

- A **joint distribution** describes two (or more) random variables *together*: a PMF $p_{X,Y}(x, y)$ on a grid (discrete), or a density $f_{X,Y}(x, y)$ as a surface with probability = volume (continuous).
- **Marginals** come from collapsing the joint – sum or integrate out the unwanted variable: $p_X(x) = \sum_y p_{X,Y}$, $f_X(x) = \int f_{X,Y}\, dy$. One-way street: joint $\to$ marginals, never back.
- **Conditionals** come from fixing one variable and renormalizing the slice: $p_{Y|X} =$

$p_{X,Y}/p_X$, $f_{Y|X} = f_{X,Y}/f_X$. The bridge is the **chain rule**: joint = marginal × conditional.

- **Independence** is the factorization $p_{X,Y} = p_X p_Y$ (resp. $f_{X,Y} = f_X f_Y$), equivalently "conditional = marginal." It must hold *everywhere*; one mismatch breaks it. Non-rectangular support guarantees dependence.
- **Uniform on a region** turns probability into an area ratio: $\mathbb{P}((X,Y) \in A) = |A|/|R|$.

Hold onto the surface-and-contours picture; it is the mental image that makes the next chapters – covariance, correlation, conditional expectation, and the multivariate normal – feel inevitable rather than abstract.

13.12 Exercises

Exercise 13.1 (Reading a table)**.** The joint PMF of (X,Y) with $X,Y \in \{0,1\}$ is $p(0,0) = 0.1$, $p(0,1) = 0.2$, $p(1,0) = 0.3$, $p(1,1) = 0.4$. (a) Find both marginal PMFs. (b) Find $p_{Y|X}(y \mid 1)$. (c) Are X and Y independent? Justify with one cell.

Exercise 13.2 (Normalizing and marginalizing)**.** Let $f_{X,Y}(x,y) = c\,x^2 y$ for $0 \le x \le 1$, $0 \le y \le 2$, and 0 otherwise. (a) Find c. (b) Find f_X and f_Y. (c) Show X and Y are independent and name each marginal's family if you can.

Exercise 13.3 (A genuinely coupled density)**.** Let $f_{X,Y}(x,y) = 6(1-y)$ for $0 \le x \le y \le 1$, and 0 otherwise. (a) Verify it integrates to 1. (b) Find $f_X(x)$ and $f_Y(y)$. (c) Find $f_{X|Y}(x \mid y)$ and identify it as a familiar distribution. (d) Explain in one sentence why X and Y *cannot* be independent, using only the shape of the support.

Exercise 13.4 (Uniform on a disk)**.** Let (X,Y) be uniform on the unit disk $\{x^2 + y^2 \le 1\}$. (a) What is the value of the density on the disk? (b) Compute $\mathbb{P}(X^2 + Y^2 \le \frac{1}{4})$ using an area ratio – no integration needed. (c) Find the marginal $f_X(x)$ for $-1 \le x \le 1$ by integrating out y, and note that X is *not* uniform.

Exercise 13.5 (Region probability)**.** With $f_{X,Y}(x,y) = x + y$ on the unit square (the density of Example 13.6), compute $\mathbb{P}(Y > X)$. (Hint: integrate over the upper triangle, or argue from symmetry of $x + y$ to get the answer almost for free.)

Exercise 13.6 (Conditional that varies)**.** For the same density $f_{X,Y}(x,y) = x+y$ on the unit square, show that $\mathbb{E}[Y \mid X = x] = \dfrac{3x+2}{6x+3}$ by computing $\int_0^1 y\, f_{Y|X}(y \mid x)\,\mathrm{d}y$. Evaluate it at $x = 0$ and $x = 1$ and comment on why the answer changes with x.

Exercise 13.7 (Independence and support, conceptual)**.** Decide (with a one-line reason each) whether independence is possible for (X,Y) uniform on: (a) the square $[0,1]^2$; (b) the triangle $0 \le y \le x \le 1$; (c) the L-shaped region that is the unit square with its upper-right quarter removed; (d) the rectangle $[0,2] \times [0,5]$.

Exercise 13.8 (Without replacement, generalized)**.** A box has r red and b blue balls; draw two without replacement, with X, Y the red-indicators as in Example 13.7. (a) Build the 2×2 joint PMF in terms of r and b. (b) Show that the marginal of Y equals the marginal of X (the symmetry of draws). (c) Show that X and Y are *always* dependent when $r, b \ge 1$, by exhibiting one cell whose

value differs from the product of its margins. (Intuition: removing one ball changes the composition for the second draw, so $p_{Y|X}(1 \mid 1) \neq p_{Y|X}(1 \mid 0)$.)

Exercise 13.9 (Proof: factored support is necessary)**.** Prove the claim used in the chapter: if X and Y are independent with marginal densities positive on intervals I_X and I_Y respectively, then the support of $f_{X,Y}$ is the rectangle $I_X \times I_Y$. (Hint: $f_{X,Y}(x, y) = f_X(x) f_Y(y) > 0$ exactly when both factors are positive.) Conclude that any joint density whose support is *not* a product of intervals must come from dependent variables.

Exercise 13.10 (Mixing discrete and continuous, challenge)**.** Let $N \sim \text{Bernoulli}(1/2)$ choose between two recipes for a continuous Y: if $N = 0$, $Y \sim \text{Uniform}(0, 1)$; if $N = 1$, $Y \sim \text{Uniform}(0, 2)$. (a) Write the conditional density $f_{Y|N}(y \mid n)$. (b) Find the marginal density $f_Y(y)$ via the law of total probability, $f_Y(y) = \sum_n p_N(n)\, f_{Y|N}(y \mid n)$, and sketch it. (c) Are N and Y independent? Explain using $f_{Y|N}$.

Chapter 14

Covariance and Correlation

So far, when we have studied two random variables together, we have mostly been content to know their *joint distribution* – the full table or surface that tells us $\mathbb{P}(X = x, Y = y)$ for every pair. That object contains everything there is to know. But "everything" is often too much. If I tell you the joint distribution of the height and weight of a random adult, I have handed you a two-dimensional cloud of probability; what you actually *wanted* was a single number that answers the question: *when one goes up, does the other tend to go up too?*

That single number is what this chapter is about. We start from a story and a picture, then sharpen it into the two standard summaries of how two random quantities move together: **covariance** and its rescaled cousin, the **correlation coefficient**. Along the way we will see why correlation is trapped forever inside the interval $[-1, 1]$ (a fact that turns out to be the Cauchy–Schwarz inequality wearing a disguise), why a correlation of zero does *not* mean independence, why correlation is not causation, and how stacking these numbers into a matrix opens the door to principal component analysis and the geometry of multivariate data.

14.1 The Question Covariance Answers

Intuition

Variance asks: *how much does a single quantity wiggle around its mean?* Covariance asks the natural two-variable sequel: *when one quantity wiggles above its mean, does the other tend to wiggle above its mean too, or below it?* Covariance is, almost literally, "co-variance" – variance shared between two things.

Let us make the story concrete. Imagine we sample students at random and record two numbers for each: X, the hours they slept the night before an exam, and Y, their exam score. Each student is a dot in the plane. If well-rested students tend to score higher, the dots will lean upward to the right. If sleep makes no difference, the dots will form a shapeless blob. If somehow sleep *hurt* scores, the dots would lean downward.

The trick that turns this picture into a number is to stop looking at the raw values X and Y and instead look at the *deviations* from their means,

$$X - \mu_X, \qquad Y - \mu_Y, \qquad \text{where } \mu_X = \mathbb{E}[X],\ \mu_Y = \mathbb{E}[Y].$$

A deviation is positive when the value is above average and negative when it is below. Now multiply the two deviations together for each student and ask what that product tends to be.

- A well-rested, high-scoring student has $X - \mu_X > 0$ and $Y - \mu_Y > 0$, so the product $(X - \mu_X)(Y - \mu_Y)$ is *positive*.
- A sleep-deprived, low-scoring student has $X - \mu_X < 0$ and $Y - \mu_Y < 0$: a negative times a negative is again *positive*.
- A student who is rested but scores low (or tired but scores high) contributes a *negative* product.

So the sign of the product $(X - \mu_X)(Y - \mu_Y)$ records whether a given point sits in the "agreeing" quadrants (both above average or both below) or the "disagreeing" quadrants. The *average* of that product over the whole population is the covariance.

Definition 14.1 (Covariance)**.** For random variables X and Y with finite variances, the **covariance** of X and Y is

$$\operatorname{Cov}(X, Y) \;=\; \mathbb{E}\big[(X - \mu_X)(Y - \mu_Y)\big], \qquad \mu_X = \mathbb{E}[X],\ \mu_Y = \mathbb{E}[Y].$$

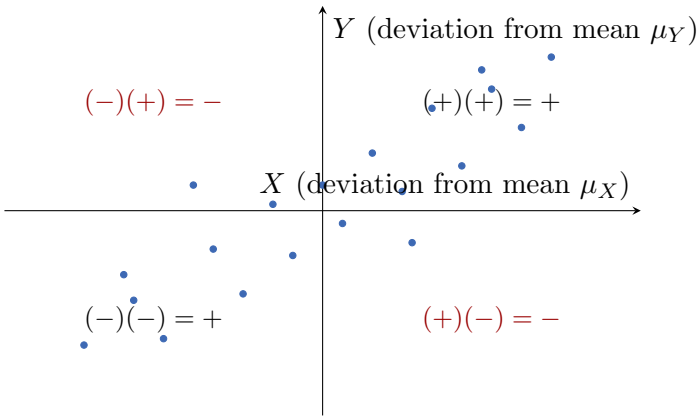


Figure 14.1: Deviations from the means split the plane into four quadrants. Points in the upper-right and lower-left contribute positive products; points in the other two contribute negative products. Covariance is the average product. This cloud leans upward, so the positive quadrants win and the covariance is positive.

What is really happening

Covariance is a *tug of war between quadrants*. Each data point casts a vote: "agree" (positive product) or "disagree" (negative product), weighted by how far from the means it sits. Far-out points in the corners shout loudly; points near the center barely whisper. The covariance is the net result of all those votes. A point exactly on either mean line ($X = \mu_X$ or $Y = \mu_Y$) contributes nothing, because one of its deviations is zero. This is why a perfectly symmetric blob averages out to zero covariance: the four quadrants cancel.

14.1.1 A computational shortcut

The definition in terms of deviations is the one to *think* with, but it is rarely the easiest one to *compute* with. Expanding the product gives a formula that needs only the three expectations $\mathbb{E}[XY]$, $\mathbb{E}[X]$, $\mathbb{E}[Y]$.

Proposition 14.1 (Shortcut formula)**.**

$$\operatorname{Cov}(X,Y) \;=\; \mathbb{E}[XY] - \mathbb{E}[X]\,\mathbb{E}[Y].$$

Why it works

Expand the product inside the expectation and use linearity. Writing $\mu_X = \mathbb{E}[X]$ and $\mu_Y = \mathbb{E}[Y]$ (these are constants),

$$\begin{aligned}
\operatorname{Cov}(X,Y) &= \mathbb{E}\left[(X-\mu_X)(Y-\mu_Y)\right] \\
&= \mathbb{E}\left[XY - \mu_X Y - \mu_Y X + \mu_X\mu_Y\right] \\
&= \mathbb{E}[XY] - \mu_X\,\mathbb{E}[Y] - \mu_Y\,\mathbb{E}[X] + \mu_X\mu_Y \\
&= \mathbb{E}[XY] - \mu_X\mu_Y - \mu_Y\mu_X + \mu_X\mu_Y \\
&= \mathbb{E}[XY] - \mathbb{E}[X]\,\mathbb{E}[Y].
\end{aligned}$$

The two middle terms collapsed because $\mathbb{E}[\mu_X Y] = \mu_X\,\mathbb{E}[Y] = \mu_X\mu_Y$, and similarly for the other. Notice the family resemblance to the variance shortcut $\operatorname{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2$ – indeed, setting $Y = X$ recovers it exactly, since $\operatorname{Cov}(X,X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2 = \operatorname{Var}(X)$.

Focus

Covariance is the *generalization of variance to two variables*. The single most useful identity to keep in your head is

$$\operatorname{Cov}(X,X) = \operatorname{Var}(X).$$

Variance is just a variable's covariance with itself. Everything we know about variance is a special case of what we are about to learn about covariance.

14.1.2 First worked example: two dice

Example 14.1 (Sum and first die)**.** Roll two fair, independent dice. Let D_1 be the first die's value and let $S = D_1 + D_2$ be the sum. Intuitively, when the first die is large the sum tends to be large, so we expect a positive covariance. Let us compute $\operatorname{Cov}(D_1, S)$.

Bilinearity (proved formally in Section 14.2) is the fast route, but doing it by hand first builds trust. We have $\mathbb{E}[D_1] = 3.5$ and $\mathbb{E}[S] = 7$. For the cross term,

$$\mathbb{E}[D_1 S] = \mathbb{E}[D_1(D_1 + D_2)] = \mathbb{E}[D_1^2] + \mathbb{E}[D_1 D_2].$$

Now $\mathbb{E}[D_1^2] = \frac{1}{6}(1+4+9+16+25+36) = \frac{91}{6}$, and because the dice are independent, $\mathbb{E}[D_1 D_2] = \mathbb{E}[D_1]\,\mathbb{E}[D_2] = 3.5 \times 3.5 = 12.25$. So

$$\mathbb{E}[D_1 S] = \tfrac{91}{6} + 12.25 = 15.1\overline{6} + 12.25 = 27.41\overline{6}.$$

Therefore

$$\mathrm{Cov}(D_1, S) = \mathbb{E}[D_1 S] - \mathbb{E}[D_1]\,\mathbb{E}[S] = 27.41\overline{6} - 3.5 \times 7 = 27.41\overline{6} - 24.5 = 2.91\overline{6} = \tfrac{35}{12}.$$

Positive, as our intuition demanded. And note $\frac{35}{12} = \mathrm{Var}(D_1)$: the covariance of a die with the sum equals the variance of that one die, because the *other* die contributes nothing to the covariance (it is independent of D_1). That is a preview of the bilinearity rules.

14.2 The Algebra of Covariance

Covariance obeys a small set of rules that make it pleasant to manipulate. They all follow from the linearity of expectation, and they will pay for themselves many times over – especially when we compute the variance of a sum.

Proposition 14.2 (Properties of covariance)**.** *For random variables X, Y, Z and constants a, b, c:*

1. ***Symmetry:*** $\mathrm{Cov}(X, Y) = \mathrm{Cov}(Y, X)$.
2. ***Self-covariance:*** $\mathrm{Cov}(X, X) = \mathrm{Var}(X)$.
3. ***Constants don't covary:*** $\mathrm{Cov}(X, c) = 0$.
4. ***Pulling out scalars and shifts:*** $\mathrm{Cov}(aX + b,\ cY + d) = ac\ \mathrm{Cov}(X, Y)$.
5. ***Bilinearity:*** $\mathrm{Cov}(aX + bY,\ Z) = a\ \mathrm{Cov}(X, Z) + b\ \mathrm{Cov}(Y, Z)$*, and likewise in the second slot.*

Why it works

Symmetry is obvious from the definition (multiplication commutes). Self-covariance we already saw. For (3), a constant has zero deviation: $c - \mathbb{E}[c] = 0$, so the product is identically zero. For (4), adding constants b, d shifts the means by the same amounts, so the deviations $X - \mu_X$ and $Y - \mu_Y$ are unchanged by b and d, while the scalars a, c factor straight out of the product. For bilinearity, deviations add: the deviation of $aX + bY$ is $a(X - \mu_X) + b(Y - \mu_Y)$, and expectation is linear, so

$$\mathrm{Cov}(aX + bY, Z) = \mathbb{E}\left[(a(X - \mu_X) + b(Y - \mu_Y))(Z - \mu_Z)\right] = a\,\mathrm{Cov}(X, Z) + b\,\mathrm{Cov}(Y, Z).$$

Intuition

Property (4) carries a crucial lesson. The shifts b and d *vanish* – moving your data left/right or up/down cannot change how two variables co-move. But the scales a and c *survive and multiply in*. This means covariance is sensitive to *units*. Measure a person's height in centimeters instead of meters and every covariance involving height multiplies by 100. Covariance has a unit problem, and that problem is exactly what correlation will fix.

14.2.1 The payoff: variance of a sum

Here is where bilinearity earns its keep. Back in the variance chapter we claimed that variances add for independent variables but warned that the general story needed covariance. Now we can tell the whole story.

Theorem 14.1 (Variance of a sum)**.** *For any random variables X and Y with finite variance,*

$$\mathrm{Var}(X+Y) = \mathrm{Var}(X) + \mathrm{Var}(Y) + 2\,\mathrm{Cov}(X,Y).$$

More generally, for $X_1, \dots, X_n$,

$$\mathrm{Var}\Big(\sum_{i=1}^{n} X_i\Big) = \sum_{i=1}^{n} \mathrm{Var}(X_i) + 2\sum_{1\le i<j\le n} \mathrm{Cov}(X_i, X_j) = \sum_{i=1}^{n}\sum_{j=1}^{n} \mathrm{Cov}(X_i, X_j).$$

Why it works

Just expand using $\mathrm{Var}(W) = \mathrm{Cov}(W,W)$ and bilinearity:

$$\mathrm{Var}(X+Y) = \mathrm{Cov}(X+Y, X+Y) = \mathrm{Cov}(X,X) + \mathrm{Cov}(X,Y) + \mathrm{Cov}(Y,X) + \mathrm{Cov}(Y,Y),$$

and the two cross terms are equal by symmetry, giving the factor of 2. For the n-term version, $\mathrm{Cov}\left(\sum_i X_i, \sum_j X_j\right)$ expands into the full double sum of all pairwise covariances; the diagonal terms $i = j$ are the variances, and the off-diagonal terms come in symmetric pairs.

Why this matters

This single identity is one of the workhorses of all of probability and statistics. It says: *the variance of a total is the sum of the variances plus a correction for how the parts move together.* If the parts are uncorrelated, the correction vanishes and variances simply add – which is why, in the chapters ahead, the standard error of an average shrinks like $1/\sqrt{n}$ when samples are independent. When the parts are positively correlated, totals are *more* volatile than you would guess from the pieces alone; this is the mathematics behind why a poorly diversified portfolio of correlated stocks is so risky, and why diversification (mixing weakly correlated assets) reduces risk. We will return to this in the application box below.

Example 14.2 (Variance of a binomial, the slick way)**.** Let $X \sim \mathrm{Binomial}(n,p)$. Write $X = \sum_{i=1}^{n} I_i$ where I_i is the indicator of a success on trial i. The trials are independent, so $\mathrm{Cov}(I_i, I_j) = 0$ for $i \neq j$ (we will prove independence kills covariance in a moment). Hence all the cross terms in Theorem 14.1 vanish and

$$\mathrm{Var}(X) = \sum_{i=1}^{n} \mathrm{Var}(I_i) = \sum_{i=1}^{n} p(1-p) = np(1-p).$$

No messy sums over the binomial pmf – the covariance bookkeeping did the work.

14.3 Independence, Covariance, and a Famous Trap

Theorem 14.2 (Independence implies zero covariance)**.** *If X and Y are independent (and have finite variances), then* $\mathrm{Cov}(X,Y) = 0$.

Why it works

Independence means the joint factors: $\mathbb{E}[XY] = \mathbb{E}[X]\,\mathbb{E}[Y]$ (this is the factorization of expectation for independent variables, proved in the joint-distributions chapter). Plug this straight into the shortcut formula:

$$\mathrm{Cov}(X,Y) = \mathbb{E}[XY] - \mathbb{E}[X]\,\mathbb{E}[Y] = \mathbb{E}[X]\,\mathbb{E}[Y] - \mathbb{E}[X]\,\mathbb{E}[Y] = 0.$$

When $\mathrm{Cov}(X,Y) = 0$ we call X and Y **uncorrelated**. So independence is a *sufficient* condition for being uncorrelated. The burning question is whether it is *necessary* – does uncorrelated imply independent? The answer is a flat **no**, and the reason rewards a careful look, because the confusion is one of the most common in applied probability.

Common trap

Uncorrelated does NOT mean independent. Covariance only detects *linear* co-movement – the straight-line tendency for one variable to rise as the other rises. A relationship can be ironclad and deterministic yet have zero covariance, simply because it is not a *straight-line* relationship. Zero covariance rules out a linear trend; it rules out nothing else.

Example 14.3 (The U-shape that fools covariance)**.** Let X be uniform on $\{-2,-1,0,1,2\}$ and let $Y = X^2$. Then Y is a *perfect, deterministic function* of X – knowing X tells you Y exactly, so they are about as far from independent as two variables can be. Yet

$$\mathbb{E}[X] = 0, \qquad \mathbb{E}[XY] = \mathbb{E}[X^3] = \tfrac{1}{5}\big((-2)^3 + (-1)^3 + 0 + 1^3 + 2^3\big) = 0,$$

so $\mathrm{Cov}(X,Y) = \mathbb{E}[XY] - \mathbb{E}[X]\,\mathbb{E}[Y] = 0 - 0 = 0$. The covariance is exactly zero. What happened? The relationship is a symmetric U (a parabola): as X goes from negative to positive, Y first falls then rises. The "rising on the right" part exactly cancels the "falling on the left" part, and covariance, which only listens for a consistent linear tilt, hears nothing.

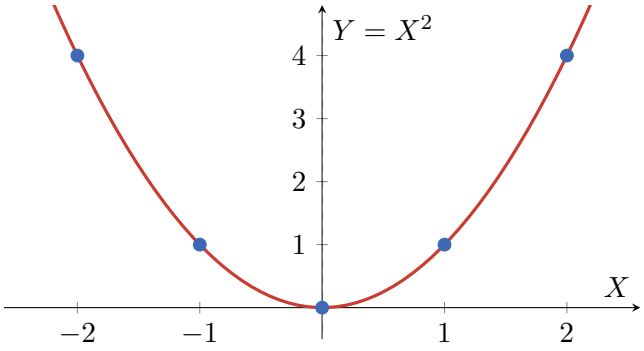


Figure 14.2: $Y = X^2$ with X symmetric about 0. The dependence is total, but it is purely curved – there is no linear tilt – so $\mathrm{Cov}(X,Y) = 0$. Covariance is blind to symmetric, nonlinear relationships.

What is really happening

The deep reason for the trap: covariance is a *single number*, and a single number can only summarize the *first-order, straight-line* part of a relationship. Think of it as the best you

could do if you were forced to predict Y from X using a straight line. If the true relationship is a symmetric curve, the best straight line is flat, and a flat line means zero covariance – even though X and Y are locked together. The lesson generalizes: *the absence of a linear signal is not the absence of a relationship.* Always plot your data; a scatterplot sees the U-shape that the covariance number erases.

There is exactly one important setting where uncorrelated *does* upgrade to independent: when the pair (X, Y) is jointly Gaussian (bivariate normal). For jointly normal variables, zero correlation is equivalent to independence. This is a special feature of the normal distribution, not a general truth, and we will meet it again when we study the multivariate normal.

14.4 Standardizing: The Correlation Coefficient

Covariance has the unit problem we flagged earlier. Its numerical value depends on the scales of X and Y, so a covariance of 2.9 (as in our two-dice example) is meaningless in isolation – is that a strong relationship or a weak one? We cannot tell without knowing how spread out the variables are. The fix is to *standardize*: divide out each variable's own spread, measured by its standard deviation. The result is the **correlation coefficient**.

Definition 14.2 (Correlation coefficient)**.** For random variables X and Y with $\mathrm{SD}(X), \mathrm{SD}(Y) > 0$, the **Pearson correlation coefficient** is

$$\mathrm{Corr}(X, Y) \;=\; \rho_{X,Y} \;=\; \frac{\mathrm{Cov}(X, Y)}{\mathrm{SD}(X)\ \mathrm{SD}(Y)} \;=\; \frac{\mathrm{Cov}(X, Y)}{\sqrt{\mathrm{Var}(X)\ \mathrm{Var}(Y)}}.$$

Intuition

The cleanest way to see what correlation is: first *standardize* each variable into a z-score by subtracting its mean and dividing by its standard deviation,

$$\tilde{X} = \frac{X - \mu_X}{\mathrm{SD}(X)}, \qquad \tilde{Y} = \frac{Y - \mu_Y}{\mathrm{SD}(Y)}.$$

Each $\tilde{X}$, $\tilde{Y}$ now has mean 0 and variance 1 – pure, unit-free numbers stripped of their original scale. Then

$$\mathrm{Corr}(X, Y) = \mathrm{Cov}(\tilde{X}, \tilde{Y}) = \mathbb{E}[\tilde{X}\tilde{Y}].$$

Correlation is just the covariance of the *standardized* variables. It asks the same quadrant tug-of-war question as covariance, but now both axes are measured in the universal currency of standard deviations, so the answer is comparable across any pair of variables in the world.

Because Corr divides covariance by exactly the scale factors that covariance was sensitive to, correlation is **scale- and shift-invariant**: for positive constants a, c and any b, d,

$$\mathrm{Corr}(aX + b,\ cY + d) = \mathrm{Corr}(X, Y).$$

(If exactly one of a, c is negative, the correlation flips sign; if both are negative, the signs cancel and Corr is unchanged, because each negative scale reverses the direction of co-movement.) Switching from meters to centimeters, or from Celsius to Fahrenheit, leaves correlation untouched. That invariance is precisely why correlation – not covariance – is the number people actually report.

14.5 Why Correlation Lives in $[-1, 1]$

The central fact about correlation is that it is bounded:

$$-1 \;\le\; \mathrm{Corr}(X, Y) \;\le\; 1.$$

A correlation of $+1$ means perfect positive linear agreement; -1 means perfect negative linear agreement; 0 means no linear relationship. Values in between interpolate. But *why* can it never escape this interval? Why not a correlation of 1.4? There are two ways to see it.

14.5.1 The elementary proof: a variance can't be negative

Theorem 14.3 (Correlation bound)**.** *For any X, Y with positive, finite standard deviations, $-1 \le \mathrm{Corr}(X, Y) \le 1$. Moreover $\mathrm{Corr}(X, Y) = \pm 1$ if and only if Y is an exact linear function of X, i.e. $Y = aX + b$ for some constants $a \neq 0$ and b (with $a > 0$ for $+1$ and $a < 0$ for -1).*

Why it works

Standardize as above, so $\tilde{X}, \tilde{Y}$ each have variance 1, and let $\rho = \mathrm{Corr}(X, Y) = \mathbb{E}[\tilde{X}\tilde{Y}]$. Now consider the variance of the *sum* and the *difference* of the standardized variables – and remember that a variance can never be negative. Using Theorem 14.1 and $\mathrm{Var}(\tilde{X}) = \mathrm{Var}(\tilde{Y}) = 1$,

$$0 \le \mathrm{Var}(\tilde{X} + \tilde{Y}) = 1 + 1 + 2\rho = 2(1 + \rho) \implies \rho \ge -1,$$

$$0 \le \mathrm{Var}(\tilde{X} - \tilde{Y}) = 1 + 1 - 2\rho = 2(1 - \rho) \implies \rho \le 1.$$

Together these pin ρ inside $[-1, 1]$. And equality is forced exactly when one of those variances is *zero*. A variable with zero variance is constant, so $\rho = 1$ means $\tilde{X} - \tilde{Y}$ is constant – i.e. $\tilde{Y} = \tilde{X} + \text{const}$, which unwinds to $Y = aX + b$ with $a > 0$. Likewise $\rho = -1$ means $\tilde{X} + \tilde{Y}$ is constant, giving a perfect line with negative slope.

What is really happening

That proof repays a second look, because it explains *what the boundary means*. Correlation hits ± 1 precisely when the points stop being a cloud and collapse onto a single straight line – because that is the only way a variance can shrink to zero. The closer $|\rho|$ is to 1, the tighter the cloud hugs some line. So ρ is best read as a measure of *how close the scatter is to lying on a straight line*, with the sign telling you which way the line tilts.

14.5.2 The structural proof: Cauchy–Schwarz

The bound is no accident of probability. It is the probabilistic face of one of the most fundamental inequalities in mathematics: the Cauchy–Schwarz inequality. Seeing this connection turns the bound from a computational curiosity into a piece of geometry.

Connection

Covariance is an inner product; correlation is the cosine of an angle. Recall from linear algebra that for vectors $\mathbf{u}, \mathbf{v}$, the dot product satisfies the *Cauchy–Schwarz inequality*

$$|\langle \mathbf{u}, \mathbf{v} \rangle| \le \|\mathbf{u}\| \, \|\mathbf{v}\|,$$

with equality exactly when the vectors are parallel. Now think of mean-zero random variables as "vectors" and define their inner product to be

$$\langle U, V \rangle \;=\; \mathbb{E}[UV].$$

This genuinely *is* an inner product: it is symmetric, linear in each slot, and $\langle U, U \rangle = \mathbb{E}[U^2] \ge 0$, vanishing only when $U = 0$. Under this dictionary,

$$\langle U, V \rangle = \mathrm{Cov}(U, V), \qquad \|U\| = \sqrt{\mathbb{E}[U^2]} = \mathrm{SD}(U),$$

so Cauchy–Schwarz reads

$$|\mathrm{Cov}(U, V)| \le \mathrm{SD}(U)\ \mathrm{SD}(V) \quad \Longleftrightarrow \quad |\mathrm{Corr}(U, V)| \le 1.$$

And the geometric meaning of the ratio is exactly the cosine of the angle θ between the two vectors:

$$\mathrm{Corr}(X, Y) = \cos\theta = \frac{\langle U, V \rangle}{\|U\| \, \|V\|}.$$

Uncorrelated variables ($\rho = 0$) are *orthogonal* – they meet at a right angle. Perfectly correlated variables ($\rho = \pm 1$) are *parallel* (or anti-parallel). This is the same geometry as the angle between arrows in space; the companion linear-algebra book develops the inner-product and Cauchy–Schwarz story in full, and it is the exact engine running underneath correlation.

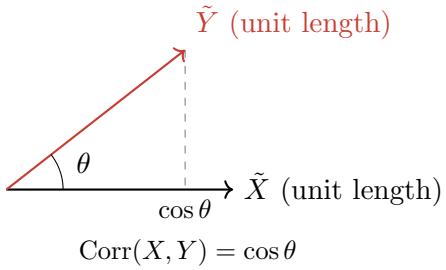


Figure 14.3: Mean-zero variables as unit vectors after standardizing. The correlation is the cosine of the angle between them: $\theta = 0°$ gives $\rho = 1$, $\theta = 90°$ gives $\rho = 0$ (uncorrelated = orthogonal), and $\theta = 180°$ gives $\rho = -1$.

History

The correlation coefficient was forged not by a mathematician but by a Victorian polymath obsessed with measuring heredity. *Sir Francis Galton*, in the 1880s, studied how the heights of children related to the heights of their parents. He plotted the data, drew an ellipse around the cloud, and noticed that tall parents tended to have tall – but on average slightly *less* tall

– children, a phenomenon he named "regression toward mediocrity" (the origin of the word *regression*). Galton sensed a number was hiding in the slope of that cloud. His protégé *Karl Pearson* made it precise around 1895, giving the formula $\rho = \mathrm{Cov}/(\mathrm{SD}\cdot\mathrm{SD})$ that still bears the name "Pearson correlation." What began as an attempt to quantify family resemblance became one of the central tools of all quantitative science. The deep geometric reading – correlation as the cosine of an angle – came later, once statisticians absorbed the language of inner-product spaces.

14.5.3 What different correlations look like

Numbers like $\rho = 0.3$ or $\rho = 0.8$ only become meaningful once you have seen the scatterplots they produce. Figure 14.4 shows clouds at several correlation levels. Train your eye: a correlation near 0 is a round blob; as $|\rho|$ climbs toward 1 the blob stretches into an ever-thinner cigar tilted along a line; the sign of ρ is the direction of the tilt.

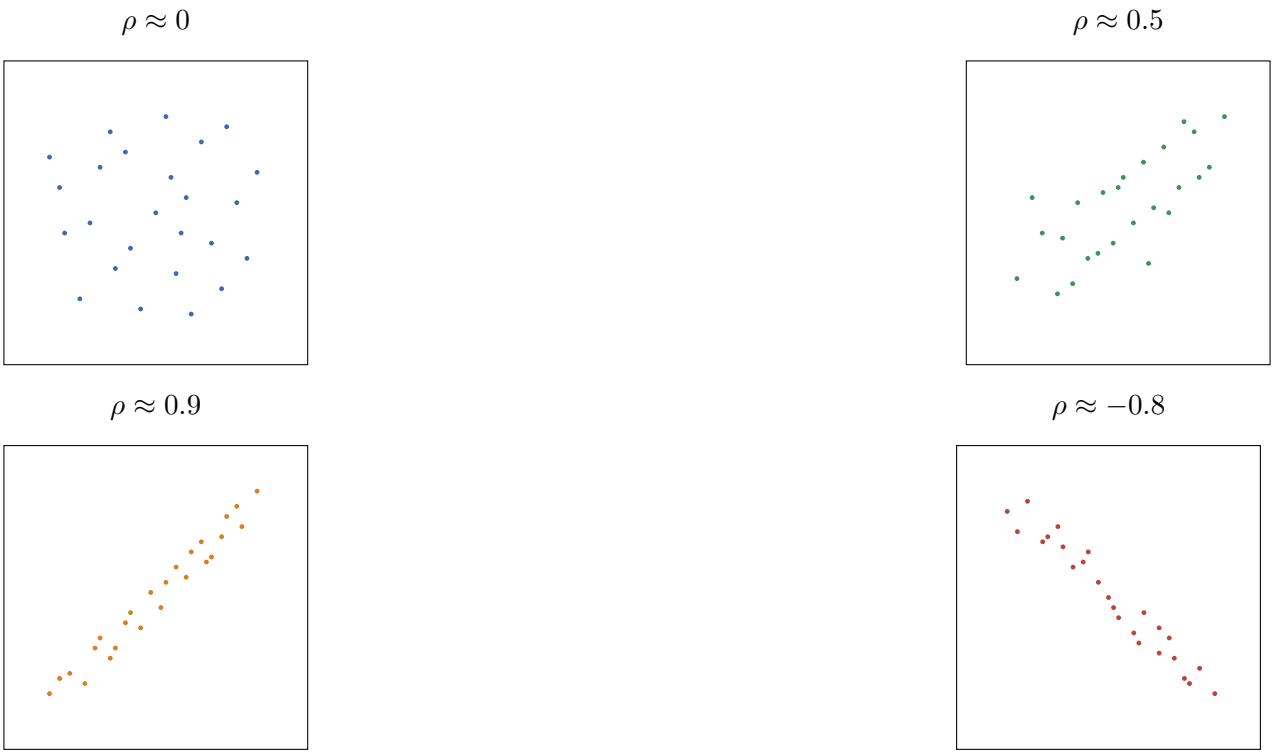


Figure 14.4: Scatter clouds at four correlation levels. Near $\rho = 0$ the cloud is a round blob; as $|\rho| \to 1$ it tightens into a tilted cigar. Positive ρ tilts up to the right; negative ρ tilts down. Note that ρ measures *tightness around a line*, not the line's steepness.

Common trap

Correlation is not slope. A common misreading is that a high correlation means a steep relationship. It does not. Correlation measures how *tightly* the points cluster around *some* line, regardless of how steep that line is. A gently sloping but razor-thin cloud can have

$\rho = 0.99$, while a steep but fat cloud can have $\rho = 0.3$. Steepness is the job of the regression slope (a different quantity); ρ only reports the strength of the linear pattern, scaled to $[-1, 1]$.

14.5.4 Worked example: correlation of two dice

Example 14.4 ($\mathrm{Corr}(D_1, S)$)**.** Return to Example 14.1. We found $\mathrm{Cov}(D_1, S) = \frac{35}{12}$. The variance of a single fair die is $\mathrm{Var}(D_1) = \frac{35}{12}$, and since the two dice are independent, $\mathrm{Var}(S) = \mathrm{Var}(D_1) + \mathrm{Var}(D_2) = \frac{35}{6}$. Therefore

$$\mathrm{Corr}(D_1, S) = \frac{\mathrm{Cov}(D_1, S)}{\sqrt{\mathrm{Var}(D_1)\,\mathrm{Var}(S)}} = \frac{35/12}{\sqrt{(35/12)(35/6)}} = \frac{35/12}{(35/12)\sqrt{2}} = \frac{1}{\sqrt{2}} \approx 0.707.$$

A clean answer. The first die "explains" the sum, but only halfway – the second die is an equal, independent contributor, so D_1 accounts for one of two equal sources of variation, and $\rho = 1/\sqrt{2}$. (If you summed n independent dice, the correlation of any one die with the total would be $1/\sqrt{n}$, fading as the crowd of other dice grows.)

14.6 Correlation Versus Causation

This is the most consequential warning in the chapter – one that escapes the mathematics entirely and reaches into how we reason about the world.

Common trap

Correlation is not causation. A nonzero correlation between X and Y tells you they move together. It says *nothing* about *why*. The co-movement could arise because X causes Y, because Y causes X, because some third variable Z drives both (a *confounder*), or by sheer coincidence in a finite sample. The formula cannot tell these apart, because it only ever saw the two columns of numbers, never the machinery that generated them.

Paradox

The ice cream and drowning paradox. Across the year, ice cream sales and drowning deaths are strongly positively correlated. Taken naively, this suggests ice cream causes drowning – ban the ice cream and save lives! Of course the real story is a lurking third variable: *summer*. Hot weather drives *both* ice cream sales *and* swimming (hence drownings). Neither one causes the other; a confounder, the season, drives both. This is the canonical illustration of a **confounded** correlation, and the cure is to control for the confounder – e.g. look at the correlation *within* a fixed month, where it largely disappears.
Statisticians collect these: the number of films Nicolas Cage appears in correlates with swimming-pool drownings; per-capita cheese consumption correlates with deaths by bedsheet entanglement. With enough variables to dredge, spurious high correlations are guaranteed to appear by chance alone. The mathematics is innocent; the interpretation is where people fall.

What is really happening

Why can't a single number distinguish causation? Because causation is a statement about *interventions* – what would happen to Y if we reached in and *forced* X to change – whereas correlation is a statement about *observation* – what Y tends to do when we passively watch X vary. Observation and intervention coincide only when there are no confounders and no reverse causation, which is precisely what we cannot verify from the data columns alone. This is why the gold standard for establishing causation is the *randomized controlled experiment*: by assigning X at random, the experimenter severs any link between X and lurking confounders, so that an observed correlation can finally be read as causation. The modern field of causal inference (Rubin, Pearl) is built to recover causal claims from observational data by modeling exactly the confounding structure that raw correlation ignores.

Application

A/B testing and why we randomize. Suppose a company notices that users who use its new feature have higher retention. Does the feature *cause* retention? Maybe – or maybe the kind of person who tries new features was already more engaged (a confounder: user type). The whole point of an **A/B test** is to assign the feature at random. Randomization makes the treatment group and control group statistically identical *in expectation* on every confounder, observed or not, so the only systematic difference is the feature itself. Then the observed difference in retention – a correlation between "got the feature" and "stayed" – can be read causally. This is correlation-versus-causation turned into an engineering discipline, and it underlies essentially all rigorous product experimentation, clinical trials, and policy evaluation.

14.7 The Covariance Matrix

So far we have correlated two variables at a time. Real data come with many columns at once – a person's height, weight, age, blood pressure; a portfolio's dozens of asset returns. We want *all* the pairwise covariances organized in one tidy object. That object is the **covariance matrix**, and it is the bridge from this chapter into the multivariate world.

Definition 14.3 (Covariance matrix)**.** For a random vector $\mathbf{X} = (X_1, X_2, \dots, X_n)^\top$, the **covariance matrix** Σ is the $n \times n$ matrix whose (i, j) entry is the covariance of the i-th and j-th components:

$$\Sigma_{ij} = \operatorname{Cov}(X_i, X_j), \qquad \Sigma = \begin{pmatrix} \operatorname{Var}(X_1) & \operatorname{Cov}(X_1, X_2) & \cdots & \operatorname{Cov}(X_1, X_n) \\ \operatorname{Cov}(X_2, X_1) & \operatorname{Var}(X_2) & \cdots & \operatorname{Cov}(X_2, X_n) \\ \vdots & \vdots & \ddots & \vdots \\ \operatorname{Cov}(X_n, X_1) & \operatorname{Cov}(X_n, X_2) & \cdots & \operatorname{Var}(X_n) \end{pmatrix}.$$

In compact vector notation, with $\boldsymbol{\mu} = \mathbb{E}[\mathbf{X}]$,

$$\Sigma = \mathbb{E}\left[(\mathbf{X} - \boldsymbol{\mu})(\mathbf{X} - \boldsymbol{\mu})^\top\right].$$

Look at the structure. The *diagonal* entries are the individual variances – each variable's covariance with itself. The *off-diagonal* entries are the pairwise covariances. The matrix is **symmetric** ($\Sigma_{ij} = \Sigma_{ji}$) because covariance is symmetric. And the correlation matrix is the same object with every entry Σ_{ij} replaced by $\Sigma_{ij}/(\operatorname{SD}(X_i)\operatorname{SD}(X_j))$, which puts 1s down the diagonal.

Proposition 14.3 (Covariance matrices are positive semidefinite)**.** *For any constant vector* $\mathbf{a} \in \mathbb{R}^n$*,*

$$\mathbf{a}^\top \Sigma\, \mathbf{a} = \mathrm{Var}\big(\mathbf{a}^\top \mathbf{X}\big) = \mathrm{Var}\Big(\sum_i a_i X_i\Big) \;\geq\; 0.$$

Hence Σ *is symmetric positive semidefinite.*

Why it works

The quadratic form $\mathbf{a}^\top \Sigma \mathbf{a}$ expands into $\sum_{i,j} a_i a_j \,\mathrm{Cov}(X_i, X_j)$, which by bilinearity (Theorem 14.1) is exactly $\mathrm{Cov}\left(\sum_i a_i X_i, \sum_j a_j X_j\right) = \mathrm{Var}\left(\sum_i a_i X_i\right)$. A variance is never negative, so the form is never negative. That is the definition of positive semidefinite.

Connection

From covariance matrix to PCA. Because Σ is symmetric and positive semidefinite, the spectral theorem from linear algebra guarantees it has an orthonormal basis of *eigenvectors* with nonnegative *eigenvalues*. That fact is the engine of **Principal Component Analysis (PCA)**, the most widely used dimensionality-reduction method in data science.

Here is the picture. Proposition 14.3 says $\mathbf{a}^\top \Sigma \mathbf{a}$ is the variance of the data projected onto direction $\mathbf{a}$. Ask: *which direction in space carries the most variance?* Maximizing $\mathbf{a}^\top \Sigma \mathbf{a}$ over unit vectors $\mathbf{a}$ is a classic eigenvalue problem, and its answer is the *top eigenvector* of Σ – the long axis of the data cloud. Its eigenvalue is the variance along that axis. The next eigenvector gives the most-spread direction orthogonal to the first, and so on. The eigenvectors are the natural axes of the ellipsoidal cloud, and the eigenvalues say how stretched the cloud is along each. PCA simply rotates your data into these axes and keeps the few with the largest eigenvalues, discarding directions where the data barely varies. So the covariance of this chapter, stacked into a matrix and handed to the spectral theorem, becomes the geometry of high-dimensional data. The companion linear-algebra book develops eigenvectors, the spectral theorem, and quadratic forms; here we see why they sit at the center of multivariate statistics.

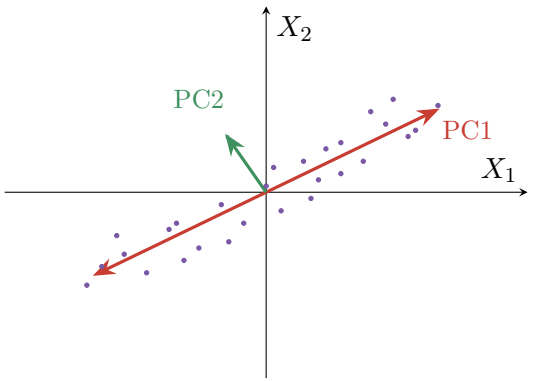


Figure 14.5: A correlated cloud and its principal axes. The first principal component (PC1, red) points along the direction of greatest variance – the top eigenvector of the covariance matrix Σ. The second (PC2, green) is orthogonal and captures the leftover spread. PCA rewrites the data in these natural axes.

Application

Portfolios and diversification. In finance, let $R_1, \dots, R_n$ be the returns of n assets and let $\mathbf{w}$ be the fractions of your money in each (so $\sum_i w_i = 1$). The portfolio return is $\mathbf{w}^\top \mathbf{R}$, and by Proposition 14.3 its risk (variance) is exactly

$$\text{Var(portfolio)} = \mathbf{w}^\top \Sigma \, \mathbf{w}.$$

Diversification is now visible in the algebra. If the assets are weakly correlated, the off-diagonal entries of Σ are small, the cross terms barely add, and spreading your money across many assets drives the portfolio variance down. If the assets are strongly positively correlated, the off-diagonal terms are large, the cross terms pile on, and "diversifying" buys you almost nothing – everything sinks together. Markowitz won a Nobel Prize for formalizing exactly this: choosing $\mathbf{w}$ to minimize $\mathbf{w}^\top \Sigma \mathbf{w}$ for a desired expected return. The entire theory of portfolio risk is the quadratic form of this chapter's covariance matrix.

14.8 Sample Covariance and Correlation

Everything above is about *true* (population) covariance, defined with expectations over the full distribution. In practice we never see the distribution; we see a finite sample $(x_1, y_1), \dots, (x_n, y_n)$ and must *estimate*. The natural estimators replace expectations with averages.

Definition 14.4 (Sample covariance and correlation)**.** Given paired data with sample means $\bar{x}, \bar{y}$, the **sample covariance** is

$$s_{xy} = \frac{1}{n-1} \sum_{i=1}^{n} (x_i - \bar{x})(y_i - \bar{y}),$$

and the **sample correlation** (Pearson's r) is

$$r = \frac{s_{xy}}{s_x \, s_y} = \frac{\sum_{i=1}^{n} (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^{n} (x_i - \bar{x})^2} \, \sqrt{\sum_{i=1}^{n} (y_i - \bar{y})^2}},$$

where s_x, s_y are the sample standard deviations.

The divisor $n - 1$ rather than n is the same *Bessel correction* you met for the sample variance: dividing by $n - 1$ makes the estimator unbiased, compensating for the fact that we used the data to estimate the means $\bar{x}, \bar{y}$ before measuring deviations from them. (Conveniently, the correction cancels in r – you get the same r whether you divide by n or $n - 1$ top and bottom.)

Example 14.5 (Sample correlation by hand)**.** Five paired observations:

x	1	2	3	4	5
y	2	1	4	3	5

Then $\bar{x} = 3$, $\bar{y} = 3$. Compute the deviation products:

$x_i - \bar{x}$	$y_i - \bar{y}$	$(x_i - \bar{x})(y_i - \bar{y})$	$(x_i - \bar{x})^2$
-2	-1	2	4
-1	-2	2	1
0	1	0	0
1	0	0	1
2	2	4	4

The sums: $\sum(x_i - \bar{x})(y_i - \bar{y}) = 8$, $\sum(x_i - \bar{x})^2 = 10$, and by the same arithmetic $\sum(y_i - \bar{y})^2 = 1 + 4 + 1 + 0 + 4 = 10$. So

$$r = \frac{8}{\sqrt{10}\sqrt{10}} = \frac{8}{10} = 0.8.$$

A strong positive linear association – the points rise together, with a little scatter. If you plot them (Figure 14.6), you will see the points hugging the rising diagonal with one small swap (the pairs $(1, 2), (2, 1)$ and $(3, 4), (4, 3)$ each cross), exactly the look of an $r = 0.8$ cloud.

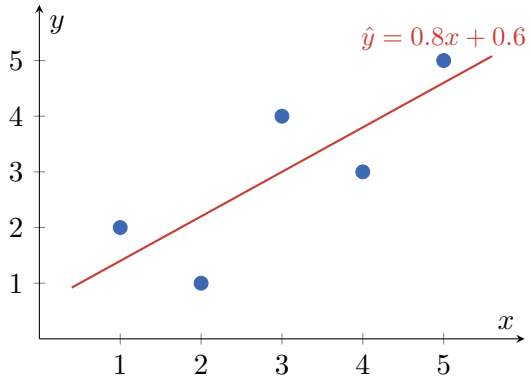


Figure 14.6: The five paired observations of Example 14.5 and their least-squares line. The points cluster tightly around a rising line, which is what $r = 0.8$ looks like.

Work it yourself

Take any dataset you can find with two numeric columns – car weight vs. fuel economy, study hours vs. grades, temperature vs. electricity use. (1) Make a scatterplot first. Guess r by eye before computing. (2) Compute r. Were you close? (3) Now try a deliberately nonlinear pair, like x and $x^2 - c$ over a symmetric range, and watch r collapse toward 0 even though the relationship is perfect. (4) Finally, find a pair with a sneaky confounder (e.g. shoe size and reading ability across all ages of children) and watch a strong correlation evaporate once you restrict to a single age. These four experiments encode the four big lessons of this chapter.

14.9 Chapter Summary

- **Covariance** $\mathrm{Cov}(X, Y) = \mathbb{E}[(X - \mu_X)(Y - \mu_Y)] = \mathbb{E}[XY] - \mathbb{E}[X]\,\mathbb{E}[Y]$ measures the average product of deviations: a quadrant tug-of-war that is positive when the variables tend to move together, negative when they move oppositely.
- It is **bilinear and symmetric**, with $\mathrm{Cov}(X, X) = \mathrm{Var}(X)$, which gives the master identity $\mathrm{Var}(X + Y) = \mathrm{Var}(X) + \mathrm{Var}(Y) + 2\,\mathrm{Cov}(X, Y)$.
- **Independence** $\Rightarrow$ zero covariance, but **not** conversely: covariance sees only *linear* co-movement and is blind to symmetric nonlinear relations like $Y = X^2$.
- **Correlation** $\mathrm{Corr}(X, Y) = \mathrm{Cov}(X, Y)/(\mathrm{SD}(X)\,\mathrm{SD}(Y))$ standardizes covariance into a unit-free number in $[-1, 1]$ – the covariance of the z-scored variables.

- The bound $|\rho| \le 1$ is **Cauchy–Schwarz**: covariance is an inner product and correlation is the *cosine of the angle* between mean-zero variables. Hitting ± 1 means the cloud collapses onto a line.
- **Correlation is not causation**: confounders, reverse causation, and coincidence all produce correlation. Randomized experiments are how we recover causal claims.
- The **covariance matrix** Σ collects all pairwise covariances; it is symmetric positive semidefinite, and its eigen-decomposition is the foundation of PCA and of portfolio risk.

14.10 Exercises

Exercise 14.1 (Warm-up computation)**.** Let X and Y have $\mathbb{E}[X] = 1$, $\mathbb{E}[Y] = 2$, $\mathbb{E}[XY] = 5$, $\mathrm{Var}(X) = 4$, and $\mathrm{Var}(Y) = 9$. Find $\mathrm{Cov}(X, Y)$, $\mathrm{Corr}(X, Y)$, and $\mathrm{Var}(2X - Y + 3)$.

Exercise 14.2 (Indicator covariance)**.** Draw a single card from a standard 52-card deck. Let $A = \mathbf{1}\{\text{card is a heart}\}$ and $B = \mathbf{1}\{\text{card is a queen}\}$. Compute $\mathrm{Cov}(A, B)$. Are A and B independent? Relate your two answers.

Exercise 14.3 (Sign of covariance from a story)**.** For each pair, state whether you expect the covariance to be positive, negative, or roughly zero, and explain in one sentence: (a) a city's daily high temperature and its ice cream sales; (b) the age of a used car and its resale price; (c) a fair coin's outcome (0/1) and an independent die's value; (d) a person's height and the last digit of their phone number.

Exercise 14.4 (Uncorrelated but dependent)**.** Let Θ be uniform on $[0, 2\pi)$ and set $X = \cos\Theta$, $Y = \sin\Theta$, so (X, Y) is a random point on the unit circle. Show that $\mathrm{Cov}(X, Y) = 0$ yet X and Y are *not* independent. (Hint for the second part: they satisfy $X^2 + Y^2 = 1$ exactly.)

Exercise 14.5 (Variance of a difference)**.** Two stocks have daily-return standard deviations $\mathrm{SD}(A) = 0.02$ and $\mathrm{SD}(B) = 0.03$, with correlation $\rho = 0.6$. You go "long A, short B" and hold the portfolio $A - B$. Compute $\mathrm{SD}(A - B)$. Compare it to the case $\rho = -0.6$ and explain, using Theorem 14.1, why negative correlation makes the spread trade so much riskier here.

Exercise 14.6 (Correlation is scale-invariant)**.** Prove from the definition that for constants $a, c > 0$ and any b, d, $\mathrm{Corr}(aX + b, cY + d) = \mathrm{Corr}(X, Y)$. Then show that if $a < 0$ the correlation flips sign. Where exactly in your derivation does the sign of a enter?

Exercise 14.7 (A correlation of exactly 1)**.** Suppose $\mathrm{Corr}(X, Y) = 1$. Using the equality case of Theorem 14.3 (or by computing $\mathrm{Var}(\tilde{X} - \tilde{Y})$), show that $Y = aX + b$ for some constants with $a > 0$, and express a in terms of the standard deviations. Interpret a geometrically.

Exercise 14.8 (Cauchy–Schwarz directly)**.** Let U, V be mean-zero random variables. Consider the function $f(t) = \mathbb{E}[(U - tV)^2] \ge 0$ as a quadratic in the real variable t. By requiring that this upward parabola never dips below zero (its discriminant is ≤ 0), derive $\big(\mathbb{E}[UV]\big)^2 \le \mathbb{E}[U^2]\,\mathbb{E}[V^2]$, and conclude $|\mathrm{Corr}(X, Y)| \le 1$. This is the slick one-line proof of Cauchy–Schwarz.

Exercise 14.9 (Confounding in numbers)**.** A confounder Z is $+1$ or -1 with equal probability. Given Z, let $X = Z + \varepsilon_1$ and $Y = Z + \varepsilon_2$ where $\varepsilon_1, \varepsilon_2$ are independent $\mathcal{N}(0, 1)$ noise terms independent of Z. Compute $\mathrm{Cov}(X, Y)$ and $\mathrm{Corr}(X, Y)$. Then argue that *conditional on* Z, X and Y are uncorrelated. Explain how this models the ice-cream-and-drowning paradox.

Exercise 14.10 (A small covariance matrix)**.** Let $\mathbf{X} = (X_1, X_2, X_3)$ have covariance matrix

$$\Sigma = \begin{pmatrix} 4 & 2 & 0 \\ 2 & 9 & -3 \\ 0 & -3 & 4 \end{pmatrix}.$$

(a) Read off $\mathrm{Var}(X_2)$ and $\mathrm{Corr}(X_2, X_3)$. (b) Compute $\mathrm{Var}(X_1 + X_2 + X_3)$ using $\mathbf{a}^\top \Sigma \mathbf{a}$ with $\mathbf{a} = (1, 1, 1)^\top$. (c) Verify Σ is symmetric, and check that the variance you found in (b) is nonnegative, as positive semidefiniteness demands.

Exercise 14.11 (Why divide by $n-1$? (conceptual))**.** Explain in words why the sample covariance uses the divisor $n-1$ rather than n. What goes wrong, in expectation, if you use n? Why does the choice of divisor not affect the sample correlation r?

Chapter 15

Conditional Distributions and Conditional Expectation

There is a single idea behind almost everything we do once randomness gets complicated, and it is deceptively simple: *learn something, then update*. You started a journey with a vague sense of how long it would take. Then you saw the first hour of traffic. Now your guess is sharper. You did not throw away probability; you sharpened it with information. That act of sharpening — replacing a distribution with a *conditional* distribution, and replacing an expectation with a *conditional* expectation — is the engine of prediction, regression, Bayesian inference, Markov chains, and martingales. Everything later in this book leans on it.

We already met conditioning on *events* back when we wrote $\mathbb{P}(A \mid B)$. This chapter goes one step further, and the step is at first slightly disorienting: we condition on a *random variable*. The result, $\mathbb{E}[Y \mid X]$, is not a number. It is itself a random variable — a function of X — and learning to think of it that way is most of the battle. Once you see it, the two main results of the chapter, the **tower property** (law of total expectation) and the **law of total variance**, become almost obvious, and you will see clearly why $\mathbb{E}[Y \mid X]$ is the *best possible prediction* of Y from X. That last fact is the mathematical seed of all of regression and supervised learning.

15.1 From conditioning on events to conditioning on a variable

Recall the definition that started it all. For events A and B with $\mathbb{P}(B) > 0$,

$$\mathbb{P}(A \mid B) \;=\; \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}.$$

This is just renormalization: throw away the part of the world where B failed, and rescale what is left so the probabilities sum to one again. The conditional probability is a genuine probability distribution living inside the smaller universe B.

Now suppose X and Y are random variables and we want "the distribution of Y once we know the value of X." If X is discrete, this is no harder than the event case: the value $\{X = x\}$ is an event, so we just condition on it.

Definition 15.1 (Conditional pmf, discrete case)**.** Let X, Y be discrete with joint pmf $p_{X,Y}(x, y) =$

$\mathbb{P}(X = x, Y = y)$. For any x with $p_X(x) = \mathbb{P}(X = x) > 0$, the *conditional pmf of Y given $X = x$* is

$$p_{Y|X}(y \mid x) \;=\; \mathbb{P}(Y = y \mid X = x) \;=\; \frac{p_{X,Y}(x, y)}{p_X(x)}.$$

For each fixed x, the function $y \mapsto p_{Y|X}(y \mid x)$ is a genuine pmf: it is nonnegative and sums to one over y, because $\sum_y p_{X,Y}(x, y) = p_X(x)$ is exactly the normalizing constant. So conditioning on $X = x$ just slices the joint table along the row $X = x$ and renormalizes that row.

Intuition

Picture the joint pmf as a grid of probabilities, one cell per (x, y) pair. The *marginal* $p_X(x)$ is the total weight in column x. The *conditional* $p_{Y|X}(\cdot \mid x)$ is that single column, blown up so its weights sum to one. Marginalizing collapses the grid; conditioning zooms into one strip of it. Same data, two different questions.

Example 15.1 (Two dice, conditioned on the sum)**.** Roll two fair dice; let X_1, X_2 be the faces and $S = X_1 + X_2$. What is the distribution of X_1 given $S = 7$? The event $\{S = 7\}$ contains the six ordered outcomes $(1, 6), (2, 5), \ldots, (6, 1)$, each with probability $1/36$, so $\mathbb{P}(S = 7) = 6/36$. For any $x_1 \in \{1, \ldots, 6\}$ there is exactly one matching pair, so

$$\mathbb{P}(X_1 = x_1 \mid S = 7) = \frac{1/36}{6/36} = \frac{1}{6}.$$

Given the sum is 7, the first die is uniform on $\{1, \ldots, 6\}$. Now condition instead on $S = 4$: the matching pairs are $(1, 3), (2, 2), (3, 1)$, so $\mathbb{P}(S = 4) = 3/36$ and $\mathbb{P}(X_1 = x_1 \mid S = 4) = 1/3$ for $x_1 \in \{1, 2, 3\}$ and 0 otherwise. The conditional distribution genuinely *changed shape* as the conditioning value changed — a fact we will turn into a random variable in a moment.

15.1.1 The continuous case: dividing densities

When X is continuous, the event $\{X = x\}$ has probability zero, and our event-based definition divides by zero. Yet intuitively the question "what is the distribution of Y given $X = x$" is perfectly sensible — think of the heights of people who weigh exactly 70 kg. The resolution is to do the same division, but with *densities* instead of probabilities.

Definition 15.2 (Conditional density, continuous case)**.** Let (X, Y) have joint density $f_{X,Y}(x, y)$ with marginal $f_X(x) = \int_{\mathbb{R}} f_{X,Y}(x, y)\,\mathrm{d}y$. For any x with $f_X(x) > 0$, the *conditional density of Y given $X = x$* is

$$f_{Y|X}(y \mid x) \;=\; \frac{f_{X,Y}(x, y)}{f_X(x)}.$$

The same sanity check works: for fixed x, integrating over y gives $\int_{\mathbb{R}} f_{Y|X}(y \mid x)\,\mathrm{d}y = f_X(x)/f_X(x) = 1$, so it is a legitimate density.

Why it works

Why is dividing densities the "right" way to condition on a measure-zero event? Condition

on the *thin slab* $\{x \le X \le x+\varepsilon\}$, which has positive probability, and let $\varepsilon \to 0$. For small ε,

$$\mathbb{P}(Y \le y \mid x \le X \le x+\varepsilon) = \frac{\int_{-\infty}^{y}\int_{x}^{x+\varepsilon} f_{X,Y}(u,v)\,du\,dv}{\int_{x}^{x+\varepsilon} f_X(u)\,du} \approx \frac{\varepsilon\int_{-\infty}^{y} f_{X,Y}(x,v)\,dv}{\varepsilon\, f_X(x)}.$$

The ε's cancel, and what survives is exactly $\int_{-\infty}^{y} f_{Y|X}(v \mid x)\,dv$. So the density quotient is the *limit* of ordinary conditional probabilities on shrinking slabs — not a trick, but the only consistent answer. (Choosing a different shape of vanishing neighborhood can give a different limit; this is the famous Borel paradox, and it is why measure-theoretic care eventually becomes necessary — see the rigorous treatment later in the book.)

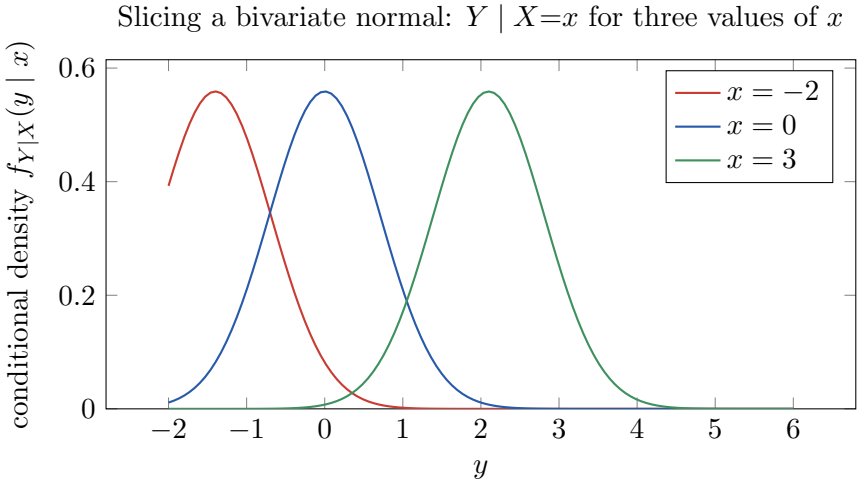


Figure 15.1: Each value of X selects a different conditional distribution of Y. For a bivariate normal with correlation $\rho = 0.7$, the slice $Y \mid X = x$ is Normal with mean ρx and the *same* variance $1-\rho^2$ for every x. The center slides with x; the spread does not. The moving center, as a function of x, is the regression line — and it is the conditional expectation $\mathbb{E}[Y \mid X = x]$ that we build next.

15.2 Conditional expectation given a value: $\mathbb{E}[Y \mid X = x]$

Once we have a conditional distribution, we can take its mean exactly as we always do — it is just an expectation computed inside the smaller universe.

Definition 15.3 (Conditional expectation given $X = x$)**.**

$$\mathbb{E}[Y \mid X = x] \;=\; \begin{cases} \displaystyle\sum_{y} y\, p_{Y|X}(y \mid x), & \text{discrete,} \\[2ex] \displaystyle\int_{\mathbb{R}} y\, f_{Y|X}(y \mid x)\,dy, & \text{continuous.} \end{cases}$$

This is a *number* for each fixed x, but a number that depends on x. Define the function

$$g(x) \;=\; \mathbb{E}[Y \mid X = x].$$

We call g the *regression function* of Y on X. It answers: "if all I knew was $X = x$, what is my best single-number guess for Y?" Figure 15.1 drew the slices; $g(x)$ is the curve traced by their centers.

Example 15.2 (Uniform on a triangle)**.** Let (X, Y) be uniform on the triangle $\{(x, y) : 0 \le x \le 1,\ 0 \le y \le x\}$. The triangle has area $1/2$, so $f_{X,Y}(x, y) = 2$ on it. The marginal of X is $f_X(x) = \int_0^x 2\,dy = 2x$ for $0 \le x \le 1$. Hence for fixed x,

$$f_{Y|X}(y \mid x) = \frac{2}{2x} = \frac{1}{x}, \qquad 0 \le y \le x,$$

which is Uniform$(0, x)$. Therefore

$$\mathbb{E}[Y \mid X = x] = \frac{x}{2}, \qquad g(x) = \frac{x}{2}.$$

The picture matches the algebra: given the cap $X = x$, the height Y is uniform up to x, so on average it sits halfway up. The regression function is the line $y = x/2$.

Example 15.3 (Mixed: a Poisson rate that is itself random)**.** Let $\Lambda \sim$ Uniform$(0, 4)$ be a random rate, and given $\Lambda = \lambda$ let $N \sim$ Poisson(λ) be a count. Then $\mathbb{E}[N \mid \Lambda = \lambda] = \lambda$ directly from the Poisson mean. So here the regression function of N on Λ is the identity, $g(\lambda) = \lambda$. We will return to this example to compute $\mathbb{E}[N]$ and $\mathrm{Var}(N)$ painlessly using the tower property and the law of total variance.

15.3 The leap: $\mathbb{E}[Y \mid X]$ as a random variable

Here is the conceptual hinge of the chapter. We have $g(x) = \mathbb{E}[Y \mid X = x]$, an ordinary function. Now feed the *random* variable X into it. The result, $g(X)$, is a new random variable. We give it a name and a notation.

Definition 15.4 (Conditional expectation as a random variable)**.** Let $g(x) = \mathbb{E}[Y \mid X = x]$. The *conditional expectation of Y given X* is the random variable

$$\mathbb{E}[Y \mid X] \;:=\; g(X).$$

It is a function of X alone. On the outcome where X took value x, it equals $g(x) = \mathbb{E}[Y \mid X = x]$.

Read that twice. $\mathbb{E}[Y \mid X = x]$ is a number (you plugged in a specific x). $\mathbb{E}[Y \mid X]$ is a random variable (you left X random). The first is a value of the second.

Common trap

The single most common error in this entire subject is treating $\mathbb{E}[Y \mid X]$ as a constant. It is *not*. It is random, because X is random. Whenever you write $\mathbb{E}[Y \mid X]$, ask yourself: "this is a function of X — which function?" If you can name g, you understand the object. Only *after* you take a further expectation over X (or plug in a specific value) do you get a number.

What is really happening

What is really happening when we form $\mathbb{E}[Y \mid X]$? We are building the *best summary of Y that is allowed to depend on X but on nothing else.* Imagine partitioning the sample space by the value of X: all outcomes with $X = x$ get lumped together. Within each lump, $\mathbb{E}[Y \mid X]$ reports the average value of Y over that lump. So $\mathbb{E}[Y \mid X]$ is a "coarsened" or "smoothed" version of Y — it forgets every detail of the outcome except what X reveals, and within each level of X it replaces Y by its local average. It is Y as seen through the blurry lens of X. This is literally the measure-theoretic definition (conditioning on the σ-algebra generated by X), which we make rigorous in a later chapter.

Example 15.4 (Back to the two dice)**.** In Example 15.1, given $S = s$ the first die X_1 is uniform on the pairs summing to s. By symmetry $\mathbb{E}[X_1 \mid S = s] = s/2$ (the two dice are exchangeable, so each carries half the sum on average). Therefore, as a random variable,

$$\mathbb{E}[X_1 \mid S] = \frac{S}{2}.$$

Notice this is genuinely random: it equals $7/2$ on outcomes with $S = 7$, equals 2 on outcomes with $S = 4$, and so on. We will check in a moment that its average, $\mathbb{E}\big[\,\mathbb{E}[X_1 \mid S]\big] = \mathbb{E}[S/2] = 7/2$, equals $\mathbb{E}[X_1] = 7/2$. That is the tower property previewing itself.

15.3.1 Two properties you will use constantly

Because $\mathbb{E}[Y \mid X]$ is a function of X, anything that is itself a function of X behaves like a *constant* inside the conditional expectation: you already know it once you know X.

Proposition 15.1 (Taking out what is known; linearity)**.** *Let h be any function and a, b constants. Then (whenever the expectations exist)*

1. ***Linearity:*** $\mathbb{E}[aY + bZ \mid X] = a\,\mathbb{E}[Y \mid X] + b\,\mathbb{E}[Z \mid X]$.
2. ***Pulling out a known factor:*** $\mathbb{E}[h(X)\,Y \mid X] = h(X)\,\mathbb{E}[Y \mid X]$.
3. ***Independence collapses it:*** *if $Y \perp\!\!\!\perp X$ then $\mathbb{E}[Y \mid X] = \mathbb{E}[Y]$ (a constant).*

Why it works

Each is intuitive once you remember that conditioning on X means "treat X as known." Linearity is just linearity of the ordinary expectation done inside each slice $\{X = x\}$. For the pull-out, once $X = x$ is known the factor $h(X) = h(x)$ is a fixed number and slides out of the average: $\mathbb{E}[h(x)Y \mid X = x] = h(x)\,\mathbb{E}[Y \mid X = x]$; now let x be random again. For independence, if Y tells you nothing about X then knowing X tells you nothing about Y, so the conditional distribution of Y equals its marginal and its mean is just $\mathbb{E}[Y]$.

15.4 The tower property (law of total expectation)

Now comes the result that makes conditional expectation indispensable. It says you can compute an expectation in two stages: first average within each level of X, then average those averages across levels of X.

Theorem 15.1 (Tower property / Law of total expectation)**.** *For random variables X, Y with $\mathbb{E}|Y| < \infty$,*

$$\mathbb{E}\left[\,\mathbb{E}[Y \mid X]\,\right] \;=\; \mathbb{E}[Y].$$

Equivalently, in the explicit two-stage form,

$$\mathbb{E}[Y] = \sum_x \mathbb{E}[Y \mid X = x]\;\mathbb{P}(X = x) \qquad \textit{or} \qquad \mathbb{E}[Y] = \int_{\mathbb{R}} \mathbb{E}[Y \mid X = x]\, f_X(x)\,\mathrm{d}x.$$

Intuition

A company has three departments. You want the average salary across the whole company. One way: list every employee and average. Another way: find the average salary *within* each department, then average those department-averages *weighted by department size*. Same answer. The tower property is exactly this: $\mathbb{E}[Y \mid X]$ is the within-group average (group = level of X), and the outer $\mathbb{E}[\cdot]$ weights groups by how likely they are. The weighting by group size is essential — a plain unweighted average of group means would be wrong (Figure 15.2).

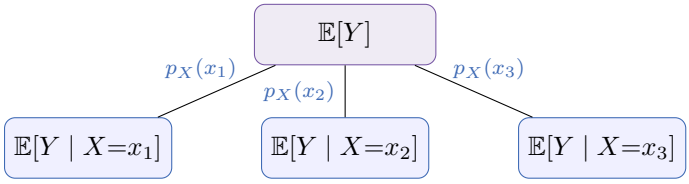


Figure 15.2: The tower property as a two-stage average. Partition the sample space by the value of X; inside each group $\{X = x_k\}$ the within-group mean is $\mathbb{E}[Y \mid X = x_k]$. Reassembling them, weighted by the group probabilities $p_X(x_k)$, returns the grand mean $\mathbb{E}[Y]$. The weights are the whole point: an unweighted average of the leaves would not equal $\mathbb{E}[Y]$.

Proof. Take the discrete case; the continuous case replaces sums by integrals verbatim. By Definition 15.4, $\mathbb{E}[Y \mid X] = g(X)$ with $g(x) = \mathbb{E}[Y \mid X = x]$. Apply the law of the unconscious statistician to the function g of X:

$$\mathbb{E}\left[g(X)\right] = \sum_x g(x)\, p_X(x) = \sum_x \Big(\sum_y y\, p_{Y|X}(y \mid x)\Big) p_X(x).$$

Now substitute $p_{Y|X}(y \mid x)\, p_X(x) = p_{X,Y}(x, y)$ (the defining identity, valid wherever $p_X(x) > 0$, and the other terms contribute nothing):

$$= \sum_x \sum_y y\, p_{X,Y}(x, y) = \sum_y y \sum_x p_{X,Y}(x, y) = \sum_y y\, p_Y(y) = \mathbb{E}[Y].$$

The swap of summation order is justified because $\mathbb{E}|Y| < \infty$ makes the double sum absolutely convergent. The continuous proof is identical with f's and integrals. □

What is really happening

The proof reveals the trick in one line: $\mathbb{E}[Y \mid X]\, p_X$ = "joint, then sum out X." The conditional expectation *undoes* the conditioning's normalization when you reweight by p_X,

and summing over x marginalizes X away, leaving plain $\mathbb{E}[Y]$. Tower property = "condition, then un-condition by averaging."

15.4.1 Why the tower property earns its keep

The tower property turns hard expectations into easy ones whenever there is a natural variable to condition on — typically a quantity that, once known, makes Y simple.

Example 15.5 (Random sum / Wald-style identity)**.** Let N be a nonnegative-integer random variable (a random count), and let $X_1, X_2, \ldots$ be i.i.d. with mean μ, independent of N. Define the *random sum* $T = \sum_{i=1}^{N} X_i$. What is $\mathbb{E}[T]$? Condition on N: once $N = n$ is known, T is a sum of exactly n i.i.d. terms, so

$$\mathbb{E}[T \mid N = n] = n\mu, \qquad \text{hence} \qquad \mathbb{E}[T \mid N] = N\mu.$$

Apply the tower property:

$$\mathbb{E}[T] = \mathbb{E}\left[\mathbb{E}[T \mid N]\right] = \mathbb{E}[N\mu] = \mu\, \mathbb{E}[N].$$

The expected total equals the expected count times the expected size — clean, and almost impossible to see without conditioning. (Think: total insurance payout = expected number of claims × expected size per claim.)

Example 15.6 (Geometric via conditioning on the first step)**.** Let N be the number of flips of a coin (heads probability p) until the first head. Condition on the first flip. With probability p the first flip is a head and $N = 1$. With probability $1 - p$ it is a tail and, by the memoryless restart, the remaining number of flips has the same distribution as a fresh N, so we have used 1 flip plus another $\mathbb{E}[N]$ on average. Writing $m = \mathbb{E}[N]$ and using the tower property over the first flip F,

$$m = \mathbb{E}[N \mid F = \mathrm{H}]\, p + \mathbb{E}[N \mid F = \mathrm{T}](1 - p) = 1 \cdot p + (1 + m)(1 - p).$$

Solve: $m = p + (1 - p) + (1 - p)m = 1 + (1 - p)m$, so $m\,p = 1$ and $\mathbb{E}[N] = 1/p$. Conditioning on one step collapses an infinite sum into a single linear equation — the prototype of "first-step analysis" that powers Markov chains.

Example 15.7 (Finishing the random Poisson rate)**.** For Example 15.3, $\mathbb{E}[N \mid \Lambda] = \Lambda$, so by the tower property

$$\mathbb{E}[N] = \mathbb{E}[\Lambda] = \frac{0 + 4}{2} = 2.$$

15.5 The law of total variance

Expectation splits cleanly under conditioning. Variance does too, but with an illuminating twist: it splits into *two* pieces, and the two pieces have a beautiful interpretation as "noise" and "signal."

Theorem 15.2 (Law of total variance)**.** *For Y with $\mathbb{E}[Y^2] < \infty$,*

$$\mathrm{Var}(Y) \;=\; \underbrace{\mathbb{E}\left[\mathrm{Var}(Y \mid X)\right]}_{\textit{within-group variance}} \;+\; \underbrace{\mathrm{Var}\left(\mathbb{E}[Y \mid X]\right)}_{\textit{between-group variance}}.$$

Here $\mathrm{Var}(Y \mid X) = \mathbb{E}[Y^2 \mid X] - \big(\mathbb{E}[Y \mid X]\big)^2$ *is itself a random variable (a function of X).*

Proof. Write $m(X) = \mathbb{E}[Y \mid X]$. Start from the conditional-variance definition and take expectations, using the tower property on the first term:

$$\mathbb{E}\left[\operatorname{Var}(Y \mid X)\right] = \mathbb{E}\left[\mathbb{E}[Y^2 \mid X]\right] - \mathbb{E}\left[m(X)^2\right] = \mathbb{E}[Y^2] - \mathbb{E}\left[m(X)^2\right].$$

Next, by definition of variance of the random variable $m(X)$ and the tower property $\mathbb{E}[m(X)] = \mathbb{E}[Y]$,

$$\operatorname{Var}\left(\mathbb{E}[Y \mid X]\right) = \mathbb{E}\left[m(X)^2\right] - \left(\mathbb{E}[m(X)]\right)^2 = \mathbb{E}\left[m(X)^2\right] - \left(\mathbb{E}[Y]\right)^2.$$

Add the two displays. The $\mathbb{E}[m(X)^2]$ terms cancel, leaving $\mathbb{E}[Y^2] - (\mathbb{E}[Y])^2 = \operatorname{Var}(Y)$. □

Intuition

The mnemonic is **EVVE**: Expectation of the Variance, plus Variance of the Expectation. The first term, $\mathbb{E}[\operatorname{Var}(Y \mid X)]$, is the *average leftover scatter* of Y once you know X — the part X *cannot* explain (call it *noise*, or *unexplained* variation). The second term, $\operatorname{Var}(\mathbb{E}[Y \mid X])$, is how much the group means themselves bounce around as X varies — the part X *does* explain (call it *signal*, or *explained* variation). Total spread = unexplained + explained. If X determines Y exactly, the first term is 0. If X is useless, the second term is 0 and all spread is noise.

Connection

This decomposition is the probabilistic skeleton of the ANOVA identity in statistics ("total sum of squares = within-group + between-group") and of the R^2 coefficient in regression: $R^2 = \operatorname{Var}(\mathbb{E}[Y \mid X])/\operatorname{Var}(Y)$ is exactly the *fraction of variance explained*, the signal share. Linear algebra fans will recognize the Pythagorean flavor — and indeed in the next section we see $Y = \mathbb{E}[Y \mid X] + (\text{residual})$ is an orthogonal decomposition, with the law of total variance as its Pythagorean theorem.

Example 15.8 (Finishing the variance of the random Poisson count)**.** Recall $\Lambda \sim \text{Uniform}(0, 4)$ and $N \mid \Lambda \sim \text{Poisson}(\Lambda)$. For a Poisson both mean and variance equal the rate, so

$$\mathbb{E}[N \mid \Lambda] = \Lambda, \qquad \operatorname{Var}(N \mid \Lambda) = \Lambda.$$

The uniform on $(0, 4)$ has mean 2 and variance $(4-0)^2/12 = 16/12 = 4/3$. Apply EVVE:

$$\operatorname{Var}(N) = \mathbb{E}[\operatorname{Var}(N \mid \Lambda)] + \operatorname{Var}(\mathbb{E}[N \mid \Lambda]) = \mathbb{E}[\Lambda] + \operatorname{Var}(\Lambda) = 2 + \frac{4}{3} = \frac{10}{3}.$$

Notice $\operatorname{Var}(N) = 10/3 > 2 = \mathbb{E}[N]$: the random rate *inflates* the variance beyond the Poisson's own Var = $\mathbb{E}$. This *overdispersion* — extra variance injected by an unknown latent rate — is exactly why mixed/hierarchical models exist in applied statistics. The $\operatorname{Var}(\Lambda) = 4/3$ is the signal (variation we could in principle explain by learning Λ); the $\mathbb{E}[\Lambda] = 2$ is the irreducible Poisson noise.

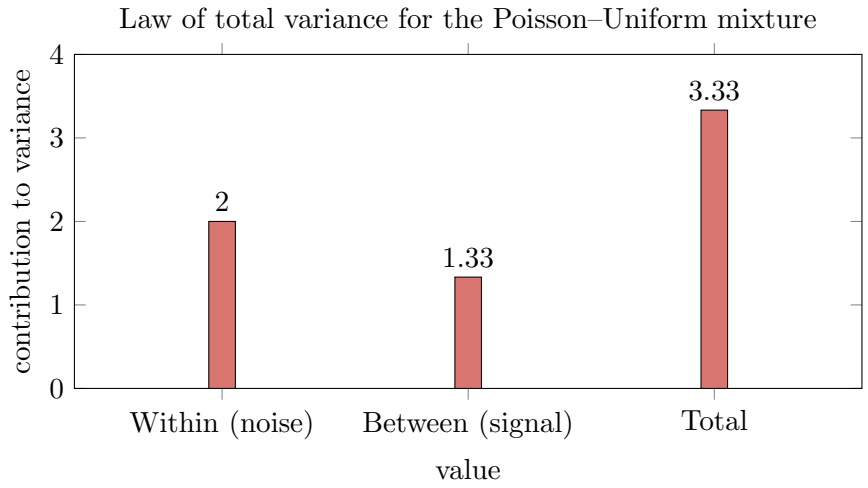


Figure 15.3: $\mathrm{Var}(N) = 10/3$ splits into within-group (unexplained) variance $\mathbb{E}[\Lambda] = 2$ and between-group (explained) variance $\mathrm{Var}(\Lambda) = 4/3$. Knowing Λ would remove the second bar but never the first.

Paradox

Simpson's paradox: when conditioning flips the story. The law of total expectation warns you, quietly, that *group averages must be weighted by group size*. Ignore that warning and you get Simpson's paradox. Suppose treatment A beats treatment B for men, *and* A beats B for women, yet B beats A overall. Impossible? No — it happens whenever the two treatments were given to very different mixes of men and women. Within each group (X = sex), A wins: $\mathbb{E}[\text{cure} \mid A, X] > \mathbb{E}[\text{cure} \mid B, X]$. But the overall average $\mathbb{E}[\text{cure} \mid A] = \sum_x \mathbb{E}[\text{cure} \mid A, X = x]\, \mathbb{P}(X = x \mid A)$ uses the *conditional group mix* $\mathbb{P}(X = x \mid A)$, which can differ wildly between A and B. If A was mostly given to the harder-to-cure group, A's good within-group rates get drowned by bad weights. The paradox is not a paradox once you internalize that conditional expectation is a *weighted* reassembly — and that the weights are part of the data, not a nuisance to be averaged away. The lesson: never average group means without their proper $\mathbb{P}(X = x)$ weights.

15.6 Conditional expectation is the best predictor

We now reach the result that justifies calling $\mathbb{E}[Y \mid X]$ the "best guess of Y from X," and that secretly underlies all of supervised learning. Suppose you must predict Y using only a function $h(X)$ of the observed X, and you will be scored by mean squared error $\mathbb{E}\left[(Y - h(X))^2\right]$. Which function h should you choose?

Theorem 15.3 ($\mathbb{E}[Y \mid X]$ minimizes mean squared error)**.** *Among all (measurable) functions h with $\mathbb{E}[h(X)^2] < \infty$, the mean squared prediction error $\mathbb{E}\left[(Y - h(X))^2\right]$ is minimized uniquely by*

$$h^\star(X) = \mathbb{E}[Y \mid X].$$

Proof. Let $m(X) = \mathbb{E}[Y \mid X]$ and let $h(X)$ be any competitor. Split and add zero:

$$Y - h(X) = \big(Y - m(X)\big) + \big(m(X) - h(X)\big).$$

Square and take expectations:

$$\mathbb{E}\left[(Y-h(X))^2\right] = \mathbb{E}\left[(Y-m(X))^2\right] + 2\,\mathbb{E}\left[(Y-m(X))(m(X)-h(X))\right] + \mathbb{E}\left[(m(X)-h(X))^2\right].$$

The cross term vanishes. To see it, note that $m(X)-h(X)$ is a function of X, so condition on X and pull it out (Proposition 15.1):

$$\mathbb{E}\left[(Y-m(X))(m(X)-h(X)) \mid X\right] = (m(X)-h(X))\,\mathbb{E}\left[Y-m(X) \mid X\right] = (m(X)-h(X))\cdot 0,$$

because $\mathbb{E}[Y \mid X] - m(X) = 0$ by definition of m. Taking the outer expectation (tower property) shows the cross term is 0. Therefore

$$\mathbb{E}\left[(Y-h(X))^2\right] = \underbrace{\mathbb{E}\left[(Y-m(X))^2\right]}_{\text{fixed, independent of } h} + \underbrace{\mathbb{E}\left[(m(X)-h(X))^2\right]}_{\geq 0}.$$

The right-most term is nonnegative and equals zero iff $h(X) = m(X)$ almost surely. So $m = \mathbb{E}[Y \mid X]$ is the unique minimizer. □

What is really happening

That "cross term vanishes" is not an algebraic accident — it is *orthogonality*. Think of random variables (with finite variance) as vectors, with inner product $\langle U, V\rangle = \mathbb{E}[UV]$. The proof shows the prediction error $Y - \mathbb{E}[Y \mid X]$ is orthogonal to *every* function of X. So $\mathbb{E}[Y \mid X]$ is the *orthogonal projection* of Y onto the subspace of all functions of X. "Best predictor" literally means "foot of the perpendicular." And the decomposition $\mathbb{E}[(Y-h)^2] = \mathbb{E}[(Y-m)^2] + \mathbb{E}[(m-h)^2]$ is the Pythagorean theorem in this geometry. The law of total variance is the same Pythagorean theorem with h taken to be the constant $\mathbb{E}[Y]$: total = residual + explained.

Why this matters

Step back and admire the unity. Conditional expectation is *simultaneously*: (i) a within-group average; (ii) a smoothing/coarsening of Y through the lens of X; (iii) the orthogonal projection of Y onto functions of X; (iv) the minimum-mean-squared-error predictor. Four descriptions, one object. When a single definition wears that many hats, you are looking at something fundamental — and indeed $\mathbb{E}[\cdot \mid X]$ is the operation that the rest of advanced probability is built around.

Application

Regression is conditional expectation. This theorem *is* the foundation of regression and supervised machine learning. In regression we seek a function that predicts a target Y (price, click, diagnosis) from features X; the gold standard, if we had unlimited data and unlimited flexibility, is the *regression function* $g(x) = \mathbb{E}[Y \mid X = x]$ — the true conditional mean, the curve that threads the center of the data cloud (Figure 15.5). Every regression method is an *estimate* of this curve under some restriction:

- **Linear regression** restricts h to the form $a+bX$ and finds the best line — the orthogonal projection of Y onto the smaller subspace of *linear* functions of X, not all functions. It recovers the true $\mathbb{E}[Y \mid X]$ only when that conditional mean happens to be linear (e.g.,

the bivariate normal of Figure 15.1, whose regression function is exactly ρx).

- **Trees, kernels, splines, neural nets** enlarge the class of allowed h to approximate the full curve $\mathbb{E}[Y \mid X = x]$, trading bias for variance.

The squared-error loss that nearly every regressor minimizes is chosen *precisely because* Theorem 15.3 guarantees its population minimizer is the conditional mean. When a data scientist says "the model predicts the expected value of Y given the features," this theorem is what makes that sentence true.

Remark 15.1 (Best *linear* predictor, for contrast). If you restrict to linear predictors $h(X) = a + bX$, minimizing $\mathbb{E}[(Y - a - bX)^2]$ gives the familiar least-squares coefficients

$$b = \frac{\operatorname{Cov}(X, Y)}{\operatorname{Var}(X)}, \qquad a = \mathbb{E}[Y] - b\,\mathbb{E}[X].$$

This is the projection onto the *line*, not onto all functions of X. It uses only the first two moments (means, variances, covariance), whereas the true $\mathbb{E}[Y \mid X]$ uses the entire joint distribution. They coincide exactly when the regression function is linear.

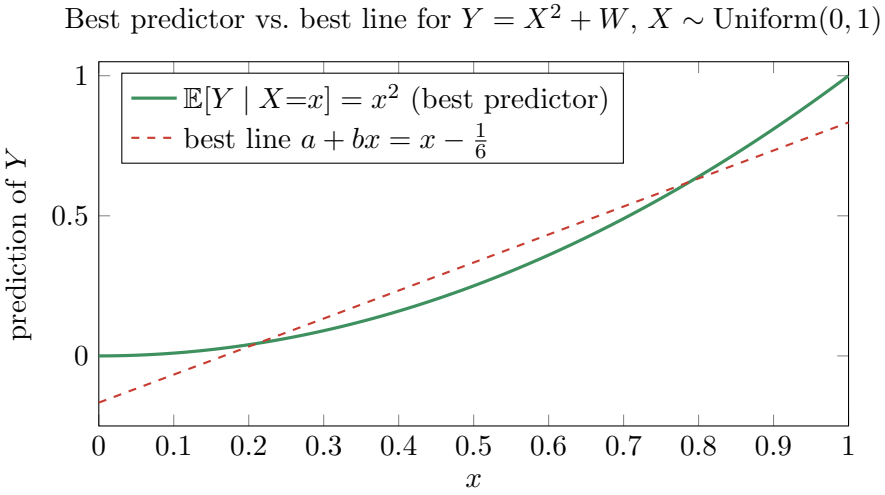


Figure 15.4: When the regression function is curved, the unconstrained best predictor $\mathbb{E}[Y \mid X = x] = x^2$ (solid) bends to follow it, while the best *linear* predictor (dashed) can only pass a straight line through the data and is forced to compromise — too low near the ends, too high in the middle. The gap between the curves is precisely the excess error a linear model pays for its rigidity; it vanishes only when $\mathbb{E}[Y \mid X]$ is already a line. This is the visual content of Theorem 15.3 and Exercise 15.5.

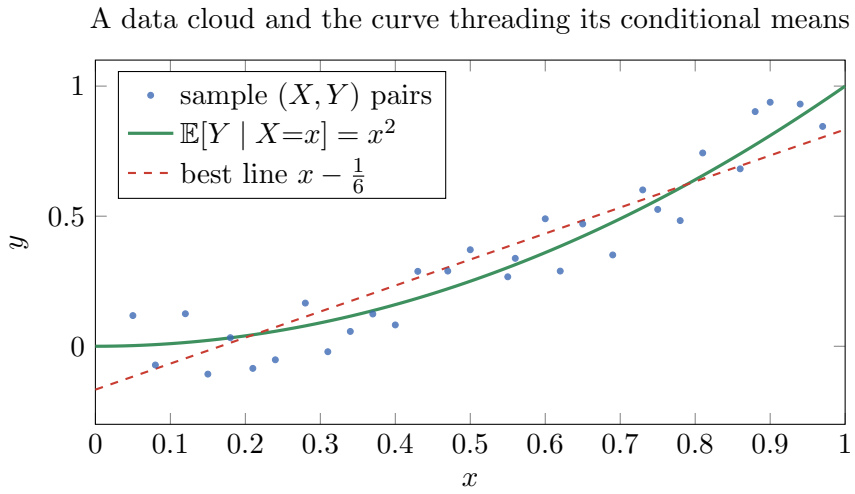


Figure 15.5: The same model $Y = X^2 + W$ shown as a scatter of draws. The conditional-mean curve $\mathbb{E}[Y \mid X{=}x] = x^2$ (solid) passes through the vertical center of the cloud at every x: that is exactly what "best mean-square predictor" means — for each x it sits at the balance point of the Y-values stacked above x. The best line (dashed) is the best straight-line compromise but misses the curvature. The vertical scatter remaining around the solid curve is the irreducible noise $\mathbb{E}[\mathrm{Var}(Y \mid X)] = \mathrm{Var}(W) = 0.01$ that no function of X can remove.

15.7 A fully worked, end-to-end example

Let us exercise every tool on one problem. A factory ships boxes. The number of items in a box is $N \sim \text{Poisson}(\lambda)$ with $\lambda = 10$. Each item independently is defective with probability $q = 0.1$. Let D be the number of *defective* items in a box. We want $\mathbb{E}[D]$, $\mathrm{Var}(D)$, the conditional distribution $D \mid N$, and the best predictor of D from N.

Step 1 — conditional distribution. Given $N = n$, each of the n items is an independent Bernoulli(q) defect, so

$$D \mid N = n \;\sim\; \text{Binomial}(n, q), \qquad \mathbb{E}[D \mid N] = qN, \qquad \mathrm{Var}(D \mid N) = Nq(1-q).$$

Step 2 — mean via the tower property.

$$\mathbb{E}[D] = \mathbb{E}[\mathbb{E}[D \mid N]] = \mathbb{E}[qN] = q\,\mathbb{E}[N] = 0.1 \times 10 = 1.$$

Step 3 — variance via EVVE. Using $\mathbb{E}[N] = \mathrm{Var}(N) = \lambda = 10$ for the Poisson:

$$\mathrm{Var}(D) = \mathbb{E}[\mathrm{Var}(D \mid N)] + \mathrm{Var}(\mathbb{E}[D \mid N]) = \mathbb{E}[Nq(1-q)] + \mathrm{Var}(qN) = q(1-q)\,\mathbb{E}[N] + q^2\,\mathrm{Var}(N).$$

Plug in: $0.1 \cdot 0.9 \cdot 10 + 0.01 \cdot 10 = 0.9 + 0.1 = 1.0$. So $\mathrm{Var}(D) = 1$.

Step 4 — recognize the answer. We found $\mathbb{E}[D] = \mathrm{Var}(D) = 1 = \lambda q$. That is no coincidence: *thinning a Poisson is Poisson.* Indeed $D \sim \text{Poisson}(\lambda q)$ exactly, a fact you can prove directly but which the tower/EVVE computation already strongly suggests. Conditioning gave us the moments with almost no algebra.

Step 5 — best predictor. By Theorem 15.3, the best predictor of D from N in mean square is $\mathbb{E}[D \mid N] = 0.1\,N$. Knowing the box held $N = 14$ items, predict $0.1 \times 14 = 1.4$ defectives. The leftover unexplained variance is $\mathbb{E}[\mathrm{Var}(D \mid N)] = q(1-q)\,\mathbb{E}[N] = 0.9$, and the explained variance is $\mathrm{Var}(qN) = q^2\lambda = 0.1$; they sum to $\mathrm{Var}(D) = 1.0$, as the law of total variance demands.

Work it yourself

Re-run Step 3 but make the defect probability itself uncertain: let $Q \sim \mathrm{Beta}(2, 18)$ (mean 0.1) independently of N, and given (N, Q) let $D \sim \mathrm{Binomial}(N, Q)$. Condition first on (N, Q), then peel off one layer at a time with the tower property and EVVE. You should find $\mathbb{E}[D]$ unchanged at 1 but $\mathrm{Var}(D)$ *larger* than before — the extra randomness in Q adds a third source of spread. Identify which term of the variance decomposition it lands in. (Hint: $\mathrm{Var}(Q) = \frac{2\cdot 18}{20^2\cdot 21} \approx 0.0043$.)

History

Conditional expectation as we use it — a *random variable*, the projection picture, conditioning on a σ-algebra rather than a single event — was forged in the 1930s. Andrey Kolmogorov's 1933 *Grundbegriffe der Wahrscheinlichkeitsrechnung* put probability on measure-theoretic foundations and *defined* conditional expectation abstractly via the Radon–Nikodym theorem, precisely to sidestep the division-by-zero that wrecks the naive definition when you condition on a continuous variable (recall the Borel paradox from the Why-box). Earlier, the two-stage averaging idea was already implicit in Laplace's and Huygens's reasoning about games and expectations, and in the actuarial mathematics of random sums. But it was Kolmogorov's leap — "define $\mathbb{E}[Y \mid X]$ by the property that it averages correctly over every event determined by X" — that turned conditional expectation from a computational recipe into the structural backbone of modern probability, martingales and stochastic calculus very much included. We give that defining property its own rigorous chapter later; everything in this chapter is the concrete, hands-on shadow of it.

15.8 The defining property, stated cleanly

One identity, in the rigorous theory, *is* the definition — and every formula above is a special case of it. It bundles the tower property and the orthogonality of the residual into a single line.

Formal

$\mathbb{E}[Y \mid X]$ is the unique (up to probability-zero events) function of X such that

$$\mathbb{E}\big[\,\mathbb{E}[Y \mid X]\cdot h(X)\,\big] \;=\; \mathbb{E}\big[\,Y\,h(X)\,\big] \qquad \text{for every bounded function } h.$$

Setting $h \equiv 1$ recovers the tower property $\mathbb{E}[\mathbb{E}[Y \mid X]] = \mathbb{E}[Y]$. Rearranged, it says $\mathbb{E}[(Y - \mathbb{E}[Y \mid X])\,h(X)] = 0$: the residual is orthogonal to every function of X, which is exactly the best-predictor theorem. One identity, all the consequences.

Pause

If only one sentence survives this chapter, let it be this: $\mathbb{E}[Y \mid X]$ **is a random variable — the function of X that averages Y within each value of X — and it is the best mean-square prediction of Y from X.** The tower property ($\mathbb{E}[\mathbb{E}[Y \mid X]] = \mathbb{E}[Y]$) and the law of total variance ($\mathrm{Var}(Y) = \mathbb{E}[\mathrm{Var}(Y \mid X)] + \mathrm{Var}(\mathbb{E}[Y \mid X])$) are the two ways this object lets you compute, and orthogonality is why it is best. Everything else is bookkeeping.

15.9 Exercises

Exercise 15.1 (Warm-up with a table)**.** The joint pmf of (X, Y) with $X \in \{0, 1\}$ and $Y \in \{0, 1, 2\}$ is

$p_{X,Y}$	$Y{=}0$	$Y{=}1$	$Y{=}2$
$X{=}0$	0.10	0.20	0.10
$X{=}1$	0.20	0.30	0.10

(a) Find the conditional pmf $p_{Y|X}(\cdot \mid 1)$. (b) Compute $\mathbb{E}[Y \mid X = 0]$ and $\mathbb{E}[Y \mid X = 1]$, and write $\mathbb{E}[Y \mid X]$ as a random variable. (c) Verify the tower property by computing $\mathbb{E}[Y]$ two ways.

Exercise 15.2 (Uniform on a triangle, continued)**.** For (X, Y) uniform on $\{0 \le y \le x \le 1\}$ as in Example 15.2: (a) confirm $\mathbb{E}[Y \mid X] = X/2$; (b) find $\mathrm{Var}(Y \mid X)$ as a function of X (recall the variance of Uniform$(0, x)$ is $x^2/12$); (c) use the law of total variance to compute $\mathrm{Var}(Y)$. You will need $\mathbb{E}[X]$, $\mathbb{E}[X^2]$ for the marginal $f_X(x) = 2x$.

Exercise 15.3 (Mixture of two coins)**.** A drawer has two coins: a fair one ($p = 1/2$) and a biased one ($p = 0.9$). You pick one uniformly at random (call the choice X) and flip it $n = 10$ times; let Y be the number of heads. (a) Find $\mathbb{E}[Y \mid X]$ and $\mathrm{Var}(Y \mid X)$ for each coin. (b) Compute $\mathbb{E}[Y]$ and $\mathrm{Var}(Y)$ via the tower property and EVVE. (c) Which term of the variance decomposition captures "not knowing which coin"?

Exercise 15.4 (Random sum variance)**.** Let $T = \sum_{i=1}^{N} X_i$ with X_i i.i.d. mean μ, variance σ^2, independent of the count N (mean $\mathbb{E}[N]$, variance $\mathrm{Var}(N)$). Using EVVE, show

$$\mathrm{Var}(T) = \mathbb{E}[N]\,\sigma^2 + \mu^2\,\mathrm{Var}(N).$$

Interpret each term. (Hint: $\mathrm{Var}(T \mid N) = N\sigma^2$ and $\mathbb{E}[T \mid N] = N\mu$.)

Exercise 15.5 (Best predictor sanity check)**.** Let $X \sim \mathrm{Uniform}(0, 1)$ and $Y = X^2 + W$ where $W \sim \mathcal{N}(0, 1)$ independent of X. (a) Find $\mathbb{E}[Y \mid X]$. (b) What is the minimum achievable mean squared error $\mathbb{E}[(Y - h(X))^2]$ over all functions h, and which h achieves it? (c) Explain why a linear predictor $a + bX$ cannot match it here.

Exercise 15.6 (Conditional expectation of an indicator is a probability)**.** Let A be an event and $\mathbf{1}_A$ its indicator. Show that $\mathbb{E}[\mathbf{1}_A \mid X] = \mathbb{P}(A \mid X)$, the conditional probability of A as a random variable. Then deduce the law of total probability $\mathbb{P}(A) = \mathbb{E}[\mathbb{P}(A \mid X)] = \sum_x \mathbb{P}(A \mid X = x)\,\mathbb{P}(X = x)$ as a special case of the tower property.

Exercise 15.7 (Overdispersion)**.** Let $\Lambda \sim \mathrm{Gamma}(\alpha, \beta)$ (mean α/β, variance α/β^2) and $N \mid \Lambda \sim \mathrm{Poisson}(\Lambda)$. (a) Find $\mathbb{E}[N]$ and $\mathrm{Var}(N)$ using the tower property and EVVE. (b) Show $\mathrm{Var}(N) > \mathbb{E}[N]$, and identify the extra variance with the "signal" term. (This mixture is the negative binomial distribution — the standard fix for overdispersed count data.)

Exercise 15.8 (Proof: pulling out a known factor)**.** Prove part (2) of Proposition 15.1: for any functions h and Y with $\mathbb{E}|h(X)Y| < \infty$, $\mathbb{E}[h(X)Y \mid X] = h(X)\,\mathbb{E}[Y \mid X]$. Do the discrete case from the definition of conditional expectation, being explicit about why $h(X)$ may leave the inner sum.

Exercise 15.9 (Proof: the defining property implies the tower property and orthogonality)**.** Assume the defining identity of the Formal box: $\mathbb{E}[\mathbb{E}[Y \mid X]\,h(X)] = \mathbb{E}[Y\,h(X)]$ for all bounded h. (a) Derive the tower property. (b) Derive that $Y - \mathbb{E}[Y \mid X]$ is uncorrelated with every $h(X)$. (c) Use (b) to give a one-line proof that $\mathbb{E}[Y \mid X]$ minimizes $\mathbb{E}[(Y - h(X))^2]$ among functions of X.

Exercise 15.10 (Conceptual: when do best and best-linear coincide?)**.** Give an example of a joint distribution where $\mathbb{E}[Y \mid X]$ is a nonlinear function of X (so the best predictor strictly beats the best line), and a second example where they are equal. For the second, explain what is special about the bivariate normal that forces $\mathbb{E}[Y \mid X]$ to be linear.

Chapter 16

Transformations of Random Variables

You already own a small zoo of random variables – uniforms, normals, exponentials, Poissons – and you can compute their means, variances, and probabilities. But the random variables you actually *care* about are almost never the ones handed to you. A physicist measures a voltage X but wants the power X^2. A statistician simulates a uniform U but needs a normal. A financier holds a log-return R but quotes the price e^R. A geometer drops a random point and asks for its distance from the origin. In every case the same question arises: *given the distribution of X, what is the distribution of $Y = g(X)$?*

Answering it is the business of this chapter. We will develop three complementary tools, each illuminating a different facet of the same problem:

1. The **CDF method** – the always-correct workhorse: chase the probability through the definition $F_Y(y) = \mathbb{P}(g(X) \le y)$.
2. The **change-of-variables (Jacobian) method** – a shortcut for smooth, invertible g that reveals *why* densities stretch and squeeze the way they do, and which is nothing other than calculus' substitution rule wearing a probabilistic hat.
3. **Order statistics** – what happens to the minimum, the maximum, and the sorted values of a sample, a transformation of *many* variables at once.

We close with the result that ties the whole chapter together: the *probability integral transform*, which says that every random variable is secretly a uniform in disguise. That single fact is the engine inside almost every simulation you will ever run, and it leads straight to the Box–Muller trick for manufacturing normals out of thin air.

16.1 The Core Question, and Why It Is Subtle

Let X be a random variable with a known distribution, and let g be a function. Then $Y = g(X)$ is a new random variable. Its randomness is entirely inherited from X – once the universe picks the value of X, the value of Y is determined. So in some sense nothing new is happening. And yet the *distribution* of Y can look wildly different from that of X.

Intuition

Picture the number line for X, and imagine its probability spread along it like a thin layer of paint – thick where the density is high, thin where it is low. The function g *relocates* every point: the paint at position x gets moved to position $g(x)$. If g stretches a region, the paint there gets spread thinner; if g compresses a region, the paint piles up. The distribution of Y is just the picture of where all the paint ended up. The whole subject is bookkeeping for *conservation of paint*: probability is never created or destroyed, only moved.

Here is the subtlety that trips everyone the first time. It is tempting to guess that the density of Y at y is simply the density of X at $x = g^{-1}(y)$. That is wrong, and understanding why is what this chapter is really about.

Common trap

The most common mistake. If $Y = g(X)$, then *it is not true* that $f_Y(y) = f_X\big(g^{-1}(y)\big)$. Densities are not heights you can just relabel; they are *amounts of probability per unit length*. When g stretches the axis, the same chunk of probability is smeared over a longer interval, so the density must *drop*. You must account for how much g stretches or compresses – that correction factor is the Jacobian. Forgetting it is the number-one error in transformation problems.

A clean way to never make that mistake is to always reason in terms of the CDF, which deals in *accumulated probability* (a dimensionless quantity that is immune to stretching) rather than density. So we start there.

16.2 The CDF Method

The cumulative distribution function $F_Y(y) = \mathbb{P}(Y \le y)$ is the most robust handle on a distribution, because it is defined for *every* random variable – discrete, continuous, mixed, weird – and because it is just a probability, so we can manipulate it with the laws of probability we already trust.

Focus

The CDF recipe. To find the distribution of $Y = g(X)$:

1. Write $F_Y(y) = \mathbb{P}(Y \le y) = \mathbb{P}\big(g(X) \le y\big)$.
2. *Solve the inequality* $g(X) \le y$ for X, turning it into a statement about X landing in some set.
3. Evaluate that probability using the known distribution of X (i.e. using F_X).
4. If Y is continuous, differentiate: $f_Y(y) = F_Y'(y)$.

The only step that takes thought is step 2 – and that is just algebra plus care about the direction of inequalities.

16.2.1 First worked example: the square of a uniform

Example 16.1 ($Y = X^2$ for a uniform)**.** Let $X \sim \text{Uniform}(0,1)$, so $f_X(x) = 1$ for $0 < x < 1$, and let $Y = X^2$. Since X lives in $(0,1)$, so does Y. For $0 < y < 1$,

$$F_Y(y) = \mathbb{P}(X^2 \le y) = \mathbb{P}(X \le \sqrt{y}) = F_X(\sqrt{y}) = \sqrt{y},$$

where we used that $X \ge 0$ so squaring is increasing on the relevant range. Differentiating,

$$f_Y(y) = \frac{\mathrm{d}}{\mathrm{d}y}\sqrt{y} = \frac{1}{2\sqrt{y}}, \qquad 0 < y < 1.$$

Stop and look at what just happened. We started with a perfectly flat density (the uniform) and ended with one that *blows up* near 0 (Figure 16.1). Squaring crushes the interval $(0,1)$ toward 0 – the values near 1 stay near 1, but the values near $\frac{1}{2}$ get pulled down to $\frac{1}{4}$, and values near 0.1 get pulled all the way to 0.01. So probability piles up near the origin, and the density spikes there. The CDF method delivered this automatically; we never had to "think" about paint piling up, because F does that accounting for free.

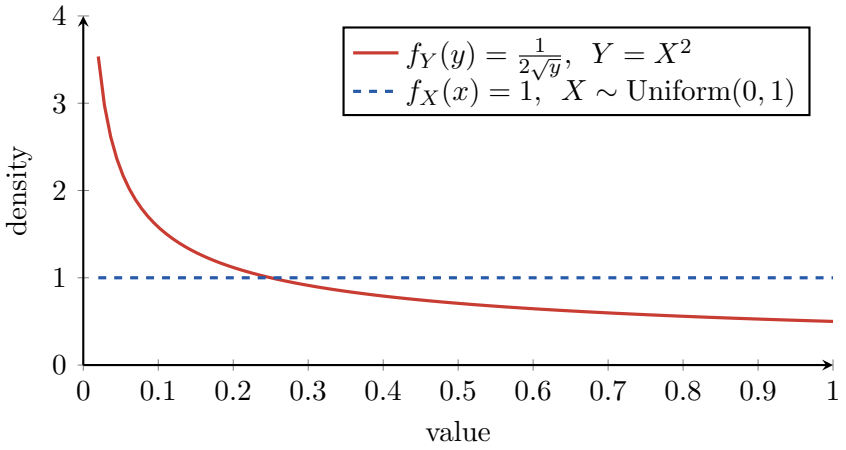


Figure 16.1: Squaring a uniform. The flat density of X (dashed) becomes the spiking density of $Y = X^2$ (solid). Probability is conserved – both curves enclose area 1 – but squaring compresses the small values together, piling density up near 0.

16.2.2 When g is not monotone: fold the probability

If g is not one-to-one, the inequality $g(X) \le y$ can correspond to *several* intervals of X, and the CDF method handles this with no extra machinery – you just add up all the pieces.

Example 16.2 ($Y = X^2$ for a standard normal – the chi-squared)**.** Let $Z \sim \mathcal{N}(0,1)$ and $Y = Z^2$. Now Z ranges over all of $\mathbb{R}$, and the event $\{Z^2 \le y\}$ for $y > 0$ is the event $\{-\sqrt{y} \le Z \le \sqrt{y}\}$ – a single interval, but a *two-sided* one. Thus

$$F_Y(y) = \mathbb{P}(-\sqrt{y} \le Z \le \sqrt{y}) = \Phi(\sqrt{y}) - \Phi(-\sqrt{y}) = 2\Phi(\sqrt{y}) - 1,$$

using the symmetry $\Phi(-a) = 1 - \Phi(a)$. Differentiate, recalling $\Phi'(z) = \varphi(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}$ and the chain rule $\frac{\mathrm{d}}{\mathrm{d}y}\sqrt{y} = \frac{1}{2\sqrt{y}}$:

$$f_Y(y) = 2\,\varphi(\sqrt{y}) \cdot \frac{1}{2\sqrt{y}} = \frac{1}{\sqrt{2\pi}}\, y^{-1/2} e^{-y/2}, \qquad y > 0.$$

This is the **chi-squared distribution with one degree of freedom**, χ^2_1 (equivalently Gamma$(\frac{1}{2}, \frac{1}{2})$). The factor of 2 came from *both* tails of the normal mapping onto the same positive y – the probability got *folded*.

Why it works

Why the factor of 2? Because $g(z) = z^2$ maps two distinct inputs, $+\sqrt{y}$ and $-\sqrt{y}$, to the same output y. Each contributes its own sliver of probability to the density at y. The general lesson: when g is many-to-one, the density of Y at y is a *sum* over all the preimages x with $g(x) = y$. The CDF method discovers this sum on its own, simply because solving $g(X) \le y$ naturally collects every relevant region of X.

16.2.3 The general monotone formula, derived

Let us extract the pattern. Suppose g is *strictly increasing* and differentiable on the range of X, with inverse g^{-1}. Then $g(X) \le y \iff X \le g^{-1}(y)$, so

$$F_Y(y) = \mathbb{P}\left(X \le g^{-1}(y)\right) = F_X\left(g^{-1}(y)\right).$$

Differentiating with the chain rule,

$$f_Y(y) = f_X\left(g^{-1}(y)\right) \cdot \frac{\mathrm{d}}{\mathrm{d}y} g^{-1}(y).$$

If instead g is *strictly decreasing*, then $g(X) \le y \iff X \ge g^{-1}(y)$, so $F_Y(y) = 1 - F_X(g^{-1}(y))$, and now $\frac{\mathrm{d}}{\mathrm{d}y} g^{-1}(y)$ is negative; differentiating flips a sign and we land on the same formula with an absolute value. Both cases collapse into one clean statement, which is important enough to box.

Formal

Change-of-variables formula (monotone case). If g is strictly monotone and differentiable with differentiable inverse, and X has density f_X, then $Y = g(X)$ has density

$$\boxed{f_Y(y) = f_X\left(g^{-1}(y)\right) \left| \frac{\mathrm{d}}{\mathrm{d}y}\, g^{-1}(y) \right|}$$

on the range of g, and 0 elsewhere. Writing $x = g^{-1}(y)$ and using $\frac{\mathrm{d}x}{\mathrm{d}y} = 1/g'(x)$, this is the same as $f_Y(y) = f_X(x)\,/\,|g'(x)|$.

That absolute-value derivative term is the **Jacobian** (in one dimension). It is exactly the stretch factor of g: where g stretches space ($|g'|$ large), the density gets divided down; where g compresses ($|g'|$ small), the density gets multiplied up. This is the precise, quantitative form of "conservation of paint."

16.3 The Change-of-Variables (Jacobian) Method

Connection

This is just u-substitution. Recall from calculus that if you substitute $y = g(x)$ in an integral, then $\mathrm{d}y = g'(x)\,\mathrm{d}x$, and the limits and the integrand both transform. The transformation formula for densities is that same substitution rule applied to the statement $\int_A f_X(x)\,\mathrm{d}x = \mathbb{P}(X \in A)$. We are literally re-expressing the probability integral in the new variable y. The absolute value on the Jacobian appears because probability is the *unsigned* area under a density – swapping the orientation of the limits (which a decreasing g does) must not flip the sign of a probability. See the companion calculus volume's chapter on substitution and the change-of-variables theorem for the multivariable Jacobian determinant we use below.

Why it works

The infinitesimal derivation. Forget formulas for a moment and think in infinitesimals. A tiny interval $[x, x + \mathrm{d}x]$ carries probability $f_X(x)\,\mathrm{d}x$. Under a smooth g it maps to the tiny interval $[y, y + \mathrm{d}y]$ with $\mathrm{d}y = |g'(x)|\,\mathrm{d}x$. *All* of that probability must end up in the image interval – paint is conserved – so

$$f_Y(y)\,\mathrm{d}y = f_X(x)\,\mathrm{d}x \quad \Longrightarrow \quad f_Y(y) = f_X(x)\,\frac{\mathrm{d}x}{\mathrm{d}y} = \frac{f_X(x)}{|g'(x)|}.$$

That is the whole theorem in two lines. Everything else is a careful version of this: "equal probability in corresponding little boxes."

16.3.1 Worked example: the lognormal

Example 16.3 (Exponentiating a normal)**.** Let $X \sim \mathcal{N}(\mu, \sigma^2)$ and $Y = e^X$. Here $g(x) = e^x$ is strictly increasing with inverse $g^{-1}(y) = \ln y$ for $y > 0$, and $\frac{\mathrm{d}}{\mathrm{d}y} \ln y = 1/y$. Plugging into the boxed formula,

$$f_Y(y) = f_X(\ln y) \cdot \frac{1}{y} = \frac{1}{y\,\sigma\sqrt{2\pi}} \exp\left(-\frac{(\ln y - \mu)^2}{2\sigma^2}\right), \qquad y > 0.$$

This is the **lognormal distribution**: a random variable whose logarithm is normal.

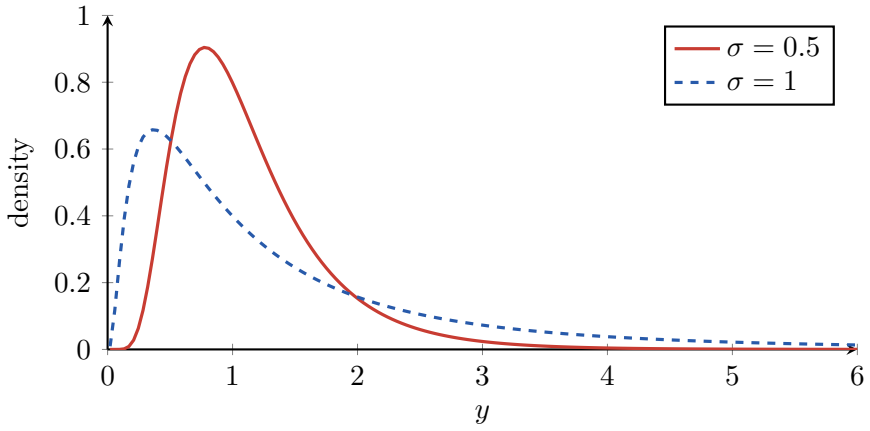


Figure 16.2: Lognormal densities ($\mu = 0$). The variable lives on $(0, \infty)$, is skewed to the right, and has a long upper tail – the signature of a quantity built from multiplicative shocks. The Gaussian exponential factor pins the density to zero at the origin, overcoming the diverging $1/y$ Jacobian factor.

Application

Why finance loves the lognormal. Stock prices cannot go negative, and over a day they tend to move by *percentages*, not by fixed dollar amounts – a \$1000 stock and a \$10 stock both "feel" a 2% move. If the log-return $\ln(S_{t+1}/S_t)$ is roughly normal (a sum of many small multiplicative shocks, blessed by the central limit theorem), then the price ratio is lognormal (Figure 16.2), and the price stays positive automatically. The celebrated Black–Scholes option-pricing model is built on exactly this transformation: model the log-price as Brownian motion (normal increments), and the price itself is lognormal. The $1/y$ Jacobian factor is the mathematical fingerprint of "returns are multiplicative."

16.3.2 The affine case, and why normals stay normal

Example 16.4 (Linear transformation of a normal)**.** Let $X \sim \mathcal{N}(\mu, \sigma^2)$ and $Y = aX + b$ with $a \neq 0$. Then $g^{-1}(y) = (y - b)/a$ and $|\frac{\mathrm{d}}{\mathrm{d}y} g^{-1}(y)| = 1/|a|$, so

$$f_Y(y) = \frac{1}{|a|} \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{((y-b)/a - \mu)^2}{2\sigma^2}\right) = \frac{1}{|a|\sigma\sqrt{2\pi}} \exp\left(-\frac{(y - (a\mu + b))^2}{2(a\sigma)^2}\right).$$

This is exactly the $\mathcal{N}(a\mu + b,\, a^2\sigma^2)$ density. So a linear function of a normal is normal, with the mean and variance transforming as $\mathbb{E}[Y] = a\mu + b$ and $\mathrm{Var}(Y) = a^2\sigma^2$ – precisely the rules you learned for $\mathbb{E}$ and Var, now confirmed at the level of the *whole distribution*. Setting $a = 1/\sigma$, $b = -\mu/\sigma$ recovers the standardization $Z = (X - \mu)/\sigma \sim \mathcal{N}(0, 1)$.

16.3.3 Several variables at once: the Jacobian determinant

When g maps a vector $(X_1, \ldots, X_n)$ to a vector $(Y_1, \ldots, Y_n)$, the one-dimensional stretch factor $|g'|$ becomes the absolute value of the *Jacobian determinant* of the inverse map. The logic is unchanged –

equal probability in corresponding little boxes – but now the "little box" is an n-dimensional volume, and a linear map scales volume by the absolute value of its determinant.

Formal

Multivariate change of variables. Let $\mathbf{X} = (X_1, \dots, X_n)$ have joint density $f_{\mathbf{X}}$, and let $\mathbf{Y} = g(\mathbf{X})$ for a smooth invertible g with inverse $\mathbf{x} = h(\mathbf{y})$. Then

$$f_{\mathbf{Y}}(\mathbf{y}) = f_{\mathbf{X}}\big(h(\mathbf{y})\big)\,\big|\det J_h(\mathbf{y})\big|, \qquad J_h(\mathbf{y}) = \left[\frac{\partial x_i}{\partial y_j}\right]_{i,j}.$$

Connection

The determinant here is the *signed volume scaling factor* of the linear map that best approximates h near a point – the same determinant from linear algebra. Its absolute value answers "how much does this map blow up or shrink volumes?" See the companion linear algebra volume's treatment of determinants as volume and the multivariable calculus chapter on the change-of-variables theorem; probability simply applies that theorem to a density.

We will put this two-dimensional version to spectacular use in the Box–Muller derivation of Section 16.6.

16.4 Order Statistics

So far g has acted on a single variable. A different and enormously useful family of transformations acts on an entire i.i.d. sample by *sorting* it. Suppose $X_1, \dots, X_n$ are i.i.d. with CDF F and density f. Sort them from smallest to largest:

$$X_{(1)} \le X_{(2)} \le \cdots \le X_{(n)}.$$

The k-th value $X_{(k)}$ is the **k-th order statistic**. In particular $X_{(1)} = \min_i X_i$ and $X_{(n)} = \max_i X_i$. These are the random variables behind record-breaking, extreme weather, system reliability (a chain fails when its *weakest* link fails – the minimum), the sample median, and the range of a dataset.

16.4.1 Maximum and minimum: independence does the work

Intuition

The maximum is $\le y$ exactly when *every single* observation is $\le y$. Because the observations are independent, that joint event factors into a product. That one observation – "all of them" becomes "each of them, multiplied" – is the entire idea behind the extreme order statistics.

For the maximum $M = X_{(n)}$:

$$F_M(y) = \mathbb{P}(\max_i X_i \le y) = \mathbb{P}(X_1 \le y, \dots, X_n \le y) = \prod_{i=1}^{n} \mathbb{P}(X_i \le y) = [F(y)]^n.$$

Differentiating, $f_M(y) = n\,[F(y)]^{n-1} f(y)$.

For the minimum $L = X_{(1)}$, it is cleaner to use the survival function, since the minimum exceeds y iff *all* observations exceed y:

$$\mathbb{P}(L > y) = \mathbb{P}(X_1 > y, \dots, X_n > y) = [1 - F(y)]^n, \qquad F_L(y) = 1 - [1 - F(y)]^n,$$

and $f_L(y) = n\,[1 - F(y)]^{n-1} f(y)$.

Example 16.5 (Minimum of exponentials – the key to Poisson processes)**.** Let $X_1, \dots, X_n$ be i.i.d. Exponential(λ), so $1 - F(y) = e^{-\lambda y}$. Then $\mathbb{P}(L > y) = (e^{-\lambda y})^n = e^{-n\lambda y}$, which is the survival function of an Exponential($n\lambda$). So the minimum of n independent exponentials is again exponential, with the rates *added*. This memorable fact – "the first of many independent alarm clocks rings at the combined rate" – is the algebraic heart of the Poisson process you will meet later.

16.4.2 The general k-th order statistic

Why it works

A density by counting arrangements. What is the density of $X_{(k)}$ at y? For $X_{(k)}$ to lie in the sliver $[y, y + \mathrm{d}y]$, we need: exactly one observation in that sliver, exactly $k-1$ observations below y, and exactly $n - k$ above $y + \mathrm{d}y$. Choosing *which* observations play these roles is a multinomial count, $\frac{n!}{(k-1)!\,1!\,(n-k)!}$. Each "below" observation contributes probability $F(y)$, each "above" contributes $1 - F(y)$, and the one in the sliver contributes $f(y)\,\mathrm{d}y$. Multiply:

$$f_{X_{(k)}}(y) = \frac{n!}{(k-1)!\,(n-k)!}\,[F(y)]^{k-1}\,[1 - F(y)]^{n-k}\,f(y).$$

Formal

Density of the k-th order statistic. For an i.i.d. sample of size n from a continuous distribution with CDF F and density f,

$$f_{X_{(k)}}(y) = \frac{n!}{(k-1)!\,(n-k)!}\,[F(y)]^{k-1}\,[1 - F(y)]^{n-k}\,f(y).$$

Setting $k = 1$ and $k = n$ recovers the minimum and maximum formulas above.

Example 16.6 (Order statistics of the uniform are Beta)**.** Let $X_1, \dots, X_n$ be i.i.d. Uniform$(0, 1)$, so $F(y) = y$ and $f(y) = 1$ on $(0, 1)$. Then

$$f_{X_{(k)}}(y) = \frac{n!}{(k-1)!\,(n-k)!}\,y^{k-1}(1-y)^{n-k}, \qquad 0 < y < 1.$$

This is exactly the Beta(k, $n - k + 1$) density! So the k-th smallest of n uniforms is Beta-distributed – which is why the Beta family is the natural language for "the distribution of a sorted uniform position," and why $\mathbb{E}[X_{(k)}] = k/(n+1)$: the order statistics divide $(0, 1)$ into $n+1$ gaps of equal expected width. The sample maximum ($k = n$) is Beta$(n, 1)$ with density ny^{n-1}, matching $n[F(y)]^{n-1}f(y)$ from before.

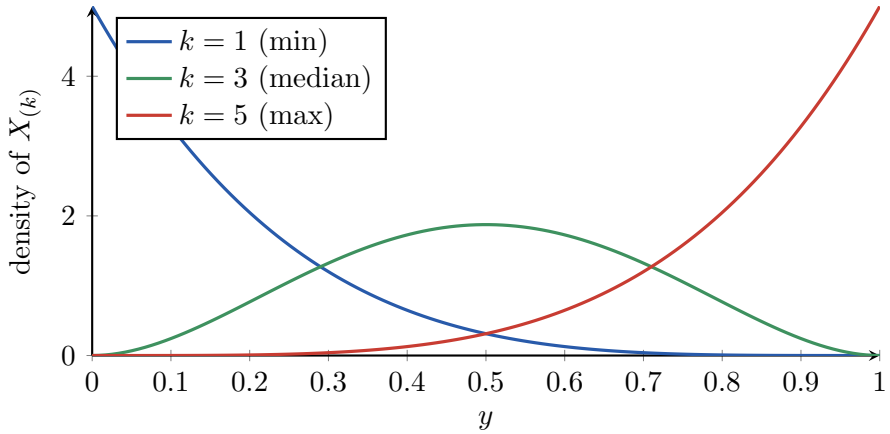


Figure 16.3: Order statistics of $n = 5$ i.i.d. uniforms, each a Beta density. The minimum (left) clusters near 0, the maximum (right) near 1, and the median (middle) peaks at $\frac{1}{2}$ – the sorted positions spread out across $(0, 1)$.

Paradox

The German Tank Problem. In World War II, the Allies wanted to estimate how many tanks Germany was producing. Captured tanks had sequential serial numbers $1, 2, \ldots, N$, and the Allies observed a sample of them; N was unknown. The largest serial number seen, $X_{(n)}$, is an order statistic, and naively it *underestimates* N – the true maximum is almost surely larger than any you have seen. But order-statistic reasoning fixes the bias: since the n observed numbers carve $\{1, \ldots, N\}$ into $n + 1$ expected-equal gaps, the gap above the maximum has expected size $N/(n+1)$, suggesting the unbiased estimate $\hat{N} = \frac{n+1}{n} X_{(n)} - 1$. Statisticians using this beat the intelligence agencies' guesses dramatically – postwar records showed the formula was off by a few percent while espionage was off by factors of several. A transformation of a sample, understood correctly, outperformed spies.

16.5 The Probability Integral Transform

We now reach the most useful transformation of all. It comes from applying one very special function g – a random variable's own CDF – to that random variable.

Intuition

The CDF F measures *how much probability lies to the left.* If you feed X into its own F, you get $U = F(X)$ – the percentile rank of X. And percentile ranks, by construction, are spread perfectly evenly: a quarter of the data has rank below 0.25, half has rank below 0.5, and so on. That "perfectly even" is the definition of the uniform distribution. So $F(X)$ is always Uniform$(0, 1)$, no matter what X was.

Theorem 16.1 (Probability Integral Transform)**.** *Let X be a continuous random variable with a strictly increasing CDF F. Then*

$$U := F(X) \sim \text{Uniform}(0, 1).$$

Conversely, if $U \sim \text{Uniform}(0,1)$*, then* $X := F^{-1}(U)$ *has CDF* F*.*

Proof. For the forward direction, take $u \in (0,1)$. Because F is strictly increasing it has an inverse, and

$$\mathbb{P}(U \le u) = \mathbb{P}\left(F(X) \le u\right) = \mathbb{P}\left(X \le F^{-1}(u)\right) = F\big(F^{-1}(u)\big) = u.$$

A random variable whose CDF is $\mathbb{P}(U \le u) = u$ on $(0,1)$ is exactly $\text{Uniform}(0,1)$.

For the converse, let $U \sim \text{Uniform}(0,1)$ and $X = F^{-1}(U)$. Then for any x,

$$\mathbb{P}(X \le x) = \mathbb{P}\left(F^{-1}(U) \le x\right) = \mathbb{P}\left(U \le F(x)\right) = F(x),$$

using that $\mathbb{P}(U \le u) = u$. So X has CDF F, as claimed. □

What is really happening

Every distribution is a uniform in disguise. The two halves of the theorem say the same thing from opposite ends. The forward half flattens *any* continuous distribution into the uniform by viewing each value through its own percentile lens. The backward half rebuilds *any* distribution from the uniform by running the percentile lens in reverse with F^{-1} (the quantile function). The uniform on $(0,1)$ is thus a universal raw material: hand me uniform randomness and a quantile function, and I can conjure any distribution you like. This is not a curiosity – it is the foundational algorithm of all stochastic simulation.

History

The idea that a CDF transforms its own variable into a uniform crystallized in the work of statisticians in the 1930s–40s – it underlies R. A. Fisher's use of p-values (a p-value is precisely F applied to a test statistic under the null, hence uniform when the null holds) and the goodness-of-fit literature. When electronic computers arrived, John von Neumann, Stanislaw Ulam, and Nicholas Metropolis at Los Alamos turned it into an engine: the *Monte Carlo method*, named for Ulam's gambling uncle, needed a way to manufacture samples from arbitrary distributions, and inverting the CDF was the first tool reached for. Von Neumann famously quipped that "anyone who considers arithmetical methods of producing random digits is, of course, in a state of sin" – but the sin was productive: every simulation you run today descends from those Los Alamos nights.

Application

Inverse-transform sampling. To generate a sample from a distribution with CDF F, when you can compute the quantile function F^{-1}:

1. Draw $U \sim \text{Uniform}(0,1)$ (every language gives you this).
2. Return $X = F^{-1}(U)$.

By Theorem 16.1, X has exactly the CDF F. For the exponential, $F(x) = 1 - e^{-\lambda x}$ inverts to $F^{-1}(u) = -\frac{1}{\lambda}\ln(1-u)$, so $X = -\frac{1}{\lambda}\ln(1-U)$ – or, since $1-U$ is also uniform, $X = -\frac{1}{\lambda}\ln U$ – is an exponential sample. Two arithmetic operations turn a uniform into an exponential. This same trick generates Cauchy ($\tan(\pi(U-\frac{1}{2}))$), logistic, Pareto, and any distribution whose quantile you can write down.

Work it yourself

Before reading on, derive an inverse-transform generator for the standard **logistic** distribution, whose CDF is $F(x) = \dfrac{1}{1+e^{-x}}$ (the sigmoid you may know from logistic regression). Solve $u = 1/(1+e^{-x})$ for x. You should find $X = \ln\big(U/(1-U)\big)$ – the *log-odds* of a uniform. Notice that the sigmoid and the log-odds (logit) are inverse functions, and that the probability integral transform is what makes them a matched encoder/decoder pair.

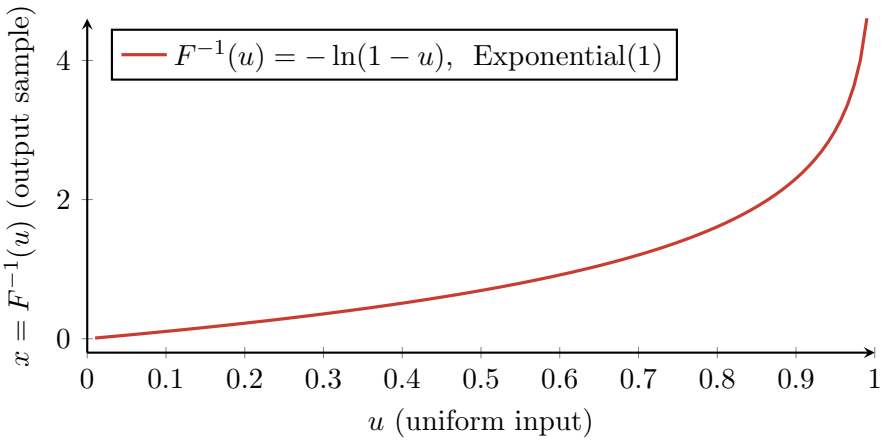


Figure 16.4: Inverse-transform sampling for the Exponential(1) distribution. Uniform inputs on the horizontal axis are pushed through the quantile function F^{-1}. Note how the steep right end stretches the rare large uniforms into the long exponential tail – the curve's slope is doing the "paint relocation" of Section 16.1.

Common trap

The probability integral transform as stated needs F *continuous* (so that $F(X)$ hits every value in $(0,1)$) and is cleanest when F is strictly increasing (so F^{-1} is a genuine function). For discrete or mixed distributions, F has jumps and flat stretches; you then use the *generalized inverse* $F^{-1}(u) = \inf\{x : F(x) \geq u\}$, and inverse-transform sampling still works – it is how you simulate a Poisson or a Bernoulli. But $F(X)$ is no longer exactly uniform when X is discrete, so do not apply the forward half blindly to a discrete variable.

16.6 Box–Muller: Manufacturing Normals

Inverse-transform sampling has one villain: the normal distribution. Its CDF Φ has *no elementary closed form*, and neither does Φ^{-1}. So the slick "$F^{-1}(U)$" recipe stalls exactly where we need it most – the normal is the most sampled distribution on Earth. In 1958, George Box and Mervin Muller found a gorgeous way around this by transforming *two* uniforms at once and exploiting the geometry of the plane.

Intuition

The trick: a single normal is hard, but a *pair* of independent normals is easy – because their joint density depends only on the distance from the origin. Picture a dartboard whose darts follow a 2D Gaussian: the cloud is perfectly circular. So instead of sampling the two Cartesian coordinates, sample in *polar* coordinates – pick a random angle (easy: uniform) and a random radius (also easy, via inverse transform on the radius's nice distribution). Then convert back to (x, y) and read off two independent normals.

Why it works

Why the radius is easy. Let $X, Y \sim \mathcal{N}(0,1)$ independently. Their joint density is

$$f(x,y) = \frac{1}{2\pi}\, e^{-(x^2+y^2)/2}.$$

It depends on (x, y) only through $r^2 = x^2 + y^2$ – the circular symmetry made visible. Switch to polar coordinates $x = r\cos\theta$, $y = r\sin\theta$. The Jacobian of the polar map is r (the standard area element $\mathrm{d}x\,\mathrm{d}y = r\,\mathrm{d}r\,\mathrm{d}\theta$), so the joint density of (R, Θ) is

$$f_{R,\Theta}(r,\theta) = \frac{1}{2\pi}\, e^{-r^2/2} \cdot r = \underbrace{\frac{1}{2\pi}}_{\Theta} \cdot \underbrace{r\, e^{-r^2/2}}_{R}, \qquad 0 < r,\ 0 \le \theta < 2\pi.$$

This *factors*: $\Theta \sim \mathrm{Uniform}(0, 2\pi)$ and, independently, R has density $r\, e^{-r^2/2}$. The angle is uniform – of course, by symmetry – and the radius has a density we can integrate in closed form, unlike Φ.

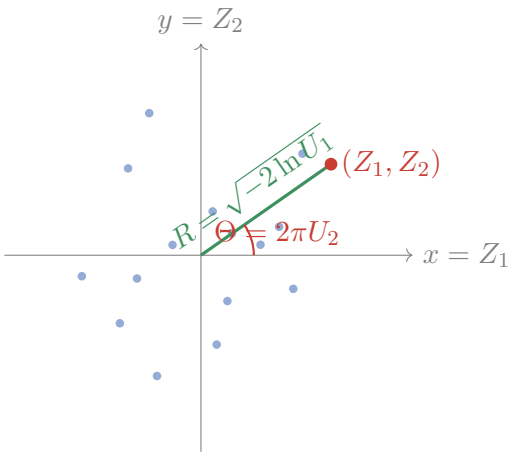


Figure 16.5: Box–Muller geometry. A pair of independent standard normals is a rotationally symmetric cloud in the plane. Rather than sampling the Cartesian coordinates directly, draw a uniform angle $\Theta = 2\pi U_2$ and a radius $R = \sqrt{-2\ln U_1}$, then read off $(Z_1, Z_2) = (R\cos\Theta,\ R\sin\Theta)$.

Now sample R by inverse transform. Its CDF is

$$F_R(r) = \int_0^r s\, e^{-s^2/2}\, \mathrm{d}s = 1 - e^{-r^2/2},$$

so $S := R^2$ has $\mathbb{P}(R^2 \le s) = 1 - e^{-s/2}$ – that is, $R^2 \sim \text{Exponential}(\frac{1}{2})$, which we already know how to sample. Setting $1 - e^{-r^2/2} = U_1$ and solving, $R = \sqrt{-2\ln(1-U_1)}$, or equivalently $R = \sqrt{-2\ln U_1}$. Convert back:

Formal

Box–Muller transform (Figure 16.5)**.** Let U_1, U_2 be independent Uniform$(0,1)$. Define

$$R = \sqrt{-2\ln U_1}, \qquad \Theta = 2\pi U_2,$$

and set

$$\boxed{Z_1 = R\cos\Theta = \sqrt{-2\ln U_1}\,\cos(2\pi U_2), \qquad Z_2 = R\sin\Theta = \sqrt{-2\ln U_1}\,\sin(2\pi U_2).}$$

Then Z_1 and Z_2 are *independent* $\mathcal{N}(0,1)$ random variables. Two uniforms in, two independent normals out.

What is really happening

What just happened is the multivariate change-of-variables theorem (Section 16.3) run *in reverse* and read as an algorithm. We could not invert Φ in one dimension, so we lifted to two dimensions where the Gaussian's rotational symmetry hands us a factorization the one-dimensional problem hid. The Jacobian r from polar coordinates was the secret ingredient – it converted the stubborn $e^{-r^2/2}$ into the friendly $r\,e^{-r^2/2}$, whose integral is elementary. Symmetry plus a Jacobian defeated a non-elementary integral.

Example 16.7 (Box–Muller by hand)**.** Suppose your random-number generator hands you $U_1 = 0.135$ and $U_2 = 0.250$. Then

$$R = \sqrt{-2\ln 0.135} = \sqrt{-2(-2.002)} = \sqrt{4.005} \approx 2.001, \qquad \Theta = 2\pi(0.250) = \tfrac{\pi}{2}.$$

So $Z_1 = 2.001\cos(\pi/2) \approx 0$ and $Z_2 = 2.001\sin(\pi/2) \approx 2.001$. We have produced two standard-normal draws, 0.00 and 2.00, from two uniforms with nothing but a log, a square root, and a sine/cosine. (Here $U_2 = 0.25$ landed the angle exactly on the y-axis, which is why Z_1 came out essentially zero – a coincidence of the chosen numbers, not a bias.)

Application

Where Box–Muller lives today. Every time a deep-learning framework initializes a weight matrix with Gaussian noise, every time a Monte Carlo option pricer simulates a stock path, every time a physics engine adds thermal jitter, normal samples are being generated – and for decades Box–Muller (or its rejection-based cousin, the *polar/Marsaglia* method, which avoids the trig calls) did the job. Modern libraries often use the even faster Ziggurat algorithm, but Box–Muller remains the one everyone *derives*, because it shows so cleanly how a clever transformation turns the cheap and universal uniform into the expensive and indispensable normal.

16.7 A Unifying View

Why this matters

Step back and notice that all three methods are one method seen from three angles.
The **CDF method** tracks accumulated probability and is always valid because probabilities never lie. The **Jacobian method** is the CDF method differentiated – it trades global accounting for a local stretch factor, and it is fastest when g is smooth and invertible. **Order statistics** are a multivariate transformation (sorting) handled by either method, plus a combinatorial count of arrangements. And the **probability integral transform** is the special case where the transforming function is the CDF itself, collapsing every distribution onto the uniform and back.
The single thread running through all of it is *conservation of probability*: probability is a fixed total of one unit, and a transformation can only *move* it, never make or destroy it. Master that one principle and you never need to memorize a transformation formula again – you can re-derive any of them from "where did the probability go?"

Exercises

Exercise 16.1 (CDF method, monotone)**.** Let $X \sim \text{Uniform}(0,1)$ and $Y = X^3$. Find F_Y and f_Y by the CDF method, then re-derive f_Y using the Jacobian formula and check they agree. Where does the density blow up, and why?

Exercise 16.2 (Folding a symmetric variable)**.** Let $X \sim \mathcal{N}(0,1)$ and $Y = |X|$ (the "half-normal"). Show that $f_Y(y) = \sqrt{2/\pi}\, e^{-y^2/2}$ for $y > 0$, and explain the factor of 2 in terms of preimages.

Exercise 16.3 (Exponential to uniform and back)**.** Let $X \sim \text{Exponential}(\lambda)$. Verify directly that $U = 1 - e^{-\lambda X}$ is $\text{Uniform}(0,1)$, illustrating the forward probability integral transform. Then start from $U \sim \text{Uniform}(0,1)$ and recover an exponential, confirming the converse.

Exercise 16.4 (Inverse-transform design)**.** The Pareto distribution has CDF $F(x) = 1 - (x_m/x)^\alpha$ for $x \geq x_m > 0$ and $\alpha > 0$. Derive the inverse-transform sampler $X = F^{-1}(U)$, and write a one-line formula that turns a uniform draw into a Pareto draw.

Exercise 16.5 (Order statistics computation)**.** Five sensors have independent $\text{Uniform}(0,1)$ failure thresholds. (a) Find the density of the *second-smallest* threshold $X_{(2)}$ and identify it as a Beta distribution. (b) Compute $\mathbb{E}[X_{(2)}]$. (c) Find the probability that the largest threshold exceeds 0.9.

Exercise 16.6 (Minimum of exponentials)**.** Let $X_1, \ldots, X_n$ be independent with $X_i \sim \text{Exponential}(\lambda_i)$ (possibly different rates). Show that $\min_i X_i \sim \text{Exponential}\big(\sum_i \lambda_i\big)$, and that the probability that X_j is the minimum equals $\lambda_j / \sum_i \lambda_i$. (Hint: use the survival function for the first part; condition on X_j and integrate for the second.)

Exercise 16.7 (A two-variable Jacobian)**.** Let X, Y be independent Exponential(1). Define $S = X + Y$ and $T = X/(X+Y)$. Using the two-dimensional change-of-variables formula, show that $S \sim \text{Gamma}(2,1)$ and $T \sim \text{Uniform}(0,1)$, and that S and T are independent. (This is a famous and surprising decoupling of "total" from "proportion.")

Exercise 16.8 (Box–Muller verification)**.** Carry out the Box–Muller change of variables in full: starting from the joint density of (U_1, U_2), with (Z_1, Z_2) as the map of (U_1, U_2), compute the 2×2 Jacobian determinant and confirm that the resulting joint density of (Z_1, Z_2) is $\frac{1}{2\pi} e^{-(z_1^2+z_2^2)/2}$, hence two independent standard normals.

Exercise 16.9 (Conceptual: why the absolute value?)**.** Explain, without computation, why the change-of-variables formula has $|g'(x)|$ rather than $g'(x)$. What would go wrong for a strictly decreasing g if the absolute value were omitted? Relate your answer to "densities are nonnegative" and "probabilities are nonnegative."

Exercise 16.10 (Proof: PIT for general continuous F)**.** Prove the forward probability integral transform without assuming F is strictly increasing – only continuous. (Hint: define the generalized inverse $F^{-1}(u) = \inf\{x : F(x) \geq u\}$ and show $\mathbb{P}(F(X) \leq u) = u$ by relating the event $\{F(X) \leq u\}$ to $\{X \leq F^{-1}(u)\}$.)

Exercise 16.11 (Median of an odd sample)**.** For an i.i.d. sample of size $n = 2m+1$ from Uniform$(0, 1)$, the sample median is $X_{(m+1)}$. Write its density, identify the Beta distribution, and argue from symmetry that its mean is exactly $\frac{1}{2}$. How does its variance behave as $n \to \infty$, and what does that say about the sample median as an estimator?

Exercise 16.12 (Simulating a triangular distribution)**.** The symmetric triangular density on $(0, 1)$ is $f(x) = 4x$ for $0 < x \leq \frac{1}{2}$ and $f(x) = 4(1 - x)$ for $\frac{1}{2} < x < 1$. (a) Find its CDF. (b) Derive the piecewise inverse-transform sampler. (c) Show that the same distribution arises as the average $(U_1 + U_2)/2$ of two independent uniforms, giving a second, slicker simulation recipe.

Chapter 17

Sums of Random Variables and Convolutions

If there is one operation that probability cannot live without, it is *addition*. The total of two dice. The sum of a hundred daily returns on a stock. The number of clicks across a thousand visitors. The position of a particle after a million tiny random kicks. Again and again, the quantity we actually care about is not a single random variable but a *sum* of them, and the central question becomes: if I know the distribution of each piece, what is the distribution of the whole?

This chapter answers that question. The central object is the **convolution**: a single formula that builds the distribution of $X + Y$ out of the distributions of X and Y when they are independent. We derive convolution twice – once by counting, once by integrating – and each time it will feel inevitable rather than imposed. We will also see which families of distributions are "closed under addition" (add two Poissons and you get a Poisson; add two normals and you get a normal), revisit why expectation and variance behave so cleanly under sums, and set the stage for the limit theorems that crown the subject.

17.1 The Question: What Is the Distribution of a Sum?

Start with the simplest possible story.

Intuition

You roll two fair four-sided dice (a Uniform$\{1, 2, 3, 4\}$ each). You do not care about the individual faces; you care about the *total* $S = X + Y$. What is $\mathbb{P}(S = 5)$?
You already know how to do this by brute force: list the 16 equally likely outcomes (x, y) and count those with $x + y = 5$. They are $(1, 4), (2, 3), (3, 2), (4, 1)$ – four of them – so $\mathbb{P}(S = 5) = 4/16 = 1/4$.
Stare at that list. Each pair has the form $(k, 5 - k)$. We summed over all the ways the two pieces can "conspire" to produce the target total 5: the first die is k, the second die is $5 - k$, and we add up the probability of each such split. That summing-over-all-splits is exactly what a convolution is. The rest of this chapter is just that idea, made precise and pushed into the continuous world.

Before naming the tool, we should be clear about *when* it applies and what could go wrong.

Focus

The distribution of $S = X + Y$ depends, in general, on the *joint* distribution of (X, Y), not just on the two marginals. Two random variables with the same individual distributions can produce very different sums depending on how they are coupled.

Extreme example. Let $X \sim \text{Uniform}\{0, 1\}$ (a fair coin coded 0/1).

- If $Y = X$ (perfectly dependent), then $S = 2X \in \{0, 2\}$, each with probability 1/2. It can never equal 1.
- If $Y = 1 - X$ (perfectly anti-dependent), then $S \equiv 1$ always.
- If Y is an independent copy of X, then $S \in \{0, 1, 2\}$ with probabilities $\frac{1}{4}, \frac{1}{2}, \frac{1}{4}$.

Same marginals, three completely different sums. So the clean formulas of this chapter will require an assumption, and that assumption is **independence**.

Throughout, unless we say otherwise, $X \perp\!\!\!\perp Y$.

17.2 Discrete Convolution: Summing Over All the Ways

Suppose X and Y are independent integer-valued random variables with probability mass functions (PMFs) p_X and p_Y. We want the PMF of $S = X + Y$. Fix a target value s. The event $\{S = s\}$ is the union, over every integer k, of the event "$X = k$ *and* $Y = s - k$." These are disjoint (different k means a different value of X), so we add their probabilities, and independence lets us factor each one:

$$\mathbb{P}(S = s) \;=\; \sum_k \mathbb{P}(X = k,\, Y = s - k) \;=\; \sum_k \mathbb{P}(X = k)\; \mathbb{P}(Y = s - k).$$

That is the whole derivation. We give it a name.

Definition 17.1 (Discrete convolution)**.** For independent integer-valued X, Y with PMFs p_X, p_Y, the PMF of $S = X + Y$ is the *convolution* $p_X * p_Y$:

$$p_S(s) \;=\; (p_X * p_Y)(s) \;=\; \sum_{k=-\infty}^{\infty} p_X(k)\, p_Y(s - k).$$

What is really happening

Read the formula as a recipe. To land the total at s, you must split s into a part contributed by X (call it k) and the leftover $s - k$ contributed by Y. There are many possible splits; you weight each split by how likely it is ($p_X(k)\, p_Y(s - k)$) and add up. The index k slides across all values while $s - k$ slides in the opposite direction – one counter goes up exactly as the other goes down so their sum is pinned at s. That "one up, one down" sliding is the visual signature of convolution, and it is why engineers describe convolution as "flip one sequence and slide it past the other."

Connection

If you have met convolution in the companion calculus book (or in signal processing), this is the *same* operation applied to PMFs. Multiplying polynomials is convolution in disguise: if $A(z) = \sum_k a_k z^k$ and $B(z) = \sum_k b_k z^k$, then the coefficient of z^s in $A(z)B(z)$ is exactly $\sum_k a_k b_{s-k}$. We will exploit this through the idea of *generating functions*: turn a PMF into a polynomial (or power series), and adding independent random variables becomes *multiplying* their generating functions. That turns hard convolutions into easy products – the central trick of the next chapter.

17.2.1 Worked example: two four-sided dice

Let $p_X(k) = p_Y(k) = \frac{1}{4}$ for $k \in \{1, 2, 3, 4\}$ and 0 otherwise. By Definition 17.1,

$$p_S(s) \;=\; \sum_{k=1}^{4} \tfrac{1}{4} \cdot p_Y(s-k) \;=\; \tfrac{1}{16}\,\#\{k \in \{1,2,3,4\} : s-k \in \{1,2,3,4\}\}.$$

So $p_S(s)$ is just 1/16 times the number of valid splits. Counting them for $s = 2, \ldots, 8$ gives $1, 2, 3, 4, 3, 2, 1$ – the famous triangular "pyramid" of dice sums. Figure 17.1 plots it.

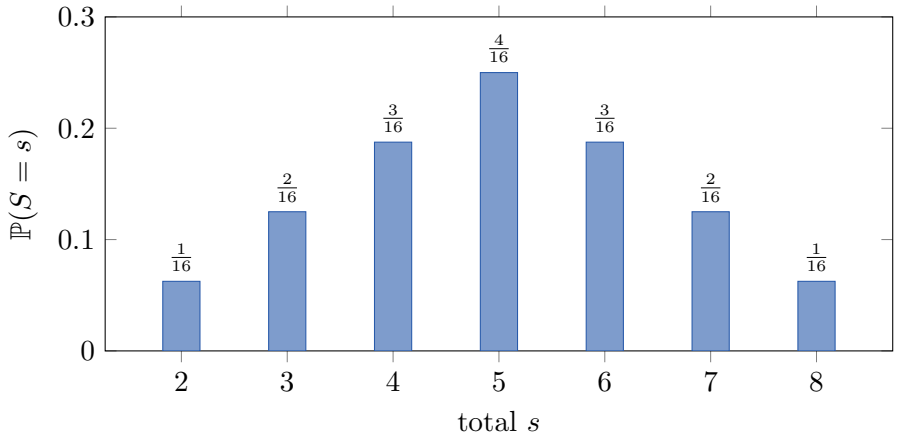


Figure 17.1: The PMF of the sum of two fair four-sided dice. Two flat (uniform) distributions convolve into a triangle. This "smoothing" – flat becomes peaked – is the first whisper of the central limit theorem.

Why this matters

Notice what convolution did to the *shape*. Each die alone is flat: every face equally likely. Their sum is a triangle, peaked in the middle. Convolve a third die in and you get a bell-ish bump; a fourth, smoother still. Adding independent copies *pulls probability toward the center* and rounds off the corners. Keep this picture in your pocket: it is the mechanism behind the bell curve, and we will cash it out fully when we reach the central limit theorem.

17.2.2 A nontrivial discrete example: two Poissons

Convolution is not just for finite supports. Let $X \sim \text{Poisson}(\lambda)$ and $Y \sim \text{Poisson}(\mu)$ be independent, supported on $\{0, 1, 2, \ldots\}$. For $s \geq 0$,

$$\mathbb{P}(S = s) = \sum_{k=0}^{s} e^{-\lambda} \frac{\lambda^k}{k!}\, e^{-\mu} \frac{\mu^{\,s-k}}{(s-k)!} = e^{-(\lambda+\mu)} \sum_{k=0}^{s} \frac{\lambda^k \mu^{\,s-k}}{k!\,(s-k)!}.$$

Multiply and divide by $s!$ inside the sum to manufacture binomial coefficients:

$$\mathbb{P}(S = s) = \frac{e^{-(\lambda+\mu)}}{s!} \sum_{k=0}^{s} \binom{s}{k} \lambda^k \mu^{\,s-k} = e^{-(\lambda+\mu)} \frac{(\lambda+\mu)^s}{s!},$$

where the last step is the binomial theorem. The right-hand side is the PMF of $\text{Poisson}(\lambda + \mu)$. We have just proved one of the most-used facts in applied probability.

Theorem 17.1 (Poissons add)**.** *If $X \sim \text{Poisson}(\lambda)$ and $Y \sim \text{Poisson}(\mu)$ are independent, then $X + Y \sim \text{Poisson}(\lambda + \mu)$.*

Intuition

This should feel *right* before you even see the algebra. The Poisson counts rare events over an interval at rate λ. If one phone line gets calls at rate λ and another, independently, at rate μ, then the combined switchboard gets calls at rate $\lambda + \mu$. Merging two independent streams of rare events just adds their rates. The convolution computation is merely the bookkeeping that confirms the story.

17.3 Continuous Convolution: From Sums to Integrals

Now let X and Y be independent *continuous* random variables with densities f_X and f_Y. We want the density of $S = X + Y$. The route that actually works runs through the cumulative distribution function (CDF), because the probability of a single point is zero and cannot be added the way PMFs were.

17.3.1 Derivation via the CDF

Fix a number s. We compute $F_S(s) = \mathbb{P}(X + Y \leq s)$ by integrating the joint density over the half-plane $\{(x, y) : x + y \leq s\}$. By independence the joint density factors, $f_{X,Y}(x, y) = f_X(x) f_Y(y)$:

$$F_S(s) = \iint_{x+y \leq s} f_X(x)\, f_Y(y)\, \mathrm{d}x\, \mathrm{d}y = \int_{-\infty}^{\infty} f_X(x) \left(\int_{-\infty}^{s-x} f_Y(y)\, \mathrm{d}y \right) \mathrm{d}x.$$

The inner integral is $F_Y(s - x)$. Now differentiate in s to recover the density. Differentiation passes under the outer integral, and $\frac{\mathrm{d}}{\mathrm{d}s} F_Y(s - x) = f_Y(s - x)$:

$$f_S(s) = \frac{\mathrm{d}}{\mathrm{d}s} F_S(s) = \int_{-\infty}^{\infty} f_X(x)\, f_Y(s - x)\, \mathrm{d}x.$$

Definition 17.2 (Continuous convolution)**.** For independent continuous X, Y with densities f_X, f_Y, the density of $S = X + Y$ is the *convolution*

$$f_S(s) = (f_X * f_Y)(s) = \int_{-\infty}^{\infty} f_X(x)\, f_Y(s - x)\, \mathrm{d}x.$$

Connection

Compare Definitions 17.1 and 17.2: the sum $\sum_k$ has become an integral $\int \mathrm{d}x$, and the PMFs have become densities. This is exactly the discrete-to-continuous translation you saw for expectation in the calculus companion – a Riemann sum passing to its integral. Convolution is one operation wearing two outfits.

Why it works

Why does differentiating "under the integral sign" give the density? Because for nice (continuous, integrable) f_X, f_Y the map $s \mapsto \int f_X(x) F_Y(s - x)\mathrm{d}x$ is differentiable and its derivative is $\int f_X(x)\, f_Y(s - x)\mathrm{d}x$, by the Leibniz rule for differentiating an integral with respect to a parameter. The regularity conditions hold for every density we use in this book; the rigorous statement lives in the measure-theory chapters. For now, trust that the CDF is smooth enough to differentiate term by term.

17.3.2 Worked example: sum of two uniforms is a triangle

Let X, Y be independent Uniform$(0, 1)$, so $f_X = f_Y = \mathbf{1}_{[0,1]}$. Then

$$f_S(s) = \int_{-\infty}^{\infty} \mathbf{1}_{[0,1]}(x)\, \mathbf{1}_{[0,1]}(s - x)\, \mathrm{d}x = \int_0^1 \mathbf{1}_{[0,1]}(s - x)\, \mathrm{d}x.$$

The integrand is 1 exactly when $0 \le s - x \le 1$, i.e. when $s - 1 \le x \le s$. Intersecting that with $[0, 1]$:

- For $0 \le s \le 1$: the overlap is $[0, s]$, length s. So $f_S(s) = s$.
- For $1 \le s \le 2$: the overlap is $[s - 1, 1]$, length $2 - s$. So $f_S(s) = 2 - s$.
- Otherwise $f_S(s) = 0$.

The result is the *triangular* (or "tent") density on $[0, 2]$, peaking at $s = 1$. Figure 17.2 shows it.

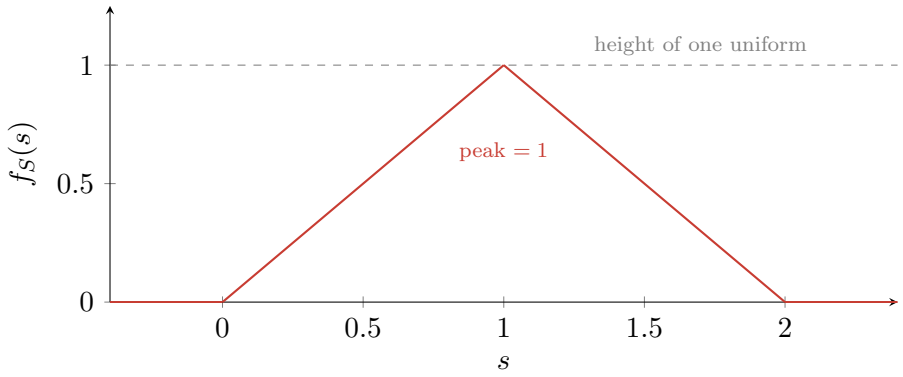


Figure 17.2: Two independent Uniform$(0,1)$ variables sum to a triangular density on $[0,2]$. Geometrically, $f_S(s)$ is the length of overlap when one unit box slides past another; the overlap grows, peaks at $s=1$, then shrinks. Flat plus flat equals peaked – the continuous twin of the dice picture.

Walking question

Walking question. Before reading on: what do you expect the density of the sum of *three* independent Uniform$(0,1)$ variables to look like? Sketch it.
Answer. Convolving the tent with one more box rounds the two straight sides into curves; you get a smooth bell-shaped piece made of three parabolic arcs (the Irwin–Hall density for $n=3$), supported on $[0,3]$ and peaking at 1.5. Each extra uniform you add makes the result smoother and more bell-like. By $n=12$ the sum is so close to a normal that the old "add twelve uniforms and subtract six" trick was a standard way to fake a standard normal before fast Gaussian samplers existed.

History

The word "convolution" (German *Faltung*, literally "a folding") spread through analysis in the early twentieth century, but the *operation* is much older in probability. Laplace, around 1810, was already adding independent errors by exactly this integral while developing what we now call the central limit theorem, and he used a multiply-the-transforms shortcut (characteristic functions in modern language) to avoid grinding through the integrals directly. The "folding" name captures the picture in Figure 17.2: to evaluate $(f*g)(s)$ you reflect g about the vertical axis and slide it, folding one function back across the other.

17.4 Sums Within a Family: Closure Under Addition

Some families of distributions have a magical property: add two members and you stay inside the family, with parameters that simply combine. We already proved it for the Poisson. Here is the roster you should know cold.

17.4.1 Binomial plus binomial (same p)

Theorem 17.2 (Binomials add when p matches)**.** *If $X \sim \text{Binomial}(m, p)$ and $Y \sim \text{Binomial}(n, p)$ are independent* with the same success probability p*, then $X + Y \sim \text{Binomial}(m + n, p)$.*

Why it works

The slick proof needs no algebra. A Binomial(m, p) counts successes in m independent coin flips with bias p; a Binomial(n, p) counts successes in another n such flips. Lay all $m + n$ flips in a row – they are independent and all have bias p – and $X + Y$ is the total number of successes among $m + n$ flips, which is by definition Binomial$(m + n, p)$. The constraint "same p" is essential: if the coins had different biases, the pooled flips would not be identically distributed and the count would not be binomial.

If you prefer to see the convolution machinery confirm it, the algebraic heart is Vandermonde's identity $\sum_k \binom{m}{k}\binom{n}{s-k} = \binom{m+n}{s}$, which collects the ways to get s total successes from the two groups.

17.4.2 Normal plus normal

Theorem 17.3 (Normals add)**.** *If $X \sim \mathcal{N}(\mu_1, \sigma_1^2)$ and $Y \sim \mathcal{N}(\mu_2, \sigma_2^2)$ are independent, then*

$$X + Y \sim \mathcal{N}\big(\mu_1 + \mu_2,\ \sigma_1^2 + \sigma_2^2\big)\,.$$

The means add and the variances *add (so the standard deviations add in quadrature, $\sqrt{\sigma_1^2 + \sigma_2^2}$, not linearly).*

Why it works

Here is the convolution computation for the clean case $\mu_1 = \mu_2 = 0$ (a shift handles general means). With $f_X(x) = \frac{1}{\sqrt{2\pi}\sigma_1} e^{-x^2/2\sigma_1^2}$ and similarly for Y,

$$f_S(s) = \frac{1}{2\pi\sigma_1\sigma_2} \int_{-\infty}^{\infty} \exp\!\left(-\frac{x^2}{2\sigma_1^2} - \frac{(s-x)^2}{2\sigma_2^2}\right) \mathrm{d}x.$$

The exponent is a quadratic in x. "Completing the square" in x splits it into a part that depends only on s and a part that integrates to a constant Gaussian integral over x. Carrying out the algebra (group the x^2 terms, the x terms, the constant terms; the x-integral becomes a standard $\int e^{-a(x-b)^2}\mathrm{d}x = \sqrt{\pi/a}$) leaves

$$f_S(s) = \frac{1}{\sqrt{2\pi(\sigma_1^2 + \sigma_2^2)}} \exp\!\left(-\frac{s^2}{2(\sigma_1^2 + \sigma_2^2)}\right),$$

which is exactly the $\mathcal{N}(0, \sigma_1^2 + \sigma_2^2)$ density. The whole miracle is that a Gaussian times a Gaussian, integrated, is a Gaussian – because the product of two exponentials-of-quadratics is an exponential-of-a-quadratic, and quadratics complete the square. (Using characteristic functions in the next chapter, the same fact takes one line.)

Why this matters

The normal family is the *fixed point* of convolution. Almost every other family changes shape when you add copies (uniform becomes triangle becomes bell), forever crawling toward the normal. The normal is already there. It does not move. That self-reproducing stability is the deep reason the normal is the universal limit: it is the only "shape" that adding cannot disturb (among distributions with finite variance). We make this precise in the central limit theorem.

17.4.3 Gamma plus gamma (same rate)

The Gamma family with shape α and rate β has density $f(x) = \frac{\beta^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\beta x}$ for $x > 0$. It too is closed under addition when the rate matches.

Theorem 17.4 (Gammas add when the rate matches)**.** *If $X \sim \text{Gamma}(\alpha_1, \beta)$ and $Y \sim \text{Gamma}(\alpha_2, \beta)$ are independent with the same rate β, then $X + Y \sim \text{Gamma}(\alpha_1 + \alpha_2, \beta)$.*

Why it works

Convolve the densities for $s > 0$ (constants pulled out, β shared):

$$f_S(s) = \frac{\beta^{\alpha_1+\alpha_2}}{\Gamma(\alpha_1)\Gamma(\alpha_2)} e^{-\beta s} \int_0^s x^{\alpha_1-1}(s-x)^{\alpha_2-1}\,\mathrm{d}x.$$

Substitute $x = su$ ($\mathrm{d}x = s\,\mathrm{d}u$, limits 0 to 1):

$$\int_0^s x^{\alpha_1-1}(s-x)^{\alpha_2-1}\mathrm{d}x = s^{\alpha_1+\alpha_2-1} \int_0^1 u^{\alpha_1-1}(1-u)^{\alpha_2-1}\mathrm{d}u = s^{\alpha_1+\alpha_2-1}\, B(\alpha_1, \alpha_2),$$

where $B(\alpha_1, \alpha_2) = \Gamma(\alpha_1)\Gamma(\alpha_2)/\Gamma(\alpha_1+\alpha_2)$ is the Beta function. Substituting back, the Gamma functions telescope:

$$f_S(s) = \frac{\beta^{\alpha_1+\alpha_2}}{\Gamma(\alpha_1+\alpha_2)}\, s^{\alpha_1+\alpha_2-1} e^{-\beta s},$$

the Gamma$(\alpha_1 + \alpha_2, \beta)$ density. The Beta integral is doing all the work; this is the same combinatorics-of-the-continuum that made the uniforms convolve into a tent.

Corollary 17.1 (Exponentials sum to an Erlang)**.** *Since* Exponential(β) = Gamma$(1, \beta)$*, the sum of n independent* Exponential(β) *variables is* Gamma(n, β) *(the* Erlang *distribution). Concretely, the waiting time until the nth event in a Poisson process of rate β is* Gamma(n, β) *– you wait for n independent exponential inter-arrival gaps and add them up.*

Application

Convolution in machine learning, two ways. The same word shows up in deep learning, and the connection is exact. A convolutional layer slides a small kernel across an input and, at each position, computes $\sum_k w_k\, x_{s-k}$ – the discrete convolution of Definition 17.1 with the kernel w playing the role of one PMF. The probabilistic reading also pays off directly: when a system adds independent random delays – a request hops through several microservices,

each adding an Exponential(β)-like latency – the end-to-end latency is the convolution of the per-hop densities, an Erlang by Corollary 17.1. Engineers who size timeouts from the *tail* of that convolved distribution (the 99.9th percentile) rather than the mean are using exactly this chapter's machinery to budget for the rare slow request.

Common trap

The matching-parameter trap. Each closure theorem comes with a *shared* parameter that you may not violate:

- Binomials add only with the **same** p.
- Gammas add only with the **same rate** β.
- Normals add for **any** parameters (no matching needed) – that is part of what makes them special.
- Poissons add for any rates (the rates simply sum).

A common error is to "add" two binomials with different p, or two gammas with different rates, and report a member of the same family. The true sum exists, of course, but it is *not* in the family and has no tidy closed form. When the shared parameter does not match, you must fall back to the raw convolution.

Work it yourself

Work it yourself. Convince yourself the closures are mutually consistent. Start from $X_i \sim$ Bernoulli(p) i.i.d.

1. Each X_i is Binomial$(1, p)$. Use Theorem 17.2 n times to conclude $\sum_{i=1}^{n} X_i \sim$ Binomial(n, p). (This is the very *definition* of the binomial – nice that the convolution machinery reproduces it.)
2. Each Exponential(β) is Gamma$(1, \beta)$. Use Theorem 17.4 to get the Erlang of Corollary 17.1.
3. Each Poisson(λ/n) is independent; their sum over $i = 1, \ldots, n$ is Poisson(λ) by Theorem 17.1. So a Poisson of rate λ is an aggregate of many tiny Poissons – the "infinitely divisible" property.

17.5 Expectation and Variance of Sums, Revisited

Convolution gives the *entire* distribution of a sum, which is more than we often need. Frequently we only want two numbers: the mean and the variance. Those obey rules so clean they almost feel like cheating, and – crucially – the mean rule needs *no* independence at all.

Theorem 17.5 (Linearity of expectation; additivity of variance)**.** *For* any *random variables* $X_1, \ldots, X_n$ *with finite means,*

$$\mathbb{E}\left[\sum_{i=1}^{n} X_i\right] = \sum_{i=1}^{n} \mathbb{E}[X_i].$$

For the variance, in general

$$\operatorname{Var}\left(\sum_{i=1}^{n} X_i\right) = \sum_{i=1}^{n} \operatorname{Var}(X_i) + 2 \sum_{1 \le i < j \le n} \operatorname{Cov}(X_i, X_j),$$

and if the X_i are (pairwise) uncorrelated *– in particular if they are independent – the covariance terms vanish:*

$$\operatorname{Var}\left(\sum_{i=1}^{n} X_i\right) = \sum_{i=1}^{n} \operatorname{Var}(X_i).$$

What is really happening

The contrast here is one of the most important lessons in the subject. *Means always add.* It does not matter whether your variables are independent, correlated, tangled, adversarial – expectation is linear, full stop. *Variances add only when there is no correlation.* If your variables move together ($\operatorname{Cov} > 0$), the sum swings more wildly than the individual variances would suggest; if they move oppositely ($\operatorname{Cov} < 0$), they cancel and the sum is calmer. This single asymmetry – mean is always linear, spread is not – underlies portfolio diversification, the design of experiments, and the $\sqrt{n}$ that governs the law of large numbers.

Focus

The $\sqrt{n}$ that runs through all of probability. Let $X_1, \dots, X_n$ be i.i.d. with mean μ and variance σ^2, and form $S_n = \sum X_i$. Then

$$\mathbb{E}[S_n] = n\mu, \qquad \operatorname{Var}(S_n) = n\sigma^2, \qquad \operatorname{SD}(S_n) = \sigma\sqrt{n}.$$

The sum's *typical size* grows like n, but its *typical wobble* grows only like $\sqrt{n}$. So the wobble *relative to* the total shrinks like $\sqrt{n}/n = 1/\sqrt{n} \to 0$. For the average $\bar{X}_n = S_n/n$ this reads $\operatorname{Var}(\bar{X}_n) = \sigma^2/n$, with standard deviation $\sigma/\sqrt{n}$. That vanishing $\sigma/\sqrt{n}$ is the engine of the law of large numbers (the average concentrates on μ) and, once you rescale by exactly $\sqrt{n}$, the engine of the central limit theorem. Remember the slogan: **sums grow like n, fluctuations like $\sqrt{n}$.**

Application

Diversification in a portfolio. Put your money equally in n assets, each with return of variance σ^2. If the assets are *independent*, the portfolio return $\bar{X}_n$ has variance σ^2/n – risk falls toward zero as you add names. This is the mathematical content of "don't put all your eggs in one basket." But the *covariance* terms are the catch: in a market crash assets fall *together* ($\operatorname{Cov} > 0$), the cross terms in Theorem 17.5 do not vanish, and diversification stalls. The reason a diversified portfolio still carries irreducible "systematic" risk is precisely those non-zero covariances. The clean $1/n$ decay is a fantasy that requires independence the real world does not grant.

Example 17.1 (Mean and variance without finding the full distribution)**.** You sum $n = 100$ i.i.d. rolls of a fair six-sided die. A single die has $\mathbb{E}[X] = 3.5$ and $\operatorname{Var}(X) = \frac{35}{12} \approx 2.9167$. Without ever computing the (genuinely messy) convolution of 100 uniforms, we read off

$$\mathbb{E}[S_{100}] = 100 \cdot 3.5 = 350, \qquad \operatorname{Var}(S_{100}) = 100 \cdot \tfrac{35}{12} \approx 291.7, \qquad \operatorname{SD}(S_{100}) \approx 17.1.$$

So a total near 350 ± 17 is typical; a total of 450 would be about 6 standard deviations out – effectively impossible. We extracted the practical content of the distribution from two one-line rules, no convolution required. (And by the central limit theorem the full distribution is very nearly $\mathcal{N}(350, 291.7)$.)

17.6 A Paradox of Sums: The Cauchy That Refuses to Average

Everything above whispered the same comforting story: add many independent pieces, divide by n, and the average settles down near a fixed number. It is so intuitive it feels like a law of nature. It is not. There is a distribution that breaks it, and the break is instructive.

Paradox

The standard Cauchy distribution has density $f(x) = \dfrac{1}{\pi(1+x^2)}$ on all of $\mathbb{R}$. It looks like a tame, symmetric bell. But it has *heavy tails*: the density only decays like $1/x^2$, so heavily that $\int |x|\, f(x)\, dx = \infty$. The Cauchy has **no mean** (and hence no variance).

Now the shock. If $X_1, \ldots, X_n$ are i.i.d. standard Cauchy, their *average* $\bar{X}_n = \frac{1}{n}\sum X_i$ is again *exactly* standard Cauchy – the *same* distribution, for every n. Averaging a million Cauchy samples is statistically no better than taking one. The sample mean does not converge to anything; it just keeps wandering with the same spread forever.

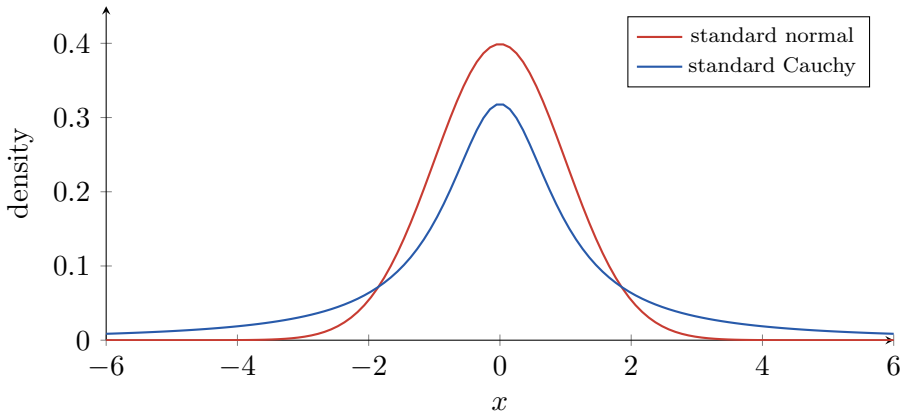


Figure 17.3: The standard normal (red) and standard Cauchy (blue) both look like symmetric bells near the center, but the Cauchy's tails decay only like $1/x^2$ – far slower than the normal's $e^{-x^2/2}$. Those fat tails carry enough probability far out that $\int |x|\, f(x)\, dx$ diverges, so the Cauchy has no mean. A single extreme draw can dominate any average, which is why averaging Cauchy samples never settles down.

Why it works

How can the average not improve? Two facts collide. First, the Cauchy is closed under addition like the normal: the convolution of two standard Cauchys is a Cauchy of scale 2, and in general the sum of n standard Cauchys is a Cauchy of scale n. (One slick proof: the characteristic function of a standard Cauchy is $e^{-|t|}$; multiplying n of them gives $e^{-n|t|}$, the

characteristic function of a Cauchy of scale n – exactly the convolution-becomes-multiplication trick of the next chapter.) Dividing that sum by n rescales by $1/n$, turning scale n back into scale 1. So the average is standard Cauchy again.
Second, the law of large numbers *requires a finite mean.* Its proof leans on $\mathrm{Var}(\bar{X}_n) = \sigma^2/n \to 0$, and the Cauchy has neither a finite variance nor even a finite mean to converge to. The $\sqrt{n}$ slogan silently assumed finite variance. When the tails are heavy enough to destroy the mean, the whole averaging machine seizes up.

Why this matters

The Cauchy is the cautionary twin of the normal. Both are stable under addition; both are fixed points of convolution. But the normal pulls probability *inward* and tames sums, while the Cauchy's fat tails (Figure 17.3) let a single freak sample dominate the whole average no matter how many other samples you collect. The lesson for data science is blunt: before you trust an average, ask whether the thing you are averaging even *has* a mean. Heavy-tailed data – file sizes, city populations, financial losses, network traffic – can behave far more like Cauchy than like the normal your instincts default to.

17.7 Setting Up the Limit Theorems

The convolution machinery points straight at the summit of the book. We have all the pieces; what remains is to see how they assemble.

Intuition

Take i.i.d. $X_1, X_2, \ldots$ with mean μ and finite variance σ^2, and let $S_n = \sum_{i=1}^{n} X_i$. The PMF/PDF of S_n is the n-fold convolution of the common distribution with itself:

$$f_{S_n} = f * f * \cdots * f \quad (n \text{ copies}).$$

We watched this in miniature: one uniform (flat) convolves into a triangle (n=2), then a bell (n=3), smoother and smoother. *Convolving a distribution with itself repeatedly drives it toward a single universal shape.* Two clean facts pin that shape down:

- **Center.** $\mathbb{E}[S_n] = n\mu$, so the bump marches off to the right (or left). Re-center by subtracting $n\mu$.
- **Spread.** $\mathrm{Var}(S_n) = n\sigma^2$, so the bump widens like $\sqrt{n}$. Re-scale by dividing by $\sigma\sqrt{n}$.

The exactly-tuned object

$$Z_n = \frac{S_n - n\mu}{\sigma\sqrt{n}}$$

has mean 0 and variance 1 for every n – centered and standardized so it can neither run away nor blow up nor collapse. What shape does its distribution converge to?

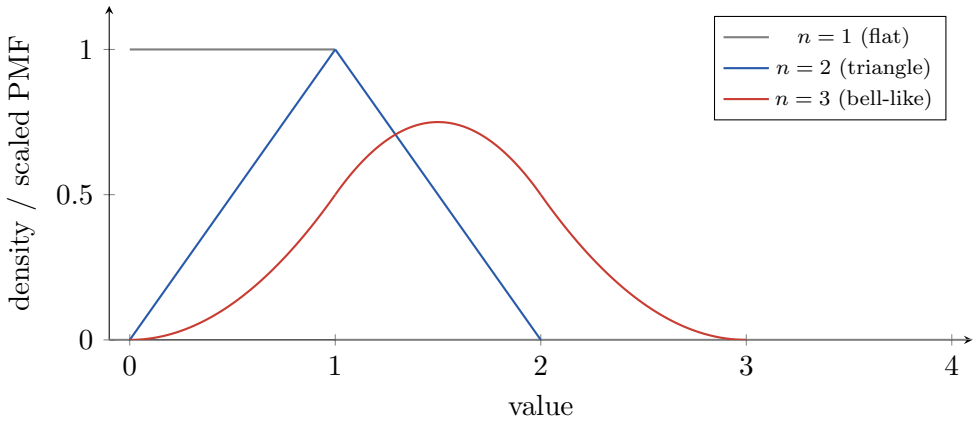


Figure 17.4: Repeatedly convolving a Uniform$(0,1)$ density with itself. The flat box (n=1) becomes a triangle (n=2) becomes a smooth, bell-shaped curve (n=3). Each extra summand smooths and centers the shape. Recenter and rescale, and the limit is the normal curve – this picture *is* the central limit theorem in slow motion.

Why this matters

The central limit theorem (preview). For i.i.d. summands with finite mean μ and variance σ^2,

$$Z_n = \frac{S_n - n\mu}{\sigma\sqrt{n}} \xrightarrow{d} \mathcal{N}(0,1) \qquad \text{as } n \to \infty,$$

no matter what distribution the X_i came from (uniform, Bernoulli, exponential, your own homemade lump – anything with finite variance). This chapter built the gears: convolution is the operation that adds independent variables; repeated convolution smooths any starting shape (Figure 17.4); linearity of expectation and additivity of variance supply the exact centering $n\mu$ and scaling $\sigma\sqrt{n}$; and the normal is the unique shape that convolution leaves unchanged, so it is the only possible destination (Figure 17.5). The Cauchy detour warned us why the "finite variance" clause cannot be dropped. A few chapters from now we prove it.

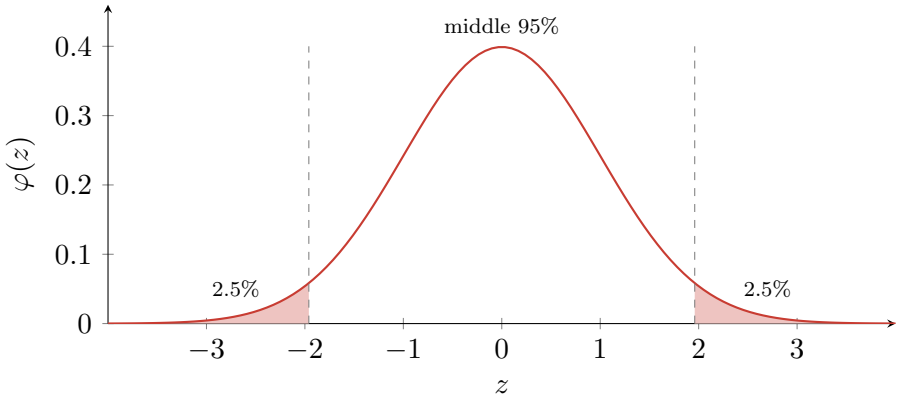


Figure 17.5: The destination of the standardized sum Z_n: the standard normal density $\varphi(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}$. The shaded tails beyond ± 1.96 each hold about 2.5% of the probability, so roughly 95% of the mass lies within two standard deviations of the mean – the rule of thumb that powers confidence intervals once the central limit theorem licenses the normal approximation.

Connection

The leap from "convolutions are hard integrals" to "the limit is normal" goes through **generating functions** and **characteristic functions**, the subject of the very next chapter. The key identity: if $\varphi_X(t) = \mathbb{E}[e^{itX}]$ is the characteristic function, then independence turns the convolution of densities into the *product* of characteristic functions,

$$\varphi_{X+Y}(t) = \varphi_X(t)\,\varphi_Y(t),$$

and the n-fold convolution into the nth *power* $\varphi_{S_n}(t) = \varphi_X(t)^n$. Hard convolutions collapse into easy algebra, and a Taylor expansion of φ around $t = 0$ – whose coefficients are exactly the mean and variance from this chapter – makes the Gaussian limit fall out almost by itself. Convolution is the *what*; the transform is the *how*.

17.8 Summary

Formal

The distribution of a sum of independent variables.

- **Discrete:** $p_{X+Y}(s) = \sum_k p_X(k)\,p_Y(s-k)$.
- **Continuous:** $f_{X+Y}(s) = \int f_X(x)\,f_Y(s-x)\,\mathrm{d}x$.
- Both are the *convolution* $p_X * p_Y$ / $f_X * f_Y$: sum (or integrate) over every way the two pieces split the target total.

Closure under addition (independent summands):

- $\text{Poisson}(\lambda) + \text{Poisson}(\mu) = \text{Poisson}(\lambda + \mu)$ (any rates).
- $\text{Binomial}(m, p) + \text{Binomial}(n, p) = \text{Binomial}(m + n, p)$ (same p).
- $\mathcal{N}(\mu_1, \sigma_1^2) + \mathcal{N}(\mu_2, \sigma_2^2) = \mathcal{N}(\mu_1 + \mu_2, \sigma_1^2 + \sigma_2^2)$ (any parameters).
- $\text{Gamma}(\alpha_1, \beta) + \text{Gamma}(\alpha_2, \beta) = \text{Gamma}(\alpha_1 + \alpha_2, \beta)$ (same rate); n exponentials sum to an Erlang.

Two numbers, two rules: $\mathbb{E}[\sum X_i] = \sum \mathbb{E}[X_i]$ *always*; $\text{Var}(\sum X_i) = \sum \text{Var}(X_i)$ *when uncorrelated*. For i.i.d. summands, sums grow like n and fluctuations like $\sqrt{n}$. **Caveat:** heavy tails (Cauchy) can destroy the mean and break averaging entirely.

17.9 Exercises

Exercise 17.1 (Convolve by hand)**.** Let $X \sim \text{Uniform}\{1, 2, 3\}$ and, independently, Y take value 0 with probability 1/2 and value 3 with probability 1/2. Use the discrete convolution formula to write out the full PMF of $S = X + Y$. Verify the probabilities sum to 1, and compute $\mathbb{E}[S]$ two ways: directly from your PMF, and via linearity of expectation.

Exercise 17.2 (Poisson merging)**.** A help desk receives emails as a Poisson(8) count per hour and phone calls as an independent Poisson(5) count per hour. (a) What is the distribution of the total

number of contacts per hour? (b) Compute $\mathbb{P}(\text{total} = 0)$ in one hour. (c) Over an 8-hour shift, what is the distribution and the expected total?

Exercise 17.3 (Uniform sum density)**.** Let $X \sim \text{Uniform}(0, 2)$ and $Y \sim \text{Uniform}(0, 1)$ be independent. Carefully carry out the convolution integral $f_S(s) = \int f_X(x) f_Y(s - x)\mathrm{d}x$, paying attention to the overlap of the two boxes, and show that f_S is a *trapezoid*: it ramps up, stays flat, then ramps down. Give the breakpoints and the flat height.

Exercise 17.4 (Variance with correlation)**.** Let X and Y each have variance 4, with $\text{Corr}(X, Y) = \rho$. (a) Express $\text{Var}(X + Y)$ in terms of ρ. (b) For which ρ is $\text{Var}(X + Y)$ largest? Smallest? Interpret both extremes in words (perfectly aligned vs. perfectly opposing variables). (c) For what value of ρ does $X + Y$ have the same variance as a single variable, namely 4?

Exercise 17.5 (Erlang waiting time)**.** Buses arrive as a Poisson process at rate $\beta = 3$ per hour, so inter-arrival times are i.i.d. Exponential(3). (a) Name the distribution of the waiting time T until the *fourth* bus, with its parameters. (b) Compute $\mathbb{E}[T]$ and $\text{Var}(T)$ using the additivity rules (do not integrate). (c) Explain in one sentence why $\text{SD}(T)$ is smaller than four times the SD of a single gap.

Exercise 17.6 (The $\sqrt{n}$ law, numerically)**.** A single fair coin scored $+1$ for heads and -1 for tails has mean 0 and variance 1. Let S_n be the sum of n such i.i.d. scores (a simple random walk). (a) Give $\mathbb{E}[S_n]$ and $\text{SD}(S_n)$. (b) For $n = 10{,}000$, roughly how far from 0 is S_n typically? Is $S_n = 500$ surprising? Is $S_n = 5000$? (c) Now consider the *fraction* of heads, $(S_n/n + 1)/2$. How does its standard deviation behave as $n \to \infty$, and what theorem does that foreshadow?

Exercise 17.7 (Why same-p matters)**.** Let $X \sim \text{Binomial}(1, 0.2)$ and $Y \sim \text{Binomial}(1, 0.8)$ be independent. (a) Write the full PMF of $S = X + Y$ on $\{0, 1, 2\}$. (b) If S were $\text{Binomial}(2, p)$ for some p, the value $\mathbb{P}(S = 2)$ would force a particular p; find it, then check whether that same p reproduces $\mathbb{P}(S = 0)$. Conclude that S is *not* binomial, and explain which hypothesis of Theorem 17.2 fails.

Exercise 17.8 (Normal closure and standardization)**.** Let $X \sim \mathcal{N}(3, 4)$ and $Y \sim \mathcal{N}(-1, 9)$ be independent. (a) State the distribution of $X + Y$. (b) State the distribution of $X - Y$ (be careful: the variance still adds). (c) Find a constant $a \in [0, 1]$ so that $aX + (1 - a)Y$ has the smallest possible variance, and report that minimum variance. (This is the seed of inverse-variance weighting in statistics.)

Exercise 17.9 (Proof: convolution is commutative)**.** Show directly from Definition 17.2 that $f_X * f_Y = f_Y * f_X$, i.e. $\int f_X(x) f_Y(s - x)\mathrm{d}x = \int f_Y(x) f_X(s - x)\mathrm{d}x$, using the substitution $u = s - x$. Then explain in one sentence why this is obvious *probabilistically*, with no integral at all (hint: what is $X + Y$ versus $Y + X$?).

Exercise 17.10 (Heavy tails break averaging)**.** Two independent standard Cauchy variables X_1, X_2 have $\bar{X}_2 = (X_1 + X_2)/2$. (a) Using the stated fact that $X_1 + X_2$ is Cauchy of scale 2, argue that $\bar{X}_2$ is standard Cauchy again. (b) Explain why this does not contradict the law of large numbers – which hypothesis of that law does the Cauchy violate? (c) In one sentence, what practical advice does this give a data analyst facing heavy-tailed measurements?

Chapter 18

Generating Functions

In the last chapter you learned what happens when you add independent random variables: their distributions *convolve*. Convolution is exact, but it is also a chore. To find the distribution of $X+Y$ you slide one pmf across another and sum, term by term; to find the distribution of $X_1 + \cdots + X_n$ you do this $n-1$ times, each pass smearing the answer a little further out. By the time you reach a sum of fifty dice the bookkeeping has buried the idea.

This chapter is about a change of language that makes the chore disappear. Instead of carrying a distribution around as a list of probabilities or as a density curve, we will encode the *entire* distribution inside a single function — a *generating function*. The encoding is chosen so cleverly that the operations we care about become easy: **convolution turns into ordinary multiplication**, **moments fall out by differentiation**, and **two distributions are equal exactly when their generating functions agree**. A hard problem about sums of random variables becomes an easy problem about products of functions, and at the end you translate back.

There are three such encodings, and they are really the same idea wearing three costumes. The *probability generating function* (PGF) is built for nonnegative integers and is the gentlest to meet first. The *moment generating function* (MGF) works for any random variable and is the workhorse of applied probability — it is how we will identify distributions and, in a later chapter, prove the Central Limit Theorem. The *characteristic function* (CF) is the MGF's bulletproof cousin: it exists for *every* random variable without exception, which is exactly why the deepest theorems are proved with it. We will build all three from the ground up, see why each one does what it does, and end with the punchline that makes the whole machine trustworthy: *a distribution is determined by its transform.*

18.1 The Idea: Hang Probabilities on a Function

Let us start with a single die and a deliberately small question, so the trick is visible before it is useful.

A fair die produces a value in $\{1,2,3,4,5,6\}$, each with probability $1/6$. Here is an unusual way to write that distribution down. Introduce a bookkeeping symbol s and form the polynomial

$$G(s) \;=\; \tfrac{1}{6}s^1 + \tfrac{1}{6}s^2 + \tfrac{1}{6}s^3 + \tfrac{1}{6}s^4 + \tfrac{1}{6}s^5 + \tfrac{1}{6}s^6.$$

Read it like a coat rack. The coefficient hanging on the hook labelled s^k is the probability that the die shows k. Nothing has been added or lost — the six numbers $1/6$ are all still there — but now they ride on a single function of s. The exponent records the *value*, the coefficient records the *probability*.

Intuition

A distribution on $\{0, 1, 2, \ldots\}$ is just a sequence of numbers $p_0, p_1, p_2, \ldots$. A generating function is the trick of bundling a whole sequence into one object by using powers of a dummy variable as labelled shelves: $G(s) = p_0 + p_1 s + p_2 s^2 + \cdots$. The shelf s^k holds p_k. What makes this worth doing is that natural operations on *sequences* (like convolution) become natural operations on *functions* (like multiplication).

Why would you ever want this? Watch what happens when you roll *two* independent dice and ask for the distribution of the total. The total is k when the first die shows i and the second shows $k - i$, for every way that can happen, so the probability of total k is the convolution

$$\mathbb{P}(\text{sum} = k) \;=\; \sum_{i} \mathbb{P}(\text{first} = i)\; \mathbb{P}(\text{second} = k - i).$$

Now look at the product of the two coat racks. Multiplying $G(s) \cdot G(s)$ and collecting the coefficient of s^k, you pick up exactly $\sum_i (\text{coeff of } s^i)(\text{coeff of } s^{k-i})$ — because $s^i \cdot s^{k-i} = s^k$. *That is the convolution sum, performed for free by the exponent arithmetic.* The exponents add precisely because random variables add. Convolution of probabilities and multiplication of polynomials are the same operation seen from two sides.

What is really happening

The whole subject lives in one observation: $s^i \cdot s^j = s^{i+j}$. When you multiply two generating functions, every cross term s^i-from-the-first times s^j-from-the-second lands on the shelf s^{i+j}, carrying the product of its two probabilities. Summing all the ways to land on shelf k *is* the convolution that gives $\mathbb{P}(X + Y = k)$. So "add independent random variables" $\longleftrightarrow$ "multiply their generating functions." Everything else in this chapter is a consequence and a refinement of that single line.

18.2 The Probability Generating Function

Let us make the coat rack official, for random variables that take values in $\{0, 1, 2, \ldots\}$.

Definition 18.1 (Probability generating function)**.** Let X be a random variable taking values in the nonnegative integers, with pmf $p_k = \mathbb{P}(X = k)$. Its *probability generating function* is

$$G_X(s) \;=\; \mathbb{E}\big[s^X\big] \;=\; \sum_{k=0}^{\infty} p_k\, s^k, \qquad |s| \le 1.$$

The two faces of the definition — "$\mathbb{E}[s^X]$" and "$\sum p_k s^k$" — are worth holding side by side. The first is compact and is how you will usually *compute* it; the second is the coat rack and is how you will usually *read* it. They agree because $\mathbb{E}[s^X] = \sum_k s^k\, \mathbb{P}(X = k)$ is just the definition of expectation of

the function $s \mapsto s^X$ (this is LOTUS, the law of the unconscious statistician, from the expectation chapter).

The series converges at least for $|s| \le 1$, because there the terms are dominated by the probabilities: $\sum_k p_k |s|^k \le \sum_k p_k = 1$. In particular $G_X(1) = \sum_k p_k = 1$ always, which is a handy sanity check — a correct PGF must pass through the point $(1, 1)$.

Example 18.1 (The usual suspects)**.** A few PGFs you should be able to write on sight.

- **Bernoulli**(p): X is 1 w.p. p and 0 w.p. $q = 1 - p$, so
$$G_X(s) = q\, s^0 + p\, s^1 = q + ps.$$
- **Binomial**(n, p): think of $X = \sum_{i=1}^n X_i$ as a sum of n independent Bernoulli(p)'s. We will see in a moment that PGFs of independent sums multiply, so
$$G_X(s) = (q + ps)^n.$$
(Notice the binomial theorem expands this into $\sum_k \binom{n}{k} p^k q^{n-k} s^k$ — the pmf, recovered as coefficients.)
- **Geometric**(p) (number of failures before the first success, $k = 0, 1, 2, \dots$): $p_k = q^k p$, so
$$G_X(s) = \sum_{k=0}^{\infty} q^k p\, s^k = p \sum_{k=0}^{\infty} (qs)^k = \frac{p}{1 - qs}, \qquad |s| < 1/q.$$
- **Poisson**(λ): $p_k = e^{-\lambda}\lambda^k / k!$, so
$$G_X(s) = \sum_{k=0}^{\infty} e^{-\lambda} \frac{\lambda^k}{k!} s^k = e^{-\lambda} \sum_{k=0}^{\infty} \frac{(\lambda s)^k}{k!} = e^{-\lambda} e^{\lambda s} = e^{\lambda(s-1)}.$$
The Taylor series of e^x does the summing for you, as it so often does with the Poisson.

18.2.1 The multiplication theorem

Now the property the whole construction was built to have.

Theorem 18.1 (PGF of an independent sum)**.** *If X and Y are independent nonnegative-integer random variables, then*
$$G_{X+Y}(s) \;=\; G_X(s)\, G_Y(s).$$
More generally, if $X_1, \dots, X_n$ are independent, then $G_{X_1+\cdots+X_n}(s) = \prod_{i=1}^n G_{X_i}(s)$.

Why it works

The proof is one line, and it is the most important line in the chapter:
$$G_{X+Y}(s) = \mathbb{E}\big[s^{X+Y}\big] = \mathbb{E}\big[s^X s^Y\big] \overset{\perp\!\!\!\perp}{=} \mathbb{E}\big[s^X\big]\, \mathbb{E}\big[s^Y\big] = G_X(s)\, G_Y(s).$$
The single starred step uses independence: the expectation of a product of independent quantities is the product of their expectations. Everything else is exponent arithmetic, $s^{X+Y} = s^X s^Y$.

This is the convolution-becomes-product fact from §18.1, now stated cleanly in the language of expectation. The n-fold version follows by induction.

Example 18.2 (Sum of independent Poissons, in one breath)**.** Let $X \sim \text{Poisson}(\lambda)$ and $Y \sim \text{Poisson}(\mu)$ be independent. By Theorem 18.1,

$$G_{X+Y}(s) = e^{\lambda(s-1)}\, e^{\mu(s-1)} = e^{(\lambda+\mu)(s-1)}.$$

But that last expression is exactly the PGF of a Poisson$(\lambda+\mu)$ variable. Once we know (next section) that the PGF determines the distribution, we may conclude $X+Y \sim \text{Poisson}(\lambda+\mu)$ — no convolution sum, no $\sum_j \binom{k}{j}\lambda^j\mu^{k-j}/k!$ to wrestle with. The exponents just added. This is the payoff promised in the chapter opening, delivered in two lines.

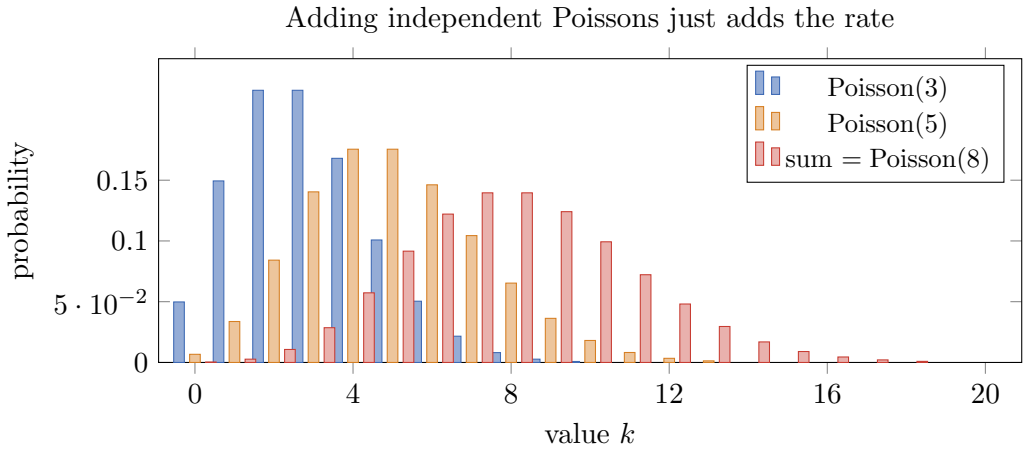


Figure 18.1: The red bars are the distribution of $X+Y$ where $X \sim \text{Poisson}(3)$ and $Y \sim \text{Poisson}(5)$ independently. It is exactly Poisson(8). The PGF proof (18.2) says this with no arithmetic: the exponents $\lambda(s-1)$ and $\mu(s-1)$ add.

18.2.2 Reading moments off the PGF

The PGF does double duty: it not only multiplies sums together, it also coughs up moments when you differentiate it. Differentiate the series term by term:

$$G'_X(s) = \sum_{k\ge 1} k\, p_k\, s^{k-1}, \qquad G''_X(s) = \sum_{k\ge 2} k(k-1)\, p_k\, s^{k-2}.$$

Now evaluate at $s=1$. The first gives $G'_X(1) = \sum_k k\,p_k = \mathbb{E}[X]$. The second gives $G''_X(1) = \sum_k k(k-1)p_k = \mathbb{E}[X(X-1)] = \mathbb{E}[X^2] - \mathbb{E}[X]$. These $\mathbb{E}[X(X-1)\cdots]$ quantities are called *factorial moments*, and the PGF is their natural home.

Proposition 18.1 (Mean and variance from the PGF)**.** *If X has PGF G_X, then*

$$\mathbb{E}[X] = G'_X(1), \qquad \operatorname{Var}(X) = G''_X(1) + G'_X(1) - \big(G'_X(1)\big)^2.$$

Example 18.3 (Poisson moments, instantly)**.** For Poisson(λ) we have $G_X(s) = e^{\lambda(s-1)}$, so $G'_X(s) = \lambda e^{\lambda(s-1)}$ and $G''_X(s) = \lambda^2 e^{\lambda(s-1)}$. At $s = 1$ the exponential is 1, giving $G'_X(1) = \lambda$ and $G''_X(1) = \lambda^2$. Hence $\mathbb{E}[X] = \lambda$ and

$$\mathrm{Var}(X) = \lambda^2 + \lambda - \lambda^2 = \lambda.$$

The famous fact that a Poisson's mean and variance coincide drops out of one exponential and two derivatives.

Walking question

Use the PGF $G(s) = p/(1 - qs)$ of the Geometric(p) (failures before first success) to show $\mathbb{E}[X] = q/p$. *Hint:* differentiate $p(1 - qs)^{-1}$ to get $G'(s) = pq(1 - qs)^{-2}$, then set $s = 1$ and remember $1 - q = p$. You should find $G'(1) = pq/p^2 = q/p$. For practice, push on to $G''(1) = 2pq^2/p^3 = 2q^2/p^2$ and assemble $\mathrm{Var}(X) = q/p^2$.

Application

[Branching processes and extinction] PGFs are the native language of *branching processes*, the model behind nuclear chain reactions, surname survival, and epidemic spread. Suppose each individual independently produces a random number of offspring with PGF G. Then the size of generation n has PGF obtained by *composing* G with itself n times: $G_n(s) = G(G(\cdots G(s)\cdots))$. The probability of eventual extinction turns out to be the smallest fixed point of G, i.e. the smallest solution of $G(s) = s$ in $[0, 1]$. A whole theory — including the clean threshold "extinction is certain iff the mean offspring count is ≤ 1" — falls out of iterating one function. Composition of PGFs encodes generations the way multiplication encodes independent sums.

18.3 The Moment Generating Function

The PGF is lovely but provincial: it only speaks to nonnegative integers, because s^X needs an integer exponent to make plain sense. Real life is full of continuous quantities — waiting times, measurement errors, asset returns — and for those we need a transform that does not care whether X is integer, continuous, positive, or negative. The fix is to replace the base s by e^t. Then $s^X = (e^t)^X = e^{tX}$ makes sense for *any* real X, because the exponential eats any real exponent.

Definition 18.2 (Moment generating function)**.** The *moment generating function* of a random variable X is

$$M_X(t) \;=\; \mathbb{E}\big[e^{tX}\big],$$

defined for those real t where the expectation is finite. We say the MGF *exists* if $M_X(t) < \infty$ for all t in some open interval $(-\delta, \delta)$ around 0.

You can think of the MGF as the PGF with $s = e^t$ substituted in: where they both exist, $M_X(t) = G_X(e^t)$. Everything good about the PGF carries over, and the domain widens to all of $\mathbb{R}$.

Theorem 18.2 (MGF of an independent sum)**.** *If X and Y are independent, then $M_{X+Y}(t) = M_X(t)\, M_Y(t)$, wherever both sides are finite.*

The proof is the same one line as before: $M_{X+Y}(t) = \mathbb{E}[e^{t(X+Y)}] = \mathbb{E}[e^{tX}e^{tY}] = \mathbb{E}[e^{tX}]\,\mathbb{E}[e^{tY}]$, the middle step by independence. Sums of independent variables become products of MGFs, now for continuous variables too.

18.3.1 Why it is called *moment* generating

Here is where the name is earned. Expand e^{tX} in its Taylor series and take the expectation term by term:

$$M_X(t) = \mathbb{E}\big[e^{tX}\big] = \mathbb{E}\left[\sum_{n=0}^{\infty} \frac{(tX)^n}{n!}\right] = \sum_{n=0}^{\infty} \frac{t^n}{n!}\,\mathbb{E}[X^n].$$

Look at what this says. The MGF, expanded as a power series in t, has the nth moment $\mathbb{E}[X^n]$ sitting on the coefficient of $t^n/n!$. The whole sequence of moments is packed into one function, exactly the way the whole pmf was packed into the PGF. To pull out the nth moment, differentiate n times and set $t = 0$:

Proposition 18.2 (Moments by differentiation)**.** *If M_X exists in a neighborhood of 0, then for every $n \geq 1$,*

$$\mathbb{E}[X^n] \;=\; M_X^{(n)}(0),$$

the nth derivative of the MGF evaluated at $t = 0$. In particular $\mathbb{E}[X] = M_X'(0)$ and $\operatorname{Var}(X) = M_X''(0) - \big(M_X'(0)\big)^2$.

Why it works

Differentiating $\sum_n \mathbb{E}[X^n]\,t^n/n!$ once knocks the series down to $\sum_{n\geq 1} \mathbb{E}[X^n]\,t^{n-1}/(n-1)!$, whose value at $t = 0$ is the $n = 1$ term, $\mathbb{E}[X]$. Differentiating n times and setting $t = 0$ similarly isolates the $\mathbb{E}[X^n]$ term — every lower term has been differentiated to zero, every higher term still carries a factor of t. The factorials in the series are placed exactly so the nth derivative cleanly recovers $\mathbb{E}[X^n]$ with no stray constants. The MGF is a moment dispenser, and $t = 0$ is the lever.

Example 18.4 (The exponential distribution)**.** Let $X \sim \text{Exponential}(\lambda)$ with density $f(x) = \lambda e^{-\lambda x}$ for $x \geq 0$. Then

$$M_X(t) = \int_0^{\infty} e^{tx}\,\lambda e^{-\lambda x}\,\mathrm{d}x = \lambda \int_0^{\infty} e^{-(\lambda - t)x}\,\mathrm{d}x = \frac{\lambda}{\lambda - t}, \qquad t < \lambda.$$

The integral converges only when $\lambda - t > 0$, which is why the MGF lives on the half-line $t < \lambda$ and not all of $\mathbb{R}$ — a first hint that MGFs can be fragile. To read off moments, expand as a geometric series:

$$M_X(t) = \frac{1}{1 - t/\lambda} = \sum_{n=0}^{\infty} \frac{t^n}{\lambda^n} = \sum_{n=0}^{\infty} \frac{n!}{\lambda^n} \cdot \frac{t^n}{n!}.$$

Matching against $\sum_n \mathbb{E}[X^n]\,t^n/n!$ gives $\mathbb{E}[X^n] = n!/\lambda^n$ for all n at once. In particular $\mathbb{E}[X] = 1/\lambda$ and $\mathbb{E}[X^2] = 2/\lambda^2$, so $\operatorname{Var}(X) = 2/\lambda^2 - 1/\lambda^2 = 1/\lambda^2$. One series, all the moments.

Example 18.5 (The normal distribution)**.** The MGF of the standard normal $Z \sim \mathcal{N}(0,1)$ is the keystone of the CLT proof, so it is worth doing carefully. With density $\varphi(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}$,

$$M_Z(t) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{tz}\, e^{-z^2/2}\,\mathrm{d}z.$$

Complete the square in the exponent: $tz - \frac{1}{2}z^2 = -\frac{1}{2}(z-t)^2 + \frac{1}{2}t^2$. Pull the t-only piece out of the integral:

$$M_Z(t) = e^{t^2/2} \cdot \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-(z-t)^2/2}\, \mathrm{d}z = e^{t^2/2} \cdot 1 = e^{t^2/2},$$

since the remaining integral is just a normal density (centered at t) integrating to 1. So $M_Z(t) = e^{t^2/2}$, finite for *all* real t — the normal's tails are so thin that e^{tZ} has finite expectation no matter how large t is. For a general $X \sim \mathcal{N}(\mu, \sigma^2)$, write $X = \mu + \sigma Z$ and use the scaling rule below to get

$$M_X(t) = e^{\mu t + \frac{1}{2}\sigma^2 t^2}.$$

18.3.2 How the MGF responds to shifting and scaling

Two small rules make the MGF a pleasure to manipulate.

Proposition 18.3 (Affine transformations)**.** *For constants a, b, $M_{aX+b}(t) = e^{bt}\, M_X(at)$.*

Why it works

$M_{aX+b}(t) = \mathbb{E}[e^{t(aX+b)}] = \mathbb{E}[e^{(at)X}\, e^{bt}] = e^{bt}\, \mathbb{E}[e^{(at)X}] = e^{bt} M_X(at)$. The constant b slides outside as the factor e^{bt}; the multiplier a reaches inside and rescales the argument. This is exactly how you turn the standard normal's MGF into the general normal's: with $a = \sigma$, $b = \mu$, $M_{\mu+\sigma Z}(t) = e^{\mu t} M_Z(\sigma t) = e^{\mu t} e^{\sigma^2 t^2/2}$, recovering Example 18.5.

Example 18.6 (Sum of independent normals, by transform)**.** Let $X \sim \mathcal{N}(\mu_1, \sigma_1^2)$ and $Y \sim \mathcal{N}(\mu_2, \sigma_2^2)$ be independent. Multiply their MGFs:

$$M_{X+Y}(t) = e^{\mu_1 t + \frac{1}{2}\sigma_1^2 t^2}\, e^{\mu_2 t + \frac{1}{2}\sigma_2^2 t^2} = e^{(\mu_1+\mu_2)t + \frac{1}{2}(\sigma_1^2+\sigma_2^2)t^2}.$$

That is the MGF of a $\mathcal{N}(\mu_1 + \mu_2,\ \sigma_1^2 + \sigma_2^2)$. So the sum of independent normals is normal, with means and variances adding — a fact that costs a serious convolution integral to prove directly, but here is just "add the exponents." The normal family is *closed under addition*, and the MGF makes that visible at a glance.

Common trap

The product rule needs **independence**. If X and Y are dependent, $M_{X+Y}(t)$ is generally *not* $M_X(t)M_Y(t)$ — the step $\mathbb{E}[e^{tX}e^{tY}] = \mathbb{E}[e^{tX}]\,\mathbb{E}[e^{tY}]$ fails. A common slip is to "add variances" or "multiply MGFs" for correlated variables. Both are independence facts. For dependent sums you must keep the joint distribution (or use the *joint* MGF $\mathbb{E}[e^{t_1X+t_2Y}]$ and its mixed partials).

Application

[Chernoff bounds, the workhorse of modern ML theory] The MGF is not only for identifying distributions; it is the engine behind the sharpest elementary tail bounds in all of probability.

For any $t > 0$, Markov's inequality applied to e^{tX} gives

$$\mathbb{P}(X \geq a) = \mathbb{P}\big(e^{tX} \geq e^{ta}\big) \leq \frac{\mathbb{E}[e^{tX}]}{e^{ta}} = e^{-ta}\, M_X(t).$$

Because this holds for *every* $t > 0$, you may optimize: the *Chernoff bound* $\mathbb{P}(X \geq a) \leq \min_{t>0} e^{-ta} M_X(t)$. For a sum of independent variables the MGF factorizes, so the bound on the sum is a product of one-variable MGFs — which is exactly why Chernoff bounds give *exponentially small* failure probabilities for averages. This is the backbone of generalization bounds in statistical learning, the analysis of randomized algorithms and hashing, and concentration inequalities like Hoeffding's. Whenever a machine-learning paper writes "with probability at least $1 - \delta$," a Chernoff/MGF argument is usually doing the work underneath.

18.4 Uniqueness: Why the Transform Is Trustworthy

Every trick in this chapter ends with the phrase "... and therefore $X + Y$ is Poisson" or "... and therefore $X + Y$ is normal." Each such conclusion secretly relies on one deep fact: that the MGF *determines* the distribution. If two random variables have the same MGF, they have the same distribution — so recognizing the MGF lets you name the distribution. Without this, computing M_{X+Y} would be a dead end; with it, recognition is proof.

Theorem 18.3 (Uniqueness of the MGF)**.** *Suppose X and Y have MGFs that are finite and equal on some open interval $(-\delta, \delta)$ around 0: $M_X(t) = M_Y(t)$ for all $|t| < \delta$. Then X and Y have the same distribution.*

We will not prove this in full — the real proof routes through the characteristic function and Fourier inversion, which we meet next. But the *reason* it is plausible is worth stating, because it demystifies what would otherwise look like a magic trick.

Why it works

A finite MGF on an interval around 0 is real-analytic there, and its Taylor coefficients are the moments $\mathbb{E}[X^n]/n!$. So equal MGFs near 0 means *all moments agree*. The remaining content of the theorem is that, when the MGF exists in a neighborhood of 0, the moments pin the distribution down uniquely. Intuitively, knowing every moment is knowing the average of every power $X, X^2, X^3, \ldots$, hence the average of every polynomial, hence (by approximating indicators with polynomials) the probability of every event. The existence of the MGF is what guarantees the tails are thin enough for this program to close — which is also exactly the fine print, addressed in the trap below.

Common trap

[Moments alone are *not* always enough] It is tempting to believe "same moments $\Rightarrow$ same distribution" unconditionally. It is false. There exist genuinely *different* distributions with *identical* moments of every order. The classic example is the lognormal distribution: if $\log X \sim \mathcal{N}(0, 1)$, one can perturb its density by a wiggle that leaves every moment $\mathbb{E}[X^n]$ unchanged, producing a different distribution with exactly the same moment sequence. The catch is that

the lognormal's moments grow so fast (its MGF is infinite for every $t > 0$) that they fail to determine the distribution. The escape hatch is precisely the hypothesis of Theorem 18.3: if the *MGF* is finite in an open interval around 0, the tails are controlled, the moment series converges, and uniqueness holds. This is why we are careful to say "the MGF exists," not merely "the moments exist."

History

The moment problem — when does a sequence of moments determine a unique distribution? — occupied Chebyshev, Markov, and Stieltjes in the late nineteenth century, and it is where modern probability cut its teeth on analysis. Stieltjes (1894) gave the first careful treatment, inventing along the way the Stieltjes integral. Hamburger and Hausdorff later pinned down exact conditions. The upshot, in the language we use: a distribution is determined by its moments when those moments do not grow too fast (Carleman's condition is the sharp form), and finiteness of the MGF near 0 is a clean sufficient condition. The lognormal counterexample, that two different curves can share every moment, was a genuine shock — a reminder that infinitely many numbers can still leave something undetermined.

Example 18.7 (Recognition in practice)**.** Suppose you compute, for some X built out of a complicated experiment, that $M_X(t) = (1-2t)^{-3}$ for $t < 1/2$. You do not need to know how X was built. You recall that a Gamma(α, β) with rate β has MGF $(1 - t/\beta)^{-\alpha}$. Matching, $\beta = 1/2$ and $\alpha = 3$. By uniqueness, $X \sim \text{Gamma}(3, 1/2)$ — equivalently a chi-squared with 6 degrees of freedom. The MGF acted as a fingerprint: identify the fingerprint, identify the distribution.

18.5 Characteristic Functions: The Transform That Always Exists

The MGF has one flaw, and it is fatal often enough to matter: *it may not exist.* We saw the exponential's MGF die for $t \geq \lambda$, and the lognormal's die for every $t > 0$. Heavy-tailed distributions — the Cauchy distribution being the headline case — have $\mathbb{E}[e^{tX}] = \infty$ for every $t \neq 0$, so their MGF is useless. Since the MGF is built from e^{tX}, which explodes when X is large and $t > 0$, any distribution with fat enough tails breaks it.

The repair is one of the most elegant moves in analysis: put an imaginary unit in the exponent. Replace e^{tX} by e^{itX}, where $i = \sqrt{-1}$.

Definition 18.3 (Characteristic function)**.** The *characteristic function* of a random variable X is

$$\varphi_X(t) \;=\; \mathbb{E}\big[e^{itX}\big] \;=\; \mathbb{E}[\cos(tX)] + i\,\mathbb{E}[\sin(tX)], \qquad t \in \mathbb{R}.$$

What is really happening

Why does the i rescue everything? Because e^{itX} does not grow. By Euler's formula, $e^{itX} = \cos(tX) + i\sin(tX)$, a point on the unit circle in the complex plane. Its magnitude is $|e^{itX}| = 1$ for every value of X, no matter how huge. Where the real exponential e^{tX} shoots off to infinity for large X, the complex exponential just *spins* around the circle. So $|\varphi_X(t)| = |\mathbb{E}[e^{itX}]| \leq \mathbb{E}|e^{itX}| = \mathbb{E}[1] = 1 < \infty$ *always.* The integral defining φ_X converges for every distribution on Earth, because we are averaging bounded quantities. The CF is the MGF with its dangerous

growth tamed into harmless rotation.

Theorem 18.4 (Properties of the characteristic function)**.** *For every random variable X:*

1. (Always exists, bounded.) $\varphi_X(t)$ *is defined for all* $t \in \mathbb{R}$ *and* $|\varphi_X(t)| \le 1$*, with* $\varphi_X(0) = 1$*.*
2. (Sums.) *If* $X \perp\!\!\!\perp Y$*, then* $\varphi_{X+Y}(t) = \varphi_X(t)\,\varphi_Y(t)$*.*
3. (Affine.) $\varphi_{aX+b}(t) = e^{ibt}\varphi_X(at)$*.*
4. (Moments, when they exist.) *If* $\mathbb{E}|X|^n < \infty$*, then* φ_X *is* n *times differentiable and* $\varphi_X^{(n)}(0) = i^n\,\mathbb{E}[X^n]$*.*
5. (Uniqueness.) *If* $\varphi_X = \varphi_Y$ *as functions on* $\mathbb{R}$*, then* X *and* Y *have the same distribution.*

Properties (2) and (3) have the identical one-line proofs as for the MGF — nothing changed but a letter i. Property (4) is the same Taylor expansion, producing factors of i. Property (5), uniqueness, is the deep one, and unlike the MGF version it comes with *no fine print*: because the CF always exists, it *always* determines the distribution. There is even an explicit recipe to get the distribution back.

Theorem 18.5 (Fourier inversion)**.** *If* X *has a density* f *and* φ_X *is integrable, then*

$$f(x) \;=\; \frac{1}{2\pi}\int_{-\infty}^{\infty} e^{-itx}\,\varphi_X(t)\,\mathrm{d}t.$$

This is the punchline of the whole chapter, stated as a formula: the characteristic function is the *Fourier transform of the density*, and the density is the *inverse Fourier transform of the characteristic function*. Nothing is lost in the round trip. The transform is not a lossy summary like the mean or the variance; it is an exact re-encoding of the entire distribution, and the inversion integral is the proof.

Connection

[This is Fourier analysis] If you have met Fourier series or transforms (in the companion calculus book, or in any signal-processing course), the characteristic function is an old friend in disguise. Writing $\varphi_X(t) = \int e^{itx} f(x)\,\mathrm{d}x$, it is literally the Fourier transform of the density f (with a sign and 2π convention chosen by probabilists). The convolution theorem of Fourier analysis — "the transform of a convolution is the product of the transforms" — is exactly our Theorem 18.4(2): the density of $X + Y$ is the convolution $f_X * f_Y$, and its transform is the product $\varphi_X\varphi_Y$. The whole machinery of this chapter is probability borrowing the deepest tool of harmonic analysis. The reason transforms turn convolution into multiplication is the same reason here as there: $e^{it(x+y)} = e^{itx}e^{ity}$.

Paradox

[Averaging Cauchys gets you nowhere] The standard Cauchy density is $f(x) = \frac{1}{\pi(1+x^2)}$. Its tails are so heavy that $\mathbb{E}|X| = \infty$: the mean does not even exist, let alone the MGF. Yet its characteristic function is perfectly well behaved:

$$\varphi_X(t) = e^{-|t|}.$$

(The derivation is a contour integral; take the result on faith for now.) Watch what the product rule does. If $X_1, \dots, X_n$ are independent standard Cauchys, then $\overline{X}_n = \frac{1}{n}\sum X_i$ has CF

$$\varphi_{\overline{X}_n}(t) = \prod_{k=1}^{n} \varphi_{X_k}(t/n) = \left(e^{-|t|/n}\right)^n = e^{-|t|} = \varphi_X(t).$$

By uniqueness, $\overline{X}_n$ has the *same* distribution as a single X. Averaging a thousand independent Cauchy observations gives you something distributed exactly like one observation — the sample mean does not concentrate, and the Law of Large Numbers fails. This famous failure is invisible to the MGF (which does not exist here) but laid bare by the CF in three lines. It is also a vivid demonstration of why the CF is the tool of choice: it works where everything else has given up. The resolution of the "paradox" is that the LLN secretly requires a finite mean, which the Cauchy lacks; the heavy tails mean any average is dominated by its largest term.

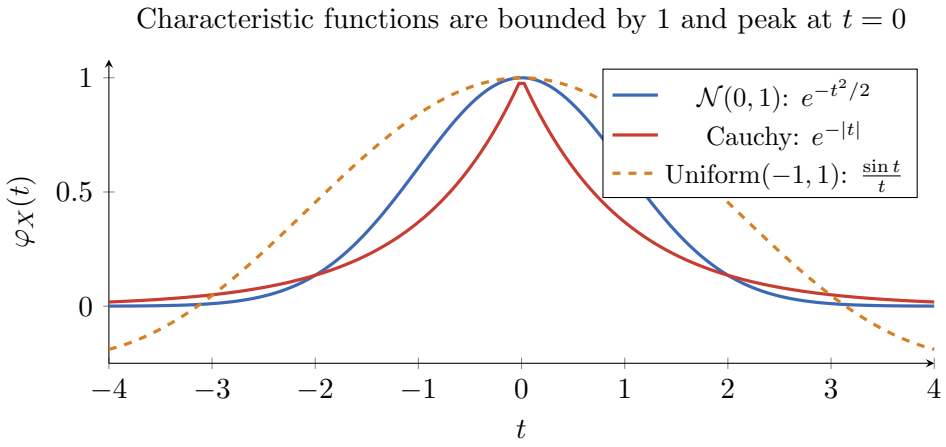


Figure 18.2: Three characteristic functions. All start at $\varphi(0) = 1$ and stay within ± 1 — a guarantee that holds for every distribution. The normal's CF is again a Gaussian bell; the Cauchy's is a sharp tent $e^{-|t|}$; the uniform's is the oscillating $\sin(t)/t$, dipping below zero, which a real MGF could never do.

Why this matters

Step back and admire the ladder we climbed. The PGF handles counts and turns convolution into polynomial multiplication. The MGF widens the door to all real random variables and dispenses moments by differentiation — but it can crash on heavy tails. The CF puts an i in the exponent, trading explosive growth for harmless rotation on the unit circle, and thereby *always* exists, always determines the distribution, and comes with an exact inversion formula. Three transforms, one idea, increasing robustness. Each is the right tool for a different job, and together they convert hard questions about adding randomness into easy questions about multiplying functions.

18.6 A Preview of the Central Limit Theorem

We close with the result that makes this entire chapter feel inevitable. The Central Limit Theorem says that the sum (suitably standardized) of many independent, identically distributed pieces is approximately normal, almost regardless of what the pieces look like. The proof, which we give in full in the CLT chapter, is essentially a one-page computation *with characteristic functions* — and now you can see its skeleton.

Focus

The CLT in transform language. Let $X_1, X_2, \dots$ be i.i.d. with mean 0 and variance 1 (subtract the mean and divide by the SD to arrange this), and let φ be their common characteristic function. Form the standardized sum

$$S_n = \frac{X_1 + \cdots + X_n}{\sqrt{n}}.$$

By the sum rule (18.4, part 2) and the affine rule (part 3),

$$\varphi_{S_n}(t) = \prod_{k=1}^{n} \varphi\!\left(\frac{t}{\sqrt{n}}\right) = \left[\varphi\!\left(\frac{t}{\sqrt{n}}\right)\right]^n.$$

Now Taylor-expand φ near 0 using the moment property: since $\mathbb{E}[X] = 0$ and $\mathbb{E}[X^2] = 1$,

$$\varphi(u) = 1 + i\,\mathbb{E}[X]\,u - \tfrac{1}{2}\,\mathbb{E}[X^2]\,u^2 + o(u^2) = 1 - \tfrac{1}{2}u^2 + o(u^2).$$

Substitute $u = t/\sqrt{n}$:

$$\varphi_{S_n}(t) = \left[1 - \frac{t^2}{2n} + o\!\left(\frac{1}{n}\right)\right]^n \xrightarrow[n\to\infty]{} e^{-t^2/2},$$

using the limit $(1 + a_n/n)^n \to e^a$ when $a_n \to a$. But $e^{-t^2/2}$ is exactly the characteristic function of the standard normal (Example 18.5, with the same calculation under the i). By uniqueness and a continuity theorem, S_n converges in distribution to $\mathcal{N}(0,1)$. *That is the Central Limit Theorem.*

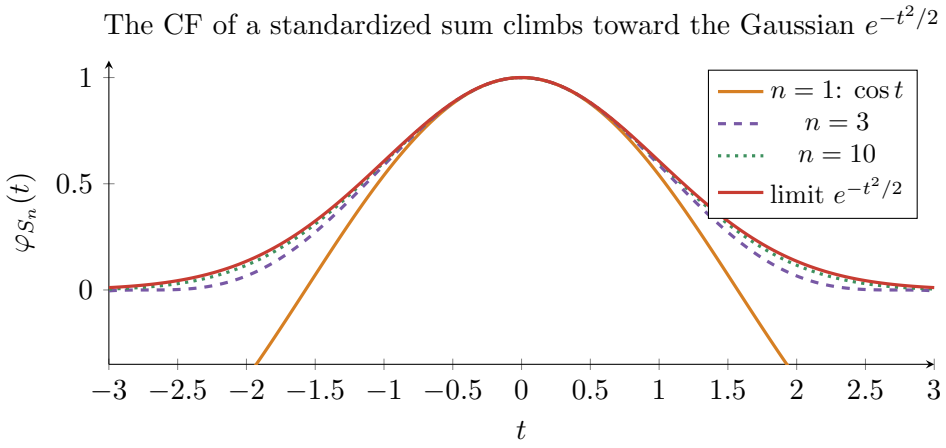


Figure 18.3: Take the X_i to be ± 1 fair coin flips (CF $\cos t$). As n grows, the CF of the standardized sum $S_n = (X_1 + \cdots + X_n)/\sqrt{n}$, namely $[\cos(t/\sqrt{n})]^n$, climbs toward the standard-normal CF $e^{-t^2/2}$ (solid red). This is the Central Limit Theorem watched in transform space: multiplication of CFs, then a limit.

Pause on how much heavy lifting the transform just did. The distribution of a sum of n convolved-together pieces would be a nightmare to track directly. In transform space it is a single power $[\varphi(t/\sqrt{n})]^n$ (watch it converge in Figure 18.3), the limit is a one-line Taylor expansion, and the answer $e^{-t^2/2}$ recognizes itself as the normal. *Why is the bell curve universal?* Because near zero every characteristic function looks like $1 - \frac{1}{2}(\text{variance})u^2$, and that quadratic — the only surviving term after the mean is killed and the scale is fixed — exponentiates into the Gaussian. The normal is universal because $-\frac{1}{2}t^2$ is what is left when you keep only the first two moments. The generating function is what let us see it.

History

Laplace, around 1810, found the bell-curve limit using what were essentially characteristic-function calculations (he worked with the Fourier-style integral transforms of the day). The modern proof was perfected by Lyapunov and then by Paul Lévy in the 1920s, who turned the characteristic function into a precise instrument and proved the *continuity theorem* — the missing link above: if characteristic functions converge pointwise to a function continuous at 0, then the distributions converge. Lévy's theorem is what licenses the leap from "$\varphi_{S_n}(t) \to e^{-t^2/2}$" to "$S_n \xrightarrow{d} \mathcal{N}(0,1)$." It is no exaggeration to say the characteristic function *is* the modern theory of the CLT.

18.7 Summary

Three transforms, one principle. Each encodes a whole distribution in a single function so that adding independent random variables becomes multiplying their transforms, and moments become derivatives.

	PGF	**MGF**	**CF**
Definition	$\mathbb{E}[s^X]$	$\mathbb{E}[e^{tX}]$	$\mathbb{E}[e^{itX}]$
For which X	$\mathbb{N}$-valued	needs thin tails	*any* X
Always exists?	on $\lvert s\rvert \le 1$	not always	**yes**
Independent sum	multiply	multiply	multiply
Moments	$G^{(n)}(1)$ (factorial)	$M^{(n)}(0) = \mathbb{E}[X^n]$	$\varphi^{(n)}(0) = i^n\,\mathbb{E}[X^n]$
Determines dist.?	yes	yes (if exists)	**always**

- The **PGF** $G_X(s) = \mathbb{E}[s^X]$ hangs the pmf on powers of s. Convolution becomes polynomial multiplication; factorial moments come from derivatives at $s = 1$.
- The **MGF** $M_X(t) = \mathbb{E}[e^{tX}] = G_X(e^t)$ extends the idea to all real X. Its Taylor coefficients are the moments ($\mathbb{E}[X^n] = M^{(n)}(0)$); it determines the distribution *when it exists in an interval around* 0.
- The **CF** $\varphi_X(t) = \mathbb{E}[e^{itX}]$ puts an i in the exponent, bounding $\lvert e^{itX}\rvert = 1$ so the transform always exists. It is the Fourier transform of the density, always determines the distribution, and is the engine of the CLT proof.

The single fact under all of it: $e^{(\cdot)(X+Y)} = e^{(\cdot)X}e^{(\cdot)Y}$, so independence turns "sum of variables" into "product of transforms." That one identity is why generating functions exist, why convolutions simplify, and why the normal distribution is universal.

18.8 Exercises

Exercise 18.1 (Warm-up PGFs)**.** Write the probability generating function of each, and check that it equals 1 at $s = 1$.

(a) A single fair four-sided die, values $\{1, 2, 3, 4\}$.

(b) $X \sim \text{Binomial}(3, 1/2)$, by direct summation, and confirm it matches $(\frac{1}{2} + \frac{1}{2}s)^3$.

(c) The constant random variable $X = 7$.

Exercise 18.2 (Sum of dice via PGF)**.** Let $G(s) = \frac{1}{6}(s + s^2 + s^3 + s^4 + s^5 + s^6)$ be the PGF of one fair die. The sum of two independent dice has PGF $G(s)^2$. Expand $G(s)^2$ and read off $\mathbb{P}(\text{sum} = 7)$ as the coefficient of s^7. Confirm it equals $6/36$.

Exercise 18.3 (Negative binomial closure)**.** The Geometric(p) (failures before first success) has PGF $p/(1 - qs)$. Using the product rule, find the PGF of a sum of r independent Geometric(p) variables. Identify the resulting distribution (it is Negative Binomial). What is its mean, via $G'(1)$?

Exercise 18.4 (MGF moment extraction)**.** Let $X \sim \text{Exponential}(\lambda)$ with $M_X(t) = \lambda/(\lambda - t)$ for $t < \lambda$.

(a) Compute $M_X'(t)$ and $M_X''(t)$ and evaluate at $t = 0$ to recover $\mathbb{E}[X] = 1/\lambda$ and $\mathbb{E}[X^2] = 2/\lambda^2$.

(b) Use these to confirm $\text{Var}(X) = 1/\lambda^2$.

(c) From the series $\sum_n (n!/\lambda^n)\, t^n/n!$, write down $\mathbb{E}[X^5]$ with no further calculus.

Exercise 18.5 (Identify by MGF)**.** A random variable Y is found to have $M_Y(t) = e^{3(e^t-1)}$ for all t. By recognizing this MGF (substitute $s = e^t$ into a PGF you know), name the distribution of Y and state its mean and variance. Then explain in one sentence which theorem of this chapter licenses the conclusion.

Exercise 18.6 (Sum of independent normals)**.** Let $X \sim \mathcal{N}(1, 4)$ and $Y \sim \mathcal{N}(-2, 9)$ be independent. Using $M_{\mathcal{N}(\mu,\sigma^2)}(t) = e^{\mu t + \frac{1}{2}\sigma^2 t^2}$ and the product rule, find the MGF of $2X - Y + 5$, and hence its distribution. (Use the affine rule on each piece first.)

Exercise 18.7 (Why the i matters)**.** For the standard Cauchy density $f(x) = 1/[\pi(1 + x^2)]$:

(a) Show that $\mathbb{E}[e^{tX}] = \infty$ for every $t \neq 0$, so the MGF does not exist. (Compare the integrand's tail e^{tx}/x^2 to a divergent integral for $t > 0$; handle $t < 0$ by symmetry.)

(b) Given $\varphi_X(t) = e^{-|t|}$, use the product and affine rules to find the characteristic function of $\overline{X}_n = \frac{1}{n}\sum_{i=1}^n X_i$ for i.i.d. standard Cauchys, and conclude that $\overline{X}_n$ has the same distribution as a single X_i.

Exercise 18.8 (CF basics)**.** (a) Show directly from the definition that $|\varphi_X(t)| \leq 1$ and $\varphi_X(0) = 1$ for every random variable.

(b) Show that $\varphi_X(-t) = \overline{\varphi_X(t)}$ (complex conjugate), and deduce that if X is symmetric about 0 (i.e. X and $-X$ have the same distribution) then $\varphi_X(t)$ is real-valued.

Exercise 18.9 (Uniqueness, carefully)**.** Two students argue. Student A says: "If X and Y have all the same moments, they have the same distribution." Student B says: "Not necessarily." Who is right, and under exactly what extra hypothesis does Student A's claim become true? Name the standard counterexample and explain in one or two sentences why its moments fail to determine it.

Exercise 18.10 (CLT skeleton, filled in)**.** Let $X_1, \ldots, X_n$ be i.i.d. with mean 0, variance 1, and characteristic function φ. Starting from $\varphi(u) = 1 - \frac{1}{2}u^2 + o(u^2)$, fill in every step showing that the CF of $S_n = (X_1 + \cdots + X_n)/\sqrt{n}$ converges to $e^{-t^2/2}$ as $n \to \infty$. State precisely where you use (i) independence, (ii) the affine rule, (iii) the second-moment Taylor expansion, and (iv) the limit $(1 + a_n/n)^n \to e^a$.

Exercise 18.11 (Branching extinction)**.** In a branching process each individual independently has 0, 1, or 2 offspring with probabilities $\frac{1}{4}, \frac{1}{4}, \frac{1}{2}$.

(a) Write the offspring PGF $G(s)$ and compute the mean offspring count $G'(1)$. Is it above or below 1?

(b) The extinction probability is the smallest root in $[0, 1]$ of $G(s) = s$. Solve the resulting quadratic and report the extinction probability. Does it match your expectation from part (a)?

Part IV

Limit Theorems

Chapter 19

Inequalities and Concentration

Up to now we have spent most of our energy computing things *exactly*. We found exact probabilities, exact expectations, exact variances, exact distributions of sums. That is a valuable skill, but it has a quiet limitation: real problems rarely hand you a distribution clean enough to compute with. You will often know the mean of a quantity, maybe its variance, perhaps that it is a sum of independent pieces — and almost nothing else. The full distribution is a mystery. And yet you are asked, again and again, a very practical question:

> *How likely is this random quantity to land far from where I expect it to be?*

This is the question of **tail probabilities**, and the tools that answer it without knowing the whole distribution are called **inequalities**. An inequality trades exactness for robustness. Instead of saying "the chance of a big deviation is exactly 0.0137," it says "the chance is *at most* 0.02, no matter what the distribution is, as long as it has this mean and this variance." That "no matter what" is where the leverage comes from. It is a guarantee that holds across an entire universe of distributions you never had to identify.

The real payoff of these inequalities is a phenomenon called **concentration**: when you average many independent random things, the average stops being random in any practical sense. It *concentrates* near a single number, tighter and tighter as you add more terms, and the probability of any meaningful deviation collapses — often *exponentially* fast. Concentration is the reason polls work, the reason casinos are never surprised, the reason a machine-learning model trained on a big sample generalizes to data it has never seen. It is, with very little exaggeration, the mathematical engine underneath all of statistics and much of modern data science.

The chapter builds in stages. We start with Markov's inequality, the humblest and most universal bound, squeezed out of nothing but non-negativity. Chebyshev's inequality comes next; it adds the variance and gives our first real concentration result. From there we make the jump to *exponential* bounds — Chernoff and Hoeffding — which are what make concentration genuinely powerful. Along the way we meet Jensen's inequality, the one bound in this chapter that is about *shape* rather than tails, and which quietly underwrites half the proofs in the subject.

19.1 The Problem: Bounding the Unknown

Let us start with a story that has no clean distribution in it at all.

A factory ships boxes of bolts. You are told one fact and one fact only: the *average* number of defective bolts per box is 2. A buyer asks: "What fraction of boxes contain 10 or more defective bolts?" You do not know the distribution of defects. It could be Poisson, it could be lumpy, it could be anything. Can you say *anything* useful?

It feels like you cannot. With only the mean, the distribution is wide open. And yet there is a hard ceiling hiding in plain sight. Defect counts cannot be negative. That single constraint — non-negativity — combined with the known average, is enough to corner the answer.

Intuition

Here is the whole idea in one sentence. *If the average is small but a value is large, then that large value cannot happen too often — because if it did, it would drag the average up past where we know it sits.*
Suppose, for contradiction, that more than 1/5 of boxes had 10 or more defects. Those boxes alone would contribute more than $\frac{1}{5} \times 10 = 2$ to the average. Every other box contributes a non-negative amount on top of that. So the overall average would have to exceed 2. But the average *is* 2. Contradiction. Therefore at most 1/5 of boxes can have 10 or more defects.

That little argument is the entire content of Markov's inequality. No distribution required. Just "values are non-negative" and "here is the mean." Let us make it a theorem.

19.2 Markov's Inequality: Everything From Non-Negativity

Theorem 19.1 (Markov's inequality)**.** *Let X be a random variable with $X \geq 0$, and let $a > 0$. Then*

$$\mathbb{P}(X \geq a) \;\leq\; \frac{\mathbb{E}[X]}{a}.$$

Why it works

The proof is the sand-on-a-seesaw picture turned into one line. The trick is to build a crude "staircase" function that lies underneath X but is easy to average.
Fix $a > 0$ and look at the indicator $\mathbf{1}\{X \geq a\}$, which is 1 when $X \geq a$ and 0 otherwise. We claim the simple inequality

$$a \cdot \mathbf{1}\{X \geq a\} \;\leq\; X \qquad \text{(always, for every outcome)}.$$

Check the two cases. If $X \geq a$, the left side is a and the right side is $X \geq a$, so the inequality holds. If $X < a$, the left side is 0 and the right side is $X \geq 0$ (here we use non-negativity!), so it holds again. The inequality is true outcome by outcome, so it survives taking expectations:

$$a\,\mathbb{E}[\mathbf{1}\{X \geq a\}] \;\leq\; \mathbb{E}[X].$$

But the expectation of an indicator is just the probability of the event, $\mathbb{E}[\mathbf{1}\{X \geq a\}] = \mathbb{P}(X \geq a)$. Divide by $a > 0$ and you are done. Figure 19.1 shows the staircase-under-the-line picture that the algebra is tracking.

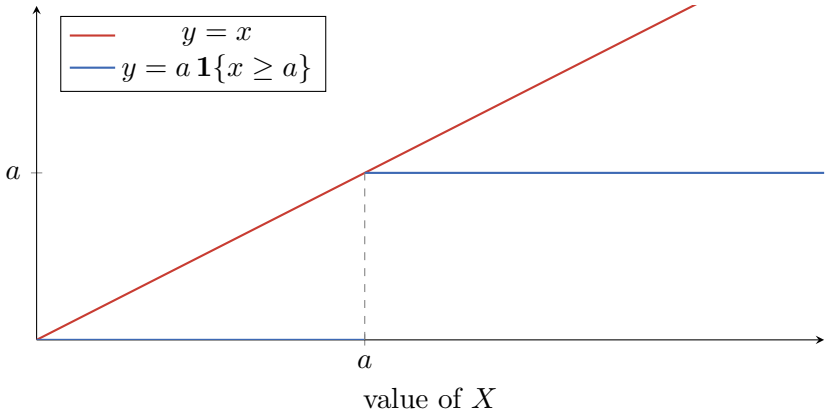


Figure 19.1: Markov's inequality in one picture. The blue staircase $a\,\mathbf{1}\{x \geq a\}$ never rises above the red line $y = x$ as long as $x \geq 0$. Averaging the staircase gives $a\,\mathbb{P}(X \geq a)$; averaging the line gives $\mathbb{E}[X]$. Since the staircase sits underneath, $a\,\mathbb{P}(X \geq a) \leq \mathbb{E}[X]$.

What is really happening

Markov's inequality looks almost too weak to be useful — and used naively, it often is. Its real power is that it is a *template*. The proof never cared that X was the original quantity of interest. If you can find *any* clever non-negative function $g(X) \geq 0$ that blows up when X strays, you can apply Markov to $g(X)$ and get a bound on X straying. Feed it $g(X) = (X - \mu)^2$ and out pops Chebyshev. Feed it $g(X) = e^{tX}$ and out pops the Chernoff bound, the most powerful tail inequality in this book. Every concentration result we prove is Markov's inequality wearing a different costume.

Example 19.1 (The bolt boxes, finished)**.** With $X =$ defects per box, $\mathbb{E}[X] = 2$, and $a = 10$:

$$\mathbb{P}(X \geq 10) \;\leq\; \frac{\mathbb{E}[X]}{10} = \frac{2}{10} = \frac{1}{5}.$$

At most 20% of boxes have 10 or more defective bolts — guaranteed, with no knowledge of the distribution. If the truth happened to be Poisson with mean 2, the real probability is about 0.0000465 (that is, 4.6×10^{-5}), so Markov is wildly loose here. That looseness is the price of universality: the bound has to hold even for the worst-case distribution, which in this case is a nasty one that puts a tiny sliver of mass exactly at 10 and the rest at 0.

Common trap

Markov needs $X \geq 0$. Forgetting this is the classic blunder. If X can be negative, the staircase argument breaks at the step "$X < a$ implies $X \geq 0$," and the bound is simply false. To bound a two-sided quantity like $X - \mu$, do not apply Markov to it directly — apply Markov to a non-negative *function* of it, such as $(X - \mu)^2$. That detour is exactly how we get Chebyshev next.

History

The inequality is named for **Andrey Markov** (1856–1922), the St. Petersburg mathematician who also gave us Markov chains. He published it in the early 1900s, though the core idea — that a non-negative quantity with a small average cannot often be large — was used earlier by his teacher **Pafnuty Chebyshev** in the proof of the inequality that now bears Chebyshev's name. The two results are a matched pair: Markov is the raw lemma, Chebyshev the polished theorem built from it. It is a small historical irony that the cruder bound carries the student's name and the sharper one the teacher's.

19.3 Chebyshev's Inequality: Bringing In the Variance

Markov used only the mean. But if you also know the *variance* — how spread out X is — you should be able to say much more about how far X wanders. Variance is literally a measure of typical squared distance from the mean, so it is exactly the right currency for buying a tail bound.

The move is the one the happeningbox promised: apply Markov not to X, but to the non-negative quantity $(X-\mu)^2$.

Theorem 19.2 (Chebyshev's inequality)**.** *Let X have mean $\mu = \mathbb{E}[X]$ and finite variance $\sigma^2 = \mathrm{Var}(X)$. Then for any $k > 0$,*

$$\mathbb{P}\big(|X-\mu| \geq k\big) \;\leq\; \frac{\sigma^2}{k^2}.$$

Equivalently, writing the deviation in units of standard deviations ($k = c\,\sigma$),

$$\mathbb{P}\big(|X-\mu| \geq c\,\sigma\big) \;\leq\; \frac{1}{c^2}.$$

Why it works

Let $Y = (X-\mu)^2$. This Y is non-negative, so Markov applies. Its mean is the variance: $\mathbb{E}[Y] = \mathbb{E}[(X-\mu)^2] = \sigma^2$. Now notice that the events $\{|X-\mu| \geq k\}$ and $\{Y \geq k^2\}$ are the *same event* — $|X-\mu| \geq k$ exactly when $(X-\mu)^2 \geq k^2$. So

$$\mathbb{P}\big(|X-\mu| \geq k\big) = \mathbb{P}\big(Y \geq k^2\big) \overset{\text{Markov}}{\leq} \frac{\mathbb{E}[Y]}{k^2} = \frac{\sigma^2}{k^2}.$$

That is the whole proof. Squaring turned a two-sided deviation into a one-sided statement about a non-negative variable, and Markov finished the job.

The "standard deviations" form is the one worth memorizing, because it is distribution-free and strikingly clean:

- At least $1 - 1/2^2 = 75\%$ of any distribution's mass lies within 2 standard deviations of the mean.
- At least $1 - 1/3^2 \approx 88.9\%$ lies within 3 standard deviations.
- At least $1 - 1/4^2 \approx 93.75\%$ lies within 4.

No bell-curve assumption, no symmetry, nothing. Any random variable with a finite variance obeys these; Figure 19.2 plots the ceiling $1/c^2$ against the much smaller Gaussian tail. (The Normal distribution does far better — about 95% within two standard deviations and 99.7% within three — but that requires *knowing* you have a Normal. Chebyshev is the floor that holds for everyone.)

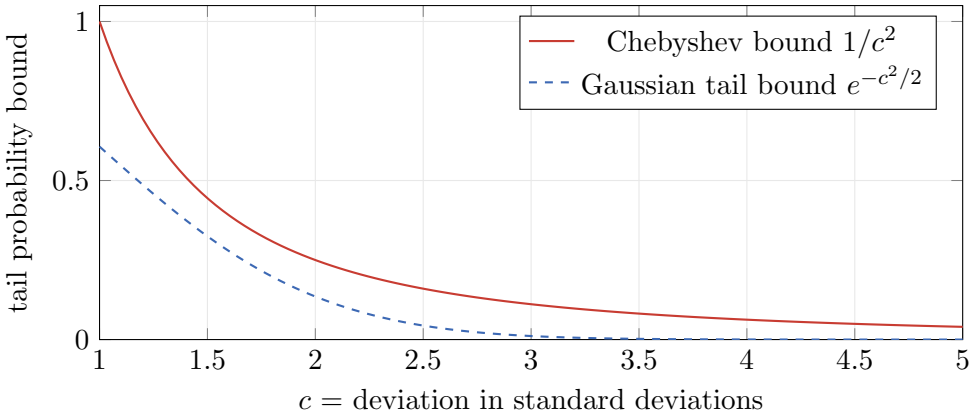


Figure 19.2: The Chebyshev ceiling $1/c^2$ (red) bounds the probability of being at least c standard deviations from the mean, for *any* distribution. For comparison the Gaussian tail bound $e^{-c^2/2}$ (blue dashed), which controls the Normal, is far smaller — Chebyshev pays for its universality with looseness, but it never lies.

Example 19.2 (Polling, the crude version)**.** You poll n people and let $\hat{p}$ be the fraction who say "yes." Each response is Bernoulli(p), so $\hat{p}$ has mean p and variance $p(1-p)/n \le 1/(4n)$ (the variance of a Bernoulli is largest at $p = 1/2$). Chebyshev says

$$\mathbb{P}\big(|\hat{p} - p| \ge \varepsilon\big) \;\le\; \frac{\mathrm{Var}(\hat{p})}{\varepsilon^2} \;\le\; \frac{1}{4n\varepsilon^2}.$$

To be within $\varepsilon = 0.03$ of the truth with probability at least 95%, set $1/(4n\varepsilon^2) \le 0.05$, giving $n \ge 1/(4 \cdot 0.05 \cdot 0.0009) \approx 5556$. So a few thousand respondents pin down a national opinion to within three points — and this is the *pessimistic* Chebyshev count. The exponential bounds later in the chapter will sharpen it, but the headline is already here: accuracy depends on n, not on the size of the population.

What is really happening

Look at what Chebyshev does to an *average*. Let $\bar{X}_n = \frac{1}{n}\sum_{i=1}^{n} X_i$ be the average of n independent copies, each with mean μ and variance σ^2. Variances of independent things add, and the $1/n$ out front gets squared, so $\mathrm{Var}(\bar{X}_n) = \sigma^2/n$. Plug into Chebyshev:

$$\mathbb{P}\big(|\bar{X}_n - \mu| \ge \varepsilon\big) \;\le\; \frac{\sigma^2}{n\,\varepsilon^2} \;\xrightarrow[n\to\infty]{}\; 0.$$

The right-hand side melts to zero as n grows. The average concentrates on μ. This single line is a complete proof of the **Weak Law of Large Numbers** — the formal statement that sample averages converge in probability to the true mean. We will revisit it in its own chapter, but you have just watched it fall out of one inequality. *This* is concentration: randomness in the average does not merely shrink, it provably shrinks, and we can say how fast.

19.4 What "Concentration" Really Means

Pause to appreciate the shift in worldview. A single fair coin flip is maximally uncertain — you cannot predict it at all. But the *average* of a million fair coin flips is almost perfectly predictable: it will be 0.5 to within a whisker, essentially every time. The randomness has not disappeared; it has been *diluted*. Each flip still surprises you, but their collective average has nowhere to hide. Figure 19.3 shows the distribution of the sample average tightening around 0.5 as n grows.

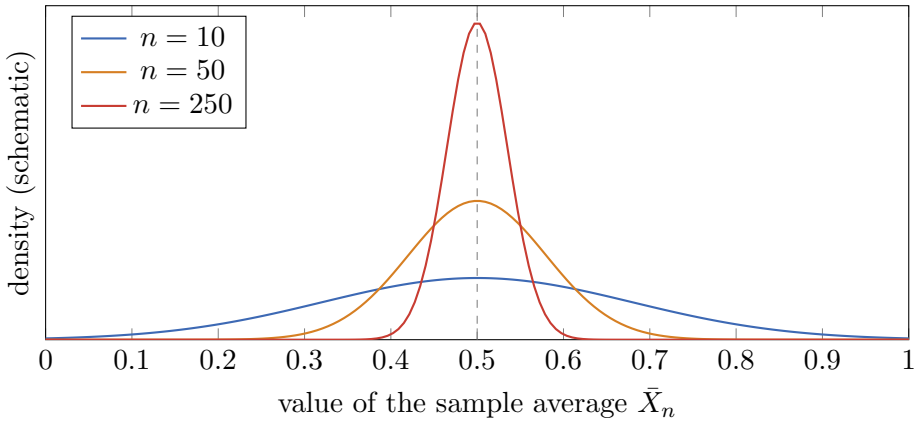


Figure 19.3: Concentration in action. As the number of averaged coin flips n grows, the distribution of the sample average $\bar{X}_n$ piles up ever more tightly around the true mean 0.5. The spread shrinks like $1/\sqrt{n}$; the *tail probability* of any fixed deviation shrinks far faster, as the next section shows.

Why this matters

Chebyshev's $\sigma^2/(n\varepsilon^2)$ tells us the tail probability shrinks at least as fast as $1/n$. That is genuinely useful, and for a long time it was the best general tool available. But $1/n$ is, for many purposes, embarrassingly slow. To drive a failure probability down to one in a million, $1/n$ decay demands about a million samples. The remarkable discovery of the twentieth century is that for *sums of independent bounded variables*, the true decay is *exponential* in n: the tail behaves like e^{-cn}, not $1/n$. One in a million then costs only a few dozen samples' worth of exponent. The leap from polynomial to exponential decay is the difference between statistics that "eventually works" and statistics that works *decisively*. The tool that captures it is the Chernoff method.

19.5 The Chernoff Method: Exponential Tails

Why is Chebyshev slow? Because $(X - \mu)^2$ grows only quadratically, so a single far-out outlier is not penalized very harshly inside the expectation. To get an exponential tail bound we need an exponentially harsh penalty. So we feed Markov's inequality the most aggressive non-negative function we have: the exponential e^{tX}.

Here is the trick in full. For any $t > 0$, the event $\{X \geq a\}$ is identical to $\{e^{tX} \geq e^{ta}\}$, because $e^{t\,\cdot}$ is

increasing. Apply Markov to the non-negative variable e^{tX}:

$$\mathbb{P}(X \geq a) \;=\; \mathbb{P}\big(e^{tX} \geq e^{ta}\big) \;\leq\; \frac{\mathbb{E}[e^{tX}]}{e^{ta}} \;=\; e^{-ta}\, M_X(t),$$

where $M_X(t) = \mathbb{E}[e^{tX}]$ is the **moment generating function** we met in the chapter on generating functions. This holds for *every* $t > 0$ simultaneously, so we are free to choose the t that makes the bound tightest:

$$\boxed{\mathbb{P}(X \geq a) \;\leq\; \min_{t>0}\; e^{-ta}\, M_X(t).}$$

This is the **Chernoff bound**. The minimization over t is what gives it its power.

Why it works

Why does optimizing over t unlock exponential decay? Think of $e^{-ta}M_X(t)$ as a family of bounds, one per t. A small t barely penalizes large values, giving a weak bound; a huge t penalizes them ferociously but blows up $M_X(t)$ itself. Somewhere in between is a sweet spot where the two effects balance, and at that sweet spot the exponent $-ta$ wins out, scaling linearly in a inside an exponential. Because $M_X(t)$ for a *sum* of n independent variables factorizes — $M_{S_n}(t) = M_X(t)^n$ — the whole bound becomes $\big(e^{-ta/n}M_X(t)\big)^n$, an n-th power. A number less than one raised to the n is exponential decay in n. That factorization, inherited from independence, is the engine. Chebyshev never exploited it because squaring does not turn products into powers the way exponentials do.

Example 19.3 (Chernoff for a sum of fair coins)**.** Let $S_n = X_1 + \cdots + X_n$ be the number of heads in n fair flips, so $\mathbb{E}[S_n] = n/2$. Each X_i has MGF $\mathbb{E}[e^{tX_i}] = \frac{1}{2}(1 + e^t)$, and by independence $M_{S_n}(t) = \big(\frac{1}{2}(1 + e^t)\big)^n$. The Chernoff bound on the chance of seeing at least $(\frac{1}{2} + \delta)n$ heads is

$$\mathbb{P}\big(S_n \geq (\tfrac{1}{2} + \delta)n\big) \;\leq\; \min_{t>0}\; e^{-t(\frac{1}{2}+\delta)n}\Big(\tfrac{1}{2}(1 + e^t)\Big)^n.$$

Carrying out the minimization (set the derivative of the exponent to zero) and simplifying yields the clean exponential

$$\mathbb{P}\big(S_n \geq (\tfrac{1}{2} + \delta)n\big) \;\leq\; e^{-2\delta^2 n}.$$

Compare to Chebyshev, which here gives only $\mathbb{P}(|S_n - n/2| \geq \delta n) \leq 1/(4\delta^2 n)$, a measly $1/n$. For a 10% excess ($\delta = 0.1$) with $n = 1000$ flips, Chebyshev allows up to $1/40 = 2.5\%$; Chernoff caps it at $e^{-20} \approx 2 \times 10^{-9}$. That is the chasm between polynomial and exponential, drawn on a log scale in Figure 19.4.

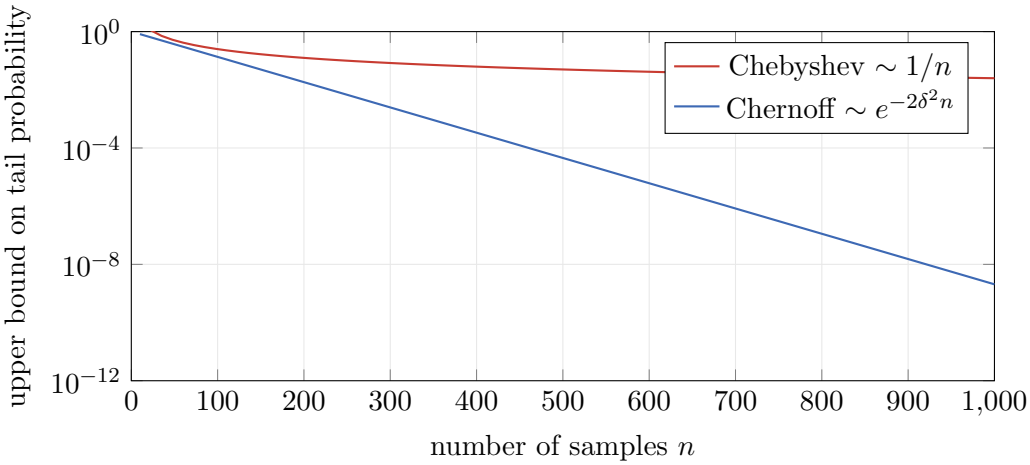


Figure 19.4: Tail bound on $\mathbb{P}(S_n \geq (\frac{1}{2}+\delta)n)$ for $\delta = 0.1$, on a log scale. Chebyshev (red) crawls down like $1/n$; Chernoff (blue) plunges exponentially. By $n = 1000$ they differ by many orders of magnitude. This gap is why concentration is taken seriously.

History

The method is named for **Herman Chernoff**, who published it in 1952 in a paper on the asymptotic efficiency of hypothesis tests — though Chernoff himself generously credited his colleague **Herman Rubin** with the key idea of optimizing the exponential moment. The underlying technique, bounding a tail by an optimized exponential moment, was already present in the work of **Sergei Bernstein** in the 1920s and traces back to the exponential-moment methods of **Cramér** in large-deviation theory. What Chernoff's name now labels is less a single inequality than a *recipe*: take the log of the optimized exponential bound and you obtain the *rate function* of large-deviations theory, the precise exponent at which rare events decay. The recipe has become so central to theoretical computer science and machine learning that "a Chernoff bound" is spoken of the way an engineer speaks of a wrench.

19.6 Hoeffding's Inequality: Concentration You Can Use

The Chernoff recipe is powerful but demands the MGF, which you may not know. **Hoeffding's inequality** packages the recipe into a ready-to-use form that needs only one mild assumption: each variable is *bounded*. This is the concentration inequality you will reach for most often in practice.

Theorem 19.3 (Hoeffding's inequality)**.** *Let $X_1, \dots, X_n$ be independent with $a_i \leq X_i \leq b_i$ almost surely. Let $S_n = \sum_{i=1}^{n} X_i$. Then for any $\varepsilon > 0$,*

$$\mathbb{P}\big(S_n - \mathbb{E}[S_n] \geq \varepsilon\big) \;\leq\; \exp\!\left(-\frac{2\varepsilon^2}{\sum_{i=1}^{n}(b_i - a_i)^2}\right),$$

and the same bound holds for the lower tail $\mathbb{P}(\mathbb{E}[S_n] - S_n \geq \varepsilon)$. In the common case where each $X_i \in [a,b]$ and we look at the average *$\bar{X}_n = S_n/n$,*

$$\mathbb{P}\big(|\bar{X}_n - \mu| \geq \varepsilon\big) \;\leq\; 2\exp\!\left(-\frac{2n\varepsilon^2}{(b-a)^2}\right).$$

Why it works

Hoeffding is Chernoff plus one technical lemma. The Chernoff step gives, for a centered bounded variable $Y_i = X_i - \mathbb{E}[X_i]$,

$$\mathbb{P}(S_n - \mathbb{E}[S_n] \geq \varepsilon) \leq e^{-t\varepsilon} \prod_{i=1}^{n} \mathbb{E}\big[e^{tY_i}\big],$$

using independence to split the expectation of the product into a product of expectations. The missing piece is a bound on each factor $\mathbb{E}[e^{tY_i}]$. **Hoeffding's lemma** supplies it: if Y has mean 0 and lies in $[a, b]$, then $\mathbb{E}[e^{tY}] \leq e^{t^2(b-a)^2/8}$. (Intuitively, among all mean-zero variables confined to an interval, the one with the fattest exponential moment is the two-point distribution sitting at the endpoints, and a short convexity argument pins down the $(b-a)^2/8$ constant.) Plugging in,

$$\mathbb{P}(S_n - \mathbb{E}[S_n] \geq \varepsilon) \leq \exp\left(-t\varepsilon + \tfrac{t^2}{8} \sum_i (b_i - a_i)^2 \right).$$

This is a parabola in t inside the exponent; minimizing over t (vertex at $t^\star = 4\varepsilon / \sum_i (b_i - a_i)^2$) gives exactly the stated $-2\varepsilon^2 / \sum_i (b_i - a_i)^2$. The whole proof is: Chernoff splits, Hoeffding's lemma bounds each piece, calculus finds the best t.

What is really happening

Stare at the exponent of the average form, $-2n\varepsilon^2/(b-a)^2$. Two features matter enormously. First, n sits in the exponent. Double your sample size and you do not halve the failure probability — you *square* it. This is the exponential concentration that Chebyshev's $1/n$ could only dream of.

Second, the bound depends on the range $(b-a)$ but *not* on the variance, the shape, or anything else about the distribution. Bounded plus independent is the entire price of admission. That distribution-freeness, married to exponential decay, is precisely the combination that statistical learning theory needs — which is why Hoeffding is the first inequality in every machine-learning theory course.

Example 19.4 (Polling, the sharp version)**.** Return to the poll. Each response $X_i \in [0, 1]$, so $b - a = 1$, and $\bar{X}_n = \hat{p}$. Hoeffding gives

$$\mathbb{P}\big(|\hat{p} - p| \geq \varepsilon\big) \;\leq\; 2e^{-2n\varepsilon^2}.$$

For $\varepsilon = 0.03$ and a 95% guarantee we need $2e^{-2n(0.03)^2} \leq 0.05$, i.e. $e^{-0.0018\,n} \leq 0.025$, giving $n \geq \ln(40)/0.0018 \approx 2050$. Recall Chebyshev demanded about 5556 for the same guarantee. Same data, same assumptions, sharper tool — less than half the sample size. Figure 19.5 traces how the guaranteed half-width shrinks with n. And unlike Chebyshev, Hoeffding hands you a confidence *interval* of half-width $\varepsilon = \sqrt{\ln(2/\alpha)/(2n)}$ at confidence $1 - \alpha$, the Hoeffding bound used throughout applied statistics.

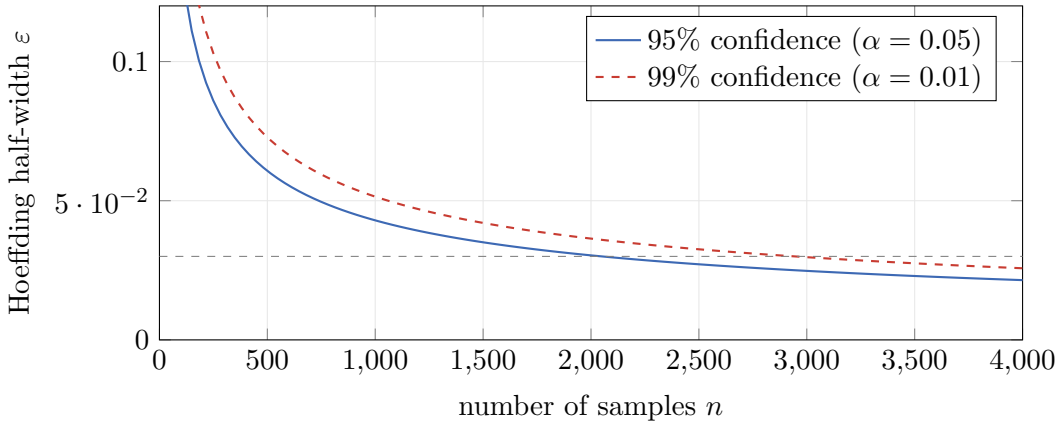


Figure 19.5: The Hoeffding confidence half-width $\varepsilon = \sqrt{\ln(2/\alpha)/(2n)}$ for a bounded average on $[0, 1]$, as a function of sample size n. It shrinks like $1/\sqrt{n}$: quadrupling the sample halves the width. The horizontal line marks $\varepsilon = 0.03$; the 95% curve crosses it near $n \approx 2050$, matching the poll calculation. Demanding higher confidence (99%, red dashed) widens the band only mildly, because confidence enters through the slow logarithm $\ln(2/\alpha)$.

Application

Why concentration is the backbone of statistics and machine learning.
Every estimate you ever compute — a sample mean, a regression coefficient, a model's accuracy on a test set — is a function of random data. The entire enterprise of statistics rests on a single hope: that this function does not swing wildly with the luck of the sample, but *concentrates* near the true value. Concentration inequalities turn that hope into a theorem with an explicit, computable failure probability.

In supervised machine learning the stakes are sharper still. You train a model to minimize error on a finite *training* sample, but you care about its error on the unseen population — its *generalization* error. The two differ by exactly the kind of deviation Hoeffding controls. For a single fixed hypothesis h, the training error is an average of n bounded $\{0, 1\}$ mistakes, so Hoeffding gives

$$\mathbb{P}\big(|\widehat{\mathrm{err}}(h) - \mathrm{err}(h)| \geq \varepsilon\big) \leq 2e^{-2n\varepsilon^2}.$$

To cover a whole hypothesis class of size M (so we may safely *choose* the best h after seeing the data) one applies the union bound, paying a factor M that becomes a mild additive $\ln M$ inside the exponent's budget: the sample size needed grows only like $\log M$. This is the seed of **PAC learning** ("probably approximately correct") and of VC theory, which extend the argument to infinite classes via cleverer counting. Strip away the formalism and the message is simple: *a model generalizes because averages concentrate, and we can bound the sample size that makes the guarantee bite.* Concentration is not a footnote to learning theory. It *is* learning theory.

Connection

The Chernoff and Hoeffding proofs lean entirely on the moment generating function $M_X(t) = \mathbb{E}[e^{tX}]$ from the chapter on generating functions, and the optimization step is plain single-variable calculus — find the t where the derivative of the exponent vanishes. The deeper

link is to *large-deviations theory*: the limit $-\frac{1}{n}\log \mathbb{P}(\bar{X}_n \geq a)$ converges to the **Legendre transform** of $\log M_X(t)$, a construction borrowed straight from convex analysis and classical mechanics. The Chernoff bound is the finite-n shadow of that exact asymptotic rate. If you enjoyed the Legendre transform in the companion calculus book, you are meeting it again here in probabilistic disguise.

19.7 Jensen's Inequality: The Bound About Shape

The inequalities so far were about *tails* — how often a variable strays. Jensen's inequality is about something different and equally fundamental: how *averaging interacts with curvature*. It answers the question, "is the average of a function the function of the average?" — and the answer, almost always, is no, with a definite direction to the error.

Intuition

Picture a function that bends upward, like $g(x) = x^2$ — a *convex* function, holding water. Take two input points and look at the chord connecting the corresponding points on the curve. Because the curve bends away below the chord, the chord sits *above* the curve. Now read this in the language of averages: the height of the chord at the midpoint is the *average of the function values*, while the curve's height at the midpoint is the *function of the average input*. Chord above curve means

$$\text{average of } g \;\geq\; g(\text{of the average}).$$

That is Jensen's inequality, and it is nothing more than "convex curves lie below their chords," generalized from two points to an entire probability distribution of points.

Theorem 19.4 (Jensen's inequality)**.** *Let g be a convex function and X a random variable with finite mean. Then*

$$g\big(\mathbb{E}[X]\big) \;\leq\; \mathbb{E}\big[g(X)\big].$$

If g is concave, the inequality reverses: $g(\mathbb{E}[X]) \geq \mathbb{E}[g(X)]$.

Why it works

The proof is the picture made rigorous with one tool: the *supporting line*. Because g is convex, at the point $\mu = \mathbb{E}[X]$ there is a line through $(\mu, g(\mu))$ that stays at or below the whole curve — a tangent if g is differentiable, a supporting line in general. Write it as $L(x) = g(\mu) + m(x-\mu)$, and convexity guarantees $g(x) \geq L(x)$ for all x. Now take expectations of both sides:

$$\mathbb{E}[g(X)] \;\geq\; \mathbb{E}[L(X)] = g(\mu) + m\;\mathbb{E}[X-\mu] = g(\mu) + m\cdot 0 = g(\mathbb{E}[X]).$$

The middle term vanished because $\mathbb{E}[X-\mu] = 0$ by definition of μ. The supporting line, anchored exactly at the mean, is what makes the linear term disappear and leaves the clean inequality behind. Figure 19.6 shows the curve, a chord, and that supporting line together.

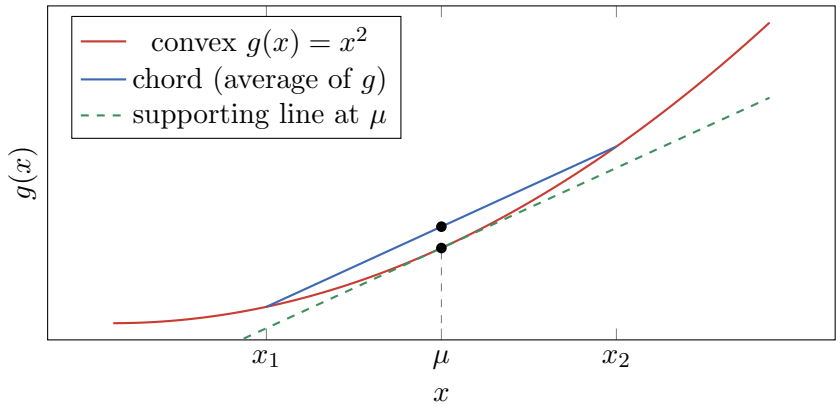


Figure 19.6: Jensen's inequality. The convex curve lies below every chord, so the chord height at μ — which equals $\mathbb{E}[g(X)]$ for a two-point variable — sits above the curve height $g(\mu) = g(\mathbb{E}[X])$. The dashed supporting line through $(\mu, g(\mu))$ is the proof: the curve never dips below it, and its expectation equals $g(\mu)$.

Example 19.5 (Why variance is non-negative, instantly)**.** Apply Jensen with the convex $g(x) = x^2$: $\big(\mathbb{E}[X]\big)^2 \le \mathbb{E}[X^2]$. Rearranged, this is $\mathbb{E}[X^2] - (\mathbb{E}[X])^2 \ge 0$ — but the left side is exactly the variance. Jensen hands you $\mathrm{Var}(X) \ge 0$ in a single line, with equality only when X is a constant (no spread). A fact you proved laboriously before is a one-step corollary of convexity.

Paradox

The flaw of averages. A consultant plans a warehouse. Demand is random with an average of 100 units, and profit is some function of demand. Surely, the consultant reasons, "planning for average demand gives average profit." This is the *flaw of averages*, and Jensen is the verdict on it. Whenever profit is a curved function of demand, $\mathbb{E}[g(\text{demand})] \neq g(\mathbb{E}[\text{demand}])$, and the gap has a guaranteed sign.

A sharp instance: a gambler holds a stock whose value next year is \$1 or \$4 with equal probability, average \$2.50. If returns *multiply* over years, the relevant growth is the logarithm, and log is *concave*, so Jensen flips: the expected log-growth $\mathbb{E}[\log V]$ is *strictly less* than $\log \mathbb{E}[V]$. Concretely $\mathbb{E}[\log V] = \frac{1}{2}(\log 1 + \log 4) = \log 2$, while $\log \mathbb{E}[V] = \log 2.5$. Over many years your wealth compounds at the rate set by $\mathbb{E}[\log V] = \log 2$ (geometric mean \$2), not the deceptively higher arithmetic mean \$2.50. Confusing the two has ruined real portfolios. Jensen's inequality is the mathematical reason the *geometric* mean never exceeds the arithmetic mean — and the reason "it averages out" is one of the most expensive phrases in finance.

Connection

Jensen is the silent workhorse behind several results elsewhere in this book. It is why **Kullback–Leibler divergence** $\mathrm{KL}(p \,\|\, q) \ge 0$ (apply Jensen to the concave log inside the divergence), the cornerstone of information theory. It is why the **EM algorithm** in machine learning provably increases the likelihood at each step (the E-step builds a Jensen lower bound on the log-likelihood). And it underlies the variational inequalities that power modern Bayesian computation. Whenever you see a proof that "moves a function inside or outside an expectation" and gains a definite direction, look for Jensen hiding underneath.

19.8 A Field Guide to the Inequalities

It helps to see the four tail bounds lined up, because they form a ladder of increasing assumptions buying increasing strength.

Inequality	**Needs**	**Gives**
Markov	$X \geq 0$, mean	$\mathbb{P}(X \geq a) \leq \mathbb{E}[X]/a$
Chebyshev	mean, variance	$\mathbb{P}(\lvert X - \mu \rvert \geq k) \leq \sigma^2/k^2$
Chernoff	MGF, independence	$\leq \min_t e^{-ta} M_X(t)$ (exponential)
Hoeffding	bounded, independent	$\leq 2e^{-2n\varepsilon^2/(b-a)^2}$ (exponential)

Read the table top to bottom as a bargain. Markov asks for almost nothing and gives a loose $1/a$ bound. Chebyshev adds the variance and earns a $1/k^2$ bound and the Weak Law. Chernoff and Hoeffding add independence and boundedness and are rewarded with *exponential* decay — the qualitative leap that makes large-scale statistics and learning theory possible. The pattern to internalize is not four separate formulas but one method: *apply Markov to a cleverly chosen non-negative function*, where each row just picks a smarter function (x, then x^2, then e^{tx}).

Pause

If you remember one thing from this chapter, remember the engine: every tail bound here is Markov's inequality applied to a non-negative function of X. The function determines the strength. Squaring buys variance-based decay; exponentiating buys exponential decay. Concentration is what happens when you apply the exponential version to a sum of many independent pieces and watch the tail collapse like e^{-cn}. That collapse is why averages are trustworthy, why polls work, and why models trained on samples generalize.

19.9 Exercises

Exercise 19.1 (Markov warm-up)**.** The average score on an exam is 65 out of 100. Using only this, give an upper bound on the fraction of students who scored at least 90. Then explain in one sentence why the true fraction is almost certainly far smaller than your bound, referring to what Markov ignores.

Exercise 19.2 (Chebyshev in standard units)**.** A random variable has mean 50 and standard deviation 4. Use Chebyshev to bound $\mathbb{P}(X \leq 38 \text{ or } X \geq 62)$. Then find the smallest interval $[\mu - c\sigma, \mu + c\sigma]$ that Chebyshev guarantees contains at least 90% of the mass.

Exercise 19.3 (Sharpness of Markov)**.** Construct an explicit non-negative random variable for which Markov's inequality holds with *equality* at a, i.e. $\mathbb{P}(X \geq a) = \mathbb{E}[X]/a$. (Hint: a two-point distribution, with mass at 0 and at a.) This shows Markov cannot be improved in general.

Exercise 19.4 (Chebyshev to the Weak Law)**.** Let $X_1, \ldots, X_n$ be i.i.d. with mean μ and variance σ^2. Using Chebyshev, find the smallest n guaranteeing $\mathbb{P}(\lvert \bar{X}_n - \mu \rvert \geq 0.1\sigma) \leq 0.01$. Then state precisely what happens to this probability as $n \to \infty$ and name the theorem.

Exercise 19.5 (Chernoff for Poisson)**.** Let $X \sim \text{Poisson}(\lambda)$, whose MGF is $M_X(t) = e^{\lambda(e^t-1)}$. Write down the Chernoff bound $\min_{t>0} e^{-ta} M_X(t)$ for $\mathbb{P}(X \geq a)$, carry out the minimization to find the optimal $t^\star = \ln(a/\lambda)$ (valid for $a > \lambda$), and show the bound simplifies to $e^{-\lambda}(e\lambda/a)^a$.

Exercise 19.6 (Hoeffding sample size)**.** You want to estimate the mean of a quantity known to lie in $[0, 10]$ to within $\varepsilon = 0.5$ with confidence 99%. Use the average form of Hoeffding to find the required number of independent samples n. Compare with the count Chebyshev would demand if you also knew $\sigma = 2$.

Exercise 19.7 (Jensen direction)**.** For each function, state whether $\mathbb{E}[g(X)]$ is at least or at most $g(\mathbb{E}[X])$, and name the convexity that decides it: (a) $g(x) = e^x$; (b) $g(x) = \sqrt{x}$ on $x \geq 0$; (c) $g(x) = 1/x$ on $x > 0$; (d) $g(x) = |x|$.

Exercise 19.8 (AM–GM from Jensen)**.** Let X take the values $x_1, \ldots, x_n > 0$ each with probability $1/n$. Apply Jensen to the concave function $g(x) = \log x$ and exponentiate the result to prove the arithmetic mean–geometric mean inequality $\frac{1}{n}\sum x_i \geq \left(\prod x_i\right)^{1/n}$. State the condition for equality.

Exercise 19.9 (Why the exponential beats the square — a proof)**.** Let S_n be the number of heads in n fair flips. (a) Use Chebyshev to bound $\mathbb{P}(S_n \geq 0.6n)$ and show the bound decays like C/n. (b) Use the Chernoff result $\mathbb{P}(S_n \geq (\frac{1}{2} + \delta)n) \leq e^{-2\delta^2 n}$ to bound the same event. (c) For $n = 500$, compute both numerically and explain in two or three sentences, in terms of the optimization over t, why the exponential method captures a decay the quadratic method structurally cannot.

Exercise 19.10 (Conceptual: what concentration is not)**.** A student claims: "Concentration means that if I flip a coin many times, I am guaranteed to get exactly half heads." Explain, using the precise statement of Hoeffding's inequality, in what sense this is wrong and in what sense it is nearly right. Distinguish carefully between the *value* concentrating and the randomness *disappearing*.

Chapter 20

The Laws of Large Numbers

There is a sentence almost everyone has heard, usually said with great confidence and almost always slightly wrong: "it all averages out in the end." Flip a fair coin enough times and you will get about half heads. Roll a die forever and each face shows up about a sixth of the time. Play a casino game long enough and the house edge is guaranteed to grind you down. Each of these claims is an appeal to the same deep fact about chance, the fact this chapter is about: *averages of many independent trials settle down to a single, predictable number.*

That fact is the Law of Large Numbers. It is the bridge between the abstract object we call "probability" and the concrete thing you can actually measure, namely the long-run frequency of an event. It is the reason a casino can publish its house edge and sleep soundly, the reason an insurance company can price a policy on a person whose future it cannot predict, and the reason a Monte Carlo simulation gives you an answer at all. It is, in a sense we will make precise, the theorem that justifies the entire frequentist picture of what probability *means*.

But the slogan "it all averages out" hides two very different theorems and at least one famous fallacy. There is a *weak* law and a *strong* law, and they claim genuinely different things. And the slogan, taken literally, tempts you into the gambler's fallacy — the belief that a coin which has come up heads ten times in a row is somehow "due" for tails. By the end of this chapter you will see exactly what averages out, what does not, and why the difference matters.

20.1 The Question: Does the Average Settle Down?

Let us be concrete before we are general. Imagine flipping a fair coin and recording 1 for heads, 0 for tails. After n flips you have a sequence $X_1, X_2, \dots, X_n$ of zeros and ones, and you compute the *sample mean* — the running fraction of heads:

$$\bar{X}_n \;=\; \frac{X_1 + X_2 + \cdots + X_n}{n}.$$

Each X_i is a random variable with $\mathbb{E}[X_i] = \frac{1}{2}$. The question is simple to state and surprisingly subtle to answer: as n grows, does $\bar{X}_n$ approach $\frac{1}{2}$? And in what *sense* of "approach"?

Intuition

A single flip is pure chaos: you cannot predict it at all. But the *average* of many flips is almost the opposite of chaos — it is nearly deterministic. The randomness does not disappear; it gets divided by n. Each new flip can still push $\bar{X}_n$ up or down, but it can only push it by $1/n$, and that lever shrinks as you go. The Law of Large Numbers is the statement that all those shrinking pushes conspire to pin the average down.

Figure 20.1 shows what this looks like in practice. Three independent simulated runs of 1000 fair-coin flips each are plotted as running averages $\bar{X}_n$ against n. Early on the runs are wild: with $n = 5$, a run can easily be at 0.8 or 0.2 just by luck. But as n climbs, the curves are squeezed inexorably toward the dashed line at 1/2. That squeezing is the phenomenon we want to understand and prove.

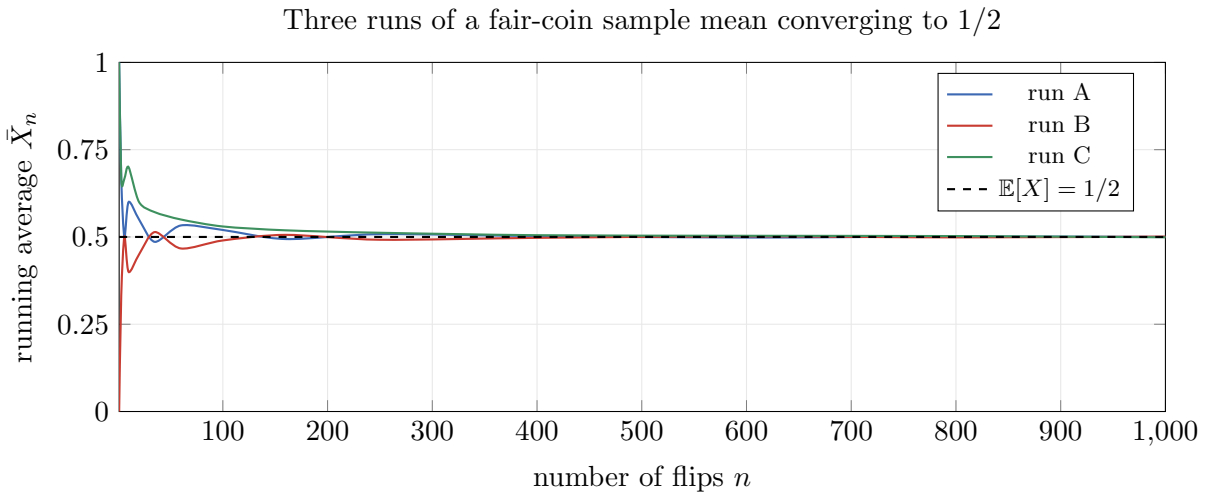


Figure 20.1: The sample mean of fair-coin flips. Three independent runs start far from 1/2 but are funneled toward it as n grows. The funnel narrows like $1/\sqrt{n}$; this chapter explains why.

Notice two features you should hold onto. First, the convergence is *not* monotone: a curve can be above 1/2, then below, then above again. Nothing forbids a long streak. Second, the spread of the curves narrows as n grows, and it narrows at a particular rate. We will see that the typical distance of $\bar{X}_n$ from 1/2 shrinks like $1/\sqrt{n}$ — a fact that is the doorway to the next chapter, the Central Limit Theorem.

20.2 The Engine Room: Mean and Variance of the Sample Mean

Before any law, we need two computations that do all the real work. Let $X_1, \ldots, X_n$ be independent and identically distributed (i.i.d.) with common mean $\mu = \mathbb{E}[X_i]$ and common variance $\sigma^2 = \mathrm{Var}(X_i) < \infty$. What are the mean and variance of $\bar{X}_n$?

The mean is immediate from linearity of expectation, which never requires independence:

$$\mathbb{E}[\bar{X}_n] \;=\; \mathbb{E}\left[\frac{1}{n}\sum_{i=1}^{n} X_i\right] \;=\; \frac{1}{n}\sum_{i=1}^{n}\mathbb{E}[X_i] \;=\; \frac{1}{n}\cdot n\mu \;=\; \mu.$$

So the sample mean is *unbiased*: on average it equals exactly the thing we are trying to estimate. Good — but "on average correct" is not the same as "reliably close." For that we need the variance.

Here independence earns its keep. Because the X_i are independent, the variance of their sum is the sum of their variances (no covariance terms survive):

$$\operatorname{Var}\left(\sum_{i=1}^{n} X_i\right) \;=\; \sum_{i=1}^{n} \operatorname{Var}(X_i) \;=\; n\sigma^2.$$

And variance scales by the square of a constant, $\operatorname{Var}(cY) = c^2 \operatorname{Var}(Y)$, so with $c = 1/n$:

$$\boxed{\operatorname{Var}(\bar{X}_n) \;=\; \frac{1}{n^2} \cdot n\sigma^2 \;=\; \frac{\sigma^2}{n}.}$$

What is really happening

This single line, $\operatorname{Var}(\bar{X}_n) = \sigma^2/n$, is the whole Law of Large Numbers in embryo. The mean of $\bar{X}_n$ stays fixed at μ, but its variance is being deflated by a factor of n. Variance measures spread; if the spread around μ shrinks to zero, the random variable is being crushed onto the single point μ. The standard deviation, the typical distance from μ, is $\sigma/\sqrt{n}$ — which is exactly the $1/\sqrt{n}$ narrowing you saw in Figure 20.1. Averaging does not make individual outcomes less random; it makes the *average* less variable, by piling more and more independent draws into a quantity that gets divided by n.

Walking question

Suppose you want to halve the typical error of your estimate of μ. Since the error scales like $\sigma/\sqrt{n}$, halving it means multiplying $\sqrt{n}$ by 2, i.e. multiplying n by 4. *To get one more digit of accuracy you pay a hundredfold in sample size.* This brutal $\sqrt{n}$ economics is why polling, simulation, and lab measurement are all expensive — and why the LLN, though it guarantees convergence, never promises it will be fast.

The deflation $\operatorname{Var}(\bar{X}_n) = \sigma^2/n$ deserves a picture in its own right, because it explains the exact shape of the funnel in Figure 20.1. The shaded envelope $\mu \pm \sigma/\sqrt{n}$ is a tube around μ whose half-width is the typical error; almost all sample paths live inside a small multiple of it. Figure 20.2 plots that envelope for the fair coin ($\mu = \frac{1}{2}$, $\sigma = \frac{1}{2}$). Two facts jump out. The tube genuinely closes onto the line $y = \mu$ — that is convergence — but it closes *slowly*: to halve the half-width you must quadruple n, the geometry of the $\sqrt{n}$ law made visible.

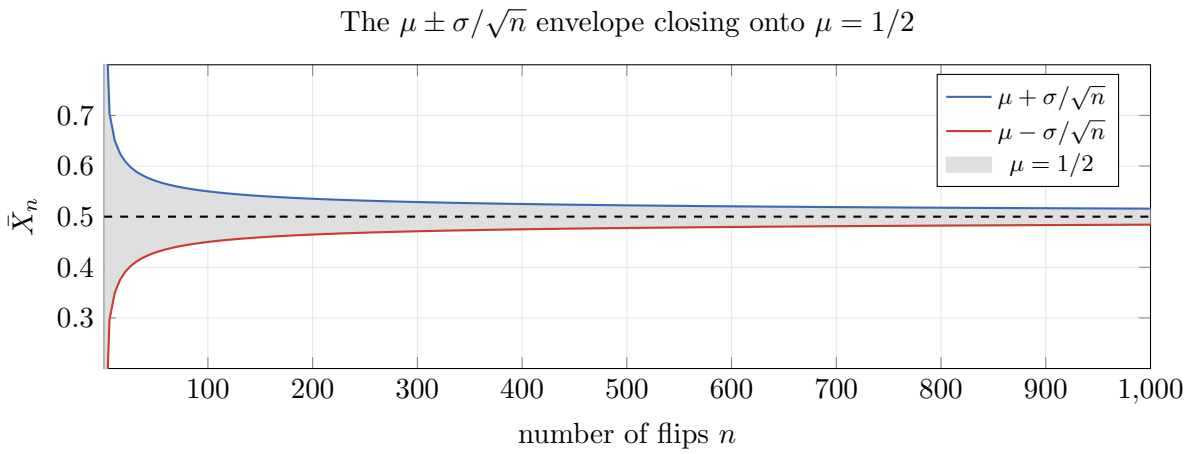


Figure 20.2: The typical-error envelope $\mu \pm \sigma/\sqrt{n}$ for fair-coin flips. The shaded tube is where sample means typically live; it pinches onto μ as $n \to \infty$ but only at rate $1/\sqrt{n}$. This is the same funnel seen in Figure 20.1, now drawn as a deterministic band rather than as random paths.

20.3 The Weak Law of Large Numbers

We now have everything we need for the first theorem. The Weak Law says that for large n, the sample mean is *probably* close to μ. "Probably close" is made precise by fixing a tolerance $\varepsilon > 0$ and asking for the probability that $\bar{X}_n$ misses μ by more than ε to vanish.

Theorem 20.1 (Weak Law of Large Numbers)**.** *Let $X_1, X_2, \ldots$ be i.i.d. with mean μ and finite variance σ^2. Then for every $\varepsilon > 0$,*

$$\mathbb{P}\big(\,|\bar{X}_n - \mu| \geq \varepsilon\,\big) \;\longrightarrow\; 0 \qquad \text{as } n \to \infty.$$

Equivalently, $\bar{X}_n \xrightarrow{p} \mu$: the sample mean converges to μ in probability.

The proof is one of the most satisfying short arguments in all of probability. It rests on a single tool, Chebyshev's inequality, which we met in the chapter on inequalities and concentration. Recall what it says: for any random variable Y with finite variance and any $t > 0$,

$$\mathbb{P}\big(|Y - \mathbb{E}[Y]| \geq t\big) \;\leq\; \frac{\operatorname{Var}(Y)}{t^2}.$$

Chebyshev is a blunt instrument — it gives only a crude bound — but bluntness is exactly what we need here, because we are about to send the right-hand side to zero anyway.

Why it works

Proof of the Weak Law. Apply Chebyshev's inequality to $Y = \bar{X}_n$, whose mean is μ and whose variance we just computed to be σ^2/n. With tolerance $t = \varepsilon$:

$$\mathbb{P}\big(|\bar{X}_n - \mu| \geq \varepsilon\big) \;\leq\; \frac{\operatorname{Var}(\bar{X}_n)}{\varepsilon^2} \;=\; \frac{\sigma^2}{n\,\varepsilon^2}.$$

Now hold ε fixed and let $n \to \infty$. The bound $\sigma^2/(n\varepsilon^2)$ is a constant over n, so it goes to 0.

Since a probability cannot be negative, the probability on the left is squeezed to 0 as well. That is exactly the claim. □

Look at how little we used. We needed linearity of expectation (always true), additivity of variance under independence (this is where "independent" entered), the scaling $\mathrm{Var}(cY) = c^2 \mathrm{Var}(Y)$, and one inequality. The entire theorem is three lines once the variance is in hand. That economy is not an accident: the Weak Law is, at bottom, just the statement that a quantity whose variance tends to zero must concentrate on its mean.

Remark 20.1. The finite-variance hypothesis is convenient but not strictly necessary. Khinchin proved the Weak Law assuming only that the mean μ exists (is finite); no variance assumption at all. That proof uses characteristic functions or a truncation argument and is more delicate. We keep the variance version because its proof *shows you the mechanism* — variance deflation — rather than hiding it inside an analytic black box.

20.3.1 A worked example: how many flips for ninety-five percent confidence?

Chebyshev does more than prove a limit; it hands you a usable, if conservative, sample-size formula. Suppose you want the sample mean of fair-coin flips ($\mu = \frac{1}{2}$, and $\sigma^2 = \frac{1}{4}$ for a Bernoulli($\frac{1}{2}$)) to be within $\varepsilon = 0.05$ of $\frac{1}{2}$ with probability at least 0.95. How large must n be?

We need $\mathbb{P}(|\bar{X}_n - \frac{1}{2}| \geq 0.05) \leq 0.05$. Chebyshev guarantees this as soon as

$$\frac{\sigma^2}{n\varepsilon^2} \leq 0.05 \quad \Longleftrightarrow \quad n \geq \frac{\sigma^2}{0.05 \cdot \varepsilon^2} = \frac{0.25}{0.05 \cdot (0.05)^2} = \frac{0.25}{0.05 \cdot 0.0025} = 2000.$$

So $n = 2000$ flips suffice. (The true number, found later via the Central Limit Theorem, is about $n = 385$ — Chebyshev overshoots by roughly fivefold because it is a worst-case bound. But it is reliable: it never fails, it just asks for more data than necessary.)

Common trap

A student computes $n \geq 2000$ and concludes "after 2000 flips, $\bar{X}_n$ *is* within 0.05 of 1/2." Not so. The guarantee is probabilistic: at least a 95% chance of being within 0.05. There remains up to a 5% chance of a larger miss, and for some unlucky sequence the average really will be off. The Weak Law trades certainty for high probability; it never promises certainty for any fixed n.

20.3.2 A worked example: polling an unknown proportion

The previous example knew $p = \frac{1}{2}$ and asked how many flips pin down a known mean. Bernoulli's deeper question runs the other way: the proportion p is *unknown* — the fraction of voters favouring a candidate, say — and we want to estimate it from a sample. Let $X_i = \mathbf{1}\{\text{voter } i \text{ favours the candidate}\}$, so the X_i are i.i.d. Bernoulli(p) with $\mathbb{E}[X_i] = p$ and $\mathrm{Var}(X_i) = p(1-p)$. The sample mean $\bar{X}_n$ is the poll's headline number.

The wrinkle is that the variance $p(1-p)$ depends on the very p we do not know. We escape with a clean worst-case bound: $p(1-p) \leq \frac{1}{4}$ for all $p \in [0,1]$, with equality at $p = \frac{1}{2}$. (A coin-flip electorate

is the hardest to poll.) So whatever p may be, Chebyshev gives

$$\mathbb{P}\big(|\bar{X}_n - p| \geq \varepsilon\big) \;\leq\; \frac{p(1-p)}{n\varepsilon^2} \;\leq\; \frac{1}{4n\varepsilon^2}.$$

To guarantee, with 95% confidence, a margin of error $\varepsilon = 0.03$ regardless of the true p, demand $1/(4n\varepsilon^2) \leq 0.05$, i.e.

$$n \;\geq\; \frac{1}{4 \cdot 0.05 \cdot (0.03)^2} \;=\; \frac{1}{4 \cdot 0.05 \cdot 0.0009} \;=\; \frac{1}{0.00018} \;\approx\; 5556.$$

So roughly 5 600 respondents suffice for a Chebyshev-certified three-point margin — and note that, just as before, the exact CLT-based requirement is far smaller (near $n = 1\,068$ for the same margin, about a fivefold overshoot), because Chebyshev refuses to use the bell shape. The structural lesson is the one Bernoulli prized: a quantity you cannot compute in advance, the unknown p, becomes knowable to any desired precision purely by collecting enough independent observations.

20.4 The Strong Law of Large Numbers

The Weak Law makes a statement about each fixed, large n: at that time, you are probably close. The Strong Law makes a statement about the entire infinite sequence of averages at once. It says that, with probability one, the sequence $\bar{X}_n$ actually *converges* to μ in the ordinary sense of calculus.

Theorem 20.2 (Strong Law of Large Numbers)**.** *Let $X_1, X_2, \ldots$ be i.i.d. with finite mean $\mu = \mathbb{E}[X_i]$ (no variance assumption needed). Then*

$$\mathbb{P}\Big(\lim_{n\to\infty} \bar{X}_n = \mu\Big) \;=\; 1.$$

Equivalently, $\bar{X}_n \xrightarrow{a.s.} \mu$: the sample mean converges to μ almost surely.

To feel the difference between the two laws, you have to be careful about *what is random*. In the Weak Law we fix n, look at the single random number $\bar{X}_n$, and ask how it is distributed. In the Strong Law we fix an entire infinite coin-flipping destiny $\omega = (X_1, X_2, X_3, \ldots)$ — one complete future — and ask whether the sequence of averages produced by *that* destiny converges to μ as a numerical limit. The Strong Law says: the set of destinies for which the limit fails to be μ has probability zero. Almost every possible future you could live produces a convergent average.

Focus

Weak vs. Strong, in one breath.

- **Weak (in probability):** for each large n, $\bar{X}_n$ is *probably* near μ. Says nothing about whether a single trajectory settles down; in principle $\bar{X}_n$ could wander out past ε infinitely often, just increasingly rarely.
- **Strong (almost surely):** a single trajectory *does* settle down. With probability one, $\bar{X}_n$ eventually enters every band $(\mu - \varepsilon, \mu + \varepsilon)$ and stays inside forever after.

The Strong Law implies the Weak Law (almost-sure convergence implies convergence in probability), but not conversely. "Strong" is a promise about whole trajectories; "weak" is a promise about snapshots.

What is really happening

Here is the operational difference in words you can picture. Draw the tolerance band $(\mu - \varepsilon, \mu + \varepsilon)$. The Weak Law says: at the n-th frame of the movie, the dot $\bar{X}_n$ is probably inside the band. It permits the dot to leave the band later — as long as such departures get rarer. The Strong Law says: for almost every movie, the dot leaves the band only finitely many times in its entire infinite life, and after some finite frame it is trapped inside forever. The Strong Law forbids the "infinitely many rare escapes" that the Weak Law tolerates.

20.4.1 Why the strong law is harder

Proving the Strong Law in full generality is genuinely harder than the Weak Law, and we will not reproduce the complete argument — the cleanest modern proofs use either a clever fourth-moment truncation (Etemadi's proof needs only pairwise independence) or the machinery of martingales, both of which belong to later chapters. But we can show you the *idea*, and even prove a clean special case, so the result does not feel like a miracle handed down from above.

The key analytic tool is the first Borel–Cantelli lemma: if $\sum_n \mathbb{P}(A_n) < \infty$, then with probability one only finitely many of the events A_n occur. The Weak Law gives us $\mathbb{P}(|\bar{X}_n - \mu| \geq \varepsilon) \leq \sigma^2/(n\varepsilon^2)$ for each n — but $\sum_n 1/n$ *diverges*, so Borel–Cantelli does not apply directly. The bound decays too slowly. To get a summable bound, we need to control higher moments, which makes the tail probability shrink faster than $1/n$.

Why it works

The strong law under a fourth-moment assumption (sketch of the real idea). Assume the stronger condition $\mathbb{E}[(X_i - \mu)^4] = K < \infty$, and write $Y_i = X_i - \mu$ so that $\mathbb{E}[Y_i] = 0$. Consider $\mathbb{E}\big[(\sum_{i=1}^n Y_i)^4\big]$. When you expand the fourth power, every term is a product $Y_a Y_b Y_c Y_d$. By independence and $\mathbb{E}[Y_i] = 0$, any term in which some index appears exactly once has expectation zero (that lone factor pulls out as $\mathbb{E}[Y_i] = 0$). The only survivors are terms where indices pair up: the n terms of the form Y_i^4, and the $3n(n-1)$ terms of the form $Y_i^2 Y_j^2$ with $i \neq j$. Hence

$$\mathbb{E}\Big[\big(\textstyle\sum_i Y_i\big)^4\Big] \;=\; nK + 3n(n-1)\sigma^4 \;\leq\; Cn^2$$

for a constant C. Dividing by n^4,

$$\mathbb{E}\big[(\bar{X}_n - \mu)^4\big] \;=\; \frac{1}{n^4}\,\mathbb{E}\Big[\big(\textstyle\sum_i Y_i\big)^4\Big] \;\leq\; \tfrac{C}{n^2}.$$

Now Markov's inequality applied to the nonnegative variable $(\bar{X}_n - \mu)^4$ gives, for any $\varepsilon > 0$,

$$\mathbb{P}\big(|\bar{X}_n - \mu| \geq \varepsilon\big) \;=\; \mathbb{P}\big((\bar{X}_n - \mu)^4 \geq \varepsilon^4\big) \;\leq\; \frac{\mathbb{E}[(\bar{X}_n - \mu)^4]}{\varepsilon^4} \;\leq\; \frac{C}{\varepsilon^4\, n^2}.$$

This bound decays like $1/n^2$, and crucially $\sum_n 1/n^2 < \infty$. By Borel–Cantelli, with probability one only finitely many of the events $\{|\bar{X}_n - \mu| \geq \varepsilon\}$ occur — which is to say $|\bar{X}_n - \mu| < \varepsilon$ eventually. Letting ε run through $1, \frac{1}{2}, \frac{1}{3}, \ldots$ gives $\bar{X}_n \to \mu$ almost surely. $\square$

The moral of that proof is exactly the difference between the laws. The Weak Law needed a $1/n$ bound (variance over n), which is just barely *not* summable. The Strong Law needed a summable bound, and we bought it by spending a fourth moment to make the tail decay like $1/n^2$. Etemadi's celebrated proof achieves the same summability with far weaker hypotheses, but the philosophy is identical: convert "probably close at each n" into "the misses are summable, hence finite," which is the very meaning of almost-sure convergence.

Connection

The split between the two laws is one instance of a general hierarchy of *modes of convergence* for random variables — in probability, almost surely, in L^p, in distribution. The Strong Law is almost-sure convergence; the Weak Law is convergence in probability; and the $1/\sqrt{n}$ funnel of Figure 20.1 hints at convergence in distribution after rescaling, which is the Central Limit Theorem. The companion calculus book's treatment of pointwise versus uniform convergence is the deterministic shadow of the same distinctions: "almost sure" is the probabilistic cousin of "pointwise," applied to functions on the sample space. The dedicated chapter on modes of convergence draws the full map.

20.5 History: Bernoulli's Golden Theorem

History

The Law of Large Numbers is the oldest deep theorem in probability, and its author knew it was deep. Jacob Bernoulli proved the first version — for coin-flip-like (Bernoulli) trials — and called it his *theorema aureum*, his "golden theorem." It was published in 1713 in *Ars Conjectandi* ("The Art of Conjecturing"), eight years after his death; he had reportedly held the manuscript back, polishing it, because he felt the result was too important to release imperfect.

Bernoulli's achievement was conceptual as much as technical. Before him, people computed probabilities for games of chance where the symmetry was given — six faces, two coin sides. Bernoulli asked the inverse, statistical question: if you do *not* know the underlying probability p of an event, can you discover it by observing the frequency of the event over many trials? His golden theorem answered yes: the observed frequency can be made as likely as you please to be as close as you please to the true p, by taking enough trials. This is precisely the Weak Law for Bernoulli variables, and it is the logical foundation of all of statistics — the license to estimate an unknown probability from data.

A century later Poisson coined the phrase *la loi des grands nombres*, "the law of large numbers," giving the theorem its enduring name. The general i.i.d. Weak Law (mean only) is due to Khinchin (1929), and the Strong Law in full generality to Kolmogorov (1930s), as part of the same measure-theoretic refounding of probability that gave us the modern axioms.

20.6 The Gambler's Fallacy: What Does *Not* Average Out

The Law of Large Numbers is so intuitive that people routinely believe a false corollary of it. This false corollary is dangerous enough — it empties wallets in casinos — that it deserves its own careful

dissection.

Common trap

The gambler's fallacy. A fair coin has come up heads ten times in a row. "Tails is overdue," says the gambler; "the law of large numbers must pull the average back, so the next flip is more likely tails." This is false. The next flip is independent of the past, so $\mathbb{P}(\text{tails}) = \frac{1}{2}$ exactly, regardless of the ten heads. The coin has no memory; it does not know it is "behind" on tails, and there is no force that pushes future flips to compensate for past ones.

How can both of these be true at once: the average converges to $\frac{1}{2}$, *and* the coin never compensates for a streak? The resolution is to see exactly *how* the average converges. It does not converge by future flips correcting past imbalances. It converges by *dilution*: the early imbalance becomes a smaller and smaller fraction of the growing total.

Paradox

Resolving the gambler's fallacy. Suppose after 10 flips you have 10 heads, an excess of 5 heads over the expected 5. The average is 1.0. Now flip a fair coin 990 more times. Those future flips, being fair, contribute about 495 heads — they do *not* undershoot to compensate. So after 1000 flips you expect about $10 + 495 = 505$ heads, an average of 0.505. The excess of 5 heads has not been cancelled; it is still there. But it has been *diluted*: 5 extra heads is enormous out of 10 but trivial out of 1000.

Write it algebraically. If the first k flips produce an excess e (number of heads minus $k/2$), and the remaining $n - k$ flips behave normally with their own excess e', then

$$\bar{X}_n - \tfrac{1}{2} \;=\; \frac{e + e'}{n}.$$

The early excess e is fixed; it never gets paid back. What saves the average is the denominator $n \to \infty$, which drives $e/n \to 0$. *The past is not corrected; it is outvoted.* The streak is permanent in absolute terms and negligible in relative terms. That is the whole secret, and it is the difference between a law about *averages* and a (nonexistent) law about *compensation*.

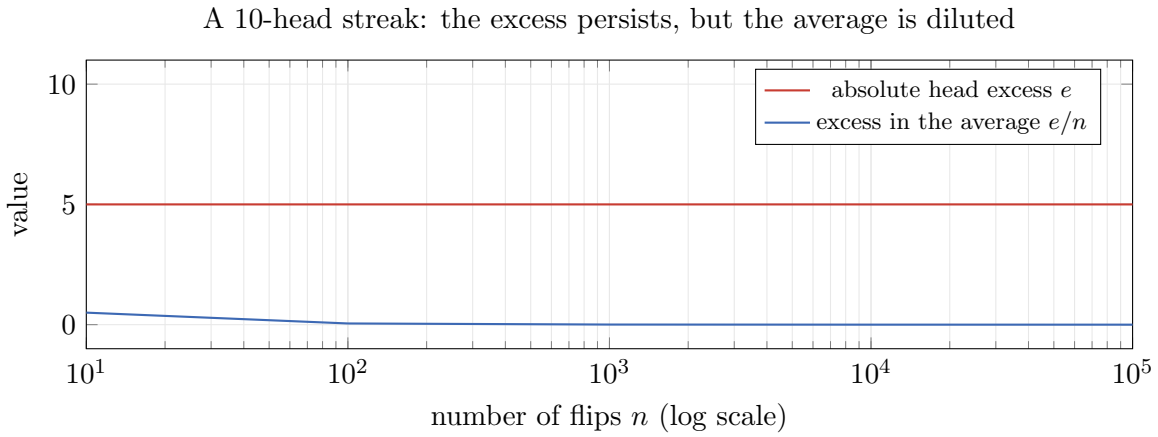


Figure 20.3: After an early run of ten heads (excess $e = 5$), the absolute surplus of heads stays put forever (red). What vanishes is its share of the average, e/n (blue), which collapses toward 0. Convergence is by dilution, not by compensation.

There is a mirror-image error worth naming too. The *hot-hand fallacy* (in its naive form) is the belief that a streak will *continue* — that a coin on a heads run is somehow "hot" and likely to flip heads again. For genuinely independent fair flips this is equally wrong: past flips tell you nothing about the next. (For human performance the story is subtler and the original "hot hand is a myth" studies have themselves been challenged — but for memoryless mechanical randomness, both the gambler's fallacy and the hot-hand fallacy are simply errors about independence.)

20.7 Why the Law Justifies the Frequentist Interpretation

We have been writing $\mathbb{P}$ and $\mathbb{E}$ as if we knew what they meant. But step back: what *is* the probability of heads? One famous answer — the *frequentist* interpretation — says that $\mathbb{P}(A)$ *is* the long-run relative frequency of A in repeated independent trials. The probability of heads is $\frac{1}{2}$ because if you flip forever, the fraction of heads tends to $\frac{1}{2}$.

This is appealing, but it threatens to be circular. We *defined* the model so that each flip has probability $\frac{1}{2}$; now we want to recover "$\frac{1}{2}$" as a frequency. Does the model actually deliver on its own definitional promise? The Law of Large Numbers is the theorem that makes the frequentist interpretation internally consistent.

Why it works

The LLN as the consistency theorem for frequencies. Fix any event A and define indicator variables $X_i = \mathbf{1}\{A \text{ occurs on trial } i\}$. These are i.i.d. Bernoulli with $\mathbb{E}[X_i] = \mathbb{P}(A) =: p$. The sample mean $\bar{X}_n$ is exactly the observed relative frequency of A in the first n trials. The Strong Law then says

$$\frac{\#\{i \le n : A \text{ occurs}\}}{n} \;=\; \bar{X}_n \;\xrightarrow{a.s.}\; p \;=\; \mathbb{P}(A).$$

In words: *with probability one, the long-run frequency of A equals the probability assigned to A by the model.* The model's internally assigned probabilities are exactly the frequencies the

model predicts you will observe. The frequentist reading is not an independent definition that might contradict the axioms; it is a *theorem* the axioms entail. That is why we can have both — an abstract measure-theoretic $\mathbb{P}$ *and* the everyday meaning of probability as frequency — without contradiction.

Why this matters

This is the quiet grandeur of the Law of Large Numbers. Probability begins as an abstraction: a number between 0 and 1 attached to an event, obeying Kolmogorov's axioms. By itself that number means nothing you can touch. The LLN is the theorem that reaches down from the abstraction and connects it to the world: it says the abstract number will, with certainty, reveal itself as a measurable frequency. Without the LLN, "probability 1/2" would be a symbol with no operational content. With it, the symbol becomes a prediction about the world — a prediction you can check by counting.

20.8 Applications: Where the Law Earns Its Living

Application

Monte Carlo integration. Want to compute $\theta = \mathbb{E}[g(X)] = \int g(x)f(x)\,dx$ but the integral is intractable? Draw i.i.d. samples $X_1, \ldots, X_n$ from f and form $\hat{\theta}_n = \frac{1}{n}\sum_i g(X_i)$. By the Strong Law, $\hat{\theta}_n \xrightarrow{a.s.} \theta$: the simulated average converges to the true integral, with probability one. This is the entire theoretical justification for Monte Carlo methods, which dominate computational physics, Bayesian statistics, and quantitative finance. The error shrinks like $1/\sqrt{n}$ regardless of the dimension of x — the reason Monte Carlo beats grid-based integration in high dimensions, where grids suffer the curse of dimensionality. The dedicated Monte Carlo chapter develops this in full.

Application

Insurance and the law of large numbers as a business model. An insurer cannot predict whether *you* will crash your car. But it can predict, very accurately, the average claim across a million policyholders, because that average is a sample mean whose variance is deflated by $n = 10^6$. The standard deviation of the per-policy cost shrinks by a factor of $\sqrt{10^6} = 1000$. This is why insurance works at scale and fails at small scale: a company insuring ten houses is gambling, while a company insuring ten million is running a near-deterministic machine. The LLN is, almost literally, the actuarial industry's founding theorem.

Application

A/B testing and machine learning. When you estimate a click-through rate from n users, or a model's accuracy from n held-out examples, or a gradient from a minibatch of n training points, you are computing a sample mean and trusting the LLN to make it close to the true quantity. The $\sigma/\sqrt{n}$ rate tells you precisely how the estimate's reliability grows with data — and warns you that the last digit of precision is the most expensive. Stochastic gradient

descent is, at heart, a bet that a noisy minibatch average points in roughly the same direction as the true gradient; the LLN is why the bet pays off on average.

20.8.1 A fully worked simulation example

Let us put numbers to the dilution picture and to the convergence rate at once. Suppose $X_i \sim \text{Uniform}(0,1)$, so $\mu = \mathbb{E}[X_i] = \frac{1}{2}$ and $\sigma^2 = \text{Var}(X_i) = \frac{1}{12}$.

(a) What does $\bar{X}_n$ converge to, and in what sense? By the Strong Law, $\bar{X}_n \xrightarrow{a.s.} \frac{1}{2}$. With probability one, the running average of uniform draws settles on one half.

(b) What is the typical error at $n = 1200$? The standard deviation of $\bar{X}_n$ is

$$\text{SD}(\bar{X}_n) = \frac{\sigma}{\sqrt{n}} = \frac{1}{\sqrt{12}\sqrt{1200}} = \frac{1}{\sqrt{14400}} = \frac{1}{120} \approx 0.0083.$$

So a typical run at $n = 1200$ sits within about 0.008 of $\frac{1}{2}$ — consistent with the kind of tight band you would see in a plot.

(c) A Chebyshev guarantee. What is the most we can say about $\mathbb{P}(|\bar{X}_{1200} - \frac{1}{2}| \geq 0.05)$? Chebyshev gives

$$\mathbb{P}\big(|\bar{X}_{1200} - \tfrac{1}{2}| \geq 0.05\big) \;\leq\; \frac{\sigma^2}{n\varepsilon^2} = \frac{1/12}{1200 \cdot (0.05)^2} = \frac{1/12}{3} = \frac{1}{36} \approx 0.028.$$

At most a 2.8% chance of missing by 0.05 or more. The true probability is far smaller (the error of 0.05 is six standard deviations out!), but Chebyshev's promise is already comfortably tight enough for many purposes, and it required no distributional details beyond the variance.

Work it yourself

Reproduce Figure 20.1 yourself. In your language of choice, draw 1000 fair coin flips, compute the cumulative average after each flip, and plot it against n. Repeat for several seeds and overlay the curves. Then try it with Uniform$(0,1)$ draws (it should funnel toward 0.5) and with Exponential(1) draws (toward 1). Finally, do the experiment that breaks the law: draw from a Cauchy distribution (e.g. the ratio of two independent standard normals) and watch the running average refuse to settle. The next section explains why.

20.9 When the Law Fails: The Cauchy Distribution

Every hypothesis in a theorem is there for a reason, and the cheapest way to respect a hypothesis is to watch what happens when it is violated. The crucial hypothesis of the Law of Large Numbers — present even in Khinchin's minimal version — is that the mean μ *exists* (is finite). When it does not, all bets are off.

The standard cautionary tale is the *standard Cauchy distribution*, with density $f(x) = \frac{1}{\pi} \cdot \frac{1}{1+x^2}$ on all of $\mathbb{R}$. It is a perfectly respectable, bell-ish, symmetric density. But its tails are so heavy that $\int_{-\infty}^{\infty} |x| f(x)\, dx = \infty$: the mean does not exist. (The symmetric "center" is 0, but the defining integral diverges, so there is no expectation to converge to.)

Paradox

The average that never settles. For i.i.d. Cauchy samples, a remarkable thing happens: the sample mean $\bar{X}_n$ has *exactly the same* standard Cauchy distribution as a single X_1, for every n. Averaging a thousand Cauchy draws is statistically no better than looking at one of them. The running-average plot does not funnel toward anything; it keeps making wild, full-scale jumps forever, because every so often a single gigantic draw from the heavy tail single-handedly hijacks the average. There is no μ for it to converge to, and no σ^2/n deflation to crush it, because σ^2 is infinite too.

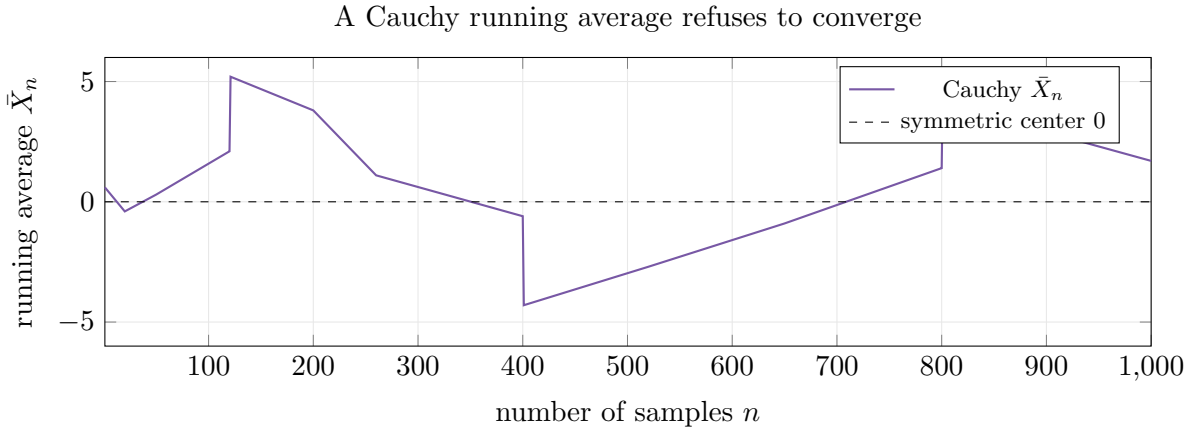


Figure 20.4: In contrast to Figure 20.1, the running average of i.i.d. Cauchy samples never settles: rare enormous draws (the jumps) repeatedly reset it. With no finite mean, the Law of Large Numbers simply does not apply.

What is really happening

The Cauchy example tells you precisely where the LLN's power comes from. The law is not magic about averaging; it is leverage about *tails*. When tails are light enough that the mean exists, no single draw can dominate the sum, so the law of dilution operates and the average concentrates. When tails are heavy enough that the mean diverges, individual draws are so unboundedly large that one of them, every so often, overwhelms all the others combined — and the average is forever at the mercy of the next monster. "Heavy tails break the averages" is one of the most important practical lessons in all of probability, from finance (where return distributions are heavier-tailed than the normal model assumes) to network traffic to insurance against catastrophes.

20.10 Summary

Focus

What to carry away.

- **The engine:** for i.i.d. X_i with mean μ and variance σ^2, the sample mean has $\mathbb{E}[\bar{X}_n] = \mu$ (unbiased) and $\mathrm{Var}(\bar{X}_n) = \sigma^2/n$ (deflating). Typical error $\sim \sigma/\sqrt{n}$.
- **Weak Law:** $\bar{X}_n \xrightarrow{p} \mu$. Proof: Chebyshev gives $\mathbb{P}(|\bar{X}_n - \mu| \geq \varepsilon) \leq \sigma^2/(n\varepsilon^2) \to 0$. A statement about each large n.
- **Strong Law:** $\bar{X}_n \xrightarrow{a.s.} \mu$ (needs only a finite mean). A statement about whole trajectories: almost every run converges in the calculus sense. Strong $\Rightarrow$ weak, not conversely.
- **Gambler's fallacy:** convergence is by *dilution* of past imbalance, never by *compensation*. Future independent trials do not pay back past streaks.
- **Meaning:** the LLN is the theorem that makes the frequentist interpretation consistent — long-run frequency equals probability.
- **The catch:** a finite mean is required. Heavy tails (Cauchy) break the law entirely. The rate is only $1/\sqrt{n}$, so precision is expensive.

The Law of Large Numbers tells you *that* the average converges and *what* it converges to. It does not tell you how the leftover randomness is shaped around μ — the precise nature of the $\sigma/\sqrt{n}$ funnel. That question, the shape of the fluctuations, is the subject of the single most celebrated theorem in probability, and it is where we turn next: the Central Limit Theorem.

20.11 Exercises

Exercise 20.1 (Sample-size by Chebyshev)**.** You measure a quantity whose single-measurement variance is $\sigma^2 = 9$. Using Chebyshev's inequality, how many independent measurements n do you need so that the sample mean is within $\varepsilon = 0.5$ of the true mean with probability at least 0.99? (Find the smallest integer n.)

Exercise 20.2 (Mean and variance of the sample mean)**.** Let $X_1, \ldots, X_n$ be i.i.d. with $\mathbb{E}[X_i] = \mu$ and $\mathrm{Var}(X_i) = \sigma^2$. Show carefully, citing where independence is used and where it is not, that $\mathbb{E}[\bar{X}_n] = \mu$ and $\mathrm{Var}(\bar{X}_n) = \sigma^2/n$. Then explain in one sentence why the first identity does not need independence but the second does.

Exercise 20.3 (Dilution, quantitatively)**.** A fair coin is flipped. The first 20 flips happen to all come up heads. Assuming every subsequent flip is fair, what is the expected value of the running average $\bar{X}_n$ after a total of (a) 40 flips, (b) 200 flips, (c) 2000 flips? Comment on how the early surplus of heads is being handled — corrected or diluted?

Exercise 20.4 (Weak does not imply trajectory convergence — a constructed counterexample)**.** Consider a sequence Y_n where $Y_n = 1$ with probability $1/n$ and $Y_n = 0$ otherwise, all independent. (a) Show $Y_n \xrightarrow{p} 0$. (b) Use the second Borel–Cantelli lemma (events independent, $\sum \mathbb{P} = \infty$) to argue that $Y_n = 1$ infinitely often with probability one, so Y_n does *not* converge to 0 almost surely.

(This is the cleanest illustration that convergence in probability is strictly weaker than almost-sure convergence.)

Exercise 20.5 (Frequencies are probabilities)**.** Let A be any event with $\mathbb{P}(A) = p$, and let $\hat{p}_n$ be the fraction of the first n independent trials on which A occurs. (a) Identify the i.i.d. random variables whose sample mean is $\hat{p}_n$, and state their mean and variance. (b) State precisely what the Strong Law says about $\hat{p}_n$. (c) In two or three sentences, explain why this makes the frequentist interpretation of probability a theorem rather than a competing definition.

Exercise 20.6 (The rate is unforgiving)**.** Using the typical-error scaling $\mathrm{SD}(\bar{X}_n) = \sigma/\sqrt{n}$: (a) by what factor must n grow to cut the typical error to one third? (b) If 1,000 samples give a typical error of 0.02, roughly how many samples are needed for a typical error of 0.001? Comment on the practical cost of high precision.

Exercise 20.7 (Why Chebyshev overshoots)**.** For fair-coin flips you found that Chebyshev requires $n = 2000$ for 95% confidence of being within 0.05 of 1/2, while the true requirement is about 385. (a) Where, conceptually, does Chebyshev throw away information? (b) Without doing the full CLT calculation, explain why a bound that uses only the variance must be more conservative than one that uses the actual (here, approximately normal) shape of $\bar{X}_n$.

Exercise 20.8 (Strong law under a fourth moment — fill in the steps)**.** Reread the fourth-moment proof sketch in Section 20.4. (a) Explain why, in the expansion of $\mathbb{E}[(\sum_i Y_i)^4]$ with $\mathbb{E}[Y_i] = 0$ and independence, any term containing a factor Y_i to the first power vanishes. (b) Count the surviving terms to confirm the coefficients n and $3n(n-1)$. (c) Explain the single role that summability of $\sum 1/n^2$ plays via Borel–Cantelli, and contrast it with the non-summability of $\sum 1/n$ that blocks the same argument for the variance bound.

Exercise 20.9 (When the law fails)**.** The standard Cauchy density is $f(x) = \frac{1}{\pi(1+x^2)}$. (a) Show that $\int_{-\infty}^{\infty} |x| f(x)\,\mathrm{d}x = \infty$, so the mean does not exist. (b) Explain, in terms of tail behavior, why no amount of averaging crushes the variability of Cauchy samples. (c) Contrast with the Exponential(1) distribution, for which the Strong Law does apply: state what $\bar{X}_n$ converges to and why the difference is entirely about the tails.

Exercise 20.10 (Conceptual: two laws, one picture)**.** In your own words, and using the tolerance-band image, explain the difference between the Weak and Strong Laws. Specifically: describe a behavior of the sequence $\bar{X}_n$ that the Weak Law would permit but the Strong Law forbids. Why is forbidding that behavior the stronger statement?

Exercise 20.11 (Polling with an unknown proportion)**.** A pollster wants to estimate the unknown fraction p of voters favouring a measure, to within $\varepsilon = 0.02$ with probability at least 0.95, no matter what p turns out to be. (a) Explain why the worst case is $p = \frac{1}{2}$ and why $p(1-p) \le \frac{1}{4}$ always holds. (b) Use the Chebyshev worst-case bound $1/(4n\varepsilon^2)$ to find the smallest n that guarantees the requirement. (c) The CLT-based sample size for the same specification is about 2,401. By what factor does Chebyshev overshoot, and why is overshooting the price of making *no* assumption about the shape of $\bar{X}_n$?

Exercise 20.12 (Reading the envelope)**.** Figure 20.2 draws the band $\mu \pm \sigma/\sqrt{n}$ for a fair coin. (a) At what n does the half-width $\sigma/\sqrt{n}$ first drop below 0.02? (b) The band is not a probability guarantee on its own; relate its half-width to the Chebyshev bound by asking: within how many half-widths $k\,\sigma/\sqrt{n}$ must $\bar{X}_n$ lie for Chebyshev to promise probability at least 0.95? (c) Explain why the band "pinches but never pinches fast," connecting your answer to the quadruple-the-data-for-half-the-error rule.

Chapter 21

The Central Limit Theorem

There is a moment in everyone's probabilistic education when a strange and beautiful fact arrives. You take a random quantity – it does not matter much which one. It could be the roll of a die, the outcome of a coin flip, the waiting time at a bus stop, the glitch in a measurement. You add up many independent copies of it. And no matter how lopsided, skewed, lumpy, or jagged the original quantity was, the *shape* of the sum settles down to the same smooth, symmetric bell curve. Every time. The bell appears whether you started with coins or dice or exponential waiting times or the strangest home-made distribution you can invent.

This is the Central Limit Theorem (CLT), and it is the closest thing probability has to a law of nature. It is why the normal distribution is everywhere. It is why your calculator's statistics functions, your A/B test, your political poll, your physics experiment, and your machine-learning confidence band all secretly run on the same engine. We will state it carefully, watch it happen in simulation, understand *why* it is true (with a near-rigorous sketch using moment generating functions), learn the practical tricks that make it usable (standardization, the continuity correction), and find out how fast it kicks in (Berry–Esseen). By the end you should not just believe the CLT – you should feel that it could hardly have been otherwise.

21.1 The phenomenon, first

Start not with a theorem but with a picture in your head. Roll a single fair six-sided die. The outcome is uniform on $\{1, 2, 3, 4, 5, 6\}$ – a flat, boxy distribution, six bars of equal height. Nothing bell-shaped about it.

Now roll *two* dice and look at the sum. The sum runs from 2 to 12, and its distribution is no longer flat: it is a triangle, peaked at 7, because there are more ways to make a 7 ($1{+}6, 2{+}5, \dots$) than to make a 2 ($1{+}1$ only). Already the act of summing has pulled probability toward the middle.

Roll three dice, four dice, ten dice, and the triangle rounds into a hump; the hump gets smoother and more symmetric; and if you rescale it so it does not just slide off to the right and spread out forever, it converges to a single fixed curve – the standard normal. The die was just the starting point. The summing did the work.

Intuition

The CLT is a statement about *averaging out idiosyncrasy*. Any one random draw carries the full personality of its distribution – its skew, its discreteness, its fat tail. But when you add many independent draws, each one's personality is one voice in a large crowd. The crowd's aggregate behavior is governed not by any individual quirk but by just two summary numbers: the mean (where the crowd centers) and the variance (how spread out it is). Everything else washes out. The normal distribution is precisely the distribution that remembers *only* a mean and a variance.

Figure 21.1 shows the dice story numerically. Watch the boxy uniform become a triangle become a bell.

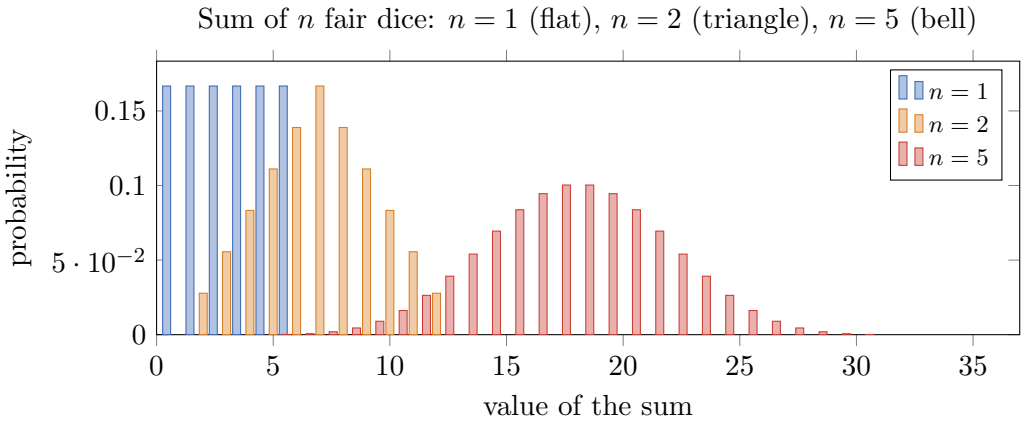


Figure 21.1: The distribution of the sum of n fair dice. With $n = 1$ it is uniform (flat). With $n = 2$ it is a perfect triangle. With $n = 5$ it is already strikingly bell-shaped, even though the underlying die has nothing bell-shaped about it.

History

The bell curve was discovered repeatedly by people chasing exactly this phenomenon. Abraham de Moivre, around 1733, was trying to approximate binomial probabilities – sums of coin flips – for large numbers of trials, because the factorials were impossible to compute by hand. He found that the binomial probabilities, suitably scaled, hugged a smooth curve involving $e^{-x^2/2}$. This was the very first central limit theorem, the *de Moivre–Laplace theorem*, though nobody called it that yet. Laplace generalized it dramatically around 1810, and over the nineteenth century Chebyshev, Markov, and Lyapunov turned it into the rigorous statement we use today. The name "central" is due to George Pólya in 1920; he meant central in the sense of *centrally important to probability theory*, not central as in middle. He was right.

21.2 Standardization: stopping the curve from running away

Before we can state the theorem we have to deal with a bookkeeping problem. If $X_1, X_2, \dots$ are independent and identically distributed (i.i.d.) with mean μ and variance σ^2, then the raw sum

$$S_n = X_1 + X_2 + \cdots + X_n$$

has mean $n\mu$ and variance $n\sigma^2$. Both of those grow without bound. The center of S_n marches off to infinity at rate n, and its spread widens at rate $\sqrt{n}$. So S_n itself has no limiting distribution at all – it just drifts away and fattens. We have to chase it with a coordinate change.

There are two natural fixes, and they are the same fix in disguise.

Fix the center: the sample mean. Divide by n:

$$\bar{X}_n = \frac{S_n}{n}, \qquad \mathbb{E}[\bar{X}_n] = \mu, \qquad \mathrm{Var}(\bar{X}_n) = \frac{\sigma^2}{n}.$$

Now the center is pinned at μ, but the variance shrinks to zero. By the Law of Large Numbers $\bar{X}_n \to \mu$. The mean stops running away, but it collapses to a point – it loses all its shape. We have over-corrected.

Fix both: standardize. Subtract the mean, divide by the standard deviation. Define

$$Z_n = \frac{S_n - n\mu}{\sigma\sqrt{n}} = \frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}}.$$

By construction $\mathbb{E}[Z_n] = 0$ and $\mathrm{Var}(Z_n) = 1$ for every n. We have removed the drift and frozen the spread. Now Z_n is neither running away nor collapsing; it is sitting still, mean 0 and variance 1, ready to converge to something with a fixed shape. That something is the standard normal.

Focus

The magic factor is $\sqrt{n}$ in the denominator. The sum's fluctuations around its mean are of size $\sigma\sqrt{n}$ – not n, not $\sqrt[3]{n}$, but exactly $\sqrt{n}$. This is the single most important number in all of statistics. It says: to halve your uncertainty about an average, you must *quadruple* your sample size. Polls, experiments, and Monte Carlo simulations all pay this $\sqrt{n}$ tax.

Why it works

Why $\sqrt{n}$ and not something else? Variance adds for independent quantities: $\mathrm{Var}(S_n) = \mathrm{Var}(X_1) + \cdots + \mathrm{Var}(X_n) = n\sigma^2$. Standard deviation is the square root of variance, so $\mathrm{SD}(S_n) = \sigma\sqrt{n}$. The independence is doing the work. If the X_i were perfectly correlated (all equal), the sum would just be nX_1, with standard deviation $n\sigma$ – linear, not square-root. The $\sqrt{n}$ savings is the dividend that independence pays you. Half of statistics is the harvest of that dividend; the other half is worrying about whether independence really holds.

21.3 The statement

Theorem 21.1 (Central Limit Theorem, i.i.d. version)**.** *Let $X_1, X_2, \ldots$ be i.i.d. random variables with finite mean $\mu = \mathbb{E}[X_1]$ and finite, nonzero variance $\sigma^2 = \mathrm{Var}(X_1)$. Let $S_n = \sum_{i=1}^n X_i$. Then the standardized sum converges in distribution to a standard normal:*

$$Z_n = \frac{S_n - n\mu}{\sigma\sqrt{n}} \xrightarrow{d} \mathcal{N}(0,1).$$

Equivalently, for every real number z,

$$\mathbb{P}\left(\frac{S_n - n\mu}{\sigma\sqrt{n}} \le z\right) \longrightarrow \Phi(z) \quad \text{as } n \to \infty,$$

where $\Phi(z) = \int_{-\infty}^{z} \frac{1}{\sqrt{2\pi}} e^{-t^2/2}\,\mathrm{d}t$ *is the standard normal cumulative distribution function.*

Read that carefully, because the precision is where the meaning lives.

- **The hypotheses are minimal.** We need i.i.d., a finite mean, and a finite nonzero variance. We need *nothing* about the shape of the X_i. They can be discrete or continuous, symmetric or skewed, bounded or unbounded. As long as the variance is finite, the bell appears.
- **The conclusion is universal.** The limit is *always* the same standard normal $\mathcal{N}(0,1)$, no matter what the X_i were. The original distribution's identity survives only through the two numbers μ and σ^2 used to standardize.
- **Convergence is in distribution.** This means the CDFs converge, $\mathbb{P}(Z_n \le z) \to \Phi(z)$, pointwise at every z (here, every z, because Φ is continuous). It does *not* say Z_n converges to any particular random number. It is a statement about probabilities and shapes, not about realized values.

Common trap

The finite-variance hypothesis is not a technicality you can wave away. If the X_i have infinite variance – like a Cauchy distribution, whose tails are so heavy that $\mathbb{E}[X^2] = \infty$ and even $\mathbb{E}[X]$ fails to exist – the CLT is *false*. The average of n independent Cauchy variables is not approximately normal; it is exactly Cauchy again, no matter how large n is. Averaging does not help at all. Heavy tails break the theorem because a single freak observation can dominate the whole sum, so the "crowd of equal voices" picture collapses. There is a beautiful generalization (the *stable laws*, with α-stable limits) for these cases, but the friendly normal limit requires finite variance.

In the language of the sample mean, dividing top and bottom by n gives the form you will actually use in statistics:

$$\frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} \xrightarrow{d} \mathcal{N}(0,1), \qquad \text{informally} \qquad \bar{X}_n \approx \mathcal{N}\left(\mu, \frac{\sigma^2}{n}\right).$$

The sample mean of a large sample is approximately normal, centered at the true mean, with standard deviation $\sigma/\sqrt{n}$ – the famous *standard error*.

21.4 Watching it happen

The fastest way to believe the CLT is to simulate it. Take a deliberately ugly starting distribution and watch its standardized sums smooth into a bell. We will use a very skewed one: the exponential distribution, which has a hard wall at 0 and a long right tail. It looks nothing like a bell.

Figure 21.2 overlays the density of the standardized sum $Z_n = (S_n - n\mu)/(\sigma\sqrt{n})$ for an exponential starting variable, at $n = 1$, $n = 3$, and $n = 30$, against the target standard-normal curve. At $n = 1$ it is the raw exponential shape (skewed, capped on the left). By $n = 3$ the skew is visibly draining away. By $n = 30$ it is nearly indistinguishable from the bell.

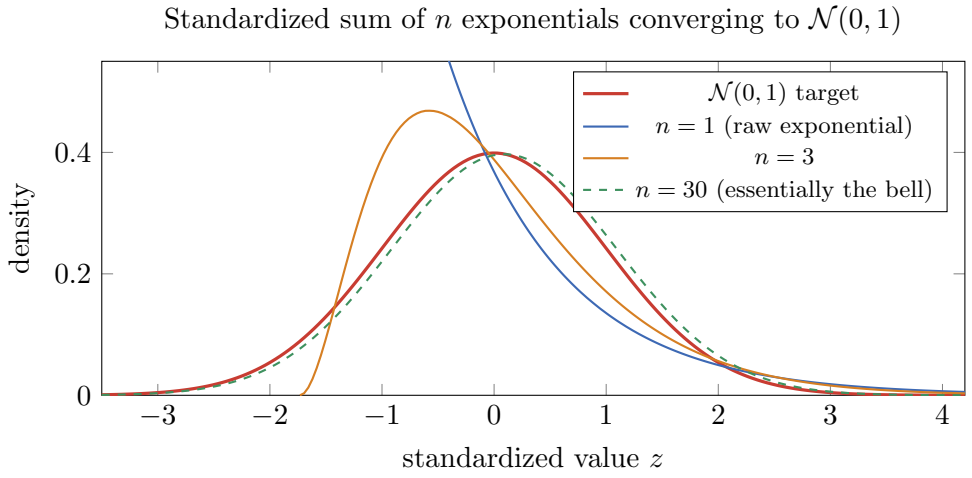


Figure 21.2: The exponential distribution is strongly right-skewed, yet its standardized sums march steadily toward the standard normal. At $n = 1$ the curve is the raw skewed exponential; by $n = 3$ most of the skew is gone; by $n = 30$ it is visually the bell. The left wall of the exponential is what takes longest to erode.

What is really happening

Look at what is physically happening as n grows in Figure 21.2. The skewed exponential has a hard floor: a single draw can never be more than 1 standard deviation below its mean (since the minimum value is 0 and the mean is 1). When you standardize a single draw, nothing can land left of $z = -1$. But the *sum* of many draws can dip below its mean by many standard deviations, because the floor on each individual term gets averaged into irrelevance. So the left tail, which was amputated at $n = 1$, slowly grows back as n increases until it matches the normal's symmetric left tail. The CLT is the story of a distribution's constraints dissolving under aggregation. The slowest part to heal is always the skewness; that is exactly what the Berry–Esseen bound in Section 21.8 will quantify.

Work it yourself

Try this yourself with a few lines of code (pseudocode; any language works):

1. Pick a wild distribution: maybe `X = (a uniform)^4`, or a 90/10 mixture of a spike at 0 and a uniform on $[0, 10]$. Compute its true μ and σ (estimate them from a million draws if you must).
2. For each of $n \in \{1, 2, 5, 30, 100\}$: simulate S_n a hundred thousand times, standardize each replicate to $z = (S_n - n\mu)/(\sigma\sqrt{n})$, and histogram the z values.
3. Overlay the standard normal density. Watch the histogram crawl toward the bell as n grows.

Then deliberately break it: use a Cauchy or a Pareto with index ≤ 2 (infinite variance) and watch the histogram *refuse* to converge. Seeing the failure case is as instructive as seeing the success.

21.5 Why it is true: the moment generating function sketch

The simulations make the CLT believable. Seeing *why* it must be true takes a short argument, and the cleanest one uses the moment generating function (MGF). It is not fully rigorous – the airtight proof uses characteristic functions, which are MGFs with an imaginary argument and which always exist – but the MGF version shows you the whole mechanism with ordinary calculus, and the upgrade to characteristic functions is purely technical.

Recall the MGF of a random variable Y is $M_Y(t) = \mathbb{E}[e^{tY}]$, when it exists in a neighborhood of $t = 0$. Two facts make MGFs the perfect tool here.

- **MGFs turn sums into products.** If $Y_1, \dots, Y_n$ are independent, then $M_{Y_1+\cdots+Y_n}(t) = \prod_i M_{Y_i}(t)$, because the expectation of a product of independent things factors. Summing random variables, which is hard, becomes multiplying their MGFs, which is easy.
- **MGFs determine distributions, and convergence of MGFs implies convergence in distribution.** This is the *continuity theorem*: if $M_{Z_n}(t) \to M_Z(t)$ for all t near 0, and M_Z is the MGF of some distribution, then $Z_n \xrightarrow{d} Z$. So if we can show the standardized sum's MGF converges to the normal's MGF, we are done.

The standard normal's MGF is the target we are aiming at:

$$M_{\mathcal{N}(0,1)}(t) = \mathbb{E}[e^{tZ}] = e^{t^2/2}.$$

Our whole job is to show $M_{Z_n}(t) \to e^{t^2/2}$.

Setup. Without loss of generality center and scale the X_i so that $\mu = 0$ and $\sigma = 1$ (just replace X_i by $(X_i - \mu)/\sigma$; this changes nothing about the shape of the limit). Then $S_n = X_1 + \cdots + X_n$ and $Z_n = S_n/\sqrt{n}$. Let $M(t) = \mathbb{E}[e^{tX_1}]$ be the common MGF of the centered, scaled variables. Because $\mathbb{E}[X_1] = 0$ and $\mathbb{E}[X_1^2] = 1$, the Taylor expansion of M around $t = 0$ begins

$$M(t) = \mathbb{E}\Big[1 + tX_1 + \tfrac{t^2}{2}X_1^2 + \cdots\Big] = 1 + t \cdot 0 + \tfrac{t^2}{2} \cdot 1 + o(t^2) = 1 + \tfrac{t^2}{2} + o(t^2).$$

This is the heart of the matter: *the first two derivatives of M at 0 are exactly the mean and the second moment*, and after centering and scaling those are 0 and 1. The personality of X_1 – everything beyond its mean and variance – has been banished into the $o(t^2)$ remainder, and that remainder is about to be crushed.

Multiply, because the terms are independent. Each $X_i/\sqrt{n}$ has MGF $M(t/\sqrt{n})$, so

$$M_{Z_n}(t) = \mathbb{E}\Big[\exp\Big(\tfrac{t}{\sqrt{n}}\textstyle\sum_i X_i\Big)\Big] = \displaystyle\prod_{i=1}^{n} M\Big(\tfrac{t}{\sqrt{n}}\Big) = \Big[M\Big(\tfrac{t}{\sqrt{n}}\Big)\Big]^n.$$

Now substitute the Taylor expansion with the small argument $t/\sqrt{n}$:

$$M\Big(\tfrac{t}{\sqrt{n}}\Big) = 1 + \frac{1}{2}\left(\frac{t}{\sqrt{n}}\right)^2 + o\left(\frac{1}{n}\right) = 1 + \frac{t^2}{2n} + o\left(\frac{1}{n}\right).$$

Raise to the n-th power and take the limit. This is the classic $\big(1 + \frac{c}{n}\big)^n \to e^c$ structure, with $c = t^2/2$:

$$M_{Z_n}(t) = \left[1 + \frac{t^2/2}{n} + o\big(\tfrac{1}{n}\big)\right]^n \longrightarrow e^{t^2/2}.$$

That is exactly the standard normal MGF. By the continuity theorem, $Z_n \xrightarrow{d} \mathcal{N}(0,1)$. $\square$

Why it works

Step back and see what just happened, because it explains the *universality*. The only features of X_1 that survived into the limit were the coefficients of t^0, t^1, and t^2 in the Taylor expansion of M – that is, 1, the mean, and the second moment. We forced the mean to 0 and the variance to 1 by standardizing. So every legal starting distribution produces the *identical* expansion $1 + t^2/(2n) + \cdots$. The higher moments (skewness lives in the t^3 term, kurtosis in the t^4 term) all entered with extra powers of $1/\sqrt{n}$ and got annihilated in the limit. The $\left(1 + \frac{c}{n}\right)^n \to e^c$ machinery does not care where c came from. *That* is why the limit is always the same bell: the limit only ever sees a mean and a variance.

Connection

The limit $\left(1 + \frac{c}{n}\right)^n \to e^c$ is the same one that defines the number e and that drives continuous compound interest in calculus – see the companion *Calculus From the Ground Up* for the careful ε–N version. It is striking that the Central Limit Theorem, the law of measurement error, and the exponential function all rest on the same little limit. The function $e^{t^2/2}$ is the unique MGF fixed by "mean 0, variance 1, and nothing else," which is another way of saying the normal is the unique distribution remembering only two moments. The full proof replaces $M(t) = \mathbb{E}[e^{tX}]$ with the characteristic function $\varphi(t) = \mathbb{E}[e^{itX}]$; since $|e^{itX}| = 1$ always, φ exists for *every* distribution, even ones with infinite MGF, so the same Taylor-then-$(1 + c/n)^n$ argument goes through with no existence worries. The complex analysis is bookkeeping; the idea above is the whole story.

21.6 The de Moivre–Laplace special case: coins

The oldest and most concrete CLT is for coin flips. Let X_i be i.i.d. Bernoulli(p), so $S_n \sim \text{Binomial}(n, p)$ counts successes in n trials. Here $\mu = p$ and $\sigma^2 = p(1-p)$, so the CLT specializes to

$$\frac{S_n - np}{\sqrt{np(1-p)}} \xrightarrow{d} \mathcal{N}(0, 1).$$

This is the de Moivre–Laplace theorem, and it is enormously useful because binomial probabilities are tedious to compute exactly for large n but trivial to approximate with a normal.

Example 21.1 (A poll)**.** You poll $n = 1000$ voters; suppose the true support for a candidate is $p = 0.52$. What is the probability your sample shows a *minority* – that is, $\bar{X}_n \le 0.50$?

Here $\mu = 0.52$, $\sigma = \sqrt{0.52 \cdot 0.48} = \sqrt{0.2496} \approx 0.4996$, and the standard error of the mean is $\sigma/\sqrt{n} = 0.4996/\sqrt{1000} \approx 0.0158$. Standardize the threshold:

$$z = \frac{0.50 - 0.52}{0.0158} \approx -1.266.$$

So $\mathbb{P}(\bar{X}_n \le 0.50) \approx \Phi(-1.266) \approx 0.103$. Even with a true majority, roughly a one-in-ten poll of 1000 people would show the candidate behind. Want to cut that to one in a hundred? You would need $z \approx -2.33$, which means shrinking the standard error by a factor of $2.33/1.266 \approx 1.84$, which by the $\sqrt{n}$ law requires about $1.84^2 \approx 3.4$ times the sample, i.e. around 3400 voters. The square-root tax in action.

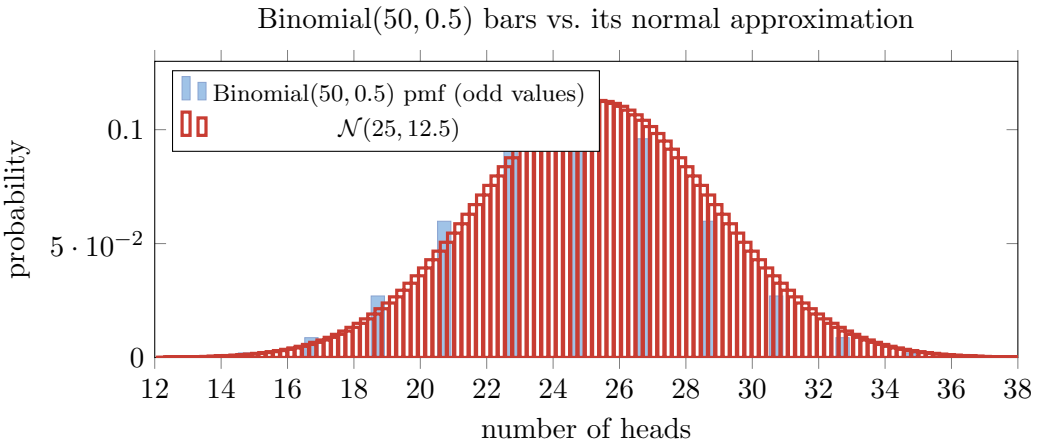


Figure 21.3: The binomial Binomial$(50, 0.5)$ probability mass function (bars, odd values shown for clarity) with its normal approximation $\mathcal{N}(25, 12.5)$ (smooth curve) laid over it. The continuous curve threads through the tops of the discrete bars; to convert between them correctly we need the continuity correction of Section 21.7.

21.7 The continuity correction

There is a subtlety when you approximate a *discrete* distribution (like the binomial, which lives on the integers) by a *continuous* one (the normal). The binomial puts a lump of probability exactly at each integer k; the normal smears probability continuously and assigns probability 0 to any single point. To translate accurately, you must decide how to split the gaps between integers.

The fix is the *continuity correction*: represent the integer k by the interval $[k - \frac{1}{2}, k + \frac{1}{2}]$ when you pass to the normal. Each integer "owns" the half-unit on either side of it. Concretely:

$$\mathbb{P}(S_n = k) \approx \mathbb{P}\big(k - \tfrac{1}{2} \le \mathcal{N} \le k + \tfrac{1}{2}\big)\,,$$

$$\mathbb{P}(S_n \le k) \approx \Phi\left(\frac{k + \frac{1}{2} - n\mu}{\sigma\sqrt{n}}\right), \qquad \mathbb{P}(S_n \ge k) \approx 1 - \Phi\left(\frac{k - \frac{1}{2} - n\mu}{\sigma\sqrt{n}}\right).$$

The half-unit shifts are easy to get backwards, so reason about which integers you mean to *include*. "At most k" includes k, so push the boundary up to $k + \frac{1}{2}$. "At least k" includes k, so push the boundary down to $k - \frac{1}{2}$.

Example 21.2 (Continuity correction pays off)**.** Flip a fair coin $n = 100$ times. What is $\mathbb{P}(S_n \le 45)$? Here $\mu = 0.5$, $\sigma = 0.5$, so $n\mu = 50$ and $\sigma\sqrt{n} = 0.5 \cdot 10 = 5$.

Without the correction: $\Phi\big(\frac{45-50}{5}\big) = \Phi(-1.0) \approx 0.1587$.

With the correction: $\Phi\big(\frac{45.5-50}{5}\big) = \Phi(-0.9) \approx 0.1841$.

The exact binomial answer is 0.1841 to four places. The corrected approximation nails it; the uncorrected one is off by nearly three percentage points. The smaller your n or the closer you are to the tails, the more the half-unit matters.

Common trap

Two common slips. First, forgetting the correction entirely when the problem is plainly discrete – this is the single most frequent CLT error on exams. Second, applying it to a genuinely continuous quantity, where it does nothing but add noise: if your X_i are continuous (waiting times, measurements), there are no integer lumps and no correction to make. The continuity correction is for discrete-to-continuous translation only.

21.8 How fast? The Berry–Esseen theorem

The CLT says $\mathbb{P}(Z_n \le z) \to \Phi(z)$, but it is silent about *how fast*. For a nearly symmetric, light-tailed starting distribution, $n = 20$ may already give an excellent fit; for a violently skewed one, even $n = 200$ may leave a visible error in the tails. We would like a guarantee. Berry (1941) and Esseen (1942) provided one.

Theorem 21.2 (Berry–Esseen)**.** *Let $X_1, \dots, X_n$ be i.i.d. with mean μ, variance $\sigma^2 > 0$, and finite third absolute moment $\rho = \mathbb{E}\left[\,|X_1 - \mu|^3\,\right] < \infty$. Let F_n be the CDF of the standardized sum Z_n. Then there is a universal constant C such that for all n and all z,*

$$\sup_{z \in \mathbb{R}} \left| F_n(z) - \Phi(z) \right| \;\le\; \frac{C\,\rho}{\sigma^3 \sqrt{n}}.$$

The best known constant is $C < 0.4748$.

Two things to read off this beautiful inequality.

The rate is $1/\sqrt{n}$. The worst-case error between the true CDF and the normal CDF shrinks like $1/\sqrt{n}$ – the same $\sqrt{n}$ that governs everything else in this chapter. Quadruple the sample, halve the approximation error.

The constant scales with skewness. The factor ρ/σ^3 is essentially the absolute skewness of the starting distribution. A symmetric, concentrated distribution has small ρ/σ^3 and converges fast; a lopsided, heavy-shouldered one has large ρ/σ^3 and converges slowly. This is the precise version of the intuition from the exponential simulation: *skewness is what takes longest to wash out.*

Example 21.3 (Berry–Esseen for a coin)**.** For a fair coin, $X_1 \in \{0, 1\}$ with $\mu = \frac{1}{2}$, $\sigma = \frac{1}{2}$, and $\rho = \mathbb{E}\,|X_1 - \frac{1}{2}|^3 = (\frac{1}{2})^3 = \frac{1}{8}$. So $\rho/\sigma^3 = (1/8)/(1/8) = 1$, and the bound reads $\sup_z |F_n - \Phi| \le 0.4748/\sqrt{n}$. At $n = 100$ that is about 0.047 – a worst-case CDF error under five percentage points, and in practice (away from the extreme tails) far better, which is why the continuity-corrected answer in Example 21.2 was essentially exact.

Why this matters

Berry–Esseen turns the CLT from a promise into a contract. The plain CLT is an asymptotic statement – true "in the limit," which by itself never tells you whether $n = 30$ is enough. Berry–Esseen gives a concrete, computable error bar for finite n, depending only on three numbers you can estimate from data (σ, ρ, and n). This is the difference between "the bridge will eventually be safe" and "the bridge holds 40 tons with this margin." Modern statistics and machine learning live on finite-sample guarantees, and Berry–Esseen is the prototype.

21.9 Why the normal, and not some other curve?

Pause on the deepest question the CLT raises: of all possible shapes, why is *this* one – $e^{-x^2/2}$ – the universal attractor? Three complementary answers.

The fixed-point answer. Suppose the limit of standardized sums exists and call its law L. If you take two independent copies of L and add them (rescaling to keep variance 1), you must get L back – because a sum of two big sums is just a bigger sum, and the limit cannot depend on how you grouped the terms. So L is a *fixed point* of the "add two independent copies and rescale" operation. The normal is exactly such a fixed point: if $A, B \sim \mathcal{N}(0,1)$ independently, then $(A+B)/\sqrt{2} \sim \mathcal{N}(0,1)$ again. The bell is the shape that reproduces itself under averaging, and the CLT says it is the stable one that everything flows toward.

The maximum-entropy answer. Among all distributions with a given mean and variance, the normal is the one of maximum entropy – the most spread-out, least committal, "most random" distribution consistent with those two constraints. Summing independent things destroys information about the individual terms while preserving the total mean and variance, so the result drifts toward the most-uncertain distribution compatible with what survives. The normal is where information about everything-but-the- mean-and-variance goes to die.

The moment answer. We saw it in the proof: the only moments that survive standardization and the $n \to \infty$ limit are the mean and variance. Any limit law is therefore pinned down by mean 0 and variance 1 alone, and the normal is the distribution determined by exactly those. There is nothing left for the limit to be.

Paradox

The paradox of the disappearing details. It feels almost paradoxical that the limit forgets so much. Two starting distributions can be wildly different – a coin and an exponential and a uniform-to-the-fourth-power – yet their standardized sums all become *the identical* curve. Where did all that distinguishing information go? It did not vanish from any finite sum: for finite n, the sum of coins and the sum of exponentials really are different distributions, distinguishable by their skewness and higher moments. What the CLT says is that those differences shrink at rate $1/\sqrt{n}$ (skewness) and faster (kurtosis), while the bell-shaped part stays $O(1)$. So in the *ratio*, the personality becomes negligible. The information is not destroyed; it is out-grown. This is also a warning: never conclude that two processes are "the same" because their averages look normal. They agree only to leading order, and the disagreements live in the tails – precisely where rare, important events hide.

21.10 Applications: the normal everywhere

Application

Why measurement error is normal. A single physical measurement is corrupted by a great many tiny, independent perturbations – thermal jitter, air currents, electronic noise, rounding, vibration. Each contributes a small random nudge; the total error is their sum. By the CLT, that sum is approximately normal, with mean 0 if the perturbations are unbiased. This is

why Gauss introduced the normal distribution precisely to model astronomical measurement error (it is still sometimes called the Gaussian), and why "assume the noise is Gaussian" is the default in physics, engineering, and signal processing. The assumption is not laziness; it is the CLT.

Application

Confidence intervals and the standard error. The single most-used consequence of the CLT in all of data analysis. Given a sample of size n with sample mean $\bar{X}_n$ and sample standard deviation s, the CLT (plus a little extra work to replace σ by s) gives the approximate 95% confidence interval

$$\bar{X}_n \pm 1.96 \cdot \frac{s}{\sqrt{n}}.$$

The 1.96 is the z-value cutting off 2.5% in each normal tail; the $s/\sqrt{n}$ is the standard error. Every poll's "margin of error," every A/B test's significance calculation, and every reported "mean $\pm$ standard error" in a scientific paper is this formula, powered by the CLT. We develop it fully in the estimation and hypothesis-testing chapters; here just notice that it requires *nothing* about the shape of the underlying data, only finite variance and a decent sample size.

Application

A/B testing. You show version A of a webpage to n_A users and version B to n_B, and measure conversion rates $\hat{p}_A$ and $\hat{p}_B$. Each rate is an average of Bernoulli outcomes, so by de Moivre–Laplace each is approximately normal. Their difference is then approximately normal too, with standard error $\sqrt{\hat{p}_A(1-\hat{p}_A)/n_A + \hat{p}_B(1-\hat{p}_B)/n_B}$. Dividing the observed difference by this standard error gives a z-statistic you can compare to the normal curve to decide whether B genuinely beats A or just got lucky. The entire online experimentation industry runs on the CLT applied to coin flips.

Application

Monte Carlo simulation. To estimate an integral or an expectation $\theta = \mathbb{E}[g(X)]$ that is too hard to compute analytically, draw n independent samples and average: $\hat{\theta}_n = \frac{1}{n}\sum_i g(X_i)$. The Law of Large Numbers says $\hat{\theta}_n \to \theta$; the CLT tells you the *error*: $\hat{\theta}_n$ is approximately $\mathcal{N}(\theta, \tau^2/n)$, so the Monte Carlo error shrinks like $\tau/\sqrt{n}$. This $1/\sqrt{n}$ rate is independent of the dimension of X – which is exactly why Monte Carlo beats grid-based numerical integration in high dimensions, and why it is the workhorse of computational physics, Bayesian statistics, and quantitative finance. See the Monte Carlo chapter for variance-reduction tricks that shrink τ (you cannot beat the $\sqrt{n}$, so you attack the constant).

Connection

Where the CLT sits among the limit theorems. The Law of Large Numbers (previous chapter) and the Central Limit Theorem are a matched pair, and the difference is the scaling. The LLN looks at $\bar{X}_n - \mu$ on its natural scale and finds it goes to 0: the average converges to the truth. The CLT zooms in by the factor $\sqrt{n}$ and asks what the residual fluctuation $\sqrt{n}\,(\bar{X}_n - \mu)$ looks like under the microscope – and finds a normal. So the LLN is the *first-*

order fact (the average is right) and the CLT is the *second-order* fact (here is the shape of how wrong it is). The modes-of-convergence chapter makes "converges in distribution" precise and relates it to the stronger convergences the LLN uses.

21.11 A worked example, end to end

Example 21.4 (Total weight on an elevator)**.** An elevator is rated for 1600 kg. Suppose adult passenger weights are i.i.d. with mean $\mu = 75$ kg and standard deviation $\sigma = 15$ kg (the distribution of weights is skewed, but we never need its shape). If 20 people board, what is the probability the total weight exceeds the limit?

The total S_{20} has mean $20 \cdot 75 = 1500$ kg and standard deviation $\sigma\sqrt{n} = 15\sqrt{20} \approx 67.08$ kg. We want $\mathbb{P}(S_{20} > 1600)$. Standardize:

$$z = \frac{1600 - 1500}{67.08} \approx 1.49.$$

Weights are continuous, so no continuity correction is needed. Then

$$\mathbb{P}(S_{20} > 1600) \approx 1 - \Phi(1.49) \approx 1 - 0.9319 = 0.0681 \approx 0.068.$$

About a 6.8% chance of overload. Notice the full power of the CLT here: we knew only the mean and standard deviation of an individual weight – not the distribution's shape at all – and yet we produced a usable probability for the total. That is the everyday miracle the theorem performs. (How trustworthy is the 6.8%? With $n = 20$ and a moderately skewed weight distribution, Berry–Esseen warns the answer could be off by a few percentage points in the tail; for a safety-critical decision you would want a larger margin or the real weight distribution.)

Figure 21.4 shows what the calculation looks like geometrically: the total weight is approximately normal, and the overload probability is the area in the right tail beyond the standardized threshold $z \approx 1.49$.

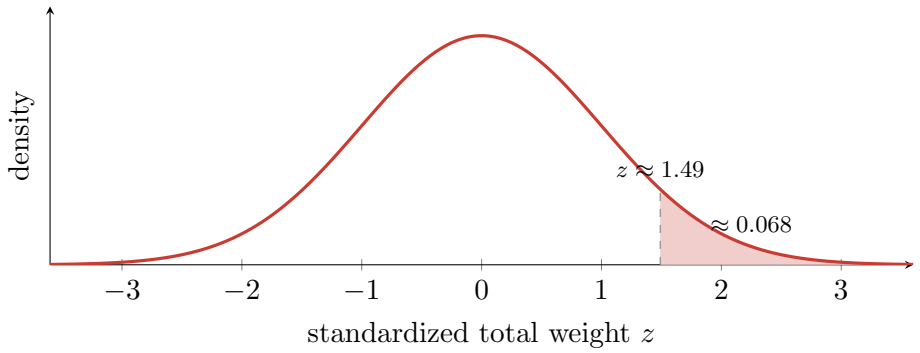


Figure 21.4: The elevator overload probability as a normal tail area. The total weight of 20 passengers is approximately $\mathcal{N}(1500, 67.08^2)$; the 1600 kg limit sits $z \approx 1.49$ standard deviations above the mean, and the shaded right tail has area $1 - \Phi(1.49) \approx 0.068$.

21.12 Summary

Focus

The chapter in one breath. Add many independent finite-variance things, subtract the mean $n\mu$, divide by $\sigma\sqrt{n}$, and you always get the same bell:

$$\frac{S_n - n\mu}{\sigma\sqrt{n}} \xrightarrow{d} \mathcal{N}(0,1).$$

- **Standardization** keeps the curve from drifting ($-n\mu$) or collapsing ($\div\,\sigma\sqrt{n}$). The $\sqrt{n}$ is the master scale of statistics.
- **Why it works:** MGFs turn sums into products; Taylor expansion leaves only the mean and variance; $(1 + \frac{t^2/2}{n})^n \to e^{t^2/2}$, the normal MGF. Higher moments die in the limit – hence universality.
- **Continuity correction:** for discrete sums, replace integer k by $[k \pm \frac{1}{2}]$ before going to the normal.
- **Berry–Esseen:** the CDF error is at most $0.4748\,\rho/(\sigma^3\sqrt{n})$ – rate $1/\sqrt{n}$, scaled by skewness.
- **Caveat:** finite variance is required. Infinite-variance (Cauchy, heavy-tail) distributions have non-normal stable limits.

Exercises

Exercise 21.1 (Standardize a sum)**.** Let $X_1, \ldots, X_{40}$ be i.i.d. with mean $\mu = 3$ and variance $\sigma^2 = 4$. Let $S = \sum_i X_i$. (a) Find $\mathbb{E}[S]$ and $\mathrm{SD}(S)$. (b) Using the CLT, estimate $\mathbb{P}(S > 140)$. (c) Estimate $\mathbb{P}(115 \le S \le 130)$.

Exercise 21.2 (Sample size and the square-root law)**.** A quantity has standard deviation $\sigma = 8$. You want to estimate its mean with a 95% confidence interval of total width at most 1 (i.e. ± 0.5). Using the interval $\bar{X}_n \pm 1.96\,\sigma/\sqrt{n}$, find the smallest n. Then find the n needed for width 0.5. By what factor did n grow, and why?

Exercise 21.3 (Continuity correction)**.** A fair die is rolled 180 times. Let K be the number of sixes. (a) Find $\mathbb{E}[K]$ and $\mathrm{SD}(K)$. (b) Use the normal approximation *with* the continuity correction to estimate $\mathbb{P}(K \ge 36)$. (c) Redo (b) *without* the correction and report the difference.

Exercise 21.4 (A poll)**.** In a population, 40% favor a proposal. You sample $n = 600$ people. (a) What is the approximate distribution of the sample proportion $\hat{p}$? (b) Estimate $\mathbb{P}(\hat{p} \ge 0.43)$. (c) How large would n need to be so that the standard error of $\hat{p}$ is at most 0.01?

Exercise 21.5 (Berry–Esseen by hand)**.** Let X_i be i.i.d. Bernoulli(0.1). (a) Compute μ, σ^2, and $\rho = \mathbb{E}\,|X_1 - \mu|^3$. (b) Compute the Berry–Esseen bound $0.4748\,\rho/(\sigma^3\sqrt{n})$ for $n = 100$ and $n = 10000$. (c) Compare to the bound for Bernoulli(0.5) from Example 21.3. Which converges to normal faster, and does that match the intuition about skewness?

Exercise 21.6 (Where the CLT fails)**.** The average of n independent standard Cauchy random variables is again standard Cauchy, for every n. (a) Explain why this does *not* contradict the CLT. (b) The Cauchy density is $f(x) = \frac{1}{\pi(1+x^2)}$. Show that $\mathbb{E}[X]$ does not exist by examining $\int |x| f(x)\,\mathrm{d}x$. (c) In one sentence, what feature of the Cauchy breaks the hypotheses of Theorem 21.1?

Exercise 21.7 (Sum of uniforms)**.** Let $U_1, \dots, U_{12}$ be i.i.d. Uniform$(0,1)$. (a) Find the mean and variance of $S = \sum_{i=1}^{12} U_i$. (b) Show that $S - 6$ has mean 0 and variance 1. (This is the classic "cheap Gaussian generator": $\sum_{i=1}^{12} U_i - 6$ is approximately $\mathcal{N}(0,1)$.) (c) Name one way this approximation differs from a true standard normal.

Exercise 21.8 (Proof: standardization arithmetic)**.** Let $X_1, \dots, X_n$ be i.i.d. with mean μ and variance σ^2. Using only linearity of expectation and the additivity of variance for independent variables, prove carefully that $Z_n = (S_n - n\mu)/(\sigma\sqrt{n})$ has $\mathbb{E}[Z_n] = 0$ and $\mathrm{Var}(Z_n) = 1$ for every n. State exactly where independence is used.

Exercise 21.9 (Proof: the MGF of a sum)**.** Let $Y_1, \dots, Y_n$ be independent with MGFs $M_{Y_i}(t)$ existing near 0. (a) Prove $M_{Y_1+\cdots+Y_n}(t) = \prod_i M_{Y_i}(t)$, naming where independence enters. (b) Specialize to the i.i.d. case to get $M_{S_n}(t) = M(t)^n$. (c) Using the substitution $t \mapsto t/(\sigma\sqrt{n})$, write down $M_{Z_n}(t)$ in terms of M for the centered case $\mu = 0$.

Exercise 21.10 (Two-sample difference)**.** An A/B test gives version A a conversion rate $\hat{p}_A = 0.118$ over $n_A = 5000$ users and version B $\hat{p}_B = 0.131$ over $n_B = 5000$ users. (a) Estimate the standard error of $\hat{p}_B - \hat{p}_A$. (b) Compute the z-statistic $(\hat{p}_B - \hat{p}_A)/\mathrm{SE}$. (c) Roughly, is the difference convincing at the two-tailed 5% level (i.e. is $|z| > 1.96$)? Explain what the CLT is doing in this calculation.

Chapter 22

Modes of Convergence

In calculus you learned exactly one way for a sequence of numbers to converge: the terms a_n get and stay arbitrarily close to a limit a. There was nothing to argue about. A number is a number; "close" means small distance on the line.

Probability is richer, and that richness is what this chapter is about. The objects that converge here are not numbers but *random variables* – whole mechanisms of chance, each carrying an entire distribution. When we say a sequence $X_1, X_2, X_3, \ldots$ "converges to" some limit X, we have to decide *what exactly* is supposed to get close to *what*. Should the distributions line up? Should the actual realized values line up? Should they line up on every possible outcome, or just with high probability, or just on average? Each answer is a genuinely different notion of convergence, and they form a strict hierarchy: some are demanding, some are gentle, and confusing them is one of the classic ways to prove a false theorem.

By the end you will hold four modes of convergence in your head at once – *almost sure*, *in probability*, *in L^p (in mean)*, and *in distribution* – you will know exactly which implies which, you will have the two power tools (the continuous mapping theorem and Slutsky's theorem) that let you manipulate them, you will know the delta method that turns the central limit theorem into a calculator for nonlinear statistics, and you will finally see the Law of Large Numbers and the Central Limit Theorem for what they really are: two statements about two different modes of convergence.

22.1 Four questions, four answers

Start with a single concrete story, and ask the four questions one at a time.

Intuition

You flip a fair coin over and over. After n flips let $\bar{X}_n$ be the fraction of heads so far. Everyone has the intuition that $\bar{X}_n \to 1/2$. Good. But pin down what that sentence means. There is not one infinite coin-flip experiment sitting in front of you; there is a whole *universe* Ω of possible infinite flip sequences ω. For each ω you get a whole numerical sequence $\bar{X}_1(\omega), \bar{X}_2(\omega), \ldots$. "$\bar{X}_n \to 1/2$" is a claim about infinitely many sequences at once. *Which* sequences, and *how*, is the substance.

Here are the four ways to make "X_n gets close to X" precise, from the most literal to the most relaxed.

1. Almost surely (the strongest, most literal). Look at each outcome ω. You get an ordinary numerical sequence $X_n(\omega)$. Demand that this sequence converges to $X(\omega)$ in the ordinary calculus sense – for *essentially every* ω. "Essentially every" means the set of bad ω where it fails has probability zero. This is pointwise convergence of functions, with a probability-zero pardon.

2. In probability. Forget tracking individual ω's for all time. Instead freeze a tolerance $\varepsilon > 0$ and ask: at stage n, how likely is X_n to be farther than ε from X? Demand that this probability shrinks to 0 as n grows. The bad set is allowed to *move around* from one n to the next; it just has to keep shrinking.

3. In L^p (in mean of order p). Measure closeness by an average size of the error: $\mathbb{E}|X_n - X|^p$. Demand that this number goes to 0. For $p = 2$ this is the familiar mean-squared error of statistics and signal processing.

4. In distribution (the weakest, most generous). Stop asking the X_n to be near X *as numbers* at all. Only ask that their *distributions* settle down: the probability of landing in any reasonable region converges to the right limiting probability. The random variables need not even live on the same probability space.

What is really happening

The deep reason there are four modes and not one: a random variable is two things glued together – a *function* $\omega \mapsto X(\omega)$, and a *distribution* (the pushforward law). Modes 1–3 compare the *functions* (with different rulers: pointwise, in-probability, in-average). Mode 4 throws the functions away and compares only the *distributions*. That is why convergence in distribution is so much weaker, and why it alone does not require the variables to share a sample space. Keep this two-faced picture in mind and the whole hierarchy becomes obvious instead of memorized.

22.2 Almost sure convergence

Definition 22.1 (Almost sure convergence)**.** A sequence of random variables X_n *converges almost surely* to X, written $X_n \xrightarrow{a.s.} X$, if

$$\mathbb{P}\Big(\{\omega : \lim_{n\to\infty} X_n(\omega) = X(\omega)\}\Big) = 1.$$

Equivalently, the set of outcomes on which the numerical sequence $X_n(\omega)$ fails to converge to $X(\omega)$ has probability 0.

The phrase "almost surely" (a.s.) – sometimes "with probability one" or "almost everywhere" in the measure-theory literature – is doing real work. We are *not* promising convergence for every single ω. We are tolerating a negligible exceptional set.

Example 22.1 (A typical bad set is genuinely nonempty)**.** In the coin-flipping story, the all-heads sequence $\omega^\star = HHHH\cdots$ is a perfectly legitimate outcome, and along it $\bar{X}_n(\omega^\star) = 1 \not\to 1/2$. So convergence really does fail somewhere. But $\{\omega^\star\}$ has probability 0 (indeed the set of *all* sequences

whose head-frequency misbehaves has probability 0, which is exactly the Strong Law of Large Numbers). Almost sure convergence shrugs off such measure-zero villains.

Why it works

Why is the exceptional-set-of-measure-zero pardon the "right" definition, rather than cheating? Because in a continuous or infinite world, demanding *literally every* outcome to behave is hopeless and useless. A single point on $[0,1]$ has probability 0; insisting on it would let one irrelevant outcome veto an otherwise perfect theorem. Probability-zero events are the things that, by the axioms, never contribute to any expectation or any probability you can actually compute. Ignoring them changes no answer you can measure. The pardon is not a loophole; it is fidelity to what probability can see.

22.2.1 Reformulating a.s. convergence with ε-tubes

There is a second, more operational way to write almost sure convergence that makes its strength visible and sets up the contrast with convergence in probability. Saying $X_n(\omega) \to X(\omega)$ means: for every $\varepsilon > 0$, eventually $|X_n - X| < \varepsilon$ *and stays there*. Translate "eventually and stays" into events:

$$X_n \xrightarrow{a.s.} X \quad \Longleftrightarrow \quad \text{for every } \varepsilon > 0, \quad \mathbb{P}\left(\bigcap_{N \geq 1} \bigcup_{n \geq N} \{|X_n - X| \geq \varepsilon\} \right) = 0.$$

The event in the middle is "$|X_n - X| \geq \varepsilon$ happens for infinitely many n" ("infinitely often"). Almost sure convergence says: for each tolerance, large deviations happen only finitely often, with probability one. Compare this with the in-probability statement below – the difference is the entire chapter in miniature.

Formal

A clean sufficient condition, by far the most useful in practice, comes from the first Borel–Cantelli lemma: if for every $\varepsilon > 0$

$$\sum_{n=1}^{\infty} \mathbb{P}(|X_n - X| \geq \varepsilon) < \infty,$$

then $X_n \xrightarrow{a.s.} X$. (The probabilities not only go to zero, they go to zero fast enough to sum.) This is the workhorse behind most strong-law-type proofs: bound the tail probabilities, show the bounds are summable, done.

22.3 Convergence in probability

Definition 22.2 (Convergence in probability)**.** X_n *converges in probability* to X, written $X_n \xrightarrow{p} X$, if for every $\varepsilon > 0$,

$$\lim_{n \to \infty} \mathbb{P}(|X_n - X| \geq \varepsilon) = 0.$$

Read it slowly against the a.s. version. There the "ε-failure" event was required to occur only *finitely often along each path*. Here we only require that, *at each fixed time* n, the failure is unlikely – and that the unlikeliness improves with n. We never look at a whole path. The bad set may be a different set of ω's for each n, marching across the sample space like a shrinking spotlight, never settling, yet always small.

Intuition

A picture for the difference. Almost sure: "pick an outcome and follow it forever; it lands and stays at the limit." In probability: "snapshot the whole population at time n; the fraction that is far away is small, and shrinks." One follows individuals through time; the other photographs the crowd at each instant. The crowd can look great in every photo even while a few unlucky individuals keep getting hit – as long as it is *different* individuals each time.

22.3.1 The typewriter sequence: in probability but not almost surely

The cleanest demonstration that these are truly different notions is the famous *typewriter* (or "sliding bump") sequence. We do it in full, because once you see it you never again confuse the two modes.

Let $\Omega = [0,1]$ with the uniform probability (length = probability). We build a sequence of indicator variables by sweeping a window of shrinking width across $[0,1]$, over and over, like a typewriter carriage returning faster each pass.

- Pass 1 (width 1): $X_1 = \mathbf{1}_{[0,1]}$.
- Pass 2 (width 1/2): $X_2 = \mathbf{1}_{[0,1/2]}$, $X_3 = \mathbf{1}_{[1/2,1]}$.
- Pass 3 (width 1/3): $X_4 = \mathbf{1}_{[0,1/3]}$, $X_5 = \mathbf{1}_{[1/3,2/3]}$, $X_6 = \mathbf{1}_{[2/3,1]}$.
- Pass k (width $1/k$): k indicators of consecutive length-$1/k$ blocks.

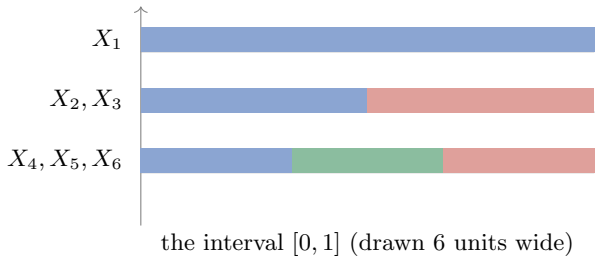


Figure 22.1: The typewriter sequence. Each colored block is where the corresponding X_n equals 1; everywhere else it is 0. The bump slides across and shrinks. At any fixed point ω, the bump keeps coming back forever, so $X_n(\omega)$ never settles – yet the width of the bump (its probability of being 1) shrinks to zero.

In probability it converges to 0. The variable X_n equals 1 only on a block of width $1/k$ (where k is the pass containing index n), so for any $\varepsilon \in (0,1)$,

$$\mathbb{P}(|X_n - 0| \geq \varepsilon) = \mathbb{P}(X_n = 1) = \tfrac{1}{k} \to 0 \quad \text{as } n \to \infty,$$

since the pass number k grows without bound. Hence $X_n \xrightarrow{p} 0$.

Almost surely it converges nowhere. Fix any point $\omega \in [0,1]$. In every pass k, exactly one block contains ω, so in every pass there is an index n with $X_n(\omega) = 1$; and there are also indices with $X_n(\omega) = 0$. Thus the sequence $X_n(\omega)$ contains infinitely many 1's and infinitely many 0's – it oscillates forever and has no limit. The convergence fails for *every* ω, the worst possible failure. So X_n does *not* converge to 0 almost surely (it fails on a set of probability 1, not 0).

What is really happening

This single example proves $\xrightarrow{p}$ does not imply $\xrightarrow{a.s.}$. The mechanism is exactly the "moving spotlight": at time n the bad set (where $X_n = 1$) is small, but it *relocates* each step, so no individual ω is spared in the long run. Almost sure convergence forbids exactly this kind of perpetual relocation – it demands the failures eventually stop happening along each path. The Borel–Cantelli connection: $\sum_n \mathbb{P}(X_n = 1) = \sum_k k \cdot \frac{1}{k} = \sum_k 1 = \infty$, so the summable-tail test is *not* satisfied, consistent with a.s. failure.

22.4 Convergence in L^p (in mean)

Definition 22.3 (Convergence in L^p)**.** Fix $p \geq 1$. We say $X_n \to X$ *in* L^p (or in pth mean), written $X_n \xrightarrow{L^p} X$, if $\mathbb{E}|X_n|^p < \infty$ for all n and

$$\lim_{n\to\infty} \mathbb{E}|X_n - X|^p = 0.$$

The case $p = 1$ is *convergence in mean*; $p = 2$ is *convergence in mean square* (or in quadratic mean), ubiquitous in statistics and signal processing.

This mode measures error with an *integrated* ruler rather than a probabilistic one. It is sensitive to how *large* the error can be, not merely how *often* it is nonnegligible. That sensitivity is exactly why it can fail when convergence in probability holds: a rare error that is enormous can keep the average error large while the probability of any error stays small.

22.4.1 Convergence in L^p implies convergence in probability

Theorem 22.1. *If $X_n \xrightarrow{L^p} X$ for some $p \geq 1$, then $X_n \xrightarrow{p} X$.*

Proof. This is a one-line consequence of Markov's inequality applied to $|X_n - X|^p$. For any $\varepsilon > 0$,

$$\mathbb{P}(|X_n - X| \geq \varepsilon) = \mathbb{P}(|X_n - X|^p \geq \varepsilon^p) \leq \frac{\mathbb{E}|X_n - X|^p}{\varepsilon^p}.$$

The right side tends to 0 by hypothesis, so the left side does too. □

Connection

Notice the engine here is Markov's inequality, the most elementary concentration bound (Chapter on inequalities). A pattern worth absorbing: *control of a moment (an average) buys control of a probability (a tail).* The whole subject of concentration of measure is built on upgrading

this trade. The factor $1/\varepsilon^p$ also explains why larger p gives a stronger statement – a higher moment penalizes large deviations more harshly, so L^p convergence for big p is harder to achieve and implies more.

22.4.2 In probability but not in L^p: the rare giant

The converse fails, and the reason is a single rare but huge excursion.

Example 22.2 (A spike that is unlikely but enormous)**.** On $\Omega = [0,1]$ with uniform probability, define

$$X_n(\omega) = \begin{cases} n & \text{if } \omega \in (0, 1/n], \\ 0 & \text{otherwise.} \end{cases}$$

Then $X_n \xrightarrow{p} 0$: for any $\varepsilon > 0$ and n large, $\mathbb{P}(|X_n| \geq \varepsilon) = \mathbb{P}(X_n = n) = 1/n \to 0$. But the mean does not vanish:

$$\mathbb{E}|X_n - 0| = \mathbb{E}\, X_n = n \cdot \tfrac{1}{n} = 1 \quad \text{for all } n,$$

so X_n does *not* converge to 0 in L^1 (and a fortiori not in L^2, where $\mathbb{E}\, X_n^2 = n^2 \cdot \frac{1}{n} = n \to \infty$). The probability of error vanishes, but the *magnitude* when it occurs grows just fast enough to keep the average error pinned at 1. The area of the spike refuses to shrink.

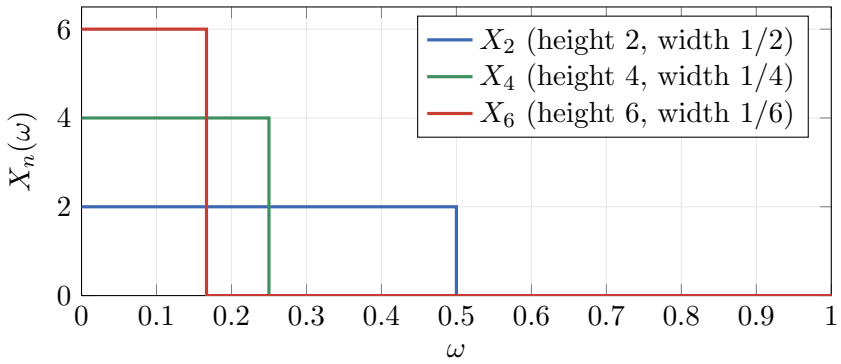


Figure 22.2: The rare giant of Example 22.2. Each spike is taller and narrower than the last, but the rectangle's area (its height n times its width $1/n$) is always exactly 1. So $\mathbb{E}\, X_n = 1$ for every n even as $\mathbb{P}(X_n \neq 0) = 1/n \to 0$: convergence in probability without convergence in L^1.

Common trap

A natural-sounding but **false** belief: "if $X_n \xrightarrow{p} X$ then surely the means converge, $\mathbb{E}\, X_n \to \mathbb{E}\, X$." Example 22.2 kills it: $X_n \xrightarrow{p} 0$ but $\mathbb{E}\, X_n = 1 \not\to 0$. Convergence in probability says nothing about averages unless you add a control on the tails. The standard such control is *uniform integrability*; under it, $\xrightarrow{p}$ does upgrade to L^1 and the means converge. Without some moment control, never pass a limit inside an expectation just because things converge in probability.

22.5 Convergence in distribution

The final mode is the most permissive and, despite that, the most famous, because the Central Limit Theorem is a statement of exactly this type.

Definition 22.4 (Convergence in distribution)**.** Let X_n have cumulative distribution functions F_n and let X have CDF F. We say X_n *converges in distribution* (or in law, or weakly) to X, written $X_n \xrightarrow{d} X$, if

$$\lim_{n\to\infty} F_n(x) = F(x)$$

at every point x where F is continuous.

Two features of this definition repay close reading. First, it speaks only of the *distributions* F_n, F – the random variables themselves may live on entirely different probability spaces, may have nothing to do with one another as functions. Second, the qualifier "at every continuity point of F" is not a technicality you may ignore; it is essential.

Why it works

Why exempt the discontinuity points of the limit? Consider X_n deterministically equal to $1/n$, converging in every sane sense to the constant 0. The limiting CDF F of the constant 0 jumps from 0 to 1 at $x = 0$, so $F(0) = 1$. But $F_n(0) = \mathbb{P}(1/n \le 0) = 0$ for every n, so $F_n(0) = 0 \not\to 1 = F(0)$ *at the jump*. Everywhere else $F_n \to F$ perfectly. If we demanded convergence at the jump too, we would absurdly conclude $1/n \not\to 0$. Exempting continuity points repairs this. The moral: convergence in distribution is about the *shape* of the distribution settling down, and a CDF's value exactly at a jump is an arbitrary bookkeeping choice that the shape should not be hostage to.

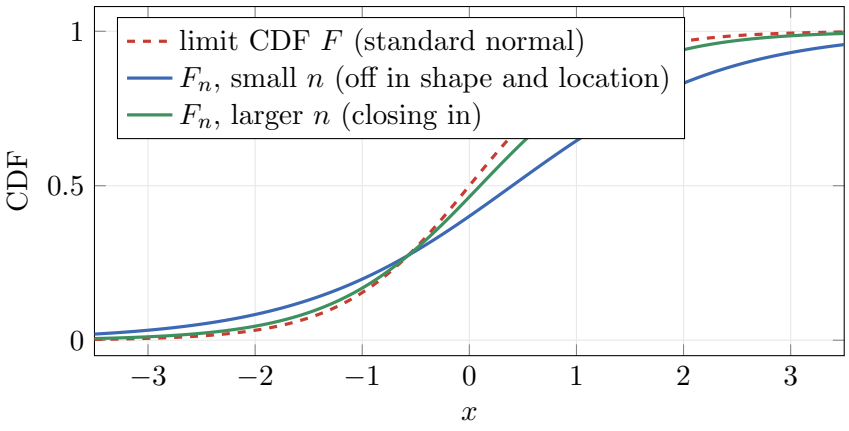


Figure 22.3: Convergence in distribution is convergence of CDFs (at continuity points). As n grows, the curve F_n relaxes onto the limiting curve F. Nothing is claimed about the underlying random values lining up – only the cumulative shapes do. (Logistic curves are used here as smooth stand-ins for normal CDFs.)

22.5.1 Equivalent faces of convergence in distribution

Convergence in distribution wears several equivalent disguises; knowing them is what makes it usable. The collection is traditionally called the *Portmanteau theorem*.

Theorem 22.2 (Portmanteau, selected equivalences)**.** *The following are equivalent to $X_n \xrightarrow{d} X$:*

1. *$F_n(x) \to F(x)$ at every continuity point of F.*
2. *$\mathbb{E}\, g(X_n) \to \mathbb{E}\, g(X)$ for every bounded continuous function g.*
3. *$\mathbb{E}\, g(X_n) \to \mathbb{E}\, g(X)$ for every bounded Lipschitz function g.*
4. *For the characteristic functions, $\varphi_{X_n}(t) \to \varphi_X(t)$ for every t (this is Lévy's continuity theorem, when the limit is a genuine characteristic function).*

Form (2) is the conceptual heart: convergence in distribution means *every bounded continuous measurement* you could take has its average converge. Form (4) is the computational workhorse – it is exactly how the Central Limit Theorem gets proved, by showing the characteristic function of the standardized sum converges to $e^{-t^2/2}$, the characteristic function of the standard normal (see the chapter on the CLT).

Connection

The phrase "test against bounded continuous functions" is the same idea that defines weak convergence of measures throughout analysis, and the same spirit as "test against smooth functions" in the theory of distributions (generalized functions). If you have met the companion calculus book's treatment of integration as pairing a function against test functions, form (2) will feel like home: a distribution is known by how it integrates against probes.

22.6 The implication diagram

Now assemble the hierarchy. Here are all the implications, with the crucial fact that *none of the missing arrows hold in general.*

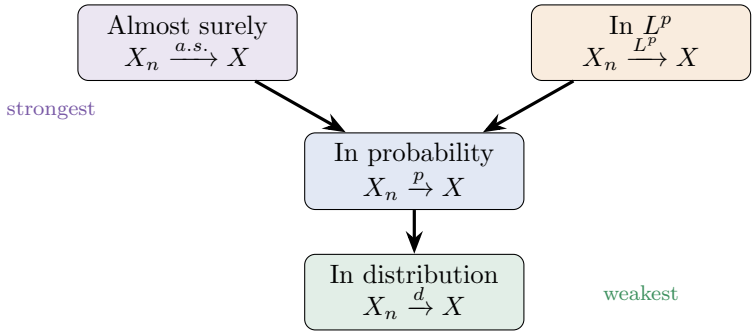


Figure 22.4: The hierarchy of stochastic convergence. Almost sure and L^p each imply convergence in probability, which implies convergence in distribution. No other arrow holds in general: a.s. and L^p do not imply each other; in probability implies neither; in distribution implies none of the upper three.

Let us nail down each arrow and each *non*-arrow.

22.6.1 Almost sure ⇒ in probability

Theorem 22.3. *If $X_n \xrightarrow{a.s.} X$ then $X_n \xrightarrow{p} X$.*

Proof. Fix $\varepsilon > 0$ and define the tail event $A_N = \bigcup_{n \geq N} \{|X_n - X| \geq \varepsilon\}$, the event that some deviation of size ε occurs at time N or later. These events decrease: $A_1 \supseteq A_2 \supseteq \cdots$. Their intersection is the event "$|X_n - X| \geq \varepsilon$ infinitely often," which has probability 0 because almost sure convergence forbids infinitely-often large deviations (Section 22.2.1). By continuity of probability from above, $\mathbb{P}(A_N) \to \mathbb{P}(\bigcap_N A_N) = 0$. But $\{|X_N - X| \geq \varepsilon\} \subseteq A_N$, so $\mathbb{P}(|X_N - X| \geq \varepsilon) \leq \mathbb{P}(A_N) \to 0$. Hence $X_n \xrightarrow{p} X$. □

The typewriter sequence (Section 22.3.1) shows the reverse fails: it converges in probability but not almost surely.

22.6.2 In probability ⇒ in distribution

Theorem 22.4. *If $X_n \xrightarrow{p} X$ then $X_n \xrightarrow{d} X$.*

Proof sketch. Let x be a continuity point of F. For any $\delta > 0$, split on whether X_n is near X:

$$F_n(x) = \mathbb{P}(X_n \leq x) \leq \mathbb{P}(X \leq x + \delta) + \mathbb{P}(|X_n - X| > \delta).$$

(If $X_n \leq x$ but $X > x + \delta$, then $|X_n - X| > \delta$.) Let $n \to \infty$; the last term vanishes by convergence in probability, giving $\limsup_n F_n(x) \leq F(x + \delta)$. Symmetrically, $\liminf_n F_n(x) \geq F(x - \delta)$. Now let $\delta \to 0$; because x is a continuity point, $F(x \pm \delta) \to F(x)$, squeezing $F_n(x) \to F(x)$. □

Common trap

The reverse, "in distribution ⇒ in probability," is false in a way that beginners trip on constantly. Let $Z \sim \mathcal{N}(0, 1)$ and set $X_n = (-1)^n Z$. By symmetry of the normal, every X_n has the *same* distribution as Z, so trivially $X_n \xrightarrow{d} Z$. But $|X_n - Z|$ is either 0 (even n) or $2|Z|$ (odd n), so for odd n, $\mathbb{P}(|X_n - Z| \geq \varepsilon) = \mathbb{P}(2|Z| \geq \varepsilon) \not\to 0$. The distributions agree perfectly while the *values* stay far apart. Distribution convergence sees the shape, never the alignment.

22.6.3 One exception: convergence in distribution to a constant

There is exactly one important case where the weak mode upgrades for free.

Theorem 22.5. *If c is a constant, then $X_n \xrightarrow{d} c$ if and only if $X_n \xrightarrow{p} c$.*

Proof. ($\Leftarrow$) is Theorem 22.4. For ($\Rightarrow$): the limit law is a point mass at c, with CDF that jumps at c. For any $\varepsilon > 0$,

$$\mathbb{P}(|X_n - c| \geq \varepsilon) = \mathbb{P}(X_n \leq c - \varepsilon) + \mathbb{P}(X_n \geq c + \varepsilon) \leq F_n(c - \varepsilon) + \big(1 - F_n(c + \tfrac{\varepsilon}{2})\big).$$

Both $c-\varepsilon$ and $c+\varepsilon/2$ are continuity points of the limit CDF (which is continuous away from c), so $F_n(c-\varepsilon) \to 0$ and $F_n(c+\varepsilon/2) \to 1$. Thus the right side $\to 0$. □

This little theorem is the secret ingredient in Slutsky's theorem below, and it is why the Weak Law of Large Numbers (convergence in probability to the constant μ) can equivalently be phrased as convergence in distribution to μ.

Pause

Pause and consolidate the four facts you now own:

- $\xrightarrow{a.s.} \Rightarrow \xrightarrow{p}$, but not conversely (typewriter).
- $\xrightarrow{L^p} \Rightarrow \xrightarrow{p}$, but not conversely (rare giant spike).
- $\xrightarrow{p} \Rightarrow \xrightarrow{d}$, but not conversely (the $(-1)^n Z$ trap).
- a.s. and L^p imply neither each other; and to a *constant* limit, $\xrightarrow{d}$ and $\xrightarrow{p}$ coincide.

Everything else in the chapter is built from these.

22.6.4 Almost sure and L^p do not imply each other

These two "strong" modes are siblings, not parent and child; neither implies the other.

- *L^p but not a.s.*: the typewriter sequence itself. With the sliding block of width $1/k$ from Section 22.3.1, we have $\mathbb{E}|X_n|^p = 1/k \to 0$, so $X_n \to 0$ in L^p, yet (as shown) X_n converges almost surely nowhere.
- *a.s. but not L^p*: the rare giant of Example 22.2 (Figure 22.2). For a fixed $\omega \in (0,1]$ we have $\omega > 1/n$ for all large n, hence $X_n(\omega) = 0$ eventually; so $X_n \xrightarrow{a.s.} 0$. Yet $\mathbb{E}\, X_n = 1 \not\to 0$, so it fails in L^1.

Both "strong" modes imply convergence in probability, but they control *different things* – one the long-run behavior of each path, the other the average magnitude of the error – and a sequence can ace one while flunking the other.

22.7 The continuous mapping theorem

Convergence would be of limited use if we could not push it through functions. The continuous mapping theorem says: a continuous transformation respects three of the modes: almost sure, in probability, and in distribution.

Theorem 22.6 (Continuous mapping)**.** *Let g be a function continuous on a set C with $\mathbb{P}(X \in C) = 1$ (continuous at every value X actually takes). Then:*

$$X_n \xrightarrow{a.s.} X \;\Rightarrow\; g(X_n) \xrightarrow{a.s.} g(X), \qquad X_n \xrightarrow{p} X \;\Rightarrow\; g(X_n) \xrightarrow{p} g(X), \qquad X_n \xrightarrow{d} X \;\Rightarrow\; g(X_n) \xrightarrow{d} g(X).$$

Why it works

Why continuity is exactly the right hypothesis. Continuity of g means: inputs that are close get mapped to outputs that are close. Each mode of convergence is, at heart, a statement about inputs getting close (pointwise, in probability, in distribution). Continuity is precisely the property that transfers "inputs close" to "outputs close," so it is no surprise it transfers these three modes. For the distributional case, the cleanest proof uses Portmanteau form (2): for bounded continuous h, the composition $h \circ g$ is bounded continuous, so $\mathbb{E}\, h(g(X_n)) = \mathbb{E}\,(h \circ g)(X_n) \to \mathbb{E}\,(h \circ g)(X) = \mathbb{E}\, h(g(X))$. The result falls right out of testing against probes.

Example 22.3 (From the CLT to a chi-square limit)**.** Suppose $\sqrt{n}(\bar{X}_n - \mu)/\sigma \xrightarrow{d} Z \sim \mathcal{N}(0,1)$ (the CLT). The map $g(x) = x^2$ is continuous, so by continuous mapping

$$\frac{n(\bar{X}_n - \mu)^2}{\sigma^2} \xrightarrow{d} Z^2 \sim \chi^2_1.$$

This is the seed of every chi-square test in statistics: square a normal limit and you get a chi-square limit, automatically, with no new probability calculation.

22.8 Slutsky's theorem

The continuous mapping theorem is about one converging sequence. Real statistics constantly combines *two*: an estimator that converges in distribution, and a nuisance quantity (an estimated variance, say) that converges in probability to a constant. Slutsky's theorem is the rule for combining them, and it may be the most-used theorem in all of asymptotic statistics.

Theorem 22.7 (Slutsky)**.** *Suppose* $X_n \xrightarrow{d} X$ *and* $Y_n \xrightarrow{p} c$ *for a* constant c*. Then:*

$$X_n + Y_n \xrightarrow{d} X + c, \qquad Y_n X_n \xrightarrow{d} c\,X, \qquad \frac{X_n}{Y_n} \xrightarrow{d} \frac{X}{c} \quad (c \neq 0).$$

Focus

Two warnings that are the whole reason Slutsky is a named theorem and not a triviality.

1. The second sequence must converge to a **constant**, not merely in distribution. If $Y_n \xrightarrow{d} Y$ for a nondegenerate Y, the conclusions are **false** in general – because convergence in distribution carries no information about the *joint* behavior of X_n and Y_n, and a sum or product depends on the joint law. Constancy is what removes that dependence: a constant has no randomness to be jointly entangled with.

2. It is genuinely surprising that you may add a $\xrightarrow{d}$ sequence and a $\xrightarrow{p}$ sequence at all, since $\xrightarrow{d}$ alone does not respect addition. The constant limit is exactly the loophole that makes it legal.

Proof idea. The trick is to convert $Y_n \xrightarrow{p} c$ into something the joint limit can absorb. Write $X_n + Y_n = (X_n + c) + (Y_n - c)$. The first piece converges in distribution to $X + c$ (continuous mapping, shift by a constant). The second piece, $Y_n - c$, converges in probability to 0. A lemma says: if $A_n \xrightarrow{d} A$ and

$B_n \xrightarrow{p} 0$ then $A_n + B_n \xrightarrow{d} A$ (a vanishing-in-probability perturbation cannot change a distributional limit – intuitively, with high probability B_n is negligible, and on the rare event it is not, it contributes vanishing probability mass). Apply the lemma. The product and quotient versions follow by the same perturbation lemma plus continuous mapping. □

Example 22.4 (The t-statistic: why you may plug in s for σ)**.** The cleanest application. By the CLT, $\sqrt{n}(\bar{X}_n - \mu)/\sigma \xrightarrow{d} \mathcal{N}(0,1)$. But in practice σ is unknown; we use the sample standard deviation s_n, which (by the LLN plus continuous mapping, since $s_n^2 \xrightarrow{p} \sigma^2$ and the square root is continuous) satisfies $s_n \xrightarrow{p} \sigma$, hence $\sigma/s_n \xrightarrow{p} 1$. Slutsky's product rule then gives

$$\frac{\sqrt{n}(\bar{X}_n - \mu)}{s_n} = \underbrace{\frac{\sqrt{n}(\bar{X}_n - \mu)}{\sigma}}_{\xrightarrow{d}\,\mathcal{N}(0,1)} \cdot \underbrace{\frac{\sigma}{s_n}}_{\xrightarrow{p}\,1} \xrightarrow{d} \mathcal{N}(0,1).$$

So replacing the unknown σ by its estimate s_n does not change the limiting distribution. This is precisely why large-sample t-tests and Wald confidence intervals are valid even though we never know the true variance.

Application

A/B testing and Wald intervals. When you compute a 95% confidence interval $\hat{\theta} \pm 1.96\,\widehat{\mathrm{SE}}$ for a conversion-rate difference, two asymptotic facts are quietly cooperating: the estimator $\hat{\theta}$ is asymptotically normal (CLT, a $\xrightarrow{d}$ statement) and the estimated standard error $\widehat{\mathrm{SE}}$ is consistent for the true SE (LLN, a $\xrightarrow{p}$ statement to a constant). Slutsky fuses them so that the studentized statistic $(\hat{\theta} - \theta)/\widehat{\mathrm{SE}}$ is asymptotically standard normal, which is what makes the ± 1.96 legitimate. Every dashboard that reports a p-value from a z- or t-statistic is running Slutsky's theorem under the hood.

22.9 The delta method

The CLT tells you the asymptotic distribution of an average. But you usually care about a *nonlinear function* of an average – a log-odds, a ratio, a variance, a correlation. The delta method is the rule that transports asymptotic normality through a smooth function. It is the first-order Taylor expansion married to Slutsky.

Theorem 22.8 (Delta method)**.** *Suppose* $\sqrt{n}\,(T_n - \theta) \xrightarrow{d} \mathcal{N}(0, \sigma^2)$ *and* g *is differentiable at* θ *with* $g'(\theta) \neq 0$*. Then*

$$\sqrt{n}\,\big(g(T_n) - g(\theta)\big) \xrightarrow{d} \mathcal{N}\big(0,\ \sigma^2\,[g'(\theta)]^2\big).$$

Intuition

Zoom in on the graph of g near θ. Because T_n is concentrating on θ (its fluctuations are of order $1/\sqrt{n}$), only the behavior of g in a tiny neighborhood of θ matters. On that tiny scale g looks like its tangent line, slope $g'(\theta)$. A straight line of slope m turns a $\mathcal{N}(0, \sigma^2)$ wiggle into a $\mathcal{N}(0, m^2\sigma^2)$ wiggle (variance scales by the square of the slope). That is the entire content: *locally linearize, then scale the variance by the slope squared.*

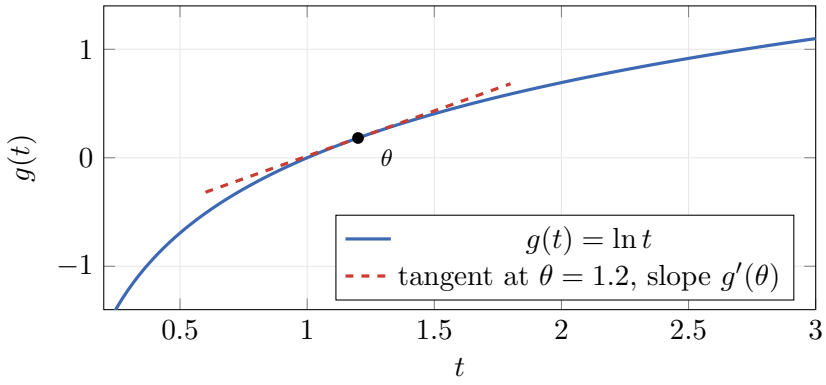


Figure 22.5: The delta method as local linearization. Where T_n concentrates (near θ), the smooth function g is indistinguishable from its tangent line. The tangent's slope $g'(\theta)$ is the only feature of g that survives into the limiting variance, which scales by $[g'(\theta)]^2$.

Proof. A first-order Taylor expansion of g about θ gives, for some $\theta_n^\star$ between T_n and θ,

$$g(T_n) = g(\theta) + g'(\theta_n^\star)\,(T_n - \theta).$$

Multiply by $\sqrt{n}$:

$$\sqrt{n}\,\big(g(T_n) - g(\theta)\big) = g'(\theta_n^\star) \cdot \sqrt{n}\,(T_n - \theta).$$

Now use the machinery of this chapter. Since $\sqrt{n}(T_n - \theta) \xrightarrow{d} \mathcal{N}(0, \sigma^2)$, the unscaled fluctuation $T_n - \theta$ is of order $1/\sqrt{n}$ and so $T_n \xrightarrow{p} \theta$. The intermediate point $\theta_n^\star$ is trapped between T_n and θ, so $\theta_n^\star \xrightarrow{p} \theta$, and by continuity of g' at θ, $g'(\theta_n^\star) \xrightarrow{p} g'(\theta)$, a constant. We now have a product of a $\xrightarrow{p}$-constant factor and a $\xrightarrow{d}$-normal factor – exactly Slutsky's product rule:

$$\sqrt{n}\,\big(g(T_n) - g(\theta)\big) \xrightarrow{d} g'(\theta) \cdot \mathcal{N}(0, \sigma^2) = \mathcal{N}\big(0, [g'(\theta)]^2\sigma^2\big). \qquad \square$$

Example 22.5 (Variance-stabilizing a proportion)**.** Let $\hat{p}_n$ be the sample proportion of successes in n Bernoulli(p) trials. The CLT gives $\sqrt{n}(\hat{p}_n - p) \xrightarrow{d} \mathcal{N}(0,\, p(1-p))$ – note the limiting variance $p(1-p)$ depends on the unknown p, an annoyance. Apply the delta method with $g(x) = \arcsin(\sqrt{x})$, whose derivative is $g'(x) = \frac{1}{2\sqrt{x(1-x)}}$. Then $[g'(p)]^2\, p(1-p) = \frac{1}{4p(1-p)} \cdot p(1-p) = \frac{1}{4}$, a constant free of p:

$$\sqrt{n}\big(\arcsin\sqrt{\hat{p}_n} - \arcsin\sqrt{p}\big) \xrightarrow{d} \mathcal{N}\big(0, \tfrac{1}{4}\big).$$

The arcsine-root transform *stabilizes the variance*. This is exactly why classical statistics uses it before plotting or modeling proportions: the delta method told us which nonlinear lens makes the noise uniform.

Connection

The single-variable delta method is a Taylor expansion; the multivariable version is a Taylor expansion with a *gradient*. If $\sqrt{n}(T_n - \theta) \xrightarrow{d} \mathcal{N}(0, \Sigma)$ for a vector parameter, then $\sqrt{n}(g(T_n) - g(\theta)) \xrightarrow{d} \mathcal{N}(0, \nabla g^\top \Sigma\, \nabla g)$. The scalar slope-squared $[g'(\theta)]^2$ becomes the quadratic form $\nabla g^\top \Sigma \nabla g$ – precisely the linear-algebra object the companion linear algebra book studies when a covariance matrix is sandwiched between a vector and its transpose. Local linearization is the bridge: calculus gives the gradient, linear algebra propagates the covariance.

22.10 Tying in the Law of Large Numbers and the CLT

You have actually met two of these modes already, under different names. The two crown jewels of the previous chapters are statements about precisely two of our four modes.

Let $X_1, X_2, \ldots$ be i.i.d. with mean μ and finite variance σ^2, and $\bar{X}_n = \frac{1}{n}\sum_{i=1}^{n} X_i$.

The Weak Law of Large Numbers is convergence in probability:

$$\bar{X}_n \xrightarrow{p} \mu.$$

The sample mean is, at each large n, very likely close to the truth. Snapshot the crowd; almost all of it is near μ.

The Strong Law of Large Numbers is almost sure convergence:

$$\bar{X}_n \xrightarrow{a.s.} \mu.$$

Pick one infinite data stream; that particular running average lands on μ and stays. This is genuinely stronger (as our hierarchy promised), which is why it takes more work to prove and why the words "weak" and "strong" are well chosen.

The Central Limit Theorem is convergence in distribution:

$$\frac{\sqrt{n}\,(\bar{X}_n - \mu)}{\sigma} \xrightarrow{d} \mathcal{N}(0,1).$$

The CLT cannot be an a.s. or in-probability statement, and seeing why is illuminating.

What is really happening

Why the CLT must be distributional and nothing stronger. The standardized mean $\sqrt{n}(\bar{X}_n - \mu)/\sigma$ does *not* converge as a sequence of numbers at all – it keeps wiggling, with standard deviation pinned at 1 forever. There is no number it homes in on, so almost-sure or in-probability convergence is simply not on the table; the quantity never settles. What *does* settle is its *histogram*: the shape of its distribution relaxes onto the bell curve. That is exactly what convergence in distribution captures and the stronger modes cannot. The LLN squeezes $\bar{X}_n$ onto a point; the CLT rescales the leftover wiggle by $\sqrt{n}$ and asks about its *shape*, which is a question only the weakest mode can answer. The two theorems are looking at the same average through different magnifications.

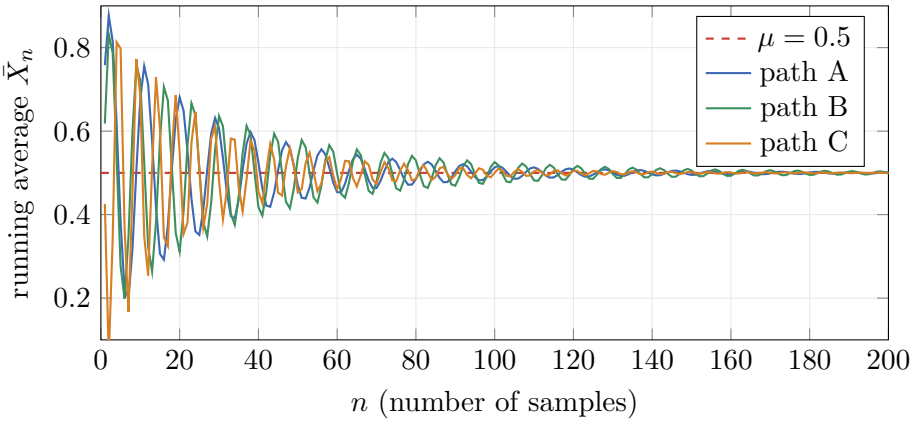


Figure 22.6: The Law of Large Numbers visualized as almost sure convergence: three individual sample paths of the running average, each homing in on and staying near $\mu = 0.5$. (Smooth deterministic curves stand in for true random paths.) The CLT lives in the *rate* at which the gap to μ closes – it shrinks like $1/\sqrt{n}$, and the rescaled gap is asymptotically normal.

History

The names "weak" and "strong" law, and the very idea of distinguishing modes of convergence, crystallized in the early twentieth century as probability was put on rigorous footing. Jacob Bernoulli proved the first law of large numbers (his *Ars Conjectandi*, published posthumously in 1713) in the in-probability sense – he had no notion of almost sure convergence, which requires the modern measure-theoretic apparatus. Émile Borel (1909) proved the first strong law (for coin flips, the "normal numbers" theorem), and Andrey Kolmogorov, in his 1933 *Grundbegriffe* that axiomatized probability via measure theory, gave the strong law its general form and clarified the whole convergence hierarchy. The reason there are four modes is that Kolmogorov's framework – random variables as measurable functions on a probability space – made it possible to even *ask* whether functions converge pointwise, in measure, in mean, or only in law. Before measure theory, the questions could not be posed.

Paradox

The St. Petersburg paradox and the limits of the LLN. A casino offers a game: flip a fair coin until the first head; if it appears on flip k, you win 2^k dollars. The expected payout is $\sum_{k\geq 1} 2^{-k} \cdot 2^k = \sum_{k\geq 1} 1 = \infty$. A "rational" player should pay any finite price to play. Yet nobody would pay a thousand dollars per round. What broke?

The Law of Large Numbers broke – or rather, its hypothesis did. The classical LLN guarantees $\bar{X}_n \to \mu$ only when the mean μ is finite. Here $\mu = \infty$, so the usual law gives no comfort. A subtler analysis (a weak law for these heavy-tailed variables) shows that after n rounds the *typical* average winnings grow only like $\log_2 n$, not like any finite constant – the average drifts upward without limit but agonizingly slowly, dominated by extremely rare colossal jackpots. So the "infinite expectation" is real but is delivered by events so rare that no finite amount of play realizes it. This is the same "rare giant" phenomenon as Example 22.2, now wearing a casino's clothes: convergence statements are only as good as their moment hypotheses, and a heavy tail can starve every mode of convergence at once.

Why this matters

Step back and admire the architecture. Four notions of "getting close," arranged in a strict order of strength, with two combining tools (continuous mapping, Slutsky) and one transport tool (delta method) on top. This is not bureaucratic taxonomy. It is what lets a statistician write down a single line – "$(\hat{\theta}-\theta)/\widehat{\mathrm{SE}} \xrightarrow{d} \mathcal{N}(0,1)$" – and have it be both true and *usable*: the CLT supplies the $\xrightarrow{d}$, the LLN supplies the $\xrightarrow{p}$ for the denominator, Slutsky fuses them, the delta method handles any nonlinear reparametrization, and continuous mapping cleans up the rest. The modes of convergence are the grammar in which all of asymptotic statistics is written.

22.11 A worked synthesis

Here is the whole toolkit run once, end to end, on a single estimator, so the pieces click together.

Example 22.6 (Asymptotics of an estimated odds, start to finish)**.** You observe n i.i.d. Bernoulli(p) trials and want the asymptotic distribution of the *estimated odds* $\widehat{\text{odds}} = \hat{p}_n/(1-\hat{p}_n)$, where $\hat{p}_n$ is the sample proportion. Watch each tool enter.

Step 1 (LLN, $\xrightarrow{p}$). By the weak law, $\hat{p}_n \xrightarrow{p} p$. So the estimator is consistent: the odds estimate concentrates on the true odds $p/(1-p)$.

Step 2 (CLT, $\xrightarrow{d}$). The standardized proportion is asymptotically normal:

$$\sqrt{n}\,(\hat{p}_n - p) \xrightarrow{d} \mathcal{N}\big(0,\ p(1-p)\big).$$

Step 3 (delta method). Let $g(x) = x/(1-x)$, so $g'(x) = 1/(1-x)^2$ and $g'(p) = 1/(1-p)^2$. The delta method transports the normality:

$$\sqrt{n}\Big(\tfrac{\hat{p}_n}{1-\hat{p}_n} - \tfrac{p}{1-p}\Big) \xrightarrow{d} \mathcal{N}\Big(0,\ \tfrac{1}{(1-p)^4}\cdot p(1-p)\Big) = \mathcal{N}\Big(0,\ \tfrac{p}{(1-p)^3}\Big).$$

Step 4 (Slutsky, for a usable interval). The asymptotic variance $p/(1-p)^3$ is unknown, but plugging in $\hat{p}_n$ gives a consistent estimate of it (continuous mapping: the variance is a continuous function of p, and $\hat{p}_n \xrightarrow{p} p$). Dividing the centered, $\sqrt{n}$-scaled estimator by the square root of the *estimated* variance, Slutsky's quotient rule cancels the unknown and leaves a standard normal limit. That yields the valid large-sample confidence interval

$$\frac{\hat{p}_n}{1-\hat{p}_n} \ \pm\ \frac{1.96}{\sqrt{n}}\sqrt{\frac{\hat{p}_n}{(1-\hat{p}_n)^3}}.$$

Every one of the four modes and three tools made an appearance, in the natural order they always appear: concentrate (LLN), find the fluctuation shape (CLT), push it through your nonlinear quantity (delta method), and make it computable by swapping unknowns for estimates (Slutsky).

22.12 Exercises

Exercise 22.1 (Reading the definitions)**.** For each statement, name the strongest mode of convergence it asserts and explain in one sentence why a weaker one would not capture it: (a) "With

probability one, the running average lands on μ and stays." (b) "At each large n, the estimator is within 0.01 of the target with probability at least 0.999." (c) "The mean-squared error tends to zero." (d) "The histogram of the standardized statistic approaches a bell curve."

Exercise 22.2 (Markov to the rescue)**.** Suppose $\mathbb{E}(X_n - X)^2 = 1/n$. Prove directly from Markov's inequality that $X_n \xrightarrow{p} X$, and find an explicit N such that $\mathbb{P}(|X_n - X| \geq 0.1) \leq 0.05$ for all $n \geq N$.

Exercise 22.3 (Build your own typewriter)**.** On $[0, 1]$ with uniform probability, construct a sequence X_n with $X_n \xrightarrow{p} 0$ but such that $X_n(\omega)$ fails to converge for *every* ω. Verify both claims. (You may reuse or modify the construction of Section 22.3.1.)

Exercise 22.4 (The rare giant, generalized)**.** On $[0, 1]$ uniform, let $X_n = a_n \mathbf{1}_{(0,1/n]}$. (a) For which sequences a_n does $X_n \xrightarrow{p} 0$? (b) For which does $X_n \to 0$ in L^1? (c) In L^2? (d) Exhibit an a_n for which $X_n \xrightarrow{p} 0$ but $X_n \not\to 0$ in L^1, and another for which $X_n \to 0$ in L^1 but not L^2.

Exercise 22.5 (Distribution does not imply probability)**.** Let $Z \sim \mathcal{N}(0, 1)$ and $X_n = (-1)^n Z$. Show $X_n \xrightarrow{d} Z$ but $X_n \not\xrightarrow{p} Z$. Then show that if instead $X_n = Z + 1/n$, we do have $X_n \xrightarrow{p} Z$. What single feature of the first example breaks convergence in probability?

Exercise 22.6 (Continuous mapping in action)**.** Suppose $\sqrt{n}(\bar{X}_n - \mu) \xrightarrow{d} \mathcal{N}(0, \sigma^2)$. Use the continuous mapping theorem to find the limiting distribution of $n(\bar{X}_n - \mu)^2/\sigma^2$, and name it.

Exercise 22.7 (Slutsky, and a trap)**.** (a) Suppose $X_n \xrightarrow{d} \mathcal{N}(0, 1)$ and $Y_n \xrightarrow{p} 3$. Find the limiting distribution of $Y_n X_n + Y_n$. (b) Now suppose instead $Y_n \xrightarrow{d} \mathcal{N}(0, 1)$ and you are told only its marginal law. Explain carefully why you *cannot* conclude $X_n + Y_n \xrightarrow{d} \mathcal{N}(0, 2)$ in general, and describe what extra information would let you.

Exercise 22.8 (Delta method computations)**.** Suppose $\sqrt{n}(T_n - \theta) \xrightarrow{d} \mathcal{N}(0, \sigma^2)$ with $\theta > 0$. Find the limiting distribution of $\sqrt{n}(\log T_n - \log \theta)$ and of $\sqrt{n}(\sqrt{T_n} - \sqrt{\theta})$. For the proportion estimator $\hat{p}_n$, derive the asymptotic variance of the log-odds $\log(\hat{p}_n/(1 - \hat{p}_n))$.

Exercise 22.9 (Variance stabilization)**.** For a Poisson(λ) count estimator $\hat{\lambda}_n$ with $\sqrt{n}(\hat{\lambda}_n - \lambda) \xrightarrow{d} \mathcal{N}(0, \lambda)$, find a transform g (a power of $\hat{\lambda}_n$) whose delta-method variance does not depend on λ. (Hint: try $g(x) = \sqrt{x}$ and compute $[g'(\lambda)]^2 \lambda$.)

Exercise 22.10 (Conceptual: why "to a constant"?)**.** Explain, with a concrete two-line counterexample, why Slutsky's theorem requires the second sequence to converge to a constant rather than merely in distribution. (Hint: consider $X_n = Z$ and $Y_n = -Z$ for a single fixed $Z \sim \mathcal{N}(0, 1)$, versus $Y_n = Z'$ an independent copy; look at $X_n + Y_n$.)

Exercise 22.11 (Proof: $L^2 \Rightarrow L^1$)**.** Using the Cauchy–Schwarz inequality (or Jensen's inequality applied to $x \mapsto x^2$), prove that convergence in L^2 implies convergence in L^1. Then explain why this, combined with Theorem 22.1, means L^2 convergence sits high in the hierarchy.

Exercise 22.12 (Putting it together)**.** You estimate a rate $\hat{r}_n$ with $\sqrt{n}(\hat{r}_n - r) \xrightarrow{d} \mathcal{N}(0, \tau^2)$ for a known functional form $\tau^2 = h(r)$. Outline, citing the specific theorem at each step, how you would (a) argue $\hat{r}_n$ is consistent, (b) obtain the asymptotic distribution of $g(\hat{r}_n)$ for a smooth g with $g'(r) \neq 0$, and (c) build an asymptotic 95% confidence interval for $g(r)$ that uses only observed quantities.

Part V

Stochastic Processes

Chapter 23

Markov Chains

So far almost every random sequence in this book has been built out of *independent* pieces. Coin flips, dice rolls, the increments of a simple random walk – in each case the next thing that happens knows nothing about the past. Independence is clean, but it is also a strong assumption, and the world rarely obliges. Tomorrow's weather depends on today's. Your next word depends on the word you just typed. Whether a gambler is up or down at the next bet depends on whether they are up or down right now. The genome you inherit at one site is correlated with the site next door.

We need a model of randomness *with memory* – but not too much memory, or there is nothing we can compute. The Markov chain strikes that balance. It is the simplest interesting way for the future to depend on the past, and it is also one of the most useful objects in applied probability. Page rankings on the web, the autocomplete in your phone, models of ion channels in neurons, queueing theory, Monte Carlo samplers that power modern Bayesian statistics, the spread of epidemics – all of these are Markov chains wearing different clothes.

The whole subject rests on a single, almost philosophical idea: *the present screens off the past.* Once you know where you are right now, the road that brought you here tells you nothing more about where you are going. This chapter is about taking that one sentence seriously and seeing how far it carries us.

23.1 The Markov property: the present is enough

Picture a frog hopping among a few lily pads. At each tick of a clock the frog jumps to a (possibly different) pad according to some probabilities that depend only on the pad it is sitting on *right now.* The frog has no diary. It does not remember the long scenic route it took to reach its current pad; it just looks around from where it is and jumps.

Intuition

A Markov chain is a random walk on a set of *states* where the only thing that matters for the next step is the current state. The history is not erased – you can still write it down – but it is *irrelevant* to the future once the present is known. "Given the present, the future and the past are independent."

Let us make this precise. We have a sequence of random variables $X_0, X_1, X_2, \ldots$, each taking values in a finite or countable set S called the *state space*. We think of X_n as the state at time n.

Definition 23.1 (Markov property)**.** The sequence $(X_n)_{n\geq 0}$ is a (discrete-time) *Markov chain* on state space S if for every $n \geq 0$ and all states $i_0, i_1, \ldots, i_{n-1}, i, j \in S$ with positive probability of the conditioning event,

$$\mathbb{P}(X_{n+1} = j \mid X_n = i,\ X_{n-1} = i_{n-1}, \ldots, X_0 = i_0) = \mathbb{P}(X_{n+1} = j \mid X_n = i).$$

The chain is *time-homogeneous* if this last quantity does not depend on n; we then write $p_{ij} = \mathbb{P}(X_{n+1} = j \mid X_n = i)$ for the *transition probability* from i to j. Unless we say otherwise, all our chains are time-homogeneous.

Read the equation slowly. On the left you condition on the *entire* history. On the right you keep only the most recent value $X_n = i$. The Markov property says these two are equal: the extra history $i_{n-1}, \ldots, i_0$ adds nothing once you know i. That is the "memoryless given the present" slogan made into mathematics.

Common trap

"Memoryless" does *not* mean the steps are independent, and it does *not* mean the chain forgets everything. The frog's current pad is highly informative about its next pad. What the Markov property says is narrower: *additional* past beyond the present contributes no *further* information. Also beware the symmetric trap – the past and future are conditionally independent *given the present*, but they are usually very much dependent if you do not condition. Knowing X_0 tells you plenty about X_5.

Why it works

Why is this the "right" weakening of independence? Independence says $\mathbb{P}(X_{n+1} = j \mid \text{past}) = \mathbb{P}(X_{n+1} = j)$ – the past is irrelevant, full stop. That is too strong for memory. The opposite extreme, letting X_{n+1} depend on the whole history, is too rich: to specify the model you would need a conditional probability for every possible history, and histories grow without bound. The Markov property is the unique middle ground where (a) the future genuinely depends on the past, yet (b) the dependence is funneled entirely through a single current state, so the model is described by a finite table. It is the most memory you can have while staying finitely specifiable.

History

The chains are named for **Andrey Andreyevich Markov** (1856–1922), a student of Chebyshev in St. Petersburg. In 1906 Markov was partly motivated by a quarrel. Pavel Nekrasov had argued that the law of large numbers *required* independence, and used this to make theological claims about free will. Markov wanted to show that the law of large numbers, and limit theorems generally, hold for *dependent* sequences too – provided the dependence is of his restricted kind. To demonstrate it on real data he did something strikingly modern: he took the text of Pushkin's verse novel *Eugene Onegin*, stripped out the first 20,000 letters, and counted how often a vowel was followed by a vowel, a vowel by a consonant, and so on. That two-state chain (vowel/consonant) was, in effect, the first statistical language model – a

direct ancestor of the autocomplete on your phone, written a century before the phone.

23.1.1 From the present to the whole future

The Markov property looks like a statement about one step, but it quietly controls the entire trajectory. The probability of any specific path factors into a product of one-step pieces.

Proposition 23.1 (Path probabilities multiply)**.** *For a Markov chain with initial distribution* $\mathbb{P}(X_0 = i_0) = \mu_{i_0}$*,*

$$\mathbb{P}(X_0 = i_0,\, X_1 = i_1,\, \dots,\, X_n = i_n) = \mu_{i_0}\, p_{i_0 i_1}\, p_{i_1 i_2} \cdots p_{i_{n-1} i_n}.$$

Why it works

Use the chain rule of probability, which always holds:

$$\mathbb{P}(X_0 = i_0, \dots, X_n = i_n) = \mathbb{P}(X_0 = i_0) \prod_{k=1}^{n} \mathbb{P}(X_k = i_k \mid X_0 = i_0, \dots, X_{k-1} = i_{k-1}).$$

Now apply the Markov property to each factor: the conditioning collapses to the immediately preceding state, $\mathbb{P}(X_k = i_k \mid X_{k-1} = i_{k-1}) = p_{i_{k-1} i_k}$. The general chain rule is unconditional truth; the Markov property is what lets us throw away all but the last condition. The trajectory's probability is just a walk along a sequence of edges, each contributing its own factor – start weight times a product of transition weights.

This factorization is the computational engine of the subject. A path is a product, and sums over paths are exactly what matrix multiplication computes – the bridge we build next.

23.2 Transition matrices and state diagrams

Collect all the one-step probabilities into a square array. If $S = \{1, 2, \dots, m\}$, the *transition matrix* is

$$P = \big(p_{ij}\big)_{i,j \in S}, \qquad p_{ij} = \mathbb{P}(X_{n+1} = j \mid X_n = i).$$

Row i of P is the probability distribution of where you go *from* state i. Because from i you must land *somewhere*, every row sums to 1:

$$\sum_{j \in S} p_{ij} = 1 \quad \text{for every } i, \qquad p_{ij} \geq 0.$$

A nonnegative square matrix whose rows each sum to 1 is called a *stochastic matrix*. Stochastic matrices and (time-homogeneous, finite) Markov chains are the same thing seen two ways.

Common trap

Rows sum to 1, not columns. Mixing this up is the single most common Markov-chain bug. The convention "state-as-row, distribution-as-row" also forces us to put probability vectors on the *left*: a row vector π updates by $\pi \mapsto \pi P$, not $P\pi$. Some books transpose everything and

multiply on the right. Pick one convention and hold it. We use rows throughout.

A picture says it faster than a matrix. Draw one node per state and an arrow $i \to j$ labeled p_{ij} for each positive transition probability. This is the *state diagram*.

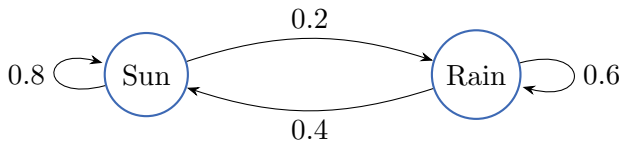


Figure 23.1: A two-state weather chain. From a sunny day, tomorrow is sunny with probability 0.8 and rainy with probability 0.2; from a rainy day, sunny with probability 0.4 and rainy with probability 0.6. Every arrow leaving a node corresponds to one entry of that node's row, and those arrows' labels sum to 1.

In matrix form, with the order (Sun, Rain),

$$P = \begin{pmatrix} 0.8 & 0.2 \\ 0.4 & 0.6 \end{pmatrix}.$$

Check the rows: $0.8 + 0.2 = 1$ and $0.4 + 0.6 = 1$. Good.

Example 23.1 (A reader who skims)**.** Model your reading mood with three states: Focused (F), Skimming (K), Distracted (D). Each paragraph you transition according to

$$P = \begin{pmatrix} 0.7 & 0.2 & 0.1 \\ 0.3 & 0.5 & 0.2 \\ 0.2 & 0.3 & 0.5 \end{pmatrix} \begin{matrix} F \\ K \\ D \end{matrix}$$

where rows and columns are ordered F, K, D. Reading off row F: if you are focused now, next paragraph you stay focused with probability 0.7, start skimming with 0.2, drift off with 0.1. The state diagram would have nine arrows (every $p_{ij} > 0$), three of them self-loops. We will return to this chain to compute long-run behavior.

Walking question

A particle does a random walk on the corners of a triangle, at each step moving to one of the two *other* corners with equal probability $1/2$, never staying put. Write its transition matrix. (Answer: label the corners $1, 2, 3$; then $p_{ii} = 0$ and $p_{ij} = 1/2$ for $i \neq j$. Each row is $(0, \frac{1}{2}, \frac{1}{2})$ in some order, summing to 1.) Notice the zero diagonal – the walk *cannot* stay. One might ask whether this forces a kind of rhythm called periodicity – a question we return to in Example 23.4.

23.3 n-step transitions are matrix powers

Now the central question. You know the one-step probabilities. What is the probability of going from i to j in *exactly* n *steps*? Write this n-step transition probability as

$$p_{ij}^{(n)} = \mathbb{P}(X_n = j \mid X_0 = i).$$

Start with two steps. To get from i to j in two steps you must pass through *some* intermediate state k at time 1. The events "go through k" are mutually exclusive and exhaustive, so we add them up:

$$p_{ij}^{(2)} = \sum_{k \in S} \mathbb{P}(X_1 = k \mid X_0 = i)\, \mathbb{P}(X_2 = j \mid X_1 = k) = \sum_{k \in S} p_{ik}\, p_{kj}.$$

The middle step used the Markov property: once we are at k at time 1, the second jump ignores that we came from i. But $\sum_k p_{ik}\, p_{kj}$ is exactly the (i, j) entry of the matrix product $P \cdot P = P^2$. The two-step probabilities *are* the entries of P^2.

Intuition

Matrix multiplication and "sum over all intermediate states" are literally the same operation. The row-times-column rule $\sum_k p_{ik} p_{kj}$ enumerates every two-step path $i \to k \to j$, multiplies the probabilities along each path (paths are products), and adds across paths (alternatives add). Probability bookkeeping *is* linear algebra.

The same argument with n split as $r + s$ gives the general law.

Theorem 23.1 (Chapman–Kolmogorov)**.** *For all $r, s \geq 0$,*

$$p_{ij}^{(r+s)} = \sum_{k \in S} p_{ik}^{(r)}\, p_{kj}^{(s)}, \qquad \textit{equivalently} \qquad P^{(r+s)} = P^{(r)} P^{(s)}.$$

Consequently $p_{ij}^{(n)} = (P^n)_{ij}$: the n-step transition matrix is the n-th power of P.

Why it works

Condition on where the chain is at the intermediate time r. To travel from i to j in $r + s$ steps, it sits at some state k at time r; the first leg ($i \to k$ in r steps) and the second leg ($k \to j$ in s steps) are governed by $P^{(r)}$ and $P^{(s)}$, and the Markov property makes the second leg independent of how the first was traveled. Summing over k is the law of total probability. Writing it as matrices, "sum over the middle index" is the definition of the product, so $P^{(r+s)} = P^{(r)} P^{(s)}$. Setting $r = s = 1$ gives $P^{(2)} = P^2$, and induction gives $P^{(n)} = P^n$.

Connection

This is where the linear algebra book pays off. The n-step behavior of a chain is the n-th power of a matrix, and powers of matrices are understood through *eigenvalues*. If P is diagonalizable as $P = V \Lambda V^{-1}$ with eigenvalues $\lambda_1, \ldots, \lambda_m$ on the diagonal of Λ, then $P^n = V \Lambda^n V^{-1}$, and Λ^n just raises each eigenvalue to the n. A stochastic matrix always has $\lambda_1 = 1$ as an eigenvalue (the all-ones vector $\mathbf{1}$ satisfies $P\mathbf{1} = \mathbf{1}$, since rows sum to 1), and all eigenvalues satisfy $|\lambda| \leq 1$. As $n \to \infty$, every $|\lambda| < 1$ term decays like $\lambda^n \to 0$, and only the $\lambda = 1$ part survives. *That surviving piece is the long-run equilibrium of the chain.* The whole theory of convergence we meet in the next chapter is, underneath, the statement "the second-largest eigenvalue controls the rate."

23.3.1 Pushing a distribution forward

If you are uncertain about the starting state, encode that uncertainty as a row vector $\mu^{(0)} = (\mu_1^{(0)}, \dots, \mu_m^{(0)})$ with $\mu_i^{(0)} = \mathbb{P}(X_0 = i)$. The distribution after one step is obtained by the law of total probability,

$$\mu_j^{(1)} = \mathbb{P}(X_1 = j) = \sum_i \mathbb{P}(X_0 = i)\, p_{ij} = \big(\mu^{(0)} P\big)_j,$$

so $\mu^{(1)} = \mu^{(0)} P$, and by induction $\mu^{(n)} = \mu^{(0)} P^n$. A distribution is a row vector; time advances by right-multiplication by P.

Example 23.2 (Two steps of the weather chain)**.** Take $P = \left(\begin{smallmatrix} 0.8 & 0.2 \\ 0.4 & 0.6 \end{smallmatrix}\right)$. Then

$$P^2 = \begin{pmatrix} 0.8 & 0.2 \\ 0.4 & 0.6 \end{pmatrix} \begin{pmatrix} 0.8 & 0.2 \\ 0.4 & 0.6 \end{pmatrix} = \begin{pmatrix} 0.72 & 0.28 \\ 0.56 & 0.44 \end{pmatrix}.$$

So if today is sunny, the probability it is sunny the day *after* tomorrow is $p_{SS}^{(2)} = 0.72$. Check the arithmetic of the top-left entry the path way: $S \to S \to S$ contributes $0.8 \times 0.8 = 0.64$, and $S \to R \to S$ contributes $0.2 \times 0.4 = 0.08$; together $0.64 + 0.08 = 0.72$. Two-step probability, two paths, products added. Suppose instead today's weather is itself uncertain, $\mu^{(0)} = (0.5, 0.5)$. Then $\mu^{(1)} = (0.5, 0.5)P = (0.6, 0.4)$ and $\mu^{(2)} = (0.6, 0.4)P = (0.64, 0.36)$.

Watch a chain forget where it started. Iterate $\mu^{(n)} = \mu^{(0)} P^n$ for the weather chain from two very different starting beliefs, and the probability of "Sun" converges to the same value no matter where we began.

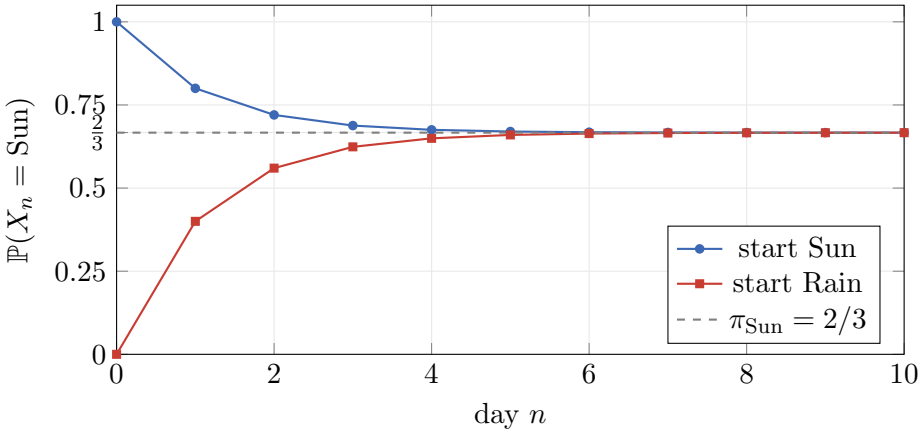


Figure 23.2: The weather chain forgets its start. Whether day 0 is certainly sunny (blue) or certainly rainy (red), the probability of sun on day n rushes to the same limit $2/3$. The gap between the two curves shrinks geometrically – it is multiplied by the second eigenvalue $\lambda_2 = 0.4$ each day. This convergence to a unique equilibrium is the subject of the next chapter; here it is a teaser for what matrix powers do.

What is really happening

Why do both curves land on $2/3$? The eigenvalues of the weather P are $\lambda_1 = 1$ and $\lambda_2 = 0.8 + 0.6 - 1 = 0.4$ (for a 2×2 stochastic matrix the second eigenvalue is always $\operatorname{tr} P - 1$).

Writing $\mu^{(0)} = c_1 v_1 + c_2 v_2$ in the eigenbasis, $\mu^{(n)} = c_1 v_1 + c_2 (0.4)^n v_2$. The $\lambda = 1$ part $c_1 v_1$ is fixed; the 0.4 part decays to nothing. What is left is the equilibrium row vector, the left eigenvector for $\lambda = 1$, which works out to $(2/3, 1/3)$. The starting distribution only enters through the dying term, so it is forgotten. The "memory" of the initial condition has a half-life of about $\log(1/2)/\log(0.4) \approx 0.76$ days.

23.4 Classifying states: where can the chain go, and does it return?

Not all states are alike. Some you can leave and never come back to; some trap you; some you visit forever. To organize this we ask two questions about every state: *what is reachable from it*, and *if you start there, do you come back*?

23.4.1 Communication and irreducibility

Definition 23.2 (Accessibility and communication)**.** State j is *accessible* from i, written $i \to j$, if $p_{ij}^{(n)} > 0$ for some $n \geq 0$ – you can get from i to j in some number of steps. States i and j *communicate*, written $i \leftrightarrow j$, if $i \to j$ and $j \to i$.

Communication is an equivalence relation (it is reflexive since $p_{ii}^{(0)} = 1$, symmetric by definition, and transitive by Chapman–Kolmogorov: if $i \to k$ in r steps and $k \to j$ in s steps then $p_{ij}^{(r+s)} \geq p_{ik}^{(r)} p_{kj}^{(s)} > 0$). It therefore partitions the state space into *communicating classes*.

Definition 23.3 (Irreducible)**.** A chain is *irreducible* if all states communicate – there is a single communicating class, and you can get from anywhere to anywhere.

On the state diagram, communicating classes are exactly the "strongly connected components": the maximal blobs in which directed arrows let you travel both ways between any two members.

23.4.2 Recurrent and transient states

Now the return question. Start the chain at state i and watch. Will it ever come back to i? Let

$$f_i = \mathbb{P}(\text{chain returns to } i \text{ at some time} \geq 1 \mid X_0 = i).$$

Definition 23.4 (Recurrent, transient)**.** State i is *recurrent* if $f_i = 1$ (return is certain) and *transient* if $f_i < 1$ (there is positive probability you never return).

Intuition

A recurrent state is a home you always come back to; a transient state is a place you pass through and, with positive probability, leave for good. The deep fact is that this is all-or-nothing over infinite time. If return is certain ($f_i = 1$), then *each* return is certain, so you return again and again, infinitely often. If return is only probable ($f_i < 1$), then the number of returns is a geometric race the chain eventually loses, so you visit i only finitely many times and then abandon it forever.

That intuition turns into a clean criterion. Let N_i be the total number of visits to i (counting time 0). Each time you are at i, the chance you ever return is f_i, and returns are independent attempts because of the Markov property. So the number of returns is geometric and

$$\mathbb{E}[N_i \mid X_0 = i] = \sum_{n=0}^{\infty} p_{ii}^{(n)} = \frac{1}{1 - f_i},$$

with the convention that the sum is $+\infty$ when $f_i = 1$. This gives a test you can run straight from the matrix powers.

Theorem 23.2 (Recurrence via the Green's function)**.** *State i is recurrent if and only if $\sum_{n=0}^{\infty} p_{ii}^{(n)} = \infty$, and transient if and only if $\sum_{n=0}^{\infty} p_{ii}^{(n)} < \infty$. Recurrence and transience are* class properties*: if $i \leftrightarrow j$, then i and j are either both recurrent or both transient.*

Why it works

The expected number of visits is $\mathbb{E}[N_i] = \mathbb{E}\left[\sum_n \mathbf{1}(X_n = i)\right] = \sum_n \mathbb{P}(X_n = i \mid X_0 = i) = \sum_n p_{ii}^{(n)}$ by linearity (swap sum and expectation, allowed for nonnegative terms). Separately, each visit independently leads to "at least one more visit" with probability f_i, so N_i is a geometric count with mean $1/(1 - f_i)$. Setting the two expressions equal: the sum is infinite exactly when $f_i = 1$. The class-property claim follows because if $i \leftrightarrow j$ you can route visits to i through j and back, so the two return sums are infinite together. The slogan: recurrence means you visit infinitely often, and infinitely many visits means an infinite expected count.

For a *finite* chain there is a satisfying shortcut: you cannot wander off to infinity, because there is no infinity to wander to. With finitely many states and infinitely many time steps, some state is visited infinitely often – and a little argument shows every state in its communicating class must be too.

Proposition 23.2 (Finite chains: recurrence is about being trapped)**.** *In a finite Markov chain, a communicating class is recurrent if and only if it is* closed *(no arrows lead out of it). At least one class is closed, so a finite chain always has at least one recurrent state. Every finite irreducible chain is recurrent.*

Intuition

Picture the communicating classes as rooms connected by one-way doors. A class with a door leading out is transient: eventually you take that door and, lacking a way back (or you'd be in the same class), you never return. A class with no exit door is a trap – once inside you bounce around its states forever, so they are recurrent. In finite chains the entire story of recurrence is just: *which rooms are sealed.*

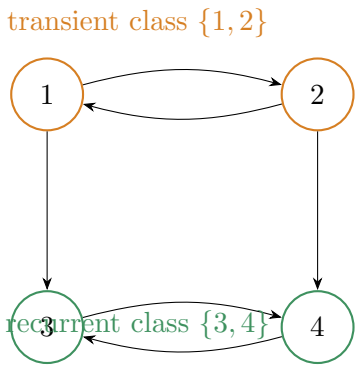


Figure 23.3: A four-state chain with two communicating classes. From $\{1, 2\}$ (orange) there are doors down into $\{3, 4\}$ (green) but none coming back up, so $\{1, 2\}$ is transient – the chain eventually falls through and never returns. The class $\{3, 4\}$ is closed: no arrow leaves it, so it is recurrent and the chain stays there forever after.

Example 23.3 (Gambler's ruin is mostly transient)**.** A gambler bets \$1 per round, winning with probability p and losing with probability $q = 1 - p$, starting with k and quitting at 0 (broke) or N (target). The state is the current fortune, $S = \{0, 1, \ldots, N\}$. The endpoints 0 and N are *absorbing*: $p_{00} = p_{NN} = 1$, each its own closed recurrent class. Every interior state $1, \ldots, N - 1$ communicates with the others but has a door out (to 0 or to N), so the whole interior is *transient*. The chain visits each fortune level only finitely often before being absorbed. This is the structural reason the game must end: with probability 1 the gambler is eventually ruined or hits the target. Computing *which* is a beautiful exercise in itself, left to the end of the chapter.

Paradox

[Polya's drunkard: recurrent in 2D, transient in 3D] Consider a symmetric random walk that starts at the origin and at each step moves to a uniformly random neighbor on the integer lattice. On the line ($\mathbb{Z}^1$) and in the plane ($\mathbb{Z}^2$) the walk is *recurrent*: with probability 1 it returns to the origin (indeed to every point) infinitely often. But in three dimensions ($\mathbb{Z}^3$) it is *transient*: there is a positive probability – about 0.66 – that it *never* comes home, and over time it drifts off to infinity. George Polya proved this in 1921. Shizuo Kakutani summarized it perfectly: "A drunk man will find his way home, but a drunk bird may get lost forever."
Why the dimension matters. Use the recurrence test $\sum_n p_{00}^{(n)}$. The return probability at time $2n$ behaves like the chance a spreading cloud of probability is back at its center. By the local central limit theorem the walk after $2n$ steps is spread over a region of "radius" $\sqrt{n}$ in each dimension, so the probability density at the origin scales like $n^{-d/2}$ in d dimensions:

$$p_{00}^{(2n)} \asymp n^{-d/2}.$$

Now sum: $\sum_n n^{-d/2}$ *diverges* for $d = 1, 2$ (the exponents $1/2$ and 1 are not big enough) but *converges* for $d \geq 3$ (exponent $3/2 > 1$). The whole dichotomy is the convergence of a p-series. Two dimensions is exactly the borderline where the harmonic series $\sum 1/n$ still diverges; one more dimension tips the sum into finiteness and lets the bird escape. Probability and the integral test, hand in hand.

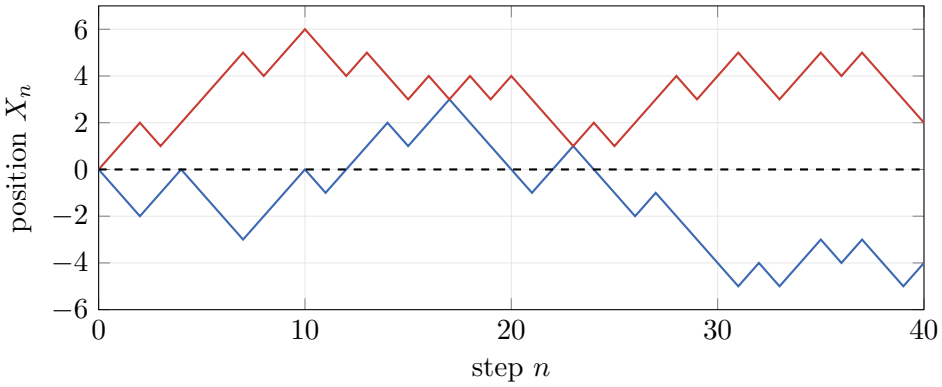


Figure 23.4: Two sample paths of the symmetric random walk on the line ($\mathbb{Z}^1$), one crossing the origin (dashed line) repeatedly and the other illustrating a long excursion away from it. In one and two dimensions every such path returns to its start infinitely often with probability 1 – the recurrence of Polya's theorem made visible. The vertical spread grows like $\sqrt{n}$, which is exactly the scaling behind the $p_{00}^{(2n)} \asymp n^{-d/2}$ estimate in the box above.

23.4.3 Periodicity

The last piece of structure is rhythm. Some states can only be revisited at multiples of a fixed beat.

Definition 23.5 (Period)**.** The *period* of state i is

$$d(i) = \gcd\{n \geq 1 : p_{ii}^{(n)} > 0\},$$

the greatest common divisor of the times at which return is possible. If $d(i) = 1$ the state is *aperiodic*; if $d(i) > 1$ it is *periodic* with period $d(i)$. Period is a class property: communicating states share a period.

Example 23.4 (The triangle walk has period... 2? No, 1)**.** Recall the walk on a triangle that must move to a different corner each step (the Walking question above). Return to corner 1 is possible in 2 steps ($1 \to 2 \to 1$) and in 3 steps ($1 \to 2 \to 3 \to 1$). So the return-time set contains 2 and 3, and $\gcd(2, 3) = 1$: the walk is *aperiodic*. Contrast a walk on a *square* (a 4-cycle) where you can only move to an adjacent corner: returns to a corner happen only at even times, gcd = 2, period 2. The triangle's odd loop breaks the rhythm; the square's bipartite structure enforces it.

What is really happening

Period is really a statement about the eigenvalues lying *on* the unit circle. An aperiodic irreducible chain has $\lambda = 1$ as its only eigenvalue of modulus 1, so P^n settles down to a single limiting matrix. A chain with period d has d eigenvalues spread evenly around the unit circle (the d-th roots of unity), and P^n never settles – it cycles through d phases forever. The bipartite square chain has eigenvalues ± 1; the $+1$ gives the equilibrium and the -1 gives the eternal odd/even flicker. Aperiodicity is exactly the condition that kills every undying oscillation and lets the chain converge, which is why the next chapter's convergence theorem assumes it.

Why this matters

We now have a complete vocabulary for the "shape" of a chain, all read off the diagram:

- **Communicating classes**: the strongly connected blobs. *Irreducible* = one blob.
- **Closed vs. open**: a class is recurrent iff no arrow leaves it (finite case); open classes are transient.
- **Period**: the gcd of possible return times; *aperiodic* = period 1.

A finite, irreducible, aperiodic chain – one blob, sealed (it must be, being everything), with no rhythm – is called *ergodic*, and these are the chains that forget their start and settle to a unique equilibrium. Every condition in that mouthful is a structural property you can verify by staring at the picture. The classification is not bureaucracy; it is exactly the set of hypotheses the limit theorems need.

23.5 A larger worked example

Time to put everything to work on the reader-mood chain of Example 23.1,

$$P = \begin{pmatrix} 0.7 & 0.2 & 0.1 \\ 0.3 & 0.5 & 0.2 \\ 0.2 & 0.3 & 0.5 \end{pmatrix}, \qquad \text{states } F, K, D.$$

Step 1: classify. Every entry of P is positive, so $i \to j$ in one step for all i, j: the chain is irreducible (one class) and, since $p_{ii} > 0$ gives return in one step, aperiodic (period gcd includes 1). It is finite and irreducible, hence recurrent. This is an ergodic chain – it will settle to a unique equilibrium.

Step 2: two- and three-step behavior. Suppose you start Focused, $\mu^{(0)} = (1, 0, 0)$. Then

$$\mu^{(1)} = (0.7,\ 0.2,\ 0.1), \qquad \mu^{(2)} = \mu^{(1)}P = (0.57,\ 0.27,\ 0.16),$$

$$\mu^{(3)} = \mu^{(2)}P = (0.512,\ 0.297,\ 0.191).$$

(For instance $\mu_F^{(2)} = 0.7 \cdot 0.7 + 0.2 \cdot 0.3 + 0.1 \cdot 0.2 = 0.49 + 0.06 + 0.02 = 0.57$.) The probability of being focused is sliding down from 1 toward its long-run value.

Step 3: the equilibrium. The long-run distribution $\pi = (\pi_F, \pi_K, \pi_D)$ is the one that does not change under a step: $\pi P = \pi$ with $\pi_F + \pi_K + \pi_D = 1$. Writing out $\pi P = \pi$:

$$0.7\pi_F + 0.3\pi_K + 0.2\pi_D = \pi_F, \quad 0.2\pi_F + 0.5\pi_K + 0.3\pi_D = \pi_K, \quad 0.1\pi_F + 0.2\pi_K + 0.5\pi_D = \pi_D.$$

Solving (any two equations plus normalization) gives the exact answer

$$\pi = \left(\tfrac{19}{41},\ \tfrac{13}{41},\ \tfrac{9}{41}\right) \approx (0.463,\ 0.317,\ 0.220).$$

In the long run you spend about 46.3% of paragraphs focused, 31.7% skimming, 22.0% distracted – regardless of how you began. Compare with $\mu^{(3)} = (0.512, 0.297, 0.191)$: after just three paragraphs the distribution is already in the right neighborhood. The figure shows the full approach.

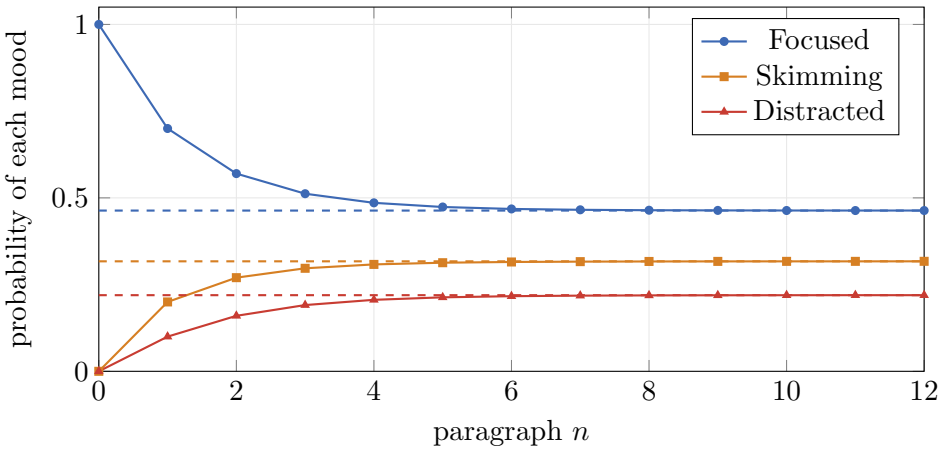


Figure 23.5: Starting Focused (the blue curve begins at probability 1), the distribution $\mu^{(n)} = \mu^{(0)}P^n$ relaxes to the equilibrium $\pi = (19/41, 13/41, 9/41) \approx (0.463, 0.317, 0.220)$ (dashed lines). Because the chain is ergodic, the same three dashed limits are reached from *any* starting mood; only the transient first few paragraphs differ.

Application

[PageRank: a Markov chain that ranks the web] Google's original PageRank algorithm is one giant Markov chain. The states are web pages; the transitions model a "random surfer" who, from a page, clicks a uniformly random outgoing link. The transition matrix P has $p_{ij} = 1/(\text{number of links out of } i)$ for each link $i \to j$. With billions of pages this matrix is astronomically large, but it is sparse (most entries zero) and we never form P^n explicitly – we just iterate $\mu^{(n+1)} = \mu^{(n)}P$, which is one sparse matrix-vector multiply, the operation of Section 23.3.

Two problems arise, and both are fixed with the classification language of this chapter. Pages with no outgoing links are absorbing dead-ends (a closed class of size one) that would soak up all the probability; and the link graph is not irreducible, so a naive surfer can get trapped in a corner of the web. Both are cured by *teleportation*: with probability $1 - \alpha$ (typically $1 - \alpha = 0.15$) the surfer jumps to a uniformly random page instead of following a link. The modified matrix

$$G = \alpha P + (1 - \alpha)\tfrac{1}{m}\mathbf{1}\mathbf{1}^\top$$

is strictly positive, hence irreducible and aperiodic – ergodic. Its unique equilibrium π exists, is unique, and is reached fast (the teleportation makes the second eigenvalue at most $\alpha = 0.85$, so the iteration converges geometrically). *That equilibrium vector π is the PageRank*: π_i is the long-run fraction of time the random surfer spends on page i, and pages where the surfer lingers are ranked higher. A multibillion-dollar idea is, at bottom, the stationary distribution of an ergodic Markov chain.

Connection

Stationary distributions and equilibrium – finding the π with $\pi P = \pi$, proving it exists and is unique, and bounding how fast $\mu^{(n)} \to \pi$ – are the whole subject of the next chapter. Everything there is built on three pillars from *this* chapter: path probabilities multiply (Proposition 23.1), n-step transitions are matrix powers (Theorem 23.1), and the structural classifi-

cation (irreducible, recurrent, aperiodic) tells you exactly when a unique equilibrium exists. Keep the state diagram in your mind's eye; it is the map for all of it.

23.6 Exercises

Exercise 23.1 (Read a matrix)**.** A mouse moves among three rooms A, B, C with $P = \begin{pmatrix} 0 & 1 & 0 \\ 1/2 & 0 & 1/2 \\ 0 & 1 & 0 \end{pmatrix}$. (a) Draw the state diagram. (b) Verify P is stochastic. (c) Is the chain irreducible? (d) Compute the period of state A and explain why this chain never settles to a single distribution.

Exercise 23.2 (Two-step probabilities by hand)**.** For the weather chain $P = \left(\begin{smallmatrix} 0.8 & 0.2 \\ 0.4 & 0.6 \end{smallmatrix}\right)$, compute P^3 and find $\mathbb{P}(X_3 = \text{Rain} \mid X_0 = \text{Sun})$. Then verify your answer for $p_{SS}^{(3)}$ by listing every length-3 path $S \to \cdot \to \cdot \to S$ and summing their probabilities.

Exercise 23.3 (Push a distribution)**.** Using the reader-mood matrix P of Section 23.5 and the starting distribution $\mu^{(0)} = (0, 0, 1)$ (you begin Distracted), compute $\mu^{(1)}$ and $\mu^{(2)}$. Does your $\mu^{(2)}$ look like it is heading to the same $\pi = (19/41, 13/41, 9/41) \approx (0.463, 0.317, 0.220)$ as the Focused start? Explain why it must, in one sentence.

Exercise 23.4 (Memoryless, carefully)**.** Let (X_n) be a Markov chain. (a) Argue from Definition 23.1 that $\mathbb{P}(X_2 = k \mid X_0 = i, X_1 = j) = \mathbb{P}(X_2 = k \mid X_1 = j)$. (b) Give a concrete chain and states for which $\mathbb{P}(X_2 = k \mid X_0 = i) \neq \mathbb{P}(X_2 = k)$, demonstrating that the past and future are *not* independent without conditioning on the present.

Exercise 23.5 (Classify a chain)**.** A chain on $\{1, 2, 3, 4, 5\}$ has transitions: $1 \to 2$ (prob 1); $2 \to 1$ or $2 \to 3$ (each $1/2$); $3 \to 3$ (prob 1); $4 \to 5$ and $5 \to 4$ (each prob 1). (a) Find all communicating classes. (b) Which are closed? (c) Which states are recurrent and which transient? (d) Find the period of each recurrent class.

Exercise 23.6 (Gambler's ruin)**.** In the gambler's ruin chain of Example 23.3 with fair bets ($p = q = 1/2$), let h_k be the probability of reaching the target N before going broke, starting from fortune k. (a) Argue by conditioning on the first bet that $h_k = \frac{1}{2}h_{k-1} + \frac{1}{2}h_{k+1}$ for $1 \le k \le N-1$, with $h_0 = 0$ and $h_N = 1$. (b) Solve this recurrence to get $h_k = k/N$. (c) Interpret: why is the chance of success just your fortune as a fraction of the target?

Exercise 23.7 (Recurrence sum)**.** A chain on $\{1, 2, 3, \dots\}$ has $p_{i,i+1} = 1$ for all i (always step right, never return). (a) Compute $\sum_{n=0}^{\infty} p_{11}^{(n)}$. (b) Using Theorem 23.2, conclude whether state 1 is recurrent or transient. (c) Reconcile this with Proposition 23.2 – why does the "finite chains always have a recurrent state" guarantee not apply here?

Exercise 23.8 (Eigenvalues and forgetting)**.** For a general two-state chain $P = \left(\begin{smallmatrix} 1-a & a \\ b & 1-b \end{smallmatrix}\right)$ with $0 < a, b < 1$: (a) Verify that $\mathbf{1} = (1, 1)^\top$ is a right eigenvector with eigenvalue 1. (b) Show the other eigenvalue is $\lambda_2 = 1 - a - b$ and note $|\lambda_2| < 1$. (c) Explain, using $P^n = V \Lambda^n V^{-1}$, why the chain forgets its start at rate $|1 - a - b|^n$, and what happens in the special case $a + b = 1$.

Exercise 23.9 (Period by gcd)**.** A chain has $p_{11}^{(n)} > 0$ exactly when $n \in \{4, 6, 8, 10, \dots\}$ (every even number ≥ 4). (a) What is the period of state 1? (b) Now suppose instead returns are possible at $\{6, 10, 15\}$ and their sums; what is the period? (c) Conceptually, why does adding a single odd return time of any length force aperiodicity once it shares no common factor with the rest?

Exercise 23.10 (Build a model, prove a property)**.** The Ehrenfest model of diffusion: N molecules are split between two chambers; each second, one molecule chosen uniformly at random switches chambers. Let X_n be the number in the left chamber, a chain on $\{0, 1, \dots, N\}$. (a) Write the transition probabilities $p_{i,i-1}$ and $p_{i,i+1}$. (b) Show the chain is irreducible. (c) Show it has period 2, and explain physically what the period means. (d) *(Proof.)* Show that the binomial distribution $\pi_i = \binom{N}{i} 2^{-N}$ satisfies the *detailed balance* equations $\pi_i p_{i,i+1} = \pi_{i+1} p_{i+1,i}$, and explain why this makes π stationary ($\pi P = \pi$).

Chapter 24

Markov Chain Steady States and Mixing

In the previous chapter you met the Markov chain: a system that hops between states, where the only thing that matters for the next step is where you *are now*, not the whole story of how you got there. That memorylessness is a strong assumption, and it buys us something remarkable. If you let a Markov chain run for a long time, it very often *forgets where it started*. The fraction of time it spends in each state settles down to a fixed set of numbers, and those numbers do not depend on your starting point at all.

This chapter is about that settling-down. We ask three questions and answer all of them:

- **Is there a long-run distribution at all?** (Stationary distributions, and how to find them as a left eigenvector.)
- **Does the chain actually converge to it, no matter where it starts?** (The convergence theorem for ergodic chains.)
- **How long do you have to wait?** (Mixing time.)

Along the way we meet *reversibility* and detailed balance — a shortcut for guessing the steady state — and we cash everything out in the two applications that made this corner of probability famous: Google's PageRank, and Markov chain Monte Carlo (MCMC), the engine behind modern Bayesian statistics.

24.1 The Long Run: A First Look

Let us anchor everything in a tiny, concrete world. Imagine the weather in a town that has only two moods: **Sunny** (state S) and **Rainy** (state R). The weather tomorrow depends only on the weather today. Suppose:

- If it is sunny today, tomorrow is sunny with probability 0.8 and rainy with probability 0.2.
- If it is rainy today, tomorrow is rainy with probability 0.6 and sunny with probability 0.4.

We collect these into a **transition matrix** P, where the entry P_{ij} is the probability of moving from state i to state j:

$$P = \begin{pmatrix} P_{SS} & P_{SR} \\ P_{RS} & P_{RR} \end{pmatrix} = \begin{pmatrix} 0.8 & 0.2 \\ 0.4 & 0.6 \end{pmatrix}.$$

Each row sums to 1, because from any state *something* must happen tomorrow. (A matrix with nonnegative entries and rows summing to one is called a **stochastic matrix**; every Markov chain is one of these and vice versa.)

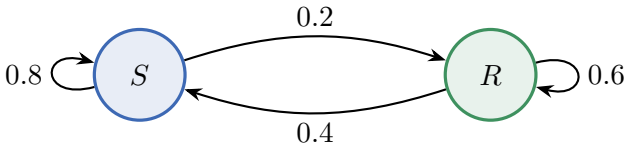


Figure 24.1: The two-state weather chain. Arrows are transition probabilities; the loops are the chances of staying put.

Now here is the move that makes matrices earn their keep. Suppose today the weather is described by a **row vector** of probabilities, $\pi^{(0)} = (\,\mathbb{P}(\text{sunny}),\ \mathbb{P}(\text{rainy})\,)$. What is the distribution tomorrow? You multiply on the right by P:

$$\pi^{(1)} = \pi^{(0)} P.$$

Why on the right, and why a row vector? Because the probability of being sunny tomorrow is "probability sunny today times P_{SS}, plus probability rainy today times P_{RS}" — exactly the dot product of the row vector with the S-column of P. Iterating, after n days,

$$\pi^{(n)} = \pi^{(0)} P^n.$$

The whole dynamics of the chain is hidden inside the powers of one matrix.

Walking question

Start certain that today is sunny, so $\pi^{(0)} = (1, 0)$. Compute the distribution for the next few days. Watch what happens.

$$\begin{aligned}
\pi^{(1)} &= (1,0)P = (0.8,\ 0.2),\\
\pi^{(2)} &= (0.8, 0.2)P = (0.8\cdot 0.8 + 0.2\cdot 0.4,\quad 0.8\cdot 0.2 + 0.2\cdot 0.6) = (0.72,\ 0.28),\\
\pi^{(3)} &= (0.72, 0.28)P = (0.688,\ 0.312),\\
\pi^{(4)} &\approx (0.6752,\ 0.3248), \qquad \pi^{(5)} \approx (0.6701,\ 0.3299).
\end{aligned}$$

Now start *certain that today is rainy*, $\pi^{(0)} = (0, 1)$:

$$\pi^{(1)} = (0.4, 0.6),\quad \pi^{(2)} = (0.56, 0.44),\quad \pi^{(3)} = (0.624, 0.376),\quad \pi^{(5)} \approx (0.6598, 0.3402).$$

The two sequences are marching toward the *same* pair of numbers, (2/3, 1/3), from opposite sides. The chain is forgetting where it began.

Figure 24.2 plots both runs. This convergence to a single distribution, regardless of the start, is the central phenomenon of the chapter. Our job now is to (a) name the limit, (b) compute it without grinding out powers, and (c) understand exactly when it is guaranteed.

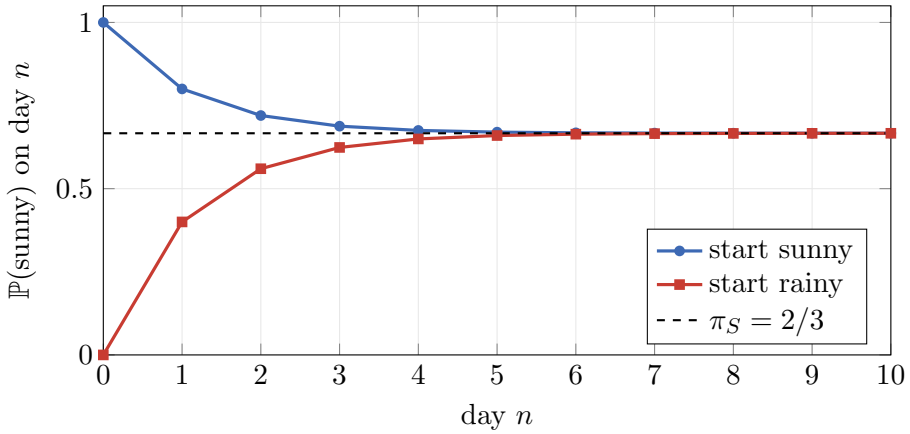


Figure 24.2: Both starting conditions converge to the same stationary probability of sun, 2/3, within a handful of days. The gap between the curves — a memory of the initial state — shrinks geometrically.

24.2 Stationary Distributions

The limit we just watched has a defining property: once you are *in* it, applying P leaves you unchanged. It is a fixed point of the dynamics.

Definition 24.1 (Stationary distribution)**.** A probability row vector $\pi = (\pi_1, \dots, \pi_m)$ (so $\pi_i \geq 0$ and $\sum_i \pi_i = 1$) is a **stationary distribution** (also called an *invariant* or *equilibrium* distribution) for a chain with transition matrix P if

$$\pi P = \pi.$$

Equivalently, for every state j, $\quad \pi_j = \sum_i \pi_i \, P_{ij}$.

Read the componentwise version as a *conservation law*. Picture a large population of independent walkers spread over the states in proportions π. In one step, each state j *receives* a flow $\pi_i P_{ij}$ from every state i. The distribution is stationary exactly when the total inflow into j equals the amount π_j that was there to begin with: probability mass sloshes around, but the overall proportions never change.

Intuition

"Stationary" does not mean nothing moves. Individual walkers keep hopping forever. What stands still is the *histogram* of where they are. It is like a steady waterfall: every droplet is in motion, yet the shape of the falling sheet is frozen. Equilibrium is dynamic, not static.

24.2.1 Finding it by hand

For the weather chain, $\pi P = \pi$ with $\pi = (\pi_S, \pi_R)$ gives

$$\begin{cases} 0.8\,\pi_S + 0.4\,\pi_R = \pi_S, \\ 0.2\,\pi_S + 0.6\,\pi_R = \pi_R. \end{cases}$$

Both equations reduce to the *same* relation $0.2\,\pi_S = 0.4\,\pi_R$, i.e. $\pi_S = 2\pi_R$. That is no accident: because the rows of P sum to one, the columns of $P - I$ sum to zero, so one of the balance equations is always redundant. The missing equation is supplied by the normalization $\pi_S + \pi_R = 1$. Solving,

$$2\pi_R + \pi_R = 1 \;\Rightarrow\; \pi_R = \tfrac{1}{3}, \qquad \pi_S = \tfrac{2}{3}.$$

So $\pi = (2/3,\ 1/3)$ — precisely the numbers Figure 24.2 crept toward. In the long run this town is sunny two days out of three.

Common trap

A stationary distribution must be *normalized*. The equation $\pi P = \pi$ is homogeneous: if π solves it, so does 5π and so does 0. Those are not distributions. You always need the extra condition $\sum_i \pi_i = 1$ to pin down the actual probability vector. In practice: solve $\pi P = \pi$ for the *ratios* between components, then scale so they add to 1.

24.2.2 It really is a left eigenvector

Connection

The equation $\pi P = \pi$ is the eigenvalue equation $\pi P = 1 \cdot \pi$. A row vector π with $\pi P = \lambda\pi$ is a **left eigenvector** of P with eigenvalue λ. *A stationary distribution is exactly a left eigenvector of P for the eigenvalue $\lambda = 1$, normalized to sum to one.*

Why is 1 always an eigenvalue? Because P has constant row sums equal to 1, the all-ones *column* vector $\mathbf{1}$ satisfies $P\mathbf{1} = \mathbf{1}$ — it is a *right* eigenvector with eigenvalue 1. A matrix and its transpose have the same eigenvalues, so 1 is a left eigenvalue too, guaranteeing a (possibly complex, but in fact real and nonnegative) left eigenvector π. That existence is half of the Perron–Frobenius theorem from linear algebra; the companion linear algebra book treats it in full. The deep fact we borrow: for a "nice" (irreducible, aperiodic) stochastic matrix, the eigenvalue 1 is *simple* (multiplicity one), every other eigenvalue has modulus strictly less than one, and the eigenvector π can be taken with all entries positive.

This eigenvalue picture is not just notation — it controls the *speed* of convergence, as we will see in Section 24.5. The second-largest eigenvalue is the rate at which memory of the starting state decays.

Example 24.1 (A three-state chain)**.** Consider three web pages A, B, C with a surfer who clicks links:

$$P = \begin{pmatrix} 0 & 1 & 0 \\ \frac{1}{4} & 0 & \frac{3}{4} \\ \frac{1}{2} & \frac{1}{2} & 0 \end{pmatrix}.$$

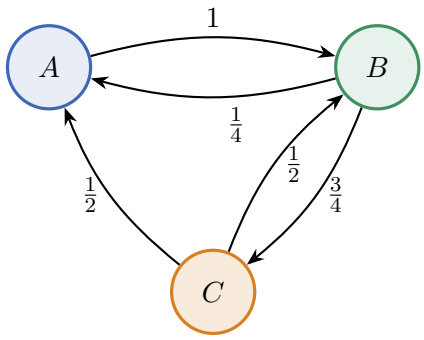


Figure 24.3: The three-page surfer chain of Example 24.1. Page A sends all its weight to B; B splits between A and C; C splits between A and B. Page B receives all of A's outbound weight, which is why it wins the largest stationary share.

We seek $\pi = (a, b, c)$ with $\pi P = \pi$:

$$\begin{aligned} a &= \tfrac{1}{4}b + \tfrac{1}{2}c, \\ b &= a + \tfrac{1}{2}c, \\ c &= \tfrac{3}{4}b. \end{aligned}$$

Substitute $c = \frac{3}{4}b$ into the first: $a = \frac{1}{4}b + \frac{1}{2}\cdot\frac{3}{4}b = \frac{1}{4}b + \frac{3}{8}b = \frac{5}{8}b$. Check the second: $a + \frac{1}{2}c = \frac{5}{8}b + \frac{3}{8}b = b$, consistent. Normalizing,

$$a + b + c = \tfrac{5}{8}b + b + \tfrac{3}{4}b = \tfrac{19}{8}b = 1 \;\Rightarrow\; b = \tfrac{8}{19}.$$

Hence $\pi = \left(\frac{5}{19}, \frac{8}{19}, \frac{6}{19}\right)$. Page B is most visited: as Figure 24.3 shows, it is the page receiving the largest net incoming weight (1 from A and $\frac{1}{2}$ from C).

24.3 Does the Chain Converge? The Ergodic Theorem

Existence of a stationary π does not by itself guarantee that the chain *approaches* it. Two failure modes block convergence, and naming them tells us exactly what to require.

24.3.1 Failure mode 1: the chain is reducible

If the state space splits into pieces that cannot reach one another, the long-run behavior depends on which piece you started in — there is no single universal limit. We need every state to be reachable from every other.

Definition 24.2 (Irreducible)**.** A chain is **irreducible** if for every pair of states i, j there is some number of steps n with $(P^n)_{ij} > 0$: you can get from anywhere to anywhere, eventually. (On the diagram: the directed graph of allowed transitions is strongly connected.)

24.3.2 Failure mode 2: the chain is periodic

Even an irreducible chain can fail to settle if it marches in lockstep. Consider two states with $P = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$: it flips $A \to B \to A \to B \to \cdots$ forever. The stationary distribution is $(1/2, 1/2)$,

but starting at A the distribution is $(1,0),(0,1),(1,0),\ldots$ — it oscillates and never converges. The trouble is *period*.

Definition 24.3 (Period; aperiodic)**.** The **period** of state i is $d(i) = \gcd\{n \geq 1 : (P^n)_{ii} > 0\}$, the greatest common divisor of the lengths of all loops returning to i. If $d(i) = 1$ the state is **aperiodic**. In an irreducible chain all states share the same period; the chain is aperiodic if that common period is 1. A single self-loop $P_{ii} > 0$ anywhere forces aperiodicity.

Intuition

Periodicity is a tyranny of timing. The flip-flop chain has plenty of "probability of being at A" on average — but the parity of the clock decides which state holds all the mass at any instant. Aperiodicity breaks the tyranny: once there is any slack in the return times (a self-loop, or two cycles of coprime length), the phases blur together and the distribution can relax.

24.3.3 The theorem

A chain that is both irreducible and aperiodic on a finite state space is called **ergodic**. For such chains everything works.

Theorem 24.1 (Fundamental theorem of Markov chains / convergence)**.** *Let P be the transition matrix of a finite, irreducible, aperiodic Markov chain. Then:*

1. *There is a* unique *stationary distribution π, with $\pi_i > 0$ for every state i.*
2. *For* every *starting distribution $\pi^{(0)}$,*
$$\pi^{(0)} P^n \longrightarrow \pi \qquad \text{as } n \to \infty.$$
Equivalently, every row of P^n converges to π: the chain forgets where it started.
3. ***(Ergodic / time-average form.)*** *With probability one, the long-run fraction of time spent in state i equals π_i:*
$$\frac{1}{n}\sum_{t=1}^{n} \mathbf{1}\{X_t = i\} \xrightarrow{a.s.} \pi_i.$$
Moreover $\pi_i = 1/\mathbb{E}_i[T_i]$, the reciprocal of the mean return time to i.

Parts (2) and (3) are different statements, and the distinction matters. Part (2) is about the *distribution* at a single far-off time n — "where is the chain likely to be on day 1000?" Part (3) is about a *single sample path* — "what fraction of its life does this one particular trajectory spend in state i?" Aperiodicity is needed for (2) (otherwise the distribution oscillates) but *not* for (3): the flip-flop chain still spends half its time in each state, even though its distribution never converges. This is why simulations work: a running average over one long run estimates π even for periodic chains.

Why it works

Why does convergence happen, intuitively? Here is the cleanest argument, the *coupling* idea. Run two copies of the chain in parallel: copy X started from your distribution of interest, copy Y started from the stationary π (so Y stays distributed as π for all time). Let them

move independently until the first moment they happen to land on the same state — call it the coupling time τ. After τ, glue them: let X copy Y's moves exactly. From that instant on, X and Y are identical, so X is also distributed as π.
Because the chain is irreducible and aperiodic, the two copies *will* meet with probability one, and in fact the chance they have *not* met by time n decays geometrically. The total-variation distance between X's distribution and π is bounded by $\mathbb{P}(\tau > n)$, which $\to 0$. So the only obstruction to convergence is the two copies never managing to meet — which is exactly what reducibility (they live in different worlds) or periodicity (they march in opposite phase and keep missing) would cause. Rule those out and convergence is forced.

History

The man behind the chains was **Andrei Andreyevich Markov** (1856–1922), a combative St. Petersburg analyst and student of Chebyshev. In 1906–1913 he introduced "chains" of dependent variables explicitly to refute a theological opponent, P. A. Nekrasov, who had claimed that the law of large numbers *required* independence and therefore (Nekrasov argued) was evidence of free will. Markov's reply was a constructive insult: here are variables that are strongly *dependent* and yet obey a law of large numbers anyway. To make the point vivid he counted the alternation of vowels and consonants across 20,000 letters of Pushkin's *Eugene Onegin* — arguably the first statistical analysis of text and a direct ancestor of every language model that followed. The general convergence theory, and the Perron–Frobenius machinery underneath it, was filled in over the following decades by Kolmogorov, Doeblin, and others.

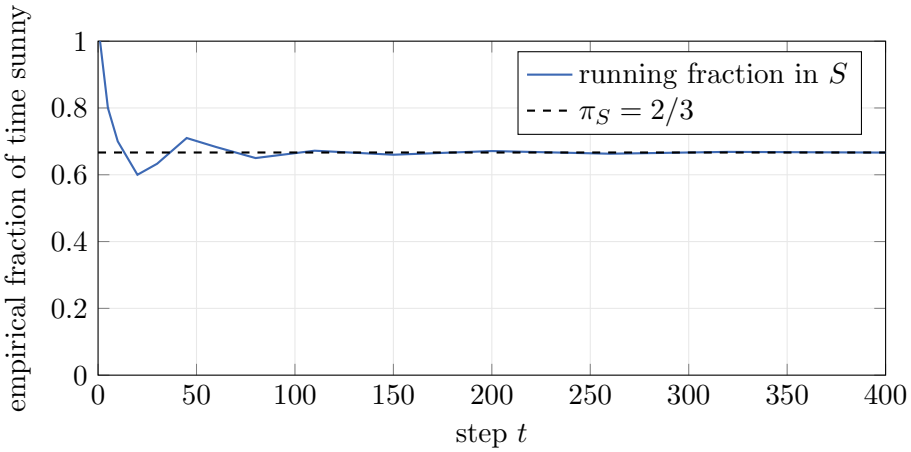


Figure 24.4: The ergodic theorem in action (a single simulated path of the weather chain). The fraction of days that are sunny wanders at first, then locks onto the stationary value 2/3. This is part (3) of Theorem 24.1: time averages converge along one trajectory.

Paradox

Equilibrium without ever being at rest. Newcomers often expect a chain "at its stationary distribution" to slow down or stop. It does the opposite — it keeps hopping just as energetically as before. The paradox dissolves once you separate two ensembles. Imagine a million towns, each running its own weather chain. If today the *collection* has 2/3 sunny

towns, then tomorrow it still has 2/3 sunny towns — even though roughly 20% of today's sunny towns turned rainy and exactly enough rainy ones turned sunny to replace them. No single town is frozen; the *population* is. Confusing the fate of one walker with the shape of the crowd is the root of nearly every Markov-chain misconception.

24.4 Reversibility and Detailed Balance

Solving $\pi P = \pi$ means solving a linear system — fine for three states, miserable for three million. There is a far easier sufficient condition that, when it holds, hands you π almost for free. It is also the single most important idea in all of MCMC.

The stationarity equation $\pi_j = \sum_i \pi_i P_{ij}$ is a statement about *total* flow: the net probability current into each state is zero. Detailed balance demands something stronger and far more rigid — that the flow balance between *every individual pair* of states, edge by edge.

Definition 24.4 (Detailed balance; reversibility)**.** A distribution π satisfies **detailed balance** for P if

$$\pi_i \, P_{ij} = \pi_j \, P_{ji} \qquad \text{for all states } i, j.$$

A chain that has such a π is called **reversible**.

The left side is the rate of $i \to j$ traffic in equilibrium; the right side is the rate of $j \to i$ traffic. Detailed balance says these are equal for each edge: no net circulation anywhere.

Theorem 24.2 (Detailed balance implies stationary)**.** *If π satisfies detailed balance for P, then π is stationary: $\pi P = \pi$.*

Proof. Sum the detailed-balance equations over i for a fixed j:

$$\sum_i \pi_i P_{ij} = \sum_i \pi_j P_{ji} = \pi_j \sum_i P_{ji} = \pi_j \cdot 1 = \pi_j,$$

using that the j-th row of P sums to one. But $\sum_i \pi_i P_{ij} = (\pi P)_j$, so $(\pi P)_j = \pi_j$ for every j, i.e. $\pi P = \pi$. □

What is really happening

What is really happening: time looks the same run backwards. The name "reversible" is literal. Record a long stationary run of the chain and play the film backwards. If detailed balance holds, the reversed film is statistically *indistinguishable* from a forward run — the same transition probabilities govern it. Check: the probability of seeing i then j forward is $\pi_i P_{ij}$; the probability of seeing j then i (which is what "i then j" looks like in reverse) is $\pi_j P_{ji}$. Detailed balance is exactly the statement that these are equal. So reversibility = time-symmetry of the equilibrium dynamics. Chains *without* it have a built-in arrow of time: a preferred direction of circulation, like a whirlpool, that you would notice if the film were reversed.

Example 24.2 (Random walk on a graph)**.** Take any connected undirected graph. A walker sits at a vertex and at each step moves to a uniformly random neighbor. If vertex i has degree d_i (number of edges), then $P_{ij} = 1/d_i$ when i, j are adjacent, else 0. Guess $\pi_i \propto d_i$, i.e. $\pi_i = d_i/(2E)$ where E is the number of edges (so $\sum_i d_i = 2E$). Check detailed balance on an edge $\{i, j\}$:

$$\pi_i P_{ij} = \frac{d_i}{2E} \cdot \frac{1}{d_i} = \frac{1}{2E} = \frac{d_j}{2E} \cdot \frac{1}{d_j} = \pi_j P_{ji}. \checkmark$$

It balances for every edge, so $\pi_i = d_i/(2E)$ is stationary. We *guessed* the answer and verified it in one line — no linear system. The lesson: a random walker lingers in proportion to how connected a place is. High-degree hubs are visited more. This single observation is the seed of PageRank.

Why this matters

Detailed balance is the probabilist's version of a conservation law in physics — indeed it *is* one. In statistical mechanics the equilibrium (Boltzmann) distribution $\pi_i \propto e^{-E_i/kT}$ satisfies detailed balance with respect to the microscopic dynamics, and that is precisely why thermal systems settle. The same equation that tells you the weather will be sunny 2/3 of the time tells a physicist how molecules distribute among energy levels. When you reach Section 24.7 you will see we run the logic *backwards*: given a target π we *design* a P that satisfies detailed balance, so that the chain is forced to equilibrate to the distribution we secretly wanted to sample.

24.5 Mixing Time: How Long Until "Long Run"?

Theorem 24.1 promises convergence but is silent about *speed*. In Figure 24.2 we were essentially done in five days. A poorly connected chain on a million states might need millions of steps. "How long?" is the question of **mixing time**, and for applications it is everything: a sampler that has not yet mixed gives you biased garbage.

To measure "how far from π," we need a distance between distributions.

Definition 24.5 (Total variation distance)**.** For two distributions μ, ν on the same states,

$$\|\mu - \nu\|_{\mathrm{TV}} = \tfrac{1}{2} \sum_i |\mu_i - \nu_i| = \max_A |\mu(A) - \nu(A)|.$$

It is 0 when the distributions coincide and 1 when their supports are disjoint; the second form says it is the largest gap in probability they assign to any event.

Definition 24.6 (Mixing time)**.** Fix a tolerance ε (commonly $\varepsilon = 1/4$). The **mixing time** is

$$t_{\mathrm{mix}}(\varepsilon) = \min\Big\{ n : \max_x \| \, \delta_x P^n - \pi \, \|_{\mathrm{TV}} \le \varepsilon \Big\},$$

the number of steps after which, *no matter the starting state* x, the chain's distribution is within ε of stationary in total variation.

24.5.1 The spectral gap controls the rate

Here the eigenvalues come back to do real work. Order the eigenvalues of P by modulus: $1 = \lambda_1 > |\lambda_2| \geq \cdots \geq |\lambda_m|$. For an ergodic chain the top one is 1 (the stationary direction) and all others are strictly smaller. Decompose any starting vector along the eigen-directions; under repeated multiplication by P, the stationary component is preserved while every other component is multiplied by its λ_k each step and *decays geometrically*. The slowest-dying piece is governed by the second eigenvalue.

Theorem 24.3 (Geometric convergence rate)**.** *For an ergodic chain (reversible, say, so the λ_k are real), there is a constant C with*

$$\| \delta_x P^n - \pi \|_{\mathrm{TV}} \leq C \, |\lambda_2|^n.$$

Writing the ***spectral gap*** *as $\gamma = 1 - |\lambda_2|$, the mixing time scales like*

$$t_{\mathrm{mix}} \approx \frac{1}{\gamma} \log \frac{1}{\varepsilon}.$$

A big gap (small $|\lambda_2|$) means fast mixing; a tiny gap means a sluggish chain.

For the weather chain the eigenvalues of P are $\lambda_1 = 1$ and $\lambda_2 = 0.8 + 0.6 - 1 = 0.4$ (the eigenvalues of a 2×2 stochastic matrix are 1 and $\operatorname{tr} P - 1$). The gap is 0.6 — large — so error falls by a factor of 0.4 per day, matching the rapid collapse in Figure 24.2. Indeed the gap between the two curves there shrinks by exactly 0.4 each step: that is $|\lambda_2|^n$ in the flesh.

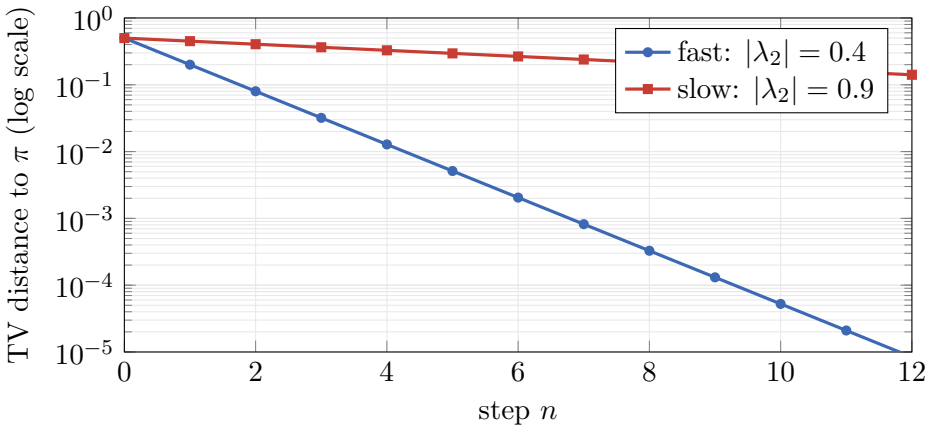


Figure 24.5: On a log scale, total-variation distance to π falls along a straight line whose slope is $\log|\lambda_2|$. The fast chain (steep line) mixes in a few steps; the slow chain (shallow line) crawls. The spectral gap governs the slope.

Intuition

What makes a chain slow to mix is a **bottleneck**: a narrow bridge between two roomy regions of state space. The walker spends ages on one side, only rarely squeezing through. Until it crosses, it cannot "know" about the far side, so its distribution stays lopsided. Mathematically, a bottleneck of small "conductance" forces $|\lambda_2|$ close to 1 and the gap close to 0. This is the central headache of MCMC: a multimodal target with widely separated modes is a chain with a vicious bottleneck, and a naive sampler can sit in one mode for billions of steps, reporting

confident nonsense.

Paradox

The cutoff phenomenon. You might guess mixing is gradual — the chain oozes toward π, getting a bit closer each step. For many natural chains the truth is startlingly different. Persi Diaconis and collaborators showed that shuffling a deck of cards by the standard riffle shuffle exhibits a sharp *cutoff*: after about 6 shuffles the deck is still glaringly nonrandom, and after about 8 it is essentially perfectly mixed. The TV distance stays near 1, then plunges to near 0 over a window much shorter than the mixing time itself — a probabilistic phase transition. The famous headline "seven shuffles" is exactly this cutoff for 52 cards: $\lceil \frac{3}{2} \log_2 52 \rceil = 9$, with the curtain falling around 7. Convergence, it turns out, can be an event rather than a trend.

24.6 Headline Application I: PageRank

Application

Google's original ranking algorithm is a stationary distribution. Model the web as a giant directed graph: pages are states, links are edges. Imagine a *random surfer* who, at each page, clicks a uniformly random outgoing link. This is a Markov chain with $P_{ij} = 1/(\text{outdegree of } i)$ for each link $i \to j$. The PageRank of a page is defined as its stationary probability π_i: the long-run fraction of time the surfer spends there. Pages that many (important) pages link to get visited often, so they score high — importance defined self-referentially and resolved by the eigenvector. Crucially, PageRank is *the left eigenvector of the web's transition matrix for eigenvalue* 1 (Section 24.2). Brin and Page's 1998 insight was that this single number, computable for the whole web, ranks pages astonishingly well.
Two repairs are needed to guarantee a unique limit (Theorem 24.1). (1) *Dangling pages* with no outgoing links are dead ends — the surfer gets stuck, breaking irreducibility. (2) The web is not strongly connected; it has many traps. Both are fixed by the **teleportation** trick. With probability $d \approx 0.85$ the surfer follows a random link as before; with probability $1 - d$ she jumps to a page chosen uniformly at random from all N pages. The modified matrix is

$$G = d\,P + (1-d)\,\tfrac{1}{N}\,\mathbf{J},$$

where $\mathbf{J}$ is the all-ones matrix (the "Google matrix"). Because G has every entry strictly positive, the chain is automatically irreducible and aperiodic, so a unique positive π exists and the surfer converges to it from any start. PageRank is just $\pi G = \pi$.

What is really happening

How is π actually computed for billions of pages? Not by solving a linear system (infeasible at web scale) and not by finding eigenvectors with standard linear-algebra routines. It is computed by *the power iteration of Section 24.1*: pick any starting vector, repeatedly multiply by G, and watch it converge. This is exactly $\pi^{(n)} = \pi^{(0)} G^n$. The damping factor $d = 0.85$ sets the second eigenvalue to about 0.85, so the spectral gap is about 0.15 and the error drops by 0.85 per iteration — roughly 50–100 iterations reach acceptable accuracy. The

whole engine that organized the early web is the convergence theorem plus a matrix-vector product run a few dozen times. This is why the eigenvalue framing in this chapter is not decoration: it *is* the algorithm.

Example 24.3 (A tiny PageRank)**.** Reuse the three-page chain of Example 24.1 (pretend it is already strongly connected, so we can skip teleportation). We found $\pi = (5/19, 8/19, 6/19) \approx (0.263, 0.421, 0.316)$. So page B ranks first, C second, A third. Notice the ranking is *not* simply "count the inbound links": it weights each inbound link by the importance of its source and splits a page's vote across its outbound links. That recursive weighting is the whole point, and the stationary distribution resolves the recursion in one shot.

24.7 Headline Application II: Markov Chain Monte Carlo

Application

Sampling from impossible distributions by building a chain on purpose. In Bayesian statistics, machine learning, and physics you constantly need to draw samples from a distribution π known only up to a constant: $\pi(x) = \tilde{\pi}(x)/Z$, where $\tilde{\pi}$ is computable but the normalizer $Z = \sum_x \tilde{\pi}(x)$ is a sum over astronomically many states (every parameter setting, every spin configuration). You cannot compute Z, so you cannot use it directly — yet you want samples from π.

MCMC turns the whole chapter inside out. Instead of being handed a chain and asked for its π, you are handed a target π and you *construct a chain whose stationary distribution is that* π. Run the chain long enough (past its mixing time) and its state is an (approximate) sample from π. Convergence (Theorem 24.1) guarantees this works; detailed balance (Section 24.4) is how you build it; mixing time (Section 24.5) tells you how long to wait.

The workhorse is the **Metropolis–Hastings** algorithm, and it is detailed balance run in reverse. We *want* $\pi_i P_{ij} = \pi_j P_{ji}$; we *design* P to make it so. The recipe: from state i, *propose* a move to a candidate j using any convenient proposal Q_{ij} (say, perturb the current parameters a little). Then *accept* the move with probability

$$\alpha_{ij} = \min\Big\{1,\ \frac{\tilde{\pi}_j\, Q_{ji}}{\tilde{\pi}_i\, Q_{ij}}\Big\};$$

otherwise stay at i. The realized transition probability is $P_{ij} = Q_{ij}\,\alpha_{ij}$ for $j \neq i$.

Why it works

Why the acceptance rule produces the right π**.** Check detailed balance directly. Suppose, without loss of generality, that the ratio $r = \tilde{\pi}_j Q_{ji}/(\tilde{\pi}_i Q_{ij}) \leq 1$, so $\alpha_{ij} = r$ and $\alpha_{ji} = 1$. Then

$$\pi_i P_{ij} = \pi_i\, Q_{ij}\, \alpha_{ij} = \frac{\tilde{\pi}_i}{Z}\, Q_{ij} \cdot \frac{\tilde{\pi}_j Q_{ji}}{\tilde{\pi}_i Q_{ij}} = \frac{\tilde{\pi}_j Q_{ji}}{Z},$$

while

$$\pi_j P_{ji} = \pi_j\, Q_{ji}\, \alpha_{ji} = \frac{\tilde{\pi}_j}{Z}\, Q_{ji} \cdot 1 = \frac{\tilde{\pi}_j Q_{ji}}{Z}.$$

They are equal, so detailed balance holds and π is stationary by Theorem 24.2. *The unknown normalizer Z cancels* — it appears in both sides and divides out, and inside α only the ratio

$\tilde{\pi}_j/\tilde{\pi}_i$ ever appears. That cancellation is the miracle that makes MCMC possible: you sample from π while being utterly unable to compute Z.

Work it yourself

Convince yourself the cancellation is real, on a baby example. Let the target be $\tilde{\pi} = (1, 3, 2)$ on three states (so the true $\pi = (1/6, 1/2, 1/3)$, but pretend you do not know the denominator 6). Use a symmetric proposal: from any state, propose each of the other two with probability $1/2$, so $Q_{ij} = Q_{ji}$ and the acceptance ratio is just $\tilde{\pi}_j/\tilde{\pi}_i$. Work out the acceptance probability for the proposal $1 \to 2$ (it is $\min\{1, 3/1\} = 1$, always accept — you are climbing toward higher probability) and for $2 \to 1$ (it is $\min\{1, 1/3\} = 1/3$ — you usually refuse to descend). Simulate a few hundred steps by hand or by code and tally the visits; the fractions should approach $(1/6, 1/2, 1/3)$ (Figure 24.6). You will have sampled from π using only the *ratios* of $\tilde{\pi}$.

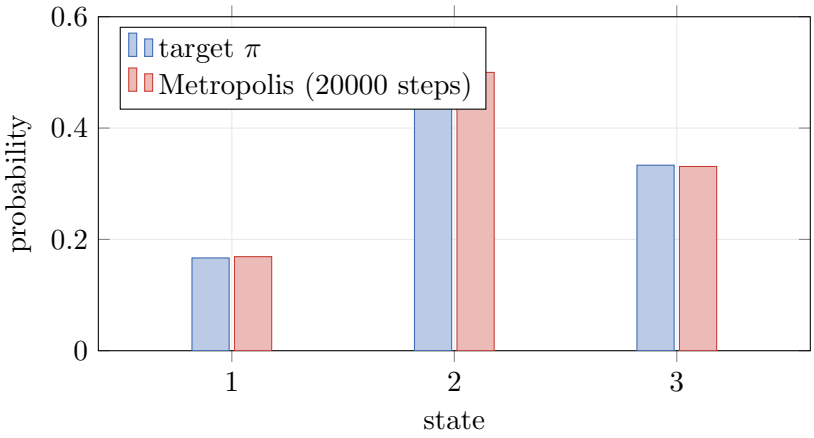


Figure 24.6: The Metropolis sampler of the lab box in action on $\tilde{\pi} = (1, 3, 2)$. Tallying the visited states over 20,000 steps reproduces the target $\pi = (1/6, 1/2, 1/3)$ to within sampling noise — all while never computing the normalizer $Z = 6$. The chain samples the right shape from the ratios alone.

Connection

MCMC is the bridge from this chapter to Bayesian inference (treated in its own chapter). There the target is the posterior $\pi(\theta) \propto \text{likelihood}(\theta) \times \text{prior}(\theta)$, whose normalizer — the "evidence" — is the very integral that makes Bayesian computation hard. Metropolis–Hastings, Gibbs sampling, and Hamiltonian Monte Carlo are all Markov chains engineered to have that posterior as their stationary distribution. Everything you have learned here — existence and uniqueness of π, the convergence theorem licensing "run it long enough," detailed balance as the construction principle, and the spectral gap warning you that a badly mixing chain (a multimodal posterior with separated modes) can lie convincingly — is exactly the theory practitioners lean on every day. The diagnostics they run (trace plots, $\hat{R}$, effective sample size) are empirical probes of the mixing time you met in Section 24.5.

Common trap

Burn-in and the perils of not-yet-mixed chains. A freshly started MCMC chain reflects its (arbitrary) starting point, not π. Samples from the early phase are biased, so practitioners discard an initial *burn-in* segment — an attempt to wait out the mixing time. But there is no reliable way to be *certain* a chain has mixed from the inside: a chain trapped in one mode of a multimodal target looks perfectly converged (flat trace, stable averages) while completely missing the other modes. Convergence diagnostics reduce the risk but cannot eliminate it. The spectral-gap intuition is the right mental safeguard: if you suspect your target has well-separated regions, suspect a bottleneck, and suspect your chain.

24.8 Worked Examples

Example 24.4 (Stationary distribution of a 3×3 chain)**.** Let

$$P = \begin{pmatrix} 0.5 & 0.5 & 0 \\ 0.25 & 0.5 & 0.25 \\ 0 & 0.5 & 0.5 \end{pmatrix}.$$

This is a random walk on a path $1-2-3$ with self-loops — irreducible (you can get anywhere) and aperiodic (self-loops). Solve $\pi P = \pi$ with $\pi = (a, b, c)$:

$$a = 0.5a + 0.25b, \qquad c = 0.25b + 0.5c.$$

The first gives $0.5a = 0.25b$, so $b = 2a$; the third gives $0.5c = 0.25b$, so $b = 2c$, hence $a = c$. Normalizing $a + 2a + a = 4a = 1$ gives $\pi = (1/4,\ 1/2,\ 1/4)$. *Sanity check via detailed balance:* this is a path graph; treating self-loops as not affecting the edge balance, the chain is reversible, and indeed $\pi_1 P_{12} = \frac{1}{4} \cdot \frac{1}{2} = \frac{1}{8} = \frac{1}{2} \cdot \frac{1}{4} = \pi_2 P_{21}$. ✓

Example 24.5 (Mean return time)**.** In the weather chain, how many days on average between rainy days? By part (3) of Theorem 24.1, the mean return time to a state is $1/\pi_i$. Since $\pi_R = 1/3$, the expected gap between consecutive rainy days (counting from one rainy day to the next return) is $1/\pi_R = 3$ days. Likewise a sunny day recurs on average every $1/\pi_S = 1.5$ days. The rarer state has the longer return time — exactly as it should.

Example 24.6 (Build a sampler with Metropolis)**.** You want samples from $\pi \propto (4, 2, 1)$ on states $\{1, 2, 3\}$ (true $\pi = (4/7, 2/7, 1/7)$, pretend 7 unknown). Use the symmetric proposal "flip a coin to propose a neighbor" on the cycle $1-2-3-1$. Acceptance for $1 \to 2$: $\min\{1, 2/4\} = 1/2$. For $2 \to 1$: $\min\{1, 4/2\} = 1$. For $2 \to 3$: $\min\{1, 1/2\} = 1/2$. For $3 \to 2$: $\min\{1, 2\} = 1$. Detailed balance check on edge $\{1, 2\}$ with $Q_{12} = Q_{21} = 1/2$:

$$\pi_1 P_{12} = \tfrac{4}{7} \cdot \tfrac{1}{2} \cdot \tfrac{1}{2} = \tfrac{1}{7}, \qquad \pi_2 P_{21} = \tfrac{2}{7} \cdot \tfrac{1}{2} \cdot 1 = \tfrac{1}{7}. \checkmark$$

Balanced, so the chain's stationary distribution is the target. The asymmetry of acceptance — always climb, sometimes descend — is what concentrates the samples on the high-probability state 1.

24.9 Exercises

Exercise 24.1 (Two-state mixing rate)**.** For the weather matrix $P = \left(\begin{smallmatrix} 0.8 & 0.2 \\ 0.4 & 0.6 \end{smallmatrix}\right)$, verify that the eigenvalues are 1 and 0.4. Starting from $(1, 0)$, find a closed form for $\pi_S^{(n)}$ and confirm it converges to $2/3$ with the error shrinking by a factor of 0.4 each step. (Hint: write $\pi^{(0)}$ as a combination of the left eigenvectors.)

Exercise 24.2 (Find the stationary distribution)**.** Compute the stationary distribution of

$$P = \begin{pmatrix} 0.2 & 0.6 & 0.2 \\ 0.3 & 0.4 & 0.3 \\ 0.5 & 0.1 & 0.4 \end{pmatrix}.$$

Confirm your answer satisfies $\pi P = \pi$ and sums to 1.

Exercise 24.3 (Reducible and periodic)**.** (a) Give an example of an irreducible chain with no self-loops that is periodic, and state its period. (b) Give an example of a chain that has *two* different stationary distributions, and explain which hypothesis of Theorem 24.1 it violates.

Exercise 24.4 (Random walk on a graph)**.** Consider the random walk on the square graph (a 4-cycle $1-2-3-4-1$). Use the degree formula from Example 24.2 to write down π without solving any linear system, then verify $\pi P = \pi$. Is this chain aperiodic? What goes wrong, and how would a single self-loop fix it?

Exercise 24.5 (Detailed balance fails)**.** Show that the cyclic chain $P = \left(\begin{smallmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{smallmatrix}\right)$ has stationary distribution $(1/3, 1/3, 1/3)$ but does *not* satisfy detailed balance. Interpret the failure in terms of the "arrow of time": what does the chain look like played backwards?

Exercise 24.6 (Metropolis by hand)**.** You want to sample from $\tilde{\pi} = (1, 4, 9, 16)$ on states $\{1, 2, 3, 4\}$ using the symmetric "propose a uniform random other state" proposal. Write down the acceptance probability for every ordered pair (i, j), $i \neq j$. Verify detailed balance on the pair $\{2, 4\}$. Which state will the sampler visit most often?

Exercise 24.7 (Spectral gap and bottlenecks)**.** Consider two "rooms" of k states each, densely connected within a room, joined by a single low-probability bridge edge. Argue qualitatively why the second eigenvalue $|\lambda_2|$ is close to 1 and the mixing time is large. Connect this to why a multimodal target is hard for MCMC.

Exercise 24.8 (PageRank with teleportation)**.** Take the two-page web $1 \to 2$, $2 \to 1$ (each page links only to the other). (a) Write the raw transition matrix P and its stationary distribution. (b) Form the Google matrix $G = 0.85\,P + 0.15 \cdot \frac{1}{2}\mathbf{J}$ and find its stationary distribution. (c) Now add a dangling page 3 with no outlinks; explain why P alone has no well-defined surfer chain and how teleportation repairs it.

Exercise 24.9 (Proof: detailed balance gives stationarity)**.** Prove Theorem 24.2 from scratch (do not just cite it): assuming $\pi_i P_{ij} = \pi_j P_{ji}$ for all i, j, show $\pi P = \pi$. Where exactly does the proof use that the rows of P sum to one?

Exercise 24.10 (Time average vs. distribution)**.** For the flip-flop chain $P = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$ started at state 1: (a) Does $\pi^{(n)}$ converge? (b) Does the time-average fraction of visits to state 1 converge, and to what? (c) Reconcile your answers with parts (2) and (3) of Theorem 24.1, identifying which hypothesis fails and which conclusion survives anyway.

Chapter 25

Poisson Processes

So far almost every random object in this book has lived at a single moment. A coin flip, a normal measurement, a Poisson count of typos on a page — each is one snapshot. But the world does not hand us snapshots. It hands us *streams*: customers walking through a door, photons striking a detector, earthquakes rattling a fault, requests hitting a server, mutations accumulating along a strand of DNA. Events arrive, one after another, scattered along the axis of time, and we want a probability model for the whole pattern at once.

The *Poisson process* is that model. It is the most important random process in applied probability — the "hydrogen atom" of stochastic processes. It is what you get when events happen completely at random in time, with no memory and no coordination, at some steady average rate. It is simple enough to compute almost everything by hand, yet rich enough to be the foundation of queueing theory, reliability engineering, insurance mathematics, particle physics, and the analysis of point patterns in space.

This chapter builds the Poisson process three times over, from three different starting points, and then proves that all three describe the *same* object. That triple viewpoint is not pedantic repetition — each lens makes a different property obvious, and the deepest understanding comes from being able to slide between them at will.

25.1 Three ways to see the same thing

Picture a long horizontal time axis and a scattering of dots on it — the instants at which events occur. There are three natural questions you can ask about that picture, and each one is a complete description of it.

1. **The counting view.** Fix a window of time, say the interval $[0, t]$, and ask: *how many* dots fall inside it? Call that number $N(t)$. As t grows, $N(t)$ ticks upward in unit jumps. The whole process is the random staircase $t \mapsto N(t)$.

2. **The interarrival view.** Forget windows. Just read off the *gaps*: how long until the first dot, then how long from the first to the second, and so on. Call these waiting times $T_1, T_2, T_3, \ldots$. Knowing all the gaps is the same as knowing where every dot is.

3. **The infinitesimal view.** Zoom in on a tiny sliver of time of width $\mathrm{d}t$. Ask only one local

question: *what is the chance a dot lands in this sliver*, and does what happens here depend on what happened elsewhere? This view describes the process by its instantaneous behavior, like a differential equation.

Intuition

These are the same information packaged three ways, exactly as a function can be described by its values, its inverse, or its derivative. The counting view is global and discrete (a staircase). The interarrival view is a list of gaps. The infinitesimal view is local and "differential." The Poisson process is the one process on which all three descriptions are as clean and as friendly as possible, and the central point of the chapter is that cleanness in any one view *forces* cleanness in the other two.

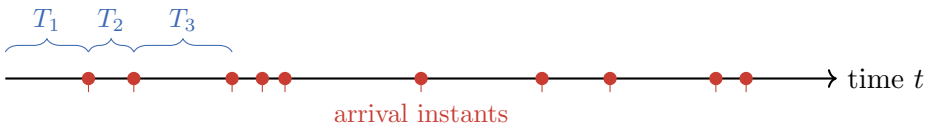


Figure 25.1: A single realization of a point process on the time line. The counting view asks how many red dots lie left of t; the interarrival view records the blue gaps $T_1, T_2, \ldots$; the infinitesimal view asks about a tiny window around one point.

25.2 The infinitesimal definition: events at random, with no memory

We start with the infinitesimal view because it captures most directly what we *mean* by "events happening at random at a steady rate." Three assumptions, each a plain-language statement about local behavior, pin the whole process down.

Definition 25.1 (Poisson process, infinitesimal form)**.** A counting process $\{N(t) : t \geq 0\}$ with $N(0) = 0$ is a *(homogeneous) Poisson process with rate* $\lambda > 0$ if, for a small time increment h,

1. **(one at a time)** the chance of exactly one event in a window of width h is $\lambda h + o(h)$;
2. **(no doubles)** the chance of two or more events in a window of width h is $o(h)$ — vanishingly small compared to h;
3. **(independence)** what happens in disjoint time windows is independent, and the rule above does not depend on *where* the window sits (stationarity).

Here $o(h)$ denotes a quantity that goes to zero faster than h, i.e. $o(h)/h \to 0$ as $h \to 0$.

Read those three lines as a story. Events come *singly* — two of them never share an instant. They come at a *constant intensity* λ: in a window of width h you expect about λh of them, so λ is events per unit time. And the future is *indifferent to the past*: knowing the process has been quiet for an hour tells you nothing about the next second.

What is really happening

The condition $\mathbb{P}(\text{one event in } h) = \lambda h + o(h)$ is the engine of the entire chapter. It says the probability of an arrival is, to first order, *proportional to the length of time you wait* — and proportional to nothing else. Not to how long since the last arrival, not to the clock reading, not to how many have already occurred. "Random" here means precisely this: each instant gets the same, independent, tiny shot at producing an event. Every clean formula below is just this local rule integrated up.

25.2.1 From the local rule to the count

We can extract the distribution of $N(t)$ directly from Definition 25.1. Chop $[0, t]$ into n equal slices of width $h = t/n$. By the "no doubles" rule, when n is enormous each slice holds either zero events or one, and the chance of one is $\lambda h + o(h) \approx \lambda t/n$. By independence, the slices behave like n independent coin flips, each landing "event" with probability $p = \lambda t/n$. So

$$N(t) \;\approx\; \text{Binomial}\big(n,\ \tfrac{\lambda t}{n}\big)\,.$$

Now let $n \to \infty$. This is exactly the limit that *defines* the Poisson distribution — a binomial with many trials and a tiny per-trial probability, holding the mean $np = \lambda t$ fixed. The law of small numbers gives

$$\mathbb{P}\left(N(t) = k\right) \;=\; \lim_{n\to\infty} \binom{n}{k} \big(\tfrac{\lambda t}{n}\big)^k \big(1 - \tfrac{\lambda t}{n}\big)^{n-k} \;=\; e^{-\lambda t}\,\frac{(\lambda t)^k}{k!}.$$

Theorem 25.1 (Counts are Poisson)**.** *In a rate-λ Poisson process, the number of events in any window of length t is*

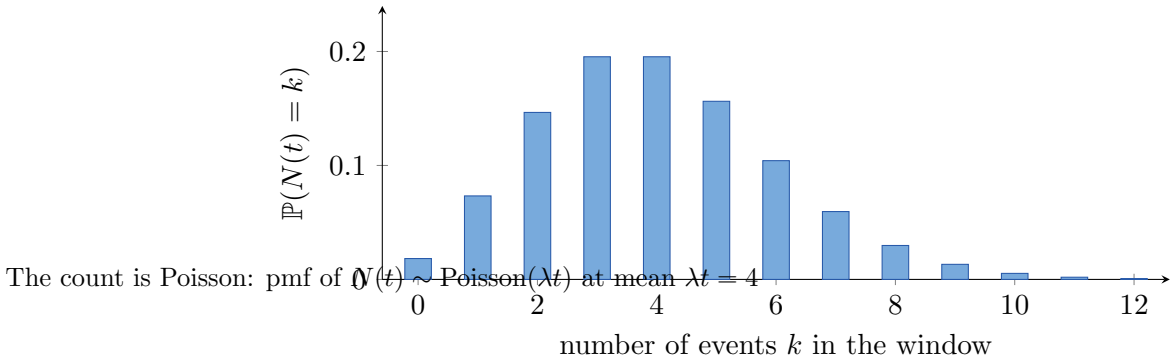
$$N(t) \sim \text{Poisson}(\lambda t), \qquad \mathbb{P}\left(N(t) = k\right) = e^{-\lambda t}\frac{(\lambda t)^k}{k!}, \quad k = 0, 1, 2, \ldots$$

More generally, the number of events in any interval $(a, b]$ is Poisson $\big(\lambda(b-a)\big)$*, and counts in disjoint intervals are independent.*

Why it works

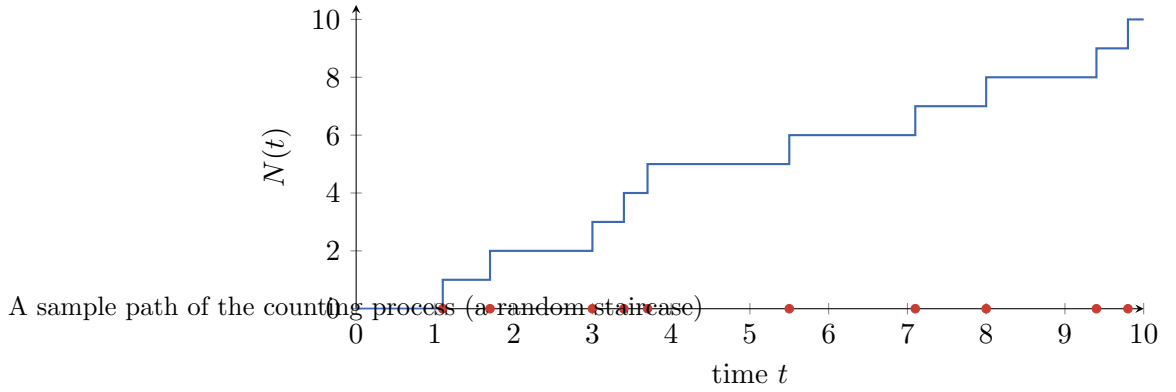
Why does the slicing argument become exact in the limit rather than merely approximate? Two reasons, and both are the $o(h)$ terms doing their job. First, the probability that any slice contains *two or more* events is $o(h)$ per slice, so across all n slices the total chance of a "double" is at most $n \cdot o(t/n) = t \cdot \frac{o(t/n)}{t/n} \to 0$. The single-event approximation is therefore not lossy in the limit. Second, the $o(h)$ correction to the single-event probability likewise washes out. What survives is precisely a Binomial$(n, \lambda t/n)$, and its limit is Poisson. The Poisson distribution is not *assumed* here — it is *forced* by "rare, independent, one-at-a-time, steady-rate."

This is why the distribution and the process share a name. The Poisson *distribution* of Chapter 7 counts rare independent events; the Poisson *process* is what you get when you let those events live on a time line and ask about every window at once. The number λt is the expected count in a window of length t, and $\mathbb{E}\,N(t) = \text{Var}\,N(t) = \lambda t$ — the count's mean and variance both equal λt, the signature of a Poisson variable (Figure 25.2).



The count is Poisson: pmf of $N(t) \sim \text{Poisson}(\lambda t)$ at mean $\lambda t = 4$

Figure 25.2: The number of events in a window of length t follows the Poisson pmf with mean λt (here $\lambda t = 4$). The distribution peaks near its mean and has the equal mean-equals-variance signature: both equal λt.



A sample path of the counting process (a random staircase)

Figure 25.3: The counting view of the same realization as Figure 25.1: $N(t)$ jumps up by one at each red arrival and is flat in between. The Poisson process *is* this staircase.

25.3 The interarrival view: why the gaps are exponential

Now switch lenses. Instead of counting, watch the clock between events. Let T_1 be the time of the first event, T_2 the gap from the first to the second, and so on. What these gaps turn out to be is the most striking fact about the Poisson process.

Theorem 25.2 (Interarrival times are i.i.d. exponential)**.** *In a rate-λ Poisson process the interarrival times $T_1, T_2, T_3, \ldots$ are independent and identically distributed, each* Exponential(λ)*:*

$$\mathbb{P}(T_i > s) = e^{-\lambda s}, \qquad f_{T_i}(s) = \lambda e^{-\lambda s}, \quad s \geq 0,$$

with mean gap $\mathbb{E}\,T_i = 1/\lambda$.

Why it works

The proof of the first gap is a one-liner once you have the counting view. The event "the first arrival happens after time s" is *exactly* the event "no arrivals in $[0, s]$." But the count $N(s)$

is Poisson(λs), so

$$\mathbb{P}(T_1 > s) = \mathbb{P}\big(N(s) = 0\big) = e^{-\lambda s}\frac{(\lambda s)^0}{0!} = e^{-\lambda s}.$$

That is the exponential survival function on the nose. So $T_1 \sim$ Exponential(λ), and differentiating gives the density $\lambda e^{-\lambda s}$. For the later gaps, independence of disjoint intervals means that once the first event lands, the process from that instant onward is a fresh Poisson process that has forgotten its past entirely — so T_2 is, by the identical argument, Exponential(λ) and independent of T_1, and so on for all the gaps.

Notice that the exponential distribution did not get assumed; it got *derived* from "no arrivals yet" plus "counts are Poisson." And the reverse holds too: if you simply *define* a process by dropping i.i.d. Exponential(λ) gaps end to end, you recover a process whose counts are Poisson(λt). This is the equivalence promised in the introduction.

Intuition

The deep reason the gaps are exponential is *memorylessness*. The infinitesimal rule says every instant is an independent fresh shot at an event, indifferent to history. A waiting time built from a memoryless local rule must itself be memoryless: $\mathbb{P}(T > s + u \mid T > s) = \mathbb{P}(T > u)$. And the exponential is the *only* continuous distribution with that property (Chapter 11). So "random events with no memory" and "exponential gaps" are two names for the same thing.

25.3.1 Where the n-th event lands: the Erlang/Gamma

The time of the n-th arrival is the sum of the first n gaps,

$$S_n = T_1 + T_2 + \cdots + T_n.$$

A sum of n independent Exponential(λ) variables is, by the convolution results of Chapter 17, a Gamma with shape n — often called the *Erlang* distribution:

$$S_n \sim \text{Gamma}(n, \lambda), \qquad f_{S_n}(t) = \frac{\lambda^n t^{n-1} e^{-\lambda t}}{(n-1)!}, \quad t \geq 0.$$

There is a consistency check hiding here, the bridge between the interarrival and counting views: the n-th event lands after time t exactly when fewer than n events have occurred by time t. In symbols, $\{S_n > t\} = \{N(t) < n\}$, so

$$\mathbb{P}(S_n > t) = \mathbb{P}\big(N(t) \leq n-1\big) = \sum_{k=0}^{n-1} e^{-\lambda t}\frac{(\lambda t)^k}{k!}.$$

Differentiating that finite sum collapses (the series telescopes) right back to the Erlang density above — the two viewpoints agreeing to the last term.

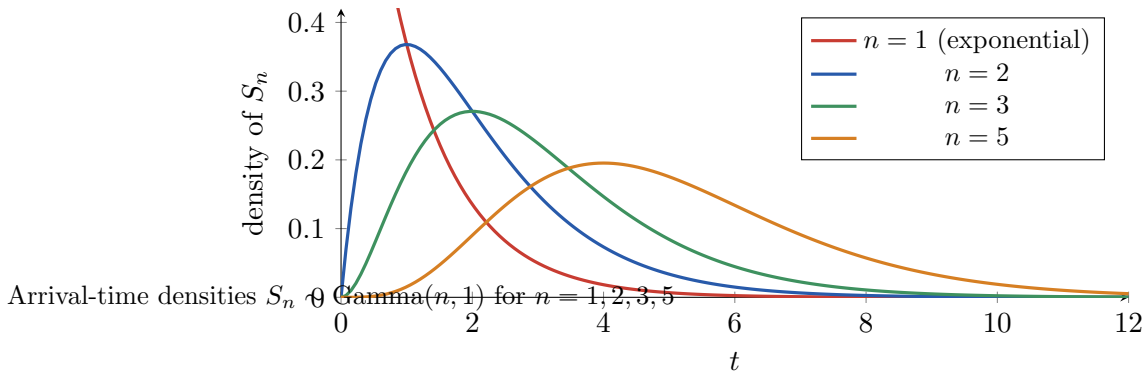


Figure 25.4: The time of the n-th arrival is a sum of n exponential gaps, so its density is Erlang/Gamma. As n grows the hump marches right (the n-th event takes longer) and grows more symmetric — a preview of the central limit theorem acting on the gaps.

Example 25.1 (Phone calls at a help desk)**.** Calls arrive as a Poisson process at $\lambda = 6$ per hour. (a) What is the chance of no calls in a given 10-minute stretch? (b) What is the expected wait until the next call? (c) What is the chance the third call of the day arrives within the first half hour?

Solution. Work in hours. (a) A 10-minute window is $t = 1/6$ hour, so the count is Poisson$(6 \cdot \frac{1}{6}) =$ Poisson(1) and $\mathbb{P}(N = 0) = e^{-1} \approx 0.368$. (b) The gap is Exponential(6) with mean $1/6$ hour $= 10$ minutes. (c) We want $\mathbb{P}(S_3 \le 0.5)$ where $S_3 \sim$ Gamma$(3, 6)$. Equivalently, by the bridge identity, we want $\mathbb{P}\left(N(0.5) \ge 3\right)$ with $N(0.5) \sim$ Poisson(3):

$$\mathbb{P}\left(N(0.5) \ge 3\right) = 1 - e^{-3}\Big(1 + 3 + \tfrac{9}{2}\Big) = 1 - 8.5\, e^{-3} \approx 0.577.$$

Notice how the counting view turned a Gamma integral into a three-term sum — that switch between lenses is the practical payoff of the equivalence.

History

The exponential-gap picture has deep engineering roots. Around 1909 the Danish mathematician and telephone engineer *Agner Krarup Erlang*, working for the Copenhagen Telephone Company, modeled incoming calls as random arrivals to size the number of circuits a switchboard needed. His insight — that call arrivals have no memory and that waiting times are exponential — launched the entire field of *queueing theory*. The Erlang distribution and the unit of telephone traffic (the *erlang*) both carry his name. The same mathematics that once dimensioned telephone exchanges now sizes data-center capacity, call centers, and hospital emergency rooms.

25.4 The two structural superpowers: superposition and thinning

Two operations make the Poisson process behave like a perfect Lego brick: you can *merge* processes and *split* a process, and in both cases you land back on Poisson processes. These are not curiosities — they are the reason the model appears everywhere, because real streams are constantly being pooled and filtered.

Theorem 25.3 (Superposition)**.** *If $N_1, N_2, \ldots, N_m$ are independent Poisson processes with rates $\lambda_1, \ldots, \lambda_m$, then their pooled process $N = N_1 + \cdots + N_m$ — the single stream you get by ignoring labels and watching all events together — is a Poisson process with rate $\lambda_1 + \cdots + \lambda_m$.*

The infinitesimal view makes this almost obvious. In a tiny window, the chance of an event from source i is $\lambda_i h + o(h)$; the chance of an event from *some* source is the sum $(\sum_i \lambda_i)h + o(h)$ (the chance two different sources fire in the same sliver is $o(h)$, negligible). Independence of disjoint windows is inherited from the parts. All three axioms of Definition 25.1 hold for the pool with the summed rate. Rates simply add.

Theorem 25.4 (Thinning)**.** *Take a rate-λ Poisson process and independently label each event "type A" with probability p and "type B" with probability $1-p$, the coin flips independent of everything else. Then the type-A events form a Poisson process of rate λp, the type-B events form a Poisson process of rate $\lambda(1-p)$, and — the surprising part — the two thinned processes are independent of each other.*

Why it works

Why are the two thinned streams independent, when they came from a single shared process? Condition on the total count $N(t) = n$. Given n events, each is independently type A with probability p, so the number of type-A events is Binomial(n, p). Summing over n against the Poisson law of $N(t)$:

$$\mathbb{P}(N_A = j,\ N_B = k) = \underbrace{e^{-\lambda t}\frac{(\lambda t)^{j+k}}{(j+k)!}}_{\mathbb{P}(N=j+k)} \cdot \underbrace{\binom{j+k}{j} p^j (1-p)^k}_{\text{labelling}}.$$

Now do the algebra: $(j+k)!$ cancels, $(\lambda t)^{j+k} = (\lambda t)^j (\lambda t)^k$ splits, and $e^{-\lambda t} = e^{-\lambda p t} e^{-\lambda(1-p)t}$ factors. The joint probability separates into

$$\left[e^{-\lambda p t}\frac{(\lambda p t)^j}{j!}\right] \cdot \left[e^{-\lambda(1-p)t}\frac{(\lambda(1-p)t)^k}{k!}\right],$$

a product of two independent Poisson pmfs with means $\lambda p t$ and $\lambda(1-p)t$. The factorization *is* the independence. This "Poissonizing the trials" trick — a random Poisson number of binomial labels splits into independent Poissons — is one of the more surprising identities in probability.

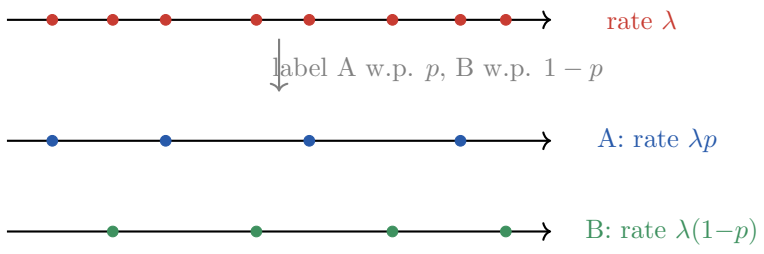


Figure 25.5: Thinning: each event of a rate-λ process is independently sorted into stream A or stream B. The two sub-streams are themselves independent Poisson processes whose rates split the original. Run the picture upward and it reads as superposition: two independent Poisson streams merge into one whose rate is the sum.

Example 25.2 (Splitting a traffic stream)**.** Cars pass a sensor as a Poisson process at $\lambda = 120$ per hour; each independently turns left with probability 0.25. Then left-turners form a Poisson process at $0.25 \cdot 120 = 30$ per hour and straight-throughs at 90 per hour, and these two streams are independent. So the number of left-turners in the next hour is Poisson(30), completely uninformative about the number going straight. The pooled-then-split structure means you can model each lane in isolation.

Application

[Modeling rare events, arrivals, and queues] The Poisson process is the default first model whenever many independent sources each rarely contribute to a common stream — precisely the setting of the law of small numbers.

- **Queues and servers.** An "M/M/1" queue (the workhorse of performance modeling) has *M*arkovian Poisson arrivals and *M*arkovian exponential service. Web-request load, database transactions, and packets on a link are all routinely modeled this way; Chapter 27 develops the queueing theory that sits on top of this chapter.
- **Reliability and insurance.** Failures of a large fleet of independent components, or claims arriving at an insurer, are classic thinned and superposed Poisson streams; the total claim amount over a year is a *compound Poisson* sum.
- **Physics and biology.** Radioactive decay clicks on a Geiger counter, photons hitting a detector, and mutations along a genome are textbook Poisson processes — the independence of disjoint windows reflects genuinely memoryless physics.
- **Networks and ML.** Arrival models for recommendation impressions, ad auctions, and event logs start from Poisson assumptions; Hawkes processes (where events *excite* more events) are the standard generalization when the independence assumption breaks.

25.5 The conditional-uniformity property

The next fact surprises almost everyone the first time, and once it sinks in it demystifies the whole process.

Theorem 25.5 (Order statistics / conditional uniformity)**.** *Condition on $N(t) = n$: that is, suppose you are told exactly n events occurred in $[0, t]$, but not when. Then the n arrival times, regarded as an unordered set, are distributed exactly like n independent* Uniform$(0, t)$ *points. Their sorted values $S_1 < \cdots < S_n$ are the order statistics of n uniform draws.*

In words: *given the count*, the Poisson process scatters its events completely uniformly and with no preference for any sub-interval. The rate λ governs *how many* events occur; once that number is fixed, λ drops out entirely and the events are just uniform confetti.

Why it works

A clean way to see it: chop $[0, t]$ into the slices of the infinitesimal view. Each event is equally likely to fall in any slice (the rate is constant), and the events are placed independently. "Equally likely across the interval, independent" is the definition of i.i.d. uniform. More formally, write the joint density of getting n events near specified ordered times $s_1 < \cdots < s_n$

and no events elsewhere; the rate factors λ^n and the empty-interval probabilities multiply to $e^{-\lambda t}$, and after dividing by $\mathbb{P}(N(t) = n) = e^{-\lambda t}(\lambda t)^n/n!$ everything λ-dependent cancels, leaving the constant density $n!/t^n$ — exactly the joint density of n ordered uniforms.

Paradox

[The inspection / waiting-time paradox] You arrive at a bus stop at a random moment. Buses come as a Poisson process with mean gap $1/\lambda$. How long is the gap of time that *you* land inside? Naively you might say $1/\lambda$, since that is the average gap. But the answer is $2/\lambda$ — on average the bus you wait for took *twice* the mean gap to arrive.

How can the gap you land in be longer than the average gap? Because your random arrival is more likely to fall inside a *long* gap than a short one — a long gap is, by definition, a bigger target. This *length-biased sampling* inflates the gap you observe. Concretely, by memorylessness the time you still wait for the next bus is Exponential(λ) with mean $1/\lambda$; but the time since the *previous* bus is *also* Exponential(λ) with mean $1/\lambda$ (the process is memoryless looking backward too). The gap containing you is the sum, mean $2/\lambda$. The lesson generalizes far beyond buses: when you sample by "whatever interval I happen to be in," you oversample the long ones. It is the same bias that makes your friends seem to have more friends than you do, and that makes class sizes feel larger than the average roster (Figure 25.6).

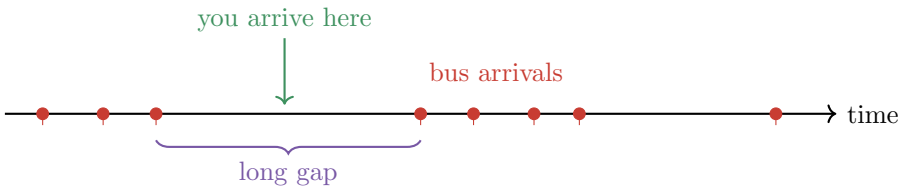


Figure 25.6: Length-biased sampling behind the inspection paradox. A uniformly random arrival time is more likely to land inside a *wide* gap than a narrow one, because wide gaps cover more of the axis. So the gap you observe is, on average, longer than a gap picked at random from the list — mean $2/\lambda$ rather than $1/\lambda$.

25.6 Nonhomogeneous Poisson processes: a rate that breathes

Constant rate is a strong assumption, and reality often violates it. A restaurant gets a lunch rush; a website spikes at 9 a.m.; earthquakes cluster after a big one. We keep the independence and one-at-a-time structure but let the rate vary in time, replacing the constant λ with an *intensity function* $\lambda(t) \geq 0$.

Definition 25.2 (Nonhomogeneous Poisson process)**.** A counting process $\{N(t)\}$ with $N(0) = 0$ is a *nonhomogeneous Poisson process* with intensity $\lambda(t) \geq 0$ if it has independent increments, events occur singly, and the chance of an event in $[t, t + h)$ is $\lambda(t)h + o(h)$. Define the *mean (cumulative) intensity*

$$m(t) = \int_0^t \lambda(u)\,\mathrm{d}u.$$

Then the count in $(a, b]$ is Poisson with mean $m(b) - m(a)$:

$$N(b) - N(a) \sim \text{Poisson}\Big(\textstyle\int_a^b \lambda(u)\,\mathrm{d}u \Big).$$

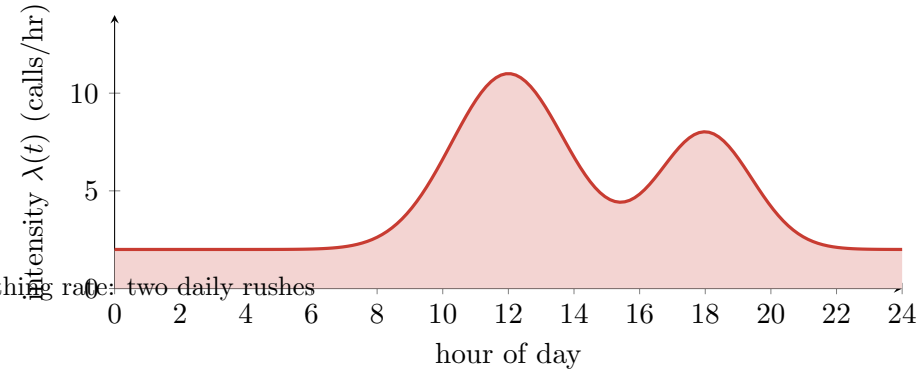
The mean count over a window is no longer rate × width; it is the *area under the intensity curve* over that window. Everything else is inherited: counts in disjoint windows are still independent Poissons, only with window-dependent means.

Connection

The leap from homogeneous to nonhomogeneous is exactly the leap from multiplication to integration. For constant rate the expected count is the rectangle $\lambda \cdot t$; for varying rate it is the integral $\int_0^t \lambda(u)\,\mathrm{d}u$, the accumulated area under the intensity. This is the same move that turns "distance = speed × time" into "distance = $\int$ speed $\mathrm{d}t$" in the companion calculus volume. The intensity $\lambda(t)$ is to the mean-count curve $m(t)$ exactly as velocity is to position: $m'(t) = \lambda(t)$.

What is really happening

There is a slick way to think of a nonhomogeneous process: it is a homogeneous rate-1 process viewed through a *warped clock*. Define new time $\tau = m(t)$. Stretching the axis so that equal amounts of *expected events* occupy equal length turns the breathing rate back into a constant rate of 1. This "time change" is why so many homogeneous results carry over: you prove them once at constant rate, then warp time. It is also exactly how you *simulate* a nonhomogeneous process — generate a rate-1 process and unwarp, or use *thinning* (next box).



A breathing rate: two daily rushes

Figure 25.7: A nonhomogeneous intensity $\lambda(t)$ with a midday and an evening peak. The expected number of events between two times is the *area* under this curve over that span, not a simple rate times width.

Application

[Simulation by thinning] To simulate a nonhomogeneous process with intensity bounded by $\lambda(t) \le \lambda^\star$, run a homogeneous rate-$\lambda^\star$ process (easy: just add up Exponential($\lambda^\star$) gaps) and then *keep* each candidate event at time t with probability $\lambda(t)/\lambda^\star$, throwing the rest away. By the thinning theorem the survivors have exactly the intensity $\lambda(t)$. This *Lewis–Shedler*

thinning algorithm is the standard way to generate arrivals for nonhomogeneous queueing simulations and for fitting point-process models in epidemiology and seismology.

25.7 A fully worked path, end to end

To pin every viewpoint to one concrete computation, run them all on a single example.

Example 25.3 (Server requests)**.** Requests hit a server as a Poisson process at $\lambda = 3$ per second.

(a) Count in a window. Requests in the next 2 seconds: $N(2) \sim \text{Poisson}(6)$, so $\mathbb{E} = 6$, $\text{Var} = 6$, and $\mathbb{P}(N(2) = 4) = e^{-6}6^4/4! = 1296\,e^{-6}/24 = 54\,e^{-6} \approx 0.134$.

(b) Waiting for the next. The next request is Exponential(3) seconds away, mean 1/3 s. Chance you wait more than 1 second: $e^{-3} \approx 0.0498$.

(c) Waiting for the fifth. $S_5 \sim \text{Gamma}(5, 3)$, mean 5/3 s. Chance the fifth request arrives within 1 second is $\mathbb{P}(N(1) \geq 5)$ with $N(1) \sim \text{Poisson}(3)$:

$$1 - e^{-3}\big(1 + 3 + \tfrac{9}{2} + \tfrac{27}{6} + \tfrac{81}{24}\big) = 1 - e^{-3} \cdot 16.375 \approx 0.185.$$

(d) Thinning. Suppose 20% of requests are writes. Writes form a Poisson process at 0.6/s, reads at 2.4/s, and the two streams are independent. Writes in the next 10 s: Poisson(6).

(e) Conditional uniformity. Told that exactly 3 requests arrived in $[0, 2]$, the chance all three came in the first second is, by uniformity, $(1/2)^3 = 1/8$ — the rate λ has vanished entirely from the question.

Common trap

A few recurring stumbles. (1) *Mismatched units.* A rate is per unit time; always convert the window to the same units before forming λt. A rate of 6/hour over 10 minutes gives mean 1, not 6. (2) *Confusing the exponential gap with the Gamma arrival time.* $T_i \sim \text{Exponential}(\lambda)$ is one gap; $S_n = \sum T_i \sim \text{Gamma}(n, \lambda)$ is when the n-th event lands. (3) *Forgetting independence after thinning.* The two thinned streams really are independent — do not double-count shared randomness. (4) *Assuming the gap you observe equals the mean gap*; the inspection paradox says the gap you land in averages $2/\lambda$.

Why this matters

Step back and look at the unity of it. A handful of words — "events at random, no memory, steady rate" — forces the Poisson count, the exponential gap, the Gamma arrival time, the additivity of rates, the splitting into independent streams, and the uniform scattering given the count. Nothing was bolted on; each fact is the same assumption seen from a new angle. That is what a *canonical* object feels like in mathematics: a single idea whose every consequence is inevitable and clean. The Poisson process is the point where "random in time" becomes fully computable.

25.8 Exercises

Exercise 25.1 (Warm-up counts)**.** Emails arrive at $\lambda = 4$ per hour as a Poisson process. Find (a) the probability of exactly 2 emails in the next half hour; (b) the probability of *no* emails in the next 15 minutes; (c) the expected number and the variance of emails in an 8-hour workday.

Exercise 25.2 (Gaps and arrival times)**.** For the email process above, find (a) the mean and standard deviation of the time until the next email; (b) the probability that the gap to the next email exceeds 20 minutes; (c) the distribution and mean of the time until the *third* email.

Exercise 25.3 (The Poisson–exponential bridge)**.** Using only the identity $\{S_n > t\} = \{N(t) < n\}$, show that $\mathbb{P}(S_2 > t) = e^{-\lambda t}(1 + \lambda t)$, and then differentiate to recover the Gamma$(2, \lambda)$ density. Explain in one sentence why this identity is the formal statement that "counting" and "waiting" are the same process.

Exercise 25.4 (Superposition in practice)**.** Three independent ticket machines sell at Poisson rates 5, 8, and 7 per minute. (a) What is the distribution of total sales in a 2-minute window? (b) Given that a sale just occurred, what is the probability it came from the second machine? (Hint: in a tiny window the chance the event is from machine i is $\lambda_i / \sum_j \lambda_j$.)

Exercise 25.5 (Thinning and independence)**.** Customers enter a shop at $\lambda = 30$ per hour; each independently makes a purchase with probability 0.4. (a) Find the distribution of the number of *buyers* in the next hour. (b) Find the distribution of the number of *non-buyers*. (c) If 20 buyers walked in this hour, what does that tell you about the number of non-buyers? Justify with the thinning theorem.

Exercise 25.6 (Conditional uniformity)**.** Exactly 5 meteors are seen during a 4-hour watch modeled as a Poisson process. (a) What is the probability that exactly 2 of them appeared in the first hour? (b) What is the expected number that appeared in the first hour? (c) Why does the rate λ not appear in your answers?

Exercise 25.7 (Nonhomogeneous process)**.** A call center has intensity $\lambda(t) = 10 + 6\sin(\pi t/12)$ calls per hour, t in hours past midnight. (a) Write the expected number of calls between 8 a.m. and noon as an integral and evaluate it. (b) What is the distribution of the number of calls in that window? (c) Sketch why a constant-rate model would mis-size staffing.

Exercise 25.8 (The inspection paradox, quantitatively)**.** Buses arrive as a Poisson process with mean gap 10 minutes. You show up at a uniformly random time. (a) What is the expected time until the next bus? (b) What is the expected time since the previous bus? (c) Hence the expected length of the gap you land in — and explain in words why it exceeds 10 minutes.

Exercise 25.9 (Proof: memorylessness forces exponential gaps)**.** Let $G(s) = \mathbb{P}(T_1 > s)$ be the survival function of the first gap. Argue from the independence and stationarity axioms that $G(s + u) = G(s)\,G(u)$ for all $s, u \geq 0$. Then, assuming G is continuous and decreasing with $G(0) = 1$, prove the only solution is $G(s) = e^{-\lambda s}$ for some $\lambda > 0$. (This is the functional-equation heart of the whole chapter.)

Exercise 25.10 (Compound Poisson, a peek ahead)**.** Claims arrive at an insurer as a Poisson process at $\lambda = 2$ per day, and each claim is an independent amount with mean \$1000 and standard deviation \$400. Let X be the total claim amount in a 30-day month. Using the laws of total expectation and total variance (Chapter 15), find $\mathbb{E}\,X$ and $\operatorname{Var} X$. (Hint: $\mathbb{E}\,X = \lambda T\ \mathbb{E}(\text{amount})$ and $\operatorname{Var} X = \lambda T\ \mathbb{E}(\text{amount}^2)$.)

Chapter 26

Random Walks

A drunk man leaves a bar at midnight. He can barely stand, and certainly cannot navigate. Each step he takes is a coin flip: east or west, with no memory of where he has been and no plan for where he is going. Will he ever stumble home? Or will he wander off into the night, never to return?

This is not an idle riddle. It is one of the most fertile questions in all of probability, and the path from it leads – astonishingly directly – to the physics of diffusing molecules, the price of a stock option, the spread of a gene through a population, the design of search algorithms, and the mathematics of Brownian motion. The humble *random walk* is the bridge between the discrete world of coin flips, which we have studied at length, and the continuous world of motion and noise.

In this chapter we build the random walk from the ground up. We start with the simplest possible version – steps of ± 1 on a line – and we ask the natural questions a curious person asks: where do you end up on average, how far do you spread, do you come back to the start, and what does a typical path actually look like? The answers are sometimes intuitive, sometimes deeply surprising, and always illuminating.

26.1 The Simple Random Walk

Intuition

Forget the formalism for a moment. Picture a token sitting on the integer 0 of a number line. A fair coin is flipped. Heads, the token moves one step right; tails, one step left. Flip again. And again. The position of the token after many flips is a random walk. Everything we do in this chapter is a question about that wandering token.

Let $X_1, X_2, X_3, \ldots$ be independent steps, each equal to $+1$ or -1 with probability $1/2$:

$$\mathbb{P}(X_i = +1) = \mathbb{P}(X_i = -1) = \tfrac{1}{2}, \qquad i = 1, 2, 3, \ldots$$

These are independent and identically distributed (i.i.d.) – the coin has no memory. The *position* of the walk after n steps is the running sum

$$S_n = X_1 + X_2 + \cdots + X_n, \qquad S_0 = 0.$$

Definition 26.1 (Simple symmetric random walk)**.** The sequence $S_0, S_1, S_2, \dots$ defined by $S_0 = 0$ and $S_n = \sum_{i=1}^{n} X_i$, where the X_i are i.i.d. with $\mathbb{P}(X_i = \pm 1) = 1/2$, is the *simple symmetric random walk* on $\mathbb{Z}$. If instead $\mathbb{P}(X_i = +1) = p$ and $\mathbb{P}(X_i = -1) = q = 1 - p$ with $p \neq 1/2$, we call it a *biased* (or asymmetric) random walk.

The word "simple" means the steps have size one; "symmetric" means the coin is fair. We will spend most of our energy on the symmetric case, because that is where the deepest surprises live, but the biased walk is exactly what we need for gambling problems, so we keep it in our pocket.

What is really happening

A random walk is a *cumulative sum of i.i.d. random variables*. That one sentence connects this entire chapter to everything you already know. The Law of Large Numbers and the Central Limit Theorem are theorems about sums S_n – so they are secretly theorems about random walks. When we ask "where is the walk after n steps?" we are asking about the distribution of a sum, and the CLT will hand us a bell curve. The new questions – recurrence, first-passage times, the reflection principle – are the ones that go *beyond* a single snapshot S_n and ask about the whole *path*.

26.1.1 A first look at a path

Before any formulas, let us just *watch*. Figure 26.1 shows several independent sample paths of the symmetric walk over 120 steps. Stare at them. Notice three things. First, the paths sprawl: by step 120 they are scattered over a wide band, not clustered near zero. Second, they are jagged and unpredictable – no path "trends." Third, and most importantly, the band of paths widens like the square root of time, not linearly. We will prove that shape precisely in the next section, but your eye can already see it.

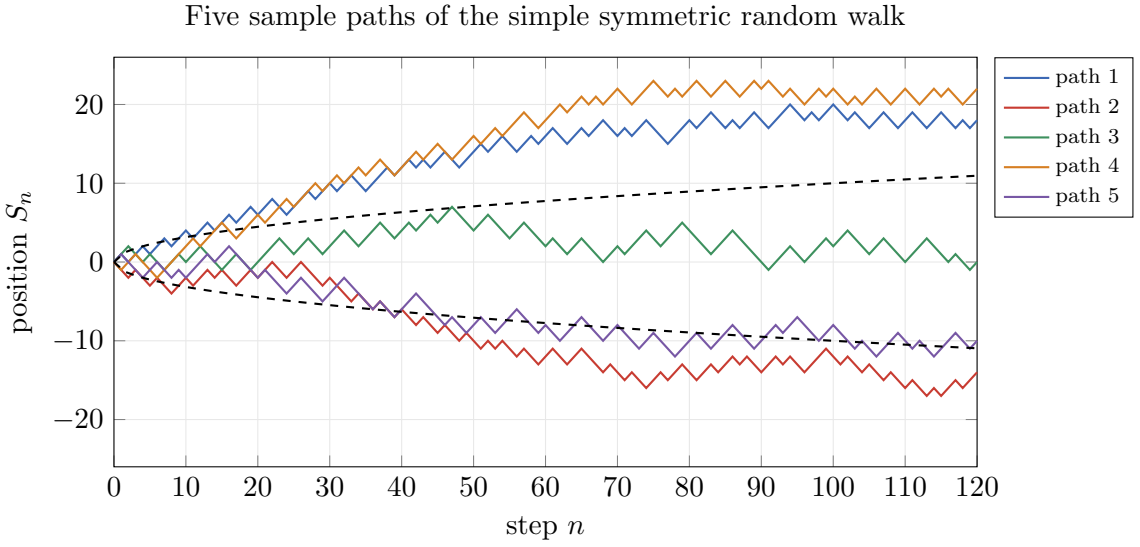


Figure 26.1: Five independent sample paths of the simple symmetric random walk. The dashed curves are $\pm\sqrt{n}$: the typical spread of the walk grows like the *square root* of the number of steps, not linearly. Most paths stay within a band of this width, though any single path can wander outside it.

26.2 Expected Position and Variance

Where is the walk, on average, after n steps? The answer feels almost like a trick.

Theorem 26.1 (Mean and variance of the walk)**.** *For the simple symmetric random walk,*

$$\mathbb{E}[S_n] = 0, \qquad \mathrm{Var}(S_n) = n, \qquad \mathrm{SD}(S_n) = \sqrt{n}.$$

For the biased walk with $\mathbb{P}(X_i = +1) = p$, $\mathbb{P}(X_i = -1) = q = 1 - p$, *writing the drift* $\mu = p - q$ *and* $\sigma^2 = 1 - \mu^2$,

$$\mathbb{E}[S_n] = n\mu = n(p - q), \qquad \mathrm{Var}(S_n) = n\sigma^2 = 4npq.$$

Why it works

Both lines follow from a single idea: expectation is linear, and for independent variables variances add.

The mean. Each step has $\mathbb{E}[X_i] = (+1)p + (-1)q = p - q = \mu$. By linearity of expectation – which needs no independence at all –

$$\mathbb{E}[S_n] = \mathbb{E}\Big[\sum_{i=1}^{n} X_i\Big] = \sum_{i=1}^{n} \mathbb{E}[X_i] = n\mu.$$

For the fair coin $p = q = 1/2$, so $\mu = 0$ and $\mathbb{E}[S_n] = 0$.

The variance. First, a single step. Since $X_i \in \{-1, +1\}$ we have $X_i^2 = 1$ always, so $\mathbb{E}[X_i^2] = 1$ and

$$\mathrm{Var}(X_i) = \mathbb{E}[X_i^2] - (\mathbb{E}[X_i])^2 = 1 - \mu^2 = \sigma^2.$$

For the fair coin $\mu = 0$ so $\mathrm{Var}(X_i) = 1$. Now because the steps are *independent*, the variance of the sum is the sum of the variances:

$$\mathrm{Var}(S_n) = \sum_{i=1}^{n} \mathrm{Var}(X_i) = n\sigma^2.$$

For the fair coin this is just n. Taking square roots gives the standard deviation $\sqrt{n}\,\sigma$, and $\sqrt{n}$ in the fair case.

That $\mathbb{E}[S_n] = 0$ is the trick the symmetric walk plays on your intuition. On average, the drunk goes nowhere. But "on average zero" does not mean "stays near zero." It means the rightward and leftward excursions cancel *in expectation*, the way the average of $+1000$ and -1000 is zero. The right measure of how far the walk roams is not the mean but the standard deviation, and that grows without bound, like $\sqrt{n}$.

Focus

The $\sqrt{n}$ law. After n steps the walk is typically a distance of order $\sqrt{n}$ from the origin. After 100 steps, about 10 away; after 10,000 steps, about 100 away. To get twice as far you must walk four times as long. This sublinear, square-root spreading is the signature of diffusion, and it reappears verbatim in the physics of heat, ink in water, and pollen grains under a microscope.

Example 26.1 (How far is "far"?)**.** A particle takes $n = 10{,}000$ symmetric unit steps. What is the probability it ends up more than 300 units from the origin?

Here $\mathrm{Var}(S_n) = 10{,}000$, so $\mathrm{SD}(S_n) = 100$. A displacement of 300 is exactly 3 standard deviations. By the Central Limit Theorem (the sum of 10,000 i.i.d. bounded steps is very nearly normal), $S_n \approx \mathcal{N}(0, 100^2)$, and

$$\mathbb{P}(|S_n| > 300) \approx \mathbb{P}(|Z| > 3) \approx 0.0027,$$

about a quarter of one percent. Ending up 3000 away (thirty standard deviations) is, for all practical purposes, impossible – even though the walk *could* in principle go straight right the entire time, with probability 2^{-10000}.

Connection

The exact distribution of S_n is a relabelled binomial. If R is the number of rightward steps, then $R \sim \mathrm{Binomial}(n, p)$ and $S_n = R - (n - R) = 2R - n$. So

$$\mathbb{P}(S_n = k) = \mathbb{P}\Big(R = \tfrac{n+k}{2}\Big) = \binom{n}{\frac{n+k}{2}} p^{\frac{n+k}{2}} q^{\frac{n-k}{2}},$$

valid when $n + k$ is even (the walk's parity must match n: after an even number of steps you sit on an even site). Everything about the position S_n is binomial in disguise – including its bell shape, which is the de Moivre–Laplace theorem, the historical seed of the CLT. See the companion *Calculus From the Ground Up* for how that bell shape emerges from Stirling's approximation to the binomial coefficient.

Figure 26.2 shows this binomial pmf for $n = 40$ together with the normal curve it converges to. The match is already excellent.

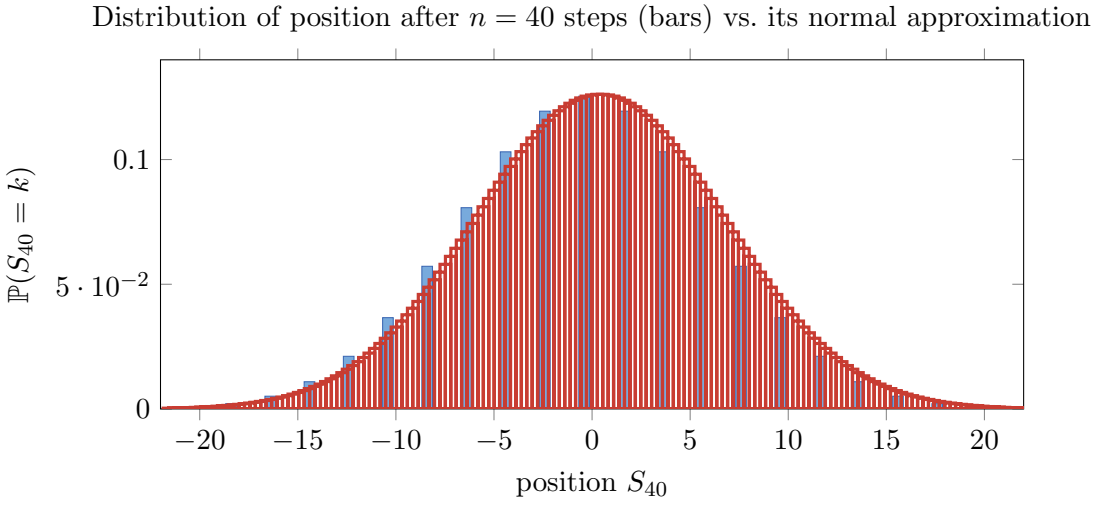


Figure 26.2: The exact distribution of S_{40} (only even sites have positive probability, since parity must match $n = 40$) overlaid with the normal density $\mathcal{N}(0, 40)$. The bars are spaced two apart, so each carries the probability of a width-2 interval; the curve is scaled to match. The CLT is visible to the naked eye.

26.3 Recurrence and Transience: Does the Drunk Get Home?

Now the question that gives this chapter its soul. The walk wanders. Does it ever come *back*? More sharply: with probability 1, will the walk return to its starting point 0?

Definition 26.2 (Recurrence and transience)**.** A random walk is *recurrent* if it returns to its starting point with probability 1 (and hence, by starting afresh each time, returns infinitely often). It is *transient* if there is positive probability it never returns – in which case it visits the origin only finitely many times and eventually drifts away forever.

The answer in one dimension is one of the loveliest facts in probability.

Theorem 26.2 (Recurrence of the 1D symmetric walk)**.** *The simple symmetric random walk on* $\mathbb{Z}$ *is recurrent: it returns to the origin with probability* 1*. In fact it visits* every *integer infinitely often, with probability* 1*.*

Why it works

The standard tool is the *expected number of visits*. Let $N = \sum_{n=0}^{\infty} \mathbf{1}\{S_n = 0\}$ count how many times (including time 0) the walk sits at the origin. By linearity (the indicators need not be independent),

$$\mathbb{E}[N] = \sum_{n=0}^{\infty} \mathbb{P}(S_n = 0).$$

There is a beautiful dichotomy: the walk is recurrent if and only if $\mathbb{E}[N] = \infty$. Why? If returns are certain, the walk returns again and again forever, so $N = \infty$ surely and $\mathbb{E}[N] = \infty$. If there is a chance $1 - u > 0$ of never returning, then the number of returns is geometric with finite mean, so $\mathbb{E}[N] < \infty$. So we just have to decide whether $\sum_n \mathbb{P}(S_n = 0)$ converges.
Return to 0 is only possible after an even number of steps, with $n/2$ rights and $n/2$ lefts. Setting $n = 2m$,

$$\mathbb{P}(S_{2m} = 0) = \binom{2m}{m} \left(\tfrac{1}{2}\right)^{2m}.$$

By Stirling's formula, $\binom{2m}{m} \sim 4^m/\sqrt{\pi m}$, so

$$\mathbb{P}(S_{2m} = 0) \sim \frac{1}{\sqrt{\pi m}}.$$

Therefore

$$\mathbb{E}[N] = \sum_{m=0}^{\infty} \mathbb{P}(S_{2m} = 0) \sim \sum_{m=1}^{\infty} \frac{1}{\sqrt{\pi m}} = \infty,$$

because $\sum 1/\sqrt{m}$ diverges (it is a p-series with $p = 1/2 \le 1$). The expected number of visits is infinite, so the walk is recurrent. □

This calculation deserves a long pause, because the *exact same machine* decides recurrence in every dimension, and the verdict flips on the razor's edge of how fast $\mathbb{P}(S_{2m} = 0)$ decays. In one dimension it decays like $m^{-1/2}$, the sum diverges, and the walk comes home. Watch what happens as we add dimensions.

26.3.1 Two dimensions: still recurrent

In two dimensions the walker stands at a lattice point and, each step, moves north, south, east, or west with probability $1/4$ each. To return to the origin after $2m$ steps, the net horizontal displacement and the net vertical displacement must both be zero. A clean way to see the count: think of the x-coordinate and y-coordinate as *nearly* two independent 1D walks (after a 45° rotation they become exactly independent), so the return probability roughly squares:

$$\mathbb{P}(S_{2m} = 0) \sim \left(\frac{1}{\sqrt{\pi m}}\right)^2 = \frac{1}{\pi m}.$$

Now the series in question is the harmonic series:

$$\mathbb{E}[N] = \sum_m \mathbb{P}(S_{2m} = 0) \sim \sum_m \frac{1}{\pi m} = \infty.$$

It *still* diverges – but just barely, since $\sum 1/m$ is the famous divergent harmonic series sitting right at the boundary. So the 2D walk is also recurrent. The drunk in a city grid *does* get home.

26.3.2 Three dimensions: transient

In three dimensions the walker moves along one of six directions $(\pm x, \pm y, \pm z)$ with probability $1/6$ each. Now the return probability behaves like the cube of the 1D rate:

$$\mathbb{P}(S_{2m} = 0) \sim \frac{C}{m^{3/2}}$$

for a constant $C > 0$. And here the series *converges*:

$$\mathbb{E}[N] = \sum_m \mathbb{P}(S_{2m} = 0) \sim \sum_m \frac{C}{m^{3/2}} < \infty,$$

a p-series with $p = 3/2 > 1$. Finite expected visits means the 3D walk is *transient*: with positive probability it never returns to the origin. The bird flying in three-dimensional space may well wander off forever.

Paradox

Pólya's theorem: "A drunk man finds his way home, but a drunk bird may not." This memorable line is due to the mathematician Shizuo Kakutani, summarizing a 1921 theorem of George Pólya. The symmetric lattice walk is recurrent in dimensions 1 and 2 but transient in dimension 3 and higher.

What makes it a genuine paradox is how *close* the calls are. The entire fate of the walker hinges on whether a single series converges, and the series exponents are $1/2$, 1, $3/2$ in dimensions 1, 2, 3. The boundary between "always returns" and "may escape" falls exactly between $d = 2$ (harmonic series, just diverges) and $d = 3$ (just converges). Two dimensions is recurrent by a hair; three is transient by a hair. Geometry, not luck, decides – there is simply *more room to get lost* in three dimensions.

A concrete number: the probability that the 3D walk ever returns to the origin is about 0.3405. So a drunk bird has roughly a 34% chance of coming home and a 66% chance of being lost to the void. The drunk man's chance is 100%.

History

George Pólya later told the story of how he discovered the theorem. While walking in the woods near Zurich, he kept running into the same young couple – again and again, no matter which path he took. He wondered: was it pure chance, or was something about the structure of paths forcing the encounters? That idle question on a walk became the recurrence theorem of 1921, one of the foundational results of the modern theory of random walks and Markov chains. The lesson is worth keeping: deep mathematics often begins with a concrete, almost childish question – "why do I keep bumping into these people?" – taken seriously.

What is really happening

Why does dimension matter so much? Recurrence is a competition between two forces. The walk *spreads* – after n steps it occupies a cloud of diameter $\sim \sqrt{n}$. In d dimensions that cloud has volume $\sim n^{d/2}$ and contains $\sim n^{d/2}$ lattice sites, so the chance of being on any one particular site (the origin) is $\sim n^{-d/2}$. Summing over all times asks whether $\sum_n n^{-d/2}$ converges, and that flips at $d = 2$. The higher the dimension, the more sites the probability is "diluted" across, and the less likely any specific site – including home – gets revisited. Recurrence is really a statement about whether the spreading cloud is dense enough to keep hitting the same point.

26.4 The Reflection Principle and the Ballot Problem

So far we have asked about the walk's position at a fixed time. The richer questions are about the *path as a whole*: How high does it ever get? When does it first hit a level? How long does it stay positive? The master tool for all of these is a single, gorgeous idea.

Intuition

The reflection principle, in one picture (Figure 26.3). Suppose a path goes from a point A above some horizontal line to a point B also above the line, but *touches* the line somewhere in between. Reflect the portion of the path *after* the first touch across the line. You get a brand-new path from A to B', the mirror image of B. This reflection is a perfect bijection between (paths from A to B that touch the line) and (all paths from A to B'). Counting one set is the same as counting the other – and the reflected set is usually far easier to count.

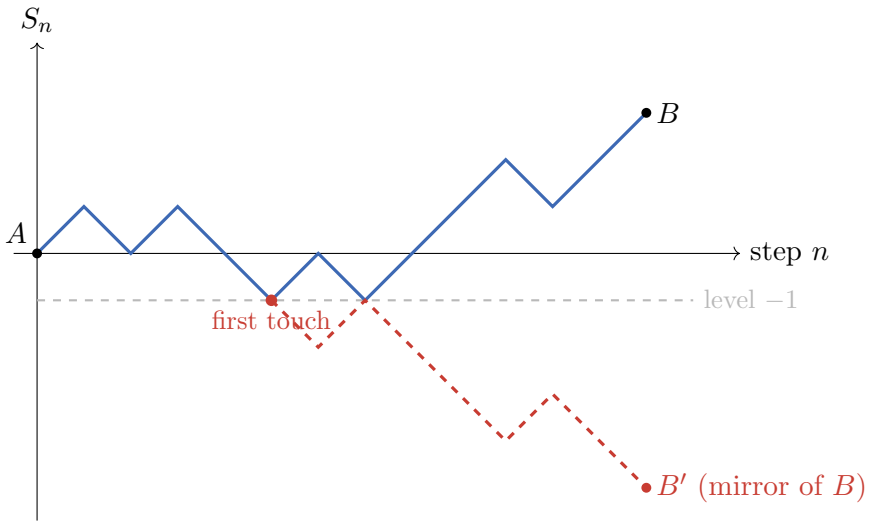


Figure 26.3: The reflection principle. A path from A to B that touches the dashed level is reflected, after its first touch, into a path from A to the mirror point B'. This pairing is exactly one-to-one, turning a hard count ("paths that touch") into an easy one ("all paths to B'").

Let us put it to work on the classic first-passage count.

Theorem 26.3 (Reflection principle / hitting count)**.** *For $a, b > 0$, the number of n-step ± 1 paths from height a to height b that touch level 0 at some time equals the total number of n-step paths from $-a$ to b.*

Why it works

Take any path from $a > 0$ to $b > 0$ that touches level 0. Let τ be the first time it hits 0. Reflect the entire path *before* time τ across the level 0: the starting height a becomes $-a$, the touch point stays at 0 (it is on the mirror), and the path after τ is left unchanged. The result is a path from $-a$ to b. Conversely, any path from $-a$ to b must cross level 0 (it starts below and ends above), and reflecting its piece up to the first crossing returns a path from a that touches 0. The map is its own inverse – a bijection. So the two counts are equal. □

This bijection is the engine behind a charming old puzzle.

Example 26.2 (The ballot problem)**.** In an election, candidate A receives α votes and candidate B receives β votes, with $\alpha > \beta$. The ballots are counted one at a time in random order. What is the probability that A is *strictly ahead* of B throughout the entire count – from the very first vote to the last?

Model the running lead $S_k = (\text{A votes so far}) - (\text{B votes so far})$ as a walk: an A-ballot is a $+1$ step, a B-ballot is a -1 step. After all $\alpha + \beta$ ballots the walk sits at $\alpha - \beta > 0$. "A always strictly ahead" means $S_k > 0$ for all $k \geq 1$, i.e. the walk never returns to (or dips below) 0 after the first step.

The first ballot must be an A (else A is not ahead at step 1). After that, starting from height 1, we need a path to $\alpha - \beta$ that never touches 0. By the reflection principle the count of paths that *do* touch 0 equals the count of paths starting from the mirror height -1 – and a short bookkeeping

with binomial coefficients collapses the whole thing into a startlingly clean answer:

$$\boxed{\mathbb{P}(\text{A always strictly ahead}) = \frac{\alpha - \beta}{\alpha + \beta}.}$$

Why this matters

Look at how clean that is. The probability that A leads wire-to-wire depends only on the *margin* $\alpha - \beta$ relative to the *total* $\alpha + \beta$ – the individual values barely matter. If A wins 600 to 400, the chance she led at every single moment of the count is exactly $200/1000 = 1/5$, no matter how large the electorate. The answer is so simple that it cries out for a slick proof, and indeed there are several (André's reflection argument above; a cyclic-shift "cycle lemma" of Dvoretzky and Motzkin). When an answer is this clean, nature is telling you that you have found the right way to count.

History

The ballot problem was posed and solved by Joseph Bertrand in 1887, and Désiré André published the elegant reflection argument the same year – which is why the technique is often called *André's reflection method*. It was one of the first deep results about the *trajectory* of a random process rather than its endpoint, and it foreshadowed the modern theory of fluctuations: the arcsine laws, the distribution of the maximum, and first-passage times all descend from this single reflection idea.

26.5 Gambler's Ruin

The reflection principle counts paths that avoid a barrier. The gambler's ruin problem asks the complementary, intensely practical question: if there are barriers on *both* sides – you stop when you go broke or when you reach a target – which barrier do you hit, and how long does it take?

Walking question

A gambler starts with $\$i$ and bets \$1 per round on a game that pays even money. She quits if she reaches a fortune of $\$N$ (her goal) or if she hits \$0 (ruin). With a fair game, what is the chance she reaches her goal before going broke? And how many rounds, on average, until the game ends?

This is a random walk on the integers $0, 1, 2, \dots, N$ with two *absorbing* barriers: once you hit 0 or N the walk stops. Let p_i be the probability of reaching N before 0, starting from i.

Theorem 26.4 (Gambler's ruin)**.** *For the fair game ($p = q = 1/2$), starting from i,*

$$\mathbb{P}(\textit{reach } N \textit{ before } 0) = \frac{i}{N}.$$

For a biased game with $p \neq q$ and ratio $r = q/p$,

$$\mathbb{P}(\textit{reach } N \textit{ before } 0) = \frac{1 - r^{\,i}}{1 - r^{\,N}}.$$

The expected duration of the fair game, starting from i, is

$$\mathbb{E}[\textit{number of rounds}] = i\,(N-i).$$

Why it works

Condition on the first step – this is the "first-step analysis" that solves almost every absorbing-walk problem. From state i (with $0 < i < N$) the walk moves to $i+1$ with probability p or to $i-1$ with probability q, so

$$p_i = p\,p_{i+1} + q\,p_{i-1}, \qquad p_0 = 0,\ p_N = 1.$$

For the fair case $p = q = 1/2$ this says $p_i = \frac{1}{2}(p_{i+1} + p_{i-1})$, i.e. each p_i is the *average* of its neighbors. A sequence whose every term is the average of its neighbors is a straight line. With boundary values $p_0 = 0$ and $p_N = 1$, the only such line is $p_i = i/N$.
For the biased case, rearrange into $p(p_{i+1} - p_i) = q(p_i - p_{i-1})$; the consecutive differences $d_i = p_i - p_{i-1}$ satisfy $d_{i+1} = r\,d_i$ with $r = q/p$, so $d_i = d_1 r^{i-1}$. Summing the geometric series of differences and applying $p_0 = 0$, $p_N = 1$ gives $p_i = (1 - r^i)/(1 - r^N)$.
For the expected duration D_i, a parallel first-step analysis gives $D_i = 1 + \frac{1}{2}D_{i+1} + \frac{1}{2}D_{i-1}$ with $D_0 = D_N = 0$ (the $+1$ counts the current round). The function $D_i = i(N-i)$ satisfies both the recurrence and the boundaries: $i(N-i)$ is a downward parabola, and one checks $1 + \frac{1}{2}(i+1)(N-i-1) + \frac{1}{2}(i-1)(N-i+1) = i(N-i)$. □

Example 26.3 (The house always wins – slowly)**.** You sit down at a fair-coin game with \$50, hoping to double to \$100 (so $i = 50$, $N = 100$). Your chance of success is $p_i = 50/100 = 1/2$ – perfectly fair, as it must be. And the expected number of rounds is $i(N-i) = 50 \times 50 = 2500$. A long evening.

Now suppose the game is slightly unfair, as every real casino game is. Take $p = 0.49$, $q = 0.51$, so $r = q/p = 0.51/0.49 \approx 1.0408$. Then

$$p_{50} = \frac{1 - 1.0408^{50}}{1 - 1.0408^{100}} \approx \frac{1 - 7.39}{1 - 54.63} \approx 0.12.$$

A house edge of a mere 2% collapses your chance of doubling up from 50% to about 12%. This is the brutal arithmetic of the casino: a tiny per-bet bias compounds, through r^N, into near-certain ruin for anyone who plays long enough.

Figure 26.4 plots the success probability against the starting stake for both a fair and a slightly unfair game, and the gap between the two curves is the whole story of the casino.

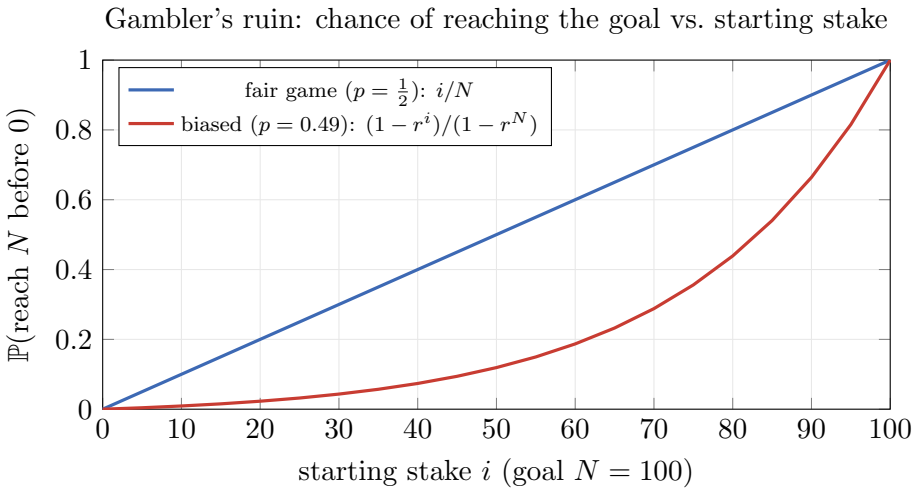


Figure 26.4: Success probability in gambler's ruin. For a fair game it is the straight line i/N. A mere 2% house edge ($p = 0.49$, $r = q/p \approx 1.041$) bends the curve sharply downward: starting at the halfway stake $i = 50$ your chance of doubling falls from 50% to about 12%, matching Example 26.3.

Paradox

The bold-play paradox. Given an unfair game and a fixed goal, what betting strategy maximizes your chance of reaching it? Intuition says "bet small, survive longer, give yourself more chances." Intuition is exactly wrong. The longer you expose your money to a negative-edge game, the more surely the edge grinds you down – duration is the enemy. The optimal strategy is *bold play*: bet as much as you can each time, racing to the goal in as few rounds as possible to minimize the house's chances to apply its edge. Against an unfair game, the rational move is to gamble *harder*, not softer. (Against a *fair* game it makes no difference – timid and bold play both leave your success probability at i/N.)

Connection

Gambler's ruin is your first encounter with a finite *Markov chain* with absorbing states, and the "each value is the average of its neighbors" equation $p_i = \frac{1}{2}(p_{i+1} + p_{i-1})$ is the discrete *Laplace equation*. Its solutions are *harmonic functions*; the absorbing barriers play the role of boundary conditions. This is the same equation that governs steady-state heat and electrostatic potential, and it is exactly why random walks turn out to be a probabilistic method for solving partial differential equations. We develop the Markov-chain machinery – transition matrices, absorption probabilities, expected hitting times – in the next chapter; the linear-algebra view of these systems lives in the companion *Linear Algebra From the Ground Up*.

26.6 From Random Walk to Brownian Motion

We end where the random walk has been pointing all along: at the continuous limit. What happens if we take smaller and smaller steps, faster and faster, in just the right balance?

Intuition

Zoom out from a random-walk path until the individual ± 1 steps blur into a continuous wiggle. To keep the picture from collapsing to a flat line or exploding to infinity, we must rescale space and time together – and the $\sqrt{n}$ law tells us exactly how. If we take n steps in one unit of time, the walk spreads by $\sqrt{n}$; to keep that spread finite as $n \to \infty$ we must shrink each step by a factor of $\sqrt{n}$. Space scales like the square root of time. The limiting object is *Brownian motion*.

Make it precise. Rescale the walk by letting n steps fit into time t and shrinking each step to size $1/\sqrt{n}$:

$$B_t^{(n)} = \frac{1}{\sqrt{n}} S_{\lfloor nt \rfloor}.$$

At time $t = 1$ this is $S_n/\sqrt{n}$, which by the CLT converges to $\mathcal{N}(0,1)$. More is true: the entire rescaled *path* converges (this is Donsker's theorem, the "functional CLT") to a random continuous function B_t with these properties.

Definition 26.3 (Brownian motion)**.** *Standard Brownian motion* $\{B_t : t \geq 0\}$ is the random process with: $B_0 = 0$; independent increments (disjoint time intervals give independent displacements); stationary Gaussian increments, $B_t - B_s \sim \mathcal{N}(0, t-s)$ for $s < t$; and continuous sample paths.

Notice the fingerprints of the random walk all over this definition. Independent increments are the i.i.d. steps. The variance $t - s$ growing linearly in elapsed time is the $\mathrm{Var}(S_n) = n$ law, survived into the limit. The Gaussian shape is the CLT doing its work on each increment. Brownian motion is the simple random walk, viewed from infinitely far away. Figure 26.5 shows one such limiting path.

A Brownian sample path as the scaling limit of a random walk

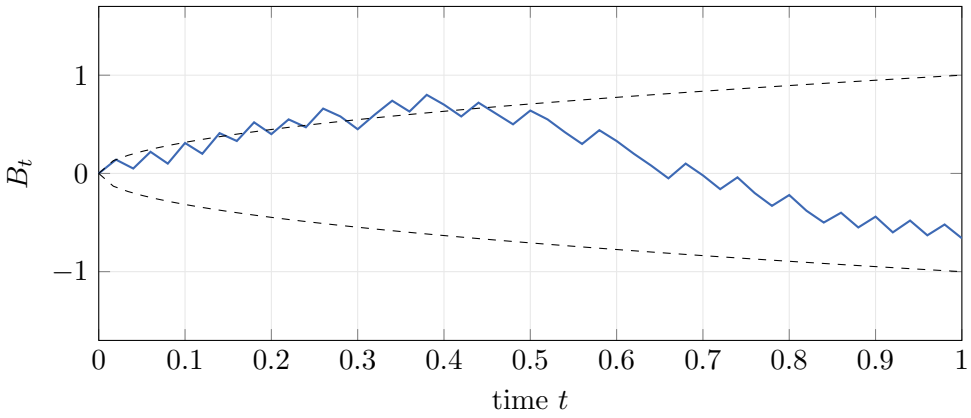


Figure 26.5: A sample path of Brownian motion on $[0, 1]$. The dashed curves $\pm\sqrt{t}$ mark one standard deviation; the path is continuous everywhere yet – as a deeper analysis shows – differentiable nowhere, infinitely jagged at every scale, exactly as the un-zoomable wiggle of the underlying random walk demands.

History

In 1827 the botanist Robert Brown peered through a microscope at pollen grains suspended in water and saw them jiggling ceaselessly, with no apparent cause. For nearly eighty years the "Brownian movement" resisted explanation. Then in his *annus mirabilis* of 1905, Albert Einstein argued that the jiggle was the pollen being bombarded by countless invisible water molecules – a visible random walk driven by molecular collisions – and derived that the mean squared displacement grows *linearly* in time, the continuous echo of $\mathrm{Var}(S_n) = n$. This gave Jean Perrin the means to measure Avogadro's number and clinched the physical reality of atoms. The rigorous mathematical object was constructed by Norbert Wiener in 1923, which is why Brownian motion is also called the *Wiener process*. A drunkard's stagger and a pollen grain's tremble obey the very same law.

Application

Brownian motion in the wild. The random-walk-to-Brownian-motion limit is one of the most useful pipelines in applied mathematics.

- **Finance.** The Black–Scholes model treats the logarithm of a stock price as Brownian motion with drift. Option pricing, hedging, and modern quantitative finance all rest on this foundation – the daily up/down ticks of a price are the random-walk steps.
- **Physics and chemistry.** Diffusion of heat, ink, perfume, and pollutants is Brownian motion of enormous numbers of particles; the $\sqrt{t}$ spreading law sets how fast a drop of dye fills a glass of water.
- **Genetics.** The frequency of a neutral allele in a finite population performs a random walk (the Wright–Fisher model), whose diffusion limit predicts how long until a gene is fixed or lost.
- **Machine learning.** Stochastic gradient descent is a noisy walk through a loss landscape; its continuous-time idealization is a Brownian-driven diffusion, and that lens explains why small-batch noise can help models generalize.
- **Algorithms.** Randomized search, Markov chain Monte Carlo (MCMC), and many randomized algorithms are analyzed as walks on enormous state spaces.

Work it yourself

Simulate it yourself. Generate $n = 1000$ i.i.d. ± 1 steps and plot the cumulative sum – you will see a jagged path like Figure 26.1. Now do it for 100 independent walks and, at each time n, compute the standard deviation across the 100 endpoints; plot that against $\sqrt{n}$ and watch them track. Finally rescale one long walk by dividing positions by $\sqrt{n}$ and squeezing time into $[0, 1]$: the result is a convincing Brownian path. Three short loops turn this whole chapter into a picture on your screen.

Pause

The thread of the chapter. A random walk is a running sum of i.i.d. steps. Its mean stays put ($\mathbb{E}[S_n] = 0$) but its spread grows like $\sqrt{n}$ ($\mathrm{Var}(S_n) = n$). Whether it returns home is decided by a single series and flips with dimension – recurrent in 1D and 2D, transient in

3D. The reflection principle turns hard path-counts into easy ones, giving the ballot problem and first-passage results for free. With absorbing barriers we get gambler's ruin, solved by first-step analysis and the discrete Laplace equation. And scaling the walk down in space and time, guided by the $\sqrt{n}$ law, produces Brownian motion – the same process that jiggles a pollen grain and prices an option. Every one of these is a question about that one wandering token we started with.

26.7 Exercises

Exercise 26.1 (Warm-up positions)**.** For the simple symmetric random walk, compute $\mathbb{P}(S_4 = 2)$ and $\mathbb{P}(S_4 = 0)$ directly by listing or counting the ± 1 step sequences. Confirm your answers against the binomial formula $\mathbb{P}(S_n = k) = \binom{n}{(n+k)/2} 2^{-n}$.

Exercise 26.2 (Mean and variance, biased)**.** A biased walk has $\mathbb{P}(X_i = +1) = 0.6$. Find $\mathbb{E}[S_{100}]$ and $\mathrm{SD}(S_{100})$. Using the normal approximation, estimate $\mathbb{P}(S_{100} > 40)$.

Exercise 26.3 (The $\sqrt{n}$ law in numbers)**.** For the symmetric walk, how many steps must you take before the standard deviation of your position first exceeds 50? Before it exceeds 500? Explain in one sentence why the second answer is one hundred times the first, not ten.

Exercise 26.4 (Return probability)**.** Compute $\mathbb{P}(S_{2m} = 0)$ exactly for $m = 1, 2, 3$ for the symmetric walk, and compare each with the Stirling approximation $1/\sqrt{\pi m}$. Comment on how fast the approximation kicks in.

Exercise 26.5 (Recurrence vs. transience, conceptual)**.** Explain, in your own words and without heavy formulas, why $\sum_n \mathbb{P}(S_n = 0) = \infty$ implies the walk returns to 0 with probability 1, while a finite sum implies positive escape probability. Then state which p-series exponent governs each dimension $d = 1, 2, 3$ and why $d = 2$ is the borderline case.

Exercise 26.6 (Ballot problem)**.** Candidate A gets 7 votes and B gets 3. (a) Using the ballot theorem, find the probability A is strictly ahead throughout the count. (b) Find the probability A is *never behind* (ties allowed). *Hint for (b): the "never behind" probability is given by the closely related formula* $(\alpha - \beta + 1)/(\alpha + 1)$.

Exercise 26.7 (Gambler's ruin, fair)**.** You start with \$30 and play a fair \$1 game, quitting at \$0 or \$100. (a) What is your probability of reaching \$100? (b) What is the expected number of rounds? (c) If instead you start at \$50, by what factor does the expected duration change, and why is it not simply proportional to your starting capital?

Exercise 26.8 (Gambler's ruin, biased)**.** With $p = 0.45$, $q = 0.55$ (so $r = q/p = 11/9$), you start with \$10 and aim for \$20. Compute your probability of success using $p_i = (1 - r^i)/(1 - r^N)$. Compare with the fair-game value $i/N = 1/2$ and comment on how a 10% edge per bet translates into the overall outcome.

Exercise 26.9 (First-step analysis from scratch)**.** Consider a walk on states $0, 1, 2, 3$ with absorbing barriers at 0 and 3, fair steps in between. Write down the equations $p_i = \frac{1}{2} p_{i+1} + \frac{1}{2} p_{i-1}$ with $p_0 = 0$, $p_3 = 1$, and solve the small linear system by hand for p_1 and p_2. Verify they match i/N.

Exercise 26.10 (Scaling to Brownian motion)**.** Define $B_t^{(n)} = S_{\lfloor nt \rfloor}/\sqrt{n}$. (a) Show that $\mathbb{E}[B_t^{(n)}] = 0$ and $\mathrm{Var}(B_t^{(n)}) \to t$ as $n \to \infty$. (b) For $s < t$, argue that $B_t^{(n)} - B_s^{(n)}$ is asymptotically independent

of $B_s^{(n)}$ and has variance approaching $t - s$. Identify which property of Brownian motion each part establishes.

Exercise 26.11 (Proof: maximum of the walk)**.** Use the reflection principle to prove that for the symmetric walk and any integer $b > 0$,

$$\mathbb{P}\Big(\max_{0 \le k \le n} S_k \ge b\Big) = 2\,\mathbb{P}(S_n > b) + \mathbb{P}(S_n = b).$$

Hint: a path reaching level b either ends at or above b, or ends below b after touching it; reflect the latter to pair it with a path ending above b.

Exercise 26.12 (Expected duration is a parabola)**.** Verify directly that $D_i = i(N - i)$ solves the duration recurrence $D_i = 1 + \tfrac{1}{2}D_{i+1} + \tfrac{1}{2}D_{i-1}$ with $D_0 = D_N = 0$. Then explain intuitively why the game lasts longest when you start in the middle ($i = N/2$) and is shortest near a barrier.

Chapter 27

Continuous-Time Markov Chains and Queues

So far our Markov chains have ticked like a clock. At each integer time $n = 0, 1, 2, \ldots$ the system looks around, rolls some dice, and jumps to a new state. That discrete metronome is a fiction of convenience. Real systems do not wait for a bell. A customer walks into a coffee shop at 10:04:17, not at "step 7." A radioactive nucleus decays at some real-valued instant. A router receives a packet, a phone call connects, a machine breaks down — all at messy, continuous times that no integer clock can pin down.

This chapter rebuilds the Markov idea in *continuous time*. A single object called the *generator matrix* captures all the dynamics, and from it we will derive the steady-state behavior of *queues* — waiting lines — which sit at the heart of every service system, computer network, hospital, and call center. We will meet the M/M/1 queue, watch a line of waiting customers settle into a geometric equilibrium, and prove Little's law, a relationship so simple and so general that it feels like a magic trick until you see why it cannot be otherwise.

We lean on two earlier ideas. From the chapters on Markov chains, keep the picture of states and transition probabilities, and the notion of a stationary distribution π that the chain settles into. From the chapter on Poisson processes, keep the *exponential distribution* and its defining magic: *memorylessness*. Continuous-time Markov chains are, in a precise sense, what you get when you glue an ordinary Markov chain together with exponential waiting times. That gluing is the whole story, and we build it up from scratch.

27.1 From Discrete Ticks to Continuous Flow

Intuition

Picture a tiny machine that is either **working** or **broken**. In discrete time we would say: each minute, if it is working it breaks with probability 0.01; if it is broken it gets fixed with probability 0.05. That is a fine model, but it has an arbitrary unit baked in — the minute. Why not the second? The half-minute? If we shrink the time step, we must shrink the

probabilities to match, and in the limit the natural object is not a *probability per step* but a *rate per unit time*. "This machine breaks at rate 0.01 per minute" makes sense at every time scale at once. Rates, not per-step probabilities, are the native language of continuous time.

Making the limit explicit shows exactly where the exponential distribution comes from. Suppose a working machine breaks, in each tiny interval of length h, with probability approximately λh (proportional to the length of the interval — twice the time, twice the chance, for small h). Cut the first t units of time into $n = t/h$ slices. The probability of *surviving* all of them, if breakdowns in disjoint slices are independent, is

$$\mathbb{P}(\text{still working at time } t) \;\approx\; (1-\lambda h)^n \;=\; \left(1-\frac{\lambda t}{n}\right)^n \;\xrightarrow[n\to\infty]{}\; e^{-\lambda t}.$$

That is the survival function of an Exponential(λ) random variable. The lifetime of the machine before its first breakdown is exponential, and the parameter λ is exactly the *rate* we started with. This is no accident; it is the engine of the entire chapter.

Why it works

Why must the holding time be exponential, and nothing else? Because we insisted on the **Markov property in continuous time**: the future depends on the present state alone, not on how long we have already been sitting in it. If the time spent in a state had any other distribution — say, "a machine that has run a long time is more likely to break soon" — then knowing the *age* of the current sojourn would tell you something about the future, violating memorylessness. The exponential distribution is the *unique* continuous distribution with no memory:

$$\mathbb{P}(T > s+t \mid T > s) = \mathbb{P}(T > t) \quad \text{for all } s, t \geq 0.$$

So "continuous-time Markov" forces "exponential holding times." The two are the same demand spoken in two languages.

27.1.1 The two-part recipe

Here is the cleanest way to picture a continuous-time Markov chain (CTMC), and we will return to it constantly. The process sits in some state i. Two things happen:

1. **It waits.** The time it spends in state i (the *holding time* or *sojourn time*) is exponential with some rate q_i that depends only on i. On average it waits $1/q_i$.

2. **It jumps.** When the holding time expires, it jumps to a new state $j \neq i$ chosen by a probability P_{ij} (an ordinary discrete transition, with $\sum_{j\neq i} P_{ij} = 1$).

Then it does the same thing from j, and so on. The embedded sequence of states it visits — ignoring how long it stayed — is just an ordinary discrete-time Markov chain, called the *jump chain* or *embedded chain*. The continuous-time part lives entirely in the exponential waiting times glued onto each visit.

What is really happening

A CTMC is a discrete Markov chain wearing exponential clothing. Strip away the random sojourn times and you have a familiar jump chain on the same states. Add them back and you have a process that lives in real time. Everything subtle about CTMCs — the generator, the forward equations, the balance conditions — is bookkeeping that keeps these two layers (where it jumps, how long it stays) consistent with one another.

27.2 Rates and the Generator Matrix

The two-part recipe (a rate q_i to leave each state, plus jump probabilities P_{ij}) is intuitive but clumsy to compute with. A single object packages both and makes the algebra manageable: the *generator matrix* Q, also called the *rate matrix* or *infinitesimal generator*.

Intuition

The transition matrix of a discrete chain answers "where will I be *next step*?" But in continuous time there is no "next step" — time flows. So instead the generator answers a derivative question: "*how fast* is probability flowing out of where I am, and into each other state, *right now*?" It is the time-derivative of transition probability at $t = 0$. Rates, not probabilities — and that shift of viewpoint is the whole definition.

Definition 27.1 (Transition rates and generator)**.** For a CTMC, define the *off-diagonal rate* from state i to state $j \neq i$ as

$$q_{ij} \;=\; \lim_{h \to 0^+} \frac{\mathbb{P}(X_{t+h} = j \mid X_t = i)}{h},$$

the instantaneous rate of jumping $i \to j$. Define the *total exit rate* from i as $q_i = \sum_{j \neq i} q_{ij}$, and set the diagonal entry $q_{ii} = -q_i$. The matrix $Q = (q_{ij})$ is the *generator*. Its defining properties are

$$q_{ij} \geq 0 \ (i \neq j), \qquad q_{ii} = -\sum_{j \neq i} q_{ij}, \qquad \text{so every row of } Q \text{ sums to } 0.$$

The two viewpoints connect cleanly. The holding-time rate in state i is the total exit rate $q_i = \sum_{j \neq i} q_{ij}$, so the sojourn time is Exponential(q_i). And when the chain does leave, it goes to j with probability

$$P_{ij} \;=\; \frac{q_{ij}}{q_i},$$

the fraction of the total outflow rate that heads toward j. So Q contains *both* parts of the recipe: the row sums of off-diagonals give the holding rates, and their proportions give the jump probabilities.

Walking question

Suppose state 1 has $q_{12} = 3$ and $q_{13} = 1$ (rates per hour). How long, on average, does the chain stay in state 1, and where does it go when it leaves?
The total exit rate is $q_1 = 3 + 1 = 4$ per hour, so the mean holding time is $1/4$ hour $= 15$ minutes. When it leaves, it goes to state 2 with probability $3/4$ and to state 3 with probability $1/4$. The faster a particular exit "ticks," the more likely it is to be the one that fires — a race

between exponential alarm clocks, won most often by the fastest.

Connection

The race-of-exponentials picture is worth keeping. Imagine that from state i, every possible target j has its own independent exponential alarm clock with rate q_{ij}. The chain jumps when the *first* clock rings. Two facts from the Poisson/exponential chapter make this work: (i) the minimum of independent exponentials with rates q_{ij} is itself exponential with rate $\sum_j q_{ij} = q_i$; and (ii) the probability that clock j is the winner is q_{ij}/q_i, independent of *when* the winner rings. So "waiting Exponential(q_i) then jumping by P_{ij}" and "racing independent clocks" are the same process. The generator is just the table of all the clock rates.

27.2.1 From the generator to transition probabilities

Let $P_{ij}(t) = \mathbb{P}(X_{t+s} = j \mid X_s = i)$ be the probability of being in j a time t later, and gather these into a matrix $P(t)$. For small h, the generator definition says

$$P(h) \;\approx\; I + Qh,$$

since the diagonal loses probability at rate q_i and each off-diagonal gains it at rate q_{ij}. Chaining two short intervals, $P(t+h) = P(t)P(h) \approx P(t)(I+Qh)$, gives

$$\frac{P(t+h) - P(t)}{h} \approx P(t)Q \quad\Longrightarrow\quad P'(t) = P(t)\,Q.$$

This is the *Kolmogorov forward equation* (the mirror image $P'(t) = QP(t)$ is the *backward equation*). Its solution is the matrix exponential

$$\boxed{P(t) = e^{Qt} = \sum_{k=0}^{\infty} \frac{(Qt)^k}{k!}\,.}$$

Connection

Compare with the scalar story. The ordinary differential equation $p'(t) = -\lambda p(t)$ has solution $p(t) = p(0)e^{-\lambda t}$ — exponential decay, the survival curve from Section 27.1. The CTMC obeys exactly the same law, only now λ is replaced by the matrix $-Q$ and the scalar exponential becomes the *matrix exponential*. If you have met matrix exponentials in a linear algebra course (eigenvalues, diagonalization, e^{Qt} acting through eigenvectors), this is precisely where they earn their keep in probability. The generator's eigenvalues control how fast the chain forgets its start and relaxes to equilibrium.

Example 27.1 (A two-state machine)**.** Let state 0 = "broken," state 1 = "working." A broken machine is repaired at rate μ, a working machine breaks at rate λ. The generator is

$$Q = \begin{pmatrix} -\mu & \mu \\ \lambda & -\lambda \end{pmatrix}.$$

Each row sums to zero, as it must. The holding time when broken is Exponential(μ) (mean repair time $1/\mu$); when working it is Exponential(λ) (mean uptime $1/\lambda$). Solving $\pi Q = 0$ with $\pi_0 + \pi_1 = 1$ (we justify this stationarity equation in Section 27.7) gives the long-run fraction of time working,

$$\pi_1 = \frac{\mu}{\lambda + \mu}, \qquad \pi_0 = \frac{\lambda}{\lambda + \mu}.$$

If repairs are fast (μ large) relative to failures (λ small), the machine is up almost all the time — the ratio $\mu/(\lambda+\mu)$ is exactly the *availability* an engineer would quote. Notice it depends only on the rates, not on the precise distribution shapes beyond their means; that robustness is a recurring theme.

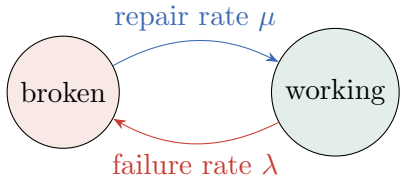


Figure 27.1: The two-state machine of Example 27.1. The chain hops between broken and working; the long-run fraction of time working is $\mu/(\lambda+\mu)$, larger when repairs (rate μ) outpace failures (rate λ). This is the smallest interesting CTMC and the building block of reliability models.

27.3 Birth-Death Processes: Counting Up and Down

A vast and useful family of CTMCs lives on the states $\{0, 1, 2, \dots\}$ — think of them as a *population size*, a *number of customers*, or a *queue length* — and can only move to a neighbor: up by one (a *birth*, or an arrival) or down by one (a *death*, or a departure). These are *birth-death processes*, and they are the workhorses of queueing theory.

Definition 27.2 (Birth-death process)**.** A birth-death process on $\{0, 1, 2, \dots\}$ has *birth rates* $\lambda_n \geq 0$ (rate of going $n \to n+1$) and *death rates* $\mu_n \geq 0$ (rate of going $n \to n-1$, with $\mu_0 = 0$ since the population cannot go negative). All other rates are zero. The generator is tridiagonal:

$$q_{n,n+1} = \lambda_n, \qquad q_{n,n-1} = \mu_n, \qquad q_{n,n} = -(\lambda_n + \mu_n).$$

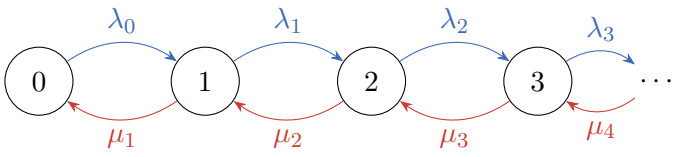


Figure 27.2: A birth-death process. Blue arrows are births (arrivals) at rates λ_n; red arrows are deaths (departures) at rates μ_n. The chain steps only to neighbors, which is what makes its equilibrium so clean.

27.3.1 Detailed balance and the product formula

What is the long-run distribution π_n over states? Here the neighbor-only structure pays off spectacularly. In equilibrium, consider the "cut" between states $\{0, \dots, n\}$ and $\{n+1, n+2, \dots\}$. The

only way to cross it upward is the birth $n \to n+1$; the only way to cross it downward is the death $n+1 \to n$. In steady state the flow across any such cut must balance — whatever goes up must come back down, or probability would pile up on one side forever. This gives the *detailed balance* equations:

$$\pi_n \lambda_n \;=\; \pi_{n+1}\,\mu_{n+1} \qquad \text{for all } n \geq 0.$$

Why it works

Why does flow have to balance across every cut? Imagine a wall between "$\leq n$" and "$\geq n+1$." Probability mass leaks across it rightward at rate $\pi_n \lambda_n$ (you must be in state n, and births fire at rate λ_n) and leftward at rate $\pi_{n+1}\mu_{n+1}$. If these were unequal, the total probability on one side would be steadily growing — but in a *stationary* distribution nothing changes with time, by definition. So the net flow across the wall is zero. For birth-death chains this cut argument gives the equilibrium directly, no matrix algebra required. (The same trick works for any chain whose graph is a tree, because every edge is a cut.)

Solving the recursion $\pi_{n+1} = \pi_n \cdot (\lambda_n/\mu_{n+1})$ telescopes into a clean product:

$$\boxed{\pi_n = \pi_0 \prod_{k=0}^{n-1} \frac{\lambda_k}{\mu_{k+1}},} \qquad \pi_0 = \left(1 + \sum_{n=1}^{\infty} \prod_{k=0}^{n-1} \frac{\lambda_k}{\mu_{k+1}}\right)^{-1}.$$

The normalizing constant π_0 just makes the π_n sum to 1; it exists (the chain has a steady state) precisely when that infinite sum converges. Hold onto this product formula — the M/M/1 queue is the simplest case, where all the birth rates are equal and all the death rates are equal, and the product collapses into a geometric distribution.

27.4 The M/M/1 Queue

Now the centerpiece. A single server — one barista, one CPU, one checkout lane — handles customers one at a time. We make the cleanest possible assumptions, encoded in the cryptic name **M/M/1**:

- **First M** (*Markovian arrivals*): customers arrive according to a Poisson process with rate λ. Equivalently, the gaps between arrivals are i.i.d. Exponential(λ).
- **Second M** (*Markovian service*): each customer's service time is Exponential(μ), independent of everything else.
- **1**: a single server.

The "M" stands for *memoryless* (sometimes said to honor Markov), and that memorylessness is exactly what makes the queue length a Markov chain. Let X_t be the number of customers in the system at time t (those waiting plus the one in service). Because arrivals come at constant rate λ and service completes at constant rate μ whenever someone is being served, X_t is a birth-death process with

$$\lambda_n = \lambda \text{ (for all } n \geq 0), \qquad \mu_n = \mu \text{ (for all } n \geq 1).$$

Figure 27.4 shows a typical realization of X_t: a staircase that jumps up at arrivals and down at service completions.

History

This model is the founding object of queueing theory, born of a very practical worry: how many telephone lines does a town need? Around 1909 the Danish engineer **Agner Krarup Erlang**, working for the Copenhagen Telephone Company, modeled incoming calls as a Poisson process and call durations as random, and computed how often callers would find every line busy. His formulas are still called *Erlang B* and *Erlang C*, and the unit of telephone traffic, the *erlang*, bears his name. Queueing theory thus began not in a mathematics department but in a phone exchange, trying to keep the citizens of Copenhagen from hearing a busy signal. The whole modern apparatus of network and server provisioning descends from Erlang's wish to size a switchboard correctly.

27.4.1 The traffic intensity and steady state

Define the *traffic intensity* (or utilization)

$$\rho \;=\; \frac{\lambda}{\mu},$$

the ratio of how fast work arrives to how fast it can be done. This single dimensionless number controls everything. Plug the constant rates into the birth-death product formula:

$$\pi_n = \pi_0 \prod_{k=0}^{n-1} \frac{\lambda}{\mu} = \pi_0\, \rho^{\,n}.$$

For the π_n to sum to one we need $\sum_n \rho^n < \infty$, i.e. $\rho < 1$: *the server must be faster than the arrivals.* When $\rho < 1$ the geometric series sums to $1/(1-\rho)$, so $\pi_0 = 1-\rho$ and

$$\boxed{\pi_n = (1-\rho)\,\rho^{\,n}, \quad n = 0, 1, 2, \ldots}$$

The number in the system has a *geometric distribution.* The probability the server is idle is $\pi_0 = 1-\rho$; the long-run fraction of time it is busy is ρ — which is why ρ is also called the utilization.

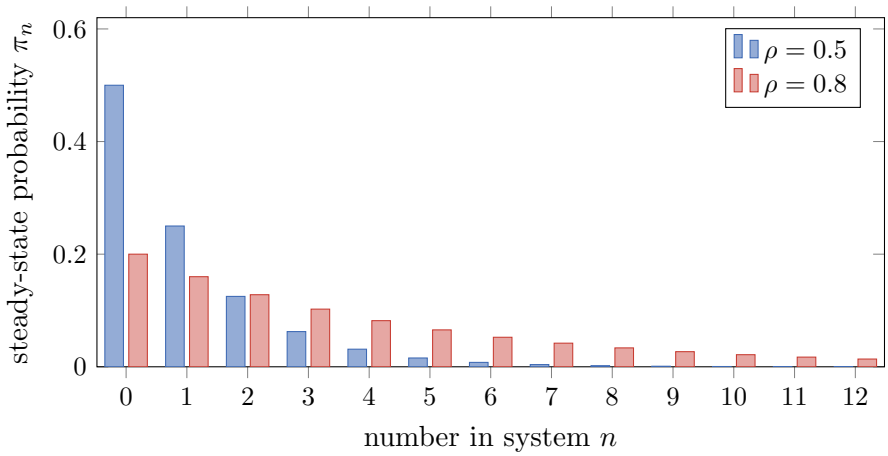


Figure 27.3: The steady-state queue length of an M/M/1 system is geometric, $\pi_n = (1-\rho)\rho^n$. At low load ($\rho = 0.5$, blue) the system is usually nearly empty. At high load ($\rho = 0.8$, red) the distribution has a long tail — long lines become common. As $\rho \to 1$ the tail stretches without bound.

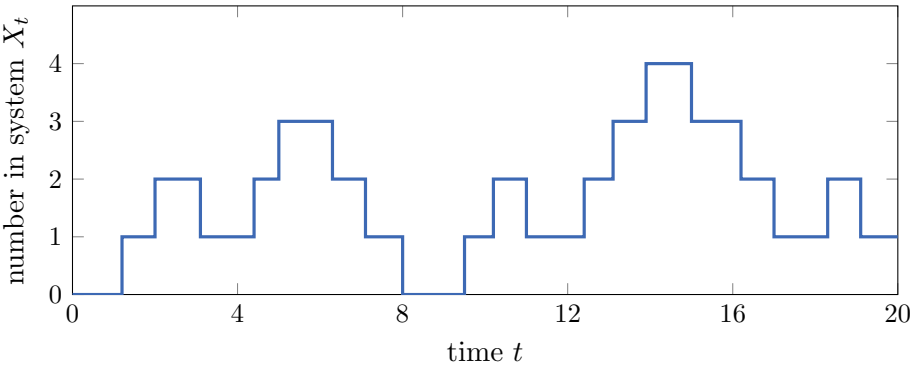


Figure 27.4: A sample path of the M/M/1 queue length X_t: a right-continuous staircase that steps up by one at each arrival and down by one at each service completion. Holding times in each level are Exponential (rate λ up, rate μ down once busy); the steady-state histogram of the heights is the geometric law of Figure 27.3.

27.4.2 Performance measures

From the geometric distribution we read off how the system performs. The mean number in the system is the mean of a geometric:

$$L \;=\; \mathbb{E}[X] \;=\; \sum_{n=0}^{\infty} n(1-\rho)\rho^{n} \;=\; \frac{\rho}{1-\rho}.$$

This is the formula that should keep every system designer awake at night. Watch what happens as the server gets busy:

$$\rho = 0.5 \Rightarrow L = 1, \qquad \rho = 0.9 \Rightarrow L = 9, \qquad \rho = 0.99 \Rightarrow L = 99.$$

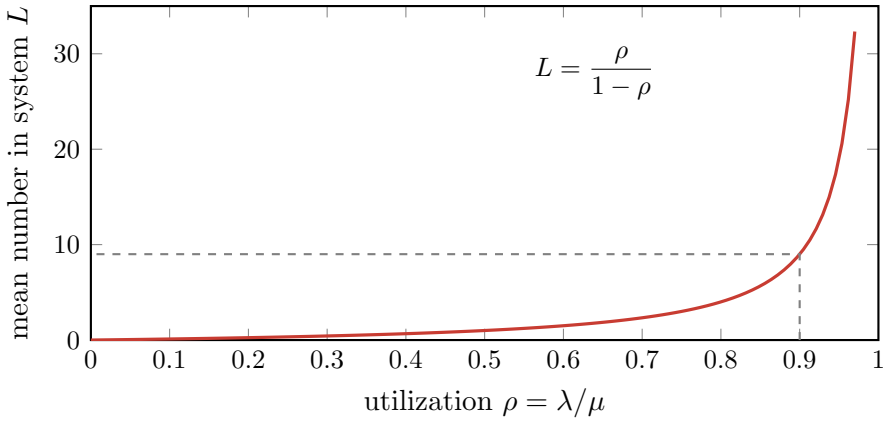


Figure 27.5: The mean number in an M/M/1 queue explodes as utilization approaches 1. The dashed lines mark $\rho = 0.9$, where $L = 9$. The blow-up is hyperbolic: pushing utilization from 90% to 99% multiplies the average backlog tenfold. This nonlinearity is the central practical lesson of queueing theory.

Common trap

The deadly intuition is that a server running at 99% utilization is "almost the same" as one at 90% — after all, it is only 9 percentage points busier, and you are squeezing nearly all the work out of your expensive hardware. The mathematics says otherwise: the queue is ten times longer. The reason is *variability*. Even though work arrives at rate λ and is served at rate $\mu > \lambda$ *on average*, the randomness in arrivals and service means temporary pile-ups happen, and the fuller the system, the longer it takes to drain them. Near $\rho = 1$ there is almost no idle capacity to absorb a burst, so backlogs linger. **Provision for the variance, not just the mean.** A system sized so that average load equals average capacity ($\rho = 1$) is not balanced — it is unstable, with an infinitely long expected queue.

Paradox

Two engineers report on a server farm. One says "utilization is 50% — we are wasting half our capacity, buy fewer machines." The other says "utilization is 99% — we are perfectly efficient, our machines are never idle." Which farm would you rather be a customer of? The high-utilization farm has, on average, 99 jobs backed up versus the relaxed farm's 1. The "efficient" system is the miserable one to wait in. The paradox dissolves once you separate two goals that pull in opposite directions: *keeping the server busy* (good for the owner's wallet) and *keeping the wait short* (good for the customer). High utilization optimizes the first and destroys the second. There is no setting that does both, because $L = \rho/(1-\rho)$ ties them together. Real capacity planning is choosing where on this curve to sit — and the right answer is almost never near the top.

The other quantities follow once we have one of them, via Little's law, which we develop next; but let us name them now. Let W be the mean time a customer spends in the system (waiting plus being served), and L_q, W_q the corresponding queue-only versions (excluding the one in service). For M/M/1:

$$W = \frac{1}{\mu - \lambda}, \qquad L_q = \frac{\rho^2}{1-\rho}, \qquad W_q = \frac{\rho}{\mu - \lambda}.$$

Example 27.2 (Sizing a help desk)**.** A help desk receives $\lambda = 18$ tickets per hour. A single agent resolves tickets at $\mu = 20$ per hour. Then $\rho = 18/20 = 0.9$. The mean number of tickets in the system is $L = 0.9/0.1 = 9$, and the mean time from submission to resolution is $W = 1/(20-18) = 0.5$ hours $= 30$ minutes. Management wants to halve the wait. Doubling staff is the obvious move, but notice the leverage in μ: if a better tool lets the agent resolve $\mu = 24$ per hour, then $\rho = 0.75$, $L = 3$, and $W = 1/6$ hour $= 10$ minutes — a 20% speedup in service time cut the wait by two-thirds, because it pushed ρ away from the cliff edge near 1. Small improvements in service rate buy outsized improvements in delay when you are highly loaded.

Work it yourself

Before reading on, predict the shape of the answer to this: in the M/M/1 system, what fraction of *arriving customers* find the server idle (and so are served immediately)? Then check: by a property of Poisson arrivals ("PASTA" — Poisson Arrivals See Time Averages), an arriving customer sees the system in its steady-state distribution, so the answer is $\pi_0 = 1 - \rho$. With $\rho = 0.9$, only 10% of customers walk straight up to a free agent; the other 90% wait. This is why a system that "feels slow" can still be only modestly utilized in the mean — most

customers experience the crowded states, not the empty ones.

27.5 Little's Law

We keep relating L (how many are in the system) to W (how long each one stays). The bridge between them is one of the most general results in applied probability, and it requires almost no assumptions.

Theorem 27.1 (Little's law)**.** *For any stable queueing system in steady state, with λ the long-run average arrival rate, L the long-run average number of customers in the system, and W the long-run average time a customer spends in the system,*

$$\boxed{L = \lambda W.}$$

Intuition

Little's law is just a statement about *accounting*, and the cleanest way to see it is to imagine each customer pays \$1 per minute for every minute they sit in the system. There are two ways to add up the total money collected per unit time. **Way one (look at the room):** at any instant there are on average L customers present, each paying \$1/minute, so money pours in at rate L dollars per minute. **Way two (look at the door):** customers arrive at rate λ per minute, and each one eventually pays W dollars total (\$1/minute for their W minutes inside), so money is committed at rate λW dollars per minute. Both count the same dollars. Therefore $L = \lambda W$. No distributions, no independence, no Markov property — just two ways of tallying the same revenue.

Why it works

Now to make the accounting argument precise. Over a long horizon $[0, T]$, let $A(T)$ be the number of arrivals and let customer i spend time W_i in the system. The *area under the occupancy curve* — the total customer-minutes accumulated — can be computed two ways. Summing over time, it is $\int_0^T X_t \, \mathrm{d}t$, which over a long horizon averages to $L \cdot T$. Summing over customers, it is $\sum_{i=1}^{A(T)} W_i$ (each customer contributes a horizontal sliver of width W_i), which equals $A(T) \cdot \overline{W}$ where $\overline{W}$ is the average sojourn. Equate and divide by T:

$$L \;=\; \frac{1}{T}\int_0^T X_t \, \mathrm{d}t \;=\; \frac{A(T)}{T} \cdot \overline{W} \;\xrightarrow[T\to\infty]{}\; \lambda \cdot W,$$

since $A(T)/T \to \lambda$. The only thing we needed was that the system is stable (everybody eventually leaves, so the area is finite per unit time) and that the long-run averages exist. That is why Little's law holds for the M/M/1 queue, the M/M/c queue, a grocery store, a hospital ward, or the number of half-finished features in a software team's pipeline.

What is really happening

Little's law is really a theorem about *areas*. Draw the occupancy X_t as a staircase over time. Its area can be sliced *vertically* (height X_t at each instant, giving $\int X_t \, dt = L \cdot T$) or *horizontally* (each customer is a bar of length W_i, giving $\sum W_i = \lambda T \cdot W$). Two ways to compute one area — that is the entire content. This is the same Fubini-style "count the same thing two ways" move that powers so much of combinatorics and the tail-sum formula for expectation. Once you see it as an area, the law's universality stops being mysterious.

Example 27.3 (Little's law as a sanity check on M/M/1)**.** We claimed $L = \rho/(1-\rho)$ and $W = 1/(\mu - \lambda)$. Do they obey $L = \lambda W$? Compute

$$\lambda W = \frac{\lambda}{\mu - \lambda} = \frac{\lambda/\mu}{1 - \lambda/\mu} = \frac{\rho}{1-\rho} = L. \checkmark$$

Indeed Little's law is *how* one usually derives W: find L from the steady-state distribution (easy, it is a geometric mean), then divide by λ. The same trick applied to the queue-only quantities gives $L_q = \lambda W_q$.

Application

[Little's law in the wild] Because it needs so few assumptions, Little's law is a Swiss army knife far beyond telephones.

- **Software / Kanban.** If a team finishes $\lambda = 5$ features per week and on average $L = 20$ features are "in progress," then each feature takes $W = L/\lambda = 4$ weeks to ship. To ship faster, either finish faster or *limit work in progress* — capping L directly caps W. This is the quantitative core of lean and agile flow management.
- **Hospitals.** With an average of $L = 60$ patients in a ward and admissions at $\lambda = 12$ per day, the average length of stay is $W = 5$ days. Planners invert it: target a stay W, multiply by admission rate, get the beds needed.
- **Inventory and supply chains.** "Average inventory equals throughput times flow time." A warehouse holding L units that move at λ units/day implies each unit sits $W = L/\lambda$ days — the basis of just-in-time analysis.
- **Web services.** If a service handles $\lambda = 2000$ requests/second and each spends $W = 50$ ms $= 0.05$ s in flight, then on average $L = 100$ requests are being processed at once — exactly the concurrency a thread pool or connection limit must support.

27.6 Beyond One Server: Networks and Variations

The single server is only a starting point. The same machinery scales up, and the generalizations are where queueing theory earns its keep in practice.

27.6.1 The M/M/c queue: many servers

With c identical servers (think c checkout lanes or c CPU cores), the death rate depends on how many are busy: with $n \le c$ customers, all n are in service and departures fire at rate $n\mu$; with $n > c$,

all servers are busy and the rate saturates at $c\mu$. This is still a birth-death process,

$$\lambda_n = \lambda, \qquad \mu_n = \begin{cases} n\mu, & n \le c, \\ c\mu, & n > c, \end{cases}$$

so the product formula applies verbatim — you just plug in the state-dependent death rates. The result is the celebrated *Erlang C* formula for the probability an arriving customer has to wait. It is what call centers use to decide how many agents to staff, and it reveals a comforting *economy of scale*.

Why this matters

There is a surprising fact here, sometimes called the resource-pooling or economy-of-scale principle. One fast server of rate $c\mu$ and c slow servers each of rate μ have the same total capacity. But *a single shared line feeding many servers beats many separate lines.* Picture a bank with one snaking queue feeding five tellers versus five separate queues. The single queue is better, because no teller ever sits idle while someone waits in another line — the dreaded "I always pick the slow lane" problem vanishes. Quantitatively, pooling c servers behind one queue lets you run each at higher utilization for the same delay, so larger systems are *relatively* more efficient. This is why cloud providers, with millions of pooled cores, can offer low latency at high utilization that no small data center can match. The mathematics of Q and the product formula is what turns this intuition into a staffing number.

27.6.2 Jackson networks

Real systems are not one queue but a web of them: a packet visits a router, then a link, then another router; a manufactured part visits station after station. A *Jackson network* is a collection of M/M/1-style queues where, on finishing service at one node, a customer is routed to another node (or leaves) by fixed probabilities. The astonishing *product-form* theorem of Jackson (1957) says that in steady state the nodes behave *as if independent*: the joint distribution of all the queue lengths factors,

$$\mathbb{P}(N_1 = n_1, \dots, N_k = n_k) = \prod_{i=1}^{k} (1 - \rho_i)\rho_i^{n_i},$$

each node a geometric in its own effective utilization ρ_i, even though the nodes clearly interact through the routing. That factorization is what makes large networks analytically tractable at all, and it underlies performance models of the Internet, of computer architectures, and of factory floors.

Application

[Where these chains run today]

- **Computer networks.** Packet buffers in routers are queues; the M/M/1 and its kin predict latency and loss, and inform buffer sizing and congestion control.
- **Cloud and data centers.** Autoscalers watch utilization ρ and the queue-length tail to add servers *before* the $\rho/(1-\rho)$ cliff is reached. Load balancers are, quite literally, trying to implement single-queue pooling across many machines.
- **Reliability engineering.** The two-state up/down machine of Example 27.1 generalizes

to CTMC models of complex systems, computing availability and mean-time-to-failure for everything from aircraft to power grids.

- **Biology and chemistry.** Population sizes, gene expression bursts, and chemical reaction networks are CTMCs; the generator Q is the *chemical master equation* of stochastic kinetics.
- **Epidemiology.** The number of infected individuals in an outbreak is a birth-death process (infections are births, recoveries are deaths); the condition $\rho < 1$ becomes the epidemic threshold $R_0 < 1$ that separates fizzle from outbreak.

27.7 Steady State, Stationarity, and the Balance Equations

We used the stationary distribution freely above; let us tie it down. In a discrete chain the stationary distribution solves $\pi = \pi P$. The continuous-time analogue replaces the one-step matrix P with the generator Q, and the fixed-point equation becomes a flow-balance equation.

Definition 27.3 (Stationary distribution of a CTMC)**.** A probability vector π (with $\pi_i \geq 0$, $\sum_i \pi_i = 1$) is *stationary* for a CTMC with generator Q if

$$\boxed{\pi Q = 0.}$$

Component-wise, for each state j, $\sum_i \pi_i q_{ij} = 0$, which rearranges to the *global balance* equation

$$\underbrace{\pi_j \sum_{i \neq j} q_{ji}}_{\text{rate of probability leaving } j} = \underbrace{\sum_{i \neq j} \pi_i \, q_{ij}}_{\text{rate of probability entering } j} .$$

Why it works

Why is $\pi Q = 0$ the right equation? Recall $P(t) = e^{Qt}$ and that the distribution at time t starting from π is $\pi P(t)$. "Stationary" means this does not change with t: $\pi P(t) = \pi$ for all t. Differentiate at $t = 0$: $\frac{\mathrm{d}}{\mathrm{d}t}\big|_0 \pi P(t) = \pi P'(0) = \pi Q$, and since the left side is the derivative of a constant, it is zero. Hence $\pi Q = 0$. The balance reading is even more intuitive: in equilibrium, the probability fluid filling each state must drain out exactly as fast as it pours in, or the state's probability would drift. $\pi Q = 0$ is the conservation law that says "nothing is accumulating anywhere."

Focus

Two equation levels, two strengths.

- **Global balance** ($\pi Q = 0$): *always* characterizes the stationary distribution. Total flow into each state equals total flow out. Solve this system and you have π.
- **Detailed balance** ($\pi_i q_{ij} = \pi_j q_{ji}$ for every pair): a *stronger* condition — flow balances across each individual edge, not just each state. When it holds the chain is called *reversible*, and π is far easier to find (telescoping products, as in the birth-death case). Detailed balance implies global balance, but not conversely.

All birth-death processes are reversible, which is exactly why the M/M/1 steady state fell out so cleanly. When you can use detailed balance, do — it turns a linear system into a simple recursion.

Example 27.4 (Solving a three-state CTMC by global balance)**.** Consider states $\{A, B, C\}$ with generator

$$Q = \begin{pmatrix} -3 & 2 & 1 \\ 1 & -2 & 1 \\ 1 & 1 & -2 \end{pmatrix} \qquad \text{(rows sum to 0)}.$$

The balance equations $\pi Q = 0$ read, column by column:

$$-3\pi_A + \pi_B + \pi_C = 0, \qquad 2\pi_A - 2\pi_B + \pi_C = 0, \qquad \pi_A + \pi_B - 2\pi_C = 0.$$

From the first, $\pi_B + \pi_C = 3\pi_A$. Combined with $\pi_A + \pi_B + \pi_C = 1$ this gives $4\pi_A = 1$, so $\pi_A = 1/4$. Now use $\pi_B + \pi_C = 3/4$ in the second equation, $2\pi_A - 2\pi_B + \pi_C = 0$, i.e. $\frac{1}{2} - 2\pi_B + \pi_C = 0$. Substituting $\pi_C = \frac{3}{4} - \pi_B$ gives $\frac{1}{2} - 2\pi_B + \frac{3}{4} - \pi_B = 0$, so $3\pi_B = \frac{5}{4}$ and $\pi_B = 5/12$, leaving $\pi_C = 3/4 - 5/12 = 1/3$. (Check: $\pi_A + \pi_B + \pi_C = \frac{1}{4} + \frac{5}{12} + \frac{1}{3} = \frac{3}{12} + \frac{5}{12} + \frac{4}{12} = 1$, and the third equation $\pi_A + \pi_B - 2\pi_C = \frac{1}{4} + \frac{5}{12} - \frac{2}{3} = 0$ holds.) The chain spends only a quarter of its time in A — the state with the fastest exit rate ($q_A = 3$, versus $q_B = q_C = 2$) — and more of its time in the slower-to-leave states B and C. Fast-to-leave states shed probability; slow-to-leave states accumulate it. (Among B and C, which exit at the same rate, the tie is broken by where the inflow comes from — here B edges out C.)

Pause

Let us collect the through-line before the exercises. (1) A CTMC is an ordinary jump chain glued to exponential holding times — "wait Exponential(q_i), then jump by $P_{ij} = q_{ij}/q_i$." (2) The generator Q packs every rate into one matrix with zero row sums; transition probabilities are $P(t) = e^{Qt}$, and the stationary law solves $\pi Q = 0$. (3) Birth-death processes move only to neighbors, so detailed balance gives the product formula for π. (4) The M/M/1 queue is the constant-rate birth-death process; its steady state is geometric, $\pi_n = (1-\rho)\rho^n$, and its mean occupancy $L = \rho/(1-\rho)$ explodes as $\rho \to 1$. (5) Little's law $L = \lambda W$ ties occupancy to delay by pure accounting, needing essentially no assumptions. Everything else in queueing — many servers, networks, priorities — is built on these five beams.

27.8 Exercises

Exercise 27.1 (Reading a generator)**.** A CTMC on states $\{1, 2, 3\}$ has $q_{12} = 2$, $q_{13} = 1$, $q_{21} = 4$, $q_{23} = 0$, $q_{31} = 1$, $q_{32} = 3$ (rates per hour). (a) Write the generator matrix Q, filling in the diagonal. (b) What is the mean holding time in each state? (c) Starting in state 1, when it leaves, what is the probability it goes to state 3?

Exercise 27.2 (Two-state availability)**.** A server fails at rate $\lambda = 1$ per week and is repaired at rate $\mu = 6$ per week. (a) Using Example 27.1, find the long-run fraction of time the server is up. (b) What is the mean uptime between failures, and the mean repair time? (c) If a second, independent identical server is added and the service is "up" whenever *at least one* server works, set up (do not fully solve) the CTMC for the number of working servers and identify it as a birth-death process.

Exercise 27.3 (M/M/1 numbers)**.** Customers arrive at a single-teller bank as a Poisson process at $\lambda = 30$ per hour; service is exponential at $\mu = 40$ per hour. Compute (a) the utilization ρ; (b) the probability the teller is idle; (c) the mean number in the system L; (d) the mean time in the system W in minutes; (e) the probability that there are at least 3 customers in the system.

Exercise 27.4 (The cliff)**.** For an M/M/1 queue, the mean number in system is $L(\rho) = \rho/(1-\rho)$. (a) Show that L doubles when you go from $\rho = 2/3$ to some ρ^*; find ρ^*. (b) Compute the derivative $\mathrm{d}L/\mathrm{d}\rho$ and explain in one sentence why it is so large near $\rho = 1$. (c) An operator runs at $\rho = 0.95$ and wants to cut L in half. By what factor must service rate μ increase if λ is fixed?

Exercise 27.5 (Little's law everywhere)**.** (a) A coffee shop serves $\lambda = 120$ customers per hour, and a customer spends on average $W = 6$ minutes inside. How many customers, on average, are inside at once? (b) A development team keeps $L = 15$ tickets in progress and closes $\lambda = 3$ per day. What is the average cycle time W? (c) The team wants to cut cycle time to 3 days without working faster (same λ). What must L become, and what policy achieves it?

Exercise 27.6 (Derive the geometric steady state)**.** Starting from the detailed-balance equations $\pi_n \lambda = \pi_{n+1}\mu$ for the M/M/1 queue, (a) show by induction that $\pi_n = \rho^n \pi_0$ with $\rho = \lambda/\mu$; (b) use $\sum_{n \geq 0} \pi_n = 1$ to show $\pi_0 = 1 - \rho$, stating clearly where you need $\rho < 1$; (c) explain in words what goes wrong physically when $\rho \geq 1$.

Exercise 27.7 (Global balance by hand)**.** For the three-state generator in Example 27.4, verify the claimed stationary distribution $\pi = (1/4,\, 5/12,\, 1/3)$ by checking that $\pi Q = 0$ directly (compute each of the three components and confirm they are zero).

Exercise 27.8 (Mean hitting time, finite chain)**.** A particle moves on states $\{0, 1, 2\}$ as a birth-death chain with birth rate $\lambda = 1$ and death rate $\mu = 2$ (so $q_{01} = 1$, $q_{10} = 2$, $q_{12} = 1$, $q_{21} = 2$). Let h_i be the expected time to first reach state 0 starting from state i. Using the first-step idea "mean holding time in i plus the expected remaining time from where you jump," set up equations for h_1 and h_2 and solve them. (Hint: $h_0 = 0$; from state i you wait $1/q_i$ then move.)

Exercise 27.9 (Proof: detailed balance implies global balance)**.** Suppose π satisfies detailed balance for a CTMC: $\pi_i q_{ij} = \pi_j q_{ji}$ for all $i \neq j$. Prove that π then satisfies global balance, i.e. $\pi Q = 0$. (Hint: fix a state j, sum the detailed-balance equations over all $i \neq j$, and recall that $q_{jj} = -\sum_{i \neq j} q_{ji}$.) Then give a one-line argument for why the converse can fail, citing a chain that is not reversible.

Exercise 27.10 (Conceptual: why exponential?)**.** Explain, in a short paragraph, why a continuous-time *Markov* chain is forced to have exponentially distributed holding times. Your answer should connect the memorylessness of the exponential distribution to the defining property that the future depends only on the present state, not on how long the chain has already occupied it.

Exercise 27.11 (Economy of scale, numerically)**.** A service must handle $\lambda = 8$ jobs per second. Option A: one fast server at $\mu = 10$/s. Option B: split the arrivals randomly into two streams of 4/s each, served by two *separate* M/M/1 servers at $\mu = 5$/s each. (a) Compute ρ, L, and W for Option A. (b) Compute L and W *per stream* for Option B and the total expected number in the whole system. (c) Which option gives shorter average wait, and which idea from Section 27.6 does this illustrate?

Chapter 28

Martingales

There is a particular fantasy that has bankrupted more people than any other in the history of gambling. It goes like this. You walk up to a roulette table and bet one dollar on red. If you win, great — pocket the dollar and start over. If you lose, you simply double your bet to two dollars. Lose again? Bet four. Lose again? Bet eight. The first time red finally comes up, you recover everything you have lost *plus* one dollar of profit. Since red is essentially certain to appear *eventually*, you have apparently discovered a way to extract guaranteed money from a game with no edge. This strategy is so famous it has a name: the **martingale**.

It does not work. And the reason it does not work — the reason *nothing* works against a fair game — is one of the deepest facts in probability. To understand it, we will spend this chapter building the mathematical object that, by an irony worth savoring, inherited the name of the betting system it destroys: the martingale.

A martingale is the precise mathematical formalization of a *fair game*: a running fortune whose expected next value, given everything you currently know, is exactly its present value. "On average, you neither gain nor lose." That one sentence, taken seriously, carries a long way. It lets us compute gambler's-ruin probabilities in a single line, explains why no clever stopping rule can beat a fair casino, prices financial options, and underpins a whole convergence theory that rivals the law of large numbers in importance. This chapter leans hard on conditional expectation, so if $\mathbb{E}[Y \mid X]$ and the tower property feel shaky, glance back at that chapter before we begin — everything here is conditional expectation wearing a story.

28.1 Fair games and the definition

Imagine watching a quantity evolve in time: your bankroll at a casino, the price of a stock, the position of a wandering particle, your running estimate of some unknown parameter as data trickles in. Call its value at time n the random variable M_n, for $n = 0, 1, 2, \ldots$. At each moment you have observed a *history* — everything that has happened up through time n. We bundle that history into the symbol $\mathcal{F}_n$, read "the information available at time n." As time passes you learn more, never less, so $\mathcal{F}_0 \subseteq \mathcal{F}_1 \subseteq \mathcal{F}_2 \subseteq \cdots$. This growing family of information is called a **filtration**, and you should picture it as the expanding contents of your own memory.

Intuition

Do not be intimidated by $\mathcal{F}_n$. In every concrete example in this chapter it means nothing more exotic than "the values of $M_0, M_1, \dots, M_n$ that you have already seen," possibly together with some other observed variables. Conditioning on $\mathcal{F}_n$ just means "given everything we know so far." When you read $\mathbb{E}[M_{n+1} \mid \mathcal{F}_n]$, hear it as: "my best forecast of tomorrow's fortune, using my entire current memory."

What does it mean for the game to be *fair*? It means that knowing the entire past, my best guess for tomorrow's fortune is exactly today's fortune. I expect to walk away with what I came with. That is the whole definition.

Definition 28.1 (Martingale)**.** A sequence of random variables $(M_n)_{n\geq 0}$ adapted to a filtration $(\mathcal{F}_n)_{n\geq 0}$ (meaning M_n is determined by $\mathcal{F}_n$) with $\mathbb{E}|M_n| < \infty$ for every n is a *martingale* if

$$\mathbb{E}[M_{n+1} \mid \mathcal{F}_n] \;=\; M_n \qquad \text{for all } n \geq 0.$$

If instead $\mathbb{E}[M_{n+1} \mid \mathcal{F}_n] \geq M_n$ for all n, the sequence is a *submartingale* (a favorable game, drifting up); if $\mathbb{E}[M_{n+1} \mid \mathcal{F}_n] \leq M_n$, it is a *supermartingale* (an unfavorable game, drifting down).

Common trap

The "super/sub" naming feels backwards to everyone at first. A *super*martingale goes *down* (it is the gambler's view of a casino: the house edge drags your fortune down). A *sub*martingale goes *up*. The mnemonic that sticks: a supermartingale is "superior for the house," a submartingale is "subpar for the house." The deeper reason is an analogy with harmonic functions, where superharmonic functions lie above their spherical averages, so the expectation drops.

The integrability condition $\mathbb{E}|M_n| < \infty$ is not bureaucratic fine print; it is what makes the conditional expectations exist as genuine finite numbers. We will keep it in the background but it is always quietly assumed.

28.1.1 The first and most important consequence

Take the defining equation and average out the conditioning. By the tower property of conditional expectation,

$$\mathbb{E}[M_{n+1}] \;=\; \mathbb{E}\left[\,\mathbb{E}[M_{n+1} \mid \mathcal{F}_n]\right] \;=\; \mathbb{E}[M_n].$$

So the expectation never moves: $\mathbb{E}[M_n] = \mathbb{E}[M_{n-1}] = \cdots = \mathbb{E}[M_0]$ for every n. A martingale has **constant expectation**. This is the seed from which the entire chapter grows.

Focus

The one fact to remember. For a martingale, $\mathbb{E}[M_n] = \mathbb{E}[M_0]$ for all n. No matter how long you play, your expected fortune equals your starting fortune. The drama of the rest of the chapter is about whether you can cheat this by choosing a clever *time to stop*. (Spoiler: under mild conditions, you cannot.)

History

The word *martingale* originally named a piece of horse tack — a strap preventing a horse from throwing its head back — and by the 18th century it had attached itself to the doubling betting system described in this chapter's opening. The mathematical object was named after the strategy by Jean Ville in 1939, who used it to repair Richard von Mises's frequentist theory of randomness. But the theory truly came alive in the hands of Joseph Leo Doob, whose 1953 book *Stochastic Processes* turned martingales into one of the central tools of modern probability. Doob's optional stopping theorem and martingale convergence theorem are load-bearing pillars of everything from stochastic calculus to the analysis of algorithms. It is a charming accident of history that the deepest theorem about why you *cannot* beat a fair game is named after the betting system that promised you could.

28.2 A gallery of martingales

Definitions only become real when you have seen enough examples that the pattern becomes visible. Here are the canonical ones; each will return later.

Example 28.1 (Symmetric random walk — the prototype)**.** Let $X_1, X_2, \dots$ be independent with $\mathbb{P}(X_i = +1) = \mathbb{P}(X_i = -1) = \frac{1}{2}$, and set $S_0 = 0$, $S_n = X_1 + \cdots + X_n$. Take $\mathcal{F}_n$ to be the history of the first n steps. Then

$$\mathbb{E}[S_{n+1} \mid \mathcal{F}_n] = \mathbb{E}[S_n + X_{n+1} \mid \mathcal{F}_n] = S_n + \mathbb{E}[X_{n+1}] = S_n + 0 = S_n.$$

So (S_n) is a martingale. This is the fair coin-flipping game: bet a dollar on each flip, win or lose a dollar, and your fortune wanders with no drift. If the coin is biased toward losing ($\mathbb{P}(X_i = +1) = p < \frac{1}{2}$), then $\mathbb{E}[S_{n+1} \mid \mathcal{F}_n] = S_n + (2p - 1) < S_n$ and you have a supermartingale — the accurate model of a real casino.

Figure 28.1 shows a few sample paths of this fair game. Each wanders without drift, and although individual paths roam far from zero, the average across paths stays pinned at the starting value — the visible signature of constant expectation.

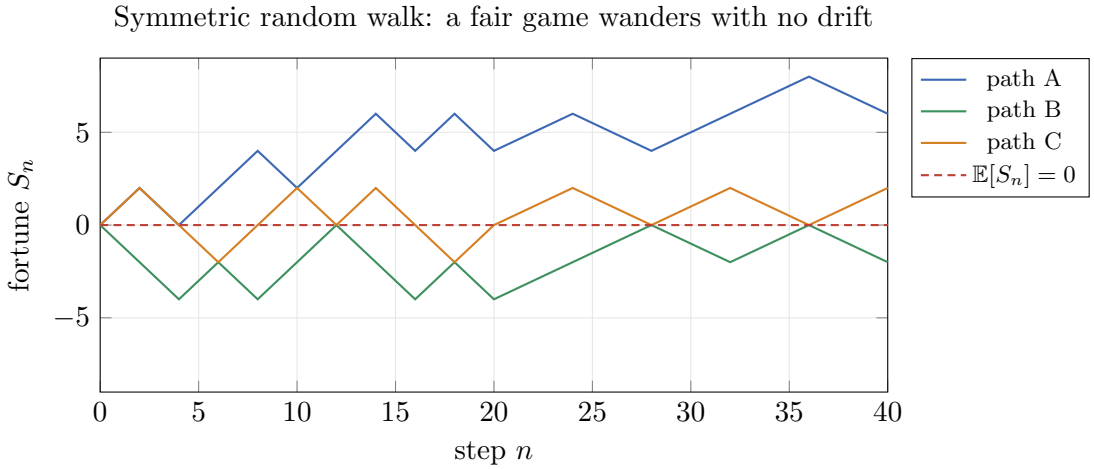


Figure 28.1: Three realizations of the symmetric simple random walk $S_n = X_1 + \cdots + X_n$. No path shows any systematic tendency to rise or fall; the expected fortune (dashed) stays exactly at the starting value of zero for all n. This is what "fair" looks like: motion without drift.

Example 28.2 (The gambler's variance martingale)**.** With the same symmetric walk, consider $M_n = S_n^2 - n$. Watch what happens:

$$\mathbb{E}[S_{n+1}^2 \mid \mathcal{F}_n] = \mathbb{E}[(S_n + X_{n+1})^2 \mid \mathcal{F}_n] = S_n^2 + 2S_n\,\mathbb{E}[X_{n+1}] + \mathbb{E}[X_{n+1}^2] = S_n^2 + 1,$$

since $\mathbb{E}[X_{n+1}] = 0$ and $X_{n+1}^2 = 1$ always. Therefore $\mathbb{E}[S_{n+1}^2 - (n+1) \mid \mathcal{F}_n] = S_n^2 + 1 - (n+1) = S_n^2 - n = M_n$. So $S_n^2 - n$ is a martingale even though S_n^2 itself drifts upward at a steady rate of one per step. The "$-n$" exactly cancels the predictable growth of the variance. Later we use this object to compute *how long* a gambler's-ruin game lasts.

Example 28.3 (Likelihood ratios)**.** Here is a martingale that does not look like gambling at all but powers all of sequential statistics. Suppose data $Y_1, Y_2, \ldots$ are i.i.d. from some true density f, but you entertain a rival hypothesis with density g. Form the running *likelihood ratio*

$$L_n = \prod_{i=1}^{n} \frac{g(Y_i)}{f(Y_i)}, \qquad L_0 = 1.$$

Then, because Y_{n+1} is independent of the past and integrates against the *true* density f,

$$\mathbb{E}[L_{n+1} \mid \mathcal{F}_n] = L_n \cdot \mathbb{E}\left[\frac{g(Y_{n+1})}{f(Y_{n+1})}\right] = L_n \int \frac{g(y)}{f(y)} f(y)\,\mathrm{d}y = L_n \int g(y)\,\mathrm{d}y = L_n.$$

So under the true model f, the likelihood ratio is a martingale with mean 1 forever — even though it is built from products, not sums. This single fact is the engine behind Wald's sequential probability ratio test, and the martingale convergence theorem will tell us that L_n converges (in fact, to 0 almost surely when $f \neq g$, which is exactly why hypothesis testing eventually discriminates between the two models).

Example 28.4 (Polya's urn)**.** An urn starts with one red and one black ball. At each step you draw a ball uniformly at random, return it, *and add one more ball of the same color*. Rich get richer. Let R_n be the number of red balls after n draws and let $M_n = R_n/(n+2)$ be the *fraction* of red balls

(there are $n+2$ balls total after n draws). Given the current composition, the chance the next ball is red is exactly the current red fraction M_n, so

$$\mathbb{E}[R_{n+1} \mid \mathcal{F}_n] = R_n + M_n = R_n + \frac{R_n}{n+2} = R_n \cdot \frac{n+3}{n+2},$$

and dividing by $n+3$ gives $\mathbb{E}[M_{n+1} \mid \mathcal{F}_n] = R_n/(n+2) = M_n$. The red fraction is a martingale! This is genuinely surprising: a process with strong positive feedback nonetheless keeps its proportion "fair" in expectation. By the convergence theorem the fraction settles to a random limit — and that limit turns out to be uniformly distributed on $[0,1]$, which is a surprising fact in its own right.

Example 28.5 (Doob martingale: revealing information)**.** Let Z be any integrable random variable and let information arrive in pieces $\mathcal{F}_0 \subseteq \mathcal{F}_1 \subseteq \cdots$. Define $M_n = \mathbb{E}[Z \mid \mathcal{F}_n]$, your best estimate of Z given what you know at time n. The tower property instantly gives $\mathbb{E}[M_{n+1} \mid \mathcal{F}_n] = \mathbb{E}[\mathbb{E}[Z \mid \mathcal{F}_{n+1}] \mid \mathcal{F}_n] = \mathbb{E}[Z \mid \mathcal{F}_n] = M_n$. So *any* sequence of sharpening estimates of a fixed quantity is automatically a martingale. This "Doob martingale" is the backbone of concentration inequalities (McDiarmid's inequality) used everywhere in machine learning to bound how far a complicated function of many independent inputs can stray from its mean.

What is really happening

Notice the unifying thread. A martingale is what you get whenever "the best prediction of the next value, given the past, is the current value." Sums of fair bets, products of mean-one factors, and running conditional expectations of a fixed target all share this property. The deep reason is that the operation $Z \mapsto \mathbb{E}[Z \mid \mathcal{F}_n]$ is a *projection*: it throws away the unpredictable part and keeps only what the current information can see. Martingale theory is, at heart, the dynamics of these projections as information grows. (This is also why the rigorous chapter on conditional expectation later in the book treats it as an orthogonal projection in a space of random variables — the geometry is real.)

28.3 Why you cannot beat a fair game

Now we confront the gambler's dream head-on. Surely *some* betting strategy can turn a fair game into a winning one? You get to choose how much to bet each round based on everything you have seen. Can cleverness manufacture an edge?

Let us formalize "a strategy." Before round n, knowing the history $\mathcal{F}_{n-1}$, you choose a stake C_n — how many units to wager. The crucial constraint is that C_n may depend only on the *past*, not on the future outcome of round n (no clairvoyance, no peeking). A sequence with this property, C_n determined by $\mathcal{F}_{n-1}$, is called **predictable**. Your winnings on round n are $C_n(M_n - M_{n-1})$, so your fortune after n rounds is the *discrete stochastic integral*, or *martingale transform*,

$$W_n = W_0 + \sum_{k=1}^{n} C_k \,(M_k - M_{k-1}).$$

Theorem 28.1 (You cannot beat a fair game)**.** *If (M_n) is a martingale and (C_n) is a bounded predictable sequence, then the transformed process (W_n) above is also a martingale. Consequently $\mathbb{E}[W_n] = W_0$ for every n: no predictable betting strategy changes your expected fortune.*

Why it works

The proof is one line of conditional expectation, and it is worth savoring because it is the mathematical core of "the house always wins on average." Condition on the past $\mathcal{F}_{n-1}$. The stake C_n is already known — it was chosen before the round — so it slides out of the conditional expectation as a known constant:

$$\mathbb{E}[W_n - W_{n-1} \mid \mathcal{F}_{n-1}] = \mathbb{E}[C_n(M_n - M_{n-1}) \mid \mathcal{F}_{n-1}] = C_n \, \mathbb{E}[M_n - M_{n-1} \mid \mathcal{F}_{n-1}] = C_n \cdot 0 = 0.$$

The last step is the martingale property itself: $\mathbb{E}[M_n \mid \mathcal{F}_{n-1}] = M_{n-1}$. So each round contributes zero expected change *no matter how you size your bet*, because you sized it before you knew the outcome. The fairness of the underlying game leaks through any strategy built only from the past. The only way to win would be to let C_n depend on the result of round n — to bet after seeing the outcome — which is precisely cheating.

This already rules out infinitely many "systems." Doubling, the Fibonacci system, anti-martingales, trend-following — all of them are predictable strategies, so all of them leave your expected fortune untouched. But the doubling system seemed to promise a *guaranteed* win, not merely a fair expectation. To finish it off we need to think about *stopping times*, and that brings us to the crown jewel of the chapter.

28.4 Stopping times and optional stopping

A gambler does not play forever; he plays until something happens — until he is up a hundred dollars, or broke, or the bar closes. The time he stops is itself random and depends on the game's history. We need a clean notion of such a time.

Definition 28.2 (Stopping time)**.** A random time T taking values in $\{0, 1, 2, \ldots\} \cup \{\infty\}$ is a *stopping time* for the filtration $(\mathcal{F}_n)$ if, for every n, the event $\{T = n\}$ is determined by $\mathcal{F}_n$.

The condition says: by time n you always *know* whether you have already stopped. "Stop the first time my fortune hits \$100" is a stopping time — you can tell at each moment whether it has happened. "Stop one step before the peak" is *not* a stopping time, because recognizing the peak requires seeing the future. The rule must be enforceable in real time, with no clairvoyance. This is the same no-peeking constraint as predictability, and it is the whole reason the next theorem holds.

Recall that $\mathbb{E}[M_n] = \mathbb{E}[M_0]$ for any *fixed* time n. The burning question is whether this survives if we replace the fixed n by a *random* stopping time T. Is $\mathbb{E}[M_T] = \mathbb{E}[M_0]$? If yes, then stopping cleverly cannot help. The optional stopping theorem says: yes, provided we rule out two specific loopholes.

Theorem 28.2 (Optional Stopping Theorem, Doob)**.** *Let (M_n) be a martingale and T a stopping time. Then $\mathbb{E}[M_T] = \mathbb{E}[M_0]$ provided any one of the following holds:*

1. *T is bounded: there is a constant N with $T \le N$ almost surely; or*
2. *$T < \infty$ almost surely and the stopped process $(M_{\min(n,T)})$ is uniformly bounded (there is a constant K with $|M_{\min(n,T)}| \le K$ for all n); or*
3. *$\mathbb{E}[T] < \infty$ and the increments are bounded, $|M_{n+1} - M_n| \le c$ for some constant c.*

For a supermartingale (unfavorable game) the corresponding conclusion is the inequality $\mathbb{E}[M_T] \leq \mathbb{E}[M_0]$*: stopping cannot raise your expected fortune above where you started.*

Intuition

The cleanest version is case 1, and you can almost prove it in your head. "Play until time T, then stop" is the same as the betting strategy "bet one unit each round while $n \leq T$, and bet zero afterward." The stake $C_n = \mathbf{1}\{T \geq n\}$ is predictable — you can tell at the start of round n whether you have already stopped — so by Theorem 28.1 the stopped fortune is a martingale, and its expectation is the starting value. The fancy conditions in cases 2 and 3 exist only to handle *unbounded* stopping times, where you must ensure the game does not run away to infinity before T arrives. That is exactly the loophole the doubling system tries to exploit.

28.4.1 The doubling system, finally killed

Return to the opening fantasy. You bet to recover all losses plus one dollar; you stop, at time T, at the first win. Since red appears eventually, $T < \infty$ almost surely, and at time T your profit is exactly \$1. So $\mathbb{E}[M_T] = 1$, while $M_0 = 0$. Optional stopping appears to be violated — has the gambler beaten the fair game?

Paradox

The martingale betting system. The profit really is \$1 with probability 1. So where is the catch? It hides in the *capital required*. If your first win comes on round k, your bets were $1, 2, 4, \ldots, 2^{k-1}$ and your peak loss just before winning was $2^{k-1} - 1$ dollars. The chance of needing exactly k rounds is 2^{-k}, so the expected size of your largest required bet is

$$\sum_{k=1}^{\infty} 2^{-k} \cdot 2^{k-1} = \sum_{k=1}^{\infty} \tfrac{1}{2} = \infty.$$

Your expected maximum exposure is *infinite*. To guarantee the \$1, you must be willing and able to lose arbitrarily large amounts. No real gambler has infinite capital, and no real casino has no table limit. The moment you impose any finite bankroll B, the strategy becomes a stopping time that violates none of the theorem's hypotheses, and the conclusion snaps back into place: $\mathbb{E}[M_T] = 0$. You win a small amount with high probability and suffer a catastrophic, capital-wiping loss with the small remaining probability, and these balance *exactly* to zero expected profit. The optional stopping theorem did not fail; the gambler's hidden assumption of infinite wealth did. This is precisely loophole case 2 of Theorem 28.2: with infinite capital the stopped process is *not* uniformly bounded, so the guarantee evaporates.

Figure 28.2 shows what this looks like in practice over many simulated nights at the table.

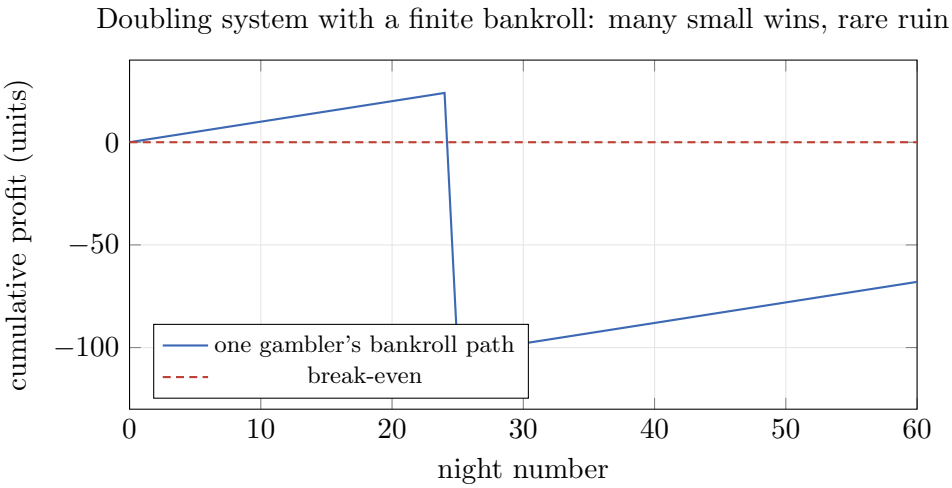


Figure 28.2: A typical realization of the doubling system under a finite bankroll. The fortune climbs steadily by one unit per night — the seductive part — until a long losing streak exceeds the bankroll and wipes out all accumulated gains and more. Average it over all possible paths and the expected profit is exactly zero, as the optional stopping theorem demands.

28.5 The gambler's ruin, in one line

Optional stopping is not merely a no-go theorem; it is also a sharp *computational* tool. The classic gambler's ruin problem normally requires solving a difference equation. Martingales reduce it to arithmetic.

Example 28.6 (Symmetric gambler's ruin)**.** You start with $\$a$ and play fair \$1 bets until you either reach a target $\$b$ (with $0 < a < b$) or go broke at \$0. Let S_n be your fortune and T the first time you hit 0 or b. What is the probability $p = \mathbb{P}(\text{reach } b \text{ before } 0)$?

The fortune S_n is a martingale (Example 28.1, shifted to start at a). The stopped process stays in $[0, b]$, hence is bounded, and $T < \infty$ almost surely (the walk cannot avoid the boundaries forever). So optional stopping case 2 applies:

$$a = \mathbb{E}[S_0] = \mathbb{E}[S_T] = b \cdot p + 0 \cdot (1 - p) = b\,p \quad \Longrightarrow \quad p = \frac{a}{b}.$$

That is the entire derivation. The probability of reaching your target before ruin is just the ratio of your starting stake to your target. Want to turn \$100 into \$200 at a fair game? Your chance is exactly 1/2. Want to turn \$100 into \$10,000? Exactly 1/100. The deck is not stacked against you — the game is perfectly fair — yet ambitious targets are punishingly unlikely, simply because you have so much more room to fall to zero than to climb to glory.

Figure 28.3 plots this success probability against the size of your target, contrasting the perfectly fair game with one carrying a tiny house edge. The fair curve decays gently as a/b; the merest bias bends it into a cliff.

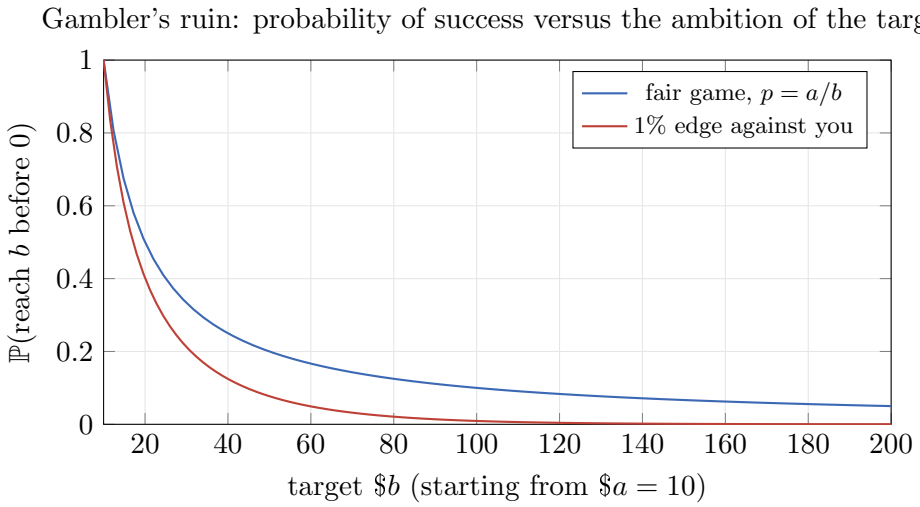


Figure 28.3: Starting from \$10, the chance of reaching a target \$$b$ before going broke. At a perfectly fair game (blue) the probability is exactly the ratio a/b, decaying gently as $1/b$. Introduce a mere one-percent house edge (red, $q/p = 51/49$) and the success probability collapses exponentially in the distance to the target — the same arithmetic that keeps casinos solvent. Both curves come straight from optional stopping applied to the appropriate martingale, with no difference equations.

Example 28.7 (How long does ruin take?)**.** Now use the variance martingale of Example 28.2. Let $D_n = S_n - a$ be the displacement from the start, so $(D_n^2 - n)$ is a martingale started at 0, and optional stopping gives $\mathbb{E}[D_T^2 - T] = 0$, that is,

$$\mathbb{E}[T] = \mathbb{E}[D_T^2].$$

At time T the displacement from the start is either $-a$ (ruin, probability $1 - p = 1 - a/b$) or $b - a$ (success, probability $p = a/b$). Therefore

$$\mathbb{E}[T] = a^2\left(1 - \frac{a}{b}\right) + (b-a)^2\,\frac{a}{b} = a^2 \cdot \frac{b-a}{b} + (b-a)^2 \cdot \frac{a}{b} = \frac{a(b-a)}{b}\big[a + (b-a)\big] = a(b-a).$$

So the expected number of bets is $\mathbb{E}[T] = a(b-a)$. With \$50 trying to reach \$100, you expect $50 \cdot 50 = 2500$ bets before the matter is settled. Fair games are not quick.

Example 28.8 (Biased ruin via an exponential martingale)**.** What if the game is unfair, $\mathbb{P}(X_i = +1) = p \neq \frac{1}{2}$, $q = 1-p$? The walk itself is no longer a martingale, but a cleverly chosen transformation is. Consider $M_n = (q/p)^{S_n}$. Then

$$\mathbb{E}[M_{n+1} \mid \mathcal{F}_n] = (q/p)^{S_n}\Big[p \cdot (q/p) + q \cdot (p/q)\Big] = (q/p)^{S_n}\big[q + p\big] = M_n,$$

so $(q/p)^{S_n}$ is a martingale. Optional stopping with starting fortune a gives $(q/p)^a = \mathbb{E}[M_T] = p_{\text{win}}(q/p)^b + (1 - p_{\text{win}})(q/p)^0$, and solving,

$$\mathbb{P}(\text{reach } b \text{ before } 0) = \frac{(q/p)^a - 1}{(q/p)^b - 1}.$$

Plug in $p = 18/38$ for an even-money bet on red at American roulette and you will see why even a tiny house edge, compounded over a long climb to a distant target, makes success vanishingly unlikely. The martingale method delivers this famous formula with no difference equations at all.

Application

Finance: fair prices and the absence of arbitrage. The same logic that forbids beating a casino is the foundation of modern option pricing. In the risk-neutral framework, a discounted asset price $\tilde{S}_n$ is modeled as a martingale: $\mathbb{E}[\tilde{S}_{n+1} \mid \mathcal{F}_n] = \tilde{S}_n$. A trading strategy is a predictable sequence of holdings C_n, and your discounted wealth is exactly the martingale transform $\sum C_k(\tilde{S}_k - \tilde{S}_{k-1})$. By Theorem 28.1 its expectation stays put — which is precisely the statement that *no self-financing strategy can manufacture a guaranteed profit from nothing*: no arbitrage. The fair price of an option that pays Z at time N is then its discounted risk-neutral expectation $\mathbb{E}[\tilde{Z}]$, computable by the same optional-stopping machinery. The Black–Scholes formula is the continuous-time limit of exactly this idea. "You cannot beat a fair game" and "you cannot get free money in an efficient market" are the *same theorem*.

28.6 Martingale convergence: where it all settles

A martingale has constant mean, but it can fluctuate. Does it eventually settle down? The answer is one of the central theorems in probability, and it asks for almost no hypotheses. A martingale that cannot grow too positive must converge.

Theorem 28.3 (Martingale Convergence Theorem, Doob)**.** *If (M_n) is a martingale (or even a supermartingale) bounded in L^1, meaning $\sup_n \mathbb{E}|M_n| < \infty$, then M_n converges almost surely to a finite limit M_∞. In particular, any* nonnegative *supermartingale converges almost surely (since for it $\mathbb{E}|M_n| = \mathbb{E}[M_n] \le \mathbb{E}[M_0] < \infty$ automatically).*

Intuition

The proof idea is delightful and is called the *upcrossing* argument. Suppose, toward a contradiction, that M_n failed to converge — then it would have to oscillate, crossing some interval $[\alpha, \beta]$ from below to above infinitely often. But each such "upcrossing" is a buy-low sell-high round trip, and Theorem 28.1 guarantees you cannot profit from a predictable strategy on a (super)martingale. So you cannot afford infinitely many upcrossings: each one would be free money, and there is no free money. Doob made this precise with an inequality bounding the expected number of upcrossings, and finitely many upcrossings of every interval forces convergence. Once again, "you cannot beat a fair game" is doing the real work.

Apply this to our gallery. The likelihood ratio L_n (Example 28.3) is nonnegative with mean 1, so it converges almost surely — and when the rival hypothesis is wrong, it converges to 0, which is exactly why accumulating data eventually crushes the false model. Figure 28.4 shows the log-likelihood-ratio sliding steadily toward $-\infty$ (so $L_n \to 0$) as data piles up against a false alternative.

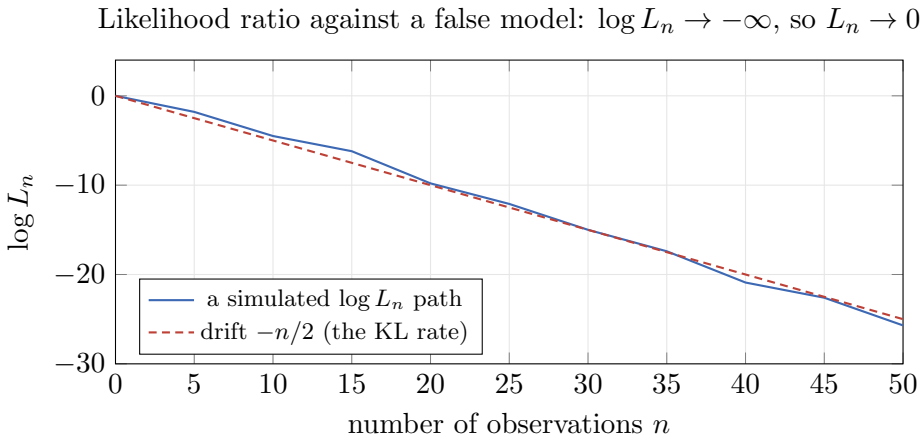


Figure 28.4: The log-likelihood ratio for $\mathcal{N}(0,1)$ truth versus an $\mathcal{N}(1,1)$ alternative. Each step adds a term of mean $-\frac{1}{2}$ under the truth (this $\frac{1}{2}$ is exactly the Kullback–Leibler divergence $\mathrm{KL}(f \,\|\, g)$ between the two models), so $\log L_n$ marches off to $-\infty$ and the likelihood ratio L_n collapses to 0. The martingale property guarantees $\mathbb{E}[L_n] = 1$ at every step, yet almost every path still tends to zero — a striking reminder that a mean-one martingale can converge to a limit with mean strictly less than one.

Polya's urn fraction (Example 28.4) is a bounded martingale, so the red proportion converges to a genuine random limit. Figure 28.5 shows several urn paths converging to their individual destinies.

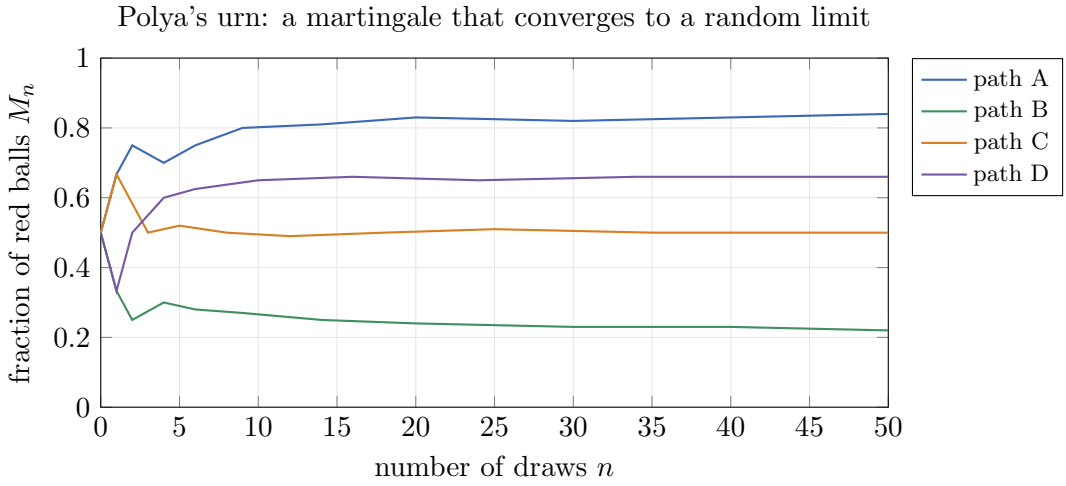


Figure 28.5: Four simulated runs of Polya's urn. Every run is a martingale with the same starting value $M_0 = 1/2$, yet each converges to a *different* random limit. Averaged across all runs the mean stays at $1/2$ forever (constant expectation), but individual paths do not converge to $1/2$ — the limit is in fact uniformly distributed on $[0,1]$.

Why this matters

The convergence theorem is the martingale world's answer to the law of large numbers, but it is far more general. The law of large numbers needs i.i.d. (or at least independent) summands;

the martingale convergence theorem needs only the fair-game property and a mild boundedness condition. Dependence, feedback, even pathological correlation structures are all permitted. This is why martingales appear wherever a quantity evolves with memory but no systematic drift — and why they are the natural language for stochastic calculus, sequential analysis, and the analysis of randomized algorithms. The single principle "you cannot profit from a fair game by stopping or strategizing" grows into a complete theory of when random processes must settle down.

28.7 A worked synthesis: the ballot problem

To see the machinery dance, here is a classic with a one-line martingale solution.

Example 28.9 (Bertrand's ballot problem)**.** In an election candidate A receives α votes and candidate B receives β votes, with $\alpha > \beta$. Ballots are counted in a uniformly random order. What is the probability that A is *strictly ahead* throughout the entire count?

Read the count *backwards* from the final tally. Let M_k be the running value (A's lead)/(votes counted) when read from the end; this turns out to be a martingale in the reverse direction (a "backward martingale," the same projection idea run in reverse). Optional stopping at the first time A's lead would touch zero, combined with the constant-expectation property, collapses the whole combinatorial problem to the clean answer

$$\mathbb{P}(\text{A always strictly ahead}) = \frac{\alpha - \beta}{\alpha + \beta}.$$

With $\alpha = 6$, $\beta = 4$, the probability A is never tied or behind is $(6-4)/(6+4) = 1/5$. A problem that looks like it demands a brutal lattice-path count yields to a single martingale identity. This is the recurring lesson of the chapter: once you spot the fair game hiding inside a problem, the answer often falls out of constant expectation and optional stopping with almost no computation.

Connection

Martingales are the discrete skeleton of the continuous theory you will meet in the next chapter on Brownian motion. Brownian motion B_t is the continuous-time limit of the symmetric random walk and is itself a martingale; $B_t^2 - t$ is the continuous analogue of Example 28.2; and the optional stopping theorem becomes Doob's optional sampling theorem, the tool that computes exit probabilities and hitting times for diffusions. The stochastic integral $\int C_t \, \mathrm{d}B_t$ is the continuous martingale transform, and Ito's formula is the rule for how functions of a martingale behave. Everything in this chapter has a continuous twin, and the discrete versions you have just seen are the cleanest way to build the intuition before the analysis gets technical. (For the geometric view of conditional expectation as a projection, and for the integration-as-averaging picture behind these densities, see the companion calculus and linear algebra readers.)

28.8 Exercises

Exercise 28.1 (Warm-up identities)**.** Let (S_n) be the symmetric simple random walk with $S_0 = 0$. Verify directly from the definition that each of the following is a martingale: (a) S_n; (b) $S_n^2 - n$; (c) $S_n^3 - 3nS_n$. For part (c) you will need $\mathbb{E}[X] = 0$, $\mathbb{E}[X^2] = 1$, and $\mathbb{E}[X^3] = 0$ for a fair $X = \pm 1$.

Exercise 28.2 (Reading the definition)**.** Classify each process as a martingale, submartingale, or supermartingale, and justify in one sentence: (a) a gambler's fortune at a fair coin game; (b) a gambler's fortune at roulette; (c) $|S_n|$ for the symmetric walk; (d) the running maximum $\max_{k \le n} S_k$. *Hint:* $|x|$ is convex, and conditional Jensen turns a martingale into a submartingale under any convex function.

Exercise 28.3 (Predictable strategies)**.** You play a fair ± 1 game. Your friend proposes: "bet \$1 normally, but after any loss, bet \$2 on the next round to make up for it." Write this as a predictable stake sequence C_n and use Theorem 28.1 to compute the expected fortune after n rounds. Explain in words why no such rule can produce a positive expected gain.

Exercise 28.4 (Gambler's ruin numbers)**.** At a fair game you start with \$30 and quit when you reach \$100 or go broke. (a) Find the probability you reach \$100. (b) Find the expected number of bets you will make. (c) Repeat (a) for a target of \$1000 and comment on how the success probability scales with the ambition of the target.

Exercise 28.5 (Biased ruin)**.** A game gives you $+\$1$ with probability $p = 0.49$ and $-\$1$ with probability $q = 0.51$. Using the exponential martingale of Example 28.8, find the probability of turning \$100 into \$110 before going broke. Compare to the symmetric-game answer a/b and remark on how much a 1% edge against you costs.

Exercise 28.6 (Likelihood ratio)**.** Data Y_i are i.i.d. $\mathcal{N}(0, 1)$ under the truth, but you track the likelihood ratio against the alternative $\mathcal{N}(1, 1)$. (a) Write L_n explicitly as an exponential of a sum. (b) Show $\mathbb{E}[L_n] = 1$ for every n by direct computation, confirming the martingale property. (c) Argue using Example 28.3 and the convergence theorem that $L_n \to 0$ almost surely, and explain why this is exactly what a good sequential test should do.

Exercise 28.7 (Polya's urn limit)**.** For Polya's urn started with one red and one black ball, let $M_n = R_n/(n + 2)$ be the red fraction. (a) Re-derive that (M_n) is a martingale. (b) Compute $\mathbb{E}[M_n]$ for all n. (c) Simulate (by hand or code) several runs to 100 draws and observe that different runs settle to different limits, consistent with the claim that the limit is Uniform$(0, 1)$.

Exercise 28.8 (Why stopping cannot help — conceptual)**.** Explain carefully why "stop one step before S_n first reaches its all-time maximum" is *not* a valid stopping time, and why "stop the first time $S_n = 10$" *is*. Then explain in your own words why the optional stopping theorem would be false if we allowed the first kind of rule. Connect this to the no-clairvoyance idea behind predictable strategies.

Exercise 28.9 (A proof to do yourself)**.** Prove that if (M_n) is a martingale and ϕ is a convex function with $\mathbb{E}|\phi(M_n)| < \infty$ for all n, then $(\phi(M_n))$ is a submartingale. *Hint:* apply the conditional form of Jensen's inequality, $\mathbb{E}[\phi(M_{n+1}) \mid \mathcal{F}_n] \ge \phi(\mathbb{E}[M_{n+1} \mid \mathcal{F}_n])$, and then use the martingale property. Deduce that S_n^2 and $|S_n|$ are submartingales for the symmetric walk.

Exercise 28.10 (Optional stopping in finance)**.** A discounted stock price is modeled as a martingale with $\tilde{S}_0 = 100$. A trader claims a strategy that, by buying and selling based only on past prices,

guarantees an expected discounted profit of \$5. Using Theorem 28.1, explain why the trader must be mistaken (or secretly using future information). What real-world feature would have to break for a genuine guaranteed profit — an arbitrage — to exist?

Chapter 29

Brownian Motion

In the summer of 1827 the Scottish botanist Robert Brown put a drop of water under his microscope. Suspended in the water were tiny grains of pollen, and inside the grains were even tinier particles. Brown watched these particles and saw something that bothered him: they would not sit still. They jittered, darted, and zig-zagged in a ceaseless, aimless dance. His first thought was that they were alive. But he tested grains from a plant pressed in a herbarium a century earlier, then specks of ground glass, then dust scraped from the Sphinx – and they all danced the same. Whatever was making them move, it was not life. It was something about the water itself.

It took most of a century to understand what Brown had seen. The answer, when it came, tied together three of the deepest ideas in science: that matter is made of molecules, that heat is molecular motion, and that the visible jitter of a pollen grain is the accumulated effect of billions of invisible water molecules pelting it from every side. The particle is a punching bag in a storm of collisions, and the storm never stops.

This chapter is about the *mathematical* object that captures that dance. We will build it the way nature builds it: as the limit of a random walk taken with smaller and smaller, faster and faster steps. The limit turns out to be a single process, at once tame and a little monstrous – *Brownian motion*, also called the *Wiener process*. It is continuous everywhere yet smooth nowhere. Its increments are normal. It carries the Central Limit Theorem over into moving, evolving randomness. It solves the heat equation. And it is the engine underneath the most famous formula in mathematical finance. By the end you should be able to *see* Brownian motion, not just compute with it.

29.1 From Random Walk to a Wild Curve

You have already met the simple random walk. A walker stands at the origin. Every unit of time she flips a fair coin and steps $+1$ or -1. After n steps her position is

$$S_n = X_1 + X_2 + \cdots + X_n, \qquad X_i = \pm 1 \text{ each with probability } \tfrac{1}{2}.$$

The steps are independent, each has mean 0 and variance 1, so

$$\mathbb{E}[S_n] = 0, \qquad \mathrm{Var}(S_n) = n.$$

The walk drifts nowhere on average, but it spreads: its typical distance from the origin grows like $\sqrt{n}$, the square root of the number of steps. That $\sqrt{n}$ is the heartbeat of this whole chapter. Hold

onto it.

Intuition

Why $\sqrt{n}$ and not n? Because the steps point in random directions and mostly cancel. If they all pointed the same way you would travel a distance n. If they were perfectly anti-correlated you would never leave home. Independent random signs land you in between, at the scale $\sqrt{n}$. The variance adds (that is the one thing variance always does for independent terms), and standard deviation is its square root. Diffusion is slow precisely because it is built out of cancellation.

Now we want a *continuous* version. The trick is a clever rescaling. Suppose we want to watch the walk over the time interval $[0, 1]$, but take n steps in that interval instead of one. Then each step lasts time $1/n$. To keep the spread finite as $n \to \infty$ we must also shrink each spatial step. How much? Variance must stay finite. After n steps the variance is n times the variance of one step. If each step has size Δx, its variance is $(\Delta x)^2$, so the total variance is $n(\Delta x)^2$. For this to equal 1 (one unit of time, one unit of variance) we need

$$(\Delta x)^2 = \frac{1}{n}, \qquad \text{i.e.} \qquad \Delta x = \frac{1}{\sqrt{n}}.$$

Here is the scaling that makes everything work: **time shrinks like $1/n$, but space shrinks only like $1/\sqrt{n}$.** Space and time scale *differently*. This asymmetry – space as the square root of time – is the signature of diffusion, and it is why Brownian motion looks the way it does.

Define the rescaled walk

$$W^{(n)}(t) \;=\; \frac{1}{\sqrt{n}}\, S_{\lfloor nt \rfloor}, \qquad 0 \le t \le 1,$$

interpolated linearly between the integer step times. At $t = 1$ this is $S_n/\sqrt{n}$, which by the Central Limit Theorem converges to a standard normal. But the CLT only tells you about the endpoint. The deep statement is that the *entire path* $t \mapsto W^{(n)}(t)$ converges, as a random function, to a limiting random function. That limit is Brownian motion.

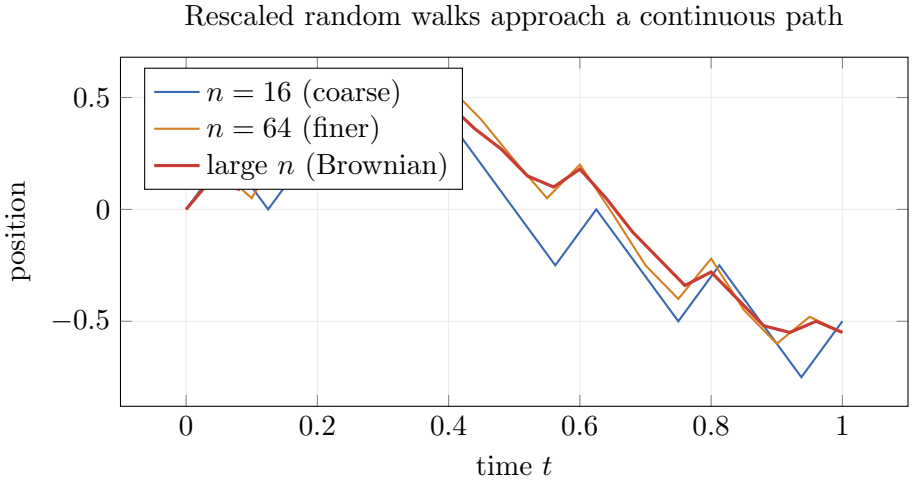


Figure 29.1: As we take more, smaller steps with the diffusive scaling $\Delta x = 1/\sqrt{n}$, the jagged random walk fills in to a continuous but infinitely wiggly curve. The endpoint distribution stays fixed (standard normal); what converges is the whole path.

History

Brown observed the dance in 1827 and published his account in 1828, but could not explain it. In 1905 – his *annus mirabilis*, the same year as special relativity – Albert Einstein derived the law $\mathbb{E}[\text{displacement}^2] = 2Dt$ from the molecular hypothesis, predicting that the mean-squared displacement grows linearly in time. This was *quantitative* evidence that atoms exist. In 1908 Jean Perrin measured the jitter under a microscope, confirmed Einstein's law, estimated Avogadro's number, and won a Nobel Prize for nailing down the reality of molecules. The rigorous mathematical object – a probability measure on the space of continuous paths – was constructed by Norbert Wiener in 1923, which is why mathematicians call Brownian motion the *Wiener process* and write it $W(t)$.

29.2 The Defining Properties

Rather than chase the limit through all its technical thickets, we do what mathematicians eventually learned to do: we *list the properties* the limit must have and take *those* as the definition. Wiener proved that a process with these properties exists and is essentially unique.

Definition 29.1 (Standard Brownian motion)**.** A *standard Brownian motion* (or Wiener process) is a random function $\{W(t) : t \geq 0\}$ with these four properties:

1. **Starts at zero:** $W(0) = 0$.
2. **Independent increments:** for any times $0 \leq t_0 < t_1 < \cdots < t_k$, the increments $W(t_1) - W(t_0)$, $W(t_2) - W(t_1)$, … are mutually independent.
3. **Gaussian, stationary increments:** for $s < t$,

$$W(t) - W(s) \;\sim\; \mathcal{N}\big(0,\; t - s\big).$$

   The increment is normal, mean zero, and its variance equals the *elapsed time*.
4. **Continuous paths:** with probability one, the function $t \mapsto W(t)$ is continuous.

Every one of these falls straight out of the random-walk limit. Independence of increments is just independence of the coin flips in disjoint time blocks. Stationarity (the increment law depends only on the gap $t-s$, not on where you start) is because every coin flip is identical. The Gaussian shape is the Central Limit Theorem applied to each block of flips. The variance equals elapsed time because we chose the scaling $\Delta x = 1/\sqrt{n}$ precisely to make it so. And continuity is the residue of the steps shrinking to zero.

Focus

The single most important formula in this chapter is

$$\boxed{\operatorname{Var}\big(W(t) - W(s)\big) = t - s} \qquad \text{and in particular} \qquad \operatorname{Var}\big(W(t)\big) = t, \quad \operatorname{SD}\big(W(t)\big) = \sqrt{t}.$$

Variance grows linearly in time; spread grows like $\sqrt{t}$. Everything else is commentary on this fact.

From the definition we can read off the *covariance structure*, which is all you ever need for a Gaussian process. Take $s \le t$. Write $W(t) = W(s) + \big(W(t) - W(s)\big)$. The second piece is independent of $W(s)$ and has mean zero, so

$$\operatorname{Cov}\big(W(s), W(t)\big) = \operatorname{Cov}\big(W(s),\, W(s)\big) + \operatorname{Cov}\big(W(s),\, W(t) - W(s)\big) = \operatorname{Var}(W(s)) + 0 = s.$$

For general s, t this reads

$$\boxed{\operatorname{Cov}\big(W(s), W(t)\big) = \min(s, t).}$$

Example 29.1 (Working with the increments)**.** Let W be standard Brownian motion. Find $\mathbb{P}\big(W(4) > 3\big)$ and $\mathbb{P}\big(W(4) - W(1) > 3\big)$.

Since $W(4) \sim \mathcal{N}(0, 4)$, its standard deviation is 2, so

$$\mathbb{P}(W(4) > 3) = \mathbb{P}\big(Z > \tfrac{3}{2}\big) = 1 - \Phi(1.5) \approx 0.0668,$$

where Z is standard normal and Φ its CDF. For the increment, $W(4) - W(1) \sim \mathcal{N}(0,\, 4-1) = \mathcal{N}(0, 3)$, with standard deviation $\sqrt{3} \approx 1.732$, so

$$\mathbb{P}\big(W(4) - W(1) > 3\big) = \mathbb{P}\Big(Z > \tfrac{3}{\sqrt{3}}\Big) = \mathbb{P}(Z > \sqrt{3}) = 1 - \Phi(1.732) \approx 0.0416.$$

The same threshold of 3 is less likely to be crossed by a 3-unit increment than by the 4-unit total, because the increment has had less time to spread.

Example 29.2 (Covariance in action)**.** Compute $\mathbb{E}\big[W(2)\, W(5)\big]$ and $\operatorname{Var}\big(W(5) - W(2)\big)$.

Means are zero, so $\mathbb{E}[W(2)W(5)] = \operatorname{Cov}(W(2), W(5)) = \min(2, 5) = 2$. For the variance, $W(5) - W(2) \sim \mathcal{N}(0, 5-2)$, hence Var $= 3$. Check it the long way: $\operatorname{Var}(W(5) - W(2)) = \operatorname{Var}(W(5)) - 2\operatorname{Cov}(W(5), W(2)) + \operatorname{Var}(W(2)) = 5 - 2(2) + 2 = 3$. Same answer.

Connection

Brownian motion is a *Gaussian process*: every finite collection $\big(W(t_1), \dots, W(t_k)\big)$ is a multivariate normal vector. Its mean is the zero vector and its covariance matrix has entries $\Sigma_{ij} = \min(t_i, t_j)$. Once you know mean and covariance, a Gaussian object is *completely* determined – a fact from the chapter on the multivariate normal and the covariance matrix in the companion linear-algebra material. So in a real sense the function $\min(s, t)$ *is* Brownian motion. Sampling a path is just drawing from a (very high-dimensional) multivariate normal.

29.3 Scaling, Symmetry, and Self-Similarity

Brownian motion has a family of symmetries that make it feel less like a specific random scribble and more like a law of nature. Each follows from the covariance $\min(s, t)$ or directly from the definition.

Proposition 29.1 (Invariances)**.** *Let W be standard Brownian motion. Then each of the following is again a standard Brownian motion:*

1. ***Scaling (self-similarity):*** *for any $c > 0$,* $\widetilde{W}(t) = \dfrac{1}{\sqrt{c}}\, W(ct)$.
2. ***Symmetry:*** $-W(t)$.

3. ***Time shift (Markov restart):*** $W(t+s) - W(s)$ *for fixed* $s \geq 0$.

4. ***Time inversion:*** $tW(1/t)$ *for* $t > 0$, *with value* 0 *at* $t = 0$.

Why it works

Take the scaling property, the most important one. We check it has the right covariance. With $\widetilde{W}(t) = c^{-1/2}W(ct)$,

$$\operatorname{Cov}\left(\widetilde{W}(s), \widetilde{W}(t)\right) = \frac{1}{c}\operatorname{Cov}\left(W(cs), W(ct)\right) = \frac{1}{c}\min(cs, ct) = \min(s, t).$$

Same covariance, still Gaussian, still mean zero, still continuous, still starts at 0 – so it is the same process in law. Read it out loud: if you zoom in on time by a factor c and on space by $\sqrt{c}$, the picture is statistically identical. Brownian motion is a *fractal*. There is no scale at which it looks smooth; magnify any tiny piece and you see the same jagged universe again. That is the $\Delta x = 1/\sqrt{n}$ scaling from §29.1 promoted to a permanent symmetry.

Why this matters

Self-similarity is why no plotter can ever truly draw Brownian motion. Whatever resolution you choose, the true path has structure finer than your pixels, with exactly the same character as what you can see. A drawn Brownian path is always a *coarsening* – which is fine, because by self-similarity the coarsening looks just like the real thing.

29.4 Continuous Everywhere, Smooth Nowhere

Here is the property that earned Brownian motion its reputation as a beautiful monster. Its paths are continuous – you can draw them without lifting your pen – yet at no single point do they have a well-defined slope. The path has no tangent line anywhere. You cannot differentiate it. Ever.

What is really happening

Why no derivative can exist. The derivative at t, if it existed, would be the limit of the difference quotient

$$\frac{W(t+h) - W(t)}{h}.$$

But the numerator is $\mathcal{N}(0, h)$, so its typical size is $\sqrt{h}$. Dividing by h gives a typical size of $\sqrt{h}/h = 1/\sqrt{h}$, which *blows up* as $h \to 0$. The increments shrink like $\sqrt{h}$, but the time gap shrinks faster, like h. The slope is forever losing the race. Over any tiny interval the path moves a distance proportional to $\sqrt{h}$, which dwarfs h – so locally the graph is far steeper than any line. Zoom in and it does not flatten out into a tangent the way a smooth curve does; it stays just as rough. That mismatch between the $\sqrt{h}$ of space and the h of time is, once again, the diffusive scaling, and it is exactly what forbids a derivative.

There is a clean way to quantify the roughness: *quadratic variation*. Chop $[0, t]$ into n equal pieces

of length $\Delta = t/n$ and add up the squared increments,

$$Q_n = \sum_{i=1}^{n} \Big(W(t_i) - W(t_{i-1})\Big)^2.$$

Each squared increment has expectation Δ (since the increment is $\mathcal{N}(0,\Delta)$ and $\mathbb{E}[\text{normal}^2]$ = variance). There are n of them, so $\mathbb{E}[Q_n] = n\Delta = t$. With a little more work one shows the variance of Q_n tends to 0, so

$$\boxed{Q_n \xrightarrow{p} t \quad \text{as } n \to \infty.}$$

This is startling. For an ordinary smooth function f, the squared increments are of order $(\Delta)^2$ and their sum vanishes; for Brownian motion the sum converges to a *nonzero* limit, the elapsed time itself. People summarize this with the formal mnemonic

$$(\mathrm{d}W)^2 = \mathrm{d}t,$$

which we will cash in shortly when we glimpse stochastic calculus. By contrast the ordinary (first-order) variation $\sum |W(t_i) - W(t_{i-1})|$ is *infinite*: the path wiggles so much that its total arc length over any interval is unbounded.

Common trap

Do not picture the velocity of a pollen grain as a Brownian path. A real grain has a real velocity at every instant – Brownian motion is an idealization that ignores the grain's inertia, valid only on time scales longer than the gap between collisions. The mathematical object $W(t)$ has *no velocity*, which is fine because it models *position*, observed at scales where the velocity has been averaged into oblivion. Confusing the model with the physical grain leads to paradoxes like "infinite kinetic energy" that are artifacts of the idealization.

Paradox

A continuous function with no derivative anywhere? Before the late 1800s, most mathematicians believed a continuous function had to be differentiable except perhaps at a few special points – it was "obvious" that a curve you can draw must have tangents almost everywhere. In 1872 Karl Weierstrass exhibited an explicit continuous-but-nowhere-differentiable function and scandalized the establishment; Charles Hermite called such functions a "lamentable plague." Brownian motion shows nature took Weierstrass's side: the *typical* continuous path – the one a fair random process actually produces – is nowhere differentiable with probability one. The smooth curves of calculus, far from being the rule, are an infinitely thin special case. The plague is the norm; the smooth functions are the freaks.

29.5 The Heat Equation: Randomness as Diffusion

Now comes a striking link: this random, jagged process is governed by a smooth deterministic partial differential equation – the *heat equation*. The same equation describes heat spreading through a metal bar, ink diffusing in still water, and the probability density of a Brownian particle.

Let $p(x,t)$ be the probability density of $W(t)$. From the definition, $W(t) \sim \mathcal{N}(0,t)$, so

$$p(x,t) = \frac{1}{\sqrt{2\pi t}} \exp\!\left(-\frac{x^2}{2t}\right).$$

This is the Gaussian, but with its variance *growing in time*. At t near 0 it is a tall thin spike at the origin (the particle is almost surely near 0); as t grows the bump flattens and spreads, exactly like a drop of dye diffusing outward.

The density of $W(t)$ spreads and flattens: a diffusing Gaussian

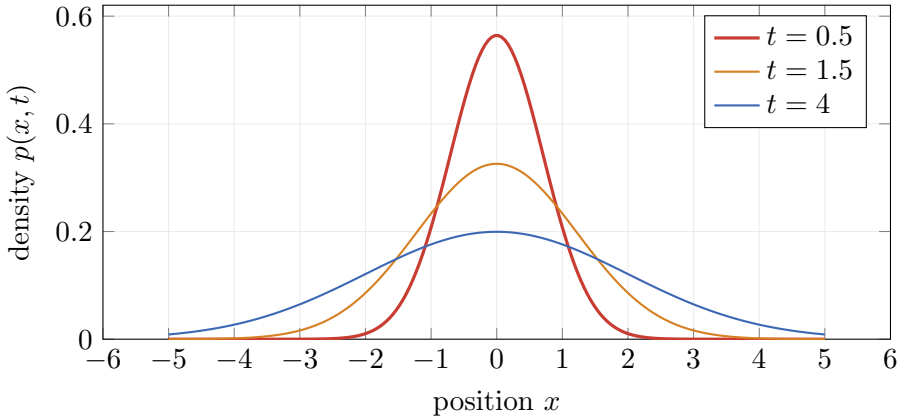


Figure 29.2: Snapshots of the Brownian density at three times. The area under each curve stays 1 (it is always a probability density); the bump just spreads. This is the same evolution heat undergoes in a conducting bar.

Now differentiate. A short calculation (done in the box below) shows that this density satisfies

$$\boxed{\frac{\partial p}{\partial t} = \frac{1}{2}\frac{\partial^2 p}{\partial x^2}.}$$

This is the *heat equation* (with diffusion constant $\frac{1}{2}$). The left side is the rate at which the density changes in time; the right side is its curvature in space. Where the density is concave down (a peak), the right side is negative and the peak sinks; where it is concave up (a valley between two bumps), probability fills in. Probability flows from high to low, smoothing itself out – the exact behavior of temperature.

Why it works

Verify $\partial_t p = \frac{1}{2}\partial_{xx} p$ directly for $p = (2\pi t)^{-1/2} e^{-x^2/2t}$. Take logs to make differentiation easy: $\ln p = -\frac{1}{2}\ln(2\pi t) - \frac{x^2}{2t}$. Then

$$\frac{\partial_t p}{p} = -\frac{1}{2t} + \frac{x^2}{2t^2}, \qquad \frac{\partial_x p}{p} = -\frac{x}{t}.$$

For the second space derivative, $\partial_{xx} p = \partial_x\big(p \cdot (-x/t)\big) = p\big(-x/t\big)\big(-x/t\big) + p\big(-1/t\big) = p\big(x^2/t^2 - 1/t\big)$. Therefore

$$\tfrac{1}{2}\,\partial_{xx} p = p\left(\frac{x^2}{2t^2} - \frac{1}{2t}\right) = \partial_t p,$$

matching the time derivative line for line. The Gaussian is not just *a* solution of the heat equation; with a point-mass initial condition it is *the* solution, the so-called heat kernel.

What is really happening

Why do randomness and heat obey the same equation? Because both are made of independent little pushes. Heat is energy random-walking from molecule to molecule; a Brownian particle is position random-walking under molecular kicks. The heat equation is just the continuum bookkeeping of "each point passes a bit of its stuff to its neighbors on each side." Whether the stuff is energy or probability, the local rule is identical, so the macroscopic law is identical. The factor of $\frac{1}{2}$ is the fingerprint of the variance-equals-time normalization in Definition 29.1; a different step variance would rescale it.

Connection

This is the gateway to the *Feynman–Kac formula* and to physics. Solving a diffusion PDE can be turned into averaging a function over Brownian paths, and vice versa. Expectation over random paths $\leftrightarrow$ solution of a deterministic PDE. The same duality, rotated into imaginary time, connects Brownian motion to the Schrödinger equation of quantum mechanics. The integral sign and the expectation sign are, here, two views of one object – a theme the companion calculus book develops as "expectation is an integral."

29.6 First Passages, the Reflection Principle, and the Maximum

A natural question: starting from 0, when does Brownian motion first reach a level $a > 0$? Call that random time $\tau_a = \inf\{t : W(t) = a\}$. By continuity the path cannot jump over a; it must actually touch it. There is a gorgeous trick, the *reflection principle*, that gives the answer almost for free.

Why it works

The reflection principle. Fix a level $a > 0$ and a time t. We want $\mathbb{P}\left(\max_{0\le s\le t} W(s) \ge a\right)$, the chance the path ever reaches a by time t. Split the event $\{\max_{0\le s\le t} W(s) \ge a\}$: any path ending above a has certainly hit a on the way (continuity again). Now take any path that hits a at some first time $\tau \le t$ but ends *below* a, and reflect its portion after τ across the line a. By the symmetry of Brownian motion (after τ it restarts fresh, and up is as likely as down), the reflected path is exactly as probable as the original, and it ends *above* a. This pairs up each "hit-then-below" path with an equally likely "hit-then-above" path. Hence

$$\mathbb{P}\left(\max_{0\le s\le t} W(s) \ge a\right) = 2\,\mathbb{P}\left(W(t) > a\right).$$

The maximum reaching a is exactly twice as likely as the endpoint exceeding a.

The reflection principle: each path below a pairs with a mirror image above

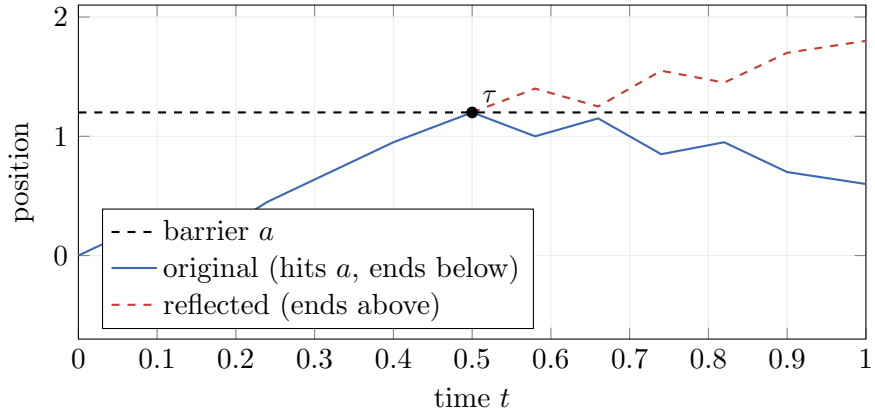


Figure 29.3: After the first hitting time τ, reflect the path across the level a. Because Brownian motion restarts fresh and symmetrically at τ, the reflected post-τ piece (red, ending above a) is exactly as probable as the original (blue, ending below a). This pairing is why $\mathbb{P}(\max_{[0,t]} W \geq a) = 2\,\mathbb{P}(W(t) > a)$.

Figure 29.3 shows the pairing in action. Since $W(t) \sim \mathcal{N}(0, t)$, this gives the clean formula

$$\mathbb{P}\left(\max_{0\leq s\leq t} W(s) \geq a\right) = 2\,\mathbb{P}\left(W(t) > a\right) = 2\Big(1 - \Phi\big(a/\sqrt{t}\big)\Big).$$

Example 29.3 (Hitting a barrier)**.** A particle does standard Brownian motion. What is the probability it reaches level $a = 2$ at some point during $[0, 4]$?

Here $a/\sqrt{t} = 2/\sqrt{4} = 1$, so

$$\mathbb{P}\left(\max_{0\leq s\leq 4} W(s) \geq 2\right) = 2\big(1 - \Phi(1)\big) = 2(0.1587) = 0.3174.$$

Even though the *endpoint* $W(4)$ exceeds 2 only with probability $1 - \Phi(1) = 0.1587$, the path *touches* 2 at some moment with twice that chance, because it can hit the barrier and wander back down.

Why this matters

The reflection principle is a rare case where a hard analytic question – the distribution of a maximum over a whole continuum of times – collapses to a one-line symmetry argument. It works because of three facts stacked together: continuity (no jumping over a), the strong Markov property (a fresh restart at the hitting time), and up-down symmetry. Lose any one and the trick breaks.

29.7 Variations: Drift and Geometric Brownian Motion

Pure Brownian motion has no preferred direction. Real systems often do: a stock tends to drift upward, a diffusing particle in a current is carried along. We add a deterministic trend and a volatility scale:

$$X(t) = \mu t + \sigma W(t).$$

Here μ is the *drift* (mean velocity) and σ the *volatility*. Then $X(t) \sim \mathcal{N}(\mu t,\ \sigma^2 t)$: the mean marches linearly while the spread still grows like $\sqrt{t}$. For small t the random $\sqrt{t}$ spread dominates the linear μt trend; for large t the trend wins. That crossover is why short-term price moves look like pure noise but long-term ones show the drift.

For prices we want something that cannot go negative and whose *percentage* changes, not absolute changes, behave nicely. The standard model is *geometric Brownian motion*,

$$S(t) = S(0)\,\exp\Big(\big(\mu - \tfrac{1}{2}\sigma^2\big)t + \sigma W(t)\Big).$$

The log-price is Brownian motion with drift, so the price itself is the exponential of a normal – a *lognormal* random variable. The stray $-\frac{1}{2}\sigma^2$ term is not a typo; it is the price of Jensen's inequality and the quadratic-variation fact $(\mathrm{d}W)^2 = \mathrm{d}t$, as we now see.

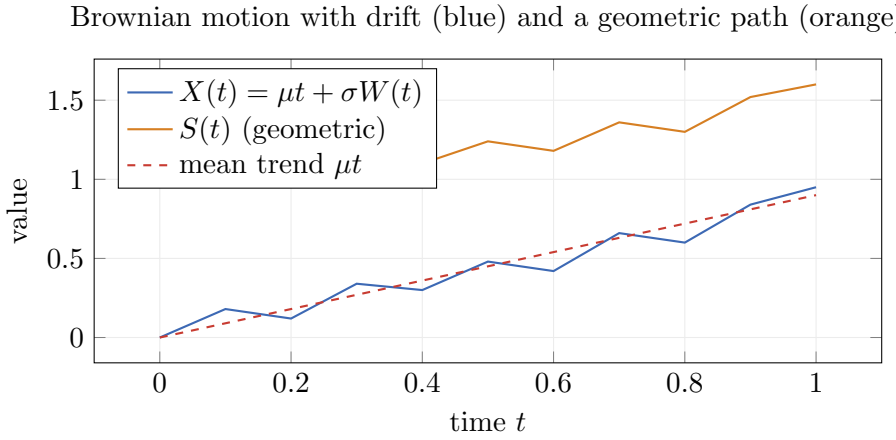


Figure 29.4: Adding a drift tilts the path; the geometric version exponentiates it so the value stays positive and compounds. Both still carry the unmistakable Brownian jitter.

29.8 A First Look at Stochastic Calculus

We have a process with no derivative. How on earth do you do calculus with it? You cannot write $\mathrm{d}W/\mathrm{d}t$. The escape is to keep everything in *integral* form and to build a new integration theory – *Itô calculus* – around the one fact that makes Brownian motion special: $(\mathrm{d}W)^2 = \mathrm{d}t$.

The central object is the Itô integral $\int_0^t H(s)\,\mathrm{d}W(s)$, which you should read as the limiting payoff of a strategy that holds $H(s)$ units of a Brownian asset and collects the increments $\mathrm{d}W$. Crucially the integrand is evaluated at the *left* endpoint of each tiny interval – you must decide your holding *before* the next random increment, not after. That "no peeking into the future" rule is what keeps the theory well behaved and tractable, and it is what links Itô integrals to martingales (the previous chapter).

The headline result is **Itô's lemma**, the chain rule of this world. For an ordinary smooth function f and ordinary calculus, $\mathrm{d}f(W) = f'(W)\,\mathrm{d}W$. But Brownian motion forces you to keep one more term, because $(\mathrm{d}W)^2$ does not vanish – it equals $\mathrm{d}t$:

$$\boxed{\mathrm{d}f\big(W(t)\big) = f'\big(W(t)\big)\,\mathrm{d}W(t) + \tfrac{1}{2}f''\big(W(t)\big)\,\mathrm{d}t.}$$

That extra $\frac{1}{2}f''\,\mathrm{d}t$ is the entire difference between ordinary and stochastic calculus, and it is the quadratic variation of §29.4 showing up to collect its due.

Why it works

Where does the extra term come from? Taylor-expand f over a tiny step:

$$f(W+\mathrm{d}W)-f(W)=f'(W)\,\mathrm{d}W+\tfrac{1}{2}f''(W)\,(\mathrm{d}W)^2+\cdots.$$

For a smooth path $(\mathrm{d}W)^2$ would be negligible compared to $\mathrm{d}W$, and you would drop it – that is ordinary calculus. But for Brownian motion $(\mathrm{d}W)^2$ is *not* negligible: summed up it converges to $\mathrm{d}t$, not to zero. So the second-order term survives into the answer while the cubic and higher terms still vanish. Itô's lemma is just Taylor's theorem with the rule $(\mathrm{d}W)^2=\mathrm{d}t$ substituted in. The nowhere-differentiability we feared turns out to be exactly what we exploit.

Example 29.4 (Why the $-\frac{1}{2}\sigma^2$ in geometric Brownian motion)**.** Apply Itô's lemma to $f(w)=e^{\sigma w}$, so $f'=\sigma e^{\sigma w}$, $f''=\sigma^2 e^{\sigma w}$. With $Y(t)=e^{\sigma W(t)}$,

$$\mathrm{d}Y=\sigma Y\,\mathrm{d}W+\tfrac{1}{2}\sigma^2 Y\,\mathrm{d}t.$$

The exponential of Brownian motion drifts *upward* at rate $\frac{1}{2}\sigma^2$ all by itself, purely from the convexity of e^x feeding on the wiggle. To model a stock whose true growth rate is μ, you must *subtract* that spurious drift, which is why $S(t)=S_0\exp\big((\mu-\frac{1}{2}\sigma^2)t+\sigma W(t)\big)$ carries the correction term. The $-\frac{1}{2}\sigma^2$ is Itô's lemma in disguise.

Application

Black–Scholes and the price of an option. In 1973 Fischer Black, Myron Scholes, and Robert Merton modeled a stock as geometric Brownian motion and asked: what is the fair price today of a *call option*, the right to buy the stock at a fixed price K at a future date T? Their insight was a kind of magic trick. At each instant you can hold a mix of the stock and cash that *exactly cancels* the random $\mathrm{d}W$ term in the option's value – a self-financing hedge. With the randomness hedged away, the portfolio must (to forbid arbitrage) earn the risk-free interest rate r. Setting those two facts equal and applying Itô's lemma turns the problem into a PDE – the Black–Scholes equation – which is the heat equation of §29.5 in disguise. Solving it gives the closed-form price

$$C=S_0\,\Phi(d_1)-Ke^{-rT}\,\Phi(d_2),\qquad d_{1,2}=\frac{\ln(S_0/K)+(r\pm\frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}.$$

There is that $\frac{1}{2}\sigma^2$ and that Φ – the Gaussian CDF – again. A nineteenth-century botanist's jittering pollen grain became, a century and a half later, the foundation of a multitrillion-dollar derivatives industry and a Nobel Prize in economics. That is the reach of one random curve.

Common trap

Geometric Brownian motion is a *model*, and a famously imperfect one. Real returns have fatter tails than the lognormal allows, volatility σ is not constant, and prices occasionally jump (Brownian paths never do). The 1987 crash and 2008 crisis both featured moves that Black–Scholes assigns probabilities so small they "should not happen in the lifetime of the

universe." Use the model for intuition and as a baseline; respect its assumptions; never mistake the elegant curve for the messy market.

29.9 Simulating Brownian Motion

The cleanest way to *see* Brownian motion is to generate one. Definition 29.1 hands you the recipe directly: increments over disjoint intervals are independent normals with variance equal to the time step. So pick a step Δt, draw independent $Z_i \sim \mathcal{N}(0,1)$, and accumulate.

Work it yourself

Build a Brownian path on $[0,1]$ with $N = 1000$ steps, so $\Delta t = 1/N$.

1. Set $W_0 = 0$.
2. For $i = 1, \dots, N$, draw $Z_i \sim \mathcal{N}(0,1)$ and set
$$W_i = W_{i-1} + \sqrt{\Delta t}\, Z_i.$$
The factor $\sqrt{\Delta t}$ is the diffusive scaling again: variance per step equals Δt, so the increment's standard deviation is $\sqrt{\Delta t}$.
3. Plot $(i\,\Delta t,\ W_i)$.

Now experiment. Halve Δt (double N) and overlay: the new path looks just as rough – self-similarity in your own plot. Replace the update with $W_i = W_{i-1} + \mu\,\Delta t + \sigma\sqrt{\Delta t}\, Z_i$ to add drift and volatility. Exponentiate to get a stock-price path. Run it a thousand times and histogram W_1: you will recover the standard normal, the Central Limit Theorem closing the loop we opened in §29.1.

Remark 29.1. That update rule, $W_i = W_{i-1} + \sqrt{\Delta t}\, Z_i$, is the *Euler–Maruyama* scheme, the workhorse of stochastic simulation. Note the $\sqrt{\Delta t}$ where an ordinary differential-equation solver would put Δt. Forget the square root and you are simulating the wrong process – the single most common bug in Monte Carlo finance code.

29.10 What to Carry Away

Focus

- Brownian motion is the scaling limit of random walks under the *diffusive scaling*: time $\sim 1/n$, space $\sim 1/\sqrt{n}$. Space is the square root of time.
- It is defined by four properties: starts at 0, independent increments, Gaussian stationary increments with $W(t) - W(s) \sim \mathcal{N}(0, t-s)$, and continuous paths. Equivalently it is the Gaussian process with covariance $\min(s,t)$.
- Its paths are continuous everywhere and differentiable nowhere; their quadratic variation over $[0,t]$ equals t, encoded as $(\mathrm{d}W)^2 = \mathrm{d}t$.

- The density of $W(t)$ is a spreading Gaussian that solves the heat equation $\partial_t p = \frac{1}{2}\partial_{xx} p$ – randomness and diffusion are the same mathematics.
- The reflection principle gives maxima and first-passage probabilities for free: $\mathbb{P}(\max_{[0,t]} W \geq a) = 2\,\mathbb{P}(W(t) > a)$.
- Itô's lemma, $\mathrm{d}f(W) = f'(W)\mathrm{d}W + \frac{1}{2}f''(W)\mathrm{d}t$, is the chain rule of a world where $(\mathrm{d}W)^2 = \mathrm{d}t$, and it powers geometric Brownian motion and the Black–Scholes formula.

The thread running through every section is a single scaling law – space grows like the square root of time – born from the cancellation of independent random steps. From that one fact flow the Gaussian increments, the fractal roughness, the missing derivative, the link to heat, and the strange extra term in stochastic calculus. Brown saw it in a microscope; Einstein explained it; Wiener made it rigorous; Black, Scholes, and Merton put a price on it. You should now see it too: a curve you can draw but never smooth, the purest picture of randomness in motion.

29.11 Exercises

Exercise 29.1 (Increment arithmetic)**.** Let W be standard Brownian motion. Compute (a) $\mathrm{Var}(W(7) - W(3))$; (b) $\mathrm{Cov}(W(3), W(7))$; (c) $\mathbb{P}(W(9) > 6)$ in terms of Φ; (d) $\mathbb{E}[W(2)^2]$.

Exercise 29.2 (Correlation between times)**.** For $s < t$, show that $\mathrm{Corr}\big(W(s), W(t)\big) = \sqrt{s/t}$. Interpret: why does the correlation between the position now and the position far in the future decay, and at what rate?

Exercise 29.3 (Scaling self-check)**.** Let $\widetilde{W}(t) = \frac{1}{2}W(4t)$. Verify directly that $\widetilde{W}(t) \sim \mathcal{N}(0, t)$ and that $\mathrm{Cov}(\widetilde{W}(s), \widetilde{W}(t)) = \min(s, t)$, confirming $\widetilde{W}$ is again a standard Brownian motion.

Exercise 29.4 (Brownian bridge)**.** Define $B(t) = W(t) - t\,W(1)$ for $0 \leq t \leq 1$ (Brownian motion "tied down" to 0 at both ends). Show $B(0) = B(1) = 0$, that $\mathbb{E}[B(t)] = 0$, and that $\mathrm{Cov}\big(B(s), B(t)\big) = \min(s, t) - st$ for $s, t \in [0, 1]$.

Exercise 29.5 (Reflection principle)**.** A standard Brownian motion runs on $[0, 9]$. (a) Find the probability it reaches level $a = 3$ at some time during $[0, 9]$. (b) Find the probability it reaches 3 but ends below 3, i.e. $\mathbb{P}(\max_{[0,9]} W \geq 3,\ W(9) < 3)$. (Hint: use the reflection pairing.)

Exercise 29.6 (Drifted Brownian motion)**.** Let $X(t) = \mu t + \sigma W(t)$ with $\mu = 1$, $\sigma = 2$. (a) Give the distribution of $X(4)$. (b) Compute $\mathbb{P}(X(4) > 10)$ in terms of Φ. (c) For what time t does the mean of $X(t)$ first exceed its standard deviation?

Exercise 29.7 (Heat equation, by hand)**.** Verify that $p(x, t) = (4\pi t)^{-1/2} \exp(-x^2/4t)$ satisfies $\partial_t p = \partial_{xx} p$ (diffusion constant 1 rather than $\frac{1}{2}$). Which variance growth rate does this correspond to, and how does the diffusion constant relate to the rate at which Var grows in time?

Exercise 29.8 (Itô's lemma practice)**.** Use Itô's lemma to find the stochastic differential of (a) $f(W) = W^2$ and (b) $f(W) = W^3$. For part (a), take expectations of both sides and integrate to re-derive $\mathbb{E}[W(t)^2] = t$. (Hint: the Itô-integral term has expectation zero.)

Exercise 29.9 (Geometric Brownian motion)**.** Let $S(t) = S_0 \exp\big((\mu - \frac{1}{2}\sigma^2)t + \sigma W(t)\big)$. (a) Show that $\ln S(t) \sim \mathcal{N}\big(\ln S_0 + (\mu - \frac{1}{2}\sigma^2)t,\ \sigma^2 t\big)$. (b) Using $\mathbb{E}[e^Z] = e^{\mathrm{Var}(Z)/2}$ for mean-zero normal Z, show that $\mathbb{E}[S(t)] = S_0 e^{\mu t}$ – so the $-\frac{1}{2}\sigma^2$ correction makes the *expected* price grow at exactly rate μ.

Exercise 29.10 (Conceptual: why no derivative)**.** Explain in your own words, using the $\sqrt{h}$ versus h scaling of increments, why $\big(W(t+h)-W(t)\big)/h$ has no finite limit as $h \to 0$. Then explain why the quadratic variation $\sum (\Delta W)^2$ converges to a nonzero limit while it would vanish for any differentiable function.

Exercise 29.11 (Simulation, proof by experiment)**.** Describe an Euler–Maruyama scheme to simulate W on $[0, T]$ with N steps. Explain why the per-step increment must use $\sqrt{\Delta t}$ and not Δt, and predict what histogram you would get from $W(T)$ over many independent runs. What does the histogram demonstrate about the connection to the Central Limit Theorem?

Part VI

Probability Meets Data

Chapter 30

Estimation: Method of Moments and Maximum Likelihood

So far this book has run in one direction. You knew the coin was fair, the machine emitted photons at a rate of three per second, the errors were Gaussian with mean zero and standard deviation two – and from those known truths you deduced what the data would look like. Probability flows from model to data.

Real life runs the other way. You are handed the data first. A new coin lands heads 61 times in 100 tosses. A call center logs the gaps between incoming calls. A drug trial reports recovery in 73 of 120 patients. Now you must swim upstream: from the data, infer the model. What is the coin's true bias? What is the call rate? Does the drug help, and by how much? This upstream journey is *statistical estimation*, and it is where probability stops being a game about dice and starts being the engine of empirical science.

This chapter introduces the two oldest and most important estimation engines: the *method of moments*, which matches the averages you see to the averages the model predicts, and *maximum likelihood*, which asks a disarmingly simple question, "which parameter value would have made my data most plausible?" We then confront the next questions. An estimate is itself random – it dances around as the data dances. Is it centered on the truth (bias)? How wildly does it dance (variance)? Does it home in on the truth as data piles up (consistency)? And is there a floor below which no estimator's dance can shrink (the Cramer–Rao bound)? By the end you should not just know the formulas; you should see estimation as a single coherent story about extracting a signal from noise.

30.1 Statistical Models and Parameters

Before you can estimate anything you must say what you are estimating. That requires a *statistical model*: a family of probability distributions, one of which we believe (or pretend) generated the data.

Definition 30.1 (Statistical model and parameter)**.** A *parametric statistical model* is a family of distributions $\{P_\theta : \theta \in \Theta\}$ indexed by a *parameter* θ living in a *parameter space* $\Theta \subseteq \mathbb{R}^k$. We assume the observed data $X_1, \dots, X_n$ are drawn from P_{θ_0} for some *true but unknown* $\theta_0 \in \Theta$. An *estimator* is any function $\hat{\theta} = g(X_1, \dots, X_n)$ of the data used to guess θ_0.

Read that definition slowly, because three distinct objects are hiding in it, and confusing them is the single most common source of muddle in all of statistics.

- The *parameter* θ_0 is a fixed, non-random number. The coin really does have some bias; we just do not know it. It does not have a distribution. It is what it is.
- The *data* $X_1, \dots, X_n$ are random – before you collect them they could come out many ways.
- The *estimator* $\hat{\theta}$ is a function of the random data, so it too is random. It is a random variable. It has a distribution, a mean, a variance. That distribution – called the *sampling distribution* of the estimator – is the central character of this chapter.

Focus

A parameter is a fixed unknown number. An estimator is a random variable that tries to pin it down. Almost everything that follows is the study of *how well a random thing can pin down a fixed thing.*

A small notational convention, used throughout statistics and throughout this book: a hat means "estimate of." So $\hat{\theta}$ estimates θ, $\hat{p}$ estimates p, $\hat{\sigma}^2$ estimates σ^2. When we want to emphasize the estimator depends on n data points we may write $\hat{\theta}_n$.

Example 30.1 (Three running models)**.** Keep these three in mind; we return to each repeatedly.

1. **Bernoulli / coin.** $X_i \sim \text{Bernoulli}(p)$, so $X_i \in \{0, 1\}$ with $\mathbb{P}(X_i = 1) = p$. Here $\theta = p \in [0, 1]$. Data: 61 heads in 100.
2. **Poisson / counts.** $X_i \sim \text{Poisson}(\lambda)$, the number of calls in successive minutes. Here $\theta = \lambda > 0$.
3. **Normal.** $X_i \sim \mathcal{N}(\mu, \sigma^2)$. Here $\theta = (\mu, \sigma^2)$ is two-dimensional, $\Theta = \mathbb{R} \times (0, \infty)$.

Why this matters

Why bother with a model at all? Because data alone cannot answer a question about the world; only data *plus* an assumption about how the world generated it can. The model is the bridge from "what I saw" to "what is true." Choose the bridge badly and even infinite data leads you to a confident wrong answer. Every estimate in this chapter is conditional on its model being roughly right – a humility worth carrying.

30.2 The Method of Moments

The oldest estimation idea is also the most intuitive, and you have been using its baby version your whole life. If you want to know the average height of adults, you measure some adults and average their heights. You are matching a *sample average* to a *population average*. The method of moments, introduced by Karl Pearson in the 1890s, simply industrializes this instinct.

Intuition

The model tells you what the theoretical moments – the mean $\mathbb{E}[X]$, the mean square $\mathbb{E}[X^2]$, and so on – ought to be, *as functions of the unknown parameters*. The data hand you empirical versions of those same moments. The method of moments sets theory equal to data and solves for the parameters. It is the statistical equivalent of "n unknowns, n equations."

Recall the k-th *population moment* $\mu_k = \mathbb{E}[X^k]$ and define the k-th *sample moment* as the obvious empirical average

$$\hat{\mu}_k = \frac{1}{n}\sum_{i=1}^{n} X_i^k.$$

The law of large numbers (a previous chapter) guarantees $\hat{\mu}_k \xrightarrow{p} \mu_k$: sample moments converge to population moments. That is the whole engine. If μ_k depends on θ, and the sample moment is close to μ_k, then solving the equation should land you near the true θ.

Definition 30.2 (Method-of-moments estimator)**.** For a model with k unknown parameters, write the first k population moments $\mu_1(\theta), \dots, \mu_k(\theta)$ as functions of θ. The *method-of-moments estimator* $\hat{\theta}_{\text{MM}}$ is the solution of the k equations

$$\mu_1(\theta) = \hat{\mu}_1, \quad \mu_2(\theta) = \hat{\mu}_2, \quad \dots, \quad \mu_k(\theta) = \hat{\mu}_k.$$

Example 30.2 (MoM for the Poisson rate)**.** For Poisson(λ) we know $\mu_1 = \mathbb{E}[X] = \lambda$. One parameter, so one equation: $\lambda = \hat{\mu}_1 = \bar{X}$. The method-of-moments estimate of the call rate is just the sample mean, $\hat{\lambda}_{\text{MM}} = \bar{X} = \frac{1}{n}\sum_i X_i$. If 200 minutes produced 560 calls, $\hat{\lambda} = 2.8$ calls per minute.

Example 30.3 (MoM for the Normal)**.** Two parameters (μ, σ^2), so two equations. The first two population moments are $\mu_1 = \mu$ and $\mu_2 = \mathbb{E}[X^2] = \sigma^2 + \mu^2$. Setting them equal to the sample moments:

$$\mu = \hat{\mu}_1 = \bar{X}, \qquad \sigma^2 + \mu^2 = \hat{\mu}_2 = \frac{1}{n}\sum_i X_i^2.$$

Substituting the first into the second,

$$\hat{\sigma}^2_{\text{MM}} = \frac{1}{n}\sum_i X_i^2 - \bar{X}^2 = \frac{1}{n}\sum_i (X_i - \bar{X})^2.$$

So MoM hands you the sample mean and the sample variance (with the $1/n$ divisor) – comfortingly familiar faces.

Example 30.4 (MoM for the Uniform – where intuition gets interesting)**.** Let $X_i \sim \text{Uniform}(0, \theta)$, so the data are spread between 0 and an unknown ceiling θ. The mean of a Uniform$(0, \theta)$ is $\theta/2$, so the moment equation is $\theta/2 = \bar{X}$, giving

$$\hat{\theta}_{\text{MM}} = 2\bar{X}.$$

Here is a delicious trap. Suppose your sample is 0.1, 0.4, 0.9. Then $\bar{X} = 0.467$ and $\hat{\theta}_{\text{MM}} = 0.933$. But you *observed* a value of 0.9! The estimated ceiling is barely above a point you actually saw, and a slightly unluckier sample could give $\hat{\theta} < \max_i X_i$ – an estimated ceiling *below* an observation, which is logically impossible under the model. The method of moments does not know that θ must be at least as large as the biggest observation. It only knows about averages. This is our first hint that intuition-by-averaging, though seductive, can be improved upon. Figure 30.1 draws the situation.

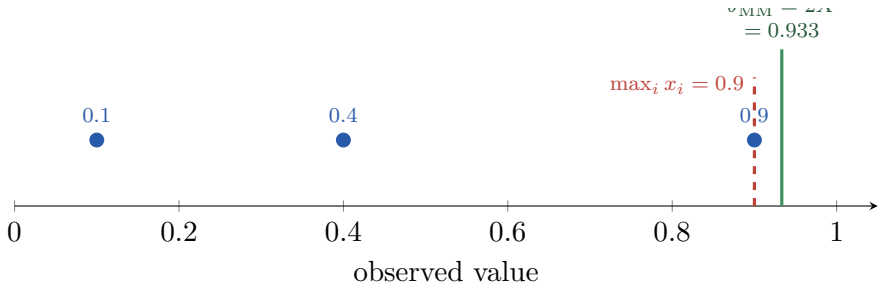


Figure 30.1: Three observations (dots) from Uniform$(0, \theta)$. The method-of-moments ceiling $2\bar{X} = 0.933$ sits barely above the largest observation 0.9. A slightly unluckier draw would push $2\bar{X}$ *below* 0.9 – an estimated ceiling lower than a value we actually saw, which the model forbids. Averaging alone is blind to this constraint.

Common trap

The method of moments is fast, requires no calculus, and gives a sensible starting guess for almost any model – which is exactly why it is still used to initialize harder optimizations. But it can produce nonsensical answers (a ceiling below an observation, a negative variance, a probability above one) because it only listens to a few summary numbers and ignores the finer structure of the likelihood. Treat it as a quick sketch, not a final portrait.

History

Karl Pearson built the method of moments around 1894 while fitting skewed biological data – crab measurements from the Bay of Naples – that refused to look Gaussian. He modeled them as mixtures and matched up to five moments, grinding through the algebra by hand. For thirty years it was *the* way to fit distributions. Then a young R. A. Fisher argued, in a series of papers from 1912 to 1922, that matching moments was wasteful: it threw away information the data contained. His alternative, maximum likelihood, eventually won the field. The friendly rivalry between Pearson's and Fisher's schools is one of the founding dramas of modern statistics – and personal: the two men famously could not stand each other.

30.3 The Likelihood Function

Fisher's complaint was that the method of moments listens to only a handful of summaries. The likelihood listens to everything. The whole idea fits in two words: *run probability backwards*. Ordinarily $\mathbb{P}(\text{data} \mid \theta)$ treats θ as fixed and asks how probable the data are. Fisher said: I have the data; let me hold *them* fixed and view that same number as a function of θ. Now it ranks the candidate parameter values by how well each would have predicted what actually happened.

Definition 30.3 (Likelihood and log-likelihood)**.** Given observed data $x_1, \ldots, x_n$ and a model with parameter θ, the *likelihood function* is the joint density (or mass) of the data, read as a function of

θ with the data held fixed:

$$L(\theta) = L(\theta; x_1, \dots, x_n) = \prod_{i=1}^{n} f(x_i \mid \theta)$$

for i.i.d. data with density/mass f. Its logarithm, the *log-likelihood*, is

$$\ell(\theta) = \log L(\theta) = \sum_{i=1}^{n} \log f(x_i \mid \theta).$$

Common trap

$L(\theta)$ is *not* a probability distribution over θ. It does not integrate to one over Θ, and "$L(\theta)$" is not "the probability that the parameter is θ." The parameter is a fixed number, not a random variable – it has no distribution to speak of. (When you *do* put a distribution on θ you have crossed into Bayesian inference, the subject of the next chapter, and the object you get is the posterior, not the likelihood.) The likelihood is a *ranking*, not a probability: it tells you which θ values make the data look plausible, in relative terms only.

Why it works

Why multiply the densities? Because the observations are independent, and independence makes joint probabilities multiply. The probability of seeing *this* value and *that* value and *the other* value is the product of the individual probabilities. Why take the log? Three reasons, all practical. First, products of n small numbers underflow to zero on a computer; sums of logs do not. Second, sums are far easier to differentiate than products – and we are about to differentiate. Third, the log turns the i.i.d. product into a *sum* over data points, which is exactly the form the law of large numbers and the central limit theorem love. Because log is strictly increasing, the θ that maximizes L also maximizes ℓ, so we lose nothing.

30.4 Maximum Likelihood Estimation

Definition 30.4 (Maximum likelihood estimator)**.** The *maximum likelihood estimator* (MLE) is the parameter value that makes the observed data most probable:

$$\hat{\theta}_{\text{MLE}} = \arg\max_{\theta \in \Theta} L(\theta) = \arg\max_{\theta \in \Theta} \ell(\theta).$$

The recipe, when ℓ is smooth, is the familiar calculus drill: differentiate ℓ, set the derivative (the *score*) to zero, solve, and check it is a maximum.

$$\text{score:} \qquad s(\theta) = \frac{\partial \ell}{\partial \theta} = 0.$$

(In Section 30.7 we will use the word *score* for the derivative of a *single* observation's log-density, $\partial_\theta \log f(X \mid \theta)$. The two usages are consistent: because ℓ is a sum over observations, the full-data score is the sum of the per-observation scores. We name both "score" as the field does, but the distinction matters when we take variances.)

Example 30.5 (The MLE of a coin's bias – the canonical case)**.** You toss a coin n times and see k heads. With $X_i \sim \text{Bernoulli}(p)$ the mass function is $f(x \mid p) = p^x(1-p)^{1-x}$, so the likelihood is

$$L(p) = \prod_{i=1}^{n} p^{x_i}(1-p)^{1-x_i} = p^k(1-p)^{n-k}, \qquad k = \sum_i x_i.$$

Take logs:

$$\ell(p) = k \log p + (n-k)\log(1-p).$$

Differentiate and set to zero:

$$\ell'(p) = \frac{k}{p} - \frac{n-k}{1-p} = 0 \implies k(1-p) = (n-k)p \implies k = np \implies \boxed{\hat{p}_{\text{MLE}} = \frac{k}{n}.}$$

The maximum likelihood estimate of the coin's bias is the observed fraction of heads. For 61 heads in 100, $\hat{p} = 0.61$. The MLE recovers the most natural estimate there is – but now you know *why* it is natural: it is the value that makes your particular run of 61 heads the most probable outcome it could have been.

Figure 30.2 shows the log-likelihood for this coin. Notice its single smooth peak at 0.61, and notice how the curve gets *sharper* as the sample grows – a foreshadowing of Fisher information in Section 30.7.

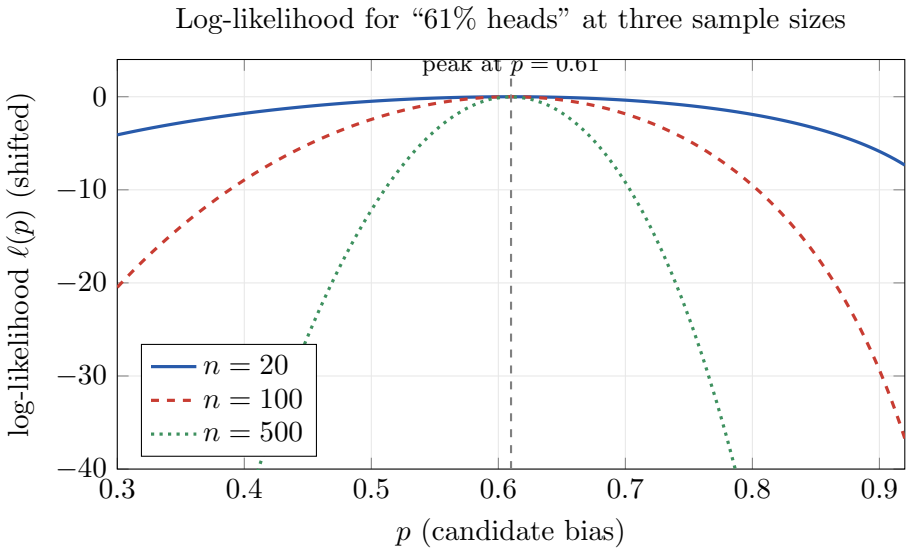


Figure 30.2: The log-likelihood for a coin showing 61% heads, shifted to peak at zero. All three curves peak at $\hat{p} = 0.61$ regardless of n, but more data makes the peak dramatically sharper – the data "insist" on 0.61 more strongly. That sharpness is precisely what makes a large sample informative.

Example 30.6 (The MLE of the Poisson rate)**.** With $X_i \sim \text{Poisson}(\lambda)$, $f(x \mid \lambda) = e^{-\lambda}\lambda^x/x!$, so

$$\ell(\lambda) = \sum_i (-\lambda + x_i \log\lambda - \log x_i!) = -n\lambda + \Big(\sum_i x_i\Big)\log\lambda - \sum_i \log x_i!.$$

The last term does not involve λ, so it falls away under differentiation:

$$\ell'(\lambda) = -n + \frac{\sum_i x_i}{\lambda} = 0 \implies \hat{\lambda}_{\text{MLE}} = \frac{\sum_i x_i}{n} = \bar{X}.$$

Here the MLE and the method-of-moments estimate coincide – both equal $\bar{X}$. They will not always agree, but when the model is "nice" (an exponential family, matched at its natural moments) they often do.

Example 30.7 (The MLE of the Normal – two parameters at once)**.** For $X_i \sim \mathcal{N}(\mu, \sigma^2)$,

$$\ell(\mu, \sigma^2) = -\frac{n}{2}\log(2\pi) - \frac{n}{2}\log\sigma^2 - \frac{1}{2\sigma^2}\sum_i (x_i - \mu)^2.$$

Set the two partial derivatives to zero. With respect to μ:

$$\frac{\partial \ell}{\partial \mu} = \frac{1}{\sigma^2}\sum_i (x_i - \mu) = 0 \implies \hat{\mu} = \bar{X}.$$

With respect to σ^2 (treat $v = \sigma^2$ as the variable):

$$\frac{\partial \ell}{\partial v} = -\frac{n}{2v} + \frac{1}{2v^2}\sum_i (x_i - \mu)^2 = 0 \implies \hat{\sigma}^2 = \frac{1}{n}\sum_i (x_i - \bar{X})^2.$$

Again the MLE matches the method of moments. Hold onto that $1/n$ divisor: it will turn out to be biased, and the fix is one of the most famous footnotes in statistics (Section 30.5).

Example 30.8 (The MLE of the Uniform ceiling – calculus fails, and that is the lesson)**.** Return to $X_i \sim \text{Uniform}(0, \theta)$. The density is $f(x \mid \theta) = 1/\theta$ for $0 \le x \le \theta$ and 0 otherwise. The likelihood is

$$L(\theta) = \prod_{i=1}^{n} \frac{1}{\theta}\,\mathbf{1}\{0 \le x_i \le \theta\} = \frac{1}{\theta^n}\,\mathbf{1}\{\theta \ge \max_i x_i\}.$$

Here differentiating is a trap. As a function of θ, $1/\theta^n$ is *decreasing*, so we want θ as small as possible – but the indicator forbids θ from dropping below the largest observation. The likelihood is therefore maximized at the boundary:

$$\boxed{\hat{\theta}_{\text{MLE}} = \max_i X_i.}$$

Compare this to the method-of-moments answer $2\bar{X}$. The MLE can never be less than an observed value – it respects the logic of the model. This is the clearest single demonstration in the book that maximum likelihood "uses more of the data": it knows the maximum matters, while moment-matching only ever saw the average.

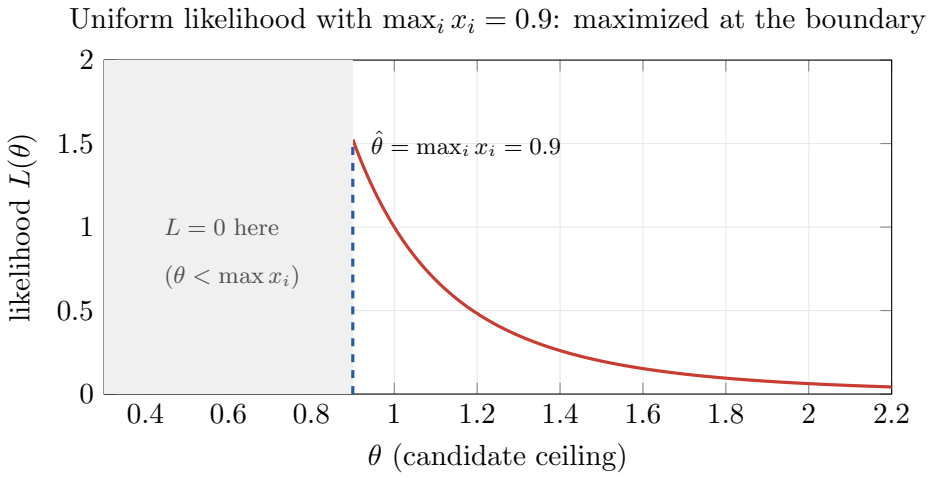


Figure 30.3: For Uniform$(0, \theta)$ with $n = 4$ and largest observation 0.9, the likelihood is zero whenever $\theta < 0.9$ (an impossible ceiling) and decays as θ^{-4} above it. The maximum sits exactly at the boundary $\hat{\theta} = 0.9$. No derivative is set to zero – the optimum is a corner.

What is really happening

What is maximum likelihood *really* doing? Picture the parameter space as a landscape and the log-likelihood as its elevation. Each data point tilts the landscape a little, pushing the summit toward parameter values that explain that point well. With i.i.d. data these little tilts *add up* (because logs turn the product into a sum), so the summit settles at the consensus of all the data points. As n grows, more votes accumulate and the summit grows sharper and more decisively located – the law of large numbers acting on the score function. Deeper still: maximizing the likelihood is the same as minimizing the Kullback–Leibler divergence from the true distribution to your model. The MLE is the member of your model family that is, in an information-theoretic sense, *closest to the truth* given the evidence. We make this precise in the information-theory chapter; here, just feel the shape of it.

Connection

Maximum likelihood is an optimization problem, so the whole machinery of calculus comes to bear: the first-order condition (score = 0) is just "gradient equals zero," and checking it is a maximum uses the second derivative being negative. For multi-parameter models the score becomes a gradient *vector* and the curvature becomes a Hessian *matrix* – so linear algebra enters too, and the matrix of expected curvatures is the Fisher information matrix of Section 30.7. When the score equation has no closed form (logistic regression, mixtures, most of modern machine learning), you climb the likelihood numerically with gradient ascent or Newton's method – literally the same algorithms that train neural networks. See the companion calculus and linear algebra readers for the optimization background.

30.4.1 The invariance property

Maximum likelihood has a property the method of moments lacks, and it saves real work.

Theorem 30.1 (Invariance of the MLE)**.** *If $\hat{\theta}$ is the MLE of θ and g is any function, then the MLE of $g(\theta)$ is $g(\hat{\theta})$.*

Why it works

The reason is almost a tautology, which is why it is so reliable. The MLE is the θ that sits at the top of the likelihood hill. If you re-label the horizontal axis by a transformation g, the hill does not move – the same *point* is still the summit, it just wears a new name $g(\hat{\theta})$. Maximization cares about *where* the peak is, not what you call the location.

Example 30.9 (Invariance in action)**.** The MLE of the Bernoulli parameter is $\hat{p} = k/n$. Suppose you actually care about the *odds* $g(p) = p/(1-p)$. By invariance, no new calculus is needed: the MLE of the odds is simply $\hat{p}/(1-\hat{p}) = k/(n-k)$. Likewise, the MLE of the Normal standard deviation σ is $\sqrt{\hat{\sigma}^2}$, the square root of the MLE of the variance. You never have to re-derive anything.

30.5 Judging an Estimator: Bias, Variance, Consistency

We now have two machines for producing estimates. But producing an estimate is not the same as producing a *good* estimate. How do we grade an estimator? The estimator is random, so we grade its sampling distribution. Three properties matter most.

Definition 30.5 (Bias)**.** The *bias* of an estimator $\hat{\theta}$ of θ is

$$\text{Bias}(\hat{\theta}) = \mathbb{E}[\hat{\theta}] - \theta.$$

The estimator is *unbiased* if its bias is zero, i.e. $\mathbb{E}[\hat{\theta}] = \theta$ for every θ: on average, across many hypothetical repetitions of the experiment, it lands on the truth.

Intuition

Bias is a systematic error – a scope that is sighted off-center. Even with infinite patience (infinitely many repeated experiments), the average of a biased estimator misses the bullseye. Unbiasedness says: *no* systematic lean. It does not say any single estimate is correct, only that the errors cancel on average.

Example 30.10 (The sample mean is unbiased)**.** If X_i have mean μ then by linearity of expectation,

$$\mathbb{E}[\bar{X}] = \mathbb{E}\left[\frac{1}{n}\sum_i X_i\right] = \frac{1}{n}\sum_i \mathbb{E}[X_i] = \frac{1}{n}\cdot n\mu = \mu.$$

The sample mean is unbiased for the population mean, full stop – it needs no assumption beyond a finite mean. This is why $\bar{X}$ is the workhorse of statistics.

Example 30.11 (The MLE of the variance is biased – and the famous fix)**.** Recall the MLE $\hat{\sigma}^2 = \frac{1}{n}\sum_i (X_i - \bar{X})^2$. Is it unbiased? Let us compute its expectation. Write σ^2 for the true variance. A standard identity expands the sum of squared deviations:

$$\sum_i (X_i - \bar{X})^2 = \sum_i (X_i - \mu)^2 - n(\bar{X} - \mu)^2.$$

Take expectations. Each $\mathbb{E}[(X_i - \mu)^2] = \sigma^2$, giving $n\sigma^2$ for the first term. And $\mathbb{E}[(\bar{X} - \mu)^2] = \mathrm{Var}(\bar{X}) = \sigma^2/n$, so the second term contributes $n \cdot \sigma^2/n = \sigma^2$. Hence

$$\mathbb{E}\left[\sum_i (X_i - \bar{X})^2\right] = n\sigma^2 - \sigma^2 = (n-1)\sigma^2,$$

and therefore

$$\mathbb{E}[\hat{\sigma}^2] = \frac{n-1}{n}\,\sigma^2 < \sigma^2.$$

The MLE *systematically underestimates* the variance. The cure is to divide by $n-1$ instead of n:

$$S^2 = \frac{1}{n-1}\sum_i (X_i - \bar{X})^2, \qquad \mathbb{E}[S^2] = \sigma^2.$$

This S^2 is the *sample variance* you have seen everywhere with its mysterious $n-1$. Now the mystery is gone.

Why it works

Why $n-1$ and not n? Because you used the data twice. To measure spread around the center you first had to *estimate* the center with $\bar{X}$ – and $\bar{X}$ is, by construction, the point that sits closest to your particular sample. Deviations measured from $\bar{X}$ are therefore a touch too small, because $\bar{X}$ was pulled toward the data. You have spent one "degree of freedom" locating the center, leaving only $n-1$ independent pieces of information about spread. Dividing by $n-1$ compensates exactly. The phrase *degrees of freedom* is just bookkeeping for "how many independent residuals are really left."

Common trap

Maximum likelihood does *not* promise unbiasedness, and the variance MLE is the standard counterexample. The MLE's virtues are large-sample virtues (consistency, efficiency); in small samples it can lean. Do not assume "MLE" means "unbiased." Conversely, do not over-worship unbiasedness either – as we are about to see, a little bias can be a bargain.

Definition 30.6 (Variance and mean squared error)**.** The *variance* of an estimator is $\mathrm{Var}(\hat{\theta})$, the spread of its sampling distribution. The *mean squared error* combines both kinds of error:

$$\mathrm{MSE}(\hat{\theta}) = \mathbb{E}\left[(\hat{\theta} - \theta)^2\right] = \underbrace{\mathrm{Var}(\hat{\theta})}_{\text{wobble}} + \underbrace{\mathrm{Bias}(\hat{\theta})^2}_{\text{lean}}.$$

Why it works

The MSE decomposition is just the "add and subtract the mean" trick. Write $\hat{\theta} - \theta = (\hat{\theta} - \mathbb{E}\,\hat{\theta}) + (\mathbb{E}\,\hat{\theta} - \theta)$. Square it: the first piece squared has expectation $\mathrm{Var}(\hat{\theta})$, the second piece is the constant Bias^2, and the cross term is $2\,\mathbb{E}[\hat{\theta} - \mathbb{E}\,\hat{\theta}] \cdot \mathrm{Bias} = 0$ because the first factor has mean zero. So total expected squared error cleanly splits into *wobble plus lean*. This little identity is the seed of the entire bias–variance tradeoff.

Definition 30.7 (Consistency)**.** An estimator is *consistent* if it converges in probability to the truth as data accumulate: $\hat{\theta}_n \xrightarrow{p} \theta$, meaning for every $\varepsilon > 0$, $\mathbb{P}(|\hat{\theta}_n - \theta| > \varepsilon) \to 0$ as $n \to \infty$.

Intuition

Consistency is the bare-minimum sanity check: *if you collect enough data, you get the right answer.* An inconsistent estimator is broken at the foundation – no amount of data rescues it. A clean sufficient condition: if $\text{Bias}(\hat{\theta}_n) \to 0$ *and* $\text{Var}(\hat{\theta}_n) \to 0$, then $\text{MSE} \to 0$, and convergence in mean square forces convergence in probability (Chebyshev). So *shrinking bias plus shrinking variance* $\Rightarrow$ consistency.

Example 30.12 (Both variance estimators are consistent)**.** The biased MLE $\hat{\sigma}^2$ has bias $-\sigma^2/n \to 0$ and variance of order $1/n \to 0$, so it is consistent despite being biased. The unbiased S^2 is also consistent. As $n \to \infty$ the factor $(n-1)/n \to 1$ and the two estimators become indistinguishable. *The $n-1$ versus n debate is a small-sample debate*; asymptotically it evaporates. This is the recurring moral of the chapter: bias usually fades with data, so the real long-run battle is over variance.

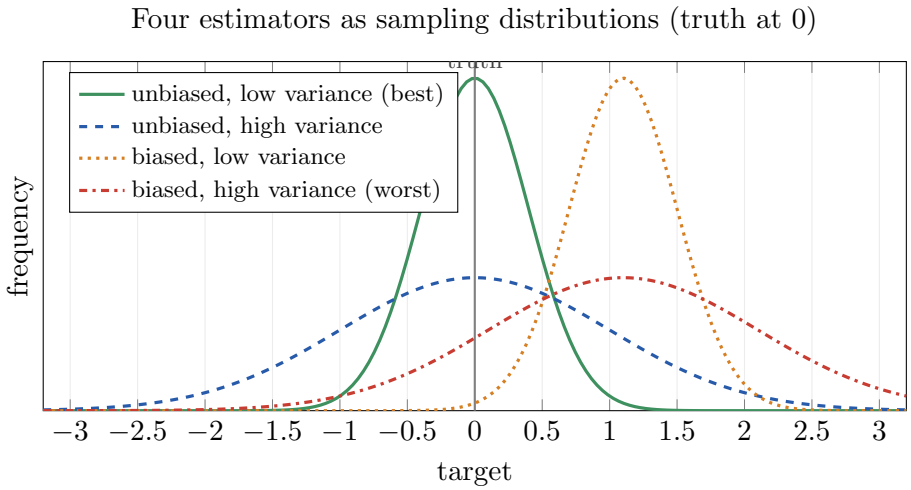


Figure 30.4: Bias is where the distribution is *centered* relative to the truth (horizontal shift); variance is how *wide* it is. A good estimator is both centered on the truth and tight around it. The mean squared error is the average squared distance from the truth, capturing both the shift and the width at once.

30.6 The Bias–Variance Tradeoff

Decompose MSE into wobble plus lean and a strategic possibility leaps out: maybe you can *accept* a little lean (bias) in exchange for far less wobble (variance), and come out ahead on total error. Unbiasedness is not sacred; *small MSE* is the real goal.

Example 30.13 (Deliberately biasing the coin estimate)**.** The unbiased estimate of a coin's bias is $\hat{p} = k/n$. But near the extremes it behaves badly: with 0 heads in 5 tosses it shouts "$p = 0$, this coin *never* lands heads," a wildly overconfident claim from five tosses. Consider instead the gently shrunk estimator

$$\tilde{p} = \frac{k+1}{n+2},$$

which nudges every estimate toward $1/2$ (it is, not coincidentally, the Bayesian posterior mean under a uniform prior – the next chapter). With 0 heads in 5 it returns $1/7 \approx 0.14$ instead of 0: still

"probably tail-heavy," but not absurdly certain. $\tilde{p}$ is *biased* – it leans toward $1/2$ – yet for moderate n and p not too near 0 or 1 it often has *smaller MSE* than the unbiased $\hat{p}$, because the variance reduction near the boundaries outweighs the small bias. A pinch of bias bought a calmer, more reliable estimator.

Application

[The bias–variance tradeoff is the soul of machine learning] This single idea – trade a little bias for a lot less variance – organizes the entire practice of supervised learning. A model's expected prediction error decomposes (exactly as our MSE did) into

$$\text{error} = \text{bias}^2 + \text{variance} + \text{irreducible noise}.$$

- A *too-simple* model (a straight line for a curvy world) has high bias: it cannot represent the truth no matter how much data you give it. It *underfits*.
- A *too-flexible* model (a degree-30 polynomial, a giant decision tree) has high variance: it contorts itself to chase the noise in this particular training set and changes wildly with a new sample. It *overfits*.

Every regularization knob in machine learning – ridge and lasso penalties, shrinkage, dropout, early stopping, pruning, limiting tree depth – is *deliberately introduced bias* that buys a large reduction in variance. Ridge regression, for instance, shrinks coefficients toward zero exactly as $\tilde{p}$ shrank toward $1/2$: biased, but lower MSE on fresh data. When a practitioner says a model is overfitting, they are saying its variance is too high; the cure is almost always to add bias. Much of applied machine learning is the humble formula $\text{MSE} = \text{Var} + \text{Bias}^2$ worked out in practice.

Paradox

[Stein's paradox: the sample mean is not always best] Here is one of the most startling results in all of statistics, and it lives exactly at the bias–variance frontier. Suppose you estimate three or more *unrelated* means at once – say the true batting averages of several players, or the means of three independent $\mathcal{N}(\mu_i, 1)$ variables. The obvious estimator is to use each sample mean for its own parameter, and each sample mean is individually unbiased and efficient. Yet Charles Stein proved in 1956 that for dimension ≥ 3 this obvious estimator is *inadmissible*: the James–Stein estimator, which deliberately *shrinks* all the sample means toward a common point, has *strictly smaller total MSE* for every true value of the parameters. Borrowing strength across problems that have nothing to do with each other beats handling each alone. The resolution is the chapter's thesis taken to its extreme: a dose of bias (shrinkage) can buy enough variance reduction to win even when the unbiased estimator was already efficient. Modern hierarchical and regularized models – and much of machine learning's success – are Stein's insight industrialized.

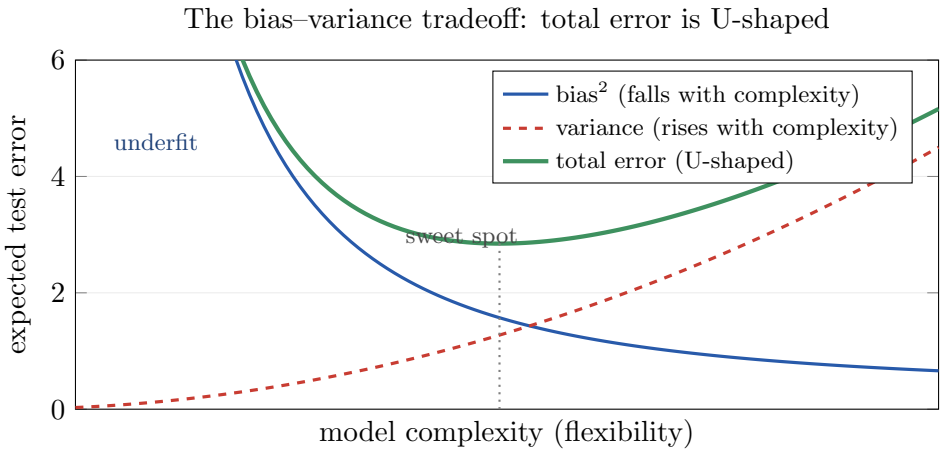


Figure 30.5: As model flexibility rises, squared bias falls but variance climbs. Their sum is U-shaped, and the best model lives at the bottom of the U – not the simplest, not the most complex, but the one that balances lean against wobble. Choosing that point (via cross-validation) is most of applied modeling.

30.7 Fisher Information and the Cramer–Rao Bound

We have seen that variance is the long-run battleground: bias usually fades, but how small can variance get? Remarkably, there is a *floor* – a hard limit, set by the model itself, below which no unbiased estimator's variance can drop. The floor is governed by a single quantity that measures how much a single observation can possibly tell you about the parameter: the *Fisher information*.

Intuition

Look again at Figure 30.2. With lots of data the log-likelihood is a *sharp* peak; with little data it is a *gentle* mound. A sharp peak means the data discriminate strongly between nearby parameter values – shift θ a hair and the likelihood plummets, so you know θ precisely. A gentle mound means many θ values explain the data nearly equally well, so you are uncertain. *Information is curvature.* Fisher information measures how sharply curved the log-likelihood is at its peak, on average.

Definition 30.8 (Fisher information)**.** For a single observation X with log-density $\log f(X \mid \theta)$, the *score* is $s(\theta) = \partial_\theta \log f(X \mid \theta)$ and the *Fisher information* is its variance,

$$I(\theta) = \operatorname{Var}\big(s(\theta)\big) = \mathbb{E}\left[\left(\frac{\partial}{\partial\theta}\log f(X \mid \theta)\right)^2\right].$$

Under mild regularity conditions this equals the expected curvature,

$$I(\theta) = -\,\mathbb{E}\left[\frac{\partial^2}{\partial\theta^2}\log f(X \mid \theta)\right].$$

For n i.i.d. observations the information adds: $I_n(\theta) = n\,I(\theta)$.

Why it works

Two facts make this definition click. First, the score has *mean zero*: $\mathbb{E}[s(\theta)] = \int (\partial_\theta f / f)\, f \, \mathrm{d}x = \partial_\theta \int f \, \mathrm{d}x = \partial_\theta 1 = 0$. So the variance of the score is just $\mathbb{E}[s^2]$ – the first formula. Second, differentiating the identity $\mathbb{E}[s] = 0$ once more and applying the product rule yields $\mathbb{E}[\partial_\theta s] + \mathbb{E}[s^2] = 0$, i.e. $\mathbb{E}[s^2] = -\mathbb{E}[\partial_\theta s] = -\mathbb{E}[\partial_\theta^2 \log f]$ – the second formula. So "variance of the score" and "expected curvature of the log-likelihood" are the same number, viewed from two sides. And information *adds* over independent observations because the total score is a sum of independent mean-zero scores, and variances of independent things add: every fresh data point contributes its own $I(\theta)$ to the pile. That additivity is why "more data" so reliably means "more precision."

Theorem 30.2 (Cramer–Rao lower bound)**.** *Under regularity conditions,* any *unbiased estimator* $\hat{\theta}$ *of* θ *based on* n *i.i.d. observations satisfies*

$$\operatorname{Var}(\hat{\theta}) \;\geq\; \frac{1}{n\, I(\theta)}.$$

No unbiased estimator can do better than the inverse of the total Fisher information. An estimator attaining the bound is called efficient*.*

What is really happening

The Cramer–Rao bound says that precision is a finite resource handed out by the model, not by your cleverness. The model fixes $I(\theta)$ – how much one observation can reveal – and then *no* unbiased procedure, however ingenious, can extract more than $nI(\theta)$ worth of certainty. Two consequences worth absorbing. First, variance falls like $1/n$: to halve your standard error you must *quadruple* your sample. That is the iron $\sqrt{n}$ law behind every "margin of error" and the reason large studies are expensive. Second, the bound is a *benchmark*: compute $1/(nI)$, compare it to your estimator's actual variance, and you know at once whether your estimator is wasting information or squeezing the model dry.

Example 30.14 (The CRLB for a coin, and why $\hat{p} = k/n$ is efficient)**.** For Bernoulli(p), $\log f(x \mid p) = x \log p + (1 - x) \log(1 - p)$. Differentiate twice:

$$\partial_p^2 \log f = -\frac{x}{p^2} - \frac{1 - x}{(1 - p)^2}.$$

Take $-\mathbb{E}[\cdot]$ using $\mathbb{E}[X] = p$:

$$I(p) = \frac{p}{p^2} + \frac{1 - p}{(1 - p)^2} = \frac{1}{p} + \frac{1}{1 - p} = \frac{1}{p(1 - p)}.$$

So the Cramer–Rao floor is

$$\operatorname{Var}(\hat{p}) \geq \frac{1}{nI(p)} = \frac{p(1 - p)}{n}.$$

Now check the actual MLE $\hat{p} = k/n = \bar{X}$. Since $k \sim \text{Binomial}(n, p)$, $\operatorname{Var}(\hat{p}) = \operatorname{Var}(k)/n^2 = np(1 - p)/n^2 = p(1 - p)/n$. It *exactly* attains the bound. The natural "fraction of heads" is not merely reasonable – it is *efficient*: no unbiased estimator of p can ever beat it. Notice too that $I(p) = 1/[p(1 - p)]$ blows up as $p \to 0$ or 1: a near-deterministic coin is, per toss, enormously informative when it does surprise you, which is why rare-event estimation behaves so differently from coin-flip estimation.

Example 30.15 (CRLB for the Poisson rate)**.** For Poisson(λ), $\log f = -\lambda + x\log\lambda - \log x!$, so $\partial^2_\lambda \log f = -x/\lambda^2$ and $I(\lambda) = -\mathbb{E}[-X/\lambda^2] = \lambda/\lambda^2 = 1/\lambda$. The floor is $\mathrm{Var}(\hat\lambda) \geq \lambda/n$. The MLE $\hat\lambda = \bar X$ has variance $\mathrm{Var}(\bar X) = \lambda/n$ (since $\mathrm{Var}(X) = \lambda$ for a Poisson). Efficient again. Maximum likelihood keeps hitting the floor – and that is no accident.

Why this matters

[Why maximum likelihood is the default] The examples are not coincidences. Under regularity conditions the MLE is, for large n, *asymptotically* the best estimator possible:

$$\hat\theta_{\mathrm{MLE}} \approx \mathcal{N}\left(\theta, \frac{1}{nI(\theta)}\right) \quad \text{for large } n.$$

In words: it is asymptotically unbiased, asymptotically normal, and its variance asymptotically attains the Cramer–Rao floor (asymptotic efficiency). That is three things at once – it lands on the truth, its errors are bell-shaped (so you can build confidence intervals, two chapters from now), and it wrings every drop of information from the model. This is why maximum likelihood is the default estimation method across statistics, econometrics, and machine learning: in the large-sample regime it is, provably, hard to beat. Fisher's thirty-year argument with Pearson ended here.

Paradox

[When maximum likelihood goes wrong] Maximum likelihood is the hero of this chapter, so it is only fair to show its scars; "asymptotically optimal" hides real failures.

- **The unbounded-likelihood catastrophe.** Fit a mixture of two Normals by maximum likelihood. Center one component exactly on a single data point and shrink its variance toward zero: that point's density blows up, and $L \to \infty$. The global maximum is a useless spike, not the sensible fit. The MLE, taken literally, does not exist.
- **Too many parameters (Neyman–Scott).** Suppose each pair of observations has its own mean but all share a common variance σ^2. As the number of pairs grows, the MLE of σ^2 converges to $\sigma^2/2$ – consistently *wrong* by a factor of two. When parameters proliferate as fast as data, the large-sample guarantees collapse.
- **Boundary trouble.** For Uniform$(0,\theta)$ the MLE $\max_i X_i$ is biased (it is always below θ) and not asymptotically normal – its error shrinks like $1/n$, faster than the usual $1/\sqrt{n}$, and follows an exponential, not a bell. The clean asymptotic theory simply does not apply at a boundary.

The resolution is not to abandon maximum likelihood but to respect its fine print: the optimality theorems require a smooth, bounded likelihood, a fixed number of parameters, and an interior optimum. When those hold – which is most of the time – the MLE is excellent. When they fail, reach for penalization, Bayesian priors (next chapter), or a bias correction. A method this useful is worth knowing at its edges, not just in its comfort zone.

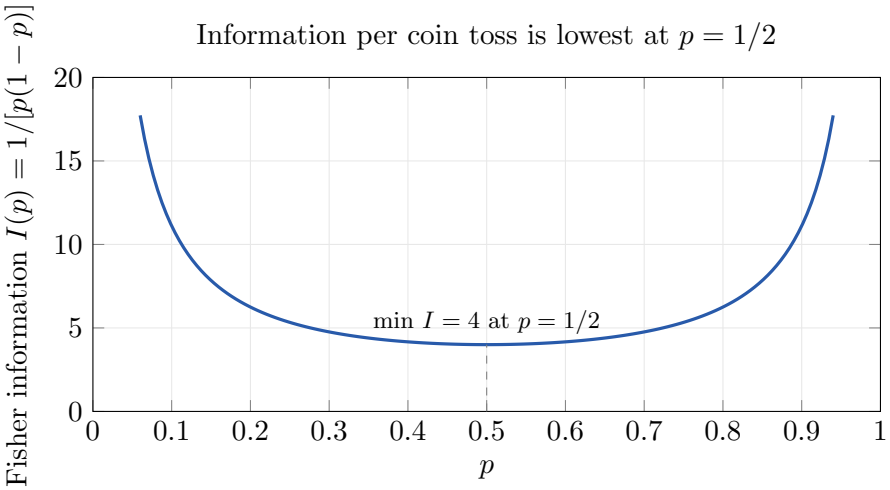


Figure 30.6: Fisher information for a single Bernoulli toss. It is *smallest* at $p = 1/2$ (a fair coin is the hardest bias to pin down – each toss is maximally unpredictable, so individual outcomes say least) and rises steeply near the extremes, where the rare surprising outcome is very telling. Each toss therefore varies in how much it teaches you, depending on where the truth lies.

30.8 Putting It Together: A Full Worked Estimation

Here is one example carried end to end, the way you would handle it in practice, so you can watch all the pieces interlock.

Example 30.16 (Estimating an exponential rate from waiting times)**.** A web server logs the gaps between requests (in seconds):

$$0.5, \quad 1.2, \quad 0.3, \quad 2.1, \quad 0.9, \quad 1.5.$$

Model them as i.i.d. Exponential(λ), density $f(x \mid \lambda) = \lambda e^{-\lambda x}$ for $x > 0$, where λ is the request *rate*. The sum is $\sum x_i = 6.5$, so $\bar{x} = 6.5/6 \approx 1.083$.

Step 1: Method of moments. $\mathbb{E}[X] = 1/\lambda$, so the moment equation is $1/\lambda = \bar{x}$, giving $\hat{\lambda}_{\text{MM}} = 1/\bar{x} \approx 0.923$ requests per second.

Step 2: Maximum likelihood. The log-likelihood is

$$\ell(\lambda) = \sum_i (\log \lambda - \lambda x_i) = n \log \lambda - \lambda \sum_i x_i.$$

Differentiate and solve:

$$\ell'(\lambda) = \frac{n}{\lambda} - \sum_i x_i = 0 \implies \hat{\lambda}_{\text{MLE}} = \frac{n}{\sum_i x_i} = \frac{1}{\bar{x}} \approx 0.923.$$

For the exponential, MoM and MLE coincide – both give the reciprocal of the mean gap. (By invariance, the MLE of the *mean* gap $1/\lambda$ is just $\bar{x} \approx 1.083$ seconds.)

Step 3: Fisher information and the precision floor. With $\log f = \log \lambda - \lambda x$ we get $\partial_\lambda^2 \log f = -1/\lambda^2$, so $I(\lambda) = 1/\lambda^2$ and the Cramer–Rao floor is

$$\text{Var}(\hat{\lambda}) \geq \frac{1}{nI(\lambda)} = \frac{\lambda^2}{n}.$$

Plugging in $\hat{\lambda} \approx 0.923$ and $n = 6$ gives an approximate standard error $\sqrt{\lambda^2/n} = \lambda/\sqrt{n} \approx 0.923/2.449 \approx 0.38$. So our estimate is roughly 0.92 ± 0.38 requests/second – usefully informative, but with only six observations, not precise. To halve that 0.38 we would need about 24 observations: the $\sqrt{n}$ law again.

Step 4: Bias check. It turns out $1/\bar{x}$ is slightly biased upward for small n (by Jensen's inequality, $\mathbb{E}[1/\bar{X}] > 1/\mathbb{E}[\bar{X}] = \lambda$); a common correction is $\hat{\lambda}_{\text{corrected}} = \frac{n-1}{n} \cdot \frac{1}{\bar{x}}$. With $n = 6$ the correction multiplies by $5/6$, giving ≈ 0.77. As $n \to \infty$ the correction vanishes – the same moral one last time: *bias fades, variance is forever.*

Figure 30.7 previews what the lab below will let you see for yourself: the sampling distribution of $\hat{\lambda} = \bar{X}$ is centred on the truth and tightens as n grows.

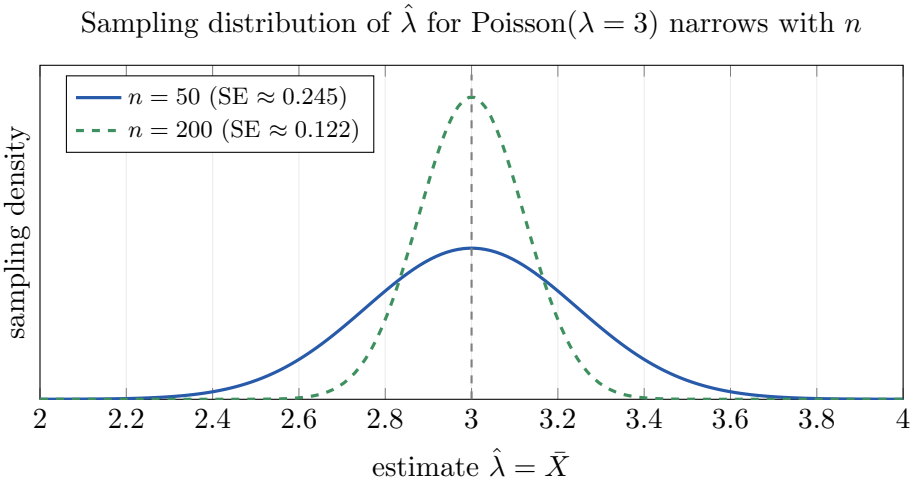


Figure 30.7: Two sampling distributions of the Poisson MLE $\hat{\lambda} = \bar{X}$, by the asymptotic-normality approximation $\hat{\lambda} \approx \mathcal{N}(\lambda, \lambda/n)$. Both centre on the truth 3 (consistency); quadrupling n from 50 to 200 halves the standard error from ≈ 0.245 to ≈ 0.122 (the $\sqrt{n}$ law) and so doubles the peak height. The bell shape is exactly what the histogram in the lab below should reproduce.

Work it yourself

Before the exercises, simulate the whole story yourself. Draw 1000 datasets of size $n = 50$ from Poisson($\lambda = 3$). For each, compute $\hat{\lambda} = \bar{X}$. Histogram the 1000 estimates: it should look bell-shaped (asymptotic normality), centered near 3 (consistency), with spread close to $\sqrt{\lambda/n} = \sqrt{3/50} \approx 0.245$ (the Cramer–Rao floor). Then redo it with $n = 200$ and watch the histogram narrow by a factor of 2 as n quadruples. You will have *seen*, in one afternoon, every theorem in this chapter breathe.

30.9 Exercises

Exercise 30.1 (Geometric MLE)**.** Let $X_1, \ldots, X_n \sim \text{Geometric}(p)$ on $\{1, 2, 3, \ldots\}$ with $\mathbb{P}(X = k) = (1-p)^{k-1}p$. Write down the log-likelihood, and show the MLE is $\hat{p} = 1/\bar{X}$. Interpret: why is the estimated success probability the reciprocal of the average number of trials?

Exercise 30.2 (Method of moments vs. MLE for the uniform)**.** For $X_i \sim \text{Uniform}(0, \theta)$ you found $\hat{\theta}_{\text{MM}} = 2\bar{X}$ and $\hat{\theta}_{\text{MLE}} = \max_i X_i$. (a) Show $\mathbb{E}[2\bar{X}] = \theta$, so the MoM estimator is unbiased. (b) It is a fact that $\mathbb{E}[\max_i X_i] = \frac{n}{n+1}\theta$; find the bias of the MLE and propose a multiplicative correction making it unbiased. (c) Given the data $0.1, 0.4, 0.9$, compute both estimates and explain which one you would trust and why.

Exercise 30.3 (The $n-1$ in your bones)**.** Using $\mathbb{E}[(\bar{X}-\mu)^2] = \sigma^2/n$, re-derive $\mathbb{E}\left[\frac{1}{n}\sum_i (X_i - \bar{X})^2\right] = \frac{n-1}{n}\sigma^2$ from scratch, and explain in one sentence, without formulas, why dividing by n undershoots the true variance.

Exercise 30.4 (Fisher information for the Normal mean)**.** For $X \sim \mathcal{N}(\mu, \sigma^2)$ with σ^2 known, show the Fisher information about μ is $I(\mu) = 1/\sigma^2$. Deduce the Cramer–Rao floor for estimating μ from n observations, and verify that $\bar{X}$ attains it. What does $I(\mu) = 1/\sigma^2$ say about how noise affects how much each observation teaches you?

Exercise 30.5 (A bias–variance numerical experiment)**.** Consider estimating p from $n = 5$ Bernoulli tosses using either $\hat{p} = k/5$ or the shrunk $\tilde{p} = (k+1)/7$. For the true value $p = 0.2$: (a) list the six possible values of k and their binomial probabilities; (b) compute $\mathbb{E}[\hat{p}]$, $\mathbb{E}[\tilde{p}]$ and confirm $\hat{p}$ is unbiased while $\tilde{p}$ is biased toward $1/2$; (c) compute the MSE of each and report which estimator wins at $p = 0.2$.

Exercise 30.6 (Invariance)**.** Let $X_i \sim \text{Exponential}(\lambda)$ with MLE $\hat{\lambda} = 1/\bar{X}$. Using the invariance property, write down the MLE of (a) the mean waiting time $1/\lambda$, (b) the median waiting time $(\log 2)/\lambda$, and (c) the probability $e^{-\lambda t}$ that the next gap exceeds a fixed time t. No new differentiation allowed.

Exercise 30.7 (Two-parameter score, conceptual)**.** For $\mathcal{N}(\mu, \sigma^2)$ with both parameters unknown, you derived $\hat{\mu} = \bar{X}$ and $\hat{\sigma}^2 = \frac{1}{n}\sum (X_i - \bar{X})^2$. Explain why the estimate of μ did not depend on σ^2 but the estimate of σ^2 depended on μ (through $\bar{X}$). What does this asymmetry say about the shape of the likelihood surface, and which parameter is "easier" to pin down?

Exercise 30.8 (Consistency from MSE – a proof)**.** Prove that if $\text{Bias}(\hat{\theta}_n) \to 0$ and $\text{Var}(\hat{\theta}_n) \to 0$ then $\hat{\theta}_n \xrightarrow{p} \theta$. (Hint: write $\text{MSE} = \text{Var} + \text{Bias}^2$ and apply Chebyshev's / Markov's inequality to $(\hat{\theta}_n - \theta)^2$.) Then give one example of a consistent estimator that is biased at every finite n.

Exercise 30.9 (Why the likelihood is not a probability over θ)**.** For Bernoulli(p) with data "1 head in 1 toss," the likelihood is $L(p) = p$. (a) Compute $\int_0^1 L(p)\,dp$ and confirm it is not 1. (b) Explain in your own words why this does not violate any axiom of probability, and what would have to change (philosophically and mathematically) for an expression like "the probability that p lies in $[0.4, 0.6]$" to make sense. (This is the doorway to the next chapter.)

Exercise 30.10 (Maximum likelihood as closeness to the truth)**.** Argue informally that maximizing the average log-likelihood $\frac{1}{n}\sum_i \log f(X_i \mid \theta)$ is, for large n, the same as minimizing the Kullback–Leibler divergence $\text{KL}(f_{\theta_0} \,\|\, f_\theta)$ from the true distribution to the candidate. (Hint: by the law of large numbers $\frac{1}{n}\sum_i \log f(X_i \mid \theta) \to \mathbb{E}_{\theta_0}[\log f(X \mid \theta)]$; subtract the θ-free quantity $\mathbb{E}_{\theta_0}[\log f(X \mid \theta_0)]$ and recognize what remains.) What does this say about what the MLE is "really" doing?

Chapter 31

Bayesian Inference

In the chapter on maximum likelihood we learned to point at the single value of a parameter that makes the data look least surprising, and to call that value our estimate. It is a sharp, powerful idea, and it has one quietly unsatisfying feature: it hands you a number and walks away. Ask it "how sure are you?" and it can only mumble about standard errors computed from a curvature it pretends to trust. Ask it "what did you believe *before* you saw the data?" and it shrugs — it has no notion of belief at all, only of likelihood.

Bayesian inference is the other half of statistics, the half that takes the word *belief* seriously and treats it as something you can do arithmetic with. The idea is breathtakingly simple, and you have already met its engine: Bayes' theorem, the rule for running the arrow of cause and effect backwards. In the diagnosis chapter we used it to update from a positive test to a disease. Here we do exactly the same thing, but the "hidden cause" is now a *parameter* — the bias of a coin, the rate of a Poisson process, the mean of a normal — and the "evidence" is a whole dataset. The output is not a single number but an entire probability distribution over the parameter: a complete, quantitative description of everything the data has taught us, uncertainty included.

That distribution is called the *posterior*, and learning to compute it, read it, summarize it, and trust it is the business of this chapter. We will start with the three-part incantation — prior, likelihood, posterior — and watch a real belief sharpen as coin flips arrive. We will discover the small miracle of *conjugate priors*, where the posterior comes out in the same family as the prior and the update is a one-line bookkeeping step. We will work three of these in full — Beta–Binomial, Gamma–Poisson, Normal–Normal — and plot the posterior tightening with every data point. We will learn to draw *credible intervals* that mean exactly what a confidence interval is so often mistaken to mean. We will contrast the single-point *MAP* summary with the full posterior, set the Bayesian and frequentist worldviews fairly side by side, and finish in the engine room of modern machine learning, where these ideas run billions of times a day.

31.1 The Three Words: Prior, Likelihood, Posterior

Let us fix a concrete question and never let go of it. A friend hands you a thumbtack and asks: when I toss this and let it land, what is the probability θ that it lands point-up? You do not know θ. A coin would be $\theta = 1/2$, but a thumbtack is lopsided, and really it could be anything. You will toss

it a number of times, count the point-ups, and try to learn θ.

The frequentist instinct is to treat θ as a fixed but unknown constant and estimate it. The Bayesian instinct is different and, once you see it, hard to unsee: treat your *uncertainty about* θ as itself a probability distribution. Before any tosses, you have some vague sense of what θ might be — maybe centered near $1/2$, maybe flat across $[0, 1]$. That sense is a distribution, the *prior* $p(\theta)$. After the tosses, your sense has sharpened into a new distribution, the *posterior* $p(\theta \mid \text{data})$. The whole of Bayesian inference is the bridge between those two.

Intuition

A prior is not a guess at the answer. It is a description of *your ignorance*, written in the language of probability. "I have no idea, it could be anything in $[0, 1]$" is the uniform prior. "It is probably near a half but I would not bet the house" is a gentle hump around 0.5. "I have tossed thousands of these tacks before and they land up about 60% of the time" is a tight spike near 0.6. Each is a legitimate starting belief, and — this is the point — each will be *dragged toward the truth by data*. The disagreement between two reasonable priors shrinks as evidence accumulates. The prior is where you start; the data decides where you end up.

The bridge is Bayes' theorem, written now for a continuous parameter. Let the data be D and the parameter θ. Then

$$\underbrace{p(\theta \mid D)}_{\text{posterior}} = \frac{\overbrace{p(D \mid \theta)}^{\text{likelihood}}\ \overbrace{p(\theta)}^{\text{prior}}}{\underbrace{p(D)}_{\text{evidence}}}. \tag{31.1}$$

Three of these four objects you already know intimately. The *prior* $p(\theta)$ is your starting belief. The *likelihood* $p(D \mid \theta)$ is the same function the maximum-likelihood chapter built its whole world on: for each candidate value of θ, how probable is the data we actually saw? The *posterior* $p(\theta \mid D)$ is the updated belief we are after. The fourth, the *evidence* $p(D) = \int p(D \mid \theta)\, p(\theta)\, \mathrm{d}\theta$, is just the number that makes the posterior integrate to one. It does not depend on θ at all. That single fact — that the denominator is a constant in θ — is the most liberating thing in the subject.

Focus

Because the evidence is constant in θ, almost all Bayesian reasoning runs on the *proportionality* form of Bayes' theorem:

$$p(\theta \mid D) \ \propto\ p(D \mid \theta)\, p(\theta), \qquad \text{in words:} \qquad \text{posterior} \ \propto\ \text{likelihood} \times \text{prior}.$$

Compute the right-hand side as a function of θ, recognize what distribution it is (or normalize it at the end), and you are done. The phrase "posterior is proportional to likelihood times prior" is the single most useful sentence in this chapter. Say it until it is reflex.

Why it works

Why is the posterior *proportional* to likelihood times prior, and not equal to it? Because the product $p(D \mid \theta)p(\theta)$ is a perfectly good shape but the wrong height: its total area is $p(D)$, not 1. Dividing by $p(D)$ rescales the area to 1 without touching the shape. So the prior $\times$ likelihood product already *is* the posterior up to a vertical rescaling, and the rescaling

is forced — it is the unique constant that makes a valid density. This is why we can so often ignore the denominator, do the algebra on the shape, and read off the answer by pattern-matching to a known family. The normalizer was never carrying information about θ; it was only ever carrying the bill for making probabilities sum to one.

31.1.1 A First Update by Hand

Let us toss the tack three times and get *up, up, down*. Suppose, to keep the first example arithmetic-free, that you have decided θ can only be one of three values: 0.3, 0.5, or 0.7, and before tossing you considered them equally plausible, so the prior puts mass $1/3$ on each. The likelihood of two ups and one down for a given θ is $\theta^2(1-\theta)$. Multiply prior by likelihood:

θ	prior $p(\theta)$	likelihood $\theta^2(1-\theta)$	prior$\times$lik.
0.3	$1/3$	0.063	0.0210
0.5	$1/3$	0.125	0.0417
0.7	$1/3$	0.147	0.0490
		sum (the evidence):	0.1117

Now normalize by dividing each product by the sum 0.1117. The posterior is $\mathbb{P}(\theta = 0.3 \mid D) = 0.188$, $\mathbb{P}(\theta = 0.5 \mid D) = 0.373$, and $\mathbb{P}(\theta = 0.7 \mid D) = 0.439$. We started believing all three equally and now lean toward 0.7, with 0.5 a close second and 0.3 trailing (Figure 31.1) — exactly the ordering of the likelihoods, because here the flat prior contributed nothing to the *shape*. That is the whole machine. Everything else in this chapter is the same machine run on a continuum of θ values instead of three, where the sums become integrals and the bookkeeping becomes algebra.

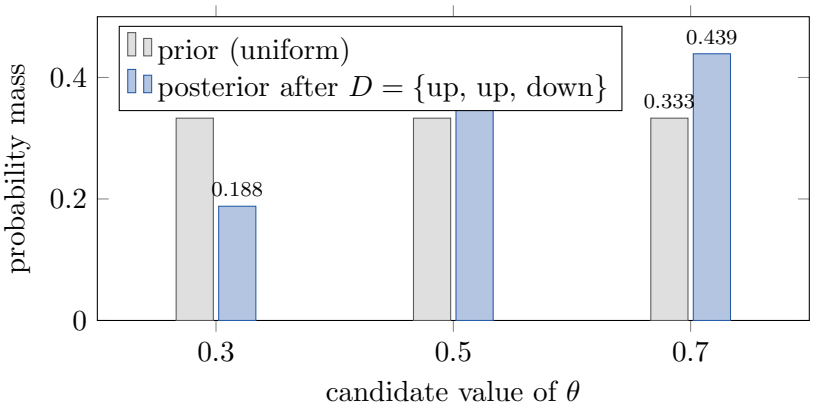


Figure 31.1: The first update made visible. The flat prior puts mass $1/3$ on each candidate; after seeing two ups and one down the posterior tilts toward larger θ, ranking the candidates in exactly the order of their likelihoods $\theta^2(1-\theta)$.

History

The reverend Thomas Bayes (c. 1701–1761) never published the theorem that bears his name; it was found in his papers after his death and read to the Royal Society in 1763 by his friend Richard Price. Bayes' own example was, in spirit, exactly our thumbtack: a ball rolled onto a table at an unknown position, with later balls reported only as landing left or right of it, and the question of what could be inferred about the first ball's place. It was Pierre-Simon Laplace, working independently a few years later, who turned the idea into a general tool and used it on everything from the masses of planets to the ratio of male to female births in Paris. For most of the next century the method was simply called "inverse probability," and it was the dominant mode of statistical reasoning. The name "Bayesian" — and the fierce twentieth-century rivalry with the frequentist school we will meet later — came much later still.

31.2 The Beta–Binomial Conjugate Pair

Now let $\theta \in [0,1]$ range over the whole interval. We toss the tack n times, independently, and see k point-ups. The likelihood is the binomial probability, viewed as a function of θ:

$$p(D \mid \theta) \;=\; \binom{n}{k}\theta^{k}(1-\theta)^{\,n-k} \;\propto\; \theta^{k}(1-\theta)^{\,n-k},$$

where we dropped $\binom{n}{k}$ because it does not involve θ and so disappears into the proportionality. We need a prior on $[0,1]$. We could pick any density there, but one family is magic for this likelihood: the *Beta* distribution. Recall $\theta \sim \mathrm{Beta}(\alpha,\beta)$ has density

$$p(\theta) \;=\; \frac{1}{B(\alpha,\beta)}\,\theta^{\alpha-1}(1-\theta)^{\beta-1}, \qquad 0 \le \theta \le 1,$$

where $B(\alpha,\beta)$ is the constant that makes it integrate to one. Look at the shape: $\theta^{\alpha-1}(1-\theta)^{\beta-1}$. It is *the same algebraic form as the binomial likelihood*, a power of θ times a power of $1-\theta$. Multiply them and the powers simply add:

$$\begin{aligned} p(\theta \mid D) \;&\propto\; \underbrace{\theta^{k}(1-\theta)^{n-k}}_{\text{likelihood}} \cdot \underbrace{\theta^{\alpha-1}(1-\theta)^{\beta-1}}_{\text{prior}} \\ &=\; \theta^{\,(\alpha+k)-1}\,(1-\theta)^{\,(\beta+n-k)-1}. \end{aligned}$$

The right-hand side is, up to its normalizing constant, exactly the density of a $\mathrm{Beta}(\alpha+k,\ \beta+n-k)$. We did not have to compute a single integral. The posterior is Beta because the prior was Beta and the likelihood was binomial; we say the Beta family is *conjugate* to the binomial likelihood.

Formal

Beta–Binomial update. If the prior is $\theta \sim \mathrm{Beta}(\alpha,\beta)$ and we observe k successes in n independent Bernoulli trials, then the posterior is

$$\theta \mid D \;\sim\; \mathrm{Beta}(\alpha+k,\ \beta+n-k). \tag{31.2}$$

In words: *add your successes to α and your failures to β.* That is the entire computation.

What is really happening

The update rule (31.2) tells you exactly what a Beta prior *is*: a record of imaginary past data. A Beta(α, β) prior behaves as though, before any real tosses, you had already seen $\alpha - 1$ successes and $\beta - 1$ failures. (The Beta$(1, 1)$ prior, which is flat, corresponds to zero imaginary data: total openness.) Seen this way, α and β are *pseudocounts*, and Bayesian updating is nothing but continuing to count, starting from a head start you chose. Real successes and imaginary successes are added in exactly the same way. This is why a strong prior (large α, β) resists the data — it is outvoted only slowly — while a weak prior is overwhelmed almost at once. The prior's strength is literally measured in units of data.

31.2.1 Watching the Posterior Tighten

Suppose we begin with the flat Beta$(1, 1)$ prior — complete openness about the tack — and the true bias is $\theta = 0.7$. We toss it and update after 5, 20, and 80 tosses, getting close to 70% ups each time (as close as the discreteness of small samples allows). The posteriors are Beta$(1 + 4,\ 1 + 1) =$ Beta$(5, 2)$ after 5 tosses (4 up, 1 down), Beta$(1 + 14,\ 1 + 6) =$ Beta$(15, 7)$ after 20 (14 up, 6 down), and Beta$(1 + 56,\ 1 + 24) =$ Beta$(57, 25)$ after 80 (56 up, 24 down). Figure 31.2 plots all three against the flat prior. Each new batch of data pulls the hump toward 0.7 and squeezes it narrower: the belief is becoming both more accurate and more confident at the same time.

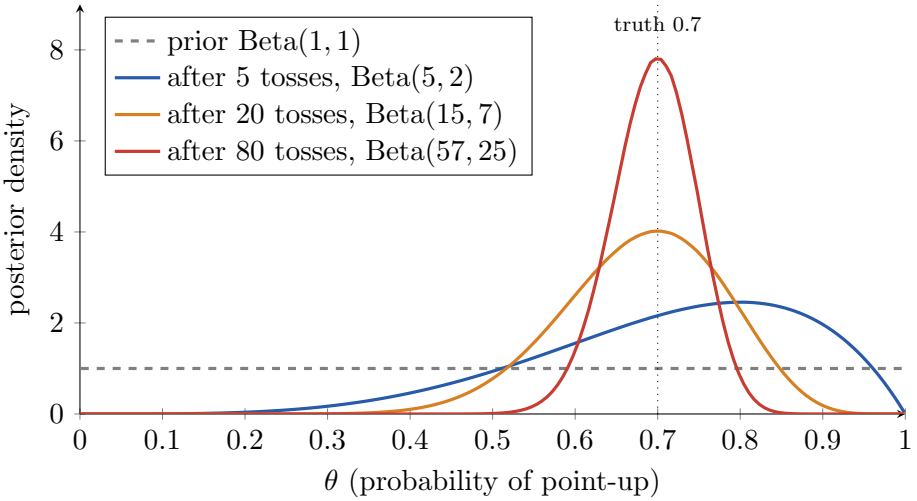


Figure 31.2: The posterior tightening as data arrives. Starting from a flat prior, each batch of tosses (all roughly 70% point-up) pulls the density toward the true value 0.7 and narrows it. More data means a taller, thinner posterior — more confident and more accurate at once. (Curves are drawn from closed-form Beta densities; exact heights are illustrative.)

The posterior mean of a Beta(a, b) is $a/(a + b)$. After 80 tosses that is $57/82 \approx 0.695$, gratifyingly close to the truth, and after only 5 tosses it was $5/7 \approx 0.714$ — already in the right neighborhood but, as the wide curve warns, not yet to be trusted. This is the great virtue of carrying the whole distribution rather than a point: the *width* of the posterior is a frank measure of how much you do not yet know.

Walking question

Notice a subtle thing about the posterior mean. We can write it as

$$\frac{\alpha+k}{\alpha+\beta+n} = \underbrace{\frac{\alpha+\beta}{\alpha+\beta+n}}_{w}\cdot\underbrace{\frac{\alpha}{\alpha+\beta}}_{\text{prior mean}} + \underbrace{\frac{n}{\alpha+\beta+n}}_{1-w}\cdot\underbrace{\frac{k}{n}}_{\text{data mean}}.$$

The posterior mean is a *weighted average* of the prior mean and the sample proportion k/n, with the weight on the data being $n/(\alpha+\beta+n)$. When n is small the prior dominates; as $n \to \infty$ the weight on the data tends to 1 and the Bayesian estimate converges to the frequentist one. The two schools agree in the limit of abundant data; they differ most exactly when data is scarce, which is precisely when being candid about uncertainty matters most.

Example 31.1 (A/B testing a button color)**.** You change a checkout button from gray to green and show the green version to 40 visitors; 12 buy. With no strong prior belief, take $\theta \sim \text{Beta}(1,1)$ for the green conversion rate. The posterior is $\text{Beta}(1+12,\,1+28) = \text{Beta}(13,29)$. Its mean is $13/42 \approx 0.310$, and we can ask any question we like of the full distribution: the probability that the true rate exceeds 0.25 is $\int_{0.25}^{1} p(\theta \mid D)\,\mathrm{d}\theta \approx 0.79$. If the old gray button is known to convert at 0.25, we can say "there is about a 79% chance the green button is genuinely better," a sentence a frequentist p-value cannot legally utter. That is what has made Bayesian A/B testing popular in industry: the output is a direct probability about the thing you care about.

31.3 The Gamma–Poisson Conjugate Pair

Counts in fixed windows — emails per hour, goals per match, mutations per genome — are the natural home of the Poisson distribution with rate λ. Suppose we observe counts $x_1, \dots, x_n$ from $\text{Poisson}(\lambda)$ and want to learn $\lambda > 0$. The likelihood is

$$p(D \mid \lambda) \;=\; \prod_{i=1}^{n} \frac{\lambda^{x_i} e^{-\lambda}}{x_i!} \;\propto\; \lambda^{\sum_i x_i}\, e^{-n\lambda},$$

again dropping factors free of λ. We need a prior on the positive reals whose shape matches $\lambda^{(\cdot)} e^{-(\cdot)\lambda}$. That is exactly the *Gamma* distribution: $\lambda \sim \text{Gamma}(\alpha, \beta)$ (shape α, rate β) has density

$$p(\lambda) \;\propto\; \lambda^{\alpha-1} e^{-\beta\lambda}, \qquad \lambda > 0.$$

Multiply prior by likelihood and, as before, the powers and the exponents simply combine:

$$p(\lambda \mid D) \;\propto\; \lambda^{\sum_i x_i} e^{-n\lambda} \cdot \lambda^{\alpha-1} e^{-\beta\lambda} \;=\; \lambda^{(\alpha+\sum_i x_i)-1}\, e^{-(\beta+n)\lambda},$$

which is the kernel of a $\text{Gamma}(\alpha + \sum_i x_i,\; \beta + n)$.

Formal

Gamma–Poisson update. With prior $\lambda \sim \text{Gamma}(\alpha, \beta)$ and observed counts summing to $S = \sum_i x_i$ over n windows,

$$\lambda \mid D \;\sim\; \text{Gamma}(\alpha + S,\; \beta + n). \tag{31.3}$$

Add the total count to the shape; add the number of observations to the rate.

Here too the parameters read as pseudo-data: β is an imaginary number of prior observation-windows and α an imaginary number of prior events. The posterior mean is $(\alpha+S)/(\beta+n)$, again a weighted blend of the prior mean α/β and the data mean S/n.

Example 31.2 (Goals per match)**.** A football analyst models a team's goals per match as Poisson(λ). From long experience she takes a Gamma$(2,1)$ prior — prior mean 2 goals, fairly broad. The team then scores $3,1,2,4,0$ in five matches, so $S=10$, $n=5$. The posterior is Gamma$(2+10,\,1+5)$ = Gamma$(12,6)$, with mean $12/6=2.0$ goals per match and standard deviation $\sqrt{12}/6\approx 0.58$. Five matches later the team scores $2,2,3,1,2$, adding $S'=10$, $n'=5$; updating again gives Gamma$(22,11)$, mean 2.0, standard deviation $\sqrt{22}/11\approx 0.43$. Same central estimate, tighter belief — the posterior has narrowed because evidence has doubled (Figure 31.3). Notice we updated in two steps and got the same answer as one combined step would: *Bayesian updating is associative*, a property we examine next.

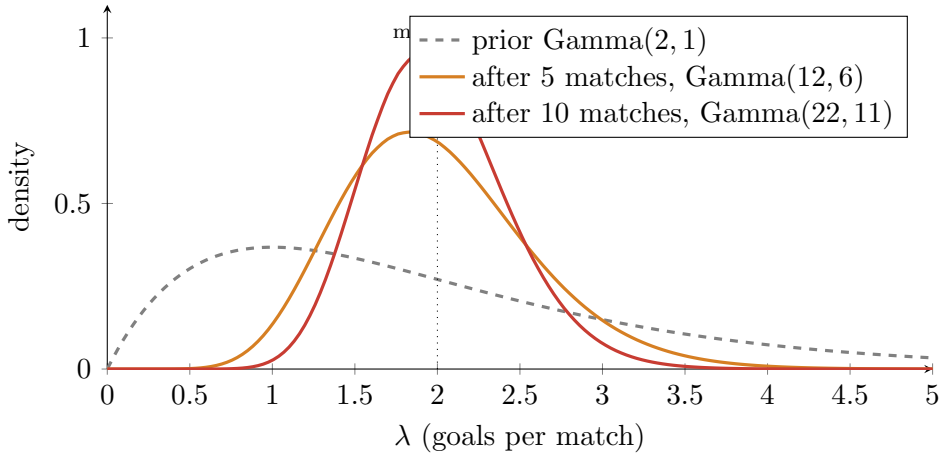


Figure 31.3: Gamma–Poisson updating for the goals example. The broad Gamma$(2,1)$ prior sharpens to Gamma$(12,6)$ after five matches and to Gamma$(22,11)$ after ten. The mean stays at 2.0 goals per match while the density grows taller and narrower: doubling the evidence roughly halves the variance, so the standard deviation falls from 0.58 to 0.43.

Why this matters

The fact that you can feed data in one observation at a time, or all at once, or in any grouping you like, and always reach the same posterior, is not a lucky coincidence. It is forced by the structure of (31.1): yesterday's posterior is exactly today's prior. Belief flows forward through time like a ledger that only ever gets appended to. This is why Bayesian methods are the natural language of *online learning* and *streaming data* — a spam filter, a recommendation engine, a robot's map of its surroundings. None of them can afford to re-process all of history at every step; all of them can afford to update a posterior. The conjugate families make that update a handful of additions.

31.4 The Normal–Normal Conjugate Pair

The third classic conjugate pair governs measurement. Suppose we measure a quantity whose true value is μ, with each measurement $x_i \sim \mathcal{N}(\mu, \sigma^2)$ where the measurement noise σ^2 is known and only the mean μ is unknown. Take a normal prior on the mean, $\mu \sim \mathcal{N}(\mu_0, \tau_0^2)$, encoding a prior best guess μ_0 with prior uncertainty τ_0^2.

The algebra is slightly heavier because of the exponentials, but the punchline is clean and worth deriving. With n observations of sample mean $\bar{x}$, the likelihood is (dropping constants)

$$p(D \mid \mu) \;\propto\; \exp\!\left(-\frac{n(\mu - \bar{x})^2}{2\sigma^2}\right),$$

and the prior is $\propto \exp\!\big(-(\mu-\mu_0)^2/(2\tau_0^2)\big)$. Their product is the exponential of a sum of two quadratics in μ, which is again a quadratic in μ, which is again (the kernel of) a normal. Completing the square gives the posterior parameters.

Formal

Normal–Normal update (known variance). With prior $\mu \sim \mathcal{N}(\mu_0, \tau_0^2)$ and n observations of mean $\bar{x}$ from $\mathcal{N}(\mu, \sigma^2)$, the posterior is $\mu \mid D \sim \mathcal{N}(\mu_n, \tau_n^2)$ where it is cleanest in terms of *precision* (the reciprocal of variance):

$$\frac{1}{\tau_n^2} = \frac{1}{\tau_0^2} + \frac{n}{\sigma^2}, \qquad \mu_n = \tau_n^2\left(\frac{\mu_0}{\tau_0^2} + \frac{n\,\bar{x}}{\sigma^2}\right). \tag{31.4}$$

Stare at (31.4) until it speaks. *Precisions add.* Your prior carries precision $1/\tau_0^2$; each of the n data points carries precision $1/\sigma^2$; the posterior precision is simply their sum. And the posterior mean μ_n is a precision-weighted average of the prior mean and the data mean — each source pulls the answer toward itself in proportion to how precise it is. A confident prior (large $1/\tau_0^2$) and a noisy measurement (small $1/\sigma^2$) leave the posterior near the prior; abundant precise data overwhelm the prior. It is the same weighted-average story as Beta–Binomial and Gamma–Poisson, now told in the currency of precision.

Why it works

Why do *precisions* add rather than variances? Because in the exponent of a Gaussian, the coefficient of μ^2 is $-\frac{1}{2} \times$ (precision), and multiplying two Gaussians adds the exponents, hence adds those coefficients, hence adds precisions. Variance is the inverse, so it does the opposite of adding — combining two equally precise sources *halves* the variance. This is the deep reason the standard error of a mean falls like $\sigma/\sqrt{n}$: each independent observation contributes one unit of precision, n of them give precision n/σ^2, and the posterior standard deviation is the reciprocal square root. The familiar $1/\sqrt{n}$ is precision-addition wearing a disguise.

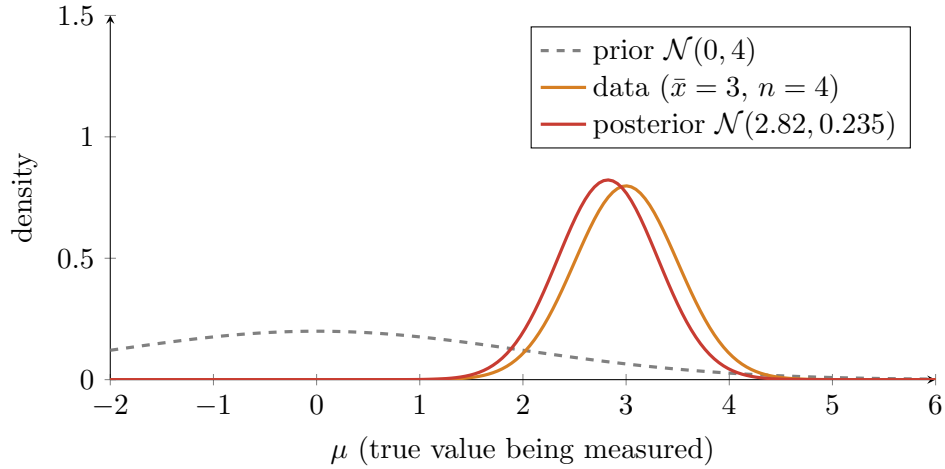


Figure 31.4: Normal–Normal updating. A broad prior centered at 0 meets data centered at $\bar{x} = 3$. The posterior (red) sits between them — pulled most of the way toward the data because the data are far more precise — and is narrower than either input, since precisions add. The compromise is automatic and quantitative.

Connection

The precision-weighted average in (31.4) is the one-dimensional seed of an idea that grows into the *Kalman filter*, the algorithm steering spacecraft, phones, and self-driving cars. There the "prior" is your prediction of where a moving object is, the "data" is a noisy sensor reading, and the posterior is the fused estimate — exactly a precision-weighted blend, run once per time step, in many dimensions with the scalar variances replaced by covariance matrices. If you want to see where the linear algebra enters, the companion linear-algebra reader's treatment of positive-definite matrices and quadratic forms is the language in which the multivariate version is written: the exponent $-(\mu - \mu_0)^2/(2\tau_0^2)$ becomes $-\frac{1}{2}(\boldsymbol{\mu} - \boldsymbol{\mu}_0)^\top \Sigma^{-1}(\boldsymbol{\mu} - \boldsymbol{\mu}_0)$, and "precisions add" becomes "inverse-covariance matrices add."

31.5 Summarizing a Posterior: Credible Intervals and MAP

The posterior is the complete answer, but humans want a headline. Two summaries dominate.

31.5.1 Credible Intervals

A *credible interval* is an interval that contains the parameter with a stated posterior probability. A 95% credible interval $[L, U]$ is any pair with

$$\int_L^U p(\theta \mid D)\, \mathrm{d}\theta = 0.95.$$

Read that integral literally: *given the data, there is a* 95% *probability that* θ *lies in* $[L, U]$. That is the sentence everyone wishes a confidence interval meant. It is the sentence a credible interval actually means.

Common trap

A frequentist 95% *confidence* interval does *not* say there is a 95% probability the parameter lies inside it. To a frequentist the parameter is a fixed constant, so it is either in the interval or not — no probability about it. The 95% refers to the *procedure*: if you repeated the whole experiment many times, 95% of the intervals so constructed would cover the true value. The Bayesian credible interval makes the natural-language statement directly, at the cost of having required a prior. Do not mix the two interpretations; the difference is the entire frequentist–Bayesian fault line in miniature.

There are two common ways to pin down the interval, because the equation above has many solutions. The *equal-tailed* interval chops 2.5% off each end, taking $[L, U]$ to be the 2.5th and 97.5th posterior percentiles — easy and the usual default. The *highest posterior density* (HPD) interval instead takes the shortest interval with 95% mass, equivalently the set of θ whose posterior density exceeds some threshold. For a symmetric unimodal posterior the two coincide; for a skewed one the HPD is shorter and always contains the mode, which the equal-tailed one need not.

Example 31.3 (Credible interval for the A/B test)**.** Return to the green button of Example 31.1, posterior $\text{Beta}(13, 29)$. Its equal-tailed 95% credible interval runs from about 0.181 to 0.455 (the 2.5th and 97.5th percentiles of that Beta, read from software or tables). So we report: "given the data, the green conversion rate is 95% likely to lie between roughly 18% and 46%, with posterior mean 31%." Compact, accurate, and directly interpretable.

31.5.2 MAP versus the Full Posterior

The *maximum a posteriori* estimate is the single most probable value of θ under the posterior — the location of the peak:

$$\hat{\theta}_{\text{MAP}} \;=\; \arg\max_{\theta}\; p(\theta \mid D) \;=\; \arg\max_{\theta}\; p(D \mid \theta)\, p(\theta).$$

The second equality (we dropped the constant evidence) reveals MAP as maximum likelihood *with a prior attached*: maximize likelihood times prior instead of likelihood alone. Take a flat prior and MAP collapses exactly to the MLE. So MAP is the Bayesian's bridge to the familiar point estimate, and the prior term acts as a *regularizer* that pulls the estimate away from extreme values the data alone might suggest.

What is really happening

MAP throws away almost everything. It reports the tallest point of the posterior and discards the width, the skew, the tails — all the information about how *sure* you are. For a sharply peaked posterior that is a fine summary; for a broad or skewed or multimodal one it can be badly misleading, even reporting a value of essentially zero density elsewhere as if it were the answer. A notorious case: if a posterior has two separated peaks, MAP names one and pretends the other does not exist. The full posterior is the object that actually contains your knowledge; MAP, the posterior mean, and credible intervals are all just lossy summaries of it. Whenever you can carry the whole thing, do.

Walking question

For the green-button posterior Beta(13, 29), compare the summaries. The MAP (mode of a Beta is $(\alpha-1)/(\alpha+\beta-2)$) is $12/40 = 0.300$. The posterior mean is $13/42 \approx 0.310$. They differ because the Beta is mildly right-skewed — the mode sits a touch left of the mean, with a longer tail to the right. Which should you report? If you must give one number and downstream calculations average over many such estimates, the posterior mean is usually the better choice (it minimizes expected squared error); if you want the single most plausible value, MAP. But the most informative report is the interval from Example 31.3, which contains both and shows the spread.

31.6 Bayesian versus Frequentist: An Even-Handed Comparison

It would be a disservice to present Bayesian inference as simply correct and frequentist inference as confused. They answer different questions, and the practitioner's job is to know which question is being asked.

	Frequentist	**Bayesian**
What is θ?	A fixed unknown constant.	A random variable with a distribution.
What is random?	The data (over hypothetical repetitions).	Your belief about θ, given the one dataset.
Need a prior?	No.	Yes — this is the cost and the power.
Main output	Point estimate + confidence interval; p-values.	Full posterior distribution.
"95% interval" means	95% of such intervals cover θ.	$\mathbb{P}(\theta \in \text{interval} \mid \text{data}) = 0.95$.
Strength	No prior needed; strong long-run guarantees.	Direct probability statements; uses prior knowledge; clean with small data.

Paradox

The same data, two different intervals — and the trouble with optional stopping. Here is a genuinely unsettling fact. A frequentist analysis can depend on your *intentions*, not just your data. Suppose you flip a coin and see 9 heads in 12 flips. A frequentist who planned to flip exactly 12 times computes one p-value; a frequentist who planned to flip "until I get 3 tails" (and happened to stop at the twelfth flip) computes a *different* p-value from the *same observed sequence*, because the sample space of hypothetical repetitions is different. The data are identical; the conclusion shifts with the experimenter's private stopping rule. The Bayesian is untroubled: the likelihood of the observed flips is $\theta^9(1-\theta)^3$ either way, the stopping rule contributes only a constant that the proportionality washes out, and the posterior is identical. This is the *likelihood principle*, and Bayesian inference obeys it automatically. To some this

is the decisive argument for the Bayesian view; to others it is a curiosity. Either way it shows the two frameworks are genuinely, not cosmetically, different.

History

The twentieth century saw a long cold war between the camps. Ronald Fisher, Jerzy Neyman, and Egon Pearson built the frequentist edifice — p-values, confidence intervals, hypothesis tests — partly in explicit reaction against "inverse probability," which they distrusted for its reliance on subjective priors. Harold Jeffreys kept the Bayesian flame through mid-century; Leonard Savage and Bruno de Finetti gave it rigorous decision- and belief-theoretic foundations. For decades Bayesian methods were admired in theory and avoided in practice, for one brutally simple reason: the evidence integral $\int p(D \mid \theta)p(\theta)\,\mathrm{d}\theta$ is usually impossible to do by hand outside the conjugate families. What broke the stalemate was not a new theorem but cheap computers and the *Markov chain Monte Carlo* revolution of the late 1980s and early 1990s, which let us *sample* from any posterior without ever computing that integral. Today the distinction in practice is less a war than a toolkit; most working statisticians use both.

Why this matters

The conjugate families of this chapter are beautiful precisely because they sidestep the hard integral — the posterior pops out in closed form. But they are a small, special island. The reason Bayesian inference is now everywhere is that we learned to leave the island: when the posterior has no closed form, draw samples from it with Markov chain Monte Carlo (the companion to this chapter on Monte Carlo methods builds exactly this machinery) and summarize the samples. Conjugacy is the place to build intuition; sampling is the place where modern Bayesian work actually happens.

31.7 Bayesian Ideas in Machine Learning

The vocabulary of this chapter is the hidden skeleton of a great deal of machine learning. Three of these connections deserve to be spelled out.

Application

Regularization is a prior in disguise. Training a model by minimizing a loss plus a penalty $\lambda\|w\|^2$ ("ridge" or weight decay) is *exactly* MAP estimation with a Gaussian prior $w \sim \mathcal{N}(0, \sigma^2 I)$ on the weights. Take the log of posterior $\propto$ likelihood $\times$ prior: the log-likelihood is the usual fit term, and the log of a Gaussian prior is $-\|w\|^2/(2\sigma^2)$ plus a constant — precisely the squared penalty, with $\lambda = 1/(2\sigma^2)$. Likewise an L_1 penalty ("lasso") is MAP under a Laplace prior, whose sharp peak at zero is what drives weights exactly to zero. So every time a practitioner "adds regularization to prevent overfitting," they are, whether they know it or not, declaring a prior belief that the weights are small. The strength λ is the prior's confidence. Overfitting, in this light, is what happens when you let a flexible model believe the data with no prior skepticism at all.

Application

Naïve Bayes and posterior class probabilities. A spam filter, or any probabilistic classifier, computes $\mathbb{P}(\text{class} \mid \text{features}) \propto \mathbb{P}(\text{features} \mid \text{class})\,\mathbb{P}(\text{class})$ — prior times likelihood, exactly (31.1) with the class playing the role of θ. The class prior $\mathbb{P}(\text{spam})$ is your base rate; the likelihood is the word model; the posterior is the calibrated probability that the message is spam. The same shape powers medical-diagnosis nets and topic models. When a modern neural network ends in a softmax and you read its outputs as "class probabilities," you are reading an approximate posterior.

Application

Calibrated uncertainty in deep learning. A standard neural network gives a single weight vector — a point estimate, the MAP of an implicit posterior. *Bayesian deep learning* instead tries to carry a distribution over weights, so the network can say "I do not know" on inputs unlike anything it trained on, by reporting a wide posterior predictive. This matters enormously in self-driving cars and medical AI, where a confident wrong answer is far more dangerous than a candid "uncertain." Exact posteriors over millions of weights are hopeless, so the field approximates — variational inference, Monte Carlo dropout, deep ensembles — but the goal is pure Bayes: replace the single best guess with a calibrated distribution of guesses.

Connection

There is also a clean bridge to information theory, the subject of a later chapter. The amount a single data point changes your belief — how far the posterior moves from the prior — is naturally measured by the Kullback–Leibler divergence $\mathrm{KL}(p(\theta \mid D) \,\|\, p(\theta))$, the "information gain." Experimental design can then be framed as choosing the experiment expected to yield the largest information gain, a principle called *Bayesian optimal design*. Belief updating and information are two views of one thing: to learn is to reduce uncertainty, and uncertainty is measured in bits.

31.8 The Posterior Predictive: Forecasting the Next Observation

We have been learning about θ, but often what we truly want is to predict the *next data point* x_{new}. The Bayesian answer does not plug in a single estimate of θ; it averages the prediction over the entire posterior, weighting each value of θ by how plausible the data have made it. This is the *posterior predictive distribution*:

$$p(x_{\text{new}} \mid D) \;=\; \int p(x_{\text{new}} \mid \theta)\, p(\theta \mid D)\, \mathrm{d}\theta. \tag{31.5}$$

Read it as a vote: every candidate θ predicts the future, and we tally the predictions weighted by posterior belief. This automatically folds parameter uncertainty into the forecast — the predictive is wider than if we had pretended to know θ exactly, which is exactly right, because we do not.

Example 31.4 (Will the next tack land up?)**.** After the 80 tosses with posterior $\mathrm{Beta}(57, 25)$, what is the probability the next toss lands point-up? Plug the Bernoulli likelihood $p(x_{\text{new}} = 1 \mid \theta) = \theta$

into (31.5):

$$\mathbb{P}(x_{\text{new}} = 1 \mid D) = \int_0^1 \theta\, p(\theta \mid D)\, \mathrm{d}\theta = \mathbb{E}[\theta \mid D] = \frac{57}{57+25} = \frac{57}{82} \approx 0.695.$$

The predictive probability is just the posterior *mean* of θ — a tidy result special to the Bernoulli case. Note we did *not* use the MAP value $56/80 = 0.700$; averaging over the posterior, not maximizing, is the principled forecast. With few data the gap between the two matters; with much data it vanishes.

Work it yourself

Try this with pencil and a tiny prior. Start with $\text{Beta}(2, 2)$ (a gentle belief that a coin is fair) and feed it the sequence H, H, H, T one flip at a time, writing the posterior after each flip using the rule "add a success to α, a failure to β." You should pass through $\text{Beta}(3, 2), \text{Beta}(4, 2), \text{Beta}(5, 2), \text{Beta}(5, 3)$. After each step compute the posterior-predictive probability of the *next* flip being heads, $\alpha/(\alpha + \beta)$, and watch it climb $0.6, 0.667, 0.714, 0.625$ as the evidence accumulates and then the lone tail pulls it back. Feel how the gentle prior keeps the early estimates from lurching all the way to 1 after three straight heads — that restraint is the prior doing its job.

31.9 Conclusion

Bayesian inference is one idea applied relentlessly: *posterior* $\propto$ *likelihood* $\times$ *prior*. The prior states what you believed before; the likelihood states what the data say; the posterior is the disciplined combination, a full distribution over the unknown rather than a lonely point. We saw that for the three great conjugate pairs — Beta–Binomial, Gamma–Poisson, Normal–Normal — the combination is a one-line update of counts or precisions, and we watched the posterior tighten toward the truth as data arrived, its narrowing width a true accounting of remaining doubt. We learned to read off credible intervals that mean what people wish confidence intervals meant, to summarize with MAP while remembering how much MAP discards, and to predict the future by averaging over the posterior rather than betting on its peak.

The deepest lesson is one of attitude. A frequentist asks, "what procedure has good long-run behavior?" A Bayesian asks, "what should I believe, right now, given exactly what I have seen?" Both are legitimate; the second happens to be the question we ask in ordinary life, and it is the one that, scaled up on fast computers, now quietly powers spam filters, recommendation engines, robot navigation, and the calibrated uncertainty of modern AI. You began this book learning to compute the probability of outcomes. You end this chapter doing the inverse and far more human thing: computing, and rationally updating, your beliefs about the hidden causes behind what you see.

31.10 Exercises

Exercise 31.1 (Discrete update)**.** A box contains a coin that is either fair ($\theta = 0.5$) or two-headed ($\theta = 1$), with prior probabilities 0.9 and 0.1. You flip it once and get heads. Find the posterior probability that the coin is two-headed. Then flip again, get heads again, and update once more. How many consecutive heads before you are more than 90% sure it is two-headed?

Exercise 31.2 (Beta–Binomial by hand)**.** You hold a Beta$(2, 3)$ prior on a website's click-through rate θ. You then serve the ad to 50 users and observe 20 clicks. (a) Write the posterior distribution. (b) Compute the posterior mean and the MAP estimate. (c) Explain in one sentence why they differ.

Exercise 31.3 (The prior as pseudocounts)**.** Show that the posterior mean $\dfrac{\alpha + k}{\alpha + \beta + n}$ of a Beta–Binomial update equals the sample proportion k/n when $\alpha = \beta = 0$ (the improper "Haldane" prior), and equals $1/2$ regardless of data when $n = 0$ and $\alpha = \beta$. Interpret both limits in words.

Exercise 31.4 (Gamma–Poisson forecast)**.** A call center receives calls as Poisson(λ) per hour. With prior $\lambda \sim \text{Gamma}(3, 1)$ you observe $4, 7, 5$ calls in three successive hours. (a) Give the posterior. (b) Give the posterior mean and standard deviation. (c) The posterior-predictive number of calls in the next hour is Negative Binomial; without deriving it, give its mean using (31.5) and $\mathbb{E}[x_{\text{new}} \mid D] = \mathbb{E}[\lambda \mid D]$.

Exercise 31.5 (Precisions add)**.** You measure a length with two instruments. The first reads 10.2 with standard deviation 0.5; the second reads 10.0 with standard deviation 0.2. Treating the first reading as a prior and the second as data (both normal, known variances), use (31.4) to find the combined posterior mean and standard deviation. Which reading does the answer favor, and why?

Exercise 31.6 (Conceptual: two interpretations)**.** A newspaper reports a "95% confidence interval of $[0.41, 0.47]$ for the candidate's vote share." (a) State precisely what this means to a frequentist. (b) State what a Bayesian 95% credible interval of $[0.41, 0.47]$ would mean. (c) Explain which one matches the way most readers will interpret the sentence, and what extra ingredient the Bayesian statement required.

Exercise 31.7 (Regularization as a prior)**.** Consider linear regression with weights w and Gaussian observation noise. Show that maximizing the posterior under the prior $w \sim \mathcal{N}(0, \sigma^2 I)$ is equivalent to minimizing (squared error) $+ \lambda \|w\|^2$, and find λ in terms of σ and the noise variance. (Hint: take $-\log$ of likelihood times prior.)

Exercise 31.8 (Proof: order does not matter)**.** Prove that updating a prior with data D_1 and then updating the result with independent data D_2 yields the same posterior as updating once with the combined data (D_1, D_2). Identify exactly where you use the conditional independence of D_1 and D_2 given θ, and explain why this justifies calling yesterday's posterior today's prior.

Exercise 31.9 (Sensitivity to the prior)**.** For the Beta–Binomial model with data $k = 3$ successes in $n = 4$ trials, compute the posterior mean under three priors: Beta$(1, 1)$, Beta$(10, 10)$, and Beta$(1, 10)$. Comment on how much each prior moves the estimate away from the raw proportion $3/4$, and relate the size of the movement to the "pseudocount" strength $\alpha + \beta$ of each prior.

Exercise 31.10 (MAP can mislead)**.** Sketch (or describe) a posterior density on $[0, 1]$ that has two well-separated peaks of equal height, say near 0.2 and 0.8, with a deep valley between. Explain why reporting the MAP estimate alone would be seriously misleading here, what the posterior mean would report, and why neither single number is an adequate summary. What would you report instead?

Chapter 32

Confidence Intervals and Hypothesis Testing

You have spent the last two chapters learning how to point at a number. Given data, the method of moments and maximum likelihood hand you an *estimate* — a single value $\hat{\theta}$ that is your best guess for some unknown quantity θ. But a single number, offered with no hint of how much you should trust it, is nearly worthless. If a pollster tells you "52% of voters favor the measure" and stops there, the right response is not "thank you" but "52% *plus or minus what*?" A guess without a margin is a guess pretending to be a fact.

This chapter is about attaching a measure of uncertainty to estimates. There are two great traditions for doing it, and they are really two answers to two different questions.

- **Confidence intervals** answer: *how wide a net should I cast so that the truth is plausibly inside it?* The output is a *range* of values.
- **Hypothesis tests** answer: *is the effect I see real, or could plain randomness have produced it?* The output is a *decision* (or a number, the p-value, that informs one).

Both are built on one idea you already own from Chapter 21: the *sampling distribution*. Everything in this chapter — every ±, every p-value, every t-test — is an accountant's report on how much an estimate wobbles when you redraw the data. We will build that idea carefully, then watch it grow into the two most-used (and two most-*misused*) tools in all of applied statistics, paying close attention to what they do and, just as importantly, what they do *not* do.

32.1 The sampling distribution: the engine underneath

Imagine the unknown quantity is the mean height μ of adult women in a country. You cannot measure everyone, so you draw a random sample of n women and compute the sample mean

$$\bar{X}_n = \frac{1}{n}\sum_{i=1}^{n} X_i.$$

Here is the conceptual leap that makes all of inference work. Your particular sample gave you, say, $\bar{X}_n = 164.3$ cm. But that number was an accident of *which* n women happened to be chosen. A

different random sample would have given a slightly different mean. So $\bar{X}_n$ is not a fixed number — it is itself a random variable, with its own distribution. That distribution is called the **sampling distribution** of the estimator.

Intuition

Picture a giant, hypothetical experiment: draw a sample of size n, write down $\bar{X}_n$, throw the sample back, and repeat forever. The histogram of all those $\bar{X}_n$ values *is* the sampling distribution. We never actually run this experiment — we have only the one sample we paid for. But probability theory lets us *predict* the shape of that histogram without running it. That prediction is the whole game.

For the sample mean we know the prediction exactly. If the X_i are i.i.d. with mean μ and variance σ^2, then by linearity and independence (Chapters 8–9),

$$\mathbb{E}[\bar{X}_n] = \mu, \qquad \mathrm{Var}(\bar{X}_n) = \frac{\sigma^2}{n}.$$

The estimator is centered on the truth (it is *unbiased*) and its variance shrinks like $1/n$. The standard deviation of an estimator has a special name: the **standard error**.

$$\mathrm{SD}(\bar{X}_n) = \frac{\sigma}{\sqrt{n}} =: \mathrm{SE}.$$

Common trap

Do not confuse the *standard deviation* σ (how spread out individual people are) with the *standard error* $\sigma/\sqrt{n}$ (how spread out the *sample mean* is across hypothetical repetitions). Heights vary by maybe 7 cm person to person; the mean of 1000 heights varies by only $7/\sqrt{1000} \approx 0.22$ cm. As n grows, people stay just as different but the *average* becomes razor-sharp. This is the single most common confusion in introductory statistics.

And the shape? This is where the Central Limit Theorem (Chapter 21) does its magic. No matter what the distribution of an individual X_i looks like — skewed, bimodal, discrete — the sampling distribution of $\bar{X}_n$ is approximately Normal for large n:

$$\bar{X}_n \overset{\text{approx}}{\sim} \mathcal{N}\left(\mu, \frac{\sigma^2}{n}\right), \qquad \text{equivalently} \qquad \frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} \xrightarrow{d} \mathcal{N}(0,1).$$

Why it works

Why is the Normal curve the universal shape of sampling distributions? Because $\bar{X}_n$ is an *average of many small independent contributions*, and the CLT says such averages forget the idiosyncrasies of their ingredients and converge to the same bell curve. This is the deep reason confidence intervals and z-tests work for almost any underlying population: we are never really using the population's shape, only the shape of the *average*, and that shape is always Normal in the limit.

Figure 32.1 shows the punchline visually: a wildly non-Normal population (a skewed exponential) whose sample means nonetheless pile up into a tidy bell as n grows, narrowing all the while.

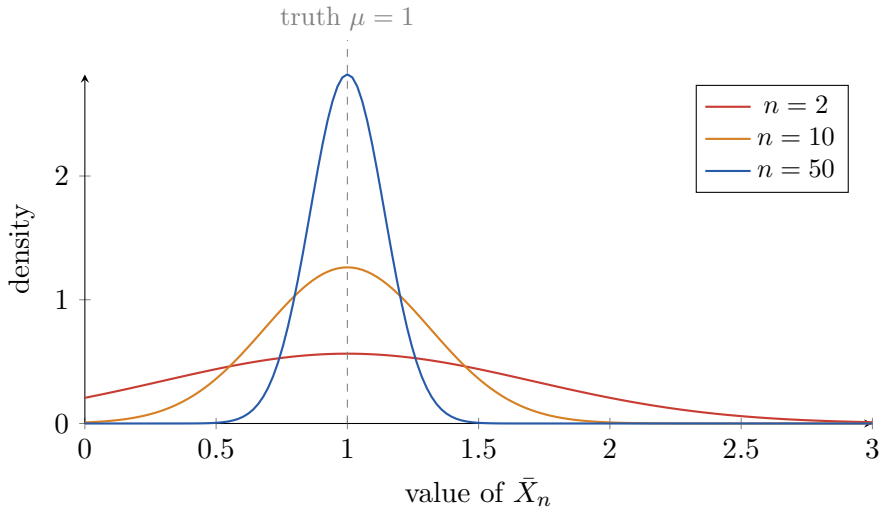


Figure 32.1: Sampling distributions of $\bar{X}_n$ for samples drawn from a skewed population with mean $\mu = 1$. Two things happen as n grows: the curves become more Normal (CLT) and narrower (standard error $\sigma/\sqrt{n} \to 0$). All are centered on the truth.

What is really happening

Inference is a kind of reverse engineering. We observe one point drawn from the sampling distribution (our single $\bar{X}_n$) and we ask: which values of the unknown μ would make this observation unsurprising? A confidence interval is the set of such μ's. A hypothesis test fixes one candidate μ and asks whether our observation is surprising under it. Same engine, two readouts.

32.2 Confidence intervals

32.2.1 Building one from the sampling distribution

Start with the clean case: i.i.d. data, large n, and suppose for a moment we know σ. Standardize the sample mean:

$$Z = \frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} \approx \mathcal{N}(0,1).$$

From the standard Normal we know that 95% of the probability lies between -1.96 and $+1.96$. Write $z_{\alpha/2}$ for the value with $\alpha/2$ probability in the upper tail; for 95% confidence, $\alpha = 0.05$ and $z_{0.025} = 1.96$. Then

$$\mathbb{P}\left(-z_{\alpha/2} \le \frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} \le z_{\alpha/2}\right) \approx 1 - \alpha.$$

Now do nothing but algebra — multiply through by the standard error and rearrange to isolate μ in the middle:

$$\mathbb{P}\left(\bar{X}_n - z_{\alpha/2}\frac{\sigma}{\sqrt{n}} \;\le\; \mu \;\le\; \bar{X}_n + z_{\alpha/2}\frac{\sigma}{\sqrt{n}}\right) \approx 1 - \alpha.$$

Formal

A $(1-\alpha)$ **confidence interval** for μ (known σ, large n) is

$$\boxed{\bar{X}_n \;\pm\; z_{\alpha/2}\,\frac{\sigma}{\sqrt{n}}}.$$

The half-width $z_{\alpha/2}\,\sigma/\sqrt{n}$ is the **margin of error**. It shrinks with $\sqrt{n}$: to halve the margin you must *quadruple* the sample.

Example 32.1 (A polling margin)**.** A poll of $n = 1000$ voters finds $\hat{p} = 0.52$ in favor. For a proportion the variance of a single response is $p(1-p)$, so the standard error is $\sqrt{p(1-p)/n}$; plugging in $\hat{p}$ gives $\sqrt{0.52 \cdot 0.48/1000} \approx 0.0158$. The 95% interval is

$$0.52 \pm 1.96 \times 0.0158 = 0.52 \pm 0.031 = (0.489,\ 0.551).$$

This is the famous "margin of error of about 3 points." Note the interval includes 0.50: on this evidence we cannot rule out a tie. The headline "leading 52 to 48" oversells what the data support.

32.2.2 What a confidence interval really means — and does not

This is the most misunderstood concept in statistics, so let us be slow and exact.

Common trap

The interval $(0.489, 0.551)$ **does NOT mean "there is a** 95% **probability that** μ **lies in** $(0.489, 0.551)$**."** In the frequentist framework μ is a fixed, unknown constant — not random. It is either in that particular interval or it is not; the probability is 0 or 1, we just do not know which. The randomness was in the *data*, which produced the *endpoints*, not in μ.

So what is the 95% attached to? It is a property of the *procedure*, not of any single interval.

Focus

The correct interpretation. The recipe "compute $\bar{X}_n \pm 1.96\,\mathrm{SE}$" is a machine that spits out a random interval every time you feed it a fresh sample. *Across many independent samples,* 95% *of the intervals this machine produces will contain the true* μ. Your one interval is a single draw from that machine. You do not know if it is one of the lucky 95% or the unlucky 5% — but you trust the machine because it is right 95% of the time.

Figure 32.2 makes this tangible. We simulate twenty samples, draw the twenty intervals, and mark the fixed truth with a vertical line. Most intervals cover it; a stray one or two miss. The "95%" is the long-run *coverage rate*, the fraction of horizontal bars that cross the line.

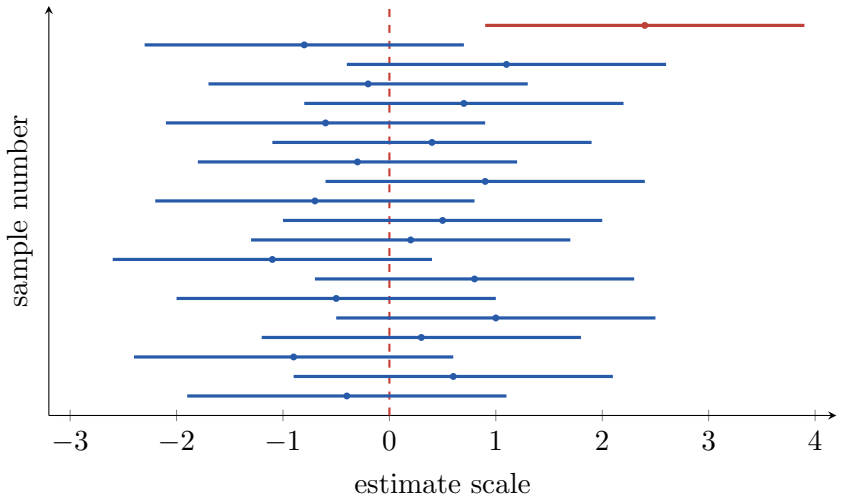


Figure 32.2: Twenty 95% confidence intervals from twenty independent samples. The truth μ (red dashed line) is fixed; the intervals jiggle. Nineteen of the twenty cover the truth (blue); one misses (red). Over the long run, 95% cover. The confidence level describes the *procedure*, not your particular interval.

History

The confidence interval was introduced by Jerzy Neyman in a landmark 1937 paper, "Outline of a Theory of Statistical Estimation Based on the Classical Theory of Probability." Neyman was deliberately, almost stubbornly, careful: he insisted the probability statement attaches to the *random interval before the data are seen*, never to the fixed parameter after. He knew the temptation to say "95% chance μ is in here" and built the whole theory to resist it. The misinterpretation has persisted ever since, a sign of how badly people *want* the Bayesian statement — which, as Chapter 31 shows, is exactly what a Bayesian credible interval delivers, at the price of a prior.

Connection

The frequentist confidence interval and the Bayesian credible interval (Chapter 31) often produce nearly identical numbers but answer different questions. The credible interval *does* let you say "95% probability μ is in here," because in the Bayesian world μ has a probability distribution. The confidence interval refuses that statement and instead guarantees long-run coverage. With a flat prior and a Normal likelihood the two intervals coincide algebraically — a reassuring meeting point of the two great schools.

32.2.3 When you do not know σ: the t-interval

We assumed we knew the population standard deviation σ. In real life we never do — we estimate it from the same data with the *sample standard deviation*

$$s = \sqrt{\frac{1}{n-1}\sum_{i=1}^{n}(X_i - \bar{X}_n)^2}.$$

The $n-1$ (rather than n) makes s^2 an unbiased estimator of σ^2; we lose one degree of freedom because we already spent the data to estimate $\bar{X}_n$. Substituting s for σ introduces extra uncertainty — now *both* the numerator and the denominator of our standardized statistic are random. William Gosset worked out the consequence: the statistic

$$T = \frac{\bar{X}_n - \mu}{s/\sqrt{n}}$$

no longer follows a Normal distribution but **Student's t-distribution** with $n-1$ degrees of freedom. The t looks like the Normal but with heavier tails — fatter, because the extra randomness in s makes large deviations more likely.

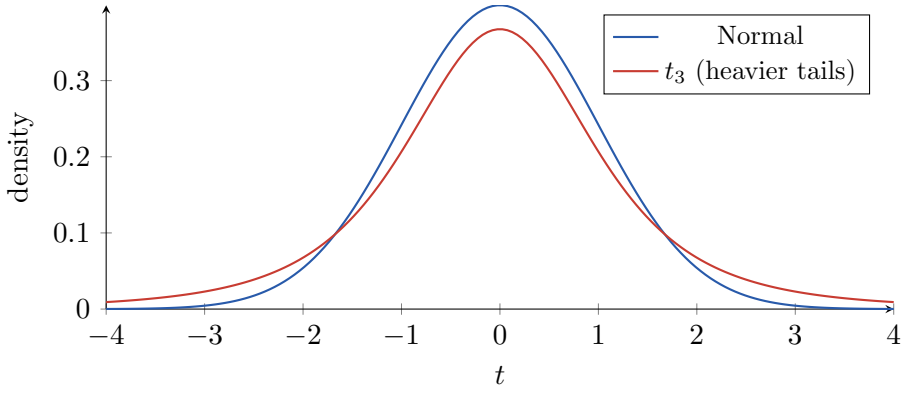


Figure 32.3: Student's t with 3 degrees of freedom versus the standard Normal. The t has heavier tails: estimating σ from the data costs you, so you must reach further out for the same confidence. As $n \to \infty$, $s \to \sigma$ and $t \to$ Normal.

The confidence interval is identical in form, with the critical value taken from the t-table instead of the Normal:

$$\boxed{\bar{X}_n \;\pm\; t_{n-1,\,\alpha/2}\,\frac{s}{\sqrt{n}}}\,.$$

Because $t_{n-1,\alpha/2} > z_{\alpha/2}$, the t-interval is always a little *wider* — the price of not knowing σ. For $n = 10$, the 95% critical value is $t_{9,0.025} = 2.262$, noticeably more than 1.96. By $n = 100$ it has fallen to 1.984, almost the Normal value. For large samples the distinction vanishes, which is why the z- and t-intervals merge.

History

Student's t is named for a pseudonym. William Sealy Gosset derived it in 1908 while working as a chemist at the Guinness brewery in Dublin, where he needed to draw conclusions from *small* batches of barley and yeast — samples far too small for the large-n Normal approximation. Guinness forbade employees from publishing (a rival had leaked trade secrets), so Gosset published under the modest pen name "Student." The most-used distribution in applied statistics is thus named after a man hiding from his employer. R. A. Fisher later put the result on rigorous footing and gave it the letter t.

Example 32.2 (A small-sample t-interval)**.** Six measurements of a chemical yield: $48, 51, 49, 53, 50, 52$ (grams). The sample mean is $\bar{X} = 50.5$. The deviations are $-2.5, 0.5, -1.5, 2.5, -0.5, 1.5$, whose

squares sum to $6.25 + 0.25 + 2.25 + 6.25 + 0.25 + 2.25 = 17.5$, so

$$s^2 = \frac{17.5}{6-1} = 3.5, \qquad s = 1.871, \qquad \frac{s}{\sqrt{6}} = 0.764.$$

With 5 degrees of freedom, $t_{5,0.025} = 2.571$. The 95% interval is

$$50.5 \pm 2.571 \times 0.764 = 50.5 \pm 1.96 = (48.54,\ 52.46) \text{ grams}.$$

Had we naively used 1.96 instead of 2.571, we would have reported a falsely narrow interval — overstating our precision, which is exactly the error the t corrects.

32.3 Hypothesis testing

A confidence interval casts a net. A hypothesis test asks a sharper, binary question: *is there any effect at all, or could what I see be pure chance?* The logic is a courtroom's — presumption of innocence, evidence, a verdict "beyond reasonable doubt" — dressed in the language of probability.

32.3.1 The null, the alternative, and the logic of contradiction

We set up two competing claims:

- The **null hypothesis** H_0: the boring, status-quo, "nothing is going on" claim. The drug has no effect; the coin is fair; the two web pages convert at the same rate. H_0 is the defendant, presumed innocent.
- The **alternative hypothesis** H_1 (or H_a): the interesting claim we would believe only if the evidence forces us. The drug works; the coin is biased; page B is better.

The strategy is *proof by contradiction, softened by probability.* We provisionally assume H_0 is true. We then ask: *if the null were true, how surprising would my data be?* If the data are wildly unlikely under H_0, we take that as evidence against H_0 and *reject* it. If the data are unremarkable under H_0, we have no grounds to abandon it and we *fail to reject.*

Common trap

We say "fail to reject H_0," never "accept H_0." Absence of evidence is not evidence of absence. A jury that does not convict declares "not guilty," not "innocent." Failing to find an effect could mean there is none — or that your sample was too small to see it. The asymmetry is deliberate and important.

32.3.2 Test statistics and the null distribution

To measure "how surprising," we compress the data into a single **test statistic** whose distribution we know *assuming H_0 is true.* For testing a mean with known σ, the natural statistic is again the standardized sample mean,

$$Z = \frac{\bar{X}_n - \mu_0}{\sigma/\sqrt{n}},$$

where μ_0 is the value claimed by H_0. Under H_0 this Z is approximately $\mathcal{N}(0,1)$ — this is the **null distribution**. A Z near 0 is what the null predicts; a Z far out in the tails is the kind of thing the null almost never produces, and so it indicts the null.

32.3.3 The p-value

Formal

The **p-value** is the probability, *computed assuming H_0 is true*, of observing a test statistic at least as extreme as the one you actually got:

$$p = \mathbb{P}\left(\text{statistic at least as extreme as observed} \;\middle|\; H_0\right).$$

Small p means your data would be a rare fluke if the null held — evidence *against* H_0.

For a two-sided test with observed value z_{obs}, "as extreme" means as far from 0 in either direction, so

$$p = \mathbb{P}(|Z| \geq |z_{\text{obs}}|) = 2\big(1 - \Phi(|z_{\text{obs}}|)\big),$$

where Φ is the standard Normal CDF. Figure 32.4 shows the p-value as the shaded tail area.

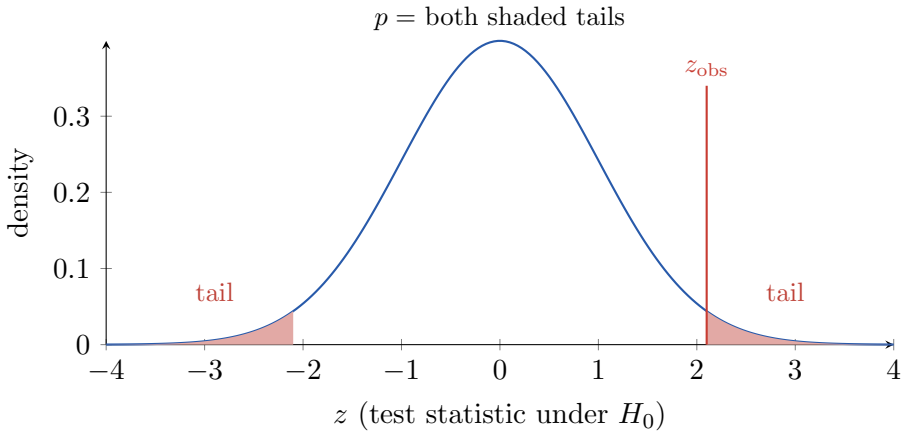


Figure 32.4: The p-value for a two-sided z-test is the total tail area beyond the observed statistic (here $z_{\text{obs}} = 2.1$), in *both* directions. It is the probability the null would generate something at least this extreme by chance alone.

We compare p to a pre-chosen threshold α (the **significance level**, conventionally 0.05). If $p \leq \alpha$, we reject H_0 and call the result "statistically significant." If $p > \alpha$, we fail to reject.

Example 32.3 (A z-test for a fair coin)**.** You flip a coin $n = 100$ times and see 62 heads. Is it fair? Let $H_0 : p = 0.5$ versus $H_1 : p \neq 0.5$. Under H_0 the number of heads has mean 50 and standard deviation $\sqrt{100 \cdot 0.5 \cdot 0.5} = 5$. The standardized statistic is

$$z_{\text{obs}} = \frac{62 - 50}{5} = 2.4.$$

The two-sided p-value is $p = 2(1 - \Phi(2.4)) = 2(0.0082) = 0.0164$. Since $0.0164 < 0.05$, we reject H_0: this coin looks biased. Had we seen 58 heads, $z = 1.6$, $p = 0.11 > 0.05$, and we would not reject — 58 heads is the sort of thing a fair coin throws up often enough.

32.3.4 What the p-value is not

The p-value may be the single most abused number in science. Three corrections:

Common trap

The p-value is NOT the probability that H_0 is true. It is computed *assuming* H_0 is true: $p = \mathbb{P}(\text{data} \mid H_0)$, not $\mathbb{P}(H_0 \mid \text{data})$. Confusing these two is the *prosecutor's fallacy* — swapping the direction of a conditional probability (Chapter 5). A small p-value says "the data are surprising if H_0 holds," which is genuine evidence against H_0, but it is not a posterior probability. Getting from $\mathbb{P}(\text{data} \mid H_0)$ to $\mathbb{P}(H_0 \mid \text{data})$ requires Bayes' theorem and a prior — which the p-value does not have.

Common trap

Statistical significance is not practical importance. With a large enough n, *any* effect, however trivially small, will eventually produce $p < 0.05$ — because the standard error $\sigma/\sqrt{n}$ shrinks to zero and even a microscopic difference lands many standard errors from the null. A drug that lowers blood pressure by 0.1 mmHg can be "highly significant" in a study of a million people and utterly useless in a clinic. Always report the *effect size* (and a confidence interval), not just the p-value.

Common trap

$p = 0.049$ and $p = 0.051$ are essentially the same evidence. The 0.05 cutoff is a convention, not a law of nature (Fisher proposed it almost offhandedly). Treating 0.05 as a bright line between "real" and "not real" — and worse, running analyses until you crawl under it ("p-hacking") — has fueled a replication crisis across the sciences. The p-value is a continuous measure of surprise; report it as such.

Paradox

The base-rate trap behind "significant" findings. Suppose only 1 in 10 hypotheses a lab tests is actually true, and tests run at $\alpha = 0.05$ with power 0.8 (defined below). Out of 1000 tests, 100 effects are real and 900 are null. The tests flag $0.8 \times 100 = 80$ true effects and $0.05 \times 900 = 45$ false ones. So among the 125 "significant" results, $45/125 = 36\%$ are *false positives* — despite every individual test having only a 5% error rate! This is Bayes' theorem again (Chapter 5): a low false-positive *rate* per test does not imply a low false-discovery *proportion* when true effects are rare. It explains why so many published "significant" findings fail to replicate.

32.4 Two kinds of error, and the power to detect

Any decision rule that says "reject" or "do not reject" can be wrong in two ways. The truth (H_0 or H_1) crossed with our verdict gives a 2×2 table:

	H_0 **true**	H_1 **true**
Reject H_0	Type I error (α)	correct (power)
Fail to reject	correct	Type II error (β)

- A **Type I error** is convicting the innocent: rejecting H_0 when it is true. We *control* its probability directly — it is exactly α, the significance level we chose. Setting $\alpha = 0.05$ means "I am willing to cry wolf 5% of the time when there is no wolf."
- A **Type II error** is acquitting the guilty: failing to reject H_0 when H_1 is true. Its probability is called β.
- The **power** of a test is $1 - \beta$: the probability of correctly detecting a real effect. More power is better.

Intuition

Think of α as the sensitivity of a smoke alarm. Crank it up (large α, quick to reject) and you catch every fire (high power) but also scream at burnt toast (many Type I errors). Quiet it down (small α) and you stop the false alarms but risk sleeping through a real blaze (Type II errors). For *fixed* n there is a genuine trade-off: pushing α down pushes β up. The only way to beat the trade-off — to shrink both errors at once — is to gather more data.

Figure 32.5 shows the two error regions as overlapping bells: the null distribution centered at μ_0 and the (true) alternative centered at some μ_1. The cutoff c slices both. The right tail of the null beyond c is α; the left tail of the alternative below c is β.

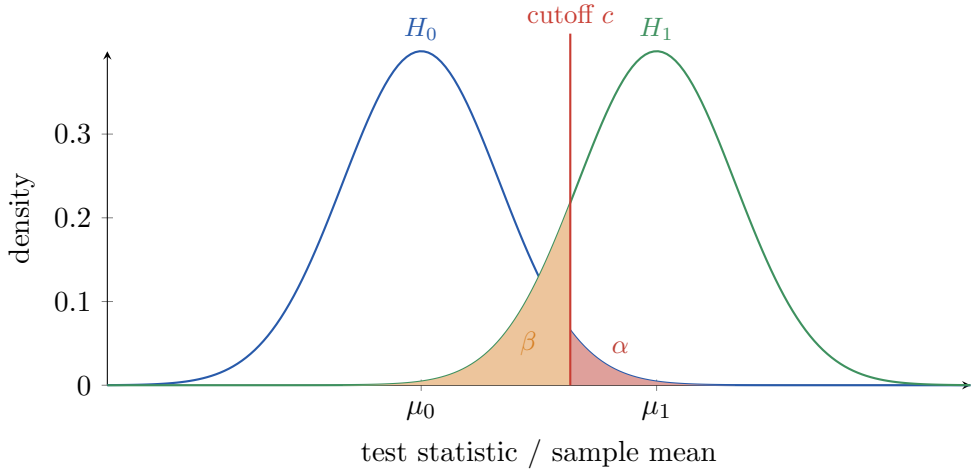


Figure 32.5: Type I and Type II errors. Reject when the statistic exceeds the cutoff c. The red area α is the chance of a false alarm under H_0; the orange area β is the chance of missing a real effect under H_1. Power is the area under the green curve to the *right* of c, i.e. $1 - \beta$. Moving c left trades β for α.

Why it works

Why does power rise with sample size? Larger n shrinks the standard error $\sigma/\sqrt{n}$, so *both* bells in Figure 32.5 narrow. Two narrow bells centered at μ_0 and μ_1 overlap far less than two wide ones. Less overlap means the cutoff can keep α fixed while slicing off much less of the alternative — so β falls and power climbs. This is the mathematical reason "get more data" is the universal cure: it separates the hypotheses.

Example 32.4 (Power and sample-size planning)**.** You want to detect whether a new process raises mean yield from $\mu_0 = 50$ to $\mu_1 = 52$, with $\sigma = 4$, at $\alpha = 0.05$ (one-sided). The cutoff in raw units is $c = \mu_0 + 1.645\,\sigma/\sqrt{n}$. To achieve power 0.8 you need μ_1 to sit 0.84 standard errors above c (since $\Phi(0.84) = 0.80$). Setting $\mu_1 = c + 0.84\,\sigma/\sqrt{n}$ and solving,

$$\mu_1 - \mu_0 = (1.645 + 0.84)\frac{\sigma}{\sqrt{n}} \;\Rightarrow\; n = \left(\frac{(1.645 + 0.84)\,\sigma}{\mu_1 - \mu_0}\right)^2 = \left(\frac{2.485 \times 4}{2}\right)^2 \approx 24.7.$$

So $n = 25$ samples give roughly 80% power. Halve the effect you wish to detect and n *quadruples* — the same $\sqrt{n}$ law that governs confidence intervals. Power analysis done *before* collecting data is how careful experimenters avoid the heartbreak of an underpowered study that could never have found the effect.

32.5 The duality of intervals and tests

Confidence intervals and hypothesis tests are not two separate tools — they are two views of the same picture, and seeing the link makes both clearer.

Focus

Interval–test duality. A two-sided level-α test of $H_0 : \mu = \mu_0$ rejects exactly when μ_0 falls *outside* the $(1 - \alpha)$ confidence interval. Equivalently: *the confidence interval is the set of all null values μ_0 that would not be rejected.*

The algebra is immediate. The test fails to reject when $\left|\frac{\bar{X}_n - \mu_0}{\sigma/\sqrt{n}}\right| \le z_{\alpha/2}$, which rearranges to $\bar{X}_n - z_{\alpha/2}\,\sigma/\sqrt{n} \le \mu_0 \le \bar{X}_n + z_{\alpha/2}\,\sigma/\sqrt{n}$ — precisely the statement "μ_0 is inside the interval." Figure 32.6 draws the equivalence: a null value lands outside the bar exactly when the test rejects it.

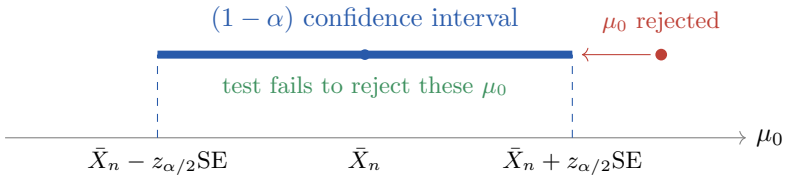


Figure 32.6: Interval–test duality. The confidence interval (blue bar) is exactly the set of null values a two-sided level-α test would *not* reject. A candidate μ_0 inside the bar survives the test; one outside it (red) is rejected. One picture, two tools.

This duality pays off in practice. Reporting a confidence interval is strictly more informative than

reporting a single test: it tells you *every* null you would reject at once, plus the plausible range and the effect size — which is exactly why statisticians increasingly prefer intervals to bare p-values.

32.6 The t-test in practice

Combining the pieces: when σ is unknown (always, in practice) and we test a mean, we use the **t-test**. The statistic is

$$T = \frac{\bar{X}_n - \mu_0}{s/\sqrt{n}},$$

compared against the t_{n-1} distribution. There are three common flavors:

- **One-sample t-test**: is the mean of one group equal to a fixed μ_0?
- **Two-sample t-test**: do two independent groups have equal means? The statistic is $T = (\bar{X} - \bar{Y})/\mathrm{SE}_{\mathrm{diff}}$, where the standard error of the difference combines both groups' variability.
- **Paired t-test**: when observations come in matched pairs (before/after on the same subject), test whether the mean *difference* is zero — a one-sample test on the differences, which cancels person-to-person variation.

Example 32.5 (Two-sample t-test)**.** A field gets $\bar{X} = 52$ bushels/acre with fertilizer ($n_X = 10$, $s_X = 4$); a control gets $\bar{Y} = 48$ ($n_Y = 10$, $s_Y = 5$). Does fertilizer help? With $H_0 : \mu_X = \mu_Y$, the standard error of the difference (assuming independent groups) is

$$\mathrm{SE}_{\mathrm{diff}} = \sqrt{\frac{s_X^2}{n_X} + \frac{s_Y^2}{n_Y}} = \sqrt{\frac{16}{10} + \frac{25}{10}} = \sqrt{4.1} = 2.025.$$

The statistic is $T = (52 - 48)/2.025 = 1.98$. With roughly 17 degrees of freedom (a standard approximation for unequal variances), the two-sided critical value is about 2.11. Since $1.98 < 2.11$, $p \approx 0.064$: we (just barely) fail to reject at $\alpha = 0.05$. The fertilizer *looks* helpful, but with only ten plots per group we cannot rule out chance. The accurate report is: "estimated gain 4 bushels, 95% CI roughly $(-0.3, 8.3)$, $p = 0.06$" — not "no effect."

Application

A/B testing: hypothesis testing at internet scale. Every major website runs hypothesis tests constantly, under the name *A/B testing*. To decide whether a new checkout button (variant B) beats the current one (variant A), you randomly assign incoming users to A or B and measure a conversion rate for each. This is exactly a two-sample test of proportions:

$$H_0 : \; p_A = p_B \quad \text{vs} \quad H_1 : \; p_A \neq p_B, \qquad Z = \frac{\hat{p}_B - \hat{p}_A}{\sqrt{\hat{p}(1-\hat{p})\left(\frac{1}{n_A} + \frac{1}{n_B}\right)}},$$

where $\hat{p}$ pools both groups under the null. *Randomization* is what makes it work: by assigning users at random, A/B testing turns observational guesswork into a genuine controlled experiment, so a significant difference can be *causally* attributed to the button, not to confounders. Companies run thousands of these simultaneously — which is exactly why the base-rate and multiple-comparisons traps below bite hard, and why disciplined teams fix α, compute required sample size *in advance* (power analysis, Example 32.4), and resist the urge to "peek."

Common trap

Peeking and the optional-stopping trap. If you watch an A/B test's p-value live and stop the moment it dips below 0.05, your true Type I error is far above 5% — sometimes 30% or more. The reason: the p-value wanders randomly over time, and a wandering quantity will eventually cross any fixed line by chance even if H_0 is true. You have effectively run many tests and reported only the lucky one. The fix is to fix the sample size in advance, or to use sequential methods (alpha-spending, sequential probability ratio tests) explicitly designed to permit valid peeking.

Application

Is model B really better than model A? When a machine-learning team reports that a new model beats the old one by 0.4 accuracy points on a benchmark, the very first question is the one this chapter is built around: *is that gap real, or test-set noise?* A held-out test set is itself a finite sample, so accuracy is an estimate with a standard error. For a test set of m examples and accuracy $\hat{a}$, the standard error is $\sqrt{\hat{a}(1-\hat{a})/m}$, and a paired test (the *McNemar test*, a chi-square on the examples where the two models *disagree*) checks whether the difference exceeds chance. With $m = 10{,}000$ the standard error of an accuracy near 0.9 is about $\sqrt{0.9 \cdot 0.1/10000} = 0.003$, so a 0.4-point gain (0.004) is barely one standard error — not significant. Leaderboards that crown winners by hundredths of a percent are usually reporting noise. The same $\sqrt{n}$ humility that tames pollsters tames benchmark hype.

Common trap

Multiple comparisons. Test 20 independent true nulls at $\alpha = 0.05$ and the chance that *at least one* comes up "significant" is $1 - (0.95)^{20} = 0.64$. Test enough things and you will find "significant" nonsense. The cartoon of the scientist who tests 20 jelly-bean colors for acne and triumphantly reports the one with $p < 0.05$ is not a joke about bad scientists — it is a mathematical inevitability. Corrections like Bonferroni (use α/m for m tests) or controlling the false discovery rate keep the error rate where you want it.

32.7 Putting it together: a complete worked study

Walking question

A clinic claims its weight-loss program produces an average loss of more than 5 kg over 12 weeks. You enroll $n = 16$ volunteers and record their losses; the sample mean is $\bar{X} = 6.2$ kg with sample standard deviation $s = 3.0$ kg. Does the data support the claim, and how big might the true effect be?

Step 1 — frame the hypotheses. The claim is "more than 5," so this is one-sided: $H_0 : \mu \leq 5$ versus $H_1 : \mu > 5$. We test at the boundary $\mu_0 = 5$.

Step 2 — choose the test. σ unknown, small sample, testing a mean: one-sample, one-sided **t-test** with $n - 1 = 15$ degrees of freedom.

Step 3 — compute the statistic.

$$\text{SE} = \frac{s}{\sqrt{n}} = \frac{3.0}{4} = 0.75, \qquad T = \frac{\bar{X} - \mu_0}{\text{SE}} = \frac{6.2 - 5}{0.75} = 1.60.$$

Step 4 — the p-value. For t_{15}, $\mathbb{P}(T \geq 1.60) \approx 0.065$ (one-sided).

Step 5 — decide and report. At $\alpha = 0.05$, since $0.065 > 0.05$, we *fail to reject* H_0: the data do not quite establish that the true mean loss exceeds 5 kg, though they lean that way. The 95% confidence interval for μ is

$$6.2 \pm t_{15,0.025}\,\text{SE} = 6.2 \pm 2.131 \times 0.75 = 6.2 \pm 1.60 = (4.6,\ 7.8).$$

The interval includes 5 — consistent with failing to reject — but also stretches to 7.8, so a clinically meaningful effect remains very plausible. The right conclusion is not "the program does not work" but "promising; the study was too small to confirm the specific 5-kg claim — run a larger trial." This is the disciplined, careful voice the whole chapter has been building toward.

32.8 Summary

Why this matters

Every tool in this chapter is a reading of one instrument: the sampling distribution, the predicted wobble of an estimate across hypothetical repetitions, made Normal by the CLT and scaled by the standard error $\sigma/\sqrt{n}$.

- A **confidence interval** inverts the wobble into a range; its "95%" is long-run *coverage of the procedure*, not a probability about your one interval.
- A **hypothesis test** asks whether the data are surprising under a null; the **p-value** measures that surprise — and is *not* $\mathbb{P}(H_0 \mid \text{data})$, not effect size, and not a sacred line at 0.05.
- Use **z** when σ is known, **t** when it is estimated (wider, heavier tails, the price of not knowing the spread).
- **Type I** (α, false alarm) and **Type II** (β, missed effect) trade off at fixed n; **power** $= 1 - \beta$ rises only with more data.
- Intervals and tests are **dual**; A/B testing is hypothesis testing with randomization; peeking, multiple comparisons, and the base-rate trap are where sound practice is won or lost.

The throughline is humility wearing the clothes of mathematics: never a number alone, always a number with its uncertainty attached.

32.9 Exercises

Exercise 32.1 (Margin and sample size)**.** A survey of $n = 400$ people estimates a proportion $\hat{p} = 0.35$. (a) Give the 95% confidence interval. (b) How large must n be to shrink the margin of

error to ± 0.0234? (c) Explain in one sentence why the answer to (b) is roughly four times (a)'s n when you halve the margin.

Exercise 32.2 (z versus t)**.** You have $n = 9$ observations with $\bar{X} = 20$ and $s = 6$. (a) Build the 95% t-interval (use $t_{8,0.025} = 2.306$). (b) Build the (incorrect) interval you would get by using 1.96. (c) By what percentage does using z understate the half-width, and why does that misrepresent your precision?

Exercise 32.3 (Coin test)**.** A coin gives 44 heads in 80 flips. (a) Compute the two-sided z-test statistic and p-value for $H_0 : p = 0.5$. (b) State your conclusion at $\alpha = 0.05$. (c) Give the 95% confidence interval for p and confirm it is consistent with your test via duality.

Exercise 32.4 (Interpreting the interval — conceptual)**.** A study reports a 95% confidence interval of $(2.1, 4.7)$ for a treatment effect. For each statement, say whether it is correct and why: (a) "There is a 95% probability the true effect is between 2.1 and 4.7." (b) "If we repeated the study many times, about 95% of the intervals would contain the true effect." (c) "We can reject H_0: effect $= 0$ at the 5% level." (d) "95% of treated subjects had an effect between 2.1 and 4.7."

Exercise 32.5 (The p-value is not what you think)**.** A test on a true null (H_0 holds) produces $p = 0.03$, and a colleague concludes "there is only a 3% chance the null is true." (a) Identify the precise error. (b) Explain what the 3% actually refers to. (c) Using the base-rate paradox, sketch why even a genuine $p = 0.03$ can correspond to a much higher probability that H_0 is true when true effects are rare.

Exercise 32.6 (Power)**.** A one-sided z-test at $\alpha = 0.05$ tests $H_0 : \mu = 0$ against $H_1 : \mu = \delta > 0$ with $\sigma = 10$ and $n = 25$. (a) Find the rejection cutoff for $\bar{X}$. (b) Compute the power when $\delta = 4$. (c) What n would raise the power to 0.9 for $\delta = 4$? (Use $\Phi(1.28) = 0.90$.)

Exercise 32.7 (Two-sample A/B test)**.** Variant A converts 180 of 3000 visitors; variant B converts 230 of 3000. (a) Compute $\hat{p}_A$, $\hat{p}_B$, and the pooled $\hat{p}$. (b) Compute the two-sided z-statistic and p-value for $H_0 : p_A = p_B$. (c) State the business conclusion, and name one trap (peeking, multiple comparisons, base rate) that could still make this result misleading.

Exercise 32.8 (Paired versus unpaired — conceptual)**.** Twelve runners are timed before and after a training regimen. Explain why a *paired* t-test on the within-runner differences is far more powerful than an unpaired two-sample test treating "before" and "after" as independent groups. What source of variation does pairing remove?

Exercise 32.9 (Duality, proved)**.** For the known-σ case, prove from the definitions that a value μ_0 lies in the $(1 - \alpha)$ confidence interval $\bar{X}_n \pm z_{\alpha/2}\,\sigma/\sqrt{n}$ if and only if the two-sided level-α z-test of $H_0 : \mu = \mu_0$ fails to reject. (This is the interval–test duality of Section 32.5.)

Exercise 32.10 (Why $n - 1$? — proof)**.** Let $X_1, \ldots, X_n$ be i.i.d. with variance σ^2. Show that $\mathbb{E}\big[\sum_{i=1}^{n}(X_i - \bar{X}_n)^2\big] = (n-1)\sigma^2$, and hence that dividing by $n-1$ (not n) makes s^2 unbiased. *Hint:* write $X_i - \bar{X}_n = (X_i - \mu) - (\bar{X}_n - \mu)$, expand the square, and use $\mathrm{Var}(\bar{X}_n) = \sigma^2/n$.

Exercise 32.11 (Significance versus importance)**.** A study of $n = 1{,}000{,}000$ users finds that a redesign increases average session length from 300.0 to 300.5 seconds, with $s = 120$ seconds, $p < 10^{-4}$. (a) Verify the result is "highly significant." (b) Compute the 95% confidence interval for the increase. (c) Argue, using the interval, why "statistically significant" here does not mean "worth shipping."

Chapter 33

Information Theory

There is a moment, early in learning probability, when you start to feel that a probability distribution is somehow *carrying* something. A fair coin and a coin that lands heads 99.9% of the time are both "random," yet they feel profoundly different. Watching the fair coin, you are genuinely in suspense every flip. Watching the lopsided coin, you can predict "heads" over and over and be right almost always; the outcomes barely tell you anything you did not already expect. One distribution is rich with uncertainty; the other is nearly empty of it.

Information theory is the branch of probability that takes this feeling and turns it into arithmetic. It gives us a single number—the *entropy*—that measures how much uncertainty a distribution holds, how surprised we should expect to be, and (astonishingly) how many bits we would need, on average, to record a draw from it. From that one idea grows a whole web of quantities: conditional entropy, mutual information, KL divergence, cross-entropy. These are not exotic curiosities. Cross-entropy is the loss function that trains essentially every modern classifier. KL divergence is the engine of variational inference. Mutual information is how we measure dependence when correlation is too crude. And, as we will see, maximizing likelihood—the workhorse of all of statistics—is secretly the very same thing as minimizing a cross-entropy.

This chapter was born in 1948, in a single paper by Claude Shannon, and it has the rare quality of being both deep and *usable on the first day*. Let us build it from the ground up, starting with a single question: how surprised should you be?

33.1 Surprise: measuring how much an outcome tells you

Imagine a friend texts you the result of an event. Some texts are boring; some are jolting. The text "the sun rose this morning" carries essentially no information—you already knew it. The text "it snowed in the Sahara today" is electrifying precisely because it was so unlikely. The lesson is immediate and intuitive:

Intuition

Information is the resolution of uncertainty. An outcome is *informative* exactly to the degree that it was *surprising*—that is, to the degree that it was *improbable*. Certain events tell you

nothing. Rare events tell you a lot.

So we want a function $S(p)$, the "surprise" of an event of probability p, that is large when p is small and small when p is large. But there are infinitely many decreasing functions. What pins down the *right* one? Shannon's move was to write down a few properties that any sensible measure of surprise *must* have, and then discover that those properties leave essentially no freedom.

Focus

We demand three things of surprise $S(p)$, where $p \in (0, 1]$ is the probability of an event:

1. **Certainty is unsurprising.** $S(1) = 0$. Being told something you already knew for sure carries no information.
2. **Rarer is more surprising.** S is decreasing in p: if $p < q$ then $S(p) > S(q)$.
3. **Independent surprises add.** If two *independent* events have probabilities p and q, learning that both happened should surprise you by the sum of the individual surprises: $S(pq) = S(p) + S(q)$.

The third axiom is the heart of it, and it deserves a sentence of defense. If I tell you two *unrelated* pieces of news—say, that a particular fair coin came up heads (probability $1/2$) and that a particular fair die came up six (probability $1/6$)—the total "newness" should be the information in the coin *plus* the information in the die. There is no double-counting and no discount, because the events have nothing to do with each other. Their joint probability is $1/2 \cdot 1/6 = 1/12$, so we are demanding $S(1/12) = S(1/2) + S(1/6)$. Additivity over multiplication of probabilities: that is a functional equation, and it has a famous solution.

Why it works

Why surprise must be a logarithm. We want $S(pq) = S(p) + S(q)$ for all $p, q \in (0, 1]$. Turn products into sums and sums stay sums—that is the defining property of the logarithm. Let us prove it is forced, not just convenient.
Substitute $p = e^{-x}$, $q = e^{-y}$ (so $x, y \geq 0$), and define $g(x) = S(e^{-x})$. The functional equation becomes

$$g(x+y) \;=\; S(e^{-(x+y)}) \;=\; S(e^{-x}e^{-y}) \;=\; S(e^{-x}) + S(e^{-y}) \;=\; g(x) + g(y).$$

This is *Cauchy's functional equation*, $g(x + y) = g(x) + g(y)$. Together with monotonicity (which we imposed in axiom 2), its only solutions are the linear ones, $g(x) = c\,x$ for a constant c. Unwinding, $S(p) = g(-\ln p) = -c \ln p$. Axiom 2 (decreasing) forces $c > 0$, and axiom 1 ($S(1) = 0$) is then automatic since $\ln 1 = 0$. So

$$\boxed{S(p) = -c \ln p = c \ln \frac{1}{p}}$$

is the *unique* measure of surprise, up to the positive scale constant c.

The constant c is just a choice of *units*, exactly like choosing meters versus feet. Two choices dominate practice:

- Take the logarithm in base 2: $S(p) = \log_2(1/p)$. The unit is the **bit**. A fair coin ($p = 1/2$) carries $\log_2 2 = 1$ bit of surprise—the canonical "one yes/no question" worth of information. This is the data-science default, and we use it throughout.
- Take the natural logarithm: $S(p) = \ln(1/p)$. The unit is the **nat**. This is common in statistics and machine learning, because ln plays nicely with calculus and with the likelihoods we will meet later.

To convert, recall $\log_2 x = \ln x / \ln 2$, so one nat $= 1/\ln 2 \approx 1.4427$ bits. We write log without a base when the base does not matter, and pin it down when it does.

Example 33.1 (How surprised is a die?)**.** Roll a fair six-sided die. Each face has probability $1/6$, so seeing any particular face carries

$$S(1/6) = \log_2 6 \approx 2.585 \text{ bits.}$$

Seeing a "6 then a 6" on two independent rolls carries $\log_2 36 = 2\log_2 6 \approx 5.17$ bits, exactly twice as much—additivity in action. Now load the die so that "six" has probability $1/2$. Seeing a six now carries only $\log_2 2 = 1$ bit (it is half-expected), while seeing the rare "one," say with probability $1/10$, carries $\log_2 10 \approx 3.32$ bits. Surprising outcomes shout; expected ones whisper.

33.2 Entropy: the average surprise of a distribution

Surprise is a property of a single *outcome*. But we usually want to characterize the whole *distribution*: how uncertain is this random variable, on average, before we see it? The natural answer is to average the surprise over all outcomes, weighting each by how often it occurs. That average is the **entropy**.

Definition 33.1 (Shannon entropy)**.** Let X be a discrete random variable with pmf $p(x) = \mathbb{P}(X = x)$ over a finite or countable alphabet $\mathcal{X}$. The *entropy* of X is the expected surprise,

$$H(X) \;=\; \mathbb{E}[\,S(p(X))\,] \;=\; \sum_{x\in\mathcal{X}} p(x)\,\log\frac{1}{p(x)} \;=\; -\sum_{x\in\mathcal{X}} p(x)\,\log p(x).$$

We adopt the convention $0\log 0 = 0$ (justified because $p\log p \to 0$ as $p \to 0^+$), so zero-probability outcomes contribute nothing. With $\log = \log_2$ the units are bits; with ln, nats.

A crucial point of language: entropy depends only on the *probabilities*, not on the labels of the outcomes. The faces of a die could be marked with colors, animals, or prime numbers; relabeling does not change how uncertain the die is. Entropy is a functional of the probability vector $(p_1, \ldots, p_n)$, and we sometimes write $H(p_1, \ldots, p_n)$ to emphasize this.

What is really happening

Entropy is an expectation in disguise. Notice the formula is literally $\mathbb{E}[\log(1/p(X))]$—the expectation of the random variable "surprise of whatever I drew." Everything you know about expectations applies. This is why entropy behaves so much like a center of mass of surprise. When one outcome dominates, almost all the weight sits on a small surprise, and the average is tiny. When the weight is spread out, every outcome is moderately surprising, and the average is large. The whole subject is, at bottom, expectations of logarithms of

probabilities.

33.2.1 The binary entropy function

The simplest nontrivial case is a single biased coin: $X = 1$ with probability p, $X = 0$ with probability $1 - p$. Its entropy is so important it gets its own name and symbol:

$$H_b(p) \;=\; -p \log_2 p - (1-p) \log_2 (1-p).$$

Look at the shape of this function (Figure 33.1). It is zero at $p = 0$ and $p = 1$—a coin that always lands the same way has no uncertainty, and indeed carries zero information per flip. It rises to a single maximum of exactly 1 bit at $p = 1/2$, the fair coin, which is the most uncertain a binary outcome can be. And it is symmetric: a coin biased 90/10 is exactly as uncertain as one biased 10/90, because uncertainty does not care which side we call "heads."

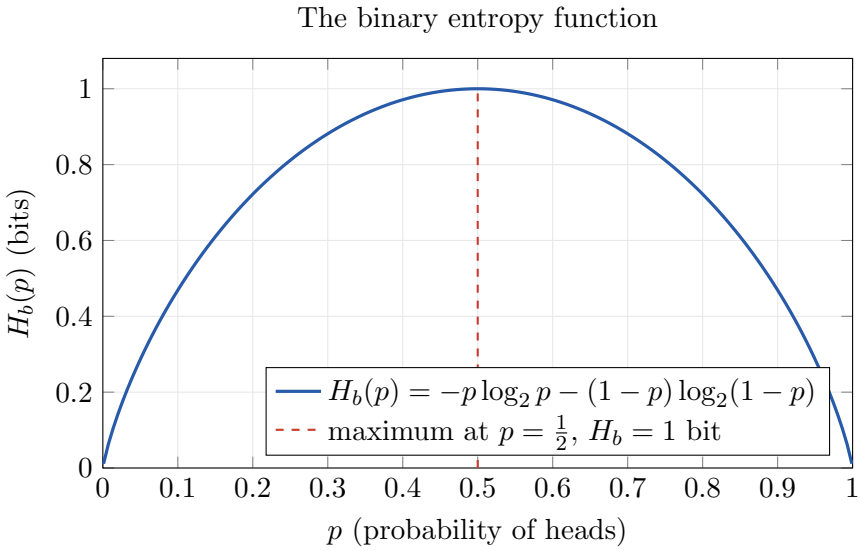


Figure 33.1: Binary entropy. A fair coin ($p = \frac{1}{2}$) is maximally uncertain at 1 bit; near-deterministic coins ($p \to 0$ or $p \to 1$) carry almost no uncertainty. The curve is steep near the ends—moving from $p = 0.5$ to $p = 0.6$ barely changes entropy, but moving from $p = 0.99$ to $p = 0.999$ matters a great deal.

Example 33.2 (Entropy of a fair vs. loaded die)**.** A fair six-sided die has $p(x) = 1/6$ for each face, so

$$H = \sum_{x=1}^{6} \tfrac{1}{6} \log_2 6 = \log_2 6 \approx 2.585 \text{ bits}.$$

Now load it: $p = (0.5, 0.1, 0.1, 0.1, 0.1, 0.1)$. Then

$$H = -0.5 \log_2 0.5 - 5\,(0.1 \log_2 0.1) = 0.5 + 5(0.1 \cdot 3.3219) = 0.5 + 1.661 = 2.161 \text{ bits}.$$

The loaded die is *less* uncertain than the fair one, as it should be—you can guess "six" and be right half the time. Concentrating probability always lowers entropy.

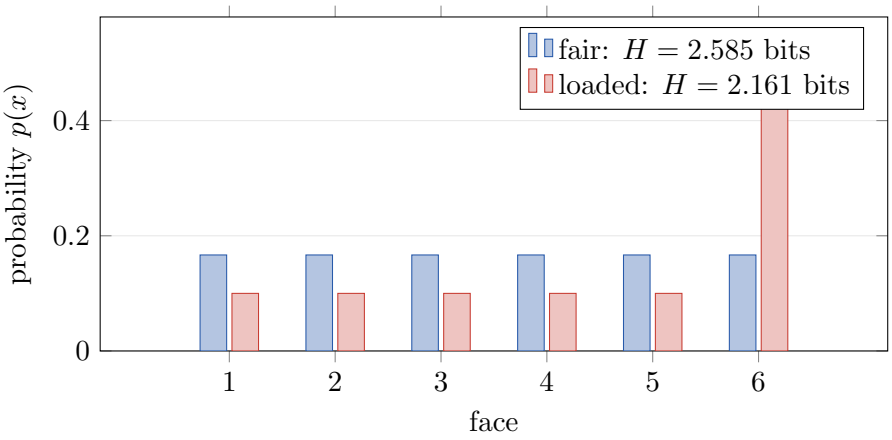


Figure 33.2: Probability mass functions of the fair die (flat, uniform) and the loaded die from Example 33.2, which dumps half its mass on the face "6." Concentrating mass on one outcome makes the die more predictable and lowers its entropy from 2.585 to 2.161 bits.

33.2.2 The bounds: why uniform is the most uncertain

Two facts about entropy are so basic that they anchor all intuition. First, entropy is never negative. Second, on an alphabet of n symbols it never exceeds $\log n$, with equality exactly for the uniform distribution.

Theorem 33.1 (Bounds on entropy)**.** *For a discrete random variable X on an alphabet of size n,*

$$0 \;\le\; H(X) \;\le\; \log n,$$

where the lower bound holds with equality iff X is deterministic (some outcome has probability 1), and the upper bound holds with equality iff X is uniform ($p(x) = 1/n$ for all x).

Why it works

Why uniform maximizes entropy. The lower bound is easy: each term $p(x)\log(1/p(x)) \ge 0$ since $0 \le p(x) \le 1$, so the sum is nonnegative, and it vanishes exactly when every term does, i.e. when some $p(x) = 1$.

For the upper bound, the slick proof uses *Jensen's inequality* for the concave function log. Write the entropy as an expectation and pull the log outside:

$$H(X) = \sum_x p(x)\log\frac{1}{p(x)} = \mathbb{E}\left[\log\frac{1}{p(X)}\right] \;\le\; \log\mathbb{E}\left[\frac{1}{p(X)}\right] = \log\left(\sum_x p(x)\cdot\frac{1}{p(x)}\right)$$
$$= \log\left(\sum_{x:\,p(x)>0} 1\right) \le \log n.$$

The single inequality is Jensen's, valid because log is concave; equality in Jensen requires $1/p(X)$ to be constant, i.e. all the $p(x)$ equal, i.e. uniform. The final $\le \log n$ counts the support. So entropy is maximized by spreading probability as evenly as possible—uncertainty is greatest when no outcome is favored.

Why this matters

This is the "maximum entropy principle" in its purest form: among all distributions on n symbols, the one that assumes the least—commits to nothing beyond "these n things can happen"—is the uniform one, and it has the most entropy. Maximizing entropy subject to constraints is a deep modeling principle: it produces the distribution that is maximally noncommittal given what you actually know. With a fixed mean it yields the exponential; with a fixed mean and variance it yields the Gaussian. The bell curve is, in this sense, the least presumptuous distribution with a given spread.

History

Shannon, 1948. Claude Shannon was a research mathematician at Bell Labs when he published *A Mathematical Theory of Communication* in the *Bell System Technical Journal*. In one stroke it created the field. The story goes that Shannon was unsure what to call his $-\sum p \log p$ quantity. John von Neumann reportedly advised him to call it *entropy*—both because the identical formula appears in statistical mechanics (where Boltzmann and Gibbs had written $S = -k \sum p \log p$ decades earlier), and, von Neumann supposedly added with a wink, because "nobody knows what entropy really is, so in a debate you will always have the advantage." The pun is real: thermodynamic entropy and Shannon entropy are the *same* mathematical object, measuring missing information about microscopic states.

33.3 Source coding: entropy is the cost of description

So far entropy is "average surprise," which is poetic but abstract. Shannon's first great theorem makes it concrete and a little miraculous: *entropy is the minimum average number of bits needed to encode draws from a distribution.* It is not a metaphor that entropy is measured in bits. It is the literal, unbeatable cost of writing the outcomes down.

Intuition

Suppose you must transmit a long sequence of draws from X over a channel that only speaks 0s and 1s, using as few symbols as possible. The trick is to give *short codewords to frequent outcomes and long codewords to rare ones*—exactly what Morse code does ("E" is a single dot; "Q" is dash-dash-dot-dash). The surprise $\log_2(1/p(x))$ turns out to be the ideal length, in bits, for the codeword of an outcome that occurs with probability $p(x)$. Averaging those ideal lengths gives the entropy.

Theorem 33.2 (Shannon's source coding theorem, informally)**.** *To encode i.i.d. draws from X with a uniquely decodable binary code, the average codeword length L must satisfy $L \geq H(X)$ (in bits). Conversely, you can always build a code with $L < H(X) + 1$, and by coding long blocks of draws together you can push the average bits-per-symbol arbitrarily close to $H(X)$. The entropy is the fundamental limit of lossless compression.*

Walking question

Why can't a code beat the assignment "length $\approx \log_2(1/p(x))$"? Because of a budget. In any prefix-free binary code (no codeword is the start of another, so decoding is unambiguous), the codeword lengths $\ell(x)$ must satisfy *Kraft's inequality* $\sum_x 2^{-\ell(x)} \le 1$—there is only so much "room" in the tree of binary strings. Minimizing the average length $\sum_x p(x)\ell(x)$ subject to that budget (treat the ℓ as continuous and use Lagrange multipliers) gives the optimum $\ell(x) = \log_2(1/p(x))$, and the minimized average is exactly $\sum_x p(x)\log_2(1/p(x)) = H(X)$. The integer constraint on real codeword lengths costs at most 1 extra bit, which is the "+1" in the theorem, and it washes out when you code blocks.

Example 33.3 (A code that hits the bound)**.** Let X take four values with probabilities $\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{8}$. The entropy is

$$H(X) = \tfrac{1}{2}\cdot 1 + \tfrac{1}{4}\cdot 2 + \tfrac{1}{8}\cdot 3 + \tfrac{1}{8}\cdot 3 = 0.5 + 0.5 + 0.375 + 0.375 = 1.75 \text{ bits}.$$

Now assign codewords by surprise: $a \mapsto \texttt{0}$ (length $1 = \log_2 2$), $b \mapsto \texttt{10}$ (length 2), $c \mapsto \texttt{110}$, $d \mapsto \texttt{111}$ (length 3 each). This is prefix-free, and its average length is $\frac{1}{2}(1) + \frac{1}{4}(2) + \frac{1}{8}(3) + \frac{1}{8}(3) = 1.75$ bits—it *exactly* matches the entropy, because every probability is a power of $1/2$ so the ideal lengths are integers. A naive fixed-length code would spend 2 bits per symbol ($\log_2 4$). Matching code lengths to surprise saves 0.25 bits per symbol, forever.

Connection

The Lagrange-multiplier optimization in the walking question is the same machinery you met in multivariable calculus: minimize $\sum_x p(x)\ell(x)$ on the constraint surface $\sum_x 2^{-\ell(x)} = 1$. The gradient condition $\nabla(\text{objective}) = \lambda\,\nabla(\text{constraint})$ delivers $\ell(x) \propto \log(1/p(x))$ immediately. See the companion calculus volume's chapter on constrained optimization; the "shadow price" λ here is literally the conversion between probability and bits.

33.4 Joint and conditional entropy

Real problems involve several random variables at once, and the questions become relational: how much total uncertainty is in the pair (X, Y)? Once I learn Y, how much uncertainty about X remains? These are answered by joint and conditional entropy, and they obey a beautifully simple bookkeeping.

Definition 33.2 (Joint and conditional entropy)**.** For discrete X, Y with joint pmf $p(x, y)$, the *joint entropy* is

$$H(X, Y) = -\sum_{x,y} p(x, y)\log p(x, y),$$

simply the entropy of the pair viewed as one combined variable. The *conditional entropy* of X given Y is the average, over y, of the remaining uncertainty about X:

$$H(X \mid Y) = \sum_y p(y)\, H(X \mid Y = y) = -\sum_{x,y} p(x, y)\log p(x \mid y).$$

The conditional entropy $H(X \mid Y)$ answers exactly the question "after I am told Y, how surprised will I still be by X, on average?" Note the averaging: for a *particular* value $Y = y$ the residual uncertainty $H(X \mid Y = y)$ could go up or down, but *averaged* over Y, conditioning can only help—more on that shortly.

Theorem 33.3 (Chain rule for entropy)**.**

$$H(X, Y) = H(X) + H(Y \mid X) = H(Y) + H(X \mid Y).$$

Why it works

Why the chain rule is just log **of the multiplication rule.** The whole identity falls out of $p(x, y) = p(x)\, p(y \mid x)$, the multiplication rule for probabilities, after taking $-\log$ and averaging. Watch:

$$H(X, Y) = -\sum_{x,y} p(x, y) \log p(x, y) = -\sum_{x,y} p(x, y) \big[\log p(x) + \log p(y \mid x) \big].$$

Split the sum. The first piece is $-\sum_{x,y} p(x, y) \log p(x) = -\sum_x p(x) \log p(x) = H(X)$ (we summed out y). The second piece is exactly $-\sum_{x,y} p(x, y) \log p(y \mid x) = H(Y \mid X)$. So $H(X, Y) = H(X) + H(Y \mid X)$. The symmetric version comes from factoring $p(x, y) = p(y)p(x \mid y)$ instead.

What is really happening

The chain rule says uncertainty is additive in the right currency. The total surprise of the pair equals the surprise of X *plus* whatever surprise about Y is left after X is known. It mirrors exactly how you would describe the pair to someone: first tell them X (costs $H(X)$ bits), then tell them Y given that they now know X (costs $H(Y \mid X)$ bits). Information bookkeeping is just careful double-entry accounting of surprise.

Theorem 33.4 (Conditioning reduces entropy)**.** $H(X \mid Y) \le H(X)$*, with equality iff* $X \perp\!\!\!\perp Y$*.*

This is the formal version of "information never hurts." On average, learning Y cannot increase your uncertainty about X. We will prove it in a moment as a one-line corollary of the nonnegativity of mutual information. First we need the tool that makes everything click: KL divergence.

33.5 Relative entropy (KL divergence): the cost of being wrong

Here is a question that turns out to organize the entire subject. Suppose the data really come from a distribution p, but you, mistakenly, believe they come from q. You design your code, your bets, your model around q. How much do you pay for the discrepancy? The answer is the **Kullback–Leibler divergence**, also called *relative entropy*.

Definition 33.3 (KL divergence)**.** For two pmfs p, q on the same alphabet $\mathcal{X}$, the KL divergence *from* q *to* p (the extra cost of using q when the truth is p) is

$$\mathrm{KL}(p \,\|\, q) \;=\; \sum_x p(x) \, \log \frac{p(x)}{q(x)} \;=\; \mathbb{E}_{X \sim p}\left[\log \frac{p(X)}{q(X)} \right].$$

We require $q(x) = 0 \implies p(x) = 0$ (else the divergence is $+\infty$): you cannot afford a model that assigns zero probability to something that actually happens.

Intuition

Read it as a difference of surprises. Under your wrong model q, an outcome x surprises you by $\log(1/q(x))$ bits. Under the truth p, it "should" surprise you by $\log(1/p(x))$. The per-outcome overpayment is $\log(1/q(x)) - \log(1/p(x)) = \log(p(x)/q(x))$, and KL is the average overpayment, weighted by how often each outcome *really* occurs (weighted by p, not q, because reality runs on p).

The single most important fact about KL divergence is that it is never negative, and is zero only when your model is exactly right. This is *Gibbs' inequality*, and it is the load-bearing wall of the whole theory.

Theorem 33.5 (Gibbs' inequality / nonnegativity of KL)**.** $\mathrm{KL}(p \,\|\, q) \geq 0$, *with equality iff* $p = q$ *everywhere.*

Why it works

Why KL is nonnegative. Again Jensen, applied to the concave log. Write the *negative* divergence and pull the log out:

$$-\mathrm{KL}(p \,\|\, q) = \sum_x p(x) \log \frac{q(x)}{p(x)} = \mathbb{E}_{X\sim p}\left[\log \frac{q(X)}{p(X)}\right] \;\leq\; \log \mathbb{E}_{X\sim p}\left[\frac{q(X)}{p(X)}\right] = \log \sum_x q(x) = \log 1 = 0.$$

So $-\mathrm{KL} \leq 0$, i.e. $\mathrm{KL} \geq 0$. Equality in Jensen for the strictly concave log requires the ratio $q(X)/p(X)$ to be constant; since both are distributions summing to 1, that constant must be 1, forcing $p = q$. A perfectly correct model pays nothing; any error costs strictly positive divergence.

Common trap

KL is not a distance. It is tempting to read $\mathrm{KL}(p \,\|\, q)$ as "how far apart p and q are," but it is *not symmetric*: $\mathrm{KL}(p \,\|\, q) \neq \mathrm{KL}(q \,\|\, p)$ in general, and it violates the triangle inequality. The asymmetry is meaningful, not a defect. $\mathrm{KL}(p \,\|\, q)$ weights errors by the *true* p, so it punishes you for assigning low q to outcomes that truly occur, but barely notices if you waste q-mass on outcomes that never happen. Swapping the arguments swaps which mistake is expensive. In variational inference, choosing to minimize $\mathrm{KL}(q \,\|\, p)$ versus $\mathrm{KL}(p \,\|\, q)$ gives qualitatively different approximations ("mode-seeking" versus "mean-seeking"). Always mind the order.

Example 33.4 (Two coins)**.** Let the truth be a fair coin $p = (0.5, 0.5)$ and your model a biased guess $q = (0.9, 0.1)$. Then

$$\mathrm{KL}(p \,\|\, q) = 0.5 \log_2 \frac{0.5}{0.9} + 0.5 \log_2 \frac{0.5}{0.1} = 0.5(-0.848) + 0.5(2.322) = 0.737 \text{ bits.}$$

You overpay about 0.74 bits per flip for believing the coin is biased when it is fair. Reverse the roles—truth $q = (0.9, 0.1)$, model $p = (0.5, 0.5)$:

$$\mathrm{KL}(q \,\|\, p) = 0.9 \log_2 \frac{0.9}{0.5} + 0.1 \log_2 \frac{0.1}{0.5} = 0.9(0.848) + 0.1(-2.322) = 0.531 \text{ bits.}$$

Different number—concrete proof of asymmetry.

33.6 Cross-entropy: the loss that trains your models

KL divergence has a twin that is even more famous in practice, because it is the literal loss function minimized by nearly every classifier you have ever heard of. It is the **cross-entropy**.

Definition 33.4 (Cross-entropy)**.** The cross-entropy of the model q relative to the truth p is the average surprise you experience when reality is p but you reason with q:

$$H(p, q) \;=\; -\sum_{x} p(x) \log q(x) \;=\; \mathbb{E}_{X\sim p}\left[\log \frac{1}{q(X)}\right].$$

The three quantities entropy, cross-entropy, and KL are bound by one short, illuminating identity. Split $\log(p/q) = \log(1/q) - \log(1/p)$ inside the KL sum:

$$\mathrm{KL}(p \,\|\, q) = \underbrace{-\sum_{x} p(x) \log q(x)}_{H(p,q)} \;-\; \underbrace{\Big(-\sum_{x} p(x) \log p(x)\Big)}_{H(p)},$$

so

$$\boxed{H(p, q) = H(p) + \mathrm{KL}(p \,\|\, q)}.$$

Why this matters

Read this identity aloud and you understand all of supervised learning. The cost of coding reality with your model equals the *irreducible* cost (the true entropy $H(p)$, which no model can avoid) plus the *avoidable* cost (the divergence $\mathrm{KL}(p \,\|\, q)$, which your error inflicts). Since $H(p)$ does not depend on your model q, *minimizing cross-entropy over q is exactly the same as minimizing KL divergence over q.* Training a model to minimize cross-entropy is training it to make q match the true p as closely as KL can measure. And because $\mathrm{KL} \geq 0$, cross-entropy is bounded below by $H(p)$, with the floor reached only by a perfect model.

Application

Why cross-entropy is *the* classification loss. In multiclass classification, each training example i has a true label, which we write as a one-hot vector $p^{(i)}$ (all mass on the correct class y_i). The network outputs predicted class probabilities $q^{(i)}$ (the softmax layer). The cross-entropy loss on example i is

$$H(p^{(i)}, q^{(i)}) = -\sum_{c} p_c^{(i)} \log q_c^{(i)} = -\log q_{y_i}^{(i)},$$

because the one-hot $p^{(i)}$ kills every term except the true class. So the loss is simply the surprise the model assigns to the correct answer: $-\log$(predicted probability of the right class). Confident and correct $\Rightarrow$ near 0 loss; confident and wrong $\Rightarrow$ enormous loss. Averaged over the dataset, minimizing this is minimizing the empirical cross-entropy between the data distribution and the model—and, by the identity above and the next section, it is *exactly maximum likelihood*. Every time you call `CrossEntropyLoss` or `categorical_crossentropy`, this is the math running underneath.

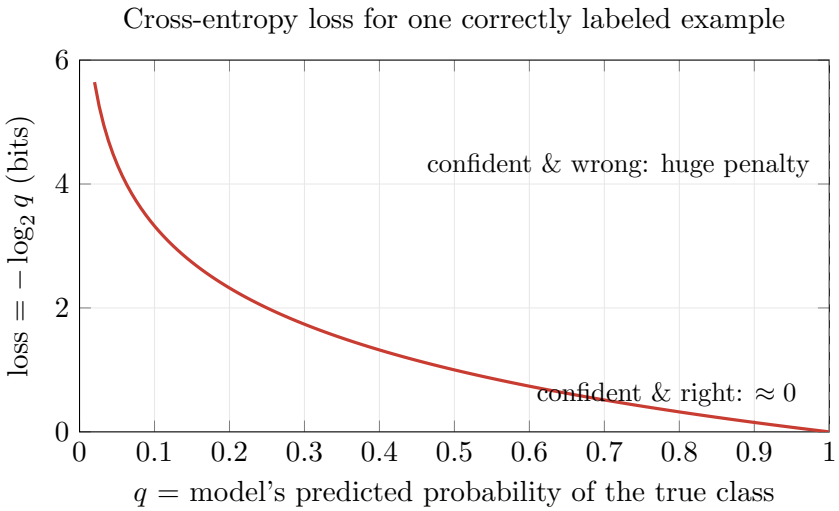


Figure 33.3: The per-example cross-entropy loss $-\log q$ as a function of the probability q the model assigned to the correct class. The penalty is gentle near $q = 1$ but explodes as $q \to 0$: a model that is confidently wrong is punished savagely, which is exactly the gradient signal that drives learning.

33.7 Maximum likelihood is minimum cross-entropy

We promised that the workhorse of statistics is secretly an information-theoretic quantity. Here is the payoff. Suppose you observe i.i.d. data $x_1, \dots, x_N$ and fit a model q_θ by maximum likelihood. The log-likelihood, scaled by $1/N$, is

$$\frac{1}{N}\sum_{i=1}^{N} \log q_\theta(x_i) \;=\; \sum_{x} \hat{p}(x)\, \log q_\theta(x),$$

where $\hat{p}(x) = \frac{1}{N}\#\{i : x_i = x\}$ is the *empirical distribution* of the data. The right-hand side is precisely $-H(\hat{p}, q_\theta)$, the negative cross-entropy between the data distribution and the model. Therefore

$$\underbrace{\min_\theta \Big(-\frac{1}{N}\sum_i \log q_\theta(x_i) \Big)}_{\text{negative log-likelihood}} \;=\; \underbrace{\min_\theta \; H(\hat{p}, q_\theta)}_{\text{minimum cross-entropy}} \;=\; \underbrace{\min_\theta \; \mathrm{KL}(\hat{p} \,\|\, q_\theta)}_{\text{best fit in KL}} + \text{const.}$$

What is really happening

What maximizing likelihood is really doing. You are choosing the model q_θ that is *closest in KL divergence* to the empirical distribution of your data. As the sample grows, $\hat{p} \to p$ (the true distribution) by the law of large numbers, so maximum likelihood drives your model toward the q_θ that minimizes $\mathrm{KL}(p \,\|\, q_\theta)$—the best possible approximation to reality within your model family. This is exactly why MLE is consistent: if the truth is *in* your family, the KL-closest model is the truth itself, and you find it. Likelihood, cross-entropy, and KL are three views of one optimization.

Connection

This recasts the estimation chapter (Maximum Likelihood) in information-theoretic clothing. There you saw MLE as "the parameter that makes the data most probable." Here you see it as "the parameter whose model distribution is informationally closest to the data." Both are correct; the second explains *why* MLE generalizes, why cross-entropy is the right loss, and why a misspecified model still converges to its KL-projection onto the truth. The Bayesian-inference chapter's variational methods minimize a KL divergence directly, making the link explicit.

33.8 Mutual information: how much Y tells you about X

Now for the quantity that ties the chapter together. Correlation measures linear co-movement and misses everything nonlinear. *Mutual information* measures *all* statistical dependence—it is zero exactly when two variables are independent, no matter how twisted their relationship.

Definition 33.5 (Mutual information)**.** The mutual information between X and Y is the KL divergence between their joint distribution and the product of their marginals:

$$I(X;Y) \;=\; \mathrm{KL}\left(p(x,y)\,\middle\|\,p(x)p(y)\right) \;=\; \sum_{x,y} p(x,y)\,\log\frac{p(x,y)}{p(x)\,p(y)}.$$

The definition rewards a second look. Independence *means* $p(x,y) = p(x)p(y)$. So mutual information literally measures how far the joint distribution is from the "independent" distribution it would be if X and Y ignored each other. By Gibbs' inequality, $I(X;Y) \geq 0$, with equality iff $p(x,y) = p(x)p(y)$, i.e. iff $X \perp\!\!\!\perp Y$. *Mutual information is zero exactly when the variables are independent*—a clean "yes/no" test for dependence of any kind.

Theorem 33.6 (Mutual information in terms of entropy)**.**

$$I(X;Y) = H(X) - H(X \mid Y) = H(Y) - H(Y \mid X) = H(X) + H(Y) - H(X,Y).$$

Why it works

Why these forms agree. Start from the definition and split the log using $p(x,y) = p(x \mid y)p(y)$:

$$I(X;Y) = \sum_{x,y} p(x,y)\log\frac{p(x \mid y)}{p(x)} = \sum_{x,y} p(x,y)\log p(x \mid y) - \sum_{x,y} p(x,y)\log p(x).$$

The first sum is $-H(X \mid Y)$; the second is $-H(X)$ (sum out y). Hence $I(X;Y) = H(X) - H(X \mid Y)$. By symmetry of the definition in x and y, also $I(X;Y) = H(Y) - H(Y \mid X)$. Finally, substitute the chain rule $H(X \mid Y) = H(X,Y) - H(Y)$ into the first form to get $I(X;Y) = H(X) + H(Y) - H(X,Y)$.

The middle form $I(X;Y) = H(X) - H(X \mid Y)$ is the one to memorize as a sentence: *mutual information is the reduction in your uncertainty about X once you learn Y.* It is the average number of bits that knowing Y saves you when you have to describe X. And since $I(X;Y) \geq 0$, this

rearranges instantly to $H(X \mid Y) \le H(X)$—the promised proof that *conditioning reduces entropy*. Information never hurts, on average.

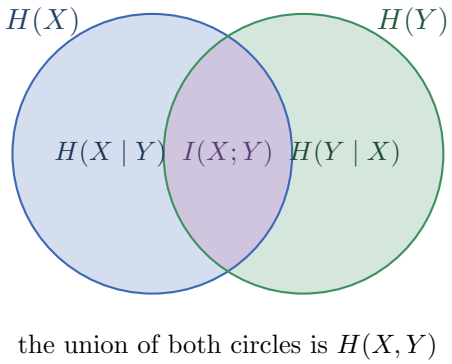


Figure 33.4: The information "Venn diagram." Each circle is the entropy of one variable. Their overlap is the mutual information $I(X;Y)$, the shared uncertainty resolved by either variable. The crescents are the conditional entropies—what remains uncertain about one variable after the other is known. The total area of the union is the joint entropy $H(X,Y) = H(X) + H(Y) - I(X;Y)$.

Example 33.5 (A noisy channel)**.** Send a fair bit $X \in \{0,1\}$ through a channel that flips it with probability $f = 0.1$ (the "binary symmetric channel," sketched in Figure 33.5). The receiver sees Y. By symmetry Y is also a fair bit, so $H(Y) = 1$. Given the input, the only uncertainty in the output is the flip, so $H(Y \mid X) = H_b(0.1) = -0.1 \log_2 0.1 - 0.9 \log_2 0.9 \approx 0.469$ bits. Therefore

$$I(X;Y) = H(Y) - H(Y \mid X) \approx 1 - 0.469 = 0.531 \text{ bits.}$$

Each use of this noisy channel conveys about half a bit of genuine information about the input. A noiseless channel ($f = 0$) would convey the full bit; a useless channel that flips with probability 0.5 would convey $1 - H_b(0.5) = 0$ bits—pure noise tells you nothing. This number, maximized over input distributions, is the channel's *capacity*, the ceiling on reliable communication rate that Shannon's second great theorem says you can actually approach.

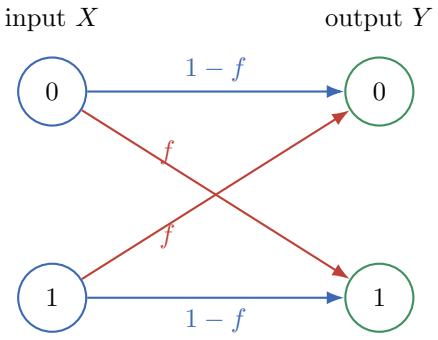


Figure 33.5: The binary symmetric channel. Each transmitted bit arrives unchanged with probability $1 - f$ and flipped with probability f. With $f = 0.1$ and a fair input, each use carries $I(X;Y) = 1 - H_b(0.1) \approx 0.531$ bits; at $f = 0.5$ the two red and blue arrows are equally likely, the output is independent of the input, and the channel conveys nothing.

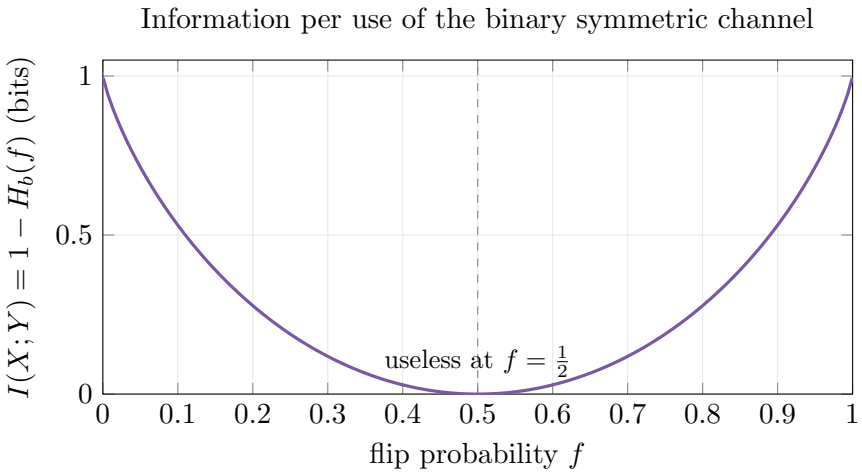


Figure 33.6: Mutual information of the binary symmetric channel as the flip probability f varies. It is a full bit at the noiseless ends $f = 0$ and $f = 1$ (a guaranteed flip is just as informative as no flip, since the receiver can invert it), and drops to 0 exactly at $f = \frac{1}{2}$, where output and input are independent.

Application

Mutual information in data science. Because $I(X;Y) = 0$ iff $X \perp\!\!\!\perp Y$ and it captures nonlinear dependence, mutual information is a go-to tool for *feature selection*: rank candidate features by how much information they carry about the target, keeping high-I features even when their *correlation* with the target is zero (think of $Y = X^2$ with symmetric X: correlation 0, mutual information large). Decision trees split on the feature with the highest *information gain*, which is exactly a conditional mutual information. And in modern representation learning, objectives like InfoNCE *maximize* a lower bound on the mutual information between an input and its learned embedding, forcing the representation to retain what matters.

Paradox

Can knowing more ever make you more uncertain? We proved $H(X \mid Y) \leq H(X)$: conditioning reduces entropy. Yet everyday experience says "the more I learned, the more confused I got!" Both are true, and the resolution is the word *average*. The theorem bounds the *expected* residual uncertainty $H(X \mid Y) = \sum_y p(y) H(X \mid Y = y)$. For a *particular* observed value $Y = y$, the specific conditional entropy $H(X \mid Y = y)$ can absolutely *exceed* $H(X)$. Example: you are fairly sure a die is fair (low uncertainty about which loading it has), then you observe three sixes in a row (an unusual y); now you genuinely do not know whether it is fair or loaded—your uncertainty has gone *up*. The theorem guarantees only that across all possible observations, weighted by how likely they are, the surprising-and-confusing y's are outnumbered by the clarifying ones. Information never hurts *on average*; any single message certainly can.

33.9 Continuous variables: differential entropy in one breath

Everything above was discrete. For a continuous random variable with density f, the natural analogue is the *differential entropy*

$$h(X) = -\int_{-\infty}^{\infty} f(x)\log f(x)\,\mathrm{d}x = \mathbb{E}[-\log f(X)],$$

replacing the sum by an integral. It is genuinely useful—among all densities with a fixed variance σ^2, the Gaussian uniquely maximizes differential entropy, at $h = \frac{1}{2}\log(2\pi e\,\sigma^2)$, which is the maximum-entropy justification for the bell curve. But beware: differential entropy is not a clean limit of discrete entropy (finely discretizing a continuous variable gives entropy that blows up like $\log(1/\Delta)$), and it can be negative or change under a change of variables. The quantities that *do* behave well in the continuous world are the *relative* ones—KL divergence and mutual information—because the troublesome infinities cancel between the two distributions being compared. That is one more reason KL and mutual information, not entropy itself, are the durable currency of modern machine learning.

Connection

The differential entropy is an ordinary Riemann/Lebesgue integral $\mathbb{E}[-\log f(X)]$; see the companion calculus volume for integration of densities, and the measure-theory chapters here for the careful statement that KL divergence is well defined whenever p is absolutely continuous with respect to q, with the integrand $\log(\mathrm{d}p/\mathrm{d}q)$ being the log of the Radon–Nikodym derivative.

33.10 Bringing it together

One idea generated this entire chapter: *surprise is* $\log(1/p)$*, and information theory is the study of its averages.*

- **Entropy** $H(X) = \mathbb{E}[\log(1/p(X))]$ is average surprise—uncertainty, and the unbeatable cost in bits of describing X. It is maximized by the uniform distribution.
- **Conditional entropy** $H(X \mid Y)$ is leftover uncertainty after seeing Y; the **chain rule** $H(X,Y) = H(X) + H(Y \mid X)$ is log of the multiplication rule.
- **KL divergence** $\mathrm{KL}(p\|q) = \mathbb{E}_p[\log(p/q)] \geq 0$ is the cost of modeling truth p with wrong model q; it is zero iff $p = q$ (Gibbs).
- **Cross-entropy** $H(p,q) = H(p) + \mathrm{KL}(p\|q)$ is the ML loss; minimizing it is minimizing KL is maximizing likelihood.
- **Mutual information** $I(X;Y) = \mathrm{KL}(p_{XY}\|p_X p_Y) = H(X) - H(X \mid Y) \geq 0$ measures all dependence and is zero iff $X \perp\!\!\!\perp Y$.

Every modern loss function, dependence measure, and compression scheme you will meet is one of these five quantities wearing a costume. Learn to recognize $\log(1/p)$ underneath, and information theory stops being a list of formulas and becomes a single, simple story about surprise.

33.11 Exercises

Exercise 33.1 (Warm-up entropies)**.** Compute the entropy in bits of: (a) a fair coin; (b) a fair 8-sided die; (c) a coin with $p = 0.25$; (d) the distribution $(\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{16}, \frac{1}{16})$. For (d), compare your answer to $\log_2 5$ and explain why it is smaller.

Exercise 33.2 (Surprise adds)**.** You draw a card from a shuffled standard deck. Express in bits the surprise of the events "it is a spade," "it is an ace," and "it is the ace of spades." Verify numerically that the third surprise equals the sum of the first two, and explain which property of independence makes this work.

Exercise 33.3 (The maximum is uniform)**.** Using Lagrange multipliers, maximize $H(p_1, \dots, p_n) = -\sum_i p_i \log p_i$ subject to $\sum_i p_i = 1$. Show the optimum is $p_i = 1/n$ for all i and that the maximal value is $\log n$. (This re-derives the upper bound of Theorem 33.1 by calculus rather than Jensen.)

Exercise 33.4 (A Huffman-style code)**.** A source emits symbols A, B, C, D, E with probabilities $0.4, 0.2, 0.2, 0.1, 0.1$. (a) Compute the entropy. (b) Build a prefix-free binary code by repeatedly merging the two least-probable symbols (Huffman's algorithm) and report the average codeword length. (c) By how many bits per symbol does it beat the naive 3-bit fixed-length code, and how close is it to the entropy lower bound?

Exercise 33.5 (KL is not symmetric)**.** Let $p = (0.7, 0.3)$ and $q = (0.5, 0.5)$. Compute $\mathrm{KL}(p\|q)$ and $\mathrm{KL}(q\|p)$ in bits, confirm both are positive, and confirm they differ. Then verify the identity $H(p, q) = H(p) + \mathrm{KL}(p\|q)$ numerically for this pair.

Exercise 33.6 (Cross-entropy as a loss)**.** A three-class classifier outputs probabilities $q = (0.7, 0.2, 0.1)$ on a particular example whose true label is class 2 (the middle one). (a) Write the one-hot true distribution p. (b) Compute the cross-entropy loss $H(p, q)$ in bits. (c) Show that the loss equals $-\log_2 q_{\text{true}}$ and explain why only the true-class probability appears. (d) What loss would a perfectly confident correct prediction incur?

Exercise 33.7 (Mutual information of a noisy channel)**.** For the binary symmetric channel with flip probability f and a fair input bit, show that $I(X; Y) = 1 - H_b(f)$. Tabulate I for $f \in \{0, 0.1, 0.25, 0.5\}$ and interpret the endpoints. At what f is the channel useless, and why?

Exercise 33.8 (Conditioning can increase specific uncertainty)**.** Construct an explicit joint distribution of two binary variables X, Y for which there is a value y with $H(X \mid Y = y) > H(X)$, yet the *average* $H(X \mid Y) \le H(X)$ still holds. Verify both inequalities numerically. (This makes the paradox box concrete.)

Exercise 33.9 (MLE equals minimum cross-entropy — proof)**.** Let $x_1, \dots, x_N$ be i.i.d. draws over a finite alphabet, with empirical distribution $\hat{p}$. Prove that $\frac{1}{N}\sum_i \log q_\theta(x_i) = -H(\hat{p}, q_\theta)$, and conclude that maximizing the log-likelihood over θ is equivalent to minimizing $\mathrm{KL}(\hat{p}\|q_\theta)$. State precisely which term in the decomposition does not depend on θ and why that makes the two optimizations identical.

Exercise 33.10 (Independence and mutual information)**.** Let $X \in \{-1, 0, 1\}$ be uniform and let $Y = X^2$. (a) Compute $\mathrm{Corr}(X, Y)$ and observe it is zero. (b) Compute $I(X; Y)$ and show it is strictly positive. (c) Explain in one sentence why mutual information detects the dependence that correlation completely misses.

Exercise 33.11 (Grouping / recursivity of entropy — proof)**.** Show that entropy satisfies the grouping property: if outcomes are partitioned into groups, the total entropy equals the entropy of the group choice plus the average within-group entropy,

$$H(p_1, \dots, p_n) = H(\text{group probabilities}) + \sum_{\text{groups } g} \mathbb{P}(g)\, H(\text{conditional dist. within } g).$$

Verify it on $(\frac{1}{2}, \frac{1}{4}, \frac{1}{4})$ grouped as $\{1\}$ and $\{2, 3\}$, and explain how this is the chain rule in disguise.

Chapter 34

Probability in Machine Learning

There is a temptation, when you first meet machine learning, to picture it as a bag of clever tricks: a decision tree here, a neural network there, a gradient descent loop turning the crank. Useful, yes – but it leaves you with a hundred disconnected recipes and no story. The goal of this chapter is to give you the story. And the story is short enough to state in one sentence:

Focus

Almost every supervised and unsupervised learning method you will ever meet is, underneath, a *probabilistic model* together with a rule for choosing its parameters from data. Learning *is* estimation. Prediction *is* computing a conditional distribution.

Once you believe that sentence, machine learning stops being a zoo and becomes a single landscape with familiar landmarks. Maximum likelihood (Chapter on estimation), Bayes' theorem (the conditioning chapter), the normal distribution, the law of large numbers, KL divergence from information theory – they all reappear here wearing work clothes. This is the flagship application chapter: it ties the book together. We will move slowly, build each idea from a concrete story, and stay clear-eyed about what works, what fails, and why.

34.1 The likelihood view of learning

34.1.1 A story: the biased coin and the spam filter

Suppose a friend hands you a bent coin and asks, "what's the probability it lands heads?" You flip it 20 times and see 14 heads. You would not hesitate: your gut says about $14/20 = 0.7$. Where does that number come from? Not from a formula you memorized – from a deeper instinct. You are reaching for the value of p that makes the data you actually saw *least surprising*. A coin with $p = 0.7$ produces "14 out of 20" far more readily than a coin with $p = 0.1$ or $p = 0.95$. You are, without naming it, doing maximum likelihood.

Now zoom out. A spam filter sees an email and must decide `spam` or `ham`. A medical model sees a scan and outputs a probability of disease. A recommender sees your clicks and predicts the next one. In every case there is an input x, an output y, and a family of probability distributions $p(y \mid x; \theta)$ indexed by parameters θ. *Learning* is choosing θ from a training set. *Prediction* is plugging in a new

x and reading off the distribution over y. The bent coin is the whole subject in miniature.

Intuition

A model is a *machine for assigning probabilities to data*. A good model assigns high probability to the things that actually happened and low probability to things that did not. Training tunes the knobs θ to do exactly that on the data you have, in the hope it generalizes to data you do not.

34.1.2 Datasets, models, and the i.i.d. assumption

Fix the setup we will use all chapter. We observe a dataset

$$\mathcal{D} = \{(x_1, y_1), \ldots, (x_n, y_n)\},$$

n examples, each a pair of an input (or *feature vector*) x_i and a label y_i. We posit a parametric model: a family of distributions $p(\cdot \mid \theta)$. The single most common assumption – so common it is often left unstated – is that the examples are drawn *independently from one fixed distribution*, i.i.d.:

$$(x_i, y_i) \overset{\text{i.i.d.}}{\sim} p^\star, \qquad i = 1, \ldots, n.$$

Here $p^\star$ is the true, unknown data-generating distribution. We never see $p^\star$ directly; we see n samples from it and try to approximate it with our model $p(\cdot \mid \theta)$.

Common trap

The i.i.d. assumption is a *modeling choice*, not a law of nature. Time series are not i.i.d. (today depends on yesterday). Data with distribution shift – the test set drawn from a different $p^\star$ than the training set – breaks every guarantee in this chapter. When a model fails mysteriously in production, "the i.i.d. assumption was violated" is the first suspect.

34.1.3 The likelihood and its logarithm

The *likelihood* of the parameters θ given the data is the probability the model assigns to the data, read as a function of θ:

$$L(\theta) = p(\mathcal{D} \mid \theta) = \prod_{i=1}^{n} p(x_i, y_i \mid \theta),$$

the product because of independence. We choose θ to make this large. But a product of n small numbers underflows to zero on any real computer, and products are awkward to differentiate. So we take logarithms. Because log is strictly increasing, maximizing L and maximizing $\log L$ give the *same* θ, and the product becomes a sum:

$$\ell(\theta) = \log L(\theta) = \sum_{i=1}^{n} \log p(x_i, y_i \mid \theta).$$

This is the *log-likelihood*, the central object of the chapter.

Why it works

Why does the maximizer not change when we take logs? If $\theta^\star$ maximizes L, then $L(\theta^\star) \geq L(\theta)$ for all θ. Apply the increasing function log to both sides: $\log L(\theta^\star) \geq \log L(\theta)$. So $\theta^\star$ also maximizes $\log L$. The logarithm relabels the heights of the landscape but never moves a peak. We get numerical stability and additive structure for free.

Example 34.1 (The coin, made precise)**.** Model: $y_i \sim \text{Bernoulli}(p)$ i.i.d., so $p(y_i \mid p) = p^{y_i}(1-p)^{1-y_i}$. With $k = \sum y_i$ heads out of n,

$$\ell(p) = \sum_{i=1}^{n} \big[y_i \log p + (1 - y_i) \log(1 - p)\big] = k \log p + (n - k) \log(1 - p).$$

Differentiate and set to zero:

$$\ell'(p) = \frac{k}{p} - \frac{n-k}{1-p} = 0 \implies k(1-p) = (n-k)p \implies \hat{p} = \frac{k}{n}.$$

With $k = 14$, $n = 20$ we get $\hat{p} = 0.7$, precisely the gut answer. The second derivative is negative, confirming a maximum. Your instinct *was* calculus on a log-likelihood all along.

Figure 34.1 plots $\ell(p)$ for this data. Notice it is a single smooth hill – one peak, no traps. That concavity is what makes the coin (and, later, logistic regression) so well-behaved.

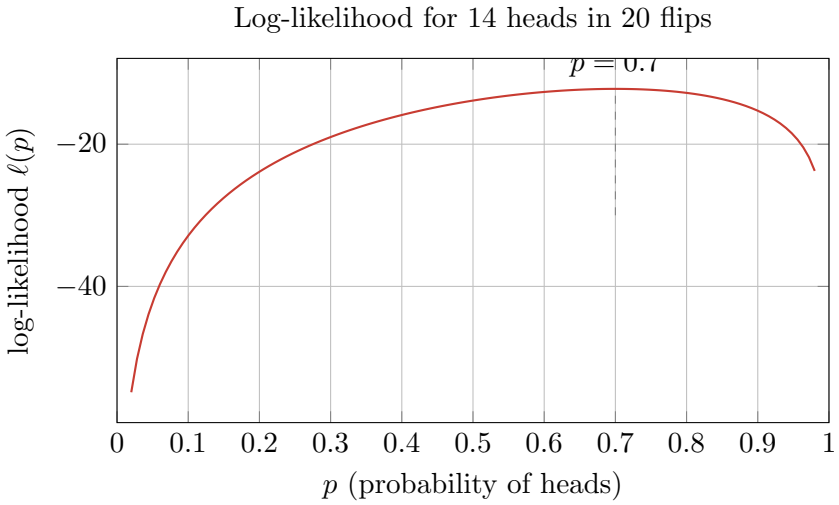


Figure 34.1: The log-likelihood is a single concave hill peaking exactly at the sample fraction $\hat{p} = k/n = 0.7$. Maximum likelihood walks uphill to this peak.

History

The likelihood as a tool for inference is largely the creation of Ronald A. Fisher in the 1910s and 1920s. He insisted on a sharp distinction that still trips people up: probability runs forward from a fixed parameter to random data; *likelihood* runs backward from fixed data to a varying parameter. They use the same formula $p(\text{data} \mid \theta)$ but read it along different axes. Fisher's word "likelihood" was deliberate – he did not want people to think a parameter has

a probability (that is the Bayesian move, coming later).

34.2 MLE and MAP as training

34.2.1 Maximum likelihood is minimizing surprise

Define the maximum likelihood estimate

$$\hat{\theta}_{\text{MLE}} = \arg\max_{\theta} \ell(\theta) = \arg\max_{\theta} \sum_{i=1}^{n} \log p(\text{datum}_i \mid \theta).$$

Flip the sign and you get a quantity machine learners love:

$$\hat{\theta}_{\text{MLE}} = \arg\min_{\theta} \underbrace{\Big[-\tfrac{1}{n} \sum_{i=1}^{n} \log p(\text{datum}_i \mid \theta) \Big]}_{\text{average negative log-likelihood}}.$$

The bracketed term is a *loss function*. So "maximize likelihood" and "minimize loss" are the same sentence said two ways. This is the bridge between the probability vocabulary and the optimization vocabulary of deep learning.

What is really happening

What is the average negative log-likelihood really measuring? It is the *cross-entropy* between the empirical data distribution and your model (see the information theory chapter). Minimizing it is the same as minimizing the KL divergence $\text{KL}(\hat{p}_{\text{data}} \,\|\, p_\theta)$: you are pulling your model as close as possible, in the information-theoretic sense, to the histogram of the data you saw. Every cross-entropy loss in every neural network is this exact quantity. "Train the network" = "do maximum likelihood."

Connection

The leap from "minimize KL to the empirical distribution" to "minimize KL to the true $p^\star$" is justified by the law of large numbers: as $n \to \infty$, the empirical average $\frac{1}{n}\sum -\log p_\theta(\text{datum}_i)$ converges to its expectation $\mathbb{E}_{p^\star}[-\log p_\theta]$, which is minimized at the θ closest to $p^\star$. So MLE is consistent precisely *because* of the LLN you proved earlier. The whole edifice rests on that one limit theorem.

34.2.2 Overfitting and the case for a prior

MLE has a famous weakness. Flip a coin three times, see three heads. MLE says $\hat{p} = 3/3 = 1$: the coin *never* lands tails. That is absurd, and worse, it is dangerous – the model now assigns probability zero to an outcome that can certainly happen, and a single tail in the future makes the likelihood collapse. With little data, maximum likelihood believes the data too literally. It overfits.

The fix is to bring in prior knowledge. Before seeing any flips you already believe a coin is roughly fair; you should need real evidence to be talked out of that. Bayes' theorem is exactly the machinery

for updating a belief with evidence:

$$\underbrace{p(\theta \mid \mathcal{D})}_{\text{posterior}} \propto \underbrace{p(\mathcal{D} \mid \theta)}_{\text{likelihood}} \underbrace{p(\theta)}_{\text{prior}}.$$

34.2.3 MAP: maximum a posteriori

If, instead of the value that maximizes the likelihood, we take the value that maximizes the *posterior*, we get the maximum a posteriori estimate:

$$\hat{\theta}_{\text{MAP}} = \arg\max_{\theta} \big[\log p(\mathcal{D} \mid \theta) + \log p(\theta)\big] = \arg\max_{\theta} \big[\ell(\theta) + \log p(\theta)\big].$$

The prior contributes an *additive penalty* $\log p(\theta)$ to the log-likelihood. This single line unifies two worlds.

Example 34.2 (The coin, regularized)**.** Put a Beta(α, β) prior on p: $p(p) \propto p^{\alpha-1}(1-p)^{\beta-1}$. The Beta is the conjugate prior for the Bernoulli (the posterior is again Beta), which is why it is the natural choice. The posterior is Beta$(\alpha + k,\ \beta + n - k)$, and its mode (the MAP estimate) is

$$\hat{p}_{\text{MAP}} = \frac{k + \alpha - 1}{n + \alpha + \beta - 2}.$$

With a mild fair-coin prior $\alpha = \beta = 2$ and the three-heads disaster $k = n = 3$,

$$\hat{p}_{\text{MAP}} = \frac{3 + 1}{3 + 2} = 0.8,$$

not the reckless 1.0 of MLE. The prior acts like two phantom flips, one head and one tail, added to the data. The Beta hyperparameters are "pseudo-counts": prior evidence measured in the same currency as real evidence.

Why this matters

MAP reveals that *regularization is a prior in disguise*. The famous L^2 penalty (ridge regression, weight decay) $\lambda\|\theta\|^2$ is exactly $-\log p(\theta)$ for a Gaussian prior $\theta \sim \mathcal{N}(0, \sigma^2 I)$ with $\lambda = 1/(2\sigma^2)$. The L^1 penalty (lasso) is a Laplace prior. So every time a practitioner "adds regularization to prevent overfitting," they are – whether they know it or not – declaring a prior belief that the parameters are small. Weight decay is Bayes.

Why it works

Why does L^2 regularization equal a Gaussian prior? A Gaussian density is $p(\theta) \propto \exp(-\|\theta\|^2/(2\sigma^2))$, so $\log p(\theta) = -\|\theta\|^2/(2\sigma^2) + \text{const}$. The MAP objective $\ell(\theta) + \log p(\theta)$ therefore equals $\ell(\theta) - \frac{1}{2\sigma^2}\|\theta\|^2 + \text{const}$. Maximizing this is the same as minimizing $-\ell(\theta) + \lambda\|\theta\|^2$ with $\lambda = 1/(2\sigma^2)$: ordinary penalized loss. The penalty strength and the prior's tightness are two names for the same dial.

Common trap

MAP gives a single point, the *peak* of the posterior. The full Bayesian answer is the entire posterior *distribution*, which also carries the width (your uncertainty). For a skewed posterior the mode can sit far from the mean and be a poor summary. MAP is Bayes' theorem used as a regularizer, not Bayes' theorem used to quantify uncertainty. Do not confuse the two.

34.3 Naive Bayes: a generative classifier

34.3.1 Turning Bayes' theorem into a classifier

Our first full-fledged learning algorithm. We want $p(y \mid x)$ – the probability of class y given features x. Bayes' theorem flips it:

$$p(y \mid x) = \frac{p(x \mid y)\, p(y)}{p(x)} \;\propto\; p(x \mid y)\, p(y).$$

To classify, pick the class with the largest right-hand side; the denominator $p(x)$ is the same for every class and drops out. We need two ingredients: the *class prior* $p(y)$ (how common each class is) and the *class-conditional* $p(x \mid y)$ (what features look like within each class).

The catch: x is often high-dimensional – thousands of word counts in an email – and $p(x \mid y)$ is a joint distribution over all of them, hopeless to estimate.

34.3.2 The naive assumption

Here is the bold, "naive" move: assume the features are *conditionally independent given the class*:

$$p(x \mid y) = \prod_{j=1}^{d} p(x_j \mid y).$$

A monstrous joint distribution collapses into a product of d one-dimensional distributions, each trivial to estimate by counting. For spam: assume that, once you know an email is spam, the appearance of "viagra" tells you nothing extra about whether "free" also appears. This is false – those words travel together – yet the classifier works astonishingly well in practice.

Figure 34.2 shows the generative story as a tree: nature first picks the class with probability $p(y)$, then – treating the words as independent given the class – emits each word with its own class-conditional probability.

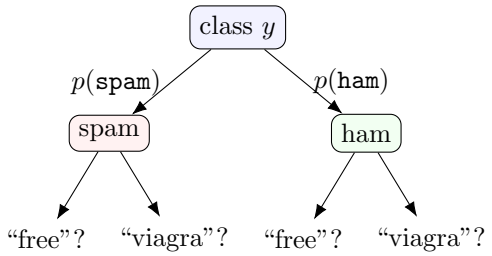


Figure 34.2: The generative tree behind naive Bayes. First a class is drawn from the prior $p(y)$; then, *given* the class, each word is emitted independently with probability $p(x_j \mid y)$. The independence is the "naive" part: the branches for "free" and "viagra" do not talk to each other once the class is fixed.

Example 34.3 (Spam filtering by counting)**.** Bag-of-words model: $x_j = 1$ if word j appears. By MLE the parameters are pure counts:

$$\hat{p}(y = \texttt{spam}) = \frac{\#\text{spam emails}}{\#\text{emails}}, \qquad \hat{p}(x_j = 1 \mid y) = \frac{\#\{y\text{-emails with word } j\}}{\#\, y\text{-emails}}.$$

A new email with word-presence vector x is scored by

$$\log \frac{p(\texttt{spam} \mid x)}{p(\texttt{ham} \mid x)} = \log \frac{p(\texttt{spam})}{p(\texttt{ham})} + \sum_j \log \frac{p(x_j \mid \texttt{spam})}{p(x_j \mid \texttt{ham})}.$$

Each word casts an additive vote on the log-odds scale, and the votes simply add up. Classify as spam when the total exceeds zero.

Common trap

The zero-frequency trap. If a word never appeared in spam during training, $\hat{p}(x_j = 1 \mid \texttt{spam}) = 0$, and a single such word drives the whole product – and the spam probability – to exactly zero, vetoing all other evidence. The cure is *Laplace smoothing*: add a pseudo-count to every tally,

$$\hat{p}(x_j = 1 \mid y) = \frac{\#\{\dots\} + 1}{\#\, y\text{-emails} + 2}.$$

This is not an ad hoc patch: it is precisely the MAP estimate under a Beta$(2, 2)$ prior. The same phantom counts from Example 34.2 return to save us. Theory and the engineering hack are the same thing.

Paradox

Naive Bayes is wrong yet wins. Its independence assumption is almost never true, so its probability *estimates* are badly miscalibrated – typically pushed toward 0 or 1 with false confidence. How can a model with wrong probabilities classify well? Because classification needs only the *ranking* of classes to be correct, not the probabilities. As long as the true class ends up with the largest score, the inflated magnitudes do not matter for the decision. Naive Bayes can be a great classifier and a terrible probability estimator at the same time – a sharp reminder to separate "which class" from "how sure."

34.4 Logistic regression as a probabilistic model

34.4.1 From a line to a probability

Naive Bayes models how the features are generated. Logistic regression makes the opposite bet: model $p(y \mid x)$ *directly*, without ever describing how x arises. Start with a linear score $z = w^\top x + b$, which ranges over all of $\mathbb{R}$, and squash it into $(0, 1)$ with the *logistic* (sigmoid) function:

$$\sigma(z) = \frac{1}{1 + e^{-z}}, \qquad p(y = 1 \mid x) = \sigma(w^\top x + b).$$

The sigmoid (Figure 34.3) is a smooth switch: very negative scores map near 0, very positive scores near 1, and the steep middle is the decision boundary.

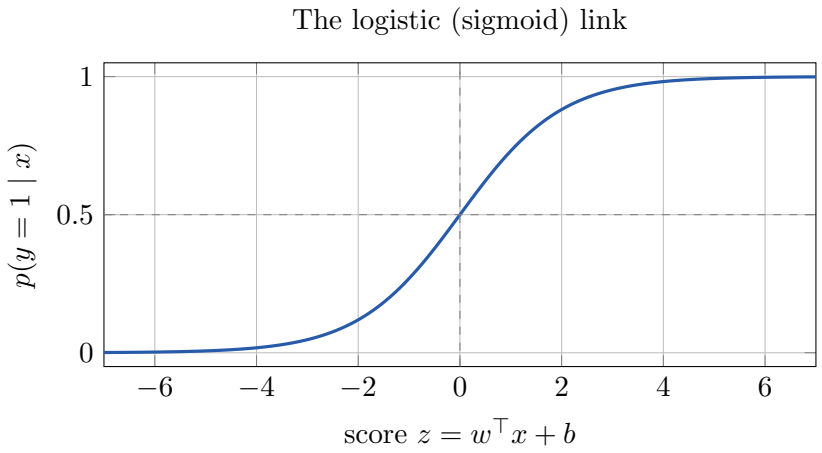


Figure 34.3: The sigmoid turns an unbounded linear score into a probability. At $z = 0$ the model is maximally uncertain ($p = 0.5$); the slope there is steepest, so small changes near the boundary swing the prediction most.

34.4.2 Why the sigmoid in particular?

It is not arbitrary. The log-odds (logit) is the natural inverse:

$$\log \frac{p(y = 1 \mid x)}{p(y = 0 \mid x)} = w^\top x + b.$$

Logistic regression is the assumption that the *log-odds are linear in the features*. And recall the naive-Bayes spam score: it too was linear in the features on the log-odds scale. That is no coincidence.

Connection

If the class-conditionals $p(x \mid y)$ are Gaussians with a shared covariance, Bayes' theorem produces *exactly* a logistic form for $p(y \mid x)$. So Gaussian naive Bayes and logistic regression can describe the same boundary – but they fit it differently. Naive Bayes estimates the Gaussians (generative); logistic regression optimizes the boundary directly (discriminative). This is the cleanest example of the generative/discriminative pair in Section 34.5.

34.4.3 Training: maximum likelihood again

With $y_i \in \{0, 1\}$ and $\hat{p}_i = \sigma(w^\top x_i + b)$, each label is Bernoulli, so the log-likelihood is

$$\ell(w, b) = \sum_{i=1}^{n} \big[y_i \log \hat{p}_i + (1 - y_i) \log(1 - \hat{p}_i) \big].$$

Negate it and you have the *binary cross-entropy* loss – the workhorse loss of binary classifiers everywhere. Maximizing this likelihood is training the classifier; once more, learning is MLE.

There is no closed form for $\hat{w}$ this time, but the news is good: ℓ is *concave* in (w, b), so gradient ascent cannot get stuck in a bad local optimum. The gradient is strikingly clean:

$$\frac{\partial \ell}{\partial w} = \sum_{i=1}^{n} (y_i - \hat{p}_i)\, x_i.$$

Each example pushes the weights in the direction of its features, scaled by the *residual* $y_i - \hat{p}_i$ – how wrong the current prediction was. Confident-and-correct examples ($\hat{p}_i \approx y_i$) contribute almost nothing; the surprises do the teaching.

Why it works

Why is that gradient so simple? Two derivatives conspire. The sigmoid satisfies $\sigma'(z) = \sigma(z)(1 - \sigma(z))$, and the cross-entropy's derivative with respect to $\hat{p}$ has a matching $\hat{p}(1 - \hat{p})$ in the denominator. They cancel exactly, leaving the bare residual $y_i - \hat{p}_i$. This cancellation is not luck – it is the signature of *matching the loss to the link* via the exponential-family/canonical-link theory of generalized linear models. Use the right loss for your link and the gradient always reduces to (truth minus prediction) times input.

Application

Logistic regression is still, in the 2020s, one of the most deployed models on Earth: credit scoring, click-through prediction, clinical risk scores, fraud detection. Why, in the age of deep nets? It is fast, its coefficients are interpretable (each w_j is a log-odds change per unit feature), it produces genuine probabilities, and it is the final layer of essentially every neural classifier – a deep net is logistic regression on top of learned features. Master this and you have mastered the output end of modern deep learning.

34.5 Generative vs. discriminative

We have now met both species, so let us name the divide explicitly. A *generative* model learns the full joint $p(x, y) = p(x \mid y)\, p(y)$ – it can, in principle, dream up new (x, y) pairs. A *discriminative* model learns only the conditional $p(y \mid x)$ – it can answer questions about y but cannot generate x.

	Generative	**Discriminative**
Models	$p(x, y)$	$p(y \mid x)$
Examples	naive Bayes, GMM, LDA	logistic regression, neural nets
Can sample new x?	yes	no
Small-data behavior	often better	needs more data
Large-data accuracy	can plateau	usually higher

What is really happening

There is a deep result behind this table (Ng and Jordan, 2002). A generative–discriminative pair like Gaussian naive Bayes and logistic regression converge to *different* answers, and the generative model reaches its (higher) asymptotic error faster – it needs fewer examples – while the discriminative model reaches a lower asymptotic error but needs more data to get there. The curves cross. Generative wins in the small-data regime; discriminative wins once data is plentiful. This is not folklore; it is a theorem about the price you pay for the generative model's stronger assumptions.

34.6 Gaussian mixtures and the EM algorithm

34.6.1 When the labels are hidden

Everything so far was *supervised*: we saw labels y_i. Now suppose we see only the features – a cloud of points – and suspect they come from a few distinct groups (customer segments, cell types, signal sources). This is *clustering*, and the probabilistic way to think about it is a *mixture model*: each point was generated by first secretly choosing a group $z \in \{1, \dots, K\}$, then drawing x from that group's distribution.

For continuous data the natural choice is the *Gaussian mixture*:

$$p(x) = \sum_{k=1}^{K} \pi_k \, \mathcal{N}(x \mid \mu_k, \Sigma_k),$$

a weighted blend of K bell curves with mixing weights $\pi_k \geq 0$, $\sum_k \pi_k = 1$. The hidden group label z is a *latent variable*: it was part of the generative story but is missing from the data.

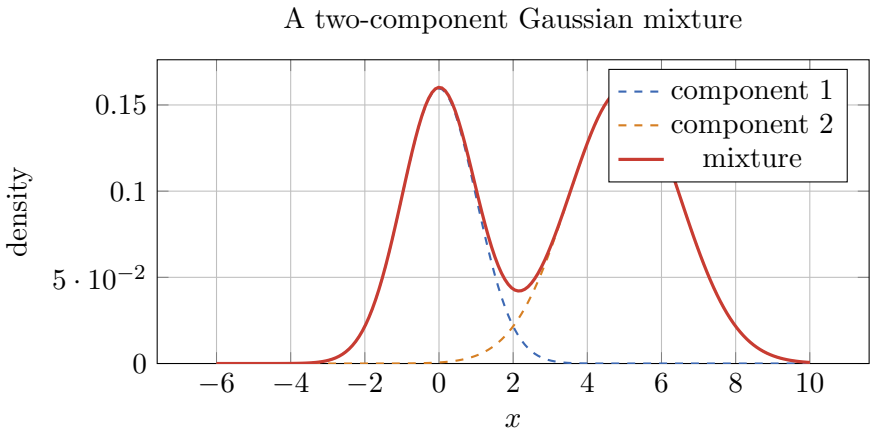


Figure 34.4: The solid curve is the observed density; the dashed curves are the hidden components. Clustering is the problem of recovering the dashed curves – and which point came from which – when you can see only the solid one.

In two dimensions the same picture becomes a cloud of points (Figure 34.5). Our eyes group it instantly into two blobs; the EM algorithm below is a precise procedure for doing what the eye does, and for attaching a probability to each membership rather than a hard verdict.

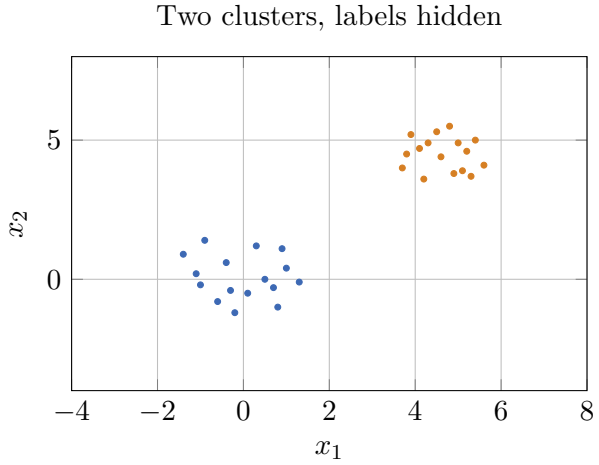


Figure 34.5: The same mixture in two dimensions: a cloud the eye splits into two groups at a glance. Clustering automates that judgment and, with a Gaussian mixture, reports *how sure* it is about each point near the gap.

34.6.2 The chicken-and-egg problem and the EM idea

Fitting a GMM by maximum likelihood faces a circular obstacle. *If* we knew each point's group, estimating each Gaussian would be trivial averaging. *If* we knew the Gaussians, assigning each point to its most probable group would be easy. But we know neither. Expectation–Maximization (EM) breaks the circle by guessing, then alternating:

- **E-step.** Given the current Gaussians, compute for each point a *soft* assignment – the posterior

probability it belongs to group k, called the *responsibility*

$$r_{ik} = \frac{\pi_k \mathcal{N}(x_i \mid \mu_k, \Sigma_k)}{\sum_{j=1}^{K} \pi_j \mathcal{N}(x_i \mid \mu_j, \Sigma_j)}.$$

This is just Bayes' theorem for the hidden label z.

- **M-step.** Given the responsibilities, re-estimate each Gaussian by *weighted* maximum likelihood:

$$\mu_k = \frac{\sum_i r_{ik} x_i}{\sum_i r_{ik}}, \quad \Sigma_k = \frac{\sum_i r_{ik}(x_i - \mu_k)(x_i - \mu_k)^\top}{\sum_i r_{ik}}, \quad \pi_k = \frac{1}{n}\sum_i r_{ik}.$$

Iterate E, M, E, M until the parameters stop moving.

Intuition

EM is a guided dance between belief and evidence. The E-step says "given what I currently believe the clusters are, how should I split the credit for each point?" The M-step says "given that split of credit, what clusters best explain it?" Each step can only *increase* the likelihood, so the dance always climbs – though, unlike logistic regression, possibly to a local rather than the global peak.

Why it works

Why must each round increase the likelihood? EM does not maximize the awkward log-likelihood $\log \sum_z p(x, z \mid \theta)$ directly. Instead it maximizes a *lower bound* $\mathcal{Q}$ built using the current responsibilities (the bound comes from Jensen's inequality applied to the concave log). The E-step makes the bound *touch* the true log-likelihood at the current θ; the M-step pushes the bound – and hence the touching log-likelihood – uphill. Climbing a bound that touches the true objective guarantees the true objective never decreases. EM is hill-climbing where you cannot see the hill, only a tangent surface you keep re-drawing.

Connection

The familiar k-means algorithm is EM's stern cousin. Replace the soft responsibilities with *hard* assignments (each point to its single nearest center) and fix all $\Sigma_k = \sigma^2 I$ with $\sigma \to 0$, and the EM updates *become* the k-means updates. So k-means is a Gaussian mixture with spherical clusters and zero uncertainty about membership. Knowing this tells you exactly when k-means fails: elongated or overlapping clusters, where the hard, spherical assumption is plainly wrong.

34.7 A first look at sampling and MCMC

34.7.1 Why sampling is the workhorse of Bayesian ML

The proper Bayesian answer to any question is an expectation over the posterior: the predictive mean is $\mathbb{E}_{\theta \sim p(\theta \mid \mathcal{D})}[\,\cdot\,]$, a credible interval is a posterior quantile, and so on. But the posterior

$$p(\theta \mid \mathcal{D}) = \frac{p(\mathcal{D} \mid \theta)\, p(\theta)}{p(\mathcal{D})}, \qquad p(\mathcal{D}) = \int p(\mathcal{D} \mid \theta)\, p(\theta)\, \mathrm{d}\theta,$$

hides a usually intractable integral in its denominator (the *evidence*). For all but a handful of conjugate models we cannot write the posterior down. The escape: never compute the integral; instead draw *samples* $\theta^{(1)}, \dots, \theta^{(m)}$ from the posterior and approximate any expectation by an average,

$$\mathbb{E}[g(\theta) \mid \mathcal{D}] \approx \frac{1}{m} \sum_{t=1}^{m} g(\theta^{(t)}),$$

which is valid by the law of large numbers. The whole problem becomes: how do you sample from a distribution you can only evaluate *up to* the unknown constant $p(\mathcal{D})$?

34.7.2 Metropolis–Hastings: a random walk that obeys

Markov chain Monte Carlo (MCMC) builds a Markov chain (Chapter on Markov chains) whose stationary distribution *is* the posterior. Run the chain long enough and its states become samples from the target. The Metropolis–Hastings recipe:

1. From the current θ, *propose* a nearby θ' (e.g. $\theta' = \theta + \mathcal{N}(0, \tau^2)$).

2. Compute the acceptance ratio $\alpha = \min\left(1,\ \dfrac{p(\mathcal{D} \mid \theta')\, p(\theta')}{p(\mathcal{D} \mid \theta)\, p(\theta)}\right)$.

3. With probability α move to θ'; otherwise stay at θ. Record the current state and repeat.

What is really happening

The trick is in step 2: the troublesome evidence $p(\mathcal{D})$ appears in both numerator and denominator of the *ratio* and cancels. We only ever need the unnormalized posterior $p(\mathcal{D} \mid \theta)p(\theta)$, which we can always evaluate. The rule "always climb to higher-probability proposals, and sometimes climb down" is what makes the chain's stationary distribution match the target exactly – this is the *detailed balance* condition from the Markov chain chapter. The occasional downhill moves are essential: a pure hill-climber would collapse to the mode and never explore, never sample.

Figure 34.6 shows what such a chain looks like in practice: a jagged walk that drifts from its arbitrary start into the bulk of the posterior and then wanders there, each recorded state a (correlated) draw.

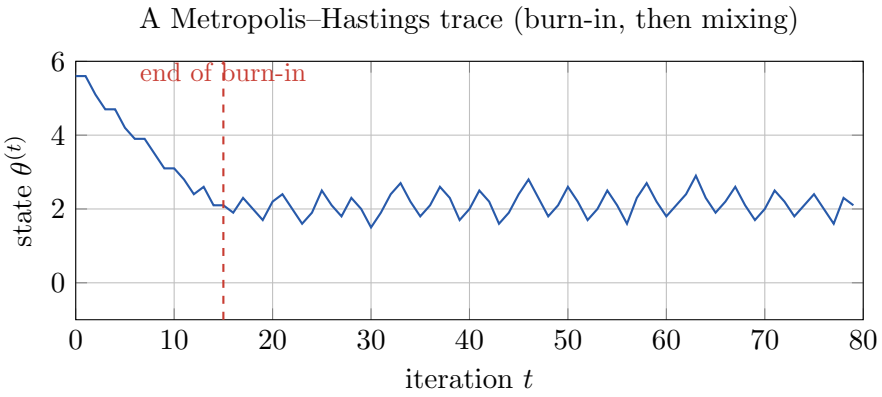


Figure 34.6: A one-dimensional Metropolis–Hastings trace. The chain starts high (an arbitrary initial guess), descends through a transient *burn-in*, then oscillates around the posterior bulk near $\theta \approx 2$. Only the post-burn-in states (right of the dashed line) are kept as samples, and even those are correlated neighbors.

Common trap

MCMC samples are *correlated* (each state is a small step from the last) and the early "burn-in" states reflect the arbitrary start, not the posterior. So you discard an initial segment and remember that m correlated draws carry less information than m independent ones. The proposal width τ is a delicate dial: too small and the chain inches along, exploring nothing; too large and almost every proposal is rejected, so the chain freezes in place. Diagnosing whether a chain has "mixed" – truly explored the posterior – is a craft unto itself. The Monte Carlo chapter develops these ideas in full.

History

The Metropolis algorithm was born in 1953 at Los Alamos, on the MANIAC computer, in a paper by Metropolis, the Rosenbluths, and the Tellers – aimed at simulating the equilibrium of interacting particles, not statistics at all. W.K. Hastings generalized it in 1970. For decades it lived in physics; only in the early 1990s, when Gelfand, Smith, and others connected it to Bayesian inference, did it detonate across statistics and machine learning. It is routinely listed among the most influential algorithms of the twentieth century – a physics trick that quietly became the engine of modern Bayesian computation.

34.8 Calibration and uncertainty

34.8.1 A confident model can be a liar

A model that outputs probabilities makes an implicit promise. When it says "70% chance of rain," it is promising that across all the days it makes that call, it rains about 70% of the time. A model that keeps this promise is *calibrated*. Calibration is a property of the probabilities, entirely separate from accuracy: a model can be accurate but overconfident (think of naive Bayes pinning probabilities to 0 or 1), or modest but perfectly accurate in its stated probabilities.

Focus

Accuracy asks: is the top-scoring class usually right? **Calibration** asks: do the probabilities mean what they say? A self-driving car or a cancer screen needs both – a wrong-and-confident model is far more dangerous than a wrong-and-uncertain one, because the confidence is what downstream decisions trust.

34.8.2 Reading a reliability diagram

To check calibration, bin predictions by their stated probability and, in each bin, plot the *actual* fraction of positives against the *average predicted* probability. Perfect calibration lies on the 45° line. An overconfident model pushes probabilities toward 0 and 1, producing an S-shaped curve (bulging above the diagonal for low probabilities and sagging below for high probabilities); an underconfident model does the reverse (Figure 34.7).

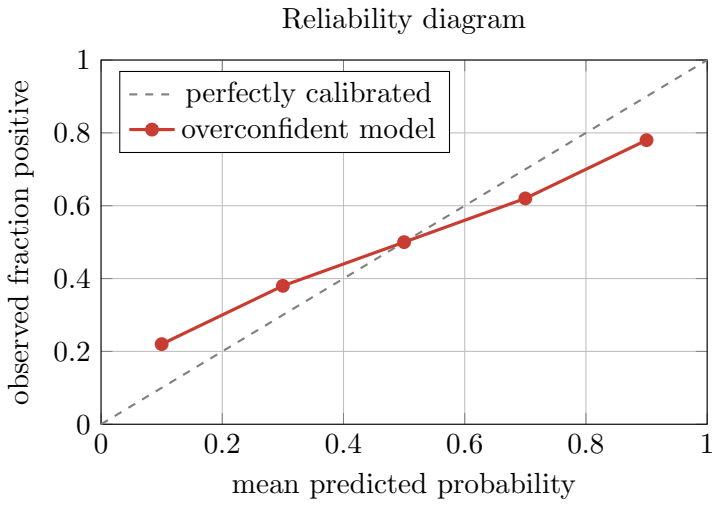


Figure 34.7: The overconfident model's curve sits below the diagonal: when it claims 0.9, positives occur only ≈ 0.78 of the time. Such a curve is the fingerprint of a model that needs recalibrating.

Application

Modern deep neural networks are notoriously overconfident – far more so than the simpler logistic regression they end in (Guo et al., 2017). Practical fixes are cheap. *Temperature scaling* divides the pre-sigmoid scores by a single learned constant $T > 1$, softening the probabilities without changing the predicted class or the accuracy at all – it only adjusts the confidence. *Platt scaling* (a tiny logistic regression on the scores) and *isotonic regression* do the same job nonparametrically. In any setting where the probability itself drives a decision – medicine, finance, autonomy – calibration is not optional polish; it is a safety requirement.

34.8.3 Two kinds of uncertainty

Finally, a distinction that matters whenever a model must say "I don't know." *Aleatoric* uncertainty is irreducible noise in the world – a fair coin is genuinely 50/50 no matter how much data you gather. *Epistemic* uncertainty is your ignorance about the model itself, and *shrinks* with more data. The Bayesian posterior tracks epistemic uncertainty automatically: a parameter seen by little data has a wide posterior, and that width flows through into wide predictive intervals – the model knows what it does not know.

Why this matters

This is the deepest reason to think probabilistically about machine learning. A point-estimate model (plain MLE, an ordinary neural net) gives an answer and no real sense of how much to trust it. A probabilistic model gives a *distribution* – it can report "I am confident" or "I am guessing," and out-of-distribution inputs, where it should be least sure, are exactly where a calibrated probabilistic model widens its intervals. In high-stakes deployment, knowing *when not to trust the model* is worth as much as the prediction itself. That self-knowledge is what probability buys you.

34.9 The thread, pulled tight

Step back and look at the single thread running through the chapter. We posit a probabilistic model $p(\text{data} \mid \theta)$. We score parameters by the likelihood, optionally tempered by a prior. Maximizing that score – by calculus when we are lucky, by gradient ascent or EM when we are not – is what everyone calls "training." Bayes' theorem turns a generative model into a classifier (naive Bayes) and turns priors into the regularizers practitioners reach for by instinct (MAP). When integrals defeat us we sample our way around them (MCMC). And throughout we insist the probabilities mean what they claim (calibration) and report uncertainty that means what it says. Maximum likelihood, Bayes, the normal distribution, KL divergence, the law of large numbers, Markov chains – every tool you built earlier in this book reappears here, doing the real work of modern machine learning. The methods change with the decade; this probabilistic skeleton does not.

Exercises

Exercise 34.1 (Poisson MLE)**.** You record the number of customer arrivals per hour over n hours, $y_1, \ldots, y_n$, modeled as i.i.d. Poisson(λ). Write the log-likelihood $\ell(\lambda)$, differentiate, and show $\hat{\lambda}_{\text{MLE}} = \bar{y}$, the sample mean. Verify it is a maximum via the second derivative.

Exercise 34.2 (Regularization is a prior)**.** Show that minimizing the ridge objective $-\ell(\theta) + \lambda\|\theta\|^2$ is exactly MAP estimation under a Gaussian prior $\theta \sim \mathcal{N}(0, \sigma^2 I)$. Express λ in terms of σ^2. Then state, in one sentence, what prior belief the lasso (L^1 penalty) encodes and why it tends to drive coefficients exactly to zero.

Exercise 34.3 (Naive Bayes by hand)**.** A spam corpus has 60 spam and 40 ham emails. The word "free" appears in 45 of the spam and 8 of the ham. Using Laplace smoothing (add one to each count, two to each denominator), compute $\hat{p}(\text{spam})$, $\hat{p}(\text{free} \mid \text{spam})$, and $\hat{p}(\text{free} \mid \text{ham})$. For an email

containing "free" (and treating it as the only feature), compute the log-odds in favor of spam and give the classification.

Exercise 34.4 (Sigmoid and log-odds)**.** Starting from $p(y = 1 \mid x) = \sigma(w^\top x + b)$ with $\sigma(z) = 1/(1 + e^{-z})$, prove algebraically that $\log \frac{p(y=1|x)}{p(y=0|x)} = w^\top x + b$. Then show the identity $\sigma'(z) = \sigma(z)(1 - \sigma(z))$ and explain in one sentence how it makes the logistic-regression gradient collapse to $(y_i - \hat{p}_i)x_i$.

Exercise 34.5 (Cross-entropy is negative log-likelihood)**.** For binary labels $y_i \in \{0, 1\}$ with predictions $\hat{p}_i$, write the Bernoulli log-likelihood and show that its negative is the binary cross-entropy $-\sum_i [y_i \log \hat{p}_i + (1-y_i) \log(1-\hat{p}_i)]$. Explain in one sentence the sense in which minimizing it minimizes a KL divergence to the empirical distribution.

Exercise 34.6 (One EM update)**.** You fit a two-component GMM in one dimension. The current parameters are $\pi_1 = \pi_2 = 0.5$, $\mu_1 = 0$, $\mu_2 = 4$, and $\sigma_1 = \sigma_2 = 1$. For the data point $x = 1$, compute the responsibilities r_{11} and r_{12} (the E-step). Which component currently claims more of the point, and does that match your intuition from the distance to each mean?

Exercise 34.7 (Why the evidence cancels)**.** In Metropolis–Hastings the acceptance ratio uses $p(\mathcal{D} \mid \theta')p(\theta')$ over $p(\mathcal{D} \mid \theta)p(\theta)$. Explain precisely why we never need the normalizing constant $p(\mathcal{D}) = \int p(\mathcal{D} \mid \theta)p(\theta)\, \mathrm{d}\theta$, and why this is what makes MCMC practical for models where that integral is impossible.

Exercise 34.8 (Calibration vs. accuracy, conceptual)**.** Construct a small concrete example of a binary classifier that is 90% accurate yet badly miscalibrated, and a second classifier that is only 60% accurate yet perfectly calibrated. (Describe the predicted probabilities and outcomes; numbers, not prose.) Use your examples to explain why accuracy alone is the wrong metric when a downstream decision depends on the probability itself.

Exercise 34.9 (Generative vs. discriminative, proof-flavored)**.** Assume two classes with class-conditional densities $p(x \mid y = c) = \mathcal{N}(x \mid \mu_c, \Sigma)$ sharing one covariance Σ, and class prior $p(y = 1) = \pi$. Using Bayes' theorem, show that $p(y = 1 \mid x)$ has the logistic form $\sigma(w^\top x + b)$, and give w and b in terms of $\mu_0, \mu_1, \Sigma, \pi$. Comment on what this reveals about the relationship between Gaussian naive Bayes and logistic regression.

Exercise 34.10 (Aleatoric vs. epistemic)**.** Classify each source of uncertainty as primarily aleatoric or epistemic, with a one-line justification: (a) the outcome of a future fair die roll; (b) a drug's effect size estimated from only 12 patients; (c) sensor readout noise on a thermometer; (d) a self-driving model's prediction on a kind of road sign absent from its training set. For the epistemic ones, state what would reduce the uncertainty.

Chapter 35

Monte Carlo Methods

There is an old joke that the fastest way to compute a hard integral is to ask a casino. Behind the joke is a deep idea: when you cannot solve a problem with pencil and paper, you can often *simulate* it. Roll dice, watch what happens, and let the law of large numbers do the work. The answer arrives not as a clean formula but as an estimate – a number with an error bar that shrinks the longer you play.

These are the *Monte Carlo methods*, named after the famous casino in Monaco. The name was a wartime code word, but it could hardly be more apt: the entire subject is built on the idea that controlled randomness is a tool, not a nuisance. We will learn how to estimate quantities by simulation, how to integrate functions that no formula can touch, how accurate those estimates are (the central limit theorem will tell us, almost for free), and how to be cleverer than brute force through variance reduction. The chapter ends at the summit of the subject: Markov Chain Monte Carlo, the engine that powers modern Bayesian statistics and much of statistical physics.

Everything in this chapter rests on machinery you have already built. Monte Carlo integration is the law of large numbers wearing a work uniform. Its error bars are the central limit theorem. Its most advanced form, MCMC, is the theory of Markov chains turned into an algorithm. So in a real sense this chapter is a victory lap: it shows you that the abstract limit theorems you proved earlier are not just true, they are *useful*.

35.1 The Core Idea: Estimating by Simulation

Start with the simplest question there is. Suppose I want to know the probability that a particular event happens. Maybe it is a complicated event in a complicated system – the probability that a randomly shuffled deck has no card in its "correct" position, or that a random network stays connected, or that a stock portfolio loses more than ten percent in a year. The exact calculation might be a nightmare. But if I can *simulate* the system, I can just run it many times and count.

Intuition

To estimate a probability, run the experiment n times and report the fraction of runs in which the event happened. To estimate an average, run the experiment n times and report the average of what you observed. That is the whole idea. The mathematics is about how good

these estimates are and how to make them better.

The simplest classic example is estimating π by throwing darts. Draw the unit square $[0,1]\times[0,1]$ and inscribe the quarter circle $x^2+y^2\le 1$. Throw a dart uniformly at random into the square. The probability it lands inside the quarter circle is the ratio of areas: $(\pi/4)/1=\pi/4$. So if I throw n darts and a fraction $\hat{p}$ land inside, then $\hat{p}\approx\pi/4$, which means $4\hat{p}\approx\pi$.

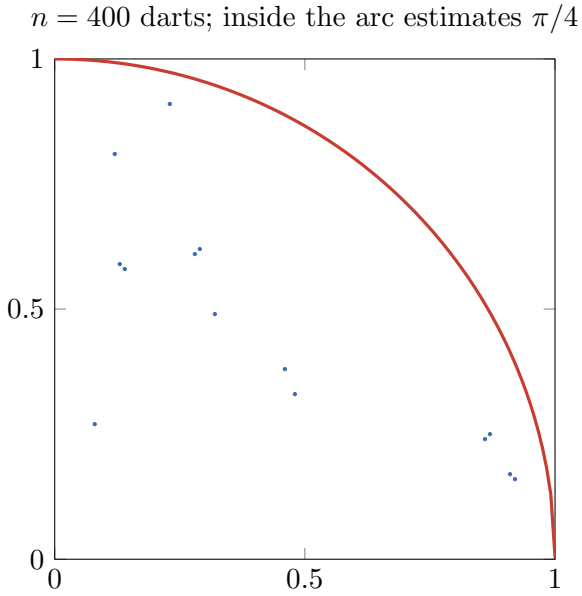


Figure 35.1: Estimating π by Monte Carlo. Each dart is uniform on the square; the fraction below the red arc estimates $\pi/4$. With 400 darts you get one or two correct digits – accuracy improves only like $1/\sqrt{n}$.

Example 35.1 (Estimating π)**.** Suppose I throw $n=10{,}000$ darts and 7,839 land inside the arc. Then $\hat{p}=0.7839$ and my estimate is $\hat{\pi}=4\hat{p}=3.1356$. The true value is $3.14159\ldots$, so I am off by about 0.006. Not bad – but notice how much work bought how little accuracy. Ten thousand darts gave two correct digits. To get one more digit I would need roughly a hundred times as many darts. This slow $1/\sqrt{n}$ convergence is the central fact, and the central frustration, of Monte Carlo.

Why does this work at all? Because each dart is an independent trial, and the indicator "did this dart land inside?" is a Bernoulli$(\pi/4)$ random variable. The fraction inside is a sample mean of i.i.d. Bernoullis, and the law of large numbers guarantees it converges to the true mean $\pi/4$. We are watching a theorem run.

History

The modern Monte Carlo method was born at Los Alamos in the late 1940s. Stanislaw Ulam, recovering from illness and playing solitaire, wondered what fraction of Canfield solitaire games are winnable. The combinatorics were hopeless, so he realized he could simply *deal many hands and count*. He took the idea to John von Neumann, who saw immediately that the new electronic computers could run such simulations for the brutal neutron-diffusion calculations of the hydrogen bomb. Nicholas Metropolis suggested the code name "Monte Carlo," after

the casino where Ulam's uncle borrowed money to gamble. The method that now underlies weather forecasting, drug discovery, and machine learning began as a card game and a bomb.

35.2 Monte Carlo Integration

The dart game is really a disguised integral. The area under the arc is $\int_0^1 \sqrt{1-x^2}\,\mathrm{d}x = \pi/4$, and our darts estimated it. Let us strip away the disguise and state the general principle, because integration is where Monte Carlo earns its keep.

Suppose we want to compute

$$I = \int_{\mathcal{X}} g(x)\, f(x)\,\mathrm{d}x = \mathbb{E}[g(X)],$$

where X is a random variable (or vector) with density f, and g is some function. *Any* expectation is an integral of this form, and conversely any integral can be written as an expectation by choosing a convenient f. The Monte Carlo recipe is about as simple as a recipe gets:

Formal

Basic Monte Carlo estimator. Draw $X_1, \dots, X_n$ independently from the density f. Estimate

$$\hat{I}_n = \frac{1}{n}\sum_{i=1}^{n} g(X_i).$$

By the law of large numbers, $\hat{I}_n \xrightarrow{a.s.} I$ as $n \to \infty$, provided $\mathbb{E}|g(X)| < \infty$.

If you want a plain integral $\int_a^b h(x)\,\mathrm{d}x$ with no probability in sight, just rewrite it. Let $X \sim \mathrm{Uniform}(a,b)$, whose density is $1/(b-a)$ on $[a,b]$. Then

$$\int_a^b h(x)\,\mathrm{d}x = (b-a)\int_a^b h(x)\,\frac{1}{b-a}\,\mathrm{d}x = (b-a)\,\mathbb{E}[h(X)],$$

so the estimator is $\hat{I}_n = \frac{b-a}{n}\sum_{i=1}^{n} h(X_i)$ with the X_i uniform on $[a,b]$. You sample points, evaluate the function, average, and scale by the width. Done.

Connection

Monte Carlo integration *is* the law of large numbers, dressed for work. Recall from Chapter 20 that for i.i.d. $Y_1, \dots, Y_n$ with mean μ, the sample mean $\bar{Y}_n \xrightarrow{a.s.} \mu$. Here $Y_i = g(X_i)$ and $\mu = \mathbb{E}[g(X)] = I$. Nothing more is happening. The reason Monte Carlo works in 1 dimension or in 1000 dimensions, for smooth functions or jagged ones, is that the LLN never asked about dimension or smoothness. It only asked for a finite mean. That indifference is exactly what makes the method so powerful.

35.2.1 The Error of Monte Carlo

How wrong is $\hat{I}_n$? Here the central limit theorem hands us an answer for free, and it is one of the most important formulas in scientific computing.

Let $\sigma^2 = \mathrm{Var}(g(X))$. The estimator $\hat{I}_n$ is a sample mean of i.i.d. terms, so it is unbiased, $\mathbb{E}[\hat{I}_n] = I$, and its variance is

$$\mathrm{Var}(\hat{I}_n) = \frac{\sigma^2}{n}.$$

The standard error – the typical size of the error – is therefore

$$\mathrm{SD}(\hat{I}_n) = \frac{\sigma}{\sqrt{n}}.$$

And by the CLT, for large n,

$$\hat{I}_n \approx \mathcal{N}\left(I, \frac{\sigma^2}{n}\right),$$

so we can even put confidence intervals on our answer: roughly 95% of the time the truth lies within $\hat{I}_n \pm 1.96\,\sigma/\sqrt{n}$. In practice we do not know σ, so we estimate it from the same sample,

$$\hat{\sigma}^2 = \frac{1}{n-1}\sum_{i=1}^{n}\big(g(X_i) - \hat{I}_n\big)^2,$$

and report $\hat{I}_n \pm 1.96\,\hat{\sigma}/\sqrt{n}$. *The simulation tells you its own error bar.* This is a gift no deterministic integration scheme gives you so cheaply.

Why this matters

Stare at $\mathrm{SD} = \sigma/\sqrt{n}$ and notice what is *not* there: the dimension. A deterministic grid in d dimensions with m points per axis costs m^d evaluations – the "curse of dimensionality." Its error shrinks like $m^{-2} = (m^d)^{-2/d}$, which becomes hopeless as d grows. Monte Carlo's error shrinks like $n^{-1/2}$ *regardless of* d. In high dimensions this is not a small advantage; it is the difference between possible and impossible. This is why Monte Carlo, not quadrature, integrates the thousand-dimensional integrals of Bayesian statistics and statistical physics.

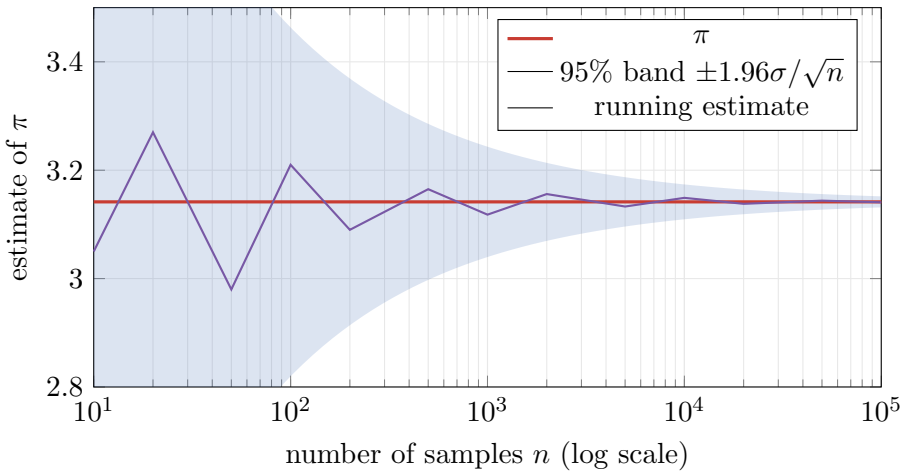


Figure 35.2: A running Monte Carlo estimate of π settling toward the truth. The shaded $1/\sqrt{n}$ band narrows agonizingly slowly: a hundredfold increase in samples buys only a tenfold reduction in error.

Common trap

The error formula $\sigma/\sqrt{n}$ assumes $\sigma^2 = \mathrm{Var}(g(X))$ is finite. If the integrand has heavy tails so that $\mathrm{Var}(g(X)) = \infty$, the CLT does not apply, the estimator's fluctuations do not shrink like $1/\sqrt{n}$, and your tidy confidence interval is a lie. Always ask whether the thing you are averaging has a finite variance. A symptom in practice: the running estimate keeps making large jumps that never settle, and a few enormous samples dominate the average.

Example 35.2 (A high-dimensional integral)**.** Estimate $I = \int_{[0,1]^{10}} \cos\!\big(\tfrac{\pi}{2}\,\overline{x}\big)\,\mathrm{d}x$, where $\overline{x} = \frac{1}{10}\sum_{j=1}^{10} x_j$ is the average coordinate. No grid can touch this: even a coarse 10 points per axis is 10^{10} evaluations. But Monte Carlo does not care about the 10 dimensions. Draw n points uniformly in the cube, evaluate the cosine of the (scaled) coordinate average at each, and average. With $n = 10^5$ samples you get three reliable digits and an automatic error bar of size $\approx \hat{\sigma}/\sqrt{n}$, all in a fraction of a second. The dimension never entered the error.

35.3 Variance Reduction: Working Smarter

The error is $\sigma/\sqrt{n}$. We can shrink it two ways: increase n (brute force, expensive, slow) or decrease σ (cleverness, often free). The art of *variance reduction* is to find an estimator with the same mean I but a smaller variance. A method that halves σ is worth quadrupling n – and costs nothing per sample. We study two classic techniques: importance sampling and antithetic variates.

35.3.1 Importance Sampling

The idea behind it is this. When you sample uniformly and average g, you waste effort sampling regions where g is tiny and contributes almost nothing to the integral. It would be smarter to draw more samples where g is *large* and correct for the oversampling afterward. That correction is importance sampling.

Intuition

Imagine estimating the average wealth of a country by polling. If you sample people uniformly, almost none of them will be billionaires, yet billionaires move the average enormously. You will get a wildly variable estimate. Instead, *deliberately oversample the rich*, then *down-weight* their answers to correct for it. You spent your samples where the action is. That is importance sampling.

Mathematically, suppose we want $I = \mathbb{E}_f[g(X)] = \int g(x)f(x)\,\mathrm{d}x$ but we sample from a different density q, called the *proposal* or *importance distribution*. As long as $q(x) > 0$ wherever $g(x)f(x) \neq 0$, we can write

$$I = \int g(x)\,f(x)\,\mathrm{d}x = \int \underbrace{\frac{g(x)\,f(x)}{q(x)}}_{\text{weighted integrand}} q(x)\,\mathrm{d}x = \mathbb{E}_q\!\left[\frac{g(X)\,f(X)}{q(X)}\right].$$

So the estimator becomes

$$\hat{I}_n^{\mathrm{IS}} = \frac{1}{n}\sum_{i=1}^{n} g(X_i)\,w(X_i), \qquad w(x) = \frac{f(x)}{q(x)}, \qquad X_i \sim q.$$

The weights $w(X_i)$ undo the fact that we drew from q instead of f. The estimator is still unbiased: $\mathbb{E}_q[g(X)w(X)] = I$. What changes is the variance.

Why it works

Why can importance sampling help? The variance of the new estimator is governed by $\mathbb{E}_q[(gw)^2] = \int \frac{g(x)^2 f(x)^2}{q(x)}\,\mathrm{d}x$. We get to choose q. If we pick q proportional to $|g(x)|f(x)$ – putting samples exactly where the integrand is big – this quantity shrinks dramatically. In the idealized limit $q(x) \propto g(x)f(x)$ (assuming $g \geq 0$), the weighted integrand $g(x)f(x)/q(x)$ becomes a *constant*, so its variance is *zero*. One sample would give the exact answer. We can never achieve this perfectly (it requires knowing I already), but it tells us the goal: **match the proposal to the shape of the integrand.**

Common trap

Importance sampling is a double-edged sword. Choose q *badly* – too narrow, missing a region where gf is large – and a rare sample lands in the missed region with an enormous weight f/q. A handful of giant weights then dominate the average, the variance explodes (possibly to infinity), and the estimator becomes worse than plain Monte Carlo. The rule of thumb: the proposal q must have *heavier tails* than the target f, never lighter. Light tails on q are how importance sampling silently fails.

Example 35.3 (Estimating a rare-event probability)**.** Suppose $X \sim \mathcal{N}(0,1)$ and we want $I = \mathbb{P}(X > 5)$. The true value is about 2.87×10^{-7}. Plain Monte Carlo draws standard normals and counts how many exceed 5; but in ten million draws you expect about *three* hits, so your estimate is dominated by counting noise – the relative error is dreadful.

Importance sampling rescues us. Sample instead from a shifted normal, $q = \mathcal{N}(5,1)$, which lives right where the action is (Figure 35.3), so roughly half the samples exceed 5. Each sample carries weight $w(x) = \phi(x)/\phi(x-5)$, where ϕ is the standard normal density. The estimator $\frac{1}{n}\sum \mathbf{1}\{X_i > 5\}\, w(X_i)$ now converges with a tiny relative error using only thousands of samples instead of billions. We moved the sampler to the rare region and let the weights pay it back.

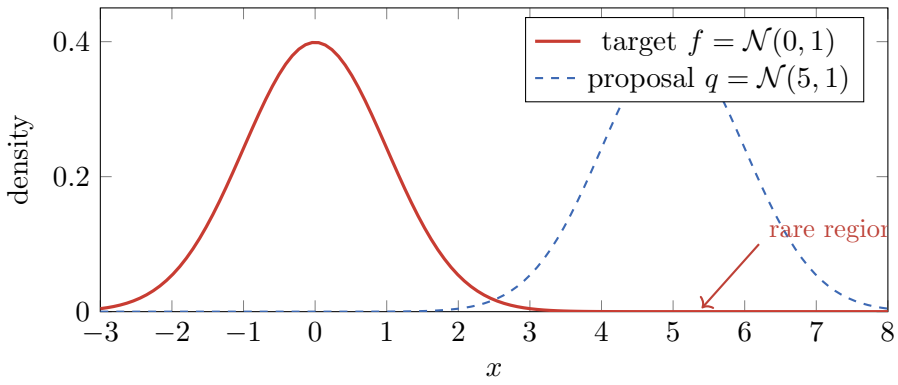


Figure 35.3: Importance sampling for a rare event. Plain Monte Carlo draws from f (red) and almost never lands in the rare region $\{x > 5\}$, so the shaded tail probability is estimated by counting a handful of hits. The proposal q (blue, dashed) is centred on the rare region, so about half its draws land there; the weights $w = f/q$ correct for the shift and keep the estimator unbiased.

35.3.2 Antithetic Variates

The second trick exploits *negative correlation*. Recall that for two random variables,

$$\operatorname{Var}\left(\frac{Y_1+Y_2}{2}\right) = \frac{1}{4}\big(\operatorname{Var} Y_1 + \operatorname{Var} Y_2 + 2\operatorname{Cov}(Y_1, Y_2)\big).$$

If Y_1 and Y_2 are independent, the covariance is zero and we get the usual variance reduction by averaging. But if we can arrange $\operatorname{Cov}(Y_1, Y_2) < 0$ – so that when one estimate is too high the other tends to be too low – the errors *cancel*, and the average is more accurate than independence would allow.

Intuition

Think of two pollsters who make opposite mistakes: when one over-counts, the other under-counts. Average them and the mistakes cancel. Antithetic variates manufacture exactly this by pairing each random draw with its "mirror image."

The standard construction: if you sample $U \sim \text{Uniform}(0,1)$, then $1-U$ is also $\text{Uniform}(0,1)$, and the two are perfectly negatively correlated. Use the pair $(U, 1-U)$ to generate two evaluations $g(U)$ and $g(1-U)$, average them, and average those averages over many pairs. If g is monotone increasing (or decreasing), then $g(U)$ and $g(1-U)$ move in opposite directions, the covariance is negative, and the variance drops – sometimes by a large factor, always for free.

Example 35.4 (Antithetic pairs in action)**.** Estimate $I = \int_0^1 e^x \, dx = e - 1 \approx 1.71828$. Plain Monte Carlo with $U \sim \text{Uniform}(0,1)$ averages e^U; the variance of e^U is $\operatorname{Var}(e^U) = \mathbb{E}[e^{2U}] - (\mathbb{E}\, e^U)^2 = \frac{e^2-1}{2} - (e-1)^2 \approx 0.2420$.

Now use antithetic pairs: average $\frac{1}{2}(e^U + e^{1-U})$. Because e^x is increasing, e^U and e^{1-U} are negatively correlated. A computation gives $\operatorname{Var}\big(\frac{1}{2}(e^U + e^{1-U})\big) \approx 0.0039$ – about *sixty times* smaller than the plain estimator's per-sample variance. But because each antithetic pair requires two function evaluations, the variance reduction for the same number of function evaluations is about *thirty times* (comparing 0.0039 to $0.2420/2 \approx 0.1210$). Thirty times less variance is the same accuracy as thirty times more brute-force samples, for no extra cost.

What is really happening

Both variance-reduction tricks are really about *where you spend your randomness*. Plain Monte Carlo spreads samples blindly. Importance sampling concentrates them where the integrand is large. Antithetic variates pairs samples so their errors cancel. Stratified sampling (a cousin we mention only in passing) forces samples into every region of the space. In each case you are not changing the question, only spending your finite budget of random draws more wisely. This is the real intellectual content of Monte Carlo beyond "just average a lot of samples."

35.4 Generating Samples: Rejection Sampling

Everything above assumed we could *draw* samples from the distribution we care about. For uniforms and normals, library routines handle it. But what about a weird density f of your own – a posterior

distribution, say, that you can write down only up to a constant? How do you sample from *that*? The first general-purpose answer is rejection sampling, and it has a beautiful geometric soul.

Intuition

To sample a point uniformly from *under the curve* of a density, throw darts uniformly at a box that contains the curve, and keep only the darts that land under the curve. The x-coordinates of the kept darts are distributed exactly according to the density. Sampling from a density is the same as sampling uniformly from the area beneath it.

This rests on a clean fact worth stating on its own.

Theorem 35.1 (Fundamental theorem of simulation)**.** *Let f be a density on $\mathbb{R}$. If (X, Y) is uniformly distributed on the region $\{(x, y) : 0 \le y \le f(x)\}$ beneath the graph of f, then the marginal distribution of X has density f.*

Proof. The region under the graph has area $\int f(x)\,\mathrm{d}x = 1$, so the uniform density there is just 1 on the region. The marginal density of X is obtained by integrating out y:

$$f_X(x) = \int_0^{f(x)} 1\,\mathrm{d}y = f(x). \qquad \square$$

So if we can sample uniformly under the curve, we are done. Rejection sampling does this when the curve is awkward by enclosing it in something easy.

Formal

Rejection sampling. Suppose we can evaluate a target density f and we have a *proposal* density q that we *can* sample, together with a constant M with $f(x) \le M\,q(x)$ for all x (the scaled proposal "envelopes" the target). Repeat:

1. Draw $X \sim q$ and an independent $U \sim \mathrm{Uniform}(0, 1)$.
2. If $U \le \dfrac{f(X)}{M\,q(X)}$, **accept** X; otherwise reject and try again.

The accepted values are exact draws from f. The probability of accepting on a given try is $1/M$, so on average it takes M tries per accepted sample.

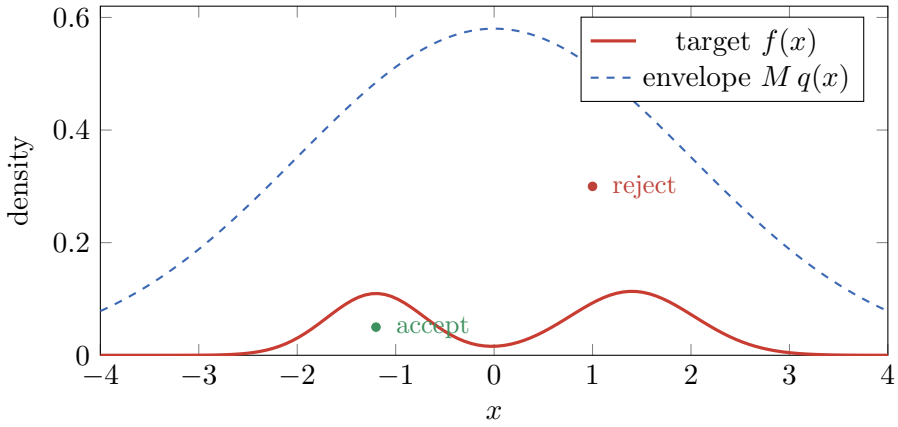


Figure 35.4: Rejection sampling. We throw points uniformly under the easy envelope $M\,q$. Points landing under f (green) are kept; points in the gap between f and $M\,q$ (red) are thrown away. Kept points are exact samples from f.

Why it works

Why are the accepted points distributed as f? Drawing $X \sim q$ and $U \sim \text{Uniform}(0,1)$ and forming $(X,\, U \cdot Mq(X))$ produces a point distributed *uniformly under the envelope* Mq. The acceptance test $U \le f(X)/(Mq(X))$ is exactly the condition that this point lies under f. So accepted points are uniform under f, and by Theorem 35.1 their x-coordinates have density f. The geometry *is* the proof.

Common trap

The cost of rejection sampling is M proposals per accepted sample, and M must satisfy $f \le Mq$ *everywhere*. If the proposal q is a poor match for the target – especially in the tails – then M must be huge, and you reject almost everything. In high dimensions this is fatal: the envelope's volume typically exceeds the target's by a factor that grows exponentially with dimension, so the acceptance probability $1/M$ becomes astronomically small. Rejection sampling is wonderful in one or two dimensions and useless in fifty. This failure is exactly the gap that MCMC was invented to cross.

35.5 Markov Chain Monte Carlo

Now for the hardest and most useful case. Suppose you have a complicated, high-dimensional distribution – a Bayesian posterior over a hundred parameters, or the Boltzmann distribution over the configurations of a magnet – and you want to sample from it or compute expectations under it. Rejection sampling drowns in the curse of dimensionality. Direct sampling is impossible. What now?

The astonishing answer of Markov Chain Monte Carlo (MCMC): *do not try to draw independent samples at all.* Instead, build a Markov chain that wanders through the space, designed so that its long-run stationary distribution is exactly the target π you want. Run the chain for a long time, and the states it visits are (correlated) samples from π. Average a function over the visited states and

you get the expectation you wanted.

Connection

This is the payoff of everything you learned about Markov chains (Chapters 23–24). There you started with a transition matrix P and *found* the stationary distribution π satisfying $\pi P = \pi$. MCMC runs the logic *backwards*: you are *given* the distribution π you want, and you must *design* a transition rule P whose stationary distribution is that π. The ergodic theorem for Markov chains then guarantees that time averages along a single long trajectory converge to expectations under π:

$$\frac{1}{n}\sum_{t=1}^{n} g(X_t) \xrightarrow{a.s.} \mathbb{E}_\pi[g(X)].$$

MCMC is the engineering inverse of the Markov chain theory you already own.

The key design tool is *detailed balance* (reversibility), which you met when studying stationary distributions. A chain with transition density $P(x \to y)$ has π as its stationary distribution if

$$\pi(x)\,P(x \to y) = \pi(y)\,P(y \to x) \qquad \text{for all } x, y.$$

Read it as a flow-conservation law: in equilibrium, the probability current from x to y exactly balances the current from y to x. If detailed balance holds, no net probability moves anywhere, so π stays put – it is stationary. Every MCMC algorithm is a recipe for cooking up a transition rule that satisfies detailed balance with respect to your chosen π.

35.5.1 The Metropolis–Hastings Algorithm

Metropolis–Hastings is the master algorithm. Its genius is that it needs π only *up to a constant* – which is exactly the situation in Bayesian inference, where the posterior is (likelihood)×(prior) divided by an intractable normalizing integral. The algorithm sidesteps that integral entirely, because it only ever looks at *ratios* $\pi(y)/\pi(x)$, in which the unknown constant cancels.

Formal

Metropolis–Hastings. Choose a proposal density $q(y \mid x)$ from which you can sample. Start at x_0. For $t = 0, 1, 2, \ldots$:

1. **Propose** a candidate $y \sim q(\cdot \mid x_t)$.
2. **Compute the acceptance ratio**

$$\alpha = \min\left(1,\ \frac{\pi(y)\,q(x_t \mid y)}{\pi(x_t)\,q(y \mid x_t)}\right).$$

3. **Accept or reject**: draw $U \sim \text{Uniform}(0,1)$. If $U \le \alpha$, set $x_{t+1} = y$ (move); otherwise set $x_{t+1} = x_t$ (stay).

The resulting chain has stationary distribution π.

Intuition

Picture a hiker exploring a landscape whose height is π, looking for the high country but wanting to visit everywhere in proportion to altitude. From the current spot she takes a tentative step (the proposal). If the new spot is *higher* ($\pi(y) > \pi(x)$), she always goes – climb toward probability mass. If it is *lower*, she still goes sometimes, with probability equal to the ratio of heights. That occasional willingness to step downhill is what stops her from getting stuck on one peak and lets her map the whole terrain. Greedy enough to find the mass, humble enough to explore all of it.

Why it works

Why does this leave π stationary? We check detailed balance. For $x \neq y$ the transition density is $P(x \to y) = q(y \mid x)\,\alpha(x, y)$. Suppose, say, that $\pi(y)q(x \mid y) \leq \pi(x)q(y \mid x)$, so that $\alpha(x, y) = \frac{\pi(y)q(x|y)}{\pi(x)q(y|x)}$ and, in the reverse direction, $\alpha(y, x) = 1$. Then

$$\pi(x)\,P(x \to y) = \pi(x)\,q(y \mid x)\,\frac{\pi(y)q(x \mid y)}{\pi(x)q(y \mid x)} = \pi(y)\,q(x \mid y),$$

while

$$\pi(y)\,P(y \to x) = \pi(y)\,q(x \mid y) \cdot 1 = \pi(y)\,q(x \mid y).$$

The two sides are equal, so detailed balance holds, and π is stationary. The min and the ratio are engineered precisely to make this equality fall out. And because only the ratio $\pi(y)/\pi(x)$ appears, any normalizing constant in π cancels – the reason MCMC conquers intractable posteriors.

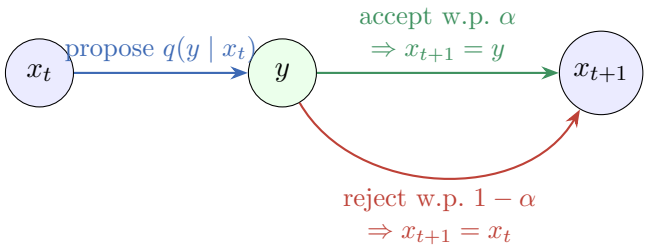


Figure 35.5: One step of Metropolis–Hastings: propose a move, then accept it with probability α or stay put. Staying put is a legitimate move – it is how the chain spends extra time in high-probability regions.

Example 35.5 (Sampling a standard normal with a random walk)**.** Let the target be $\pi(x) \propto e^{-x^2/2}$ (a standard normal, but pretend we only know it up to the constant). Use a symmetric random-walk proposal $y = x_t + \varepsilon$ with $\varepsilon \sim \mathcal{N}(0, s^2)$. Symmetry means $q(y \mid x) = q(x \mid y)$, so those terms cancel and the acceptance ratio simplifies to the pure Metropolis form

$$\alpha = \min\left(1, \frac{\pi(y)}{\pi(x_t)}\right) = \min\left(1, e^{-(y^2 - x_t^2)/2}\right).$$

Steps toward 0 (where π is large) are always accepted; steps into the tails are accepted only sometimes. Run for ten thousand iterations and a histogram of the visited states reproduces the bell curve. The step size s matters: too small and the chain inches along, exploring slowly; too large and almost

every bold proposal is rejected, so the chain sticks. A classic rule of thumb targets an acceptance rate near 23% for high-dimensional random-walk Metropolis.

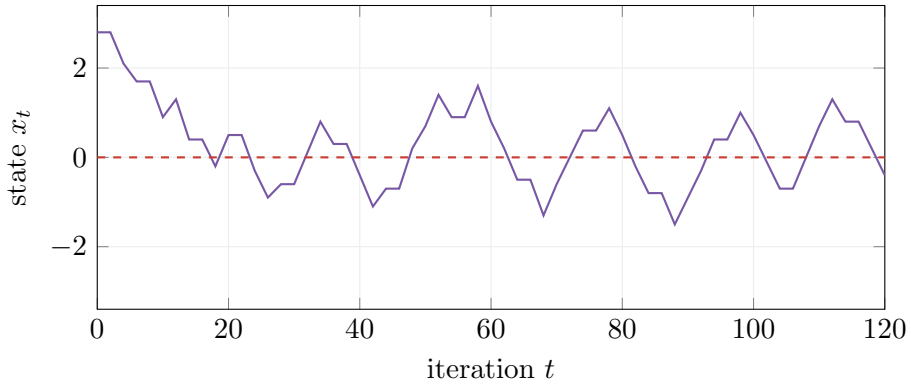


Figure 35.6: A trace of a random-walk Metropolis chain targeting $\mathcal{N}(0, 1)$. The chain wanders around the high-probability center, occasionally venturing into the tails. Flat segments are rejected proposals – the chain stayed put. Time averages along this single path estimate expectations under the normal.

35.5.2 Gibbs Sampling

When the target lives in many dimensions, proposing good joint moves is hard. Gibbs sampling is a clever special case of Metropolis–Hastings that sidesteps the problem: instead of moving all coordinates at once, update them *one at a time*, each from its *conditional* distribution given all the others.

Intuition

Suppose you and a partner are decorating a room and cannot agree on everything at once. So you take turns: you fix the wall color given the current furniture, then your partner picks furniture given the wall color, then you re-pick the wall given the new furniture, and so on. Each step you solve a small one-variable problem, holding everything else fixed. Remarkably, this back-and-forth converges to a draw from the full *joint* distribution.

Formal

Gibbs sampling. To sample a joint distribution $\pi(x_1, \ldots, x_d)$, cycle through the coordinates. At each step, replace coordinate j by a fresh draw from its full conditional:

$$x_j^{\text{new}} \sim \pi\big(x_j \,\big|\, x_1, \ldots, x_{j-1}, x_{j+1}, \ldots, x_d\big).$$

Repeat for $j = 1, 2, \ldots, d$ and loop. The stationary distribution of the resulting chain is exactly π.

Gibbs sampling is Metropolis–Hastings with a proposal that is the exact conditional distribution; one can check that its acceptance probability is always 1. Nothing is ever rejected – every proposed move is taken. This makes Gibbs attractive whenever the full conditionals have nice closed forms, which happens constantly in hierarchical Bayesian models built from conjugate pieces.

Example 35.6 (Gibbs for a bivariate normal)**.** Let (X, Y) be bivariate normal with standard marginals and correlation ρ. The full conditionals are themselves normal – a fact you can derive from the joint density:

$$X \mid Y = y \;\sim\; \mathcal{N}(\rho y,\; 1-\rho^2), \qquad Y \mid X = x \;\sim\; \mathcal{N}(\rho x,\; 1-\rho^2).$$

Gibbs sampling alternates: draw X given the current Y, then Y given the new X, and repeat. After a short burn-in the pairs (X_t, Y_t) trace out the elliptical correlation cloud of the true bivariate normal (Figure 35.7). We sampled a two-dimensional distribution using only one-dimensional normal draws – no matrix square roots, no rejection.

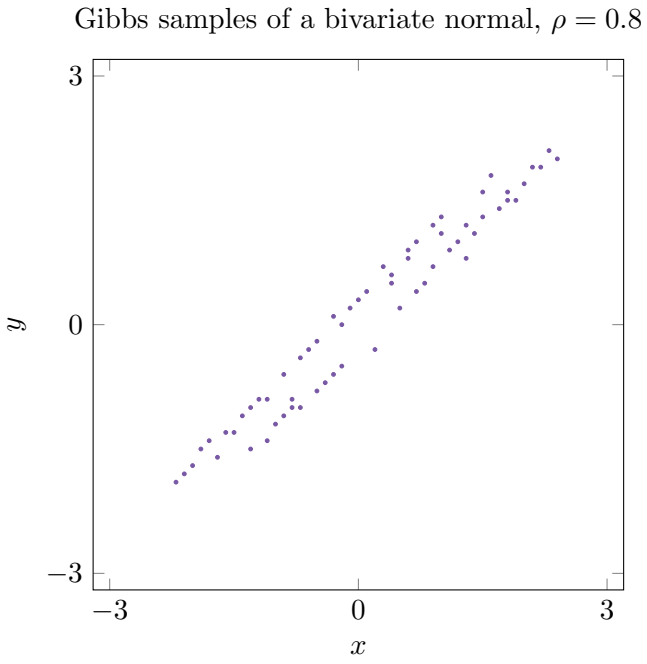


Figure 35.7: Output of Gibbs sampling for a bivariate normal with correlation $\rho = 0.8$. Each point is one (X_t, Y_t) pair produced by alternately drawing one coordinate from its conditional given the other. The cloud is the tilted correlation ellipse of the target – recovered using only one-dimensional normal draws.

Common trap

MCMC samples are *correlated*, and this changes how you read the output. Consecutive states are not independent, so the effective sample size is smaller than the number of iterations – sometimes far smaller. Two practical consequences. First, *burn-in*: the chain starts at an arbitrary point and needs time to "forget" it and reach equilibrium; discard the early states. Second, the naive error bar $\hat{\sigma}/\sqrt{n}$ is *too optimistic* because it assumes independence; the true variance of an MCMC average involves the autocorrelation and is larger. And if the target has well-separated modes, a random-walk chain can get stuck in one mode for a very long time, silently missing the others – a failure that no single-chain diagnostic reliably catches. Run multiple chains from different starts and check that they agree.

Paradox

The convergence you cannot see. Here is the unsettling part of MCMC. The theory guarantees that *eventually* time averages converge to the right answer – but "eventually" can mean longer than the age of the universe, and the output gives you few reliable warnings. A chain trapped in one mode of a multimodal distribution looks perfectly healthy: the trace wanders, the histogram is smooth, the running average settles. It has simply settled to the *wrong* value, because it never discovered the other modes. This is the deep difference between i.i.d. Monte Carlo, whose error the CLT pins down exactly, and MCMC, whose error depends on an unknown mixing time. The resolution is not a formula but a discipline: run several independent chains, start them far apart, compare their answers (the Gelman–Rubin diagnostic formalizes this), and stay humble. MCMC is correct in the limit and treacherous in practice – which is exactly why diagnosing convergence is a serious craft.

35.6 Applications Across Science and Machine Learning

Application

Bayesian inference. The single most important use of MCMC. A Bayesian posterior is $\pi(\theta \mid \text{data}) \propto p(\text{data} \mid \theta)\, p(\theta)$, and the constant of proportionality is an integral over all of parameter space that is almost never computable. Metropolis–Hastings and Gibbs need only the unnormalized product, so they let you draw from the posterior anyway. Tools like Stan, PyMC, and JAGS are, underneath, automated MCMC engines. Every modern hierarchical model – in ecology, astronomy, pharmacology, sports analytics – is fit this way. When you read "we used 4 chains of 2000 iterations," that is MCMC.

Application

Statistical physics. Monte Carlo was born for physics and still lives there. The Boltzmann distribution $\pi(s) \propto e^{-E(s)/kT}$ over the configurations s of a system (say, the spins of the Ising model of magnetism) has an utterly intractable normalizing constant, the partition function. The Metropolis algorithm – proposed in that very 1953 paper – samples configurations to estimate magnetization, heat capacity, and phase transitions. Simulating how a magnet behaves near its critical temperature is a Monte Carlo calculation.

History

The 1953 paper by Metropolis, the Rosenbluths, and the Tellers introduced what we now call the Metropolis algorithm to compute equations of state for interacting particles. In 1970 W. K. Hastings generalized it to asymmetric proposals, giving the Metropolis–Hastings form we use today. For decades the method lived mostly in physics. Then in the early 1990s statisticians (notably Gelfand and Smith, who popularized Gibbs sampling for Bayesian models) realized it was the missing key to practical Bayesian inference. MCMC promptly transformed statistics: problems that had been theoretically Bayesian but computationally hopeless became routine. A physics tool for bombs and magnets became the workhorse of modern data analysis.

Application

Machine learning. Monte Carlo is everywhere in modern ML. Variational inference and the "reparameterization trick" in variational autoencoders are Monte Carlo estimates of intractable expectations over latent variables. Policy-gradient methods in reinforcement learning estimate the gradient of expected reward by sampling trajectories – pure Monte Carlo, often with variance reduction via "baselines" (a control-variate trick). Bayesian neural networks quantify uncertainty with MCMC or its scalable cousins. Diffusion models, which generate images, are built on a Markov chain that gradually denoises noise into data. Dropout at test time can be read as a Monte Carlo estimate of predictive uncertainty. The thread connecting all of these: an expectation no one can compute, estimated by averaging samples.

Application

Finance and risk. Pricing a complex financial derivative means computing $\mathbb{E}[\text{payoff}]$ under a model of how prices move – usually a high-dimensional, path-dependent expectation with no closed form. Banks estimate it by simulating thousands of price paths and averaging the discounted payoffs, with antithetic variates and importance sampling to tighten the estimate. Value at Risk and stress testing simulate portfolio losses to estimate the probability and size of rare, catastrophic outcomes – exactly the rare-event regime where importance sampling earns its keep (Example 35.3).

Why this matters

Step back and admire the unity. A single idea – replace a quantity you cannot compute with an average over samples, and let a limit theorem guarantee the answer – spans card games, hydrogen bombs, the magnetism of iron, the price of options, the posterior of a hierarchical model, and the images conjured by a diffusion model. The law of large numbers promises the average converges; the central limit theorem sizes the error; the theory of Markov chains lets us sample when nothing else can. Monte Carlo is where the abstract probability of the earlier chapters becomes a practical engine for computing the otherwise incomputable. That is as good as applied mathematics gets.

35.7 Exercises

Exercise 35.1 (Estimating π, and its error)**.** You throw $n = 4000$ darts at the unit square and 3,131 land inside the quarter circle. (a) Give the Monte Carlo estimate of π. (b) Using $\hat{p} = 3131/4000$, estimate the standard error of $\hat{\pi} = 4\hat{p}$. (*Hint:* $\hat{p}$ is a sample mean of Bernoullis, so its variance is $\hat{p}(1-\hat{p})/n$, and $\hat{\pi} = 4\hat{p}$.) (c) Roughly how many darts would you need to cut the standard error in half?

Exercise 35.2 (Monte Carlo integration by hand)**.** Consider $I = \int_0^2 x^3\,\mathrm{d}x$ (whose true value is 4). (a) Write I as $(b-a)\,\mathbb{E}[h(U)]$ for $U \sim \text{Uniform}(0,2)$ and identify h. (b) Given five uniform draws on $[0,2]$ – $0.4, 1.7, 0.9, 1.2, 1.9$ – compute the Monte Carlo estimate $\hat{I}_5$. (c) Is your estimate above or below the truth? Such small-sample estimates are noisy; explain what the law of large numbers promises as n grows.

Exercise 35.3 (Why the error does not depend on dimension)**.** A friend claims that integrating in 50 dimensions must be vastly less accurate, per sample, than integrating in 1 dimension. Explain precisely why the Monte Carlo standard error formula $\sigma/\sqrt{n}$ contains no dimension. Then explain the catch: in what sense *can* high dimension still hurt a Monte Carlo estimate? (*Hint:* think about what controls σ.)

Exercise 35.4 (Antithetic variates)**.** You want $I = \int_0^1 \frac{1}{1+x}\,\mathrm{d}x = \ln 2$. (a) Explain why $g(x) = 1/(1+x)$ is monotone on $[0,1]$, so that antithetic pairing $(U, 1-U)$ should reduce variance. (b) With a single uniform draw $U = 0.3$, compute both the plain one-sample estimate $g(U)$ and the antithetic estimate $\frac{1}{2}\big(g(U) + g(1-U)\big)$. (c) Compare each to the true value $\ln 2 \approx 0.6931$ and comment on which is closer.

Exercise 35.5 (Importance sampling weights)**.** You estimate $I = \mathbb{E}_f[g(X)]$ by sampling from a proposal q and weighting by $w(x) = f(x)/q(x)$. (a) Show the estimator $\frac{1}{n}\sum g(X_i)w(X_i)$ with $X_i \sim q$ is unbiased for I. (b) Explain in words why a proposal q with *lighter* tails than f can produce occasional gigantic weights, and why this inflates the variance. (c) What property should a good importance proposal have relative to the integrand $g \cdot f$?

Exercise 35.6 (Rejection sampling efficiency)**.** You sample from a target f using a proposal q with envelope constant $M = 8$ (that is, $f(x) \le 8\,q(x)$ for all x, and 8 is the smallest such constant). (a) What is the probability of accepting on a given attempt? (b) On average, how many proposals are needed per accepted sample? (c) If a tighter envelope with $M = 1.5$ were available, by what factor would the method speed up? Relate this to why rejection sampling fails in high dimensions.

Exercise 35.7 (Metropolis acceptance ratio)**.** Consider Metropolis–Hastings targeting $\pi(x) \propto e^{-x^2/2}$ with a symmetric proposal. The chain is currently at $x_t = 1.0$ and proposes $y = -0.5$. (a) Since the proposal is symmetric, write the acceptance ratio α in terms of π only. (b) Compute α numerically. (c) Is the move to y accepted with certainty, and why does the direction of the move (toward 0) make this unsurprising?

Exercise 35.8 (Detailed balance, conceptual)**.** State the detailed-balance equation for a Markov chain with stationary distribution π and transition rule $P(x \to y)$. (a) Explain in words why detailed balance implies π is stationary – interpret it as a statement about probability "currents" between states. (b) Is detailed balance necessary for π to be stationary, or only sufficient? (*Hint:* recall that not every stationary chain is reversible.)

Exercise 35.9 (Gibbs for the bivariate normal)**.** Using the full conditionals from Example 35.6 with correlation $\rho = 0.8$, suppose the chain is currently at $(X, Y) = (1.5, 0.0)$. (a) Write the distribution from which you draw the new Y given $X = 1.5$. (b) Give its mean and variance numerically. (c) Explain why, as $\rho \to 1$, Gibbs sampling mixes more and more slowly. (*Hint:* how big are the conditional steps when $1 - \rho^2$ is tiny?)

Exercise 35.10 (Diagnosing MCMC, proof-flavored)**.** Let $X_1, X_2, \ldots$ be a stationary chain with $\mathrm{Var}(g(X_t)) = \sigma^2$ and lag-k autocorrelation $\rho_k = \mathrm{Corr}(g(X_t), g(X_{t+k}))$. (a) Show that

$$\mathrm{Var}\left(\frac{1}{n}\sum_{t=1}^{n} g(X_t)\right) = \frac{\sigma^2}{n}\left(1 + 2\sum_{k=1}^{n-1}\Big(1 - \tfrac{k}{n}\Big)\rho_k\right).$$

(*Hint:* expand the variance of the sum and collect terms by lag.) (b) When all $\rho_k > 0$ (positive autocorrelation, typical of MCMC), is the true variance larger or smaller than the i.i.d. value σ^2/n? (c) Explain why this means the naive standard error $\hat{\sigma}/\sqrt{n}$ is overconfident for MCMC output, and define what is meant by the "effective sample size."

Part VII

Rigorous Foundations

Chapter 36

Measure-Theoretic Probability

For thirty-five chapters we have done probability, and we have done a great deal of it: dice and coins, densities and expectations, Markov chains and Brownian motion, the law of large numbers and the central limit theorem. We have computed, proved, simulated, and applied. So you may reasonably ask: if we got this far without measure theory, why do we need it at all? Why end the book by tearing up the floorboards to look at the joists?

The truthful answer is that the floor has been holding all along, but in a few places it has been holding by luck rather than by carpentry. We wrote $\mathbb{P}(A)$ for "the probability of an event A" and quietly assumed that *every* subset A of the sample space could be assigned a probability. For finite and countable sample spaces that assumption is harmless. For the continuum — the interval $[0, 1]$, the real line, the space of infinite coin-flip sequences — it is, astonishingly, *false*. There exist subsets of $[0, 1]$ so pathological that no consistent notion of "length" or "probability" can be attached to them at all. If we pretend otherwise, we can derive flat contradictions: a set that is simultaneously assigned two different probabilities, or a way to cut a region into pieces and reassemble them into something twice as big.

So the continuum forces a choice. Either we abandon the dream of a single clean theory that covers discrete and continuous probability at once, or we are more careful about *which* subsets we allow ourselves to talk about. Measure theory makes the careful choice, and the reward is large: a single set of definitions — σ-algebra, measure, measurable function — under which a coin flip and Brownian motion are *the same kind of object*, governed by the *same* theorems. The sum $\sum_x x\, \mathbb{P}(X = x)$ and the integral $\int x\, f(x)\, \mathrm{d}x$ stop being two separate constructions and become two faces of one integral. Almost everything we proved twice, once for the discrete case and once for the continuous, gets proved once.

This chapter motivates the machinery rather than drowning you in it. We *see* why naive probability on the continuum breaks, build the three central objects with as much intuition as rigor, pin down what "almost surely" really means, and watch the discrete and continuous worlds fuse. The two chapters that follow build the Lebesgue integral and rigorous conditional expectation on this foundation; here we lay the foundation itself.

36.1 The crack in the foundation

Intuition

Probability is, at heart, the assignment of numbers to subsets of a sample space. For dice, the subsets are finite and you can list them, so assigning numbers is no trouble. But when the sample space is something like "a point chosen uniformly at random from $[0, 1]$," the subsets form a wild, sprawling collection. The hidden difficulty is that *not every one of those subsets can carry a sensible probability.* We have to find out which ones can, and restrict our attention to those — without losing anything we actually care about.

Let us make the trouble concrete. Pick a point U uniformly at random from the interval $[0, 1]$. "Uniformly" should mean: the probability of landing in a sub-interval is its length. So $\mathbb{P}(U \in [0.2, 0.5]) = 0.3$, and more generally $\mathbb{P}(U \in [a, b]) = b - a$. This is the most natural continuous experiment there is, the source of every density we ever drew. We would like a function $\mathbb{P}$ that eats *any* subset $A \subseteq [0, 1]$ and returns its probability, agreeing with length on intervals.

What properties must $\mathbb{P}$ have? Three feel non-negotiable, and they are exactly Kolmogorov's axioms from Chapter 3 in disguise:

1. **Non-negativity and normalization.** $\mathbb{P}(A) \geq 0$ for every A, and $\mathbb{P}([0, 1]) = 1$.
2. **Translation invariance.** Sliding a set sideways (wrapping around the circle, so $[0, 1]$ becomes a loop) should not change its probability. A uniform point does not prefer the left half to the right.
3. **Countable additivity.** If $A_1, A_2, \ldots$ are disjoint, then $\mathbb{P}(\bigcup_n A_n) = \sum_n \mathbb{P}(A_n)$. The probability of a countable disjoint union is the sum of the probabilities. We *relied* on this in every continuous-distribution argument in the book.

These three look mild. The shocking fact, discovered by Giuseppe Vitali in 1905, is that *no function $\mathbb{P}$ defined on all subsets of $[0, 1]$ can satisfy all three at once.* Something has to give.

Theorem 36.1 (There is no uniform probability on all subsets)**.** *There is no function $\mathbb{P}$ assigning a number to* every *subset of $[0, 1]$ that is non-negative, normalized with $\mathbb{P}([0, 1]) = 1$, translation-invariant (mod 1), and countably additive.*

Why it works

The proof is one of the most elegant "you can't have it all" arguments in mathematics. We build a single set V — a *Vitali set* — and show that whatever number $\mathbb{P}(V)$ might be leads to a contradiction.

Work on the circle $[0, 1)$ with addition mod 1. Declare two points x, y *equivalent* if their difference $x - y$ is rational. This chops $[0, 1)$ into disjoint equivalence classes; each class is a copy of the rationals shifted by some amount. Using the axiom of choice, pick exactly one representative from each class and collect them into a set V.

Now list the rationals in $[0, 1)$ as $q_0, q_1, q_2, \ldots$ (they are countable), and form the shifted copies $V_n = V + q_n \pmod 1$. Two facts hold:

- The V_n are *pairwise disjoint.* If a point landed in both V_n and V_m, two representatives would differ by a rational, contradicting "one representative per class."

- The V_n *cover everything*: $\bigcup_n V_n = [0,1)$. Any point x is equivalent to the representative of its class, so x differs from that representative by some rational q_n, putting x in V_n.

So $[0,1)$ is the disjoint union of the countably many sets V_n, each a sideways slide of V. By translation invariance, $\mathbb{P}(V_n) = \mathbb{P}(V)$ for all n — call this common value c. By countable additivity,

$$1 = \mathbb{P}([0,1)) = \sum_{n=0}^{\infty} \mathbb{P}(V_n) = \sum_{n=0}^{\infty} c.$$

Now stare at $\sum_n c$. If $c = 0$, the sum is $0 \neq 1$. If $c > 0$, the sum is $+\infty \neq 1$. There is *no* value of c that makes a constant series add to 1. Contradiction. Therefore no such $\mathbb{P}$ exists, and in particular V itself cannot be assigned any probability consistent with the axioms.

Sit with what just happened. We did not assume anything exotic about $\mathbb{P}$ — only the three properties any reasonable "uniform probability" must have. The contradiction means that the set V is *non-measurable*: it is a genuine subset of $[0,1]$ to which no probability can be assigned at all. Not 0, not 1, not anything. The question "what is the chance that U lands in V?" has no answer, and the fault is not ours — it is the set's.

History

The axiom of choice, which we used to pick one representative from each of uncountably many classes, was isolated by Ernst Zermelo in 1904 amid fierce controversy. Vitali's construction (1905) was among the first signs that the axiom has *consequences* — it conjures sets you can prove exist but can never write down or draw. The unease deepened in 1924 when Stefan Banach and Alfred Tarski showed something even wilder: using non-measurable pieces, a solid ball in three dimensions can be cut into finitely many parts and reassembled, by rigid motions alone, into *two* balls each the same size as the original. Volume is not conserved because the pieces have no volume to conserve. The Banach–Tarski "paradox" is not a trick; it is a theorem, and it is the price of pretending every set has a size. Measure theory is the discipline that refuses to pay that price, by simply declining to talk about such sets.

Paradox

"But I can choose a uniform random point — so surely it has *some* chance of landing anywhere?" The intuition feels airtight, and yet it is exactly what fails. The resolution is that "the event $\{U \in V\}$" is not an event at all. Events are special subsets — the ones we can build from intervals by countably many unions, intersections, and complements. The Vitali set is built by a wholly different, non-constructive process (the axiom of choice) and escapes that family entirely. Your random point U does land *somewhere*; it is the *question* "did it land in V?" that is ill-posed, because V is not the kind of subset to which the experiment assigns odds. The lesson is liberating: once we restrict to legitimate events, every paradox above evaporates, and ordinary probability proceeds untroubled.

The way out is now forced. We cannot define $\mathbb{P}$ on *all* subsets. So we will define it on a carefully chosen *collection* of subsets — big enough to contain every set we will ever need (intervals, and everything you can build from them), but small enough to exclude the monsters. That collection is the σ-algebra, and it is the first of our three central objects.

36.2 σ-algebras: the family of legitimate events

Intuition

A σ-algebra is a list of "allowed questions" about a random outcome, closed under the logical operations you actually use. If "did A happen?" is an allowed question, so is "did A *not* happen?" If you can ask about A and about B, you can ask about "A *or* B" and "A *and* B" — and crucially, about countably many such combinations at once. It is the closure of common sense under countable logic.

Let Ω be a sample space — the set of all outcomes. We want to single out a family $\mathcal{F}$ of subsets of Ω, to be called the *events*, such that we can manipulate events the way probability requires.

Definition 36.1 (σ-algebra)**.** A collection $\mathcal{F}$ of subsets of Ω is a *σ-algebra* (or *σ-field*) if:

1. $\Omega \in \mathcal{F}$ (the certain event is allowed);
2. if $A \in \mathcal{F}$ then $A^c = \Omega \setminus A \in \mathcal{F}$ (closed under complement);
3. if $A_1, A_2, \dots \in \mathcal{F}$ then $\bigcup_{n=1}^{\infty} A_n \in \mathcal{F}$ (closed under *countable* unions).

The pair $(\Omega, \mathcal{F})$ is called a *measurable space*, and the members of $\mathcal{F}$ are the *measurable sets* or *events*.

Three short axioms, and they automatically deliver everything else you would want.

Proposition 36.1 (What a σ-algebra contains for free)**.** *If $\mathcal{F}$ is a σ-algebra then: (i) $\emptyset \in \mathcal{F}$; (ii) $\mathcal{F}$ is closed under* countable intersections*; (iii) $\mathcal{F}$ is closed under set differences $A \setminus B$.*

Why it works

Each follows by combining complement with union, the way De Morgan's laws let you trade "and" for "or." (i) $\Omega \in \mathcal{F}$, so $\emptyset = \Omega^c \in \mathcal{F}$. (ii) Given $A_n \in \mathcal{F}$, each complement $A_n^c \in \mathcal{F}$, their union $\bigcup_n A_n^c \in \mathcal{F}$, and its complement is, by De Morgan, $\bigcap_n A_n \in \mathcal{F}$. (iii) $A \setminus B = A \cap B^c$, an intersection of two members. Notice the recurring theme: we only *demanded* complements and countable unions, but those two operations generate the whole comfortable toolkit. That is the point of choosing the axioms with care — a minimal list that closes under everything.

Focus

Why *countable* unions, not finite (too weak) and not arbitrary (too strong)? *Finite* is too weak: continuous probability lives on limits. We constantly write events like "$X \le x$" as $\bigcap_n \{X < x + 1/n\}$, a countable intersection; a finitely-closed family could not even express a half-line as a limit of open ones. *Arbitrary* unions are too strong: every single point $\{x\}$ is a set, and an *uncountable* union of points is any set whatever — including the Vitali monster. Allowing arbitrary unions would drag every pathological subset back in and resurrect Theorem 36.1. Countable is the Goldilocks choice: rich enough for analysis, restrained enough to stay consistent. The single letter "σ" in "σ-algebra" *is* the word "countable."

36.2.1 Examples, from trivial to indispensable

Example 36.1 (The two extremes)**.** The smallest σ-algebra on any Ω is the *trivial* one $\mathcal{F} = \{\emptyset, \Omega\}$: you can only ask "did anything happen?" The largest is the *power set* $\mathcal{F} = 2^{\Omega}$, every subset allowed. For finite or countable Ω the power set is the natural choice and no monsters appear — which is exactly why elementary discrete probability never needed this chapter. The whole difficulty is the continuum.

Example 36.2 (Generating from a question)**.** Often you have a few sets you definitely want to be events — say a single event A — and you want the smallest σ-algebra containing them. For one set A it is $\{\emptyset, A, A^c, \Omega\}$: the four answers to "what is the status of A?" In general, given any collection $\mathcal{C}$ of subsets, the *σ-algebra generated by $\mathcal{C}$*, written $\sigma(\mathcal{C})$, is the intersection of all σ-algebras containing $\mathcal{C}$. (An intersection of σ-algebras is again a σ-algebra — check the three axioms — so this is well defined and is the smallest one that works.)

The single most important generated σ-algebra is the one that tames the real line.

Definition 36.2 (Borel σ-algebra)**.** The *Borel σ-algebra* $\mathcal{B}(\mathbb{R})$ is the σ-algebra on $\mathbb{R}$ generated by the open intervals (equivalently, by the half-lines $(-\infty, x]$). Its members are the *Borel sets*.

What is really happening

The Borel sets are precisely "everything you can reach from intervals by countably many ands, ors, and nots." That includes every open set, every closed set, every single point, the rationals, the irrationals, the Cantor set, and any combination of these you could possibly describe with ordinary mathematical language. It is genuinely hard — impossible without the axiom of choice — to exhibit a subset of $\mathbb{R}$ that is *not* Borel. So when we restrict events to Borel sets, we lose nothing you will ever want and we shed every pathology. The Borel σ-algebra is the precise formalization of "the reasonable subsets of the line." Every density, every CDF, every "$\mathbb{P}(a \le X \le b)$" you wrote in earlier chapters was secretly a statement about Borel sets.

Connection

There is a clean analogy with linear algebra (see the companion linear-algebra volume). A σ-algebra generated by a collection $\mathcal{C}$ is the *closure* of $\mathcal{C}$ under the allowed operations, exactly as the *span* of a set of vectors is the closure under linear combinations, or as the *subgroup generated* by some elements is the closure under the group operation. "Generated by" always means "the smallest structure of this type containing your seeds," obtained by intersecting all candidates. Once you see generation as closure, $\sigma(\mathcal{C})$ stops being mysterious: it is the event-space spanned by your questions.

36.3 Measures and probability measures

We now have the stage (Ω) and the cast of allowed events ($\mathcal{F}$). The last ingredient is the assignment of sizes. A *measure* is the abstract notion of "size" — length, area, volume, mass, or probability — that behaves the way size should.

Definition 36.3 (Measure and probability measure)**.** Let $(\Omega, \mathcal{F})$ be a measurable space. A *measure* is a function $\mu : \mathcal{F} \to [0, \infty]$ such that

1. $\mu(\emptyset) = 0$;
2. **Countable additivity (σ-additivity):** for disjoint $A_1, A_2, \dots \in \mathcal{F}$,

$$\mu\left(\bigcup_{n=1}^{\infty} A_n\right) = \sum_{n=1}^{\infty} \mu(A_n).$$

If in addition $\mu(\Omega) = 1$, we call μ a *probability measure* and usually write it $\mathbb{P}$. The triple $(\Omega, \mathcal{F}, \mathbb{P})$ is a *probability space* — the complete description of a random experiment.

That is it. Kolmogorov's three axioms, which we met in Chapter 3 as rules of thumb, are *exactly* the definition of a probability measure: non-negativity ($\mu \geq 0$ is built into the codomain), normalization ($\mathbb{P}(\Omega) = 1$), and countable additivity. The entire abstract apparatus is just those axioms, said once, for any space at all.

History

Andrey Kolmogorov published *Grundbegriffe der Wahrscheinlichkeitsrechnung* ("Foundations of the Theory of Probability") in 1933, a slim book of about 60 pages that did for probability what Euclid did for geometry. Before it, probability was a collection of brilliant techniques without an agreed foundation; mathematicians argued about what a random variable even *was*. Kolmogorov's insight was that the measure theory developed by Émile Borel and Henri Lebesgue for "length" and "area" around 1900 was *the same theory* needed for probability — you simply normalize total mass to 1 and rename "measurable set" as "event." At a stroke, every theorem about Lebesgue integration became a theorem about expectation. The unification we celebrate in this chapter is his.

36.3.1 Three measures you already know

Example 36.3 (Counting measure: discrete probability)**.** Take Ω countable, $\mathcal{F} = 2^{\Omega}$, and a function $p : \Omega \to [0, 1]$ with $\sum_{\omega} p(\omega) = 1$ — a pmf. Define $\mathbb{P}(A) = \sum_{\omega \in A} p(\omega)$. Countable additivity is just the fact that you can split a sum over a disjoint union into separate sums. This is *all of discrete probability*, repackaged. A fair die is $\Omega = \{1, \dots, 6\}$, $p(\omega) = 1/6$.

Example 36.4 (Lebesgue measure: length, area, the continuum)**.** On $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$ there is a unique measure λ with $\lambda([a, b]) = b - a$ for every interval: *Lebesgue measure*, the rigorous notion of length. It assigns $\lambda(\{x\}) = 0$ to every point (a point has no length), and — a fact that surprises everyone — $\lambda(\mathbb{Q}) = 0$: the rationals, dense though they are, have total length zero, because they are countable and you can cover the nth rational by an interval of length $\varepsilon/2^n$, for total length ε, as small as you like. Restricting λ to $[0, 1]$ gives the uniform probability measure we tried and failed to define on *all* subsets in Section 36.1 — it works perfectly once we confine it to the Borel sets.

Example 36.5 (Dirac measure: a point mass)**.** Fix a point a. The *Dirac measure* δ_a assigns $\delta_a(A) = 1$ if $a \in A$ and 0 otherwise — all the mass piled at a single point. As a probability measure it is a degenerate random variable, "equals a with certainty." Mixing Dirac measures with Lebesgue measure is how a single framework handles random variables that are part-discrete, part-continuous (Section 36.6).

36.3.2 Properties that fall out for free

The same countable additivity that defines a measure forces the everyday identities you have used all along. We prove a representative few; they are templates for the rest.

Proposition 36.2 (Basic measure properties)**.** *For a probability measure $\mathbb{P}$ and events A, B:*

1. ***Complement:*** $\mathbb{P}(A^c) = 1 - \mathbb{P}(A)$.
2. ***Monotonicity:*** *if* $A \subseteq B$ *then* $\mathbb{P}(A) \le \mathbb{P}(B)$.
3. ***Inclusion–exclusion:*** $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B)$.
4. ***Continuity from below:*** *if* $A_1 \subseteq A_2 \subseteq \cdots$ *then* $\mathbb{P}(\bigcup_n A_n) = \lim_{n\to\infty} \mathbb{P}(A_n)$.

Why it works

For (1), $\Omega = A \cup A^c$ disjointly, so additivity gives $1 = \mathbb{P}(\Omega) = \mathbb{P}(A) + \mathbb{P}(A^c)$. For (2), write $B = A \cup (B \setminus A)$ disjointly; then $\mathbb{P}(B) = \mathbb{P}(A) + \mathbb{P}(B \setminus A) \ge \mathbb{P}(A)$ because the extra term is ≥ 0. For (3), split $A \cup B$ into the three disjoint pieces $A \setminus B$, $A \cap B$, $B \setminus A$ and add.
Property (4), *continuity*, is the genuinely new gift of *countable* (rather than merely finite) additivity, and it is what lets probability commute with limits — the engine behind the convergence theorems of Chapters 20–22. Set $B_1 = A_1$ and $B_n = A_n \setminus A_{n-1}$ for $n \ge 2$. The B_n are disjoint, $\bigcup_{k\le n} B_k = A_n$, and $\bigcup_k B_k = \bigcup_n A_n$. By countable additivity,

$$\mathbb{P}\left(\bigcup_n A_n\right) = \sum_{k=1}^{\infty} \mathbb{P}(B_k) = \lim_{n\to\infty} \sum_{k=1}^{n} \mathbb{P}(B_k) = \lim_{n\to\infty} \mathbb{P}(A_n).$$

The limit *is* the infinite sum; that equality is countable additivity wearing a different hat.

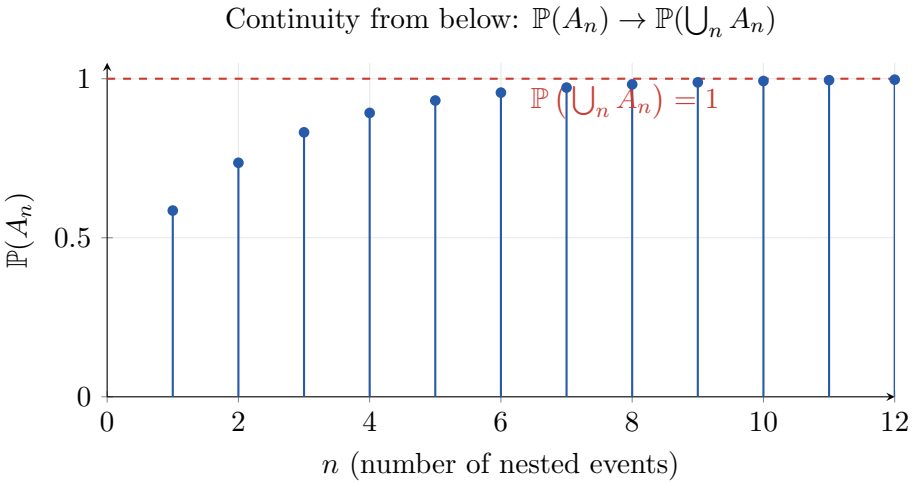


Figure 36.1: As the events A_n swell upward toward their union, their probabilities climb to the probability of the union. Countable additivity is exactly the statement that this limit is attained — probability is *continuous* along nested sequences. Finite additivity alone would not guarantee it.

Figure 36.1 pictures property (4): nested events filling up their union, with probabilities marching to

the limit. Every "in the limit" argument in earlier chapters — the law of large numbers, convergence of CDFs — quietly cashed in this continuity.

36.4 Random variables, done right

In Chapter 6 we said a random variable is "a function from the sample space to the real numbers." That was the right picture, but it was missing one word, and the missing word is the whole content of this section: a random variable must be *measurable*.

Intuition

The issue, in brief. A random variable X turns an outcome into a number. We constantly want to ask "what is the probability that X lands in some range B?" — that is, $\mathbb{P}(X \in B)$. But $\mathbb{P}$ only knows how to score *events* (members of $\mathcal{F}$). So the set of outcomes that X sends into B, namely $X^{-1}(B) = \{\omega : X(\omega) \in B\}$, had better be a legitimate event. Measurability is exactly the requirement that asking a numerical question about X always pulls back to a question $\mathbb{P}$ can answer.

Definition 36.4 (Measurable function / random variable)**.** Let $(\Omega, \mathcal{F})$ be a measurable space. A function $X : \Omega \to \mathbb{R}$ is *measurable* (a *random variable*) if for every Borel set $B \in \mathcal{B}(\mathbb{R})$,

$$X^{-1}(B) = \{\omega \in \Omega : X(\omega) \in B\} \in \mathcal{F}.$$

Equivalently — and this is the version you check in practice — it suffices that $\{\omega : X(\omega) \le x\} \in \mathcal{F}$ for every $x \in \mathbb{R}$.

Why it works

Why is checking the half-lines $\{X \le x\}$ enough to guarantee *all* Borel sets pull back into $\mathcal{F}$? Because preimages respect set operations perfectly: $X^{-1}(B^c) = (X^{-1}(B))^c$ and $X^{-1}(\bigcup_n B_n) = \bigcup_n X^{-1}(B_n)$. So the collection of sets B whose preimage lands in $\mathcal{F}$ is *itself* a σ-algebra. If that collection contains the half-lines $(-\infty, x]$, it must contain the entire σ-algebra they generate — which is $\mathcal{B}(\mathbb{R})$. One family of simple checks propagates to every Borel set automatically. This is the same "generation" principle as before, now doing real work.

What is really happening

Measurability is the mathematical encoding of *information*. The σ-algebra $\mathcal{F}$ is the catalogue of questions you are allowed to ask about the outcome; a function is measurable when every numerical question about it is answerable from that catalogue. This reading becomes powerful with *sub-σ-algebras*: a smaller $\mathcal{G} \subseteq \mathcal{F}$ represents *partial* information — what you would know having seen only some of the experiment. A variable is $\mathcal{G}$-measurable exactly when it can be computed from that partial information. This is the true meaning of conditioning, of filtrations in martingale theory (Chapter 28), and of "what the market knows at time t" in finance. Measurability is not red tape; it is information made precise.

Example 36.6 (Almost everything is measurable)**.** The good news, mirroring the Borel situation, is that measurability is hard to violate by accident. Constants are measurable. Indicators $\mathbf{1}_A$ are

measurable iff $A \in \mathcal{F}$. Sums, products, maxima, minima, and limits of measurable functions are measurable. Every continuous function $\mathbb{R} \to \mathbb{R}$ is Borel measurable, as is every monotone function and every piecewise-defined function you would write down. To build a *non*-measurable function you essentially have to invoke a non-measurable set like Vitali's. In all of applied probability you will never meet one; the definition is there to fence out monsters, not to trip you.

36.4.1 The distribution: pushing the measure forward

A random variable carries the probability measure on Ω over to a probability measure on $\mathbb{R}$, and *that* is the object earlier chapters actually studied.

Definition 36.5 (Distribution / law)**.** The *distribution* (or *law*) of a random variable X is the probability measure $\mathbb{P}_X$ on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$ defined by

$$\mathbb{P}_X(B) = \mathbb{P}\left(X^{-1}(B)\right) = \mathbb{P}(X \in B), \qquad B \in \mathcal{B}(\mathbb{R}).$$

This is the *pushforward* of $\mathbb{P}$ by X. It is a bona fide probability measure on $\mathbb{R}$ — measurability is exactly what makes $X^{-1}(B)$ a legal event so the right side makes sense, and countable additivity transfers because preimages respect disjoint unions.

Why this matters

Now comes the striking part. Two random variables can live on *completely different* probability spaces — one built from coin flips, one from a spinning pointer — and yet have the *same* distribution $\mathbb{P}_X$. When we said in earlier chapters "let $X \sim \mathcal{N}(0,1)$," we never specified the underlying Ω, and it never mattered. The reason it never mattered is precisely this: all the probabilistic content of X lives in its pushforward law on $\mathbb{R}$, not in the scaffolding Ω it was built from. Measure theory explains *why* the underlying sample space could always be left politely unmentioned. The distribution is the shadow X casts on the real line, and that shadow is all we ever compute with.

Connection

The cumulative distribution function $F_X(x) = \mathbb{P}_X((-\infty, x]) = \mathbb{P}(X \le x)$ is just the law evaluated on half-lines. Because the half-lines generate $\mathcal{B}(\mathbb{R})$, the CDF *determines the entire law* — knowing $\mathbb{P}(X \le x)$ for every x pins down $\mathbb{P}(X \in B)$ for every Borel B. That is the rigorous reason the CDF was so useful a summary back in Chapters 6 and 10: it is a complete fingerprint of the distribution, and measure theory is what certifies the fingerprint is complete.

36.5 "Almost surely": events of probability one

You have seen the phrase "almost surely" (a.s.) and "with probability 1" throughout the convergence chapters. Measure theory gives it an exact and surprisingly subtle meaning.

Definition 36.6 (Null sets and almost-sure events)**.** An event N is a *null set* if $\mathbb{P}(N) = 0$. A statement holds *almost surely* (or *almost everywhere*, in pure measure theory) if the set of outcomes where it *fails* is contained in a null set — i.e. it holds except on a set of probability 0.

The subtlety is that *"probability zero" is not the same as "impossible."* On $\Omega = [0,1]$ with uniform measure, the event $\{U = 0.5\}$ is a single point, so $\mathbb{P}(U = 0.5) = 0$ — yet it is a perfectly possible outcome; the point 0.5 is right there in the sample space. Indeed *every* particular value the random point takes had probability 0 before the experiment, and yet some value occurs. Probability 0 means "negligible in aggregate," not "cannot happen."

Paradox

The dartboard paradox. Throw a dart at the interval $[0,1]$; it lands at a uniformly random point U. The probability it hits any specific point is 0. So the probability it hits *some* point should be "a sum of zeros" $= 0$ — yet the dart certainly lands somewhere. Where did the probability come from? *Resolution:* the union $[0,1] = \bigcup_{x\in[0,1]}\{x\}$ is an *uncountable* union, and additivity — even countable additivity — says *nothing* about uncountable sums of zeros. Countable additivity gives $\mathbb{P}(\mathbb{Q}\cap[0,1]) = \sum_q 0 = 0$, correctly, because the rationals are countable. But you may not add up the uncountably many points of $[0,1]$ that way. The dartboard paradox is, at root, a misuse of additivity across an uncountable index set — the very abuse the σ in σ-algebra was designed to forbid. The continuum holds positive probability precisely because it cannot be disassembled into countably many zero-probability atoms.

Focus

Why "almost surely" is the natural mode of truth in continuous probability: because changing a random variable on a null set changes *nothing* you can measure. If $X = Y$ except on a set of probability 0, then $\mathbb{P}(X \in B) = \mathbb{P}(Y \in B)$ for every B, so X and Y have identical distributions, identical expectations, identical everything observable. Measure theory therefore treats functions that agree almost surely as the same object. This is why theorems are stated "$X_n \to X$ almost surely" rather than "everywhere": demanding convergence at *every* outcome, including a harmless null set of exceptions, would be both too strong and pointless. Truth up to a null set is the right grain of resolution.

Example 36.7 (Strong law, reread)**.** The strong law of large numbers (Chapter 20) says $\bar{X}_n \xrightarrow{a.s.} \mu$. Decoded: the set of infinite outcome sequences $\omega = (\omega_1, \omega_2, \dots)$ for which the running average $\bar{X}_n(\omega)$ *fails* to converge to μ is a null set. Sequences that misbehave certainly *exist* — "always heads forever" is one — but together they form a set of probability 0. The strong law is, word for word, an almost-sure statement, and it can only even be *phrased* on a probability space rich enough to host infinite sequences, namely $\Omega = \{H, T\}^{\mathbb{N}}$ with a product σ-algebra. Measure theory is what lets us build that space and put a coherent probability on it.

36.6 One framework, two worlds: unifying discrete and continuous

We promised that measure theory dissolves the wall between discrete and continuous probability. Now we can make that promise explicit.

In Chapter 8 expectation was a *sum* for discrete variables, $\mathbb{E}[X] = \sum_x x\,\mathbb{P}(X = x)$, and an *integral* for continuous ones, $\mathbb{E}[X] = \int x\, f(x)\,\mathrm{d}x$. We proved every property (linearity, LOTUS, etc.) twice, once in each language. The Lebesgue integral of the next chapter erases the duplication: *both* are the single object

$$\mathbb{E}[X] = \int_\Omega X \,\mathrm{d}\,\mathbb{P},$$

the integral of the function X against the probability measure $\mathbb{P}$. The two familiar formulas are what this one integral *becomes* when $\mathbb{P}$ is a counting measure (you get a sum) versus when $\mathbb{P}$ has a density (you get a Riemann-style integral). One definition, two appearances.

The bridge that names the density rigorously is the Radon–Nikodym theorem, and we can state its meaning now even before the next chapter proves it.

Focus

A distribution $\mathbb{P}_X$ on $\mathbb{R}$ has a *density* f exactly when it is "compatible with length" — when every set of *zero length* also has *zero probability* (this compatibility is called *absolute continuity* with respect to Lebesgue measure). In that case f is the local exchange rate between probability and length: $\mathbb{P}_X(B) = \int_B f \, \mathrm{d}\lambda$. A distribution is *discrete* when, by contrast, all its mass sits on a countable set of atoms (a sum of Dirac measures). And nothing forces a distribution to be one or the other.

Example 36.8 (A variable that is neither discrete nor continuous)**.** Let T be the time you wait at a crosswalk. With probability $1/2$ the light is already green and $T = 0$; otherwise T is uniform on $(0, 1)$ minutes. Then

$$\mathbb{P}_T = \tfrac{1}{2}\,\delta_0 + \tfrac{1}{2}\,\lambda|_{(0,1)},$$

half a point mass at 0 plus half a stretch of uniform density. This T has no pmf (it spreads mass over an interval) and no density (it puts a positive lump on the single point 0, where any density would integrate to 0). Pre-measure theory, T fell between the cracks — not in our discrete chapters, not in our continuous chapters. As a measure, T is no trouble at all: a measure can be any mixture of atoms and spread, and $\mathbb{E}[T] = \int_\Omega T \, \mathrm{d}\,\mathbb{P}$ handles it in one stroke. *This* is the unification working.

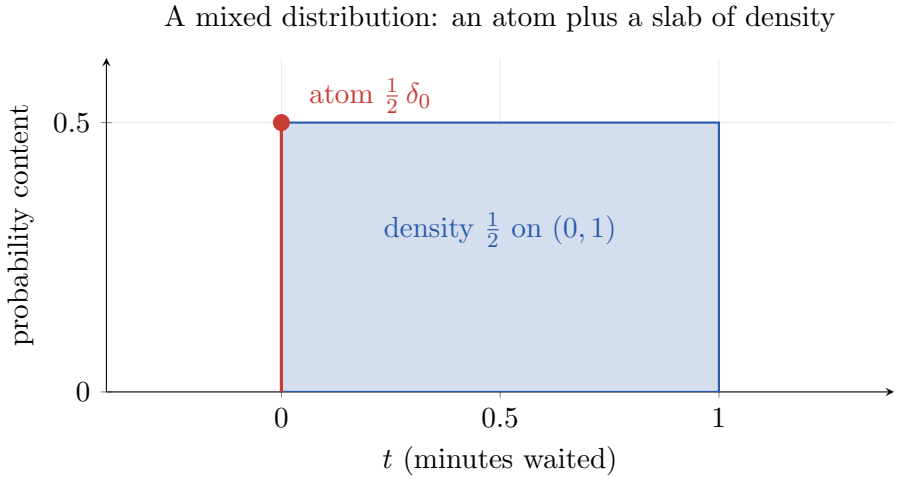


Figure 36.2: The waiting time T of Example 36.8. Half the probability is a point mass (the red spike at $t = 0$: "light already green"), half is spread as a uniform density over $(0, 1)$ (the blue slab). Neither a pmf nor a pdf can describe this, but a single *measure* does so effortlessly — which is the whole point of the measure-theoretic viewpoint.

The same mixed law looks even cleaner through its CDF, where the atom and the spread leave unmistakable signatures.

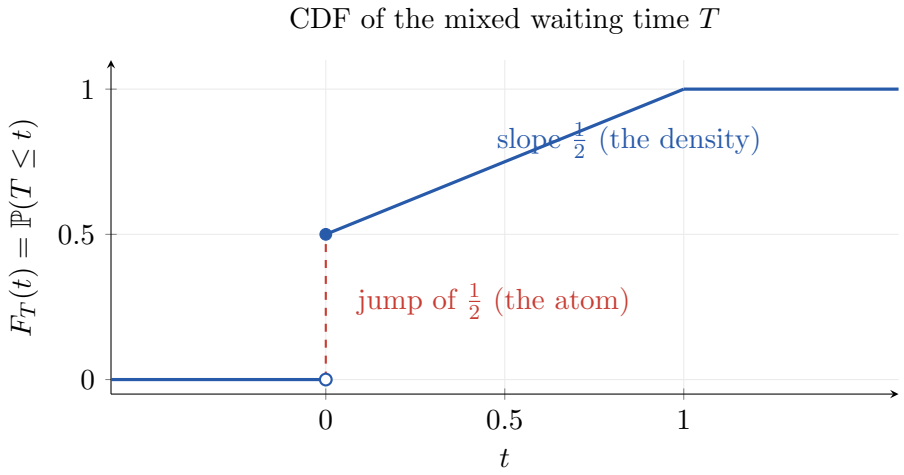


Figure 36.3: The cumulative distribution function of T. The jump of height $\frac{1}{2}$ at $t = 0$ is the point mass $\frac{1}{2}\,\delta_0$; the steady rise of slope $\frac{1}{2}$ across $(0, 1)$ is the uniform density. A pure pmf would give a staircase of jumps; a pure density would give a smooth curve with no jumps. A mixed law shows both at once, and the CDF — which always exists — is the one description that copes with every distribution. (Exercise 36.8 asks you to derive this curve.)

Application

Why machine learning quietly runs on measures. A modern generative model defines a probability distribution over a high-dimensional space (images, text embeddings, sensor readings). Loss functions like the KL divergence $\mathrm{KL}(P \,\|\, Q) = \int \log \frac{\mathrm{d}P}{\mathrm{d}Q}\,\mathrm{d}P$ are written with exactly the Radon–Nikodym derivative $\mathrm{d}P/\mathrm{d}Q$ of this chapter; it only makes sense when P is absolutely continuous with respect to Q (else the divergence is $+\infty$, the model's way of saying "Q rules out something P insists on"). Diffusion models, normalizing flows, and variational inference all manipulate densities-relative-to-a-reference — pushforwards of a base measure through a learned map, scored by change-of-variables. Importance sampling reweights one measure to estimate expectations under another via the same derivative. The engineer rarely says "measure," but every change-of-distribution step is a measure-theoretic operation, and the framework of this chapter is what guarantees the manipulations are legal. The same goes for signal processing (see the companion volume): a power spectrum is a measure on frequency, and white noise is rigorously a random *measure*, not a function.

36.7 Putting it together: the anatomy of a probability space

Let us assemble the three objects into the single picture they were always meant to form. Every random experiment in this book — and in all of probability — is a triple

$$(\Omega, \mathcal{F}, \mathbb{P}),$$

the *probability space*, on which random variables live as measurable functions:

- Ω — the *sample space*, the set of all outcomes ω. Often left unspecified, because only the induced laws matter.

- $\mathcal{F}$ — the *σ-algebra* of events, the catalogue of answerable yes/no questions, closed under complement and countable union. It carves out the well-behaved subsets and bars the non-measurable monsters.
- $\mathbb{P}$ — the *probability measure*, assigning each event a number in $[0, 1]$, countably additive, with $\mathbb{P}(\Omega) = 1$.

A random variable $X : \Omega \to \mathbb{R}$ is a measurable function; its law $\mathbb{P}_X = \mathbb{P} \circ X^{-1}$ is a probability measure on $\mathbb{R}$; its CDF, density or pmf, expectation, and variance are all read off from $\mathbb{P}_X$.

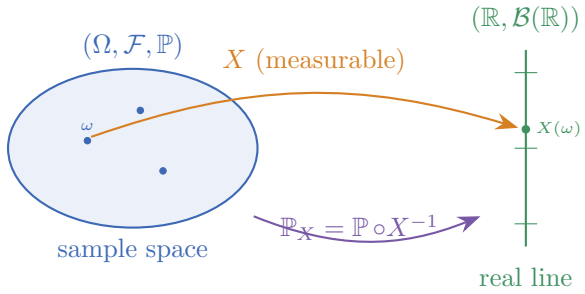


Figure 36.4: The anatomy of probability. A random variable X is a measurable map from the probability space $(\Omega, \mathcal{F}, \mathbb{P})$ to the real line. Each outcome ω is sent to a number $X(\omega)$; measurability ensures every Borel question about the number pulls back to an event $\mathbb{P}$ can score. The pushforward $\mathbb{P}_X = \mathbb{P} \circ X^{-1}$ is the distribution — the only part we ever compute with, which is why Ω may stay anonymous.

Pause

Step back and notice how little we actually had to invent. Three definitions — σ-algebra, measure, measurable function — and a single example of a non-measurable set to motivate them. From those, the entire scaffolding of rigorous probability stands, and every theorem of the previous chapters acquires a precise home. The machinery is not a new subject; it is the *justification* of the old one. You already knew the music; this chapter showed you the instrument it was being played on.

36.8 What measure theory buys you, in one paragraph

Why endure the abstraction? Concretely, measure theory gives four things you cannot get without it. *First*, consistency: it guarantees the continuum can carry a probability at all, which Section 36.1 showed is not automatic. *Second*, unification: discrete sums and continuous integrals, pmfs and densities and mixtures, all become special cases of one integral against one measure (Section 36.6). *Third*, limits: countable additivity makes probability continuous along sequences (Proposition 36.2), licensing every "$\to$" in the convergence theorems and the construction of infinite-dimensional spaces like the path space of Brownian motion. *Fourth*, information: sub-σ-algebras give a precise language for "what is known so far," which is the backbone of conditional expectation (Chapter 38), martingales, and filtrations. None of these are luxuries; they are the load-bearing beams under everything you have already used.

Exercises

Exercise 36.1 (Warm-up with the axioms)**.** Let $\Omega = \{1, 2, 3, 4\}$. Decide which of the following are σ-algebras on Ω, and for those that are not, name the failing axiom. (a) $\{\emptyset, \Omega\}$. (b) $\{\emptyset, \{1\}, \{2, 3, 4\}, \Omega\}$. (c) $\{\emptyset, \{1\}, \{2\}, \Omega\}$. (d) The collection of all subsets with an even number of elements.

Exercise 36.2 (Generated σ-algebra)**.** On $\Omega = \{1, 2, 3, 4, 5, 6\}$ (a die), write down the smallest σ-algebra containing the events $A = \{1, 2\}$ and $B = \{2, 3\}$. List all its members. How many are there? (Hint: it is generated by the "atoms" you can no longer split using A, B, and complements.)

Exercise 36.3 (Intersections preserve structure)**.** Prove that if $\mathcal{F}_1$ and $\mathcal{F}_2$ are σ-algebras on the same Ω, then $\mathcal{F}_1 \cap \mathcal{F}_2$ is a σ-algebra. Then give an example showing $\mathcal{F}_1 \cup \mathcal{F}_2$ need *not* be. (This is why "generated by" is defined via intersection, not union.)

Exercise 36.4 (Countable sets are null)**.** Using Lebesgue measure λ on $\mathbb{R}$, prove $\lambda(\mathbb{Q}) = 0$ by covering the nth rational with an interval of length $\varepsilon/2^n$ and using countable subadditivity. Then explain in one sentence why this does *not* let you conclude $\lambda([0, 1]) = 0$, even though $[0, 1]$ is also a union of single points.

Exercise 36.5 (Continuity from above)**.** State and prove the "continuity from above" analogue of Proposition 36.2(4): if $A_1 \supseteq A_2 \supseteq \cdots$ and $\mathbb{P}(A_1) < \infty$, then $\mathbb{P}(\bigcap_n A_n) = \lim_n \mathbb{P}(A_n)$. (Hint: apply continuity from below to the complements A_n^c.) Give an example on $\mathbb{R}$ with Lebesgue measure where dropping the finiteness assumption makes the conclusion fail.

Exercise 36.6 (Measurability via half-lines)**.** Let $X : \Omega \to \mathbb{R}$ satisfy $\{X \le x\} \in \mathcal{F}$ for every rational x. Prove that $\{X \le x\} \in \mathcal{F}$ for every *real* x, hence X is a random variable. (Hint: write $\{X \le x\}$ as a countable intersection over rationals decreasing to x, and use closure.)

Exercise 36.7 (Pushforward is a measure)**.** Let X be a random variable on $(\Omega, \mathcal{F}, \mathbb{P})$ and define $\mathbb{P}_X(B) = \mathbb{P}(X^{-1}(B))$ for Borel B. Verify directly that $\mathbb{P}_X$ satisfies the two axioms of a probability measure (Definition 36.3 plus normalization). Which property of preimages do you need for countable additivity?

Exercise 36.8 (A mixed distribution)**.** For the crosswalk waiting time T of Example 36.8 with law $\mathbb{P}_T = \frac{1}{2}\delta_0 + \frac{1}{2}\lambda|_{(0,1)}$, compute (a) $\mathbb{P}(T = 0)$; (b) $\mathbb{P}(T > 1/2)$; (c) the CDF $F_T(t)$ for all t; and (d) $\mathbb{E}[T]$, treating the expectation as $\frac{1}{2} \cdot 0 + \frac{1}{2}\int_0^1 t \, dt$. Sketch F_T and identify the jump.

Exercise 36.9 (Almost surely versus surely)**.** Let $U \sim \text{Uniform}(0, 1)$. Give an event that is (a) impossible (empty); (b) possible but of probability 0; (c) certain to hold (all of Ω); (d) of probability 1 but *not* all of Ω. Explain why (b) and (d) are two sides of the same phenomenon, and why no analogue of (b) or (d) exists when Ω is finite.

Exercise 36.10 (Why countable, not arbitrary — a proof)**.** Suppose, for contradiction, that we strengthened Definition 36.1 to require closure under *arbitrary* unions. Show that the only such "σ-algebra" on $\mathbb{R}$ containing every singleton $\{x\}$ is the full power set $2^{\mathbb{R}}$. Combine this with Theorem 36.1 to explain in a sentence why arbitrary-union closure is incompatible with a translation-invariant uniform probability — i.e. why the word "countable" in the definition is not negotiable.

Exercise 36.11 (Conceptual: information as a σ-algebra)**.** Two fair coins are flipped; $\Omega = \{HH, HT, TH, TT\}$ with the power set and uniform $\mathbb{P}$. Let $\mathcal{G}$ be the smallest σ-algebra making the random variable "number of heads" measurable. (a) List the members of $\mathcal{G}$. (b) Is the random variable "did the *first* coin land heads?" measurable with respect to $\mathcal{G}$? Explain what this says about which questions "knowing only the number of heads" lets you answer.

Exercise 36.12 (Proof: agreement a.s. implies equal law)**.** Let X, Y be random variables on the same space with $\mathbb{P}(X = Y) = 1$. Prove that $\mathbb{P}_X = \mathbb{P}_Y$, i.e. $\mathbb{P}(X \in B) = \mathbb{P}(Y \in B)$ for every Borel set B. (Hint: the events $\{X \in B\}$ and $\{Y \in B\}$ differ only within the null set $\{X \neq Y\}$; bound the symmetric difference.) Conclude that X and Y have the same CDF, expectation, and variance — the precise sense in which measure theory "cannot tell them apart."

Chapter 37

The Lebesgue Integral and Expectation

You have been computing expectations for many chapters now. For a discrete random variable you summed $\sum_x x\,\mathbb{P}(X = x)$; for a continuous one you integrated $\int x f(x)\,\mathrm{d}x$. Two formulas, two worlds. Sometimes you met a variable that was part discrete and part continuous — a payout that is zero with probability $1/2$ and otherwise exponential — and you patched the two formulas together by hand. It worked, but it felt like a confession: we have no single definition of expectation, only a family of recipes.

This chapter fixes that. There *is* one definition of $\mathbb{E}[X]$, valid for every random variable on every probability space, discrete, continuous, mixed, or stranger still. It is the *Lebesgue integral*. Once you have it, the sum and the Riemann integral both fall out as special cases, the convergence theorems you have been quietly using become real theorems, and the deepest limit results of probability — the laws of large numbers, the martingale theorems — rest on solid ground.

The story is also one of the great ideas of twentieth-century mathematics, and it begins with a change of perspective so simple it can be drawn on a napkin.

37.1 Why the Riemann Integral Is Not Enough

Intuition

Riemann slices the *domain*: chop the x-axis into thin vertical strips and add up their areas. Lebesgue slices the *range*: chop the y-axis into thin horizontal layers and ask, for each height, *how much x lives above that height?* Same area, opposite cut. The Lebesgue cut is the one probability wants, because "how much x lives there" is exactly a measure — a probability.

The familiar Riemann integral, for all its virtues in calculus, buckles under the demands of probability. Here is why.

37.1.1 The cashier's parable

Imagine you must total a pile of coins. The Riemann cashier sweeps the coins left to right in a line and adds them one by one: $5 + 25 + 1 + 10 + 25 + \dots$ The Lebesgue cashier instead *sorts* the coins into stacks — all the pennies, all the nickels, all the quarters — counts each stack, and computes $1 \cdot (\#\text{ pennies}) + 5 \cdot (\#\text{ nickels}) + 25 \cdot (\#\text{ quarters})$. Both get the right total. But the second method does not care about the *order* the coins came in, and it generalizes to wildly irregular piles. That sorting-by-value is the Lebesgue idea: group the points of the domain by the *value* of the function, then weigh each group.

37.1.2 A function Riemann cannot touch

Example 37.1 (The Dirichlet function)**.** Define $D\colon [0,1] \to \mathbb{R}$ by $D(x) = 1$ if x is rational and $D(x) = 0$ if x is irrational. Try to Riemann-integrate it. In every subinterval, no matter how small, there are both rationals and irrationals, so the lower Riemann sum is always 0 and the upper sum is always 1. They never meet: D is *not Riemann integrable*.

Yet morally the answer is obvious. The rationals in $[0,1]$ are countable, hence negligible — they take up no length. The function is 0 "almost everywhere," so its integral should be 0. Lebesgue's theory delivers exactly this: $\int_0^1 D\,\mathrm{d}x = 0$. Riemann cannot, because it is forced to resolve D strip by strip, and every strip is hopelessly mixed.

Why it works

The deeper failure of Riemann is not about pathological functions — it is about *limits*. Probability is the study of limits: laws of large numbers, central limit theorems, convergence of estimators. You constantly want to say "the limit of the expectations is the expectation of the limit," $\lim_n \mathbb{E}[X_n] = \mathbb{E}[\lim_n X_n]$. With Riemann integrals this exchange can fail badly, and there is no clean theorem telling you when it is safe. A pointwise limit of Riemann-integrable functions need not even be Riemann integrable (build D as a limit of finite indicator functions of rationals and you have an example). Lebesgue's integral is *built to survive limits* — that is its whole reason for existing, and the convergence theorems of Section 37.4 are the payoff.

History

Henri Lebesgue presented his theory in his 1902 thesis, *Intégrale, longueur, aire*. He described the change of viewpoint with his own merchant's metaphor: a shopkeeper totalling the day's takings can add the bills in the order they were received (Riemann), or first sort the banknotes by denomination and count each kind (Lebesgue). Émile Borel had already built the notion of measure for sets; Lebesgue's leap was to use measure to integrate *functions*, and to prove that the resulting integral behaves beautifully under limits. Within three decades Kolmogorov would take Lebesgue's machinery and, in his 1933 *Grundbegriffe*, declare that a probability is just a measure of total mass one and an expectation is just a Lebesgue integral — the foundation the rest of this book stands on.

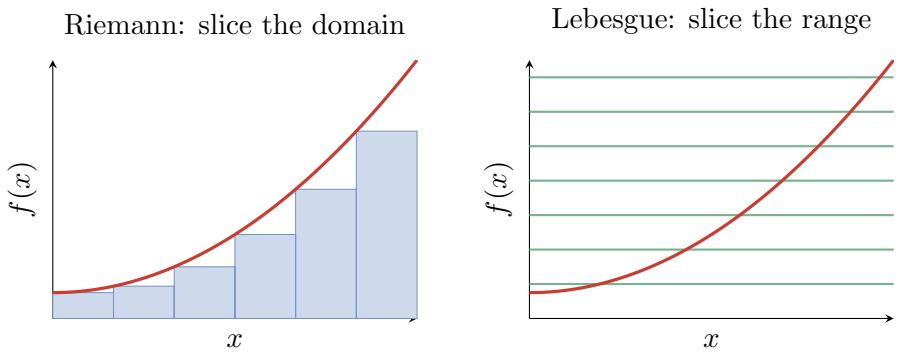


Figure 37.1: Two ways to compute the same area. Left, Riemann partitions the x-axis into vertical strips. Right, Lebesgue partitions the y-axis into horizontal layers and measures the set of x lying above each height. The right-hand cut is the one that turns "area" into "probability times value."

37.2 The Building Blocks: Simple Functions

We build the integral in four stages, each one extending the last: *indicators*, then *simple functions*, then *nonnegative measurable functions*, then *general (signed) functions*. This is the ladder essentially every textbook climbs, and it is worth climbing once yourself so the convergence theorems later feel inevitable rather than magical.

Throughout, fix a measure space $(\Omega, \mathcal{F}, \mu)$: a set Ω, a σ-algebra $\mathcal{F}$ of "measurable" subsets, and a measure μ assigning sizes to them. For probability we will later take $\mu = \mathbb{P}$ with $\mathbb{P}(\Omega) = 1$, but nothing in the construction needs that.

37.2.1 Step 1: indicators

The atom of the whole theory is the indicator $\mathbf{1}_A$ of a measurable set $A \in \mathcal{F}$, which is 1 on A and 0 off it. Its integral is, by *decree*, the size of A:

$$\int_\Omega \mathbf{1}_A \, d\mu \;=\; \mu(A).$$

There is nothing to prove; this is the seed. "The integral of a flat unit-height plateau over A is the measure of A." In probability, where $\mu = \mathbb{P}$, this reads $\mathbb{E}[\mathbf{1}_A] = \mathbb{P}(A)$ — the expectation of an indicator is the probability of the event, a fact you have used since the very first chapters on expectation. Now you see it is not a coincidence but the *definition* restricted to its simplest case.

37.2.2 Step 2: simple functions

Definition 37.1 (Simple function)**.** A *simple function* is a measurable function $s\colon \Omega \to \mathbb{R}$ that takes only finitely many distinct values $a_1, \dots, a_n$. Writing $A_k = \{\omega : s(\omega) = a_k\}$ for the (measurable) level sets, we have the canonical representation

$$s \;=\; \sum_{k=1}^{n} a_k \, \mathbf{1}_{A_k},$$

where the A_k are disjoint and cover Ω.

A simple function is a finite, irregular staircase — but a staircase laid out in the *range*, not the domain. Each A_k can be any measurable set, no matter how scattered; it need not be an interval. This freedom is precisely what let Lebesgue tame the Dirichlet function.

Definition 37.2 (Integral of a nonnegative simple function)**.** For a nonnegative simple function $s = \sum_{k=1}^{n} a_k \mathbf{1}_{A_k}$ (canonical form, $a_k \geq 0$), define

$$\int_\Omega s \, \mathrm{d}\mu \;=\; \sum_{k=1}^{n} a_k \, \mu(A_k).$$

Read it aloud: *value times the measure of the set where that value is taken, summed over values.* That is the sorted-coins total. Every later integral is a limit of these.

Walking question

A subtlety worth pausing on: a simple function can be written in many ways, not just canonically. Is $\sum a_k \mu(A_k)$ the same number regardless of representation? It is, and the proof is the everyday fact that overlapping descriptions can be refined to a common one. If $s = \sum a_k \mathbf{1}_{A_k} = \sum b_j \mathbf{1}_{B_j}$, intersect the pieces into $C_{kj} = A_k \cap B_j$; on each C_{kj} both representations agree that the value is $a_k = b_j$, and finite additivity of μ makes the two sums collapse to the same total. So the integral is *well defined*. Convince yourself of this; it is the only "bookkeeping" lemma in the whole construction, and everything downstream inherits its linearity and monotonicity from it.

Two properties of the simple integral are immediate from finite additivity and will propagate up the ladder:

- **Linearity:** $\int (\alpha s + \beta t)\, \mathrm{d}\mu = \alpha \int s \, \mathrm{d}\mu + \beta \int t \, \mathrm{d}\mu$ for nonnegative simple s, t and $\alpha, \beta \geq 0$.
- **Monotonicity:** if $s \leq t$ pointwise then $\int s \, \mathrm{d}\mu \leq \int t \, \mathrm{d}\mu$.

37.2.3 Step 3: nonnegative measurable functions

Here is the master stroke. To integrate an arbitrary nonnegative measurable function $f \geq 0$, we do not approximate it by staircases from the *outside* (which is what dooms Riemann); we squeeze it from *below* by simple functions and take the best lower estimate.

Definition 37.3 (Lebesgue integral, nonnegative case)**.** For measurable $f \colon \Omega \to [0, \infty]$,

$$\int_\Omega f \, \mathrm{d}\mu \;=\; \sup\Big\{ \int_\Omega s \, \mathrm{d}\mu \;:\; s \text{ simple},\; 0 \leq s \leq f \Big\}.$$

The supremum is over *all* simple functions sitting under f. It always exists in $[0, \infty]$ (it may be $+\infty$).

Why does taking the sup of lower estimates work here when it failed for Riemann? Because Lebesgue's simple functions are allowed to use *arbitrary measurable level sets*, so they can hug any measurable f from below as tightly as we like. The following standard construction makes "as tightly as we like" concrete and is the engine behind the Monotone Convergence Theorem.

Theorem 37.1 (Approximation by simple functions)**.** *For every measurable $f\colon \Omega \to [0,\infty]$ there is an increasing sequence of nonnegative simple functions $s_1 \le s_2 \le \cdots$ with $s_n(\omega) \to f(\omega)$ for every ω.*

Why it works

The construction is the "dyadic staircase." At stage n, slice the range $[0,\infty)$ into layers of height 2^{-n} up to level n, and round f *down* to the nearest layer (capping at n):

$$s_n \;=\; \sum_{k=0}^{n\,2^n-1} \frac{k}{2^n}\,\mathbf{1}_{\{\frac{k}{2^n} \le f < \frac{k+1}{2^n}\}} \;+\; n\,\mathbf{1}_{\{f \ge n\}}.$$

Each level set is measurable because f is, so each s_n is a legitimate simple function. As n grows, the layers thin out (height 2^{-n}) and the cap rises ($n \to \infty$), so $s_n \uparrow f$ at every point. The picture is a staircase of horizontal — not vertical — treads creeping up under the curve. This single construction is why "slice the range" is not a slogan but an algorithm.

Dyadic simple-function staircase rising under f

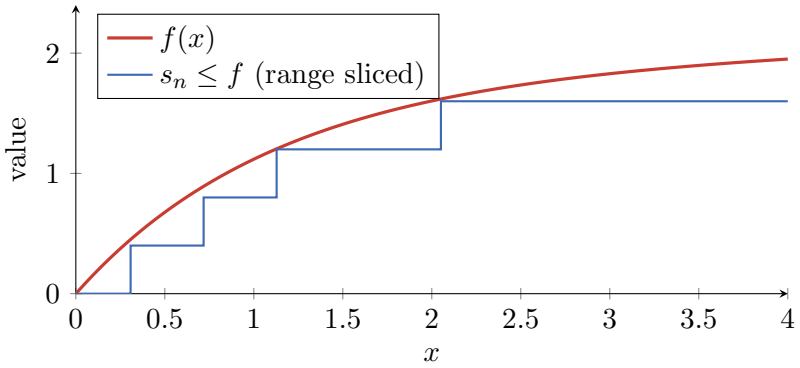


Figure 37.2: The Lebesgue staircase s_n rounds f *down* to a grid of values on the y-axis. Refining the grid (smaller steps, higher cap) pushes $s_n \uparrow f$ everywhere. Each tread sits over a measurable level set, which is all the construction ever requires.

37.2.4 Step 4: general functions

Finally, any measurable $f\colon \Omega \to \mathbb{R}$ splits into its positive and negative parts,

$$f^+ = \max(f,0), \qquad f^- = \max(-f,0), \qquad f = f^+ - f^-,$$

both nonnegative and measurable. Define

$$\int_\Omega f\,d\mu \;=\; \int_\Omega f^+\,d\mu \;-\; \int_\Omega f^-\,d\mu,$$

provided at least one of the two pieces is finite (otherwise we would be writing $\infty - \infty$, which is meaningless). When both are finite — equivalently when $\int |f|\,d\mu = \int f^+\,d\mu + \int f^-\,d\mu < \infty$ — we call f *integrable* and write $f \in L^1(\mu)$.

Formal

The full Lebesgue integral, in one breath:

$$\int_\Omega f \,\mathrm{d}\mu \;=\; \underbrace{\sup_{0\le s\le f^+} \sum_k a_k\,\mu(A_k)}_{\text{positive part}} \;-\; \underbrace{\sup_{0\le s\le f^-} \sum_k a_k\,\mu(A_k)}_{\text{negative part}}.$$

Build from indicators ($\int \mathbf{1}_A = \mu(A)$), sum them into simple functions, take suprema to reach nonnegative functions, and subtract parts for signed functions. Linearity and monotonicity, established for simple functions, climb the ladder rung by rung. That is the entire integral.

37.3 Expectation Is a Lebesgue Integral

Now specialize to a probability space $(\Omega, \mathcal{F}, \mathbb{P})$, where the measure is a probability and $\mathbb{P}(\Omega) = 1$. A random variable X is a measurable function $X\colon \Omega \to \mathbb{R}$. Its *expectation* is, simply and finally,

$$\boxed{\mathbb{E}[X] \;=\; \int_\Omega X \,\mathrm{d}\,\mathbb{P}}$$

the Lebesgue integral of X against the probability measure. There is no longer a discrete formula and a separate continuous formula — there is one definition, and the old formulas are theorems about how to *compute* it.

What is really happening

What is really happening when we write $\int_\Omega X \,\mathrm{d}\,\mathbb{P}$? We are sorting the outcomes ω by the value $X(\omega)$ takes, weighing each group of outcomes by its probability, and adding up value-times-probability. That is the sorted-coins cashier again, with "how many coins" replaced by "how much probability." Whether X is discrete, continuous, or mixed is irrelevant to the *definition*; it only changes how you *evaluate* the resulting integral. The unification you wanted is built into the architecture, not glued on afterward.

37.3.1 Recovering the discrete sum

Let X be discrete, taking values $x_1, x_2, \ldots$ with $\mathbb{P}(X = x_i) = p_i$. Then X is exactly a (possibly countable) simple function: $X = \sum_i x_i \mathbf{1}_{A_i}$ with $A_i = \{X = x_i\}$ and $\mathbb{P}(A_i) = p_i$. The definition of the simple integral gives, directly,

$$\mathbb{E}[X] \;=\; \int_\Omega X \,\mathrm{d}\,\mathbb{P} \;=\; \sum_i x_i\,\mathbb{P}(A_i) \;=\; \sum_i x_i p_i.$$

The discrete formula is not a different rule — it is the simple-function integral, the very first rung of the ladder.

37.3.2 Recovering the continuous integral

Let X be continuous with density f. The change-of-variables theorem (the "law of the unconscious statistician" in its measure-theoretic form) says that to integrate against the *distribution* of X on the line we may use the density:

$$\mathbb{E}[X] \;=\; \int_{\Omega} X \,\mathrm{d}\,\mathbb{P} \;=\; \int_{\mathbb{R}} x \,\mathrm{d}\mu_X(x) \;=\; \int_{\mathbb{R}} x\, f(x)\,\mathrm{d}x,$$

where μ_X is the law of X (the pushforward of $\mathbb{P}$). The middle equality moves the integral from the abstract space Ω to the real line; the last equality holds because $\mathrm{d}\mu_X = f\,\mathrm{d}x$. The continuous formula, too, is a special case.

Why this matters

Look at what one definition bought you. The same symbol $\int X \,\mathrm{d}\,\mathbb{P}$ computes a sum when X is discrete, a Riemann-style density integral when X is continuous, and *something sensible* when X is neither. You no longer prove the linearity of expectation twice (once for sums, once for integrals); you prove it once, for the Lebesgue integral, and it holds universally. Every theorem about expectation in this book — linearity, the law of total expectation, Jensen's inequality, the convergence theorems — is now a single statement rather than a discrete version plus a continuous version stapled together.

37.3.3 The mixed case, painlessly

Example 37.2 (A payout that is part atom, part density)**.** Let X be the payout of an option: with probability $\frac{1}{2}$ it expires worthless ($X = 0$), and otherwise it pays an amount that is Exponential(1). There is no density at 0 (a point mass sits there), and no point mass on $(0, \infty)$ (a density lives there); see Figure 37.3 for the tell-tale jump-plus-ramp distribution function. Neither the pure-sum nor the pure-integral formula applies on its own. But the Lebesgue integral does not blink. Split by the positive/negative-part machinery, or just integrate against the law:

$$\mathbb{E}[X] = \underbrace{0 \cdot \tfrac{1}{2}}_{\text{atom at } 0} + \underbrace{\tfrac{1}{2}\int_0^{\infty} x\, e^{-x}\,\mathrm{d}x}_{\text{density part}} = 0 + \tfrac{1}{2}\cdot 1 = \tfrac{1}{2}.$$

The decomposition into an atomic piece and a continuous piece is automatic — it is just $\mathbb{E}[X\mathbf{1}_{X=0}] + \mathbb{E}[X\mathbf{1}_{X>0}]$, two Lebesgue integrals over complementary events, added by linearity. No special-case formula needed.

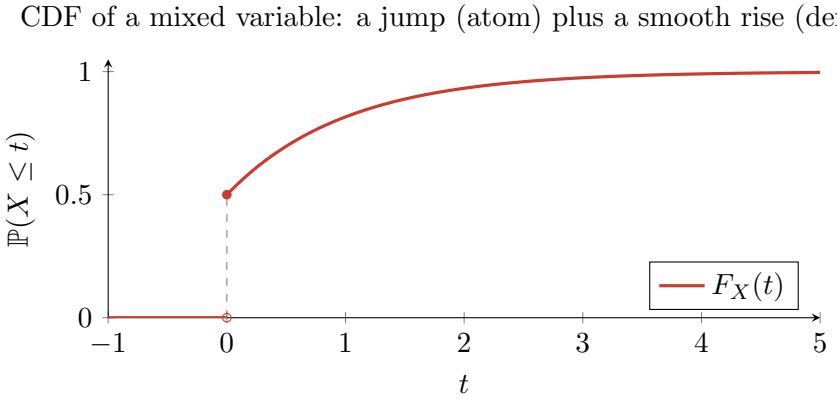


Figure 37.3: The distribution function of the option payout X: an atom of mass $\frac{1}{2}$ at 0 shows up as a *jump* of height $\frac{1}{2}$ in the CDF, while the Exponential(1) part contributes the *smooth* climb from $\frac{1}{2}$ toward 1. A pure pmf would be all jumps; a pure density would be all smooth rise. The Lebesgue integral handles both halves with one definition.

Application

This single-definition viewpoint is not pedantry — it is what makes modern probability *computable in code*. When you write a Monte Carlo estimator $\frac{1}{N}\sum_{i=1}^{N} g(X_i)$ for $\mathbb{E}[g(X)]$, you are approximating one Lebesgue integral, and you never ask whether X is discrete, continuous, or mixed; you just sample and average. In machine learning, the expected loss $\mathbb{E}_{(x,y)\sim\mathcal{D}}[\ell(f(x),y)]$ is a Lebesgue integral over a data distribution $\mathcal{D}$ that is almost always mixed (discrete labels, continuous features). Empirical risk minimization replaces the true measure $\mathbb{P}$ by the empirical measure $\hat{\mathbb{P}}_N = \frac{1}{N}\sum \delta_{(x_i,y_i)}$ — a sum of point masses — and minimizes $\int \ell \, \mathrm{d}\hat{\mathbb{P}}_N$. The reason you can swap the true distribution for a pile of data points and still call both "expectation" is precisely that both are Lebesgue integrals against a measure.

37.4 The Convergence Theorems

We promised that Lebesgue's integral was built to survive limits. Here is the payoff: three theorems telling you exactly when $\lim_n \int f_n = \int \lim_n f_n$. They are the workhorses behind almost every limit theorem in probability, and you have been implicitly trusting them for chapters.

37.4.1 Monotone Convergence

Theorem 37.2 (Monotone Convergence Theorem, MCT)**.** *Let $0 \leq f_1 \leq f_2 \leq \cdots$ be measurable with $f_n \uparrow f$ pointwise. Then*

$$\int_\Omega f_n \, \mathrm{d}\mu \longrightarrow \int_\Omega f \, \mathrm{d}\mu.$$

The limit may be $+\infty$ on both sides; the equality still holds.

Why it works

MCT is almost a restatement of how we *defined* the integral. We set $\int f = \sup\{\int s : s \text{ simple}, s \le f\}$, and Theorem 37.1 produced an increasing sequence of simple functions climbing to f. MCT says any increasing sequence climbing to f — not just the dyadic one — has integrals that climb to $\int f$. The "$\le$" direction is monotonicity (each $f_n \le f$, so $\int f_n \le \int f$, so the limit is $\le \int f$). The "$\ge$" direction is the heart: take any simple $s \le f$ and any $c \in (0,1)$; the sets $E_n = \{f_n \ge cs\}$ increase to Ω, and continuity of measure from below gives $\int f_n \ge c \int_{E_n} s \to c \int s$. Let $c \uparrow 1$, then sup over s, and the limit of $\int f_n$ dominates $\int f$. The two inequalities pin the limit to $\int f$.

Connection

MCT is the rigorous engine behind a move you have made since the calculus book: *integrating a series term by term.* If $g_k \ge 0$ are measurable and you want $\int \sum_k g_k = \sum_k \int g_k$, apply MCT to the partial sums $f_n = \sum_{k \le n} g_k$, which increase to $\sum_k g_k$. So the Tonelli-style interchange "sum of integrals equals integral of the sum" for nonnegative terms is just MCT in disguise. The calculus book swapped $\sum$ and $\int$ for nice power series and asked you to take it on faith; here is the theorem that licenses it for *any* nonnegative measurable terms. The same logic, applied to expectations, proves $\mathbb{E}\left[\sum_k Y_k\right] = \sum_k \mathbb{E}[Y_k]$ for nonnegative Y_k — countable linearity of expectation, used constantly in renewal theory and branching processes.

37.4.2 Fatou's Lemma

A stepping stone, but a useful one in its own right.

Lemma 37.1 (Fatou)**.** *For any measurable $f_n \ge 0$,*

$$\int_\Omega \liminf_n f_n \, d\mu \;\le\; \liminf_n \int_\Omega f_n \, d\mu.$$

In words: passing a lim inf inside the integral can only *lose* mass, never gain it. Mass can escape to infinity (a bump that slides off to the right, or grows ever taller and thinner), and Fatou says that is the only thing that can go wrong — the limit integral cannot exceed what the integrals were doing. It follows from MCT applied to the increasing sequence $g_n = \inf_{k \ge n} f_k$.

37.4.3 Dominated Convergence

This is the one you will reach for most often, because it handles *signed* functions and gives a clean equality.

Theorem 37.3 (Dominated Convergence Theorem, DCT)**.** *Suppose $f_n \to f$ pointwise (almost everywhere) and there is a single integrable "dominating" function $g \in L^1(\mu)$ with $|f_n| \le g$ for all n. Then $f \in L^1$ and*

$$\int_\Omega f_n \, d\mu \;\longrightarrow\; \int_\Omega f \, d\mu, \qquad \textit{indeed} \qquad \int_\Omega |f_n - f| \, d\mu \to 0.$$

Why it works

The proof is a clever double application of Fatou. Since $|f_n| \le g$ we have $g + f_n \ge 0$ and $g - f_n \ge 0$, both nonnegative, so Fatou applies to each. Applying it to $g + f_n$ gives $\int (g+f) \le \liminf \int (g+f_n)$, i.e. $\int f \le \liminf \int f_n$. Applying it to $g - f_n$ gives $\int (g-f) \le \liminf \int (g-f_n)$, i.e. $\int f \ge \limsup \int f_n$. Sandwiching, $\limsup \int f_n \le \int f \le \liminf \int f_n$, which forces $\int f_n \to \int f$. The dominating g is the fence that stops mass from escaping to infinity — the single hypothesis that rules out Fatou's strict-inequality disasters.

Common trap

The domination hypothesis is not decoration — drop it and the theorem is false. The classic counterexample is the "escaping bump." On $[0,1]$ with Lebesgue measure let $f_n = n\,\mathbf{1}_{(0,1/n)}$: a spike of height n on a base of width $1/n$. Then $f_n(x) \to 0$ for every $x > 0$, so $f = 0$ and $\int f = 0$. But $\int f_n = n \cdot \frac{1}{n} = 1$ for every n, so $\int f_n \to 1 \neq 0$. The limit of the integrals is not the integral of the limit. Where did the mass go? It escaped to infinite height as the base shrank. No integrable g can dominate all the f_n (you would need $g \ge n$ near 0 for every n, so $g = \infty$ there). DCT correctly refuses to apply, and Fatou's inequality holds *strictly*: $0 = \int \liminf f_n < \liminf \int f_n = 1$.

The escaping bump: $f_n = n\,\mathbf{1}_{(0,1/n)}$, each with integral 1

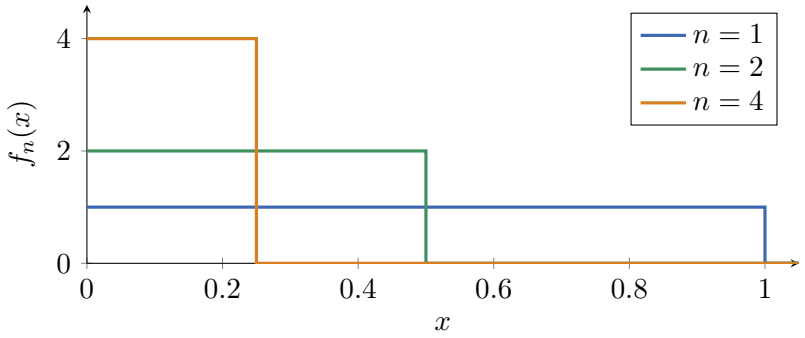


Figure 37.4: Each bump has area exactly 1, yet $f_n(x) \to 0$ at every fixed $x > 0$. Mass escapes upward as the base narrows, so $\lim \int f_n = 1 \neq 0 = \int \lim f_n$. Domination by an integrable g is exactly the fence that prevents this, and DCT requires it.

37.4.4 Why probability cannot live without them

Why this matters

These theorems are the silent partners in results you have already seen.

- **Linearity of expectation for infinite sums** ($\mathbb{E}\sum Y_k = \sum \mathbb{E}\,Y_k$ for $Y_k \ge 0$) is MCT.
- **Differentiating under the expectation/integral sign** — computing $\frac{\mathrm{d}}{\mathrm{d}\theta}\,\mathbb{E}[g(X,\theta)] = \mathbb{E}[\partial_\theta g(X,\theta)]$ in maximum-likelihood and Fisher-information arguments — is DCT applied to difference quotients dominated by an integrable bound.

- **The continuity of characteristic functions** and the proof of the Central Limit Theorem via $\mathbb{E}[e^{itX_n}] \to \mathbb{E}[e^{itX}]$ is DCT (the integrand e^{itX_n} is bounded by the constant 1, an integrable dominator on a probability space).
- **Martingale and renewal limit theorems** repeatedly swap lim and $\mathbb{E}$; each swap is a citation to MCT or DCT.

Whenever you have written "$\mathbb{E}$ of the limit is the limit of $\mathbb{E}$," you were spending one of these theorems. Now you know the bank they are drawn on.

Paradox

Here is a puzzle that feels like a contradiction until the convergence theorems dissolve it. Consider random variables X_n that equal n with probability $1/n$ and 0 otherwise. As $n \to \infty$, $\mathbb{P}(X_n \neq 0) = 1/n \to 0$, so $X_n \to 0$ in probability — the variable is 0 with probability approaching 1. You might expect $\mathbb{E}[X_n] \to 0$. But $\mathbb{E}[X_n] = n \cdot \frac{1}{n} = 1$ for every n: the expectation stubbornly refuses to vanish.

This is the escaping-bump paradox wearing probabilistic clothes. The *value* n races to infinity exactly as fast as the *probability* $1/n$ races to zero, and their product stays pinned at 1. Convergence in probability does *not* imply convergence of expectations — you need a dominating variable (uniform integrability, the probabilist's name for the DCT hypothesis) to license the interchange. The paradox is not a flaw in probability; it is the precise warning that expectation is a global, mass-based quantity that can be hijacked by rare-but-enormous events. Insurance against catastrophe, heavy-tailed financial risk, and the St. Petersburg game all live in this gap between "almost never happens" and "contributes to the mean."

37.5 Almost Everywhere, and What the Integral Cannot See

One more idea makes the whole theory click: the integral is *blind to sets of measure zero*. A set $N \in \mathcal{F}$ is *null* if $\mu(N) = 0$. A property holds *almost everywhere* (a.e.), or in probability *almost surely* (a.s.), if it holds outside some null set.

Proposition 37.1 (Null sets are invisible)**.** *If $f = g$ almost everywhere, then $\int f \, d\mu = \int g \, d\mu$. In particular, if $f = 0$ a.e. then $\int f \, d\mu = 0$, and conversely if $f \geq 0$ with $\int f \, d\mu = 0$ then $f = 0$ a.e.*

This is why the Dirichlet function of Example 37.1 integrates to 0: it differs from the constant 0 only on the rationals, a null set. It is also why probabilists say "almost surely" so often. A statement like "the sample mean converges to μ" (the Strong Law) is an almost-sure statement: it can fail, but only on a set of outcomes of probability zero, and the integral — the expectation — never notices that set.

What is really happening

The right way to think about L^1 is that its objects are not really functions but *equivalence classes* of functions that agree almost everywhere. Two random variables that differ only on a probability-zero event are, for every purpose the integral cares about, the same random variable. This is liberating: you can redefine X arbitrarily on a null set — to fix a division by zero, to make it continuous, to tidy a formula — without changing a single expectation,

variance, or distribution. The integral sees *mass*, and null sets carry none.

37.6 The Bridge to Riemann

We close the circle by relating the new integral to the old one from the calculus book, so you never have to choose between them.

Theorem 37.4 (Lebesgue dominates Riemann)**.** *If $f\colon [a,b] \to \mathbb{R}$ is bounded and Riemann integrable, then it is Lebesgue integrable on $[a,b]$ and the two integrals agree:*

$$\int_a^b f(x)\,\mathrm{d}x \ \text{(Riemann)} \ = \int_{[a,b]} f\,\mathrm{d}\lambda \ \text{(Lebesgue)},$$

where λ is Lebesgue measure. Moreover a bounded f on $[a,b]$ is Riemann integrable if and only if *its set of discontinuities is a null set.*

Connection

This is the formal handshake with the companion calculus book. Everything you computed there — areas under curves, the Fundamental Theorem of Calculus, $\int_0^\infty xe^{-x}\mathrm{d}x = 1$ — is *also* a Lebesgue integral, with the identical numerical value. So you lose nothing: every Riemann computation you know transfers verbatim. What you *gain* is (1) functions Riemann cannot handle (Dirichlet, and limits of nice functions), (2) the convergence theorems with no analogue in Riemann's world, and (3) a single integral that works on abstract spaces Ω, not just on intervals of $\mathbb{R}$. The Riemann integral is the special case "$\Omega = [a,b]$, $\mu =$ length, f continuous enough"; the Lebesgue integral is the genuine article. The second sentence of the theorem — Riemann integrability is *exactly* "continuous almost everywhere" — is the sharpest possible statement of why Riemann fails on D: D is discontinuous at *every* point, the opposite of null.

Common trap

The two theories *disagree* for improper integrals. The function $f(x) = \frac{\sin x}{x}$ has a convergent improper Riemann integral on $[0,\infty)$ (the famous $\int_0^\infty \frac{\sin x}{x}\mathrm{d}x = \frac{\pi}{2}$), because the positive and negative areas cancel as you let the upper limit grow. But f is *not* Lebesgue integrable there, because $\int_0^\infty \frac{|\sin x|}{x}\mathrm{d}x = \infty$ — and the Lebesgue integral of a signed function requires *absolute* integrability ($\int |f| < \infty$) so that $\int f^+ - \int f^-$ is not $\infty - \infty$. The lesson: Lebesgue insists on absolute convergence; conditional "cancellation" convergence is a Riemann-only trick. For probability this is no loss — an expectation $\mathbb{E}[X]$ that depends on the *order* of summation is not a meaningful average anyway, and we always demand $\mathbb{E}|X| < \infty$ before speaking of $\mathbb{E}[X]$.

37.7 A Fully Worked Computation

We run the entire machine once, end to end, on a single problem, watching the four stages and the convergence theorems cooperate.

Example 37.3 (Expectation of a geometric variable, from the ground up)**.** Let X take value k with probability $\mathbb{P}(X = k) = 2^{-k}$ for $k = 1, 2, 3, \dots$ (a Geometric(1/2) counting trials to first success). We compute $\mathbb{E}[X]$ as a Lebesgue integral and watch the theory work.

As a simple-function limit. $X = \sum_{k \geq 1} k\, \mathbf{1}_{A_k}$ with $A_k = \{X = k\}$, $\mathbb{P}(A_k) = 2^{-k}$ (Figure 37.5). This is a *countable* simple function, so it is not literally simple (finitely many values), but its truncations $X_n = \sum_{k=1}^{n} k\, \mathbf{1}_{A_k}$ are, and $X_n \uparrow X$. Each truncation integrates by the simple-function rule:

$$\mathbb{E}[X_n] = \int X_n \, \mathrm{d}\, \mathbb{P} = \sum_{k=1}^{n} k\, 2^{-k}.$$

Pass the limit by MCT. Because $0 \leq X_1 \leq X_2 \leq \cdots \uparrow X$, the Monotone Convergence Theorem licenses

$$\mathbb{E}[X] = \lim_{n \to \infty} \mathbb{E}[X_n] = \sum_{k=1}^{\infty} k\, 2^{-k}.$$

No hand-waving: the swap of lim and $\int$ is a theorem here, not a hope.

Evaluate the series. Using $\sum_{k \geq 1} k r^k = \frac{r}{(1-r)^2}$ at $r = \frac{1}{2}$,

$$\mathbb{E}[X] = \frac{1/2}{(1 - 1/2)^2} = \frac{1/2}{1/4} = 2.$$

So $\mathbb{E}[X] = 2$, matching the Geometric(p) mean $1/p$ with $p = \frac{1}{2}$, but now *derived* from the integral rather than recited from a table. Every step was a rung of the ladder or a convergence theorem.

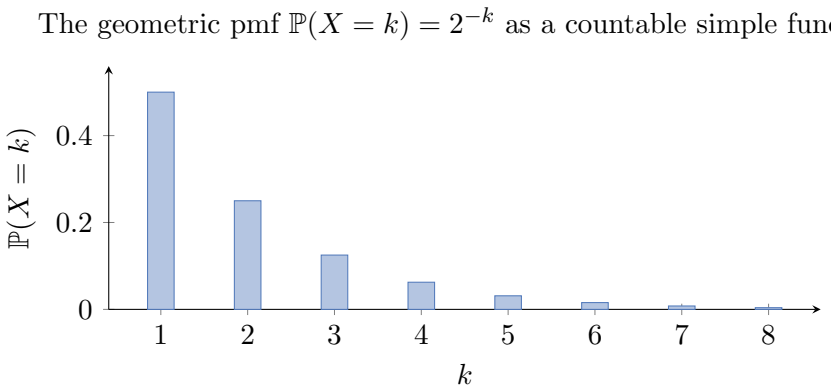


Figure 37.5: Each bar is a level set $A_k = \{X = k\}$ with measure $\mathbb{P}(A_k) = 2^{-k}$. The truncations $X_n = \sum_{k \leq n} k\, \mathbf{1}_{A_k}$ keep only the first n bars and rise to X; the Monotone Convergence Theorem lets their integrals $\sum_{k \leq n} k\, 2^{-k}$ climb to $\mathbb{E}[X] = 2$.

Work it yourself

Re-run the machine yourself on the *tail-sum formula*. For a nonnegative integer-valued X, prove

$$\mathbb{E}[X] = \sum_{k=1}^{\infty} \mathbb{P}(X \geq k)$$

by writing $X = \sum_{k\geq 1} \mathbf{1}_{\{X\geq k\}}$ (check this identity pointwise: if $X(\omega) = m$, exactly m of the indicators are 1), then integrating term by term. Which convergence theorem licenses the interchange of $\sum$ and $\mathbb{E}$? Now verify it numerically on the X of Example 37.3: here $\mathbb{P}(X \geq k) = 2^{-(k-1)}$, so $\sum_{k\geq 1} 2^{-(k-1)} = 2$, agreeing with the direct computation. The tail-sum formula is one of the most useful tricks in applied probability, and it is pure MCT.

37.8 What to Carry Forward

Focus

The one-paragraph summary of a hundred pages of measure theory:

- **One integral.** $\int f \, \mathrm{d}\mu$ is built in four rungs — indicators ($\int \mathbf{1}_A = \mu(A)$), simple functions ($\int \sum a_k \mathbf{1}_{A_k} = \sum a_k \mu(A_k)$), nonnegative functions (sup of simple functions underneath), signed functions (positive minus negative part). It slices the *range*, not the domain.
- **One expectation.** $\mathbb{E}[X] = \int_\Omega X \, \mathrm{d}\,\mathbb{P}$. The discrete sum and the continuous density-integral are both special cases; mixed variables need no special handling.
- **Limits are safe.** MCT (increasing nonnegative), Fatou (lower bound), and DCT (dominated) tell you exactly when $\lim \int = \int \lim$. The escaping bump shows what goes wrong without domination.
- **Null sets are invisible.** Almost-everywhere / almost-surely is the natural language, and the integral never sees a measure-zero set.
- **Riemann lives inside.** Every Riemann integral is a Lebesgue integral of the same value; Lebesgue adds the limit theorems and the abstract domain.

With this in hand, the next chapter can finally define conditional expectation $\mathbb{E}[X \mid \mathcal{G}]$ rigorously — as a Lebesgue integral relative to a sub-σ-algebra — and the martingale theory rests on bedrock.

37.9 Exercises

Exercise 37.1 (Indicators and probabilities)**.** Let A, B be events. Using only $\mathbb{E}[\mathbf{1}_E] = \mathbb{P}(E)$ and linearity of the integral, prove $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B)$ by integrating the pointwise identity $\mathbf{1}_{A\cup B} = \mathbf{1}_A + \mathbf{1}_B - \mathbf{1}_{A\cap B}$. (This is inclusion– exclusion as a one-line integral.)

Exercise 37.2 (Well-definedness of the simple integral)**.** Suppose a simple function has two representations $s = \sum_k a_k \mathbf{1}_{A_k} = \sum_j b_j \mathbf{1}_{B_j}$ with each family disjoint and covering Ω. Prove $\sum_k a_k \mu(A_k) = \sum_j b_j \mu(B_j)$ by refining with $C_{kj} = A_k \cap B_j$ and using finite additivity. (This is the only bookkeeping lemma the whole construction needs.)

Exercise 37.3 (Dirichlet, made precise)**.** Let $D = \mathbf{1}_{\mathbb{Q}\cap[0,1]}$. (a) Explain why D is a simple function on the measure space $([0,1], \mathcal{B}, \lambda)$. (b) Compute its Lebesgue integral directly from Definition 37.2, using that $\mathbb{Q}\cap[0,1]$ is countable and hence λ-null. (c) Show D is not Riemann integrable by exhibiting its upper and lower Darboux sums.

Exercise 37.4 (Tail-sum formula)**.** For a nonnegative integer-valued random variable X, prove $\mathbb{E}[X] = \sum_{k=1}^{\infty} \mathbb{P}(X \geq k)$. State clearly which convergence theorem justifies exchanging $\mathbb{E}$ and the infinite sum, and verify that its hypotheses hold. Then apply it to compute $\mathbb{E}[X]$ when $X \sim$ Geometric(p) with $\mathbb{P}(X = k) = (1-p)^{k-1}p$.

Exercise 37.5 (The escaping bump, your version)**.** On $([0,1], \lambda)$ construct measurable $f_n \geq 0$ with $f_n \to 0$ pointwise but $\int f_n \, \mathrm{d}\lambda = 5$ for every n. (a) Why does this not contradict DCT? Identify precisely which hypothesis fails. (b) Compute both $\int \liminf f_n$ and $\liminf \int f_n$ and confirm Fatou's inequality is strict.

Exercise 37.6 (Monotone vs. dominated)**.** For each sequence on $([0,\infty), \lambda)$, decide whether MCT, DCT, both, or neither applies, and find $\lim_n \int f_n$. (a) $f_n(x) = \mathbf{1}_{[0,n]}(x)\,(1 - x/n)_+$. (b) $f_n(x) = \frac{1}{n}\mathbf{1}_{[0,n]}(x)$. (c) $f_n(x) = e^{-x/n}\, e^{-x}$. Justify each answer by naming a dominating function or an increasing structure where one exists.

Exercise 37.7 (Mixed random variable)**.** Let $X = 0$ with probability p and $X \sim$ Uniform$(0,1)$ with probability $1-p$. (a) Write $\mathbb{E}[X]$ as a sum of two Lebesgue integrals over complementary events. (b) Compute $\mathbb{E}[X]$ and $\mathbb{E}[X^2]$, and hence $\mathrm{Var}(X)$, in terms of p. (c) Explain why neither the pure discrete nor the pure continuous expectation formula computes this directly, but the Lebesgue definition does.

Exercise 37.8 (Term-by-term integration)**.** Let $g_k(x) = x^k(1-x)$ on $[0,1]$ for $k = 0, 1, 2, \ldots$ (all nonnegative). (a) Use MCT (via partial sums) to justify $\int_0^1 \sum_{k\geq 0} g_k \, \mathrm{d}x = \sum_{k \geq 0} \int_0^1 g_k \, \mathrm{d}x$. (b) Evaluate both sides ($\sum_{k\geq 0} g_k(x) = 1$ for $x \in [0,1)$) and confirm they agree. (c) Where, exactly, did nonnegativity of the g_k matter?

Exercise 37.9 (Riemann inside Lebesgue)**.** (a) State the criterion from Theorem 37.4 for a bounded function on $[a,b]$ to be Riemann integrable. (b) Use it to decide whether $f(x) = \mathbf{1}_{\{x \text{ is a dyadic rational}\}}$ on $[0,1]$ is Riemann integrable, and compute its Lebesgue integral. (c) Give a bounded function on $[0,1]$ that is Lebesgue integrable but not Riemann integrable, and one (improper) example that is Riemann integrable but not Lebesgue integrable, explaining the difference in one sentence.

Exercise 37.10 (Differentiating under the integral, conceptual)**.** Suppose X has density f and $g(x,\theta)$ is differentiable in θ with $|\partial_\theta g(x,\theta)| \leq h(x)$ where $\mathbb{E}[h(X)] < \infty$. Explain, citing the relevant convergence theorem, why $\frac{\mathrm{d}}{\mathrm{d}\theta}\mathbb{E}[g(X,\theta)] = \mathbb{E}[\partial_\theta g(X,\theta)]$. (Hint: write the derivative as a limit of difference quotients and dominate them with h via the mean value theorem.) Why is this the workhorse behind score functions and Fisher information in maximum-likelihood estimation?

Chapter 38

Conditional Expectation, Rigorously

38.1 Why we are doing this again

You already know what conditional expectation *is*, at least in the friendly cases. Back when we conditioned on a variable, we wrote $g(x) = \mathbb{E}[Y \mid X = x]$, an ordinary function of x, and then promoted it to a random variable $\mathbb{E}[Y \mid X] = g(X)$. We proved the tower property, the pull-out rule, the law of total variance, and we used them happily. So why return to the topic with the word *rigorously* stapled to the title?

Three reasons, and each one is a crack in the elementary picture that only widens as you go deeper.

- **The dividing-densities recipe breaks.** The definition $\mathbb{E}[Y \mid X = x] = \int y\, f_{Y|X}(y \mid x)\, \mathrm{d}y$ silently assumed a joint density existed. But many of the most important random objects — a Brownian path, a mixture of a coin flip and a Gaussian, the information carried by a whole filtration — have *no* joint density. The recipe has nothing to say about them.

- **Conditioning on a probability-zero event is genuinely ambiguous.** The expression $\mathbb{E}[Y \mid X = x]$ conditions on $\{X = x\}$, an event of probability zero when X is continuous. We swept this under the rug. The Borel–Kolmogorov paradox (Section 38.8) shows that the rug was hiding a real contradiction.

- **Martingales and Bayesian updating want to condition on *information*, not on a value.** A martingale is defined by $\mathbb{E}[X_{n+1} \mid \mathcal{F}_n] = X_n$, where $\mathcal{F}_n$ is the sigma-algebra of everything known by time n. There is no single variable being plugged in. We need to condition on a sigma-algebra directly.

Kolmogorov's resolution, from his 1933 *Grundbegriffe*, is a striking change of tactics. He stopped trying to *construct* conditional expectation by a formula and instead *characterized* it by the one property it must have. The payoff is a single definition that covers densities, masses, Brownian motion, and everything in between, and that turns out to be exactly the geometric notion of *orthogonal projection*. That equivalence is the heart of this chapter, so we start there, with a picture you already trust.

Focus

This chapter assumes the measure-theoretic backbone from the previous two chapters: a probability space $(\Omega, \mathcal{F}, \mathbb{P})$, sigma-algebras as "bundles of information," the Lebesgue integral $\mathbb{E}[X] = \int_\Omega X \, \mathrm{d}\,\mathbb{P}$, and the Radon–Nikodym theorem. If any of those feel hazy, reread them first. Everything here is built on them.

38.2 The best-guess picture

Forget formulas for a moment. Conditional expectation answers a forecasting question: *given some information, what is my best guess for Y?* "Best" has to mean something, and the cleanest meaning is mean-squared error. Among all guesses you are *allowed* to make, which one minimizes the expected squared miss?

What does "allowed" mean? If your information is the value of X, then a guess is allowed exactly when it is a function of X — you cannot use information you do not have. So the question becomes:

$$\text{minimize} \quad \mathbb{E}\big[(Y - g(X))^2\big] \quad \text{over all (measurable) functions } g. \tag{38.1}$$

Intuition

Imagine Y is a person's height and X is their shoe size. You will see many people with shoe size $X = 9$; their heights scatter around some center. Your single allowed guess for all of them is $g(9)$. To minimize squared error *within that group*, you should aim at the group's mean height. Do this for every shoe size and you have built the function $g(x) = \mathbb{E}[Y \mid X = x]$. The conditional expectation is just "the within-group average, computed group by group." The whole rigorous theory is a precise way to say exactly this when the groups are infinitesimally thin or do not exist at all.

Here is the calculus that confirms the intuition, before we abstract it. Fix a value x and treat $g(x)$ as a single number c we get to choose. Conditionally on $X = x$,

$$\mathbb{E}\big[(Y - c)^2 \mid X = x\big] = \mathrm{Var}(Y \mid X = x) + \big(\mathbb{E}[Y \mid X = x] - c\big)^2,$$

by the same "mean minimizes squared deviation" identity you proved for plain expectation. The first term does not depend on c; the second is a square, minimized (at zero) by $c = \mathbb{E}[Y \mid X = x]$. Minimizing pointwise in x minimizes the average over x, so the best g is exactly $g(x) = \mathbb{E}[Y \mid X = x]$. *Conditional expectation is the least-squares predictor.* Hold onto that.

38.2.1 A picture worth more than the algebra

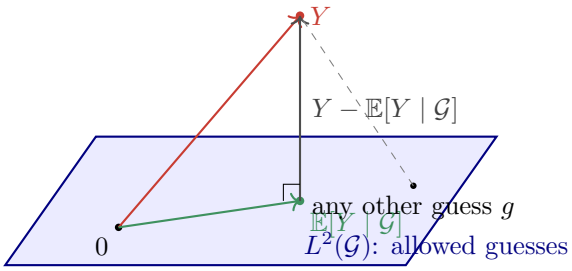


Figure 38.1: Conditional expectation as orthogonal projection. The plane is the space of guesses you are allowed to make (functions of your information $\mathcal{G}$). The random variable Y lives above it. The closest point in the plane — the foot of the perpendicular — is $\mathbb{E}[Y \mid \mathcal{G}]$. The error vector $Y - \mathbb{E}[Y \mid \mathcal{G}]$ is orthogonal to the entire plane. Any other guess g is farther from Y (Pythagoras).

If Figure 38.1 looks exactly like the picture of projecting a vector onto a subspace from linear algebra, that is not a coincidence; it *is* that picture. Making the analogy exact is the cleanest way to understand existence and uniqueness at once.

Connection

In the companion linear algebra book you projected a vector $\mathbf{v} \in \mathbb{R}^n$ onto a subspace W. The projection $\hat{\mathbf{v}}$ is the unique point of W closest to $\mathbf{v}$, and it is characterized by orthogonality: $\mathbf{v} - \hat{\mathbf{v}} \perp \mathbf{w}$ for every $\mathbf{w} \in W$, where $\perp$ means the inner product $\langle \cdot, \cdot \rangle$ vanishes.
The probabilistic dictionary:

Linear algebra	Probability
vectors $\mathbf{v}$	random variables Y with $\mathbb{E}[Y^2] < \infty$
inner product $\langle \mathbf{u}, \mathbf{v} \rangle$	$\langle U, V \rangle = \mathbb{E}[UV]$
length^2 $\|\mathbf{v}\|^2$	$\mathbb{E}[Y^2]$
subspace W	$\mathcal{G}$-measurable variables $L^2(\mathcal{G})$
projection onto W	conditional expectation $\mathbb{E}[\cdot \mid \mathcal{G}]$
$\mathbf{v} - \hat{\mathbf{v}} \perp W$	error is uncorrelated with every $\mathcal{G}$-variable

The space L^2 of square-integrable random variables, with $\langle U, V \rangle = \mathbb{E}[UV]$, is a genuine (infinite-dimensional) inner-product space — a *Hilbert space*. Hilbert spaces have a projection theorem: onto any closed subspace there is a unique nearest point. Conditional expectation is that projection. Existence is not magic; it is the Hilbert-space projection theorem.

38.3 Conditioning on a sigma-algebra: the definition

We now make the leap that the elementary theory could not. Instead of conditioning on a value or even on a variable, we condition on a sigma-algebra $\mathcal{G} \subseteq \mathcal{F}$. Think of $\mathcal{G}$ as the collection of all events whose outcome you are allowed to know — your information.

Intuition

A sigma-algebra is a precise way to say "how much can you resolve." If $\mathcal{G}$ is generated by a variable X, then knowing $\mathcal{G}$ means knowing X: you can tell apart two outcomes exactly when they give different X values. A coarse $\mathcal{G}$ blurs many outcomes together; the finest $\mathcal{G} = \mathcal{F}$ resolves everything. "Conditioning on $\mathcal{G}$" means "give the best forecast computable from the information $\mathcal{G}$."

Two demands pin down conditional expectation. First, your forecast must be *computable from your information*: it must be a $\mathcal{G}$-measurable random variable. Second, it must be *right on average over every event you can see*: over any $G \in \mathcal{G}$, the forecast and the true Y must have the same integral. The second demand is the rigorous form of "correct group averages."

Definition 38.1 (Conditional expectation given a sigma-algebra)**.** Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space, Y an integrable random variable ($\mathbb{E}|Y| < \infty$), and $\mathcal{G} \subseteq \mathcal{F}$ a sub-sigma-algebra. A **conditional expectation of Y given $\mathcal{G}$** is any random variable, written $\mathbb{E}[Y \mid \mathcal{G}]$, satisfying:

1. **(Measurability)** $\mathbb{E}[Y \mid \mathcal{G}]$ is $\mathcal{G}$-measurable.
2. **(Averaging / partial-integral identity)** For every $G \in \mathcal{G}$,

$$\int_G \mathbb{E}[Y \mid \mathcal{G}] \, \mathrm{d}\,\mathbb{P} \;=\; \int_G Y \, \mathrm{d}\,\mathbb{P}, \qquad \text{equivalently} \qquad \mathbb{E}\big[\mathbf{1}_G \, \mathbb{E}[Y \mid \mathcal{G}]\big] = \mathbb{E}[\mathbf{1}_G \, Y]. \tag{38.2}$$

Common trap

$\mathbb{E}[Y \mid \mathcal{G}]$ is a *random variable*, not a number. It is only defined "almost surely": any two versions agree with probability one. The notation hides this, so be careful — statements about $\mathbb{E}[Y \mid \mathcal{G}]$ are statements that hold a.s. Also note we never divide by $\mathbb{P}(G)$ anywhere in (38.2). That division — the source of all the trouble with probability-zero conditioning — has vanished. We compare *integrals*, which behave perfectly even when $\mathbb{P}(G) = 0$ pieces are involved.

38.3.1 Reading the definition as orthogonality

The averaging property (38.2) says $\mathbb{E}[\mathbf{1}_G(Y - \mathbb{E}[Y \mid \mathcal{G}])] = 0$ for all $G \in \mathcal{G}$. Indicators $\mathbf{1}_G$ are the simplest $\mathcal{G}$-measurable variables; by approximating any bounded $\mathcal{G}$-measurable Z with simple functions and taking limits, the property upgrades to

$$\mathbb{E}\big[Z\,(Y - \mathbb{E}[Y \mid \mathcal{G}])\big] = 0 \qquad \text{for every bounded } \mathcal{G}\text{-measurable } Z. \tag{38.3}$$

In inner-product language: the error $Y - \mathbb{E}[Y \mid \mathcal{G}]$ is orthogonal to every variable in $L^2(\mathcal{G})$. That is precisely the orthogonality condition of Figure 38.1. **Kolmogorov's definition and the projection picture are literally the same equation.**

38.4 Why it exists: two routes to the same object

A definition by axioms is only worth something if such an object actually exists, and is essentially unique. We give the intuition for both. Uniqueness is short; existence has two proofs depending on how much you assume about Y.

38.4.1 Uniqueness (almost surely)

Proposition 38.1 (Uniqueness)**.** *If V and W both satisfy Definition 38.1, then $V = W$ almost surely.*

Why it works

Both are $\mathcal{G}$-measurable and both integrate to $\int_G Y$ over every $G \in \mathcal{G}$, so $\int_G (V - W)\,\mathrm{d}\,\mathbb{P} = 0$ for all $G \in \mathcal{G}$. Now $D := V - W$ is $\mathcal{G}$-measurable. Take $G = \{D > \varepsilon\} \in \mathcal{G}$; then $0 = \int_G D \geq \varepsilon\,\mathbb{P}(G)$, forcing $\mathbb{P}(D > \varepsilon) = 0$. Let $\varepsilon \downarrow 0$ to get $\mathbb{P}(D > 0) = 0$; by symmetry $\mathbb{P}(D < 0) = 0$. So $D = 0$ a.s. The trick — "test the difference against the very event where it would be positive" — is the workhorse of the whole subject. It is how you prove two conditional expectations are equal without ever computing them.

38.4.2 Existence the easy way: the L^2 projection

Suppose first that $\mathbb{E}[Y^2] < \infty$. Then Y is a vector in the Hilbert space $L^2(\Omega, \mathcal{F}, \mathbb{P})$, and $L^2(\mathcal{G})$ — the square-integrable $\mathcal{G}$-measurable variables — is a *closed* subspace of it. The Hilbert-space projection theorem hands us a unique nearest point $\hat{Y} \in L^2(\mathcal{G})$, characterized by $\langle Y - \hat{Y}, Z\rangle = 0$ for all $Z \in L^2(\mathcal{G})$. Taking $Z = \mathbf{1}_G$ recovers exactly (38.2). So for square-integrable Y, conditional expectation simply *is* the orthogonal projection onto $L^2(\mathcal{G})$, and existence is the projection theorem. This is the route worth remembering; it is clean, geometric, and complete.

What is really happening

The reason the subspace must be *closed* for the projection theorem to apply is the same reason L^2 must be *complete* (a Hilbert space, not just an inner-product space): you take limits. Completeness of L^2 is itself a consequence of the dominated convergence machinery from the Lebesgue chapter. So the abstract-looking phrase "conditional expectation exists" unwinds, layer by layer, into the concrete fact that the Lebesgue integral plays nicely with limits. Every rigorous comfort in probability eventually cashes out as a convergence theorem.

38.4.3 Existence the general way: Radon–Nikodym

What if Y is integrable but not square-integrable (so it is not a vector in L^2)? The projection argument no longer applies directly. Kolmogorov's original construction handles every integrable Y at once, using the Radon–Nikodym theorem. The idea is elegant: build a *measure* out of Y, notice it has a density with respect to $\mathbb{P}$, and *that density is the conditional expectation.*

Theorem 38.1 (Existence via Radon–Nikodym)**.** *For any integrable Y and any sub-sigma-algebra $\mathcal{G}$, a conditional expectation $\mathbb{E}[Y \mid \mathcal{G}]$ exists.*

Why it works

Assume first $Y \geq 0$. Define a set function on the smaller sigma-algebra $\mathcal{G}$:

$$\nu(G) := \int_G Y\,\mathrm{d}\,\mathbb{P}, \qquad G \in \mathcal{G}.$$

This ν is a (finite) measure on $(\Omega, \mathcal{G})$: it is countably additive because the integral is, by monotone convergence. Crucially, ν is *absolutely continuous* with respect to $\mathbb{P}$ restricted to $\mathcal{G}$: if $\mathbb{P}(G) = 0$ then $\int_G Y \, \mathrm{d}\,\mathbb{P} = 0$, so $\nu(G) = 0$. Whenever one measure is absolutely continuous with respect to another, the Radon–Nikodym theorem produces a density — a $\mathcal{G}$-measurable function $h \geq 0$ with

$$\nu(G) = \int_G h \, \mathrm{d}\,\mathbb{P} \quad \text{for all } G \in \mathcal{G}.$$

Compare with the definition of ν: $\int_G h \, \mathrm{d}\,\mathbb{P} = \int_G Y \, \mathrm{d}\,\mathbb{P}$ for every $G \in \mathcal{G}$, and h is $\mathcal{G}$-measurable. That is precisely Definition 38.1. So $h = \mathbb{E}[Y \mid \mathcal{G}]$. For general integrable Y, split $Y = Y^+ - Y^-$ into positive and negative parts, apply the above to each, and subtract.

Why this matters

Take stock of what just happened. The Radon–Nikodym density $\mathrm{d}\nu/\mathrm{d}\,\mathbb{P}$ — a concept from pure measure theory with no probabilistic content — *is* the conditional expectation. Conditional expectation is a change-of-measure derivative. This is the same Radon–Nikodym derivative that defines likelihood ratios in statistics and the change of measure in mathematical finance (Girsanov). The fact that one theorem simultaneously delivers conditional expectation, likelihood ratios, and risk-neutral pricing is one of those moments where the architecture of probability shows through.

History

Before Kolmogorov, "conditional probability given a continuous variable" was a folklore notion, computed by dividing densities and hoping for the best. The contradictions (the Borel–Kolmogorov paradox below) were known and embarrassing. In his 1933 monograph *Grundbegriffe der Wahrscheinlichkeitsrechnung*, Andrey Kolmogorov reframed the whole subject on measure theory and *defined* conditional expectation by the partial-integral property, invoking Radon–Nikodym (proved by Otton Nikodym in 1930, building on Johann Radon's 1913 work) for existence. He turned a shaky recipe into a theorem. It is hard to overstate how much modern probability — martingales, stochastic calculus, filtering theory — rests on this single reframing.

38.5 It really does recover everything you knew

A new definition is only trustworthy if the old correct answers fall out of it. They do.

38.5.1 Conditioning on a discrete partition

Let $\mathcal{G} = \sigma(B_1, B_2, \dots)$ be generated by a countable partition of Ω into events B_i with $\mathbb{P}(B_i) > 0$. A $\mathcal{G}$-measurable variable must be constant on each B_i (it cannot distinguish points inside a block), so $\mathbb{E}[Y \mid \mathcal{G}] = \sum_i c_i \mathbf{1}_{B_i}$ for some constants c_i. The averaging property over $G = B_i$ reads $c_i \, \mathbb{P}(B_i) = \int_{B_i} Y \, \mathrm{d}\,\mathbb{P}$, hence

$$c_i = \frac{1}{\mathbb{P}(B_i)} \int_{B_i} Y \, \mathrm{d}\,\mathbb{P} = \mathbb{E}[Y \mid B_i].$$

So on the block B_i, the conditional expectation equals the elementary $\mathbb{E}[Y \mid B_i]$ — the within-group average from the height-and-shoe-size story. The new definition has reproduced the old one exactly, with no density assumption.

38.5.2 Conditioning on a continuous variable

Let $\mathcal{G} = \sigma(X)$ and suppose (X, Y) has a joint density. A theorem (the Doob–Dynkin lemma) says every $\sigma(X)$-measurable variable is a function of X, so $\mathbb{E}[Y \mid \sigma(X)] = g(X)$ for some measurable g. Plug into the averaging property over $G = \{X \in A\}$:

$$\int_{\{X\in A\}} g(X)\,\mathrm{d}\,\mathbb{P} = \int_{\{X\in A\}} Y\,\mathrm{d}\,\mathbb{P} \implies \int_A g(x) f_X(x)\,\mathrm{d}x = \int_A\!\int y\, f_{X,Y}(x,y)\,\mathrm{d}y\,\mathrm{d}x.$$

Since this holds for every Borel set A, the integrands match for almost every x:

$$g(x) f_X(x) = \int y\, f_{X,Y}(x,y)\,\mathrm{d}y \implies g(x) = \int y\,\frac{f_{X,Y}(x,y)}{f_X(x)}\,\mathrm{d}y = \int y\, f_{Y|X}(y \mid x)\,\mathrm{d}y.$$

The dividing-densities formula re-emerges — but now as a *consequence*, valid where the densities exist, rather than as a shaky starting point. We write $\mathbb{E}[Y \mid \sigma(X)]$ and $\mathbb{E}[Y \mid X]$ interchangeably; they mean the same random variable.

Walking question

Convince yourself why the constant "$\mathbb{E}[Y]$" is the right answer for the trivial sigma-algebra $\mathcal{G} = \{\emptyset, \Omega\}$. A $\mathcal{G}$-measurable variable must be constant (it cannot resolve any event), say $\equiv c$. The averaging property over $G = \Omega$ gives $c = \int_\Omega Y\,\mathrm{d}\,\mathbb{P} = \mathbb{E}[Y]$. With *no* information, your best guess is the overall mean. The projection picture agrees: $L^2(\mathcal{G})$ is the line of constants, and the nearest constant to Y is $\mathbb{E}[Y]$.

38.6 The properties, re-proved from the definition

Every property you used in the elementary chapter now follows cleanly from measurability plus the averaging identity. We prove the most important ones; the pattern is always the same — propose a candidate, check it is $\mathcal{G}$-measurable, verify the partial-integral identity, and invoke uniqueness.

Proposition 38.2 (Core properties)**.** *Let Y, Y_1, Y_2 be integrable and $\mathcal{G} \subseteq \mathcal{F}$.*

1. ***Linearity:*** $\mathbb{E}[aY_1 + bY_2 \mid \mathcal{G}] = a\,\mathbb{E}[Y_1 \mid \mathcal{G}] + b\,\mathbb{E}[Y_2 \mid \mathcal{G}]$.
2. ***Taking out what is known:*** *if Z is $\mathcal{G}$-measurable and ZY integrable, then* $\mathbb{E}[ZY \mid \mathcal{G}] = Z\,\mathbb{E}[Y \mid \mathcal{G}]$.
3. ***Tower / smaller wins:*** *if $\mathcal{H} \subseteq \mathcal{G}$, then* $\mathbb{E}\big[\mathbb{E}[Y \mid \mathcal{G}] \,\big|\, \mathcal{H}\big] = \mathbb{E}[Y \mid \mathcal{H}]$.
4. ***Independence drops:*** *if Y is independent of $\mathcal{G}$, then* $\mathbb{E}[Y \mid \mathcal{G}] = \mathbb{E}[Y]$.
5. ***Monotonicity & Jensen:*** *$Y_1 \le Y_2$ a.s. implies $\mathbb{E}[Y_1 \mid \mathcal{G}] \le \mathbb{E}[Y_2 \mid \mathcal{G}]$ a.s.; and $\varphi(\mathbb{E}[Y \mid \mathcal{G}]) \le \mathbb{E}[\varphi(Y) \mid \mathcal{G}]$ for convex φ.*

Why it works

[Taking out what is known] We want to show $Z\,\mathbb{E}[Y \mid \mathcal{G}]$ is the conditional expectation of ZY. It is $\mathcal{G}$-measurable (product of two $\mathcal{G}$-measurable things). For the averaging property, we must check $\int_G Z\,\mathbb{E}[Y \mid \mathcal{G}]\,\mathrm{d}\,\mathbb{P} = \int_G ZY\,\mathrm{d}\,\mathbb{P}$ for every $G \in \mathcal{G}$. Start with $Z = \mathbf{1}_B$, $B \in \mathcal{G}$: the left side is $\int_{G\cap B} \mathbb{E}[Y \mid \mathcal{G}]\,\mathrm{d}\,\mathbb{P}$, which by the defining property of $\mathbb{E}[Y \mid \mathcal{G}]$ (applied to the set $G \cap B \in \mathcal{G}$) equals $\int_{G\cap B} Y\,\mathrm{d}\,\mathbb{P} = \int_G \mathbf{1}_B Y\,\mathrm{d}\,\mathbb{P}$. So it holds for indicators; by linearity it holds for simple $\mathcal{G}$-measurable Z; by monotone convergence it holds for all $\mathcal{G}$-measurable $Z \geq 0$, then for general Z by splitting. Uniqueness finishes the proof. The intuition: variables you already know act like *constants* once you have conditioned — there is nothing left to average about them.

38.6.1 The tower property, re-proved

The tower property is the engine of martingale theory, so it deserves a clean proof from scratch.

Theorem 38.2 (Tower property)**.** *If $\mathcal{H} \subseteq \mathcal{G} \subseteq \mathcal{F}$ and Y is integrable, then*

$$\mathbb{E}\left[\,\mathbb{E}[Y \mid \mathcal{G}]\,\middle|\,\mathcal{H}\,\right] = \mathbb{E}[Y \mid \mathcal{H}] \qquad \text{a.s.}$$

Why it works

Let $W := \mathbb{E}[Y \mid \mathcal{G}]$ and let us show $\mathbb{E}[W \mid \mathcal{H}] = \mathbb{E}[Y \mid \mathcal{H}]$. By uniqueness it suffices to show that $\mathbb{E}[Y \mid \mathcal{H}]$ *itself* qualifies as the conditional expectation of W given $\mathcal{H}$. It is $\mathcal{H}$-measurable by construction. For the averaging property, take any $H \in \mathcal{H}$. Because $\mathcal{H} \subseteq \mathcal{G}$, we also have $H \in \mathcal{G}$, so we may use the defining property of $W = \mathbb{E}[Y \mid \mathcal{G}]$ at the set H:

$$\int_H W\,\mathrm{d}\,\mathbb{P} \;=\; \int_H Y\,\mathrm{d}\,\mathbb{P} \;=\; \int_H \mathbb{E}[Y \mid \mathcal{H}]\,\mathrm{d}\,\mathbb{P},$$

where the last equality is the defining property of $\mathbb{E}[Y \mid \mathcal{H}]$ at the same $H \in \mathcal{H}$. Thus $\int_H W\,\mathrm{d}\,\mathbb{P} = \int_H \mathbb{E}[Y \mid \mathcal{H}]\,\mathrm{d}\,\mathbb{P}$ for all $H \in \mathcal{H}$, and $\mathbb{E}[Y \mid \mathcal{H}]$ is $\mathcal{H}$-measurable, so it is the conditional expectation of W. Done.

What is really happening

Notice how short and inevitable that proof is. The entire content is the single line "$H \in \mathcal{H} \subseteq \mathcal{G}$, so the averaging set is legal for both," — coarse information is a sub-collection of fine information. The tower property is just *transitivity of averaging*: averaging finely and then coarsely is the same as averaging coarsely once, because the coarse blocks are unions of fine ones. In the projection picture (Figure 38.1), $L^2(\mathcal{H})$ is a subspace of $L^2(\mathcal{G})$, and projecting onto the big plane and then onto the smaller line inside it lands in the same place as projecting onto the line directly. The tower property is "projection of a projection is a projection onto the smaller space" — a fact you already trust about shadows.

Setting $\mathcal{H} = \{\emptyset, \Omega\}$ recovers the law of total expectation $\mathbb{E}\left[\,\mathbb{E}[Y \mid \mathcal{G}]\right] = \mathbb{E}[Y]$, the special case you have been using all along.

38.7 A picture: the conditional expectation as a step function

Let us see the projection concretely. Take $X \sim \text{Uniform}(0,1)$ and $Y = X^2 + \text{noise}$, so the true regression curve is $\mathbb{E}[Y \mid X = x] = x^2$. If instead of knowing X exactly we only know *which quarter* X falls in — the coarse sigma-algebra $\mathcal{G}$ generated by the partition $[0, \frac{1}{4}), [\frac{1}{4}, \frac{1}{2}), [\frac{1}{2}, \frac{3}{4}), [\frac{3}{4}, 1)$ — then $\mathbb{E}[Y \mid \mathcal{G}]$ must be constant on each quarter, equal to the average of x^2 over that quarter. It is the best step-function approximation to x^2.

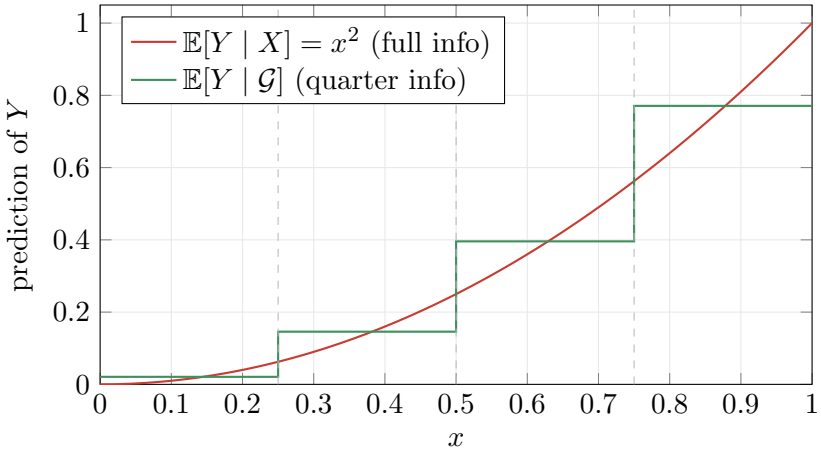


Figure 38.2: Conditional expectation given coarser information. The smooth curve is $\mathbb{E}[Y \mid X] = x^2$ (you know X exactly). The step function is $\mathbb{E}[Y \mid \mathcal{G}]$ when you know only which quarter X lies in: on each block it is the *average* of the curve, the best constant fit there. As the partition refines, the steps shrink toward the curve — finer information, sharper forecast. This is the projection of Figure 38.1 drawn as a function.

Why this matters

Figure 38.2 is the whole subject in one image. More information = a finer partition = a richer subspace to project onto = a prediction that can hug the truth more closely. The tower property says: average the fine step function over the coarse blocks and you get the coarse step function. The conditional variance — the expected squared gap between the steps and the curve — is exactly the part of $\text{Var}(Y)$ that your information cannot explain. Everything we call "learning from data" is somewhere on this picture.

38.8 The paradox that forced this rigor

Why insist on conditioning on sigma-algebras instead of just writing $\mathbb{E}[Y \mid X = x]$? Because the latter, taken literally, is *ambiguous* when $\mathbb{P}(X = x) = 0$ — and the ambiguity is not academic.

Paradox

[Borel–Kolmogorov] Pick a point uniformly at random on the surface of the Earth (a sphere). Now condition on the event that the point lies *on a particular great circle*. What is the

conditional distribution of the point along that circle?
Answer 1. Treat the great circle as a line of longitude (a meridian). The two-dimensional uniform density on the sphere, expressed in latitude θ and longitude ϕ, carries a factor $\cos\theta$ (rings near the equator are bigger than rings near the poles). Conditioning on a fixed longitude and normalizing the latitude, you get a distribution *proportional to* $\cos\theta$ — points bunch toward the equator.
Answer 2. Treat the same great circle as the equator (rotate your coordinates). Now by symmetry the conditional distribution is *uniform* along the circle.
Same circle, same uniform point on the sphere, two different conditional distributions. A contradiction — unless conditioning on a measure-zero event is not actually well defined by the event alone.

Why it works

[Resolving it] The resolution is exactly Kolmogorov's reframing. The event $\{X = x\}$ has probability zero, and the symbol $\mathbb{E}[Y \mid X = x]$ has *no meaning in isolation*. What is well defined is $\mathbb{E}[Y \mid \sigma(X)]$, a single function whose values on probability-zero slices are pinned down only up to a null set. To get a number on one particular slice you must say *which family of thinning events* converges to it — meridians of longitude versus rings of latitude are two different limiting families, and the Jacobian factor ($\cos\theta$ versus 1) differs between them. The "paradox" is the failure to specify the limiting procedure. Definition 38.1 sidesteps it entirely: it never conditions on a single null event, only ever integrates over positive-measure sets $G \in \mathcal{G}$, where no ambiguity exists. The price is that the conditional expectation is defined only a.s.; the reward is that it is defined *consistently*.

Common trap

The practical lesson: do not assign meaning to $\mathbb{E}[Y \mid X = x]$ at an isolated point without a density or a stated limiting scheme. With a joint density, the dividing-densities formula gives the canonical answer (it secretly fixes the limiting family — the natural Cartesian thinning). Without one, ask "conditional with respect to which sigma-algebra?" before you compute anything.

38.9 Why this is the right notion for martingales

Now the payoff. A **filtration** is an increasing chain $\mathcal{F}_0 \subseteq \mathcal{F}_1 \subseteq \cdots$ of sigma-algebras: $\mathcal{F}_n$ is everything known by time n, and information only accumulates. A process (M_n) is a **martingale** (adapted to the filtration) if each M_n is $\mathcal{F}_n$-measurable, integrable, and

$$\mathbb{E}[M_{n+1} \mid \mathcal{F}_n] = M_n \qquad \text{for all } n. \tag{38.4}$$

This definition is *impossible to state* with the elementary $\mathbb{E}[Y \mid X = x]$ machinery: the "information at time n" is a whole sigma-algebra, generated by the entire past $M_0, \ldots, M_n$ (and possibly more), not a single value. Conditioning on a sigma-algebra is not a luxury here; it is the only language in which a martingale can be written down.

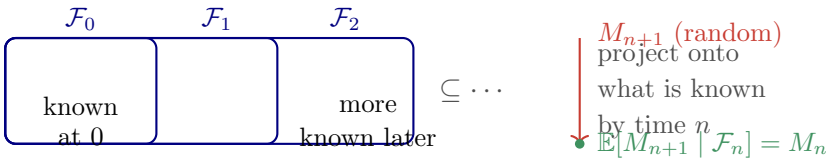


Figure 38.3: A filtration is a growing stack of information: $\mathcal{F}_0 \subseteq \mathcal{F}_1 \subseteq \cdots$. The martingale condition projects the unknown future value M_{n+1} onto the information $\mathcal{F}_n$ available now; for a fair game the projection lands back on the current value M_n. Each step is one application of Definition 38.1.

Application

[The tower property is the fairness of a fair game] Read (38.4) as: given all the history up to now, the expected next value equals the current value — a fair game, no drift (Figure 38.3). The tower property (Theorem 38.2) immediately gives the long-range version. For any $m < n$,

$$\mathbb{E}[M_n \mid \mathcal{F}_m] = \mathbb{E}\left[\mathbb{E}[M_n \mid \mathcal{F}_{n-1}] \,\middle|\, \mathcal{F}_m\right] = \mathbb{E}[M_{n-1} \mid \mathcal{F}_m] = \cdots = M_m,$$

peeling off one step at a time via the tower. A fair game is fair over any horizon, not just one step. And taking $m = 0$ with $\mathcal{F}_0$ trivial, $\mathbb{E}[M_n] = \mathbb{E}[M_0]$ for all n: the expected fortune never changes. This single consequence underlies optional stopping, the gambler's-ruin computations, and the no-arbitrage pricing of derivatives, where the discounted asset price is a martingale under the risk-neutral measure.

What is really happening

Why must the *whole* sigma-algebra $\mathcal{F}_n$ appear, rather than just M_n? Because fairness must hold given *everything* a gambler could know, not merely the current price. A process can satisfy $\mathbb{E}[M_{n+1} \mid M_n] = M_n$ (fair given only the last value) yet fail $\mathbb{E}[M_{n+1} \mid \mathcal{F}_n] = M_n$ (using the full past lets you detect a pattern and beat the game). The sigma-algebra formulation closes that loophole. "You cannot beat a martingale" is a statement about *all* usable information, which is exactly what a sigma-algebra encodes.

38.10 Why this is the right notion for Bayesian updating

Bayesian inference is conditioning made into a worldview: you hold a prior over an unknown Θ, observe data D, and update to the posterior. The posterior *mean* $\mathbb{E}[\Theta \mid D]$ is a conditional expectation in exactly our sense, and the projection picture explains why it is the optimal estimate.

Application

[The posterior mean is the optimal point estimate] Suppose you must report a single number $\hat{\theta}(D)$ to estimate Θ, and you will be scored by squared error $\mathbb{E}[(\Theta - \hat{\theta}(D))^2]$. Your estimate may depend only on the data, i.e. it must be $\sigma(D)$-measurable. This is precisely the least-squares problem (38.1) with $Y = \Theta$ and information $\mathcal{G} = \sigma(D)$. Its solution is the projection:

$$\hat{\theta}(D) = \mathbb{E}[\Theta \mid D] = \text{posterior mean}.$$

Under squared-error loss, the Bayes-optimal estimator is the posterior mean — not by convention, but because it is the orthogonal projection of the truth onto the space of data-driven guesses. The data filtration grows as observations arrive; the tower property says yesterday's posterior is the prior for today, and updating in one batch or sequentially gives the same answer. Bayesian learning *is* a sequence of projections onto ever-finer information.

Connection

The same projection idea, restricted to *linear* functions of the data instead of all functions, gives linear regression and the Kalman filter. Linear regression projects Y onto the subspace spanned by $\{1, X_1, \ldots, X_p\}$; conditional expectation projects onto the much larger subspace of *all* functions of the data. So $\mathbb{E}[Y \mid X]$ is the "true" regression function and ordinary least squares is its best linear approximation. When (Y, X) is jointly Gaussian the two coincide — the conditional expectation is already linear — which is the secret reason the Kalman filter, built from linear projections, is exactly optimal in the Gaussian world. See the linear algebra companion for the projection machinery and the estimation chapters here for the statistics.

38.11 A fully worked example

Example 38.1 (Conditioning a sum on one of its parts, three ways)**.** Let X, Y be independent with $\mathbb{E}[X] = \mu_X$, $\mathbb{E}[Y] = \mu_Y$, and let $S = X + Y$. We compute $\mathbb{E}[X \mid S]$, $\mathbb{E}[S \mid X]$, and verify the tower property, using only the rules of Proposition 38.2.

(a) $\mathbb{E}[S \mid X]$. Write $S = X + Y$ and use linearity: $\mathbb{E}[S \mid X] = \mathbb{E}[X \mid X] + \mathbb{E}[Y \mid X]$. Now X is $\sigma(X)$-measurable so $\mathbb{E}[X \mid X] = X$ (take out what is known, with $Y = 1$). And $Y \perp\!\!\!\perp X$, so $\mathbb{E}[Y \mid X] = \mathbb{E}[Y] = \mu_Y$ (independence drops). Hence

$$\mathbb{E}[S \mid X] = X + \mu_Y.$$

Sensible: knowing X, the only randomness left in S is Y, whose average is μ_Y.

(b) $\mathbb{E}[X \mid S]$ when X, Y are i.i.d. Now suppose X and Y have the *same* distribution. By symmetry $\mathbb{E}[X \mid S]$ and $\mathbb{E}[Y \mid S]$ are a.s. equal as functions of S (the symmetric roles give identical $\sigma(S)$-measurable answers). Their sum is forced:

$$\mathbb{E}[X \mid S] + \mathbb{E}[Y \mid S] = \mathbb{E}[X + Y \mid S] = \mathbb{E}[S \mid S] = S,$$

so each equals $S/2$:

$$\boxed{\mathbb{E}[X \mid S] = \tfrac{S}{2}.}$$

No densities, no integrals — pure structure. If $X, Y \sim \text{Poisson}(\lambda)$ then given $S = n$, X is Binomial$(n, \frac{1}{2})$ with mean $n/2$, matching. If $X, Y \sim \mathcal{N}(0, \sigma^2)$, then given S, X is Gaussian with mean $S/2$, matching again. One symmetry argument, every distribution at once.

(c) Tower check. We verify $\mathbb{E}\big[\mathbb{E}[X \mid S]\big] = \mathbb{E}[X]$ in the i.i.d. case: $\mathbb{E}[S/2] = \frac{1}{2}(\mu_X + \mu_Y) = \mu_X$ since $\mu_X = \mu_Y$. The law of total expectation holds.

Work it yourself

Redo Example 38.1(b) without symmetry, when $X \sim \text{Poisson}(\alpha)$, $Y \sim \text{Poisson}(\beta)$ are independent with $\alpha \neq \beta$. Show that given $S = n$, $X \sim \text{Binomial}(n, \frac{\alpha}{\alpha+\beta})$ and therefore $\mathbb{E}[X \mid S] = \frac{\alpha}{\alpha+\beta} S$. Notice the answer is again *linear* in S even though no Gaussian is in sight — and check it reduces to $S/2$ when $\alpha = \beta$.

38.12 Summary

Pause

- Conditional expectation $\mathbb{E}[Y \mid \mathcal{G}]$ is the unique (a.s.) random variable that is (i) $\mathcal{G}$-measurable and (ii) has the same integral as Y over every $G \in \mathcal{G}$. No division by probabilities, so it survives null events.
- Geometrically it is the *orthogonal projection* of Y onto $L^2(\mathcal{G})$ — the best mean-squared predictor using the information $\mathcal{G}$. The error is uncorrelated with everything you know.
- Existence: for square-integrable Y, the Hilbert-space projection theorem; for any integrable Y, the Radon–Nikodym theorem, which makes $\mathbb{E}[Y \mid \mathcal{G}]$ a change-of-measure density.
- It recovers the elementary definitions — group averages over a partition, dividing densities for a continuous X, and $\mathbb{E}[Y]$ for no information — as special cases.
- Linearity, taking out known factors, independence-drops, monotonicity, Jensen, and the *tower property* all fall out by the candidate-and-uniqueness method. The tower property is transitivity of averaging, or projection-of-a-projection.
- The Borel–Kolmogorov paradox shows why conditioning on a bare null event is ill-posed; the sigma-algebra formulation is the cure.
- This is the native language of martingales ($\mathbb{E}[M_{n+1} \mid \mathcal{F}_n] = M_n$) and of Bayesian updating (the posterior mean $\mathbb{E}[\Theta \mid D]$ is the squared-error-optimal estimator).

38.13 Exercises

Exercise 38.1 (Definition drill)**.** Let $\Omega = \{1, 2, 3, 4\}$ with $\mathbb{P}$ uniform, and let $\mathcal{G} = \sigma(\{1, 2\}, \{3, 4\})$. Let $Y(\omega) = \omega^2$. Compute $\mathbb{E}[Y \mid \mathcal{G}]$ explicitly as a function on Ω, and verify the averaging property (38.2) for $G = \{1, 2\}$ and for $G = \{3, 4\}$.

Exercise 38.2 (Smoothing reduces variance)**.** Assume $\mathbb{E}[Y^2] < \infty$. Using the orthogonality (38.3) with $Z = \mathbb{E}[Y \mid \mathcal{G}]$ (here a bounded approximation argument lets you take Z square-integrable), prove the Pythagorean identity $\mathbb{E}[Y^2] = \mathbb{E}\left[\mathbb{E}[Y \mid \mathcal{G}]^2\right] + \mathbb{E}\left[(Y - \mathbb{E}[Y \mid \mathcal{G}])^2\right]$. Deduce that $\text{Var}(\mathbb{E}[Y \mid \mathcal{G}]) \leq \text{Var}(Y)$: conditioning never increases variance. When is it an equality?

Exercise 38.3 (Take-out rule by hand)**.** Let $X \sim \text{Uniform}(0, 1)$ and $Y = XU$ where $U \sim \text{Uniform}(0, 1)$ is independent of X. Find $\mathbb{E}[Y \mid X]$ using "take out what is known" plus "independence drops," then verify your answer by direct integration against $f_{Y|X}$.

Exercise 38.4 (Conditioning on a coarser sigma-algebra)**.** For the setup of Figure 38.2 ($\mathbb{E}[Y \mid X] = X^2$, $X \sim \text{Uniform}(0,1)$), compute the four step heights of $\mathbb{E}[Y \mid \mathcal{G}]$ for the quarter-partition by averaging x^2 over each quarter, and confirm they match the values plotted. Then compute $\mathbb{E}\big[\,\mathbb{E}[Y \mid \mathcal{G}]\big]$ two ways (from the steps, and as $\mathbb{E}[X^2] = 1/3$) to check the tower property.

Exercise 38.5 (Tower in action)**.** Let $N \sim \text{Poisson}(\lambda)$ and, given $N = n$, let $S = \sum_{i=1}^{n} X_i$ with the X_i i.i.d. mean μ, independent of N. Using the tower property with $\mathcal{F} = \sigma(N)$, prove $\mathbb{E}[S] = \lambda\mu$. Then identify the sigma-algebra you conditioned on and explain in one sentence why $\mathbb{E}[S \mid N] = N\mu$ qualifies as a conditional expectation per Definition 38.1.

Exercise 38.6 (Martingale from a fixed variable)**.** Let $\mathcal{F}_0 \subseteq \mathcal{F}_1 \subseteq \cdots$ be a filtration and Y an integrable variable. Define $M_n := \mathbb{E}[Y \mid \mathcal{F}_n]$ (a "Doob martingale"). Use the tower property to prove (M_n) is a martingale, i.e. $\mathbb{E}[M_{n+1} \mid \mathcal{F}_n] = M_n$. Interpret M_n as "the current best estimate of Y given what is known so far," and explain why a fair-game property is exactly what you would expect of a well-behaved running estimate.

Exercise 38.7 (Posterior mean is optimal)**.** Let $\Theta \sim \text{Beta}(a, b)$ and, given $\Theta = \theta$, let $D \sim \text{Binomial}(n, \theta)$. Find the posterior $\Theta \mid D$ and its mean $\mathbb{E}[\Theta \mid D]$. Then argue from the least-squares characterization (38.1) that no other function of D achieves smaller expected squared error against Θ.

Exercise 38.8 (Why measurability cannot be dropped — proof)**.** Show by a counterexample that the averaging property (38.2) *alone*, without the $\mathcal{G}$-measurability requirement, fails to determine a unique object: exhibit two distinct variables that integrate to $\int_G Y$ over every $G \in \mathcal{G}$ but are not both $\mathcal{G}$-measurable. Conclude that measurability is doing essential work in pinning down "what your information can compute."

Exercise 38.9 (Borel–Kolmogorov in coordinates — proof)**.** On the unit disk with uniform distribution, let (R, Θ) be polar coordinates. Conditioning on $\Theta = \theta_0$ (a radius), show the conditional density of R is proportional to r on $[0, 1]$, not uniform. Then re-derive the "conditional distribution along a radius" as the limit of thin angular wedges and confirm the same answer; finally explain in one paragraph how choosing a different thinning family would change it, mirroring the sphere paradox of Section 38.8.

Part VIII

Mathematical Background

Chapter 39

Appendix: Sets, Sums, and Combinatorial Identities

This appendix is the toolbox you keep on the workbench. Probability rests on three plain foundations: *sets* (the language of events), *sums* (the way we add up probabilities and expectations), and *counting* (the way we measure how many outcomes there are). None of this is glamorous, but every theorem in the main text leans on it. If a step in some derivation ever felt like a magic trick — "and now we recognize this as a geometric series," or "by the binomial theorem this collapses" — this is where the trick gets explained slowly, with no rush.

You do not need to read this front to back. Treat it like a reference you can trust: each result is stated cleanly, motivated by a picture or a story, and then carefully derived, so that when you use it you know exactly why it is allowed.

39.1 Sets: the language of events

Before probability assigns numbers to anything, it has to say *what* it is assigning numbers to. The answer is always a set. A *sample space* Ω is the set of all possible outcomes of an experiment, and an *event* is just a subset $A \subseteq \Omega$. "The die shows an even number" is the set $\{2, 4, 6\}$. "At least one head in three flips" is a subset of the eight possible flip-sequences. Once you see that events *are* sets, the algebra of probability becomes the algebra of sets, and the strange-looking laws of probability inherit their shape from much older, much simpler laws about unions and intersections.

Intuition

Think of Ω as a big rectangle — the universe of everything that could happen. An event is a region drawn inside it. "And" is the overlap of two regions, "or" is their combined area, and "not" is everything outside a region. Probability later puts *weight* on these regions, but the geometry of and/or/not is fixed before any weights arrive. Almost every identity in this section is something you can literally see in a Venn diagram.

39.1.1 The basic operations

Let $A, B \subseteq \Omega$. The three workhorse operations are:

- **Union** $A \cup B = \{\omega : \omega \in A \text{ or } \omega \in B\}$ — everything in at least one of them. The "or" is *inclusive*: outcomes in both count too.
- **Intersection** $A \cap B = \{\omega : \omega \in A \text{ and } \omega \in B\}$ — everything in both at once.
- **Complement** $A^c = \{\omega \in \Omega : \omega \notin A\}$ — everything *not* in A. The complement always lives inside the fixed universe Ω.

Two more derived notions appear constantly. The *difference* $A \setminus B = A \cap B^c$ is "in A but not B." Two events are *disjoint* (or *mutually exclusive*) if $A \cap B = \emptyset$ — they cannot both happen. A collection $A_1, A_2, \ldots$ *partitions* Ω if the sets are pairwise disjoint and their union is all of Ω; this is exactly the setup behind the law of total probability.

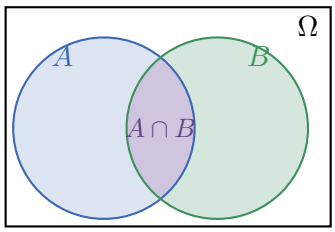


Figure 39.1: Two events as overlapping regions inside the universe Ω. The purple lens is $A \cap B$; the entire shaded area is $A \cup B$; everything outside both circles is $(A \cup B)^c = A^c \cap B^c$.

39.1.2 The identities you actually use

The following laws are the grammar of set algebra. Each one has a one-line "why" that you can check by chasing a single outcome ω through both sides — this technique, *element-chasing*, proves every set identity: to show $S = T$, show that $\omega \in S$ if and only if $\omega \in T$.

Formal

For all events $A, B, C \subseteq \Omega$:

Commutative:	$A \cup B = B \cup A, \qquad A \cap B = B \cap A.$
Associative:	$(A \cup B) \cup C = A \cup (B \cup C), \quad (A \cap B) \cap C = A \cap (B \cap C).$
Distributive:	$A \cap (B \cup C) = (A \cap B) \cup (A \cap C),$
	$A \cup (B \cap C) = (A \cup B) \cap (A \cup C).$
Identity:	$A \cup \emptyset = A, \qquad A \cap \Omega = A.$
Complement:	$A \cup A^c = \Omega, \qquad A \cap A^c = \emptyset, \qquad (A^c)^c = A.$
De Morgan:	$(A \cup B)^c = A^c \cap B^c, \qquad (A \cap B)^c = A^c \cup B^c.$

The last line is the famous pair of *De Morgan's laws*, and they deserve a moment because probability uses them constantly — often to turn a hard "at least one" question into an easy "none" question.

Why it works

De Morgan, in words and then in symbols. Saying "it is *not* the case that (A or B)" is exactly the same as saying "not-A *and* not-B." If neither rained-on-Monday nor rained-on-Tuesday happened, then it failed to rain on Monday *and* it failed to rain on Tuesday.
Element-chasing makes it airtight. Take $\omega \in (A \cup B)^c$. By definition of complement, $\omega \notin A \cup B$, which means ω is in neither A nor B, i.e. $\omega \in A^c$ and $\omega \in B^c$, i.e. $\omega \in A^c \cap B^c$. Every step is reversible, so the two sets contain exactly the same outcomes and are therefore equal. The general form for any collection is

$$\Big(\bigcup_i A_i\Big)^c = \bigcap_i A_i^c, \qquad \Big(\bigcap_i A_i\Big)^c = \bigcup_i A_i^c.$$

What is really happening

Why probabilists love the complement trick. Suppose you want the probability that *at least one* of n independent events occurs. Computed directly, "at least one" is a sprawling union with many overlaps. But its complement, by De Morgan, is "none of them occur" — a single clean intersection. So

$$\mathbb{P}\left(\bigcup_{i=1}^{n} A_i\right) = 1 - \mathbb{P}\left(\bigcap_{i=1}^{n} A_i^c\right),$$

and for independent events the right-hand side factors into a product. This is the engine behind the birthday problem (Chapter on counting) and behind any "probability that the system fails" reliability calculation. The set identity does the heavy lifting; probability just rides on top.

39.1.3 Cartesian products and power sets

Two constructions show up whenever we model several things at once or talk about "all possible events."

The *Cartesian product* $A \times B = \{(a, b) : a \in A,\ b \in B\}$ is the set of ordered pairs. If you roll a die and flip a coin, the joint sample space is $\{1, \dots, 6\} \times \{H, T\}$, which has $6 \times 2 = 12$ outcomes. This multiplication of sizes is no accident — it is the *multiplication principle* of counting, which we meet again in Section 39.3. For finite sets, $|A \times B| = |A| \cdot |B|$.

The *power set* $\mathcal{P}(\Omega)$ is the set of *all* subsets of Ω, including $\emptyset$ and Ω itself. When Ω is finite with $|\Omega| = n$, a subset is built by an independent yes/no choice for each element — "in or out" — so there are 2^n subsets:

$$|\mathcal{P}(\Omega)| = 2^{|\Omega|}.$$

This is why a finite probability model on n outcomes has 2^n possible events. (For infinite Ω we cannot use *all* subsets; we restrict to a σ-algebra. That subtlety belongs to the measure-theory discussion in the main text, but it is the reason the careful definition of an event is a little fussier than "any subset.")

Connection

The "in-or-out" bijection between subsets of an n-element set and binary strings of length n is the same idea that powers the binomial theorem in Section 39.2: choosing which factors contribute is exactly choosing a subset. Counting subsets by their size will hand us $\sum_k \binom{n}{k} = 2^n$ for free. Keep that picture in mind — sets and sums are two views of one object.

39.2 Sigma-notation and the sums we cannot live without

39.2.1 Reading and bending $\sum$

The symbol $\sum_{k=m}^{n} a_k$ means "add up a_k as the index k runs from m to n." The index is a *dummy*: $\sum_{k=1}^{n} a_k$ and $\sum_{j=1}^{n} a_j$ are the same number. Three manipulations cover nearly every move you will ever make.

- **Linearity.** $\displaystyle\sum_k (c\,a_k + b_k) = c\sum_k a_k + \sum_k b_k$. Constants pull out; sums split apart. This is exactly why expectation is linear — $\mathbb{E}$ is a weighted sum.
- **Reindexing (shifting).** Let $j = k - m$ to slide the limits: $\sum_{k=m}^{n} a_k = \sum_{j=0}^{n-m} a_{j+m}$. Useful for lining up two sums so their terms match.
- **Swapping double sums.** $\sum_i \sum_j a_{ij} = \sum_j \sum_i a_{ij}$ when the terms are nonnegative or absolutely summable. This is the discrete twin of Fubini's theorem and underlies "$\mathbb{E}[\sum X_i] = \sum \mathbb{E}[X_i]$" even when the X_i are dependent.

Common trap

Off-by-one is the national sport of summation. The sum $\sum_{k=0}^{n}$ has $n+1$ terms, not n. The sum $\sum_{k=1}^{n}$ has n. When you reindex, change *both* the limits and the summand together, and sanity-check by writing out the first and last term explicitly. A single misplaced endpoint has wrecked more probability calculations than any deep misunderstanding.

39.2.2 The arithmetic series

Adding the first n positive integers gives a result every probabilist has memorized, and its proof is the prettiest argument in elementary mathematics.

Proposition 39.1 (Gauss's sum)**.** *For every $n \geq 1$,*

$$\sum_{k=1}^{n} k = 1 + 2 + \cdots + n = \frac{n(n+1)}{2}.$$

Why it works

Write the sum forwards and backwards and add columnwise:

$$\begin{array}{rcccccc} S & = & 1 & + & 2 & +\cdots+ & n, \\ S & = & n & + & (n-1) & +\cdots+ & 1. \end{array}$$

Each of the n columns adds to $n+1$, so $2S = n(n+1)$, giving $S = n(n+1)/2$. There is no induction, no cleverness to memorize — just a reflection. A standard telescoping argument gives $\sum_{k=1}^{n} k^2 = n(n+1)(2n+1)/6$, which you meet whenever you compute the variance of a discrete uniform distribution.

History

The legend is that a schoolboy named Carl Friedrich Gauss, told to add the numbers from 1 to 100 to keep him quiet, produced 5050 almost at once by pairing $1{+}100, 2{+}99, \ldots$ into fifty pairs of 101. Whether or not the classroom story is embroidered, the pairing trick is genuinely how the identity is best seen, and Gauss really did grow into the mathematician who, among a hundred other things, gave us the normal distribution at the heart of this book.

39.2.3 The geometric series: the most important sum in probability

If one sum deserves to be tattooed on your wrist, it is this one. It governs the geometric distribution, the expected number of trials until a success, the present value of a perpetuity, and the steady state of a Markov chain.

Proposition 39.2 (Finite and infinite geometric series)**.** *For $r \neq 1$,*

$$\sum_{k=0}^{n-1} r^k = \frac{1-r^n}{1-r}.$$

If $|r| < 1$, then $r^n \to 0$ and the infinite series converges:

$$\sum_{k=0}^{\infty} r^k = \frac{1}{1-r}.$$

Why it works

The telescoping trick. Let $S = 1 + r + r^2 + \cdots + r^{n-1}$. Multiply by r: $rS = r + r^2 + \cdots + r^n$. Subtract:

$$S - rS = 1 - r^n \quad \Longrightarrow \quad S(1-r) = 1 - r^n \quad \Longrightarrow \quad S = \frac{1-r^n}{1-r}.$$

Almost everything cancels — the middle terms *telescope*, leaving only the first term of S and the last term of rS. For $|r| < 1$ the tail r^n shrinks to nothing, so letting $n \to \infty$ leaves $1/(1-r)$. The condition $|r| < 1$ is not a technicality you may ignore: if $|r| \geq 1$ the terms do not shrink and the sum is infinite or undefined.

A relative that appears whenever we compute the *mean* of a geometric random variable is the differentiated series. Differentiating $\sum r^k = 1/(1-r)$ term by term with respect to r (legal inside

the radius of convergence) gives

$$\sum_{k=1}^{\infty} k\, r^{k-1} = \frac{1}{(1-r)^2}, \qquad \text{hence} \qquad \sum_{k=1}^{\infty} k\, r^{k} = \frac{r}{(1-r)^2} \quad (|r|<1).$$

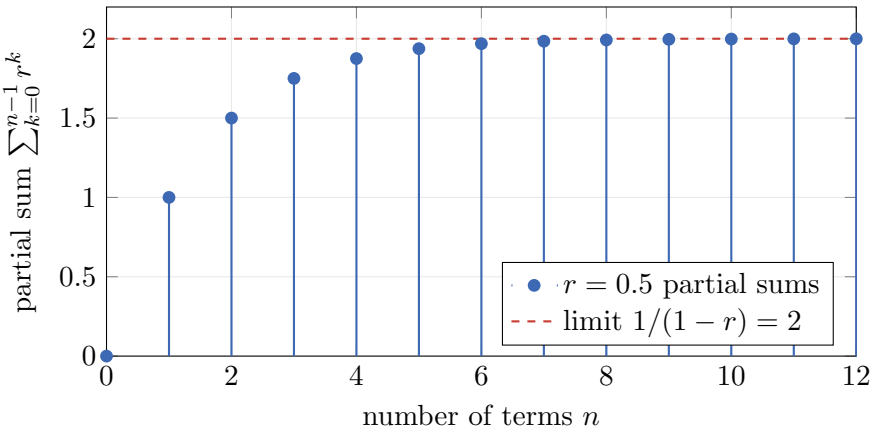


Figure 39.2: Partial sums of the geometric series with $r = 0.5$ marching up to their limit $1/(1-r) = 2$. The gap remaining after n terms is exactly $r^n/(1-r)$, which halves at every step.

Application

Where does $1/(1-r)$ become a probability statement? Let X be the number of independent trials, each succeeding with probability p, up to and including the first success (a geometric random variable). Then $\mathbb{P}(X = k) = (1-p)^{k-1}p$ for $k = 1, 2, \ldots$. For this to be a valid distribution the probabilities must sum to 1, and indeed, with $r = 1-p$,

$$\sum_{k=1}^{\infty} (1-p)^{k-1}p = p\sum_{j=0}^{\infty} r^j = p \cdot \frac{1}{1-r} = \frac{p}{p} = 1.$$

The geometric series is what *certifies* the geometric distribution. And the differentiated series gives its mean: $\mathbb{E}[X] = \sum_k k(1-p)^{k-1}p = p/(1-r)^2 = 1/p$. So if a success has probability $1/6$, you wait 6 trials on average — a fact you can now derive rather than recall.

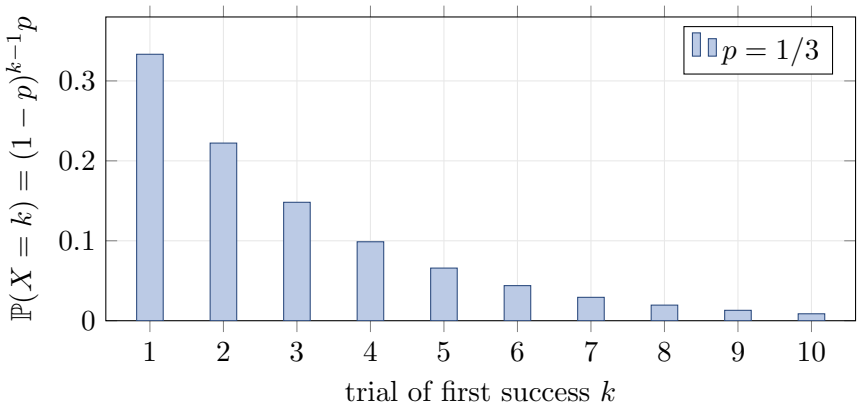


Figure 39.3: The geometric pmf with $p = 1/3$. The bar heights are exactly the terms of the geometric series $p\sum_{k\geq 1} r^{k-1}$ with $r = 1-p = 2/3$; each bar is r times the one before it, and the whole row of bars sums to 1. The mean $\mathbb{E}[X] = 1/p = 3$ sits to the right of the tall first bar because the long thin tail pulls the balance point outward.

39.2.4 The exponential series

The number e and its series are the bridge from discrete to continuous probability — they convert the binomial into the Poisson and define the exponential and normal densities.

Proposition 39.3 (Exponential series)**.** *For every real (indeed complex) x,*

$$e^x = \sum_{k=0}^{\infty} \frac{x^k}{k!} = 1 + x + \frac{x^2}{2} + \frac{x^3}{6} + \cdots,$$

and equivalently $e^x = \lim_{n\to\infty} \left(1 + \frac{x}{n}\right)^n$.

What is really happening

Why two formulas for the same thing, and why probability needs both. The series $\sum x^k/k!$ is the Taylor expansion of e^x; it is what you differentiate and integrate. The limit $(1 + x/n)^n$ is the "compound-interest" face of e: split a rate x into n tiny increments and compound. Probability uses the second form to derive the Poisson distribution as the limit of a binomial: take n trials each with tiny success probability λ/n, and watch

$$\left(1 - \frac{\lambda}{n}\right)^n \longrightarrow e^{-\lambda}.$$

That single limit is why rare events — radioactive decays, typos per page, mutations per genome — follow a Poisson law. The two faces of e are the discrete world and the continuous world shaking hands.

Connection

Setting $x = -\lambda$ shows $\sum_{k=0}^{\infty} e^{-\lambda}\lambda^k/k! = e^{-\lambda}e^{\lambda} = 1$, which is exactly the statement that the Poisson pmf sums to one. As with the geometric distribution, a series identity is what makes

a distribution legal. The calculus you need to differentiate these series term by term — power series, radius of convergence, term-by-term differentiation — is developed in the companion calculus volume; here we use it as a trusted tool.

39.2.5 The binomial theorem

This is the algebraic identity that ties counting to sums. It is the reason the binomial *distribution* is named after it.

Theorem 39.1 (Binomial theorem)**.** *For any numbers x, y and any integer $n \geq 0$,*

$$(x+y)^n = \sum_{k=0}^{n} \binom{n}{k} x^k y^{n-k},$$

where $\binom{n}{k} = \dfrac{n!}{k!\,(n-k)!}$ is the number of k-element subsets of an n-element set.

Why it works

The proof is a counting argument, not an algebra grind. Write $(x+y)^n = (x+y)(x+y)\cdots(x+y)$ with n factors. To expand, you pick either x or y from *each* factor and multiply your picks; the full product is the sum over all 2^n such choices. A term equals $x^k y^{n-k}$ precisely when you chose x from k of the factors and y from the other $n-k$. How many ways are there to choose *which* k factors give the x? Exactly $\binom{n}{k}$. Collecting like terms gives the coefficient $\binom{n}{k}$ in front of $x^k y^{n-k}$. The binomial coefficients are not decorations; they are literally counting the ways the term arises.

Two specializations are worth keeping at your fingertips. Setting $x = y = 1$:

$$2^n = \sum_{k=0}^{n} \binom{n}{k},$$

the count of all subsets, matching the power-set count from Section 39.1. Setting $x = 1,\ y = -1$ (for $n \geq 1$):

$$0 = \sum_{k=0}^{n} (-1)^k \binom{n}{k},$$

so the alternating sum of a row of Pascal's triangle vanishes — the even- and odd-sized subsets exactly balance.

Application

The binomial distribution sums to one. If $X \sim \text{Binomial}(n, p)$ counts successes in n independent trials, then $\mathbb{P}(X = k) = \binom{n}{k} p^k (1-p)^{n-k}$. Summing over k and applying the binomial theorem with $x = p$, $y = 1 - p$:

$$\sum_{k=0}^{n} \binom{n}{k} p^k (1-p)^{n-k} = \big(p + (1-p)\big)^n = 1^n = 1.$$

Once again an algebraic identity guarantees a probability distribution is well-defined. The pattern across this whole section is unmistakable: every named distribution is licensed by a named sum.

39.3 Combinatorial identities: counting the same thing two ways

39.3.1 The two principles everything rests on

All of counting comes from two ideas so simple they feel like cheating.

Multiplication principle. If a process has two stages, with m ways to do the first and (for each of those) n ways to do the second, there are $m \cdot n$ ways total. Generalizes to any number of stages. This is why an n-element set has $n!$ orderings: n choices for first place, $n-1$ for second, and so on.

Addition principle. If the outcomes split into disjoint cases, the total count is the sum of the case counts. This is just $|A \cup B| = |A| + |B|$ for disjoint A, B — the set law from Section 39.1 reappearing as a counting rule.

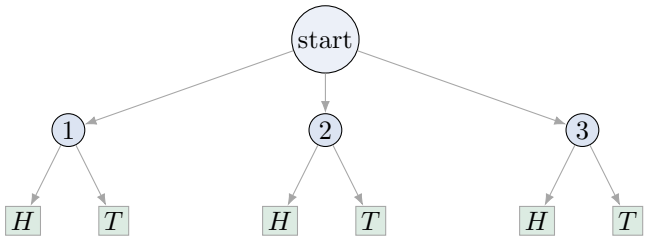


Figure 39.4: The multiplication principle as a tree. Rolling a 3-sided die (blue, 3 choices) and then flipping a coin (green, 2 choices) yields $3 \times 2 = 6$ leaves, one per outcome. Each layer of branching multiplies the count; this is the same product that makes $|A \times B| = |A|\,|B|$ in Section 39.1.

From these we get the four basic counts. Choosing k items from n:

	order matters	**order ignored**
no repetition	$\dfrac{n!}{(n-k)!}$	$\dbinom{n}{k} = \dfrac{n!}{k!(n-k)!}$
with repetition	n^k	$\dbinom{n+k-1}{k}$

Intuition

The binomial coefficient $\binom{n}{k}$ — read "n choose k" — is the star of the table. It counts the k-element subsets of an n-set. The formula $n!/(k!(n-k)!)$ has a clean story: order all n items ($n!$ ways), then agree not to care about the order of the k you keep ($k!$ ways) nor the order of the $n-k$ you discard ($(n-k)!$ ways). Dividing out the orderings you do not care about leaves the count of subsets.

39.3.2 Identities and how to read them

The most satisfying way to prove a combinatorial identity is a *combinatorial proof*: find a set of objects, count it two different ways, and conclude that the two formulas are equal. No algebra, just two counts of the same pile. Here are the identities the book uses, each with its two-counts story.

Formal

For integers $0 \le k \le n$:

$$\binom{n}{k} = \binom{n}{n-k} \qquad \text{(symmetry)} \tag{39.1}$$

$$\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k} \qquad \text{(Pascal's rule)} \tag{39.2}$$

$$\sum_{k=0}^{n} \binom{n}{k} = 2^n \qquad \text{(row sum)} \tag{39.3}$$

$$\sum_{k=0}^{r} \binom{m}{k}\binom{n}{r-k} = \binom{m+n}{r} \qquad \text{(Vandermonde)} \tag{39.4}$$

$$k\binom{n}{k} = n\binom{n-1}{k-1} \qquad \text{(absorption / committee–chair)} \tag{39.5}$$

Why it works

Symmetry (39.1)**.** Choosing which k people to *invite* is the same as choosing which $n-k$ to *leave out*. Each choice of an invited group corresponds to exactly one excluded group, so the two counts must match. (Algebraically, swap k and $n-k$ in the factorial formula and nothing changes.)

Pascal's rule (39.2)**.** Count the k-subsets of $\{1,\dots,n\}$ by whether they contain element n. If they *do*, the remaining $k-1$ members come from the other $n-1$ elements: $\binom{n-1}{k-1}$ ways. If they *do not*, all k come from the other $n-1$: $\binom{n-1}{k}$ ways. The two cases are disjoint and exhaustive, so by the addition principle they sum to $\binom{n}{k}$. This single recurrence *is* Pascal's triangle.

Vandermonde's identity (39.4)**.** You have m women and n men and want to form a committee of r people: $\binom{m+n}{r}$ ways. Alternatively, decide first that the committee has k women (chosen $\binom{m}{k}$ ways) and $r-k$ men ($\binom{n}{r-k}$ ways), then sum over all possible k. Same committees, counted two ways.

Absorption (39.5)**.** Count ways to pick a k-person committee *and* designate one of its members as chair. Pick the committee, then the chair: $\binom{n}{k}\cdot k$. Or pick the chair from all n people first, then fill the remaining $k-1$ seats from the other $n-1$: $n\binom{n-1}{k-1}$. Equal counts. This identity is exactly the algebraic step that produces the mean of the binomial distribution, $\mathbb{E}[X] = np$.

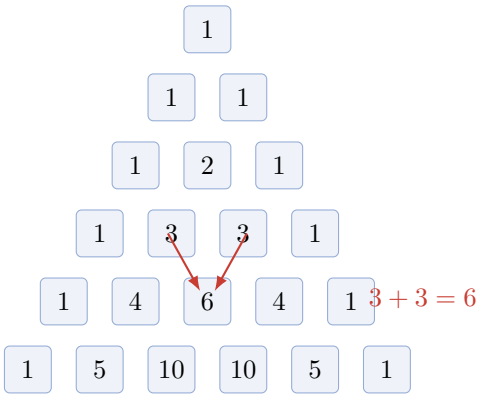


Figure 39.5: Pascal's triangle. Each entry is the sum of the two above it (Pascal's rule, (39.2)); row n holds $\binom{n}{0}, \dots, \binom{n}{n}$ and sums to 2^n.

Paradox

The birthday "paradox" is pure counting. In a room of just 23 people, the probability that two share a birthday already exceeds one half — shocking, because 23 feels tiny against 365 days. The resolution lives entirely in this section. Use the complement trick (Section 39.1): the chance that all 23 birthdays are *distinct* is, by the multiplication principle,

$$\frac{365}{365} \cdot \frac{364}{365} \cdots \frac{343}{365} = \frac{365!/(365-23)!}{365^{23}} \approx 0.493,$$

so a match has probability ≈ 0.507. The trap is that we instinctively compare ourselves to everyone else, but the relevant quantity is the number of *pairs*, $\binom{23}{2} = 253$ — and 253 chances at a 1/365 event is no longer rare. Nothing here is paradoxical; it is the gap between linear intuition (people) and quadratic reality (pairs).

39.3.3 Inclusion–exclusion

When events overlap, you cannot just add their sizes — you would double-count the overlap. Fixing this systematically gives one of the most useful counting (and probability) formulas in the book.

Theorem 39.2 (Inclusion–exclusion)**.** *For any events $A_1, \dots, A_n$,*

$$\Big|\bigcup_{i=1}^{n} A_i\Big| = \sum_i |A_i| - \sum_{i<j} |A_i \cap A_j| + \sum_{i<j<k} |A_i \cap A_j \cap A_k| - \cdots + (-1)^{n+1} \Big|\bigcap_{i=1}^{n} A_i\Big|.$$

The identical statement holds with $|\cdot|$ replaced by $\mathbb{P}(\cdot)$.

Why it works

For two sets it is the picture in Figure 39.1: $|A \cup B| = |A| + |B| - |A \cap B|$, because adding the two circles counts the lens twice, so we subtract it once. For three sets you add back the triple overlap you over-subtracted, and so on — each correction fixes the previous over- or under-count. The clean proof tracks a single element belonging to exactly m of the sets and

checks that the alternating sum counts it

$$\binom{m}{1} - \binom{m}{2} + \binom{m}{3} - \cdots = 1 - (1-1)^m = 1$$

time (using the binomial theorem with $x = -1, y = 1!$), exactly as it should. The combinatorial identity (39.3)'s alternating cousin is doing the bookkeeping.

Application

Inclusion–exclusion in the data center. A search or analytics engine constantly needs the size of a union: "how many distinct users visited page A *or* page B *or* page C?" Storing every user id is expensive, so systems keep a tiny probabilistic sketch (a Bloom filter or a HyperLogLog counter) of each set. To estimate the union from the individual sketches, engineers apply exactly Theorem 39.2: add the single-set cardinalities, subtract the pairwise intersections, add back the triples. The same alternating signs that fix double-counting on a Venn diagram fix it at the scale of billions of rows. When you read about "set cardinality estimation" in a modern database, you are reading inclusion– exclusion wearing an engineering hat.

39.4 A little asymptotics: how big is $n!$?

Factorials explode. By the time $n = 70$, $n!$ exceeds the number of atoms in the observable universe, and the exact value is useless for intuition. What we usually need is not the exact factorial but its *size and shape* as n grows — and for that we have one of the most beautiful approximations in mathematics.

Theorem 39.3 (Stirling's approximation)**.** *As $n \to \infty$,*

$$n! \sim \sqrt{2\pi n}\left(\frac{n}{e}\right)^n,$$

meaning the ratio of the two sides tends to 1*. A more precise version with error bounds is*

$$n! = \sqrt{2\pi n}\left(\frac{n}{e}\right)^n \exp\!\Big(\frac{1}{12n} + O(n^{-3})\Big).$$

What is really happening

Where the pieces come from. Take logs: $\ln n! = \sum_{k=1}^{n} \ln k$. A sum of a slowly varying function is well approximated by an integral, and $\int_1^n \ln x \, dx = n \ln n - n + 1$. That already explains the dominant factor $(n/e)^n = e^{n \ln n - n}$: the $n \ln n$ is the area under $\ln x$, and the $-n$ is the $-n$ in the integral. The leftover $\sqrt{2\pi n}$ comes from the correction to replacing a sum by an integral (the trapezoid/Euler–Maclaurin correction), and the 2π is the same 2π that appears in the normal density — not a coincidence, as we will see.

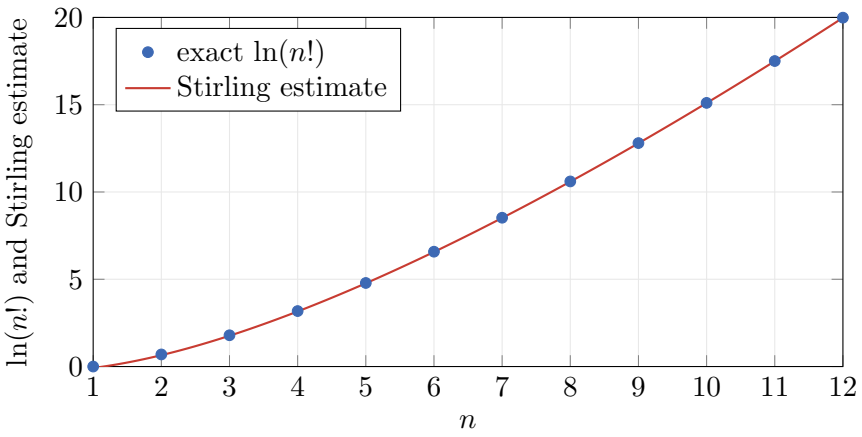


Figure 39.6: Stirling's formula tracks $\ln(n!)$ astonishingly well even for small n. At $n = 10$ the relative error in $n!$ itself is already under one percent.

Application

Stirling builds the bell curve. The central limit theorem's bell shape can be seen directly by applying Stirling to the binomial coefficient $\binom{2m}{m}$, the central term of the symmetric binomial. Stirling gives

$$\binom{2m}{m} = \frac{(2m)!}{(m!)^2} \sim \frac{4^m}{\sqrt{\pi m}}.$$

Normalizing the binomial Binomial$(n, \frac{1}{2})$ and letting n grow, those $\sqrt{\cdot}$ factors assemble into $\frac{1}{\sqrt{2\pi}} e^{-x^2/2}$ — the standard normal density. The 2π inside Stirling *is* the 2π inside the Gaussian. This is the most concrete route to "why is the bell curve shaped like that," and it runs straight through this appendix.

History

The formula is named for James Stirling, who published it in *Methodus Differentialis* (1730), but the leading behavior and the crucial constant $\sqrt{2\pi}$ were nailed down in correspondence with Abraham de Moivre, who needed exactly this estimate to approximate the binomial distribution. De Moivre's binomial-to-bell calculation, powered by Stirling's factorial estimate, is the historical first appearance of the normal curve in probability — decades before Gauss. The toolbox in this appendix is, quite literally, where the most important distribution in the book was born.

Why this matters

Step back and notice the recurring theme of this appendix. A set identity (De Morgan, complement) turns a hard probability into an easy one. A sum identity (geometric, exponential, binomial) certifies that a distribution is legal and computes its mean. A counting identity (Pascal, Vandermonde, absorption) proves itself by counting one pile two ways. And an asymptotic identity (Stirling) turns the discrete binomial into the continuous bell. None of these are isolated facts; they are four dialects of one language for measuring possibility. Fluency in that language is most of what it means to be good at probability.

39.5 Exercises

Exercise 39.1 (Element-chasing)**.** Prove the distributive law $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$ by showing each side is a subset of the other. Then state the dual law obtained by swapping $\cup$ and $\cap$, and explain why De Morgan's laws make the two duals equivalent.

Exercise 39.2 (Complement trick)**.** A machine has 5 components that fail independently, each with probability 0.1. The machine works only if *all* components work. Using De Morgan and the complement, write the probability that the machine fails as $1 - \mathbb{P}(\text{none fail})$ and compute it. Why is this far easier than summing over the ways at least one component fails?

Exercise 39.3 (Geometric series)**.** Sum each series, stating any convergence condition you use: (a) $\sum_{k=0}^{\infty}(2/3)^k$; (b) $\sum_{k=2}^{\infty}(2/3)^k$; (c) $\sum_{k=1}^{\infty} k(2/3)^k$. For (c), identify which derived identity from Section 39.2 you are using.

Exercise 39.4 (Distributions from sums)**.** Verify by a series identity that each is a valid probability distribution (probabilities are nonnegative and sum to 1): (a) Poisson, $\mathbb{P}(X = k) = e^{-\lambda}\lambda^k/k!$ for $k = 0, 1, 2, \ldots$; (b) Geometric, $\mathbb{P}(X = k) = (1-p)^{k-1}p$ for $k = 1, 2, \ldots$. Name the identity used in each case.

Exercise 39.5 (Binomial theorem in action)**.** Use the binomial theorem to evaluate $\sum_{k=0}^{n}\binom{n}{k}2^k$ in closed form. Then give a one-sentence combinatorial interpretation of your answer in terms of choosing a subset and then doing something extra to its elements.

Exercise 39.6 (Combinatorial proof)**.** Give a committee-counting (two-ways) proof of Vandermonde's identity $\sum_{k=0}^{r}\binom{m}{k}\binom{n}{r-k} = \binom{m+n}{r}$. As a special case, set $m = n$ and $r = n$ to deduce $\sum_{k=0}^{n}\binom{n}{k}^2 = \binom{2n}{n}$, and explain in words what is being counted.

Exercise 39.7 (Pascal and the triangle)**.** Prove Pascal's rule $\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}$ both (a) combinatorially (condition on whether a fixed element is chosen) and (b) algebraically from the factorial formula. Which proof would you rather have shown a beginner, and why?

Exercise 39.8 (Absorption and the binomial mean)**.** Starting from $\mathbb{E}[X] = \sum_{k=0}^{n} k\binom{n}{k}p^k(1-p)^{n-k}$ for $X \sim \text{Binomial}(n, p)$, apply the absorption identity (39.5) to rewrite $k\binom{n}{k}$, reindex the sum, and conclude $\mathbb{E}[X] = np$. State clearly where the binomial theorem reappears to collapse the leftover sum.

Exercise 39.9 (Inclusion–exclusion)**.** At a party, 70% like tea, 80% like coffee, and 60% like both. What fraction like at least one of the two? Now suppose a third drink is added with its own overlaps; write the general three-set inclusion–exclusion formula you would use, and explain why the signs alternate.

Exercise 39.10 (Stirling)**.** (a) Use Stirling's approximation to show $\binom{2m}{m} \sim 4^m/\sqrt{\pi m}$. (b) Estimate $\binom{100}{50}$ to one significant figure and compare the *exponent of growth* (4^{50}) with the polynomial correction ($1/\sqrt{50\pi}$). What does the relative smallness of the correction tell you about why the central binomial probability still shrinks as m grows?

Exercise 39.11 (Conceptual synthesis)**.** This appendix repeatedly used one identity to license a probability fact. Pick any *three* distributions from the main text and, for each, name the single sum or counting identity that proves its probabilities total 1. Write two sentences on why "the probabilities sum to one" is never an accident but always a disguised version of an identity from this chapter.

Chapter 40

Appendix: The Calculus and Linear Algebra You Need

Probability is built on two older subjects, and it leans on them constantly. Calculus is the language of *continuous* chance: the moment you write a probability density, an expectation becomes an integral, a tail probability becomes an area, and the normal curve becomes a function you actually have to integrate. Linear algebra is the language of *many variables at once*: a vector of random quantities, a covariance *matrix*, a Markov chain's transition matrix, a principal component, a least-squares prediction — all of these are linear-algebra objects wearing probabilistic clothes.

This appendix is not a course. It is a focused, intuition-first refresher on exactly the calculus and linear algebra the rest of this book uses, and nothing more. Where a topic deserves a full treatment, we point you to the companion volumes — *Calculus From the Ground Up* and *Linear Algebra From the Ground Up* — which develop these ideas with the same care we bring to probability here. The aim is narrower: to make sure that when the chapter on expectation writes $\mathbb{E}[X] = \int x f(x)\,\mathrm{d}x$, or the chapter on the multivariate normal writes $\Sigma = Q\Lambda Q^{\top}$, you see exactly what those symbols are doing and why they are allowed to.

The appendix comes in two halves. The calculus half covers derivatives as rates and slopes, integrals as accumulated area, the Fundamental Theorem that ties them together, multiple integrals over regions of the plane, and gradients for functions of several variables. The linear-algebra half covers vectors and the dot product, matrices as linear maps, eigenvectors and eigenvalues, the special role of symmetric and positive-definite matrices, and orthogonal projection. Throughout, we keep one eye on probability and flag the exact spot in the main text where each tool gets used.

Focus

You do not need to be fluent in calculus and linear algebra to read this book. You need to be *comfortable* with a specific short list of ideas, and this appendix is that list. If a section feels obvious, skim it; if a section feels new, slow down — it will pay off the next time you meet an integral or a covariance matrix in the wild.

40.1 Part I: The Calculus You Need

40.1.1 Functions, slopes, and the derivative

Start with the most basic object: a function f that takes a number x and returns a number $f(x)$. Picture its graph, the curve $y = f(x)$. The first question calculus asks is: *how fast is f changing* as x moves? That "rate of change" is the slope of the curve, and the derivative measures it.

The trick is to look at a tiny step. Move from x to $x+h$. The output changes from $f(x)$ to $f(x+h)$. The average rate of change over that step is the rise over the run,

$$\frac{f(x+h) - f(x)}{h}.$$

This is the slope of the straight line through the two nearby points $(x, f(x))$ and $(x+h, f(x+h))$ — a *secant* line. Now shrink the step. As $h \to 0$ the secant pivots until it just kisses the curve at x: it becomes the *tangent* line, and its slope is the derivative,

$$f'(x) \;=\; \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}.$$

We also write it $\frac{\mathrm{d}f}{\mathrm{d}x}$, read "d$f$ by dx," a notation that deliberately looks like a ratio of tiny changes, because morally that is what it is.

Intuition

The derivative answers "if I nudge the input by a hair, how much does the output move, per unit of nudge?" If $f'(x) = 3$, then near x the function climbs three units of output for every one unit of input. If $f'(x) = 0$, the curve is momentarily flat — a peak, a valley, or a plateau. If $f'(x) < 0$, the function is falling. Everything else in differential calculus is bookkeeping on top of this one idea.

A handful of derivatives cover almost everything we need in this book. The power rule says $\frac{\mathrm{d}}{\mathrm{d}x} x^n = n x^{n-1}$; in particular constants have derivative 0 and lines ax have derivative a. The exponential is its own derivative, $\frac{\mathrm{d}}{\mathrm{d}x} e^x = e^x$, which is the secret reason the exponential and normal distributions are so analytically friendly. The logarithm gives $\frac{\mathrm{d}}{\mathrm{d}x} \ln x = 1/x$. And two combining rules do the heavy lifting: the product rule $(fg)' = f'g + fg'$ and the chain rule $\big(f(g(x))\big)' = f'(g(x))\, g'(x)$. The chain rule is the one you will reach for most when transforming random variables — see the change-of-variables chapter, where the factor $|g'|$ that appears in density transformations is the chain rule made visible.

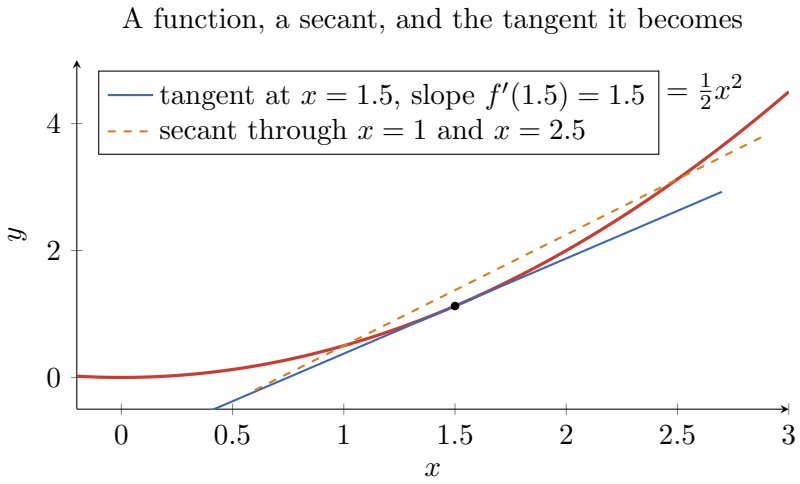


Figure 40.1: The derivative is the limiting slope. The dashed secant connects two points on the curve; as the points slide together it pivots into the solid tangent, whose slope is $f'(1.5) = 1.5$.

Connection

In probability the derivative shows up the moment a quantity is continuous. The probability density function f is literally the derivative of the cumulative distribution function F: $f(x) = F'(x)$. That is the whole content of "density" — it is the *rate* at which probability accumulates as x increases. We unpack this in the continuous-random-variables chapter; here just notice that "density = derivative of the CDF" is the same tangent-line idea applied to a curve that happens to measure probability. The companion *Calculus From the Ground Up* treats limits and derivatives in full.

40.1.2 The integral as accumulated area

Now turn the question around. Instead of "how fast is f changing," ask "how much of f has piled up between $x = a$ and $x = b$?" Geometrically: what is the area under the curve $y = f(x)$ from a to b? That area is the definite integral

$$\int_a^b f(x)\,\mathrm{d}x.$$

The construction mirrors the derivative's limiting step, but in reverse spirit. Chop the interval $[a, b]$ into many thin slices of width Δx. Over each slice the function is nearly constant, so the area of that sliver is approximately height $\times$ width, $f(x_i)\,\Delta x$. Add up all the slivers to get a *Riemann sum*, $\sum_i f(x_i)\,\Delta x$, and then let the slices get infinitely thin. In the limit the jagged staircase of rectangles becomes the smooth area under the curve, and the summation sign $\sum$ relaxes into the integral sign $\int$ — an elongated S, for "sum." The $\mathrm{d}x$ is the vanishing width left over from Δx.

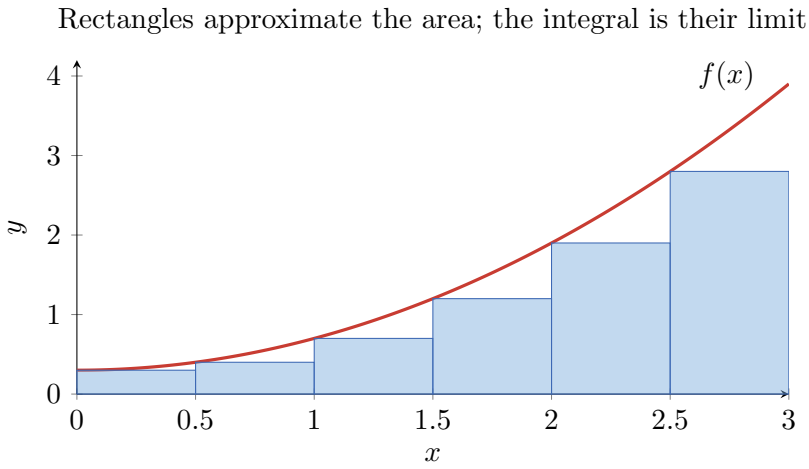


Figure 40.2: A Riemann sum. Each rectangle has area (height)×(width)= $f(x_i)\,\Delta x$. As the widths shrink, the total rectangle area converges to $\int_0^3 f(x)\,\mathrm{d}x$, the exact area under the curve.

What is really happening

An integral is a continuous sum. That single sentence is the bridge between the discrete and continuous worlds of probability. When a random variable is discrete you compute its expectation by a weighted sum, $\mathbb{E}[X] = \sum_x x\,\mathbb{P}(X = x)$. When it is continuous you compute the *same* thing as a weighted "sum" over a continuum, $\mathbb{E}[X] = \int x f(x)\,\mathrm{d}x$. Nothing conceptual changed — you are still adding up "value times how much probability sits there." Only the bookkeeping switched from $\sum$ to $\int$ because there are now uncountably many values, each carrying an infinitesimal slice of probability $f(x)\,\mathrm{d}x$. The measure-theory chapters make this unification airtight; the integral is the everyday face of it.

40.1.3 The Fundamental Theorem of Calculus

The two halves of calculus — slopes and areas — look unrelated. The miracle, and it really is one, is that they are inverse operations. That is the Fundamental Theorem of Calculus.

Theorem 40.1 (Fundamental Theorem of Calculus)**.** *Let f be continuous on $[a, b]$. Define the "area-so-far" function $A(x) = \int_a^x f(t)\,\mathrm{d}t$. Then A is differentiable and $A'(x) = f(x)$. Consequently, if F is any antiderivative of f (so $F' = f$), then*

$$\int_a^b f(x)\,\mathrm{d}x \;=\; F(b) - F(a).$$

Why it works

Why is the derivative of "area so far" equal to the height of the curve? Push the right edge of the shaded region from x to $x + h$. You add a thin strip of area. That strip is almost a

rectangle of width h and height $f(x)$, so the added area is about $f(x) \cdot h$. Therefore

$$\frac{A(x+h) - A(x)}{h} \approx f(x),$$

and as $h \to 0$ the approximation becomes exact: $A'(x) = f(x)$. The rate at which area accumulates is exactly the current height of the curve — which is just common sense once you see the strip. Knowing this, evaluating an integral becomes a search for an antiderivative: find any F with $F' = f$, then subtract its values at the endpoints.

This is exactly the relationship between a density and a CDF. Because $f = F'$, the Fundamental Theorem says $\mathbb{P}(a \le X \le b) = \int_a^b f(x)\,\mathrm{d}x = F(b) - F(a)$. Differentiating a CDF gives the density; integrating a density gives back probabilities. Calculus is the dictionary that translates between the two descriptions of a continuous random variable.

Example 40.1 (The exponential density integrates to one)**.** The Exponential(λ) density is $f(x) = \lambda e^{-\lambda x}$ for $x \ge 0$. An antiderivative of $\lambda e^{-\lambda x}$ is $F(x) = -e^{-\lambda x}$, because by the chain rule $F'(x) = \lambda e^{-\lambda x}$. Then the total probability is

$$\int_0^\infty \lambda e^{-\lambda x}\,\mathrm{d}x = \Big[-e^{-\lambda x} \Big]_0^\infty = \big(0\big) - \big(-1\big) = 1,$$

as a density must. The same antiderivative gives any tail: $\mathbb{P}(X > t) = \int_t^\infty \lambda e^{-\lambda x}\,\mathrm{d}x = e^{-\lambda t}$, the memoryless survival curve you meet in the exponential-distribution section.

40.1.4 Integration techniques you will actually meet

Two techniques recur often enough to name. The first is *substitution*, which is the chain rule read backwards. If you spot an inner function $u = g(x)$ whose derivative $g'(x)$ also appears (up to a constant), substitute and the integral simplifies:

$$\int f(g(x))\, g'(x)\,\mathrm{d}x = \int f(u)\,\mathrm{d}u, \qquad u = g(x),\ \mathrm{d}u = g'(x)\,\mathrm{d}x.$$

This is precisely the engine behind change-of-variables for densities: when $Y = g(X)$, the $\mathrm{d}u = g'(x)\,\mathrm{d}x$ bookkeeping is where the Jacobian factor $|g'|$ comes from.

The second is *integration by parts*, the product rule read backwards:

$$\int u\,\mathrm{d}v = uv - \int v\,\mathrm{d}u.$$

You use it to derive moments of distributions. For instance, $\mathbb{E}[X]$ for an exponential, or the gamma-function identity $\Gamma(n) = (n-1)!$, both fall out of one or two integrations by parts.

Example 40.2 (Mean of the exponential, by parts)**.** For $X \sim \text{Exponential}(\lambda)$,

$$\mathbb{E}[X] = \int_0^\infty x\,\lambda e^{-\lambda x}\,\mathrm{d}x.$$

Take $u = x$ (so $\mathrm{d}u = \mathrm{d}x$) and $\mathrm{d}v = \lambda e^{-\lambda x}\,\mathrm{d}x$ (so $v = -e^{-\lambda x}$). Then

$$\mathbb{E}[X] = \Big[-xe^{-\lambda x} \Big]_0^\infty + \int_0^\infty e^{-\lambda x}\,\mathrm{d}x = 0 + \Big[-\tfrac{1}{\lambda} e^{-\lambda x} \Big]_0^\infty = \frac{1}{\lambda}.$$

The boundary term vanishes because $e^{-\lambda x}$ crushes the linear x at infinity. So the mean of an exponential is $1/\lambda$ — a one-line computation once integration by parts is in hand.

History

The integral sign $\int$ and the $\mathrm{d}x$ notation are Leibniz's, introduced in the 1670s; the elongated S stood for *summa*, "sum." Newton, working at the same time, thought of the same ideas in terms of "fluxions" — quantities flowing in time — and the two were later embroiled in a bitter priority dispute. For probability the relevant later hero is Henri Lebesgue, who in 1902 reinvented the integral by slicing the *range* of f into thin horizontal bands instead of the domain into vertical strips. That seemingly small change is what lets us integrate against probability measures and unify discrete and continuous expectation, and it is the foundation of the measure-theoretic chapters near the end of this book.

40.1.5 Improper integrals and why tails can be finite

Probability densities live on infinite ranges — the normal stretches over all of $\mathbb{R}$, the exponential over $[0, \infty)$. Integrating to infinity is an *improper* integral, defined as a limit: $\int_a^\infty f = \lim_{b\to\infty} \int_a^b f$. The remarkable fact, which beginners find counterintuitive, is that an infinitely long region can enclose a finite area, provided the curve decays fast enough.

Paradox

Infinite length, finite area. Consider $f(x) = 1/x^2$ on $[1, \infty)$. The region under it runs forever to the right, yet

$$\int_1^\infty \frac{1}{x^2}\,\mathrm{d}x = \Big[-\tfrac{1}{x}\Big]_1^\infty = 0 - (-1) = 1.$$

A finite answer for an unbounded region! The resolution is that "length" and "area" are different beasts: the strips get thin so fast that their areas form a convergent sum. But beware the other direction. With $f(x) = 1/x$ the same region has *infinite* area, $\int_1^\infty \frac{1}{x}\,\mathrm{d}x = \lim_{b\to\infty} \ln b = \infty$, even though $1/x$ also decays to zero. The boundary between "decays fast enough" and "not quite" is exactly $1/x$, and it is the same boundary that decides whether a heavy-tailed distribution has a finite mean. A Cauchy distribution's density decays like $1/x^2$, so it integrates to a valid probability of 1; but x times that density decays like $1/x$, so its *mean* integral diverges — which is why the Cauchy has no mean at all. The St. Petersburg paradox and heavy-tailed risk models in the inequalities chapter are this same arithmetic of convergence.

40.1.6 Functions of several variables and multiple integrals

The moment two random variables share a story, we need functions of two inputs and integrals over two-dimensional regions. A joint density $f(x, y)$ is a surface sitting above the plane; the probability that (X, Y) lands in a region R is the *volume* under that surface over R,

$$\mathbb{P}\big((X, Y) \in R\big) = \iint_R f(x, y)\,\mathrm{d}x\,\mathrm{d}y.$$

You compute a double integral by integrating one variable at a time — an *iterated* integral. Hold y fixed, integrate over x (sweeping out the area of a single slice), then integrate the resulting function of y. Fubini's theorem guarantees that, for the well-behaved densities we use, the order does not matter: $\iint = \int(\int \mathrm{d}x)\mathrm{d}y = \int(\int \mathrm{d}y)\mathrm{d}x$. This is the engine behind *marginalization*: to get the density of X alone, integrate the joint density over all of y, $f_X(x) = \int_{-\infty}^{\infty} f(x,y)\,\mathrm{d}y$. You are summing the probability over every way the other variable could have come out — the law of total probability, written as an integral.

A joint density as a surface; probability is volume underneath

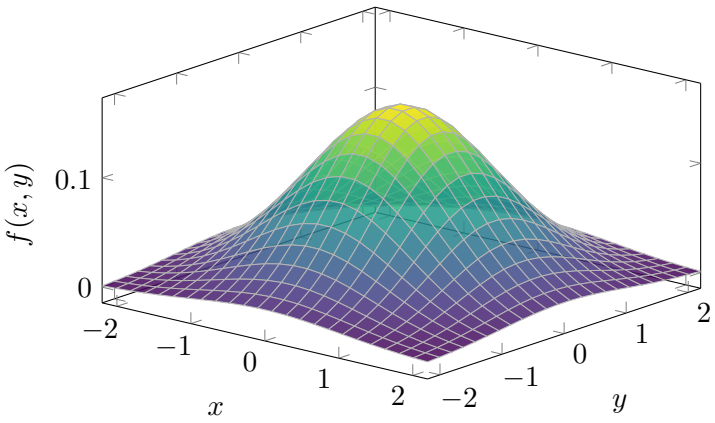


Figure 40.3: The standard bivariate normal density $f(x,y) = \frac{1}{2\pi}e^{-(x^2+y^2)/2}$. The probability of any region R in the xy-plane is the volume of the solid standing over R and below this bell-shaped surface. Marginalizing out y flattens the surface onto the x-axis.

Connection

Polar coordinates earn their keep in probability through one famous trick: the integral $\int_{-\infty}^{\infty} e^{-x^2/2}\,\mathrm{d}x$ has no elementary antiderivative, yet its *square* becomes a double integral that switches to polar coordinates and evaluates cleanly to 2π, giving $\int_{-\infty}^{\infty} e^{-x^2/2}\,\mathrm{d}x = \sqrt{2\pi}$. Its reciprocal $\frac{1}{\sqrt{2\pi}}$ is the normalizing constant of the standard normal, and the substitution $r^2 = x^2 + y^2$, $\mathrm{d}x\,\mathrm{d}y = r\,\mathrm{d}r\,\mathrm{d}\theta$ is the reason the $\sqrt{2\pi}$ appears. The normal-distribution chapter does this computation; the multiple-integral machinery here is what makes it legal.

40.1.7 Gradients: derivatives in many directions

For a function of several variables, $g(x_1, \ldots, x_n)$, there is no single "slope" — the surface tilts differently depending on which way you walk. The *partial derivative* $\partial g/\partial x_i$ measures the slope when you move along the x_i axis only, holding the others fixed; it is just an ordinary derivative with the other variables treated as constants. Collect all the partials into a vector and you get the *gradient*,

$$\nabla g = \left(\frac{\partial g}{\partial x_1},\ \ldots,\ \frac{\partial g}{\partial x_n}\right).$$

Intuition

The gradient points in the direction of steepest ascent, and its length is how steep that climb is. Stand on a hillside; the gradient is the compass bearing of "straight uphill," and its magnitude is the grade. At a peak, valley, or saddle the ground is momentarily level in every direction, so the gradient is the zero vector — that is the multivariable version of "set the derivative to zero to find the maximum."

This is the workhorse of estimation. Maximum likelihood (the MLE chapter) finds the parameter that maximizes a likelihood by setting its gradient to zero and solving. Modern machine learning (the ML chapter) does the same in reverse: *gradient descent* repeatedly steps in the direction of $-\nabla g$, downhill, to minimize a loss function. Both are the one-variable idea "optimum where the derivative is zero" lifted to many parameters at once. The companion *Calculus From the Ground Up* develops partial derivatives and gradients carefully; we only need the picture of "uphill arrow whose zero marks an optimum."

40.2 Part II: The Linear Algebra You Need

If calculus is the mathematics of *change*, linear algebra is the mathematics of *many quantities handled together*. The instant you study several random variables jointly — their spread, their correlations, how a Markov chain shuffles probability between states — you are doing linear algebra whether you call it that or not. What follows is the small toolkit the book relies on.

40.2.1 Vectors and the dot product

A vector is an ordered list of numbers, $\mathbf{v} = (v_1, \ldots, v_n)$, which you can picture as an arrow from the origin to the point $(v_1, \ldots, v_n)$ in n-dimensional space. In probability a vector might hold a random sample $(X_1, \ldots, X_n)$, a vector of means, or the probabilities of a chain's states. Two operations make vectors useful: you can add them coordinatewise, and you can scale them by a number. That is the entire definition of "linear" — additions and scalings, nothing more exotic.

The single most important vector operation for us is the *dot product*,

$$\mathbf{u} \cdot \mathbf{v} = u_1 v_1 + u_2 v_2 + \cdots + u_n v_n = \sum_{i=1}^{n} u_i v_i.$$

It eats two vectors and returns one number. Two facts give it meaning. First, a vector's length is $\|\mathbf{v}\| = \sqrt{\mathbf{v} \cdot \mathbf{v}}$ — the Pythagorean theorem in n dimensions. Second, the dot product measures alignment: $\mathbf{u} \cdot \mathbf{v} = \|\mathbf{u}\|\,\|\mathbf{v}\| \cos\theta$, where θ is the angle between the arrows. So a dot product of zero means the vectors are perpendicular — *orthogonal* — and a large positive dot product means they point much the same way (Figure 40.4).

What is really happening

Correlation *is* a dot product. Center two data vectors by subtracting their means, and the

sample correlation is exactly the cosine of the angle between the centered vectors:

$$\mathrm{Corr}(X,Y) = \frac{\sum_i (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_i (x_i - \bar{x})^2}\,\sqrt{\sum_i (y_i - \bar{y})^2}} = \cos\theta.$$

Numerator: a dot product. Denominator: the two lengths. This is why correlation always lies in $[-1, 1]$ — because a cosine does — and why "uncorrelated" is the statistical name for "orthogonal." The covariance-and-correlation chapter leans on this geometry; the whole notion of one variable being "perpendicular" to another is the dot product talking.

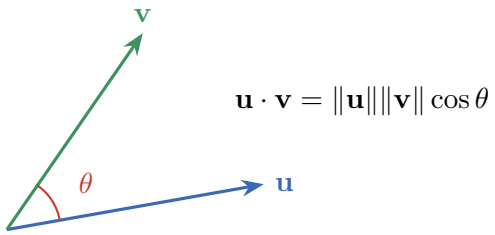


Figure 40.4: The dot product as alignment. Two vectors meet at angle θ; their dot product is the product of their lengths times $\cos\theta$. When the centered data vectors are used, this $\cos\theta$ is exactly the sample correlation, which is why correlation is pinned to $[-1, 1]$ and why "uncorrelated" means "perpendicular."

40.2.2 Matrices as linear transformations

A matrix is a rectangular grid of numbers, but the right way to think about it is as a *machine*: feed it a vector, it returns another vector. A matrix A acts on $\mathbf{x}$ by the rule that the i-th output coordinate is the dot product of the i-th row of A with $\mathbf{x}$. Matrix multiplication is just the composition of two such machines: AB means "do B, then A."

The crucial property is *linearity*: $A(\mathbf{x} + \mathbf{y}) = A\mathbf{x} + A\mathbf{y}$ and $A(c\mathbf{x}) = cA\mathbf{x}$. Geometrically, a matrix sends lines to lines and keeps the origin fixed; it can rotate, stretch, shear, and reflect space, but it cannot bend it. This is exactly the linearity that makes $\mathbb{E}$ behave well: expectation of a linear combination is the linear combination of expectations, $\mathbb{E}[A\mathbf{X}] = A\,\mathbb{E}[\mathbf{X}]$, treating the random vector $\mathbf{X}$ just like any vector the matrix acts on.

Example 40.3 (A 2×2 map in action)**.** Let $A = \begin{pmatrix} 2 & 0 \\ 0 & \frac{1}{2} \end{pmatrix}$. Acting on $\mathbf{x} = (x_1, x_2)$ it returns $(2x_1,\ \frac{1}{2}x_2)$: it stretches the horizontal direction by 2 and squashes the vertical by half. A circle of data points becomes an ellipse twice as wide and half as tall. Keep this picture — "a matrix turns a circle into an ellipse" — because it is precisely how a covariance matrix reshapes a round cloud of independent noise into the tilted elliptical cloud of correlated data.

40.2.3 Transition matrices and the stochastic special case

A Markov chain is a matrix in disguise. List the states $1, \ldots, n$ and let $P_{ij} = \mathbb{P}(\text{go to } j \mid \text{now in } i)$. The matrix P is *stochastic*: every entry is a probability in $[0, 1]$ and every row sums to 1. If $\boldsymbol{\pi}$ is a row vector holding the current distribution over states, then one step of the chain is one matrix

multiplication, $\boldsymbol{\pi} \mapsto \boldsymbol{\pi}P$, and k steps is $\boldsymbol{\pi}P^k$. The entire dynamics of the chain — where probability flows, where it pools — is encoded in powers of one matrix.

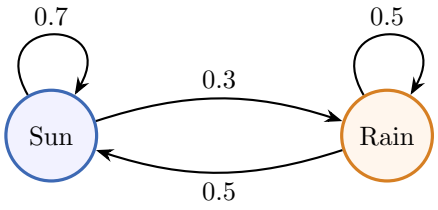


Figure 40.5: A two-state weather chain and its transition matrix $P = \left(\begin{smallmatrix} 0.7 & 0.3 \\ 0.5 & 0.5 \end{smallmatrix}\right)$. Each row sums to 1. Multiplying a distribution vector by P advances the weather one day; the steady state is the vector $\boldsymbol{\pi}$ fixed by $\boldsymbol{\pi}P = \boldsymbol{\pi}$.

That steady-state condition, $\boldsymbol{\pi}P = \boldsymbol{\pi}$, says $\boldsymbol{\pi}$ is unchanged by the matrix — it is a special vector for P, of the kind the next section names and studies.

40.2.4 Eigenvectors and eigenvalues

Most vectors get knocked off their direction when a matrix acts on them. But for any square matrix there are usually a few special directions that survive: vectors $\mathbf{v}$ that the matrix merely stretches, without rotating. Such a vector satisfies

$$A\mathbf{v} = \lambda\mathbf{v}$$

for some number λ. We call $\mathbf{v}$ an *eigenvector* and λ its *eigenvalue* (German *eigen*, "own" or "characteristic"). The eigenvector names a direction the matrix leaves alone; the eigenvalue says by how much it scales along that direction.

Intuition

Picture the matrix as a flow that pushes every arrow around. Eigenvectors are the directions where the push is purely outward or inward — no swirl, just a clean stretch by the factor λ. If $|\lambda| > 1$ that direction grows under repeated application; if $|\lambda| < 1$ it shrinks; if $\lambda = 1$ it is perfectly preserved. Repeatedly applying the matrix amplifies the largest-eigenvalue direction and washes the rest out, which is why eigenvectors govern long-run behavior.

This is the secret behind Markov-chain steady states. A stochastic matrix always has eigenvalue $\lambda = 1$, and its eigenvector is exactly the stationary distribution $\boldsymbol{\pi}$ with $\boldsymbol{\pi}P = \boldsymbol{\pi}$ — a $\lambda = 1$ direction that the chain preserves forever. All other eigenvalues have $|\lambda| < 1$, so their contributions decay; that decay is precisely why a well-behaved chain forgets its starting point and converges to $\boldsymbol{\pi}$. The steady-state chapter proves this; the eigenvalue picture is why it is true.

Why this matters

Eigenvectors are how a matrix tells you its own preferred coordinate system. In the right basis — the eigenvector basis — a complicated transformation becomes nothing but independent stretches along each axis, a diagonal matrix. Half of applied linear algebra is the art of finding that basis, because in it the hard problem falls apart into easy one-dimensional pieces. We are about to see this with covariance matrices, where the eigenvectors are the axes of the data

cloud's ellipse.

40.2.5 Symmetric and positive-definite matrices

The matrices that appear in probability are not arbitrary; they are almost always *symmetric*, meaning $A = A^\top$ (the entry in row i column j equals the entry in row j column i). A covariance matrix is the prime example: its (i, j) entry is $\mathrm{Cov}(X_i, X_j) = \mathrm{Cov}(X_j, X_i)$, manifestly symmetric. Symmetry is a gift, because it forces excellent behavior.

Theorem 40.2 (Spectral theorem, the version we use)**.** *Every real symmetric matrix A has only real eigenvalues, and its eigenvectors can be chosen mutually orthogonal and of unit length. Collecting them as the columns of an orthogonal matrix Q (so $Q^\top Q = I$) and the eigenvalues on the diagonal of Λ, we can write*

$$A = Q\,\Lambda\,Q^\top.$$

This factorization says any symmetric matrix is "rotate into the eigenvector frame ($Q^\top$), stretch each axis by its eigenvalue (Λ), rotate back (Q)." It is the rigorous form of "a matrix turns a circle into an ellipse," and it is exactly the decomposition $\Sigma = Q\Lambda Q^\top$ that the multivariate-normal chapter uses to draw the elliptical contours of a Gaussian and to sample from it.

A symmetric matrix is *positive-definite* if all its eigenvalues are strictly positive, or equivalently if $\mathbf{x}^\top A\,\mathbf{x} > 0$ for every nonzero vector $\mathbf{x}$ (it is *positive-semidefinite* if we allow ≥ 0). The quantity $\mathbf{x}^\top A\,\mathbf{x}$ is a *quadratic form* — the matrix generalization of ax^2 — and "positive-definite" is the matrix way of saying "opens upward like a bowl."

Why it works

Why must a covariance matrix be positive-semidefinite? Take any vector of weights $\mathbf{a}$ and form the scalar random variable $Y = \mathbf{a}^\top \mathbf{X} = \sum_i a_i X_i$. Its variance cannot be negative — variance is an average of squared deviations. But a direct expansion gives

$$0 \leq \mathrm{Var}(Y) = \mathrm{Var}\Big(\sum_i a_i X_i\Big) = \sum_{i,j} a_i a_j\,\mathrm{Cov}(X_i, X_j) = \mathbf{a}^\top \Sigma\,\mathbf{a}.$$

So $\mathbf{a}^\top \Sigma\,\mathbf{a} \geq 0$ for *every* $\mathbf{a}$ — which is the definition of positive-semidefinite. The abstract algebraic condition is nothing but "no combination of your variables can have negative variance." That is also why you cannot hand a statistics package an arbitrary symmetric matrix and call it a covariance matrix: if it has a negative eigenvalue, some linear combination would have negative variance, which is impossible.

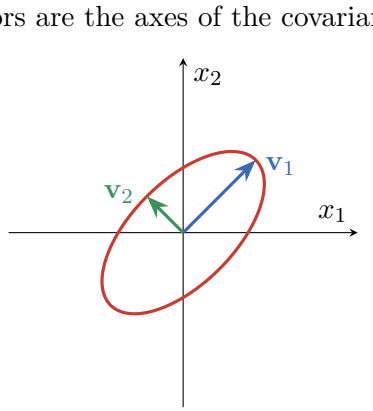


Figure 40.6: A correlated Gaussian's contour. The eigenvectors $\mathbf{v}_1, \mathbf{v}_2$ of the covariance matrix point along the ellipse's principal axes; the square roots of the eigenvalues give the axis half-lengths. This is the spectral theorem made visible, and it is exactly what principal component analysis extracts from data.

Application

Principal Component Analysis (PCA). Given a cloud of data, compute its covariance matrix Σ and find the eigenvectors. The eigenvector with the largest eigenvalue points along the direction of greatest variance — the long axis of the data ellipse — and the eigenvalue *is* the variance captured along it. Keeping the top few eigenvectors compresses high-dimensional data to a few coordinates while preserving as much variance as possible. PCA is, almost in its entirety, "apply the spectral theorem to a covariance matrix," which is why the machine-learning chapter can treat it in a couple of pages: all the heavy lifting was done the moment we knew $\Sigma = Q\Lambda Q^\top$.

History

The word *eigenvalue* is a half-translation of the German *Eigenwert* ("own value"), which entered mathematics through David Hilbert's work on integral equations around 1904. The deeper story is older: the "principal axes" of a quadratic form were studied by Cauchy in the 1820s and by Lagrange before him, in the guise of finding the axes of an ellipse or the principal stresses in a solid. Karl Pearson, one of the founders of mathematical statistics, reinvented the same axes in 1901 under the name "principal axes" (later named "principal components" by Hotelling) — the very PCA above — without realizing he was rediscovering the spectral theorem. Probability, mechanics, and pure algebra all converged on the same eigenvectors.

40.2.6 Orthogonal projection

The last tool is projection: given a vector $\mathbf{y}$ and a subspace (say, the line or plane spanned by some other vectors), find the point in the subspace closest to $\mathbf{y}$. Geometrically you drop a perpendicular from $\mathbf{y}$ onto the subspace; the foot of that perpendicular is the projection $\hat{\mathbf{y}}$, and the leftover $\mathbf{y} - \hat{\mathbf{y}}$ is orthogonal to everything in the subspace.

For projection onto the line through a unit vector $\mathbf{u}$, the formula is just a dot product times a

direction,

$$\hat{\mathbf{y}} = (\mathbf{y} \cdot \mathbf{u})\,\mathbf{u}.$$

The number $\mathbf{y} \cdot \mathbf{u}$ is "how much of $\mathbf{y}$ lies along $\mathbf{u}$," and we lay that much off in the $\mathbf{u}$ direction. The key property is that $\hat{\mathbf{y}}$ is the *closest* point in the subspace — projection minimizes squared distance.

What is really happening

Least squares is a projection, and so is conditional expectation. When you fit a line by least squares (the estimation chapter), you are projecting the vector of observed responses onto the subspace of fitted values, choosing the prediction that minimizes the sum of squared errors — exactly the closest point. More strikingly, the conditional expectation $\mathbb{E}[Y \mid X]$ is the orthogonal projection of Y onto the space of functions of X: among all predictors built from X, it is the one with the smallest mean squared error. "Best prediction" and "perpendicular drop" are the same statement in different languages. The residual $Y - \mathbb{E}[Y \mid X]$ being uncorrelated with every function of X is just "the leftover is orthogonal to the subspace," which the conditional- expectation chapter calls the orthogonality principle.

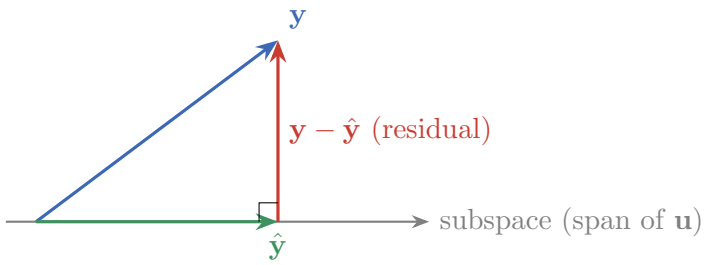


Figure 40.7: Orthogonal projection of $\mathbf{y}$ onto a subspace. The projection $\hat{\mathbf{y}}$ is the closest point; the residual $\mathbf{y} - \hat{\mathbf{y}}$ meets the subspace at a right angle. Least-squares regression and conditional expectation are both this picture.

Connection

The Pythagorean theorem returns one last time, now as a probabilistic identity. Because the residual is orthogonal to the prediction, $\|\mathbf{y}\|^2 = \|\hat{\mathbf{y}}\|^2 + \|\mathbf{y} - \hat{\mathbf{y}}\|^2$. Translated into variances, this is the law of total variance, $\mathrm{Var}(Y) = \mathrm{Var}\left(\mathbb{E}[Y \mid X]\right) + \mathbb{E}\left[\mathrm{Var}(Y \mid X)\right]$ — "explained plus unexplained." Geometry and probability are telling the same story: total spread splits into the part captured by the prediction and the part left in the residual, and they are perpendicular, so their sizes add in squares. The conditional-expectation chapter develops this; *Linear Algebra From the Ground Up* develops projections and orthogonality in full generality.

40.3 How the Pieces Fit the Book

The two halves of this appendix line up neatly with the two faces of probability.

Focus

Calculus, where it lands.

- *Derivative* = density from CDF, $f = F'$; the Jacobian $|g'|$ in change of variables.
- *Integral* = continuous expectation $\int x f(x)\,\mathrm{d}x$ and probabilities as areas $\int_a^b f$.
- *Fundamental Theorem* = the two-way street between density and CDF.
- *Multiple integrals* = joint densities, marginalization, the $\sqrt{2\pi}$ of the normal.
- *Gradient* = maximum likelihood (set $\nabla = 0$) and gradient descent in machine learning.

Linear algebra, where it lands.

- *Dot product* = correlation as a cosine; uncorrelated means orthogonal.
- *Matrix* = linear maps, $\mathbb{E}[A\mathbf{X}] = A\,\mathbb{E}[\mathbf{X}]$, Markov transition steps.
- *Eigenvectors* = Markov steady states ($\lambda = 1$) and long-run behavior.
- *Spectral theorem / positive-definite* = covariance matrices, Gaussian ellipses, PCA.
- *Projection* = least squares and conditional expectation as best prediction.

None of these tools is exotic, and you need only the working picture of each: a derivative is a slope, an integral is an area, a gradient is an uphill arrow, a dot product measures alignment, a matrix reshapes space, an eigenvector is a direction left alone, and a projection is the closest point. Carry those seven images and the calculus and linear algebra in this book will read as common sense dressed in notation. When you want the rigorous development behind any of them, the companion *Calculus From the Ground Up* and *Linear Algebra From the Ground Up* are written in the same spirit as this one.

40.4 Exercises

Exercise 40.1 (Density is the derivative of the CDF)**.** The CDF of an Exponential(λ) random variable is $F(x) = 1 - e^{-\lambda x}$ for $x \geq 0$. Differentiate F to recover the density f, and confirm it equals $\lambda e^{-\lambda x}$. Then explain in one sentence why "density is the derivative of the CDF" is just the Fundamental Theorem of Calculus.

Exercise 40.2 (A density that integrates to one)**.** A random variable has density $f(x) = c\,x^2$ on $[0, 2]$ and 0 elsewhere. Find the constant c by requiring $\int_0^2 f(x)\,\mathrm{d}x = 1$. Then compute $\mathbb{E}[X] = \int_0^2 x f(x)\,\mathrm{d}x$.

Exercise 40.3 (Integration by parts for a moment)**.** Use integration by parts to evaluate $\int_0^\infty x^2\,\lambda e^{-\lambda x}\,\mathrm{d}x$, the second moment of an Exponential(λ). (Hint: take $u = x^2$, integrate the rest; you will need to integrate by parts a second time, or reuse the mean from Example 40.2.) Confirm your answer is $2/\lambda^2$, and deduce that $\mathrm{Var}(X) = 1/\lambda^2$.

Exercise 40.4 (Convergence of a tail integral)**.** For which exponents $p > 0$ does $\int_1^\infty x^{-p}\,\mathrm{d}x$ converge to a finite value, and what is that value when it does? Relate your finding to the paradox box on infinite length but finite area, and explain why a density decaying like $1/x^2$ can be a valid density yet have an infinite mean.

Exercise 40.5 (Marginalizing a joint density)**.** Let $f(x, y) = 4xy$ on the unit square $0 \leq x \leq 1$,

$0 \le y \le 1$, and 0 elsewhere. Verify it integrates to 1 over the square. Then find the marginal density of X by computing $f_X(x) = \int_0^1 f(x,y)\,dy$, and check that the marginal itself integrates to 1.

Exercise 40.6 (A gradient set to zero)**.** Consider $g(a,b) = \sum_{i=1}^{n}(y_i - a - bx_i)^2$, the sum of squared errors of a line $a+bx$ fit to data (x_i, y_i). Compute the partial derivatives $\partial g/\partial a$ and $\partial g/\partial b$, set both to zero, and show that the resulting equations are the normal equations of least squares. Explain how this is the gradient-equals-zero idea from Section 40.1.7.

Exercise 40.7 (Correlation as a cosine)**.** Take the two centered vectors $\mathbf{u} = (1,-1,0)$ and $\mathbf{v} = (2,0,-2)$. Compute $\mathbf{u}\cdot\mathbf{v}$, the two lengths, and hence $\cos\theta$. Interpret the result as a sample correlation. What would $\mathbf{u}\cdot\mathbf{v} = 0$ have meant about the two variables?

Exercise 40.8 (Eigenvector of a transition matrix)**.** For the weather chain $P = \left(\begin{smallmatrix} 0.7 & 0.3 \\ 0.5 & 0.5 \end{smallmatrix}\right)$ of Figure 40.5, find the row vector $\boldsymbol{\pi} = (\pi_1, \pi_2)$ with $\boldsymbol{\pi}P = \boldsymbol{\pi}$ and $\pi_1 + \pi_2 = 1$. Confirm $\boldsymbol{\pi}$ is a left eigenvector of P with eigenvalue 1, and interpret it as the long-run fraction of sunny days.

Exercise 40.9 (Positive-definiteness, by hand)**.** Let $A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$. Compute the quadratic form $\mathbf{x}^\top A\,\mathbf{x}$ for $\mathbf{x} = (x_1, x_2)$, write it as a sum that is visibly nonnegative, and conclude A is positive-definite. Then find the eigenvalues of A and verify they are both positive, consistent with the criterion in Section 40.2.5.

Exercise 40.10 (Projection minimizes distance (proof))**.** Let $\mathbf{u}$ be a unit vector and $\hat{\mathbf{y}} = (\mathbf{y}\cdot\mathbf{u})\mathbf{u}$ the projection of $\mathbf{y}$ onto the line through $\mathbf{u}$. For any scalar t, expand $\|\mathbf{y} - t\mathbf{u}\|^2 = \|\mathbf{y}\|^2 - 2t(\mathbf{y}\cdot\mathbf{u}) + t^2$ and show it is minimized at $t = \mathbf{y}\cdot\mathbf{u}$. Conclude that $\hat{\mathbf{y}}$ is the closest point on the line, and that the residual $\mathbf{y} - \hat{\mathbf{y}}$ is orthogonal to $\mathbf{u}$. Explain how this is the geometric heart of least-squares regression.